ACC FOR MINIMAL LOG DISCREPANCIES IN DIMENSION THREE, I

JIHAO LIU

Abstract. We prove Shokurov's ACC conjecture for minimal log discrepancies of threefold pairs with standard coefficients, in particular for threefolds without boundary, and the part of the conjecture on minimal log discrepancies at least one for threefold pairs with coefficients in an arbitrary DCC set. The conjecture predicts that the minimal log discrepancies at closed points of log canonical pairs of a fixed dimension with coefficients in a DCC set satisfy the ACC, and this paper is the first of a series of three papers which together prove it in dimension three. We also prove the ACC for the minimal log discrepancies of quotients of klt pairs on smooth threefold germs by finite abelian groups, where the coefficients of the pairs belong to a DCC set.

Contents

1. Introduction 1
2. Preliminaries 9
3. Finite quotients, completions and transverse surfaces 20
4. Lattice Newton minima 31
5. Equivariant terminal models and covers 39
6. Reduction to quotients of terminal points 58
7. Polynomial models 76
8. Adjunction for quotients and toric models 86
9. Binary quadratic hypersurfaces 106
10. Double covers of threefold germs 124
11. Quadratic rank one: type cD 135
12. Quadratic rank one: type cE 161
13. Abelian quotients of Gorenstein threefolds 189
14. Mixed circles for smooth germs 199
15. Singular points above one 245
16. Limits of diagonal groups 269
17. Finite real and rational coefficients 332
18. Gauss valuations on split hypersurfaces 349
19. Clean (-2) segments on surface quotients 400
20. Proofs of the main theorems 417
References 420

1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Shokurov asked in [Sho88, Problem 5(c)] whether the minimal log discrepancies of (log) terminal singularities of fixed dimension, without boundary, satisfy the ACC. The following conjecture is its version for pairs with coefficients in a DCC set, in the form of [Sho04, p. 316].

Date: October 27, 2026.

2020 *Mathematics Subject Classification.* 14E30, 14B05, 14J17, 14E15.

Key words and phrases. minimal log discrepancy, ACC, threefold singularities, quotient singularities, compound Du Val singularities.

Conjecture 1.1 (cf. [Sho88, Problem 5]). *Let d be a positive integer and let $\Gamma \subset [0,1]$ be a DCC set. Then the set of minimal log discrepancies $\mathrm{mld}(X \ni x, B)$, where (X,B) is an lc pair of dimension d, $x \in X$ is a closed point, and the coefficients of B belong to Γ, satisfies the ACC.*

Shokurov showed that Conjecture 1.1, together with Ambro's lower semicontinuity conjecture for minimal log discrepancies [Amb99, Conjecture 2.4], implies the termination of flips [Sho04].

Conjecture 1.1 is known in the following cases.

(1) When $d = 2$ and $\Gamma \subset \mathbb{Q}$ [Ale93, Theorem 3.8]; see also [Sho94]. For any DCC set $\Gamma \subset [0,1]$ of real numbers, see [HL24, Theorem 1.2].
(2) When X is toric and B is torus-invariant [Amb06, Theorem 1.1(2)].
(3) When $(X \ni x, B)$ is an exceptional singularity [HLS26, Theorem 1.2].
(4) When $d = 3$ and X is smooth at x [Kaw26, Theorem 1.1]. The values in $(1,3]$ were treated earlier in [Kaw15a, Theorem 1.3].
(5) When $d = 3$ and $x \in X$ is a terminal quotient singularity whose index is at most a fixed positive integer r [Kaw26, Corollary 1.6].
(6) When $d = 3$ and X is a fixed normal threefold [Kaw24, Theorem 1.1]. In this case the minimal log discrepancies at all scheme-theoretic points of X are included.
(7) When $d = 3$, (X,B) is canonical, and the coefficients of B belong to a finite set [Nak16, Corollary 1.5]. In this case the minimal log discrepancies belong to $[1,3]$, and 1 is the only accumulation point of their set.
(8) When $d = 3$, X is terminal near x, and $\mathrm{mld}(X \ni x, B) \geq 1$ [HLL25, Theorem 4.1].
(9) When the normalized volume of $(X \ni x, B)$ is bounded from below by a fixed positive real number [HLQ24, Theorem 1.3].
(10) When $B = 0$, X is a $\mathbb{Q}$-factorial klt singularity of regularity one at x, and $\mathrm{mld}(X \ni x)$ belongs to an interval $(0, \epsilon_d)$, where $\epsilon_d > 0$ depends only on d [Mor23a, Theorem 3].
(11) When $B = 0$, $x \in X$ is a quotient singularity, and $\mathrm{mld}(X \ni x)$ belongs to an interval $(0, \epsilon_d)$, where $\epsilon_d > 0$ depends only on d [Mor23b, Theorem 1].

For $d = 3$, Conjecture 1.1 is open in general.

For a fixed Gorenstein index the minimal log discrepancies are discrete. Let $\mathfrak{a} = \prod_j \mathfrak{a}_j^{e_j}$ denote a formal product of non-zero ideal sheaves $\mathfrak{a}_j$ on X with real exponents $e_j \geq 0$ (an $\mathbb{R}$-ideal), with log discrepancies $a(E, X, \mathfrak{a}) := a(E, X, 0) - \sum_j e_j \operatorname{mult}_E \mathfrak{a}_j$. If d, a positive integer r, and a finite set $I \subset [0, +\infty)$ are fixed, then the set of minimal log discrepancies $\mathrm{mld}(X \ni x, \mathfrak{a})$, where $(X, \mathfrak{a})$ is an lc pair of dimension d with rK_X Cartier, $x \in X$ is a closed point, and the exponents e_j belong to I, is discrete in $\mathbb{R}$ [Nak16, Corollary 1.3]. By [Nak16, Remark 2.2], boundaries whose components are Cartier are special cases of such $\mathbb{R}$-ideals.

On a fixed klt germ, the ACC for minimal log discrepancies is equivalent to the ACC for a-lc thresholds, to Mustaţă's uniform ideal-adic semicontinuity, and to Nakamura's boundedness of a divisor computing the minimal log discrepancy [Kaw21, Theorem 4.6], and each of the statements involved is formulated in [Kaw21, Conjecture 4.2]. The boundedness conjecture is [MN18, Conjecture 1.1], and it implies the ACC for the minimal log discrepancies $\mathrm{mld}(X \ni x, \mathfrak{a})$, where $\mathfrak{a}$ runs over the $\mathbb{R}$-ideals on the fixed germ $X \ni x$ with exponents in a DCC set such that $(X, \mathfrak{a})$ is lc near x [MN18, Theorem 1.5], for instance the $\mathbb{R}$-ideals given by boundaries whose components are Cartier near x. A prime divisor E over X computing the minimal log discrepancy with $a(E, X, 0)$ bounded from above exists for lc surface germs with coefficients in a DCC set [HL23, Theorem 1.2], for a fixed log terminal threefold with coefficients in a DCC set [Kaw24, Theorem 1.4], and for threefold germs with X terminal and $\mathrm{mld}(X \ni x, B) \geq 1$ [HLL25, Theorem 1.2].

Several results in dimension three concern the global minimal log discrepancy

$$\mathrm{mld}(X, B) = \inf\{a(E, X, B) \mid E \text{ is a prime divisor exceptional over } X\}.$$

For any DCC set $\Gamma \subset [0,1]$ there exists $\delta > 0$ such that the set of $\mathrm{mld}(X, B)$, where $\dim X = 3$ and the coefficients of B belong to Γ, satisfies the ACC on $[1-\delta, +\infty)$ [HLL25, Theorem 1.1]. For

threefolds without boundary, there exists $\delta > 0$ such that $\mathrm{mld}(X) < 1$ implies $\mathrm{mld}(X) \leq 1 - \delta$ [Jia21, Theorem 1.3]. The optimal value is $\mathrm{mld}(X) \leq \frac{12}{13}$, attained by $\mathbb{A}^3/\mu_{13}(3,4,5)$ [LX21, Theorem 1.4 and Example 1.5]. The second largest accumulation point of the set of global minimal log discrepancies of threefolds without boundary is $\frac{5}{6}$ [LL22, Theorem 1.3], and the ACC holds for global minimal log discrepancies of threefolds without boundary on $[\frac{5}{6}, +\infty)$ [LL22, Theorem 1.5]. The ACC for global minimal log discrepancies also holds for exceptionally non-canonical threefold pairs [HL25, Theorem 1.4], for klt threefold pairs with a bounded number of distinct exceptional log discrepancies in $[0,1]$ [HL25, Theorem 1.6], and for lc threefold pairs polarized by a very ample divisor H with H^3 bounded from above [CL26, Theorem 1.3], in each case with coefficients in a DCC set.

We now explain where the hypotheses of the closed-point results in dimension three enter. The closed-point statement [HLL25, Theorem 4.1] assumes that X is terminal near x. The terminal hypothesis is used in [HLL25, Proof of Theorem 4.1, Step 1], where the index of X at x is bounded by means of [Kaw93], and then [Nak16, Corollary 1.3] is applied to the boundaries obtained by replacing the coefficients with their limits, which belong to a finite set. Indeed, by [Kaw93, Theorem] a terminal threefold point $x \in X$ of index $n \geq 2$ has a prime divisor over X with center x and log discrepancy $1 + \frac{1}{n}$, so

$$n \leq \frac{1}{\mathrm{mld}(X \ni x, B) - 1} \quad \text{whenever } \mathrm{mld}(X \ni x, B) > 1.$$

The bounded computing divisor [HLL25, Theorem 1.2] also assumes that X is terminal and $\mathrm{mld}(X \ni x, B) \geq 1$.

For a canonical threefold X, the index of K_X at x is still bounded from above by a number depending only on a positive lower bound for $\mathrm{mld}(X \ni x, B) - 1$ when $\mathrm{mld}(X \ni x, B) > 1$: in this case $\mathrm{mld}(X \ni x)$ is 3 or $1 + \frac{1}{k}$ for a positive integer $k \leq 1/(\mathrm{mld}(X \ni x, B) - 1)$, and the index of K_X at x is 1 or at most $k!$, respectively [Kaw15b, Theorem 5.3]. Without the canonical hypothesis, the index of K_X at x has no bound depending only on a positive lower bound for $\mathrm{mld}(X \ni x, B) - 1$, as the germs $X'_r \ni x$ defined below show. For an integer $r \geq 3$, let

$$X_r := (\mathbb{A}^2/\mu_r(1,1)) \times \mathbb{A}^1,$$

and let $x \in X_r$ be a closed point over the origin of $\mathbb{A}^2/\mu_r(1,1)$. The product of the exceptional curve of the minimal resolution of $\mathbb{A}^2/\mu_r(1,1)$ with $\mathbb{A}^1$ is a prime divisor over X_r with log discrepancy $\frac{2}{r} < 1$, so X_r is not canonical along the image of $\{0\} \times \mathbb{A}^1$, and $\mathrm{mld}(X_r, 0) \leq \frac{2}{r}$. On the other hand,

$$\mathrm{mld}(X_r \ni x) = 1 + \frac{2}{r},$$

and the Cartier index of K_{X_r} at x is $r/\gcd(r,2)$ (the log discrepancy $\frac{2}{r}$ and the value $1 + \frac{2}{r}$ by Lemma 3.4 applied to $\mu_r(1,1)$ on $\mathbb{A}^2$ and to $\mu_r(1,1,0)$ on $\mathbb{A}^3$, and, since $\mu_r(1,1,0)$ contains no pseudo-reflection, the index is the order of ζ^2 for a primitive r-th root of unity ζ). Thus the germs $X_r \ni x$ satisfy the hypotheses of neither [HLL25, Theorem 1.1] nor [HLL25, Theorem 4.1], and for odd r the index r of K_{X_r} at x is larger than $1/(\mathrm{mld}(X_r \ni x) - 1) = \frac{r}{2}$. For lc points that are not klt, the index is unbounded even when the minimal log discrepancy is fixed: for any positive integer m there exists a normal $\mathbb{Q}$-Gorenstein threefold X_m with a closed point x_m such that X_m is not klt at x_m, $\mathrm{mld}(X_m \ni x_m) = 1$, and the Cartier index of K_{X_m} at x_m is m [NS26a, Theorem 1.2 and the paragraph before the acknowledgements in Section 1]. Shokurov's index conjecture [NS22b, Conjecture 1.1] asserts that, for any positive integer n and any real number $a \geq 0$, the Cartier index of K_X at a closed point x of an n-dimensional $\mathbb{Q}$-Gorenstein variety X with $\mathrm{mld}(X \ni x) = a$ is bounded from above by a number depending only on n and a. It holds for quotient singularities [NS22b, Theorem 1.2] and for canonical threefold singularities [Kaw15b, Theorem 5.3], and the threefolds X_m show that it fails at points that are not klt. The threefolds X_m are not counterexamples to Conjecture 1.1.

Replacing X by a crepant terminal model does not preserve the DCC condition on the coefficients. Let $r \geq 3$ be an odd integer, let

$$X'_r := (\mathbb{A}^2/\mu_r(1, r-2)) \times \mathbb{A}^1,$$

and let $x \in X'_r$ be a closed point over the origin of $\mathbb{A}^2/\mu_r(1, r-2)$. Since $\mu_r(1, r-2, 0)$ contains no pseudo-reflection and its determinant character, which sends ζ to ζ^{r-1}, has order r, the Cartier index of $K_{X'_r}$ at x is r, while

$$\operatorname{mld}(X'_r \ni x) = 1 + \frac{r+1}{2r} > \frac{3}{2}$$

by Lemma 3.4. Thus the index of K_X at x is unbounded for klt germs $X \ni x$ with $\operatorname{mld}(X \ni x) > \frac{3}{2}$. Let E_r be the exceptional curve of the minimal resolution of $\mathbb{A}^2/\mu_r(1, r-2)$ at the end of the Hirzebruch–Jung chain corresponding to the vector $(\frac{1}{r}, \frac{r-2}{r})$, and let E'_r be the product of E_r with $\mathbb{A}^1$. Then E'_r is the toric prime divisor over X'_r corresponding to the primitive vector $(\frac{1}{r}, \frac{r-2}{r}, 0)$, the center of E'_r on X'_r is the singular curve of X'_r, and by the toric formula for log discrepancies [Amb06, Section 2, p. 360]

$$a(E'_r, X'_r, 0) = \frac{1}{r} + \frac{r-2}{r} = 1 - \frac{1}{r}.$$

Let $f : Y \to X'_r$ be a projective birational morphism with Y terminal, and let B_Y be the $\mathbb{R}$-divisor with $K_Y + B_Y = f^* K_{X'_r}$ and $f_* B_Y = 0$. Since a terminal threefold is smooth in codimension two [KM98, Corollary 5.18], the fiber Y_t of the composite $Y \to X'_r \to \mathbb{A}^1$ over a general point $t \in \mathbb{A}^1$ is a resolution of $\mathbb{A}^2/\mu_r(1, r-2)$, and over a neighborhood of the origin $Y_t \to \mathbb{A}^2/\mu_r(1, r-2)$ factors through the minimal resolution: it is an isomorphism outside the inverse image of finitely many points (cf. [Har77, Lemma V.5.1], since $\mathbb{A}^2/\mu_r(1, r-2)$ is normal), so over a neighborhood U of the origin it is an isomorphism over $U \setminus \{0\}$, that is, a resolution over U in the sense of [CLS11, Definition 10.1.7], and such a resolution of the toric surface $\mathbb{A}^2/\mu_r(1, r-2)$ factors through the minimal one [CLS11, Definition 10.4.8 and Corollary 10.4.9] (the factorization is local over the origin). Thus E'_r is a divisor on Y, and the coefficient of E'_r in B_Y is $1 - a(E'_r, X'_r, 0) = \frac{1}{r}$. Hence the coefficients $\frac{1}{r}$, for odd $r \geq 3$, do not belong to a DCC set, and [HLL25, Theorem 4.1] cannot be applied directly to the crepant terminal models of the germs $X'_r \ni x$.

The other closed-point results also restrict the index, the variety, or the volume. [Nak16, Corollary 1.3] fixes the Gorenstein index and a finite coefficient set, [Kaw24, Theorem 1.1] fixes the threefold, and [Kaw26, Corollary 1.6] assumes that the index of the terminal quotient singularities is bounded, so they cover only subfamilies with bounded index, a fixed variety, or bounded group order. In contrast, Theorem 1.5 below concerns quotients of smooth threefold germs by finite abelian groups G, with boundaries cut out by semi-invariant functions, and has no bound on the order $|G|$. [HLQ24, Theorem 1.3] assumes that the normalized volume is bounded from below by a positive real number, whereas Example 13.3 contains two families of such quotient germs whose minimal log discrepancies decrease to $\frac{2}{3}$ while their normalized volumes tend to 0. For klt germs of dimension $n \geq 2$, the normalized volume is at most n^{n-1} times the minimal log discrepancy [Han26, Theorem 1.1], and Example 13.3 shows that there is no bound in the opposite direction along such quotient germs.

When G acts diagonally on some local coordinates at the fixed point and each semi-invariant function defining the boundary is a monomial in the coordinates, the quotient in Theorem 1.5 below and the quotient of a toric pair by G have isomorphic completions at the images of the fixed point, together with the group actions and the boundaries. By Lemma 3.3(2), the quotient in Theorem 1.5 and the toric quotient have equal minimal log discrepancies at the images of the fixed point, so this case is contained in [Amb06, Theorem 1.1(2)], since the codimension one log discrepancies $\frac{1-b}{r}$ of the toric quotient, with $b < 1$ a sum of elements of Γ with non-negative integer coefficients and r a positive integer, form an ACC set (cf. [Amb06, Proof of Theorem 4.4]).

For general semi-invariant boundaries the quotient is not toric. Moreover, the prime divisors given by monomial valuations in a fixed system of such coordinates need not compute the minimal log discrepancy of the quotient: in Example 8.10 the minimal log discrepancy is $\frac{3}{7}$, while the least log discrepancy of a prime divisor given by a monomial valuation in the given coordinates is $\frac{5}{7}$.

[NS26b, Theorem 7.1] is a formula for minimal log discrepancies of quotients of semi-invariant complete intersections, a precise inversion of adjunction, and not an ACC statement. See also [NS22a, NS25]. It assumes that G contains no pseudo-reflection, that the quotient W of the complete intersection by G is klt, and that the intersection of W with the branch locus of the quotient morphism has codimension at least two in W. [NS26b, Remark 3.3] shows that, for semi-invariant equations, the codimension condition does not follow from the other hypotheses. By [NS24, Theorem 1.5], the klt condition cannot be removed: in [NS24, Example 1.4] with $n \geq 2$, the group contains no pseudo-reflection, and the complete intersection is a hypersurface, defined by an invariant equation, on which the group acts freely outside the origin, so the other two hypotheses hold.

This paper is the first of a series of three papers, which together prove Conjecture 1.1 for $d = 3$.

For a positive integer d and a set $\Gamma \subset [0,1]$, let $\mathrm{MLD}(d,\Gamma)$ be the set of minimal log discrepancies $\mathrm{mld}(X \ni x, B)$ appearing in Conjecture 1.1 (see Definition 2.3). Our first result concerns the values not smaller than one.

Theorem 1.2. *Let $\Gamma \subset [0,1]$ be a DCC set. Then $\mathrm{MLD}(3,\Gamma) \cap [1,+\infty)$ satisfies the ACC.*

For the standard coefficients we obtain all values.

Theorem 1.3. *Let $\Gamma_{\mathrm{std}} := \{1\} \cup \{1 - \frac{1}{m} \mid m \in \mathbb{Z}_{\geq 1}\}$. Then $\mathrm{MLD}(3,\Gamma_{\mathrm{std}})$ satisfies the ACC.*

We state the boundary-free $\mathbb{Q}$-factorial klt case separately, in the form used in the next paper of the series.

Corollary 1.4. *The set*

$$\{\mathrm{mld}(X \ni x) \mid X \text{ is a } \mathbb{Q}\text{-factorial klt threefold and } x \in X \text{ is a closed point}\} \cap (0,1)$$

satisfies the ACC.

Corollary 1.4 follows from the case $B = 0$ of Theorem 1.3. More generally, since $0 \in \Gamma_{\mathrm{std}}$, Theorem 1.3 implies that the minimal log discrepancies $\mathrm{mld}(X \ni x)$, where X is an lc threefold such that K_X is $\mathbb{Q}$-Cartier and $x \in X$ is a closed point, satisfy the ACC.

Let X be a normal quasi-projective variety, G a finite group acting faithfully on X, $x \in X$ a closed point fixed by G, and B a G-invariant $\mathbb{R}$-divisor on X such that $K_X + B$ is $\mathbb{R}$-Cartier. Let $\pi : X \to X/G$ be the quotient morphism. We denote by $(X \ni x, B)/G$ the germ at $\pi(x)$ of the crepant quotient $(X/G, B_{X/G})$ of (X,B) by G, where $B_{X/G}$ is the $\mathbb{R}$-divisor on X/G such that

$$K_X + B = \pi^*(K_{X/G} + B_{X/G})$$

(see Definition 2.5). The notation $(X \ni x, B)/G$ is also used in the sequels.

Theorem 1.5. *Let $\Gamma \subset [0,1]$ be a DCC set. Then the set of positive numbers $\mathrm{mld}((X \ni x, B)/G)$, where $(X \ni x)$ is a smooth threefold germ, G is a finite abelian group acting faithfully on X and fixing x, and $B = \sum_{j=1}^{s} c_j \,\mathrm{div}(h_j)$ with $c_j \in \Gamma$ and non-zero regular G-semi-invariant functions h_j near x such that (X,B) is klt near x, satisfies the ACC. In particular, the conclusion holds for G-invariant $\mathbb{R}$-divisors $B \geq 0$ with $B \in \Gamma$ and (X,B) klt near x.*

Theorem 1.5 concerns the closed-point minimal log discrepancies of quotients of smooth threefold germs by finite abelian groups. The coefficients of $B_{X/G}$ include the coefficients coming from the ramification of π, so they need not belong to Γ. The last statement of Theorem 1.5 follows from the first one, since near x the reduced union of each G-orbit of components of B is $\mathrm{div}(h)$ for a non-zero regular G-semi-invariant function h (see the proof in Section 20).

In the next paper of the series we prove that $\mathrm{MLD}(3,\Gamma)$ satisfies the ACC for any finite set $\Gamma \subset [0,1]$. In the last paper we prove that for any positive rational number a and any finite set $\Gamma \subset [0,1] \cap \mathbb{Q}$, there exists a real number $\epsilon(a,\Gamma) > 0$ such that if X is a $\mathbb{Q}$-factorial klt threefold, $x \in X$ is a closed point with $\mathrm{mld}(X \ni x) \geq a$, and $D \geq 0$ is a $\mathbb{Q}$-divisor with coefficients in Γ such that (X, D) is lc near x, then

$$\sup\{t \in [0,1] \mid \mathrm{mld}(X \ni x, tD) \geq a\} \notin (1 - \epsilon(a,\Gamma), 1),$$

and that the existence of such numbers $\epsilon(a,\Gamma)$ together with the result of the next paper implies Conjecture 1.1 for $d = 3$.

Sketch of the proof. Each theorem is proved by contradiction with a strictly increasing sequence a_i.

(1) Lemma 6.1(2) reduces the lc case to $\mathbb{Q}$-factorial klt pairs, and Construction-Theorem 6.7 writes $a_i = \mathrm{mld}((Y_i \ni y_i, C_i)/I_i)$, where $y_i \in Y_i$ is a smooth or isolated cDV point (Corollary 6.8), C_i has coefficients in a fixed DCC set, and I_i is the inertia group of a lift of a prime divisor computing a_i.

(2) For $a_i \geq 1 + \epsilon$, Theorems 4.8 and 15.5 reduce, up to quotient varieties in a finite list (Proposition 14.6(1)) and groups of bounded order (Proposition 15.6), to groups I_i containing $\mu_{r_i}(p, -q, 0)$ (acting on $\mathbb{A}^3_{x,y,t}$) or $\mu_{r_i}(1, -1, 0, 0)$ (acting on the ambient $\mathbb{A}^4$ of the cDV hypersurface) as a subgroup of index at most a fixed integer, with $r_i \to +\infty$, and then

$$b_{H_i} \leq a_i \leq b_{H_i} + o(1),$$

where H_i is the image of I_i under the characters of $u := x^q y^p$ and t, and $b_{H_i} := \mathrm{mld}((T \ni 0, B_T)/H_i)$ for the plane pair (T, B_T) on $T = \mathbb{A}^2_{u,t}$ of Set-up 14.1 (for $\mu_{r_i}(1, -1, 0, 0) = I_i \cap \mathbf{T}_{(1,-1,0,0)}$ (Definition 2.15), $H_i := I_i/\mu_{r_i}(1, -1, 0, 0)$ acting on the plane pair on $\mathbb{A}^2_{z,t}$ of Theorem 15.9); the b_{H_i} satisfy the ACC (Theorems 14.2, 15.9, and 2.18).

(3) For Theorem 1.3, Proposition 20.1 covers $[1, +\infty)$, Construction-Theorem 5.3 removes the boundary, and each value in $(0, 1)$ is a lattice Newton minimum or a weighted one (Definition 2.16, Corollary 4.6), whose ACC follows from Theorem 4.5, or arises from groups of bounded order (Theorem 13.1).

(4) For Theorem 1.5, Hausdorff limits of the groups and Theorems 16.4, 16.5, and 16.21 leave groups containing $\mu_{r_i}(p, -q, 0)$ as in (2) (Theorem 16.1), which Theorem 14.3 excludes.

The key new step is the lower bound $a_i \geq b_{H_i}$, explained in (b).

If the groups had bounded order, or the quotient varieties belonged to a finite list, [Kaw24, Theorem 1.1] and [HLL25, Theorem 4.1] would prove the rest of Theorems 1.2 and 1.5. The proofs do not determine the accumulation points of the sets in Theorems 1.3, 1.2, and 1.5.

The steps (1)–(4) use the following four ideas.

(a) Exact reduction to cyclic quotients of terminal points. [HLL25, Proof of Theorem 4.1, Step 1] bounds the index of K_X by [Kaw93], which needs X terminal, while for klt X the index is unbounded even when $\mathrm{mld}(X \ni x) > 1$ and crepant terminal models need not preserve the DCC of the coefficients. Construction-Theorem 6.7 instead passes to the index one cover X' of K_X and the components of B [Zhu24, Lemma-Definition 2.15], and extracts only prime divisors over X' with log discrepancy one with respect to $(X', 0)$ (Construction-Theorem 5.1).

On the extraction Y, K_Y stays Cartier, and since the coefficients of the j-th component of B converge to b_j and the pairs with the limit coefficients are lc, the pullback of the j-th component of B has multiplicity at most $\lfloor 1/b_j \rfloor$ along any prime divisor, so the coefficients stay in a fixed DCC set. The quotient by the inertia group of a lift of a computing divisor keeps the minimal log discrepancy, since Lemma 3.1 multiplies the log discrepancy of any competitor by a positive integer and that of the computing divisor by one.

(b) Groups and plane pairs above one. A large minimal log discrepancy does not bound the group, so [Kaw24, Theorem 1.1] for a fixed threefold does not suffice: for any $n \geq 2$, $\mathbb{A}^3/\mu_n(1, -1, 0)$

has minimal log discrepancy 2 at the origin (Example 4.2(2)). Instead, $a_i \geq 1+\epsilon$ bounds the coordinate sums of the positive vectors of N_{I_i} (Definition 2.15) from below by $1+\epsilon$. When y_i is a smooth point, by compactness (Lemma 4.4) the image of I_i in $\mathbb{R}^3/\mathbb{Z}^3$ is contained in one of finitely many closed subgroups (Theorem 4.8). When y_i is an isolated cDV point, a limit argument in $\mathbb{R}^4/\mathbb{Z}^4$ gives Theorem 15.5.

For groups containing a large $\mu_{r_i}(p,-q,0)$, the quotient is compared with the quotient $(T,B_T)/H_i$ of the plane pair of Set-up 14.1. For the lower bound $a_i \geq b_{H_i}$, which is the key new step, work in the coordinates x,y,t of Set-up 14.1, on which $\mu_{r_i}(p,-q,0)$ acts diagonally. Deleting the terms of large y-degree does not change the order along a lift F of a prime divisor computing a_i of any equation h_j of the boundary (Proposition 14.12), and in the degree-p extension $x=z^p$, $y=z^{-q}y'$ of $\mathbb{C}(x,y,t)$ (proof of Theorem 14.2(3)), the plane $(Z=0)$, $Z:=z^{r_i}$, with the restricted boundary plus a non-negative multiple of $\operatorname{div}(y')$ is a crepant cover of (T,B_T) through $(y',t)\to(y'^p,t)$.

Theorem 8.3, which compares the quotients, holds without the hypotheses on pseudo-reflections and branch loci of [NS26b, Theorem 7.1]: the proof adds, for each coordinate of the ambient affine space, a new coordinate with the character of that coordinate and with coefficient one in the boundary, both hypotheses hold for the new pair, and adding the new coordinates does not change the minimal log discrepancy of the quotient (Lemma 8.1). The formula of [NS26b, Theorem 7.1] is used only for pairs satisfying the codimension condition, which is consistent with [NS26b, Remark 3.3].

For the upper bound, Gauss valuations (Definition 2.11) over a divisor computing b_{H_i} with bounded log discrepancy [HL23, Theorem 1.2] do not see the terms of high degree, which implies $a_i \leq b_{H_i}+o(1)$ (Proposition 14.9).

(c) Exact formulas below one. Below one the group is not confined, since $\mathbb{A}^3/\mu_n(1,1,1)$ has minimal log discrepancy $\frac{3}{n}$ at the origin (Example 4.2(1)), and monomial divisors in given eigenparameters need not compute the minimal log discrepancy (Example 8.10). Instead, the cover of Construction-Theorem 5.3 absorbs the standard coefficients $1-\frac{1}{m}$ into ramification of index m, and each value is computed exactly (Theorem 13.1).

For $X=(xy+g(z,t)=0)$, the integer $a(F,X,0)-\operatorname{ord}_F z-\operatorname{ord}_F t\geq 0$ is divisible by $|I_G(F)|$ for any prime divisor F over the origin (Lemma 9.2), so it vanishes if the quotient log discrepancy of F is below one, making F monomial in x,z,t (Proposition 9.4). Analogous statements treat $x^2+y^2+g(z,t)$ and $x^2+h(y,z,t)$ (Proposition 9.5 and Lemma 10.4), and points of quadratic rank one reduce to them or to groups of bounded order through their residual cubics (Theorems 11.2 and 12.1).

Theorem 4.5 allows the monomial ideals J_l in $\psi_{J,c}(w)=\sum_i w_i-\sum_l c_l \operatorname{wt}_w(J_l)$ (Definition 2.16) to vary along the sequence: by the antichain theorem [Mac01, Theorem 1.1] one may pass to a subsequence along which each J_l is nonincreasing under inclusion and the coefficients are nondecreasing, so that the functions $\psi_{J,c}$ are pointwise nonincreasing, whereas in [Amb06, Proof of Theorem 4.4] only the coefficients vary.

(d) Generic limits of plane pairs below one. The lower bound $a_i\geq b_{H_i}$ uses a uniform $\lambda>1$ such that $(X,\lambda B_i)$ is lc, which bounds the orders of the equations along the divisors of small quotient log discrepancy (Proposition 14.12), and below one such a λ need not exist. For Theorem 1.5, the upper bound of Proposition 14.9 and the ACC for the b_{H_i} imply $a_i<L:=\lim_i b_{H_i}$, and Theorem 14.19 takes a generic limit of the plane pairs, whose quotient minimal log discrepancy is L (Lemma 14.14). If the generic limit is klt, then it yields such a λ, and Proposition 14.15 implies $a_i\geq b_{H_i}\geq L$. Otherwise it has exactly one curve with coefficient one, the curve is smooth, and adding to all the pairs (X,B_i) a fixed multiple of the pullback of a coordinate of T keeps them lc, which implies $a_i\geq L$.

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce crepant quotients of pairs by finite groups and their inertia groups,

Gauss extensions of valuations, and the lattices attached to diagonal group actions, and we state results of [Amb99], [HL24], [HL23], [Kaw24], [NS26b], and [Kaw07] that are used several times.

In Section 3 we study quotients by inertia groups, completions of germs, and surfaces transverse to a curve, and we compute the minimal log discrepancy of the quotient of a smooth germ by a finite diagonal group, with boundary supported on the coordinate hyperplanes, as a minimum over the lattice of the group (Definition 2.15 and Lemma 3.4). In Section 4 we prove the ACC for minima over the lattices of finite diagonal groups of the piecewise linear functions given by Newton polyhedra, with coefficients in a DCC set (Theorem 4.5). We also show that, for a fixed $\epsilon > 0$, if the quotient of a pair on a smooth threefold germ by a finite abelian group has minimal log discrepancy at least $1+\epsilon$, then the image of the group in $\mathbb{R}^3/\mathbb{Z}^3$ is contained in one of finitely many closed subgroups depending only on ϵ (Theorem 4.8).

In Section 5 we construct equivariant terminal models, index one covers of several divisors, covers that remove a boundary with standard coefficients (Construction-Theorem 5.3), and canonical covers of terminal points. In Section 6 we pass from lc pairs to $\mathbb{Q}$-factorial klt pairs, and we realize a strictly increasing sequence of minimal log discrepancies with DCC coefficients by quotients of terminal threefold germs with Cartier canonical divisor by cyclic groups (Construction-Theorem 6.7).

In Section 7 we replace a smooth or isolated Gorenstein terminal threefold germ, with a faithful action of a finite abelian group and a klt boundary defined by semi-invariant functions, by a polynomial germ such that the quotients of the polynomial germ and of the original germ by any subgroup of the group have equal minimal log discrepancies (Construction-Theorem 7.2). In Section 8 we prove an inversion of adjunction for quotients of complete intersections by finite diagonal groups (Theorem 8.3), and we compute the minimal log discrepancies of quotients of smooth germs and of isolated Gorenstein terminal hypersurface germs by finite abelian groups as minima over the lattice of the group, when the initial forms of the equations define lc pairs on the torus (Theorem 8.14).

In Section 9 we prove the ACC in $(0,1)$ for the minimal log discrepancies of quotients by finite abelian groups of klt threefold hypersurface singularities whose quadratic part has rank at least two (Theorem 9.1), by reducing to hypersurfaces $xy+g(z,t)=0$ and $x^2+y^2+g(z,t)=0$. In Section 10 we compute the minimal log discrepancies in $(0,1)$ of quotients of hypersurface germs $x^2+h(y,z,t)=0$ by finite diagonal groups as minima over the lattice of the group, under a log canonicity condition on the initial forms of h (Proposition 10.5).

In Section 11 we prove the ACC in $(0,1)$ for the minimal log discrepancies of quotients by finite abelian groups of isolated cDV points of type cD (Theorem 11.2). In Section 12 we treat the points of type cE, and we prove the ACC in $(0,1)$ for the minimal log discrepancies of quotients by finite abelian groups of isolated cDV points whose quadratic part has rank one (Theorem 12.1). In Section 13 we prove the ACC in $(0,1)$ for the minimal log discrepancies of crepant quotients of Gorenstein klt threefold germs by finite abelian groups (Theorem 13.1).

In Section 14 we prove the ACC in $[1,+\infty)$ for abelian quotients of smooth threefold germs (Theorem 14.2), and, for fixed coprime positive integers p,q, the ACC for quotients of $\mathbb{A}^3$ by finite diagonal groups whose intersection with the image of the line $\mathbb{R}(p,-q,0)$ in $\mathbb{R}^3/\mathbb{Z}^3$ has bounded index (Theorems 14.19 and 14.3). In Section 15 we prove the ACC in $[1,+\infty)$ for abelian quotients of singular isolated cDV points (Proposition 15.1). In Section 16 we show that the groups in a strictly increasing sequence of minimal log discrepancies of quotients of $\mathbb{A}^3$ by finite diagonal groups converge, after passing to a subsequence and permuting the coordinates, to a closed subgroup of $\mathbb{R}^3/\mathbb{Z}^3$ whose identity component is the image of $\mathbb{R}(p,-q,0)$ for coprime positive integers p,q (Theorem 16.1).

Sections 17, 18, and 19 are not used in the proofs of the main theorems. Some of their results are used in the sequels.

Section 17 carries out the reduction of Section 6 for boundaries whose coefficients belong to a finite set (Construction-Theorem 17.6). There the boundaries on the terminal threefolds have coefficients in a fixed finite set, and they are convex combinations of $\mathbb{Q}$-boundaries such that any

prime divisor has log discrepancy less than one, equal to one, or greater than one with respect to the boundary exactly when it does with respect to each of these $\mathbb{Q}$-boundaries.

Section 18 uses Gauss extensions of valuations to prove the ACC for quotients by finite diagonal groups of hypersurfaces $xy + g(z,t) = 0$ with boundaries defined by functions of z, t, for values in $(0,1)$ (Theorem 18.8), of products of a plane pair with a line (Theorem 18.16), and of hypersurfaces $x^q y^p = g(z,t)$ with coprime p, q bounded by a fixed integer (Theorem 18.17).

Section 19 shows that, for a pair on the product of a cyclic quotient surface singularity with a line, if a prime divisor F computes the minimal log discrepancy $c < 1$ at a closed point o and all other prime divisors with center o have log discrepancy at least a number $t \in (c, 1]$, then the center of F on the product of the minimal resolution with the line does not meet the middle of a long chain of (-2)-curves that the boundary does not meet (Theorem 19.2). In Section 20 we prove Theorems 1.2, 1.3, and 1.5, and Corollary 1.4.

Acknowledgements. The work was partially supported by the National Key R&D Program of China #2024YFA1014400. Jihao Liu would like to thank Bin Dong, Guoxiong Gao, Zeming Sun, Bin Wu, and Leheng Chen for AI assistance. Jihao Liu would like to thank Ruochuan Liu and Gang Tian for consistent support and encouragement.

2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

In this section we fix the conventions on pairs, minimal log discrepancies, and crepant quotients by finite groups, collect facts on DCC sets, monomial valuations, semi-invariant functions, and the lattices attached to diagonal group actions, and recall four known results that are used several times (Subsection 2.5).

2.1. Pairs and minimal log discrepancies.

Definition 2.1. A *pair* (X, B) consists of a normal quasi-projective variety X and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \operatorname{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B). For any divisorial valuation $v = c \operatorname{ord}_E$ for some real number $c > 0$, we set $a(v, X, B) := ca(E, X, B)$.

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X. For an $\mathbb{R}$-divisor D on a normal variety X and a point $x \in X$, the *Cartier index* of D at x is the smallest positive integer N such that ND is Cartier near x, if such an integer exists.

Definition 2.2. Let (X, B) be a sub-pair and $\eta \in X$ a scheme-theoretic point. The *minimal log discrepancy* of (X, B) at η is

$$\operatorname{mld}(X \ni \eta, B) := \inf\{a(E, X, B) \mid E \text{ is a prime divisor over } X \text{ with } \operatorname{center}_X E = \overline{\{\eta\}}\},$$

and $\operatorname{mld}(X \ni \eta) := \operatorname{mld}(X \ni \eta, 0)$. For non-zero ideal sheaves $\mathfrak{a}_1, \dots, \mathfrak{a}_s$ on X and real numbers $r_j > 0$, the formal product $\mathfrak{a} = \prod_j \mathfrak{a}_j^{r_j}$ is called an $\mathbb{R}$-*ideal*. We set

$$a(E, X, B, \mathfrak{a}) := a(E, X, B) - \sum_j r_j \operatorname{mult}_E \mathfrak{a}_j,$$

where $\operatorname{mult}_E \mathfrak{a}_j := \min\{\operatorname{ord}_E f \mid f \in (\mathfrak{a}_j)_\xi\}$ and ξ is the generic point of $\operatorname{center}_X E$, and we define $\operatorname{mld}(X \ni \eta, B, \mathfrak{a})$ by replacing $a(E, X, B)$ with $a(E, X, B, \mathfrak{a})$. We say that $(X, B, \mathfrak{a})$ is *(sub-)lc* (resp. *(sub-)klt*) if $a(E, X, B, \mathfrak{a}) \geq 0$ (resp. > 0) for any prime divisor E over X.

Definition 2.3. A subset of $\mathbb{R}$ satisfies the *DCC* (resp. the *ACC*) if it contains no infinite strictly decreasing (resp. increasing) sequence. For $\Gamma \subset \mathbb{R}$ we write $B \in \Gamma$ if the coefficients of B belong to Γ. For a positive integer d and $\Gamma \subset [0, 1]$ we define

$$\operatorname{MLD}(d, \Gamma) := \{\operatorname{mld}(X \ni x, B) \mid (X, B) \text{ is lc},\ \dim X = d,\ x \in X \text{ is a closed point},\ B \in \Gamma\},$$

and for $\Lambda \subset [0, 1]$ we define

$$\operatorname{MLD}_{\mathrm{Qf}}(3, \Lambda) := \{\operatorname{mld}(X \ni x, B) \mid (X, B) \text{ is klt},\ X \text{ is } \mathbb{Q}\text{-factorial},\ \dim X = 3,\ x \in X \text{ is a closed point},\ B \in \Lambda\}.$$

Every sub-pair in the sense of Definition 2.1 is a log pair in the sense of [Amb99, Section 1], where the boundary is an arbitrary $\mathbb{R}$-divisor. The log discrepancies of [Amb99, Section 1] coincide with those of Definition 2.1, and the minimal log discrepancies of [Amb99, Section 1] coincide with those of Definition 2.2 at any point of X other than the generic point of X.

Lemma 2.4 ([Amb99, Proposition 2.1])**.** *Let (X, B) be a sub-pair and $\eta \in X$ a scheme-theoretic point which is not the generic point of X. Let V be the closure of η. Assume that (X, B) is sub-lc near η. Then*

$$\operatorname{mld}(X \ni \eta, B) = \operatorname{mld}(X \ni y, B) - \dim V$$

for a general closed point $y \in V$.

2.2. **Finite quotients.**

Definition 2.5. Let X be a normal quasi-projective variety and G a finite group acting on X. For a prime divisor F over X, the *decomposition group* $\operatorname{Stab}_G(F)$ is the stabilizer of ord_F in G, and the *inertia group* $I_G(F)$ is the subgroup of $\operatorname{Stab}_G(F)$ acting trivially on the residue field of ord_F.

Assume that G acts faithfully on X. Let $\pi : X \to X/G$ be the quotient morphism, $R(\pi)$ the ramification divisor of π, and B a G-invariant $\mathbb{R}$-divisor on X such that $K_X + B$ is $\mathbb{R}$-Cartier. The *crepant quotient* of (X, B) by G is $(X, B)/G := (X/G, B_{X/G})$, where

$$B_{X/G} := \frac{1}{|G|}\pi_*(B + R(\pi)).$$

For a closed point $x \in X$ fixed by G, we write $(X \ni x, B)/G := (X/G \ni \pi(x), B_{X/G})$.

A finite morphism $f : Y \to X$ between normal varieties is *quasi-étale* if f is étale in codimension one.

The boundary of a crepant quotient carries the branch coefficients of the quotient morphism (Lemma 2.6(4)), and the minimal log discrepancy of a quotient germ is computed by the formula at the end of Lemma 2.6. Example 2.8 shows the effect of the branch coefficients.

Lemma 2.6. *Notation as in Definition 2.5. In particular, G acts faithfully on X, and B is a G-invariant $\mathbb{R}$-divisor on X such that $K_X + B$ is $\mathbb{R}$-Cartier. Let F be a prime divisor over X. Then there exist a unique prime divisor E over X/G and a unique positive integer e such that $\operatorname{ord}_F|_{\mathbb{C}(X/G)} = e \operatorname{ord}_E$. Moreover:*

- *(0) $K_X + B = \pi^*(K_{X/G} + B_{X/G})$, $K_{X/G} + B_{X/G}$ is $\mathbb{R}$-Cartier, and $B_{X/G} \geq 0$ if $B \geq 0$; in particular, $(X, B)/G$ is a sub-pair, and it is a pair if $B \geq 0$;*
- *(1) $e = |I_G(F)|$, the group $I_G(F)$ is cyclic, and $I_G(F)$ acts faithfully on $\mathfrak{m}_F/\mathfrak{m}_F^2$, where $\mathfrak{m}_F$ is the maximal ideal of the valuation ring of ord_F;*

(2) $a(F, X, B) = |I_G(F)| \cdot a(E, X/G, B_{X/G})$, $\operatorname{center}_{X/G} E = \pi(\operatorname{center}_X F)$, *and*

$$a(F, X, B, \pi^*\mathfrak{b}) = |I_G(F)| \cdot a(E, X/G, B_{X/G}, \mathfrak{b})$$

for any $\mathbb{R}$*-ideal* $\mathfrak{b} = \prod_j \mathfrak{b}_j^{r_j}$ *on* X/G*, where* $\pi^*\mathfrak{b} := \prod_j (\mathfrak{b}_j \mathcal{O}_X)^{r_j}$*;*

(3) *for any prime divisor* E' *over* X/G*, there exists a prime divisor* F' *over* X *with* $\operatorname{ord}_{F'}|_{\mathbb{C}(X/G)} \in \mathbb{N}^+ \operatorname{ord}_{E'}$*;*

(4) *if* P *is a prime divisor on* X *with* $\operatorname{mult}_P B = b$ *and ramification index* e_P *over* $\pi(P)$*, then*

$$\operatorname{mult}_{\pi(P)} B_{X/G} = 1 - \frac{1-b}{e_P};$$

(5) *if* $R \subset G$ *is a normal subgroup, then* G/R *acts faithfully on* X/R*,* $B_{X/R}$ *is* G/R*-invariant, and* $(X, B)/G = ((X, B)/R)/(G/R)$*, where* X/G *is identified with* $(X/R)/(G/R)$*.*

In particular, for a closed point $x \in X$ *fixed by* G*,*

$$\operatorname{mld}((X \ni x, B)/G) = \inf_F \frac{a(F, X, B)}{|I_G(F)|},$$

where F *runs over the prime divisors over* X *with* $\operatorname{center}_X F = \{x\}$*.*

Proof. Existence and uniqueness of E *and* e*.* Let ξ be the generic point of $\operatorname{center}_X F$ and w the restriction of ord_F to $\mathbb{C}(X/G)$. Since π is finite, $\pi(\xi)$ is not the generic point of X/G. Since $\pi^*\mathcal{O}_{X/G,\pi(\xi)} \subset \mathcal{O}_{X,\xi}$, the valuation w is non-negative on $\mathcal{O}_{X/G,\pi(\xi)}$ and positive on the non-zero maximal ideal of $\mathcal{O}_{X/G,\pi(\xi)}$. Thus $w = ev$ for a positive integer e and a discrete valuation v of $\mathbb{C}(X/G)$ with value group $\mathbb{Z}$, and v has center $\pi(\xi)$ on X/G. Let $\mathcal{O}_v$ be the valuation ring of v and S the integral closure of $\mathcal{O}_v$ in $\mathbb{C}(X)$. Then S is a Dedekind domain which is finite over $\mathcal{O}_v$ [Ser79, Chapter I, §4, Propositions 8 and 9]. Since the valuation ring $\mathcal{O}_F$ of ord_F contains $\mathcal{O}_v$ and is integrally closed, $S \subset \mathcal{O}_F$, hence $\mathcal{O}_F = S_{\mathfrak{P}}$ for the prime ideal $\mathfrak{P} := \mathfrak{m}_F \cap S$. By [Ser79, Chapter I, §4, Proposition 10], the residue field of ord_F is a finite extension of the residue field of v, so the residue field of v has transcendence degree $\dim X - 1$ over $\mathbb{C}$. By [KM98, Lemma 2.45], $v = \operatorname{ord}_E$ for a prime divisor E over X/G. Since the value group of ord_E is $\mathbb{Z}$, the value group of w is $e\mathbb{Z}$, so e is determined by w. Thus $\operatorname{ord}_E = w/e$ is determined by ord_F, and so is E.

Proof of (4). Let $P_1 = P, P_2, \dots, P_m$ be the prime divisors on X whose image under π is $\pi(P)$. They correspond to the prime ideals over the maximal ideal in the integral closure of $\mathcal{O}_{X/G,\pi(P)}$ in $\mathbb{C}(X)$, so G acts transitively on $\{P_1, \dots, P_m\}$ [Ser79, Chapter I, §7, Proposition 19]. Thus the ramification index of π along P_i is e_P and the degree of $P_i \to \pi(P)$ is a positive integer f independent of i, and

$$e_P f m = |G|$$

[Ser79, Chapter I, §7, Proposition 21 and the Corollary to Proposition 21]. Since B is G-invariant, $\operatorname{mult}_{P_i} B = b$ for any i. Since we work in characteristic zero, the ramification of π is tame, and the Hurwitz formula [Kol13, 2.41] implies that $\operatorname{mult}_{P_i} R(\pi) = e_P - 1$. Since $\pi_* P_i = f\pi(P)$, we obtain

$$\operatorname{mult}_{\pi(P)} B_{X/G} = \frac{mf(e_P - 1 + b)}{|G|} = 1 - \frac{1-b}{e_P}.$$

Proof of (0). Since π is finite and surjective, any prime divisor on X/G is $\pi(P)$ for some prime divisor P on X. By (4), for any prime divisor P on X,

$$\operatorname{mult}_P \pi^* B_{X/G} = e_P\Big(1 - \frac{1 - \operatorname{mult}_P B}{e_P}\Big) = e_P - 1 + \operatorname{mult}_P B = \operatorname{mult}_P(R(\pi) + B),$$

so $\pi^* B_{X/G} = R(\pi) + B$. By the Hurwitz formula $K_X = \pi^* K_{X/G} + R(\pi)$ [Kol13, 2.41], we have

$$\pi^*(K_{X/G} + B_{X/G}) = \pi^* K_{X/G} + R(\pi) + B = K_X + B.$$

Write $K_X + B = \sum_i r_i D_i$, where $r_i \in \mathbb{R}$ and the D_i are Cartier divisors. Since $e_P f m = |G|$ for any prime divisor P on X, we have $\pi_*\pi^* D = |G|D$ for any $\mathbb{R}$-divisor D on X/G. Thus

$$|G|(K_{X/G} + B_{X/G}) = \pi_*(K_X + B) = \sum_i r_i \pi_* D_i.$$

Each $\pi_* D_i$ is the norm of D_i, hence Cartier [Kol13, 2.40]. Therefore $K_{X/G} + B_{X/G}$ is $\mathbb{R}$-Cartier. If $B \geq 0$, then $\operatorname{mult}_P B \geq 0$ for any prime divisor P on X, and $B_{X/G} \geq 0$ by (4). Since X/G is a normal quasi-projective variety, (0) follows.

Proof of (1). Since $\mathcal{O}_{X/G} = (\pi_* \mathcal{O}_X)^G$ [Mum74, §7, Theorem] and $\operatorname{Frac}(A^G) = \operatorname{Frac}(A)^G$ for a domain A with an action of G [AM69, Chapter 5, Exercise 12], we have $\mathbb{C}(X/G) = \mathbb{C}(X)^G$. Since G acts faithfully on X, the extension $\mathbb{C}(X)/\mathbb{C}(X/G)$ is Galois with group G. For $\sigma \in G$, σ stabilizes ord_F if and only if $\sigma(S_{\mathfrak{P}}) = S_{\mathfrak{P}}$, if and only if $\sigma(\mathfrak{P}) = \mathfrak{P}$. Moreover, the residue field κ of ord_F is $S/\mathfrak{P}$. Thus $\operatorname{Stab}_G(F)$ and $I_G(F)$ are the decomposition group and the inertia group of $\mathfrak{P}$ in the sense of [Ser79, Chapter I, §7]. Since κ is a separable extension of the residue field of ord_E, we have $e = |I_G(F)|$ [Ser79, Chapter I, §7, Proposition 21 and the Corollary to Proposition 21].

By [Ser79, Chapter IV, §1, Remark 2], $\operatorname{Stab}_G(F)$ is the Galois group of the extension of the completions of $\mathbb{C}(X)$ and $\mathbb{C}(X/G)$ with respect to ord_F and ord_E, and $I_G(F)$ is the group G_0 of [Ser79, Chapter IV, §1]. Let t be a uniformizer of ord_F. Since κ has characteristic zero, $G_1 = \{1\}$, $I_G(F)$ is cyclic, and the homomorphism $I_G(F) \to \kappa^*$ sending σ to the residue class of $\sigma(t)/t$ is injective [Ser79, Chapter IV, §2, Proposition 7 and Corollary 2]. The κ-vector space $\mathfrak{m}_F/\mathfrak{m}_F^2$ is spanned by the class of t, and any $\sigma \in I_G(F)$ acts trivially on κ. Hence σ acts on $\mathfrak{m}_F/\mathfrak{m}_F^2$ as multiplication by the residue class of $\sigma(t)/t$, and $I_G(F)$ acts faithfully on $\mathfrak{m}_F/\mathfrak{m}_F^2$.

Proof of (2). Since E is a prime divisor over X/G, there exists a proper birational morphism $g : Y' \to X/G$ from a normal variety Y' such that E is a prime divisor on Y'. Let X' be the normalization of Y' in $\mathbb{C}(X)$ and $p : X' \to Y'$ the induced finite surjective morphism. Since X is the normalization of X/G in $\mathbb{C}(X)$, there exists a birational morphism $h : X' \to X$ such that $\pi \circ h = g \circ p$, and h is proper because $g \circ p$ is proper. Since $\mathcal{O}_{Y',E} = \mathcal{O}_v$, the base change of p to $\operatorname{Spec} \mathcal{O}_{Y',E}$ is $\operatorname{Spec} S \to \operatorname{Spec} \mathcal{O}_v$. Thus $\mathfrak{P}$ corresponds to a codimension one point of X' with local ring $S_{\mathfrak{P}} = \mathcal{O}_F$, so F is a prime divisor on X' with $p(F) = E$, and the ramification index of p along F is e. Hence

$$\operatorname{center}_{X/G} E = g(E) = g(p(F)) = \pi(h(F)) = \pi(\operatorname{center}_X F).$$

Let $K_{Y'} + B_{Y'} := g^*(K_{X/G} + B_{X/G})$ and $K_{X'} + B' := h^*(K_X + B)$. By (0),

$$K_{X'} + B' = h^*\pi^*(K_{X/G} + B_{X/G}) = p^*(K_{Y'} + B_{Y'}).$$

Take a log resolution $q : Z \to X'$ such that $h \circ q$ is a log resolution of (X, B), and write $K_Z + B_Z := q^*(K_{X'} + B') = (h \circ q)^*(K_X + B)$. Then $q_* B_Z = B'$, so the strict transform of F has coefficient $\operatorname{mult}_F B'$ in B_Z, and Definition 2.1 gives $a(F, X, B) = 1 - \operatorname{mult}_F B'$. Similarly, take a log resolution $r : W \to Y'$ such that $g \circ r$ is a log resolution of $(X/G, B_{X/G})$, and write $K_W + B_W := r^*(K_{Y'} + B_{Y'}) = (g \circ r)^*(K_{X/G} + B_{X/G})$. Then $r_* B_W = B_{Y'}$, so Definition 2.1 gives $a(E, X/G, B_{X/G}) = 1 - \operatorname{mult}_E B_{Y'}$ (cf. [KM98, Lemma 2.30]). Since the ramification of p along F is tame, the Hurwitz formula [Kol13, 2.41] reads $K_{X'} = p^* K_{Y'} + (e-1)F$ near the generic point of F. Comparing the coefficients of F in $K_{X'} + B' = p^*(K_{Y'} + B_{Y'})$, we obtain

$$e - 1 + \operatorname{mult}_F B' = e \operatorname{mult}_E B_{Y'}, \quad \text{that is,} \quad 1 - \operatorname{mult}_F B' = e(1 - \operatorname{mult}_E B_{Y'}).$$

Together with (1), $a(F, X, B) = |I_G(F)| \cdot a(E, X/G, B_{X/G})$. The computation only compares coefficients near the generic point of F, so it holds for $\mathbb{R}$-divisors (cf. [Kol13, 2.42], where the boundary is a $\mathbb{Q}$-divisor). The degree of $F \to E$ does not enter the formula.

Let $\mathfrak{b} = \prod_j \mathfrak{b}_j^{r_j}$ be an $\mathbb{R}$-ideal on X/G. Since $\operatorname{center}_{X/G} E = \pi(\operatorname{center}_X F)$, the generic point of $\operatorname{center}_{X/G} E$ is $\pi(\xi)$. The ideal $(\mathfrak{b}_j \mathcal{O}_X)_\xi$ is generated by $\pi^*(\mathfrak{b}_j)_{\pi(\xi)}$, and ord_F is non-negative

on $\mathcal{O}_{X,\xi}$. Hence

$$\operatorname{mult}_F(\mathfrak{b}_j\mathcal{O}_X) = \min\{\operatorname{ord}_F(\pi^*f) \mid f \in (\mathfrak{b}_j)_{\pi(\xi)}\} = e \operatorname{mult}_E \mathfrak{b}_j,$$

and the formula for $a(F, X, B, \pi^*\mathfrak{b})$ follows from the formula for $a(F, X, B)$ and $e = |I_G(F)|$.

Proof of (3). Let E' be a prime divisor over X/G. Let $g : Y' \to X/G$ be a proper birational morphism from a normal variety Y' such that E' is a prime divisor on Y'. Let X' be the normalization of Y' in $\mathbb{C}(X)$, $p : X' \to Y'$ the induced finite surjective morphism, and $h : X' \to X$ the induced proper birational morphism, as in the first paragraph of the proof of (2). Since p is finite and surjective, $p^{-1}(E')$ has an irreducible component F' with $p(F') = E'$, and $\dim F' = \dim E'$. Thus F' is a prime divisor on X', hence a prime divisor over X as h is proper and birational. Since $\mathcal{O}_{Y',E'} \subset \mathcal{O}_{X',F'}$ and $\mathcal{O}_{Y',E'}$ is a discrete valuation ring with fraction field $\mathbb{C}(X/G)$, the restriction of $\operatorname{ord}_{F'}$ to $\mathbb{C}(X/G)$ is $e' \operatorname{ord}_{E'}$ for a positive integer e' (cf. [Kol13, 2.42]), that is, $\operatorname{ord}_{F'}|_{\mathbb{C}(X/G)} \in \mathbb{N}^+ \operatorname{ord}_{E'}$.

Proof of the last formula. Since x is fixed by G and the fibers of π are G-orbits, $\pi^{-1}(\pi(x)) = \{x\}$. By (2), a prime divisor F over X satisfies $\operatorname{center}_X F = \{x\}$ if and only if the corresponding E satisfies $\operatorname{center}_{X/G} E = \{\pi(x)\}$, and then $a(E, X/G, B_{X/G}) = a(F, X, B)/|I_G(F)|$. By (3), any prime divisor E over X/G with $\operatorname{center}_{X/G} E = \{\pi(x)\}$ satisfies $\operatorname{ord}_F|_{\mathbb{C}(X/G)} = e \operatorname{ord}_E$ for a prime divisor F over X and a positive integer e, and then $\operatorname{center}_X F = \{x\}$ by (2). Taking infima, we obtain the formula.

Proof of (5). Since R is normal in G, the group G/R acts on X/R, and $X/G = (X/R)/(G/R)$. If $\sigma \in G$ acts trivially on X/R, then X is the union of the closed subsets $\{y \in X \mid \sigma(y) = \rho(y)\}$ for $\rho \in R$, so $\sigma = \rho$ on X for some $\rho \in R$ since X is irreducible, and $\sigma \in R$ since G acts faithfully on X. Thus G/R acts faithfully on X/R. Since B and the ramification divisor of $X \to X/R$ are G-invariant, $B_{X/R}$ is G/R-invariant, and $K_{X/R} + B_{X/R}$ is $\mathbb{R}$-Cartier by (0). Let $B'_{X/G}$ be the boundary of $(X/R, B_{X/R})/(G/R)$. By (0) applied to R and to G/R,

$$\pi^*(K_{X/G} + B_{X/G}) = K_X + B = \pi^*(K_{X/G} + B'_{X/G}).$$

Since $\pi_*\pi^*D = |G|D$ for any $\mathbb{R}$-divisor D on X/G, we obtain $B_{X/G} = B'_{X/G}$. □

Definition 2.7. Let X be a normal quasi-projective variety, G a finite group acting faithfully on X, and $\pi : X \to X/G$ the quotient morphism. For a prime divisor F over X, the prime divisor E over X/G with $\operatorname{ord}_F|_{\mathbb{C}(X/G)} \in \mathbb{N}^+ \operatorname{ord}_E$ (Lemma 2.6, applied with $B := -K_X$ for the G-invariant canonical divisor $K_X := \pi^*K_{X/G} + R(\pi)$ of the Hurwitz formula [Kol13, 2.41]) is called the *prime divisor under* F, and F is called a prime divisor *lying over* E, or a *lift* of E to X.

Example 2.8. Let μ_2 act on $\mathbb{A}^1_x$ by $x \to -x$, and let $u := x^2$. Then $\mathbb{A}^1_x/\mu_2 = \mathbb{A}^1_u$, the quotient morphism is étale over $\mathbb{A}^1_u \backslash \{0\}$, and the ramification index along the origin is 2. By Lemma 2.6(4) with $b = 0$, the coefficient of the origin in the boundary of the crepant quotient is $1 - \frac{1}{2} = \frac{1}{2}$, so

$$(\mathbb{A}^1_x, 0)/\mu_2 = \Big(\mathbb{A}^1_u, \frac{1}{2}\operatorname{div}(u)\Big).$$

The only prime divisor F over $\mathbb{A}^1_x$ with $\operatorname{center}_{\mathbb{A}^1_x} F = \{0\}$ is the origin itself. Since $a(F, \mathbb{A}^1_x, 0) = 1$ and $|I_{\mu_2}(F)| = 2$, the last formula of Lemma 2.6 implies that

$$\operatorname{mld}((\mathbb{A}^1_x \ni 0, 0)/\mu_2) = \frac{1}{2},$$

whereas $\operatorname{mld}(\mathbb{A}^1_u \ni 0) = 1$.

2.3. **DCC sets.**

Definition 2.9. Let $\Gamma \subset [0, 1]$. We denote by Γ_+ the set of sums $\sum_{i=1}^m \gamma_i$, where $m \geq 0$ is an integer and $\gamma_1, \dots, \gamma_m \in \Gamma$ (the empty sum is 0), and we let

$$\Psi(\Gamma) := \Big\{1 - \frac{1-c}{e} \;\Big|\; c \in \Gamma,\ e \in \mathbb{Z}_{\geq 1}\Big\}.$$

If $\Gamma\backslash\{0\} \neq \emptyset$, we let $\gamma(\Gamma) := \inf(\Gamma\backslash\{0\})$ and

$$\delta(\Gamma) := \inf\Big(\{1\} \cup \Big\{ \sum_{i=1}^{m} \gamma_i - 1 \;\Big|\; m \geq 1,\ \gamma_1, \dots, \gamma_m \in \Gamma\backslash\{0\},\ \sum_{i=1}^{m} \gamma_i > 1 \Big\}\Big).$$

Lemma 2.10. *Let $\Gamma \subset [0,1]$ be a DCC set and n a positive integer. Then:*

(1) the sets Γ_+, $\Gamma_+ \cap [0,1]$, $\Psi(\Gamma)$, and $\bigcup_{q=1}^{n} \frac{1}{q}\Gamma$ satisfy the DCC;
(2) if $\Gamma\backslash\{0\} \neq \emptyset$, then $\gamma(\Gamma) > 0$ and $\delta(\Gamma) > 0$;
(3) any sequence in Γ has a nondecreasing subsequence;
(4) a finite union of subsets of $\mathbb{R}$ satisfying the ACC satisfies the ACC;
(5) let X be a normal variety, $x \in X$ a closed point, G a finite group acting faithfully on X and fixing x, $h_1, \dots, h_k$ non-zero regular functions near x, $c_1, \dots, c_k \in \Gamma$, and $B := \sum_j c_j \operatorname{div}(h_j)$. Assume that (X, B) is lc near x and G preserves B. Then there exists a G-invariant affine open neighborhood X' of x such that each h_j is regular on X', $(X', B|_{X'})$ is lc, the coefficients of $B|_{X'}$ belong to $\Gamma_+ \cap [0,1]$, and the coefficients of $B_{X'/G}$, where $(X', B|_{X'})/G = (X'/G, B_{X'/G})$ (Definition 2.5), belong to the set $\Psi(\Gamma_+ \cap [0,1])$, which satisfies the DCC.

Proof. We first show (3). Since Γ satisfies the DCC, any non-empty subset of Γ has a minimum. Let $(a_k)_{k\geq 1}$ be a sequence in Γ and $k_0 := 0$. For $i \geq 1$, let $k_i > k_{i-1}$ be an integer such that

$$a_{k_i} = \min\{a_k \mid k > k_{i-1}\}.$$

Since $\{a_k \mid k > k_i\} \subset \{a_k \mid k > k_{i-1}\}$, we have $a_{k_{i+1}} \geq a_{k_i}$ for any $i \geq 1$. Thus $(a_{k_i})_{i\geq 1}$ is a nondecreasing subsequence of $(a_k)_{k\geq 1}$.

Next, let N be a positive integer and $(s_k)_{k\geq 1}$ a sequence of real numbers such that any s_k is a sum of at most N elements of Γ. Passing to a subsequence, we may assume that there exists an integer $0 \leq l \leq N$ such that $s_k = \sum_{i=1}^{l} \gamma_{k,i}$ with $\gamma_{k,1}, \dots, \gamma_{k,l} \in \Gamma$ for any k. Applying (3) to the sequences $(\gamma_{k,1})_{k\geq 1}, \dots, (\gamma_{k,l})_{k\geq 1}$ one after another and passing to subsequences, we may assume that $(\gamma_{k,i})_{k\geq 1}$ is nondecreasing for any i. Thus $(s_k)_{k\geq 1}$ is nondecreasing. In particular, the following holds.

($*$) For any integer $N \geq 1$, the set of sums of at most N elements of Γ satisfies the DCC.

We show (1). If $\Gamma\backslash\{0\} = \emptyset$, then $\Gamma_+ = \{0\}$. Otherwise, $\Gamma\backslash\{0\}$ has a minimum $\gamma(\Gamma) > 0$, and any element of $\Gamma_+ \cap [0,1]$ is a sum of at most $\lfloor 1/\gamma(\Gamma) \rfloor$ elements of Γ. Hence $\Gamma_+ \cap [0,1]$ satisfies the DCC by ($*$). Similarly, the elements of Γ_+ that are at most a given real number s are sums of at most $\lfloor s/\gamma(\Gamma) \rfloor$ elements of Γ. Since a strictly decreasing sequence $(s_k)_{k\geq 1}$ in Γ_+ satisfies $s_k \leq s_1$ for any k, Γ_+ satisfies the DCC by ($*$). Let $(1 - \frac{1-c_k}{e_k})_{k\geq 1}$ be a sequence in $\Psi(\Gamma)$, where $c_k \in \Gamma$ and $e_k \in \mathbb{Z}_{\geq 1}$. By (3), passing to a subsequence, we may assume that $(c_k)_{k\geq 1}$ is nondecreasing. If $(e_k)_{k\geq 1}$ is bounded, then it has a constant subsequence, and otherwise it has a strictly increasing subsequence. Thus, passing to a subsequence, we may assume that $(e_k)_{k\geq 1}$ is nondecreasing. Since $1 - c_k \geq 0$ for any k, the sequence $(\frac{1-c_k}{e_k})_{k\geq 1}$ is nonincreasing, and so $(1 - \frac{1-c_k}{e_k})_{k\geq 1}$ is nondecreasing. Hence $\Psi(\Gamma)$ contains no strictly decreasing infinite sequence, that is, $\Psi(\Gamma)$ satisfies the DCC. Finally, any sequence $(a_k)_{k\geq 1}$ in $\bigcup_{q=1}^{n} \frac{1}{q}\Gamma$ has a subsequence contained in $\frac{1}{q}\Gamma$ for some $1 \leq q \leq n$. Applying (3) to the corresponding subsequence of $(qa_k)_{k\geq 1}$, we obtain a nondecreasing subsequence of $(a_k)_{k\geq 1}$. Hence $\bigcup_{q=1}^{n} \frac{1}{q}\Gamma$ satisfies the DCC.

We show (2). We have shown that $\gamma(\Gamma) = \min(\Gamma\backslash\{0\}) > 0$. Let S be the set of sums $s = \sum_{i=1}^{m} \gamma_i$ with $m \geq 1$, $\gamma_1, \dots, \gamma_m \in \Gamma\backslash\{0\}$, and $1 < s \leq 2$. Any element of S is a sum of at most $\lfloor 2/\gamma(\Gamma) \rfloor$ elements of Γ, so S satisfies the DCC by ($*$). Since the sums $s > 2$ contribute $s - 1 > 1$ to the infimum defining $\delta(\Gamma)$, we have $\delta(\Gamma) = 1$ if $S = \emptyset$, and

$$\delta(\Gamma) = \min\{1, \min S - 1\} > 0$$

if $S \neq \emptyset$. In particular, the infima defining $\gamma(\Gamma)$ and $\delta(\Gamma)$ are minima.

We show (4). Let $A_1, \dots, A_r \subset \mathbb{R}$ satisfy the ACC. Any strictly increasing infinite sequence in $\bigcup_{i=1}^r A_i$ has a strictly increasing infinite subsequence contained in some A_i, which is impossible. Thus $\bigcup_{i=1}^r A_i$ satisfies the ACC.

We show (5). Let V be an affine open neighborhood of x such that each h_j is regular on V and $(V, B|_V)$ is lc. Since X is separated and G fixes x, $X' := \bigcap_{g\in G} gV$ is a G-invariant affine open neighborhood of x. Since G preserves B, $B|_{X'}$ is G-invariant, and $K_{X'} + B|_{X'}$ is $\mathbb{R}$-Cartier since $(X', B|_{X'})$ is lc. Thus $(X', B|_{X'})/G$ is defined (Definition 2.5). Possibly replacing X with X', we may assume that X is affine, each h_j is regular on X, and (X, B) is lc. Let P be a prime divisor on X. Since each h_j is regular on X, $\operatorname{mult}_P \operatorname{div}(h_j) \in \mathbb{Z}_{\geq 0}$, and so

$$\operatorname{mult}_P B = \sum_{j=1}^{k} c_j \operatorname{mult}_P \operatorname{div}(h_j)$$

is a sum of elements of Γ with repetitions. Since (X, B) is lc, $1 - \operatorname{mult}_P B = a(P, X, B) \geq 0$. Thus $\operatorname{mult}_P B \in \Gamma_+ \cap [0, 1]$. Let $\pi : X \to X/G$ be the quotient morphism. Since π is finite and surjective, any prime divisor on X/G is $\pi(P)$ for some prime divisor P on X, and

$$\operatorname{mult}_{\pi(P)} B_{X/G} = 1 - \frac{1 - \operatorname{mult}_P B}{e_P} \in \Psi(\Gamma_+ \cap [0, 1])$$

by Lemma 2.6(4), where e_P is the ramification index of P over $\pi(P)$. Since $\Gamma_+ \cap [0, 1] \subset [0, 1]$ satisfies the DCC by (1), $\Psi(\Gamma_+ \cap [0, 1])$ satisfies the DCC by (1) applied to $\Gamma_+ \cap [0, 1]$. □

2.4. **Monomial valuations and lattices.**

Definition-Proposition 2.11. Let L be a field, $v : L^* \to \mathbb{Z}$ a non-trivial valuation, s an element transcendental over L, and c an integer. The *Gauss extension* of v to $L(s)$ with s-value c is the valuation $v_c : L(s)^* \to \mathbb{Z}$ determined by

$$v_c\Big(\sum_i a_i s^i\Big) = \min_i \big(v(a_i) + ic\big) \quad \text{for } a_i \in L. \tag{2.1}$$

If $v(L^*) = m\mathbb{Z}$ and $\gcd(m, c) = 1$, then $v_c(L(s)^*) = \mathbb{Z}$.

Proof. By [Bou72, Chapter VI, §10, no. 1, Lemma 1], applied to the totally ordered group $\mathbb{Z} \supset v(L^*)$ and $\xi = c$, there exists a unique valuation v_c on $L(s)$ satisfying (2.1). Assume that $v(L^*) = m\mathbb{Z}$. By the formula, v_c takes values in $m\mathbb{Z} + c\mathbb{Z}$ on non-zero polynomials in s over L, hence on $L(s)^*$. Since $v_c|_{L^*} = v$ and $v_c(s) = c$, we have

$$v_c(L(s)^*) = m\mathbb{Z} + c\mathbb{Z} = \gcd(m, c)\mathbb{Z},$$

which is $\mathbb{Z}$ if $\gcd(m, c) = 1$. □

Lemma 2.12. *Let X and T be varieties and $\mathbb{C}(T) \subset \mathbb{C}(X)$ an inclusion of fields over $\mathbb{C}$ such that $\mathbb{C}(X)$ has transcendence degree one over $\mathbb{C}(T)$. Let F be a prime divisor over X such that $\operatorname{ord}_F|_{\mathbb{C}(T)}$ is non-trivial, and let $\xi \in T$ be a scheme-theoretic point such that ord_F is non-negative on $\mathcal{O}_{T,\xi}$ and positive on the maximal ideal of $\mathcal{O}_{T,\xi}$. Then $\operatorname{ord}_F|_{\mathbb{C}(T)} = \varrho \operatorname{ord}_\nu$ for a positive integer ϱ and a prime divisor ν over T with $\operatorname{center}_T \nu = \overline{\{\xi\}}$.*

Proof. The value group of $\operatorname{ord}_F|_{\mathbb{C}(T)}$ is a non-zero subgroup $\varrho\mathbb{Z}$ of $\operatorname{ord}_F(\mathbb{C}(X)^*) = \mathbb{Z}$ for a positive integer ϱ. The residue field of ord_F has transcendence degree $\dim X - 1$ over $\mathbb{C}$, and $\mathbb{Z}/\varrho\mathbb{Z}$ is finite. Hence, by [Bou72, Chapter VI, §10, no. 3, Corollary 1 to Theorem 1] applied to $\mathbb{C}(T) \subset \mathbb{C}(X)$, the residue field κ of $\operatorname{ord}_F|_{\mathbb{C}(T)}$ has transcendence degree at least $\dim X - 2 = \dim T - 1$ over $\mathbb{C}$. The valuation ring of $\operatorname{ord}_F|_{\mathbb{C}(T)}$ dominates $\mathcal{O}_{T,\xi}$, so [Abh56, Theorem 1] applied to $\mathcal{O}_{T,\xi}$ implies that κ has transcendence degree at most $\dim T - 1$ over $\mathbb{C}$. Thus the valuation ring of $\operatorname{ord}_F|_{\mathbb{C}(T)}$ is a discrete valuation ring of $\mathbb{C}(T)$ dominating $\mathcal{O}_{T,\xi}$ whose residue field has transcendence degree $\dim T - 1$ over $\mathbb{C}$. By [KM98, Lemma 2.45], it is the valuation ring of ord_ν for a prime divisor ν over T, and $\operatorname{center}_T \nu = \overline{\{\xi\}}$. Comparing the value groups, we obtain $\operatorname{ord}_F|_{\mathbb{C}(T)} = \varrho \operatorname{ord}_\nu$. □

Lemma 2.13. *Let X be a normal variety, $\eta \in X$ a scheme-theoretic point of codimension c at which X is smooth, $x_1, \dots, x_c$ regular parameters of $\mathcal{O}_{X,\eta}$, and $b_1, \dots, b_c$ real numbers. Then there exists an open neighborhood $U \subset X$ of η such that U is smooth, every x_i is regular on U, and $\sum_{i=1}^{c} \operatorname{div}(x_i)$ is an snc divisor on U. Moreover, for any such U and any prime divisor F over X with $\operatorname{center}_X F = \overline{\{\eta\}}$,*

$$a\Big(F, U, \sum_{i=1}^{c} b_i \operatorname{div}(x_i)\Big) \geq \sum_{i=1}^{c} (1 - b_i) \operatorname{ord}_F x_i.$$

Proof. We first show that such U exists. Since $x_1, \dots, x_c$ are regular parameters of $\mathcal{O}_{X,\eta}$, the divisor $\sum_i \operatorname{div}(x_i)$ is snc at η, so we may remove from a smooth open neighborhood of η on which each x_i is regular the finitely many components of $\sum_i \operatorname{div}(x_i)$ not containing η, and then take the open locus where $\sum_i \operatorname{div}(x_i)$ is snc. Now let U be any such neighborhood. Near η, any component of $\sum_i \operatorname{div}(x_i)$ contains η and has coefficient one, since $x_1, \dots, x_c$ are pairwise non-associate prime elements of the regular local ring $\mathcal{O}_{X,\eta}$, so $(U, \sum_i \operatorname{div}(x_i))$ is lc near η [KM98, Corollary 2.31(3)]. Since $\eta \in U$, F is a prime divisor over U, and $a(F, U, \sum_i \operatorname{div}(x_i)) \geq 0$, since $a(F, U, \cdot)$ can be computed on any open neighborhood of η. Since $\operatorname{mult}_F h^* \operatorname{div}(x_i) = \operatorname{ord}_F x_i$ for any log resolution h of $(U, \sum_i \operatorname{div}(x_i))$ on which F is a divisor, Definition 2.1 implies

$$a\Big(F, U, \sum_{i=1}^{c} b_i \operatorname{div}(x_i)\Big) = a\Big(F, U, \sum_{i=1}^{c} \operatorname{div}(x_i)\Big) + \sum_{i=1}^{c} (1 - b_i) \operatorname{ord}_F x_i \geq \sum_{i=1}^{c} (1 - b_i) \operatorname{ord}_F x_i.$$

□

Definition 2.14. Let G be a finite group acting on a normal variety X. For $\sigma \in G$, we denote by σ^* the pullback of rational functions and rational differential forms on X. A non-zero rational function or rational differential form u on X is *G-semi-invariant* with *character* $\chi : G \to \mathbb{C}^*$ if $\sigma^* u = \chi(\sigma) u$ for any $\sigma \in G$.

Let q be a positive integer. For a G-invariant $\mathbb{Q}$-divisor D on X such that qD is a Weil divisor, we regard $\mathcal{O}_X(q(K_X + D))$ as the subsheaf of rational sections ϑ of $\omega_X^{\otimes q}$ with $\operatorname{div}(\vartheta) + qD \geq 0$. The sheaf $\mathcal{O}_X(q(K_X + D))$ is G-invariant. Let $x \in X$ be a closed point fixed by G, and $\mathcal{L}$ a G-invariant subsheaf of the sheaf of rational functions or of the sheaf of rational sections of $\omega_X^{\otimes q}$ which is invertible near x. The *character* of $\mathcal{L}$ at x is the character by which G acts on the fiber $\mathcal{L} \otimes \kappa(x)$ through pullback. We also call it the character at x of any generator of $\mathcal{L}$ near x.

Definition 2.15. Let d be a positive integer, $\mathbb{T}^d := \mathbb{R}^d / \mathbb{Z}^d$, and $\rho : \mathbb{R}^d \to \mathbb{T}^d$ the quotient map. For a subgroup $H \subset \mathbb{T}^d$, let $N_H := \rho^{-1}(H)$. A non-zero vector w of a lattice $N \subset \mathbb{R}^d$ is *primitive* in N if $\frac{1}{k} w \notin N$ for any integer $k \geq 2$.

Let X be a variety, $x \in X$ a closed point, and G a finite abelian group acting faithfully on X and fixing x. We call G-semi-invariant regular parameters of $\mathcal{O}_{X,x}$ *G-eigenparameters* at x. For G-semi-invariant regular functions $y_1, \dots, y_d$ near x with characters $\chi_1, \dots, \chi_d$, the *image* of G in $\mathbb{T}^d$ is the subgroup of the vectors $(u_1, \dots, u_d) \bmod \mathbb{Z}^d$ for which there exists $g \in G$ with $\chi_i(g) = e^{2\pi\sqrt{-1}u_i}$ for any i, and $N_G := N_H$, where H is the image of G in $\mathbb{T}^d$. If G acts on X and fixes x, but not necessarily faithfully, we still call G-semi-invariant regular parameters of $\mathcal{O}_{X,x}$ G-eigenparameters at x, and define the image of G in $\mathbb{T}^d$ as for a faithful action.

For a positive integer r and integers $a_1, \dots, a_d$, we denote by $\mu_r(a_1, \dots, a_d)$ the cyclic subgroup of $(\mathbb{C}^*)^d$ generated by $(\zeta^{a_1}, \dots, \zeta^{a_d})$, where $\zeta = e^{2\pi\sqrt{-1}/r}$, and we let

$$\mathbf{T}_{(a_1,\dots,a_d)} := \{(s^{a_1}, \dots, s^{a_d}) \mid s \in \mathbb{C}^*\}.$$

Definition 2.16. Let d be a positive integer, and let $\mathbb{C}[x_1, \dots, x_d]$ and $\mathbb{C}[[x_1, \dots, x_d]]$ be the polynomial ring and the power series ring in variables $x_1, \dots, x_d$. For $w \in \mathbb{R}^d$, the *w-weight* of a monomial x^α is $\langle w, \alpha \rangle$. For integers $a_1, \dots, a_d$, the *$\mathbf{T}_{(a_1,\dots,a_d)}$-weight* of x^α is $\sum_i a_i \alpha_i$, and the $\mathbf{T}_{(a_1,\dots,a_d)}$-weight of a non-zero polynomial whose monomials all have $\mathbf{T}_{(a_1,\dots,a_d)}$-weight κ is κ.

For a non-zero monomial ideal $J \subset \mathbb{C}[x_1, \dots, x_d]$ and $w \in \mathbb{R}^d_{\geq 0}$, let

$$\mathrm{wt}_w(J) := \min\{\langle w, \alpha\rangle \mid x^\alpha \in J\}.$$

For a non-zero power series $f \in \mathbb{C}[[x_1, \dots, x_d]]$, let $J(f) \subset \mathbb{C}[x_1, \dots, x_d]$ be the ideal generated by the monomials occurring in f, let $\mathrm{wt}_w(f) := \mathrm{wt}_w(J(f))$, and let $\mathrm{in}_w(f)$ be the sum of the terms of f of w-weight $\mathrm{wt}_w(f)$.

For a finite subgroup $H \subset \mathbb{T}^d$, an s-tuple $J = (J_1, \dots, J_s)$ of non-zero monomial ideals of $\mathbb{C}[x_1, \dots, x_d]$, and real numbers $c = (c_1, \dots, c_s)$, let

$$\psi_{J,c}(w) := \sum_{i=1}^{d} w_i - \sum_{l=1}^{s} c_l \,\mathrm{wt}_w(J_l) \quad \text{and} \quad \psi_{\min}(H, J, c) := \inf\{\psi_{J,c}(w) \mid w \in N_H \cap \mathbb{R}^d_{>0}\},$$

and call $\psi_{\min}(H, J, c)$ a *lattice Newton minimum*.

Let X be a variety, $x \in X$ a closed point at which X is smooth, $x_1, \dots, x_d$ regular parameters of $\mathcal{O}_{X,x}$, and $w \in \mathbb{R}^d_{\geq 0}\backslash\{0\}$. The *monomial valuation* in $x_1, \dots, x_d$ with weights w is the valuation v of the function field of X such that

$$v(f) = \min\{\langle w, \alpha\rangle \mid c_\alpha \neq 0\}$$

for any non-zero $f \in \mathcal{O}_{X,x}$, where $f = \sum_\alpha c_\alpha x^\alpha$ is the expansion of f in $\widehat{\mathcal{O}}_{X,x} = \kappa(x)[[x_1, \dots, x_d]]$ [JM12, Proposition 3.1]. A prime divisor F over X with $\mathrm{ord}_F = v$, if it exists, is called the *monomial prime divisor* in $x_1, \dots, x_d$ with weights w.

Remark 2.17. Definitions 2.1, 2.2, 2.5, 2.14, 2.15, and 2.16 carry over verbatim to varieties over a field $k \supset \mathbb{C}$. Lemmas 2.6 and 2.13 hold for normal quasi-projective varieties over k, with $\mathbb{C}(X/G)$ replaced by the function field of X/G. The proofs work verbatim, since log resolutions exist over k [KM98, Notation 0.4(10)] and ramification is tame in characteristic zero. The results [KM98, Lemmas 2.27, 2.29, 2.30, and 5.17(1), Corollaries 2.31 and 2.32, and Proposition 5.20] hold for $\mathbb{R}$-divisors, also over k, with the proofs given there for $\mathbb{Q}$-divisors, since log discrepancies are affine-linear in the coefficients of the boundary, and, for [KM98, Proposition 5.20(1)], by writing $K_{X'} + \Delta'$ (in the notation there) as an $\mathbb{R}$-linear combination of Cartier divisors and taking norms, which are Cartier [Kol13, 2.40].

2.5. **Known results.** The ACC for minimal log discrepancies of surfaces is due to Alexeev [Ale93] and Shokurov [Sho94]. We use it in the form of Theorem 2.18. Part (1) of Theorem 2.18 is [HL24, Theorem 1.2] and part (2) is [HL23, Theorem 1.2]. Both papers work over an algebraically closed field of arbitrary characteristic, and pairs and minimal log discrepancies over k are given by Definitions 2.1 and 2.2 (cf. Remark 2.17). In [HL23] and [HL24] the components of the boundary are assumed to pass through x, which holds after shrinking X around x. The ACC for threefolds is reduced to Theorem 2.18(1) in Sections 6, 14, 15, 16, 17, and 18. Part (2) is used in Sections 14, 15, 16, and 18.

Theorem 2.18 ([HL24, Theorem 1.2], [HL23, Theorem 1.2])**.** *Let $\Gamma \subset [0, 1]$ be a DCC set and k an algebraically closed field containing $\mathbb{C}$.*

(1) The set of $\mathrm{mld}(X \ni x, B)$, where (X, B) is an lc pair over k with $\dim X = 2$, $x \in X$ is a closed point, and $B \in \Gamma$, satisfies the ACC.

(2) There exists a positive integer l depending only on Γ such that any lc surface germ $(X \ni x, B)$ over k with $B \in \Gamma$ has a prime divisor E over X with $\mathrm{center}_X E = \{x\}$, $a(E, X, B) = \mathrm{mld}(X \ni x, B)$, and $a(E, X, 0) \leq l$.

Part (1) of Theorem 2.19 is [Kaw24, Theorem 1.1] and part (2) is [Kaw24, Theorem 1.3]. In [Kaw24, Section 2], a pair consists of a normal variety and an $\mathbb{R}$-divisor B with non-negative coefficients such that $K + B$ is $\mathbb{R}$-Cartier, with no upper bound on the coefficients, and Definition 2.1 also requires quasi-projectivity, so (1) and (2) hold for the pairs of this paper. In part (2), the number of factors of $\mathfrak{a}$ and $\mathfrak{b}$ is arbitrary, and neither $(X, B, \mathfrak{a})$ nor $(X, B, \mathfrak{b})$ is

assumed to be lc. Theorem 2.19 is used in Proposition 14.6(1), Theorems 16.1, 16.5, 14.3, and 16.21, and Lemma 14.14.

Theorem 2.19 ([Kaw24, Theorems 1.1 and 1.3])**.** *Let X be a normal threefold and $\Gamma \subset \mathbb{R}_{>0}$ a DCC set.*

(1) The set of $\mathrm{mld}(X \ni \eta, B)$, where $\eta \in X$ is a scheme-theoretic point and (X, B) is a pair with $B \in \Gamma$, satisfies the ACC.

(2) There exists a positive integer l such that the following holds. Let $\eta \in X$ be a scheme-theoretic point, (X, B) a pair with $B \in \Gamma$, and $\mathfrak{a} = \prod_j \mathfrak{a}_j^{r_j}$, $\mathfrak{b} = \prod_j \mathfrak{b}_j^{r_j}$ two $\mathbb{R}$-ideals with $r_j \in \Gamma$ such that

$$\mathfrak{a}_j \mathcal{O}_{X,\eta} + \mathfrak{m}_\eta^l = \mathfrak{b}_j \mathcal{O}_{X,\eta} + \mathfrak{m}_\eta^l \quad \text{for any } j.$$

Then $\mathrm{mld}(X \ni \eta, B, \mathfrak{a}) = \mathrm{mld}(X \ni \eta, B, \mathfrak{b})$.

Theorem 2.22 below is a local form of [NS26b, Theorem 7.1]. We use the following notation from [NS26b, Section 2.1].

Definition 2.20. Let $h : Y \to X$ be a morphism and $\mathfrak{a} = \prod_j \mathfrak{a}_j^{r_j}$ an $\mathbb{R}$-ideal on X. The *restriction* of $\mathfrak{a}$ to Y is $\mathfrak{a}|_Y := \prod_j (\mathfrak{a}_j \mathcal{O}_Y)^{r_j}$. The product of two $\mathbb{R}$-ideals is their formal product.

Remark 2.21. Let G be a finite group of order r acting on $\mathbb{A}^n$ and f a G-semi-invariant polynomial with character χ (Definition 2.14). Since $\sigma^r = 1$ for any $\sigma \in G$, we have $\chi(\sigma)^r = 1$, so f^r is G-invariant. Thus in Theorem 2.22, $(f_1^r \cdots f_c^r)^{\frac{1}{r}}$ is an $\mathbb{R}$-ideal on $\mathbb{A}^n/G$. In [NS26b, Theorem 7.1], which holds for any $\mathbb{R}$-ideal and any closed point, X is a closed subvariety of $\mathbb{A}^n$, X/G is normal and klt, and the integer denoted d in [NS26b, Theorem 7.1] equals r here. In [NS26b, Section 2.1], a negative minimal log discrepancy on a curve is replaced by $-\infty$. This is why Theorem 2.22 assumes $n - c \geq 2$.

Theorem 2.22 (cf. [NS26b, Theorem 7.1])**.** *Let G be a finite group acting faithfully and linearly on $\mathbb{A}^n$ without pseudoreflections, and let $X \subset \mathbb{A}^n$ be the closed subscheme defined by G-semi-invariant polynomials $f_1, \ldots, f_c$. Assume that, near 0, X is a complete intersection of codimension c which is normal and klt, and that $\dim X \geq 2$ near 0, i.e. $n - c \geq 2$. Let $r = |G|$ and let $y \in X/G$ be the image of 0. Assume that, near y, the image in $\mathbb{A}^n/G$ of the locus where $\mathbb{A}^n \to \mathbb{A}^n/G$ is not étale intersects X/G in codimension at least two. Let $\mathfrak{a}$ be an $\mathbb{R}$-ideal on $\mathbb{A}^n/G$ whose restriction to X/G is non-zero near y. Then*

$$\mathrm{mld}(X/G \ni y, \mathfrak{a}|_{X/G}) = \mathrm{mld}\big(\mathbb{A}^n/G \ni y, (f_1^r \cdots f_c^r)^{\frac{1}{r}} \mathfrak{a}\big).$$

Proof. Let $\pi : \mathbb{A}^n \to \mathbb{A}^n/G$ be the quotient morphism and $Z \subset \mathbb{A}^n/G$ the image of the locus where π is not étale. For a G-invariant polynomial g, we denote by $D(g)$ the locus where $g \neq 0$, both in $\mathbb{A}^n$ and in $\mathbb{A}^n/G$.

Step 1. Since $y \in X/G$ is the image of 0, we have $0 \in X$. Since X is normal at 0, exactly one irreducible component X_0 of X passes through 0. Let W be the union of the irreducible components of X other than X_0, the closed subset of points $x \in X$ such that X is not normal and klt near x, and $\pi^{-1}(Z_1) \cap X$, where Z_1 is the union of the irreducible components of $Z \cap (X/G)$ of codimension at most one in X/G. By assumption, $y \notin Z_1$, and $\pi^{-1}(y) \cap X = \{0\}$, so $0 \notin \pi^{-1}(Z_1) \cap X$. Since G fixes 0, G preserves X_0, so W is a G-invariant closed subset of $\mathbb{A}^n$ with $0 \notin W$. Let h be a polynomial with $h|_W = 0$ and $h(0) = 1$, and let $g := \prod_{\sigma \in G} \sigma^* h$. Then g is G-invariant, $g|_W = 0$, and $g(0) = 1$. Thus $X \cap D(g) = X_0 \cap D(g)$ is irreducible, normal, and klt.

Step 2. Let G act on $\mathbb{A}^{n+1} = \mathbb{A}^n \times \mathbb{A}^1_s$ by $\sigma \cdot (x, s) = (\sigma x, s)$. The fixed locus of σ on $\mathbb{A}^{n+1}$ is the product of the fixed locus of σ on $\mathbb{A}^n$ with $\mathbb{A}^1_s$, so G acts on $\mathbb{A}^{n+1}$ without pseudoreflections. We have $\mathbb{A}^{n+1}/G = (\mathbb{A}^n/G) \times \mathbb{A}^1_s$, and the quotient morphism is $\pi \times \mathrm{id}_{\mathbb{A}^1_s}$. Hence the image of the locus where $\mathbb{A}^{n+1} \to \mathbb{A}^{n+1}/G$ is not étale is $Z \times \mathbb{A}^1_s$. Let $p : (\mathbb{A}^n/G) \times \mathbb{A}^1_s \to \mathbb{A}^n/G$ be the

projection, and let $y_1 := (y, 1)$, the image of $(0, 1)$. We regard g and $f_1^r, \dots, f_c^r$ as functions on $(\mathbb{A}^n/G) \times \mathbb{A}^1_s$ via p.

Step 3. Let $X_1 := V(f_1, \dots, f_c, sg - 1)$ and $X_2 := V(sg - 1)$ in $\mathbb{A}^{n+1}$. The projection $\mathbb{A}^{n+1} \to \mathbb{A}^n$ induces G-equivariant isomorphisms $X_1 \cong X \cap D(g)$ and $X_2 \cong D(g)$, whose inverses are $x \to (x, 1/g(x))$. Since $X \cap D(g) = X_0 \cap D(g)$ (**Step 1**) and X_0 has codimension c in $\mathbb{A}^n$, X_1 is an irreducible closed subvariety of codimension $c+1$ of $\mathbb{A}^{n+1}$ defined by the $c+1$ semi-invariant polynomials $f_1, \dots, f_c, sg - 1$, and X_2 is an irreducible closed subvariety of codimension one defined by the invariant polynomial $sg - 1$. Since $\mathcal{O}(\mathbb{A}^n)^G \to \mathcal{O}(X)^G$ is surjective (apply the Reynolds operator $\frac{1}{r}\sum_{\sigma \in G} \sigma^*$, which exists in characteristic zero), X/G is a closed subscheme of $\mathbb{A}^n/G$, and similarly X_1/G and X_2/G are closed subvarieties of $\mathbb{A}^{n+1}/G$. Since g is G-invariant, $(\mathcal{O}(X)_g)^G = (\mathcal{O}(X)^G)_g$. Thus p restricts to isomorphisms

$$X_1/G \cong (X/G) \cap D(g) \quad \text{and} \quad X_2/G \cong D(g) \subset \mathbb{A}^n/G,$$

both sending y_1 to y. Under the isomorphisms $X_1/G \cong (X/G) \cap D(g)$ and $X_2/G \cong D(g)$, $(Z \times \mathbb{A}^1_s) \cap (X_1/G)$ corresponds to $Z \cap (X/G) \cap D(g)$, and $(Z \times \mathbb{A}^1_s) \cap (X_2/G)$ corresponds to $Z \cap D(g)$.

We show that $(X/G) \cap D(g)$ is normal and klt and that $Z \cap (X/G) \cap D(g)$ has codimension at least two in $(X/G) \cap D(g)$. The variety $(X/G) \cap D(g)$ is normal, as the quotient of the normal variety $X \cap D(g)$. Since π is étale over $(\mathbb{A}^n/G)\backslash Z$, the base change $\pi^{-1}(X/G) \to X/G$ is étale over $(X/G)\backslash Z$. Since $(X/G) \cap D(g)$ is normal, hence reduced, $\pi^{-1}(X/G)$ is reduced over $((X/G) \cap D(g))\backslash Z$, and since X is G-invariant, $\pi^{-1}(X/G)$ has support X, so $\pi^{-1}(X/G)$ coincides with X over $((X/G) \cap D(g))\backslash Z$. Since g vanishes on $\pi^{-1}(Z_1) \cap X$, we have $Z_1 \cap D(g) = \emptyset$, so $Z \cap (X/G) \cap D(g)$ has codimension at least two in $(X/G) \cap D(g)$. Thus the restriction $\pi_X : X \cap D(g) \to (X/G) \cap D(g)$ of π is quasi-étale, and $K_{X \cap D(g)} = \pi_X^* K_{(X/G) \cap D(g)}$ by the Hurwitz formula. Since $X \cap D(g)$ is a complete intersection in $D(g) \subset \mathbb{A}^n$, $K_{X \cap D(g)}$ is Cartier. By [KM98, Proposition 5.20(1)], applied with zero boundaries, $K_{(X/G) \cap D(g)}$ is $\mathbb{Q}$-Cartier, and by [KM98, Proposition 5.20(4)], $(X/G) \cap D(g)$ is klt since $X \cap D(g)$ is klt.

We show that Z has codimension at least two in $\mathbb{A}^n/G$ and that $\mathbb{A}^n/G$ is klt. Let P be a prime divisor on $\mathbb{A}^n$. Any element of $I_G(P)$ acts trivially on the residue field of ord_P, so it fixes P pointwise. The fixed locus of a linear map is a linear subspace, so the fixed locus of an element of $I_G(P)$ is a hyperplane or $\mathbb{A}^n$. Since G acts faithfully and contains no pseudoreflections, $I_G(P)$ is trivial. Thus the ramification index of π along P is 1 (Lemma 2.6(1)), and π is étale at the generic point of P since the characteristic is zero. Hence π is quasi-étale, Z has codimension at least two in $\mathbb{A}^n/G$, and $K_{\mathbb{A}^n} = \pi^* K_{\mathbb{A}^n/G}$. By [KM98, Proposition 5.20(1) and (4)], $K_{\mathbb{A}^n/G}$ is $\mathbb{Q}$-Cartier and $\mathbb{A}^n/G$ is klt. In particular, $D(g) \subset \mathbb{A}^n/G$ is normal and klt, and $Z \cap D(g)$ has codimension at least two in $D(g)$.

Step 4. By **Steps 2 and 3**, the hypotheses of [NS26b, Theorem 7.1] hold for the closed subvarieties X_1 and X_2 of $\mathbb{A}^{n+1}$ and the closed point y_1, except for the non-vanishing of the $\mathbb{R}$-ideals, which we check now. Let $\mathfrak{a}' := \mathfrak{a}|_{(\mathbb{A}^n/G) \times \mathbb{A}^1_s}$ be the restriction of $\mathfrak{a}$ along the projection p (Definition 2.20), and let $\mathfrak{b} := (f_1^r \cdots f_c^r)^{\frac{1}{r}} \mathfrak{a}'$. Under the isomorphisms of **Step 3**, $\mathfrak{a}'|_{X_1/G}$ corresponds to $\mathfrak{a}|_{(X/G) \cap D(g)}$, which is non-zero since $\mathfrak{a}|_{X/G}$ is non-zero near y and $(X/G) \cap D(g)$ is irreducible, and $\mathfrak{b}|_{X_2/G}$ corresponds to $\big((f_1^r \cdots f_c^r)^{\frac{1}{r}} \mathfrak{a}\big)|_{D(g)}$, which is non-zero since $f_1, \dots, f_c$ and the factors of $\mathfrak{a}$ are non-zero. By [NS26b, Theorem 7.1] for X_1 with the $\mathbb{R}$-ideal $\mathfrak{a}'$ and for X_2 with the $\mathbb{R}$-ideal $\mathfrak{b}$, we have

$$\mathrm{mld}(X_1/G \ni y_1, \mathfrak{a}'|_{X_1/G}) = \mathrm{mld}\big((\mathbb{A}^n/G) \times \mathbb{A}^1_s \ni y_1, (f_1^r \cdots f_c^r (sg - 1)^r)^{\frac{1}{r}} \mathfrak{a}'\big) \tag{2.2}$$

and

$$\mathrm{mld}(X_2/G \ni y_1, \mathfrak{b}|_{X_2/G}) = \mathrm{mld}\big((\mathbb{A}^n/G) \times \mathbb{A}^1_s \ni y_1, ((sg - 1)^r)^{\frac{1}{r}} (f_1^r \cdots f_c^r)^{\frac{1}{r}} \mathfrak{a}'\big). \tag{2.3}$$

By Definition 2.2, a minimal log discrepancy with an $\mathbb{R}$-ideal $\prod_j \mathfrak{a}_j^{r_j}$ depends only on the function $E \to \sum_j r_j \operatorname{mult}_E \mathfrak{a}_j$. Since

$$\frac{1}{r}\operatorname{mult}_E\left(f_1^r\cdots f_c^r(sg-1)^r\right)=\frac{1}{r}\operatorname{mult}_E(f_1^r\cdots f_c^r)+\frac{1}{r}\operatorname{mult}_E\left((sg-1)^r\right)$$

for any prime divisor E over $(\mathbb{A}^n/G)\times\mathbb{A}^1_s$, the right-hand sides of (2.2) and (2.3) are equal. Since minimal log discrepancies at y only depend on a neighborhood of y, the isomorphisms of **Step 3** imply that the left-hand sides of (2.2) and (2.3) are equal to

$$\operatorname{mld}(X/G\ni y,\mathfrak{a}|_{X/G})\quad\text{and}\quad\operatorname{mld}\big(\mathbb{A}^n/G\ni y,(f_1^r\cdots f_c^r)^{\frac{1}{r}}\mathfrak{a}\big)$$

respectively. This proves the theorem. □

We also use the inversion of adjunction on log canonicity for $\mathbb{R}$-divisors. The different is defined in [FST11, 11.1]. See also [FST11, Theorem 11.3].

Theorem 2.23 ([Kaw07, Theorem]). *Let $(X,S+B)$ be a pair, where S is a reduced divisor with normalization S^ν and $B\geq 0$ has no common component with S, and let B_{S^ν} be the different, $K_{S^\nu}+B_{S^\nu}=(K_X+S+B)|_{S^\nu}$. Then $(X,S+B)$ is lc near S if and only if (S^ν,B_{S^ν}) is lc.*

3. Finite quotients, completions and transverse surfaces

In this section, we prove four lemmas which are used repeatedly in the rest of the paper to move the minimal log discrepancy at a closed point to other models. Lemma 3.1 moves it exactly to a quotient by a group containing the inertia group of a lift of a computing divisor, Lemma 3.3 moves it to the completion of the germ, Lemma 3.4 computes it on a diagonal quotient of a smooth germ as a lattice Newton minimum (Definition 2.16), which is the prototype of the lattice formulas in this paper, and Lemma 3.6 reduces minimal log discrepancies at generic points of curves to minimal log discrepancies at closed points.

3.1. Inertia quotients.

In this subsection, we compare the minimal log discrepancy at a closed point with the minimal log discrepancy of a quotient of a finite Galois cover. In general, passing to the cover and then to the quotient by a subgroup H of the Galois group only bounds the minimal log discrepancy of the quotient from below. If H contains the inertia group of a lift of a divisor computing the minimal log discrepancy, then the minimal log discrepancy of the quotient equals the minimal log discrepancy at the closed point (Lemma 3.1(1)). The comparison of the log discrepancies of individual divisors is Lemma 3.1(2).

Lemma 3.1. *Let (X,B) be a sub-pair, $x\in X$ a closed point, and E a prime divisor over X with $\operatorname{center}_X E=\{x\}$. Let $\pi: X'\to X$ be a finite Galois morphism from a normal variety with Galois group G, let B' be the $\mathbb{R}$-divisor with $K_{X'}+B'=\pi^*(K_X+B)$, let F be a prime divisor over X' with $\operatorname{ord}_F|_{\mathbb{C}(X)}\in\mathbb{N}^+\operatorname{ord}_E$, and let H be a subgroup of G containing $I_G(F)$. Let $f: Y\to X'$ be an H-equivariant birational morphism from a normal quasi-projective variety such that F has a center on Y, let B_Y be the $\mathbb{R}$-divisor with $K_Y+B_Y=f^*(K_{X'}+B')$, let $(Y/H,B_{Y/H}):=(Y,B_Y)/H$, and let η be the generic point of the image of $\operatorname{center}_Y F$ in Y/H. Assume that $\operatorname{mld}(X\ni x,B)\geq 0$ and $\operatorname{mld}(X\ni x,B)=a(E,X,B)$. Then the following hold.*

(1) $\operatorname{mld}(Y/H\ni\eta,B_{Y/H})=\operatorname{mld}(X\ni x,B)$. Moreover, for the prime divisor P_0 over Y/H with $\operatorname{ord}_F|_{\mathbb{C}(Y/H)}\in\mathbb{N}^+\operatorname{ord}_{P_0}$, we have $\operatorname{center}_{Y/H}P_0=\overline{\{\eta\}}$ and

$$a(P_0,Y/H,B_{Y/H})=\operatorname{mld}(Y/H\ni\eta,B_{Y/H}).$$

(2) For any prime divisor P over Y/H with $\operatorname{center}_{Y/H}P=\overline{\{\eta\}}$, any prime divisor F' over Y with $\operatorname{ord}_{F'}|_{\mathbb{C}(Y/H)}\in\mathbb{N}^+\operatorname{ord}_P$, and the prime divisor E' over X with $\operatorname{ord}_{F'}|_{\mathbb{C}(X)}\in\mathbb{N}^+\operatorname{ord}_{E'}$, we have $\operatorname{center}_X E'=\{x\}$ and

$$a(P,Y/H,B_{Y/H})=\frac{|I_G(F')|}{|H\cap I_G(F')|}a(E',X,B)\geq\operatorname{mld}(X\ni x,B).$$

Proof. Since X' is finite over the quasi-projective variety X, X' is quasi-projective, and G acts faithfully on X' with $X = X'/G$. Since $\pi \circ \sigma = \pi$ for any $\sigma \in G$, B' is G-invariant. By Lemma 2.6(0),

$$\pi^*(K_X + B) = K_{X'} + B' = \pi^*(K_X + B_{X'/G}).$$

Since π is finite and surjective, $B = B_{X'/G}$, that is, $(X', B')/G = (X, B)$. Thus we may use Lemma 2.6 for π and G. The group H acts faithfully on $\mathbb{C}(Y) = \mathbb{C}(X')$, hence on Y, and B_Y is H-invariant because f is H-equivariant. Let $q : Y \to Y/H$ be the quotient morphism. Since $K_Y + B_Y$ is $\mathbb{R}$-Cartier, we may use Lemma 2.6 for q and H. Since $X = X'/G$, $\mathbb{C}(Y) = \mathbb{C}(X')$, and $H \subset G$, we have $\mathbb{C}(X) = \mathbb{C}(X')^G \subset \mathbb{C}(Y)^H = \mathbb{C}(Y/H)$.

We first show that $f(q^{-1}(\overline{\{\eta\}})) \subset \pi^{-1}(x)$. Since $\operatorname{ord}_F|_{\mathbb{C}(X)} \in \mathbb{N}^+ \operatorname{ord}_E$, Lemma 2.6(2) for π implies that $\pi(\operatorname{center}_{X'} F) = \operatorname{center}_X E = \{x\}$. Thus

$$f(\operatorname{center}_Y F) \subset \operatorname{center}_{X'} F \subset \pi^{-1}(x).$$

Since q is finite and the fibers of q are H-orbits,

$$\overline{\{\eta\}} = q(\operatorname{center}_Y F) \quad \text{and} \quad q^{-1}(\overline{\{\eta\}}) = \bigcup_{\sigma \in H} \sigma(\operatorname{center}_Y F).$$

Since f is H-equivariant and $\pi^{-1}(x)$ is G-invariant, $f(\sigma(\operatorname{center}_Y F)) = \sigma(f(\operatorname{center}_Y F)) \subset \pi^{-1}(x)$ for any $\sigma \in H$. Hence $f(q^{-1}(\overline{\{\eta\}})) \subset \pi^{-1}(x)$.

We prove (2). Let P, F', and E' be as in (2). By Lemma 2.6(2) for q,

$$q(\operatorname{center}_Y F') = \operatorname{center}_{Y/H} P = \overline{\{\eta\}},$$

so $\operatorname{center}_Y F' \subset q^{-1}(\overline{\{\eta\}})$. By the previous paragraph, $f(\operatorname{center}_Y F') \subset \pi^{-1}(x)$. Since $\pi^{-1}(x)$ is closed,

$$\operatorname{center}_{X'} F' = \overline{f(\operatorname{center}_Y F')} \subset \pi^{-1}(x).$$

By Lemma 2.6(2) for π, $\operatorname{center}_X E' = \pi(\operatorname{center}_{X'} F') \subset \{x\}$, hence $\operatorname{center}_X E' = \{x\}$.

Let $e_G := |I_G(F')|$ and $e_H := |H \cap I_G(F')|$. Since $\operatorname{Stab}_H(F') = H \cap \operatorname{Stab}_G(F')$, and both inertia groups consist of the elements of the decomposition group acting trivially on the residue field of $\operatorname{ord}_{F'}$, we have $I_H(F') = H \cap I_G(F')$. By Lemma 2.6(1)(2) for π and for q,

$$\operatorname{ord}_{F'}|_{\mathbb{C}(X)} = e_G \operatorname{ord}_{E'}, \quad a(F', X', B') = e_G a(E', X, B),$$

$$\operatorname{ord}_{F'}|_{\mathbb{C}(Y/H)} = e_H \operatorname{ord}_P, \quad a(F', Y, B_Y) = e_H a(P, Y/H, B_{Y/H}).$$

Restricting the second line to the subfield $\mathbb{C}(X) \subset \mathbb{C}(Y/H)$ and comparing with the first line, we obtain

$$\operatorname{ord}_P|_{\mathbb{C}(X)} = \frac{e_G}{e_H} \operatorname{ord}_{E'}.$$

Since ord_P takes integer values and $\operatorname{ord}_{E'}$ takes the value 1, e_G/e_H is a positive integer. Since $K_Y + B_Y = f^*(K_{X'} + B')$ and f is birational, $a(F', Y, B_Y) = a(F', X', B')$. Thus

$$a(P, Y/H, B_{Y/H}) = \frac{e_G}{e_H} a(E', X, B).$$

Since $\operatorname{center}_X E' = \{x\}$, $a(E', X, B) \geq \operatorname{mld}(X \ni x, B) \geq 0$. Since $e_G/e_H \geq 1$, we obtain

$$\frac{e_G}{e_H} a(E', X, B) \geq a(E', X, B) \geq \operatorname{mld}(X \ni x, B).$$

This proves (2).

We prove (1). Let P be any prime divisor over Y/H with $\operatorname{center}_{Y/H} P = \overline{\{\eta\}}$. By Lemma 2.6(3) for q, there exists a prime divisor F' over Y with $\operatorname{ord}_{F'}|_{\mathbb{C}(Y/H)} \in \mathbb{N}^+ \operatorname{ord}_P$. Since F' is also a prime divisor over X', Lemma 2.6 for π provides the prime divisor E' over X with $\operatorname{ord}_{F'}|_{\mathbb{C}(X)} \in \mathbb{N}^+ \operatorname{ord}_{E'}$. By (2), $a(P, Y/H, B_{Y/H}) \geq \operatorname{mld}(X \ni x, B)$. Hence $\operatorname{mld}(Y/H \ni \eta, B_{Y/H}) \geq \operatorname{mld}(X \ni x, B)$.

By Lemma 2.6(2) for q, $\operatorname{center}_{Y/H} P_0 = q(\operatorname{center}_Y F) = \overline{\{\eta\}}$. We use (2) with $P = P_0$ and $F' = F$. Then $E' = E$, and $H \cap I_G(F) = I_G(F)$ because $I_G(F) \subset H$. Thus

$$a(P_0, Y/H, B_{Y/H}) = a(E, X, B) = \operatorname{mld}(X \ni x, B).$$

Together with $\operatorname{mld}(Y/H \ni \eta, B_{Y/H}) \geq \operatorname{mld}(X \ni x, B)$, this proves (1). □

3.2. **Completions.** In this subsection, we show that minimal log discrepancies of finite quotients of a germ can be computed on the completion of the germ. We use the canonical divisors of schemes over complete local rings defined in [dFEM11, Appendix A].

Definition 3.2. Let $k \supset \mathbb{C}$ be a field, $(X \ni x)$ a germ of a normal variety over k at a closed point x, $\widehat{\mathcal{O}}$ the completion of $\mathcal{O}_{X,x}$, $\widehat{X} := \operatorname{Spec} \widehat{\mathcal{O}}$, and $\hat{x}$ the closed point of $\widehat{X}$. A *formal prime divisor* over $\widehat{X}$ is a prime divisor on a normal scheme which is proper and birational over $\widehat{X}$. Let $\widehat{B} = \sum_j c_j \widehat{D}_j$ be an $\mathbb{R}$-divisor on $\widehat{X}$ such that $K_{\widehat{X}} + \widehat{B}$ is $\mathbb{R}$-Cartier, where the c_j are real numbers, the $\widehat{D}_j$ are prime divisors, and $K_{\widehat{X}}$ is the canonical divisor of [dFEM11, Appendix A]. The *log discrepancy* $a(P, \widehat{X}, \widehat{B})$ of a formal prime divisor P over $\widehat{X}$ is defined as in Definition 2.1 with the canonical divisors of [dFEM11, Appendix A] (as in [KM98, Remark 2.23], the coefficient of P may be computed on any normal model on which P is a divisor). The *minimal log discrepancy* $\operatorname{mld}(\widehat{X} \ni \hat{x}, \widehat{B})$ is the infimum of $a(P, \widehat{X}, \widehat{B})$ over the formal prime divisors P over $\widehat{X}$ with center $\hat{x}$.

Let H be a finite group acting faithfully on X and fixing x, with the induced action on $\widehat{X}$, and assume that $\widehat{B}$ is H-invariant. The decomposition group $\operatorname{Stab}_H(P)$ and the inertia group $I_H(P)$ of a formal prime divisor P are defined as in Definition 2.5. We define

$$\operatorname{mld}((\widehat{X} \ni \hat{x}, \widehat{B})/H) := \inf\{a(P, \widehat{X}, \widehat{B})/|I_H(P)| \mid P \text{ is a formal prime divisor over } \widehat{X} \text{ with center } \hat{x}\}.$$

By Lemma 2.6 and Remark 2.17, the formula defining $\operatorname{mld}((\widehat{X} \ni \hat{x}, \widehat{B})/H)$ in Definition 3.2, with prime divisors over X with center x in place of formal prime divisors, is the formula for $\operatorname{mld}((X \ni x, B)/H)$. For $\mathbb{Q}$-Gorenstein X, $\mathbb{R}$-ideals, and trivial H, the equality of the first part of Lemma 3.3(2) is noted in [NS25, Section 3 and Remark 3.1].

Lemma 3.3. *Let $k \supset \mathbb{C}$ be a field, $(X \ni x)$ a germ of a normal variety over k, and B an $\mathbb{R}$-divisor on X such that $K_X + B$ is $\mathbb{R}$-Cartier near x. Let H be a finite group acting faithfully on X, fixing x and preserving B. Let $\widehat{\mathcal{O}}$, $\widehat{X}$, and $\hat{x}$ be as in Definition 3.2, with the induced action of H, and let $\widehat{B}$ be the pullback of B to $\widehat{X}$. Then $K_{\widehat{X}} + \widehat{B}$ is the pullback of $K_X + B$ and hence $\mathbb{R}$-Cartier, and the following hold.*

(1) The completion of the local ring of X/H at the image of x is $\widehat{\mathcal{O}}^H$.
(2) $\operatorname{mld}(\widehat{X} \ni \hat{x}, \widehat{B}) = \operatorname{mld}(X \ni x, B)$ and $\operatorname{mld}((\widehat{X} \ni \hat{x}, \widehat{B})/H) = \operatorname{mld}((X \ni x, B)/H)$.
(3) Any formal prime divisor over $\widehat{X}$ with center $\hat{x}$ restricts to a prime divisor over X with center x which has isomorphic residue field, equal log discrepancy with respect to $\widehat{B}$ and B, and equal decomposition group and inertia group, and any prime divisor over X with center x arises in this way.

Proof. Let U be an affine open neighborhood of x on which $K_X + B$ is $\mathbb{R}$-Cartier. Since x is fixed by H and X is separated, $\bigcap_{\sigma \in H} \sigma(U)$ is an H-invariant affine open neighborhood of x. Possibly replacing X with $\bigcap_{\sigma \in H} \sigma(U)$, we may assume that X is affine and $K_X + B$ is $\mathbb{R}$-Cartier. In particular, X/H exists, and we may use Definitions 2.1, 2.2, and 2.5 and Lemma 2.6 for (X, B) and H over k (Remark 2.17). Write $\mathcal{O} := \mathcal{O}_{X,x}$, let $\mathfrak{m}$ be the maximal ideal of $\mathcal{O}$, and let $g : \widehat{X} \to X$ be the natural morphism. Since $\mathcal{O}$ is an excellent normal local ring [Gro65, Scholie 7.8.3(ii)(iii)], $\mathcal{O} \to \widehat{\mathcal{O}}$ is faithfully flat [Mat89, Theorem 8.14] and regular [Gro65, Scholie 7.8.3(v)], hence $\widehat{\mathcal{O}}$ is normal [Mat89, Theorem 32.2(i)], and g is flat and H-equivariant. Thus $\widehat{B} = g^*B$ is an

H-invariant $\mathbb{R}$-divisor on $\widehat{X}$. By [dFEM11, Proposition A.14], $K_{\widehat{X}} = g^*K_X$. Since pullback of $\mathbb{R}$-Cartier $\mathbb{R}$-divisors commutes with flat pullback of $\mathbb{R}$-divisors,

$$K_{\widehat{X}} + \widehat{B} = g^*(K_X + B),$$

which is $\mathbb{R}$-Cartier. We prove (1) and (3), and then deduce (2) from (3).

We prove (1). Write $X = \operatorname{Spec} A$, so $X/H = \operatorname{Spec} A^H$. Let $\mathfrak{n} := \mathfrak{m} \cap A^H$ and $\mathcal{O}' := (A^H)_{\mathfrak{n}}$, the local ring of X/H at the image of x. The fiber of $X \to X/H$ over the image of x is the H-orbit of x, that is, $\{x\}$. Hence the localization of A at the multiplicative set $A^H \backslash \mathfrak{n}$ is $\mathcal{O}$, the ring $\mathcal{O}$ is finite over $\mathcal{O}'$ with $\mathcal{O}^H = \mathcal{O}'$, and $\mathfrak{n}\mathcal{O}$ is $\mathfrak{m}$-primary. Thus the $\mathfrak{n}$-adic and $\mathfrak{m}$-adic topologies on $\mathcal{O}$ coincide, and $\widehat{\mathcal{O}} = \mathcal{O} \otimes_{\mathcal{O}'} \widehat{\mathcal{O}'}$ [Mat89, Theorem 8.7]. The Reynolds operator $\frac{1}{|H|}\sum_{\sigma \in H} \sigma$ is an $\mathcal{O}'$-linear projection of $\mathcal{O}$ onto $\mathcal{O}'$. Since $\widehat{\mathcal{O}'}$ is flat over $\mathcal{O}'$ [Mat89, Theorem 8.8], the base change of the Reynolds operator of $\mathcal{O}$ to $\widehat{\mathcal{O}'}$ is the Reynolds operator of $\widehat{\mathcal{O}}$ and is a projection of $\widehat{\mathcal{O}}$ onto $\widehat{\mathcal{O}'}$. The image of the Reynolds operator of $\widehat{\mathcal{O}}$ is $\widehat{\mathcal{O}}^H$, hence $\widehat{\mathcal{O}}^H = \widehat{\mathcal{O}'}$.

We prove (3). Let P be a formal prime divisor over $\widehat{X}$ with center $\hat{x}$. By [dFEM11, Lemma 2.21, Remark A.15], there exist a proper birational morphism $h : Y \to X$ from a smooth variety Y and a smooth prime divisor E on Y with $h(E) = \{x\}$ with the following property: for $W := Y \times_X \widehat{X}$ with projections $g_Y : W \to Y$ and $h_W : W \to \widehat{X}$, the divisor g_Y^*E is reduced and equal to the prime divisor P, and $\operatorname{ord}_P|_{k(X)} = \operatorname{ord}_E$. Conversely, let E be a prime divisor over X with $\operatorname{center}_X E = \{x\}$, and let $h : Y \to X$ be a log resolution of X on which E is a smooth prime divisor. Then W is regular [dFEM11, Remark 2.10] and proper and birational over $\widehat{X}$, and by [dFEM11, Remark A.15], g_Y^*E is a prime divisor P on W with $\operatorname{ord}_P|_{k(X)} = \operatorname{ord}_E$. Since $h_W(P) = \{\hat{x}\}$, P is a formal prime divisor over $\widehat{X}$ with center $\hat{x}$. Thus restriction to $k(X)$ maps the formal prime divisors over $\widehat{X}$ with center $\hat{x}$ onto the prime divisors over X with center x.

Let P be a formal prime divisor over $\widehat{X}$ with center $\hat{x}$, and let $h : Y \to X$, E, W, g_Y, and h_W be given for P by [dFEM11, Lemma 2.21, Remark A.15], so that $g_Y^*E = P$ and $\operatorname{ord}_P|_{k(X)} = \operatorname{ord}_E$. Let $v := \operatorname{ord}_P$, and $v_0 := \operatorname{ord}_E = v|_{k(X)}$. For $m \in \mathbb{Z}_{\geq 0}$ let

$$\mathfrak{a}_m := \{u \in \widehat{\mathcal{O}} \mid v(u) \geq m\} \quad \text{and} \quad \mathfrak{b}_m := \mathfrak{a}_m \cap \mathcal{O} = \{u \in \mathcal{O} \mid v_0(u) \geq m\}.$$

Since v is centered at $\hat{x}$, $v(u) \geq 1$ for any $u \in \mathfrak{m}\widehat{\mathcal{O}}$, so $\mathfrak{m}^m\widehat{\mathcal{O}} \subset \mathfrak{a}_m$. Since $\widehat{\mathcal{O}}/\mathfrak{m}^m\widehat{\mathcal{O}} = \mathcal{O}/\mathfrak{m}^m$, the maps $J \to J \cap \mathcal{O}$ and $I \to I\widehat{\mathcal{O}}$ are mutually inverse bijections between the ideals of $\widehat{\mathcal{O}}$ containing $\mathfrak{m}^m\widehat{\mathcal{O}}$ and the ideals of $\mathcal{O}$ containing $\mathfrak{m}^m$. Hence

$$\mathfrak{a}_m = \mathfrak{b}_m\widehat{\mathcal{O}} \quad \text{and} \quad \widehat{\mathcal{O}}/\mathfrak{a}_{m+1} = \mathcal{O}/\mathfrak{b}_{m+1}$$

for any m, and the inclusion $\mathcal{O} \subset \widehat{\mathcal{O}}$ induces $\mathfrak{b}_m/\mathfrak{b}_{m+1} = \mathfrak{a}_m/\mathfrak{a}_{m+1}$. Since $v(u) = \max\{m \mid u \in \mathfrak{a}_m\}$ for any non-zero $u \in \widehat{\mathcal{O}}$, the valuation v is determined by the ideals $\mathfrak{b}_m$, that is, by v_0. In particular, restriction to $k(X)$ is injective on the formal prime divisors over $\widehat{X}$ with center $\hat{x}$. For $\sigma \in H$, σ fixes v if and only if $\sigma(\mathfrak{a}_m) = \mathfrak{a}_m$ for any m, if and only if $\sigma(\mathfrak{b}_m) = \mathfrak{b}_m$ for any m, if and only if σ fixes v_0. Thus $\operatorname{Stab}_H(P) = \operatorname{Stab}_H(E)$.

Any element of the residue field $\kappa(v)$ of v is the residue of u_1/u_2 for some m and some $u_1, u_2 \in \mathfrak{a}_m \backslash \mathfrak{a}_{m+1}$, and the residue of u_1/u_2 depends only on the classes of u_1 and u_2 in $\mathfrak{a}_m/\mathfrak{a}_{m+1}$. Since $\mathfrak{a}_m/\mathfrak{a}_{m+1} = \mathfrak{b}_m/\mathfrak{b}_{m+1}$, we may choose $u_1, u_2 \in \mathcal{O}$. Hence the natural inclusion of the residue field $\kappa(v_0)$ of v_0 into $\kappa(v)$ is an isomorphism, and it is compatible with the actions of $\operatorname{Stab}_H(P) = \operatorname{Stab}_H(E)$. Thus $I_H(P) = I_H(E)$.

It remains to compare log discrepancies. Let B_Y be the $\mathbb{R}$-divisor with $K_Y + B_Y = h^*(K_X + B)$. By [dFEM11, Proposition A.10], taking the top exterior power of the special differentials, $K_W = g_Y^*K_Y$. Since $g \circ h_W = h \circ g_Y$,

$$K_W + g_Y^*B_Y = g_Y^*h^*(K_X + B) = h_W^*g^*(K_X + B) = h_W^*(K_{\widehat{X}} + \widehat{B}).$$

Since $g_Y^* E = P$, the coefficient of P in $g_Y^* B_Y$ is the coefficient of E in B_Y. Hence $a(P, \widehat{X}, \widehat{B}) = a(E, X, B)$ (as in [KM98, Remark 2.23], the coefficient of P may be computed on any normal model on which P is a divisor, here the regular scheme W). This proves (3).

We prove (2). By (3), restriction to $k(X)$ is a bijection from the set of formal prime divisors P over $\widehat{X}$ with center $\hat{x}$ to the set of prime divisors E over X with $\operatorname{center}_X E = \{x\}$, and it preserves log discrepancies and inertia groups. Hence

$$\{a(P, \widehat{X}, \widehat{B})/|I_H(P)|\}_P = \{a(E, X, B)/|I_H(E)|\}_E,$$

where P runs over the formal prime divisors over $\widehat{X}$ with center $\hat{x}$ and E runs over the prime divisors over X with $\operatorname{center}_X E = \{x\}$. By Definition 3.2 and Lemma 2.6, the infima of the left-hand side and the right-hand side are $\operatorname{mld}((\widehat{X} \ni \hat{x}, \widehat{B})/H)$ and $\operatorname{mld}((X \ni x, B)/H)$, respectively. If H is the trivial group, then all inertia groups are trivial, and by Definitions 3.2 and 2.2, the infima of the left-hand side and the right-hand side are $\operatorname{mld}(\widehat{X} \ni \hat{x}, \widehat{B})$ and $\operatorname{mld}(X \ni x, B)$, respectively. Hence $\operatorname{mld}(\widehat{X} \ni \hat{x}, \widehat{B}) = \operatorname{mld}(X \ni x, B)$. □

3.3. **Diagonal quotients of smooth germs.** In this subsection, we compute the minimal log discrepancy of the quotient of a smooth germ by a finite abelian group acting diagonally on eigenparameters, for a boundary supported on the coordinate hyperplanes.

In Lemma 3.4, the right-hand side of the formula is the lattice Newton minimum $\psi_{\min}(H, ((x_1), \dots, (x_d)), (b_1, \dots, b_d))$ of Definition 2.16, where H is the image of G in $\mathbb{T}^d$, since $\operatorname{wt}_w((x_i)) = w_i$. The lower bound in Lemma 3.4 follows from Lemma 2.13 and the fact that $(\operatorname{ord}_F x_i)_i/|I_G(F)|$ belongs to N_G for any prime divisor F over X with $\operatorname{center}_X F = \{x\}$, and the upper bound is attained by a monomial prime divisor. The lattice formulas of Propositions 9.4, 9.5, and 10.5 are proved by reducing to Lemma 3.4 after a change of coordinates. When $k = \mathbb{C}$ and $b_i \in [0, 1)$ for any i, compare the toric formula [Amb06, Section 2, pp. 360–361].

Lemma 3.4. *Let $k \supset \mathbb{C}$ be a field, X a normal quasi-projective variety over k, and $x \in X$ a closed point at which X is smooth, with $d := \dim \mathcal{O}_{X,x}$. Let G be a finite abelian group acting faithfully on X by k-automorphisms, fixing x and acting trivially on the residue field $\kappa(x)$. Let $x_1, \dots, x_d \in \mathcal{O}_{X,x}$ be G-eigenparameters, let N_G be the lattice of Definition 2.15 for their characters, and let $B = \sum_{i=1}^d b_i \operatorname{div}(x_i)$ with real numbers $b_i < 1$. Then*

$$\operatorname{mld}((X \ni x, B)/G) = \min\Big\{ \sum_{i=1}^d (1 - b_i) w_i \;\Big|\; w \in N_G \cap \mathbb{R}^d_{>0} \Big\}.$$

Moreover, let $w \in N_G \cap \mathbb{R}^d_{>0}$ attain the minimum, and let m be the least positive integer with $mw \in \mathbb{Z}^d$. Then w is primitive in N_G, the monomial prime divisor F in $x_1, \dots, x_d$ with weights mw exists (Definition 2.16), and the prime divisor E over X/G with $\operatorname{ord}_F|_{k(X/G)} \in \mathbb{N}^+ \operatorname{ord}_E$ satisfies $\operatorname{ord}_F|_{k(X/G)} = m \operatorname{ord}_E$ and attains the minimum, where $k(X/G)$ is the function field of X/G.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we describe N_G by invariant monomials and shrink X.

Let $\chi_1, \dots, \chi_d$ be the characters of $x_1, \dots, x_d$, so that $\sigma^* x_i = \chi_i(\sigma) x_i$ for any $\sigma \in G$ (Definition 2.14 and Remark 2.17). For $\alpha \in \mathbb{Z}^d$ let $x^\alpha := \prod_i x_i^{\alpha_i}$ and $\chi^\alpha := \prod_i \chi_i^{\alpha_i}$, and let

$$M_G := \{\alpha \in \mathbb{Z}^d \mid \chi^\alpha(\sigma) = 1 \text{ for any } \sigma \in G\}.$$

Let $k(X)$ be the function field of X. Then $x^\alpha \in k(X)^G$ for any $\alpha \in M_G$. We show that

$$N_G = \{u \in \mathbb{R}^d \mid \langle u, \alpha \rangle \in \mathbb{Z} \text{ for any } \alpha \in M_G\}. \tag{3.1}$$

If $u \in \mathbb{R}^d$ and $\sigma \in G$ satisfy $\chi_i(\sigma) = e^{2\pi\sqrt{-1}u_i}$ for any i, then $e^{2\pi\sqrt{-1}\langle u, \alpha\rangle} = \chi^\alpha(\sigma) = 1$ for any $\alpha \in M_G$. Thus the left-hand side of (3.1) is contained in the right-hand side. Conversely, let H

be the image of G in $\mathbb{T}^d$, and let Λ be the right-hand side of (3.1). Since $|G|\mathbb{Z}^d \subset M_G$, we have $\Lambda \subset \frac{1}{|G|}\mathbb{Z}^d$. The pairing

$$H \times \mathbb{Z}^d \to \tfrac{1}{|G|}\mathbb{Z}/\mathbb{Z}, \quad (h, \alpha) \to \langle h, \alpha \rangle$$

has left kernel zero and right kernel M_G, so $|\mathbb{Z}^d/M_G| = |H|$ by [Lan02, Theorem I.9.2]. The pairing

$$(\Lambda/\mathbb{Z}^d) \times (\mathbb{Z}^d/M_G) \to \tfrac{1}{|G|}\mathbb{Z}/\mathbb{Z}, \quad (u, \alpha) \to \langle u, \alpha \rangle$$

has both kernels zero, so $|\Lambda/\mathbb{Z}^d| = |\mathbb{Z}^d/M_G|$ by [Lan02, Theorem I.9.2]. By Definition 2.15, $N_G/\mathbb{Z}^d = H$, and $\mathbb{Z}^d \subset N_G \subset \Lambda$. Thus $N_G = \Lambda$. In particular, $\mathbb{Z}^d \subset N_G$ and $|G|N_G \subset \mathbb{Z}^d$. For $w \in \mathbb{R}^d$ let

$$s(w) := \sum_{i=1}^{d} (1 - b_i) w_i.$$

Let $U \subset X$ be an open neighborhood of x such that U is smooth, any x_i is regular on U, and $\sum_i \operatorname{div}(x_i)$ is an snc divisor on U. Since $\sigma^* x_i = \chi_i(\sigma) x_i$, the open subset $U' := \bigcap_{\sigma \in G} \sigma(U)$ is smooth, any x_i is regular on U', and $\sum_i \operatorname{div}(x_i)$ is an snc divisor on U'. Moreover, U' is G-invariant and quasi-projective, and U'/G is an open neighborhood of $\pi(x)$ in X/G, where $\pi : X \to X/G$ is the quotient morphism. Since $\operatorname{mld}((X \ni x, B)/G)$ only depends on a neighborhood of $\pi(x)$, we may replace X with U'. Then X is smooth, any x_i is regular on X, and $\sum_i \operatorname{div}(x_i)$ is an snc divisor on X. Moreover, B is G-invariant and $K_X + B$ is $\mathbb{R}$-Cartier, so we may use Lemma 2.6 for (X, B) and G, and Lemma 2.13 for X and $x_1, \dots, x_d$ (Remark 2.17). Finally, $k(X/G) = k(X)^G$, since $\mathcal{O}_{X/G} = (\pi_* \mathcal{O}_X)^G$ (cf. [Mum74, §7, Theorem]) and $\operatorname{Frac}(A^G) = \operatorname{Frac}(A)^G$ for a domain A with an action of G [AM69, Chapter 5, Exercise 12].

Step 2. In this step we prove that $\operatorname{mld}((X \ni x, B)/G) \geq \min\{s(w) \mid w \in N_G \cap \mathbb{R}^d_{>0}\}$ and that the minimum exists.

Let F be a prime divisor over X with $\operatorname{center}_X F = \{x\}$, let $e := |I_G(F)|$, and let E be the prime divisor over X/G with $\operatorname{ord}_F|_{k(X/G)} = e \operatorname{ord}_E$ (Lemma 2.6(1)). For any $\alpha \in M_G$ we have $x^\alpha \in k(X/G)$, hence

$$\sum_{i=1}^{d} \alpha_i \operatorname{ord}_F x_i = \operatorname{ord}_F(x^\alpha) \in e\mathbb{Z}.$$

By (3.1), $u := \frac{1}{e}(\operatorname{ord}_F x_1, \dots, \operatorname{ord}_F x_d)$ belongs to N_G, and $u \in \mathbb{R}^d_{>0}$ since $x_i \in \mathfrak{m}_x$. By Lemma 2.13,

$$a(F, X, B) \geq \sum_{i=1}^{d} (1 - b_i) \operatorname{ord}_F x_i = e \cdot s(u).$$

Since F is arbitrary, the formula for $\operatorname{mld}((X \ni x, B)/G)$ at the end of Lemma 2.6 implies that

$$\operatorname{mld}((X \ni x, B)/G) = \inf_F \frac{a(F, X, B)}{|I_G(F)|} \geq \inf\{s(w) \mid w \in N_G \cap \mathbb{R}^d_{>0}\}.$$

Since $|G|N_G \subset \mathbb{Z}^d$ and $1 - b_i > 0$ for any i, for any real number r the set

$$\{w \in N_G \cap \mathbb{R}^d_{>0} \mid s(w) \leq r\}$$

is finite. Thus the infimum on the right-hand side is a minimum, which we denote by $s_{\min}$. If $s(w) = s_{\min}$ and $w = jw'$ with $w' \in N_G$ and an integer $j \geq 2$, then $w' \in \mathbb{R}^d_{>0}$ and $s(w') = s(w)/j < s_{\min}$, which is not possible. Thus any $w \in N_G \cap \mathbb{R}^d_{>0}$ with $s(w) = s_{\min}$ is primitive in N_G.

Step 3. In this step we fix $w \in N_G \cap \mathbb{R}^d_{>0}$ attaining $s_{\min}$, which is primitive in N_G by **Step 2**, let $n := mw$, and compute the values of the monomial valuation v in $x_1, \dots, x_d$ with weights n on $k(X)$ and on $k(X/G)$.

First, we show that $n \in \mathbb{Z}^d_{>0}$ is primitive in $\mathbb{Z}^d$. Let g be the greatest common divisor of $n_1, \dots, n_d$. If a prime number p divides both g and m, then $\frac{m}{p}w = \frac{1}{p}n \in \mathbb{Z}^d$, which contradicts the minimality of m. Thus there exist integers μ, ρ with $\mu g + \rho m = 1$, and

$$\frac{1}{g}w = \mu w + \rho \cdot \frac{1}{g}n \in N_G,$$

since $\mathbb{Z}^d \subset N_G$. Since w is primitive, $g = 1$.

Second, $\{\langle n, \alpha\rangle \mid \alpha \in M_G\} = m\mathbb{Z}$. By (3.1), $\{\langle w, \alpha\rangle \mid \alpha \in M_G\} = j\mathbb{Z}$ for some positive integer j, and again by (3.1), $\frac{1}{j}w \in N_G$. Since w is primitive, $j = 1$. Thus $\{\langle n, \alpha\rangle \mid \alpha \in M_G\} = m\mathbb{Z}$.

Third, we describe the action of G on the completion $\widehat{\mathcal{O}}_{X,x}$. Since $\kappa(x)$ is a finite separable extension of k, it is 0-étale over k [Mat89, Theorem 25.3]. Thus k is a quasi-coefficient field of $\mathcal{O}_{X,x}$, and $\widehat{\mathcal{O}}_{X,x}$ has a unique coefficient field containing k [Mat89, Theorem 28.3(iv)]. Equivalently, there exists a unique homomorphism of k-algebras $\iota : \kappa(x) \to \widehat{\mathcal{O}}_{X,x}$ whose composition with the residue map is the identity. Since G fixes x, any σ^* extends to a continuous k-algebra automorphism of $\widehat{\mathcal{O}}_{X,x}$. Since G acts trivially on $\kappa(x)$, $\sigma^* \circ \iota$ is again such a homomorphism, hence $\sigma^* \circ \iota = \iota$ for any $\sigma \in G$. By [Mat89, Section 29, first paragraph], $\widehat{\mathcal{O}}_{X,x}$ is a quotient of $\iota(\kappa(x))[[X_1, \dots, X_d]]$ by the homomorphism sending X_i to x_i. Since $\widehat{\mathcal{O}}_{X,x}$ is a regular local ring of dimension d and $\iota(\kappa(x))[[X_1, \dots, X_d]]$ is a domain of dimension d, the kernel is zero, so $\widehat{\mathcal{O}}_{X,x} = \iota(\kappa(x))[[x_1, \dots, x_d]]$. For $f = \sum_\alpha c_\alpha x^\alpha$ with $c_\alpha \in \iota(\kappa(x))$ we obtain

$$\sigma^* f = \sum_\alpha \chi^\alpha(\sigma) c_\alpha x^\alpha.$$

The value $v(f)$ may be computed with the expansions in $\iota(\kappa(x))[[x_1, \dots, x_d]]$, since $v(f) \geq r$ if and only if f belongs to the ideal of $\widehat{\mathcal{O}}_{X,x}$ generated by the monomials x^α with $\langle n, \alpha\rangle \geq r$, which does not depend on the coefficient field.

Fourth, $v(k(X/G)^*) = m\mathbb{Z}$. Let $f \in \mathcal{O}^G_{X,x}$ be non-zero, with expansion $f = \sum_\alpha c_\alpha x^\alpha$ in $\iota(\kappa(x))[[x_1, \dots, x_d]]$. Since $\sigma^* f = f$ for any $\sigma \in G$, $c_\alpha \neq 0$ implies $\alpha \in M_G$. Thus $v(f) \in m\mathbb{Z}$. For any $a \in k(X)^G$, choose a non-zero $h \in \mathcal{O}_{X,x}$ with $ha \in \mathcal{O}_{X,x}$. Since G fixes x, $h' := \prod_{\sigma \in G} \sigma^* h$ belongs to $\mathcal{O}^G_{X,x}$, and $a = (h'a)/h'$ with $h'a \in \mathcal{O}^G_{X,x}$. Thus $k(X/G)$ is the fraction field of $\mathcal{O}^G_{X,x}$, and $v(k(X/G)^*) \subset m\mathbb{Z}$. Conversely, choose $\alpha \in M_G$ with $\langle n, \alpha\rangle = m$. Then $x^\alpha \in k(X/G)$ and $v(x^\alpha) = m$. Thus $v(k(X/G)^*) = m\mathbb{Z}$.

Step 4. In this step we show that $v = \mathrm{ord}_F$ for a prime divisor F over X with $\mathrm{center}_X F = \{x\}$, compute $a(F, X, B)$ and $|I_G(F)|$, and conclude the proof.

Since v takes non-negative integer values on $\mathcal{O}_{X,x}\backslash\{0\}$ and $k(X)$ is the fraction field of $\mathcal{O}_{X,x}$, $v(k(X)^*) \subset \mathbb{Z}$. Since $v(x^\beta) = \langle n, \beta\rangle$ for $\beta \in \mathbb{Z}^d$ and n is primitive in $\mathbb{Z}^d$, $v(k(X)^*) = \mathbb{Z}$. Since $v \geq 0$ on $\mathcal{O}_{X,x}$ and $v > 0$ on $\mathfrak{m}_x$, the residue field $\kappa(v)$ of v contains $\kappa(x)$. Let $\beta^1, \dots, \beta^{d-1}$ be a basis of $\{\beta \in \mathbb{Z}^d \mid \langle n, \beta\rangle = 0\}$, and let $\tau_j \in \kappa(v)$ be the residue of x^{β^j}. We show that $\tau_1, \dots, \tau_{d-1}$ are algebraically independent over $\kappa(x)$. Let $P = \sum_\lambda c_\lambda T^\lambda \in \kappa(x)[T_1, \dots, T_{d-1}]$ be non-zero, let $\tilde{c}_\lambda \in \mathcal{O}_{X,x}$ be lifts of c_λ, and choose $\gamma \in \mathbb{Z}^d$ such that $\gamma + \sum_j \lambda_j \beta^j \in \mathbb{Z}^d_{\geq 0}$ for any λ with $c_\lambda \neq 0$. Then

$$Q := x^\gamma \sum_\lambda \tilde{c}_\lambda x^{\sum_j \lambda_j \beta^j} \in \mathcal{O}_{X,x}.$$

Since $\tilde{c}_\lambda - \iota(c_\lambda) \in \mathfrak{m}_x \widehat{\mathcal{O}}_{X,x}$, the expansion of Q is the sum of $\sum_\lambda \iota(c_\lambda) x^{\gamma + \sum_j \lambda_j \beta^j}$ and a power series whose monomials have n-weight at least $\langle n, \gamma\rangle + \min_i n_i$. The exponents $\gamma + \sum_j \lambda_j \beta^j$ are pairwise distinct, since the β^j are linearly independent, and they have n-weight $\langle n, \gamma\rangle$. Thus $v(Q) = \langle n, \gamma\rangle = v(x^\gamma)$, so Q/x^γ has value zero and residue $P(\tau_1, \dots, \tau_{d-1}) \neq 0$. Hence

$$\mathrm{trdeg}_k\, \kappa(v) = \mathrm{trdeg}_{\kappa(x)}\, \kappa(v) \geq d - 1 = \dim X - 1,$$

and equality holds by Abhyankar's inequality [Bou72, Chapter VI, §10, no. 3, Corollary 1 to Theorem 1], applied to the extension $k \subset k(X)$, since v is trivial on k and $v(k(X)^*) \neq 0$. By

[KM98, Lemma 2.45], the valuation ring of v is the local ring of a prime divisor F over X. Since $v(k(X)^*) = \mathbb{Z}$, $v = \operatorname{ord}_F$, and $\operatorname{center}_X F = \{x\}$ since $v \geq 0$ on $\mathcal{O}_{X,x}$ and $v > 0$ on $\mathfrak{m}_x$. Thus F is the monomial prime divisor in $x_1, \dots, x_d$ with weights $n = mw$ (Definition 2.16).

By [JM12, Proposition 3.1, Proposition 5.1, and (5.1)], applied to the snc divisor $\sum_i \operatorname{div}(x_i)$ on X and the monomial valuation v, we have $a(F, X, 0) = \sum_i n_i$. Since X is smooth and $B = \sum_i b_i \operatorname{div}(x_i)$,

$$a(F, X, B) = a(F, X, 0) - \sum_{i=1}^{d} b_i \operatorname{ord}_F x_i = \sum_{i=1}^{d} (1 - b_i) n_i = m \cdot s(w).$$

Let E be the prime divisor over X/G with $\operatorname{ord}_F |_{k(X/G)} = e \operatorname{ord}_E$. Since $\operatorname{ord}_E(k(X/G)^*) = \mathbb{Z}$ and $v(k(X/G)^*) = m\mathbb{Z}$ by **Step 3**, $e = m$. By Lemma 2.6(1)(2), $|I_G(F)| = m$, $\operatorname{center}_{X/G} E = \{\pi(x)\}$, and

$$a(E, X/G, B_{X/G}) = \frac{a(F, X, B)}{m} = s(w) = s_{\min}.$$

Together with **Step 2**, $\operatorname{mld}((X \ni x, B)/G) = s_{\min}$, and E attains it. □

3.4. **Transverse surfaces.** In this subsection, we reduce minimal log discrepancies at generic points of curves on threefolds to minimal log discrepancies at closed points of surfaces over $\mathbb{C}(t)$ and over an algebraic closure of $\mathbb{C}(t)$, and deduce the ACC for positive minimal log discrepancies at generic points of curves on threefold pairs with coefficients in a DCC subset of $[0, 1]$ (Lemma 3.7). We state Lemma 3.6 for sub-pairs, since Proposition 16.16 uses it for a sub-pair.

Lemma 3.5. *Let $k \supset \mathbb{C}$ be a field, (X, B) a sub-pair over k, and $x \in X$ a closed point. Let $f : Y \to X$ be a projective birational morphism from a smooth variety Y such that $\mathfrak{m}_x \mathcal{O}_Y$ is invertible and $f^{-1}(x) \cup \operatorname{Exc}(f) \cup \operatorname{Supp} f_*^{-1} B$ is an snc divisor, let B_Y be the $\mathbb{R}$-divisor with $K_Y + B_Y = f^*(K_X + B)$, and let $E_1, \dots, E_N$ be the irreducible components of $f^{-1}(x)$. Assume that $\operatorname{mult}_P B_Y \leq 1$ for any prime divisor P on Y with $P \cap f^{-1}(x) \neq \emptyset$. Then*

$$\operatorname{mld}(X \ni x, B) = \min_{1 \leq j \leq N} a(E_j, X, B).$$

Proof. Any E_j has center $\{x\}$ on X, so the left-hand side is at most the right-hand side. Let D be a prime divisor over X with $\operatorname{center}_X D = \{x\}$. Then $\operatorname{center}_Y D \subset f^{-1}(x)$. If $\operatorname{center}_Y D$ is a divisor, then $\operatorname{center}_Y D = E_j$ for some j and $\operatorname{ord}_D = \operatorname{ord}_{E_j}$. Otherwise let ξ be the generic point of $\operatorname{center}_Y D$ and $c \geq 2$ its codimension in Y. Since $\operatorname{Supp} B_Y \subset \operatorname{Exc}(f) \cup \operatorname{Supp} f_*^{-1} B$, $f^{-1}(x) \cup \operatorname{Supp} B_Y$ is an snc divisor. Since $\xi \in f^{-1}(x)$, there exist regular parameters $z_1, \dots, z_c$ of $\mathcal{O}_{Y,\xi}$ and an open neighborhood $U \subset Y$ of ξ such that U is smooth, any z_i is regular on U, $\sum_{i=1}^{c} \operatorname{div}(z_i)$ is an snc divisor on U, $B_Y|_U = \sum_{i=1}^{c} b_i \operatorname{div}(z_i)$, and $\operatorname{div}(z_1) = E_j|_U$ for some j. Then $b_1 = 1 - a(E_j, X, B)$, $b_i \leq 1$ for any i by assumption, and $\operatorname{ord}_D z_i \geq 1$ for any i. By [KM98, Lemma 2.30] and Lemma 2.13 (Remark 2.17),

$$a(D, X, B) = a(D, U, B_Y|_U) \geq \sum_{i=1}^{c} (1 - b_i) \operatorname{ord}_D z_i \geq 1 - b_1 = a(E_j, X, B).$$

For $k = \mathbb{C}$, this is [Amb99, Section 1, the second paragraph after Definition 1.1]. □

Lemma 3.6. *Let (X, B) be a threefold sub-pair, $C \subset X$ an integral curve with generic point η, and $t \in \mathbb{C}(X)$ a rational function regular at η whose restriction to C is not constant. Let $U \subset X$ be the open subset on which t is regular, let $k := \mathbb{C}(t)$, let X_k be the generic fiber over k of the morphism $t : U \to \mathbb{A}^1_{\mathbb{C}}$, let $c \in X_k$ be the closed point corresponding to η, and let B_k be the restriction of B. Then:*

(1) $\operatorname{mld}(X_k \ni c, B_k) = \operatorname{mld}(X \ni \eta, B)$;

(2) let Ω be an algebraic closure of k, c' a closed point of $X_k \otimes_k \Omega$ over c, W the connected component of $X_k \otimes_k \Omega$ containing c', and B_W the restriction of $B_k \otimes_k \Omega$ to W; then $\operatorname{mld}(W \ni c', B_W) = \operatorname{mld}(X \ni \eta, B)$, and the coefficients of $B_k \otimes_k \Omega$ at c' are coefficients of B;

(3) *if a finite group H acts faithfully on X, preserving B and fixing C pointwise, and t is H-invariant, let $\pi : X \to X/H$ be the quotient morphism, $\bar{\eta} := \pi(\eta)$, and $\bar{t} \in \mathbb{C}(X/H)$ the rational function with $t = \pi^*\bar{t}$; then H acts on X_k fixing c, the generic fiber of $\bar{t}$ on the crepant quotient $(X, B)/H$ is $(X_k, B_k)/H$, and* $\mathrm{mld}((X_k \ni c, B_k)/H) = \mathrm{mld}(X/H \ni \bar{\eta}, B_{X/H})$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we compare prime divisors over X_k with prime divisors over X, and prove (1).

The rational function t defines a morphism $U \to \mathbb{A}^1_{\mathbb{C}}$, and $X_k = U \times_{\mathbb{A}^1_{\mathbb{C}}} \operatorname{Spec} k$. Let $S := \mathbb{C}[t]\backslash\{0\}$. Then X_k is the set of points $\xi \in U$ such that $p \notin \mathfrak{m}_\xi$ for any $p \in S$, with $\mathcal{O}_{X_k,\xi} = \mathcal{O}_{X,\xi}$, and X_k is a normal quasi-projective surface over k with function field $\mathbb{C}(X)$. Since $t|_C$ is not constant, $p(t|_C) \neq 0$ for any $p \in S$, hence $\eta \in X_k$. The point c is η regarded as a point of X_k. Thus $\mathcal{O}_{X_k,c} = \mathcal{O}_{X,\eta}$ and $\kappa(c) = \mathbb{C}(C)$, which is finite over k.

Let ω be a non-zero element of $\Omega^2_{\mathbb{C}(X)/k}$. Let $g : X' \to X$ be either the identity or a projective birational morphism from a normal variety, and let X'_k be the generic fiber of $t \circ g$ over k. For any prime divisor P on X'_k, $\mathcal{O}_{X',P} = \mathcal{O}_{X'_k,P}$ is a discrete valuation ring. Since X'_k is of finite type over k, its regular locus is open [Gro65, Scholie 7.8.3(ii)–(iv)], and since k has characteristic zero, X'_k is smooth over k on a neighborhood of P [Gro67, Proposition 17.15.5]. By [Gro67, Proposition 17.2.3(ii)] for $X'_k \to \operatorname{Spec} k \to \operatorname{Spec} \mathbb{C}$ and since $\Omega^1_{k/\mathbb{C}} = kdt$, the sequence

$$0 \to \mathcal{O}_{X',P}dt \to \Omega^1_{\mathcal{O}_{X',P}/\mathbb{C}} \to \Omega^1_{\mathcal{O}_{X',P}/k} \to 0$$

is split exact. Taking exterior powers, $\Omega^3_{\mathcal{O}_{X',P}/\mathbb{C}} = dt \wedge \Omega^2_{\mathcal{O}_{X',P}/k}$, so $\operatorname{ord}_P(dt \wedge \omega) = \operatorname{ord}_P(\omega)$. Thus $K_{X'_k} = K_{X'}|_{X'_k}$ for $K_{X'} := \operatorname{div}(dt \wedge \omega)$ and $K_{X'_k} := \operatorname{div}(\omega)$. In particular, $K_{X_k} + B_k = (K_X + B)|_{X_k}$ is $\mathbb{R}$-Cartier, so (X_k, B_k) is a sub-pair over k (Remark 2.17). Since pullback of $\mathbb{R}$-Cartier $\mathbb{R}$-divisors commutes with restriction to X'_k, we obtain

$$K_{X'_k} - g_k^*(K_{X_k} + B_k) = \big(K_{X'} - g^*(K_X + B)\big)|_{X'_k}, \tag{3.2}$$

where $g_k : X'_k \to X_k$ is the restriction of g.

Let E be a prime divisor over X_k, and let $\xi \in X_k$ be the generic point of $\operatorname{center}_{X_k} E$. Then ord_E is a valuation of $\mathbb{C}(X)$ which is trivial on k, whose residue field has transcendence degree one over k, hence two over $\mathbb{C}$, and whose valuation ring dominates $\mathcal{O}_{X_k,\xi} = \mathcal{O}_{X,\xi}$. By [KM98, Lemma 2.45], E is a prime divisor over X and $\operatorname{center}_X E$ is the closure of ξ in X. Conversely, let E be a prime divisor over X such that the generic point ξ of $\operatorname{center}_X E$ belongs to X_k. Any $p \in S$ is a unit of $\mathcal{O}_{X,\xi}$, so ord_E is trivial on k, and the residue field of ord_E has transcendence degree one over k. By [KM98, Lemma 2.45] over k, E is a prime divisor over X_k with center $\overline{\{\xi\}} \cap X_k$. In both cases, let $g : X' \to X$ be a projective birational morphism from a normal variety such that E is a prime divisor on X', which exists by [KM98, Lemma 2.45] after normalization. The generic point of E belongs to X'_k, so (3.2) implies (cf. [KM98, Lemma 2.30])

$$a(E, X_k, B_k) = a(E, X, B). \tag{3.3}$$

In particular, prime divisors over X_k with center c are exactly the prime divisors over X with center C, and (3.3) holds for any such E. This proves (1).

Step 2. In this step we prove (3).

Since t is H-invariant, U is H-invariant and H acts on U over $\mathbb{A}^1_{\mathbb{C}}$. Thus H acts on X_k by k-automorphisms, faithfully since H acts faithfully on $\mathbb{C}(X)$. Since H fixes C pointwise, any $\sigma \in H$ fixes η and acts trivially on $\kappa(\eta) = \mathbb{C}(C)$, so H fixes c. The quotient morphism π exists since X is quasi-projective. The open subset of X/H on which $\bar{t}$ is regular is $\pi(U) = U/H$: $\bar{t}$ is regular on U/H since t is regular on the H-invariant open subset U, and $t = \pi^*\bar{t}$ is regular at any point over a point at which $\bar{t}$ is regular. Moreover, $\bar{t}|_{\pi(C)}$ is not constant, since the composition of $\bar{t}|_{\pi(C)}$ with $C \to \pi(C)$ is $t|_C$. Since S consists of H-invariant functions, $(\mathcal{O}(U')_S)^H = (\mathcal{O}(U')^H)_S$

for any H-invariant affine open subset $U' \subset U$. Thus the generic fiber of $\bar{t}$ is X_k/H, which is quasi-projective over k, and π restricts to the quotient morphism $\pi_k : X_k \to X_k/H$. For a prime divisor P on X_k, the ramification index of π along P and the degree of $P \to \pi(P)$ are determined by the discrete valuation rings $\mathcal{O}_{X,P} \supset \mathcal{O}_{X/H,\pi(P)}$ and their residue fields. Since $\mathcal{O}_{X_k,P} = \mathcal{O}_{X,P}$ and $\mathcal{O}_{X_k/H,\pi_k(P)} = \mathcal{O}_{X/H,\pi(P)}$, the ramification index and the degree for π_k along P equal those for π. Thus $R(\pi_k) = R(\pi)|_{X_k}$ and

$$\frac{1}{|H|}\pi_{k*}(B_k + R(\pi_k)) = B_{X/H}|_{X_k/H},$$

that is, the generic fiber of $(X, B)/H$ is $(X_k, B_k)/H$. By Lemma 2.6(0), $(X/H, B_{X/H})$ is a threefold sub-pair, and the closed point of X_k/H corresponding to $\bar{\eta}$ is $\pi_k(c)$. Thus (1) for $(X/H, B_{X/H})$, $\pi(C)$, and $\bar{t}$ implies

$$\operatorname{mld}((X_k \ni c, B_k)/H) = \operatorname{mld}(X_k/H \ni \pi_k(c), B_{X/H}|_{X_k/H}) = \operatorname{mld}(X/H \ni \bar{\eta}, B_{X/H}).$$

Step 3. In this step we prove that the coefficients of $B_k \otimes_k \Omega$ at c' are coefficients of B, and prove the equality in (2) when (X, B) is sub-lc near η.

Let $X_\Omega := X_k \otimes_k \Omega$ and $B_\Omega := B_k \otimes_k \Omega$, so that W is the connected component of X_Ω containing c' and $B_W = B_\Omega|_W$. For a normal quasi-projective variety Y over k, we use the following facts. The scheme $Y_\Omega := Y \otimes_k \Omega$ is normal, since normality is preserved under separable algebraic extensions of the base field [Gro65, Corollary 6.14.2]. Thus the connected components of Y_Ω are normal quasi-projective varieties over Ω. Since $\Omega^1_{Y_\Omega/\Omega}$ is the pullback of $\Omega^1_{Y/k}$, we take K_{Y_Ω} to be the pullback of K_Y. For a prime divisor P on Y, $\kappa(P) \otimes_k \Omega$ is reduced, so the pullback of P to Y_Ω is a sum of distinct prime divisors with coefficient one. In particular, $K_W + B_W$ is the pullback of $K_{X_k} + B_k$, (W, B_W) is a sub-pair over Ω, and the coefficients of B_Ω are coefficients of B_k. The coefficient of B_k at a prime divisor P on X_k is the coefficient of B at the closure of P in X, so the coefficients of B_Ω at c' are coefficients of B. The fiber of X_Ω over c is $\operatorname{Spec}(\kappa(c) \otimes_k \Omega)$, which is reduced and finite. Let $\tau : \kappa(c) \to \Omega$ be the k-embedding corresponding to c'.

Assume that $(V, B|_V)$ is sub-lc for an open neighborhood V of η, and let $V_k := V \cap X_k$. By **Step 1**, any prime divisor over V_k is a prime divisor over V, and (3.3) holds for it, so $(V_k, B_k|_{V_k})$ is sub-lc. Let $f : T \to X_k$ be a projective log resolution of (X_k, B_k) such that $\mathfrak{m}_c\mathcal{O}_T$ is invertible and $f^{-1}(c) \cup \operatorname{Exc}(f) \cup \operatorname{Supp} f_*^{-1}B_k$ is an snc divisor (cf. [KM98, Theorem 0.2] and Remark 2.17), and write $K_T + D_T = f^*(K_{X_k} + B_k)$. Any prime divisor P on T meeting $f^{-1}(c)$ is a prime divisor over V_k, so $\operatorname{mult}_P D_T = 1 - a(P, X_k, B_k) \le 1$. Let $E_1, \dots, E_N$ be the irreducible components of $f^{-1}(c)$. By Lemma 3.5 over k,

$$\operatorname{mld}(X_k \ni c, B_k) = \min_j a(E_j, X_k, B_k).$$

Let $T_W := (T \otimes_k \Omega) \times_{X_\Omega} W$ and let $f_W : T_W \to W$ be the induced morphism. Then T_W is smooth over Ω [Gro67, Proposition 17.3.3(iii)]. The morphism f_W is projective, since f_W is the restriction over W of the base change $f \otimes_k \Omega : T \otimes_k \Omega \to X_\Omega$ of the projective morphism f, and birational, since it is an isomorphism over the inverse image of the open subset of X_k over which f is an isomorphism (every irreducible component of $T \otimes_k \Omega$ dominates T, since $T \otimes_k \Omega \to T$ is flat). The strata of the snc divisor $f^{-1}(c) \cup \operatorname{Exc}(f) \cup \operatorname{Supp} f_*^{-1}B_k$ are smooth over k, so their pullbacks are smooth over Ω [Gro67, Proposition 17.3.3(iii)], and $f_W^{-1}(c') \cup \operatorname{Exc}(f_W) \cup \operatorname{Supp} {f_W}_*^{-1}B_W$ is an snc divisor. Since the fiber of X_Ω over c is reduced and finite, $\mathfrak{m}_{c'}\mathcal{O}_{X_\Omega} = \mathfrak{m}_c\mathcal{O}_{X_\Omega}$ near c', so $\mathfrak{m}_{c'}\mathcal{O}_{T_W}$ is invertible. By the facts at the beginning of this step, $K_{T_W} + D_{T_W} = f_W^*(K_W + B_W)$, where D_{T_W} is the pullback of D_T, and any prime divisor on T_W meeting $f_W^{-1}(c')$ is a component of the pullback of a prime divisor P on T meeting $f^{-1}(c)$, with coefficient $\operatorname{mult}_P D_T \le 1$. Moreover,

$$f_W^{-1}(c') = f^{-1}(c) \times_{\operatorname{Spec}\kappa(c),\tau} \operatorname{Spec}\Omega,$$

so the irreducible components of $f_W^{-1}(c')$ are the irreducible components of the schemes $E_j \times_{\operatorname{Spec}\kappa(c),\tau} \operatorname{Spec}\Omega$, $1 \le j \le N$. For any j, the scheme $E_j \times_{\operatorname{Spec}\kappa(c),\tau} \operatorname{Spec}\Omega$ is non-empty, since

E_j is a non-empty $\kappa(c)$-scheme, and any irreducible component of $E_j \times_{\operatorname{Spec}\kappa(c),\tau} \operatorname{Spec}\Omega$ has log discrepancy $a(E_j, X_k, B_k)$ with respect to (W, B_W). By Lemma 3.5 over Ω and by (1),

$$\operatorname{mld}(W \ni c', B_W) = \min_j a(E_j, X_k, B_k) = \operatorname{mld}(X_k \ni c, B_k) = \operatorname{mld}(X \ni \eta, B).$$

Step 4. In this step we prove (2) when $(V, B|_V)$ is not sub-lc for any open neighborhood V of η. We show that both sides of (2) are $-\infty$.

Let $h : X' \to X$ be a log resolution of (X, B) [KM98, Theorem 0.2], and write $K_{X'} + B' = h^*(K_X + B)$. Let V be the complement in X of the union of the sets $h(P)$, where P runs over the components of B' with $\operatorname{mult}_P B' > 1$ and $\eta \notin h(P)$. If no component P of B' with $\operatorname{mult}_P B' > 1$ satisfies $\eta \in h(P)$, then V is an open neighborhood of η, $(h^{-1}(V), B'|_{h^{-1}(V)})$ is log smooth with coefficients at most one, hence sub-lc [KM98, Corollary 2.31(3)], and $(V, B|_V)$ is sub-lc [KM98, Lemma 2.30], which is not the case. Thus there exists a component E_0 of B' with $\eta \in h(E_0)$ and $a(E_0, X, B) = 1 - \operatorname{mult}_{E_0} B' < 0$.

Since $\eta \in h(E_0)$, the fiber $E_0 \times_X \operatorname{Spec}\kappa(\eta)$ is a non-empty scheme of finite type over $\kappa(\eta)$. Let ζ be a closed point of $E_0 \times_X \operatorname{Spec}\kappa(\eta)$, and let Γ_0 be the closure of ζ in X'. Since $\kappa(\zeta)$ is finite over $\kappa(\eta) = \mathbb{C}(C)$, $\Gamma_0 \subset E_0$ is an irreducible curve with $h(\Gamma_0) = C$. Replace X' with an open neighborhood of ζ on which Γ_0 is smooth and any component of $\operatorname{Supp} B' \cup \operatorname{Exc}(h)$ contains Γ_0. By the snc condition, at most one such component other than E_0 exists. Let E_1 be the exceptional divisor of the blow-up of Γ_0. For $j \geq 1$, let $E_0^{(j)}$ be the strict transform of E_0 after j blow-ups, let $\Gamma_j := E_j \cap E_0^{(j)}$, and let E_{j+1} be the exceptional divisor of the blow-up of Γ_j, after shrinking to an open neighborhood of the generic point of Γ_j on which Γ_j is smooth. If smooth prime divisors P_1 and P_2 meet transversally along a smooth curve Γ, then their strict transforms on the blow-up of Γ meet the exceptional divisor in two disjoint sections, so they are disjoint over a neighborhood of the generic point of Γ. For $j = 0$, we take $P_1 = E_0$ and P_2 the component of $\operatorname{Supp} B' \cup \operatorname{Exc}(h)$ other than E_0 containing Γ_0, if it exists: the strict transform of P_2 does not contain the generic point of Γ_1. For $j \geq 1$, we take $P_1 = E_0^{(j)}$ and $P_2 = E_j$: the strict transform of E_j does not contain the generic point of Γ_{j+1}. Thus, by induction on j, for $j \geq 1$, the generic point of Γ_j is contained in exactly two components of the crepant boundary, namely E_j and $E_0^{(j)}$, with coefficients $1 - a(E_j, X, B)$ and $1 - a(E_0, X, B)$. By [KM98, Lemmas 2.29 and 2.30],

$$a(E_{j+1}, X, B) = 2 - \big(1 - a(E_j, X, B)\big) - \big(1 - a(E_0, X, B)\big) = a(E_j, X, B) + a(E_0, X, B).$$

Thus $a(E_j, X, B) = a(E_1, X, B) + (j-1)a(E_0, X, B)$ for any $j \geq 1$, so $a(E_j, X, B) \to -\infty$ as $j \to +\infty$, since $a(E_0, X, B) < 0$. Since Γ_j maps onto Γ_0 for any $j \geq 1$, $\operatorname{center}_X E_j = h(\Gamma_0) = C$. Thus $\operatorname{mld}(X \ni \eta, B) = -\infty$ (compare [KM98, Corollary 2.31(1)]).

By **Step 1**, any E_j is a prime divisor over X_k with center c, and $a(E_j, X_k, B_k) = a(E_j, X, B)$. Let $g_j : T_j \to X_k$ be a projective birational morphism from a normal variety such that E_j is a prime divisor on T_j, which exists by [KM98, Lemma 2.45] after normalization, and write $K_{T_j} + D_j = g_j^*(K_{X_k} + B_k)$. By the facts at the beginning of **Step 3**, $T_j \otimes_k \Omega$ is normal and

$$K_{T_j \otimes_k \Omega} + D_{j,\Omega} = (g_j \otimes_k \Omega)^*(K_{X_\Omega} + B_\Omega),$$

where $D_{j,\Omega}$ is the pullback of D_j. Since $g_j(E_j) = \{c\}$, $E_j \times_{\operatorname{Spec}\kappa(c),\tau} \operatorname{Spec}\Omega$ is a non-empty union of prime divisors on $(T_j \otimes_k \Omega) \times_{X_\Omega} W$, which is projective and birational over W (every irreducible component of $T_j \otimes_k \Omega$ dominates T_j, since $T_j \otimes_k \Omega \to T_j$ is flat), and any irreducible component of $E_j \times_{\operatorname{Spec}\kappa(c),\tau} \operatorname{Spec}\Omega$ has center $\{c'\}$ on W and coefficient $1 - a(E_j, X, B)$ in $D_{j,\Omega}$ (cf. [KM98, Lemma 2.30]). Thus $\operatorname{mld}(W \ni c', B_W) \leq a(E_j, X, B)$ for any $j \geq 1$, so $\operatorname{mld}(W \ni c', B_W) = -\infty = \operatorname{mld}(X \ni \eta, B)$. This proves (2). □

Lemma 3.7. *Let $\Theta \subset [0,1]$ be a DCC set. Then the set of numbers $\operatorname{mld}(W \ni \xi, D) > 0$, where (W, D) is a threefold pair with $D \in \Theta$ and ξ is the generic point of an integral curve on W, satisfies the ACC.*

Proof. Suppose that there exist threefold pairs (W_i, D_i) with $D_i \in \Theta$ and integral curves $C_i \subset W_i$ with generic points ξ_i, $i \geq 1$, such that $a_i := \mathrm{mld}(W_i \ni \xi_i, D_i) > 0$ and the sequence $(a_i)_i$ is strictly increasing. For any i, choose $t_i \in \mathbb{C}(W_i)$ regular at ξ_i whose restriction to C_i is not constant. Let $k := \mathbb{C}(t)$, identify any $\mathbb{C}(t_i)$ with k by sending t_i to t, and fix an algebraic closure Ω of k. Let $W_{i,k}$ be the generic fiber of t_i over k, $c_i \in W_{i,k}$ the closed point corresponding to ξ_i, c_i' a closed point of $W_{i,k} \otimes_k \Omega$ over c_i, T_i the connected component of $W_{i,k} \otimes_k \Omega$ containing c_i', and D_{T_i} the restriction to T_i of the pullback of D_i. By Lemma 3.6(2),

$$\mathrm{mld}(T_i \ni c_i', D_{T_i}) = a_i,$$

and the coefficients of D_{T_i} at c_i' are coefficients of D_i. Since $\mathrm{mld}(T_i \ni c_i', D_{T_i})$ only depends on a neighborhood of c_i', possibly replacing T_i with an open neighborhood of c_i', we may assume that $D_{T_i} \in \Theta$.

We show that (T_i, D_{T_i}) is lc near c_i'. Let $h : T_i' \to T_i$ be a log resolution of (T_i, D_{T_i}), which exists over Ω (Remark 2.17), and write $K_{T_i'} + D' = h^*(K_{T_i} + D_{T_i})$. Let $Z \subset T_i$ be the union of the points $h(E)$, where E runs over the h-exceptional prime divisors with $a(E, T_i, D_{T_i}) < 0$. The $\mathbb{R}$-divisor $D'|_{h^{-1}(T_i \setminus Z)}$ is supported on an snc divisor with coefficients at most 1, since the coefficient of an h-exceptional prime divisor E is $1 - a(E, T_i, D_{T_i}) \leq 1$ and the other coefficients are coefficients of $D_{T_i} \in \Theta \subset [0, 1]$. Thus $(h^{-1}(T_i \setminus Z), D'|_{h^{-1}(T_i \setminus Z)})$ is sub-lc [KM98, Corollary 2.31(3)], and (T_i, D_{T_i}) is lc on $T_i \setminus Z$ [KM98, Lemma 2.30], applied to open subsets of T_i (Remark 2.17). If $h(E) = \{c_i'\}$ for an h-exceptional prime divisor E, then $a(E, T_i, D_{T_i}) \geq a_i > 0$. Thus $c_i' \notin Z$, and possibly replacing T_i with $T_i \setminus Z$, we may assume that (T_i, D_{T_i}) is lc.

Thus any a_i is the minimal log discrepancy at a closed point of an lc surface pair over Ω with coefficients in Θ. By Theorem 2.18(1), the set of such numbers satisfies the ACC, which contradicts the strict increase of $(a_i)_i$. □

4. Lattice Newton minima

In this section, we prove that lattice Newton minima satisfy the ACC (Theorem 4.5). Each minimal log discrepancy in $(0, 1)$ treated in the proof of Theorem 1.3 is either a lattice Newton minimum or a weighted one (Corollary 4.6), whose ACC follows from Theorem 4.5, or arises from groups of bounded order (Theorem 13.1). Lemma 4.4 extracts a convergent subsequence in the Hausdorff distance from any sequence of closed subgroups of $\mathbb{T}^d$, and it is used in this section and in Sections 15 and 16.

Definition 4.1. For a primitive vector $w \in \mathbb{Z}^d$, the subgroup $\rho(\mathbb{R}w) \subset \mathbb{T}^d$ is called the *circle* with *weight vector* w, and $\rho(\mathbb{R}w) = \rho(\mathbb{R}(-w))$. It is called a *coordinate circle* if exactly one coordinate of w is non-zero, a *two-coordinate circle* if exactly two coordinates of w are non-zero, *positive* if the non-zero coordinates of w are all positive or all negative, and *mixed* otherwise.

For a circle whose weight vector has no zero coordinate, whether the circle is positive or mixed decides whether the points of the circle close to the identity have lifts in $\mathbb{R}^d_{>0}$ with small coordinate sum (Example 4.2). If the weight vector has zero coordinates, then whether the circle is positive or mixed decides whether the non-zero coordinates of such lifts can have small sum, since the other coordinates of any lift in $\mathbb{R}^d_{>0}$ are positive integers (Example 4.2(1)). This is why Theorems 4.8 and 16.1 distinguish cases according to whether a circle is positive or mixed.

Example 4.2.

(1) Let w be a weight vector with $w_i > 0$ for any i. Then for small positive s the point $\rho(sw)$ has a lift in $(0, 1)^d$ with small coordinates, so N_H may contain points of $\mathbb{R}^d_{>0}$ with small coordinate sum for a finite subgroup $H \subset \rho(\mathbb{R}w)$. For example, for any positive integer n, the minimal log discrepancy of the quotient of $\mathbb{A}^3$ by $\mu_n(1, 1, 1)$ at the image of the origin is $3/n$, which tends to zero as n grows. If instead w is a weight vector with $w_j = 0$ for some j, as for a coordinate circle when $d \geq 2$, then the j-th coordinate of any lift in $\mathbb{R}^d_{>0}$ of a point of $\rho(\mathbb{R}w)$ is a positive integer.

(2) Let w be the weight vector of a mixed circle and t a real number with $0 < |t| \max_i |w_i| < 1$. Then tw has a negative coordinate tw_i, so the i-th coordinate of any lift of $\rho(tw)$ in $\mathbb{R}^d_{>0}$ is at least $1 + tw_i \geq 1 - |t| \max_i |w_i|$. Thus, for t close to 0, the points $\rho(tw)$ have no lifts in $\mathbb{R}^d_{>0}$ with small coordinate sum. For example, for any integer n at least two, the quotient of $\mathbb{A}^3$ by $\mu_n(1, -1, 0)$ is the product of an A_{n-1} surface singularity and a line, and the minimal log discrepancy of the quotient at the image of the origin is two.

Lemma 4.3. *Let d be a positive integer.*

(1) Let $H \subset \mathbb{T}^d$ be a closed subgroup and H^0 the connected component of H containing 0. Then H/H^0 is finite, and $H^0 = \rho(V)$ for a linear subspace $V \subset \mathbb{R}^d$ spanned by $V \cap \mathbb{Z}^d$. If moreover $\dim V = 1$, then H^0 is a circle.

(2) Let $G \subset (\mathbb{C}^*)^d$ be a finite subgroup and $w \in \mathbb{Z}^d \backslash \{0\}$, and identify G with the image of G in $\mathbb{T}^d$ under the characters of the coordinates of $\mathbb{A}^d$. Then the image of $G \cap \mathbf{T}_w$ in $\mathbb{T}^d$ is $G \cap \rho(\mathbb{R}w)$.

Proof. (1) The subgroup $N_H = \rho^{-1}(H)$ of $\mathbb{R}^d$ is closed and contains $\mathbb{Z}^d$. By the structure of closed subgroups of $\mathbb{R}^d$ [Bou98, Chapter VII, §1, no. 2, Theorem 2], $N_H = V \oplus L$, where V is the largest linear subspace of $\mathbb{R}^d$ contained in N_H, and L is a discrete subgroup of a linear subspace $W \subset \mathbb{R}^d$ with $\mathbb{R}^d = V \oplus W$. Let $p : \mathbb{R}^d \to \mathbb{R}^d/V$ be the projection. Since $p|_W$ is a linear isomorphism, $p(N_H) = p(L)$ is discrete. Thus $p(\mathbb{Z}^d) \subset p(L)$ is a discrete subgroup spanning $\mathbb{R}^d/V$, so by [Bou98, Chapter VII, §1, no. 2, Theorem 2] $p(\mathbb{Z}^d)$ is generated by $d - \dim V$ linearly independent vectors. Hence the kernel $V \cap \mathbb{Z}^d$ of $p|_{\mathbb{Z}^d}$ has rank $\dim V$. Since elements of $\mathbb{Z}^d$ linearly independent over $\mathbb{Z}$ are linearly independent over $\mathbb{R}$, a subgroup of $\mathbb{Z}^d$ of rank $\dim V$ spans a linear subspace of dimension $\dim V$, so V is spanned by $V \cap \mathbb{Z}^d$.

Since $p(\mathbb{Z}^d)$ is discrete, the subgroup $V + \mathbb{Z}^d = p^{-1}(p(\mathbb{Z}^d))$ is closed. Thus $\rho(V) = \rho(V + \mathbb{Z}^d)$ is a closed connected subgroup of H. Moreover, $p(L)$ is a discrete subgroup of $\mathbb{R}^d/V$ containing the lattice $p(\mathbb{Z}^d)$, so $p(L)/p(\mathbb{Z}^d)$ is a discrete subgroup of the compact group $(\mathbb{R}^d/V)/p(\mathbb{Z}^d)$, hence finite. The homomorphism $N_H \to p(L)/p(\mathbb{Z}^d)$ induced by p is surjective with kernel $N_H \cap (V + \mathbb{Z}^d) = V + \mathbb{Z}^d$, hence

$$H/\rho(V) \cong N_H/(V + \mathbb{Z}^d) \cong p(L)/p(\mathbb{Z}^d)$$

is finite. Therefore $\rho(V)$ is a closed subgroup of finite index in H, so $\rho(V)$ is also open in H. Since $\rho(V)$ is connected and contains 0, $H^0 = \rho(V)$, and H/H^0 is finite.

If $\dim V = 1$, let w be a generator of $V \cap \mathbb{Z}^d \cong \mathbb{Z}$. Then $V = \mathbb{R}w$. If $\frac{1}{k}w \in \mathbb{Z}^d$ for an integer $k \geq 2$, then $\frac{1}{k}w \in V \cap \mathbb{Z}^d$, which contradicts the choice of w. Thus w is primitive in $\mathbb{Z}^d$, and $H^0 = \rho(\mathbb{R}w)$ is the circle with weight vector w.

(2) Under the identification, $g = (g_1, \ldots, g_d) \in G$ corresponds to $(u_1, \ldots, u_d) \bmod \mathbb{Z}^d$, where $g_i = e^{2\pi\sqrt{-1}u_i}$ for any i, and the map $G \to \mathbb{T}^d$ is injective. Let $g = (s^{w_1}, \ldots, s^{w_d}) \in G \cap \mathbf{T}_w$ with $s \in \mathbb{C}^*$. Since g has finite order, $|s|^{w_i} = |s^{w_i}| = 1$ for any i. Since $w \neq 0$, $|s| = 1$. Write $s = e^{2\pi\sqrt{-1}t}$ with $t \in \mathbb{R}$. Then $g_i = e^{2\pi\sqrt{-1}tw_i}$ for any i, so the image of g in $\mathbb{T}^d$ is $\rho(tw) \in \rho(\mathbb{R}w)$. Conversely, if the image of $g \in G$ is $\rho(tw)$ with $t \in \mathbb{R}$, then $g = (s^{w_1}, \ldots, s^{w_d})$ with $s = e^{2\pi\sqrt{-1}t}$, so $g \in \mathbf{T}_w$. □

Lemma 4.4. *Let $(H_i)_{i \geq 1}$ be a sequence of closed subgroups of $\mathbb{T}^d$, where $\mathbb{T}^d$ carries the metric induced by the Euclidean metric of $\mathbb{R}^d$. Then, after passing to a subsequence, there exists a closed subgroup $H \subset \mathbb{T}^d$ such that the H_i converge to H in the Hausdorff distance and, for any $i \gg 0$, $H_i \subset H$ and H_i intersects any connected component of H. Moreover, any subgroup of $\mathbb{T}^1$ contained in $\rho((-\frac{1}{4}, \frac{1}{4}))$ is trivial.*

Proof. We write dist for the metrics of $\mathbb{T}^d$ and $\mathbb{T}^1$, and $\mathrm{dist_H}$ for the Hausdorff distance between non-empty closed subsets of $\mathbb{T}^d$. Both metrics are translation invariant.

We first prove the last statement. Let $H' \subset \mathbb{T}^1$ be a subgroup contained in $\rho((-\frac{1}{4}, \frac{1}{4}))$, and suppose that H' contains a non-zero element $\rho(t)$ with $t \in (-\frac{1}{4}, \frac{1}{4})$. Since $t \neq 0$, there exists a

least positive integer n such that $|nt| \geq \frac{1}{4}$. Then $n \geq 2$ and $|(n-1)t| < \frac{1}{4}$, so

$$\frac{1}{4} \leq |nt| < \frac{1}{4} + |t| < \frac{1}{2}.$$

Thus $\mathrm{dist}(\rho(nt), 0) = |nt| \geq \frac{1}{4}$, so $\rho(nt) \notin \rho((-\frac{1}{4}, \frac{1}{4}))$. Since $\rho(nt) = n\rho(t) \in H'$, this is a contradiction. Thus H' is trivial.

By [BBI01, Theorem 7.3.8], the non-empty closed subsets of $\mathbb{T}^d$ form a compact metric space under $\mathrm{dist}_{\mathrm{H}}$. Since $0 \in H_i$ for any i, possibly replacing $(H_i)_{i\geq 1}$ with a subsequence, we may assume that $\lim_{i\to+\infty} \mathrm{dist}_{\mathrm{H}}(H_i, H) = 0$ for a non-empty closed subset $H \subset \mathbb{T}^d$. Let $a, b \in H$. For any i there exist $a_i, b_i \in H_i$ such that $\mathrm{dist}(a_i, a) \leq \mathrm{dist}_{\mathrm{H}}(H_i, H)$ and $\mathrm{dist}(b_i, b) \leq \mathrm{dist}_{\mathrm{H}}(H_i, H)$. Since $a_i - b_i \in H_i$, there exists $y_i \in H$ with $\mathrm{dist}(a_i - b_i, y_i) \leq \mathrm{dist}_{\mathrm{H}}(H_i, H)$. By translation invariance, $a_i - b_i$ converges to $a - b$, so y_i converges to $a - b$. Since H is closed, $a - b \in H$. Thus H is a closed subgroup of $\mathbb{T}^d$.

Next we show that $H_i \subset H$ for $i \gg 0$. For $\lambda \in \mathbb{Z}^d$, let $\bar{\lambda} : \mathbb{T}^d \to \mathbb{T}^1$ be the homomorphism defined by $\bar{\lambda}(\rho(a)) = \rho(\langle \lambda, a\rangle)$ for $a \in \mathbb{R}^d$. For $a, b \in \mathbb{R}^d$ and $n \in \mathbb{Z}^d$, we have $\langle \lambda, a - b - n\rangle \in \langle \lambda, a - b\rangle + \mathbb{Z}$, so

$$\mathrm{dist}(\bar{\lambda}(\rho(a)), \bar{\lambda}(\rho(b))) \leq |\langle \lambda, a - b - n\rangle| \leq |\lambda| \cdot |a - b - n|.$$

Taking the minimum over $n \in \mathbb{Z}^d$, we obtain

$$\mathrm{dist}(\bar{\lambda}(\rho(a)), \bar{\lambda}(\rho(b))) \leq |\lambda| \cdot \mathrm{dist}(\rho(a), \rho(b)).$$

Let $H^{\perp} := \{\lambda \in \mathbb{Z}^d \mid \bar{\lambda}(H) = 0\}$. By [HR79, (24.10) Theorem], H is the intersection of the kernels of $\bar{\lambda}$ for $\lambda \in H^{\perp}$. The subgroup $H^{\perp} \subset \mathbb{Z}^d$ has a finite basis $\lambda_1, \ldots, \lambda_t$, and $\overline{\lambda + \lambda'} = \bar{\lambda} + \bar{\lambda}'$ for $\lambda, \lambda' \in \mathbb{Z}^d$, so

$$H = \bigcap_{j=1}^{t} \mathrm{Ker}\, \bar{\lambda}_j,$$

where the right-hand side is $\mathbb{T}^d$ if $t = 0$. Pick a real number $\epsilon > 0$ such that $\epsilon|\lambda_j| < \frac{1}{4}$ for any j. For $i \gg 0$, $\mathrm{dist}_{\mathrm{H}}(H_i, H) < \epsilon$. Then for any $h \in H_i$ there exists $h' \in H$ with $\mathrm{dist}(h, h') < \epsilon$, so

$$\mathrm{dist}(\bar{\lambda}_j(h), 0) = \mathrm{dist}(\bar{\lambda}_j(h), \bar{\lambda}_j(h')) < \frac{1}{4},$$

that is, $\bar{\lambda}_j(h) \in \rho((-\frac{1}{4}, \frac{1}{4}))$. Thus $\bar{\lambda}_j(H_i)$ is a subgroup of $\mathbb{T}^1$ contained in $\rho((-\frac{1}{4}, \frac{1}{4}))$, so $\bar{\lambda}_j(H_i) = \{0\}$ by the last statement. Hence $H_i \subset H$ for $i \gg 0$.

Finally, by Lemma 4.3(1), H has finitely many connected components, namely the cosets $y + H^0$ with $y \in H$. They are compact and pairwise disjoint, so there exists a real number $r > 0$ such that $\mathrm{dist}(y, y') \geq r$ for any $y, y' \in H$ in distinct connected components. Let $y \in H$. For $i \gg 0$, $H_i \subset H$ and $\mathrm{dist}_{\mathrm{H}}(H_i, H) < r$, so there exists $h \in H_i \subset H$ with $\mathrm{dist}(h, y) < r$, and hence $h \in y + H^0$. Since H has finitely many connected components, H_i intersects any connected component of H for $i \gg 0$. □

In [Amb06, Theorem 4.4] the functions are linear and only their coefficients vary. Its proof passes to a subsequence with nonincreasing coefficients and then uses [Amb06, Theorem 4.3]. In Theorem 4.5 the ideals J_l vary as well, so **Step 2** first passes to a subsequence by [Mac01, Theorem 1.1] and Ramsey's theorem [Ram30, Theorem A] on which the functions $\psi_{J,c}$ are pointwise nonincreasing, and Lemma 4.4 takes the place of [Amb06, Theorem 4.3].

Theorem 4.5. *Let d, s be positive integers and $\Lambda \subset [0, +\infty)$ a DCC set. Then the set of finite numbers $\psi_{\min}(H, J, c)$, where $H \subset \mathbb{T}^d$ is a finite subgroup, $J = (J_1, \ldots, J_s)$ is an s-tuple of non-zero monomial ideals of $\mathbb{C}[x_1, \ldots, x_d]$, and $c \in \Lambda^s$, satisfies the ACC, and $\psi_{\min}(H, J, c) \geq 0$ whenever $\psi_{\min}(H, J, c)$ is finite. In particular, for fixed d and a positive integer m, the finite numbers*

$$\inf\left\{ \sum_i w_i - \frac{1}{m}\,\mathrm{wt}_w(f) \;\middle|\; w \in N_H \cap \mathbb{R}^d_{>0} \right\}$$

satisfy the ACC, where f runs over the non-zero power series in d variables and H over the finite subgroups of $\mathbb{T}^d$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we show that the functions $\psi_{J,c}$ are continuous and positively homogeneous, that wt_w decreases when the ideal grows, and that $\psi_{\min}(H,J,c) \geq 0$ whenever $\psi_{\min}(H,J,c)$ is finite.

Let $\mathfrak{a} \subset \mathbb{C}[x_1,\dots,x_d]$ be a non-zero monomial ideal. Since $\mathbb{C}[x_1,\dots,x_d]$ is Noetherian and $\mathfrak{a}$ is generated by monomials, $\mathfrak{a}$ is generated by finitely many monomials $x^{\alpha^1},\dots,x^{\alpha^r}$, and any monomial in $\mathfrak{a}$ is divisible by some x^{α^k}. Thus for $w \in \mathbb{R}^d_{\geq 0}$,

$$\mathrm{wt}_w(\mathfrak{a}) = \min_{1\leq k\leq r} \langle w, \alpha^k\rangle \geq 0.$$

Hence $\mathrm{wt}_w(\mathfrak{a})$ is continuous in w and $\mathrm{wt}_{\tau w}(\mathfrak{a}) = \tau\,\mathrm{wt}_w(\mathfrak{a})$ for any real number $\tau > 0$. If $\mathfrak{a}' \subset \mathfrak{a}$ is another non-zero monomial ideal, then $\mathrm{wt}_w(\mathfrak{a}') \geq \mathrm{wt}_w(\mathfrak{a})$ by definition. In particular, any $\psi_{J,c}$ is continuous on $\mathbb{R}^d_{\geq 0}$ and $\psi_{J,c}(\tau w) = \tau\psi_{J,c}(w)$.

Suppose that $\psi_{J,c}(w_0) < 0$ for some $w_0 \in N_H \cap \mathbb{R}^d_{>0}$. Then $kw_0 \in N_H \cap \mathbb{R}^d_{>0}$ for any positive integer k, and $\psi_{J,c}(kw_0) = k\psi_{J,c}(w_0)$ is unbounded from below, so $\psi_{\min}(H,J,c) = -\infty$. Thus $\psi_{\min}(H,J,c) \geq 0$ whenever $\psi_{\min}(H,J,c)$ is finite.

Step 2. In this step, assuming that the ACC fails, we construct a strictly increasing sequence of lattice Newton minima whose functions are pointwise nonincreasing.

Suppose that the first statement does not hold. Then there exist finite subgroups $H_i \subset \mathbb{T}^d$, s-tuples $J_i = (J_{i,1},\dots,J_{i,s})$ of non-zero monomial ideals of $\mathbb{C}[x_1,\dots,x_d]$, and $c_i = (c_{i,1},\dots,c_{i,s}) \in \Lambda^s$, such that the numbers $a_i := \psi_{\min}(H_i,J_i,c_i)$ are finite and $a_1 < a_2 < \cdots$. We write $\psi_i := \psi_{J_i,c_i}$.

Fix $l \in \{1,\dots,s\}$. We color any pair of indices $i < k$ by one of four colors, according to whether $J_{i,l} = J_{k,l}$, $J_{i,l} \subsetneq J_{k,l}$, $J_{i,l} \supsetneq J_{k,l}$, or $J_{i,l}$ and $J_{k,l}$ are incomparable under inclusion. By Ramsey's theorem [Ram30, Theorem A], there exists an infinite set of indices on whose pairs the coloring is constant. The color $J_{i,l} \subsetneq J_{k,l}$ is impossible, since an ascending chain of ideals of $\mathbb{C}[x_1,\dots,x_d]$ stabilizes. The incomparable color is impossible by [Mac01, Theorem 1.1], since the ideals indexed by such a set are pairwise distinct and no two of them are comparable. Thus $(J_{i,l})_{i\geq 1}$ has a subsequence that is nonincreasing under inclusion. Moreover, any real sequence has a nondecreasing subsequence or a strictly decreasing subsequence, and the DCC set Λ contains no strictly decreasing sequence. Thus $(c_{i,l})_{i\geq 1}$ has a nondecreasing subsequence.

Since any subsequence of $(a_i)_{i\geq 1}$ is strictly increasing, possibly replacing $(H_i,J_i,c_i)_{i\geq 1}$ with a subsequence $2s$ times, we may assume that $J_{i+1,l} \subset J_{i,l}$ and $c_{i+1,l} \geq c_{i,l}$ for any i and l. By **Step 1**, $\mathrm{wt}_w(J_{i+1,l}) \geq \mathrm{wt}_w(J_{i,l}) \geq 0$ for $w \in \mathbb{R}^d_{>0}$. Since $c_{i+1,l} \geq 0$,

$$c_{i+1,l}\,\mathrm{wt}_w(J_{i+1,l}) \geq c_{i+1,l}\,\mathrm{wt}_w(J_{i,l}) \geq c_{i,l}\,\mathrm{wt}_w(J_{i,l}).$$

Thus $\psi_{i+1}(w) \leq \psi_i(w)$ for any i and any $w \in \mathbb{R}^d_{>0}$.

Step 3. In this step, we construct an increasing sequence of open subsets $U_i \subset \mathbb{T}^d$ and a closed subgroup $H \subset \mathbb{T}^d$ disjoint from all U_i.

By Lemma 4.4, possibly replacing $(H_i,J_i,c_i)_{i\geq 1}$ with a subsequence and removing finitely many terms, we may assume that the H_i converge to a closed subgroup $H \subset \mathbb{T}^d$ in the Hausdorff distance and $H_i \subset H$ for any i. **Step 2** still holds for the new sequence. Let

$$V_i := \{w \in \mathbb{R}^d_{>0} \mid \psi_i(w) < a_i\} \quad \text{and} \quad U_i := \rho(V_i).$$

Since ψ_i is continuous (**Step 1**), V_i is open in $\mathbb{R}^d$. Since $\rho^{-1}(U_i) = \bigcup_{n\in\mathbb{Z}^d}(V_i + n)$ is open, U_i is open in $\mathbb{T}^d$. Since $\psi_{i+1} \leq \psi_i$ on $\mathbb{R}^d_{>0}$ (**Step 2**) and $a_i < a_{i+1}$, we have $V_i \subset V_{i+1}$, so $U_i \subset U_{i+1}$.

We show that $H_k \cap U_k = \emptyset$ for any k. Otherwise, there exists $w \in V_k$ with $\rho(w) \in H_k$, that is, $w \in N_{H_k} \cap \mathbb{R}^d_{>0}$ and $\psi_k(w) < a_k$. This contradicts $a_k = \psi_{\min}(H_k,J_k,c_k)$. Since $U_i \subset U_k$ for $k \geq i$, we obtain $H_k \cap U_i = \emptyset$ for any $k \geq i$.

We show that $H \cap U_i = \emptyset$ for any i. Suppose that $h \in H \cap U_i$. Since U_i is open, there exists a real number $r > 0$ such that U_i contains the ball of radius r around h. For $k \gg 0$, the Hausdorff distance between H_k and H is less than r, so H_k contains a point of the ball of radius r around h. For such $k \geq i$, $H_k \cap U_i \neq \emptyset$, a contradiction.

Step 4. In this step, we conclude the proof of the first statement.

Let $j < k$ be positive integers. Since $a_j < a_k$ and a_j is the infimum of ψ_j on the non-empty set $N_{H_j} \cap \mathbb{R}^d_{>0}$, there exists $w \in N_{H_j} \cap \mathbb{R}^d_{>0}$ with $\psi_j(w) < a_k$. By **Step 2**,

$$\psi_k(w) \leq \psi_j(w) < a_k,$$

so $w \in V_k$ and $\rho(w) \in U_k$. On the other hand, $\rho(w) \in H_j \subset H$. This contradicts $H \cap U_k = \emptyset$ (**Step 3**). Thus the first statement holds, and $\psi_{\min}(H, J, c) \geq 0$ whenever $\psi_{\min}(H, J, c)$ is finite by **Step 1**.

Step 5. In this step, we prove the second statement.

Let f be a non-zero power series in d variables. The ideal $J(f) \subset \mathbb{C}[x_1, \dots, x_d]$ generated by the monomials occurring in f is a non-zero monomial ideal, finitely generated by the Hilbert basis theorem, and $\mathrm{wt}_w(f) = \mathrm{wt}_w(J(f))$ by Definition 2.16. Thus the infimum in the second statement is $\psi_{\min}(H, (J(f)), (\frac{1}{m}))$, and the second statement follows from the first statement with $s = 1$, $J_1 = J(f)$, and $\Lambda = \{\frac{1}{m}\}$. □

Corollary 4.6. *Let d, s be positive integers, $\mathcal{V} \subset \mathbb{Q}^d_{>0}$ a finite set and $\Lambda \subset [0, +\infty)$ a DCC set. Then the set of finite numbers*

$$\inf \Big\{ \sum_{i=1}^{d} v_i w_i - \sum_{l=1}^{s} \beta_l \,\mathrm{wt}_w(J_l) \;\Big|\; w \in N_H \cap \mathbb{R}^d_{>0} \Big\},$$

where $v \in \mathcal{V}$, $H \subset \mathbb{T}^d$ is a finite subgroup, $J_1, \dots, J_s$ are non-zero monomial ideals and $\beta_l \in \Lambda$, satisfies the ACC.

Proof. Since $\mathcal{V}$ is finite, by Lemma 2.10(4), it suffices to show that, for any $v \in \mathcal{V}$, the finite numbers in the statement with the fixed vector v satisfy the ACC. Fix $v \in \mathcal{V}$ and a positive integer q such that $qv_i \in \mathbb{Z}$ for any i. Then qv_i is a positive integer for any i. Let $d' := \sum_{i=1}^d qv_i$. We index the coordinates of $\mathbb{R}^{d'}$ and the variables of $\mathbb{C}[x_{i,k}]$ by the pairs (i, k) with $1 \leq i \leq d$ and $1 \leq k \leq qv_i$, that is, we duplicate the i-th coordinate qv_i times. Let $g : \mathbb{R}^d \to \mathbb{R}^{d'}$ be the linear map with $g(w)_{i,k} = w_i$. Since $g(\mathbb{Z}^d) \subset \mathbb{Z}^{d'}$, g induces a homomorphism $\bar{g} : \mathbb{T}^d \to \mathbb{T}^{d'}$.

Let $H \subset \mathbb{T}^d$ be a finite subgroup, $J_1, \dots, J_s$ non-zero monomial ideals, and $\beta_1, \dots, \beta_s \in \Lambda$. Let $H' := \bar{g}(H)$, a finite subgroup of $\mathbb{T}^{d'}$. For any l, let $J'_l \subset \mathbb{C}[x_{i,k}]$ be the monomial ideal generated by the monomials $\prod_{i=1}^d x_{i,k_i}^{\alpha_i}$, where $x^\alpha \in J_l$ and $1 \leq k_i \leq qv_i$ for any i. Then J'_l is a non-zero monomial ideal. For $w' \in \mathbb{R}^{d'}_{\geq 0}$, let $u \in \mathbb{R}^d_{\geq 0}$ be the vector with $u_i = \min_k w'_{i,k}$. Any monomial in J'_l is divisible by one of the generators above, and for fixed α the least w'-weight of the monomials $\prod_i x_{i,k_i}^{\alpha_i}$ is $\langle u, \alpha \rangle$. Thus

$$\mathrm{wt}_{w'}(J'_l) = \mathrm{wt}_u(J_l). \tag{4.1}$$

Let $J' := (J'_1, \dots, J'_s)$ and $q\beta := (q\beta_1, \dots, q\beta_s)$. We show that

$$q \inf \Big\{ \sum_{i=1}^{d} v_i w_i - \sum_{l=1}^{s} \beta_l \,\mathrm{wt}_w(J_l) \;\Big|\; w \in N_H \cap \mathbb{R}^d_{>0} \Big\} = \psi_{\min}(H', J', q\beta).$$

Let $w \in N_H \cap \mathbb{R}^d_{>0}$. Then $g(w) \in N_{H'} \cap \mathbb{R}^{d'}_{>0}$, and $u = w$ for $w' = g(w)$. By (4.1),

$$\psi_{J', q\beta}(g(w)) = \sum_{i=1}^{d} qv_i w_i - \sum_{l=1}^{s} q\beta_l \,\mathrm{wt}_w(J_l) = q\Big(\sum_{i=1}^{d} v_i w_i - \sum_{l=1}^{s} \beta_l \,\mathrm{wt}_w(J_l) \Big).$$

Thus the right-hand side of the equality is at most the left-hand side.

Conversely, let $w' \in N_{H'} \cap \mathbb{R}^{d'}_{>0}$ and let $u \in \mathbb{R}^d_{\geq 0}$ be the vector with $u_i = \min_k w'_{i,k}$. Since $\rho(w') \in \bar{g}(H)$, there exists $a \in N_H$ with $w' - g(a) \in \mathbb{Z}^{d'}$. Thus $w'_{i,k} - a_i \in \mathbb{Z}$ for any (i,k), so $u - a \in \mathbb{Z}^d$ and $u \in N_H \cap \mathbb{R}^d_{>0}$. Moreover, $\sum_{i,k} w'_{i,k} \geq \sum_i q v_i u_i$. Since $q\beta_l \geq 0$, (4.1) implies

$$\psi_{J',q\beta}(w') \geq \sum_{i=1}^{d} q v_i u_i - \sum_{l=1}^{s} q\beta_l \operatorname{wt}_u(J_l) = q\Big(\sum_{i=1}^{d} v_i u_i - \sum_{l=1}^{s} \beta_l \operatorname{wt}_u(J_l)\Big).$$

Thus the left-hand side of the equality is at most the right-hand side, and the equality holds.

The set $q\Lambda \subset [0,+\infty)$ is a DCC set, and d' and s do not depend on H, J_l, and β_l. By Theorem 4.5 in dimension d' with the DCC set $q\Lambda$, the finite numbers $\psi_{\min}(H', J', q\beta)$ satisfy the ACC. Dividing by q, the finite numbers in the statement with the fixed v satisfy the ACC. □

In Corollary 4.6, some J_l may be the unit ideal and some β_l may be zero, since the unit ideal is a non-zero monomial ideal and Λ may contain zero. Corollary 4.7 treats quotients of $\mathbb{A}^3$ with boundaries on the coordinate hyperplanes, and it is not used in the proofs of the main theorems.

Corollary 4.7. *Let $\Gamma \subset [0,1)$ be a DCC set. Then the set of numbers $\operatorname{mld}((\mathbb{A}^3 \ni 0, B)/G)$, where $G \subset (\mathbb{C}^*)^3$ is a finite subgroup acting faithfully on $\mathbb{A}^3$ and $B = \sum_{i=1}^3 b_i \operatorname{div}(x_i)$ with $b_i \in \Gamma \cup \{0\}$, satisfies the ACC.*

Proof. Let $G \subset (\mathbb{C}^*)^3$ be a finite subgroup acting faithfully on $\mathbb{A}^3$, and $B = \sum_{i=1}^3 b_i \operatorname{div}(x_i)$ with $b_i \in \Gamma \cup \{0\}$. Then G is abelian, fixes 0, and acts trivially on the residue field $\mathbb{C}$ of 0, and x_1, x_2, x_3 are G-eigenparameters at 0. Since $b_i < 1$ for any i, Lemma 3.4 with $k = \mathbb{C}$ and $X = \mathbb{A}^3$ implies

$$\operatorname{mld}((\mathbb{A}^3 \ni 0, B)/G) = \min\Big\{ \sum_{i=1}^{3} (1 - b_i) w_i \;\Big|\; w \in N_G \cap \mathbb{R}^3_{>0} \Big\}.$$

Since $\operatorname{wt}_w((x_i)) = w_i$, the right-hand side is $\psi_{\min}(H, J, b)$, where H is the image of G in $\mathbb{T}^3$, $J = ((x_1), (x_2), (x_3))$, and $b = (b_1, b_2, b_3)$. Since $\Gamma \cup \{0\} \subset [0,+\infty)$ is a DCC set, Theorem 4.5 with $d = s = 3$ and the DCC set $\Gamma \cup \{0\}$ implies that the numbers $\psi_{\min}(H, J, b)$, and hence the numbers $\operatorname{mld}((\mathbb{A}^3 \ni 0, B)/G)$, satisfy the ACC. □

4.1. **Subgroups avoiding small coordinate sums.** In this subsection, we prove Theorem 4.8. A lower bound on the minimal log discrepancy of the quotient forces the image of the acting group in $\mathbb{T}^3$ to avoid the points which have a lift with positive coordinates and small coordinate sum, and hence to be contained in one of finitely many closed subgroups of $\mathbb{T}^3$.

Theorem 4.8. *For any $\epsilon > 0$ there exist a positive integer $N(\epsilon)$ and finitely many closed subgroups $H_1, \dots, H_m$ of $\mathbb{T}^3$ with the following properties.*

1. *Any H_a is disjoint from $\rho(\{u \in \mathbb{R}^3_{>0} \mid u_1+u_2+u_3 < 1+\epsilon\})$ and has a coordinate projection $\mathbb{T}^3 \to \mathbb{T}^1$ with finite image. Either H_a is finite, or the identity component of H_a is a coordinate circle, or, after a permutation of the coordinates, the identity component of H_a is the circle with weight vector $(p, -q, 0)$ for coprime positive integers $p, q \leq 1/\epsilon$.*
2. *Let $(X \ni x)$ be a smooth threefold germ, G a finite abelian group acting faithfully on X and fixing x, and $B \geq 0$ a G-invariant $\mathbb{R}$-divisor such that (X, B) is klt and $\operatorname{mld}((X \ni x, B)/G) \geq 1 + \epsilon$. Let x_1, x_2, x_3 be G-eigenparameters at x. Then the image of G in $\mathbb{T}^3$ under the characters of x_1, x_2, x_3 is contained in some H_a. In particular, one of the characters of x_1, x_2, x_3 has order at most $N(\epsilon)$.*
3. *In (2), assume that the character of $t := x_3$ has order at most $N(\epsilon)$. After replacing X by a G-invariant open neighborhood of x on which x_1, x_2, x_3 are regular, let C be the irreducible component of $(x_1 = x_2 = 0)$ containing x. Then, after shrinking X further, C is a smooth G-invariant curve, the kernel of the character of t fixes C pointwise and is the stabilizer of a general point of C, and the image of G in $\operatorname{Aut}(C)$ has order at most $N(\epsilon)$.*

Proof. The proof is long, so we divide it into several steps. Let

$$U := \{u \in \mathbb{R}^3_{>0} \mid u_1 + u_2 + u_3 < 1 + \epsilon\},$$

let $\mathcal{A}$ be the set of closed subgroups of $\mathbb{T}^3$ disjoint from $\rho(U)$, and for $j \in \{1,2,3\}$ let $\mathrm{pr}_j : \mathbb{T}^3 \to \mathbb{T}^1$ be the j-th coordinate projection. We equip $\mathbb{T}^3$ with the metric induced by the Euclidean metric of $\mathbb{R}^3$.

Step 1. In this step we construct $H_1, \dots, H_m$ as the maximal members of $\mathcal{A}$ and show that there exist only finitely many of them.

Since $\rho^{-1}(\rho(U)) = U + \mathbb{Z}^3$ is open in $\mathbb{R}^3$, $\rho(U)$ is open in $\mathbb{T}^3$. Let $(H_i)_{i\geq 1}$ be a sequence in $\mathcal{A}$ which converges in the Hausdorff distance to a closed subgroup H of $\mathbb{T}^3$. Suppose that there exists $h \in H \cap \rho(U)$. Since $\rho(U)$ is open, there exists $r > 0$ such that any point of $\mathbb{T}^3$ at distance less than r from h belongs to $\rho(U)$. For $i \gg 0$, H_i contains such a point, which is not possible. Thus $H \in \mathcal{A}$.

Let $\mathcal{C} \subset \mathcal{A}$ be a non-empty chain under inclusion and let H' be the union of the members of $\mathcal{C}$. Then H' is a subgroup of $\mathbb{T}^3$ disjoint from $\rho(U)$. Since $\mathbb{T}^3 \backslash \rho(U)$ is closed and contains H', the closure $\overline{H'}$ is a closed subgroup disjoint from $\rho(U)$, so $\overline{H'} \in \mathcal{A}$ is an upper bound of $\mathcal{C}$. By Zorn's lemma, any member of $\mathcal{A}$ is contained in a maximal member of $\mathcal{A}$.

Suppose that $\mathcal{A}$ has infinitely many maximal members. Then there exists a sequence $(H_i)_{i\geq 1}$ of pairwise distinct maximal members of $\mathcal{A}$. By Lemma 4.4, after passing to a subsequence, the H_i converge in the Hausdorff distance to a closed subgroup H such that $H_i \subset H$ for any $i \gg 0$. By the second paragraph of this step, $H \in \mathcal{A}$. By the maximality of the H_i, $H_i = H$ for any $i \gg 0$, which contradicts the assumption that the H_i are pairwise distinct. Thus $\mathcal{A}$ has finitely many maximal members, and we denote them by $H_1, \dots, H_m$.

If $\epsilon > 2$, then $(1,1,1) \in U$, so $0 = \rho(1,1,1) \in \rho(U)$. Thus no subgroup of $\mathbb{T}^3$ is disjoint from $\rho(U)$, $\mathcal{A} = \emptyset$, and $m = 0$.

Step 2. In this step we show that for any $H \in \mathcal{A}$ there exists $j \in \{1,2,3\}$ such that $\mathrm{pr}_j(H)$ is finite.

Let $H \in \mathcal{A}$ and let H^0 be the identity component of H. By Lemma 4.3(1), H/H^0 is finite and $H^0 = \rho(V)$ for a linear subspace $V \subset \mathbb{R}^3$ spanned by $V \cap \mathbb{Z}^3$.

Suppose that for any $j \in \{1,2,3\}$ the j-th coordinate function does not vanish identically on V. Since a real vector space is not a finite union of proper subspaces, there exists $v \in V$ with $v_1 v_2 v_3 \neq 0$. Possibly replacing v with $-v$, we may assume that the number n of negative coordinates of v satisfies $n \leq 1$. For $0 < s < 1/\max_i |v_i|$, let $u(s) \in \mathbb{R}^3$ be the vector with $u_i(s) = sv_i$ if $v_i > 0$ and $u_i(s) = 1 + sv_i$ if $v_i < 0$. Then $u(s) \in (0,1)^3$, $u(s) - sv \in \mathbb{Z}^3$, and

$$u_1(s) + u_2(s) + u_3(s) = n + s(v_1 + v_2 + v_3) \leq 1 + s(v_1 + v_2 + v_3).$$

Thus $u(s) \in U$ for $0 < s \ll 1$. Since $\rho(u(s)) = \rho(sv) \in \rho(V) = H^0 \subset H$, this contradicts $H \in \mathcal{A}$. Therefore, there exists $j \in \{1,2,3\}$ such that $v_j = 0$ for any $v \in V$. Then $\mathrm{pr}_j(H^0) = 0$, and $\mathrm{pr}_j(H)$ is the image of the finite group H/H^0, hence finite.

Step 3. In this step we determine the identity component of any $H \in \mathcal{A}$, which completes the proof of (1).

Let $H \in \mathcal{A}$, and let $H^0 = \rho(V)$ and j be as in **Step 2**. Since U is invariant under the permutations of the coordinates, so is $\mathcal{A}$. Possibly permuting the coordinates, we may assume that $j = 3$, that is, $V \subset \{v \in \mathbb{R}^3 \mid v_3 = 0\}$.

If $\dim V = 0$, then $H = H/H^0$ is finite. Suppose that $\dim V = 2$. Then $V = \{v \in \mathbb{R}^3 \mid v_3 = 0\}$. For $0 < s < \epsilon/2$, we have $\rho(s,s,0) \in H^0$, $\rho(s,s,0) = \rho(s,s,1)$, and $(s,s,1) \in U$, which contradicts $H \in \mathcal{A}$. Thus we may assume that $\dim V = 1$. Since $\mathbb{Z}^3 \cap V$ spans V (Lemma 4.3(1)), $V = \mathbb{R}w$ for a primitive vector $w = (w_1, w_2, 0) \in \mathbb{Z}^3$, and H^0 is the circle with weight vector w (Definition 4.1).

If $w_1 w_2 = 0$, then H^0 is a coordinate circle. Suppose that $w_1 w_2 > 0$. Possibly replacing w with $-w$, we may assume that $w_1, w_2 > 0$. For $0 < s < \epsilon/(w_1 + w_2)$, we have $\rho(sw) = \rho(sw_1, sw_2, 1)$ and $(sw_1, sw_2, 1) \in U$, which contradicts $H \in \mathcal{A}$. Thus we may assume that $w_1 w_2 < 0$. Possibly

replacing w with $-w$ and swapping the first two coordinates, we may assume that $w = (p, -q, 0)$ for positive integers $p \le q$, which are coprime since w is primitive.

Since p, q are coprime, there exists an integer l with $pl \equiv 1 \pmod q$. Let $0 < s < 1/(pq)$ and

$$u(s) := \Big(\frac{1}{q} - ps,\ qs,\ 1\Big).$$

Since $\frac{pl-1}{q} \in \mathbb{Z}$, we have

$$\rho\Big(\Big(\frac{l}{q} - s\Big)w\Big) = \rho\Big(\frac{pl}{q} - ps,\ -l + qs,\ 0\Big) = \rho(u(s)),$$

so $\rho(u(s)) \in H^0 \subset H$. Since $ps < \frac{1}{q}$, $u(s) \in \mathbb{R}^3_{>0}$. Since $H \in \mathcal{A}$, $u(s) \notin U$, that is,

$$1 + \frac{1}{q} + (q - p)s \ge 1 + \epsilon.$$

Since $s \in (0, 1/(pq))$ is arbitrary, $\frac{1}{q} \ge \epsilon$. Thus $p \le q \le 1/\epsilon$.

Applying **Steps 2 and 3** to $H = H_1, \dots, H_m$, we obtain (1).

Step 4. In this step we define $N(\epsilon)$ and prove (2).

By **Step 2**, for any $a \in \{1, \dots, m\}$ there exists $j_a \in \{1, 2, 3\}$ such that $\mathrm{pr}_{j_a}(H_a)$ is finite. We let

$$N(\epsilon) := \max_{1 \le a \le m} |\,\mathrm{pr}_{j_a}(H_a)| \quad \text{if } m \ge 1,$$

and $N(\epsilon) := 1$ if $m = 0$.

Let $(X \ni x)$, G, B, and x_1, x_2, x_3 be as in (2). Let W be an affine open neighborhood of x. Since X is separated and G fixes x, $\bigcap_{g \in G} g(W)$ is a G-invariant affine open neighborhood of x. Replacing X with $\bigcap_{g \in G} g(W)$ changes neither the characters of x_1, x_2, x_3 nor $\mathrm{mld}((X \ni x, B)/G)$, since the new quotient is an open neighborhood of the image of x in X/G. Moreover, G still acts faithfully on $\bigcap_{g \in G} g(W)$, since X is irreducible and separated (cf. [Har77, Exercise II.4.2]). Thus we may assume that X is a smooth quasi-projective threefold. Let χ_1, χ_2, χ_3 be the characters of x_1, x_2, x_3, and let H_G be the image of G in $\mathbb{T}^3$ under χ_1, χ_2, χ_3, so that $N_G = \rho^{-1}(H_G)$. By Lemma 3.4, applied with $k = \mathbb{C}$ and the coordinate boundary 0 (here G acts by $\mathbb{C}$-automorphisms and trivially on $\kappa(x) = \mathbb{C}$),

$$\mathrm{mld}((X \ni x, 0)/G) = \min\{w_1 + w_2 + w_3 \mid w \in N_G \cap \mathbb{R}^3_{>0}\}.$$

Since $B \ge 0$, $a(F, X, B) \le a(F, X, 0)$ for any prime divisor F over X. Thus, by Lemma 2.6 applied to B and to 0,

$$\mathrm{mld}((X \ni x, B)/G) = \inf_F \frac{a(F, X, B)}{|I_G(F)|} \le \inf_F \frac{a(F, X, 0)}{|I_G(F)|} = \mathrm{mld}((X \ni x, 0)/G),$$

where F runs over the prime divisors over X with $\mathrm{center}_X F = \{x\}$. Therefore,

$$w_1 + w_2 + w_3 \ge \mathrm{mld}((X \ni x, B)/G) \ge 1 + \epsilon \quad \text{for any } w \in N_G \cap \mathbb{R}^3_{>0}.$$

Suppose that there exists $h \in H_G \cap \rho(U)$. Then $h = \rho(u)$ for some $u \in U$. Since $u \in N_G \cap \mathbb{R}^3_{>0}$ and $u_1 + u_2 + u_3 < 1 + \epsilon$, this is not possible. Thus H_G is disjoint from $\rho(U)$. Since H_G is finite, $H_G \in \mathcal{A}$, so by **Step 1**, $m \ge 1$ and $H_G \subset H_a$ for some a.

The homomorphism $\mathbb{T}^1 \to \mathbb{C}^*$, $u \to e^{2\pi\sqrt{-1}u}$, is injective and maps $\mathrm{pr}_{j_a}(H_G)$ onto $\chi_{j_a}(G)$. Thus the order of χ_{j_a} is

$$|\chi_{j_a}(G)| = |\,\mathrm{pr}_{j_a}(H_G)| \le |\,\mathrm{pr}_{j_a}(H_a)| \le N(\epsilon).$$

This implies (2).

Step 5. In this step we prove (3).

Let χ_1, χ_2, χ_t be the characters of x_1, x_2, t. Since x_1, x_2, x_3 are regular parameters of $\mathcal{O}_{X,x}$, the morphism $(x_1, x_2, x_3) : X \to \mathbb{A}^3$ is étale at x, hence étale on an open neighborhood W of x.

Since G fixes x, $\bigcap_{g\in G} g(W)$ is a G-invariant open neighborhood of x. Possibly replacing X with $\bigcap_{g\in G} g(W)$, we may assume that (x_1, x_2, x_3) is étale on X. Then $(x_1 = x_2 = 0)$ is the preimage of a line in $\mathbb{A}^3$, hence a smooth curve, and the irreducible components of $(x_1 = x_2 = 0)$ are pairwise disjoint. Thus C is a smooth curve, C is the only irreducible component of $(x_1 = x_2 = 0)$ containing x, and the ideal of C in $\mathcal{O}_{X,x}$ is (x_1, x_2). In particular, $t|_C$ generates the maximal ideal of $\mathcal{O}_{C,x}$, and $\widehat{\mathcal{O}}_{C,x} = \mathbb{C}[[t|_C]]$.

Let $g \in G$. For $i = 1, 2$, $g^*x_i = \chi_i(g)x_i$ (Definition 2.14), so g maps $(x_1 = x_2 = 0)$ onto itself. Since $g(x) = x$, $g(C)$ is an irreducible component of $(x_1 = x_2 = 0)$ containing x, so $g(C) = C$. Thus C is G-invariant.

Let $g \in \operatorname{Ker}\chi_t$. Then $g^*(t|_C) = t|_C$. Since g fixes x, g^* induces a local $\mathbb{C}$-algebra automorphism of $\widehat{\mathcal{O}}_{C,x} = \mathbb{C}[[t|_C]]$ which fixes $t|_C$, so g^* is the identity on $\widehat{\mathcal{O}}_{C,x}$. Since $\mathcal{O}_{C,x} \subset \widehat{\mathcal{O}}_{C,x}$, g^* is the identity on the function field of C, so $g|_C$ and id_C coincide on a dense open subset of C. Since C is a separated reduced curve, $g|_C = \mathrm{id}_C$. Conversely, if $g \in G$ fixes C pointwise, then

$$t|_C = g^*(t|_C) = \chi_t(g)t|_C,$$

and $\chi_t(g) = 1$ since $t|_C \neq 0$. Thus $\operatorname{Ker}\chi_t$ is the kernel of the homomorphism $G \to \operatorname{Aut}(C)$, and the image of G in $\operatorname{Aut}(C)$ is isomorphic to $G/\operatorname{Ker}\chi_t \cong \chi_t(G)$. Since the order of χ_t is at most $N(\epsilon)$, the image of G in $\operatorname{Aut}(C)$ has order at most $N(\epsilon)$.

Since $t|_C \neq 0$, the closed points $c \in C$ with $t(c) \neq 0$ form a non-empty open subset of C. Let c be such a point and $g \in G\setminus\operatorname{Ker}\chi_t$. Then

$$t(g(c)) = (g^*t)(c) = \chi_t(g)t(c) \neq t(c),$$

so $g(c) \neq c$. Since $\operatorname{Ker}\chi_t$ fixes C pointwise, the stabilizer of c in G is $\operatorname{Ker}\chi_t$. This implies (3). □

5. Equivariant terminal models and covers

In this section, we prove Construction-Theorem 5.1 on equivariant terminal models whose log discrepancies belong to a fixed semigroup, Construction-Theorem 5.2 on index one covers of several divisors and their inertia characters, Construction-Theorem 5.3 on covers for boundaries with standard coefficients, Lemma 5.4 on equivariant smooth or cDV hypersurface embeddings of Gorenstein terminal threefold germs, Construction-Theorem 5.5 on canonical covers of terminal points, and Proposition 5.6 on semi-invariant equations of the boundary on a canonical cover.

The first two results move the minimal log discrepancy of a $\mathbb{Q}$-factorial klt threefold pair at a closed point to a cyclic quotient of a terminal point. On an index one cover, the Weil divisors $q(K_X + B^{(j)})$ attached to rational boundaries $B^{(j)}$ near B become Cartier for a fixed integer q, so that all log discrepancies belong to a fixed semigroup. An equivariant terminal extraction then extracts only divisors of log discrepancy at most 1. Together they replace the passage to a terminal model of X, which does not preserve the DCC of the coefficients. In the proofs of the main theorems we use Construction-Theorems 5.1 and 5.3 and Lemma 5.4, while Construction-Theorems 5.2 and 5.5 and Proposition 5.6 are, apart from the use of Construction-Theorem 5.5 in Proposition 5.6, used only in Construction-Theorem 6.6, Corollary 6.8(2), and Section 17, which are not used in the proofs of the main theorems.

5.1. **Equivariant terminal models.** In this subsection, we construct equivariant terminal models of klt threefold pairs whose log discrepancies belong to a finitely generated semigroup containing 1. The extraction is performed on the quotient by the group and then normalized in the function field of X, which yields the equivariance, and the semigroup bounds the coefficients of the new boundary and the Cartier index of the canonical divisor. The case of the semigroup $\mathbb{Z}_{\geq 0}$ is used in the proofs of Construction-Theorem 6.7 and Theorem 13.1.

Construction-Theorem 5.1. Let $\alpha_1, \ldots, \alpha_N$ be positive real numbers such that

$$1 \in \mathcal{S} := \Big\{ \sum_{j=1}^{N} \alpha_j n_j \;\Big|\; n_j \in \mathbb{Z}_{\geq 0} \Big\},$$

and let

$$\Lambda_{\mathcal{S}} := \{1 - s \mid s \in \mathcal{S},\ 0 < s \leq 1\}, \quad \eta_{\mathcal{S}} := \min\{s \in \mathcal{S} \mid s > 1\} - 1,$$
$$R_{\mathcal{S}} := \lfloor 1/\eta_{\mathcal{S}} \rfloor, \quad M_{\mathcal{S}} := \operatorname{lcm}(1, \dots, R_{\mathcal{S}}).$$

Let (X, B) be a threefold pair, $x \in X$ a closed point, and H a finite group, and assume that:

(i) (X, B) is klt near x;
(ii) H acts faithfully on a neighborhood of x, fixing x and preserving B;
(iii) $a(E, X, B) \in \mathcal{S}$ for any prime divisor E over the neighborhood of x on which H acts.

Then $\Lambda_{\mathcal{S}} \subset [0, 1)$ is finite, $0 < \eta_{\mathcal{S}} \leq 1$, and, after shrinking to an H-invariant neighborhood of x, there exist an H-equivariant projective birational morphism $g : Y \to X$ and an $\mathbb{R}$-divisor $B_Y \geq 0$ on Y with the following properties.

(1) $K_Y + B_Y = g^*(K_X + B)$, the g-exceptional prime divisors are exactly the exceptional prime divisors E over X with $a(E, X, B) \leq 1$, and $\operatorname{mult}_E B_Y = 1 - a(E, X, B)$ for them.
(2) $B_Y \in \Lambda_{\mathcal{S}}$, the pair (Y, B_Y) is terminal, and all log discrepancies of (Y, B_Y) belong to $\mathcal{S}$.
(3) Any H-invariant Weil divisor on Y is $\mathbb{Q}$-Cartier, that is, Y is H-$\mathbb{Q}$-*factorial*. Each reduced H-orbit sum of components of B_Y is $\mathbb{Q}$-Cartier, and B_Y is a non-negative real linear combination of such orbit sums.
(4) K_Y is $\mathbb{Q}$-Cartier, Y is terminal, and for any closed point $y \in Y$ we have
$$\operatorname{mld}(Y \ni y) \geq \operatorname{mld}(Y \ni y, B_Y) \geq 1 + \eta_{\mathcal{S}},$$
and the Cartier index of K_Y at y is at most $R_{\mathcal{S}}$. In particular $M_{\mathcal{S}} K_Y$ is Cartier.
(5) Let F be a prime divisor over X with $\operatorname{center}_X F = \{x\}$. If $a(F, X, B) \leq 1$, then F is a prime divisor on Y. If $a(F, X, B) > 1$, then $\operatorname{center}_Y F$ is a closed point or a complete curve in $g^{-1}(x)$, which is H-invariant if H stabilizes F.

Moreover, let q be a positive integer, $\lambda_1, \dots, \lambda_s$ positive real numbers with $\sum_j \lambda_j = 1$, and $B^{(1)}, \dots, B^{(s)} \geq 0$ H-invariant $\mathbb{Q}$-divisors such that $B = \sum_j \lambda_j B^{(j)}$, $(X, B^{(j)})$ is lc for any j, and $q(K_X + B^{(j)})$ is Cartier for any j. Define $B_Y^{(j)}$ by $K_Y + B_Y^{(j)} = g^*(K_X + B^{(j)})$. Then $B_Y = \sum_j \lambda_j B_Y^{(j)}$, the coefficients of $B_Y^{(j)}$ belong to

$$\{1 - \tfrac{n}{q} \mid n \in \mathbb{Z},\ 0 \leq n \leq \lfloor q/\lambda_j \rfloor\},$$

and

$$\mathcal{O}_Y(q(K_Y + B_Y^{(j)})) \cong g^* \mathcal{O}_X(q(K_X + B^{(j)}))$$

as H-linearized invertible sheaves. In particular, for any H-fixed closed point $y \in g^{-1}(x)$ and any j, the character of $\mathcal{O}_Y(q(K_Y + B_Y^{(j)}))$ at y equals the character of $\mathcal{O}_X(q(K_X + B^{(j)}))$ at x.

Proof. Let X_0 be an affine open neighborhood of x, contained in the neighborhood of x on which H acts, such that $(X_0, B|_{X_0})$ is klt. Then $\bigcap_{\sigma \in H} \sigma(X_0)$ is an H-invariant affine open neighborhood of x, and (i)–(iii) hold on it. Possibly replacing X with $\bigcap_{\sigma \in H} \sigma(X_0)$, we may assume that X is affine, (X, B) is klt, H acts faithfully on X, and $a(E, X, B) \in \mathcal{S}$ for any prime divisor E over X.

Step 1. In this step, we prove the assertions on $\Lambda_{\mathcal{S}}$ and $\eta_{\mathcal{S}}$, and show that $B \in \Lambda_{\mathcal{S}}$.

Let $L \geq 0$ be a real number. If $\sum_j \alpha_j n_j \leq L$ with $n_j \in \mathbb{Z}_{\geq 0}$, then $0 \leq n_j \leq L/\alpha_j$ for any j. Thus $\mathcal{S} \cap [0, L]$ is finite for any $L \geq 0$. In particular, $\mathcal{S} \cap (0, 1]$ is finite, and hence $\Lambda_{\mathcal{S}}$ is a finite subset of $[0, 1)$. Since $1 \in \mathcal{S}$, we have $2 \in \mathcal{S}$, so $\mathcal{S} \cap (1, 2]$ is a non-empty finite set. Thus $\min\{s \in \mathcal{S} \mid s > 1\} = \min(\mathcal{S} \cap (1, 2])$ exists, $0 < \eta_{\mathcal{S}} \leq 1$, $R_{\mathcal{S}} \geq 1$, and

$$s \geq 1 + \eta_{\mathcal{S}} \quad \text{for any } s \in \mathcal{S} \text{ with } s > 1. \tag{5.1}$$

For any prime divisor P on X, we have $1 - \operatorname{mult}_P B = a(P, X, B) \in \mathcal{S}$, and $0 < 1 - \operatorname{mult}_P B \leq 1$ since (X, B) is klt and $B \geq 0$. Thus $B \in \Lambda_{\mathcal{S}}$.

Step 2. In this step, we show that the set $\mathcal{E}$ of exceptional prime divisors E over X with $a(E, X, B) \leq 1$ is finite and that H permutes the members of $\mathcal{E}$.

By [KM98, Proposition 2.36(1)], there exists a log resolution $f : X'' \to X$ of (X, B) such that, writing $K_{X''} + B'' = f^*(K_X + B)$, the components of B'' with positive coefficients are pairwise disjoint. Here and below, the proofs of [KM98, 2.31, 2.36] work for $\mathbb{R}$-divisors. Since (X, B) is klt, any coefficient of B'' is less than 1. If two components of B'' intersect, then at most one of them has a positive coefficient in B''. Thus [KM98, Corollary 2.31(3)] implies that $\operatorname{discrep}(X'', B'') > 0$, that is,

$$a(F, X'', B'') > 1 \quad \text{for any exceptional prime divisor } F \text{ over } X''.$$

Since $a(F, X, B) = a(F, X'', B'')$ for any prime divisor F over X'', any member of $\mathcal{E}$ is a prime divisor on X'' which is exceptional over X. Thus $\mathcal{E}$ is finite. Since H preserves B, H permutes the members of $\mathcal{E}$.

Step 3. In this step, we construct g and B_Y, and prove (1) and (2).

Let $\pi : X \to W := X/H$ be the quotient morphism and $(W, B_W) := (X, B)/H$. For a prime divisor E over X, let E_W be the prime divisor over W such that $\operatorname{ord}_E|_{\mathbb{C}(W)} \in \mathbb{N}^+ \operatorname{ord}_{E_W}$. By Lemma 2.6(0)(2)(3), $K_X + B = \pi^*(K_W + B_W)$, $K_W + B_W$ is $\mathbb{R}$-Cartier, $B_W \geq 0$,

$$a(E, X, B) = |I_H(E)| \cdot a(E_W, W, B_W) \quad \text{and} \quad \operatorname{center}_W E_W = \pi(\operatorname{center}_X E)$$

for any prime divisor E over X, and any prime divisor over W is of the form E_W. Thus (W, B_W) is klt. Since π is finite, E is exceptional over X if and only if E_W is exceptional over W. Let $\mathcal{E}_W := \{E_W \mid E \in \mathcal{E}\}$. Any member of $\mathcal{E}_W$ is exceptional over W, and $a(E_W, W, B_W) \leq a(E, X, B) \leq 1$ for any $E \in \mathcal{E}$. For a prime divisor P_W over W, let R be the valuation ring of ord_{P_W} in $\mathbb{C}(W)$. By [Ser79, Chapter I, §7, Proposition 19], applied to R and the integral closure of R in $\mathbb{C}(X)$, the prime divisors E over X with $E_W = P_W$ form one H-orbit. Since H permutes the members of $\mathcal{E}$ (**Step 2**), we obtain

$$\mathcal{E} = \{E \mid E \text{ is a prime divisor over } X \text{ with } E_W \in \mathcal{E}_W\}. \tag{5.2}$$

By [KM98, Lemma 2.45], after normalizing the last blow-up, each member P_W of $\mathcal{E}_W$ is a prime divisor on some normal variety V_{P_W} projective and birational over W, and a log resolution of (W, B_W) dominating all V_{P_W} (cf. [KM98, Theorem 0.2]) has every member of $\mathcal{E}_W$ as an exceptional prime divisor. By [BCHM10, Corollary 1.4.3], there exists a projective birational morphism $h : W' \to W$ from a $\mathbb{Q}$-factorial normal variety W' whose exceptional prime divisors are exactly the members of $\mathcal{E}_W$. Let Y be the normalization of W' in $\mathbb{C}(X)$, $\pi_Y : Y \to W'$ the induced finite morphism, and $g : Y \to X$ the induced morphism. Since Y is the normalization of the irreducible component of $X \times_W W'$ dominating X, g is projective and birational. The action of H on $\mathbb{C}(X)$ fixes $\mathbb{C}(W') = \mathbb{C}(W)$, hence induces an action of H on Y such that g is H-equivariant, π_Y is H-invariant, and $Y/H = W'$.

Let P be a prime divisor on Y. Since π and π_Y are finite, P is g-exceptional if and only if $\pi_Y(P)$ is h-exceptional. Since Y is normal and finite over W', the prime divisors on Y over a prime divisor P_W on W' are exactly the prime divisors E over X with $E_W = P_W$. By (5.2), the g-exceptional prime divisors are exactly the members of $\mathcal{E}$. Let B_Y be the $\mathbb{R}$-divisor with $K_Y + B_Y = g^*(K_X + B)$ and $g_*B_Y = B$. For $E \in \mathcal{E}$, we have $\operatorname{mult}_E B_Y = 1 - a(E, X, B)$, where $a(E, X, B) \in \mathcal{S} \cap (0, 1]$, so $\operatorname{mult}_E B_Y \in \Lambda_{\mathcal{S}}$. Together with **Step 1**, $B_Y \geq 0$ and $B_Y \in \Lambda_{\mathcal{S}}$. This proves (1) and $B_Y \in \Lambda_{\mathcal{S}}$ in (2).

Since $K_Y + B_Y = g^*(K_X + B)$, we have $a(F, Y, B_Y) = a(F, X, B) \in \mathcal{S}$ for any prime divisor F over Y. If F is an exceptional prime divisor over Y with $a(F, Y, B_Y) \leq 1$, then F is exceptional over X and $a(F, X, B) \leq 1$, so $F \in \mathcal{E}$ is a prime divisor on Y, a contradiction. Thus (Y, B_Y) is terminal, and (2) holds.

Step 4. In this step, we prove (3) and (4).

Let L be an H-invariant Weil divisor on Y. For a prime divisor P on Y, let e_P be the ramification index of π_Y along P. By [Ser79, Chapter I, §7, Proposition 19], H acts transitively on the prime divisors on Y over $\pi_Y(P)$. Since L is H-invariant, $\operatorname{mult}_P L/e_P$ depends only on $\pi_Y(P)$. Let $L_{W'}$ be the $\mathbb{Q}$-divisor on W' with $\operatorname{mult}_{\pi_Y(P)} L_{W'} = \operatorname{mult}_P L/e_P$ for any prime divisor

P on Y. Then $L = \pi_Y^* L_{W'}$. Since W' is $\mathbb{Q}$-factorial, $mL_{W'}$ is Cartier for some positive integer m, so $mL = \pi_Y^*(mL_{W'})$ is Cartier. Thus Y is H-$\mathbb{Q}$-factorial. Let $D_1, \dots, D_l$ be the reduced H-orbit sums of the components of B_Y. They are H-invariant, hence $\mathbb{Q}$-Cartier. Since B_Y is H-invariant, $B_Y = \sum_k b_k D_k$ for some real numbers $b_k \geq 0$. This proves (3).

Let θ be a non-zero rational section of $\omega_{W'}$. Then $\pi_Y^*\theta$ is an H-invariant rational section of ω_Y, so $K_Y := \operatorname{div}(\pi_Y^*\theta)$ is an H-invariant Weil divisor. By (3), K_Y is $\mathbb{Q}$-Cartier. For any exceptional prime divisor F over Y, let $\operatorname{mult}_F D_k$ be the coefficient along F of the pullback of D_k to a higher model on which F is a prime divisor. Since $D_k \geq 0$ is $\mathbb{Q}$-Cartier, $\operatorname{mult}_F D_k \geq 0$, and hence

$$a(F, Y, 0) = a(F, Y, B_Y) + \sum_k b_k \operatorname{mult}_F D_k \geq a(F, Y, B_Y) > 1.$$

Thus Y is terminal, and $\operatorname{mld}(Y \ni y) \geq \operatorname{mld}(Y \ni y, B_Y)$ for any closed point $y \in Y$. Since $\dim Y = 3$, any prime divisor F over Y with $\operatorname{center}_Y F = \{y\}$ is exceptional over Y, so $a(F, Y, B_Y) \in \mathcal{S}$ and $a(F, Y, B_Y) > 1$ by (2). By (5.1), $a(F, Y, B_Y) \geq 1 + \eta_{\mathcal{S}}$. Thus $\operatorname{mld}(Y \ni y, B_Y) \geq 1 + \eta_{\mathcal{S}}$.

Since Y is terminal, Y is smooth in codimension two by [KM98, Corollary 5.18], so the singular points of Y are isolated closed points. Let $y \in Y$ be a closed point and r the Cartier index of K_Y at y. If $r > 1$, then by [Kaw93, Theorem], there exists a prime divisor F over Y with $\operatorname{center}_Y F = \{y\}$ and discrepancy $\frac{1}{r}$, that is, $a(F, Y, 0) = 1 + \frac{1}{r}$. Thus

$$1 + \eta_{\mathcal{S}} \leq \operatorname{mld}(Y \ni y) \leq 1 + \frac{1}{r},$$

so $r \leq 1/\eta_{\mathcal{S}}$, and hence $r \leq R_{\mathcal{S}}$. If $r = 1$, then $r \leq R_{\mathcal{S}}$ since $R_{\mathcal{S}} \geq 1$. Thus r divides $M_{\mathcal{S}}$, and $M_{\mathcal{S}} K_Y$ is Cartier at any closed point of Y. Since the locus where $M_{\mathcal{S}} K_Y$ is not Cartier is closed and contains no closed point, it is empty. This proves (4).

Step 5. In this step, we prove (5) and the assertions on the $B_Y^{(j)}$.

Let F be a prime divisor over X with $\operatorname{center}_X F = \{x\}$. Since $\dim X = 3$, F is exceptional over X. If $a(F, X, B) \leq 1$, then $F \in \mathcal{E}$ is a prime divisor on Y by **Step 3**. Assume that $a(F, X, B) > 1$. If F were a prime divisor on Y, then F would be g-exceptional, hence $F \in \mathcal{E}$, a contradiction. Thus F is exceptional over Y, and $\operatorname{center}_Y F$ is an irreducible closed subset of $g^{-1}(x)$ with $\dim \operatorname{center}_Y F \leq 1$. Since g is projective, $g^{-1}(x)$ is complete, so $\operatorname{center}_Y F$ is a closed point or a complete curve in $g^{-1}(x)$. If $\sigma \in H$ stabilizes F, then $\sigma(\operatorname{center}_Y F) = \operatorname{center}_Y F$, since g is H-equivariant and the center of ord_F on Y is unique. This proves (5).

Now let q, λ_j, and $B^{(j)}$ be as in the last part of the theorem. Since $\sum_j \lambda_j = 1$ and $B = \sum_j \lambda_j B^{(j)}$,

$$K_Y + B_Y = g^*(K_X + B) = \sum_j \lambda_j g^*(K_X + B^{(j)}) = K_Y + \sum_j \lambda_j B_Y^{(j)},$$

so $B_Y = \sum_j \lambda_j B_Y^{(j)}$. Rational sections of $\omega_X^{\otimes q}$ and of $\omega_Y^{\otimes q}$ are identified through g^*, as g is birational. By the definition of $B_Y^{(j)}$,

$$\operatorname{div}_Y(g^*\vartheta) + qB_Y^{(j)} = g^*\big(\operatorname{div}_X(\vartheta) + qB^{(j)}\big) \tag{5.3}$$

for any non-zero rational section ϑ of $\omega_X^{\otimes q}$.

Since $q(K_X + B^{(j)})$ is Cartier, $qB^{(j)}$ is a Weil divisor, and the right-hand side of (5.3) is a Cartier divisor. Thus $qB_Y^{(j)}$ is a Weil divisor. Let P be a prime divisor on Y and $n_j := qa(P, X, B^{(j)})$. Then $n_j = q - q \operatorname{mult}_P B_Y^{(j)} \in \mathbb{Z}$, and $n_j \geq 0$ since $(X, B^{(j)})$ is lc. Since log discrepancies are affine in the boundary and $B_Y \geq 0$,

$$\frac{1}{q} \sum_j \lambda_j n_j = \sum_j \lambda_j a(P, X, B^{(j)}) = a(P, X, B) = 1 - \operatorname{mult}_P B_Y \leq 1.$$

Since $\lambda_j > 0$ for any j, we have $n_j \le \lfloor q/\lambda_j \rfloor$, and $\mathrm{mult}_P B_Y^{(j)} = 1 - \frac{n_j}{q}$. Thus the coefficients of $B_Y^{(j)}$ belong to

$$\left\{1 - \frac{n}{q} \;\middle|\; n \in \mathbb{Z},\ 0 \le n \le \lfloor q/\lambda_j \rfloor\right\}.$$

Let ϑ be a generator of $\mathcal{O}_X(q(K_X + B^{(j)}))$ (Definition 2.14) on an open subset $X_1 \subset X$. Then $\mathrm{div}_X(\vartheta) + qB^{(j)} = 0$ on X_1, so $\mathrm{div}_Y(g^*\vartheta) + qB_Y^{(j)} = 0$ on $g^{-1}(X_1)$ by (5.3). Thus $g^*\vartheta$ generates $\mathcal{O}_Y(q(K_Y + B_Y^{(j)}))$ (Definition 2.14) on $g^{-1}(X_1)$. Hence $\mathcal{O}_Y(q(K_Y + B_Y^{(j)}))$ is invertible, and g^* induces an isomorphism $g^*\mathcal{O}_X(q(K_X + B^{(j)})) \cong \mathcal{O}_Y(q(K_Y + B_Y^{(j)}))$. Since g is H-equivariant, $\sigma^* g^*\vartheta = g^*\sigma^*\vartheta$ for any $\sigma \in H$, so the isomorphism respects the H-linearizations.

Let $y \in g^{-1}(x)$ be an H-fixed closed point, χ the character of $\mathcal{O}_X(q(K_X + B^{(j)}))$ at x, ϑ_0 a generator of $\mathcal{O}_X(q(K_X + B^{(j)}))$ near x, and

$$\vartheta := \frac{1}{|H|}\sum_{\sigma \in H} \chi(\sigma)^{-1}\sigma^*\vartheta_0.$$

Then $\sigma^*\vartheta = \chi(\sigma)\vartheta$ for any $\sigma \in H$. Since $\sigma^*\vartheta_0 - \chi(\sigma)\vartheta_0$ has zero image in $\mathcal{O}_X(q(K_X + B^{(j)})) \otimes \kappa(x)$ for any $\sigma \in H$, so does $\vartheta - \vartheta_0$. By Nakayama's lemma [Mat89, Theorem 2.2], ϑ generates $\mathcal{O}_X(q(K_X + B^{(j)}))$ near x. Thus $g^*\vartheta$ generates $\mathcal{O}_Y(q(K_Y + B_Y^{(j)}))$ near y, and $\sigma^* g^*\vartheta = \chi(\sigma) g^*\vartheta$ for any $\sigma \in H$. Hence the character of $\mathcal{O}_Y(q(K_Y + B_Y^{(j)}))$ at y is χ. □

5.2. **Index one covers of several divisors.** In this subsection, we construct, for klt threefold pairs with coefficients in a fixed finite set, the index one cover of the Weil divisors $q(K_X + B^{(j)})$ for rational boundaries $B^{(j)}$ near B, and compute the characters of inertia groups of divisors over the cover. Part (3) below shows that along a strictly increasing sequence of minimal log discrepancies, the inertia groups of lifts of computing divisors have unbounded orders, and hence the degrees of the covers are unbounded. Thus a method with a fixed index does not work, and later sections deal with quotients by cyclic groups of unbounded order.

Construction-Theorem 5.2. Let $\Gamma \subset [0,1]$ be a finite set and $\gamma_1, \dots, \gamma_t$ the elements of $\Gamma \cap (0,1)$. Then there exist positive integers s, q, positive real numbers $\lambda_1, \dots, \lambda_s$ with $\sum_j \lambda_j = 1$, and rational numbers $v_{jk} \in [0,1]$, depending only on Γ, such that the following hold. The numbers $\lambda_1, \dots, \lambda_s$ are linearly independent over $\mathbb{Q}$. The points $v_j := (v_{j1}, \dots, v_{jt})$ of $\mathbb{R}^t$ are affinely independent, and the affine hull of $v_1, \dots, v_s$ is the smallest affine subspace of $\mathbb{R}^t$ defined over $\mathbb{Q}$ containing $(\gamma_1, \dots, \gamma_t)$. The set $\mathcal{S} := \{\frac{1}{q}\sum_j \lambda_j n_j \mid n_j \in \mathbb{Z}_{\ge 0}\}$ has finite intersection with any bounded interval. Let (X, B), x, D_k, and $B^{(j)}$ be as follows:

(i) (X, B) is a $\mathbb{Q}$-factorial klt threefold pair with $B \in \Gamma$;
(ii) $x \in X$ is a closed point;
(iii) after shrinking X near x, $B = \sum_k \gamma_k D_k$, where the D_k are reduced divisors without common components, and $B^{(j)} := \sum_k v_{jk} D_k$ for $1 \le j \le s$.

Then $B = \sum_j \lambda_j B^{(j)}$, $(X, B^{(j)})$ is lc, and $Q_j := q(K_X + B^{(j)})$ is a $\mathbb{Q}$-Cartier Weil divisor for any j. Moreover:

(1) Let $A \subset \mathrm{Cl}(\mathcal{O}_{X,x})$ be the subgroup generated by the classes of $Q_1, \dots, Q_s$, and let $\pi : X' \to X$ be the index one cover of $Q_1, \dots, Q_s$ [Zhu24, Lemma-Definition 2.15]: it is a finite quasi-étale abelian Galois morphism with group $G = \mathrm{Hom}(A, \mathbb{C}^*)$, and $\pi^{-1}(x)$ is a single point x'. Let $B' := \pi^* B$ and $B'^{(j)} := \pi^* B^{(j)}$. Then $K_{X'} + B' = \pi^*(K_X + B)$, $K_{X'} + B'^{(j)} = \pi^*(K_X + B^{(j)})$, the pair (X', B') is klt, and $(X', B'^{(j)})$ is lc for any j. The sheaf $\mathcal{O}_{X'}(q(K_{X'} + B'^{(j)}))$ is invertible near x', the character χ_j of $\mathcal{O}_{X'}(q(K_{X'} + B'^{(j)}))$ at x' satisfies $\chi_j(\sigma) = \sigma([Q_j])^{-1}$ for any $\sigma \in G$, and $\bigcap_j \mathrm{Ker}\,\chi_j = \{1\}$. Any prime divisor F over X' satisfies $a(F, X', B') \in \mathcal{S}\backslash\{0\}$. The number $\mathrm{mld}(X \ni x, B)$ is attained.

(2) Let E be a prime divisor over X with $\operatorname{center}_X E = \{x\}$, F a prime divisor over X' lying over E, $e := |I_G(F)|$, and ρ_F the character of $I_G(F)$ on $\mathfrak{m}_F/\mathfrak{m}_F^2$, where $\mathfrak{m}_F$ is the maximal ideal of the valuation ring of ord_F (Lemma 2.6(1)). Let $n_j := qa(F, X', B'^{(j)}) \in \mathbb{Z}_{\geq 0}$. Then $\chi_j|_{I_G(F)} = \rho_F^{n_j}$, $\gcd(e, n_1, \dots, n_s) = 1$, $a(E, X, B^{(j)}) = n_j/(qe)$, and e is the least common multiple of the denominators of the rational numbers $qa(E, X, B^{(j)})$. If E computes $a := \operatorname{mld}(X \ni x, B)$, then $qea = \sum_j \lambda_j n_j$.

(3) For a strictly increasing sequence $a_i = \operatorname{mld}(X_i \ni x_i, B_i)$ in an interval $[\epsilon, M]$ with $0 < \epsilon < M$, where (X_i, B_i) and $x_i \in X_i$ satisfy (i)–(iii), and for prime divisors E_i computing a_i with lifts F_i as in (2), we have $e_i \to +\infty$, $e_i a_i \in \mathcal{S}$, and $a(F_i, X_i', B_i') > 1$ for any $i \gg 0$.

(4) Let F be as in (2), $I := I_G(F)$, and $f : Y \to X'$ an I-equivariant proper birational morphism from a normal quasi-projective variety. Define B_Y and $B_Y^{(j)}$ by $K_Y + B_Y = f^*(K_{X'} + B')$ and $K_Y + B_Y^{(j)} = f^*(K_{X'} + B'^{(j)})$. Then

$$f^*\mathcal{O}_{X'}(q(K_{X'} + B'^{(j)})) = \mathcal{O}_Y(q(K_Y + B_Y^{(j)})),$$

the characters of $\mathcal{O}_Y(q(K_Y + B_Y^{(j)}))$ at any I-fixed point of Y over x' are $\chi_j|_I$, with trivial joint kernel, and $qB_Y^{(j)}$ is a Weil divisor. If $B_Y \geq 0$, then the coefficient of $B_Y^{(j)}$ along an f-exceptional prime divisor P is $1 - \frac{n_j(P)}{q}$ with integers $n_j(P) \geq 0$ satisfying $\frac{1}{q}\sum_j \lambda_j n_j(P) \leq 1$, the coefficients of $B_Y^{(j)}$ along the strict transforms of the components of $\pi^* D_k$ are v_{jk}, and $B_Y \in \Gamma \cup \{0\} \cup \{1 - u \mid u \in \mathcal{S} \cap (0, 1]\}$.

(5) In (4), assume that $\operatorname{center}_Y F$ is a closed point y. Let $h : Y' \to Y$ be a finite quasi-étale Galois morphism from a variety, with group J, let $y' \in Y'$ be a point over y, and assume that Y' is Gorenstein near y'. Let $\hat{I}$ be a finite group acting on Y', fixing y', with an exact sequence $1 \to J \to \hat{I} \to I \to 1$ compatible with h. Then $qh^* B_Y^{(j)}$ is Cartier near y', there exist a nowhere vanishing semi-invariant generator ω of $\omega_{Y'}$ near y' and semi-invariant rational functions g_j with $\operatorname{div}(g_j) = qh^* B_Y^{(j)}$, the character of $\omega^{\otimes q}/g_j$ is the pullback of $\chi_j|_I$ to $\hat{I}$, and the joint kernel of the characters of the $\omega^{\otimes q}/g_j$ is J. If a further character of $\hat{I}$ is faithful on J, then the joint kernel of the characters of the $\omega^{\otimes q}/g_j$ together with the further character is trivial.

Proof of Construction-Theorem 5.2. **Step 1.** In this step, we fix s, q, λ_j, and v_{jk}, and prove the properties of λ_j, v_j, and $\mathcal{S}$.

If $t = 0$, let $s = q = \lambda_1 = 1$. Then $\mathbb{R}^0$ is a point, $\mathcal{S} = \mathbb{Z}_{\geq 0}$, and the properties of the data hold. Assume that $t \geq 1$, let $\gamma := (\gamma_1, \dots, \gamma_t)$, and let $V \subset \mathbb{R}^t$ be the smallest affine subspace defined over $\mathbb{Q}$ containing γ. We work in V because the uniform neighborhood on which [HLS26, Theorem 5.6] preserves the lc property is a neighborhood in V: by [HLS26, Theorem 5.6], there exists an open neighborhood U of γ in V, depending only on t and γ, such that the following holds: for any $\mathbb{Q}$-factorial normal threefold X and reduced divisors $D_1, \dots, D_t$ on X without common components such that $(X, \sum_k \gamma_k D_k)$ is lc, the pair $(X, \sum_k w_k D_k)$ is lc for any $(w_1, \dots, w_t) \in U$. Since $\gamma \in (0, 1)^t$, possibly shrinking U, we may assume that $U \subset (0, 1)^t$. Since V is defined over $\mathbb{Q}$, the rational points of V are dense in V. Thus there exist affinely independent points $v_1, \dots, v_s \in U \cap \mathbb{Q}^t$ with $s - 1 = \dim V$ such that γ is contained in the relative interior of the convex hull of $v_1, \dots, v_s$. If $\dim V = 0$, then $s = 1$ and $v_1 = \gamma$. Write $v_j = (v_{j1}, \dots, v_{jt})$, and write

$$\gamma = \sum_j \lambda_j v_j, \quad \lambda_j > 0, \quad \sum_j \lambda_j = 1.$$

Let q be a positive integer such that $qv_{jk} \in \mathbb{Z}$ for any j, k. The numbers s, q, λ_j, and v_{jk} depend only on Γ, $v_{jk} \in (0, 1) \cap \mathbb{Q}$, and the affine hull of $v_1, \dots, v_s$ is V, as both have dimension $s - 1$.

Suppose that $\sum_j c_j\lambda_j = 0$ for rational numbers $c_1, \dots, c_s$. Since $v_1, \dots, v_s$ are affinely independent rational points with affine hull V, there exists a unique affine function $\ell : V \to \mathbb{R}$ defined over $\mathbb{Q}$ such that $\ell(v_j) = c_j$ for any j. Since $\gamma = \sum_j \lambda_j v_j$ and $\sum_j \lambda_j = 1$,

$$\ell(\gamma) = \sum_j \lambda_j \ell(v_j) = \sum_j \lambda_j c_j = 0.$$

If ℓ is not constant, then $\ell^{-1}(0)$ is a proper affine subspace of V defined over $\mathbb{Q}$ containing γ, which contradicts the minimality of V. Thus ℓ is constant, and $\ell = \ell(\gamma) = 0$. Therefore $c_j = \ell(v_j) = 0$ for any j, and $\lambda_1, \dots, \lambda_s$ are linearly independent over $\mathbb{Q}$. If $n_j \in \mathbb{Z}_{\geq 0}$ and $\sum_j \lambda_j n_j = q$, then $\sum_j \lambda_j(n_j - q) = 0$, so $n_1 = \cdots = n_s = q$ by the linear independence. Construction-Proposition 17.1 uses the equalities $n_1 = \cdots = n_s = q$.

The set $\mathcal{S}$ is the semigroup of Construction-Theorem 5.1 with $N = s$ and $\alpha_j = \lambda_j/q$, and $1 \in \mathcal{S}$ by taking $n_j = q$ for any j. If $\frac{1}{q}\sum_j \lambda_j n_j \leq L$ with $n_j \in \mathbb{Z}_{\geq 0}$, then $n_j \leq qL/\lambda_j$ for any j. Thus $\mathcal{S} \cap [0, L]$ is finite for any $L \geq 0$, and $\mathcal{S}$ has finite intersection with any bounded interval.

Step 2. In this step, we prove the assertions on the $B^{(j)}$.

If $t = 0$, then $B = 0 = B^{(1)}$ and $Q_1 = K_X$ is $\mathbb{Q}$-Cartier. Assume that $t \geq 1$. Since (X, B) is klt and $B \in \Gamma$, any non-zero coefficient of B is one of $\gamma_1, \dots, \gamma_t$. Then

$$\sum_j \lambda_j B^{(j)} = \sum_k \Big(\sum_j \lambda_j v_{jk}\Big) D_k = \sum_k \gamma_k D_k = B.$$

Since $v_j \in U$ and (X, B) is lc, $(X, B^{(j)})$ is lc by **Step 1**. Since X is $\mathbb{Q}$-factorial and $qv_{jk} \in \mathbb{Z}$, Q_j is a $\mathbb{Q}$-Cartier Weil divisor.

Step 3. In this step, we prove (1).

Let θ be a non-zero rational section of ω_X. We take $K_X := \operatorname{div}(\theta)$ in the definition of the Q_j. Since each Q_j is $\mathbb{Q}$-Cartier, the class $[Q_j] \in \operatorname{Cl}(\mathcal{O}_{X,x})$ has finite order, so A is a finite abelian group. By [Zhu24, Lemma-Definition 2.15], π is a finite quasi-étale abelian Galois morphism with group G, $\pi^{-1}(x)$ is a single point x', and x' is fixed by G. We recall the definition of π from [Zhu24, Lemma-Definition 2.15]. After shrinking X to an affine neighborhood of x, fix a Weil divisor L_a on X with class a for any $a \in A$, with $L_0 = 0$. Then $\pi_*\mathcal{O}_{X'} = \bigoplus_{a\in A} \mathcal{O}_X(L_a)$, with the multiplication given by fixed rational functions, and we let $\sigma \in G$ act on X' so that σ^* multiplies the summand $\mathcal{O}_X(L_a)$ by $\sigma(a)$. Let w_a be the rational function on X' given by the rational section 1 of the summand $\mathcal{O}_X(L_a)$. The multiplication $\mathcal{O}_X(L_a) \otimes \mathcal{O}_X(L_b) \to \mathcal{O}_X(L_{a+b})$ is multiplication by a rational function $c_{a,b}$ with $\operatorname{div}(c_{a,b}) = L_a + L_b - L_{a+b}$. Hence $\sigma^* w_a = \sigma(a) w_a$, and if r is the order of a, then $w_a^r \in \mathbb{C}(X)$ and $\operatorname{div}_X(w_a^r) = rL_a$, so $\operatorname{div}(w_a) = \frac{1}{r}\pi^* \operatorname{div}_X(w_a^r) = \pi^* L_a$. For any j, there exists $k_j \in \mathbb{C}(X)^*$ with $Q_j = L_{[Q_j]} + \operatorname{div}(k_j)$ near x. Possibly shrinking X near x, the rational function $u_j := k_j w_{[Q_j]}$ satisfies

$$\operatorname{div}(u_j) = \pi^* Q_j \quad \text{and} \quad \sigma^* u_j = \sigma([Q_j]) u_j \ \text{ for any } \sigma \in G.$$

Let $K_{X'} := \operatorname{div}(\pi^*\theta)$. Since π is quasi-étale, $K_{X'} = \pi^* K_X$ by the Hurwitz formula. Thus $K_{X'} + B' = \pi^*(K_X + B)$ and $K_{X'} + B'^{(j)} = \pi^*(K_X + B^{(j)})$. By [KM98, Proposition 5.20], (X', B') is klt and $(X', B'^{(j)})$ is lc for any j. Let

$$\Theta_j := \frac{(\pi^*\theta)^{\otimes q}}{u_j}.$$

Then

$$\operatorname{div}(\Theta_j) = q\pi^* K_X - \pi^* Q_j = -qB'^{(j)},$$

so Θ_j is a nowhere vanishing generator of the sheaf $\mathcal{O}_{X'}(q(K_{X'} + B'^{(j)}))$ (Definition 2.14). Thus $\mathcal{O}_{X'}(q(K_{X'} + B'^{(j)}))$ is invertible. Since $\pi^*\theta$ is G-invariant, $\sigma^*\Theta_j = \sigma([Q_j])^{-1}\Theta_j$ for any $\sigma \in G$. Thus $\chi_j(\sigma) = \sigma([Q_j])^{-1}$ (Definition 2.14). If $\sigma \in \bigcap_j \operatorname{Ker}\chi_j$, then $\sigma([Q_j]) = 1$ for any j, and $\sigma = 1$ since the $[Q_j]$ generate A.

Let F be a prime divisor over X', and $f_1 : X'' \to X'$ a proper birational morphism from a normal variety on which F is a prime divisor. Write $K_{X''} + B''^{(j)} = f_1^*(K_{X'} + B'^{(j)})$. Since $q(K_{X'} + B'^{(j)})$ is Cartier and $qK_{X''}$ is a Weil divisor, $qB''^{(j)}$ is a Weil divisor. Thus $qa(F, X', B'^{(j)}) = q - q \operatorname{mult}_F B''^{(j)} \in \mathbb{Z}$, and $qa(F, X', B'^{(j)}) \geq 0$ since $(X', B'^{(j)})$ is lc. Since $B' = \sum_j \lambda_j B'^{(j)}$ and $\sum_j \lambda_j = 1$,

$$a(F, X', B') = \sum_j \lambda_j a(F, X', B'^{(j)}) = \frac{1}{q} \sum_j \lambda_j \cdot qa(F, X', B'^{(j)}) \in \mathcal{S}, \tag{5.4}$$

and $a(F, X', B') > 0$ since (X', B') is klt.

Since π is quasi-étale, $R(\pi) = 0$, so $(X', B')/G = (X, B)$ and $(X', B'^{(j)})/G = (X, B^{(j)})$. Let E be a prime divisor over X with $\operatorname{center}_X E = \{x\}$. By Lemma 2.6(3), there exists a prime divisor F over X' lying over E, and by Lemma 2.6(2), $a(F, X', B') = |I_G(F)| \cdot a(E, X, B)$. By (5.4), $a(E, X, B) \in \bigcup_{e=1}^{|G|} \frac{1}{e}\mathcal{S}$, which has finite intersection with any bounded interval by **Step 1**. Let E_0 be a component of the exceptional divisor of the normalized blow-up of x. Then $\operatorname{center}_X E_0 = \{x\}$, and the set of the $a(E, X, B) \leq a(E_0, X, B)$, where E runs over the prime divisors over X with $\operatorname{center}_X E = \{x\}$, is finite and non-empty. Thus $\operatorname{mld}(X \ni x, B)$ is attained. This proves (1).

Step 4. In this step, we prove (2).

Let E, F, e, ρ_F, and n_j be as in (2), and let $I := I_G(F)$. By **Step 3**, $n_j \in \mathbb{Z}_{\geq 0}$. By Lemma 2.6(1), I is cyclic of order e and acts faithfully on $\mathfrak{m}_F/\mathfrak{m}_F^2$. Thus ρ_F is faithful, where $\sigma^*\varpi \equiv \rho_F(\sigma)\varpi \bmod \mathfrak{m}_F^2$ for $\varpi \in \mathfrak{m}_F$ and $\sigma \in I$. Let $\mathcal{O}_F$ be the valuation ring of ord_F and $\kappa(F)$ the residue field. The group I preserves $\mathcal{O}_F$ and acts trivially on $\kappa(F)$.

Let ϖ_0 be a uniformizer of $\mathcal{O}_F$ and

$$\varpi := \frac{1}{e} \sum_{\sigma \in I} \rho_F(\sigma)^{-1} \sigma^* \varpi_0.$$

Then $\varpi \equiv \varpi_0 \bmod \mathfrak{m}_F^2$, so ϖ is a uniformizer, and $\sigma^*\varpi = \rho_F(\sigma)\varpi$ for any $\sigma \in I$. Since $\kappa(F)$ has transcendence degree 2 over $\mathbb{C}$, there exist $\bar{z}_1, \bar{z}_2 \in \kappa(F)$ forming a separating transcendence basis of $\kappa(F)$ over $\mathbb{C}$. Let $z_1^0, z_2^0 \in \mathcal{O}_F$ be lifts of $\bar{z}_1, \bar{z}_2$, and

$$z_i := \frac{1}{e} \sum_{\sigma \in I} \sigma^* z_i^0 \quad (i = 1, 2).$$

Since I acts trivially on $\kappa(F)$, z_i is I-invariant with residue $\bar{z}_i$. Let $\omega_F := d\varpi \wedge dz_1 \wedge dz_2$. Then $\sigma^*\omega_F = \rho_F(\sigma)\omega_F$ for any $\sigma \in I$. For any normal variety on which F is a prime divisor, $\mathcal{O}_F$ is the local ring of that variety at the generic point of F, so $\Omega_{\mathcal{O}_F/\mathbb{C}}$ is a finite $\mathcal{O}_F$-module. Since $\kappa(F)$ is separable over $\mathbb{C}$, [Mat89, Theorems 25.2 and 26.9] give an exact sequence

$$0 \to \mathfrak{m}_F/\mathfrak{m}_F^2 \to \Omega_{\mathcal{O}_F/\mathbb{C}} \otimes \kappa(F) \to \Omega_{\kappa(F)/\mathbb{C}} \to 0.$$

The class of ϖ spans $\mathfrak{m}_F/\mathfrak{m}_F^2$, and $d\bar{z}_1, d\bar{z}_2$ form a basis of $\Omega_{\kappa(F)/\mathbb{C}}$ [Mat89, Theorem 26.5]. Thus $d\varpi, dz_1, dz_2$ generate $\Omega_{\mathcal{O}_F/\mathbb{C}}$ by Nakayama's lemma [Mat89, Theorem 2.2], and they form a basis since $\dim_{\mathbb{C}(X')} \Omega_{\mathbb{C}(X')/\mathbb{C}} = 3$. Hence ω_F generates $\Omega^3_{\mathcal{O}_F/\mathbb{C}}$, which is the stalk of the canonical sheaf at the generic point of F on any normal variety on which F is a prime divisor.

Let $f_1 : X'' \to X'$ and $B''^{(j)}$ be as in **Step 3**, and write $\Theta_j = h_j \omega_F^{\otimes q}$ with $h_j \in \mathbb{C}(X')^*$. By the definition of $B''^{(j)}$,

$$\operatorname{div}_{X''}(\Theta_j) + qB''^{(j)} = f_1^*\big(\operatorname{div}_{X'}(\Theta_j) + qB'^{(j)}\big) = 0.$$

Since ω_F generates $\omega_{X''}$ at the generic point of F,

$$\operatorname{ord}_F(h_j) = \operatorname{mult}_F \operatorname{div}_{X''}(\Theta_j) = -q \operatorname{mult}_F B''^{(j)} = -q\big(1 - a(F, X', B'^{(j)})\big) = n_j - q.$$

Write $h_j = \varpi^{n_j - q}\xi_j$ with $\xi_j \in \mathcal{O}_F^*$. For $\sigma \in I$, we have $\sigma^*\Theta_j = \chi_j(\sigma)\Theta_j$ and $\sigma^*\omega_F^{\otimes q} = \rho_F(\sigma)^q\omega_F^{\otimes q}$, so $\sigma^* h_j = \chi_j(\sigma)\rho_F(\sigma)^{-q}h_j$, and hence

$$\sigma^*\xi_j = \chi_j(\sigma)\rho_F(\sigma)^{-n_j}\xi_j.$$

Since I acts trivially on $\kappa(F)$, the residues of $\sigma^*\xi_j$ and ξ_j in $\kappa(F)$ are equal and non-zero. Thus $\chi_j(\sigma) = \rho_F(\sigma)^{n_j}$, that is, $\chi_j|_I = \rho_F^{n_j}$.

By **Step 3**, $\bigcap_j \operatorname{Ker}(\chi_j|_I) = \{1\}$. Since I is cyclic of order e and ρ_F is faithful, $\bigcap_j \operatorname{Ker}\rho_F^{n_j}$ has order $\gcd(e, n_1, \dots, n_s)$. Thus $\gcd(e, n_1, \dots, n_s) = 1$. By Lemma 2.6(2) applied to $(X', B'^{(j)})$ and G, we have $a(F, X', B'^{(j)}) = ea(E, X, B^{(j)})$. Hence, by the definition of n_j,

$$a(E, X, B^{(j)}) = \frac{n_j}{qe}.$$

The denominator of $qa(E, X, B^{(j)}) = \frac{n_j}{e}$ is $e/\gcd(e, n_j)$, and the least common multiple of $e/\gcd(e, n_1), \dots, e/\gcd(e, n_s)$ is $e/\gcd(e, n_1, \dots, n_s) = e$. If E computes a, then by Lemma 2.6(2) and (5.4),

$$ea = a(F, X', B') = \frac{1}{q}\sum_j \lambda_j n_j.$$

Step 5. In this step, we prove (3).

The numbers s, q, λ_j and the set $\mathcal{S}$ depend only on Γ, so they do not depend on i. By (2) and (5.4), $e_i a_i = a(F_i, X_i', B_i') \in \mathcal{S}$. Let $R \geq 1$ be an integer. If $e_i \leq R$, then

$$a_i \in \bigcup_{e=1}^{R} \frac{1}{e}\big(\mathcal{S} \cap [0, RM]\big),$$

which is a finite set by **Step 1**. Since the a_i are pairwise distinct, $e_i \leq R$ for only finitely many i. Thus $e_i \to +\infty$. Hence $a(F_i, X_i', B_i') = e_i a_i \geq \epsilon e_i > 1$ for any $i \gg 0$.

Step 6. In this step, we prove (4) and (5).

Let F, I, f, B_Y, and $B_Y^{(j)}$ be as in (4). By the definition of $B_Y^{(j)}$,

$$\operatorname{div}_Y(\Theta_j) + qB_Y^{(j)} = f^*\big(\operatorname{div}_{X'}(\Theta_j) + qB'^{(j)}\big) = 0.$$

Thus Θ_j is a nowhere vanishing generator of $\mathcal{O}_Y(q(K_Y + B_Y^{(j)}))$, so

$$f^*\mathcal{O}_{X'}(q(K_{X'} + B'^{(j)})) = \mathcal{O}_Y(q(K_Y + B_Y^{(j)}))$$

as subsheaves of the sheaf of rational sections of $\omega_Y^{\otimes q}$, and $qB_Y^{(j)} = -\operatorname{div}_Y(\Theta_j)$ is a Weil divisor. Since Θ_j is I-semi-invariant with character $\chi_j|_I$, the character of $\mathcal{O}_Y(q(K_Y + B_Y^{(j)}))$ at any I-fixed point of Y over x' is $\chi_j|_I$. By (1), $\bigcap_j \operatorname{Ker}(\chi_j|_I) = \{1\}$.

Assume that $B_Y \geq 0$. Let P be an f-exceptional prime divisor on Y and $n_j(P) := qa(P, X', B'^{(j)})$. By **Step 3**, $n_j(P) \in \mathbb{Z}_{\geq 0}$, and $\operatorname{mult}_P B_Y^{(j)} = 1 - \frac{n_j(P)}{q}$. By (5.4),

$$\frac{1}{q}\sum_j \lambda_j n_j(P) = a(P, X', B') = 1 - \operatorname{mult}_P B_Y \leq 1.$$

Since π is quasi-étale, π^*D_k is reduced, so the coefficient of $B_Y^{(j)}$ along the strict transform of any component of π^*D_k is v_{jk}. Thus the coefficient of B_Y along a prime divisor on Y which is not f-exceptional is 0 or one of the γ_k, and the coefficient along an f-exceptional prime divisor P is $1 - u$ with $u = a(P, X', B') \in \mathcal{S} \cap (0, 1]$ by (1). Hence $B_Y \in \Gamma \cup \{0\} \cup \{1 - u \mid u \in \mathcal{S} \cap (0, 1]\}$. This proves (4).

Let y, h, Y', y', J, and $\hat{I}$ be as in (5). For $\tau \in \hat{I}$, let $\bar{\tau}$ be the image of τ in I, so that $h \circ \tau = \bar{\tau} \circ h$. Since h is quasi-étale, $\operatorname{div}_{Y'}(h^*\vartheta) = h^* \operatorname{div}_Y(\vartheta)$ for any non-zero rational section ϑ of $\omega_Y^{\otimes q}$ by the Hurwitz formula. Thus

$$q(K_{Y'} + h^* B_Y^{(j)}) = h^*\big(q(K_Y + B_Y^{(j)})\big)$$

is Cartier by (4). Since Y' is Gorenstein near y', $K_{Y'}$ is Cartier near y', so $qh^*B_Y^{(j)}$ is Cartier near y'.

Let ψ be the character of $\omega_{Y'}$ at y', ω_0 a generator of $\omega_{Y'}$ on an $\hat{I}$-invariant neighborhood of y', and

$$\omega := \frac{1}{|\hat{I}|} \sum_{\tau \in \hat{I}} \psi(\tau)^{-1} \tau^* \omega_0.$$

Then $\tau^*\omega = \psi(\tau)\omega$ for any $\tau \in \hat{I}$, and $\omega - \omega_0$ has zero image in $\omega_{Y'} \otimes \kappa(y')$. By Nakayama's lemma [Mat89, Theorem 2.2], ω is a nowhere vanishing generator of $\omega_{Y'}$ near y'. For $\tau \in \hat{I}$,

$$\tau^* h^* \Theta_j = h^* \bar{\tau}^* \Theta_j = \chi_j(\bar{\tau}) h^* \Theta_j.$$

Let $g_j := \omega^{\otimes q}/h^*\Theta_j$. Then g_j is a semi-invariant rational function, and

$$\operatorname{div}(g_j) = q \operatorname{div}(\omega) - h^* \operatorname{div}_Y(\Theta_j) = qh^* B_Y^{(j)}$$

near y'. The rational section $\omega^{\otimes q}/g_j = h^*\Theta_j$ has character $\tau \to \chi_j(\bar{\tau})$, which is the pullback of $\chi_j|_I$ to $\hat{I}$. An element $\tau \in \hat{I}$ is in the kernels of the characters $\tau \to \chi_j(\bar{\tau})$ for all j if and only if $\bar{\tau} \in \bigcap_j \operatorname{Ker}(\chi_j|_I) = \{1\}$, that is, $\tau \in J$. If a further character ψ' of $\hat{I}$ is faithful on J, then the joint kernel of the characters of the $\omega^{\otimes q}/g_j$ and ψ' is contained in $J \cap \operatorname{Ker} \psi' = \{1\}$. This proves (5). □

5.3. **Covers for standard coefficients.** In this subsection, we construct a cyclic cover of a klt threefold germ with standard coefficients which removes the boundary, together with a lift of a finite cyclic group action, such that the ratios $a(F, X, B)/|I_G(F)|$, hence the minimal log discrepancy of the crepant quotient, do not change.

In the proof of Theorem 1.3, Construction-Theorem 5.3 absorbs the boundary into the ramification of the cover: a coefficient $1 - \frac{1}{m}$ corresponds to ramification index m. For a coefficient $b \in [0, 1) \backslash \Gamma_{\mathrm{std}}$, the corresponding index $\frac{1}{1-b}$ is not an integer (an integer value m would give $b = 1 - \frac{1}{m} \in \Gamma_{\mathrm{std}}$), for example $\frac{3}{2}$ for $b = \frac{1}{3}$.

Construction-Theorem 5.3. Let $(X \ni x, B)$ be a klt threefold germ, where $B \geq 0$ is a $\mathbb{Q}$-divisor whose non-zero coefficients belong to Γ_{std}, and let G be a finite cyclic group acting faithfully on X, fixing x and preserving B. Let N be the Cartier index of $K_X + B$ at x. Then the denominator m of any coefficient $1 - \frac{1}{m}$ of a component of B through x divides N, and there exists a finite morphism $p : X' \to X$ of degree N from a normal connected variety with the following properties.

(1) $p^{-1}(x)$ is a single point x', p is Galois with group μ_N, the group of N-th roots of unity, X' is klt and $K_{X'}$ is Cartier, $K_{X'} = p^*(K_X + B)$, the ramification index of p is m over a component of B with coefficient $1 - \frac{1}{m}$ and 1 over any other prime divisor, and X' is terminal if (X, B) is terminal.

(2) There exists a finite abelian group $\hat{G}$ of order $N|G|$ acting faithfully on X', fixing x', with an exact sequence $1 \to \mu_N \to \hat{G} \to G \to 1$ compatible with p, such that $X'/\mu_N = X$, $X'/\hat{G} = X/G$, and $(X, B)/G = (X', 0)/\hat{G}$.

(3) For any prime divisor E over X/G, any prime divisor F over X lying over E and any prime divisor F' over X' lying over F, with $e(F'/F)$ the ramification index of $\operatorname{ord}_{F'}$ over ord_F,

$$|I_{\hat{G}}(F')| = e(F'/F)|I_G(F)|, \quad a(F', X', 0) = e(F'/F)a(F, X, B),$$

and

$$\frac{a(F',X',0)}{|I_{\hat{G}}(F')|} = \frac{a(F,X,B)}{|I_G(F)|} = a(E,X/G,B_{X/G}).$$

In particular,

$$\operatorname{mld}((X \ni x, B)/G) = \min_{F'} \frac{a(F',X',0)}{|I_{\hat{G}}(F')|} = \min_{F} \frac{a(F,X,B)}{|I_G(F)|},$$

where F' and F run over the prime divisors with center x' and x, respectively.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove that $m \mid N$, construct p, and prove (1).

Since B is a $\mathbb{Q}$-divisor and $K_X + B$ is $\mathbb{R}$-Cartier, $K_X + B$ is $\mathbb{Q}$-Cartier (being $\mathbb{R}$-Cartier is a system of linear equations with rational coefficients, which has a rational solution if it has a real one), so N exists. Since $N(K_X + B)$ is Cartier near x, the coefficients of $N(K_X + B)$ at the components through x are integers. Since K_X has integer coefficients, $N(1 - \frac{1}{m}) \in \mathbb{Z}$ for any component of B through x with coefficient $1 - \frac{1}{m}$. Thus $m \mid N$.

For any integer $j \geq 0$, let $\mathcal{A}_j$ be the sheaf of rational sections ω of $\omega_X^{\otimes j}$ with $\operatorname{div}(\omega) + jB \geq 0$. Then $\mathcal{A}_j = \mathcal{O}_X(\lfloor j(K_X + B) \rfloor)$, $\mathcal{A}_0 = \mathcal{O}_X$, and $\mathcal{A}_N = \mathcal{O}_X(N(K_X + B))$ is invertible near x. Since G preserves B, G acts on any $\mathcal{A}_j$ by pullback. Let χ be the character of $\mathcal{A}_N$ at x (Definition 2.14), let s_0 be a generator of $\mathcal{A}_N$ near x, and let

$$s := \frac{1}{|G|} \sum_{g \in G} \chi(g)^{-1} g^* s_0.$$

Then $\sigma^* s = \chi(\sigma) s$ for any $\sigma \in G$, and $s - s_0$ vanishes in the fiber of $\mathcal{A}_N$ at x. By Nakayama's lemma [Mat89, Theorem 2.2], s is a G-semi-invariant generator of $\mathcal{A}_N$ near x. Since the statement only concerns the germ at x and G fixes x, possibly replacing X with a G-invariant affine open neighborhood of x (the intersection of the G-translates of an affine open neighborhood of x is G-invariant, and it is affine since X is separated), we may assume that s generates $\mathcal{A}_N$ on X and that any component of B contains x. In particular, $\operatorname{div}(s) = -NB$.

Let

$$\mathcal{A} := \bigoplus_{j=0}^{N-1} \mathcal{A}_j,$$

where the product of $\omega \in \mathcal{A}_i$ and $\omega' \in \mathcal{A}_j$ is $\omega\omega' \in \mathcal{A}_{i+j}$ if $i + j < N$ and $\omega\omega'/s \in \mathcal{A}_{i+j-N}$ if $i + j \geq N$, and let $p : X' := \operatorname{Spec}_X \mathcal{A} \to X$. Then p is the index one cover of (X, B) defined by s [Kol13, Definition 2.49]. Hence X' is normal, $p^{-1}(x)$ is a single point x', and $K_{X'}$ is Cartier [Kol13, Definitions 2.39 and 2.49] and [Kol13, 2.44]. Moreover, the ramification index of p over a component of B with coefficient $1 - \frac{1}{m}$ is m [Kol13, 2.48(3)].

The group μ_N of N-th roots of unity acts on $\mathcal{A}$ by $\mathcal{O}_X$-algebra automorphisms, where $\zeta \in \mu_N$ acts on $\mathcal{A}_j$ by multiplication by ζ^j. The action is faithful since $\mathcal{A}_1 \neq 0$, and $\mathcal{A}^{\mu_N} = \mathcal{A}_0 = \mathcal{O}_X$. Thus $X'/\mu_N = X$. Since $\mathcal{A}$ is a torsion-free $\mathcal{O}_X$-module, no non-zero element of $\mathcal{O}_X$ is a zero divisor of $\mathcal{A}$. Hence any irreducible component of X' dominates X, and so maps onto X since p is finite. Thus any irreducible component of X' contains x'. Since X' is normal, only one irreducible component of X' passes through x'. Therefore X' is a normal connected variety, $\deg p = \operatorname{rank} \mathcal{A} = N$, and p is Galois with group μ_N.

Let P be a prime divisor on X which is not a component of B. Near the generic point of P, X is smooth and $B = 0$, so ω_X has a generator γ there, $\mathcal{A}_j$ is generated by $\gamma^{\otimes j}$ there, and

$$\mathcal{A} \otimes \mathcal{O}_{X,P} \cong \mathcal{O}_{X,P}[T]/(T^N - \gamma^{\otimes N}/s),$$

where $\gamma^{\otimes N}/s$ is a unit of $\mathcal{O}_{X,P}$. Hence p is étale over the generic point of P, and the ramification index of p over P is 1.

By Lemma 2.6(4) applied to the action of μ_N on X', the coefficient of the boundary of $(X',0)/\mu_N$ along a prime divisor P on X is $1-\frac{1}{e}$, where e is the ramification index of p over P. Thus $(X',0)/\mu_N=(X,B)$, and $K_{X'}=p^*(K_X+B)$ by Lemma 2.6(0).

Let F' be a prime divisor over X' and F the prime divisor over X with $\operatorname{ord}_{F'}|_{\mathbb{C}(X)}\in\mathbb{N}^+\operatorname{ord}_F$. By Lemma 2.6(2),

$$a(F',X',0)=|I_{\mu_N}(F')|a(F,X,B)>0.$$

Thus X' is klt (cf. [Kol13, Proposition 2.50(3)]). If (X,B) is terminal and $\operatorname{codim}\operatorname{center}_{X'}F'\geq 2$, then $\operatorname{codim}\operatorname{center}_X F\geq 2$ since $\operatorname{center}_X F=p(\operatorname{center}_{X'}F')$ and p is finite, so $a(F,X,B)>1$ and $a(F',X',0)>1$. Thus X' is terminal if (X,B) is terminal. This proves (1).

Step 2. In this step we lift the action of G to X' and prove (2).

Let σ be a generator of G and $\epsilon\in\mathbb{C}$ a number with $\epsilon^N=\chi(\sigma)^{-1}$. Define $\hat\sigma:\mathcal{A}\to\mathcal{A}$ by $\hat\sigma(\omega):=\epsilon^j\sigma^*\omega$ for $\omega\in\mathcal{A}_j$. For $\omega\in\mathcal{A}_i$ and $\omega'\in\mathcal{A}_j$ with $i+j\geq N$, since $\sigma^*s=\chi(\sigma)s$ and $\epsilon^{-N}=\chi(\sigma)$,

$$\hat\sigma\Big(\frac{\omega\omega'}{s}\Big)=\epsilon^{i+j-N}\frac{\sigma^*\omega\sigma^*\omega'}{\chi(\sigma)s}=\epsilon^{i+j}\frac{\sigma^*\omega\sigma^*\omega'}{s}=\hat\sigma(\omega)\hat\sigma(\omega'),$$

and $\hat\sigma(\omega\omega')=\epsilon^{i+j}\sigma^*\omega\sigma^*\omega'=\hat\sigma(\omega)\hat\sigma(\omega')$ for $i+j<N$. Since σ^* is bijective on any $\mathcal{A}_j$, $\hat\sigma$ is a ring automorphism of $\mathcal{A}$ whose restriction to $\mathcal{A}_0=\mathcal{O}_X$ is σ^*. Hence $\hat\sigma$ is the pullback by an automorphism of X' over σ, which we also denote by $\hat\sigma$.

Let $\hat G$ be the group of automorphisms of X' generated by μ_N and $\hat\sigma$. Since both $\hat\sigma$ and the elements of μ_N act on any $\mathcal{A}_j$ by a constant multiple of σ^* or of the identity, $\hat\sigma$ commutes with μ_N, and $\hat G$ is abelian. Since $(\sigma^{|G|})^*$ is the identity, $\hat\sigma^{|G|}$ acts on $\mathcal{A}_j$ by multiplication by $\epsilon^{j|G|}$, where

$$(\epsilon^{|G|})^N=\chi(\sigma^{|G|})^{-1}=1.$$

Thus $\hat\sigma^{|G|}$ is the element $\epsilon^{|G|}$ of μ_N, and $\hat G=\{\zeta\hat\sigma^k\mid\zeta\in\mu_N,\ k\in\mathbb{Z}\}$. The automorphism of $X=X'/\mu_N$ induced by $\zeta\hat\sigma^k$ is σ^k. Hence the induced homomorphism $\hat G\to G$ is surjective, and the kernel of $\hat G\to G$ is $\{\zeta\hat\sigma^k\mid\zeta\in\mu_N,\ |G|\text{ divides }k\}=\mu_N$. Thus we have an exact sequence $1\to\mu_N\to\hat G\to G\to 1$ compatible with p, and $|\hat G|=N|G|$. Since $p^{-1}(x)=\{x'\}$ and G fixes x, $\hat G$ fixes x'. Moreover, $X'/\hat G=(X'/\mu_N)/G=X/G$.

Let Q be a prime divisor on X/G, P a prime divisor on X over Q with ramification index e_P over Q, and P' a prime divisor on X' over P with ramification index e' over P. By (1), $e'(1-\operatorname{mult}_P B)=1$, and the ramification index of $X'\to X/G$ along P' is $e'e_P$. By Lemma 2.6(4), the coefficients of $B_{X/G}$ and of the boundary of $(X',0)/\hat G$ along Q are

$$1-\frac{1-\operatorname{mult}_P B}{e_P}=1-\frac{1}{e'e_P}$$

respectively. Thus $(X,B)/G=(X',0)/\hat G$. This proves (2).

Step 3. In this step we prove (3).

Let E, F, and F' be as in (3). By **Step 1**, $(X',0)/\mu_N=(X,B)$, so Lemma 2.6(1)(2) applied to μ_N implies that $e(F'/F)=|I_{\mu_N}(F')|$ and $a(F',X',0)=e(F'/F)a(F,X,B)$. By Lemma 2.6(1)(2) applied to G and to $\hat G$, using $(X,B)/G=(X',0)/\hat G$,

$$\operatorname{ord}_F|_{\mathbb{C}(X/G)}=|I_G(F)|\operatorname{ord}_E,\quad \operatorname{ord}_{F'}|_{\mathbb{C}(X/G)}=|I_{\hat G}(F')|\operatorname{ord}_E,$$

$$a(F,X,B)=|I_G(F)|a(E,X/G,B_{X/G}),\quad a(F',X',0)=|I_{\hat G}(F')|a(E,X/G,B_{X/G}).$$

Restricting $\operatorname{ord}_{F'}$ to $\mathbb{C}(X/G)$ in two steps, using $\operatorname{ord}_{F'}|_{\mathbb{C}(X)}=e(F'/F)\operatorname{ord}_F$, gives

$$|I_{\hat G}(F')|=e(F'/F)|I_G(F)|.$$

The formulas in (3) follow.

By the last sentence of Lemma 2.6 applied to (X, B), G, x and to $(X', 0)$, $\hat{G}$, x', and since $(X, B)/G = (X', 0)/\hat{G}$,

$$\operatorname{mld}((X \ni x, B)/G) = \inf_F \frac{a(F, X, B)}{|I_G(F)|} = \inf_{F'} \frac{a(F', X', 0)}{|I_{\hat{G}}(F')|},$$

where F and F' run over the prime divisors with center x and x', respectively. Since $K_{X'}$ is Cartier and X' is klt, $a(F', X', 0)$ is a positive integer, and $|I_{\hat{G}}(F')|$ divides $|\hat{G}| = N|G|$. Thus any ratio $a(F', X', 0)/|I_{\hat{G}}(F')|$ belongs to $\frac{1}{N|G|}\mathbb{N}^+$, and the second infimum is attained by some F'_0. Let F_0 be the prime divisor over X with $\operatorname{ord}_{F'_0}|_{\mathbb{C}(X)} \in \mathbb{N}^+ \operatorname{ord}_{F_0}$. Then $\operatorname{center}_X F_0 = p(\{x'\}) = \{x\}$, and F_0 attains the first infimum by the formulas in (3). This proves (3). □

5.4. **Canonical covers of terminal points.** In this subsection, we embed Gorenstein terminal threefold germs with a finite abelian group action equivariantly into smooth germs with eigenparameters, and we construct the canonical cover of a terminal threefold point together with a lift of a finite cyclic group action.

Lemma 5.4. *Let $(X \ni x)$ be a terminal threefold germ with K_X Cartier, and G a finite abelian group acting faithfully on X and fixing x. Then one of the following two cases occurs.*

(1) X is smooth at x, and there exist G-semi-invariant regular parameters x_1, x_2, x_3 of X at x defining a morphism $X \to \mathbb{A}^3$ étale at x, in which case let $Y := X$, $n := 3$, and $d_f := 1$.

(2) $(X \ni x)$ is an isolated cDV hypersurface singularity [KM98, Definition 5.32], and there exist a smooth fourfold germ $(Y \ni x)$ with a faithful action of G, a G-equivariant closed embedding $X \subset Y$ with $X = (f = 0)$ for a regular G-semi-invariant function f, and G-semi-invariant regular parameters $x_1, \dots, x_4$ of Y at x defining a morphism $Y \to \mathbb{A}^4$ étale at x, in which case let $n := 4$ and let d_f be the character of f.

In both cases any regular G-semi-invariant function on X near x lifts to a regular G-semi-invariant function on Y with equal character, and the character of a generator of ω_X at x is $\prod_{i=1}^n c_i/d_f$, where c_i is the character of x_i.

Proof. Since K_X is Cartier, $(X \ni x)$ is terminal of index 1. Thus either X is smooth at x, or $(X \ni x)$ is an isolated cDV singularity [KM98, Corollary 5.38] (cf. [Rei83, Theorem (1.1)]).

Assume first that X is smooth at x. We show that (1) occurs. Since G is finite abelian and fixes x, $\mathfrak{m}_x/\mathfrak{m}_x^2$ has a basis $\bar{x}_1, \bar{x}_2, \bar{x}_3$ such that $\sigma^*\bar{x}_i = c_i(\sigma)\bar{x}_i$ for any $\sigma \in G$, where c_i is a character of G. Let $x_i^0 \in \mathfrak{m}_x$ be a lift of $\bar{x}_i$ and

$$x_i := \frac{1}{|G|} \sum_{g \in G} c_i(g)^{-1} g^* x_i^0.$$

Then x_i is G-semi-invariant with character c_i, and the class of x_i in $\mathfrak{m}_x/\mathfrak{m}_x^2$ is $\bar{x}_i$. Thus x_1, x_2, x_3 are regular parameters of X at x, and the morphism $X \to \mathbb{A}^3$ defined by them is étale at x. Let $Y := X$ and $d_f := 1$. Then any regular G-semi-invariant function on X near x lifts to itself on $Y = X$, and $dx_1 \wedge dx_2 \wedge dx_3$ is a generator of ω_X near x with character $c_1c_2c_3$, which is the character of ω_X at x by Definition 2.14.

Assume now that X is singular at x. We show that (2) occurs. A general hypersurface section $(H \ni x)$ of $(X \ni x)$ is a Du Val singularity [KM98, Definition 5.32], so it is analytically isomorphic to a hypersurface singularity in $\mathbb{A}^3$ [KM98, Theorem 4.20]. Thus $\dim \mathfrak{m}_{H,x}/\mathfrak{m}_{H,x}^2 \le 3$. Since $\mathfrak{m}_{H,x}$ is the quotient of $\mathfrak{m}_x$ by an equation of H, $\dim \mathfrak{m}_x/\mathfrak{m}_x^2 \le 4$. Since $\dim X = 3$ and X is singular at x, $\dim \mathfrak{m}_x/\mathfrak{m}_x^2 = 4$.

Possibly replacing X with a G-invariant affine open neighborhood of x (the intersection of the G-translates of an affine open neighborhood of x is G-invariant, and it is affine since X is separated), we may assume that X is affine. Let $q_1, \dots, q_k$ be generators of the $\mathbb{C}$-algebra $\mathcal{O}(X)$, and let $W \subset \mathcal{O}(X)$ be the linear span of the functions $g^*q_i - q_i(x)$ for $g \in G$ and $1 \le i \le k$. Since

G fixes x, $(g^*q_i)(x) = q_i(x)$, so any element of W vanishes at x. Moreover, W is G-stable and $\dim W < +\infty$. Since G is finite abelian, W has a basis $y_1, \dots, y_M$ of G-semi-invariant functions. Since $q_1, \dots, q_k$ belong to $W + \mathbb{C}$, $y_1, \dots, y_M$ define a G-equivariant closed embedding $X \subset \mathbb{A}^M$ with x mapped to the origin, where G acts on $\mathbb{A}^M$ such that the pullback of the coordinate y_i by $\sigma \in G$ is a constant multiple of y_i.

Let $\mathfrak{n} \subset \mathbb{C}[y_1, \dots, y_M]$ be the maximal ideal of the origin and $\mathfrak{a}$ the ideal of X. Since $\mathfrak{n}/(\mathfrak{a} + \mathfrak{n}^2) \cong \mathfrak{m}_x/\mathfrak{m}_x^2$, $\dim(\mathfrak{a} + \mathfrak{n}^2)/\mathfrak{n}^2 = M - 4$. The space $(\mathfrak{a} + \mathfrak{n}^2)/\mathfrak{n}^2$ is G-stable, so it has a basis of G-semi-invariant classes. Lift each class to an element of $\mathfrak{a}$, and apply the projection $\frac{1}{|G|}\sum_{g\in G} \lambda(g)^{-1} g^*$, where λ is the character of the class. Since $\mathfrak{a}$ is G-stable and the class is fixed by the projection, we obtain G-semi-invariant elements $\ell_1, \dots, \ell_{M-4} \in \mathfrak{a}$ whose differentials at the origin are linearly independent. By the Jacobian criterion, the common zero locus Y_0 of $\ell_1, \dots, \ell_{M-4}$ is smooth of dimension 4 near the origin, and Y_0 is G-stable since any ℓ_i is G-semi-invariant. Let U be a G-invariant affine open neighborhood of the origin such that $Y := Y_0 \cap U$ is smooth and irreducible (the intersection of the G-translates of an affine open neighborhood of the origin is G-invariant, and it is affine since $\mathbb{A}^M$ is separated). Since $X \subset Y_0$, replacing X with $X \cap U$, we obtain a G-equivariant closed embedding $X \subset Y$. The action of G on Y is faithful since the action of G on X is faithful.

The ideal of X in the regular local ring $\mathcal{O}_{Y,x}$ is a prime ideal of height 1, and a regular local ring is a unique factorization domain by the theorem of Auslander and Buchsbaum [Mat89, Theorem 20.3]. Thus the ideal of X in $\mathcal{O}_{Y,x}$ is principal [Mat89, Theorem 20.1], and the ideal sheaf $\mathcal{I}$ of X in Y is a G-invariant subsheaf of $\mathcal{O}_Y$ which is invertible near x. Let d_f be the character of $\mathcal{I}$ at x, f_0 a generator of $\mathcal{I}$ near x, and

$$f := \frac{1}{|G|}\sum_{g\in G} d_f(g)^{-1} g^* f_0.$$

Then f is G-semi-invariant with character d_f, and $f - f_0$ vanishes in the fiber of $\mathcal{I}$ at x. By Nakayama's lemma, f generates $\mathcal{I}$ near x. Thus, after shrinking Y to a G-invariant open neighborhood of x (the intersection of the G-translates of an open neighborhood of x), $X = (f = 0)$.

Since G fixes x and is finite abelian, $\mathfrak{m}_{Y,x}/\mathfrak{m}_{Y,x}^2$ has a basis of G-semi-invariant classes $\bar{x}_1, \dots, \bar{x}_4$ with characters $c_1, \dots, c_4$. Let $x_i^0 \in \mathfrak{m}_{Y,x}$ be a lift of $\bar{x}_i$ and

$$x_i := \frac{1}{|G|}\sum_{g\in G} c_i(g)^{-1} g^* x_i^0.$$

Then $x_1, \dots, x_4$ are G-semi-invariant regular parameters of Y at x with characters $c_1, \dots, c_4$, and the morphism $Y \to \mathbb{A}^4$ defined by them is étale at x since their differentials form a basis of $\mathfrak{m}_{Y,x}/\mathfrak{m}_{Y,x}^2$.

Let ψ be a regular G-semi-invariant function on X near x with character χ_ψ. Since $\mathcal{O}_{Y,x} \to \mathcal{O}_{X,x}$ is surjective, there exists a regular function $\tilde{\psi}$ on Y near x with $\tilde{\psi}|_X = \psi$. Since the restriction $\mathcal{O}_{Y,x} \to \mathcal{O}_{X,x}$ is G-equivariant,

$$\Big(\frac{1}{|G|}\sum_{g\in G} \chi_\psi(g)^{-1} g^* \tilde{\psi}\Big)\Big|_X = \frac{1}{|G|}\sum_{g\in G} \chi_\psi(g)^{-1} g^* \psi = \psi.$$

Thus ψ lifts to a regular G-semi-invariant function on Y near x with character χ_ψ.

Finally, $dx_1 \wedge \cdots \wedge dx_4$ generates ω_Y near x, so $dx_1 \wedge \cdots \wedge dx_4/f$ generates $\omega_Y(X)$ near x. Since $(X \ni x)$ is an isolated singularity, $X\backslash\{x\}$ is smooth near x, and on $X\backslash\{x\}$ the isomorphism $\omega_Y(X)|_X \cong \omega_X$ of [Har77, Proposition II.8.20] is given by the Poincaré residue. Hence the residue

$$\theta := \mathrm{Res}\Big(\frac{dx_1 \wedge \cdots \wedge dx_4}{f}\Big)$$

generates ω_X on $X\backslash\{x\}$ near x. Since X is normal, $\{x\}$ has codimension 3 in X, and ω_X is reflexive, θ is a generator of ω_X near x. The residue does not depend on the choice of the equation

of X: replacing f with uf for a unit u changes $d(uf) = u(df) + f(du)$ from $u(df)$ only by $f(du)$, which vanishes on X. Hence the residue map $\omega_Y(X) \to \omega_X$ commutes with automorphisms of Y preserving X. Since

$$\sigma^*\Big(\frac{dx_1 \wedge \cdots \wedge dx_4}{f}\Big) = \frac{c_1c_2c_3c_4}{d_f}(\sigma) \cdot \frac{dx_1 \wedge \cdots \wedge dx_4}{f}$$

for any $\sigma \in G$, the character of θ is $\prod_{i=1}^4 c_i/d_f$, which is the character of ω_X at x by Definition 2.14. □

Construction-Theorem 5.5. Let $(X \ni x)$ be a terminal threefold germ, H a finite cyclic group acting faithfully on X and fixing x, and r the Cartier index of K_X at x. Let $B \geq 0$ be an H-invariant $\mathbb{R}$-divisor such that (X, B) is klt, let $a := \mathrm{mld}((X \ni x, B)/H) > 0$, and let F be a prime divisor over X with $\mathrm{center}_X F = \{x\}$, $I_H(F) = H$, and $a(F, X, B) = |H|a$. Then, after shrinking H-invariant neighborhoods, the following hold.

(1) There exist a finite quasi-étale morphism $h : X' \to X$ of degree r, the canonical cover [KM98, Definition 5.19], and a finite abelian group $\hat{H}$ acting on X', with the following properties.
  (a) $h^{-1}(x)$ is a single point x'.
  (b) X' is Gorenstein and terminal.
  (c) (X', B') is klt, where $B' := h^*B$.
  (d) h is Galois with group μ_r, the group of r-th roots of unity, and $X'/\mu_r = X$.
  (e) $\hat{H}$ acts faithfully on X' and fixes x'.
  (f) There exists an exact sequence $1 \to \mu_r \to \hat{H} \to H \to 1$ compatible with h, and $X'/\hat{H} = X/H$.
  (g) The character of a generator of $\omega_{X'}$ at x' is faithful on μ_r.

(2) Let F' be a prime divisor over X' lying over F, u the ramification index of $\mathrm{ord}_{F'}$ over ord_F, and $H' := I_{\hat{H}}(F')$. Then the following hold.
  (a) u divides r.
  (b) u is the denominator of the rational number $a(F, X, 0)$.
  (c) H' is cyclic of order $u|H|$.
  (d) H' maps onto H.
  (e) $|H' \cap \mu_r| = u$.
  (f) $\mathrm{mld}((X' \ni x', B')/H') = a$, attained by the prime divisor under F'.
  (g) The morphism $X'/H' \to X/H$ has degree r/u.
  (h) For any prime divisor P over X'/H' with center the image of x', any prime divisor P' over X' lying over P, and the prime divisor P'' over X/H under P', the number $|I_{\hat{H}}(P')|/|H' \cap I_{\hat{H}}(P')|$ is a positive integer and

$$a(P, X'/H', B'_{X'/H'}) = \frac{|I_{\hat{H}}(P')|}{|H' \cap I_{\hat{H}}(P')|} a(P'', X/H, B_{X/H}) \geq a.$$

(3) One of the following two cases occurs.
  (a) X' is smooth at x', in which case let $Y := X'$, $n := 3$, and $d_f := 1$.
  (b) $(X' \ni x')$ is an isolated cDV singularity, and there exists an $\hat{H}$-equivariant closed embedding of X' into a smooth fourfold germ $(Y \ni x')$ with $X' = (f = 0)$ for a regular $\hat{H}$-semi-invariant function f, in which case let $n := 4$ and let d_f be the character of f.

  In both cases $\hat{H}$ acts faithfully on Y, there exist $\hat{H}$-semi-invariant regular parameters $x_1, \ldots, x_n$ of Y at x', they define a morphism $Y \to \mathbb{A}^n$ étale at x', any regular $\hat{H}$-semi-invariant function on X' near x' lifts to a regular semi-invariant function on Y with equal character, and $\kappa = \prod_i c_i/d_f$, where c_i is the character of x_i and κ is the character of a generator of $\omega_{X'}$ at x'.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove (1).

Since X is terminal, $(X \ni x, 0)$ is a klt threefold germ, and the Cartier index of $K_X + 0$ at x is r. Apply Construction-Theorem 5.3 to $(X \ni x, 0)$ and H, and let $h : X' \to X$, x', and $\hat{H}$ be the morphism p, the point x', and the group $\hat{G}$ obtained there. The index one cover of $(X, 0)$ defined by an H-semi-invariant generator of $\mathcal{O}_X(rK_X)$ at x [Kol13, Definition 2.49] is the canonical cover of X at x [KM98, Definition 5.19], [Kol13, Example 3.7]. By Construction-Theorem 5.3(1)(2), h is a finite morphism of degree r, Galois with group μ_r, $h^{-1}(x) = \{x'\}$, the ramification index of h over any prime divisor on X is 1, $K_{X'} = h^*K_X$ is Cartier, X' is terminal, $\hat{H}$ is a finite abelian group acting faithfully on X' and fixing x', and there exists an exact sequence $1 \to \mu_r \to \hat{H} \to H \to 1$ compatible with h such that $X'/\mu_r = X$ and $X'/\hat{H} = X/H$. In particular, h is quasi-étale. Since X' is klt, X' is Cohen–Macaulay [KM98, Theorem 5.22, Theorem 5.10], so $\omega_{X'}$ is the dualizing sheaf of X'. Since $K_{X'}$ is Cartier, X' is Gorenstein. Moreover, μ_r acts faithfully on the fiber of $\omega_{X'}$ at x' (see the last sentence of [Kol13, Definition 2.49]), so the character of a generator of $\omega_{X'}$ at x' is faithful on μ_r.

Since h is quasi-étale, $K_{X'} + B' = h^*(K_X + B)$, and B' is $\hat{H}$-invariant since B is H-invariant. By Lemma 2.6(4) applied to μ_r, $(X', B')/\mu_r = (X, B)$. Let E' be a prime divisor over X' and E the prime divisor over X with $\mathrm{ord}_{E'}|_{\mathbb{C}(X)} \in \mathbb{N}^+ \mathrm{ord}_E$. By Lemma 2.6(2), $a(E', X', B') = |I_{\mu_r}(E')|a(E, X, B) > 0$. Thus (X', B') is klt. This proves (1).

Step 2. In this step we compute the character of a generator of $\omega_{X'}$ at x' on H'.

Let κ be the character of $\omega_{X'}$ at x', and θ an $\hat{H}$-semi-invariant generator of $\omega_{X'}$ near x' with character κ, which exists by applying the projection $\frac{1}{|\hat{H}|}\sum_{g\in\hat{H}} \kappa(g)^{-1}g^*$ to a generator and by Nakayama's lemma. Let $\mathcal{O}_{F'}$ be the valuation ring of $\mathrm{ord}_{F'}$, $\mathfrak{m}_{F'}$ the maximal ideal of $\mathcal{O}_{F'}$, and $k_{F'} := \mathcal{O}_{F'}/\mathfrak{m}_{F'}$. Since H' stabilizes $\mathrm{ord}_{F'}$ and acts trivially on $k_{F'}$, H' acts on the one-dimensional $k_{F'}$-vector space $\mathfrak{m}_{F'}/\mathfrak{m}_{F'}^2$ by a character $\rho : H' \to k_{F'}^*$. The values of ρ are roots of unity, so they belong to $\mathbb{C}$, and ρ is faithful by Lemma 2.6(1).

Let t_0 be a uniformizer of $\mathcal{O}_{F'}$ and

$$t := \frac{1}{|H'|}\sum_{g\in H'} \rho(g)^{-1}g^*t_0.$$

Since $g^*t_0 - \rho(g)t_0 \in \mathfrak{m}_{F'}^2$ for any $g \in H'$, $t - t_0 \in \mathfrak{m}_{F'}^2$. Thus t is a uniformizer of $\mathcal{O}_{F'}$ with character ρ. Let $z_1^0, z_2^0 \in \mathcal{O}_{F'}$ be lifts of a transcendence basis of $k_{F'}$ over $\mathbb{C}$, and

$$z_i := \frac{1}{|H'|}\sum_{g\in H'} g^*z_i^0 \quad (i = 1, 2).$$

Since H' acts trivially on $k_{F'}$, z_i is an H'-invariant element of $\mathcal{O}_{F'}$ whose residue in $k_{F'}$ equals the residue of z_i^0. The ring $\mathcal{O}_{F'}$ is the local ring at the generic point of F' of a normal variety on which F' is a prime divisor, and a normal variety is smooth at the generic point of any prime divisor. Since t is a uniformizer and the residues of z_1, z_2 form a transcendence basis of $k_{F'}$, the conormal sequence of $\mathcal{O}_{F'} \to k_{F'}$ and Nakayama's lemma imply that dt, dz_1, dz_2 form a basis of $\Omega_{\mathcal{O}_{F'}/\mathbb{C}}$. Hence

$$\theta = wt^{a(F',X',0)-1}dt \wedge dz_1 \wedge dz_2$$

for some unit w of $\mathcal{O}_{F'}$, since θ generates $\omega_{X'}$ near x' and the order of θ along F' is $a(F', X', 0) - 1$ (Definition 2.1). For $\sigma \in H'$, $\sigma^*\theta = \kappa(\sigma)\theta$, $\sigma^*t = \rho(\sigma)t$, and $\sigma^*z_i = z_i$, so

$$\sigma^*w = \kappa(\sigma)\rho(\sigma)^{-a(F',X',0)}w.$$

Since σ acts trivially on $k_{F'}$ and the residue of w is non-zero, we obtain

$$\kappa|_{H'} = \rho^{a(F',X',0)}.$$

Step 3. In this step we prove (2).

By definition, $I_{\mu_r}(F') = H' \cap \mu_r$. By Lemma 2.6(1) applied to μ_r, $u = |H' \cap \mu_r|$, so u divides r. By Construction-Theorem 5.3(3) applied as in **Step 1**,

$$|H'| = u|I_H(F)| = u|H| \quad \text{and} \quad a(F', X', 0) = ua(F, X, 0).$$

The group H' is cyclic by Lemma 2.6(1). The image of H' in H has order $|H'|/|H' \cap \mu_r| = |H|$, so H' maps onto H.

Since $K_{X'}$ is Cartier, $a(F', X', 0)$ is an integer. By (1), κ is faithful on μ_r, so κ is faithful on the cyclic group $H' \cap \mu_r$ of order u. Since ρ is faithful on H', **Step 2** implies that $\rho^{a(F',X',0)}$ is faithful on $H' \cap \mu_r$, so $a(F', X', 0)$ is coprime to u. Since $a(F, X, 0) = a(F', X', 0)/u$, the denominator of $a(F, X, 0)$ is u.

Let $\pi : X' \to X/H$ be the quotient morphism by $\hat{H}$, which is finite Galois with group $\hat{H}$, and let E_F be the prime divisor over X/H with $\operatorname{ord}_F|_{\mathbb{C}(X/H)} \in \mathbb{N}^+ \operatorname{ord}_{E_F}$. By Lemma 2.6(0), $K_X + B$ is the pullback of $K_{X/H} + B_{X/H}$, so

$$K_{X'} + B' = h^*(K_X + B) = \pi^*(K_{X/H} + B_{X/H}).$$

By Lemma 2.6(2), $\operatorname{center}_{X/H} E_F$ is the image of x, and

$$a(E_F, X/H, B_{X/H}) = \frac{a(F, X, B)}{|I_H(F)|} = \frac{|H|a}{|H|} = a = \operatorname{mld}((X \ni x, B)/H) > 0.$$

Moreover, $\operatorname{ord}_{F'}|_{\mathbb{C}(X/H)} \in \mathbb{N}^+ \operatorname{ord}_{E_F}$, and $\operatorname{center}_{X'} F' = \{x'\}$ since $\operatorname{center}_X F = \{x\}$ and $h^{-1}(x) = \{x'\}$. Since $H' = I_{\hat{H}}(F')$, Lemma 3.1, with $(X/H, B_{X/H})$, the image of x, E_F, π, $\hat{H}$, F', H', and $\mathrm{id}_{X'}$ in place of (X, B), x, E, π, G, F, H, and f, implies that $\operatorname{mld}(X'/H' \ni \bar{x}', B'_{X'/H'}) = a$, where $\bar{x}'$ is the image of x', attained by the prime divisor under F', and that for P, P', and P'' as in (2),

$$a(P, X'/H', B'_{X'/H'}) = \frac{|I_{\hat{H}}(P')|}{|H' \cap I_{\hat{H}}(P')|} a(P'', X/H, B_{X/H}) \geq a.$$

By Definition 2.5, $(X' \ni x', B')/H' = (X'/H' \ni \bar{x}', B'_{X'/H'})$, so $\operatorname{mld}((X' \ni x', B')/H') = a$. Since $H' \cap I_{\hat{H}}(P')$ is a subgroup of $I_{\hat{H}}(P')$, the ratio $|I_{\hat{H}}(P')|/|H' \cap I_{\hat{H}}(P')|$ is the index of a subgroup, hence a positive integer. Finally, $X'/H' \to X'/\hat{H} = X/H$ has degree

$$[\hat{H} : H'] = \frac{r|H|}{u|H|} = \frac{r}{u}.$$

This proves (2).

Step 4. In this step we prove (3).

By (1), $(X' \ni x')$ is a terminal threefold germ with $K_{X'}$ Cartier, and $\hat{H}$ is a finite abelian group acting faithfully on X' and fixing x'. Thus (3) follows from Lemma 5.4 applied to $(X' \ni x')$ and $\hat{H}$. □

5.5. **Equations of boundaries.** In this subsection, we write the boundary B' of Construction-Theorem 5.5 with regular semi-invariant equations on X', and we express a in terms of the orders of the equations along F'. Proposition 5.6 is used only in Corollary 6.8(2), which is not used in the proofs of the main theorems.

Proposition 5.6. *Notation and conditions as in Construction-Theorem 5.5. Let $\Lambda \subset [0, 1)$ be a finite set, ℓ a non-negative integer, and $\Sigma \subset [0, 1)$ a DCC set. Assume that any H-invariant Weil divisor on X is $\mathbb{Q}$-Cartier, and that $B = B^0 + \sum_{j=1}^{\ell} t_j P_j$, where $B^0 \geq 0$ is H-invariant with $B^0 \in \Lambda$ and (X, B^0) terminal, $P_j \geq 0$ are H-invariant Cartier divisors, and $t_j \in \Sigma$. For $\beta \in \Lambda \cap (0, 1)$, let V_β be the reduced sum of the components of B^0 with coefficient β. Then h^*V_β is Cartier on X', and there exist regular $\hat{H}$-semi-invariant functions φ_β and g_j on X' such that*

$\operatorname{div}(\varphi_\beta) = h^*V_\beta$, $\operatorname{div}(g_j) = h^*P_j$, *the functions* g_j *are pullbacks of* H*-semi-invariant equations of* P_j*, and*

$$B' = \sum_\beta \beta \operatorname{div}(\varphi_\beta) + \sum_j t_j \operatorname{div}(g_j).$$

Moreover, if ρ *is the character of* H' *on* $\mathfrak{m}_{F'}/\mathfrak{m}_{F'}^2$*, where* $\mathfrak{m}_{F'}$ *is the maximal ideal of the valuation ring of* $\operatorname{ord}_{F'}$*, then* $\kappa|_{H'} = \rho^{a(F',X',0)}$*, the characters of* φ_β *and* g_j *restrict to* $\rho^{\operatorname{ord}_{F'}\varphi_\beta}$ *and* $\rho^{\operatorname{ord}_{F'} g_j}$ *on* H'*, and*

$$a = \frac{a(F',X',0) - \sum_\beta \beta \operatorname{ord}_{F'}\varphi_\beta - \sum_j t_j \operatorname{ord}_{F'} g_j}{u|H|}.$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we show that h^*V_β is Cartier.

Since B^0 is H-invariant, any element of H permutes the components of B^0 with coefficient β. Thus V_β is an H-invariant Weil divisor, hence V_β is $\mathbb{Q}$-Cartier by assumption. Since h is quasi-étale, the coefficient of any prime divisor of X' in h^*V_β is 0 or 1. Thus h^*V_β is a $\mathbb{Q}$-Cartier Weil divisor on X'. Since X' is Gorenstein and terminal (Construction-Theorem 5.5(1)), $(X' \ni x')$ is a terminal threefold singularity of index one, hence h^*V_β is Cartier near x' by [Kaw88, Lemma 5.1]. That lemma concerns analytic germs, and the algebraic statement follows since $\mathcal{O}_{X',x'} \to \mathcal{O}^{\mathrm{an}}_{X',x'}$ is faithfully flat. Shrinking the $\hat{H}$-invariant neighborhood of x', we may assume that h^*V_β is Cartier on X' for any β.

Step 2. In this step, we construct φ_β and g_j, and we prove the formula for B'.

Since the actions of $\hat{H}$ on X' and of H on X are compatible with h and V_β is H-invariant, h^*V_β is $\hat{H}$-invariant. Thus $\mathcal{L}_\beta := \mathcal{O}_{X'}(-h^*V_\beta)$ is an $\hat{H}$-invariant subsheaf of the sheaf of rational functions on X', and $\mathcal{L}_\beta$ is invertible near x' by **Step 1**. Let χ_β be the character of $\mathcal{L}_\beta$ at x' (Definition 2.14), let ψ_β be a generator of $\mathcal{L}_\beta$ near x', and let

$$\varphi_\beta := \frac{1}{|\hat{H}|} \sum_{\sigma\in\hat{H}} \chi_\beta(\sigma)^{-1}\sigma^*\psi_\beta.$$

Since χ_β is a character, $\tau^*\varphi_\beta = \chi_\beta(\tau)\varphi_\beta$ for any $\tau \in \hat{H}$. Since $\sigma^*\psi_\beta - \chi_\beta(\sigma)\psi_\beta$ vanishes in the fiber of $\mathcal{L}_\beta$ at x' for any $\sigma \in \hat{H}$, the images of φ_β and ψ_β in the fiber are equal and non-zero. By Nakayama's lemma, φ_β generates $\mathcal{L}_\beta$ near x'. Shrinking the neighborhood of x', φ_β is a regular $\hat{H}$-semi-invariant function on X' with $\operatorname{div}(\varphi_\beta) = h^*V_\beta$. If $V_\beta = 0$, we take $\varphi_\beta = 1$.

Since P_j is an H-invariant Cartier divisor, the formula defining φ_β, with H, $\mathcal{O}_X(-P_j)$, and x in place of $\hat{H}$, $\mathcal{L}_\beta$, and x', implies that there exists a regular H-semi-invariant function p_j near x with $\operatorname{div}(p_j) = P_j$. If $P_j = 0$, we take $p_j = 1$. Let $g_j := h^*p_j$. Then $\operatorname{div}(g_j) = h^*P_j$. If $\hat{\sigma} \in \hat{H}$ maps to $\sigma \in H$, then $h \circ \hat{\sigma} = \sigma \circ h$, hence $\hat{\sigma}^*g_j = h^*\sigma^*p_j$. Thus g_j is $\hat{H}$-semi-invariant, and the character of g_j is the composition of $\hat{H} \to H$ with the character of p_j.

Since $B^0 \in \Lambda$, we have $B^0 = \sum_\beta \beta V_\beta$. Since $B' = h^*B$, we obtain

$$B' = \sum_\beta \beta h^*V_\beta + \sum_j t_j h^*P_j = \sum_\beta \beta \operatorname{div}(\varphi_\beta) + \sum_j t_j \operatorname{div}(g_j).$$

Step 3. In this step, we show that any non-zero rational function ξ on X' which is H'-semi-invariant with character χ satisfies

$$\chi = \rho^{\operatorname{ord}_{F'}\xi}. \tag{5.5}$$

Let $\mathcal{O}_{F'}$ be the valuation ring of $\operatorname{ord}_{F'}$ and $k_{F'} := \mathcal{O}_{F'}/\mathfrak{m}_{F'}$ the residue field. Since $H' = I_{\hat{H}}(F')$, any $\sigma \in H'$ preserves $\mathcal{O}_{F'}$ and $\mathfrak{m}_{F'}$, and σ acts trivially on $k_{F'}$. Let s_0 be a generator of $\mathfrak{m}_{F'}$ and

$$s := \frac{1}{|H'|} \sum_{\sigma\in H'} \rho(\sigma)^{-1}\sigma^*s_0.$$

By the definition of ρ, $\sigma^* s_0 - \rho(\sigma) s_0 \in \mathfrak{m}_{F'}^2$ for any $\sigma \in H'$. Thus $s - s_0 \in \mathfrak{m}_{F'}^2$, hence s generates $\mathfrak{m}_{F'}$. Since ρ is a character, $\tau^* s = \rho(\tau) s$ for any $\tau \in H'$.

Let $N := \operatorname{ord}_{F'} \xi$ and write $\xi = s^N v$, where v is a unit of $\mathcal{O}_{F'}$. For any $\sigma \in H'$,

$$\chi(\sigma) s^N v = \sigma^* \xi = \rho(\sigma)^N s^N \sigma^* v,$$

hence $\sigma^* v = \chi(\sigma)\rho(\sigma)^{-N} v$. Since σ acts trivially on $k_{F'}$, the image of $\sigma^* v$ in $k_{F'}$ equals the image of v, which is non-zero. Thus $\chi(\sigma) = \rho(\sigma)^N$, which is (5.5).

Applying (5.5) to $\xi = \varphi_\beta$ and $\xi = g_j$, the characters of φ_β and g_j restrict to $\rho^{\operatorname{ord}_{F'} \varphi_\beta}$ and $\rho^{\operatorname{ord}_{F'} g_j}$ on H'.

Step 4. In this step, we show that $\kappa|_{H'} = \rho^{a(F',X',0)}$.

Since $K_{X'}$ is Cartier, the argument of **Step 2**, applied to the $\hat{H}$-invariant sheaf $\omega_{X'}$, implies that there exists an $\hat{H}$-semi-invariant generator θ of $\omega_{X'}$ near x' with character κ. Let $z_1^0, z_2^0 \in \mathcal{O}_{F'}$ be elements whose images in $k_{F'}$ form a transcendence basis of $k_{F'}$ over $\mathbb{C}$, and let

$$z_i := \frac{1}{|H'|} \sum_{\sigma \in H'} \sigma^* z_i^0$$

for $i = 1, 2$. Since H' acts trivially on $k_{F'}$, the image of z_i in $k_{F'}$ equals the image of z_i^0, and z_i is H'-invariant. Since $\mathcal{O}_{F'}$ is the local ring of a smooth variety at the generic point of a prime divisor, the $\mathcal{O}_{F'}$-module $\Omega_{\mathcal{O}_{F'}/\mathbb{C}}$ is free of rank 3. The kernel of the map

$$\Omega_{\mathcal{O}_{F'}/\mathbb{C}} \otimes k_{F'} \to \Omega_{k_{F'}/\mathbb{C}}$$

is spanned by the image of ds. Since $k_{F'}$ is a finite extension of $\mathbb{C}(\bar{z}_1, \bar{z}_2)$, where $\bar{z}_i$ is the image of z_i in $k_{F'}$, we have $\Omega_{k_{F'}/\mathbb{C}(\bar{z}_1,\bar{z}_2)} = 0$. Thus the images of dz_1, dz_2 form a basis of $\Omega_{k_{F'}/\mathbb{C}}$ [Har77, Propositions II.8.3A and II.8.4A, Theorem II.8.6A]. By Nakayama's lemma, ds, dz_1, dz_2 form a basis of $\Omega_{\mathcal{O}_{F'}/\mathbb{C}}$, hence

$$\delta := ds \wedge dz_1 \wedge dz_2$$

generates $\Omega^3_{\mathcal{O}_{F'}/\mathbb{C}}$. Let $\xi := \theta/\delta \in \mathbb{C}(X')^*$. Since θ generates $\omega_{X'}$ near x', we have $\operatorname{ord}_{F'} \xi = a(F', X', 0) - 1$.

For any $\sigma \in H'$, we have

$$\sigma^* \delta = d(\sigma^* s) \wedge d(\sigma^* z_1) \wedge d(\sigma^* z_2) = \rho(\sigma)\delta$$

and $\sigma^* \theta = \kappa(\sigma)\theta$. Thus ξ is H'-semi-invariant with character $\kappa|_{H'}\rho^{-1}$. By (5.5),

$$\kappa|_{H'}\rho^{-1} = \rho^{a(F',X',0)-1},$$

hence $\kappa|_{H'} = \rho^{a(F',X',0)}$.

Step 5. In this step, we prove the formula for a.

By Construction-Theorem 5.5(2), $|H'| = u|H|$, and $\operatorname{mld}((X' \ni x', B')/H') = a$ is attained by the prime divisor E' over X'/H' under F' (Definition 2.7). Since $\hat{H}$ acts faithfully on X', so does H'. Since (X', B') is klt, $K_{X'} + B'$ is $\mathbb{R}$-Cartier. Since $H' = I_{\hat{H}}(F')$, we have $I_{H'}(F') = H'$. By Lemma 2.6(2) for $G = H'$,

$$a(F', X', B') = |H'| a(E', X'/H', B'_{X'/H'}) = u|H|a.$$

For any non-zero rational function φ on X' and any higher model of X' on which F' is a divisor, the coefficient of F' in the pullback of $\operatorname{div}(\varphi)$ is $\operatorname{ord}_{F'} \varphi$. Thus the formula for B' in **Step 2** implies

$$a(F', X', B') = a(F', X', 0) - \sum_\beta \beta \operatorname{ord}_{F'} \varphi_\beta - \sum_j t_j \operatorname{ord}_{F'} g_j.$$

Dividing by $u|H|$, we obtain

$$a = \frac{a(F', X', 0) - \sum_\beta \beta \operatorname{ord}_{F'} \varphi_\beta - \sum_j t_j \operatorname{ord}_{F'} g_j}{u|H|}.$$

□

6. Reduction to quotients of terminal points

In this section, we prove Lemma 6.1, Proposition 6.3, and Construction-Theorems 6.6 and 6.7. Lemma 6.1 reduces Theorems 1.2 and 1.3 to $\mathbb{Q}$-factorial klt pairs, and Construction-Theorem 6.7, which starts from Part (1) of Proposition 6.3, realizes the sequences arising in Theorem 1.2 by minimal log discrepancies of quotients of terminal threefold points by cyclic groups.

6.1. **From lc pairs to klt pairs.** In this subsection, we prove Lemmas 6.1 and 6.2, which reduce closed-point minimal log discrepancies of lc threefold pairs to those of $\mathbb{Q}$-factorial klt threefold pairs.

On a crepant dlt modification the reduced part of the boundary is non-zero in general, while the minimal log discrepancy at a closed point may still be positive. We replace the reduced part by the average of two general members of a linear subsystem of the complete linear system of the reduced part, obtained by perturbing the reduced part by functions vanishing to high order at the point. The linear subsystem is finite-dimensional, contains the reduced part itself, and is base point free outside the point, and the order of a general member along a divisor computing the minimal log discrepancy equals the order of the reduced part along that divisor. The replacement removes all lc places without changing the minimal log discrepancy at the point, at the cost of one extra coefficient one half and a shift by the dimension of a center.

Lemma 6.1. *Let $\Gamma \subset [0,1]$ be a set and $\Lambda := (\Gamma \cap [0,1)) \cup \{\frac{1}{2}\}$. Then the following hold.*

1. *For any $\mathbb{Q}$-factorial dlt threefold pair (Y, D) with $D \in [0,1]$ and any closed point $y \in Y$ with $b := \operatorname{mld}(Y \ni y, D) > 0$, there exist an affine neighborhood U of y and reduced divisors S_1, S_2 on U, linearly equivalent to $\lfloor D \rfloor|_U$ and without common components with each other or with $(D - \lfloor D \rfloor)|_U$, such that (U, D') is klt and $\operatorname{mld}(U \ni y, D') = b$, where $D' := (D - \lfloor D \rfloor)|_U + \frac{1}{2}(S_1 + S_2)$. If $\lfloor D \rfloor = 0$, then one can take $D' = D|_U$.*
2. *Any positive element $a \in \operatorname{MLD}(3, \Gamma)$ is of the form $a = b - d$ with $b \in \operatorname{MLD}_{\mathrm{Qf}}(3, \Lambda)$ and $d \in \{0, 1\}$.*
3. *If Γ is finite and $\operatorname{MLD}_{\mathrm{Qf}}(3, \Lambda')$ satisfies the ACC for any finite set $\Lambda' \subset [0,1]$, then $\operatorname{MLD}(3, \Gamma)$ satisfies the ACC.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we derive from Lemma 2.13 a lower bound for log discrepancies on a smooth variety.

Let W be a smooth variety, $\Sigma = \sum_k P_k$ a reduced snc divisor on W, $\Theta = \sum_k \theta_k P_k$ an $\mathbb{R}$-divisor with $\theta_k \leq 1$ for all k, F a prime divisor over W, ξ the generic point of $\operatorname{center}_W F$, and $c := \operatorname{codim}_W \overline{\{\xi\}}$. After renumbering, let $P_1, \dots, P_r$ be the components of Σ containing ξ, and $x_1, \dots, x_r \in \mathcal{O}_{W,\xi}$ local equations of $P_1, \dots, P_r$. Since Σ is snc, $x_1, \dots, x_r$ extend to regular parameters $x_1, \dots, x_c$ of $\mathcal{O}_{W,\xi}$. By Lemma 2.13, applied with the coefficients $\theta_1, \dots, \theta_r, 0, \dots, 0$ on an open neighborhood of ξ on which $\Theta = \sum_{j=1}^{r} \theta_j \operatorname{div}(x_j)$ and $\sum_{j=1}^{c} \operatorname{div}(x_j)$ is snc, we have

$$a(F, W, \Theta) \geq \sum_{j=1}^{r} (1 - \theta_j) \operatorname{ord}_F x_j + \sum_{j=r+1}^{c} \operatorname{ord}_F x_j. \tag{6.1}$$

Since $\operatorname{ord}_F x_j \geq 1$ and $1 - \theta_j \geq 0$ for any j, each summand on the right-hand side is non-negative. Thus $a(F, W, \Theta) \geq 1 - \theta_j$ for any $1 \leq j \leq r$, and $a(F, W, \Theta) \geq 0$. Moreover, since $c \geq 1$, if $\theta_k < 1$ for all k, then $a(F, W, \Theta) > 0$.

Step 2. In this step, we show that (Y, D) is lc, prove (1) when $\lfloor D \rfloor = 0$, and describe the centers of the lc places of (Y, D).

By assumption, the coefficients of D are at most one. By [KM98, Theorem 2.44] (cf. [Fuj11, Definition 10.1]), there exists a resolution $g : Y'' \to Y$ such that $\operatorname{Exc}(g)$ and $\operatorname{Exc}(g) \cup \operatorname{Supp} g_*^{-1} D$

are snc divisors and $a(E,Y,D)>0$ for any g-exceptional prime divisor E on Y''. Let $K_{Y''}+D'':=g^*(K_Y+D)$. Then

$$D''=g_*^{-1}D+\sum_E(1-a(E,Y,D))E,$$

where E runs over the g-exceptional prime divisors. Thus $\operatorname{Supp}D''$ is contained in the snc divisor $\operatorname{Exc}(g)\cup\operatorname{Supp}g_*^{-1}D$, the coefficients of D'' are at most one, and the coefficients of the g-exceptional prime divisors in D'' are less than one. By **Step 1** with $W=Y''$ and $\Theta=D''$, we have $a(F,Y,D)=a(F,Y'',D'')\geq 0$ for any prime divisor F over Y, so (Y,D) is lc. If $\lfloor D\rfloor=0$, then all coefficients of D'' are less than one, so (Y,D) is klt by **Step 1**. In this case, for any affine neighborhood U of y, the pair $(U,D|_U)$ is klt and $\operatorname{mld}(U\ni y,D|_U)=b$, which proves (1).

Now let P be a prime divisor over Y with $a(P,Y,D)=0$, ξ the generic point of $\operatorname{center}_{Y''}P$, and $\eta:=g(\xi)$, which is the generic point of $\operatorname{center}_Y P$. Apply (6.1) with $W=Y''$, $\Theta=D''$, and $F=P$. Since the left-hand side is zero and each summand on the right-hand side is non-negative, we have $r=c\geq 1$ and $\theta_1=\cdots=\theta_r=1$. In particular, no component of $\operatorname{Supp}D''$ containing ξ is g-exceptional. Thus $\xi\notin\operatorname{Exc}(g)$, g is an isomorphism near ξ, and $P_1,\dots,P_r$ are the strict transforms of components of $\lfloor D\rfloor$ containing η. Therefore, Y is smooth at η, and $\lfloor D\rfloor$ is defined near η by one function φ in the maximal ideal $\mathfrak{m}_{Y,\eta}$ of $\mathcal{O}_{Y,\eta}$. Since $\operatorname{center}_Y P=\overline{\{\eta\}}$, we have $\operatorname{ord}_P\varphi>0$.

Step 3. In this step, we assume that $\lfloor D\rfloor\neq 0$, choose a prime divisor E over Y computing b, and construct a vector space $\mathcal{V}$ of rational functions.

Let U be an affine neighborhood of y, $S:=\lfloor D\rfloor|_U$, and $\mathfrak{m}_y\subset\mathcal{O}_U$ the ideal sheaf of y. By **Step 2**, $(U,D|_U)$ is lc.

By [KM98, Theorem 0.2], applied to the product of $\mathfrak{m}_y$ and the ideal sheaf of $\operatorname{Supp}D|_U$, there exists a log resolution $h_0:Y_0\to U$ of $(U,D|_U)$ such that $\mathfrak{m}_y\mathcal{O}_{Y_0}$ is invertible and $\operatorname{Exc}(h_0)\cup\operatorname{Supp}h_{0*}^{-1}D|_U\cup h_0^{-1}(y)$ is an snc divisor. Let Q be a prime divisor over U with $\operatorname{center}_U Q=\{y\}$. Then $\operatorname{center}_{Y_0}Q$ is contained in a component T of $h_0^{-1}(y)$. By **Step 1**, applied to Y_0 and the crepant pullback of $D|_U$, whose coefficients are at most one because $(U,D|_U)$ is lc, we have $a(Q,U,D|_U)\geq a(T,U,D|_U)$. Since $\operatorname{center}_U T=\{y\}$, there exists a prime divisor E over U such that $\operatorname{center}_U E=\{y\}$ and $a(E,U,D|_U)=b$.

Since U is affine, the $\mathcal{O}(U)$-module

$$H^0(U,\mathcal{O}_U(S))=\{s\in\mathbb{C}(U)\backslash\{0\}\mid S+\operatorname{div}(s)\geq 0\}\cup\{0\}$$

is generated by finitely many elements $s_1,\dots,s_t$. Moreover, $1\in H^0(U,\mathcal{O}_U(S))$ because $S\geq 0$. Since $\operatorname{mult}_E\mathfrak{m}_y\geq 1$, there exists a positive integer N such that

$$N\operatorname{mult}_E\mathfrak{m}_y+\operatorname{ord}_E s_j>0\quad\text{for any } 1\leq j\leq t.$$

Let $u_1,\dots,u_q\in\mathcal{O}(U)$ be generators of the ideal $H^0(U,\mathfrak{m}_y^N)$, and let $\mathcal{V}\subset\mathbb{C}(U)$ be the $\mathbb{C}$-linear span of 1 and u_ks_j for $1\leq k\leq q$ and $1\leq j\leq t$. Then $\dim\mathcal{V}<+\infty$, and the following hold.

(a) $S+\operatorname{div}(s)\geq 0$ for any $s\in\mathcal{V}\backslash\{0\}$.
(b) $\mathcal{V}_E:=\{s\in\mathcal{V}\backslash\{0\}\mid\operatorname{ord}_E s>0\}\cup\{0\}$ is a proper linear subspace of $\mathcal{V}$.
(c) For any scheme-theoretic point $p\neq y$ of U, the stalk $\mathcal{O}_U(S)_p$ is generated by $\mathcal{V}$ as an $\mathcal{O}_{U,p}$-module.
(d) For any prime divisor P_U on U, $\min_{s\in\mathcal{V}\backslash\{0\}}\operatorname{ord}_{P_U}s=-\operatorname{mult}_{P_U}S$.

Property (a) holds because any u_k is regular on U. For (b), $\mathcal{V}_E$ is a linear subspace because $\operatorname{ord}_E(s+s')\geq\min\{\operatorname{ord}_E s,\operatorname{ord}_E s'\}$, it contains u_ks_j for any k and j because

$$\operatorname{ord}_E(u_ks_j)\geq N\operatorname{mult}_E\mathfrak{m}_y+\operatorname{ord}_E s_j>0,$$

and it does not contain 1. For (c), since $p\neq y$, the ideal $\mathfrak{m}_y^N\mathcal{O}_{U,p}$ is the unit ideal, so some u_k is a unit in $\mathcal{O}_{U,p}$, and $s_j=u_k^{-1}(u_ks_j)$ for any j. For (d), let p be the generic point of P_U and $m:=\operatorname{mult}_{P_U}S$. Then $\mathcal{O}_{U,p}$ is a discrete valuation ring, $\mathcal{O}_U(S)_p=\pi^{-m}\mathcal{O}_{U,p}$ for a uniformizer π, and $p\neq y$ because $\dim U=3$. Thus (d) follows from (a) and (c).

Step 4. In this step, we construct a resolution $h : Y' \to U$ and a divisor $R \geq 0$ on Y', and show that the coefficients of $\Delta_0 := \Delta - R$ are less than one, where $K_{Y'} + \Delta := h^*(K_U + D|_U)$.

Since $\dim \mathcal{V} < +\infty$, there exists $w \in \mathcal{O}(U)\backslash\{0\}$ such that $ws \in \mathcal{O}(U)$ for any $s \in \mathcal{V}$. By [KM98, Theorem 0.2], applied to the product of $\mathfrak{m}_y$, the ideal sheaf of $\operatorname{Supp} D|_U$, the ideal sheaf $w\mathcal{O}_U$, and the ideal sheaf generated by $w\mathcal{V}$, there exist a smooth variety Y', a projective birational morphism $h : Y' \to U$, and a reduced snc divisor Σ on Y' such that the pullback $\mathcal{I}\mathcal{O}_{Y'}$ of any of the four ideal sheaves $\mathcal{I}$ above is invertible, and Σ contains $\operatorname{Exc}(h)$ and the support of the divisor defined by $\mathcal{I}\mathcal{O}_{Y'}$ for any such $\mathcal{I}$. Here $\mathcal{I}\mathcal{O}_{Y'}$ is invertible because the pullback of the product is invertible and Y' is integral. Moreover, $\operatorname{Supp} h_*^{-1}D|_U \subset \Sigma$. In particular, $\mathfrak{m}_y\mathcal{O}_{Y'}$ is invertible, so $h^{-1}(y) \subset \Sigma$ is a divisor, and the $\mathcal{O}_{Y'}$-submodule $\mathcal{V}\mathcal{O}_{Y'}$ of $\mathbb{C}(Y')$ generated by $\mathcal{V}$ is invertible. Let R be the divisor on Y' with $\mathcal{V}\mathcal{O}_{Y'} = \mathcal{O}_{Y'}(R)$, that is,

$$\operatorname{mult}_P R = -\min_{s\in\mathcal{V}\backslash\{0\}} \operatorname{ord}_P s$$

for any prime divisor P on Y'. Then $\operatorname{Supp} R \subset \Sigma$, and $R \geq 0$ because $1 \in \mathcal{V}$. By **Step 3**(d), $h_*R = S$. For $s \in \mathcal{V}\backslash\{0\}$, let $H_s := \operatorname{div}(s) + R$. Then $H_s \geq 0$, and the divisors H_s form a base point free linear system, because $\mathcal{V}$ generates $\mathcal{O}_{Y'}(R)$.

Since $(U, D|_U)$ is lc, the coefficients of Δ are at most one. Since $R \geq 0$, any prime divisor P with $\operatorname{mult}_P \Delta < 1$ satisfies $\operatorname{mult}_P \Delta_0 < 1$. Let P be a prime divisor on Y' with $\operatorname{mult}_P \Delta = 1$, that is, $a(P, U, D|_U) = 0$, and let η be the generic point of $\operatorname{center}_U P$. Since $b > 0$, we have $\operatorname{center}_U P \neq \{y\}$, so $\eta \neq y$. By **Step 2**, Y is smooth at η, and S is defined near η by a function $\varphi \in \mathfrak{m}_{Y,\eta}$ with $\operatorname{ord}_P \varphi > 0$. Thus $\mathcal{O}_U(S)_\eta = \varphi^{-1}\mathcal{O}_{U,\eta}$. By **Step 3**(c), there exists $s \in \mathcal{V}$ with $s = \varphi^{-1}v$ for a unit $v \in \mathcal{O}_{U,\eta}$. Then $\operatorname{ord}_P s = -\operatorname{ord}_P \varphi$, so

$$\operatorname{mult}_P R \geq \operatorname{ord}_P \varphi > 0 \quad \text{and} \quad \operatorname{mult}_P \Delta_0 = 1 - \operatorname{mult}_P R < 1.$$

Hence all coefficients of Δ_0 are less than one.

Step 5. In this step, we choose S_1 and S_2 and finish the proof of (1).

By **Step 3**(b) and Bertini's theorem [Har77, Corollary III.10.9 and Remark III.10.9.2], applied to the restrictions of the base point free linear system $\{H_s\}$ to the strata of Σ, and then to the strata of $\Sigma + H_1$, there exist $s^{(1)}, s^{(2)} \in \mathcal{V}\backslash\{0\}$ such that $\operatorname{ord}_E s^{(1)} = \operatorname{ord}_E s^{(2)} = 0$, the divisors $H_i := H_{s^{(i)}}$, $i = 1, 2$, are smooth and contain no component of Σ, and $\Sigma + H_1 + H_2$ is a reduced snc divisor. Let

$$S_i := S + \operatorname{div}(s^{(i)}), \quad i = 1, 2.$$

By **Step 3**(a), $S_i \geq 0$ and $S_i \sim S$. Since S is $\mathbb{Q}$-Cartier,

$$h^*S_i = h^*S + \operatorname{div}(s^{(i)}) = h^*S - R + H_i,$$

so $S_i = h_*h^*S_i = S - h_*R + h_*H_i = h_*H_i$. Since $H_1 + H_2$ is reduced and contains neither an h-exceptional prime divisor nor a component of $h_*^{-1}D|_U$, the divisors S_1 and S_2 are reduced and have no common components with each other or with $(D - \lfloor D \rfloor)|_U$. Let $D' := (D - \lfloor D \rfloor)|_U + \frac{1}{2}(S_1 + S_2)$. Then

$$D' - D|_U = \frac{1}{2}\big(\operatorname{div}(s^{(1)}) + \operatorname{div}(s^{(2)})\big),$$

so $K_U + D'$ is $\mathbb{R}$-Cartier and

$$h^*(K_U + D') = K_{Y'} + \Delta + \frac{1}{2}\sum_{i=1}^{2}(H_i - R) = K_{Y'} + \Delta_0 + \frac{1}{2}(H_1 + H_2). \tag{6.2}$$

The support of $\Delta_0 + \frac{1}{2}(H_1 + H_2)$ is contained in the snc divisor $\Sigma + H_1 + H_2$, and the coefficients of $\Delta_0 + \frac{1}{2}(H_1 + H_2)$ are less than one by **Step 4**. By **Step 1** and (6.2), $a(F, U, D') > 0$ for any prime divisor F over U, so (U, D') is klt.

Let $T_1, \ldots, T_m$ be the components of $h^{-1}(y)$. Then $\operatorname{center}_U T_j = \{y\}$, so

$$a(T_j, U, D') = 1 - \operatorname{mult}_{T_j} \Delta_0 = a(T_j, U, D|_U) + \operatorname{mult}_{T_j} R \geq b$$

for any j, where the first equality holds because $T_j \not\subset H_1 \cup H_2$. Let Q be a prime divisor over U with $\operatorname{center}_U Q = \{y\}$. Then $\operatorname{center}_{Y'} Q$ is contained in some T_j, so **Step 1**, applied to Y' and $\Delta_0 + \frac{1}{2}(H_1 + H_2)$, implies $a(Q, U, D') \geq a(T_j, U, D') \geq b$. Hence $\operatorname{mld}(U \ni y, D') \geq b$. On the other hand, since $\operatorname{ord}_E s^{(1)} = \operatorname{ord}_E s^{(2)} = 0$,

$$a(E, U, D') = a(E, U, D|_U) - \frac{1}{2}\big(\operatorname{ord}_E s^{(1)} + \operatorname{ord}_E s^{(2)}\big) = b.$$

Thus $\operatorname{mld}(U \ni y, D') = b$, which proves (1).

Step 6. In this step, we prove (2).

Let $a \in \operatorname{MLD}(3, \Gamma)$ with $a > 0$. By Definition 2.3, $a = \operatorname{mld}(X \ni x, B)$ for an lc threefold pair (X, B) with $B \in \Gamma$ and a closed point $x \in X$. By [Amb99, Section 1, the second paragraph after Definition 1.1], there exists a prime divisor E over X with $\operatorname{center}_X E = \{x\}$ and $a(E, X, B) = a$.

By [Fuj11, Theorem 10.4], there exists a crepant dlt modification of (X, B), that is, a projective birational morphism $f : Y \to X$ such that Y is $\mathbb{Q}$-factorial, (Y, D) is dlt, and $K_Y + D = f^*(K_X + B)$, where $D = f_*^{-1}B + \sum_{E'} E'$ and E' runs over the f-exceptional prime divisors. Then $D \in \Gamma \cup \{1\}$. Let $V := \operatorname{center}_Y E$ and η the generic point of V. Since f is proper, $f(V) = \{x\}$. Thus any prime divisor F over Y with $\operatorname{center}_Y F = V$ satisfies $\operatorname{center}_X F = \{x\}$, so

$$a(F, Y, D) = a(F, X, B) \geq a,$$

with equality for $F = E$. Hence $\operatorname{mld}(Y \ni \eta, D) = a$.

We show that $d := \dim V \in \{0, 1\}$. Suppose that $\dim V = 2$. Then V is a prime divisor on Y, and $\mathcal{O}_{Y,\eta}$ is a discrete valuation ring, so $\operatorname{ord}_E = \operatorname{ord}_V$. Since $f(V) = \{x\}$, V is f-exceptional, so $\operatorname{mult}_V D = 1$ and $a = a(V, Y, D) = 0$, a contradiction. Thus $d \in \{0, 1\}$.

Since (Y, D) is lc and η is not the generic point of Y, Lemma 2.4 implies $\operatorname{mld}(Y \ni y, D) = a + d$ for a general closed point $y \in V$. Let $b := a + d > 0$. By (1), there exist an affine neighborhood U of y and a klt pair (U, D') with $\operatorname{mld}(U \ni y, D') = b$, where either $D' = D|_U$ or $D' = (D - \lfloor D \rfloor)|_U + \frac{1}{2}(S_1 + S_2)$ for reduced divisors S_1, S_2 without common components with each other or with $(D - \lfloor D \rfloor)|_U$. The variety U is $\mathbb{Q}$-factorial because Y is. The coefficients of $D - \lfloor D \rfloor$ belong to $\Gamma \cap [0, 1)$, and the coefficients of the components of S_1 and S_2 in D' are $\frac{1}{2}$. Thus $D' \in \Lambda$, $b \in \operatorname{MLD}_{\mathrm{Qf}}(3, \Lambda)$, and $a = b - d$.

Step 7. In this step, we prove (3).

Since Γ is finite, Λ is finite. Since $\operatorname{MLD}(3, \Gamma) \subset [0, +\infty)$, (2) implies

$$\operatorname{MLD}(3, \Gamma) \subset \{0\} \cup \operatorname{MLD}_{\mathrm{Qf}}(3, \Lambda) \cup \{c - 1 \mid c \in \operatorname{MLD}_{\mathrm{Qf}}(3, \Lambda)\}.$$

By assumption, $\operatorname{MLD}_{\mathrm{Qf}}(3, \Lambda)$ satisfies the ACC, hence so does $\{c - 1 \mid c \in \operatorname{MLD}_{\mathrm{Qf}}(3, \Lambda)\}$. By Lemma 2.10(4), $\operatorname{MLD}(3, \Gamma)$ satisfies the ACC. □

Lemma 6.2 is not used in this paper. It is kept for a later paper of the series. It is the analogue of Lemma 6.1 for DCC coefficient sets, in which the coefficient one half is replaced by a scaling of the boundary.

Lemma 6.2. *Let $\Gamma \subset [0, 1]$ be a DCC set, (X_i, B_i), $i \geq 1$, lc threefold pairs with $B_i \in \Gamma$, and $x_i \in X_i$ closed points. Assume that $a_i := \operatorname{mld}(X_i \ni x_i, B_i)$ is strictly increasing. Then, after passing to a subsequence, there exist $d \in \{0, 1, 2\}$, real numbers $0 < t_i < t_{i+1} < 1$ with $\lim_{i \to +\infty} t_i = 1$, projective birational morphisms $f_i : Y_i \to X_i$, and closed points $y_i \in Y_i$ such that Y_i is $\mathbb{Q}$-factorial, (Y_i, D_i) is dlt, $K_{Y_i} + D_i = f_i^*(K_{X_i} + B_i)$, where D_i is the sum of the strict transform of B_i and the reduced f_i-exceptional divisor, $(Y_i, t_i D_i)$ is klt, and*

$$a_i + d \leq \operatorname{mld}(Y_i \ni y_i, t_i D_i) < a_{i+1} + d.$$

The coefficients of $t_i D_i$ belong to the DCC set $\{t_i \gamma \mid i \geq 1,\ \gamma \in \Gamma \cup \{1\}\}$. In particular, if $\operatorname{MLD}_{\mathrm{Qf}}(3, \Lambda)$ satisfies the ACC for any DCC set $\Lambda \subset [0, 1]$, then $\operatorname{MLD}(3, \Gamma')$ satisfies the ACC for any DCC set $\Gamma' \subset [0, 1]$.

Proof. By [Amb99, Section 1, the second paragraph after Definition 1.1], for any i there exists a prime divisor E_i over X_i with $\operatorname{center}_{X_i} E_i = \{x_i\}$ and $a(E_i, X_i, B_i) = a_i$.

By [Fuj11, Theorem 10.4], for any i there exists a crepant dlt modification $f_i : Y_i \to X_i$ of (X_i, B_i) as in the statement. Then $D_i \in \Gamma \cup \{1\}$. Let $V_i := \operatorname{center}_{Y_i} E_i$ and η_i the generic point of V_i. Since f_i is proper, $f_i(V_i) = \{x_i\}$, so $d_i := \dim V_i \in \{0, 1, 2\}$, and any prime divisor F over Y_i with $\operatorname{center}_{Y_i} F = V_i$ satisfies $a(F, Y_i, D_i) = a(F, X_i, B_i) \geq a_i$, with equality for $F = E_i$. Possibly passing to a subsequence, we may assume that $d_i = d$ for any i.

By [Fuj11, Definition 10.1], there exists a resolution $g_i : W_i \to Y_i$ such that $\operatorname{Exc}(g_i)$ is an snc divisor and $a(E, Y_i, D_i) > 0$ for any g_i-exceptional prime divisor E on W_i. Let $K_{W_i} + \Theta_i := g_i^* K_{Y_i}$, which is defined because Y_i is $\mathbb{Q}$-factorial. Then $\operatorname{Supp} \Theta_i \subset \operatorname{Exc}(g_i)$, and for any g_i-exceptional prime divisor E on W_i, $\operatorname{mult}_E \Theta_i \leq \operatorname{mult}_E(\Theta_i + g_i^* D_i) = 1 - a(E, Y_i, D_i) < 1$ because $D_i \geq 0$. By Lemma 2.13, applied with local equations of the components of the snc divisor $\operatorname{Exc}(g_i)$ as the first regular parameters, Y_i is klt. For $0 \leq t < 1$ and any prime divisor F over Y_i,

$$a(F, Y_i, tD_i) = ta(F, Y_i, D_i) + (1 - t)a(F, Y_i, 0) > 0,$$

because (Y_i, D_i) is lc and Y_i is klt. Thus (Y_i, tD_i) is klt.

Let $c_i := \operatorname{mult}_{E_i} D_i \geq 0$ and $b_i(t) := \operatorname{mld}(Y_i \ni \eta_i, tD_i)$ for $0 < t < 1$. For any prime divisor F over Y_i with $\operatorname{center}_{Y_i} F = V_i$, we have $a(F, Y_i, tD_i) = a(F, Y_i, D_i) + (1 - t) \operatorname{mult}_F D_i \geq a_i$, with $a(E_i, Y_i, tD_i) = a_i + (1 - t)c_i$. Thus

$$a_i \leq b_i(t) \leq a_i + (1 - t)c_i.$$

Let $t_0 := 0$, and choose recursively $t_i \in (0, 1)$ such that $t_i > t_{i-1}$, $t_i > 1 - \frac{1}{i+1}$, and $(1 - t_i)c_i < a_{i+1} - a_i$. Then $0 < t_i < t_{i+1} < 1$, $\lim_{i \to +\infty} t_i = 1$, and $a_i \leq b_i(t_i) < a_{i+1}$. Since $(Y_i, t_i D_i)$ is klt and η_i is not the generic point of Y_i, Lemma 2.4 implies that $\operatorname{mld}(Y_i \ni y_i, t_i D_i) = b_i(t_i) + d$ for a general closed point $y_i \in V_i$. Hence

$$a_i + d \leq \operatorname{mld}(Y_i \ni y_i, t_i D_i) < a_{i+1} + d.$$

The coefficients of $t_i D_i$ belong to

$$\Lambda := \{t_i \gamma \mid i \geq 1,\ \gamma \in \Gamma \cup \{1\}\} \subset [0, 1].$$

Suppose that Λ contains a strictly decreasing sequence $t_{i_k} \gamma_k$, $k \geq 1$. Since $\{t_i\}_{i \geq 1}$ and $\Gamma \cup \{1\}$ satisfy the DCC, by Lemma 2.10(3), after passing to a subsequence twice, both t_{i_k} and γ_k are nondecreasing. Since both are non-negative, $t_{i_k} \gamma_k$ is nondecreasing, a contradiction. Thus Λ satisfies the DCC.

Finally, assume that $\operatorname{MLD}_{\mathrm{Qf}}(3, \Lambda)$ satisfies the ACC for any DCC set $\Lambda \subset [0, 1]$, and suppose that $\operatorname{MLD}(3, \Gamma')$ does not satisfy the ACC for some DCC set $\Gamma' \subset [0, 1]$. Then there exist lc threefold pairs (X_i, B_i) with $B_i \in \Gamma'$ and closed points $x_i \in X_i$ such that $a_i := \operatorname{mld}(X_i \ni x_i, B_i)$ is strictly increasing. By the first part of the lemma with $\Gamma = \Gamma'$, after passing to a subsequence, $m_i := \operatorname{mld}(Y_i \ni y_i, t_i D_i)$ satisfies $m_i < a_{i+1} + d \leq m_{i+1}$, and $m_i \in \operatorname{MLD}_{\mathrm{Qf}}(3, \Lambda)$ for the DCC set Λ above, which contradicts the assumption. □

6.2. **Limits of boundary coefficients.** In this subsection, we extract fixed components with nondecreasing coefficients from the boundaries of a strictly increasing sequence of minimal log discrepancies, and we compare the log discrepancies before and after replacing the coefficients by their limits. Part (1) of Proposition 6.3 provides the nondecreasing sequences of coefficients and the lc pairs $(X_i, \bar{B}_i)$ used in Set-up 6.4. Part (3) is not used in this paper. It is used in the last paper of the series, and its hypothesis is proved in the next one.

Proposition 6.3. *Let $\Gamma \subset [0, 1]$ be a DCC set, and let $(X_i \ni x_i, B_i)$ be $\mathbb{Q}$-factorial klt threefold germs with $B_i \in \Gamma$. Assume that $a_i := \operatorname{mld}(X_i \ni x_i, B_i)$ is strictly increasing. Then, after shrinking near x_i and passing to a subsequence, there exist a non-negative integer r and expressions $B_i = \sum_{j=1}^{r} b_{ij} D_{ij}$, where the D_{ij} are distinct prime divisors through x_i and b_{ij} is nondecreasing in i with positive limit b_j for any j, such that the following hold.*

(1) The pair $(X_i, \bar{B}_i)$, $\bar{B}_i := \sum_j b_j D_{ij}$, is lc for any i.

(2) *Let* $\boldsymbol{b} := (b_1, \dots, b_r)$, $\boldsymbol{b}_i := (b_{i1}, \dots, b_{ir})$, *and let* $V(\boldsymbol{b}) \subset \mathbb{R}^r$ *be the smallest affine subspace defined over* $\mathbb{Q}$ *containing* $\boldsymbol{b}$. *There exists* $\varrho > 0$ *depending only on* $\boldsymbol{b}$ *and* r *such that if* $\boldsymbol{b}_i \in V(\boldsymbol{b})$ *and* $\epsilon_i := \|\boldsymbol{b}_i - \boldsymbol{b}\|/\varrho < 1$, *where* $\|\cdot\|$ *is the Euclidean norm on* $\mathbb{R}^r$, *then*

$$(1-\epsilon_i)a(E, X_i, \bar{B}_i) \le a(E, X_i, B_i) \le (1+\epsilon_i)a(E, X_i, \bar{B}_i)$$

for any prime divisor E *over* X_i. *In particular,* $(X_i, \bar{B}_i)$ *is klt and* $m_i := \mathrm{mld}(X_i \ni x_i, \bar{B}_i)$ *satisfies* $m_i \le a_i \le (1+\epsilon_i)m_i$.

(3) *Assume that* $r \ge 1$ *and that* $\mathrm{MLD}_{\mathrm{Qf}}(3, \Lambda)$ *satisfies the ACC for any finite set* $\Lambda \subset [0,1]$. *Then only finitely many* $\boldsymbol{b}_i$ *belong to* $V(\boldsymbol{b})$. *In particular, after passing to a further subsequence, there exist rational numbers* $q_0, \dots, q_r$, *not all zero, such that* $q_0 + \sum_j q_j b_j = 0$ *and* $q_0 + \sum_j q_j b_{ij} \ne 0$ *for any* i.

Proof. **Step 1.** In this step we construct the expressions $B_i = \sum_j b_{ij} D_{ij}$.

Restricting X_i to an open neighborhood of x_i changes neither the prime divisors E over X_i with $\mathrm{center}_{X_i} E = \{x_i\}$ nor their log discrepancies $a(E, X_i, B_i)$. Thus, possibly replacing X_i with an open neighborhood of x_i, we may assume that any component of B_i contains x_i. If $\Gamma \backslash \{0\} = \emptyset$, then $B_i = 0$ for any i. Otherwise $\gamma(\Gamma) > 0$ by Lemma 2.10(2). Since X_i is $\mathbb{Q}$-factorial, K_{X_i} and the components of B_i are $\mathbb{Q}$-Cartier. By [Kol92, Theorem 18.22] (cf. [Liu18, Theorem 5.6]), the sum of the coefficients of B_i is at most 3. Since any coefficient of B_i is at least $\gamma(\Gamma)$, the number of components of B_i is at most $3/\gamma(\Gamma)$. In both cases, passing to a subsequence, we may assume that the number r of components of B_i does not depend on i. We label the components of B_i arbitrarily and write $B_i = \sum_{j=1}^r b_{ij} D_{ij}$.

By Lemma 2.10(3) for the sequences $(b_{i1})_i, \dots, (b_{ir})_i$ in turn, passing to a subsequence each time, we may assume that b_{ij} is nondecreasing in i for any j. Since $b_{ij} \le 1$, the sequence $(b_{ij})_i$ converges to a limit b_j, and

$$1 \ge b_j \ge b_{1j} \ge \gamma(\Gamma) > 0.$$

If $r = 0$, then $\bar{B}_i = B_i = 0$, and (1) and (2) hold, with $\boldsymbol{b}_i = \boldsymbol{b}$ and $\epsilon_i = 0$, since (X_i, B_i) is klt. Thus from now on we assume that $r \ge 1$.

Step 2. In this step we prove (1).

By [HLS26, Theorem 5.11] (applied with $d = 3$ and $\alpha = 1$), there exist a finite set Γ' and a nondecreasing function g from the closure of Γ to Γ', with $g(c) \ge c$ for any $c \in \Gamma$, such that the pair (X_i, B_i^+), $B_i^+ := \sum_j g(b_{ij}) D_{ij}$, is lc for any i. For any j, the sequence $(g(b_{ij}))_i$ is nondecreasing and takes values in the finite set Γ'. Hence it is constant for $i \gg 0$. Passing to a subsequence, we may assume that $g(b_{ij}) = b_j^+$ for any i and j, where $b_j^+ \in \Gamma'$ does not depend on i. Since $b_j^+ = g(b_{ij}) \ge b_{ij}$ for any i, we have $b_j^+ \ge b_j$. Thus

$$B_i^+ - \bar{B}_i = \sum_j (b_j^+ - b_j) D_{ij}, \qquad b_j^+ - b_j \ge 0.$$

Since each D_{ij} is $\mathbb{Q}$-Cartier, by [KM98, Lemma 2.27] for each D_{ij} and by adding the inequalities with non-negative real coefficients, we obtain $a(E, X_i, \bar{B}_i) \ge a(E, X_i, B_i^+) \ge 0$ for any prime divisor E over X_i. Thus $(X_i, \bar{B}_i)$ is lc for any i.

Step 3. In this step we prove (2).

By [HLS26, Theorem 5.6] (applied with $d = 3$, $m = r$, and $v = \boldsymbol{b}$), there exists an open subset U of $V(\boldsymbol{b})$ containing $\boldsymbol{b}$ and depending only on $\boldsymbol{b}$ and r, such that for any normal threefold X and any Weil divisors $D_1, \dots, D_r \ge 0$ on X, if $(X, \sum_j b_j D_j)$ is lc, then $(X, \sum_j u_j D_j)$ is lc for any $u = (u_1, \dots, u_r) \in U$. Let $\varrho > 0$ be a real number such that the closed ball in $V(\boldsymbol{b})$ with center $\boldsymbol{b}$ and radius ϱ is contained in U. Then ϱ depends only on $\boldsymbol{b}$ and r. Since X_i is $\mathbb{Q}$-factorial, $K_{X_i} + \sum_j u_j D_{ij}$ is $\mathbb{R}$-Cartier for any $u \in \mathbb{R}^r$. By (1), $(X_i, \sum_j u_j D_{ij})$ is lc for any u in the closed ball.

Fix an index i such that $\boldsymbol{b}_i \in V(\boldsymbol{b})$ and $\epsilon_i < 1$, and fix a prime divisor E over X_i. For $u \in \mathbb{R}^r$, let $\varphi(u) := a(E, X_i, \sum_j u_j D_{ij})$. By Definition 2.1 and the linearity of pullback,

$$\varphi(u) = a(E, X_i, 0) - \sum_j u_j \operatorname{mult}_E D_{ij},$$

where $\operatorname{mult}_E D_{ij}$ is the coefficient of E in the pullback of D_{ij} to a higher model of X_i on which E is a divisor. Moreover, $\varphi(u) \geq 0$ for any u in the closed ball. We have $\varphi(\boldsymbol{b}_i) = a(E, X_i, B_i)$ and $\varphi(\boldsymbol{b}) = a(E, X_i, \bar{B}_i)$. If $\boldsymbol{b}_i = \boldsymbol{b}$, then the two-sided estimate in (2) holds. Suppose that $\boldsymbol{b}_i \neq \boldsymbol{b}$, and let $w = (w_1, \ldots, w_r) := (\boldsymbol{b}_i - \boldsymbol{b})/\|\boldsymbol{b}_i - \boldsymbol{b}\|$. Since $\boldsymbol{b}_i, \boldsymbol{b} \in V(\boldsymbol{b})$, the points $\boldsymbol{b} \pm \varrho w$ belong to the closed ball. Thus

$$\varphi(\boldsymbol{b} \pm \varrho w) = \varphi(\boldsymbol{b}) \mp \varrho \sum_j w_j \operatorname{mult}_E D_{ij} \geq 0,$$

and hence

$$\varrho \Big| \sum_j w_j \operatorname{mult}_E D_{ij} \Big| \leq \varphi(\boldsymbol{b}).$$

Since $\boldsymbol{b}_i = \boldsymbol{b} + \|\boldsymbol{b}_i - \boldsymbol{b}\| w$, we obtain

$$|\varphi(\boldsymbol{b}_i) - \varphi(\boldsymbol{b})| = \|\boldsymbol{b}_i - \boldsymbol{b}\| \cdot \Big| \sum_j w_j \operatorname{mult}_E D_{ij} \Big| \leq \frac{\|\boldsymbol{b}_i - \boldsymbol{b}\|}{\varrho} \varphi(\boldsymbol{b}) = \epsilon_i \varphi(\boldsymbol{b}).$$

Thus the two-sided estimate in (2) holds for E.

Since (X_i, B_i) is klt, $a(E, X_i, B_i) > 0$ for any prime divisor E over X_i. If $a(E, X_i, \bar{B}_i) = 0$ for some E, then the two-sided estimate implies $a(E, X_i, B_i) = 0$, a contradiction. Thus $(X_i, \bar{B}_i)$ is klt. Taking the infimum over the prime divisors E over X_i with $\operatorname{center}_{X_i} E = \{x_i\}$ in $a(E, X_i, B_i) \leq (1 + \epsilon_i) a(E, X_i, \bar{B}_i)$, we obtain $a_i \leq (1 + \epsilon_i) m_i$. Since $\bar{B}_i \geq B_i$ and each D_{ij} is $\mathbb{Q}$-Cartier, [KM98, Lemma 2.27] implies $a(E, X_i, \bar{B}_i) \leq a(E, X_i, B_i)$ for any prime divisor E over X_i. Hence $m_i \leq a_i$.

Step 4. In this step we prove (3).

Suppose that infinitely many $\boldsymbol{b}_i$ belong to $V(\boldsymbol{b})$. Since $\boldsymbol{b}_i$ converges to $\boldsymbol{b}$, we have $\epsilon_i \to 0$. Passing to a subsequence, we may assume that $\boldsymbol{b}_i \in V(\boldsymbol{b})$ and $\epsilon_i < 1$ for any i. Let $\Lambda := \{b_1, \ldots, b_r\} \subset [0, 1]$. By (2), $(X_i, \bar{B}_i)$ is a $\mathbb{Q}$-factorial klt threefold pair with $\bar{B}_i \in \Lambda$, so $m_i \in \mathrm{MLD}_{\mathrm{Qf}}(3, \Lambda)$ for any i. Since $\mathrm{MLD}_{\mathrm{Qf}}(3, \Lambda)$ satisfies the ACC, it is bounded from above, as otherwise it contains a strictly increasing sequence. Let c be an upper bound of $\mathrm{MLD}_{\mathrm{Qf}}(3, \Lambda)$. By (2),

$$a_i \leq (1 + \epsilon_i) m_i \leq 2c$$

for any i. Since a_i is strictly increasing and bounded from above, a_i converges to a limit a_∞, and $a_i < a_\infty$ for any i. Since $m_i \geq 0$, (2) implies

$$0 \leq a_i - m_i \leq \epsilon_i m_i \leq \epsilon_i c,$$

and the right-hand side tends to 0. Thus $m_i \to a_\infty$, and $m_i \leq a_i < a_\infty$ for any i. Hence, for any index i, there exists an index $i' > i$ such that $m_i < m_{i'} < a_\infty$. Repeating the choice, we obtain a strictly increasing subsequence of $(m_i)_i$, which contradicts the ACC for $\mathrm{MLD}_{\mathrm{Qf}}(3, \Lambda)$. Thus only finitely many $\boldsymbol{b}_i$ belong to $V(\boldsymbol{b})$.

Passing to a subsequence, we may assume that $\boldsymbol{b}_i \notin V(\boldsymbol{b})$ for any i. Since $V(\boldsymbol{b})$ is defined over $\mathbb{Q}$, there exist rational numbers q_{kl}, $1 \leq k \leq s$, $0 \leq l \leq r$, such that $(q_{k0}, \ldots, q_{kr}) \neq 0$ for any k and

$$V(\boldsymbol{b}) = \Big\{ u \in \mathbb{R}^r \;\Big|\; q_{k0} + \sum_j q_{kj} u_j = 0 \text{ for any } 1 \leq k \leq s \Big\}.$$

For any i, since $\boldsymbol{b}_i \notin V(\boldsymbol{b})$, there exists k such that $q_{k0} + \sum_j q_{kj} b_{ij} \neq 0$. Hence there exists k such that $q_{k0} + \sum_j q_{kj} b_{ij} \neq 0$ for infinitely many i. Passing to a subsequence, we may assume that $q_{k0} + \sum_j q_{kj} b_{ij} \neq 0$ for any i. Since $\boldsymbol{b} \in V(\boldsymbol{b})$, we have $q_{k0} + \sum_j q_{kj} b_j = 0$. Thus $q_l := q_{kl}$, $0 \leq l \leq r$, satisfy (3). □

6.3. **Cyclic quotients of terminal points for DCC coefficients.** In this subsection, we show that a strictly increasing sequence of minimal log discrepancies of $\mathbb{Q}$-factorial klt threefold pairs with DCC coefficients is a sequence of minimal log discrepancies, at closed points, of quotients of terminal threefolds by cyclic groups, with boundary coefficients in a fixed DCC set (Construction-Theorem 6.6). Construction-Theorem 6.6 is used only in Corollary 6.8(2), which is not used in the proofs of the main theorems. The proofs of the main theorems use Construction-Theorem 6.7 instead.

Set-up 6.4. Let Γ, $(X_i \ni x_i, B_i)$, a_i, r, D_{ij}, b_{ij}, b_j, $\bar{B}_i$, and E_i be as follows.

(1) $\Gamma \subset [0,1]$ is a DCC set, and $(X_i \ni x_i, B_i)$, $i \geq 1$, are $\mathbb{Q}$-factorial klt threefold germs with X_i affine and $B_i \in \Gamma$, such that $a_i := \mathrm{mld}(X_i \ni x_i, B_i) > 0$ is strictly increasing.
(2) $r \geq 0$ is an integer and $B_i = \sum_{j=1}^{r} b_{ij} D_{ij}$, where $D_{i1}, \dots, D_{ir}$ are distinct prime divisors through x_i and, for any j, the numbers b_{ij} are nondecreasing in i with limit $b_j \in (0,1]$ and $b_{ij} \geq \frac{b_j}{2}$.
(3) $(X_i, \bar{B}_i)$ is lc for $\bar{B}_i := \sum_j b_j D_{ij}$.
(4) E_i is a prime divisor over X_i computing a_i.

Lemma 6.5. *Let $\Gamma \subset [0,1]$ be a DCC set, and let $(X_i \ni x_i, B_i)$, $i \geq 1$, be $\mathbb{Q}$-factorial klt threefold germs with $B_i \in \Gamma$. Assume that $a_i := \mathrm{mld}(X_i \ni x_i, B_i)$ is strictly increasing. Then, after shrinking X_i to an affine neighborhood of x_i and passing to a subsequence, there exist an integer r, prime divisors D_{ij} on X_i, real numbers b_{ij} and b_j, and prime divisors E_i over X_i such that the conditions of Set-up 6.4 hold.*

Proof. By Proposition 6.3(1), after shrinking X_i near x_i and passing to a subsequence, there exist an integer $r \geq 0$, distinct prime divisors $D_{i1}, \dots, D_{ir}$ through x_i, and real numbers b_{ij} such that $B_i = \sum_j b_{ij} D_{ij}$, the numbers b_{ij} are nondecreasing in i with limit $b_j > 0$, and $(X_i, \bar{B}_i)$ is lc for any i, where $\bar{B}_i = \sum_j b_j D_{ij}$. Moreover, $b_j \leq 1$ since $b_{ij} \in \Gamma \subset [0,1]$, and, since $b_{ij} \to b_j > 0$, after deleting finitely many terms we have $b_{ij} \geq \frac{b_j}{2}$ for any i and j. Since (X_i, B_i) is klt, $a_i > 0$, and a_i is attained by a prime divisor E_i over X_i [Amb99, Section 1, the second paragraph after Definition 1.1]. Finally, we replace X_i with an affine neighborhood of x_i. The conditions of Set-up 6.4 are preserved, since they only concern X_i near x_i. □

In Construction-Theorem 6.6, we work with $\frac{1}{2}\bar{B}_i$ in place of $\bar{B}_i$, since $(X_i, \bar{B}_i)$ is only lc, while Construction-Theorems 5.2 and 5.1 require klt pairs. On a quasi-étale cover of X_i, Construction-Theorem 5.1 then gives terminal threefolds whose boundaries have coefficients in a fixed finite set, and the remaining part of B_i contributes coefficients in a fixed DCC set.

Construction-Theorem 6.6. Notation and conditions as in Set-up 6.4. Then, after passing to a subsequence, there exist positive integers s, q, positive real numbers $\lambda_1, \dots, \lambda_s$ with $\sum_\ell \lambda_\ell = 1$, and $\mathbb{Q}$-divisors $B_i^{(\ell)} \geq 0$ supported on the D_{ij} with fixed coefficient vectors, with $qB_i^{(\ell)}$ Weil divisors, $(X_i, B_i^{(\ell)})$ lc, and $\frac{1}{2}\bar{B}_i = \sum_\ell \lambda_\ell B_i^{(\ell)}$. Let $\mathcal{S}$, $\Lambda := \Lambda_{\mathcal{S}}$, $R := R_{\mathcal{S}}$, and $M := M_{\mathcal{S}}$ be as in Construction-Theorem 5.1 for $\alpha_\ell = \lambda_\ell / q$, let $L_j := \lfloor 2/b_j \rfloor$, and let

$$\Sigma := \left\{ c + \sum_j n_j (t_j - \tfrac{b_j}{2}) \;\middle|\; c \in \Lambda,\ n_j \in \{0, \dots, L_j\},\ t_j \in \Gamma \cap [\tfrac{b_j}{2}, b_j] \right\} \cap [0,1).$$

Then Σ and $\Psi(\Sigma) \subset [0,1)$ satisfy the DCC, and, after shrinking X_i near x_i and deleting finitely many terms, the following hold.

(1) Let $A_i \subset \mathrm{Cl}(\mathcal{O}_{X_i, x_i})$ be the subgroup generated by the classes of $K_{X_i}, D_{i1}, \dots, D_{ir}$. There exists a finite quasi-étale abelian Galois morphism $\pi_i : X_i' \to X_i$ with group $G_i = \mathrm{Hom}(A_i, \mathbb{C}^*)$ such that the following hold.
   (a) $\pi_i^{-1}(x_i)$ is a single point x_i'.
   (b) $K_{X_i'}$ is Cartier.
   (c) $Q_{ij} := \pi_i^* D_{ij}$ are reduced Cartier divisors.

(d) The pair $(X_i', \frac{1}{2}\pi_i^*\bar{B}_i)$ is klt.
(e) $q(K_{X_i'} + \pi_i^* B_i^{(\ell)})$ is Cartier.
(f) For any prime divisor T over X_i', the numbers $k_i(T) := a(T, X_i', 0)$ and $n_{ij}(T) := \operatorname{mult}_T Q_{ij}$ are non-negative integers.
(g) $k_i(T) \geq \max\{1, \sum_j b_j n_{ij}(T)\}$ for any prime divisor T over X_i'.
(h) $a(T, X_i', \frac{1}{2}\pi_i^*\bar{B}_i) \geq \frac{1}{2}k_i(T)$ for any prime divisor T over X_i'.
(i) The log discrepancies of $(X_i', \frac{1}{2}\pi_i^*\bar{B}_i)$ belong to $\mathcal{S}$.

(2) Let F_i be a prime divisor over X_i' lying over E_i, and let $I_i := I_{G_i}(F_i)$ and $e_i := |I_i|$. There exist an I_i-equivariant projective birational morphism $g_i : Y_i \to X_i'$ and an $\mathbb{R}$-divisor $C_i^0 \geq 0$ with $K_{Y_i} + C_i^0 = g_i^*(K_{X_i'} + \frac{1}{2}\pi_i^*\bar{B}_i)$ such that the following hold.
(a) I_i is cyclic.
(b) $a(F_i, X_i', \pi_i^* B_i) = e_i a_i$.
(c) The pair (Y_i, C_i^0) is terminal.
(d) $C_i^0 \in \Lambda$.
(e) The g_i-exceptional prime divisors are exactly the exceptional prime divisors over X_i' with log discrepancy at most 1 with respect to $(X_i', \frac{1}{2}\pi_i^*\bar{B}_i)$.
(f) Y_i is terminal.
(g) Y_i is I_i-$\mathbb{Q}$-factorial.
(h) The Cartier index of K_{Y_i} at any closed point is at most R.
(i) MK_{Y_i} is Cartier.

(3) Let $P_{ij} := g_i^* Q_{ij}$, $\beta_{ij} := b_{ij} - \frac{b_j}{2} \geq 0$, $C_i := C_i^0 + \sum_j \beta_{ij} P_{ij}$, and $\bar{C}_i := C_i^0 + \frac{1}{2}\sum_j b_j P_{ij}$. Then the following hold.
(a) P_{ij} is a Cartier divisor with coefficients in $\{0, \dots, L_j\}$.
(b) $P_{ij} = \operatorname{div}(p_{ij})$ for an I_i-semi-invariant regular function p_{ij} on Y_i, pulled back from a G_i-semi-invariant equation of Q_{ij}.
(c) $K_{Y_i} + C_i = g_i^*\pi_i^*(K_{X_i} + B_i)$.
(d) $K_{Y_i} + \bar{C}_i = g_i^*\pi_i^*(K_{X_i} + \bar{B}_i)$.
(e) The pair (Y_i, C_i) is klt.
(f) The pair $(Y_i, \bar{C}_i)$ is lc.
(g) $C_i \leq \bar{C}_i$.
(h) $C_i \in \Sigma$.

(4) Let $(Z_i, \Delta_i) := (Y_i, C_i)/I_i$. Then the following hold.
(a) $\operatorname{center}_{Y_i} F_i$ is a closed point y_i fixed by I_i.
(b) $\Delta_i \in \Psi(\Sigma)$.
(c) The pair (Z_i, Δ_i) is klt.
(d) For the image z_i of y_i, $\operatorname{mld}(Z_i \ni z_i, \Delta_i) = a_i$, attained by the prime divisor under F_i.
(e) For any prime divisor P over Z_i with $\operatorname{center}_{Z_i} P = \{z_i\}$, any prime divisor F' over X_i' lying over P, and the prime divisor E' over X_i under F', we have $\operatorname{center}_{X_i} E' = \{x_i\}$ and

$$a(P, Z_i, \Delta_i) = \frac{|I_{G_i}(F')|}{|I_i \cap I_{G_i}(F')|} a(E', X_i, B_i) \geq a_i.$$

The data s, q, λ_ℓ, Λ, R, M, L_j, and Σ are fixed along the subsequence, and they may depend on Γ and $(b_1, \dots, b_r)$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we construct s, q, λ_ℓ, and $B_i^{(\ell)}$, and we show that Σ and $\Psi(\Sigma)$ satisfy the DCC.

Since X_i is $\mathbb{Q}$-factorial, any D_{ij} is $\mathbb{Q}$-Cartier, so $a(T, X_i, 0) \geq a(T, X_i, B_i) > 0$ for any prime divisor T over X_i [KM98, Lemma 2.27]. Thus X_i is klt. Since $(X_i, \bar{B}_i)$ is lc,

$$a(T, X_i, \tfrac{1}{2}\bar{B}_i) = \tfrac{1}{2}\big(a(T, X_i, 0) + a(T, X_i, \bar{B}_i)\big) > 0$$

for any prime divisor T over X_i, so $(X_i, \frac{1}{2}\bar{B}_i)$ is klt.

We apply Construction-Theorem 5.2 to the finite set $\{\frac{b_1}{2}, \dots, \frac{b_r}{2}\} \subset (0, \frac{1}{2}]$ and to the pairs $(X_i, \frac{1}{2}\bar{B}_i)$, where the divisor D_k corresponding to an element γ_k of the set is the sum of the D_{ij} with $\frac{b_j}{2} = \gamma_k$. We obtain positive integers s, q and positive real numbers $\lambda_1, \dots, \lambda_s$ with $\sum_\ell \lambda_\ell = 1$, depending only on $(b_1, \dots, b_r)$, and $\mathbb{Q}$-divisors $B_i^{(\ell)} \geq 0$ such that $\frac{1}{2}\bar{B}_i = \sum_\ell \lambda_\ell B_i^{(\ell)}$, $(X_i, B_i^{(\ell)})$ is lc, and $q(K_{X_i} + B_i^{(\ell)})$ is a Weil divisor. The coefficient of $B_i^{(\ell)}$ along D_{ij} is the rational number $v_{\ell k}$ of Construction-Theorem 5.2 for the index k with $\gamma_k = \frac{b_j}{2}$, which does not depend on i. Thus $B_i^{(\ell)}$ is supported on the D_{ij} with a fixed coefficient vector, and $qB_i^{(\ell)}$ is a Weil divisor.

Let $\alpha_\ell := \lambda_\ell/q$. Taking $n_\ell = q$ for any ℓ, we have $\sum_\ell \alpha_\ell n_\ell = \sum_\ell \lambda_\ell = 1$, so $1 \in \mathcal{S}$. By Construction-Theorem 5.1, $\Lambda \subset [0, 1)$ is finite. Moreover, $0 = 1 - 1 \in \Lambda$, hence $0 \in \Sigma$ (take $n_j = 0$ and $t_j = b_{1j} \in \Gamma \cap [\frac{b_j}{2}, b_j]$, Set-up 6.4(1)(2)).

Let $(\sigma_m)_{m\geq 1}$ be a sequence in Σ, and write $\sigma_m = c_m + \sum_j n_{mj}(t_{mj} - \frac{b_j}{2})$ with $c_m \in \Lambda$, $n_{mj} \in \{0, \dots, L_j\}$, and $t_{mj} \in \Gamma \cap [\frac{b_j}{2}, b_j]$. Since Λ is finite, after passing to a subsequence, $c_m = c$ and $n_{mj} = n_j$ do not depend on m. Since Γ satisfies the DCC, by Lemma 2.10(3) applied to $j = 1, \dots, r$ in turn, after passing to a further subsequence, t_{mj} is nondecreasing in m for any j. Since $n_j \geq 0$, the sequence σ_m is then nondecreasing. Thus Σ contains no strictly decreasing sequence, that is, Σ satisfies the DCC. Since $\Sigma \subset [0, 1)$, the set $\Psi(\Sigma)$ satisfies the DCC by Lemma 2.10(1), and $\Psi(\Sigma) \subset [0, 1)$ since $0 \leq 1 - \frac{1-c}{e} < 1$ for any $c \in [0, 1)$ and any positive integer e.

Step 2. In this step we construct π_i and prove (1).

Since X_i is $\mathbb{Q}$-factorial, the classes of $K_{X_i}, D_{i1}, \dots, D_{ir}$ in $\mathrm{Cl}(\mathcal{O}_{X_i,x_i})$ are torsion, so A_i is a finite abelian group. Let $\pi_i : X_i' \to X_i$ be the index one cover of $K_{X_i}, D_{i1}, \dots, D_{ir}$ [Zhu24, Lemma-Definition 2.15]. Then π_i is a finite quasi-étale abelian Galois morphism with group $G_i = \mathrm{Hom}(A_i, \mathbb{C}^*)$, $\pi_i^{-1}(x_i)$ is a single point x_i', and the pullbacks of $K_{X_i}, D_{i1}, \dots, D_{ir}$ are Cartier near x_i'.

Let W be an open neighborhood of x_i'. Since π_i is finite, $\pi_i(X_i' \backslash W)$ is a closed subset of X_i, and it does not contain x_i since $\pi_i^{-1}(x_i) = \{x_i'\}$. Let U be an affine open neighborhood of x_i disjoint from $\pi_i(X_i' \backslash W)$. Then $\pi_i^{-1}(U)$ is a G_i-invariant open neighborhood of x_i' contained in W. Thus any shrinking of X_i' near x_i' in this proof is obtained by shrinking X_i to an affine neighborhood of x_i, which preserves the conditions of Set-up 6.4 and the properties proved before, since they only concern X_i near x_i. In particular, after shrinking X_i, the pullbacks of K_{X_i} and of the D_{ij} are Cartier on X_i'.

Let D be an $\mathbb{R}$-divisor on X_i. Since X_i is $\mathbb{Q}$-factorial, $K_{X_i} + D$ is $\mathbb{R}$-Cartier. Since π_i is quasi-étale, $R(\pi_i) = 0$, and by the Hurwitz formula [Kol13, 2.41], $K_{X_i'} = \pi_i^* K_{X_i}$, so

$$K_{X_i'} + \pi_i^* D = \pi_i^*(K_{X_i} + D) \tag{6.3}$$

is $\mathbb{R}$-Cartier. Moreover, $\pi_{i*}\pi_i^* D = |G_i| D$, since for any prime divisor Q on X_i the ramification indices e_P and the residue degrees f_P of the prime divisors P on X_i' over Q satisfy $\sum_P e_P f_P = \deg \pi_i = |G_i|$ [Ser79, Chapter I, §4, Proposition 10], so $(X_i', \pi_i^* D)/G_i = (X_i, D)$ by Definition 2.5, and by [KM98, Proposition 5.20], $(X_i', \pi_i^* D)$ is klt (resp. lc) if and only if (X_i, D) is klt (resp. lc). With $D = 0$, $K_{X_i'} = \pi_i^* K_{X_i}$ is Cartier and X_i' is klt. The divisor $Q_{ij} = \pi_i^* D_{ij}$ is Cartier, and it is reduced since π_i is étale over the generic point of D_{ij}. By **Step 1**, $(X_i, \frac{1}{2}\bar{B}_i)$ and (X_i, B_i) are klt, and $(X_i, \bar{B}_i)$ and $(X_i, B_i^{(\ell)})$ are lc. Hence $(X_i', \frac{1}{2}\pi_i^*\bar{B}_i)$ and $(X_i', \pi_i^* B_i)$ are klt, and $(X_i', \pi_i^*\bar{B}_i)$ and $(X_i', \pi_i^* B_i^{(\ell)})$ are lc. Since $qB_i^{(\ell)}$ is a Weil divisor supported on the D_{ij}, $q\pi_i^* B_i^{(\ell)}$ is an integral combination of the Q_{ij}, so $q(K_{X_i'} + \pi_i^* B_i^{(\ell)})$ is Cartier.

Let T be a prime divisor over X_i'. Since $K_{X_i'}$ is Cartier, $k_i(T) = a(T, X_i', 0)$ is an integer, and $k_i(T) \geq 1$ since X_i' is klt. Since $Q_{ij} \geq 0$ is a Cartier divisor, $n_{ij}(T) = \mathrm{mult}_T Q_{ij}$ is a non-negative

integer. Since $\pi_i^* B_i = \sum_j b_{ij} Q_{ij}$ and $\pi_i^* \bar{B}_i = \sum_j b_j Q_{ij}$,

$$\begin{aligned} a(T, X_i', \pi_i^* B_i) &= k_i(T) - \sum_j b_{ij} n_{ij}(T), \\ a(T, X_i', \pi_i^* \bar{B}_i) &= k_i(T) - \sum_j b_j n_{ij}(T), \\ a(T, X_i', \tfrac{1}{2}\pi_i^* \bar{B}_i) &= k_i(T) - \tfrac{1}{2}\sum_j b_j n_{ij}(T). \end{aligned} \tag{6.4}$$

Since $(X_i', \pi_i^* \bar{B}_i)$ is lc, the second formula of (6.4) implies $k_i(T) \geq \sum_j b_j n_{ij}(T)$, and then the third formula implies $a(T, X_i', \frac{1}{2}\pi_i^* \bar{B}_i) \geq \frac{1}{2} k_i(T)$. Since $(X_i', \pi_i^* B_i^{(\ell)})$ is lc and $q(K_{X_i'} + \pi_i^* B_i^{(\ell)})$ and $K_{X_i'}$ are Cartier, $m_\ell := qa(T, X_i', \pi_i^* B_i^{(\ell)})$ is a non-negative integer. Since $\frac{1}{2}\bar{B}_i = \sum_\ell \lambda_\ell B_i^{(\ell)}$ and $\sum_\ell \lambda_\ell = 1$,

$$a(T, X_i', \tfrac{1}{2}\pi_i^* \bar{B}_i) = \sum_\ell \lambda_\ell a(T, X_i', \pi_i^* B_i^{(\ell)}) = \sum_\ell \alpha_\ell m_\ell \in \mathcal{S}.$$

This proves (1).

Step 3. In this step we prove (2).

By (6.3) with $D = B_i$, we have $(X_i, B_i) = (X_i', \pi_i^* B_i)/G_i$. By Lemma 2.6(3), there exists a prime divisor F_i over X_i' lying over E_i, and by Lemma 2.6(1)(2), $\operatorname{ord}_{F_i}|_{\mathbb{C}(X_i)} = e_i \operatorname{ord}_{E_i}$, the group I_i is cyclic, and

$$a(F_i, X_i', \pi_i^* B_i) = e_i a(E_i, X_i, B_i) = e_i a_i.$$

The prime divisors over X_i' lying over E_i form one G_i-orbit [Ser79, Chapter I, §7, Proposition 19], applied to the integral closure in $\mathbb{C}(X_i')$ of the valuation ring of ord_{E_i}. In particular, the set of prime divisors over X_i' lying over E_i is finite. Since G_i is abelian, for any $\sigma \in G_i$ the inertia group of the conjugate of F_i by σ is $\sigma I_i \sigma^{-1} = I_i$. Thus I_i and e_i do not depend on the choice of F_i.

Since $\operatorname{center}_{X_i} E_i = \{x_i\}$ and $\pi_i^{-1}(x_i) = \{x_i'\}$, we have $\operatorname{center}_{X_i'} F_i = \{x_i'\}$. The group I_i acts faithfully on X_i', fixes x_i', and preserves the G_i-invariant divisor $\frac{1}{2}\pi_i^* \bar{B}_i$. By (1), $(X_i', \frac{1}{2}\pi_i^* \bar{B}_i)$ is klt and the log discrepancies of $(X_i', \frac{1}{2}\pi_i^* \bar{B}_i)$ belong to $\mathcal{S}$. By Construction-Theorem 5.1 with $H = I_i$, after shrinking X_i (**Step 2**), there exist an I_i-equivariant projective birational morphism $g_i : Y_i \to X_i'$ and an $\mathbb{R}$-divisor $C_i^0 \geq 0$ with $K_{Y_i} + C_i^0 = g_i^*(K_{X_i'} + \frac{1}{2}\pi_i^* \bar{B}_i)$ such that the g_i-exceptional prime divisors are exactly the exceptional prime divisors over X_i' with log discrepancy at most 1 with respect to $(X_i', \frac{1}{2}\pi_i^* \bar{B}_i)$, $C_i^0 \in \Lambda$, (Y_i, C_i^0) is terminal, Y_i is terminal and I_i-$\mathbb{Q}$-factorial, the Cartier index of K_{Y_i} at any closed point is at most R, and MK_{Y_i} is Cartier. This proves (2).

Step 4. In this step we prove (3).

Let T be a prime divisor on Y_i. Since $C_i^0 \geq 0$,

$$a(T, X_i', \tfrac{1}{2}\pi_i^* \bar{B}_i) = a(T, Y_i, C_i^0) = 1 - \operatorname{mult}_T C_i^0 \leq 1.$$

By (1), $\frac{1}{2} k_i(T) \leq 1$, so $k_i(T) \leq 2$, and, since $(X_i', \pi_i^* \bar{B}_i)$ is lc, the second formula of (6.4) implies

$$b_j n_{ij}(T) \leq \sum_{j'} b_{j'} n_{ij'}(T) \leq k_i(T) \leq 2$$

for any j. Since $n_{ij}(T)$ is an integer, $n_{ij}(T) \leq \lfloor 2/b_j \rfloor = L_j$. Since $P_{ij} = g_i^* Q_{ij}$ is Cartier and $\operatorname{mult}_T P_{ij} = n_{ij}(T)$, the coefficients of P_{ij} belong to $\{0, \ldots, L_j\}$.

Let χ_{ij} be the character of G_i on the fiber at x_i' of the G_i-invariant invertible sheaf $\mathcal{O}_{X_i'}(-Q_{ij})$ (Definition 2.14). Let u be a local equation of Q_{ij} at x_i', and let

$$u_{ij} := \frac{1}{|G_i|}\sum_{\sigma\in G_i}\chi_{ij}(\sigma)^{-1}\sigma^* u.$$

Since Q_{ij} is G_i-invariant and x_i' is fixed by G_i, for any $\sigma\in G_i$ the function σ^*u is a local equation of Q_{ij} at x_i' whose image in the fiber $\mathcal{O}_{X_i'}(-Q_{ij})\otimes\mathbb{C}(x_i')$ is $\chi_{ij}(\sigma)$ times the image of u. Thus the images of u_{ij} and u in the fiber coincide and are non-zero, so u_{ij} generates $\mathcal{O}_{X_i'}(-Q_{ij})$ near x_i' by Nakayama's lemma. Moreover, $\sigma^*u_{ij}=\chi_{ij}(\sigma)u_{ij}$ for any $\sigma\in G_i$. After shrinking X_i (**Step 2**), u_{ij} is a G_i-semi-invariant regular function on X_i' with $\operatorname{div}(u_{ij})=Q_{ij}$. Let $p_{ij}:=g_i^*u_{ij}$. Then p_{ij} is a regular function on Y_i with $\operatorname{div}(p_{ij})=g_i^*Q_{ij}=P_{ij}$, and p_{ij} is I_i-semi-invariant since g_i is I_i-equivariant.

Since $\frac{1}{2}b_j+\beta_{ij}=b_{ij}$ and $\pi_i^*B_i=\sum_j b_{ij}Q_{ij}$, by (6.3),

$$K_{Y_i}+C_i=g_i^*\Big(K_{X_i'}+\tfrac{1}{2}\pi_i^*\bar{B}_i+\sum_j\beta_{ij}Q_{ij}\Big)=g_i^*(K_{X_i'}+\pi_i^*B_i)=g_i^*\pi_i^*(K_{X_i}+B_i),$$

and, since $\frac{1}{2}\pi_i^*\bar{B}_i+\frac{1}{2}\sum_j b_jQ_{ij}=\pi_i^*\bar{B}_i$, by (6.3) with $D=\bar{B}_i$,

$$K_{Y_i}+\bar{C}_i=g_i^*(K_{X_i'}+\pi_i^*\bar{B}_i)=g_i^*\pi_i^*(K_{X_i}+\bar{B}_i).$$

Since $C_i^0\ge 0$, $\beta_{ij}\ge 0$, $b_j>0$, and $P_{ij}\ge 0$, we have $C_i\ge 0$ and $\bar{C}_i\ge 0$. The log discrepancies of (Y_i,C_i) (resp. $(Y_i,\bar{C}_i)$) are those of $(X_i',\pi_i^*B_i)$ (resp. $(X_i',\pi_i^*\bar{B}_i)$) [KM98, Lemma 2.30], so (Y_i,C_i) is klt and $(Y_i,\bar{C}_i)$ is lc by **Step 2**. Since $b_{ij}\le b_j$, we have $\beta_{ij}\le\frac{b_j}{2}$, so $C_i\le\bar{C}_i$. Finally, for any prime divisor T on Y_i,

$$\operatorname{mult}_T C_i=\operatorname{mult}_T C_i^0+\sum_j n_{ij}(T)\big(b_{ij}-\tfrac{b_j}{2}\big),$$

where $\operatorname{mult}_T C_i^0\in\Lambda$, $n_{ij}(T)\in\{0,\dots,L_j\}$, and $b_{ij}\in\Gamma\cap[\frac{b_j}{2},b_j]$. Since (Y_i,C_i) is klt and $C_i\ge 0$, $\operatorname{mult}_T C_i\in[0,1)$. Thus $\operatorname{mult}_T C_i\in\Sigma$, that is, $C_i\in\Sigma$. This proves (3).

Step 5. In this step we show that, after deleting finitely many terms, $\operatorname{center}_{Y_i}F_i$ is not a divisor for any choice of F_i.

Suppose that, for infinitely many i, some prime divisor over X_i' lying over E_i is a prime divisor on Y_i, and choose such an F_i for these i. We derive a contradiction. For any such i, since $\operatorname{center}_{X_i'}F_i=\{x_i'\}$, the divisor F_i is g_i-exceptional, so $a(F_i,X_i',\frac{1}{2}\pi_i^*\bar{B}_i)\le 1$ by (2). By **Step 4**, $k_i(F_i)\in\{1,2\}$ and $n_{ij}(F_i)\in\{0,\dots,L_j\}$. Since $b_{ij}\ge\frac{b_j}{2}$ and $a_i\ge a_1>0$, by (2) and (6.4),

$$\begin{aligned}0<e_ia_1\le e_ia_i=a(F_i,X_i',\pi_i^*B_i)&=k_i(F_i)-\sum_j b_{ij}n_{ij}(F_i)\\&\le k_i(F_i)-\tfrac{1}{2}\sum_j b_jn_{ij}(F_i)=a(F_i,X_i',\tfrac{1}{2}\pi_i^*\bar{B}_i)\le 1.\end{aligned}$$

Thus $e_i\le 1/a_1$. Hence, after passing to an infinite subset of such indices, $e_i=e$, $k_i(F_i)=k$, and $n_{ij}(F_i)=n_j$ do not depend on i. Then

$$a_i=\frac{1}{e}\Big(k-\sum_j b_{ij}n_j\Big)$$

is nonincreasing in i, since b_{ij} is nondecreasing in i and $n_j\ge 0$. This contradicts the strict increase of a_i.

Thus, after deleting finitely many terms, F_i is exceptional over Y_i for any choice of F_i. Since $\operatorname{center}_{X_i'}F_i=\{x_i'\}$, $V_i:=\operatorname{center}_{Y_i}F_i$ is an irreducible closed subset of $g_i^{-1}(x_i')$ of codimension at least two in Y_i, that is, a closed point or a curve. Since I_i fixes ord_{F_i} and g_i is I_i-equivariant, V_i is I_i-invariant.

Step 6. In this step we prove that (Z_i, Δ_i) is a klt pair with $\Delta_i \in \Psi(\Sigma)$ and compute the minimal log discrepancy of (Z_i, Δ_i) at the generic point η_i of the image of V_i.

Since I_i acts faithfully on X_i' and g_i is I_i-equivariant and birational, I_i acts faithfully on Y_i. Since g_i is I_i-equivariant and $\frac{1}{2}\pi_i^*\bar{B}_i$ and the Q_{ij} are G_i-invariant, C_i^0 and the P_{ij} are I_i-invariant, so C_i is I_i-invariant. Let $\tau_i : Y_i \to Z_i$ be the quotient morphism. By (3), $K_{Y_i}+C_i = g_i^*\pi_i^*(K_{X_i}+B_i)$ is $\mathbb{R}$-Cartier, so by Lemma 2.6(0), $K_{Z_i}+\Delta_i$ is $\mathbb{R}$-Cartier, $K_{Y_i}+C_i = \tau_i^*(K_{Z_i}+\Delta_i)$, and $\Delta_i \geq 0$. Since (Y_i, C_i) is klt, (Z_i, Δ_i) is klt by [KM98, Proposition 5.20] applied to τ_i.

Any prime divisor on Z_i is $\tau_i(P)$ for a prime divisor P on Y_i. By Lemma 2.6(4), $\operatorname{mult}_{\tau_i(P)}\Delta_i = 1 - \frac{1-\operatorname{mult}_P C_i}{e_P}$, where e_P is the ramification index of τ_i along P. Since $\operatorname{mult}_P C_i \in \Sigma$ by (3), $\operatorname{mult}_{\tau_i(P)}\Delta_i \in \Psi(\Sigma)$. Thus $\Delta_i \in \Psi(\Sigma)$.

The hypotheses of Lemma 3.1 hold for (X_i, B_i), x_i, E_i, the Galois morphism π_i with group G_i, $B' = \pi_i^*B_i$, $F = F_i$, $H = I_i$, and $f = g_i$:

(a) $K_{X_i'} + \pi_i^*B_i = \pi_i^*(K_{X_i}+B_i)$ by (6.3);
(b) $\operatorname{ord}_{F_i}|_{\mathbb{C}(X_i)} = e_i \operatorname{ord}_{E_i}$ by **Step 3**;
(c) Y_i is normal and quasi-projective, since g_i is projective and X_i' is affine;
(d) F_i has the center V_i on Y_i, since g_i is proper;
(e) $K_{Y_i}+C_i = g_i^*(K_{X_i'}+\pi_i^*B_i)$ by (3);
(f) $\operatorname{mld}(X_i \ni x_i, B_i) = a(E_i, X_i, B_i) = a_i > 0$.

Let η_i be the generic point of $\tau_i(V_i)$. By Lemma 3.1,

$$\operatorname{mld}(Z_i \ni \eta_i, \Delta_i) = a_i, \tag{6.5}$$

attained by the prime divisor over Z_i under F_i. More precisely, for any prime divisor P over Z_i with $\operatorname{center}_{Z_i} P = \overline{\{\eta_i\}}$, any prime divisor F' over X_i' lying over P, and the prime divisor E' over X_i under F', we have $\operatorname{center}_{X_i} E' = \{x_i\}$ and

$$a(P, Z_i, \Delta_i) = \frac{|I_{G_i}(F')|}{|I_i \cap I_{G_i}(F')|} a(E', X_i, B_i) \geq a_i.$$

Step 7. In this step we show that, after deleting finitely many terms, V_i is a closed point for any choice of F_i, and we conclude the proof of (4).

As in **Step 5**, suppose that, for infinitely many i, $\operatorname{center}_{Y_i} F$ is a curve for some prime divisor F over X_i' lying over E_i, and choose such an F as F_i for these i. Let J be the set of these i. Let $i \in J$. Then $\tau_i(V_i)$ is an integral curve on Z_i with generic point η_i, and $\operatorname{mld}(Z_i \ni \eta_i, \Delta_i) = a_i > 0$ by (6.5). By **Step 6**, (Z_i, Δ_i) is a threefold pair with $\Delta_i \in \Psi(\Sigma)$, and $\Psi(\Sigma) \subset [0, 1)$ satisfies the DCC by **Step 1**. By Lemma 3.7 for $\Theta = \Psi(\Sigma)$, the set $\{a_i\}_{i\in J}$ satisfies the ACC. This contradicts the strict increase of a_i, since J is infinite.

Hence, after deleting finitely many terms, V_i is a closed point y_i for any choice of F_i. By **Step 5**, y_i is I_i-invariant, that is, y_i is fixed by I_i. Then η_i is the image z_i of y_i, and **Step 6** implies that $\Delta_i \in \Psi(\Sigma)$, that (Z_i, Δ_i) is klt, and that $\operatorname{mld}(Z_i \ni z_i, \Delta_i) = a_i$, attained by the prime divisor under F_i, together with the formula for $a(P, Z_i, \Delta_i)$ in (4). This proves (4).

The data s, q, λ_ℓ depend only on $(b_1, \dots, b_r)$ by **Step 1**, and Λ, R, M, L_j, and Σ are determined by s, q, λ_ℓ, b_j, and Γ. Thus they are fixed along the subsequence. □

6.4. **Terminal models of index one covers.** In this subsection, we prove Construction-Theorem 6.7 and Corollary 6.8.

Unlike Construction-Theorem 6.6, Construction-Theorem 6.7 extracts over the index one cover X_i' only the exceptional prime divisors with log discrepancy 1, so the canonical divisor stays Cartier, the multiplicities of the boundary equations are at most $\lfloor 1/b_j \rfloor$, and the quotient by the inertia group of a lift of the computing divisor keeps the minimal log discrepancy (Lemma 3.1). Along the increasing sequence, the lifts of the computing divisors have divisorial centers only finitely many times, and curve centers only finitely many times by Lemma 3.7. The remaining centers are smooth points or isolated cDV points (Corollary 6.8(1)), and their quotients are

the objects of Sections 14 and 15. Construction-Theorem 6.7 is used in the proofs of the main theorems in Section 20, while Construction-Theorem 6.6 is used only in Corollary 6.8(2), which is not used in the proofs of the main theorems.

Construction-Theorem 6.7. Notation and conditions as in Set-up 6.4. Let

$$\Sigma := \Big\{ \sum_j n_j t_j \;\Big|\; n_j \in \{0, \dots, L_j\},\ t_j \in \Gamma \cap [\tfrac{b_j}{2}, b_j] \Big\} \cap [0, 1),$$

where $L_j := \lfloor 1/b_j \rfloor$, with $\Sigma = \{0\}$ if $r = 0$. Then Σ and $\Psi(\Sigma)$ satisfy the DCC, and, after shrinking X_i near x_i, the following hold.

(1) Let $A_i \subset \mathrm{Cl}(\mathcal{O}_{X_i, x_i})$ be the subgroup generated by the classes of $K_{X_i}, D_{i1}, \dots, D_{ir}$. There exists a finite quasi-étale abelian Galois morphism $\pi_i : X_i' \to X_i$ with group $G_i = \mathrm{Hom}(A_i, \mathbb{C}^*)$ such that the following hold.
  (a) $\pi_i^{-1}(x_i)$ is a single point x_i'.
  (b) $K_{X_i'}$ is Cartier.
  (c) X_i' is klt.
  (d) $Q_{ij} := \pi_i^* D_{ij}$ is a reduced Cartier divisor with a regular G_i-semi-invariant equation q_{ij}.
  (e) Fix a prime divisor F_i over X_i' lying over E_i. Then the group $I_i := I_{G_i}(F_i)$ is cyclic.
  (f) For the prime divisor F_i fixed in (e), we have $a(F_i, X_i', \pi_i^* B_i) = |I_i| a_i$.

(2) There exists an I_i-equivariant projective birational morphism $g_i : Y_i \to X_i'$ whose exceptional prime divisors are exactly the exceptional prime divisors T over X_i' with $a(T, X_i', 0) = 1$ such that the following hold, where $p_{ij} := g_i^* q_{ij}$, $P_{ij} := \mathrm{div}(p_{ij})$, $C_i := \sum_j b_{ij} P_{ij}$, and $\bar{C}_i := \sum_j b_j P_{ij}$.
  (a) Y_i is terminal.
  (b) Y_i is I_i-$\mathbb{Q}$-factorial.
  (c) $K_{Y_i} = g_i^* K_{X_i'}$ is Cartier.
  (d) $K_{Y_i} + C_i = g_i^* \pi_i^* (K_{X_i} + B_i)$.
  (e) $K_{Y_i} + \bar{C}_i = g_i^* \pi_i^* (K_{X_i} + \bar{B}_i)$.
  (f) The pair (Y_i, C_i) is klt.
  (g) The pair $(Y_i, \bar{C}_i)$ is lc.
  (h) The coefficients of P_{ij} belong to $\{0, \dots, L_j\}$.
  (i) $C_i \in \Sigma$.
  (j) $(Y_i, t\bar{C}_i)$ is terminal for any $0 \le t < \frac{1}{2}$.

(3) After deleting finitely many terms, for the prime divisor F_i fixed in (1)(e), $\mathrm{center}_{Y_i} F_i$ is a closed point y_i, and the following hold, where $(Z_i, \Delta_i) := (Y_i, C_i)/I_i$ and z_i is the image of y_i.
  (a) The point y_i is fixed by I_i.
  (b) $\Delta_i \in \Psi(\Sigma)$.
  (c) $\mathrm{mld}(Z_i \ni z_i, \Delta_i) = a_i$, attained by the prime divisor under F_i.
  (d) For any prime divisor P over Z_i with $\mathrm{center}_{Z_i} P = \{z_i\}$, any prime divisor F' over X_i' lying over P, and the prime divisor E' over X_i under F', we have $\mathrm{center}_{X_i} E' = \{x_i\}$.
  (e) For P, F', and E' as in (d), we have

$$a(P, Z_i, \Delta_i) = \frac{|I_{G_i}(F')|}{|I_i \cap I_{G_i}(F')|} a(E', X_i, B_i) \ge a_i.$$

(4) There exists a nowhere vanishing semi-invariant generator of $\omega_{X_i'}$ near x_i' whose pullback generates ω_{Y_i}, and the following hold, where κ_i is the character of such a generator on I_i and d_{ij} is the character of p_{ij} on I_i.
  (a) The characters $\kappa_i, d_{i1}, \dots, d_{ir}$ have trivial joint kernel.
  (b) Any p_{ij} vanishes at y_i.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove the DCC for Σ and $\Psi(\Sigma)$, construct π_i, and prove (1).

Any element of Σ is a finite sum of elements of Γ and belongs to $[0,1)$, hence $\Sigma \subset \Gamma_+ \cap [0,1]$. Thus Σ satisfies the DCC by Lemma 2.10(1). By Lemma 2.10(1), the set $\Psi(\Sigma)$ also satisfies the DCC. Since $\Sigma \subset [0,1)$, we have $\Psi(\Sigma) \subset [0,1)$. Taking all $n_j = 0$, we have $0 \in \Sigma$.

Since X_i is $\mathbb{Q}$-factorial, K_{X_i} and the D_{ij} are $\mathbb{Q}$-Cartier, so their classes in $\mathrm{Cl}(\mathcal{O}_{X_i,x_i})$ are torsion. Hence A_i is a finite abelian group. Let $[T_1],\dots,[T_m]$ be invariant-factor generators of A_i, where the T_ν are divisors on X_i and $[T_\nu]$ has order r_ν. Let ω be a non-zero rational differential 3-form on X_i. Since $[\mathrm{div}(\omega)] = [K_{X_i}]$ and $[D_{ij}]$ belong to A_i, possibly replacing X_i with an affine neighborhood of x_i, we may assume that there exist $h_\nu, v_K, v_j \in \mathbb{C}(X_i)^*$ and integers $c_\nu, a_{j\nu}$ such that

$$r_\nu T_\nu = \mathrm{div}(h_\nu), \quad \mathrm{div}(\omega) = \sum_\nu c_\nu T_\nu + \mathrm{div}(v_K), \quad D_{ij} = \sum_\nu a_{j\nu} T_\nu + \mathrm{div}(v_j).$$

By [Zhu24, Lemma-Definition 2.15], after shrinking X_i near x_i, the index one cover of $K_{X_i}, D_{i1},\dots,D_{ir}$ is a finite quasi-étale abelian Galois morphism $\pi_i : X_i' \to X_i$ with group $G_i = \mathrm{Hom}(A_i, \mathbb{C}^*)$, such that $\pi_i^{-1}(x_i)$ is a single point x_i' and the pullbacks of K_{X_i} and of the D_{ij} are Cartier. Here $X_i' = \mathrm{Spec} \bigoplus_{D \in A_i} \mathcal{O}_{X_i}(D)$, where each element of A_i is represented by $\sum_\nu a_\nu T_\nu$ with $0 \le a_\nu < r_\nu$, the products are induced by the isomorphisms $\mathcal{O}_{X_i}(r_\nu T_\nu) \to \mathcal{O}_{X_i}$, $f \to f h_\nu$, and $\sigma \in G_i$ acts on the summand of degree D by $\sigma(D)$. At the generic point every summand is $\mathbb{C}(X_i)$. Let u_ν be the element 1 of the summand of degree $[T_\nu]$. Then $u_\nu^{r_\nu} = h_\nu$, $\sigma^* u_\nu = \sigma([T_\nu]) u_\nu$, and the generic fiber of π_i is

$$\mathrm{Spec}\, \mathbb{C}(X_i)[u_1,\dots,u_m]/(u_1^{r_1} - h_1, \dots, u_m^{r_m} - h_m).$$

Since $\pi_{i*}\mathcal{O}_{X_i'}$ is a torsion-free $\mathcal{O}_{X_i}$-module and π_i is finite, any connected component of X_i' maps onto X_i. Since X_i' is normal and $\pi_i^{-1}(x_i)$ is a single point, X_i' is irreducible. Thus $\mathbb{C}(X_i') = \mathbb{C}(X_i)(u_1,\dots,u_m)$. Since π_i is finite and $\pi_i^{-1}(x_i) = \{x_i'\}$, any open neighborhood of x_i' contains $\pi_i^{-1}(V)$ for an affine open neighborhood V of x_i. Thus any shrinking of X_i' near x_i' below is induced by shrinking X_i near x_i, and it keeps the conditions of Set-up 6.4.

Since $r_\nu \,\mathrm{div}(u_\nu) = \pi_i^* \,\mathrm{div}(h_\nu) = r_\nu \pi_i^* T_\nu$, we have $\mathrm{div}(u_\nu) = \pi_i^* T_\nu$. Let $q_{ij} := v_j \prod_\nu u_\nu^{a_{j\nu}}$. Then

$$\mathrm{div}(q_{ij}) = \pi_i^* D_{ij} = Q_{ij} \ge 0,$$

so q_{ij} is regular because X_i' is normal, and $\sigma^* q_{ij} = \sigma([D_{ij}]) q_{ij}$ for any $\sigma \in G_i$. Since π_i is quasi-étale, Q_{ij} is reduced, and $K_{X_i'} = \pi_i^* K_{X_i}$ by the Hurwitz formula. In particular $K_{X_i'}$ is Cartier.

Since (X_i, B_i) is klt, $B_i \ge 0$, and K_{X_i} is $\mathbb{Q}$-Cartier, X_i is klt [KM98, Lemma 2.27]. Since π_i is quasi-étale,

$$K_{X_i'} = \pi_i^* K_{X_i}, \quad K_{X_i'} + \pi_i^* B_i = \pi_i^*(K_{X_i} + B_i), \quad K_{X_i'} + \pi_i^* \bar{B}_i = \pi_i^*(K_{X_i} + \bar{B}_i).$$

By [KM98, Proposition 5.20], X_i' is klt, $(X_i', \pi_i^* B_i)$ is klt, and $(X_i', \pi_i^* \bar{B}_i)$ is lc.

For any i we fix a prime divisor F_i over X_i' lying over E_i (Lemma 2.6(3)), and we let $I_i := I_{G_i}(F_i)$ and $e_i := |I_i|$. Since π_i is quasi-étale, $(X_i', \pi_i^* B_i)/G_i = (X_i, B_i)$, so by Lemma 2.6(1) and (2), the group I_i is cyclic, $\mathrm{ord}_{F_i}|_{\mathbb{C}(X_i)} = e_i \,\mathrm{ord}_{E_i}$, and

$$a(F_i, X_i', \pi_i^* B_i) = e_i a(E_i, X_i, B_i) = e_i a_i.$$

This proves (1). Since E_i computes $\mathrm{mld}(X_i \ni x_i, B_i)$, we have $\mathrm{center}_{X_i} E_i = \{x_i\}$, hence $\mathrm{center}_{X_i'} F_i = \{x_i'\}$.

For any prime divisor T over X_i', let $k_i(T) := a(T, X_i', 0)$ and $n_{ij}(T) := \mathrm{mult}_T Q_{ij}$. Since $K_{X_i'}$ is Cartier and X_i' is klt, $k_i(T)$ is a positive integer. Since $Q_{ij} \ge 0$ is Cartier, $n_{ij}(T)$ is a non-negative integer. Since $(X_i', \pi_i^* \bar{B}_i)$ is lc and $\pi_i^* B_i = \sum_j b_{ij} Q_{ij}$,

$$a(T, X_i', \pi_i^* B_i) = k_i(T) - \sum_j b_{ij} n_{ij}(T), \qquad k_i(T) - \sum_j b_j n_{ij}(T) = a(T, X_i', \pi_i^* \bar{B}_i) \ge 0. \tag{6.6}$$

Step 2. In this step we construct g_i and prove (2).

The group G_i acts faithfully on X_i' and fixes x_i', hence so does I_i. We apply Construction-Theorem 5.1 to $N = 1$, $\alpha_1 = 1$, the klt pair $(X_i', 0)$, the point x_i', and $H = I_i$. Here $\mathcal{S} = \mathbb{Z}_{\geq 0}$ contains 1, and $a(T, X_i', 0) = k_i(T) \in \mathcal{S}$ for any prime divisor T over X_i' by **Step 1**. Moreover $\Lambda_{\mathcal{S}} = \{0\}$. After shrinking X_i near x_i (see **Step 1**), we obtain an I_i-equivariant projective birational morphism $g_i : Y_i \to X_i'$ and an $\mathbb{R}$-divisor $B_{Y_i} \geq 0$ with $K_{Y_i} + B_{Y_i} = g_i^* K_{X_i'}$. By Construction-Theorem 5.1(1), the g_i-exceptional prime divisors are exactly the exceptional prime divisors T over X_i' with $k_i(T) \leq 1$, that is, with $k_i(T) = 1$, and $\operatorname{mult}_T B_{Y_i} = 1 - k_i(T) = 0$ for them. Since $g_{i*} B_{Y_i} = 0$, we have $B_{Y_i} = 0$, so

$$K_{Y_i} = g_i^* K_{X_i'} \tag{6.7}$$

is Cartier. By Construction-Theorem 5.1(2) and (3), Y_i is terminal and I_i-$\mathbb{Q}$-factorial.

Since g_i is I_i-equivariant, $p_{ij} = g_i^* q_{ij}$ is a regular I_i-semi-invariant function on Y_i whose character is the restriction of the character of q_{ij} to I_i, and $P_{ij} = g_i^* Q_{ij} \geq 0$. Pulling back $\pi_i^* B_i = \sum_j b_{ij} Q_{ij}$ and $\pi_i^* \bar{B}_i = \sum_j b_j Q_{ij}$ and using (6.7) and $K_{X_i'} = \pi_i^* K_{X_i}$, we obtain

$$K_{Y_i} + C_i = g_i^* \pi_i^* (K_{X_i} + B_i), \qquad K_{Y_i} + \bar{C}_i = g_i^* \pi_i^* (K_{X_i} + \bar{B}_i).$$

Since $C_i, \bar{C}_i \geq 0$, the pair (Y_i, C_i) is klt and the pair $(Y_i, \bar{C}_i)$ is lc [KM98, Lemma 2.30].

Let T be a prime divisor on Y_i. Then $a(T, Y_i, 0) = 1$, so $k_i(T) = 1$ by (6.7). By (6.6),

$$\sum_j b_j n_{ij}(T) \leq 1.$$

Since any summand is non-negative, $b_j n_{ij}(T) \leq 1$, so $\operatorname{mult}_T P_{ij} = n_{ij}(T) \leq \lfloor 1/b_j \rfloor = L_j$. Since b_{ij} is nondecreasing in i with limit b_j and $b_{ij} \geq \frac{b_j}{2}$ (Set-up 6.4(1)(2)), we have $b_{ij} \in \Gamma \cap [\frac{b_j}{2}, b_j]$. Since (Y_i, C_i) is klt, $\operatorname{mult}_T C_i = \sum_j b_{ij} n_{ij}(T) < 1$. Hence $\operatorname{mult}_T C_i \in \Sigma$, that is, $C_i \in \Sigma$.

Let T be a prime divisor exceptional over Y_i and $k := a(T, Y_i, 0)$. Since Y_i is terminal and K_{Y_i} is Cartier, k is an integer and $k \geq 2$. Since $(Y_i, \bar{C}_i)$ is lc, $0 \leq \operatorname{ord}_T \bar{C}_i \leq k$. For $0 \leq t < \frac{1}{2}$,

$$a(T, Y_i, t\bar{C}_i) = k - t \operatorname{ord}_T \bar{C}_i \geq (1-t)k \geq 2(1-t) > 1.$$

Thus $(Y_i, t\bar{C}_i)$ is terminal. This proves (2).

Step 3. In this step we show that, after deleting finitely many terms, F_i is exceptional over Y_i.

Assume that F_i is a prime divisor on Y_i. Since $\operatorname{center}_{X_i'} F_i = \{x_i'\}$, F_i is g_i-exceptional, so $k_i(F_i) = 1$ by (2). By (1) and (6.6),

$$0 < e_i a_1 \leq e_i a_i = a(F_i, X_i', \pi_i^* B_i) = 1 - \sum_j b_{ij} n_{ij}(F_i) \leq 1.$$

Hence $e_i \leq 1/a_1$, and $n_{ij}(F_i) \in \{0, \dots, L_j\}$ by **Step 2**. Thus the vector $(e_i, n_{i1}(F_i), \dots, n_{ir}(F_i))$ takes only finitely many values. Suppose that F_i is a prime divisor on Y_i for infinitely many i. Then there exist e and $n_1, \dots, n_r$ such that $e_i = e$ and $n_{ij}(F_i) = n_j$ for infinitely many i. For any such i,

$$a_i = \frac{1 - \sum_j b_{ij} n_j}{e}$$

is nonincreasing in i, because b_{ij} is nondecreasing in i and $n_j \geq 0$. This contradicts the strict increase of a_i. Deleting finitely many terms, we may assume that F_i is exceptional over Y_i for any i.

Step 4. In this step we form (Z_i, Δ_i) and compute the minimal log discrepancy at the image of $\operatorname{center}_{Y_i} F_i$.

Let $V_i := \operatorname{center}_{Y_i} F_i$. Since $V_i \subset g_i^{-1}(x_i')$ and F_i is exceptional over Y_i, V_i is a closed point or a curve. Since $\sigma^* \operatorname{ord}_{F_i} = \operatorname{ord}_{F_i}$ for any $\sigma \in I_i$ and g_i is I_i-equivariant, V_i is I_i-invariant.

The group I_i acts faithfully on Y_i, since G_i acts faithfully on X_i' and g_i is I_i-equivariant and birational. Since Y_i is projective over the affine variety X_i', Y_i is quasi-projective, so the

quotient $\tau_i : Y_i \to Z_i := Y_i/I_i$ exists [Mum74, §7, Theorem]. Since the p_{ij} are semi-invariant, $C_i = \sum_j b_{ij} \operatorname{div}(p_{ij})$ is I_i-invariant. Let $(Z_i, \Delta_i) := (Y_i, C_i)/I_i$. By Lemma 2.6(0), $K_{Z_i} + \Delta_i$ is $\mathbb{R}$-Cartier, $K_{Y_i} + C_i = \tau_i^*(K_{Z_i} + \Delta_i)$, and $\Delta_i \geq 0$. By Lemma 2.6(4), if a prime divisor P on Y_i has ramification index u over $\tau_i(P)$ and $\operatorname{mult}_P C_i = c$, then $\operatorname{mult}_{\tau_i(P)} \Delta_i = 1 - \frac{1-c}{u}$. Since any prime divisor on Z_i is $\tau_i(P)$ for some prime divisor P on Y_i, $C_i \in \Sigma$, and $0 \in \Sigma$, we have $\Delta_i \in \Psi(\Sigma)$. By Lemma 2.6(2) and (3), any prime divisor E over Z_i has a lift F to Y_i, and $a(E, Z_i, \Delta_i) = a(F, Y_i, C_i)/|I_{I_i}(F)| > 0$ because (Y_i, C_i) is klt by (2). Thus (Z_i, Δ_i) is klt.

Let η_i be the generic point of $\tau_i(V_i)$. The assumptions of Lemma 3.1 hold for the pair (X_i, B_i), the point x_i, the prime divisor E_i, the morphism π_i with group G_i and $\pi_i^* B_i$, the prime divisor F_i, $H = I_i$, and the morphism g_i with C_i, by (1), (2), and $\operatorname{mld}(X_i \ni x_i, B_i) = a(E_i, X_i, B_i) = a_i > 0$. By Lemma 3.1, we have

$$\operatorname{mld}(Z_i \ni \eta_i, \Delta_i) = a_i, \tag{6.8}$$

attained by the prime divisor under F_i. More precisely, for any prime divisor P over Z_i with $\operatorname{center}_{Z_i} P = \overline{\{\eta_i\}}$, any prime divisor F' over X_i' lying over P, and the prime divisor E' over X_i under F', we have $\operatorname{center}_{X_i} E' = \{x_i\}$ and

$$a(P, Z_i, \Delta_i) = \frac{|I_{G_i}(F')|}{|I_i \cap I_{G_i}(F')|} a(E', X_i, B_i) \geq a_i. \tag{6.9}$$

Step 5. In this step we show that V_i is a curve only for finitely many i.

Assume that V_i is a curve. Then $\tau_i(V_i)$ is an integral curve on Z_i with generic point η_i, and $\operatorname{mld}(Z_i \ni \eta_i, \Delta_i) = a_i > 0$ by (6.8). By **Step 4**, (Z_i, Δ_i) is a threefold pair with $\Delta_i \in \Psi(\Sigma)$, and $\Psi(\Sigma) \subset [0, 1)$ satisfies the DCC by **Step 1**. By Lemma 3.7 for $\Theta = \Psi(\Sigma)$, the numbers a_i for which V_i is a curve satisfy the ACC. Since a_i is strictly increasing, V_i is a curve only for finitely many i.

Step 6. In this step we prove (3).

By **Steps 3** and 5, after deleting finitely many terms, $V_i = \operatorname{center}_{Y_i} F_i$ is a closed point y_i for any i. By **Step 4**, y_i is fixed by I_i and $\Delta_i \in \Psi(\Sigma)$. In this case $\eta_i = z_i$, and (6.8) and (6.9) prove the rest of (3).

Step 7. In this step we prove (4).

Let v_K, c_ν, u_ν, and ω be as in **Step 1**, and let $u_K := v_K \prod_\nu u_\nu^{c_\nu}$. Then $\operatorname{div}(u_K) = \pi_i^* \operatorname{div}(\omega)$. Since π_i is quasi-étale, $\operatorname{div}(\pi_i^* \omega) = \pi_i^* \operatorname{div}(\omega)$ by the Hurwitz formula. Let $\theta_i := \pi_i^* \omega / u_K$. Then $\operatorname{div}(\theta_i) = 0$, and $K_{X_i'}$ is Cartier, hence θ_i is a nowhere vanishing generator of $\omega_{X_i'}$ near x_i'. Since $\sigma^* \pi_i^* \omega = \pi_i^* \omega$ and $\sigma^* u_K = \sigma([K_{X_i}]) u_K$ for any $\sigma \in G_i$,

$$\sigma^* \theta_i = \sigma([K_{X_i}])^{-1} \theta_i.$$

For any g_i-exceptional prime divisor T, we have $\operatorname{ord}_T(g_i^* \theta_i) = a(T, X_i', 0) - 1 = 0$ by (2). Thus $g_i^* \theta_i$ generates ω_{Y_i}, and the character of $g_i^* \theta_i$ on I_i is the character of θ_i on I_i, because g_i is I_i-equivariant.

Two nowhere vanishing semi-invariant generators of $\omega_{X_i'}$ near x_i' differ by a semi-invariant unit w. Since I_i fixes x_i' and $w(x_i') \neq 0$, evaluating $\sigma^* w$ at x_i' shows that the character of w on I_i is trivial. Hence κ_i is the restriction of $\sigma \to \sigma([K_{X_i}])^{-1}$ to I_i. By **Step 2**, d_{ij} is the restriction of $\sigma \to \sigma([D_{ij}])$ to I_i. Let $\sigma \in I_i$ with $\kappa_i(\sigma) = d_{i1}(\sigma) = \cdots = d_{ir}(\sigma) = 1$. Then $\sigma([K_{X_i}]) = \sigma([D_{ij}]) = 1$ for any j. Since $[K_{X_i}], [D_{i1}], \ldots, [D_{ir}]$ generate A_i, σ is the trivial element of $\operatorname{Hom}(A_i, \mathbb{C}^*)$. Thus $\kappa_i, d_{i1}, \ldots, d_{ir}$ have trivial joint kernel.

Since $x_i \in D_{ij}$ and $\pi_i^{-1}(x_i) = \{x_i'\}$, we have $q_{ij}(x_i') = 0$. Hence $p_{ij} = g_i^* q_{ij}$ vanishes on $g_i^{-1}(x_i')$. Since $\operatorname{center}_{X_i'} F_i = \{x_i'\}$, we have $y_i \in g_i^{-1}(x_i')$, so $p_{ij}(y_i) = 0$. This proves (4). □

The next corollary describes the point quotients $(Y_i \ni y_i, C_i)/I_i$ of Construction-Theorems 6.7(3) and 6.6(4) by equivariant embeddings and semi-invariant equations.

Corollary 6.8.

(1) *In the situation of Construction-Theorem 6.7(3), with Y_i, y_i, I_i, p_{ij}, κ_i, and d_{ij} as in Construction-Theorem 6.7, for any $i \gg 0$, either Y_i is smooth at y_i, or there exists an I_i-equivariant closed embedding of $(Y_i \ni y_i)$ into a smooth fourfold germ $(W_i \ni y_i)$ with a faithful action of I_i and I_i-semi-invariant regular parameters, such that $Y_i = (f_i = 0)$ for a regular semi-invariant function f_i and $(Y_i \ni y_i)$ is an isolated cDV singularity. In the smooth case there exist I_i-semi-invariant regular parameters of Y_i at y_i. In both cases the functions p_{ij} lift to regular semi-invariant functions on the ambient germ ($W_i = Y_i$ in the smooth case), κ_i and the d_{ij} have trivial joint kernel on I_i, and $\kappa_i = \prod_k c_k/d_{f_i}$, where the c_k are the characters of the I_i-semi-invariant regular parameters of the ambient germ and d_{f_i} is the character of f_i, with $d_{f_i} := 1$ in the smooth case.*

(2) *In the situation of Construction-Theorem 6.6(4), with Y_i, y_i, I_i, F_i, C_i^0, C_i, P_{ij}, β_{ij}, Λ, and Σ as in Construction-Theorem 6.6, Construction-Theorem 5.5 and Proposition 5.6 hold for $(Y_i \ni y_i, C_i)$, $H = I_i$, the prime divisor F_i, $B^0 = C_i^0$, $P_j = P_{ij}$, and $t_j = \beta_{ij}$. In particular, the numbers a_i are minimal log discrepancies of crepant quotients of smooth or isolated Gorenstein terminal cDV threefold germs by cyclic groups, with boundaries of the form $\sum_k w_k \operatorname{div}(\phi_k)$ for at most $|\Lambda \cap (0,1)| + r$ regular semi-invariant functions ϕ_k and weights w_k in the DCC set $\Lambda \cup \Sigma$.*

Proof. (1) By Construction-Theorem 6.7(2) and (3), for any $i \gg 0$, $(Y_i \ni y_i)$ is a terminal threefold germ with K_{Y_i} Cartier, and the cyclic group I_i fixes y_i. As a subgroup of G_i, I_i acts faithfully on X_i', hence on Y_i. Thus $(Y_i \ni y_i)$ and $G = I_i$ satisfy the assumptions of Lemma 5.4 without a further cover. We denote by W_i the smooth fourfold germ of Lemma 5.4, called Y there, and by f_i the semi-invariant equation of Y_i in W_i. Lemma 5.4 implies the dichotomy, the semi-invariant regular parameters in both cases, the isolated cDV property in the singular case, and the lifts of the regular I_i-semi-invariant functions p_{ij} with equal characters. The trivial joint kernel of $\kappa_i, d_{i1}, \dots, d_{ir}$ is Construction-Theorem 6.7(4).

By Construction-Theorem 6.7(4), there exists an I_i-semi-invariant generator of $\omega_{X_i'}$ near x_i' with character κ_i whose pullback generates ω_{Y_i} near the I_i-fixed point y_i, so κ_i is the character of ω_{Y_i} at y_i (Definition 2.14). Lemma 5.4 implies that the character of ω_{Y_i} at y_i is $\prod_k c_k/d_{f_i}$. Hence $\kappa_i = \prod_k c_k/d_{f_i}$.

(2) We check the hypotheses of Construction-Theorem 5.5 for $(Y_i \ni y_i)$, $H = I_i$, $B = C_i$, and $F = F_i$. By Construction-Theorem 6.6(2), $(Y_i \ni y_i)$ is a terminal threefold germ and I_i is cyclic. As a subgroup of G_i, I_i acts faithfully on X_i', hence on Y_i because g_i is I_i-equivariant and birational, and I_i fixes y_i by Construction-Theorem 6.6(4). Since g_i is I_i-equivariant, C_i^0 is I_i-invariant. Since the p_{ij} are I_i-semi-invariant, the P_{ij} are I_i-invariant. Thus $C_i = C_i^0 + \sum_j \beta_{ij} P_{ij} \geq 0$ is I_i-invariant, and (Y_i, C_i) is klt by Construction-Theorem 6.6(3). By Construction-Theorem 6.6(4), $\operatorname{center}_{Y_i} F_i = \{y_i\}$ and

$$\operatorname{mld}((Y_i \ni y_i, C_i)/I_i) = \operatorname{mld}(Z_i \ni z_i, \Delta_i) = a_i > 0.$$

By Definition 2.5, $I_{I_i}(F_i) = I_i \cap I_{G_i}(F_i) = I_i$. Since $K_{Y_i} + C_i = g_i^*(K_{X_i'} + \pi_i^* B_i)$, Construction-Theorem 6.6(2) implies

$$a(F_i, Y_i, C_i) = a(F_i, X_i', \pi_i^* B_i) = e_i a_i = |I_i| a_i.$$

Thus the hypotheses of Construction-Theorem 5.5 hold.

We check the remaining hypotheses of Proposition 5.6 with $\ell = r$. Any I_i-invariant Weil divisor on Y_i is $\mathbb{Q}$-Cartier, because Y_i is I_i-$\mathbb{Q}$-factorial by Construction-Theorem 6.6(2). The set $\Lambda = \Lambda_{\mathcal{S}} \subset [0,1)$ is finite by Construction-Theorem 5.1, and $\Sigma \subset [0,1)$ satisfies the DCC by Construction-Theorem 6.6. By Construction-Theorem 6.6(2) and (3), $C_i^0 \geq 0$ is I_i-invariant, $C_i^0 \in \Lambda$, (Y_i, C_i^0) is terminal, and $P_{ij} = \operatorname{div}(p_{ij}) \geq 0$ is an I_i-invariant Cartier divisor. It remains to show that $\beta_{ij} \in \Sigma$. Since $\sum_\ell q\alpha_\ell = \sum_\ell \lambda_\ell = 1$, we have $1 \in \mathcal{S}$, hence $0 = 1 - 1 \in \Lambda$. By Set-up 6.4(1)(2), $b_{ij} \in \Gamma \cap [\frac{b_j}{2}, b_j]$. Since $b_j \leq 1$, we have $L_j = \lfloor 2/b_j \rfloor \geq 1$. Taking $c = 0$, $t_j = b_{ij}$,

$n_j = 1$, and $n_{j'} = 0$ for $j' \neq j$ in the definition of Σ, we obtain

$$\beta_{ij} = b_{ij} - \frac{b_j}{2} \in [0, \tfrac{1}{2}] \subset [0, 1),$$

so $\beta_{ij} \in \Sigma$. Thus the hypotheses of Proposition 5.6 hold.

By Construction-Theorem 5.5(1) and (3), after shrinking Y_i near y_i, the canonical cover $h : X' \to Y_i$ satisfies $h^{-1}(y_i) = \{x'\}$ for a closed point x', the finite abelian group $\hat{H}$ of Construction-Theorem 5.5(1), which is an extension of I_i by the Galois group of h, acts faithfully on X' and fixes x', and $(X' \ni x')$ is smooth or an isolated Gorenstein terminal cDV threefold germ. Let $B' := h^*C_i$, let F' be a prime divisor over X' lying over F_i, and let $H' := I_{\hat{H}}(F')$. By Construction-Theorem 5.5(2), H' is cyclic and

$$a_i = \mathrm{mld}((X' \ni x', B')/H').$$

By Proposition 5.6, there exist $|\Lambda \cap (0,1)| + r$ regular $\hat{H}$-semi-invariant functions φ_β, $\beta \in \Lambda \cap (0,1)$, and $\psi_1, \dots, \psi_r$ on X' (ψ_j is the function g_j of Proposition 5.6), which are in particular H'-semi-invariant, such that

$$B' = \sum_{\beta \in \Lambda \cap (0,1)} \beta \operatorname{div}(\varphi_\beta) + \sum_{j=1}^{r} \beta_{ij} \operatorname{div}(\psi_j).$$

The weights belong to $\Lambda \cup \Sigma$, which satisfies the DCC because Λ is finite and Σ satisfies the DCC. □

7. Polynomial models

In this section, we prove Construction-Theorem 7.2. Construction-Theorem 7.2 replaces an equivariant smooth or isolated cDV threefold germ, together with semi-invariant boundary equations, by a polynomial model, and keeps the minimal log discrepancy of the crepant quotient by any subgroup.

7.1. **Equivariant contact determinacy.** In this subsection, we prove an equivariant version of finite contact determinacy. A power series f in the maximal ideal $\mathfrak{m}$ of $\mathbb{C}[[x_1, \dots, x_n]]$ with finite Tjurina number is finitely contact determined [GP19, Theorem 4.8]. The theorem in [GP19] does not involve a group action. In Lemma 7.1, each correction is projected onto an eigenspace of the group action. The hypothesis of Lemma 7.1 involves only the derivatives $\partial_i F$ for i in a subset S of the indices, and φ fixes x_i for any index i outside S. When S does not contain the index of t, Lemma 7.1(2) implies that φ fixes t, so φ preserves $\operatorname{div}(t)$ and any multiple of it. Proposition 12.11 uses Lemma 7.1 with such S for a boundary which is a non-positive rational multiple of $\operatorname{div}(t)$.

Lemma 7.1. *Let a finite abelian group G act diagonally on $\mathfrak{R} := \mathbb{C}[[x_1, \dots, x_n]]$, let $\mathfrak{m}$ be the maximal ideal, let $F \in \mathfrak{R}$ be G-semi-invariant, let $S \subset \{1, \dots, n\}$, and let b be a positive integer. For a positive integer L, let $F_{<L}$ be the sum of the terms of F of degree less than L. Assume that*

$$\mathfrak{m}^b \subset (F) + \mathfrak{m}(\partial_i F \mid i \in S).$$

Then for any integer $L > 2b$ there exist a G-equivariant automorphism φ of $\mathfrak{R}$ and a G-invariant unit v of $\mathfrak{R}$ such that the following hold.

(1) $v\varphi(F) = F_{<L}$, and $\varphi(x_i) - x_i \in \mathfrak{m}^{L-b+1}$ for any $i \in S$.
(2) $\varphi(x_i) = x_i$ for any $i \notin S$.

In particular, if $n = 4$, $\mathfrak{R} = \mathbb{C}[[x, y, z, t]]$, and S consists of the indices of x, y, z, then $\varphi(t) = t$.

Proof. For any $\sigma \in G$, write $\sigma^* x_i = \chi_i(\sigma) x_i$ and $\sigma^* F = \chi_F(\sigma) F$. For $g \in \mathfrak{R}$, let

$$T_S(g) := (g) + \mathfrak{m}(\partial_i g \mid i \in S).$$

Since σ^* multiplies any monomial by a scalar, any monomial occurring in F has character χ_F. Thus $F_{<L}$ is a semi-invariant polynomial with character χ_F, and $\partial_i F_{<L}$ is semi-invariant with character $\chi_F \chi_i^{-1}$.

Step 1. In this step, we show that $\mathfrak{m}^b \subset T_S(F_{<L})$.

Since $F - F_{<L} \in \mathfrak{m}^L$ and $\mathfrak{m}\partial_i(F - F_{<L}) \subset \mathfrak{m}^L$, we have $T_S(F) \subset T_S(F_{<L}) + \mathfrak{m}^L$. As $L > b$, we obtain

$$\mathfrak{m}^b \subset T_S(F_{<L}) + \mathfrak{m}^{b+1}.$$

Let $N := \mathfrak{m}^b \cap T_S(F_{<L})$. Since $\mathfrak{m}^{b+1} \subset \mathfrak{m}^b$, the inclusion $\mathfrak{m}^b \subset T_S(F_{<L}) + \mathfrak{m}^{b+1}$ implies $\mathfrak{m}^b = N + \mathfrak{m} \cdot \mathfrak{m}^b$. Since $\mathfrak{R}$ is Noetherian, $\mathfrak{m}^b/N$ is a finite $\mathfrak{R}$-module. Thus $\mathfrak{m}^b = N \subset T_S(F_{<L})$ by Nakayama's lemma [Mat89, Theorem 2.2, Corollary].

Step 2. In this step, we introduce the projections onto the eigenspaces.

For a character χ of G and $g \in \mathfrak{R}$, let

$$P_\chi(g) := \frac{1}{|G|} \sum_{\sigma \in G} \chi(\sigma)^{-1} \sigma^* g.$$

Since G is abelian, $P_\chi(g)$ is semi-invariant with character χ, and $P_\chi(g) = g$ if g is semi-invariant with character χ. Since $\sigma^* \mathfrak{m}^k = \mathfrak{m}^k$, we have $P_\chi(\mathfrak{m}^k) \subset \mathfrak{m}^k$ for any $k \geq 0$. If $h \in \mathfrak{R}$ is semi-invariant with character χ_h, then

$$P_\chi(gh) = P_{\chi\chi_h^{-1}}(g)h.$$

We write $\mathbf{1}$ for the trivial character, so that $P_{\mathbf{1}}(g)$ is G-invariant.

Step 3. In this step, we construct the corrections φ_j.

Let $L_0 := L$ and $L_{j+1} := 2L_j - 2b$ for $j \geq 0$. Then $L_j - 2b = 2^j(L - 2b)$. Since $L > 2b$, we have $L_j \geq L$ for any j and $L_j \to +\infty$. We construct, for any $j \geq 0$, a G-equivariant automorphism Φ_j of $\mathfrak{R}$ and a G-invariant unit v_j of $\mathfrak{R}$ such that $v_j - 1 \in \mathfrak{m}$, and such that

$$\epsilon_j := v_j \Phi_j(F) - F_{<L} \in \mathfrak{m}^{L_j}, \qquad \Phi_j(x_i) - x_i \in \mathfrak{m}^{L-b+1} \text{ for } i \in S, \qquad \Phi_j(x_i) = x_i \text{ for } i \notin S.$$

Let $\Phi_0 := \mathrm{id}_{\mathfrak{R}}$ and $v_0 := 1$. Then $\epsilon_0 = F - F_{<L} \in \mathfrak{m}^{L_0}$.

Suppose that Φ_j and v_j are constructed. Since Φ_j is G-equivariant and v_j is G-invariant, ϵ_j is semi-invariant with character χ_F. Since $L_j \geq b$, **Step 1** implies

$$\epsilon_j \in \mathfrak{m}^{L_j - b}\mathfrak{m}^b \subset \mathfrak{m}^{L_j - b} T_S(F_{<L}) = \mathfrak{m}^{L_j - b}(F_{<L}) + \mathfrak{m}^{L_j - b + 1}(\partial_i F_{<L} \mid i \in S).$$

Thus there exist $a_j \in \mathfrak{m}^{L_j - b}$ and $u_{i,j} \in \mathfrak{m}^{L_j - b + 1}$ for $i \in S$ such that

$$\epsilon_j = a_j F_{<L} + \sum_{i \in S} u_{i,j} \partial_i F_{<L}.$$

We apply P_{χ_F} to both sides. By **Step 2**,

$$P_{\chi_F}(\epsilon_j) = \epsilon_j, \qquad P_{\chi_F}(a_j F_{<L}) = P_{\mathbf{1}}(a_j) F_{<L}, \qquad P_{\chi_F}(u_{i,j}\partial_i F_{<L}) = P_{\chi_i}(u_{i,j})\partial_i F_{<L}.$$

Possibly replacing a_j with $P_{\mathbf{1}}(a_j)$ and $u_{i,j}$ with $P_{\chi_i}(u_{i,j})$, we may assume that a_j is G-invariant and that $u_{i,j}$ is semi-invariant with character χ_i for any $i \in S$.

Let φ_j be the endomorphism of $\mathfrak{R}$ such that $\varphi_j(x_i) = x_i - u_{i,j}$ for $i \in S$ and $\varphi_j(x_i) = x_i$ for $i \notin S$. Since $x_i - u_{i,j}$ is semi-invariant with character χ_i, the continuous endomorphisms $\sigma^* \circ \varphi_j$ and $\varphi_j \circ \sigma^*$ coincide on $x_1, \ldots, x_n$ for any $\sigma \in G$. Thus φ_j is G-equivariant. Since $L_j - b + 1 \geq 2$, we have $u_{i,j} \in \mathfrak{m}^2$. Hence φ_j induces the identity on $\mathfrak{m}^k/\mathfrak{m}^{k+1}$ for any $k \geq 0$, so φ_j is bijective on $\mathfrak{R}/\mathfrak{m}^k$ for any k and is an automorphism of $\mathfrak{R}$. A direct calculation on monomials, writing $\varphi_j(x^\alpha) - x^\alpha$ as a telescoping sum in which each term has one factor $u_{i,j}$ and $|\alpha| - 1$ factors in $\mathfrak{m}$, shows that

$$\varphi_j(g) - g \in \mathfrak{m}^{k + L_j - b} \quad \text{for any } k \geq 1 \text{ and any } g \in \mathfrak{m}^k. \tag{$*$}$$

In particular, $\varphi_j(g) - g \in \mathfrak{m}^{L_j - b + 1}$ for any $g \in \mathfrak{R}$.

Let

$$\Phi_{j+1} := \varphi_j \circ \Phi_j, \qquad v_{j+1} := (1-a_j)\varphi_j(v_j).$$

Since $a_j \in \mathfrak{m}$ is G-invariant, v_{j+1} is a G-invariant unit, and $v_{j+1} - 1 \in \mathfrak{m}$. For $i \notin S$, we have $\Phi_{j+1}(x_i) = \varphi_j(x_i) = x_i$. For $i \in S$, $(*)$ implies

$$\Phi_{j+1}(x_i) - \Phi_j(x_i) = \varphi_j(\Phi_j(x_i)) - \Phi_j(x_i) \in \mathfrak{m}^{L_j-b+1} \subset \mathfrak{m}^{L-b+1},$$

hence $\Phi_{j+1}(x_i) - x_i \in \mathfrak{m}^{L-b+1}$. It remains to bound ϵ_{j+1}. Since $F_{<L}$ is a polynomial, the Taylor formula implies

$$\varphi_j(F_{<L}) = F_{<L} - \sum_{i\in S} u_{i,j}\partial_i F_{<L} + Q_j, \qquad Q_j \in (u_{i,j}u_{k,j} \mid i,k \in S) \subset \mathfrak{m}^{2L_j-2b+2}.$$

By $(*)$ with $k = L_j$, we have $\varphi_j(\epsilon_j) - \epsilon_j \in \mathfrak{m}^{2L_j-b}$. Since $\epsilon_j - \sum_{i\in S} u_{i,j}\partial_i F_{<L} = a_j F_{<L}$, we obtain

$$\varphi_j(v_j\Phi_j(F)) = \varphi_j(F_{<L} + \epsilon_j) = (1+a_j)F_{<L} + Q_j + \big(\varphi_j(\epsilon_j) - \epsilon_j\big).$$

Multiplying by $1 - a_j$, we have

$$\epsilon_{j+1} = v_{j+1}\Phi_{j+1}(F) - F_{<L} = -a_j^2 F_{<L} + (1-a_j)\big(Q_j + \varphi_j(\epsilon_j) - \epsilon_j\big) \in \mathfrak{m}^{2L_j-2b} = \mathfrak{m}^{L_{j+1}}.$$

By induction on j, Φ_j and v_j exist for any $j \geq 0$.

Step 4. In this step, we take the limit and conclude the proof.

By **Step 3**, $\Phi_{j+1}(x_i) - \Phi_j(x_i) \in \mathfrak{m}^{L_j-b+1}$ for any i, and

$$v_{j+1} - v_j = -a_j\varphi_j(v_j) + \big(\varphi_j(v_j) - v_j\big) \in \mathfrak{m}^{L_j-b}.$$

Since $L_j \to +\infty$, the sequences $\Phi_j(x_i)$ and v_j converge in the $\mathfrak{m}$-adic topology. Let $y_i := \lim_j \Phi_j(x_i)$, let $v := \lim_j v_j$, and let φ be the endomorphism of $\mathfrak{R}$ such that $\varphi(x_i) = y_i$. Since the ideals $\mathfrak{m}^k$ are closed, we have $y_i - x_i \in \mathfrak{m}^{L-b+1}$ for $i \in S$, $y_i = x_i$ for $i \notin S$, $v - 1 \in \mathfrak{m}$, and $\Phi_j(x_i) - y_i \in \mathfrak{m}^{L_j-b+1}$ for any i and j. Since σ^* is continuous, y_i is semi-invariant with character χ_i and v is G-invariant. Thus φ is G-equivariant, and v is a G-invariant unit. Since $L - b + 1 \geq 2$, φ induces the identity on $\mathfrak{m}^k/\mathfrak{m}^{k+1}$ for any $k \geq 0$, hence φ is an automorphism of $\mathfrak{R}$.

For any j, $\Phi_j(F) - \varphi(F)$ belongs to the ideal generated by $\Phi_j(x_i) - y_i$ for $1 \leq i \leq n$, hence $\Phi_j(F) - \varphi(F) \in \mathfrak{m}^{L_j-b+1}$. Thus $v_j\Phi_j(F)$ converges to $v\varphi(F)$. Since $v_j\Phi_j(F) - F_{<L} = \epsilon_j \in \mathfrak{m}^{L_j}$, $v_j\Phi_j(F)$ also converges to $F_{<L}$. Therefore $v\varphi(F) = F_{<L}$. This implies (1) and (2).

If $n = 4$, $\mathfrak{R} = \mathbb{C}[[x,y,z,t]]$, and S consists of the indices of x, y, z, then the index of t does not belong to S, so $\varphi(t) = t$ by (2). □

7.2. **Polynomial models with equal quotient minima.** In this subsection, we replace a smooth or isolated Gorenstein terminal threefold germ $(X \ni x)$ with boundary B and with an action of a finite abelian group H by a polynomial germ $(X_0 \ni 0, B_0)$ such that

$$\mathrm{mld}((X_0 \ni 0, B_0)/H') = \mathrm{mld}((X \ni x, B)/H')$$

for any subgroup $H' \subset H$ (Construction-Theorem 7.2).

We truncate the boundary equations at a high degree. By the continuity of log canonical thresholds, the truncated boundary keeps a uniform klt margin, so any prime divisor at which the order of some boundary equation changes has log discrepancy larger than $|H|$ times a constant exceeding the minimal log discrepancies of the quotients by all subgroups of H. Thus the minimal log discrepancy of the quotient by any subgroup of H does not change. Later we use one polynomial model for different subgroups of H, for example H itself and the inertia group of a prime divisor over X. Here smooth threefold germs also count as isolated Gorenstein terminal threefold germs, and condition (i) below names them separately because for n equal to three the germ X is its own smooth ambient germ Y.

Construction-Theorem 7.2. Let H be a finite abelian group and $n \in \{3,4\}$, and assume the following.

(i) $(X \ni x)$ is a smooth threefold germ or an isolated Gorenstein terminal threefold germ, and H acts faithfully on X and fixes x.
(ii) X has an H-equivariant closed embedding into a smooth germ $(Y \ni x)$ of dimension n, with $Y = X$ if $n = 3$ and $X = (f = 0)$ for a regular H-semi-invariant function f if $n = 4$, and $x_1, \dots, x_n$ are H-semi-invariant regular parameters of Y at x.
(iii) $h_1, \dots, h_s$ are regular H-semi-invariant functions on Y, non-zero on X, $\beta_1, \dots, \beta_s$ are positive real numbers, and (X, B) is klt near x for $B := \sum_k \beta_k \operatorname{div}(h_k|_X)$ ($s = 0$ is allowed, in which case $B = 0$).
(iv) $\beta^{(1)}, \dots, \beta^{(r)} \in \mathbb{R}^s_{\ge 0}$ are finitely many vectors such that $(X, \sum_k \beta_k^{(j)} \operatorname{div}(h_k|_X))$ is klt near x for any j ($r = 0$ is allowed).
(v) $P_1, \dots, P_l$ are finitely many prime divisors over X with center x ($l = 0$ is allowed).

Let $\hat{h}_k$, and $\hat{f}$ if $n = 4$, be the expansions of h_k and f in $\mathfrak{R} := \mathbb{C}[[x_1, \dots, x_n]]$ with the induced diagonal action of H. Then there exist a faithful diagonal action of H on $\mathbb{A}^n$ in which the coordinates have the characters of the x_i, a germ $X_0 = \mathbb{A}^3$ at 0 if $n = 3$, or $X_0 = (f_0 = 0) \subset \mathbb{A}^4$ at 0 for a semi-invariant polynomial f_0 with the character of f such that X_0 is an isolated Gorenstein terminal germ if $n = 4$, and semi-invariant polynomials $h_{1,0}, \dots, h_{s,0}$ with the characters of $h_1, \dots, h_s$, non-zero on X_0, such that, for $B_0 := \sum_k \beta_k \operatorname{div}(h_{k,0}|_{X_0})$:

(1) (X_0, B_0) is klt near 0, H acts faithfully on X_0, and
$$\operatorname{mld}((X_0 \ni 0, B_0)/H') = \operatorname{mld}((X \ni x, B)/H')$$
for any subgroup $H' \subset H$;
(2) there exists an H-equivariant isomorphism between the completions of the local rings of X at x and of X_0 at 0, under which each P_i corresponds to a prime divisor $P_{i,0}$ over X_0 with center 0 having isomorphic residue field and equal decomposition group, inertia group, log discrepancy $a(\cdot, \cdot, 0)$, and orders of the functions h_k and $h_{k,0}$;
(3) $(X_0, \sum_k \beta_k^{(j)} \operatorname{div}(h_{k,0}|_{X_0}))$ is klt near 0 for any j;
(4) for any H-equivariant automorphism ψ of $\mathfrak{R}$ and any H-invariant unit v of $\mathfrak{R}$ (for example $\psi = \mathrm{id}_{\mathfrak{R}}$ and $v = 1$), there exists a positive integer N_1 such that for any $N \ge N_1$ there exists L_1 such that for any $L \ge L_1$ one can take $h_{k,0} = \psi(\hat{h}_k)_{<N}$ for any k and, if $n = 4$, $f_0 = (v\psi(\hat{f}))_{<L}$, where $g_{<M}$ denotes the sum of the terms of g of degree less than M.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we reduce Construction-Theorem 7.2 to a statement for fixed ψ and v, and we fix the klt margin ϵ, a reference prime divisor P_*, and the integer N_1.

Fix ψ and v as in (4). We show that there exists N_1 with the following property: for any $N \ge N_1$ there exists L_1 such that for any $L \ge L_1$, the polynomials
$$h_{k,0} := \psi(\hat{h}_k)_{<N} \ (1 \le k \le s) \quad \text{and, if } n = 4, \quad f_0 := (v\psi(\hat{f}))_{<L}$$
have the properties of $h_{k,0}$, f_0, and X_0 listed in the statement and satisfy (1), (2), and (3). This proves (4), and taking $\psi = \mathrm{id}_{\mathfrak{R}}$, $v = 1$, $N = N_1$, and $L = L_1$ proves the existence of X_0 and $h_{1,0}, \dots, h_{s,0}$.

We use $x_1, \dots, x_n$ also for the coordinates of $\mathbb{A}^n$, and we let H act on $\mathbb{A}^n$ by $\sigma^* x_i = \chi_i(\sigma) x_i$, where χ_i is the character of the function x_i on Y (Definition 2.14). Then the inclusion $\mathbb{C}[x_1, \dots, x_n] \subset \mathfrak{R}$ is H-equivariant.

Since $x_1, \dots, x_n$ are regular parameters of the smooth germ Y at x, $\widehat{\mathcal{O}}_{Y,x} = \mathfrak{R}$ [Mat89, Theorem 29.7], and $\widehat{\mathcal{O}}_{X,x} = \mathfrak{R}$ if $n = 3$ and $\widehat{\mathcal{O}}_{X,x} = \mathfrak{R}/(\hat{f})$ if $n = 4$ [Mat89, Theorem 8.7]. Let $\widehat{X} := \operatorname{Spec} \widehat{\mathcal{O}}_{X,x}$ with closed point $\hat{x}$, with the induced action of H, and denote the images of $\hat{h}_1, \dots, \hat{h}_s$ in $\widehat{\mathcal{O}}_{X,x}$ again by $\hat{h}_1, \dots, \hat{h}_s$. Since $\mathcal{O}_{X,x} \to \widehat{\mathcal{O}}_{X,x}$ is injective and $h_k|_X \ne 0$, we have $\hat{h}_k \ne 0$ in $\widehat{\mathcal{O}}_{X,x}$. For a formal prime divisor P over $\widehat{X}$ with center $\hat{x}$ (Definition 3.2), we write $\operatorname{mult}_P \mathfrak{m}$ for the order along P of the maximal ideal $\mathfrak{m}\widehat{\mathcal{O}}_{X,x}$. It is a positive integer.

By Lemma 3.3, $K_{\widehat{X}}$ is the pullback of K_X, and $\widehat{B} := \sum_k \beta_k \operatorname{div}(\hat{h}_k)$ is the pullback of B. Since X is Gorenstein, $K_{\widehat{X}}$ is Cartier, and for any formal prime divisor P over $\widehat{X}$ and any non-zero $g_1, \dots, g_s \in \widehat{\mathcal{O}}_{X,x}$,

$$a\Big(P, \widehat{X}, \sum_k \beta_k \operatorname{div}(g_k)\Big) = a(P, \widehat{X}, 0) - \sum_k \beta_k \operatorname{ord}_P(g_k). \tag{7.1}$$

Since X is terminal and K_X is Cartier, any prime divisor E over X with center x has $a(E, X, 0) \in \mathbb{Z}$ and $a(E, X, 0) > 1$, so $a(E, X, 0) \geq 2$. Hence, by Lemma 3.3(3),

$$a(P, \widehat{X}, 0) \geq 2 \quad \text{for any formal prime divisor } P \text{ over } \widehat{X} \text{ with center } \hat{x}. \tag{7.2}$$

Let $\beta^{(0)} := (\beta_1, \dots, \beta_s)$, let $\mathcal{K} := \{k \mid h_k(x) = 0\}$, and for $0 \leq j \leq r$ let $\mathcal{K}^{(j)} := \{k \in \mathcal{K} \mid \beta_k^{(j)} > 0\}$. For $k \notin \mathcal{K}$, $\hat{h}_k$ is a unit and $\operatorname{div}(\hat{h}_k) = 0$. For any j with $\mathcal{K}^{(j)} \neq \emptyset$, let

$$\mathfrak{a}^{(j)} := \prod_{k \in \mathcal{K}^{(j)}} \big(\hat{h}_k \widehat{\mathcal{O}}_{X,x}\big)^{\beta_k^{(j)}},$$

an $\mathbb{R}$-ideal on $\widehat{X}$ whose factors are non-zero proper ideals. By (iii) and (iv), $(X, \sum_k \beta_k^{(j)} \operatorname{div}(h_k|_X))$ is klt near x, so $(\widehat{X}, \sum_k \beta_k^{(j)} \operatorname{div}(\hat{h}_k))$ is klt [dFEM11, Remark 2.10 and Proposition 2.11(i)]. By [dFEM11, Section 2, formula (5)], the log canonical threshold $\operatorname{lct}(\mathfrak{a}^{(j)})$ on $\widehat{X}$ is the minimum of the ratios

$$\frac{a(E, \widehat{X}, 0)}{\sum_k \beta_k^{(j)} \operatorname{ord}_E(\hat{h}_k)}$$

over the finitely many prime divisors E on a log resolution of $(\widehat{X}, \mathfrak{a}^{(j)})$ with positive denominator, and each ratio is larger than 1 by (7.1) and the klt property. Thus we may fix a real number $\epsilon > 0$ such that

$$\operatorname{lct}(\mathfrak{a}^{(j)}) > 1 + 2\epsilon \quad \text{for any } j \text{ with } \mathcal{K}^{(j)} \neq \emptyset. \tag{7.3}$$

Let P_* be a prime divisor over X with center x, for example a prime divisor on the normalization of the blow-up of X at x mapping to x. By Lemma 3.3(3), we may fix formal prime divisors $\widehat{P}_*, \widehat{P}_1, \dots, \widehat{P}_l$ over $\widehat{X}$ with center $\hat{x}$ such that the restrictions of $\operatorname{ord}_{\widehat{P}_*}, \operatorname{ord}_{\widehat{P}_1}, \dots, \operatorname{ord}_{\widehat{P}_l}$ to $\mathbb{C}(X)$ are $\operatorname{ord}_{P_*}, \operatorname{ord}_{P_1}, \dots, \operatorname{ord}_{P_l}$, respectively. Lemma 3.3(3), applied to (X, B) and to $(X, 0)$, implies that $\widehat{P}_i$ and P_i have isomorphic residue fields, equal decomposition groups, equal inertia groups, $a(\widehat{P}_i, \widehat{X}, 0) = a(P_i, X, 0)$, and $a(\widehat{P}_*, \widehat{X}, \widehat{B}) = a(P_*, X, B)$. Moreover, $\operatorname{ord}_{\widehat{P}_i}(\hat{h}_k) = \operatorname{ord}_{P_i}(h_k)$ for any i and k. Fix a real number C with

$$C > a(P_*, X, B) = a(\widehat{P}_*, \widehat{X}, \widehat{B}) > 0,$$

where the last inequality holds since (X, B) is klt near x. Let N_1 be a positive integer such that any integer $N \geq N_1$ satisfies the following.

(a) $N \geq 3/(\epsilon \beta_k^{(j)})$ for any $0 \leq j \leq r$ and any $k \in \mathcal{K}^{(j)}$.
(b) $\epsilon \beta_k N > C|H|$ for any $k \in \mathcal{K}$.
(c) $N \operatorname{mult}_P \mathfrak{m} > \operatorname{ord}_P(\hat{h}_k)$ for any $P \in \{\widehat{P}_*, \widehat{P}_1, \dots, \widehat{P}_l\}$ and any $1 \leq k \leq s$.

Such an N_1 exists since each condition is a lower bound on N. In (a), the number $3 = \dim X$ is the dimension in [dFEM11, Proposition 2.19(ii)].

Step 2. In this step, we fix $N \geq N_1$, define L_1, and for any $L \geq L_1$ we construct an H-equivariant automorphism ψ' of $\mathfrak{R}$ such that $\psi'(\hat{h}_k) - h_{k,0} \in \mathfrak{m}^N$ for any k and, if $n = 4$, $\psi'((\hat{f})) = (f_0)$.

For $\sigma \in H$,

$$\sigma^* \psi(\hat{h}_k) = \psi(\sigma^* \hat{h}_k) = \chi(\sigma) \psi(\hat{h}_k),$$

where χ is the character of h_k. Since H acts diagonally, any monomial is H-semi-invariant, so any monomial occurring in $\psi(\hat{h}_k)$ has the character of h_k. Hence $h_{k,0}$ is a semi-invariant polynomial with the character of h_k once $h_{k,0} \neq 0$, which we prove in **Step 4**.

If $n = 3$, let $L_1 := 1$, $\varphi := \mathrm{id}_{\mathfrak{R}}$, and $\psi' := \psi$.

Assume that $n = 4$, and let $\hat{f}' := v\psi(\hat{f})$. Since ψ is H-equivariant and v is H-invariant, $\hat{f}'$ is semi-invariant with the character of f. After shrinking Y, the differentials $dx_1, \dots, dx_4$ form a basis of Ω_Y. Let $\partial_1, \dots, \partial_4$ be the dual derivations, which extend to the derivations $\partial/\partial x_i$ of $\mathfrak{R}$. By (i), X is smooth outside x near x, so, by [Mat89, Theorem 14.2] and [Har77, Proposition II.8.7], the ideal $(f, \partial_1 f, \dots, \partial_4 f)$ of $\mathcal{O}_{Y,x}$ contains a power of the maximal ideal. Thus there exists an integer $p \geq 0$ such that

$$\mathfrak{m}^p \subset (\hat{f}, \partial_1 \hat{f}, \dots, \partial_4 \hat{f}),$$

and $p = 0$ if X is smooth at x. For $g \in \mathfrak{R}$ we have $\psi(g) = g(\psi(x_1), \dots, \psi(x_4))$, so, by the chain rule,

$$\partial_i \psi(g) = \sum_{j=1}^{4} \psi(\partial_j g)\partial_i \psi(x_j).$$

The matrix $(\partial_i \psi(x_j))_{i,j}$ is invertible over $\mathfrak{R}$, since the value of $(\partial_i \psi(x_j))_{i,j}$ at the closed point is the matrix of the automorphism of $\mathfrak{m}/\mathfrak{m}^2$ induced by ψ. Since v is a unit and $\partial_i \hat{f}' = v\partial_i \psi(\hat{f}) + \psi(\hat{f})\partial_i v$, we obtain

$$(\hat{f}', \partial_1 \hat{f}', \dots, \partial_4 \hat{f}') = \psi\big((\hat{f}, \partial_1 \hat{f}, \dots, \partial_4 \hat{f})\big) \supset \psi(\mathfrak{m}^p) = \mathfrak{m}^p. \tag{7.4}$$

Let $b := p + 1$. Then

$$\mathfrak{m}^b \subset (\hat{f}') + \mathfrak{m}(\partial_1 \hat{f}', \dots, \partial_4 \hat{f}').$$

Let $L_1 := \max\{2b + 3, N + b - 1\}$ and fix an integer $L \geq L_1$. By Lemma 7.1, applied to $\hat{f}'$ with $S = \{1, 2, 3, 4\}$, there exist an H-equivariant automorphism φ of $\mathfrak{R}$ and an H-invariant unit v' of $\mathfrak{R}$ such that

$$v'\varphi(\hat{f}') = \hat{f}'_{<L} = f_0 \quad \text{and} \quad \varphi(x_i) - x_i \in \mathfrak{m}^{L-b+1} \subset \mathfrak{m}^N \text{ for any } i.$$

As for $h_{k,0}$, the polynomial f_0 is semi-invariant with the character of f. Since $x \in X$ and $X \neq Y$, we have $\hat{f} \in \mathfrak{m}\backslash\{0\}$, hence $f_0 = v'\varphi(\hat{f}') \in \mathfrak{m}\backslash\{0\}$. Let $\psi' := \varphi \circ \psi$. Then

$$\psi'(\hat{f}) = \varphi(v^{-1}\hat{f}') = \varphi(v)^{-1}v'^{-1}f_0,$$

so $\psi'((\hat{f})) = (f_0)$.

In both cases, ψ' is an H-equivariant automorphism of $\mathfrak{R}$. We show that

$$\psi'(\hat{h}_k) - h_{k,0} \in \mathfrak{m}^N \quad \text{for any } k. \tag{7.5}$$

Since $\psi(\hat{h}_k) - h_{k,0} = \psi(\hat{h}_k) - \psi(\hat{h}_k)_{<N} \in \mathfrak{m}^N$, it suffices to show that $\varphi(g) - g \in \mathfrak{m}^N$ for any $g \in \mathfrak{R}$. The set of $g \in \mathfrak{R}$ with $\varphi(g) - g \in \mathfrak{m}^N$ is closed under sums and under products, since

$$\varphi(gg') - gg' = \varphi(g)\big(\varphi(g') - g'\big) + \big(\varphi(g) - g\big)g',$$

and it contains $\mathbb{C}$ and $x_1, \dots, x_n$. Thus it contains $\mathbb{C}[x_1, \dots, x_n]$. For any $g \in \mathfrak{R}$, $g - g_{<N} \in \mathfrak{m}^N$ and $\varphi(\mathfrak{m}^N) = \mathfrak{m}^N$, hence $\varphi(g) - g \in \mathfrak{m}^N$.

Step 3. In this step, we show that X_0 is an isolated Gorenstein terminal threefold germ with a faithful action of H, and we construct the isomorphism ι of completions.

If $n = 3$, then $X_0 = \mathbb{A}^3$ is smooth, hence Gorenstein terminal, $\widehat{\mathcal{O}}_{X_0,0} = \mathfrak{R}$, and we let

$$\iota := \psi' : \widehat{\mathcal{O}}_{X,x} = \mathfrak{R} \to \mathfrak{R} = \widehat{\mathcal{O}}_{X_0,0}.$$

Assume that $n = 4$. Since $f_0 = v'\varphi(\hat{f}')$, the argument of **Step 2** for (7.4), with φ and v' in place of ψ and v, implies

$$(f_0, \partial_1 f_0, \dots, \partial_4 f_0) = \varphi\big((\hat{f}', \partial_1 \hat{f}', \dots, \partial_4 \hat{f}')\big) \supset \mathfrak{m}^p.$$

Since $\mathfrak{R}$ is faithfully flat over $\mathcal{O}_{\mathbb{A}^4,0}$ [Mat89, Theorem 8.14], the ideal $(f_0, \partial_1 f_0, \dots, \partial_4 f_0)$ of $\mathcal{O}_{\mathbb{A}^4,0}$ contains the p-th power of the maximal ideal. By [Mat89, Theorem 14.2] and [Har77, Proposition II.8.7], after shrinking, X_0 is smooth outside 0, and X_0 is smooth at 0 if $p = 0$, that is, if X is smooth at x. Since $f_0 \in \mathfrak{m}\backslash\{0\}$, X_0 is a hypersurface germ with $\dim X_0 = 3$, so X_0 is Gorenstein, and X_0 is regular in codimension 2. Since X_0 is a hypersurface in $\mathbb{A}^4$ and is regular in codimension 1, it is normal [Har77, Proposition II.8.23]. By [Mat89, Theorem 8.7], $\widehat{\mathcal{O}}_{X_0,0} = \mathfrak{R}/(f_0)$. Since $\psi'((\hat{f})) = (f_0)$, the automorphism ψ' induces an H-equivariant isomorphism

$$\iota : \widehat{\mathcal{O}}_{X,x} = \mathfrak{R}/(\hat{f}) \to \mathfrak{R}/(f_0) = \widehat{\mathcal{O}}_{X_0,0}.$$

In both cases, let $\widehat{X}_0 := \operatorname{Spec} \widehat{\mathcal{O}}_{X_0,0}$ with closed point $\hat{0}$. Since ι is an H-equivariant isomorphism of $\mathbb{C}$-algebras, ι induces a bijection $P \to \iota_* P$ from the formal prime divisors over $\widehat{X}$ with center $\hat{x}$ to the formal prime divisors over $\widehat{X}_0$ with center $\hat{0}$, such that $\operatorname{ord}_{\iota_* P}(\iota(g)) = \operatorname{ord}_P(g)$ for any $g \in \widehat{\mathcal{O}}_{X,x}$, the residue fields of P and $\iota_* P$ are isomorphic, $\operatorname{Stab}_H(\iota_* P) = \operatorname{Stab}_H(P)$, $I_H(\iota_* P) = I_H(P)$, and $a(\iota_* P, \widehat{X}_0, 0) = a(P, \widehat{X}, 0)$, because ι carries the canonical divisor of [dFEM11, Appendix A] on $\widehat{X}$ to the one on $\widehat{X}_0$ (the relative canonical divisors do not depend on the presentation [dFEM11, Remark 2.12]).

Assume that $n = 4$ and that X_0 is singular at 0. By Lemma 3.3(3), applied to X_0 with boundary 0 and the trivial group, by the bijection $P \to \iota_* P$ of the previous paragraph, which preserves log discrepancies, and by (7.2), any prime divisor E over X_0 with center 0 satisfies $a(E, X_0, 0) \geq 2$. Any exceptional prime divisor E over X_0 whose center is not $\{0\}$ has center meeting the smooth locus $X_0\backslash\{0\}$. Such E is exceptional over the smooth variety $X_0\backslash\{0\}$, and its log discrepancy over $X_0\backslash\{0\}$ equals $a(E, X_0, 0)$, so $a(E, X_0, 0) \geq 2$ by [KM98, Corollary 2.31(2)] applied to $X_0\backslash\{0\}$. Thus X_0 is terminal near 0, and X_0 is an isolated Gorenstein terminal threefold germ. If X_0 is smooth at 0, then X_0 is also Gorenstein terminal.

Let $\sigma \in H$ act trivially on X_0. Then σ acts trivially on $\widehat{\mathcal{O}}_{X_0,0}$, so, since ι is H-equivariant, σ acts trivially on $\widehat{\mathcal{O}}_{X,x} \supset \mathcal{O}_{X,x}$ and hence on $\mathbb{C}(X)$. Since H acts faithfully on X, $\sigma = 1$. Thus H acts faithfully on X_0, and hence on $\mathbb{A}^n \supset X_0$. This proves the faithfulness of the action of H on X_0 in (1).

Step 4. In this step, we show that the truncated equations keep the klt margin on $\widehat{X}$, and we prove (3) and the klt property of (X_0, B_0) in (1).

For $1 \leq k \leq s$, let $h'_k := \iota^{-1}(h_{k,0}) \in \widehat{\mathcal{O}}_{X,x}$, where $h_{k,0}$ also denotes the image of $h_{k,0}$ in $\widehat{\mathcal{O}}_{X_0,0}$. Then h'_k is the image of $\psi'^{-1}(h_{k,0})$ in $\widehat{\mathcal{O}}_{X,x}$, so, by (7.5) and $\psi'^{-1}(\mathfrak{m}^N) = \mathfrak{m}^N$,

$$h'_k - \hat{h}_k \in \mathfrak{m}^N \widehat{\mathcal{O}}_{X,x}. \tag{7.6}$$

Let P be a formal prime divisor over $\widehat{X}$ with center $\hat{x}$ and fix k. If $\operatorname{ord}_P(\hat{h}_k) < N \operatorname{mult}_P \mathfrak{m}$, then $\operatorname{ord}_P(h'_k - \hat{h}_k) \geq N \operatorname{mult}_P \mathfrak{m} > \operatorname{ord}_P(\hat{h}_k)$ by (7.6), so $\operatorname{ord}_P(h'_k) = \operatorname{ord}_P(\hat{h}_k)$. Similarly, if $\operatorname{ord}_P(h'_k) < N \operatorname{mult}_P \mathfrak{m}$, then $\operatorname{ord}_P(\hat{h}_k - h'_k) \geq N \operatorname{mult}_P \mathfrak{m} > \operatorname{ord}_P(h'_k)$ by (7.6), so $\operatorname{ord}_P(\hat{h}_k) = \operatorname{ord}_P(h'_k)$. Thus

$$\operatorname{ord}_P(h'_k) \neq \operatorname{ord}_P(\hat{h}_k) \implies \operatorname{ord}_P(h'_k) \geq N \operatorname{mult}_P \mathfrak{m} \geq N \text{ and } \operatorname{ord}_P(\hat{h}_k) \geq N \operatorname{mult}_P \mathfrak{m} \geq N. \tag{7.7}$$

In particular, by (c), $\operatorname{ord}_P(h'_k) = \operatorname{ord}_P(\hat{h}_k)$ for any $P \in \{\widehat{P}_*, \widehat{P}_1, \dots, \widehat{P}_l\}$ and any k. Since $\operatorname{ord}_{\widehat{P}_*}(\hat{h}_k) < +\infty$, we have $h'_k \neq 0$, so $h_{k,0} = \iota(h'_k)$ is non-zero on X_0. For $k \notin \mathcal{K}$, h'_k is a unit by (7.6), and for $k \in \mathcal{K}$, h'_k is a non-zero element of $\mathfrak{m}\widehat{\mathcal{O}}_{X,x}$.

Fix j with $\mathcal{K}^{(j)} \neq \emptyset$, and let

$$\mathfrak{a}_0^{(j)} := \prod_{k \in \mathcal{K}^{(j)}} \left(h'_k \widehat{\mathcal{O}}_{X,x}\right)^{\beta_k^{(j)}}.$$

By (7.6), $(h'_k) + \mathfrak{m}^N = (\hat{h}_k) + \mathfrak{m}^N$ in $\widehat{\mathcal{O}}_{X,x}$ for any k. Since X is klt near x, $\dim X = 3$, and $N \geq 3/(\epsilon\beta_k^{(j)})$ for any $k \in \mathcal{K}^{(j)}$ by (a), [dFEM11, Proposition 2.19(ii)] implies

$$|\mathrm{lct}(\mathfrak{a}_0^{(j)}) - \mathrm{lct}(\mathfrak{a}^{(j)})| \leq \epsilon.$$

By (7.3), $\mathrm{lct}(\mathfrak{a}_0^{(j)}) > 1 + \epsilon$.

We show that, for any $0 \leq j \leq r$ and any formal prime divisor P over $\widehat{X}$ with center $\hat{x}$,

$$a(P, \widehat{X}, 0) > (1+\epsilon) \sum_k \beta_k^{(j)} \mathrm{ord}_P(\hat{h}_k) \quad \text{and} \quad a(P, \widehat{X}, 0) > (1+\epsilon) \sum_k \beta_k^{(j)} \mathrm{ord}_P(h'_k). \tag{7.8}$$

Let $(g_1, \dots, g_s)$ be $(\hat{h}_1, \dots, \hat{h}_s)$ or $(h'_1, \dots, h'_s)$, and correspondingly let $\mathfrak{b}$ be $\mathfrak{a}^{(j)}$ or $\mathfrak{a}_0^{(j)}$ if $\mathcal{K}^{(j)} \neq \emptyset$. If $\sum_k \beta_k^{(j)} \mathrm{ord}_P(g_k) = 0$, then $a(P, \widehat{X}, 0) > (1+\epsilon) \sum_k \beta_k^{(j)} \mathrm{ord}_P(g_k)$ by (7.2). Otherwise $\mathcal{K}^{(j)} \neq \emptyset$, and let $c := \mathrm{lct}(\mathfrak{b})$, so $c > 1+\epsilon$ by (7.3) and the previous paragraph. Since $(\widehat{X}, \mathfrak{b}^c)$ is lc, where log canonicity can be checked on one log resolution and any formal prime divisor over $\widehat{X}$ is a divisor on some log resolution [dFEM11, Section 2 and Theorems 2.3 and 2.4], and since $\mathrm{ord}_P(g_k) = 0$ for $k \notin \mathcal{K}$,

$$a(P, \widehat{X}, 0) - c \sum_k \beta_k^{(j)} \mathrm{ord}_P(g_k) \geq 0,$$

so

$$a(P, \widehat{X}, 0) \geq c \sum_k \beta_k^{(j)} \mathrm{ord}_P(g_k) > (1+\epsilon) \sum_k \beta_k^{(j)} \mathrm{ord}_P(g_k).$$

This proves (7.8).

Now we show that $(\widehat{X}, \sum_k \beta_k^{(j)} \mathrm{div}(h'_k))$ is klt for any j. If $\mathcal{K}^{(j)} = \emptyset$, then $\sum_k \beta_k^{(j)} \mathrm{div}(h'_k) = 0$, and $\widehat{X}$ is klt [dFEM11, Remark 2.10 and Proposition 2.11(i)]. Otherwise let $c := \mathrm{lct}(\mathfrak{a}_0^{(j)}) > 1$. For any formal prime divisor E over $\widehat{X}$, since $(\widehat{X}, (\mathfrak{a}_0^{(j)})^c)$ is lc and E is a divisor on some log resolution [dFEM11, Section 2 and Theorems 2.3 and 2.4],

$$a(E, \widehat{X}, 0) - \sum_k \beta_k^{(j)} \mathrm{ord}_E(h'_k) \geq (c-1) \sum_k \beta_k^{(j)} \mathrm{ord}_E(h'_k),$$

which is positive if $\sum_k \beta_k^{(j)} \mathrm{ord}_E(h'_k) > 0$, and equals $a(E, \widehat{X}, 0) > 0$ otherwise since $\widehat{X}$ is klt. By (7.1), the pair is klt. Through ι, the pair $(\widehat{X}_0, \sum_k \beta_k^{(j)} \mathrm{div}(h_{k,0}))$ is klt. Since K_{X_0} is Cartier and $\sum_k \beta_k^{(j)} \mathrm{div}(h_{k,0}|_{X_0})$ is a combination of principal divisors, [dFEM11, Remark 2.10 and Proposition 2.11(i)] implies that $(X_0, \sum_k \beta_k^{(j)} \mathrm{div}(h_{k,0}|_{X_0}))$ is klt near 0. For $1 \leq j \leq r$ this is (3), and for $j = 0$ this is the klt property of (X_0, B_0) in (1).

Step 5. In this step, we prove the equality of minimal log discrepancies in (1).

Let $\widehat{B}' := \sum_k \beta_k \mathrm{div}(h'_k)$ on $\widehat{X}$ and $\widehat{B}_0 := \sum_k \beta_k \mathrm{div}(h_{k,0})$ on $\widehat{X}_0$. Then $\widehat{B}_0$ is the pullback of B_0, and ι carries $\widehat{B}'$ to $\widehat{B}_0$. Fix a subgroup $H' \subset H$. The $\mathbb{R}$-divisor B_0 is H-invariant since the $h_{k,0}$ are semi-invariant, $K_{X_0} + B_0$ is $\mathbb{R}$-Cartier, and H' acts faithfully on X_0 by **Step 3**. By Lemma 3.3(2), applied to (X, B) and to (X_0, B_0), and by **Step 3**,

$$\mathrm{mld}((X \ni x, B)/H') = \mathrm{mld}((\widehat{X} \ni \hat{x}, \widehat{B})/H')$$

and

$$\mathrm{mld}((X_0 \ni 0, B_0)/H') = \mathrm{mld}((\widehat{X}_0 \ni \hat{0}, \widehat{B}_0)/H') = \mathrm{mld}((\widehat{X} \ni \hat{x}, \widehat{B}')/H').$$

Thus it suffices to show that

$$\mathrm{mld}((\widehat{X} \ni \hat{x}, \widehat{B})/H') = \mathrm{mld}((\widehat{X} \ni \hat{x}, \widehat{B}')/H').$$

By Definition 3.2, the left-hand side and the right-hand side are the infima of

$$\frac{a(P, \widehat{X}, \widehat{B})}{|I_{H'}(P)|} \quad \text{and} \quad \frac{a(P, \widehat{X}, \widehat{B}')}{|I_{H'}(P)|}$$

over the formal prime divisors P over $\widehat{X}$ with center $\hat{x}$. Let $\mathcal{P}_=$ be the set of such P with $\mathrm{ord}_P(h'_k) = \mathrm{ord}_P(\hat{h}_k)$ for any k. Let $\mathcal{P}_{\neq}$ be the set of the other such P.

For $P \in \mathcal{P}_=$, (7.1) implies $a(P, \widehat{X}, \widehat{B}) = a(P, \widehat{X}, \widehat{B}')$. By **Step 4**, $\widehat{P}_* \in \mathcal{P}_=$, and since $|I_{H'}(\widehat{P}_*)| \geq 1$,

$$\frac{a(\widehat{P}_*, \widehat{X}, \widehat{B}')}{|I_{H'}(\widehat{P}_*)|} = \frac{a(\widehat{P}_*, \widehat{X}, \widehat{B})}{|I_{H'}(\widehat{P}_*)|} \leq a(P_*, X, B) < C.$$

Hence both infima are smaller than C.

Let $P \in \mathcal{P}_{\neq}$, and fix k with $\mathrm{ord}_P(h'_k) \neq \mathrm{ord}_P(\hat{h}_k)$. Then $k \in \mathcal{K}$, since $\mathrm{ord}_P(h'_{k'}) = \mathrm{ord}_P(\hat{h}_{k'}) = 0$ for $k' \notin \mathcal{K}$, and $\mathrm{ord}_P(\hat{h}_k) \geq N$ and $\mathrm{ord}_P(h'_k) \geq N$ by (7.7). Let $(g_1, \ldots, g_s)$ be $(\hat{h}_1, \ldots, \hat{h}_s)$ or $(h'_1, \ldots, h'_s)$. By (7.1), (7.8) for $j = 0$, and (b),

$$a\Big(P, \widehat{X}, \sum_{k'} \beta_{k'} \operatorname{div}(g_{k'})\Big) = a(P, \widehat{X}, 0) - \sum_{k'} \beta_{k'} \mathrm{ord}_P(g_{k'}) > \epsilon \sum_{k'} \beta_{k'} \mathrm{ord}_P(g_{k'}) \geq \epsilon \beta_k N > C|H|.$$

Thus $a(P, \widehat{X}, \widehat{B}) > C|H|$ and $a(P, \widehat{X}, \widehat{B}') > C|H|$. Since $|I_{H'}(P)| \leq |H|$, we obtain $a(P, \widehat{X}, \widehat{B})/|I_{H'}(P)| > C$ and $a(P, \widehat{X}, \widehat{B}')/|I_{H'}(P)| > C$.

Therefore, since $\widehat{P}_* \in \mathcal{P}_=$, both infima can be taken over $\mathcal{P}_=$, and

$$\mathrm{mld}((\widehat{X} \ni \hat{x}, \widehat{B})/H') = \inf_{P \in \mathcal{P}_=} \frac{a(P, \widehat{X}, \widehat{B})}{|I_{H'}(P)|} = \inf_{P \in \mathcal{P}_=} \frac{a(P, \widehat{X}, \widehat{B}')}{|I_{H'}(P)|} = \mathrm{mld}((\widehat{X} \ni \hat{x}, \widehat{B}')/H').$$

Thus (1) holds.

Step 6. In this step, we prove (2).

The isomorphism in (2) is ι. Fix $1 \leq i \leq l$. The formal prime divisor $\iota_* \widehat{P}_i$ over $\widehat{X}_0$ has center $\hat{0}$. By Lemma 3.3(3), applied to $(X_0, 0)$ with the action of H, the restriction of $\mathrm{ord}_{\iota_* \widehat{P}_i}$ to $\mathbb{C}(X_0)$ is $\mathrm{ord}_{P_{i,0}}$ for a prime divisor $P_{i,0}$ over X_0 with center 0, and $P_{i,0}$ and $\iota_* \widehat{P}_i$ have isomorphic residue fields, equal decomposition groups, equal inertia groups, and $a(P_{i,0}, X_0, 0) = a(\iota_* \widehat{P}_i, \widehat{X}_0, 0)$. Combining **Step 1** and **Step 3**, P_i and $P_{i,0}$ have isomorphic residue fields,

$$\mathrm{Stab}_H(P_i) = \mathrm{Stab}_H(P_{i,0}), \quad I_H(P_i) = I_H(P_{i,0}), \quad a(P_i, X, 0) = a(P_{i,0}, X_0, 0).$$

By **Step 4**, for any k,

$$\mathrm{ord}_{P_{i,0}}(h_{k,0}) = \mathrm{ord}_{\iota_* \widehat{P}_i}(h_{k,0}) = \mathrm{ord}_{\widehat{P}_i}(h'_k) = \mathrm{ord}_{\widehat{P}_i}(\hat{h}_k) = \mathrm{ord}_{P_i}(h_k).$$

This proves (2). □

Hypotheses (iv) and (v) are not used for the main theorems of this paper. They are kept for later use. By (iv) and (3), the polynomial model also keeps klt finitely many other boundaries $\sum_k \beta_k^{(j)} \operatorname{div}(h_k|_X)$ built from $h_1, \ldots, h_s$, for example the rational proxy boundaries of Construction-Proposition 17.1. By (v) and (2), finitely many given prime divisors, for example a divisor computing the minimal log discrepancy of the quotient by a subgroup of H, are carried to the model together with their residue fields, decomposition groups, inertia groups, and the orders of the h_k, so they still compute the corresponding minimal log discrepancies of quotients on the model.

Definition-Lemma 7.3. Let $g \in \mathbb{C}[[x_1, \ldots, x_d]]$ be a non-zero power series. An exponent $\alpha \in \mathbb{Z}^d_{\geq 0}$ of a monomial occurring in g is *componentwise minimal* if no other exponent of a monomial occurring in g is componentwise smaller than or equal to α. For an integer M, let $g_{<M}$ be the sum of the terms of g of degree less than M. Then g has finitely many componentwise minimal exponents, and any exponent of a monomial occurring in g is componentwise larger than or equal to a componentwise minimal exponent of g. Moreover, for any $u \in \mathbb{R}^d_{>0}$, the exponent of any term of $\mathrm{in}_u(g)$ is componentwise minimal, and if M is larger than the degrees of all componentwise minimal exponents of g, then $\mathrm{in}_u(g_{<M}) = \mathrm{in}_u(g)$ and $\mathrm{wt}_u(g_{<M}) = \mathrm{wt}_u(g)$.

Proof. By Dickson's lemma [CLO15, Chapter 2, Section 4, Theorem 5], g has finitely many componentwise minimal exponents, and any exponent of a monomial occurring in g is componentwise larger than or equal to a componentwise minimal exponent of g.

Fix $u \in \mathbb{R}^d_{>0}$, and let x^α be a monomial occurring in g with $\langle u, \alpha\rangle = \mathrm{wt}_u(g)$. If α were not componentwise minimal, there would exist an exponent $\alpha' \neq \alpha$ of a monomial occurring in g with $\alpha'_i \leq \alpha_i$ for any i, and $\langle u, \alpha'\rangle < \langle u, \alpha\rangle$ since $u \in \mathbb{R}^d_{>0}$, a contradiction. Thus the exponent of any term of $\mathrm{in}_u(g)$ is componentwise minimal.

Let M be an integer larger than the degrees of all componentwise minimal exponents of g. The monomials occurring in $g_{<M}$ occur in g, and all monomials with componentwise minimal exponents of g occur in $g_{<M}$. Hence the monomial ideals $J(g_{<M})$ and $J(g)$ of Definition 2.16 are both generated by the monomials with componentwise minimal exponents of g, so $J(g_{<M}) = J(g)$. By the previous paragraph, any term of g of u-weight $\mathrm{wt}_u(g)$ has a componentwise minimal exponent, hence has degree less than M and is a term of $g_{<M}$. Conversely, any term of $g_{<M}$ is a term of g. Thus $\mathrm{in}_u(g_{<M}) = \mathrm{in}_u(g)$ and $\mathrm{wt}_u(g_{<M}) = \mathrm{wt}_u(g)$. □

Corollary 7.4. *In Construction-Theorem 7.2, the $h_{k,0}$, and f_0 if $n = 4$, can be taken so that, in addition, $\mathrm{in}_u(h_{k,0}) = \mathrm{in}_u(\hat{h}_k)$ for any k and any $u \in \mathbb{R}^n_{>0}$, and, if $n = 4$, $\mathrm{in}_u(f_0) = \mathrm{in}_u(\hat{f})$ for any $u \in \mathbb{R}^n_{>0}$.*

Proof. Let $\psi = \mathrm{id}_{\mathfrak{R}}$, $v = 1$, and let N_1 be as in Construction-Theorem 7.2(4). Fix $N \geq N_1$ larger than the degrees of all componentwise minimal exponents of $\hat{h}_1, \dots, \hat{h}_s$ (Definition-Lemma 7.3), let L_1 be as in Construction-Theorem 7.2(4) for N, and, if $n = 4$, fix $L \geq L_1$ larger than the degrees of all componentwise minimal exponents of $\hat{f}$. By Construction-Theorem 7.2(4), we can take $h_{k,0} = (\hat{h}_k)_{<N}$ and, if $n = 4$, $f_0 = \hat{f}_{<L}$, and the equalities of leading forms hold by Definition-Lemma 7.3. □

Corollary 7.5. *Let $(X \ni o)$ be an isolated Gorenstein terminal threefold singularity, G a finite abelian group acting faithfully on X and fixing o, and $(Y \ni o)$ a smooth fourfold germ with an action of G such that $X \subset Y$ is a G-equivariant closed embedding. Let x, y, z, t be G-semi-invariant elements of the maximal ideal of $\widehat{\mathcal{O}}_{Y,o}$ with $\widehat{\mathcal{O}}_{Y,o} = \mathbb{C}[[x, y, z, t]]$, and $F \in \mathbb{C}[[x, y, z, t]]$ a G-semi-invariant generator of the kernel of $\widehat{\mathcal{O}}_{Y,o} \to \widehat{\mathcal{O}}_{X,o}$. Let ψ be a G-equivariant automorphism of $\mathbb{C}[[x, y, z, t]]$, v a G-invariant unit of $\mathbb{C}[[x, y, z, t]]$, and, for a power series P and an integer L, let $P_{<L}$ be the sum of the terms of P of degree less than L in x, y, z, t. Then there exists a positive integer L_1 such that for any integer $L \geq L_1$ the following hold for the polynomial $f_0 := (v\psi(F))_{<L}$ in the coordinates x, y, z, t of $\mathbb{A}^4$, on which G acts diagonally with the characters of x, y, z, t.*

(1) The action of G on $\mathbb{A}^4$ is faithful, and f_0 is G-semi-invariant with the character of F.

(2) $X_0 := (f_0 = 0) \subset \mathbb{A}^4$ is an isolated Gorenstein terminal threefold germ at 0, G acts faithfully on X_0, and $\mathrm{mld}((X_0 \ni 0, 0)/H) = \mathrm{mld}((X \ni o, 0)/H)$ for any subgroup $H \subset G$.

Proof. Let $\mathfrak{m}_{Y,o}$ be the maximal ideal of $\mathcal{O}_{Y,o}$. We first find G-semi-invariant regular parameters of Y at o. Since G is finite abelian and fixes o, the vector space $\mathfrak{m}_{Y,o}/\mathfrak{m}^2_{Y,o}$ has a basis $\bar{q}_1, \dots, \bar{q}_4$ of eigenvectors of G. For each i, let χ_i be the character of $\bar{q}_i$, let $q_i \in \mathfrak{m}_{Y,o}$ be a lift of $\bar{q}_i$, and let $x_i := \frac{1}{|G|}\sum_{\sigma\in G} \chi_i(\sigma)^{-1}\sigma^* q_i$. Then x_i is G-semi-invariant with the character χ_i, and the image of x_i in $\mathfrak{m}_{Y,o}/\mathfrak{m}^2_{Y,o}$ is $\bar{q}_i$ since $\sigma^*\bar{q}_i = \chi_i(\sigma)\bar{q}_i$. After shrinking Y, $x_1, \dots, x_4$ are G-semi-invariant regular parameters of Y at o.

Next we find a G-semi-invariant equation of X. Since $\mathcal{O}_{Y,o}$ is a regular local ring, it is a UFD [Mat89, Theorem 20.3]. Thus the ideal I of X in $\mathcal{O}_{Y,o}$, a prime ideal of height one, is principal [Mat89, Theorem 20.1], and $I/\mathfrak{m}_{Y,o}I$ is a one-dimensional vector space on which G acts, since I is G-stable. Let χ_I be the character of G on $I/\mathfrak{m}_{Y,o}I$, let $g \in I$ be a lift of a non-zero element of $I/\mathfrak{m}_{Y,o}I$, and let $f := \frac{1}{|G|}\sum_{\sigma\in G} \chi_I(\sigma)^{-1}\sigma^* g$. Then $f \in I$ is G-semi-invariant and the images of f and g in $I/\mathfrak{m}_{Y,o}I$ are equal, so f generates I [Mat89, Theorem 2.2]. After shrinking Y, f is a regular function on Y and $X = (f = 0)$.

The expansion $\hat{f}$ of f in $\widehat{\mathcal{O}}_{Y,o} = \mathbb{C}[[x,y,z,t]]$ generates the kernel of $\widehat{\mathcal{O}}_{Y,o} \to \widehat{\mathcal{O}}_{X,o}$ [Mat89, Theorem 8.7], and so does F. Since $\mathbb{C}[[x,y,z,t]]$ is a domain, $\hat{f} = wF$ for a unit w. The unit w is G-semi-invariant since $\hat{f}$ and F are. Write $\sigma^* w = \chi(\sigma)w$ for $\sigma \in G$. Since G fixes o, comparing constant terms implies $w(o) = \chi(\sigma)w(o)$ with $w(o) \neq 0$, so $\chi(\sigma) = 1$ for any $\sigma \in G$, and w is G-invariant.

The images of x,y,z,t and of $x_1,\dots,x_4$ in $\mathfrak{m}_{Y,o}/\mathfrak{m}_{Y,o}^2$ are two bases of eigenvectors of G, so the characters of x,y,z,t, counted with multiplicity, coincide with the characters of $x_1,\dots,x_4$. After renumbering, x_1,x_2,x_3,x_4 have the characters of x,y,z,t, respectively. Since the expansions of $x_1,\dots,x_4$ form a regular system of parameters of $\mathbb{C}[[x,y,z,t]]$, the homomorphism ρ of $\mathbb{C}[[x,y,z,t]]$ sending $P(x,y,z,t)$ to $P(x_1,x_2,x_3,x_4)$ is an automorphism, and $\mathbb{C}[[x,y,z,t]] = \mathbb{C}[[x_1,\dots,x_4]]$ [Mat89, Theorem 29.7]. Since x_i has the character of the i-th variable, ρ is G-equivariant. Moreover, $\rho(P_{<L}) = \rho(P)_{<L}$ for any P and L, where on the right-hand side the degree is taken in $x_1,\dots,x_4$.

Now we apply Construction-Theorem 7.2 with $H = G$, $n = 4$, the embedding $X = (f = 0) \subset Y$, the parameters $x_1,\dots,x_4$, and $s = r = l = 0$. Its hypotheses hold: $(X,0)$ is klt near o since X is terminal. Then $\mathfrak{R} = \mathbb{C}[[x_1,\dots,x_4]]$. Let $\psi' := \rho \circ \psi$ and $v' := \rho(v)\rho(\psi(w))^{-1}$. Then ψ' is a G-equivariant automorphism of $\mathfrak{R}$, v' is a G-invariant unit of $\mathfrak{R}$, and

$$v'\psi'(\hat{f}) = \rho(v)\rho(\psi(w))^{-1}\rho(\psi(w))\rho(\psi(F)) = \rho(v\psi(F)).$$

By Construction-Theorem 7.2(4) for ψ' and v', applied with $N = N_1$, there exists a positive integer L_1 such that for any $L \geq L_1$ one can take

$$(v'\psi'(\hat{f}))_{<L} = \rho(v\psi(F))_{<L} = \rho\big((v\psi(F))_{<L}\big)$$

as the polynomial f_0 of Construction-Theorem 7.2. Renaming the coordinates $x_1,\dots,x_4$ of $\mathbb{A}^4$ to x,y,z,t turns $\rho((v\psi(F))_{<L})$ into $(v\psi(F))_{<L}$, and the diagonal action of G on $\mathbb{A}^4$ there into the action with the characters of x,y,z,t. Thus Construction-Theorem 7.2 and its (1), with $B = 0$ and $B_0 = 0$, imply that G acts faithfully on $\mathbb{A}^4$ and on X_0, that X_0 is an isolated Gorenstein terminal threefold germ at 0, and that $\mathrm{mld}((X_0 \ni 0,0)/H) = \mathrm{mld}((X \ni o,0)/H)$ for any subgroup $H \subset G$. Finally, $v\psi(F)$ is G-semi-invariant with the character of F, since ψ is G-equivariant and v is G-invariant. Since G acts diagonally, any monomial is G-semi-invariant, so any monomial occurring in $v\psi(F)$ has the character of F, and f_0 is G-semi-invariant with the character of F. □

8. Adjunction for quotients and toric models

In this section, we prove an inversion of adjunction for quotients of semi-invariant complete intersections (Theorem 8.3), and we show that the minimal log discrepancy of a quotient of a smooth or isolated Gorenstein terminal threefold germ is a lattice Newton minimum when all weighted leading boundaries are log canonical on the torus (Theorem 8.14). Theorem 8.3 is used repeatedly in later sections to pass from a hypersurface to the ambient space, and from a threefold to a surface section.

8.1. Products with a coefficient-one coordinate.

In this subsection, we show that adding a coordinate with coefficient one in the boundary, on which the group acts by a character, does not change the minimal log discrepancy of the quotient (Lemma 8.1). Then we prove Theorem 8.3.

Lemma 8.1. *Let X be a normal quasi-projective variety of positive dimension, $x \in X$ a closed point, B an $\mathbb{R}$-divisor on X with $K_X + B$ $\mathbb{R}$-Cartier, H a finite group acting faithfully on X, fixing x and preserving B, and χ a character of H. Let H act on $X \times \mathbb{A}^1_t$ by $g\cdot(p,t) = (g\cdot p, \chi(g)t)$, so that $g^*t = \chi(g)t$, where g^* denotes the pullback of functions, and let $B^+ := \mathrm{pr}_1^* B + (t = 0)$. Then the following hold.*

(1) For any prime divisor F over $X \times \mathbb{A}^1$ with center $(x,0)$, there exist a positive integer ϱ and a prime divisor E over X with center x such that $\mathrm{ord}_F|_{\mathbb{C}(X)} = \varrho\,\mathrm{ord}_E$, $I_H(F) \subset I_H(E)$, and $a(F, X \times \mathbb{A}^1, B^+) \geq \varrho a(E,X,B)$.

(2) *For any prime divisor E over X with center x, let λ be the character by which $I_H(E)$ acts on the line $\mathfrak{m}_E/\mathfrak{m}_E^2$ (Lemma 2.6(1)) through the pullback of functions. Let $q \in \{1, \dots, |I_H(E)|\}$ be the integer with $\chi|_{I_H(E)} = \lambda^q$. Then the Gauss extension (Definition-Proposition 2.11) of ord_E to $\mathbb{C}(X)(t)$ with t-value q is ord_F for a prime divisor F over $X \times \mathbb{A}^1$ with center $(x,0)$, $I_H(F) = I_H(E)$, and $a(F, X \times \mathbb{A}^1, B^+) = a(E, X, B)$.*

In particular, if $\operatorname{mld}((X \ni x, B)/H) \geq 0$, *then*

$$\operatorname{mld}(((X \times \mathbb{A}^1) \ni (x,0), B^+)/H) = \operatorname{mld}((X \ni x, B)/H).$$

Proof. We regard $\mathbb{C}(X)$ as a subfield of $\mathbb{C}(X \times \mathbb{A}^1) = \mathbb{C}(X)(t)$ through pr_1^*. For a prime divisor F over $X \times \mathbb{A}^1$, we denote by $\mathcal{O}_F$ the valuation ring of ord_F and by $\kappa(F)$ the residue field of ord_F. Let $d := \dim X$.

For a prime divisor E over X, we use the following data. Let $f : Y \to X$ be a proper birational morphism from a normal variety Y such that E is a prime divisor on Y, and let B_Y be the $\mathbb{R}$-divisor with $K_Y + B_Y = f^*(K_X + B)$ and $f_* B_Y = B$. Then $\operatorname{mult}_E B_Y = 1 - a(E, X, B)$, and

$$K_{Y \times \mathbb{A}^1} + \operatorname{pr}_1^* B_Y + (t = 0) = (f \times \mathrm{id}_{\mathbb{A}^1})^*(K_{X \times \mathbb{A}^1} + B^+).$$

Thus $a(F, X \times \mathbb{A}^1, B^+) = a(F, Y \times \mathbb{A}^1, \operatorname{pr}_1^* B_Y + (t = 0))$ for any prime divisor F over $X \times \mathbb{A}^1$ [KM98, Lemma 2.30]. Let $\tau \in \mathbb{C}(X)$ with $\operatorname{ord}_E \tau = 1$, and let $U_0 \subset Y$ be an open subset with $U_0 \cap E \neq \emptyset$ such that U_0 and $E \cap U_0$ are smooth, τ is regular on U_0, $\operatorname{div}(\tau)|_{U_0} = E \cap U_0$, and $\operatorname{Supp} B_Y \cap U_0 \subset E$. Then $\operatorname{div}(\tau) + \operatorname{div}(t)$ is an snc divisor on $U_0 \times \mathbb{A}^1$, and

$$(\operatorname{pr}_1^* B_Y + (t = 0))|_{U_0 \times \mathbb{A}^1} = (1 - a(E, X, B)) \operatorname{div}(\tau) + \operatorname{div}(t).$$

(1) Let $w := \operatorname{ord}_F |_{\mathbb{C}(X)}$. Since $\dim X \geq 1$, there exists a non-zero element $u \in \mathfrak{m}_x \subset \mathcal{O}_{X,x}$, and $w(u) > 0$ because $\operatorname{center}_{X \times \mathbb{A}^1} F = \{(x,0)\}$. Thus w is a non-trivial valuation and $w(\mathbb{C}(X)^*) = \varrho\mathbb{Z}$ for some positive integer ϱ. The field $\kappa(F)$ has transcendence degree d over $\mathbb{C}$, and $\mathbb{C}(X)(t)$ has transcendence degree one over $\mathbb{C}(X)$. Hence, by the Abhyankar inequality for the extension $\mathbb{C}(X) \subset \mathbb{C}(X)(t)$ [Bou72, Chapter VI, §10, no. 3, Corollary 1 to Theorem 1], the residue field $\kappa(w)$ of w satisfies

$$\operatorname{trdeg}_{\mathbb{C}} \kappa(w) \geq \operatorname{trdeg}_{\mathbb{C}} \kappa(F) - 1 = d - 1.$$

Moreover, w is non-negative on $\mathcal{O}_{X,x}$ and positive on $\mathfrak{m}_x$, so the Abhyankar inequality [Abh56, Theorem 1] for $\mathcal{O}_{X,x}$ implies $\operatorname{trdeg}_{\mathbb{C}} \kappa(w) \leq d - 1$. Thus $\operatorname{trdeg}_{\mathbb{C}} \kappa(w) = d - 1$, and by [KM98, Lemma 2.45], $w = \varrho \operatorname{ord}_E$ for a prime divisor E over X. Since ord_E is non-negative on $\mathcal{O}_{X,x}$ and positive on $\mathfrak{m}_x$, $\operatorname{center}_X E = \{x\}$.

Let $g \in I_H(F)$. Since g^* preserves $\mathbb{C}(X)$ and ord_F, it preserves ord_E. The valuation ring of ord_E is $\mathcal{O}_F \cap \mathbb{C}(X)$, so $\kappa(E)$ is a subfield of $\kappa(F)$ stable under g^*, and g^* acts on $\kappa(E)$ by the restriction of the action on $\kappa(F)$. Since g acts trivially on $\kappa(F)$, it acts trivially on $\kappa(E)$. Thus $g \in I_H(E)$, and $I_H(F) \subset I_H(E)$.

Let $V := \operatorname{center}_{Y \times \mathbb{A}^1} F$. Since $\operatorname{ord}_F |_{\mathbb{C}(Y)} = \varrho \operatorname{ord}_E$, the image in Y of the generic point of V is the generic point of E. Since $\operatorname{ord}_F(t) > 0$, $V \subset (t = 0)$. Thus $V \subset E \times \{0\}$, and since $\dim V \geq \dim E = \dim(E \times \{0\})$, we have $V = E \times \{0\}$. Let η be the generic point of $E \times \{0\}$. Then τ, t are regular parameters of $\mathcal{O}_{Y \times \mathbb{A}^1, \eta}$, and $U_0 \times \mathbb{A}^1$ is an open neighborhood of η satisfying the conditions of Lemma 2.13. By Lemma 2.13,

$$\begin{aligned} a(F, X \times \mathbb{A}^1, B^+) &= a\big(F, U_0 \times \mathbb{A}^1, (1 - a(E, X, B)) \operatorname{div}(\tau) + \operatorname{div}(t)\big) \\ &\geq a(E, X, B) \operatorname{ord}_F \tau = \varrho a(E, X, B). \end{aligned}$$

(2) Let $e := |I_H(E)|$. By Lemma 2.6(1), $I_H(E)$ is cyclic of order e and λ is faithful. Thus λ generates the character group of $I_H(E)$, and the integer q exists. Let v be the coordinate of $\mathbb{A}^1_v$, let $h : U_0 \times \mathbb{A}^1_v \to U_0 \times \mathbb{A}^1_t$ be the birational morphism given by $t = \tau^q v$, and let F be the prime divisor $(E \cap U_0) \times \mathbb{A}^1_v$ on $U_0 \times \mathbb{A}^1_v$. Since F is the pullback of $E \cap U_0$ under the smooth projection $U_0 \times \mathbb{A}^1_v \to U_0$, we have $\operatorname{ord}_F |_{\mathbb{C}(X)} = \operatorname{ord}_E$ and $\kappa(F) = \kappa(E)(\bar{v})$, where $\bar{v}$ is the residue of $v = t/\tau^q$ and is transcendental over $\kappa(E)$. Since $\kappa(F)$ has transcendence degree d, F is a prime divisor over $X \times \mathbb{A}^1$ (cf. [KM98, Lemma 2.45]).

Let $a_0, \dots, a_m \in \mathbb{C}(X)$, not all zero, and let $\mu := \min_i(\operatorname{ord}_E a_i + iq)$. Then

$$\tau^{-\mu} \sum_i a_i t^i = \sum_i a_i \tau^{iq-\mu} v^i,$$

where $a_i \tau^{iq-\mu} \in \mathcal{O}_F$ for any i, and $a_i \tau^{iq-\mu}$ is a unit of $\mathcal{O}_F$ if and only if $\operatorname{ord}_E a_i + iq = \mu$. Thus the residue of the right-hand side is a non-zero polynomial in $\bar{v}$ with coefficients in $\kappa(E)$, and $\operatorname{ord}_F(\sum_i a_i t^i) = \mu$. Hence ord_F is the Gauss extension of ord_E with t-value q (Definition-Proposition 2.11). Since $q \geq 1$ and $\operatorname{center}_X E = \{x\}$, ord_F is non-negative on $\mathcal{O}_{X,x}[t]$ and positive on $\mathfrak{m}_x$ and on t. Thus $\operatorname{center}_{X\times\mathbb{A}^1} F = \{(x,0)\}$. Moreover, since $t = \tau^q v$ vanishes along F, the center of F on $Y \times \mathbb{A}^1$ is $E \times \{0\}$.

We compute $a(F, X \times \mathbb{A}^1, B^+)$. Let $b := 1 - a(E, X, B)$. Then

$$a(F, X \times \mathbb{A}^1, B^+) = a\big(F, U_0 \times \mathbb{A}^1_t, b \operatorname{div}(\tau) + \operatorname{div}(t)\big).$$

Let $y_2, \dots, y_d$ be regular functions which together with τ form local coordinates of U_0 at a general point of $E \cap U_0$. Since $t = \tau^q v$, we have

$$h^*(d\tau \wedge dy_2 \wedge \cdots \wedge dy_d \wedge dt) = \tau^q d\tau \wedge dy_2 \wedge \cdots \wedge dy_d \wedge dv,$$
$$h^* \operatorname{div}(\tau) = F, \qquad h^* \operatorname{div}(t) = qF + \operatorname{div}(v)$$

near a general point of F, and τ is a local equation of F there. Thus

$$a(F, X \times \mathbb{A}^1, B^+) = 1 + q - b - q = 1 - b = a(E, X, B).$$

It remains to prove $I_H(F) = I_H(E)$. Applying (1) to F, since $\operatorname{ord}_F|_{\mathbb{C}(X)} = \operatorname{ord}_E$, we obtain $I_H(F) \subset I_H(E)$. Conversely, let $g \in I_H(E)$. For $a_i \in \mathbb{C}(X)$,

$$\operatorname{ord}_F\Big(g^* \sum_i a_i t^i\Big) = \operatorname{ord}_F\Big(\sum_i g^*(a_i)\chi(g)^i t^i\Big) = \min_i(\operatorname{ord}_E(g^* a_i) + iq)$$
$$= \min_i(\operatorname{ord}_E a_i + iq) = \operatorname{ord}_F\Big(\sum_i a_i t^i\Big),$$

so g^* preserves ord_F. By the definition of λ, $g^*\tau - \lambda(g)\tau \in \mathfrak{m}_E^2$, so $g^*\tau = \lambda(g)\tau(1+\epsilon)$ for some $\epsilon \in \mathfrak{m}_E$. Thus

$$g^*\Big(\frac{t}{\tau^q}\Big) = \chi(g)\lambda(g)^{-q}(1+\epsilon)^{-q}\frac{t}{\tau^q}.$$

Since $\operatorname{ord}_F \epsilon = \operatorname{ord}_E \epsilon > 0$ and $\chi(g) = \lambda(g)^q$, the residue of $g^*(t/\tau^q)$ in $\kappa(F)$ is $\bar{v}$. Since g acts trivially on $\kappa(E)$ and fixes $\bar{v}$, and $\kappa(F) = \kappa(E)(\bar{v})$, g acts trivially on $\kappa(F)$. Thus $g \in I_H(F)$, and $I_H(F) = I_H(E)$.

Finally, we prove the last statement of the lemma. Since H acts faithfully on $X \times \mathbb{A}^1$, fixes $(x,0)$, and preserves B^+, and since $K_{X\times\mathbb{A}^1} + B^+ = \operatorname{pr}_1^*(K_X + B) + (t = 0)$ is $\mathbb{R}$-Cartier, the crepant quotient $(X \times \mathbb{A}^1, B^+)/H$ is defined. Let $m := \operatorname{mld}((X \ni x, B)/H)$ and $m^+ := \operatorname{mld}(((X \times \mathbb{A}^1) \ni (x,0), B^+)/H)$. By Lemma 2.6,

$$m = \inf_E \frac{a(E, X, B)}{|I_H(E)|}, \qquad m^+ = \inf_F \frac{a(F, X \times \mathbb{A}^1, B^+)}{|I_H(F)|},$$

where E runs over the prime divisors over X with center x, and F runs over the prime divisors over $X \times \mathbb{A}^1$ with center $(x,0)$. By (2), $m^+ \leq m$. Conversely, let F be a prime divisor over $X \times \mathbb{A}^1$ with center $(x,0)$. By (1), there exist a positive integer ϱ and a prime divisor E over X with center x such that $a(F, X \times \mathbb{A}^1, B^+) \geq \varrho a(E, X, B)$ and $I_H(F) \subset I_H(E)$. Since $a(E, X, B) \geq |I_H(E)| m \geq 0$, $\varrho \geq 1$, and $|I_H(F)| \leq |I_H(E)|$, we have

$$\frac{a(F, X \times \mathbb{A}^1, B^+)}{|I_H(F)|} \geq \frac{\varrho a(E, X, B)}{|I_H(F)|} \geq \frac{a(E, X, B)}{|I_H(E)|} \geq m.$$

Thus $m^+ \geq m$, and $m^+ = m$. □

Theorem 2.22 requires that the group acts without pseudoreflections and that, near the image of 0, the image of the non-étale locus intersects the quotient of the complete intersection in codimension at least two. For semi-invariant equations, the second hypothesis may fail [NS26b, Remark 3.3], as the following example shows.

Example 8.2. Let $\langle \mathrm{diag}(-1,-1)\rangle$ act on $\mathbb{A}^2_{x,y}$, and let $X = (x = 0)$. Then the action has no pseudoreflections, and the non-étale locus is the origin. The image of the origin in the curve $X/\langle \mathrm{diag}(-1,-1)\rangle$ has codimension one, so the second hypothesis of Theorem 2.22 does not hold.

After we add coordinates with coefficient one on which the group acts by the characters of the coordinates of $\mathbb{A}^n$, both hypotheses hold, and Lemma 8.1 shows that the minimal log discrepancies of the quotients do not change.

Theorem 8.3. *Let $n \geq 2$ and $1 \leq c < n$ be integers, G a finite abelian group acting diagonally on $\mathbb{A}^n$, $f_1, \dots, f_c$ G-semi-invariant polynomials vanishing at 0 with common zero scheme X, $h_1, \dots, h_s$ G-semi-invariant polynomials with non-zero restrictions to X, and $b_1, \dots, b_s$ positive real numbers. Let $B := \sum_j b_j \operatorname{div}(h_j|_X)$ and $B_{\mathbb{A}^n} := \sum_i \operatorname{div}(f_i) + \sum_j b_j \operatorname{div}(h_j)$. Assume that X is, near 0, a normal integral complete intersection of codimension c with klt singularities, that G acts faithfully on X near 0, and that (X, B) is klt near 0. Then, for any subgroup $H \subset G$,*

$$\operatorname{mld}((X \ni 0, B)/H) = \operatorname{mld}((\mathbb{A}^n \ni 0, B_{\mathbb{A}^n})/H) > 0.$$

In particular, any prime divisor over $\mathbb{A}^n/H$ with center the image of 0 has log discrepancy at least $\operatorname{mld}((X \ni 0, B)/H)$ with respect to $(\mathbb{A}^n, B_{\mathbb{A}^n})/H$.

Proof. **Step 1.** In this step, we choose a G-invariant affine neighborhood of 0 on which the hypotheses on X and (X, B) hold.

There exists a closed subset $Z_0 \subset \mathbb{A}^n$ with $0 \notin Z_0$ such that, on $\mathbb{A}^n \backslash Z_0$, X is a normal integral complete intersection of codimension c with klt singularities and (X, B) is klt. Let $Z := \bigcup_{g \in G} g(Z_0)$. Since X and B are G-invariant, on $\mathbb{A}^n \backslash g(Z_0)$ for any $g \in G$, hence on $\mathbb{A}^n \backslash Z$, X is a normal integral complete intersection of codimension c with klt singularities and (X, B) is klt, and $0 \notin Z$ since G fixes 0. Let ϕ be a regular function on $\mathbb{A}^n$ with $\phi|_Z = 0$ and $\phi(0) = 1$, and let $\psi := \prod_{g \in G} g^*\phi$. Then ψ is G-invariant, $\psi|_Z = 0$, and $\psi(0) = 1$. Let $U := (\psi \neq 0)$ and $X_U := X \cap U$. Then U is a G-invariant affine open neighborhood of 0, X_U is a normal integral complete intersection of codimension c in U with klt singularities, and (X_U, B) is klt. Since G acts faithfully on X near 0 and X_U is integral, G acts faithfully on X_U. Since X_U is a complete intersection in U, K_{X_U} is Cartier, and $K_{X_U} + B$ is $\mathbb{R}$-Cartier. Moreover, G acts faithfully on $\mathbb{A}^n$ and preserves $B_{\mathbb{A}^n}$. Thus the crepant quotients $(X_U, B)/H$ and $(\mathbb{A}^n, B_{\mathbb{A}^n})/H$ are defined. The minimal log discrepancies in the theorem only depend on neighborhoods of the images of 0 in X/H and in $\mathbb{A}^n/H$, and X_U/H is an open neighborhood of the image of 0 in X/H. Thus $\operatorname{mld}((X \ni 0, B)/H) = \operatorname{mld}((X_U \ni 0, B)/H)$. In **Step 3**, Theorem 2.22 is applied to a closed subscheme of $\mathbb{A}^n \times \mathbb{A}^n$, while the neighborhood U is used only for computations and verifications near 0 in **Step 1**–**Step 4**.

Step 2. In this step, we add n auxiliary coordinates and check the hypotheses of Theorem 2.22.

Let $x_1, \dots, x_n$ be the coordinates of $\mathbb{A}^n$ and $\chi_1, \dots, \chi_n$ the characters of G with $g^*x_i = \chi_i(g)x_i$. Let H act on $\mathbb{A}^n \times \mathbb{A}^n_{t_1,\dots,t_n}$ by $g^*x_i = \chi_i(g)x_i$ and $g^*t_i = \chi_i(g)t_i$. Let $X' := X \times \mathbb{A}^n$, the closed subscheme $V(f_1, \dots, f_c) \times \mathbb{A}^n$ of $\mathbb{A}^n \times \mathbb{A}^n$, a complete intersection of codimension c near 0, let $X'_U := X' \cap (U \times \mathbb{A}^n) = X_U \times \mathbb{A}^n$, and let

$$B' := \mathrm{pr}_1^* B + \sum_{i=1}^n \operatorname{div}(t_i), \qquad B'_{\mathbb{A}^n \times \mathbb{A}^n} := \mathrm{pr}_1^* B_{\mathbb{A}^n} + \sum_{i=1}^n \operatorname{div}(t_i).$$

Let $1 \neq g \in H$, and let $\mathrm{Fix}(g) \subset \mathbb{A}^n$ be the fixed locus of g on $\mathbb{A}^n$. Since H acts faithfully on $X_U \subset \mathbb{A}^n$, there exists i with $\chi_i(g) \neq 1$. Thus g moves both x_i and t_i, and the fixed locus $\mathrm{Fix}(g) \times \mathrm{Fix}(g)$ of g on $\mathbb{A}^n \times \mathbb{A}^n$ has codimension at least two. In particular, H acts on $\mathbb{A}^n \times \mathbb{A}^n$ without pseudoreflections.

The subscheme $X' \subset \mathbb{A}^n \times \mathbb{A}^n$ is defined by the H-semi-invariant polynomials $f_1, \dots, f_c$, and X'_U is a complete intersection of codimension c in $U \times \mathbb{A}^n$. Since X_U is normal with klt singularities and the projection $X'_U \to X_U$ is smooth and surjective, X'_U is normal with klt singularities [Kol13, Proposition 2.15]. In particular, X' is normal and klt near 0.

The fixed locus of g on X'_U is $(X_U \cap \operatorname{Fix}(g)) \times \operatorname{Fix}(g)$. Since X_U is integral and g acts non-trivially on X_U, $\dim(X_U \cap \operatorname{Fix}(g)) \leq \dim X_U - 1$. Since $\dim \operatorname{Fix}(g) \leq n-1$,

$$\dim\big((X_U \cap \operatorname{Fix}(g)) \times \operatorname{Fix}(g)\big) \leq \dim X'_U - 2.$$

Let $\pi : \mathbb{A}^n \times \mathbb{A}^n \to (\mathbb{A}^n \times \mathbb{A}^n)/H$ be the quotient morphism, and let $y \in X'/H$ be the image of 0. Since π is étale at the points with trivial stabilizer [Gro71, Exp. V, Proposition 2.2] (applied with the trivial subgroup), the image $Z' \subset (\mathbb{A}^n \times \mathbb{A}^n)/H$ of the non-étale locus of π is contained in $\bigcup_{g \neq 1} \pi(\operatorname{Fix}(g) \times \operatorname{Fix}(g))$. Since X'_U is H-invariant, $Z' \cap \pi(X'_U)$, where $\pi(X'_U) = X'_U/H$, is contained in $\bigcup_{g\neq 1} \pi((X_U \cap \operatorname{Fix}(g)) \times \operatorname{Fix}(g))$, which has codimension at least two in X'_U/H because π is finite. Since $U \times \mathbb{A}^n$ is H-invariant, X'_U/H is an open neighborhood of y in X'/H. Thus, near y, Z' intersects X'/H in codimension at least two. Since, for any $1 \neq g \in H$, $\operatorname{Fix}(g) \times \operatorname{Fix}(g)$ has codimension at least two in $\mathbb{A}^n \times \mathbb{A}^n$ and $(X_U \cap \operatorname{Fix}(g)) \times \operatorname{Fix}(g)$ has codimension at least two in X'_U, the quotient morphisms $\mathbb{A}^n \times \mathbb{A}^n \to (\mathbb{A}^n \times \mathbb{A}^n)/H$ and $X'_U \to X'_U/H$ are étale in codimension one, and their ramification divisors are zero.

Step 3. In this step, we prove

$$\operatorname{mld}((X'_U \ni 0, B')/H) = \operatorname{mld}\big(((\mathbb{A}^n \times \mathbb{A}^n) \ni 0, B'_{\mathbb{A}^n \times \mathbb{A}^n})/H\big). \tag{$*$}$$

Let $r := |H|$. The functions h_j^r and t_i^r are H-invariant, and we let

$$\mathfrak{a} := \prod_{j=1}^{s} (h_j^r)^{\frac{b_j}{r}} \prod_{i=1}^{n} (t_i^r)^{\frac{1}{r}}$$

be the corresponding $\mathbb{R}$-ideal on $(\mathbb{A}^n \times \mathbb{A}^n)/H$. Since X'_U is integral, $h_j|_{X_U} \neq 0$ for any j, and $t_i|_{X'_U} \neq 0$ for any i, the restriction $\mathfrak{a}|_{X'/H}$ is non-zero near y. Consider the action of H on $\mathbb{A}^n \times \mathbb{A}^n$, the closed subscheme $X' = V(f_1, \dots, f_c) \times \mathbb{A}^n$ of $\mathbb{A}^n \times \mathbb{A}^n$, a complete intersection of codimension c near 0, and $\mathfrak{a}$. By **Step 2**, H acts faithfully and linearly on $\mathbb{A}^n \times \mathbb{A}^n$ without pseudoreflections, since any $1 \neq g \in H$ satisfies $\chi_i(g) \neq 1$ for some i, X' is normal and klt near 0 with $\dim X' = 2n - c \geq 2$ near 0, since $c < n$, and, near y, the image of the non-étale locus of π intersects X'/H in codimension at least two (**Step 2** proves the last two on the H-invariant neighborhood $U \times \mathbb{A}^n$ of 0). Thus, by Theorem 2.22,

$$\operatorname{mld}(X'/H \ni y, \mathfrak{a}|_{X'/H}) = \operatorname{mld}\big((\mathbb{A}^n \times \mathbb{A}^n)/H \ni y, (f_1^r \cdots f_c^r)^{\frac{1}{r}}\mathfrak{a}\big).$$

Let F be a prime divisor over X'_U, and let E be the prime divisor over X'_U/H and e the positive integer such that $\operatorname{ord}_F|_{\mathbb{C}(X'_U/H)} = e \operatorname{ord}_E$. By Lemma 2.6(1), $e = |I_H(F)|$. Since the ramification divisor of $X'_U \to X'_U/H$ is zero, the crepant quotient of $(X'_U, 0)$ by H is $(X'_U/H, 0)$. By Lemma 2.6(2), applied to $(X'_U, 0)$ with the $\mathbb{R}$-ideal $\mathfrak{a}|_{X'_U/H}$ and to (X'_U, B'),

$$a(F, X'_U, 0, \pi^*(\mathfrak{a}|_{X'_U/H})) = |I_H(F)| \cdot a(E, X'_U/H, 0, \mathfrak{a}|_{X'_U/H}),$$
$$a(F, X'_U, B') = |I_H(F)| \cdot a(E, (X'_U, B')/H).$$

Since

$$\operatorname{mult}_F \pi^*(\mathfrak{a}|_{X'_U/H}) = \sum_j b_j \operatorname{ord}_F h_j + \sum_i \operatorname{ord}_F t_i = \operatorname{mult}_F B',$$

we have

$$a(F, X'_U, 0, \pi^*(\mathfrak{a}|_{X'_U/H})) = a(F, X'_U, B'),$$
$$a(E, X'_U/H, 0, \mathfrak{a}|_{X'_U/H}) = a(E, (X'_U, B')/H),$$

where the second equality follows from the first one and from the equalities given by Lemma 2.6(2). By Lemma 2.6(2)(3), any prime divisor over X'_U/H with center y is obtained in this way from a prime divisor over X'_U with center 0. Since X'_U/H is an open neighborhood of y in X'/H,

$$\operatorname{mld}(X'/H \ni y, \mathfrak{a}|_{X'/H}) = \operatorname{mld}((X'_U \ni 0, B')/H).$$

Now let F be a prime divisor over $\mathbb{A}^n \times \mathbb{A}^n$, and let E be the prime divisor over $(\mathbb{A}^n \times \mathbb{A}^n)/H$ and e the positive integer such that $\operatorname{ord}_F|_{\mathbb{C}((\mathbb{A}^n \times \mathbb{A}^n)/H)} = e \operatorname{ord}_E$. By Lemma 2.6(1), $e = |I_H(F)|$. Since the ramification divisor of π is zero, the crepant quotient of $(\mathbb{A}^n \times \mathbb{A}^n, 0)$ by H is $((\mathbb{A}^n \times \mathbb{A}^n)/H, 0)$. Since

$$\operatorname{mult}_F \pi^*\big((f_1^r \cdots f_c^r)^{\frac{1}{r}}\mathfrak{a}\big) = \sum_i \operatorname{ord}_F f_i + \sum_j b_j \operatorname{ord}_F h_j + \sum_i \operatorname{ord}_F t_i = \operatorname{mult}_F B'_{\mathbb{A}^n \times \mathbb{A}^n},$$

Lemma 2.6(2), applied to $(\mathbb{A}^n \times \mathbb{A}^n, 0)$ with the $\mathbb{R}$-ideal $(f_1^r \cdots f_c^r)^{\frac{1}{r}}\mathfrak{a}$ and to $(\mathbb{A}^n \times \mathbb{A}^n, B'_{\mathbb{A}^n \times \mathbb{A}^n})$, implies

$$a\big(E, (\mathbb{A}^n \times \mathbb{A}^n)/H, 0, (f_1^r \cdots f_c^r)^{\frac{1}{r}}\mathfrak{a}\big) = a\big(E, (\mathbb{A}^n \times \mathbb{A}^n, B'_{\mathbb{A}^n \times \mathbb{A}^n})/H\big).$$

By Lemma 2.6(2)(3), any prime divisor over $(\mathbb{A}^n \times \mathbb{A}^n)/H$ with center y is obtained in this way from a prime divisor over $\mathbb{A}^n \times \mathbb{A}^n$ with center 0. Hence

$$\operatorname{mld}\big((\mathbb{A}^n \times \mathbb{A}^n)/H \ni y, (f_1^r \cdots f_c^r)^{\frac{1}{r}}\mathfrak{a}\big) = \operatorname{mld}\big(((\mathbb{A}^n \times \mathbb{A}^n) \ni 0, B'_{\mathbb{A}^n \times \mathbb{A}^n})/H\big).$$

Thus $(*)$ holds.

Step 4. In this step, we remove the auxiliary coordinates and conclude the proof.

Let $a := \operatorname{mld}((X \ni 0, B)/H) = \operatorname{mld}((X_U \ni 0, B)/H)$. Since (X_U, B) is klt, $\operatorname{mld}(X \ni 0, B) > 0$ (by [KM98, Corollary 2.31(3)] on a log resolution of (X, B) near 0, together with [KM98, Lemma 2.30]). By Lemma 2.6, $a \geq \operatorname{mld}(X \ni 0, B)/|H| > 0$. For $0 \leq k \leq n$, let $X^{(k)} := X_U \times \mathbb{A}^k_{t_1,\dots,t_k}$ and $B^{(k)} := \operatorname{pr}_1^* B + \sum_{i=1}^k \operatorname{div}(t_i)$, so that $(X^{(0)}, B^{(0)}) = (X_U, B)$ and $(X^{(n)}, B^{(n)}) = (X'_U, B')$. Since $\dim X_U = n - c \geq 1$, Lemma 8.1, applied to $(X^{(k-1)}, B^{(k-1)})$, the point 0, and the character $\chi_k|_H$, implies by induction on k that $\operatorname{mld}((X^{(k)} \ni 0, B^{(k)})/H) = a$ for any k. In particular,

$$\operatorname{mld}((X'_U \ni 0, B')/H) = a.$$

Let $a_{\mathbb{A}} := \operatorname{mld}((\mathbb{A}^n \ni 0, B_{\mathbb{A}^n})/H)$. The last statement of Lemma 8.1 requires $a_{\mathbb{A}} \geq 0$, so we first show $a_{\mathbb{A}} \geq a$. For $0 \leq k \leq n$, let $D_k := \operatorname{pr}_1^* B_{\mathbb{A}^n} + \sum_{i=1}^k \operatorname{div}(t_i)$ on $\mathbb{A}^n \times \mathbb{A}^k_{t_1,\dots,t_k}$, so that $D_0 = B_{\mathbb{A}^n}$ and $D_n = B'_{\mathbb{A}^n \times \mathbb{A}^n}$. Let E be a prime divisor over $\mathbb{A}^n$ with center 0. By Lemma 8.1(2), applied to $(\mathbb{A}^n \times \mathbb{A}^{k-1}, D_{k-1})$ and the character $\chi_k|_H$ for $k = 1, \dots, n$ in turn, there exists a prime divisor F over $\mathbb{A}^n \times \mathbb{A}^n$ with center 0 such that $I_H(F) = I_H(E)$ and $a(F, \mathbb{A}^n \times \mathbb{A}^n, B'_{\mathbb{A}^n \times \mathbb{A}^n}) = a(E, \mathbb{A}^n, B_{\mathbb{A}^n})$. By Lemma 2.6 and $(*)$,

$$\frac{a(E, \mathbb{A}^n, B_{\mathbb{A}^n})}{|I_H(E)|} = \frac{a(F, \mathbb{A}^n \times \mathbb{A}^n, B'_{\mathbb{A}^n \times \mathbb{A}^n})}{|I_H(F)|} \geq \operatorname{mld}\big(((\mathbb{A}^n \times \mathbb{A}^n) \ni 0, B'_{\mathbb{A}^n \times \mathbb{A}^n})/H\big) = a.$$

Since E is arbitrary, Lemma 2.6 implies $a_{\mathbb{A}} \geq a$. In particular $a_{\mathbb{A}} \geq 0$ since $a > 0$, so Lemma 8.1, applied to $(\mathbb{A}^n \times \mathbb{A}^{k-1}, D_{k-1})$, the point 0, and the character $\chi_k|_H$, implies by induction on k that $\operatorname{mld}(((\mathbb{A}^n \times \mathbb{A}^k) \ni 0, D_k)/H) = a_{\mathbb{A}}$ for any k. In particular,

$$\operatorname{mld}\big(((\mathbb{A}^n \times \mathbb{A}^n) \ni 0, B'_{\mathbb{A}^n \times \mathbb{A}^n})/H\big) = a_{\mathbb{A}}.$$

By $(*)$, $a_{\mathbb{A}} = a$, and the equality holds. The last statement of Theorem 8.3 follows from the definition of $\operatorname{mld}((\mathbb{A}^n \ni 0, B_{\mathbb{A}^n})/H)$. □

The klt hypotheses in Theorem 8.3 cannot be removed: for the cubic cones X of dimension at least four in [NS24, Example 1.4], which are not klt, with no polynomials h_j and the diagonal action of μ_3 there, which is free outside the vertex on X and has no pseudoreflections on the ambient affine space, the equality of minimal log discrepancies in Theorem 8.3 fails [NS24, Theorem 1.5].

8.2. **Toric models of quotients.** In this subsection, we compare the quotient of a semi-invariant boundary on $\mathbb{A}^d$ by a finite diagonal group with a toric model (Lemmas 8.5 and 8.6), and then we prove two consequences and discuss an example.

Recall that the Newton fan of a non-zero polynomial $f \in \mathbb{C}[x_1, \dots, x_d]$ is a rational fan with support $\mathbb{R}^d_{\geq 0}$ such that $\mathrm{in}_w(f)$ is constant for w in the relative interior of each cone and $w \to \mathrm{wt}_w(f)$ is linear on each cone [Amb03, Section 1-B, p. 45].

Set-up 8.4. Let d, G, M_G, s, f_j, b_j, B, $B_{\mathbb{A}^d/G}$, ψ, B_w, Σ, V_Σ, μ_Σ, Δ_Σ, and B'_Σ be as follows.

(1) $d \in \{3, 4\}$, G is a finite diagonal subgroup of $(\mathbb{C}^*)^d$ acting faithfully on $\mathbb{A}^d$, N_G is the lattice of Definition 2.15 for the coordinates $x_1, \dots, x_d$ of $\mathbb{A}^d$, and $M_G := \{\alpha \in \mathbb{Z}^d \mid \langle v, \alpha\rangle \in \mathbb{Z} \text{ for any } v \in N_G\}$.

(2) $s \geq 0$ is an integer, $f_1, \dots, f_s$ are non-zero G-semi-invariant polynomials vanishing at 0, $b_1, \dots, b_s$ are positive real numbers, $B := \sum_j b_j \operatorname{div}(f_j)$, and $(\mathbb{A}^d/G, B_{\mathbb{A}^d/G}) := (\mathbb{A}^d, B)/G$.

(3) $\psi(w) := \sum_i w_i - \sum_j b_j \mathrm{wt}_w(f_j)$ for $w \in \mathbb{R}^d_{\geq 0}$, and $B_w := \sum_j b_j \operatorname{div}(\mathrm{in}_w(f_j))$ on $(\mathbb{C}^*)^d$ for $w \in \mathbb{R}^d_{>0}$, with all multiplicities. We call B_w the *weighted leading boundary* of B at w.

(4) Σ is a regular subdivision of $\mathbb{R}^d_{\geq 0}$ with respect to N_G refining the Newton fans of all f_j, $V_\Sigma := X_{\Sigma, N_G}$ is the toric variety of Σ with respect to N_G, with the toric morphism $\mu_\Sigma : V_\Sigma \to \mathbb{A}^d/G$, Δ_Σ is the reduced toric boundary of V_Σ, and B'_Σ is the strict transform on V_Σ of the closure in $\mathbb{A}^d/G$ of the restriction of $B_{\mathbb{A}^d/G}$ to $(\mathbb{C}^*)^d/G$. We call $(V_\Sigma, \Delta_\Sigma + B'_\Sigma)$ the *toric model* of $(\mathbb{A}^d, B)/G$ for Σ.

Lemma 8.5. *Notation as in Set-up 8.4. Then*

$$\mu_\Sigma^*(K_{\mathbb{A}^d/G} + B_{\mathbb{A}^d/G}) = K_{V_\Sigma} + \Delta_\Sigma + B'_\Sigma - \sum_v \psi(v) E_v,$$

where v runs over the primitive generators in N_G of the rays of Σ and E_v is the corresponding toric prime divisor of V_Σ. Moreover, any primitive vector $v \in N_G \cap \mathbb{R}^d_{\geq 0} \setminus \{0\}$ defines a toric prime divisor E_v over $\mathbb{A}^d/G$ with $a(E_v, \mathbb{A}^d/G, B_{\mathbb{A}^d/G}) = \psi(v)$. If ev is the primitive vector of $\mathbb{Z}^d$ on $\mathbb{R}_{\geq 0} v$ and F is the monomial prime divisor over $\mathbb{A}^d$ with weights ev, then $\operatorname{ord}_F|_{\mathbb{C}(\mathbb{A}^d)^G} = e \operatorname{ord}_{E_v}$, $e = |I_G(F)|$, and $a(F, \mathbb{A}^d, B) = e\psi(v)$.

Proof. For $g \in G$, write $g = (e^{2\pi\sqrt{-1}u_1}, \dots, e^{2\pi\sqrt{-1}u_d})$ with $u \in \mathbb{R}^d$. Then $g^* x^\alpha = e^{2\pi\sqrt{-1}\langle u, \alpha\rangle} x^\alpha$ for any $\alpha \in \mathbb{Z}^d$, so a Laurent monomial x^α is G-invariant if and only if $\alpha \in M_G$. In particular, $\mathbb{C}[x_1, \dots, x_d]^G = \mathbb{C}[M_G \cap \mathbb{R}^d_{\geq 0}]$, $\mathbb{A}^d/G$ is the affine toric variety of the cone $\mathbb{R}^d_{\geq 0}$ with respect to N_G, and $(\mathbb{C}^*)^d/G$ is the dense torus of $\mathbb{A}^d/G$.

Let $v \in N_G \cap \mathbb{R}^d_{\geq 0} \setminus \{0\}$ be primitive. Let E_v be the toric prime divisor of the ray $\mathbb{R}_{\geq 0} v$ on a toric variety W over $\mathbb{A}^d/G$ whose fan contains $\mathbb{R}_{\geq 0} v$, for example the toric variety of the star subdivision of $\mathbb{R}^d_{\geq 0}$ along v [Amb06, Section 2, p. 360]. Since v is primitive in N_G, the affine open subset of W corresponding to the cone $\mathbb{R}_{\geq 0} v$ is isomorphic to $\mathbb{A}^1 \times (\mathbb{C}^*)^{d-1}$, where E_v is defined by the coordinate t of $\mathbb{A}^1$, and for $\alpha \in M_G$ the character x^α is $t^{\langle v, \alpha\rangle}$ times a character of $(\mathbb{C}^*)^{d-1}$. Distinct $\alpha, \alpha' \in M_G$ with $\langle v, \alpha\rangle = \langle v, \alpha'\rangle$ correspond to distinct characters of $(\mathbb{C}^*)^{d-1}$. Hence

$$\operatorname{ord}_{E_v}\Big(\sum_\alpha c_\alpha x^\alpha\Big) = \min\{\langle v, \alpha\rangle \mid c_\alpha \neq 0\} \quad \text{for any } \sum_\alpha c_\alpha x^\alpha \in \mathbb{C}[x_1, \dots, x_d]^G. \tag{8.1}$$

In particular, E_v does not depend on the fan as a prime divisor over $\mathbb{A}^d/G$. Since $\mathbb{Z}^d \subset N_G$, the primitive vector of $\mathbb{Z}^d$ on $\mathbb{R}_{\geq 0} v$ is ev for a positive integer e. Let F be the monomial prime divisor over $\mathbb{A}^d$ in $x_1, \dots, x_d$ with weights ev (Definition 2.16). By Definition 2.16 and (8.1), $\operatorname{ord}_F(\phi) = e \operatorname{ord}_{E_v}(\phi)$ for any $\phi \in \mathbb{C}[x_1, \dots, x_d]^G$. Since $\mathbb{C}(\mathbb{A}^d/G) = \mathbb{C}(\mathbb{A}^d)^G$ is the fraction field of $\mathbb{C}[x_1, \dots, x_d]^G$, we have $\operatorname{ord}_F|_{\mathbb{C}(\mathbb{A}^d)^G} = e \operatorname{ord}_{E_v}$. By Lemma 2.6(1)(2), $e = |I_G(F)|$ and

$$a(F, \mathbb{A}^d, B) = e a(E_v, \mathbb{A}^d/G, B_{\mathbb{A}^d/G}).$$

On the other hand, $a(F, \mathbb{A}^d, 0) = \sum_i ev_i$ [Amb06, Section 2, p. 360], and by Definition 2.16,

$$\operatorname{ord}_F(f_j) = \operatorname{wt}_{ev}(f_j) = e \operatorname{wt}_v(f_j).$$

Thus

$$a(F, \mathbb{A}^d, B) = e \sum_i v_i - \sum_j b_j e \operatorname{wt}_v(f_j) = e\psi(v),$$

and $a(E_v, \mathbb{A}^d/G, B_{\mathbb{A}^d/G}) = \psi(v)$. If v is on the ray of the i-th coordinate, then F is the hyperplane $P = (x_i = 0)$, e is the ramification index of P over $\mathbb{A}^d/G$, $\psi(v) = \frac{1-\operatorname{mult}_P B}{e}$, and the equality $a(E_v, \mathbb{A}^d/G, B_{\mathbb{A}^d/G}) = \psi(v)$ is Lemma 2.6(4).

We prove the formula for $\mu_\Sigma^*(K_{\mathbb{A}^d/G} + B_{\mathbb{A}^d/G})$. Let $\pi : \mathbb{A}^d \to \mathbb{A}^d/G$ be the quotient morphism. Since G acts diagonally and faithfully, a non-identity element of G has no fixed point on $(\mathbb{C}^*)^d$, so π is étale over $(\mathbb{C}^*)^d/G$. As B is G-invariant, Definition 2.5 implies

$$\pi^*\left(B_{\mathbb{A}^d/G}|_{(\mathbb{C}^*)^d/G}\right) = B|_{(\mathbb{C}^*)^d}.$$

Since $b_j > 0$ for any j, $B \geq 0$, so the display gives $B_{\mathbb{A}^d/G}|_{(\mathbb{C}^*)^d/G} \geq 0$ and hence $B'_\Sigma \geq 0$. Since B'_Σ is the strict transform of the closure of a divisor on the dense torus $(\mathbb{C}^*)^d/G$, it has no toric component. By Lemma 2.6(0), $K_{\mathbb{A}^d/G} + B_{\mathbb{A}^d/G}$ is $\mathbb{R}$-Cartier, so we may write $\mu_\Sigma^*(K_{\mathbb{A}^d/G} + B_{\mathbb{A}^d/G}) = K_{V_\Sigma} + \Theta$, where $\operatorname{mult}_P \Theta = 1 - a(P, \mathbb{A}^d/G, B_{\mathbb{A}^d/G})$ for any prime divisor P on V_Σ. A prime divisor P on V_Σ is either E_v for a primitive ray generator v of Σ, or the closure of a prime divisor P° on the dense torus $(\mathbb{C}^*)^d/G$. In the first case, $\operatorname{mult}_P \Theta = 1 - \psi(v)$ by the previous paragraph. In the second case, μ_Σ is an isomorphism near the generic point of P, so $\operatorname{mult}_P \Theta = \operatorname{mult}_{P^\circ} B_{\mathbb{A}^d/G} = \operatorname{mult}_P B'_\Sigma$. Since $\Delta_\Sigma = \sum_v E_v$ and B'_Σ has no toric component, $\Theta = \Delta_\Sigma + B'_\Sigma - \sum_v \psi(v) E_v$. □

Propositions 8.8, 8.11, and 10.5 and Theorem 8.14 are all reduced to the following lemma.

Lemma 8.6. *Notation as in Set-up 8.4, and let $\bar{0} \in \mathbb{A}^d/G$ be the image of 0.*

(1) $((\mathbb{C}^*)^d, B_w)$ is lc for any $w \in \mathbb{R}^d_{>0}$ if and only if $(V_\Sigma, \Delta_\Sigma + B'_\Sigma)$ is lc near $\mu_\Sigma^{-1}(\bar{0})$.

(2) Assume that $(V_\Sigma, \Delta_\Sigma + B'_\Sigma)$ is lc near $\mu_\Sigma^{-1}(\bar{0})$ and that $(\mathbb{A}^d, B)$ is lc near 0. Then $\operatorname{mld}((\mathbb{A}^d \ni 0, B)/G) = \min_{v \in N_G \cap \mathbb{R}^d_{>0}} \psi(v)$, the minimum is attained at a primitive vector of N_G, any primitive $v \in N_G \cap \mathbb{R}^d_{>0}$ defines a toric prime divisor E_v over $\mathbb{A}^d/G$ with center $\bar{0}$ and $a(E_v, \mathbb{A}^d/G, B_{\mathbb{A}^d/G}) = \psi(v)$, and any prime divisor E over $\mathbb{A}^d/G$ with center $\bar{0}$ satisfies $a(E, \mathbb{A}^d/G, B_{\mathbb{A}^d/G}) \geq \psi(v)$ for some $v \in N_G \cap \mathbb{R}^d_{>0}$.

Proof. For a cone τ of Σ, let $U_\tau \subset V_\Sigma$ be the affine open subset and $O(\tau) \subset U_\tau$ the torus orbit of τ. By the orbit–cone correspondence [CLS11, Theorem 3.2.6] and [CLS11, Lemma 3.3.21(b)], μ_Σ maps $O(\tau)$ into the torus orbit of $\mathbb{A}^d/G$ of the smallest face of $\mathbb{R}^d_{\geq 0}$ containing τ, and $\bar{0}$ is the torus orbit of $\mathbb{R}^d_{\geq 0}$ itself. Hence $\mu_\Sigma^{-1}(\bar{0})$ is the union of the orbits $O(\tau)$ for the cones τ of Σ whose relative interiors are contained in $\mathbb{R}^d_{>0}$. For such a cone τ, let $v_1, \dots, v_k$ be the primitive generators in N_G of the rays of τ. Since Σ is regular, $v_1, \dots, v_k$ is a part of a basis of N_G, and the toric prime divisors of V_Σ containing $O(\tau)$ are $E_{v_1}, \dots, E_{v_k}$.

(1) Fix a cone τ of Σ whose relative interior is contained in $\mathbb{R}^d_{>0}$, a point $y \in O(\tau)$, and a vector w in the relative interior of τ. Write $f_j = \sum_\alpha c_{j,\alpha} x^\alpha$, and take a_j with $c_{j,a_j} \neq 0$ and $\langle w, a_j \rangle = \operatorname{wt}_w(f_j)$. The function $u \to \langle u, a_j \rangle - \operatorname{wt}_u(f_j)$ is non-negative on τ, linear on τ because τ is contained in a cone of the Newton fan of f_j, and vanishes at the relative interior point w of τ. Thus $\langle u, a_j \rangle = \operatorname{wt}_u(f_j)$ for any $u \in \tau$. Since f_j is G-semi-invariant, $x^{\alpha - a_j}$ is G-invariant for any α with $c_{j,\alpha} \neq 0$, that is, $\alpha - a_j \in M_G$, and $\langle u, \alpha - a_j \rangle \geq 0$ for any $u \in \tau$. Hence $g_j := f_j / x^{a_j}$ is a regular function on $U_\tau = \operatorname{Spec} \mathbb{C}[\tau^\vee \cap M_G]$. Since $O(\tau) = \operatorname{Spec} \mathbb{C}[\tau^\perp \cap M_G]$ is defined in U_τ by the characters in $(\tau^\vee \setminus \tau^\perp) \cap M_G$ [CLS11, Lemma 3.2.5], and $\langle u, \alpha - a_j \rangle$ vanishes for any $u \in \tau$

if and only if it vanishes at $u = w$, we obtain

$$g_j|_{O(\tau)} = \sum_{\alpha - a_j \in \tau^\perp} c_{j,\alpha} x^{\alpha - a_j} = \frac{\mathrm{in}_w(f_j)}{x^{a_j}} \neq 0. \tag{8.2}$$

In particular, g_j does not vanish identically on $E_{v_1}, \dots, E_{v_k}$, so $\mathrm{div}(g_j)$ has no toric component on U_τ.

Let $\pi : \mathbb{A}^d \to \mathbb{A}^d/G$ be the quotient morphism. A non-identity element of G has no fixed point on $(\mathbb{C}^*)^d$, so π is étale over $(\mathbb{C}^*)^d/G$, and $\pi^*(B_{\mathbb{A}^d/G}|_{(\mathbb{C}^*)^d/G}) = B|_{(\mathbb{C}^*)^d}$ by Definition 2.5. Since x^{a_j} is a unit on $(\mathbb{C}^*)^d$, $\pi^* \mathrm{div}(g_j) = \mathrm{div}(f_j)$ on $(\mathbb{C}^*)^d$. Hence $B_{\mathbb{A}^d/G}|_{(\mathbb{C}^*)^d/G} = \sum_j b_j \mathrm{div}(g_j)|_{(\mathbb{C}^*)^d/G}$, and since $\mathrm{div}(g_j)$ has no toric component on U_τ,

$$B'_\Sigma|_{U_\tau} = \sum_j b_j \mathrm{div}(g_j).$$

Let $Z_0 := U_\tau$ and $Z_l := E_{v_1} \cap \cdots \cap E_{v_l} \cap U_\tau$ for $1 \le l \le k$. Since Σ is regular, $U_\tau \cong \mathbb{A}^k \times (\mathbb{C}^*)^{d-k}$, where $E_{v_1} \cap U_\tau, \dots, E_{v_k} \cap U_\tau$ are the coordinate hyperplanes of $\mathbb{A}^k$. Thus Z_l is smooth and irreducible, Z_{l+1} is a smooth Cartier divisor on Z_l, $Z_k = O(\tau)$, and $\Delta_\Sigma|_{U_\tau} = \sum_i E_{v_i}|_{U_\tau}$. By (8.2), $g_j|_{Z_l}$ does not vanish identically on $Z_{l+1} \supset O(\tau)$. Hence $B'_\Sigma|_{Z_l} = \sum_j b_j \mathrm{div}(g_j|_{Z_l}) \ge 0$ is an $\mathbb{R}$-divisor with no common component with Z_{l+1}, and $E_{v_i}|_{Z_l}$ with $i > l+1$ has no common component with Z_{l+1} either. By the adjunction formula, the different of $(Z_l, Z_{l+1} + \sum_{i>l+1} E_{v_i}|_{Z_l} + B'_\Sigma|_{Z_l})$ on Z_{l+1} is the restriction $\sum_{i>l+1} E_{v_i}|_{Z_{l+1}} + B'_\Sigma|_{Z_{l+1}}$. By Theorem 2.23 for $l = k-1, k-2, \dots, 0$, shrinking U_τ around $O(\tau)$ at each step, $(V_\Sigma, \Delta_\Sigma + B'_\Sigma)$ is lc near $O(\tau)$ if $(O(\tau), B'_\Sigma|_{O(\tau)})$ is lc.

Let $q : (\mathbb{C}^*)^d \to O(\tau)$ be the morphism with $q^* x^m = x^m$ for $m \in \tau^\perp \cap M_G \subset \mathbb{Z}^d$. Since $\tau^\perp \cap M_G \to \mathbb{Z}^d$ is injective, q is a surjective homomorphism of tori, hence q is smooth. By (8.2),

$$q^*\big(B'_\Sigma|_{O(\tau)}\big) = \sum_j b_j \,\mathrm{div}\left(\frac{\mathrm{in}_w(f_j)}{x^{a_j}}\right) = \sum_j b_j \,\mathrm{div}(\mathrm{in}_w(f_j)) = B_w.$$

Assume that $((\mathbb{C}^*)^d, B_{w'})$ is lc for any $w' \in \mathbb{R}^d_{>0}$. Then $((\mathbb{C}^*)^d, B_w)$ is lc, so $(O(\tau), B'_\Sigma|_{O(\tau)})$ is lc [Kol13, Proposition 2.15]. Hence $(V_\Sigma, \Delta_\Sigma + B'_\Sigma)$ is lc near y. Since $y \in \mu_\Sigma^{-1}(\bar{0})$ is arbitrary, $(V_\Sigma, \Delta_\Sigma + B'_\Sigma)$ is lc near $\mu_\Sigma^{-1}(\bar{0})$.

Conversely, assume that $(V_\Sigma, \Delta_\Sigma + B'_\Sigma)$ is lc near $\mu_\Sigma^{-1}(\bar{0})$, and let $w \in \mathbb{R}^d_{>0}$. Let τ be the cone of Σ containing w in the relative interior. The coordinates are non-negative on τ, so a coordinate vanishing at a point of the relative interior of τ vanishes on τ, in particular at w. Thus the relative interior of τ is contained in $\mathbb{R}^d_{>0}$, so $O(\tau) \subset \mu_\Sigma^{-1}(\bar{0})$. Choose a_j and define g_j, Z_l, and q for τ and w as at the beginning of the proof of (1). Since Theorem 2.23 is an equivalence, Theorem 2.23 for $l = 0, 1, \dots, k-1$, shrinking U_τ around $O(\tau)$ at each step, shows that $(O(\tau), B'_\Sigma|_{O(\tau)})$ is lc. Since q is a smooth surjection with $q^*(B'_\Sigma|_{O(\tau)}) = B_w$, $((\mathbb{C}^*)^d, B_w)$ is lc by [Kol13, Proposition 2.15]. This proves (1).

(2) Since $(\mathbb{A}^d, B)$ is lc near 0, $(\mathbb{A}^d/G, B_{\mathbb{A}^d/G})$ is lc near $\bar{0}$ by Lemma 2.6(2)(3). For a ray of Σ with primitive generator v, the center of E_v on $\mathbb{A}^d/G$ is the closure of the torus orbit of the smallest face of $\mathbb{R}^d_{\ge 0}$ containing v, which contains $\bar{0}$. Thus $\psi(v) = a(E_v, \mathbb{A}^d/G, B_{\mathbb{A}^d/G}) \ge 0$ by Lemma 8.5. If $v \in N_G \cap \mathbb{R}^d_{>0}$ is primitive, then the center of E_v is $\bar{0}$, and $a(E_v, \mathbb{A}^d/G, B_{\mathbb{A}^d/G}) = \psi(v)$ by Lemma 8.5.

Let E be a prime divisor over $\mathbb{A}^d/G$ with center $\bar{0}$. Since μ_Σ is proper [CLS11, Theorem 3.4.11], E has a center on V_Σ. Let $\eta \in \mu_\Sigma^{-1}(\bar{0})$ be the generic point of the center of E on V_Σ. Let τ be the cone of Σ with $\eta \in O(\tau)$, and let $v_1, \dots, v_k$ be the primitive generators in N_G of the rays of τ. Then $c_i := \mathrm{mult}_E E_{v_i}$ is a positive integer for any i, and $\mathrm{mult}_E E_v = 0$ for the other primitive ray generators v of Σ. The vector $v_E := c_1 v_1 + \cdots + c_k v_k \in N_G$ is in the relative interior of τ, so $v_E \in \mathbb{R}^d_{>0}$. Since Σ refines the Newton fans of all f_j, ψ is linear on τ [Amb03, Section 1-B, p. 45].

By Lemma 8.5,

$$a(E,\mathbb{A}^d/G,B_{\mathbb{A}^d/G})=a(E,V_\Sigma,\Delta_\Sigma+B'_\Sigma)+\sum_{i=1}^k c_i\psi(v_i)\geq\sum_{i=1}^k c_i\psi(v_i)=\psi(v_E),$$

where the inequality holds because $(V_\Sigma,\Delta_\Sigma+B'_\Sigma)$ is lc near $\mu_\Sigma^{-1}(\bar{0})\ni\eta$.

Let $v\in N_G\cap\mathbb{R}^d_{>0}$, and let τ be the cone of Σ containing v in the relative interior, with primitive ray generators $v_1,\dots,v_k$. Then the relative interior of τ is contained in $\mathbb{R}^d_{>0}$. Since $v_1,\dots,v_k$ is a part of a basis of N_G, $v=n_1v_1+\cdots+n_kv_k$ for positive integers $n_1,\dots,n_k$, and $v_1+\cdots+v_k$ is a primitive vector of N_G in the relative interior of τ. Since $\psi(v_i)\geq 0$ and ψ is linear on τ,

$$\psi(v)=\sum_{i=1}^k n_i\psi(v_i)\geq\sum_{i=1}^k\psi(v_i)=\psi(v_1+\cdots+v_k).$$

As Σ has finitely many cones, $\inf_{v\in N_G\cap\mathbb{R}^d_{>0}}\psi(v)$ is the minimum of $\psi(v_1+\cdots+v_k)$ over the finitely many cones τ of Σ whose relative interiors are contained in $\mathbb{R}^d_{>0}$, and it is attained at a primitive vector of N_G. Combining with the previous two paragraphs, we obtain (2). □

The following proposition is not used in the proofs of the main theorems of this paper, and we include it for use in subsequent work.

Proposition 8.7. *Let m be a positive integer, and let either (S) $X=\mathbb{A}^3$, or (H) $X=(f=0)\subset\mathbb{A}^4$ for a polynomial f such that X is a normal klt hypersurface germ at 0. Let G be a finite abelian group acting diagonally and faithfully on $\mathbb{A}^d$, where $d=3$ in case (S) and $d=4$ in case (H), fixing 0 and acting faithfully on X near 0, with f G-semi-invariant in case (H). Let h be a G-semi-invariant polynomial with $h|_X\neq 0$ such that (X,B) is klt for $B:=\frac{1}{m}\operatorname{div}(h|_X)$. Let $C:=\frac{1}{m}\operatorname{div}(h)$ in case (S), $C:=\operatorname{div}(f)+\frac{1}{m}\operatorname{div}(h)$ in case (H), let $(U,C_U):=(\mathbb{A}^d,C)/G$, and let $\bar{0}\in U$ be the image of 0. Assume that there exists a proper birational toric morphism $\mu:V\to U$ from a smooth variety such that $\mu^{-1}(\bar{0})$ is a divisor and the union of the toric boundary and the strict transforms of the components of $\operatorname{Supp}C_U$ is a simple normal crossing divisor near $\mu^{-1}(\bar{0})$. Then $\operatorname{mld}((X\ni 0,B)/G)$ is attained and equals*

$$\min_{w\in N_G\cap\mathbb{R}^d_{>0}}\Big(\sum_i w_i-\tfrac{1}{m}\operatorname{wt}_w(h)\Big)\quad\text{in case (S)},$$

$$\min_{w\in N_G\cap\mathbb{R}^d_{>0}}\Big(\sum_i w_i-\operatorname{wt}_w(f)-\tfrac{1}{m}\operatorname{wt}_w(h)\Big)\quad\text{in case (H)},$$

and the minima are attained at primitive vectors of N_G. In particular, for fixed m, the set of the numbers $\operatorname{mld}((X\ni 0,B)/G)$ for all such X, G, h (and f in case (H)) satisfies the ACC.

Proof. If $h(0)\neq 0$, then $\operatorname{wt}_w(h)=0$ for any $w\in\mathbb{R}^d_{\geq 0}$, and $\operatorname{div}(h)$ does not contain 0. Since all the statements concern neighborhoods of 0 and of $\bar{0}$, possibly replacing h with 1, we may assume that either $h=1$ or $h(0)=0$. We identify G with the image of G in $(\mathbb{C}^*)^d$ and use the notation of Set-up 8.4 with the following data: in case (S), $s=1$, $f_1=h$, and $b_1=\frac{1}{m}$ if $h\neq 1$, and $s=0$ if $h=1$; in case (H), $s=2$, $f_1=f$, $b_1=1$, $f_2=h$, and $b_2=\frac{1}{m}$ if $h\neq 1$, and $s=1$, $f_1=f$, and $b_1=1$ if $h=1$. We have $f(0)=0$ since $0\in X$. Here Σ is any fan as in Set-up 8.4(4), which exists by [CLS11, Theorem 11.1.9]. Then $B_{\mathbb{A}^d/G}=C_U$, and ψ is the function minimized in the statement.

In case (H), we also fix a neighborhood of 0 in X on which the quotient by G is defined. Let $Z\subset X$ be the union of the irreducible components of X not containing 0 and of the complement of the largest open subset of X that is normal with klt singularities. Since G fixes 0, Z is a G-invariant closed subset with $0\notin Z$. Take a polynomial ϕ_0 vanishing on Z with $\phi_0(0)=1$, and let $U'\subset\mathbb{A}^4$ be the complement of the zero locus of $\prod_{g\in G}g^*\phi_0$. Then U' is a G-invariant affine open neighborhood of 0, $X\cap U'$ is a normal variety with klt singularities, and G acts faithfully on

$X \cap U'$, since a non-identity element of G acting trivially on $X \cap U'$ would act trivially on X near 0. Thus $(X \ni 0, B)/G$ is formed on $X \cap U'$, and we write $(X/G, B_{X/G}) := (X \cap U', B|_{X\cap U'})/G$. The hypotheses of Theorem 8.3 used below concern only a neighborhood of 0, so they hold for X.

We first show that

$$a(E', U, C_U) > 0 \quad \text{for any prime divisor } E' \text{ over } U \text{ with center } \bar{0}. \tag{8.3}$$

In case (S), $(U, C_U) = (X, B)/G$ is klt near $\bar{0}$ by Lemma 2.6(2)(3). In case (H), Theorem 8.3, with $n = 4$, $c = 1$, $f_1 = f$, $h_1 = h$, $b_1 = \frac{1}{m}$, and $H = G$, implies that

$$\operatorname{mld}((X \ni 0, B)/G) = \operatorname{mld}(U \ni \bar{0}, C_U) > 0$$

and that $a(E', U, C_U) \geq \operatorname{mld}((X \ni 0, B)/G)$ for any E' as in (8.3).

Since μ is proper, birational, and toric, and V is smooth, V is the toric variety of a regular fan with support $\mathbb{R}^d_{\geq 0}$ with respect to N_G [CLS11, Theorems 3.1.19(a), 3.3.4(b), and 3.4.11]. The divisor $\mu^{-1}(\bar{0})$ is torus invariant, so by the orbit–cone correspondence [CLS11, Theorem 3.2.6] and [CLS11, Lemma 3.3.21(b)] the components of $\mu^{-1}(\bar{0})$ are the toric prime divisors E_v with v a primitive ray generator in $\mathbb{R}^d_{>0}$, and $a(E_v, U, C_U) = \psi(v)$ by Lemma 8.5. Write $K_V + B_V := \mu^*(K_U + C_U)$, and let L_0 be the union of the toric boundary of V and the strict transforms of the components of $\operatorname{Supp} C_U$. Then $\operatorname{Supp} B_V \subset L_0$, and L_0 is a simple normal crossing divisor near $\mu^{-1}(\bar{0})$.

We claim that $a(P, U, C_U) \geq 0$ for any prime divisor P on V with $P \cap \mu^{-1}(\bar{0}) \neq \emptyset$. If $P \subset \mu^{-1}(\bar{0})$, then this follows from (8.3). Otherwise, suppose that $a(P, U, C_U) < 0$. Then $P \subset \operatorname{Supp} B_V \subset L_0$, and $P \cap E \neq \emptyset$ for a component E of $\mu^{-1}(\bar{0})$. Let ξ be the generic point of a component of $P \cap E$ near $\mu^{-1}(\bar{0})$. Near ξ, the only components of L_0 through ξ are P and E. Blow up the closure of ξ, and then blow up $N - 1$ times the intersection of the strict transform of P with the last exceptional divisor. By [KM98, Lemma 2.29], the last exceptional divisor E_N satisfies

$$a(E_N, U, C_U) = Na(P, U, C_U) + a(E, U, C_U),$$

and the center of E_N on V is contained in $E \subset \mu^{-1}(\bar{0})$. For $N \gg 0$, this contradicts (8.3). Thus the claim holds.

Let E' be a prime divisor over U with center $\bar{0}$, and let η be the generic point of the center of E' on V. Then η is contained in a component E of $\mu^{-1}(\bar{0})$. Choose regular parameters $y_1, \dots, y_{c'}$ of $\mathcal{O}_{V,\eta}$ containing local equations of the components of L_0 through η, with y_1 a local equation of E, and let $\beta_i := \operatorname{mult}_{(y_i=0)} B_V$. By the claim, $1 - \beta_i \geq 0$ for any i, and $1 - \beta_1 = a(E, U, C_U) > 0$. By Lemma 2.13, applied on a neighborhood of η on which $B_V = \sum_i \beta_i \operatorname{div}(y_i)$,

$$a(E', U, C_U) = a(E', V, B_V) \geq \sum_{i=1}^{c'} (1 - \beta_i) \operatorname{ord}_{E'}(y_i) \geq (1 - \beta_1) \operatorname{ord}_{E'}(y_1) \geq a(E, U, C_U).$$

Thus $\operatorname{mld}(U \ni \bar{0}, C_U)$ is the minimum of $\psi(v)$ over the finitely many primitive $v \in N_G \cap \mathbb{R}^d_{>0}$ with E_v a component of $\mu^{-1}(\bar{0})$. Conversely, for any primitive $v \in N_G \cap \mathbb{R}^d_{>0}$, the center of E_v is $\bar{0}$, so $\psi(v) \geq \operatorname{mld}(U \ni \bar{0}, C_U) > 0$ by Lemma 8.5. Any $w \in N_G \cap \mathbb{R}^d_{>0}$ is kv for a primitive $v \in N_G$ and a positive integer k, and $\psi(kv) = k\psi(v) \geq \psi(v)$. Hence

$$\operatorname{mld}(U \ni \bar{0}, C_U) = \min_{w \in N_G \cap \mathbb{R}^d_{>0}} \psi(w),$$

and the minimum is attained at a primitive vector of N_G.

In case (S), $(X, B)/G = (U, C_U)$, so we are done. In case (H), $\operatorname{mld}((X \ni 0, B)/G)$ equals $\operatorname{mld}(U \ni \bar{0}, C_U)$ by Theorem 8.3, as shown above. It remains to show that $\operatorname{mld}((X \ni 0, B)/G)$ is attained in case (H). Let $\bar{x} \in X/G$ be the image of 0. By Lemma 2.6(0)(2)(3), $(X/G, B_{X/G})$ is klt near $\bar{x}$. Let $g : Y' \to X/G$ be a log resolution of $(X/G, B_{X/G})$ such that $g^{-1}(\bar{x})$ is a divisor and $\operatorname{Exc}(g) \cup g^{-1}(\bar{x}) \cup g_*^{-1} \operatorname{Supp} B_{X/G}$ is a simple normal crossing divisor [KM98, Theorem 0.2], and write $K_{Y'} + B_{Y'} := g^*(K_{X/G} + B_{X/G})$. Near $g^{-1}(\bar{x})$, the coefficients of $B_{Y'}$ are less than

one. For a prime divisor E' over X/G with center $\bar{x}$, let E'' be a component of $g^{-1}(\bar{x})$ containing the center of E' on Y'. Lemma 2.13, applied at the generic point of the center of E' with regular parameters containing local equations of the components of $\operatorname{Supp} B_{Y'} \cup g^{-1}(\bar{x})$ through it, implies

$$a(E', X/G, B_{X/G}) \geq a(E'', X/G, B_{X/G}) \operatorname{mult}_{E'} E'' \geq a(E'', X/G, B_{X/G}).$$

Thus $\operatorname{mld}((X \ni 0, B)/G)$ is the minimum of the finitely many $a(E'', X/G, B_{X/G})$, so it is attained.

Finally, fix m, and let $\bar{G}$ be the image of G in $\mathbb{T}^d$, so $N_G = N_{\bar{G}}$. With $J(\cdot)$ as in Definition 2.16, the minimum in case (S) is $\psi_{\min}(\bar{G}, J(h), \frac{1}{m})$, and the minimum in case (H) is $\psi_{\min}(\bar{G}, (J(f), J(h)), (1, \frac{1}{m}))$. By Theorem 4.5, with $d = 3$, $s = 1$, $\Lambda = \{\frac{1}{m}\}$ in case (S) and with $d = 4$, $s = 2$, $\Lambda = \{1, \frac{1}{m}\}$ in case (H), the values of the minimum in case (S) form a set satisfying the ACC, and so do the values in case (H). By Lemma 2.10(4), the union of the set of the values in case (S) and the set of the values in case (H) satisfies the ACC. □

Proposition 8.8. *Notation as in Set-up 8.4 with $d = 3$, and assume that $f_1, \dots, f_s$ are reduced and pairwise coprime as germs at 0, that $b_j \leq 1$ for any j, and that $(\mathbb{A}^3, B)$ is lc near 0. Let $\bar{0} \in \mathbb{A}^3/G$ be the image of 0. Assume one of the following:*

(a) for any $w \in \mathbb{R}^3_{>0}$ and any non-empty $J \subset \{1, \dots, s\}$, the differentials of the polynomials $\operatorname{in}_w(f_j)$, $j \in J$, are linearly independent at any common zero of the $\operatorname{in}_w(f_j)$, $j \in J$, in $(\mathbb{C}^*)^3$ (equivalently, the subscheme of $(\mathbb{C}^*)^3$ defined by the $\operatorname{in}_w(f_j)$, $j \in J$, is empty or smooth of codimension $|J|$);

(b) $(V_\Sigma, \Delta_\Sigma + B'_\Sigma)$ is lc near $\mu_\Sigma^{-1}(\bar{0})$ for some choice of the fan Σ in Set-up 8.4(4).

Then (a) implies (b), and under (b),

$$\operatorname{mld}((\mathbb{A}^3 \ni 0, B)/G) = \min_{v \in N_G \cap \mathbb{R}^3_{>0}} \psi(v),$$

the minimum is attained at a primitive vector of N_G, and any primitive $v \in N_G \cap \mathbb{R}^3_{>0}$ defines a toric prime divisor E_v over $\mathbb{A}^3/G$ with center $\bar{0}$ and $a(E_v, \mathbb{A}^3/G, B_{\mathbb{A}^3/G}) = \psi(v)$. If ev is the primitive vector of $\mathbb{Z}^3$ on the ray of v, then $e = |I_G(F)|$ for the monomial prime divisor F over $\mathbb{A}^3$ with weights ev, $\operatorname{Stab}_G(F) = G$, and $a(F, \mathbb{A}^3, B) = e\psi(v)$. In particular, for any DCC set $\Gamma \subset [0, 1]$, the numbers $\operatorname{mld}((\mathbb{A}^3 \ni 0, B)/G)$ for such data with $b_j \in \Gamma$ satisfy the ACC.

Proof. Assume (a), and let $w \in \mathbb{R}^3_{>0}$. By (a) with $|J| = 1$, the differential of $\operatorname{in}_w(f_j)$ does not vanish at any zero of $\operatorname{in}_w(f_j)$ in $(\mathbb{C}^*)^3$, so $\operatorname{div}(\operatorname{in}_w(f_j))$ is a reduced smooth divisor on $(\mathbb{C}^*)^3$. For a point $p \in (\mathbb{C}^*)^3$, let J_p be the set of the j with $\operatorname{in}_w(f_j)(p) = 0$. By (a) with $J = J_p$, the differentials of the $\operatorname{in}_w(f_j)$, $j \in J_p$, at p are linearly independent. Thus $\sum_j \operatorname{div}(\operatorname{in}_w(f_j))$ is a simple normal crossing divisor on $(\mathbb{C}^*)^3$. Since $b_j \leq 1$ for any j, Lemma 2.13, applied at the generic point of the center of any prime divisor over $(\mathbb{C}^*)^3$ with regular parameters containing local equations of the components of B_w through it, shows that $((\mathbb{C}^*)^3, B_w)$ is lc. By Lemma 8.6(1), (b) holds for any fan Σ as in Set-up 8.4(4), and such a fan exists [CLS11, Theorem 11.1.9].

Assume (b) for a fan Σ. Then Lemma 8.6(2), applied to Σ, implies the formula for $\operatorname{mld}((\mathbb{A}^3 \ni 0, B)/G)$, the attainment at a primitive vector of N_G, and the properties of E_v.

Let v, e, and F be as in the statement. For $g \in G$, g^* multiplies any monomial by a non-zero constant, so $\operatorname{ord}_F(g^*\phi) = \operatorname{ord}_F(\phi)$ for any polynomial ϕ by Definition 2.16. Hence $\operatorname{Stab}_G(F) = G$. By Lemma 8.5, $e = |I_G(F)|$ and $a(F, \mathbb{A}^3, B) = e\psi(v)$.

Finally, let $\Gamma \subset [0, 1]$ be a DCC set. If $\Gamma \backslash \{0\} = \emptyset$, then $s = 0$, and $\operatorname{mld}((\mathbb{A}^3 \ni 0, B)/G) = \psi_{\min}(\bar{G}, (1), 0)$, where $\bar{G}$ is the image of G in $\mathbb{T}^3$ and (1) is the unit ideal, and we let $s_0 := 1$. Otherwise, $\gamma(\Gamma) > 0$ by Lemma 2.10(2). Let E_0 be the exceptional divisor of the blow-up of $0 \in \mathbb{A}^3$. Since $(\mathbb{A}^3, B)$ is lc near 0, $\operatorname{mult}_0 f_j \geq 1$, and $b_j \geq \gamma(\Gamma)$ for any j, [KM98, Lemma 2.29] implies

$$0 \leq a(E_0, \mathbb{A}^3, B) = 3 - \sum_{j=1}^{s} b_j \operatorname{mult}_0 f_j \leq 3 - s\gamma(\Gamma).$$

Thus $s \le s_0 := \lfloor 3/\gamma(\Gamma) \rfloor$. Let $J_j := J(f_j)$ and $c_j := b_j$ for $j \le s$, and $J_j := (1)$ and $c_j := 0$ for $s < j \le s_0$. Since $\mathrm{wt}_w((1)) = 0$ and $\mathrm{wt}_w(J(f_j)) = \mathrm{wt}_w(f_j)$, we have $\psi = \psi_{J,c}$ and

$$\mathrm{mld}((\mathbb{A}^3 \ni 0, B)/G) = \psi_{\min}(\bar{G}, (J_1, \dots, J_{s_0}), (c_1, \dots, c_{s_0})).$$

Since $\Gamma \cup \{0\}$ is a DCC set, Theorem 4.5 with $d = 3$ and the fixed number s_0 of ideals implies the ACC. □

Remark 8.9. For $s = 1$, condition (a) of Proposition 8.8 is the non-degeneracy of f_1 with respect to the compact faces of the Newton polyhedron [Amb03, Definition 1.1]. The version for all faces and the corresponding embedded toric resolution are in [IP01, Definition 2.6, Proposition 2.7]. For $s \ge 2$, condition (a) also requires the leading forms of different f_j to intersect transversally in $(\mathbb{C}^*)^3$. In Proposition 8.8, the pair $(\mathbb{A}^3, B)$ is only assumed to be lc and s is arbitrary. Theorem 16.4 and Example 16.10 use Proposition 8.8 for lc pairs whose boundaries contain coordinate hyperplanes.

The lattice formula depends on the choice of eigenparameters. In the following example, the minimum of $a(F, X, B)/|I_G(F)|$ over the monomial prime divisors F in x, y, z is $\frac{5}{7}$, while the minimal log discrepancy $\frac{3}{7}$ of the quotient is attained in the eigenparameters $x + y$, $x - y$, z, in which the equation satisfies condition (a) of Proposition 8.8. The normal forms and the translations in the later sections are used to choose such eigenparameters (cf. the introduction).

Example 8.10. Let $X = \mathbb{A}^3$ with coordinates x, y, z, $h := (x + y)^3 + z^4 + (x - y)^{24}$, $B := \frac{1}{2}\,\mathrm{div}(h)$, and $G := \mu_7(1, 1, 6)$. Then (X, B) is terminal, $2B$ is Cartier, G acts on the generator $(dx \wedge dy \wedge dz)^{\otimes 2}/h$ of $\mathcal{O}_X(2(K_X + B))$ by a faithful character, G acts freely on $X\backslash\{0\}$, and $\mathrm{mld}((X \ni 0, B)/G) = \frac{3}{7}$. The minimum is attained by the prime divisor under the monomial valuation with weights $(8, 1, 6)$ in the eigenparameters $x+y$, $x-y$, z, whose inertia group is G and whose log discrepancy with respect to $(X, 0)$ is 15. However, the minimum of $a(F, X, B)/|I_G(F)|$ over the monomial prime divisors F in x, y, z with center 0 is $\frac{5}{7}$, attained by the weights $(4, 4, 3)$.

Proof of Example 8.10. The proof is long, so we divide it into several steps.

Step 1. In this step we compute the characters and the singular locus of $\mathrm{div}(h)$.

Let $\zeta := e^{2\pi\sqrt{-1}/7}$ and let $g := (\zeta, \zeta, \zeta^6)$ be the generator of G. Then $g^*(x + y) = \zeta(x + y)$, $g^*(x - y) = \zeta(x - y)$, and $g^*z = \zeta^6 z$, so $x + y$, $x - y$, z are G-eigenparameters at 0, on which G acts as $\mu_7(1, 1, 6)$. The monomials $(x+y)^3$, z^4, $(x-y)^{24}$ of h in $x+y$, $x-y$, z have the characters $g \to \zeta^3$, $g \to \zeta^{24} = \zeta^3$, and $g \to \zeta^{24} = \zeta^3$, so $g^*h = \zeta^3 h$. Since $g^*(dx\wedge dy\wedge dz) = \zeta^{1+1+6} dx\wedge dy\wedge dz$ and $2(1 + 1 + 6) - 3 = 13 \equiv 6 \bmod 7$,

$$g^*\Big(\frac{(dx \wedge dy \wedge dz)^{\otimes 2}}{h}\Big) = \zeta^6 \frac{(dx \wedge dy \wedge dz)^{\otimes 2}}{h}.$$

The form $(dx \wedge dy \wedge dz)^{\otimes 2}/h$ generates $\mathcal{O}_X(2(K_X + B)) = \mathcal{O}_X(2K_X + \mathrm{div}(h))$, $2B = \mathrm{div}(h)$ is Cartier, and the character $g \to \zeta^6$ is faithful since $\gcd(6, 7) = 1$. For $1 \le k \le 6$, g^k multiplies x, y, z by ζ^k, ζ^k, ζ^{6k}, which are all different from 1, so the fixed locus of g^k is $\{0\}$ and G acts freely on $X\backslash\{0\}$. The partial derivatives of h with respect to $x + y$, $x - y$, z are $3(x + y)^2$, $24(x - y)^{23}$, $4z^3$, which vanish simultaneously only at 0. Thus $\mathrm{div}(h)$ is reduced, and smooth on $X\backslash\{0\}$.

Step 2. In this step we show that (X, B) is terminal.

For $w = (w_1, w_2, w_3) \in \mathbb{R}^3_{\ge 0}$, the w-weights of $(x + y)^3$, $(x - y)^{24}$, z^4 in the eigenparameters $x + y$, $x - y$, z are $3w_1$, $24w_2$, $4w_3$. Let

$$\psi(w) := w_1 + w_2 + w_3 - \tfrac{1}{2}\min\{3w_1, 24w_2, 4w_3\},$$

which is the function ψ of Set-up 8.4 for $d = 3$, the coordinates $x + y$, $x - y$, z, $s = 1$, $f_1 = h$, and $b_1 = \frac{1}{2}$. Since $\min\{3w_1, 24w_2, 4w_3\} \le \min\{3w_1, 4w_3\}$, we have $\psi(w) \ge w_2 + w_1 + w_3 -$

$\frac{1}{2}\min\{3w_1, 4w_3\}$. If $3w_1 \le 4w_3$, then $w_1 + w_3 - \frac{3}{2}w_1 = w_3 - \frac{1}{2}w_1 \ge \frac{1}{3}w_3$, and if $4w_3 \le 3w_1$, then $w_1 + w_3 - 2w_3 = w_1 - w_3 \ge \frac{1}{3}w_3$. Thus

$$\psi(w) \ge w_2 + \tfrac{1}{3}w_3 \quad \text{for any } w \in \mathbb{R}^3_{\ge 0}. \tag{8.4}$$

For a primitive $w \in \mathbb{Z}^3_{\ge 0}\backslash\{0\}$, let F_w be the monomial prime divisor over X in $x+y$, $x-y$, z with weights w. By [Amb06, Section 2, p. 360] and Definition 2.16,

$$a(F_w, X, B) = a(F_w, X, 0) - \tfrac{1}{2}\operatorname{ord}_{F_w}(h) = w_1 + w_2 + w_3 - \tfrac{1}{2}\operatorname{wt}_w(h) = \psi(w).$$

If $w \in \mathbb{Z}^3_{>0}$, then $\psi(w) \ge 1 + \frac{1}{3} = \frac{4}{3}$ by (8.4). If w has a zero coordinate and is not on a coordinate ray, then $\operatorname{wt}_w(h) = 0$ and w has at least two positive integral coordinates, so $\psi(w) \ge 2$. If w is on a coordinate ray, then $\psi(w) = 1$.

In the eigenparameters $x+y$, $x-y$, z, any face polynomial of the Newton polyhedron of h, compact or not, is a non-empty sum of some of $(x+y)^3$, $(x-y)^{24}$, z^4. On $(\mathbb{C}^*)^3$, such a face polynomial has no zero if it has one term, and otherwise one of the partial derivatives $3(x+y)^2$, $24(x-y)^{23}$, $4z^3$ of the face polynomial has no zero. By [IP01, Definition 2.6, Proposition 2.7], there exist a regular subdivision Σ_0 of $\mathbb{R}^3_{\ge 0}$ with respect to $\mathbb{Z}^3$ and a neighborhood X_0 of 0 such that the toric morphism $\mu_0 : Y_0 \to X$ of Σ_0 in the coordinates $x+y$, $x-y$, z is an embedded resolution of $(X, \operatorname{div}(h))$ over X_0. Here X_0 exists since the locus where the snc condition fails is closed, and its image under the proper morphism μ_0 does not contain 0. Thus $\mu_0^{-1}(\operatorname{div}(h)) \cup \operatorname{Exc}(\mu_0)$ is a simple normal crossing divisor over X_0. Write $K_{Y_0} + B_{Y_0} := \mu_0^*(K_X + B)$. The toric prime divisors of Y_0 are the F_w for the primitive ray generators w of Σ_0, and $\operatorname{mult}_{F_w} B_{Y_0} = 1 - \psi(w)$, which is 0 if w is on a coordinate ray and at most $-\frac{1}{3}$ otherwise. Hence

$$B_{Y_0} = \tfrac{1}{2}h' + \sum_w (1 - \psi(w))F_w,$$

where h' is the strict transform of $\operatorname{div}(h)$ and w runs over the primitive ray generators of Σ_0 not on coordinate rays. All the coefficients of B_{Y_0} are at most $\frac{1}{2}$.

Let E be an exceptional prime divisor over X, and let $c \ge 2$ be the codimension of the center of E on X. If the center of E on X does not contain 0, then near the generic point of the center, $\operatorname{div}(h)$ is smooth by **Step 1**, and Lemma 2.13 implies $a(E, X, B) \ge \frac{1}{2} + (c-1) \ge \frac{3}{2}$. Assume that the center of E on X contains 0, and let η be the generic point of the center of E on Y_0, so $\mu_0(\eta) \in X_0$. If η is the generic point of a divisor, then $E = F_w$ for w not on a coordinate ray, and $a(E, X, B) = \psi(w) \ge \frac{4}{3}$. Otherwise, choose regular parameters $y_1, \dots, y_{c'}$ of $\mathcal{O}_{Y_0,\eta}$, $c' \ge 2$, containing local equations of the components of $\operatorname{Supp} B_{Y_0}$ through η, and let $\beta_i := \operatorname{mult}_{(y_i=0)} B_{Y_0} \le \frac{1}{2}$. By Lemma 2.13,

$$a(E, X, B) = a(E, Y_0, B_{Y_0}) \ge \sum_{i=1}^{c'} (1 - \beta_i)\operatorname{ord}_E(y_i).$$

If $\eta \in F_w$ for some w not on a coordinate ray, then the term of a local equation of F_w is at least $\psi(w) \ge \frac{4}{3}$, and the other terms are non-negative. Otherwise, $\beta_i \in \{0, \frac{1}{2}\}$ for any i, and $\beta_i = \frac{1}{2}$ for at most one i since h' is smooth near η. Thus $a(E, X, B) \ge \frac{1}{2} + (c' - 1) \ge \frac{3}{2}$. Hence $a(E, X, B) > 1$ for any exceptional prime divisor E over X, and (X, B) is terminal.

Step 3. In this step we show that $\operatorname{mld}((X \ni 0, B)/G) = \frac{3}{7}$ and that the monomial prime divisor with weights $(8, 1, 6)$ in $x+y$, $x-y$, z attains it.

We check the hypotheses of Proposition 8.8 in the coordinates $x+y$, $x-y$, z, with $s = 1$, $f_1 = h$, and $b_1 = \frac{1}{2}$. By **Step 1**, $G = \mu_7(1, 1, 6)$ in $x+y$, $x-y$, z, and h is a reduced G-semi-invariant polynomial vanishing at 0, and (X, B) is lc by **Step 2**. For $w \in \mathbb{R}^3_{>0}$, $\operatorname{in}_w(h)$ is a non-empty sum of some of $(x+y)^3$, $(x-y)^{24}$, z^4, so condition (a) of Proposition 8.8 holds by **Step 2**. Since $N_G = \mathbb{Z}^3 + \mathbb{Z} \cdot \frac{1}{7}(1, 1, 6)$, Proposition 8.8 implies

$$\operatorname{mld}((X \ni 0, B)/G) = \min_{w \in N_G \cap \mathbb{R}^3_{>0}} \psi(w).$$

Let $w \in N_G \cap \mathbb{R}^3_{>0}$. If $w \in \mathbb{Z}^3$, then $\psi(w) \geq \frac{4}{3}$ by **Step 2**. Otherwise, $w \equiv \frac{k}{7}(1,1,6) \bmod \mathbb{Z}^3$ for some $k \in \{1,\dots,6\}$. Since $6k \equiv -k \bmod 7$ and $w \in \mathbb{R}^3_{>0}$, we have $w_2 \geq \frac{k}{7}$ and $w_3 \geq \frac{7-k}{7}$. By (8.4),

$$\psi(w) \geq \frac{k}{7} + \frac{7-k}{21} = \frac{1}{3} + \frac{2k}{21} \geq \frac{3}{7}.$$

Let $v := \frac{1}{7}(8,1,6) = (1,0,0) + \frac{1}{7}(1,1,6) \in N_G \cap \mathbb{R}^3_{>0}$. Since $N_G \subset \frac{1}{7}\mathbb{Z}^3$ and $\gcd(8,1,6) = 1$, v is primitive in N_G and $7v = (8,1,6)$ is primitive in $\mathbb{Z}^3$. Moreover,

$$\psi(v) = \frac{8+1+6}{7} - \frac{1}{2} \cdot \frac{\min\{24,24,24\}}{7} = \frac{15}{7} - \frac{12}{7} = \frac{3}{7}.$$

Thus $\mathrm{mld}((X \ni 0, B)/G) = \frac{3}{7}$. Let F be the monomial prime divisor in $x+y$, $x-y$, z with weights $(8,1,6)$. By Lemma 8.5, $|I_G(F)| = 7$, that is, $I_G(F) = G$, and $a(F,X,B) = 7\psi(v) = 3$. Thus $a(F,X,B)/|I_G(F)| = \frac{3}{7}$, so F attains the minimum by Lemma 2.6. Moreover, $a(F,X,0) = 8+1+6 = 15$ [Amb06, Section 2, p. 360].

Step 4. In this step we compute the minimum over the monomial prime divisors in x, y, z.

Let $w = (w_1, w_2, w_3) \in \mathbb{R}^3_{>0}$ be weights of x, y, z. The monomials of $(x+y)^3$ are x^3, x^2y, xy^2, y^3, with non-zero coefficients, the monomials of $(x-y)^{24}$ have w-weight at least $24\min\{w_1, w_2\}$, and no monomial occurs in two of $(x+y)^3$, $(x-y)^{24}$, z^4. Hence $\mathrm{wt}_w(h) = \min\{3\min\{w_1,w_2\}, 4w_3\}$. Let

$$\psi'(w) := w_1 + w_2 + w_3 - \tfrac{1}{2}\min\{3\min\{w_1,w_2\}, 4w_3\}.$$

Let F be a monomial prime divisor over X in x, y, z with center 0, and let $w' \in \mathbb{Z}^3_{>0}$ be the weights of F, which are primitive in $\mathbb{Z}^3$. Let v be the primitive vector of N_G on $\mathbb{R}_{\geq 0}w'$, and e the positive integer with $w' = ev$. Lemma 8.5 with $s = 0$ in the coordinates x, y, z implies $e = |I_G(F)|$. By [Amb06, Section 2, p. 360] and Definition 2.16, $a(F,X,B) = \sum_i w'_i - \frac{1}{2}\mathrm{wt}_{w'}(h)$, so

$$\frac{a(F,X,B)}{|I_G(F)|} = \psi'(v).$$

Conversely, any primitive $v \in N_G \cap \mathbb{R}^3_{>0}$ arises in this way. Thus it suffices to show that $\min_{v \in N_G \cap \mathbb{R}^3_{>0}} \psi'(v) = \frac{5}{7}$, attained at $\frac{1}{7}(4,4,3)$.

Let $w \in N_G \cap \mathbb{R}^3_{>0}$, and let $t := \min\{w_1, w_2\}$. Replacing the larger of w_1, w_2 with t does not change the last term of $\psi'(w)$, so

$$\psi'(w) \geq 2t + w_3 - \tfrac{1}{2}\min\{3t, 4w_3\} = \max\{\tfrac{1}{2}t + w_3, 2t - w_3\}.$$

If $w \in \mathbb{Z}^3$, then $t \geq 1$ and $w_3 \geq 1$, so $\psi'(w) \geq \frac{3}{2}$. Otherwise, $w \equiv \frac{k}{7}(1,1,6) \bmod \mathbb{Z}^3$ for some $k \in \{1,\dots,6\}$. Let $p := \frac{k}{7}$. Then $w_1 \equiv w_2 \equiv p$ and $w_3 \equiv 1 - p \bmod \mathbb{Z}$, so $t \geq p$ and $w_3 \in \{1-p, 2-p, \dots\}$. If $w_3 \geq 2-p$, then $\psi'(w) \geq \frac{1}{2}t + w_3 > 1$. If $w_3 = 1-p$, then, since both $\frac{1}{2}t + w_3$ and $2t - w_3$ are increasing in t,

$$\psi'(w) \geq \max\{1 - \tfrac{1}{2}p, 3p - 1\} \geq \frac{5}{7},$$

because $1 - \frac{1}{2}p \geq \frac{5}{7}$ for $p \leq \frac{4}{7}$ and $3p - 1 \geq \frac{5}{7}$ for $p \geq \frac{4}{7}$. Thus $\psi'(w) \geq \frac{5}{7}$ for any $w \in N_G \cap \mathbb{R}^3_{>0}$.

Finally, $\frac{1}{7}(4,4,3) \equiv \frac{4}{7}(1,1,6) \bmod \mathbb{Z}^3$, so $\frac{1}{7}(4,4,3) \in N_G \cap \mathbb{R}^3_{>0}$. Since $N_G \subset \frac{1}{7}\mathbb{Z}^3$ and $\gcd(4,4,3) = 1$, $\frac{1}{7}(4,4,3)$ is primitive in N_G and $(4,4,3)$ is primitive in $\mathbb{Z}^3$. For the monomial prime divisor F in x, y, z with weights $(4,4,3)$, we obtain $|I_G(F)| = 7$, that is, $I_G(F) = G$, and

$$\frac{a(F,X,B)}{|I_G(F)|} = \frac{4+4+3-\frac{1}{2}\min\{12,12\}}{7} = \frac{5}{7}.$$

□

8.3. **Log canonical leading forms.** In this subsection, we apply Lemma 8.6 to quotients of smooth and isolated Gorenstein terminal threefold germs. The results of this subsection are not used in the proofs of the main theorems of this paper.

The following proposition replaces condition (a) of Proposition 8.8 by log canonicity of the pairs $((\mathbb{C}^*)^3, B_w)$.

Proposition 8.11. *Notation as in Set-up 8.4 with $d = 3$, and assume that $f_1, \dots, f_s$ are reduced and pairwise coprime as germs at 0 and that $b_j \le 1$ for any j. Let $\bar{0} \in \mathbb{A}^3/G$ be the image of 0.*

(1) $((\mathbb{C}^*)^3, B_w)$ is lc for any $w \in \mathbb{R}^3_{>0}$ if and only if $(V_\Sigma, \Delta_\Sigma + B'_\Sigma)$ is lc near $\mu_\Sigma^{-1}(\bar{0})$ for one Σ, if and only if $(V_\Sigma, \Delta_\Sigma + B'_\Sigma)$ is lc near $\mu_\Sigma^{-1}(\bar{0})$ for any Σ.

(2) Assume that $((\mathbb{C}^*)^3, B_w)$ is lc for any $w \in \mathbb{R}^3_{>0}$. Then $(\mathbb{A}^3, B)$ is lc near 0 if and only if $(\mathbb{A}^3/G, B_{\mathbb{A}^3/G})$ is lc near $\bar{0}$, if and only if $\psi \ge 0$ on $\mathbb{R}^3_{\ge 0}$.

(3) Assume that $((\mathbb{C}^*)^3, B_w)$ is lc for any $w \in \mathbb{R}^3_{>0}$ and that $(\mathbb{A}^3, B)$ is lc near 0. Then $\operatorname{mld}((\mathbb{A}^3 \ni 0, B)/G) = \min_{v \in N_G \cap \mathbb{R}^3_{>0}} \psi(v)$, attained by a toric prime divisor, and for any DCC set $\Gamma \subset [0,1]$ the set of such numbers with $b_j \in \Gamma$ satisfies the ACC.

Proof of Proposition 8.11. (1) Regular subdivisions Σ as in Set-up 8.4(4) exist [CLS11, Theorem 11.1.9]. By Lemma 8.6(1), for any Σ, $((\mathbb{C}^*)^3, B_w)$ is lc for any $w \in \mathbb{R}^3_{>0}$ if and only if $(V_\Sigma, \Delta_\Sigma + B'_\Sigma)$ is lc near $\mu_\Sigma^{-1}(\bar{0})$. Hence the three conditions in (1) are equivalent.

(2) By (1), $(V_\Sigma, \Delta_\Sigma + B'_\Sigma)$ is lc near $\mu_\Sigma^{-1}(\bar{0})$ for any Σ, and we fix one Σ. Let $\pi : \mathbb{A}^3 \to \mathbb{A}^3/G$ be the quotient morphism. Since π is finite, $\pi^{-1}(\bar{0}) = \{0\}$, and $K_{\mathbb{A}^3} + B = \pi^*(K_{\mathbb{A}^3/G} + B_{\mathbb{A}^3/G})$, the pair $(\mathbb{A}^3, B)$ is lc near 0 if and only if $(\mathbb{A}^3/G, B_{\mathbb{A}^3/G})$ is lc near $\bar{0}$ by Lemma 2.6(2)(3).

Assume that $(\mathbb{A}^3, B)$ is lc near 0. By Lemma 8.6(2), any primitive $v \in N_G \cap \mathbb{R}^3_{>0}$ defines a toric prime divisor E_v over $\mathbb{A}^3/G$ with center $\bar{0}$ and $a(E_v, \mathbb{A}^3/G, B_{\mathbb{A}^3/G}) = \psi(v)$. Since $(\mathbb{A}^3/G, B_{\mathbb{A}^3/G})$ is lc near $\bar{0}$, $\psi(v) \ge 0$. Since $\psi(tv) = t\psi(v)$ for any $t \ge 0$, $\psi \ge 0$ on $\mathbb{Q}^3_{>0} \subset \mathbb{Q}_{>0} \cdot (N_G \cap \mathbb{R}^3_{>0})$, which is dense in $\mathbb{R}^3_{\ge 0}$. Each $\operatorname{wt}_w(f_j)$ is the minimum of finitely many linear functions of w, so ψ is continuous, and $\psi \ge 0$ on $\mathbb{R}^3_{\ge 0}$.

Conversely, assume that $\psi \ge 0$ on $\mathbb{R}^3_{\ge 0}$. By Lemma 8.5,

$$\mu_\Sigma^*(K_{\mathbb{A}^3/G} + B_{\mathbb{A}^3/G}) = K_{V_\Sigma} + B_{V_\Sigma}, \quad B_{V_\Sigma} := \Delta_\Sigma + B'_\Sigma - \sum_v \psi(v) E_v \le \Delta_\Sigma + B'_\Sigma,$$

where v runs over the primitive generators in N_G of the rays of Σ. There exists an open neighborhood W of $\mu_\Sigma^{-1}(\bar{0})$ such that $(W, \Delta_\Sigma + B'_\Sigma)$ is lc. Since V_Σ is smooth, each E_v is Cartier, so $a(E, V_\Sigma, B_{V_\Sigma}) \ge a(E, V_\Sigma, \Delta_\Sigma + B'_\Sigma) \ge 0$ for any prime divisor E over V_Σ whose center intersects W [KM98, Lemma 2.27]. Since μ_Σ is proper, $\mu_\Sigma(V_\Sigma \backslash W)$ is closed and does not contain $\bar{0}$. Let $U' := (\mathbb{A}^3/G) \backslash \mu_\Sigma(V_\Sigma \backslash W)$. For any prime divisor E over $\mathbb{A}^3/G$ whose center intersects U', the generic point of the center of E on V_Σ belongs to $\mu_\Sigma^{-1}(U') \subset W$, so

$$a(E, \mathbb{A}^3/G, B_{\mathbb{A}^3/G}) = a(E, V_\Sigma, B_{V_\Sigma}) \ge 0$$

by [KM98, Lemma 2.30]. Thus $(U', B_{\mathbb{A}^3/G}|_{U'})$ is lc, and $\bar{0} \in U'$.

(3) By (1), $(V_\Sigma, \Delta_\Sigma + B'_\Sigma)$ is lc near $\mu_\Sigma^{-1}(\bar{0})$. By Lemma 8.6(2),

$$\operatorname{mld}((\mathbb{A}^3 \ni 0, B)/G) = \min_{v \in N_G \cap \mathbb{R}^3_{>0}} \psi(v),$$

and the minimum is attained at a primitive $v \in N_G \cap \mathbb{R}^3_{>0}$, whose toric prime divisor E_v has center $\bar{0}$ and $a(E_v, \mathbb{A}^3/G, B_{\mathbb{A}^3/G}) = \psi(v)$.

We prove the ACC. Let $\Gamma \subset [0,1]$ be a DCC set. If $\Gamma \backslash \{0\} = \emptyset$, then $s = 0$ since $b_j > 0$, and we let $m := 1$. Otherwise $\gamma(\Gamma) > 0$, and we let $m := \lfloor 3/\gamma(\Gamma) \rfloor$. If $\Gamma \backslash \{0\} \ne \emptyset$, let E_0 be the exceptional divisor of the blow-up of $\mathbb{A}^3$ at 0. Since $(\mathbb{A}^3, B)$ is lc near 0 and $\operatorname{mult}_0 f_j \ge 1$ as

$f_j(0) = 0$,

$$0 \le a(E_0, \mathbb{A}^3, B) = 3 - \sum_{j=1}^{s} b_j \operatorname{mult}_0 f_j \le 3 - s\gamma(\Gamma)$$

[KM98, Lemma 2.29], so $s \le m$. Let $J_l := J(f_l)$ and $c_l := b_l$ for $1 \le l \le s$, and let $J_l := (1)$ and $c_l := 0$ for $s < l \le m$. Since $\operatorname{wt}_w((1)) = 0$, we have $\psi = \psi_{J,c}$ on $\mathbb{R}^3_{>0}$ for $J := (J_1, \dots, J_m)$ and $c := (c_1, \dots, c_m)$. Thus

$$\operatorname{mld}((\mathbb{A}^3 \ni 0, B)/G) = \psi_{\min}(\bar{G}, J, c),$$

where $\bar{G} \subset \mathbb{T}^3$ is the image of G, so that $N_{\bar{G}} = N_G$. Since $c \in (\Gamma \cup \{0\})^m$, $\Gamma \cup \{0\}$ is a DCC set, and m depends only on Γ, the ACC follows from Theorem 4.5 with $d = 3$ and m ideals. □

Remark 8.12. In Proposition 8.11, the assumptions that $f_1, \dots, f_s$ are reduced and pairwise coprime and that $b_j \le 1$ are a normalization of B. Let Σ be as in Set-up 8.4(4), $\pi : \mathbb{A}^3 \to \mathbb{A}^3/G$ the quotient morphism, and, for each j, D_j the strict transform on V_Σ of the closure in $\mathbb{A}^3/G$ of $\frac{1}{|G|}\pi_*(\operatorname{div}(f_j)|_{(\mathbb{C}^*)^3})$. Then, since a non-identity element of G has no fixed point on $(\mathbb{C}^*)^3$, so that $R(\pi)$ does not meet $(\mathbb{C}^*)^3$ and $\frac{1}{|G|}\pi_*$ maps the sum of the prime divisors in a G-orbit on $(\mathbb{C}^*)^3$ to the image of the orbit with coefficient one (Definition 2.5),

$$B'_\Sigma = \sum_j b_j D_j,$$

and under the assumptions of Proposition 8.11 each b_j is at most 1, and the D_j are reduced and have no common components near $\mu_\Sigma^{-1}(\bar{0})$, since a component of D_j meeting $\mu_\Sigma^{-1}(\bar{0})$ is the strict transform of the image of a component of $\operatorname{div}(f_j)$ through 0.

Let $\Gamma \subset [0,1]$ be a DCC set, and let B be as in Set-up 8.4 with $d = 3$ and $b_j \in \Gamma$, such that $(\mathbb{A}^3, B)$ is lc near 0. The products $g_1, \dots, g_t$ over the G-orbits of the irreducible factors of the f_j vanishing at 0 are reduced, pairwise coprime, G-semi-invariant polynomials vanishing at 0, and $B = \sum_l c_l \operatorname{div}(g_l)$ near 0, where $c_1, \dots, c_t \in \Gamma_+ \cap [0,1]$ (Lemma 2.10(5)). Since in_w is multiplicative, wt_w is additive, and the factors not vanishing at 0 have constant leading forms for $w \in \mathbb{R}^3_{>0}$, the $\mathbb{R}$-divisors B_w and the function ψ for the data g_l, c_l are equal to those for the data f_j, b_j. Since $\Gamma_+ \cap [0,1]$ satisfies the DCC by Lemma 2.10(1), the ACC in Proposition 8.11(3) also holds for such B.

The polynomial f of the following example satisfies the assumptions of Proposition 8.11(3) but not condition (a) of Proposition 8.8.

Example 8.13. Let $r \ge 3$ be an integer, $0 < b < \frac{1}{2}$ a real number, $f := (x+y)^2 + z^r$, and $G \subset (\mathbb{C}^*)^3$ a finite diagonal subgroup acting faithfully on $\mathbb{A}^3$ such that f is G-semi-invariant. Then $(\mathbb{A}^3, b\operatorname{div}(f))$ is terminal, the data $s = 1$, $f_1 = f$, and $b_1 = b$ of Set-up 8.4 with $d = 3$ satisfy the assumptions of Proposition 8.11(3), and

$$\operatorname{mld}((\mathbb{A}^3 \ni 0, b\operatorname{div}(f))/G) = \min_{v \in N_G \cap \mathbb{R}^3_{>0}} \left(v_1 + v_2 + v_3 - b\min\{2v_1, 2v_2, rv_3\}\right),$$

while the differential of $\operatorname{in}_{(1,1,1)}(f) = (x+y)^2$ vanishes along $(x+y=0) \cap (\mathbb{C}^*)^3$.

Proof of Example 8.13. The proof is long, so we divide it into several steps.

Step 1. In this step we compute the leading forms of f and show that the pairs $((\mathbb{C}^*)^3, b\operatorname{div}(\operatorname{in}_w(f)))$ are klt.

Write χ_x, χ_y, χ_z for the characters of x, y, z. Since G is diagonal, f is G-semi-invariant if and only if xy, y^2, and z^r have the character χ_x^2 of x^2, that is, $\chi_x = \chi_y$ and $\chi_x^2 = \chi_z^r$. In particular, such groups G exist: for integers $n \ge 1$ and k with $rk \equiv 2 \bmod n$, the group $\mu_n(1,1,k)$ acts faithfully on $\mathbb{A}^3$ and f is $\mu_n(1,1,k)$-semi-invariant, for example $n = r-2$ and $k = 1$.

Let $x' := x + y$ and $y' := x - y$. Then $f = x'^2 + z^r$, so $\partial f/\partial x' = 2x'$, $\partial f/\partial y' = 0$, and $\partial f/\partial z = rz^{r-1}$. Thus the singular locus of $\operatorname{div}(f)$ is the line

$$C := (x' = z = 0) \subset \mathbb{A}^3.$$

Since $\operatorname{codim}_{\mathbb{A}^3} C = 2$, f is reduced, and $f(0) = 0$.

Let $w \in \mathbb{R}^3_{>0}$. Since $w_1 + w_2 \geq \min\{2w_1, 2w_2\}$, with equality if and only if $w_1 = w_2$, we have

$$\operatorname{wt}_w(f) = \min\{2w_1, 2w_2, rw_3\},$$

and $\operatorname{in}_w(f)$ is a monomial, or $x^2 + z^r$, or $y^2 + z^r$ (if $w_1 \neq w_2$ and $rw_3 = \min\{2w_1, 2w_2\}$), or $(x + y)^2$ (if $w_1 = w_2$ and $2w_1 < rw_3$), or f (if $2w_1 = 2w_2 = rw_3$). The divisor on $(\mathbb{C}^*)^3$ of a monomial is zero. Since $\partial(x^2 + z^r)/\partial x = 2x$, $\partial(y^2 + z^r)/\partial y = 2y$, and $\partial f/\partial z = rz^{r-1}$ do not vanish on $(\mathbb{C}^*)^3$, the divisors on $(\mathbb{C}^*)^3$ of $x^2 + z^r$, $y^2 + z^r$, and f are smooth and reduced. The divisor of $(x + y)^2$ is $2\operatorname{div}(x + y)$ with $\operatorname{div}(x + y)$ smooth and reduced. Since $2b < 1$, $((\mathbb{C}^*)^3, b\operatorname{div}(\operatorname{in}_w(f)))$ is klt [KM98, Corollary 2.31(3)].

We also consider Set-up 8.4 in the coordinates x', y', z, with the trivial group, $s = 1$, $f_1 = f = x'^2 + z^r$, and $b_1 = b$, and we denote by ψ' the function ψ of Set-up 8.4(3) for the data in the coordinates x', y', z. For $w \in \mathbb{R}^3_{>0}$, the leading form of $x'^2 + z^r$ with respect to w is x'^2, z^r, or f, and the divisor of f on $(\mathbb{C}^*)^3$ is smooth and reduced. Thus $((\mathbb{C}^*)^3, b\operatorname{div}(\operatorname{in}_w(x'^2 + z^r)))$ is klt for any $w \in \mathbb{R}^3_{>0}$, where in_w is taken in the coordinates x', y', z. Moreover, for any $w \in \mathbb{R}^3_{\geq 0}\backslash\{0\}$,

$$\psi'(w) = w_1 + w_2 + w_3 - b\min\{2w_1, rw_3\} \geq (1 - 2b)w_1 + w_2 + w_3 > 0.$$

Step 2. In this step we show that $(\mathbb{A}^3, b\operatorname{div}(f))$ is terminal.

Let E be an exceptional prime divisor over $\mathbb{A}^3$. We show that $a(E, \mathbb{A}^3, b\operatorname{div}(f)) > 1$.

Suppose that the center of E on $\mathbb{A}^3$ is not contained in C. On $\mathbb{A}^3\backslash C$, the divisor $\operatorname{div}(f)$ is smooth and $b < 1$, so $a(E, \mathbb{A}^3, b\operatorname{div}(f)) \geq 2 - b > 1$ [KM98, Corollary 2.31(3)].

Suppose that the center of E on $\mathbb{A}^3$ is a closed point $p \in C$, and write $p = (0, c, 0)$ in the coordinates x', y', z. The translation $(x', y', z) \to (x', y' + c, z)$ preserves f and maps 0 to p, so we may assume that $p = 0$. Since $\psi' \geq 0$ on $\mathbb{R}^3_{\geq 0}$, Proposition 8.11(2) for the data in the coordinates x', y', z implies that $(\mathbb{A}^3, b\operatorname{div}(f))$ is lc near 0. By Proposition 8.11(3), with $N_{\{1\}} = \mathbb{Z}^3$,

$$\operatorname{mld}(\mathbb{A}^3 \ni 0, b\operatorname{div}(f)) = \min_{v \in \mathbb{Z}^3_{>0}} \psi'(v) \geq \min_{v \in \mathbb{Z}^3_{>0}} \big((1 - 2b)v_1 + v_2 + v_3\big) = 3 - 2b > 1.$$

Suppose that the center of E on $\mathbb{A}^3$ is C. Let Σ be a regular subdivision of $\mathbb{R}^3_{\geq 0}$ with respect to $\mathbb{Z}^3$ refining the Newton fan of $x'^2 + z^r$ [CLS11, Theorem 11.1.9], and let V_Σ, μ_Σ, Δ_Σ, and B'_Σ be as in Set-up 8.4 for the data in the coordinates x', y', z. By Lemma 8.6(1), there exists an open neighborhood W of $\mu_\Sigma^{-1}(0)$ such that $(W, \Delta_\Sigma + B'_\Sigma)$ is lc. Let V_E be the center of E on V_Σ and η the generic point of V_E. Since μ_Σ is proper, $\mu_\Sigma(V_E) = C \ni 0$, so $V_E \cap W \neq \emptyset$, $\eta \in W$, and $a(E, V_\Sigma, \Delta_\Sigma + B'_\Sigma) \geq 0$. By Lemma 8.5,

$$a(E, \mathbb{A}^3, b\operatorname{div}(f)) = a(E, V_\Sigma, \Delta_\Sigma + B'_\Sigma) + \sum_v \psi'(v)\operatorname{mult}_E E_v \geq \sum_v \psi'(v)\operatorname{mult}_E E_v,$$

where v runs over the primitive generators of the rays of Σ, and each term on the right is non-negative. Let $\tau \in \Sigma$ be the cone with $\eta \in O(\tau)$. Then $\operatorname{mult}_E E_v \geq 1$ for the primitive generator v of any ray of τ, since $O(\tau) \subset E_v$. The point $\mu_\Sigma(\eta)$ is the generic point of C, so x' and z vanish at $\mu_\Sigma(\eta)$ and y' does not. For $m \in \mathbb{Z}^3 \cap \tau^\vee$, the restriction of $x'^{m_1}y'^{m_2}z^{m_3}$ to $O(\tau)$ is non-zero at every point of $O(\tau)$ if $m \in \tau^\perp$, and is zero otherwise [CLS11, Lemma 3.2.5]. Since y' does not vanish at $\eta \in O(\tau)$, we have $(0, 1, 0) \in \tau^\perp$, so $\tau \subset (w_2 = 0)$. Since x' and z vanish at η, we have $(1, 0, 0) \notin \tau^\perp$ and $(0, 0, 1) \notin \tau^\perp$, so w_1 and w_3 are not identically zero on τ. Since they are non-negative on $\tau \subset \mathbb{R}^3_{\geq 0}$, they are positive on the relative interior of τ. Thus $\tau \subset \operatorname{Cone}(e_1, e_3)$, and τ is neither the ray of e_1 nor the ray of e_3. Since Σ refines the Newton fan of $x'^2 + z^r$, the function $\operatorname{wt}_w(x'^2 + z^r) = \min\{2w_1, rw_3\}$ is linear on τ. The function

$\min\{2w_1, rw_3\}$ vanishes at e_1 and at e_3 and equals 2 at e_1+e_3, so $\tau \neq \mathrm{Cone}(e_1, e_3)$. Hence τ has a ray with primitive generator $v=(v_1,0,v_3)$, where v_1, v_3 are positive integers, and

$$a(E, \mathbb{A}^3, b\,\mathrm{div}(f)) \geq \psi'(v)\,\mathrm{mult}_E E_v \geq (1-2b)v_1 + v_3 \geq 2-2b > 1.$$

Thus $(\mathbb{A}^3, b\,\mathrm{div}(f))$ is terminal.

Step 3. In this step we prove the formula for the quotient and the statement on $\mathrm{in}_{(1,1,1)}(f)$.

The polynomial f is reduced, $b \leq 1$, $((\mathbb{C}^*)^3, b\,\mathrm{div}(\mathrm{in}_w(f)))$ is lc for any $w \in \mathbb{R}^3_{>0}$ in the coordinates x, y, z, and $(\mathbb{A}^3, b\,\mathrm{div}(f))$ is terminal, hence lc. Thus the data $s=1$, $f_1=f$, $b_1=b$ satisfy the assumptions of Proposition 8.11(3), and Proposition 8.11(3) implies the formula for $\mathrm{mld}((\mathbb{A}^3 \ni 0, b\,\mathrm{div}(f))/G)$, since

$$\psi(v) = v_1+v_2+v_3 - b\,\mathrm{wt}_v(f) = v_1+v_2+v_3 - b\min\{2v_1, 2v_2, rv_3\}.$$

Finally, $r \geq 3$ implies $\mathrm{wt}_{(1,1,1)}(f)=2$ and $\mathrm{in}_{(1,1,1)}(f) = x^2+2xy+y^2 = (x+y)^2$, whose differential $2(x+y)d(x+y)$ vanishes along $(x+y=0)\cap(\mathbb{C}^*)^3 \ni (1,-1,1)$. □

The proof of the following theorem combines Corollary 7.4, Theorem 8.3 with $c=1$, and Lemma 8.6, and its ACC statement follows from Theorem 4.5.

Theorem 8.14. *Let H be a finite abelian group fixing a point x of a smooth germ Y, and $x_1, \dots, x_d$ H-eigenparameters at x defining an étale morphism $(Y \ni x) \to (\mathbb{A}^d \ni 0)$. Assume either (i) $d=3$ and $X=Y$, or (ii) $d=4$ and $X=(f=0)$ is an isolated Gorenstein terminal threefold germ for a regular H-semi-invariant function f. Assume that H acts faithfully on X. Let $h_1, \dots, h_s$ be regular H-semi-invariant functions on Y, non-zero on X, and $\beta_1, \dots, \beta_s$ positive real numbers such that (X, B) is klt for $B := \sum_k \beta_k\,\mathrm{div}(h_k|_X)$. Let $\hat{h}_k$, and $\hat{f}$ in case (ii), be the expansions of h_k and f in $\mathbb{C}[[x_1, \dots, x_d]]$. Assume that for any $u \in \mathbb{R}^d_{>0}$ the pair $((\mathbb{C}^*)^d, C_u)$ is lc, where*

$$C_u := \sum_k \beta_k\,\mathrm{div}(\mathrm{in}_u(\hat{h}_k)) \quad \textit{in case (i)},$$

$$C_u := \mathrm{div}(\mathrm{in}_u(\hat{f})) + \sum_k \beta_k\,\mathrm{div}(\mathrm{in}_u(\hat{h}_k)) \quad \textit{in case (ii)}.$$

Let

$$\psi(u) := \sum_i u_i - \sum_k \beta_k\,\mathrm{wt}_u(\hat{h}_k) \quad \textit{in case (i)},$$

$$\psi(u) := \sum_i u_i - \mathrm{wt}_u(\hat{f}) - \sum_k \beta_k\,\mathrm{wt}_u(\hat{h}_k) \quad \textit{in case (ii)}.$$

Then

$$\mathrm{mld}((X \ni x, B)/H) = \min_{u \in N_H \cap \mathbb{R}^d_{>0}} \psi(u) > 0,$$

where N_H is the lattice of Definition 2.15 for the characters of $x_1, \dots, x_d$, and the minimum is attained at a primitive vector of N_H. The equality also holds for any subgroup $H' \subset H$ with N_H replaced by the lattice $N_{H'}$ of Definition 2.15 for the characters of $x_1, \dots, x_d$ restricted to H'. In particular, for a fixed s and a DCC set $\Lambda \subset \mathbb{R}_{>0}$, the set of such numbers $\mathrm{mld}((X \ni x, B)/H)$ with $\beta_k \in \Lambda$ satisfies the ACC.

Proof. The proof is long, so we divide it into several steps. Fix a subgroup $H' \subset H$.

Step 1. In this step we reduce to the case when $h_k(x)=0$ for any k.

Let k be an index with $h_k(x) \neq 0$. Then h_k is a unit near x, so $\mathrm{div}(h_k|_X)$ does not contain x, and $\mathrm{div}(h_k|_X)$ is H-invariant. Moreover, $J(\hat{h}_k)$ is the unit ideal, so for any $u \in \mathbb{R}^d_{>0}$ we have $\mathrm{wt}_u(\hat{h}_k)=0$, $\mathrm{in}_u(\hat{h}_k) = h_k(x)$ is a non-zero constant, and $\mathrm{div}(\mathrm{in}_u(\hat{h}_k))=0$ on $(\mathbb{C}^*)^d$. Thus removing h_k from the list $h_1, \dots, h_s$ and replacing Y by the H-invariant open subset $Y \setminus \mathrm{div}(h_k)$

does not change $\mathrm{mld}((X \ni x, B)/H')$, the function ψ, or the $\mathbb{R}$-divisors C_u. Possibly removing all such h_k, we may assume that $h_k(x) = 0$ for any k.

Step 2. In this step we pass to polynomials on $\mathbb{A}^d$.

We apply Construction-Theorem 7.2 with $n = d$, $r = 0$, and $l = 0$. Assumptions (i)–(v) of Construction-Theorem 7.2 hold with $r = l = 0$: in case (i) of the theorem, $X = Y$ is étale over $\mathbb{A}^3$, hence a smooth threefold germ, and in case (ii) X is an isolated Gorenstein terminal threefold germ by assumption. By Corollary 7.4, there exist a faithful diagonal action of H on $\mathbb{A}^d$ in which the coordinates have the characters of $x_1, \dots, x_d$, a germ X_0 at 0, with $X_0 = \mathbb{A}^3$ in case (i) and $X_0 = (f_0 = 0) \subset \mathbb{A}^4$ an isolated Gorenstein terminal germ for a semi-invariant polynomial f_0 in case (ii), and semi-invariant polynomials $h_{1,0}, \dots, h_{s,0}$, non-zero on X_0, satisfying the following, where $B_0 := \sum_k \beta_k \operatorname{div}(h_{k,0}|_{X_0})$.

- (X_0, B_0) is klt near 0, and $\mathrm{mld}((X_0 \ni 0, B_0)/H') = \mathrm{mld}((X \ni x, B)/H')$.
- $\mathrm{in}_u(h_{k,0}) = \mathrm{in}_u(\hat{h}_k)$ for any k and any $u \in \mathbb{R}^d_{>0}$, and $\mathrm{in}_u(f_0) = \mathrm{in}_u(\hat{f})$ for any $u \in \mathbb{R}^d_{>0}$ in case (ii).

By Construction-Theorem 7.2(2), the completions of the local rings of X at x and of X_0 at 0 are H-equivariantly isomorphic. Since H acts faithfully on X, H acts faithfully on X_0. Since the coordinates of $\mathbb{A}^d$ have the characters of $x_1, \dots, x_d$, the lattice $N_{H'}$ defined by the coordinates of $\mathbb{A}^d$ is the lattice $N_{H'}$ in the theorem. Since $h_k(x) = 0$, the series $\hat{h}_k$ has no constant term, so $\mathrm{in}_u(h_{k,0}) = \mathrm{in}_u(\hat{h}_k)$ is not a constant for $u \in \mathbb{R}^d_{>0}$, and hence $h_{k,0}(0) = 0$. In case (ii), $f_0(0) = 0$ since $0 \in X_0$.

We identify H' with the image of H' in $(\mathbb{C}^*)^d$, and we consider Set-up 8.4 with $G = H'$ for the following data: the polynomials $h_{1,0}, \dots, h_{s,0}$ with the coefficients $\beta_1, \dots, \beta_s$ in case (i), and the polynomials $f_0, h_{1,0}, \dots, h_{s,0}$ with the coefficients $1, \beta_1, \dots, \beta_s$ in case (ii). Then the $\mathbb{R}$-divisor B of Set-up 8.4 is

$$B_{\mathbb{A}^d} := \sum_k \beta_k \operatorname{div}(h_{k,0}) \text{ in case (i)}, \qquad B_{\mathbb{A}^d} := \operatorname{div}(f_0) + \sum_k \beta_k \operatorname{div}(h_{k,0}) \text{ in case (ii)}.$$

For $u \in \mathbb{R}^d_{>0}$ and a non-zero series g, $\mathrm{wt}_u(g)$ is the u-weight of $\mathrm{in}_u(g)$, so the equalities of leading forms above imply that the function ψ and the $\mathbb{R}$-divisors B_u of Set-up 8.4 are, on $\mathbb{R}^d_{>0}$, the function ψ and the $\mathbb{R}$-divisors C_u of the theorem.

Step 3. In this step we show that $(\mathbb{A}^d, B_{\mathbb{A}^d})$ is lc near 0 and that $\mathrm{mld}((X \ni x, B)/H') = \mathrm{mld}((\mathbb{A}^d \ni 0, B_{\mathbb{A}^d})/H')$.

In case (i), $X_0 = \mathbb{A}^3$ and $B_0 = B_{\mathbb{A}^3}$, so $(\mathbb{A}^3, B_{\mathbb{A}^3})$ is klt near 0 and the equality of minimal log discrepancies holds by **Step 2**.

In case (ii), X_0 is near 0 a normal integral hypersurface with klt singularities, H acts faithfully on X_0, and (X_0, B_0) is klt near 0. By Theorem 8.3 with $n = 4$, $c = 1$, $G = H$, $f_1 = f_0$, and the polynomials $h_{k,0}$ with the coefficients β_k,

$$\mathrm{mld}((X_0 \ni 0, B_0)/H') = \mathrm{mld}((\mathbb{A}^4 \ni 0, B_{\mathbb{A}^4})/H') > 0.$$

Together with **Step 2**, this proves the equality of minimal log discrepancies. It remains to show that $(\mathbb{A}^4, B_{\mathbb{A}^4})$ is lc near 0. Let $S := \operatorname{div}(f_0)$. Near 0, $S = X_0$ is a normal Cartier divisor on the smooth variety $\mathbb{A}^4$, and $\sum_k \beta_k \operatorname{div}(h_{k,0})$ has no common component with S since each $h_{k,0}$ is non-zero on X_0. Thus the different of $(\mathbb{A}^4, B_{\mathbb{A}^4})$ on S is B_0 [FST11, 11.1]. Since (X_0, B_0) is klt near 0, Theorem 2.23, applied on a neighborhood of 0, implies that $(\mathbb{A}^4, B_{\mathbb{A}^4})$ is lc near 0.

Step 4. In this step we prove the formula for $\mathrm{mld}((X \ni x, B)/H')$.

Let $\bar{0} \in \mathbb{A}^d/H'$ be the image of 0. By assumption and **Step 2**, $((\mathbb{C}^*)^d, B_u) = ((\mathbb{C}^*)^d, C_u)$ is lc for any $u \in \mathbb{R}^d_{>0}$. By Lemma 8.6(1), $(V_\Sigma, \Delta_\Sigma + B'_\Sigma)$ is lc near $\mu_\Sigma^{-1}(\bar{0})$ for any Σ as in Set-up 8.4(4) (such Σ exist [CLS11, Theorem 11.1.9]). By **Step 3** and Lemma 8.6(2),

$$\mathrm{mld}((\mathbb{A}^d \ni 0, B_{\mathbb{A}^d})/H') = \min_{u \in N_{H'} \cap \mathbb{R}^d_{>0}} \psi(u),$$

and the minimum is attained at a primitive vector v of $N_{H'}$ whose toric prime divisor E_v has center $\bar{0}$ and $a(E_v, \mathbb{A}^d/H', B_{\mathbb{A}^d/H'}) = \psi(v)$. By **Step 3**, $\mathrm{mld}((X \ni x, B)/H') = \min_{u \in N_{H'} \cap \mathbb{R}^d_{>0}} \psi(u)$. Moreover, $\psi(v) > 0$: in case (ii) by **Step 3**, and in case (i) because $(\mathbb{A}^3, B_{\mathbb{A}^3})$ is klt near 0, so $(\mathbb{A}^3/H', B_{\mathbb{A}^3/H'})$ is klt near $\bar{0}$ by Lemma 2.6(2)(3), and $\psi(v) = a(E_v, \mathbb{A}^3/H', B_{\mathbb{A}^3/H'}) > 0$. Since $H' \subset H$ is arbitrary, the formula holds for H and for any subgroup of H.

Step 5. In this step we prove the ACC.

By **Step 1**, the formula holds for the functions $h_1, \dots, h_s$ in the theorem, also when $h_k(x) \neq 0$ for some k. Let $\Lambda' := \Lambda \cup \{0, 1\}$, which is a DCC set, and let $\bar{H} \subset \mathbb{T}^d$ be the image of H, so that $N_{\bar{H}} = N_H$. In case (i) let

$$J := ((1), J(\hat{h}_1), \dots, J(\hat{h}_s)), \qquad c := (0, \beta_1, \dots, \beta_s),$$

and in case (ii) let

$$J := (J(\hat{f}), J(\hat{h}_1), \dots, J(\hat{h}_s)), \qquad c := (1, \beta_1, \dots, \beta_s).$$

In both cases J consists of $s+1$ non-zero monomial ideals of $\mathbb{C}[x_1, \dots, x_d]$, where $J(\hat{h}_k) = (1)$ if $h_k(x) \neq 0$, and $c \in \Lambda'^{s+1}$. Since $\mathrm{wt}_u(\hat{h}_k) = \mathrm{wt}_u(J(\hat{h}_k))$, $\mathrm{wt}_u(\hat{f}) = \mathrm{wt}_u(J(\hat{f}))$, and $\mathrm{wt}_u((1)) = 0$ (Definition 2.16), we have $\psi = \psi_{J,c}$ on $\mathbb{R}^d_{>0}$. Thus

$$\mathrm{mld}((X \ni x, B)/H) = \psi_{\min}(\bar{H}, J, c).$$

By Theorem 4.5 with $d = 3$ and with $d = 4$, and with $s+1$ ideals and the coefficient set Λ', the numbers $\psi_{\min}(\bar{H}, J, c)$ with $d = 3$ and those with $d = 4$ satisfy the ACC. Hence the numbers $\mathrm{mld}((X \ni x, B)/H)$ in cases (i) and (ii) together satisfy the ACC. □

When H is trivial and the boundary B is zero, the equality between the minimal log discrepancy of X at x and that of the pair (Y, X) at x, which enters the proof of case (ii) through Theorem 8.3, was proved for $\hat{f}$ non-degenerate with respect to its Newton polyhedron [Amb03, Definition 1.1] in [Amb03, Main Theorem (i)].

9. Binary quadratic hypersurfaces

In this section, we prove Theorem 9.1, whose proof is in Subsection 9.2. A cDV point whose quadratic part has rank at least two becomes a binary quadratic hypersurface after an equivariant étale change of coordinates, and we show that the minimal log discrepancy in $(0, 1)$ of an abelian quotient of such a hypersurface is a lattice Newton minimum. Corollary 9.6 in Subsection 9.2, which allows fixed negative coefficients on the divisors of the eigenparameters z and t, is used in Sections 11 and 12 for cDV points whose quadratic part has rank one.

Theorem 9.1. *Let $(Y \ni o)$ be a smooth fourfold germ, G a finite abelian group acting on Y and fixing o, f a regular G-semi-invariant function on Y vanishing to order at least two at o, $X := (f = 0)$, and $\Gamma \subset [0, 1)$ a DCC set. Assume that X is a normal klt threefold germ at o, that G acts faithfully on X, and that the quadratic part of f has rank at least two. Then the following hold.*

1. *The set of numbers $\mathrm{mld}((X \ni o, 0)/G) \in (0, 1)$ for such data satisfies the ACC.*
2. *If there exist G-eigenparameters u_1, u_2, z, t of Y at o such that the restriction of the quadratic part of f to the (u_1, u_2)-plane is non-degenerate, and real numbers $b_z, b_t < 1$ such that (X, B) is sub-klt for $B := b_z \operatorname{div}_X(z) + b_t \operatorname{div}_X(t)$, then the set of numbers $\mathrm{mld}((X \ni o, B)/G) \in (0, 1)$ for such data with $b_z, b_t \in \Gamma \cup \{0\}$ satisfies the ACC.*
3. *In the situation of (2), or of (1) with $B = 0$ (in which case G-eigenparameters u_1, u_2, z, t as in (2) exist), there exist a smooth surface germ $(V \ni v)$ with an action of G fixing v and G-eigenparameters z, t of V at v, a diagonal action of G on $\mathbb{A}^2_{x,y}$, $Q_2 \in \{xy, x^2 + y^2\}$, a non-zero regular G-semi-invariant function g on V vanishing to order at least two at v, and a germ $(X' \ni o')$ with an action of G and G-equivariant morphisms $(X' \ni o') \to (X \ni o)$ and $(X' \ni o') \to (X_0 \ni o_0)$ étale at o', where $X_0 := (Q_2(x, y) + g = 0) \subset \mathbb{A}^2_{x,y} \times V$ and*

$o_0 := (0,0,v)$, such that the pullbacks to X' of z and t on X coincide with those of z and t on X_0. In particular, the prime divisors over X with center o and the prime divisors over X_0 with center o_0 correspond through X', with equal log discrepancies with respect to B and $B_0 := b_z \operatorname{div}_{X_0}(z) + b_t \operatorname{div}_{X_0}(t)$, isomorphic residue fields and equal inertia groups in G. If $a := \operatorname{mld}((X \ni o, B)/G) \in (0,1)$, then

$$a = \min_{w \in N_G \cap \mathbb{R}^4_{>0}} \Big(\sum_j w_j - \operatorname{wt}_w(Q_2 + \hat{g}) - b_z w_z - b_t w_t \Big),$$

where $\hat{g} \in \mathbb{C}[[z,t]]$ is the expansion of g and N_G is the lattice of Definition 2.15 for the characters of x, y, z, t, and the minimum is attained at a primitive vector.

In particular, by Lemma 5.4, (1) holds for isolated Gorenstein terminal threefold singularities $(X \ni o)$ with a faithful action of a finite abelian group fixing o, provided that the quadratic part of a local equation of X in a smooth fourfold germ has rank at least two. Here the rank of the quadratic part does not depend on the local equation or on the embedding into a smooth fourfold germ, since two such differ by an automorphism of the completed ambient ring and a unit.

The next two lemmas compute characters on inertia groups and compare prime divisors along étale morphisms. We use them for Proposition 9.4, Proposition 9.5, Theorem 9.1, and Theorem 11.7.

Lemma 9.2. *Let X be a normal variety, G a finite abelian group acting on X, F a prime divisor over X, $I := I_G(F)$, and λ the character by which I acts on the line $\mathfrak{m}_F/\mathfrak{m}_F^2$ through the pullback of functions, where $\mathfrak{m}_F$ is the maximal ideal of the valuation ring of ord_F. Then the following hold.*

(1) If $h \in \mathbb{C}(X)$ is a non-zero G-semi-invariant rational function with character χ, then $\chi|_I = \lambda^{\operatorname{ord}_F h}$.

(2) Assume that $X = (f = 0) \subset Y$, where $(Y \ni o)$ is a smooth fourfold germ with an action of G fixing o, $x_1, \dots, x_4$ are G-eigenparameters of Y at o with characters $\chi_1, \dots, \chi_4$, and f is a regular G-semi-invariant function on Y with character χ_f, and that $\operatorname{center}_X F = \{o\}$. Then $(\chi_1\chi_2\chi_3\chi_4\chi_f^{-1})|_I = \lambda^{a(F,X,0)}$.

(3) Assume that G acts faithfully on X and fixes a closed point of X, and let $x_1, \dots, x_d \in \mathbb{C}(X)$ be non-zero G-semi-invariant rational functions. Then $\frac{1}{|I|}(\operatorname{ord}_F x_1, \dots, \operatorname{ord}_F x_d) \in N_G$, where N_G is the lattice of Definition 2.15 for the characters of $x_1, \dots, x_d$.

Proof. Let $\mathcal{O}_F$ be the valuation ring of ord_F, π a uniformizer of $\mathcal{O}_F$, and $\mathfrak{m}_F^k := \pi^k \mathcal{O}_F$ for $k \in \mathbb{Z}$. Any $\sigma \in I$ preserves $\mathcal{O}_F$ and each $\mathfrak{m}_F^k$, induces the identity on $\mathcal{O}_F/\mathfrak{m}_F$, and satisfies $\sigma^*\pi - \lambda(\sigma)\pi \in \mathfrak{m}_F^2$. Thus $\sigma^*\pi = \epsilon_\sigma \pi$ for some $\epsilon_\sigma \in \mathcal{O}_F$ with $\epsilon_\sigma - \lambda(\sigma) \in \mathfrak{m}_F$.

(1) Let $k := \operatorname{ord}_F h$ and write $h = \pi^k u$ with $u \in \mathcal{O}_F^*$. For $\sigma \in I$ we have $\sigma^* u - u \in \mathfrak{m}_F$, hence

$$\chi(\sigma)\pi^k u = \sigma^* h = \epsilon_\sigma^k \pi^k \sigma^* u \equiv \lambda(\sigma)^k \pi^k u \bmod \mathfrak{m}_F^{k+1}.$$

Since $\pi^k u \notin \mathfrak{m}_F^{k+1}$, we obtain $\chi(\sigma) = \lambda(\sigma)^k$ (cf. [Ser79, Chapter IV, §2, Proposition 7]).

(2) Let

$$\Omega := \operatorname{Res}\Big(\frac{dx_1 \wedge dx_2 \wedge dx_3 \wedge dx_4}{f}\Big)$$

be the residue, that is, the regular 3-form on the smooth locus of X near o characterized by $df \wedge \tilde{\Omega} = dx_1 \wedge dx_2 \wedge dx_3 \wedge dx_4$ for local lifts $\tilde{\Omega}$ of Ω (cf. [Kol13, Definition 4.1]). Where $\partial f/\partial x_i \neq 0$, Ω is the restriction to X of $\pm(\partial f/\partial x_i)^{-1} dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_4$, so Ω generates Ω^3_X on the smooth locus of X near o. Since X is normal, Ω generates $\omega_X = \mathcal{O}_X(K_X)$ near o. Any $\sigma \in G$ preserves X and satisfies $\sigma^* x_i = \chi_i(\sigma) x_i$ and $\sigma^* f = \chi_f(\sigma) f$, so the characterization of Ω gives $\sigma^*\Omega = \chi_1\chi_2\chi_3\chi_4\chi_f^{-1}(\sigma)\Omega$. Thus Ω is G-semi-invariant with character $\chi_\Omega := \chi_1\chi_2\chi_3\chi_4\chi_f^{-1}$. Let $a := a(F,X,0)$, an integer because ω_X is invertible. Choose $s_1, s_2 \in \mathcal{O}_F$ whose residues form a transcendence basis of the residue field $\kappa(F)$ over $\mathbb{C}$. Since $\mathcal{O}_F$ is the local ring of a variety at

the generic point of a prime divisor, $\Omega^1_{\mathcal{O}_F/\mathbb{C}}$ is free with basis $d\pi, ds_1, ds_2$, and $\Omega^3_{\mathcal{O}_F/\mathbb{C}}$ is free with basis $d\pi \wedge ds_1 \wedge ds_2$. Since $\operatorname{center}_X F = \{o\}$ and Ω generates ω_X near o, the definition of log discrepancy implies

$$\Omega = \pi^{a-1} r d\pi \wedge ds_1 \wedge ds_2 \quad \text{for some } r \in \mathcal{O}_F^*.$$

Let $\sigma \in I$. Write $\sigma^* s_i = s_i + \pi\eta_i$ with $\eta_i \in \mathcal{O}_F$. Then

$$d(\sigma^* s_i) - ds_i - \eta_i d\pi \in \mathfrak{m}_F \Omega^1_{\mathcal{O}_F/\mathbb{C}}$$

and

$$d(\sigma^*\pi) - \lambda(\sigma) d\pi = (\epsilon_\sigma - \lambda(\sigma)) d\pi + \pi d\epsilon_\sigma \in \mathfrak{m}_F \Omega^1_{\mathcal{O}_F/\mathbb{C}}.$$

Since $d\pi \wedge d\pi = 0$,

$$\sigma^*(d\pi \wedge ds_1 \wedge ds_2) - \lambda(\sigma) d\pi \wedge ds_1 \wedge ds_2 \in \mathfrak{m}_F \Omega^3_{\mathcal{O}_F/\mathbb{C}}.$$

Together with $\sigma^* \pi^{a-1} = \epsilon_\sigma^{a-1} \pi^{a-1}$ and $\sigma^* r - r \in \mathfrak{m}_F$, we obtain

$$\chi_\Omega(\sigma)\Omega = \sigma^*\Omega \equiv \lambda(\sigma)^a \Omega \bmod \pi^a \Omega^3_{\mathcal{O}_F/\mathbb{C}}.$$

Since $\Omega \notin \pi^a \Omega^3_{\mathcal{O}_F/\mathbb{C}}$, we have $\chi_\Omega(\sigma) = \lambda(\sigma)^a$.

(3) Let $e := |I|$ and let χ_i be the character of x_i. By Lemma 2.6(1), I is cyclic and λ is injective, so there exists $h_0 \in I$ with $\lambda(h_0) = \exp(2\pi\sqrt{-1}/e)$. By (1),

$$\chi_i(h_0) = \exp(2\pi\sqrt{-1} \operatorname{ord}_F x_i / e) \quad \text{for any } i.$$

Thus $\frac{1}{e}(\operatorname{ord}_F x_1, \dots, \operatorname{ord}_F x_d) \bmod \mathbb{Z}^d$ is the image of h_0 in $\mathbb{T}^d$, and $\frac{1}{e}(\operatorname{ord}_F x_1, \dots, \operatorname{ord}_F x_d) \in N_G$ by Definition 2.15. □

Lemma 9.3. *Let $\psi : X' \to X$ be a morphism between normal varieties, $x' \in X'$ a closed point, $x := \psi(x')$, and B an $\mathbb{R}$-divisor on X with $K_X + B$ $\mathbb{R}$-Cartier. Assume that ψ is étale at x' and $\psi^{-1}(x) = \{x'\}$. Then for any prime divisor E over X with center x there exists a unique prime divisor E' over X' with center x' such that $\operatorname{ord}_{E'}|_{\mathbb{C}(X)} = \operatorname{ord}_E$, and any prime divisor over X' with center x' arises in this way. Moreover, E and E' have isomorphic residue fields and $a(E', X', \psi^* B) = a(E, X, B)$, and if a finite group G acts on X and X', fixing x and x', and ψ is G-equivariant, then $\operatorname{Stab}_G(E') = \operatorname{Stab}_G(E)$ and $I_G(E') = I_G(E)$.*

Proof. The étale locus of ψ is open, and in the equivariant case it is G-invariant. Since $\psi^{-1}(x) = \{x'\}$, replacing X' with an open neighborhood of x' does not change the prime divisors over X' with center x'. Possibly replacing X' with a (G-invariant) open neighborhood of x', we may assume that ψ is étale.

Let E be a prime divisor over X with center x, and let $\mu : W \to X$ be a projective birational morphism from a normal variety W such that E is a divisor on W. Let $W' := W \times_X X'$ and $\mu' : W' \to X'$ the projection. Since $W' \to W$ is étale, W' is normal. Let $U \subset X$ be a dense open subset over which μ is an isomorphism. Then $W'_U := W' \times_{X'} \psi^{-1}(U)$ is isomorphic to $\psi^{-1}(U)$, which is a non-empty irreducible open subset of X' because ψ is étale, hence dominant. Since $W' \to W$ is étale, it is an open map, so for each irreducible component C of W' the image of C minus the other components is a non-empty open subset of W, so it intersects $\mu^{-1}(U)$. Hence C contains a non-empty open subset of the irreducible open subset W'_U disjoint from the other components, and W' is irreducible. As a base change of the projective morphism μ, the morphism μ' is projective, and it is birational. Since $\mu(E) = \{x\}$ and $\psi^{-1}(x)$ is the reduced point x', the preimage $E' := E \times_X X' = E \times_{\{x\}} \{x'\}$ of E in W' is isomorphic to E. Thus E' is a prime divisor on W' with $\mu'(E') = \{x'\}$. The morphism $W' \to W$ is étale at the generic point of E', so the local ring of W' at the generic point of E' is an unramified extension of the local ring of W at the generic point of E inducing an isomorphism of residue fields. Hence $\operatorname{ord}_{E'}|_{\mathbb{C}(X)} = \operatorname{ord}_E$, and E and E' have isomorphic residue fields. Moreover, the fiber of $W' \to W$ over the generic point ξ of E is $\xi \times_{\{x\}} \{x'\}$, so the generic point of E' is the only point of W' over ξ.

Let E'' be a prime divisor over X' with center x'. Since $\mathbb{C}(X')$ is a finite extension of $\mathbb{C}(X)$, the restriction $\operatorname{ord}_{E''}|_{\mathbb{C}(X)}$ has residue field of transcendence degree $\dim X - 1$ over $\mathbb{C}$, so $\operatorname{ord}_{E''}|_{\mathbb{C}(X)} = c \operatorname{ord}_E$ for a prime divisor E over X and a positive integer c [KM98, Lemma 2.45], and $\operatorname{center}_X E = \psi(x') = x$. Let $\mu : W \to X$ be a projective birational morphism from a normal variety W such that E is a divisor on W, let ξ be the generic point of E, and let $W' := W \times_X X'$, $\mu' : W' \to X'$ the projection, and $E' := E \times_X X'$. By the second paragraph of this proof, μ' is a projective birational morphism, E' is a prime divisor on W', and the generic point of E' is the only point of W' over ξ. Since μ' is proper, $\operatorname{ord}_{E''}$ has a center on W', and the image of the center of $\operatorname{ord}_{E''}$ on W' is the center of ord_E on W, which is ξ. Thus the center of $\operatorname{ord}_{E''}$ on W' is the generic point of E'. So the valuation ring of $\operatorname{ord}_{E''}$ dominates the local ring of W' at the generic point of E', which is a discrete valuation ring of $\mathbb{C}(X')$. Hence the valuation ring of $\operatorname{ord}_{E''}$ is equal to the local ring of W' at the generic point of E', $E'' = E'$, and $c = 1$. This proves the existence and uniqueness of E' and that any prime divisor over X' with center x' arises in this way.

Let B_W be the $\mathbb{R}$-divisor with $K_W + B_W = \mu^*(K_X + B)$ and $\mu_* B_W = B$. Since ψ and $W' \to W$ are étale, $K_{W'} + B_{W'} = \mu'^*(K_{X'} + \psi^* B)$ for the pullback $B_{W'}$ of B_W to W', and $\operatorname{mult}_{E'} B_{W'} = \operatorname{mult}_E B_W$ since $W' \to W$ is étale at the generic point of E'. Hence $a(E', X', \psi^* B) = a(E, X, B)$.

In the equivariant case, let $\sigma \in G$. Since σ fixes x' and $\sigma^* \circ \psi^* = \psi^* \circ \sigma^*$, the valuation $\operatorname{ord}_{E'} \circ \sigma^*$ is the valuation of a prime divisor over X' with center x' whose restriction to $\mathbb{C}(X)$ is $\operatorname{ord}_E \circ \sigma^*$. By uniqueness, $\sigma \in \operatorname{Stab}_G(E')$ if and only if $\sigma \in \operatorname{Stab}_G(E)$. For $\sigma \in \operatorname{Stab}_G(E)$, σ^* preserves the valuation rings $\mathcal{O}_E \subset \mathcal{O}_{E'}$ of ord_E and $\operatorname{ord}_{E'}$, and the isomorphism of residue fields induced by $\mathcal{O}_E \subset \mathcal{O}_{E'}$ commutes with σ^*. Thus σ acts trivially on the residue field of $\operatorname{ord}_{E'}$ if and only if it acts trivially on that of ord_E, and $I_G(E') = I_G(E)$. □

For the binary hypersurface $xy + g$, the excess of the log discrepancy over the orders of the base coordinates is a multiple of the inertia order, so a quotient value below one forces the excess to vanish and the computing divisor is a monomial valuation. For this reason the results of this section concern minimal log discrepancies below one.

Proposition 9.4. *Let $(V \ni v)$ be a smooth surface germ, G a finite abelian group acting on V and fixing v, and z, t G-eigenparameters of V at v. Let G act on $\mathbb{A}^2_{x,y} \times V$ by the given action on V and diagonally on x, y, let g be a non-zero regular G-semi-invariant function on V vanishing to order at least two at v such that $xy + g$ is G-semi-invariant, let $X := (xy + g = 0) \subset \mathbb{A}^2_{x,y} \times V$ and $o := (0, 0, v)$, let $\hat{g} \in \mathbb{C}[[z, t]]$ be the expansion of g, let $b_z, b_t < 1$ be real numbers, and let $B := b_z \operatorname{div}_X(z) + b_t \operatorname{div}_X(t)$. Assume that X is a normal klt threefold germ at o, that G acts faithfully on X, and that (X, B) is sub-klt. Let N_G be the lattice of Definition 2.15 for the characters of x, y, z, t, and for $w \in \mathbb{R}^4_{>0}$ let*

$$P(w) := w_x + w_y + w_z + w_t - \min\{w_x + w_y, \operatorname{wt}_{(w_z, w_t)}(\hat{g})\} - b_z w_z - b_t w_t.$$

Then the following hold.

(1) Any prime divisor F over X with center o and $a(F, X, B)/|I_G(F)| < 1$ is, through $\mathbb{C}(X) = \mathbb{C}(\mathbb{A}^1_x \times V)$, the monomial valuation in the regular parameters x, z, t of $\mathbb{A}^1_x \times V$ at $(0, v)$ with weights (u, c, d), where $u, c, d > 0$, $\operatorname{wt}_{(c,d)}(\hat{g}) > u$, $\operatorname{ord}_F y = \operatorname{wt}_{(c,d)}(\hat{g}) - u$, and $a(F, X, 0) = c + d$.

(2) If $0 < a := \operatorname{mld}((X \ni o, B)/G) < 1$, then $a = \min_{w \in N_G \cap \mathbb{R}^4_{>0}} P(w)$, attained at a primitive vector.

In particular, for a DCC set $\Gamma \subset [0, 1)$, the set of such numbers $a \in (0, 1)$ with $b_z, b_t \in \Gamma \cup \{0\}$ satisfies the ACC.

Proof. The proof is long, so we divide it into several steps. Let $\chi_x, \chi_y, \chi_z, \chi_t$ be the characters of x, y, z, t. Since $xy + g$ is G-semi-invariant and g is a non-zero semi-invariant function not depending on x, y, each of the functions xy, g, and $xy + g$ has character $\chi_x \chi_y$.

Step 1. In this step we write a as an infimum over prime divisors over X and show that the infimum is attained.

Since z and t are G-semi-invariant, B is G-invariant, and $K_X + B$ is $\mathbb{R}$-Cartier because X is a hypersurface and $\mathrm{div}_X(z)$, $\mathrm{div}_X(t)$ are Cartier. By Lemma 2.6(0), $(X, B)/G$ is a sub-pair, and by the last statement of Lemma 2.6,

$$\mathrm{mld}((X \ni o, B)/G) = \inf_F \frac{a(F, X, B)}{|I_G(F)|}, \tag{9.1}$$

where F runs over the prime divisors over X with $\mathrm{center}_X F = \{o\}$. By Lemma 2.6(2)(3), any prime divisor E over X/G satisfies $a(E, X/G, B_{X/G}) = a(F, X, B)/|I_G(F)| > 0$ for some prime divisor F over X, so $(X, B)/G$ is sub-klt. Hence the minimal log discrepancy of $(X, B)/G$ at the image of o is attained by a prime divisor [Amb99, Section 1, the second paragraph after Definition 1.1]. By Lemma 2.6(2)(3) again, the infimum in (9.1) is attained.

Step 2. In this step we show that, for any prime divisor F over X with center o, the excess $a(F, X, 0) - \mathrm{ord}_F z - \mathrm{ord}_F t$ is a non-negative integer divisible by $|I_G(F)|$.

Let F be a prime divisor over X with $\mathrm{center}_X F = \{o\}$, and let $e := |I_G(F)|$, $u := \mathrm{ord}_F x$, $c := \mathrm{ord}_F z$, and $d := \mathrm{ord}_F t$. Since x, z, t vanish at o, the numbers u, c, d are positive integers. The projection $X \to \mathbb{A}^1_x \times V$ is G-equivariant and birational, because $y = -g/x$ on X. Thus $\mathbb{C}(X) = \mathbb{C}(\mathbb{A}^1_x \times V)$, and F is a prime divisor over $\mathbb{A}^1_x \times V$ with center $(0, v)$, since the projection maps o to $(0, v)$ and x, z, t are regular parameters of $\mathbb{A}^1_x \times V$ at $(0, v)$.

Let Ω be the residue of $dx \wedge dy \wedge dz \wedge dt/(xy + g)$, that is, the regular 3-form on the smooth locus of X near o characterized by $d(xy + g) \wedge \tilde{\Omega} = dx \wedge dy \wedge dz \wedge dt$ for local lifts $\tilde{\Omega}$ of Ω (cf. [Kol13, Definition 4.1]). Where a partial derivative $\partial(xy+g)/\partial x_i$ with $x_i \in \{x, y, z, t\}$ is non-zero, Ω is the restriction to X of $\pm(\partial(xy + g)/\partial x_i)^{-1}$ times the wedge product of the differentials of the other three coordinates, so Ω generates Ω^3_X on the smooth locus of X near o. Since X is normal, Ω generates $\omega_X = \mathcal{O}_X(K_X)$ near o. Moreover, $\Omega = -x^{-1} dx \wedge dz \wedge dt$, because

$$d(xy + g) \wedge (-x^{-1} dx \wedge dz \wedge dt) = dx \wedge dy \wedge dz \wedge dt \quad \text{on } X \cap (x \neq 0).$$

The form $dx \wedge dz \wedge dt$ generates $\omega_{\mathbb{A}^1_x \times V}$ near $(0, v)$. Comparing the orders of Ω and $dx \wedge dz \wedge dt$ along F, we obtain

$$a(F, X, 0) = a(F, \mathbb{A}^1_x \times V, 0) - u. \tag{9.2}$$

By Lemma 2.13, applied to the regular parameters x, z, t at $(0, v)$ with coefficients 0,

$$a(F, \mathbb{A}^1_x \times V, 0) \geq u + c + d.$$

Let

$$\Delta_F := a(F, X, 0) - c - d = a(F, \mathbb{A}^1_x \times V, 0) - u - c - d.$$

Then $\Delta_F \geq 0$, and Δ_F is an integer because ω_X and $\omega_{\mathbb{A}^1_x \times V}$ are invertible.

Let λ be the character of $I_G(F)$ on $\mathfrak{m}_F/\mathfrak{m}^2_F$. By Lemma 9.2(2), applied to $X \subset \mathbb{A}^2_{x,y} \times V$ and the G-eigenparameters x, y, z, t at o, and since $xy + g$ has character $\chi_x \chi_y$,

$$(\chi_z \chi_t)|_{I_G(F)} = (\chi_x \chi_y \chi_z \chi_t (\chi_x \chi_y)^{-1})|_{I_G(F)} = \lambda^{a(F,X,0)}.$$

By Lemma 9.2(1), $\chi_z|_{I_G(F)} = \lambda^c$ and $\chi_t|_{I_G(F)} = \lambda^d$. Thus $\lambda^{\Delta_F} = \lambda^{a(F,X,0)-c-d} = 1$. By Lemma 2.6(1), $I_G(F)$ is cyclic of order e and λ is injective, so $e \mid \Delta_F$. Since $\mathrm{div}_X(z)$ and $\mathrm{div}_X(t)$ are Cartier with orders c and d along F,

$$\frac{a(F, X, B)}{e} = \frac{(1 - b_z)c + (1 - b_t)d}{e} + \frac{\Delta_F}{e}, \tag{9.3}$$

where the first summand is positive, because $c, d > 0$ and $b_z, b_t < 1$, and the second summand is a non-negative integer.

Step 3. In this step we prove (1).

Let F be a prime divisor over X with center o and $a(F,X,B)/|I_G(F)| < 1$, and let u,c,d be as in **Step 2**. By (9.3), $\Delta_F = 0$, that is, $a(F, \mathbb{A}^1_x \times V, 0) = u+c+d$. Equivalently, the log discrepancy of F with respect to $(\mathbb{A}^1_x \times V, \operatorname{div}(x)+\operatorname{div}(z)+\operatorname{div}(t))$, which is log smooth near $(0,v)$, is zero. The monomial valuation in x,z,t with weights (u,c,d) (Definition 2.16) is divisorial with center $(0,v)$ and log discrepancy $u+c+d$ on $\mathbb{A}^1_x \times V$ [JM12, Proposition 3.1, Remark 3.9, Proposition 5.1, and (5.1)], and it takes the values u,c,d on x,z,t. By [JM12, Corollary 5.4], the vanishing of the log discrepancy of F with respect to the reduced coordinate boundary implies that ord_F is the monomial valuation in x,z,t with weights (u,c,d).

The expansion of g in $\widehat{\mathcal{O}}_{\mathbb{A}^1_x \times V,(0,v)} = \mathbb{C}[[x,z,t]]$ is $\hat{g}$, so Definition 2.16 implies

$$\operatorname{ord}_F g = \operatorname{wt}_{(u,c,d)}(\hat{g}) = \operatorname{wt}_{(c,d)}(\hat{g}).$$

Since $y = -g/x$,

$$\operatorname{ord}_F y = \operatorname{wt}_{(c,d)}(\hat{g}) - u,$$

and $\operatorname{ord}_F y > 0$ because y vanishes at o. Thus $\operatorname{wt}_{(c,d)}(\hat{g}) > u$. Finally, $\Delta_F = 0$ means $a(F,X,0) = c+d$. This proves (1).

Step 4. In this step we assume $0 < a < 1$ and construct $w \in N_G \cap \mathbb{R}^4_{>0}$ with $P(w) = a$.

By **Step 1**, there exists a prime divisor F over X with center o and $a(F,X,B)/e = a$, where $e := |I_G(F)|$. Let u,c,d be as in **Step 2**. Since $a < 1$, **Step 3** implies $\operatorname{ord}_F y = \operatorname{wt}_{(c,d)}(\hat{g}) - u > 0$ and $\Delta_F = 0$, so $a = ((1-b_z)c + (1-b_t)d)/e$ by (9.3). Let

$$w := \frac{1}{e}(\operatorname{ord}_F x, \operatorname{ord}_F y, \operatorname{ord}_F z, \operatorname{ord}_F t) = \frac{1}{e}\big(u, \operatorname{wt}_{(c,d)}(\hat{g}) - u, c, d\big) \in \mathbb{R}^4_{>0}.$$

By Lemma 9.2(3), applied to $x,y,z,t \in \mathbb{C}(X)$, $w \in N_G$. Since $\operatorname{wt}_{(w_z,w_t)}(\hat{g}) = \frac{1}{e}\operatorname{wt}_{(c,d)}(\hat{g}) = w_x + w_y$, we obtain

$$P(w) = w_z + w_t - b_z w_z - b_t w_t = \frac{(1-b_z)c + (1-b_t)d}{e} = a. \tag{9.4}$$

Step 5. In this step we assume $0 < a < 1$ and show that $P(w) \geq a$ for any $w \in N_G \cap \mathbb{R}^4_{>0}$.

Let $w \in N_G \cap \mathbb{R}^4_{>0}$, $T := \operatorname{wt}_{(w_z,w_t)}(\hat{g})$, and $n := w_x + w_y - T$. By Definition 2.15, there exists $h \in G$ with

$$\chi_x(h) = \exp(2\pi\sqrt{-1}w_x), \quad \chi_y(h) = \exp(2\pi\sqrt{-1}w_y), \quad \chi_z(h) = \exp(2\pi\sqrt{-1}w_z), \quad \chi_t(h) = \exp(2\pi\sqrt{-1}w_t).$$

Since z,t are G-eigenparameters, G acts on $\widehat{\mathcal{O}}_{V,v} = \mathbb{C}[[z,t]]$ diagonally, and g has character $\chi_x\chi_y$. Hence any monomial $z^j t^k$ occurring in $\hat{g}$ satisfies $\chi_z^j \chi_t^k = \chi_x\chi_y$, and evaluating at h yields $jw_z + kw_t - w_x - w_y \in \mathbb{Z}$. Taking a monomial of $\hat{g}$ of (w_z,w_t)-weight T, we obtain $n \in \mathbb{Z}$, and

$$P(w) = (1-b_z)w_z + (1-b_t)w_t + \max\{n, 0\}. \tag{9.5}$$

If $P(w) \geq 1$, then $P(w) > a$. Suppose that $P(w) < 1$. Since $(1-b_z)w_z + (1-b_t)w_t > 0$, (9.5) implies $n \leq 0$, so $P(w) = (1-b_z)w_z + (1-b_t)w_t$ and $w_x < w_x + w_y \leq T$.

Since G is finite, $N_G \subset \frac{1}{|G|}\mathbb{Z}^4$. Let $\alpha = (\alpha_x, \alpha_z, \alpha_t) \in \mathbb{Z}^3_{>0}$ be the primitive integral vector on the ray $\mathbb{R}_{>0}(w_x, w_z, w_t)$, and let $s > 0$ be the rational number with $(w_x, w_z, w_t) = s\alpha$. Since α is primitive, the value group of the monomial valuation in the regular parameters x,z,t of $\mathbb{A}^1_x \times V$ at $(0,v)$ with weights α (Definition 2.16) is $\mathbb{Z}$, so the monomial valuation is $\operatorname{ord}_{F_\alpha}$ for a prime divisor F_α over $\mathbb{A}^1_x \times V$ with center $(0,v)$, and $a(F_\alpha, \mathbb{A}^1_x \times V, 0) = \alpha_x + \alpha_z + \alpha_t$ [JM12, Proposition 3.1, Remark 3.9, Proposition 5.1, and (5.1)]. Through $\mathbb{C}(X) = \mathbb{C}(\mathbb{A}^1_x \times V)$, F_α is a prime divisor over X. By Definition 2.16, $\operatorname{ord}_{F_\alpha} g = \operatorname{wt}_{(\alpha_z,\alpha_t)}(\hat{g}) = T/s$, so

$$\operatorname{ord}_{F_\alpha} y = \frac{T - w_x}{s} > 0.$$

The local ring $\mathcal{O}_{X,o}$ is a localization of a quotient of $\mathcal{O}_{\mathbb{A}^1_x\times V,(0,v)}[y]$ at the maximal ideal generated by x,y,z,t. Since $\operatorname{ord}_{F_\alpha}$ is non-negative on $\mathcal{O}_{\mathbb{A}^1_x\times V,(0,v)}$ and positive on x,y,z,t, it is non-negative on $\mathcal{O}_{X,o}$ and positive on $\mathfrak{m}_{X,o}$. Thus $\operatorname{center}_X F_\alpha = \{o\}$. By (9.2),

$$a(F_\alpha,X,0)=\alpha_x+\alpha_z+\alpha_t-\alpha_x=\alpha_z+\alpha_t,\quad \text{so}\quad a(F_\alpha,X,B)=(1-b_z)\alpha_z+(1-b_t)\alpha_t=\frac{P(w)}{s}.$$

We bound the order $e_\alpha := |I_G(F_\alpha)|$ from below. Let G act on $\mathbb{A}^3$ diagonally through χ_x,χ_z,χ_t, and let $\varphi:\mathbb{A}^1_x\times V\to\mathbb{A}^3$ be the G-equivariant morphism defined by x,z,t. Since z,t are regular parameters of V at v, possibly replacing V with a G-invariant open neighborhood of v, we may assume that φ is étale at $(0,v)$ and $\varphi^{-1}(0)=\{(0,v)\}$. Let F^0_α be the monomial prime divisor over $\mathbb{A}^3$ in the coordinates with weights α. The expansion of a polynomial in x,z,t is the polynomial itself, so $\operatorname{ord}_{F_\alpha}|_{\mathbb{C}(\mathbb{A}^3)}=\operatorname{ord}_{F^0_\alpha}$. By Lemma 9.3, $I_G(F_\alpha)=I_G(F^0_\alpha)$. Any $\sigma\in G$ multiplies each monomial in x,z,t by a constant, so σ preserves $\operatorname{ord}_{F^0_\alpha}$. The residue field of $\operatorname{ord}_{F^0_\alpha}$ is generated over $\mathbb{C}$ by the residues of the Laurent monomials $x^{\beta_1}z^{\beta_2}t^{\beta_3}$ with $\langle\alpha,\beta\rangle=0$, since for polynomials p_1,p_2 with $\operatorname{ord}_{F^0_\alpha}p_1=\operatorname{ord}_{F^0_\alpha}p_2$ the residue of p_1/p_2 is that of $\operatorname{in}_\alpha(p_1)/\operatorname{in}_\alpha(p_2)$. For such β,

$$h^*(x^{\beta_1}z^{\beta_2}t^{\beta_3})=\exp(2\pi\sqrt{-1}s\langle\alpha,\beta\rangle)x^{\beta_1}z^{\beta_2}t^{\beta_3}=x^{\beta_1}z^{\beta_2}t^{\beta_3}.$$

Thus $h\in I_G(F^0_\alpha)=I_G(F_\alpha)$. Let λ_α be the character of $I_G(F_\alpha)$ on $\mathfrak{m}_{F_\alpha}/\mathfrak{m}^2_{F_\alpha}$. Since α is primitive, there exist integers n_x,n_z,n_t with $n_x\alpha_x+n_z\alpha_z+n_t\alpha_t=1$. By Lemma 9.2(1),

$$\lambda_\alpha(h)=\lambda_\alpha(h)^{n_x\alpha_x+n_z\alpha_z+n_t\alpha_t}=\chi_x(h)^{n_x}\chi_z(h)^{n_z}\chi_t(h)^{n_t}=\exp(2\pi\sqrt{-1}s).$$

Since $\lambda_\alpha(h)^{e_\alpha}=1$, the number se_α is a positive integer, so $se_\alpha\geq 1$. By (9.1),

$$a\leq\frac{a(F_\alpha,X,B)}{e_\alpha}=\frac{P(w)}{se_\alpha}\leq P(w).$$

Step 6. In this step we prove (2) and the ACC.

Assume that $0<a<1$. By **Steps 4** and 5, $a=\min_{w\in N_G\cap\mathbb{R}^4_{>0}}P(w)$, and the minimum is attained at the vector w of **Step 4**. If a vector w attaining the minimum satisfies $w=kw'$ for some $w'\in N_G$ and some integer $k\geq 2$, then $w'\in N_G\cap\mathbb{R}^4_{>0}$ and $P(w')=\frac{1}{k}P(w)=\frac{a}{k}<a$, because P is positively homogeneous of degree one, which contradicts **Step 5**. Thus any vector attaining the minimum is primitive in N_G. This proves (2).

Let $\Gamma\subset[0,1)$ be a DCC set and assume that $b_z,b_t\in\Gamma\cup\{0\}$. Let $H\subset\mathbb{T}^4$ be the image of G, so $N_G=N_H$. The monomials occurring in $xy+\hat{g}\in\mathbb{C}[[x,y,z,t]]$ are xy and the monomials of $\hat{g}$, so for $J_1:=J(xy+\hat{g})$, $J_2:=(z)$, and $J_3:=(t)$,

$$\psi_{J,(1,b_z,b_t)}(w)=\sum_j w_j-\min\{w_x+w_y,\operatorname{wt}_{(w_z,w_t)}(\hat{g})\}-b_zw_z-b_tw_t=P(w).$$

By (2), $a=\psi_{\min}(H,J,(1,b_z,b_t))$ with $(1,b_z,b_t)\in(\Gamma\cup\{0,1\})^3$. Since $\Gamma\cup\{0,1\}$ is a DCC set, Theorem 4.5 implies that the set of such numbers a satisfies the ACC. □

9.1. **Diagonal binary hypersurfaces.** In this subsection, we prove Proposition 9.5.

If the characters of x and y are different, then $x+\sqrt{-1}y$ and $x-\sqrt{-1}y$ are not G-semi-invariant, so Proposition 9.4, with $x\pm\sqrt{-1}y$ as coordinates, cannot be used for G but only for subgroups on which the characters of x and y coincide. In the proof, it is used for the inertia group $I_G(F)$ of a prime divisor F with $\operatorname{ord}_F x=\operatorname{ord}_F y$, on which the characters of x and y coincide by Lemma 9.2(1). The valuations in Proposition 9.5(1)(2) are used in the proof of Theorem 11.7, and the prime divisors in Proposition 9.5(2) are constructed in three cases, namely $w_x=w_y$, $|w_x-w_y|=\frac{1}{2}$ with $\operatorname{wt}_{(w_z,w_t)}(\hat{g})>2\min\{w_x,w_y\}$, and $|w_x-w_y|=\frac{1}{2}$ with $\operatorname{wt}_{(w_z,w_t)}(\hat{g})=2\min\{w_x,w_y\}$.

Proposition 9.5. *Let $(V\ni v)$ be a smooth surface germ, G a finite abelian group acting on V and fixing v, and z,t G-eigenparameters of V at v. Let G act on $\mathbb{A}^2_{x,y}\times V$ by the given action on V and diagonally on x,y, let g be a non-zero regular G-semi-invariant function on V vanishing to order*

at least two at v such that x^2+y^2+g is G-semi-invariant, let $X := (x^2+y^2+g=0) \subset \mathbb{A}^2_{x,y} \times V$ and $o := (0,0,v)$, let $\hat{g} \in \mathbb{C}[[z,t]]$ be the expansion of g, let $b_z, b_t < 1$ be real numbers, and let $B := b_z \operatorname{div}_X(z) + b_t \operatorname{div}_X(t)$. Assume that X is a normal klt threefold germ at o, that G acts faithfully on X, and that (X,B) is sub-klt. Let N_G be the lattice of Definition 2.15 for the characters of x,y,z,t, and for $w \in \mathbb{R}^4_{>0}$ let

$$P(w) := w_x + w_y + w_z + w_t - \min\{2w_x, 2w_y, \operatorname{wt}_{(w_z,w_t)}(\hat{g})\} - b_z w_z - b_t w_t.$$

Then the following hold.

(1) *Let F be a prime divisor over X with center o such that $\sigma_F := a(F,X,B)/|I_G(F)| < 1$. Then $w_F := \frac{1}{|I_G(F)|}(\operatorname{ord}_F x, \operatorname{ord}_F y, \operatorname{ord}_F z, \operatorname{ord}_F t) \in N_G$ and $P(w_F) = \sigma_F$. Here, for $s \in \{x, y, x+\sqrt{-1}y\}$, $\mathbb{A}^1_s$ is the affine line with coordinate s, and $\mathbb{C}(\mathbb{A}^1_s \times V) \subset \mathbb{C}(X)$ through the projection $X \to \mathbb{A}^1_s \times V$. If $\operatorname{ord}_F x < \operatorname{ord}_F y$, then the restriction of ord_F to $\mathbb{C}(\mathbb{A}^1_y \times V)$ is a positive multiple of a monomial valuation in the regular parameters y,z,t at $(0,v)$, and*

$$a(F,X,0) = |\operatorname{ord}_F x - \operatorname{ord}_F y| + \operatorname{ord}_F z + \operatorname{ord}_F t.$$

If $\operatorname{ord}_F y < \operatorname{ord}_F x$, then the restriction of ord_F to $\mathbb{C}(\mathbb{A}^1_x \times V)$ is a positive multiple of a monomial valuation in the regular parameters x,z,t at $(0,v)$, and

$$a(F,X,0) = |\operatorname{ord}_F x - \operatorname{ord}_F y| + \operatorname{ord}_F z + \operatorname{ord}_F t.$$

If $\operatorname{ord}_F x = \operatorname{ord}_F y$, then the restriction of ord_F to $\mathbb{C}(\mathbb{A}^1_{x+\sqrt{-1}y} \times V)$ is a positive multiple of a monomial valuation in the regular parameters $x+\sqrt{-1}y, z, t$ at $(0,v)$, and

$$a(F,X,0) = \operatorname{ord}_F z + \operatorname{ord}_F t.$$

(2) *Let $w \in N_G \cap \mathbb{R}^4_{>0}$ with $P(w) < 1$, and let $\eta := \min\{w_x, w_y\}$, $\kappa := |w_x - w_y|$, and $\tau := \operatorname{wt}_{(w_z,w_t)}(\hat{g})$. Then $\tau \geq 2\eta$, $\kappa \in \{0, \frac{1}{2}\}$, $P(w) = \kappa + (1-b_z)w_z + (1-b_t)w_t$, and there exist a positive rational number e and a prime divisor F over X with center o such that $a(F,X,B)/|I_G(F)| \leq P(w)$, $\operatorname{ord}_F z = ew_z$, $\operatorname{ord}_F t = ew_t$, and: if $\kappa = 0$, then $\operatorname{ord}_F x, \operatorname{ord}_F y \geq ew_x$; if $\kappa = \frac{1}{2}$ and $\tau > 2\eta$, then $\operatorname{ord}_F x, \operatorname{ord}_F y \geq e\max\{w_x, w_y\}$; if $\kappa = \frac{1}{2}$ and $\tau = 2\eta$, then $(\operatorname{ord}_F x, \operatorname{ord}_F y) = (ew_x, ew_y)$.*

(3) *If $0 < a := \operatorname{mld}((X \ni o, B)/G) < 1$, then $a = \min_{w \in N_G \cap \mathbb{R}^4_{>0}} P(w)$, attained at a primitive vector.*

In particular, for a DCC set $\Gamma \subset [0,1)$, the set of such numbers $a \in (0,1)$ with $b_z, b_t \in \Gamma \cup \{0\}$ satisfies the ACC.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we fix notation and prove two formulas for log discrepancies.

All the statements concern the germ of X at o and prime divisors over X with center o. Hence we may shrink V, X, and the products $\mathbb{A}^1 \times V$ below to open neighborhoods of v, o, and $(0,v)$ that are invariant under the finite groups acting on them, and we do so without further mention. In particular, we may assume that X is a normal variety, that z,t are regular functions on V, that the morphism $(z,t) : V \to \mathbb{A}^2$ is étale, and that $dz \wedge dt$ generates ω_V. Let $\chi_x, \chi_y, \chi_z, \chi_t$ be the characters of x,y,z,t. Since x^2+y^2+g is G-semi-invariant and the pullback of y^2+g by any element of G does not involve x, the character of x^2+y^2+g is $\chi_x^2 = \chi_y^2$, and g also has character χ_x^2. Let

$$x_+ := x + \sqrt{-1}y \quad \text{and} \quad x_- := x - \sqrt{-1}y,$$

so that $X = (x_+x_- + g = 0)$. For a regular function $s \in \{x, y, x_+\}$, let $U_s := \mathbb{A}^1_s \times V$ with $\mathbb{A}^1_s$ as in (1). Then s,z,t are regular parameters of U_s at $(0,v)$, the completion of $\mathcal{O}_{U_s,(0,v)}$ is $\mathbb{C}[[s,z,t]]$, the expansion of g in it is $\hat{g}$, and $\operatorname{div}(s) + \operatorname{div}(z) + \operatorname{div}(t)$ is an snc divisor on U_s.

Let $p : X \to U_y$ be the projection and ι the involution $(x,y,v') \to (-x,y,v')$ of X. Since $\mathcal{O}(X) = \mathcal{O}(U_y) \oplus \mathcal{O}(U_y)x$ and $\iota^* x = -x$, the morphism p is the quotient morphism of X by

$\langle\iota\rangle \cong \mu_2$. If $y^2+g = q_1^2 q_2$ in $\mathcal{O}_{U_y,(0,v)}$ with q_1 vanishing at $(0,v)$, then $x^2+y^2+g \in (x,q_1)^2$, so X is singular along the surface $(x=q_1=0)$ through o, which contradicts the normality of X. Thus $\operatorname{div}(y^2+g)$ is reduced. The involution ι fixes $X \cap (x=0)$ pointwise, p maps $X\cap(x=0)$ isomorphically onto $\operatorname{div}(y^2+g)$, and ι acts non-trivially on the function field of any prime divisor on X not contained in $(x=0)$. Hence by Lemma 2.6(4) (with $B=0$),

$$(X,0)/\langle\iota\rangle = \Big(U_y, \frac{1}{2}\operatorname{div}(y^2+g)\Big).$$

For a prime divisor F over X, let ν_F be the prime divisor over U_y and ϱ_F the positive integer such that $\operatorname{ord}_F|_{\mathbb{C}(U_y)} = \varrho_F \operatorname{ord}_{\nu_F}$. By Lemma 2.6(1)(2), $\varrho_F = |I_{\langle\iota\rangle}(F)|$, $\operatorname{center}_{U_y}\nu_F = p(\operatorname{center}_X F)$, and

$$a(F,X,0) = \varrho_F a\Big(\nu_F, U_y, \frac{1}{2}\operatorname{div}(y^2+g)\Big) = \varrho_F a(\nu_F,U_y,0) - \frac{1}{2}\varrho_F\nu_F(y^2+g).$$

Since $\varrho_F\nu_F(y^2+g) = \operatorname{ord}_F(y^2+g) = \operatorname{ord}_F(-x^2) = 2\operatorname{ord}_F x$, we obtain

$$a(F,X,0) = \varrho_F a(\nu_F,U_y,0) - \operatorname{ord}_F x. \tag{9.6}$$

Interchanging x and y, formula (9.6) holds with U_x and $\operatorname{ord}_F y$ in place of U_y and $\operatorname{ord}_F x$.

The projection $X \to U_{x_+}$ is an isomorphism over $(x_+\neq 0)$, with inverse given by $x_- = -g/x_+$, so $\mathbb{C}(X) = \mathbb{C}(U_{x_+})$. Since the morphism $(z,t): V \to \mathbb{A}^2$ is étale, x_+, x_-, z, t are étale coordinates on $\mathbb{A}^2\times V$. Let Ω_0 be the residue of $dx_+\wedge dx_-\wedge dz\wedge dt/(x_+x_-+g)$, that is, the regular 3-form on the smooth locus of X characterized by $d(x_+x_-+g)\wedge\tilde{\Omega}_0 = dx_+\wedge dx_-\wedge dz\wedge dt$ for local lifts $\tilde{\Omega}_0$ of Ω_0 (cf. [Kol13, Definition 4.1]). Where a partial derivative of x_+x_-+g with respect to one of x_+, x_-, z, t is non-zero, Ω_0 is the restriction to X of $\pm$ the inverse of that partial derivative times the wedge product of the differentials of the other three coordinates, so Ω_0 generates Ω^3_X on the smooth locus of X. Since X is normal, Ω_0 generates $\omega_X = \mathcal{O}_X(K_X)$. Where $x_+\neq 0$, $\Omega_0 = -x_+^{-1}dx_+\wedge dz\wedge dt$, because $d(x_+x_-+g)\wedge(-x_+^{-1}dx_+\wedge dz\wedge dt) = dx_+\wedge dx_-\wedge dz\wedge dt$ on $X\cap(x_+\neq 0)$. Let $\Omega := dx_+\wedge dz\wedge dt/x_+$. On U_{x_+}, $\operatorname{div}(\Omega) = -\operatorname{div}(x_+)$. Hence for any prime divisor F over X with a center on U_{x_+}, computing $\operatorname{mult}_F\operatorname{div}(\Omega)$ on a higher model on which F is a divisor,

$$a(F,X,0) = 1+\operatorname{mult}_F\operatorname{div}(\Omega) = a(F,U_{x_+},\operatorname{div}(x_+)) = a(F,U_{x_+},0) - \operatorname{ord}_F x_+. \tag{9.7}$$

Step 2. In this step, we prove (1).

Let F be a prime divisor over X with center o and $\sigma_F<1$, and let $I := I_G(F)$. Since $\operatorname{center}_X F = \{o\}$, the numbers $\operatorname{ord}_F x, \operatorname{ord}_F y, \operatorname{ord}_F z, \operatorname{ord}_F t$ are positive integers, and $w_F \in N_G$ by Lemma 9.2(3). Let λ be the character by which I acts on $\mathfrak{m}_F/\mathfrak{m}_F^2$. By Lemma 2.6(1), I is cyclic and λ is faithful, so the order of λ is $|I|$. Since x^2+y^2+g has character χ_x^2, Lemma 9.2(1)(2) implies

$$\lambda^{a(F,X,0)} = (\chi_x\chi_y\chi_z\chi_t\chi_x^{-2})|_I = \lambda^{\operatorname{ord}_F y+\operatorname{ord}_F z+\operatorname{ord}_F t-\operatorname{ord}_F x}.$$

Hence

$$|I| \text{ divides } a(F,X,0) - \operatorname{ord}_F y - \operatorname{ord}_F z - \operatorname{ord}_F t + \operatorname{ord}_F x. \tag{9.8}$$

Since $\operatorname{div}_X(z)$ and $\operatorname{div}_X(t)$ are Cartier, $a(F,X,B) = a(F,X,0) - b_z\operatorname{ord}_F z - b_t\operatorname{ord}_F t$.

First assume that $\operatorname{ord}_F x \neq \operatorname{ord}_F y$. The hypotheses, the statement of (1), and the definition of P are symmetric in x and y, and so is the argument above: the character of x^2+y^2+g is $\chi_x^2 = \chi_y^2$, and by **Step 1**, formula (9.6) holds with U_x and $\operatorname{ord}_F y$ in place of U_y and $\operatorname{ord}_F x$. Hence, possibly interchanging x and y (and replacing U_y with U_x), we may assume that $\operatorname{ord}_F x < \operatorname{ord}_F y$. Let $\nu := \nu_F$ and $\varrho := \varrho_F$ be as in **Step 1**. Then $\operatorname{center}_{U_y}\nu = (0,v)$. By Lemma 2.13, applied to U_y, the point $(0,v)$, the parameters y,z,t, and zero coefficients, $a(\nu,U_y,0) \geq \nu(y)+\nu(z)+\nu(t)$. Thus by (9.6),

$$L_F := a(F,X,0) - (\operatorname{ord}_F y - \operatorname{ord}_F x + \operatorname{ord}_F z + \operatorname{ord}_F t) = \varrho\big(a(\nu,U_y,0) - \nu(y) - \nu(z) - \nu(t)\big) \geq 0,$$

and $L_F/|I|$ is a non-negative integer by (9.8). We have

$$\sigma_F = \frac{\operatorname{ord}_F y - \operatorname{ord}_F x + (1-b_z)\operatorname{ord}_F z + (1-b_t)\operatorname{ord}_F t}{|I|} + \frac{L_F}{|I|},$$

where the first term is positive because $\operatorname{ord}_F y > \operatorname{ord}_F x$ and $b_z, b_t < 1$. Since $\sigma_F < 1$, $L_F = 0$, that is, $a(\nu, U_y, 0) = \nu(y)+\nu(z)+\nu(t)$. Let $D := \operatorname{div}(y)+\operatorname{div}(z)+\operatorname{div}(t)$ on U_y, and let ν' be the monomial valuation in y, z, t at $(0, v)$ with weights $(\nu(y), \nu(z), \nu(t))$. Since all the components of D pass through $\operatorname{center}_{U_y}\nu = (0, v)$, ν' is the retraction of ν to $\mathrm{QM}(U_y, D)$ [JM12, §4.3], and the log discrepancy of ν' is $\nu(y)+\nu(z)+\nu(t)$ by [JM12, Proposition 5.1, formula (5.1)]. Hence ν' and ν both have log discrepancy $\nu(y)+\nu(z)+\nu(t)$, and [JM12, Corollary 5.4] implies $\nu \in \mathrm{QM}(U_y, D)$, that is, $\nu = \nu'$. Therefore $\operatorname{ord}_F|_{\mathbb{C}(U_y)} = \varrho\nu'$ is a positive multiple of a monomial valuation in y, z, t at $(0, v)$, and

$$a(F, X, 0) = \operatorname{ord}_F y - \operatorname{ord}_F x + \operatorname{ord}_F z + \operatorname{ord}_F t = |\operatorname{ord}_F x - \operatorname{ord}_F y| + \operatorname{ord}_F z + \operatorname{ord}_F t.$$

Since $x^2 = -(y^2+g)$ and $2\operatorname{ord}_F y > 2\operatorname{ord}_F x$, we have $\operatorname{ord}_F g = 2\operatorname{ord}_F x$. Since the expansion of g in $\mathbb{C}[[y, z, t]]$ is $\hat{g}$,

$$2\operatorname{ord}_F x = \operatorname{ord}_F g = \varrho\nu'(g) = \operatorname{wt}_{(\operatorname{ord}_F z, \operatorname{ord}_F t)}(\hat{g}).$$

Dividing by $|I|$, $\min\{2w_x, 2w_y, \operatorname{wt}_{(w_z, w_t)}(\hat{g})\} = 2w_x$ for $w = w_F$, so

$$P(w_F) = \frac{\operatorname{ord}_F y - \operatorname{ord}_F x + (1-b_z)\operatorname{ord}_F z + (1-b_t)\operatorname{ord}_F t}{|I|} = \frac{a(F, X, B)}{|I|} = \sigma_F.$$

Now assume that $\operatorname{ord}_F x = \operatorname{ord}_F y$. By Lemma 9.2(1), $\chi_x|_I = \lambda^{\operatorname{ord}_F x} = \chi_y|_I$. Thus x_+ and x_- are I-semi-invariant with character $\chi_x|_I$, the group I acts on $\mathbb{A}^2_{x_+,x_-} \times V$ by the given action on V and diagonally on x_+, x_-, and $x_+x_- + g = x^2 + y^2 + g$ is I-semi-invariant. Moreover, I acts faithfully on X, $I_I(F) = I$, and $a(F, X, B)/|I_I(F)| = \sigma_F < 1$. Hence by Proposition 9.4(1), applied to I in place of G and to x_+, x_- in place of x, y, ord_F is the monomial valuation in x_+, z, t at $(0, v)$ on $\mathbb{C}(X) = \mathbb{C}(U_{x_+})$,

$$\operatorname{ord}_F x_- = \operatorname{wt}_{(\operatorname{ord}_F z, \operatorname{ord}_F t)}(\hat{g}) - \operatorname{ord}_F x_+, \quad \text{and} \quad a(F, X, 0) = \operatorname{ord}_F z + \operatorname{ord}_F t.$$

Since $x = \frac{1}{2}(x_+ + x_-)$, $y = \frac{1}{2\sqrt{-1}}(x_+ - x_-)$, and $x_\pm = x \pm \sqrt{-1}y$, we have

$$\min\{\operatorname{ord}_F x_+, \operatorname{ord}_F x_-\} = \operatorname{ord}_F x = \operatorname{ord}_F y.$$

Thus

$$2\operatorname{ord}_F x \le \operatorname{ord}_F x_+ + \operatorname{ord}_F x_- = \operatorname{wt}_{(\operatorname{ord}_F z, \operatorname{ord}_F t)}(\hat{g}),$$

so for $w = w_F$ the minimum in $P(w)$ is $2w_x = w_x + w_y$, and

$$P(w_F) = \frac{(1-b_z)\operatorname{ord}_F z + (1-b_t)\operatorname{ord}_F t}{|I|} = \frac{a(F, X, B)}{|I|} = \sigma_F.$$

This proves (1).

Step 3. In this step, we prove the first part of (2) and fix a group element.

Let w, η, κ, and τ be as in (2). The statement of (2) is symmetric in x and y, so possibly interchanging x and y, we may assume that $w_x \le w_y$. Thus $\eta = w_x$ and $w_y = w_x + \kappa$. Let

$$R := (1-b_z)w_z + (1-b_t)w_t > 0.$$

By Definition 2.15, there exists $h \in G$ such that $\chi_i(h) = e^{2\pi\sqrt{-1}w_i}$ for $i \in \{x, y, z, t\}$. Let m be the order of h and $\zeta_m := e^{2\pi\sqrt{-1}/m}$. Since $\chi_i(h)^m = 1$, mw_i is a positive integer and $\chi_i(h) = \zeta_m^{mw_i}$ for any i.

The group G acts on $\mathbb{C}[[z, t]]$, the completion of $\mathcal{O}_{V,v}$, so that z and t are semi-invariant with characters χ_z and χ_t, and $\hat{g}$ is semi-invariant with character $\chi_x^2 = \chi_y^2$. Hence $\chi_z^\beta\chi_t^\gamma = \chi_x^2 = \chi_y^2$

for any monomial $z^\beta t^\gamma$ occurring in $\hat{g}$. Evaluating at h, for any monomial $z^\beta t^\gamma$ occurring in $\hat{g}$ we obtain

$$\beta w_z + \gamma w_t - 2w_x \in \mathbb{Z} \quad \text{and} \quad \beta w_z + \gamma w_t - 2w_y \in \mathbb{Z}. \tag{9.9}$$

Taking a monomial of (w_z, w_t)-weight τ, we obtain $2\kappa \in \mathbb{Z}$ and $\tau - 2\eta \in \mathbb{Z}$. Since $w_x + w_y = 2\eta + \kappa$,

$$P(w) = \kappa + \max\{2\eta - \tau, 0\} + R.$$

Here $\max\{2\eta-\tau, 0\}$ is a non-negative integer, $\kappa \geq 0$, and $R > 0$. Since $P(w) < 1$, $\max\{2\eta-\tau, 0\} = 0$, that is, $\tau \geq 2\eta$, and $P(w) = \kappa + R$. Thus $\kappa < 1$ and $2\kappa \in \mathbb{Z}$, so $\kappa \in \{0, \frac{1}{2}\}$. If $\kappa = \frac{1}{2}$, then m is even because $mw_y - mw_x = \frac{m}{2}$ is an integer, and $\tau > 2\eta$ implies $\tau \geq 2\eta + 1 = 2w_y$ because $\tau - 2\eta \in \mathbb{Z}$.

Step 4. In this step, we embed $\mathbb{C}(U_s)$ into a field of Laurent series.

Let $s \in \{y, x_+\}$, let c_s, c_z, c_t be positive integers, let $K := \mathbb{C}(S_0, Z_0, T_0)$ be the field of rational functions in three variables, and let ϖ be a variable. For any integer n, only finitely many monomials in s, z, t have (c_s, c_z, c_t)-weight n. Hence the substitution

$$s = \varpi^{c_s} S_0, \quad z = \varpi^{c_z} Z_0, \quad t = \varpi^{c_t} T_0$$

defines a continuous $\mathbb{C}$-algebra homomorphism $\varphi : \mathbb{C}[[s, z, t]] \to K[[\varpi]]$. For a non-zero power series f, the coefficient of ϖ^n in $\varphi(f)$ is a linear combination of distinct monomials in S_0, Z_0, T_0, with the coefficients of the monomials of f of weight n. Thus $\operatorname{ord}_\varpi \varphi(f) = \operatorname{wt}_{(c_s,c_z,c_t)}(f)$. In particular, φ is injective, and through $\mathcal{O}_{U_s,(0,v)} \subset \mathbb{C}[[s, z, t]]$ it induces a field embedding $\varphi : \mathbb{C}(U_s) \to K((\varpi))$ such that $\operatorname{ord}_\varpi \circ \varphi$ is the monomial valuation in s, z, t at $(0, v)$ with weights (c_s, c_z, c_t).

For a field $K' \supset K$ and a root of unity ζ, let θ_ζ be the automorphism $\sum_n a_n \varpi^n \to \sum_n a_n \zeta^n \varpi^n$ of $K'((\varpi))$. It fixes K' and preserves $\operatorname{ord}_\varpi$. Let $h' \in G$ such that $h'^* s = \zeta^{c_s} s$, $\chi_z(h') = \zeta^{c_z}$, and $\chi_t(h') = \zeta^{c_t}$. Then h' acts on U_s and fixes $(0, v)$, and h'^* extends to $\mathbb{C}[[s, z, t]]$ as the continuous substitution $s \to \zeta^{c_s} s$, $z \to \zeta^{c_z} z$, $t \to \zeta^{c_t} t$. The continuous homomorphisms $\theta_\zeta \circ \varphi$ and $\varphi \circ h'^*$ coincide on s, z, t, so

$$\theta_\zeta \circ \varphi = \varphi \circ h'^* \quad \text{on } \mathbb{C}(U_s). \tag{9.10}$$

Step 5. In this step, we show that the embeddings of **Step 4** define prime divisors over X and control their inertia.

Let s, c_s, c_z, c_t, and φ be as in **Step 4**, $K' \supset K$ a field, and $\psi : \mathbb{C}(X) \to K'((\varpi))$ a field homomorphism such that $\psi|_{\mathbb{C}(U_s)} = \varphi$, with $\mathbb{C}(U_s) \subset \mathbb{C}(X)$ as in (1), and such that $\psi(x), \psi(y) \in \varpi K'[[\varpi]]$. We claim that $\operatorname{val}_\psi := \operatorname{ord}_\varpi \circ \psi$ is $d_0 \operatorname{ord}_F$ for a positive integer d_0 and a prime divisor F over X with center o, and that if moreover h' and ζ are as in **Step 4** and $\theta_\zeta \circ \psi = \psi \circ h'^*$ on $\mathbb{C}(X)$, then $h' \in I_G(F)$.

The value group of val_ψ is a non-zero subgroup $d_0\mathbb{Z}$ of $\mathbb{Z}$. On the valuation ring of val_ψ, taking the constant term of $\psi(f)$ induces an embedding of the residue field of val_ψ into K'. The functions $s^{c_z} z^{-c_s}$ and $s^{c_t} t^{-c_s}$ have value 0, and their residues are $S_0^{c_z} Z_0^{-c_s}$ and $S_0^{c_t} T_0^{-c_s}$, which are algebraically independent over $\mathbb{C}$. Hence the residue field of val_ψ has transcendence degree $2 = \dim X - 1$ over $\mathbb{C}$, the upper bound being Abhyankar's inequality [Bou72, Chapter VI, §10, no. 3, Corollary 1 to Theorem 1], applied to $\mathbb{C}(X)$ over $\mathbb{C}$ with the trivial valuation on $\mathbb{C}$, since the value group of val_ψ has rational rank one. By [KM98, Lemma 2.45], the valuation ring of val_ψ is the local ring of a prime divisor F over X, and $\operatorname{val}_\psi = d_0 \operatorname{ord}_F$. The homomorphism ψ maps $\mathcal{O}(V) \subset \mathbb{C}[[z, t]]$ into $K'[[\varpi]]$ and the functions vanishing at v into $\varpi K'[[\varpi]]$, and $\psi(x), \psi(y) \in \varpi K'[[\varpi]]$. Thus val_ψ is non-negative on $\mathcal{O}(X)$ and positive on the maximal ideal of o, so $\operatorname{center}_X F = \{o\}$. Finally, assume that $\theta_\zeta \circ \psi = \psi \circ h'^*$. Since θ_ζ preserves $\operatorname{ord}_\varpi$, $\operatorname{val}_\psi \circ h'^* = \operatorname{val}_\psi$, so $h' \in \operatorname{Stab}_G(F)$. For f in the valuation ring of val_ψ, the constant terms of $\psi(h'^* f) = \theta_\zeta(\psi(f))$ and of $\psi(f)$ coincide, so h' acts trivially on the residue field of ord_F. Thus $h' \in I_G(F)$.

Step 6. In this step, we prove (2) when $\kappa = 0$.

Let h, m, and ζ_m be as in **Step 3**. Since $w_x = w_y = \eta$, $\chi_x(h) = \chi_y(h)$, so $h^* x_\pm = \zeta_m^{m\eta} x_\pm$. Let φ be the embedding of **Step 4** for $s = x_+$ and $(c_s, c_z, c_t) = (m\eta, mw_z, mw_t)$, and let $\psi := \varphi$ on $\mathbb{C}(X) = \mathbb{C}(U_{x_+})$. Then $\operatorname{ord}_\varpi \psi(x_+) = m\eta$ and $\operatorname{ord}_\varpi \psi(g) = \operatorname{wt}_{(mw_z, mw_t)}(\hat{g}) = m\tau$. Since $x_- = -g/x_+$ and $\tau \geq 2\eta$ (**Step 3**),

$$\operatorname{ord}_\varpi \psi(x_-) = m\tau - m\eta \geq m\eta.$$

Thus $\operatorname{ord}_\varpi \psi(x), \operatorname{ord}_\varpi \psi(y) \geq m\eta > 0$. By (9.10) for $h' = h$ and $\zeta = \zeta_m$, $\theta_{\zeta_m} \circ \psi = \psi \circ h^*$. Hence by **Step 5**, $\operatorname{ord}_\varpi \circ \psi = d_0 \operatorname{ord}_F$ for a positive integer d_0 and a prime divisor F over X with center o, and $h \in I_G(F)$, so $|I_G(F)| \geq m$. Let $e := m/d_0$. Then $\operatorname{ord}_F z = ew_z$, $\operatorname{ord}_F t = ew_t$, and $\operatorname{ord}_F x, \operatorname{ord}_F y \geq e\eta = ew_x$. By [JM12, Proposition 5.1, formula (5.1)], applied to the snc divisor $\operatorname{div}(x_+) + \operatorname{div}(z) + \operatorname{div}(t)$ on U_{x_+}, $a(F, U_{x_+}, 0) = e(\eta + w_z + w_t)$. By (9.7), $a(F, X, 0) = e(w_z + w_t)$, so $a(F, X, B) = eR$ and

$$\frac{a(F, X, B)}{|I_G(F)|} \leq \frac{eR}{m} = \frac{R}{d_0} \leq R = P(w).$$

Step 7. In this step, we prove (2) when $\kappa = \frac{1}{2}$ and $\tau > 2\eta$.

By **Step 3**, $w_y = w_x + \frac{1}{2}$, m is even, and $\tau \geq 2w_y$. Let $\zeta := \zeta_m^2$, a primitive $\frac{m}{2}$-th root of unity. Since $2w_y - 2w_x = 1$, $\chi_x(h^2) = e^{4\pi\sqrt{-1}w_y} = \chi_y(h^2)$, so $(h^2)^* x_+ = \zeta^{mw_y} x_+$, $\chi_z(h^2) = \zeta^{mw_z}$, and $\chi_t(h^2) = \zeta^{mw_t}$. Let φ be the embedding of **Step 4** for $s = x_+$ and $(c_s, c_z, c_t) = (mw_y, mw_z, mw_t)$, and let $\psi := \varphi$ on $\mathbb{C}(X) = \mathbb{C}(U_{x_+})$. Then $\operatorname{ord}_\varpi \psi(x_+) = mw_y$ and

$$\operatorname{ord}_\varpi \psi(x_-) = m\tau - mw_y \geq mw_y,$$

so $\operatorname{ord}_\varpi \psi(x), \operatorname{ord}_\varpi \psi(y) \geq mw_y > 0$. By (9.10) for $h' = h^2$, $\theta_\zeta \circ \psi = \psi \circ (h^2)^*$. Hence by **Step 5**, $\operatorname{ord}_\varpi \circ \psi = d_0 \operatorname{ord}_F$ for a positive integer d_0 and a prime divisor F over X with center o, and $h^2 \in I_G(F)$, so $|I_G(F)| \geq \frac{m}{2}$. Let $e := m/d_0$. Then $\operatorname{ord}_F z = ew_z$, $\operatorname{ord}_F t = ew_t$, and $\operatorname{ord}_F x, \operatorname{ord}_F y \geq ew_y = e \max\{w_x, w_y\}$. By [JM12, Proposition 5.1, formula (5.1)], $a(F, U_{x_+}, 0) = e(w_y + w_z + w_t)$, so $a(F, X, 0) = e(w_z + w_t)$ by (9.7) and $a(F, X, B) = eR$. Since $P(w) = \frac{1}{2} + R < 1$, we have $R < \frac{1}{2}$, so

$$\frac{a(F, X, B)}{|I_G(F)|} \leq \frac{eR}{m/2} = \frac{2R}{d_0} \leq 2R < \frac{1}{2} + R = P(w).$$

Step 8. In this step, we prove (2) when $\kappa = \frac{1}{2}$ and $\tau = 2\eta$.

By **Step 3**, $w_y = w_x + \frac{1}{2}$, m is even, and $\tau = 2w_x$. Let φ be the embedding of **Step 4** for $s = y$ and $(c_s, c_z, c_t) = (mw_y, mw_z, mw_t)$, and write Y_0 for S_0, so $K = \mathbb{C}(Y_0, Z_0, T_0)$. By (9.9), the (mw_z, mw_t)-weight of any monomial occurring in $\hat{g}$ is congruent to $2mw_x$ modulo m, and the minimal weight is $m\tau = 2mw_x$. Hence

$$\varphi(g) = \varpi^{2mw_x} \sum_{k \geq 0} \varpi^{km} g_k,$$

where $g_k \in \mathbb{C}[Z_0, T_0]$ and $g_0 = \operatorname{in}_{(w_z, w_t)}(\hat{g})(Z_0, T_0) \neq 0$. Since $\varphi(y^2) = \varpi^{2mw_y} Y_0^2 = \varpi^{2mw_x + m} Y_0^2$,

$$\varphi(y^2 + g) = \varpi^{2mw_x} \Big(g_0 + \sum_{k \geq 1} \varpi^{km} H_k \Big), \quad H_1 := g_1 + Y_0^2, \quad H_k := g_k \ (k \geq 2).$$

Let $K' := K(\xi)$, where ξ is a square root of $-g_0$ in an algebraic closure of K, and let

$$\tilde{x} := \varpi^{mw_x} \xi \sum_{n \geq 0} \binom{1/2}{n} \Big(\sum_{k \geq 1} \varpi^{km} \frac{H_k}{g_0} \Big)^n \in K'[[\varpi]].$$

Then $\tilde{x}^2 = -\varphi(y^2 + g)$ and $\operatorname{ord}_\varpi \tilde{x} = mw_x$. Since X is integral, $x^2 + y^2 + g$ is irreducible over $\mathbb{C}(U_y)$ and $\mathbb{C}(X) = \mathbb{C}(U_y)[x]/(x^2 + y^2 + g)$. Hence φ extends to a field homomorphism $\psi : \mathbb{C}(X) \to K'((\varpi))$ with $\psi(x) = \tilde{x}$. We have $\operatorname{ord}_\varpi \psi(x) = mw_x > 0$ and $\operatorname{ord}_\varpi \psi(y) = mw_y > 0$.

By (9.10) for $h' = h$ and $\zeta = \zeta_m$, $\theta_{\zeta_m} \circ \psi = \psi \circ h^*$ on $\mathbb{C}(U_y)$. Since the series defining $\tilde{x}/\varpi^{mw_x}$ involves only powers of ϖ^m and θ_{ζ_m} fixes K',

$$\theta_{\zeta_m}(\tilde{x}) = \zeta_m^{mw_x}\tilde{x} = \chi_x(h)\tilde{x} = \psi(h^*x).$$

Thus $\theta_{\zeta_m} \circ \psi = \psi \circ h^*$ on $\mathbb{C}(X)$. By **Step 5**, $\operatorname{ord}_\varpi \circ \psi = d_0 \operatorname{ord}_F$ for a positive integer d_0 and a prime divisor F over X with center o, and $h \in I_G(F)$, so $|I_G(F)| \geq m$. Let $e := m/d_0$. Then

$$(\operatorname{ord}_F x, \operatorname{ord}_F y, \operatorname{ord}_F z, \operatorname{ord}_F t) = (ew_x, ew_y, ew_z, ew_t).$$

We compute $a(F, X, B)$. Let $\nu := \nu_F$ and $\varrho := \varrho_F$ be as in **Step 1**, and let v_0 be the monomial valuation in y, z, t at $(0, v)$ with weights (mw_y, mw_z, mw_t). By **Step 4**, $\operatorname{ord}_F|_{\mathbb{C}(U_y)} = \frac{1}{d_0}v_0$, so $d_0\varrho \operatorname{ord}_\nu = v_0$. By [JM12, Proposition 5.1, formula (5.1)], the log discrepancy of v_0 is $m(w_y + w_z + w_t)$, so

$$d_0\varrho a(\nu, U_y, 0) = m(w_y + w_z + w_t), \quad \text{that is,} \quad \varrho a(\nu, U_y, 0) = e(w_y + w_z + w_t).$$

By (9.6), $a(F, X, 0) = e(w_y + w_z + w_t) - ew_x$. Thus

$$a(F, X, B) = e\big(w_y - w_x + (1 - b_z)w_z + (1 - b_t)w_t\big) = e\Big(\frac{1}{2} + R\Big) = eP(w),$$

and

$$\frac{a(F, X, B)}{|I_G(F)|} \leq \frac{eP(w)}{m} = \frac{P(w)}{d_0} \leq P(w).$$

Together with **Steps 3, 6**, and **7**, this proves (2).

Step 9. In this step, we prove (3) and the ACC.

Let $\pi : X \to X/G$ be the quotient morphism. By Lemma 2.6(0), $(X, B)/G$ is a sub-pair, and by Lemma 2.6(2)(3), any prime divisor E over X/G satisfies

$$a(E, X/G, B_{X/G}) = \frac{a(F, X, B)}{|I_G(F)|} > 0$$

for some prime divisor F over X, so $(X, B)/G$ is sub-klt. Assume that $0 < a < 1$. By [Amb99, Section 1, the second paragraph after Definition 1.1], there exists a prime divisor E over X/G with center $\pi(o)$ such that $a(E, X/G, B_{X/G}) = a$. By Lemma 2.6(3), there exist a prime divisor F over X and a positive integer r such that $\operatorname{ord}_F|_{\mathbb{C}(X/G)} = r \operatorname{ord}_E$, and by Lemma 2.6(2), $\pi(\operatorname{center}_X F) = \operatorname{center}_{X/G} E = \{\pi(o)\}$ and $\sigma_F = a(E, X/G, B_{X/G}) = a$. Since G fixes o, $\pi^{-1}(\pi(o)) = \{o\}$, so $\operatorname{center}_X F = \{o\}$. By (1), $w_F \in N_G \cap \mathbb{R}^4_{>0}$ and $P(w_F) = a$.

Let $w \in N_G \cap \mathbb{R}^4_{>0}$. If $P(w) \geq 1$, then $P(w) > a$. If $P(w) < 1$, then (2) implies that there exists a prime divisor F' over X with center o such that $\sigma_{F'} \leq P(w)$, so $a \leq P(w)$ by the last statement of Lemma 2.6. Thus $a = \min_{w \in N_G \cap \mathbb{R}^4_{>0}} P(w)$, attained at w_F. If a vector w attaining the minimum satisfies $w = kw'$ for some $w' \in N_G$ and some integer $k \geq 2$, then $w' \in N_G \cap \mathbb{R}^4_{>0}$ and $P(w') = \frac{1}{k}P(w) = \frac{a}{k} < a$, because P is homogeneous of degree one, a contradiction. Thus any vector attaining the minimum is primitive in N_G. This proves (3).

Finally, let $J_1 := J(x^2 + y^2 + \hat{g})$, $J_2 := (z)$, and $J_3 := (t)$ in $\mathbb{C}[x, y, z, t]$, and $c := (1, b_z, b_t)$. Since $\hat{g} \in (z, t)^2$ has no monomial involving x or y,

$$\operatorname{wt}_w(J_1) = \min\{2w_x, 2w_y, \operatorname{wt}_{(w_z, w_t)}(\hat{g})\}.$$

Moreover, $\operatorname{wt}_w(J_2) = w_z$ and $\operatorname{wt}_w(J_3) = w_t$. Hence $P = \psi_{(J_1,J_2,J_3),c}$, and by (3), $a = \psi_{\min}(H, (J_1, J_2, J_3), c)$, where H is the image of G in $\mathbb{T}^4$. If $b_z, b_t \in \Gamma \cup \{0\}$, then $c \in \Lambda^3$ for the DCC set $\Lambda := \Gamma \cup \{0, 1\}$. Thus the ACC follows from Theorem 4.5. □

9.2. **The Morse splitting.** In this subsection, we prove Theorem 9.1 and deduce Corollary 9.6.

Proof of Theorem 9.1. The proof is long, so we divide it into several steps. Let χ be the character of f. We shrink Y to G-invariant affine neighborhoods of o whenever necessary.

Step 1. In this step, we find G-eigenparameters u_1, u_2, z, t of Y at o such that the restriction of the quadratic part of f to the (u_1, u_2)-plane is $Q_2(u_1, u_2)$ for some $Q_2 \in \{xy, x^2 + y^2\}$.

We first show that G-eigenparameters as in (2) exist in the situation of (1). By complete reducibility [Ser77, §1.4, Theorem 2], and since the irreducible representations of a finite abelian group are one-dimensional [Ser77, §3.1, Theorem 9], the representation of G on $\mathfrak{m}_o/\mathfrak{m}_o^2$ by pullback is a direct sum of characters. Let $\xi \in \mathfrak{m}_o/\mathfrak{m}_o^2$ be a non-zero vector with $\gamma^*\xi = \alpha(\gamma)\xi$ for any $\gamma \in G$, where α is a character of G, and let $\tilde{y} \in \mathfrak{m}_o$ be a lift of ξ. Then

$$\frac{1}{|G|}\sum_{\gamma\in G}\alpha(\gamma)^{-1}\gamma^*\tilde{y}$$

is a lift of ξ which is G-semi-invariant with character α. Thus there exist G-eigenparameters $y_1, \dots, y_4$ of Y at o. Let α_i be the character of y_i. In $\widehat{\mathcal{O}}_{Y,o} = \mathbb{C}[[y_1, \dots, y_4]]$ we have $\gamma^* f = \chi(\gamma) f$ and

$$\gamma^*(y_1^{\beta_1}\cdots y_4^{\beta_4}) = \prod_i \alpha_i(\gamma)^{\beta_i} y_1^{\beta_1}\cdots y_4^{\beta_4}.$$

Hence any monomial $y_1^{\beta_1}\cdots y_4^{\beta_4}$ occurring in f satisfies $\prod_i \alpha_i^{\beta_i} = \chi$. Write the quadratic part of f as $\sum_{i\le j} a_{ij} y_i y_j$. Then $\alpha_i\alpha_j = \chi$ whenever $a_{ij} \neq 0$.

Suppose first that $a_{ij} \neq 0$ for some $i < j$ with $\alpha_i \neq \alpha_j$. Then $\alpha_i^2 \neq \chi$, since otherwise $\alpha_i^2 = \chi = \alpha_i\alpha_j$. Similarly $\alpha_j^2 \neq \chi$. Thus $a_{ii} = a_{jj} = 0$, and the restriction of the quadratic part of f to the (y_i, y_j)-plane is $a_{ij} y_i y_j$, which is non-degenerate. Suppose next that $a_{ij} = 0$ whenever $\alpha_i \neq \alpha_j$. Then the quadratic part of f is $\sum_\alpha q_\alpha$, where α runs over the characters of G and q_α is a quadratic form in the y_i with $\alpha_i = \alpha$. A constant linear combination of the y_i with $\alpha_i = \alpha$ is G-semi-invariant with character α. Thus, possibly replacing $y_1, \dots, y_4$ by constant linear combinations of the y_i with equal characters, we may assume that each q_α is a sum of squares of some of the y_i with $\alpha_i = \alpha$. Since the quadratic part of f has rank at least two, there exist $i \neq j$ with $a_{ii} = a_{jj} = 1$ and $a_{ij} = 0$, and the restriction of the quadratic part of f to the (y_i, y_j)-plane is $y_i^2 + y_j^2$. In both cases let $u_1 := y_i$ and $u_2 := y_j$, and let z, t be the remaining y_k. Then u_1, u_2, z, t are G-eigenparameters as in (2).

Now let u_1, u_2, z, t be G-eigenparameters as in (2), let χ_1, χ_2 be the characters of u_1, u_2, and let $q(u_1, u_2)$ be the restriction of the quadratic part of f to the (u_1, u_2)-plane. As for $y_1, \dots, y_4$, any monomial in u_1, u_2, z, t occurring in f has character χ. Thus the coefficients of u_1^2, u_1u_2, and u_2^2 in q vanish unless $\chi_1^2 = \chi$, $\chi_1\chi_2 = \chi$, and $\chi_2^2 = \chi$, respectively. If $\chi_1 \neq \chi_2$, then $\chi_1\chi_2 = \chi$ excludes $\chi_1^2 = \chi$ and $\chi_2^2 = \chi$. Thus, since q is non-degenerate, either $q = cu_1u_2$ or $q = c_1u_1^2 + c_2u_2^2$ with $c, c_1, c_2 \in \mathbb{C}^*$, and multiplying u_1, u_2 by non-zero constants preserves G-eigenparameters. If $\chi_1 = \chi_2$, then any invertible constant linear change of (u_1, u_2) preserves G-eigenparameters, and any non-degenerate binary quadratic form is equivalent to u_1u_2. Thus, possibly replacing u_1, u_2 by constant linear combinations of u_1 and u_2, we may assume that

$$q = Q_2(u_1, u_2) \quad \text{for some} \quad Q_2 \in \{xy, x^2 + y^2\}.$$

The replacement changes neither z, t, nor B.

Let $u := (u_1, u_2)^T$, let $\mathcal{Q}$ be the invertible symmetric 2×2 matrix with $Q_2(u_1, u_2) = u^T\mathcal{Q}u$, and let $\sigma(\gamma) := \operatorname{diag}(\chi_1(\gamma), \chi_2(\gamma))$ for $\gamma \in G$. Then $\gamma^* u = \sigma(\gamma)u$. Since any monomial occurring in $Q_2(u_1, u_2)$ occurs in f, we have $\gamma^* Q_2(u_1, u_2) = \chi(\gamma)Q_2(u_1, u_2)$, that is,

$$\sigma(\gamma)\mathcal{Q}\sigma(\gamma) = \chi(\gamma)\mathcal{Q} \quad \text{for any } \gamma \in G. \tag{9.11}$$

Step 2. In this step, we construct the germs $(V \ni v)$, $(X' \ni o')$, and $(X_0 \ni o_0)$ of (3), keeping z and t fixed (cf. [GLS07, Theorem I.2.47] for the splitting lemma without group action).

Since u_1, u_2, z, t are regular parameters of Y at o, after shrinking Y they define a morphism $Y \to \mathbb{A}^4$ which is étale, and the derivations $\frac{\partial}{\partial u_1}, \frac{\partial}{\partial u_2}$ of the basis dual to du_1, du_2, dz, dt are defined on Y. For $\gamma \in G$ and $i = 1, 2$, the derivations $\gamma^* \circ \frac{\partial}{\partial u_i} \circ (\gamma^*)^{-1}$ and $\chi_i(\gamma)^{-1} \frac{\partial}{\partial u_i}$ take equal values on u_1, u_2, z, t, so they are equal. Hence

$$\gamma^* \frac{\partial f}{\partial u_i} = \chi(\gamma)\chi_i(\gamma)^{-1} \frac{\partial f}{\partial u_i}.$$

Let $V := (\frac{\partial f}{\partial u_1} = \frac{\partial f}{\partial u_2} = 0) \subset Y$ and $v := o$. Then V is G-invariant. Since $f \in \mathfrak{m}_o^2$ and the restriction of the quadratic part of f to the (u_1, u_2)-plane is $u^T \mathcal{Q} u$, we have

$$\Big(\frac{\partial f}{\partial u_1}, \frac{\partial f}{\partial u_2}\Big)^T \equiv 2\mathcal{Q}u + \ell \mod \mathfrak{m}_o^2,$$

where ℓ is a vector of linear forms in z, t. Since $\mathcal{Q}$ is invertible, $\frac{\partial f}{\partial u_1}, \frac{\partial f}{\partial u_2}, z, t$ are regular parameters of Y at o. Thus V is a smooth surface germ at v, the restrictions of z, t (still denoted z, t) are G-eigenparameters of V at v, and the morphism $V \to \mathbb{A}^2_{z,t}$ given by z, t is étale at v. Let $g := f|_V$. Then g is a regular G-semi-invariant function on V with character χ, and $g \in \mathfrak{m}_v^2$ since $f \in \mathfrak{m}_o^2$.

Let G act diagonally on $\mathbb{A}^2_{z,t}$ through the characters of z and t, so that the morphisms $Y \to \mathbb{A}^2_{z,t}$ and $V \to \mathbb{A}^2_{z,t}$ given by z, t are G-equivariant. Let $Y_1 := Y \times_{\mathbb{A}^2_{z,t}} V$ with the diagonal action of G, let $p_1 : Y_1 \to Y$ and $p_2 : Y_1 \to V$ be the projections, and let $o_1 := (o, v)$. Since $V \to \mathbb{A}^2_{z,t}$ is étale at v, p_1 is étale at o_1. On Y_1 we write u for $p_1^* u$, z, t for $p_1^* z = p_2^* z$ and $p_1^* t = p_2^* t$, and we let $u_V = (u_{V,1}, u_{V,2})^T := p_2^*(u|_V)$. Then $\gamma^* u_V = \sigma(\gamma) u_V$. Since $u|_V$ vanishes at v and z, t are regular parameters of V at v, the components of $u - u_V$ together with z, t are regular parameters of Y_1 at o_1. Let $\Sigma \subset Y_1$ be the image of the section $V \to Y_1$, $q \to (q, q)$, of p_2. Then $\Sigma \subset (u - u_V = 0)$, and both are smooth surface germs at o_1. Thus the ideal of Σ near o_1 is generated by $u_1 - u_{V,1}$ and $u_2 - u_{V,2}$.

Let $h := p_1^* f - p_2^* g$, and let $\frac{\partial}{\partial u_i}$ also denote the derivations of Y_1 dual to du_1, du_2, dz, dt. Since $p_2^* \Omega_V$ is generated by dz, dt near o_1, we have $\frac{\partial}{\partial u_i}(p_2^* g) = 0$ and $\frac{\partial}{\partial u_i} u_{V,j} = 0$. Thus $\frac{\partial h}{\partial u_i} = p_1^* \frac{\partial f}{\partial u_i}$, which vanishes on Σ. Moreover $h|_\Sigma = 0$, since $f|_V = g$. Write $h = \sum_i (u_i - u_{V,i}) h_i$ with regular functions h_i. Then $h_i|_\Sigma = \frac{\partial h}{\partial u_i}|_\Sigma = 0$, so h belongs to the square of the ideal of Σ. Hence

$$h = (u - u_V)^T \mathcal{A} (u - u_V)$$

for a symmetric 2×2 matrix $\mathcal{A}$ of regular functions near o_1. Comparing the second derivatives $\frac{\partial^2}{\partial u_i \partial u_j}$ at o_1, we obtain $2\mathcal{A}(o_1) = 2\mathcal{Q}$. For a point p of Y_1 near o_1, replace $\mathcal{A}(p)$ by

$$\frac{1}{|G|} \sum_{\gamma \in G} \chi(\gamma)^{-1} \sigma(\gamma)^T \mathcal{A}(\gamma p) \sigma(\gamma).$$

Since $h(\gamma p) = \chi(\gamma) h(p)$ and $(u - u_V)(\gamma p) = \sigma(\gamma)(u - u_V)(p)$, the expression $(u - u_V)^T \mathcal{A} (u - u_V)$ does not change, and by (9.11) we still have $\mathcal{A}(o_1) = \mathcal{Q}$. A direct calculation, substituting $\gamma\gamma'$ for γ in the sum, shows that

$$\mathcal{A}(\gamma' p) = \chi(\gamma') \sigma(\gamma')^{-1} \mathcal{A}(p) \sigma(\gamma')^{-1} \quad \text{for any } \gamma' \in G.$$

By (9.11), $\mathcal{Q}^{-1} \sigma(\gamma)^{-1} = \chi(\gamma)^{-1} \sigma(\gamma) \mathcal{Q}^{-1}$, and hence

$$\mathcal{Q}^{-1} \mathcal{A}(\gamma p) = \sigma(\gamma) \mathcal{Q}^{-1} \mathcal{A}(p) \sigma(\gamma)^{-1} \quad \text{for any } \gamma \in G. \tag{9.12}$$

Let $\mathbb{A}^4_{\mathcal{S}}$ be the affine space of 2×2 matrices $\mathcal{S}$, and let G act on $Y_1 \times \mathbb{A}^4_{\mathcal{S}}$ by

$$\gamma \cdot (p, \mathcal{S}) := (\gamma p, \sigma(\gamma) \mathcal{S} \sigma(\gamma)^{-1}).$$

Let Y' be the germ at $o' := (o_1, \mathrm{Id})$ of the closed subscheme $(\mathcal{S}^2 = \mathcal{Q}^{-1}\mathcal{A})$ of $Y_1 \times \mathbb{A}^4_{\mathcal{S}}$. By (9.12), Y' is G-invariant and G fixes o'. The differential of $\mathcal{S} \to \mathcal{S}^2$ at Id is $\Delta \to 2\Delta$, which is invertible.

Thus the projection $Y' \to Y_1$ is étale at o' by the Jacobian criterion, and Y' is a smooth fourfold germ at o'. Let $\mathcal{R}$ be the restriction of the matrix $\mathcal{S}$ of coordinate functions to Y'. Then

$$\mathcal{R}^2 = \mathcal{Q}^{-1}\mathcal{A}, \quad \mathcal{R}(o') = \mathrm{Id}, \quad \text{and} \quad \gamma^*\mathcal{R} = \sigma(\gamma)\mathcal{R}\sigma(\gamma)^{-1} \text{ for any } \gamma \in G.$$

We show that $\mathcal{R}^T\mathcal{Q} = \mathcal{Q}\mathcal{R}$. Since $\mathcal{A}$ and $\mathcal{Q}$ are symmetric, $(\mathcal{Q}^{-1}\mathcal{S}^T\mathcal{Q})^2 = \mathcal{Q}^{-1}(\mathcal{S}^2)^T\mathcal{Q}$ equals $\mathcal{Q}^{-1}\mathcal{A}$ if $\mathcal{S}^2 = \mathcal{Q}^{-1}\mathcal{A}$. Thus $\iota(p, \mathcal{S}) := (p, \mathcal{Q}^{-1}\mathcal{S}^T\mathcal{Q})$ defines an automorphism ι of Y' over Y_1 fixing o'. Since $Y' \to Y_1$ is étale at o' and both residue fields are $\mathbb{C}$, the induced homomorphism $\widehat{\mathcal{O}}_{Y_1,o_1} \to \widehat{\mathcal{O}}_{Y',o'}$ is an isomorphism, so ι^* is the identity on $\widehat{\mathcal{O}}_{Y',o'}$, and hence on $\mathcal{O}_{Y',o'} \subset \widehat{\mathcal{O}}_{Y',o'}$. Therefore $\mathcal{R} = \mathcal{Q}^{-1}\mathcal{R}^T\mathcal{Q}$, that is, $\mathcal{R}^T\mathcal{Q} = \mathcal{Q}\mathcal{R}$, and

$$\mathcal{R}^T\mathcal{Q}\mathcal{R} = \mathcal{Q}\mathcal{R}^2 = \mathcal{A}.$$

On Y' define regular functions x, y by

$$(x, y)^T := \mathcal{R}(u - u_V),$$

where u, u_V also denote their pullbacks to Y'. Then

$$Q_2(x, y) = (u - u_V)^T\mathcal{R}^T\mathcal{Q}\mathcal{R}(u - u_V) = (u - u_V)^T\mathcal{A}(u - u_V) = h.$$

Thus the pullback of f to Y' is $Q_2(x, y)$ plus the pullback of g. Moreover

$$\gamma^*(x, y)^T = \sigma(\gamma)\mathcal{R}\sigma(\gamma)^{-1}\sigma(\gamma)(u - u_V) = \sigma(\gamma)(x, y)^T,$$

so x and y are G-semi-invariant with characters χ_1 and χ_2. Since $\mathcal{R}(o') = \mathrm{Id}$ and $Y' \to Y_1$ is étale at o', the functions x, y, z, t are regular parameters of Y' at o'.

Let G act diagonally on $\mathbb{A}^2_{x,y}$ so that the coordinates x, y are G-semi-invariant with characters χ_1, χ_2, and let G act on $\mathbb{A}^2_{x,y} \times V$ by the product action. Let $o_0 := (0, 0, v)$ and

$$X_0 := (Q_2(x, y) + g = 0) \subset \mathbb{A}^2_{x,y} \times V.$$

The morphism $Y' \to \mathbb{A}^2_{x,y} \times V$ given by x, y and the composite $Y' \to Y_1 \to V$ is G-equivariant, maps o' to o_0, and pulls back the regular parameters x, y, z, t of $\mathbb{A}^2_{x,y} \times V$ at o_0 to regular parameters of Y' at o'. Thus it is étale at o'. The composite $Y' \to Y_1 \to Y$ is G-equivariant and étale at o'. Let $X' \subset Y'$ be the inverse image of X. By the previous paragraph, X' is also the inverse image of X_0. Let $\tau : X' \to X$ and $\tau_0 : X' \to X_0$ be the induced morphisms. Then τ and τ_0 are G-equivariant and étale at o'. Possibly shrinking X', we may assume that $\tau^{-1}(o) = \tau_0^{-1}(o_0) = \{o'\}$, since τ and τ_0 are quasi-finite at o'. By construction, τ^*z and τ_0^*z are both the pullback of z on Y_1, and similarly for t. Hence

$$\tau^*B = \tau_0^*B_0 = b_z \operatorname{div}_{X'}(z) + b_t \operatorname{div}_{X'}(t), \quad \text{where } B_0 := b_z \operatorname{div}_{X_0}(z) + b_t \operatorname{div}_{X_0}(t).$$

Step 3. In this step, we show that $(V \ni v)$, z, t, g, X_0, and B_0 satisfy the assumptions of Propositions 9.4 and 9.5, and we prove (3).

The varieties X' and X_0 are hypersurfaces in smooth fourfolds, so they are Cohen–Macaulay. Since τ and τ_0 are étale at o', étale morphisms preserve and reflect regularity, and X is normal, X' and X_0 are regular in codimension one near o' and o_0. Thus X' and X_0 are normal at o' and o_0 by Serre's criterion. The base change by τ (resp. τ_0) of a log resolution of $(X, \operatorname{Supp} B)$ near o (resp. of $(X_0, \operatorname{Supp} B_0)$ near o_0) is a log resolution of $(X', \operatorname{Supp} \tau^*B)$ near o', the canonical divisors pull back to canonical divisors under étale morphisms, and the images of τ and τ_0 are open. Since X is klt at o and (X, B) is sub-klt, X_0 is klt at o_0 and (X_0, B_0) is sub-klt near o_0.

If $g = 0$, then $X_0 = (Q_2(x, y) = 0)$ near o_0, and $Q_2(x, y)$ is xy or $(x + \sqrt{-1}y)(x - \sqrt{-1}y)$, so $\mathcal{O}_{X_0,o_0}$ is not a domain, which contradicts the normality of X_0 at o_0. Thus g is non-zero. The function $Q_2(x, y) + g$ is G-semi-invariant with character χ, since x, y have the characters χ_1, χ_2 of u_1, u_2.

Let $\gamma \in G$ act trivially on X_0 near o_0. Since τ_0 is étale at o' and both residue fields are $\mathbb{C}$, $\widehat{\mathcal{O}}_{X_0,o_0} \to \widehat{\mathcal{O}}_{X',o'}$ is a G-equivariant isomorphism. Thus γ^* is the identity on $\widehat{\mathcal{O}}_{X',o'}$, hence on $\mathcal{O}_{X',o'}$ and on $\mathbb{C}(X')$. Since τ is dominant, $\mathbb{C}(X) \subset \mathbb{C}(X')$, and γ acts trivially on X. Since G acts faithfully on X, γ is the identity. Thus G acts faithfully on X_0.

By Lemma 9.3 applied to τ, for any prime divisor E over X with center o there exists a unique prime divisor E' over X' with center o' and $\mathrm{ord}_{E'}|_{\mathbb{C}(X)} = \mathrm{ord}_E$. Let E_0 be the prime divisor over X_0 with center o_0 and $\mathrm{ord}_{E_0} = \mathrm{ord}_{E'}|_{\mathbb{C}(X_0)}$. By Lemma 9.3 applied to τ and to τ_0, $E \to E_0$ is a bijection between the prime divisors over X with center o and the prime divisors over X_0 with center o_0, the residue fields of E, E', and E_0 are isomorphic, $I_G(E) = I_G(E') = I_G(E_0)$, and

$$a(E, X, B) = a(E', X', \tau^* B) = a(E', X', \tau_0^* B_0) = a(E_0, X_0, B_0).$$

Lemma 9.3 holds for any $\mathbb{R}$-divisor, so the signs of b_z and b_t play no role here. The $\mathbb{R}$-divisors B and B_0 are G-invariant since z and t are G-semi-invariant. Thus, by Lemma 2.6,

$$\mathrm{mld}((X \ni o, B)/G) = \inf_E \frac{a(E, X, B)}{|I_G(E)|} = \inf_{E_0} \frac{a(E_0, X_0, B_0)}{|I_G(E_0)|} = \mathrm{mld}((X_0 \ni o_0, B_0)/G),$$

where E (resp. E_0) runs over the prime divisors over X (resp. X_0) with center o (resp. o_0).

Assume that $a := \mathrm{mld}((X \ni o, B)/G) \in (0, 1)$. If $Q_2 = xy$, then Proposition 9.4(2) implies that $a = \min_{w \in N_G \cap \mathbb{R}^4_{>0}} P(w)$, attained at a primitive vector, where

$$P(w) = w_x + w_y + w_z + w_t - \min\{w_x + w_y, \mathrm{wt}_{(w_z, w_t)}(\hat{g})\} - b_z w_z - b_t w_t.$$

If $Q_2 = x^2 + y^2$, then Proposition 9.5(3) implies that $a = \min_{w \in N_G \cap \mathbb{R}^4_{>0}} P(w)$, attained at a primitive vector, where

$$P(w) = w_x + w_y + w_z + w_t - \min\{2w_x, 2w_y, \mathrm{wt}_{(w_z, w_t)}(\hat{g})\} - b_z w_z - b_t w_t.$$

Since $\hat{g} \in \mathbb{C}[[z, t]]$ contains no monomial divisible by x or y, in both cases

$$P(w) = \sum_j w_j - \mathrm{wt}_w(Q_2 + \hat{g}) - b_z w_z - b_t w_t.$$

This proves (3) in the situation of (2). In the situation of (1), **Step 1** provides G-eigenparameters u_1, u_2, z, t as in (2), and $b_z = b_t = 0$, so (3) holds with $B = 0$.

Step 4. In this step, we prove (1) and (2).

Let $\Lambda := \Gamma \cup \{0, 1\}$, which is a DCC set. Identify x, y, z, t with $x_1, \dots, x_4$, let $J_1 \subset \mathbb{C}[x, y, z, t]$ be the ideal generated by the monomials occurring in $Q_2 + \hat{g}$, which is a non-zero monomial ideal although $\hat{g}$ may have infinitely many terms, and let $J_2 := (z)$ and $J_3 := (t)$. Let $H \subset \mathbb{T}^4$ be the image of G under the characters of x, y, z, t, which is a finite subgroup with $N_H = N_G$. Since $\mathrm{wt}_w(J_2) = w_z$ and $\mathrm{wt}_w(J_3) = w_t$, (3) implies that any number $a = \mathrm{mld}((X \ni o, B)/G) \in (0, 1)$ in (2) satisfies

$$a = \psi_{\min}(H, (J_1, J_2, J_3), (1, b_z, b_t)), \quad (1, b_z, b_t) \in \Lambda^3.$$

Thus (2) follows from Theorem 4.5 with $d = 4$ and $s = 3$. In the situation of (1), any number $a = \mathrm{mld}((X \ni o, B)/G) \in (0, 1)$ in (1) satisfies $a = \psi_{\min}(H, (J_1, J_2, J_3), (1, 0, 0))$, so (1) follows from Theorem 4.5 as well.

Finally, let $(X \ni o)$ be an isolated Gorenstein terminal threefold singularity with a faithful action of a finite abelian group G fixing o. By Lemma 5.4, there exist a smooth fourfold germ $(Y \ni o)$ with a faithful action of G and a G-equivariant closed embedding $X \subset Y$ with $X = (f = 0)$ for a regular G-semi-invariant function f. Since X is singular at o, f vanishes to order at least two at o. Since X is terminal, X is normal and klt. The rank of the quadratic part of a local equation of X in a smooth fourfold germ does not depend on the embedding (Definition-Lemma 11.1). Thus, if the quadratic part of f has rank at least two, then (1) holds for $(X \ni o)$. □

In Propositions 11.11 and 12.13, points of quadratic rank one are reduced by fixed weighted blow-ups to points of quadratic rank at least two, and the coordinate divisors acquire negative coefficients from a fixed finite set: -2 in Proposition 11.11, and $-k$ with $k \in \{3, 4, 6\}$ in Proposition 12.13. The formula of Theorem 9.1(3) holds for any real $b_z, b_t < 1$, while the ACC below uses that (b_z, b_t) runs over a finite set of rational pairs. Example 9.7 shows that a negative

boundary which is not supported on fixed coordinate divisors destroys the ACC already for surfaces: with the single coefficient -1, the minimal log discrepancies $1-\frac{1}{r}$ strictly increase.

Corollary 9.6. *Fix a finite set $\mathcal{B}$ of pairs of rational numbers (b_z, b_t) with $b_z, b_t < 1$. Let $(Y \ni o)$ be a smooth fourfold germ with a finite abelian group G fixing o, f a regular G-semi-invariant function vanishing to order at least two at o such that $X := (f = 0)$ is a normal klt threefold germ on which G acts faithfully, and u_1, u_2, z, t G-eigenparameters of Y at o such that the restriction of the quadratic part of f to the (u_1, u_2)-plane is non-degenerate. Let $(b_z, b_t) \in \mathcal{B}$ and $B := b_z \operatorname{div}_X(z) + b_t \operatorname{div}_X(t)$ with (X, B) sub-klt. Then the set of numbers $\operatorname{mld}((X \ni o, B)/G) \in (0, 1)$ for such data satisfies the ACC.*

Proof. Since the restriction of the quadratic part of f to the (u_1, u_2)-plane is non-degenerate, the quadratic part of f has rank at least two. Let $a := \operatorname{mld}((X \ni o, B)/G) \in (0, 1)$. By Theorem 9.1(3), there exist $Q_2 \in \{xy, x^2 + y^2\}$, a non-zero power series $\hat{g} \in \mathbb{C}[[z, t]]$, and characters of G on the coordinates x, y, z, t such that, if $H \subset \mathbb{T}^4$ is the image of G under the characters of x, y, z, t, so that $N_H = N_G$ (Definition 2.15),

$$a = \min_{w \in N_H \cap \mathbb{R}^4_{>0}} \Big(w_x + w_y + (1 - b_z) w_z + (1 - b_t) w_t - \operatorname{wt}_w(Q_2 + \hat{g}) \Big).$$

The vector $(1, 1, 1 - b_z, 1 - b_t)$ belongs to the finite set $\{(1, 1, 1 - b'_z, 1 - b'_t) \mid (b'_z, b'_t) \in \mathcal{B}\} \subset \mathbb{Q}^4_{>0}$, and $\operatorname{wt}_w(Q_2 + \hat{g}) = \operatorname{wt}_w(J)$ for the non-zero monomial ideal J generated by the monomials occurring in $Q_2 + \hat{g}$. Thus the ACC follows from Corollary 4.6 with $d = 4$, $s = 1$, and $\Lambda = \{1\}$. □

Example 9.7. For a positive integer n, let $r := 3n + 1$, let the group μ_r of r-th roots of unity act on $\mathbb{A}^2_{u,v}$ by $(u, v) \to (\zeta u, \zeta v)$ for $\zeta \in \mu_r$, let $\pi : \mathbb{A}^2 \to X_r := \mathbb{A}^2/\mu_r$ be the quotient morphism with $o_r := \pi(0)$, and let $D_r := \pi(C_r)$ for $C_r := (u^{r-3} - v^{2r-3} = 0)$. Let F be the exceptional divisor of the blow-up of $\mathbb{A}^2$ at 0, and E the prime divisor over X_r such that the restriction of ord_F to $\mathbb{C}(X_r)$ is a positive multiple of ord_E. Then X_r is a $\mathbb{Q}$-factorial klt surface, D_r is a prime divisor with $\pi^* D_r = C_r$, the sub-pair $(X_r, -D_r)$ is sub-klt, and $\operatorname{mld}(X_r \ni o_r, -D_r) = 1 - \frac{1}{r}$, attained by E. In particular, the numbers $\operatorname{mld}(X_r \ni o_r, -D_r)$ form a strictly increasing sequence.

Proof. Let $h := u^{r-3} - v^{2r-3}$. Since $r = 3n + 1$, we have $r - 3 \geq 1$ and

$$\gcd(r - 3, 2r - 3) = \gcd(r - 3, r) = \gcd(3, r) = 1.$$

Thus $(r - 3, -(2r - 3))$ is a primitive vector of $\mathbb{Z}^2$, so there exists a Laurent monomial s_2 in u, v such that $\mathbb{C}[u^{\pm 1}, v^{\pm 1}] = \mathbb{C}[s_1^{\pm 1}, s_2^{\pm 1}]$ for $s_1 := u^{r-3} v^{-(2r-3)}$. Since $h = v^{2r-3}(s_1 - 1)$, the polynomial h is prime in $\mathbb{C}[u^{\pm 1}, v^{\pm 1}]$. A non-constant polynomial which is a unit in $\mathbb{C}[u^{\pm 1}, v^{\pm 1}]$ is a monomial, and neither u nor v divides h. Hence h is irreducible in $\mathbb{C}[u, v]$, and C_r is a prime divisor on $\mathbb{A}^2$.

The quotient $X_r = \operatorname{Spec} \mathbb{C}[u, v]^{\mu_r}$ is an affine surface and π is finite [Mum74, §7, Theorem]. The surface X_r is normal, since $\mathbb{C}[u, v]^{\mu_r} = \mathbb{C}[u, v] \cap \mathbb{C}(u, v)^{\mu_r}$ is integrally closed, as $\mathbb{C}[u, v]$ is integrally closed in $\mathbb{C}(u, v)$. If $\zeta \neq 1$, then $(\zeta a, \zeta b) = (a, b)$ only for $a = b = 0$. Thus $\pi^{-1}(o_r) = \{0\}$ and π is étale over $X_r \backslash \{o_r\}$, so π is quasi-étale and $R(\pi) = 0$. The surface X_r is $\mathbb{Q}$-factorial by [KM98, Lemma 5.16], since π is finite and surjective and $\mathbb{A}^2$ is smooth. The group μ_r acts faithfully on $\mathbb{A}^2$, and $(\mathbb{A}^2, 0)/\mu_r = (X_r, 0)$. By Lemma 2.6(2)(3), any prime divisor E' over X_r satisfies $a(E', X_r, 0) = a(F', \mathbb{A}^2, 0)/|I_{\mu_r}(F')| > 0$ for some prime divisor F' over $\mathbb{A}^2$. Thus X_r is klt.

Since $\zeta^{2r-3} = \zeta^{r-3}$, we have $h(\zeta u, \zeta v) = \zeta^{r-3} h(u, v)$. Thus C_r is μ_r-invariant, D_r is a prime divisor, and $\pi^{-1}(D_r) = C_r$. Since π is étale at the generic point of C_r, $\pi^* D_r = C_r$. The function h^r is μ_r-invariant, so $h^r = \pi^* \varphi$ for a regular function φ on X_r. Since $\pi^* \operatorname{div}(\varphi) = r C_r = \pi^*(r D_r)$ and π^* is injective on divisors, $\operatorname{div}(\varphi) = r D_r$. By Lemma 2.6(4) with $b = -1$ and $e_{C_r} = 1$, we have $(\mathbb{A}^2, -C_r)/\mu_r = (X_r, -D_r)$. For any prime divisor E' over X_r,

$$a(E', X_r, -D_r) = a(E', X_r, 0) + \frac{1}{r} \operatorname{ord}_{E'}(\varphi) > 0,$$

since X_r is klt and φ is regular. Thus $(X_r, -D_r)$ is sub-klt.

Let E' be a prime divisor over X_r with center o_r. By Lemma 2.6(2)(3), there exists a prime divisor F' over $\mathbb{A}^2$ with $\pi(\mathrm{center}_{\mathbb{A}^2} F') = \{o_r\}$ and

$$a(F', \mathbb{A}^2, -C_r) = |I_{\mu_r}(F')| \cdot a(E', X_r, -D_r).$$

Since $\pi^{-1}(o_r) = \{0\}$, $\mathrm{center}_{\mathbb{A}^2} F' = \{0\}$, so $w_u := \mathrm{ord}_{F'} u \geq 1$ and $w_v := \mathrm{ord}_{F'} v \geq 1$. Since $(\mathbb{A}^2, \mathrm{div}(u)+\mathrm{div}(v))$ is lc [KM98, Corollary 2.31(3)], we have $a(F', \mathbb{A}^2, 0) \geq w_u + w_v \geq 2$. Moreover $\mathrm{ord}_{F'} h \geq \min\{(r-3)w_u, (2r-3)w_v\} \geq r-3$. Hence

$$a(F', \mathbb{A}^2, -C_r) = a(F', \mathbb{A}^2, 0) + \mathrm{ord}_{F'} h \geq r - 1.$$

Since $|I_{\mu_r}(F')| \leq |\mu_r| = r$, we obtain $a(E', X_r, -D_r) \geq \frac{r-1}{r}$.

We have $\mathrm{ord}_F u = \mathrm{ord}_F v = 1$, $a(F, \mathbb{A}^2, 0) = 2$ [KM98, Lemma 2.29], and $\mathrm{ord}_F h = r-3$, since the lowest-degree homogeneous part of h is u^{r-3}. Thus $a(F, \mathbb{A}^2, -C_r) = r-1$. On the affine open subset of the blow-up with coordinates u, v_1, where $v = uv_1$, we have $F = (u = 0)$, $\zeta^* u = \zeta u$, and $\zeta^* v_1 = v_1$. Thus μ_r stabilizes ord_F and acts trivially on the residue field $\mathbb{C}(v_1)$ of ord_F, so $I_{\mu_r}(F) = \mu_r$ and $\mathrm{ord}_F|_{\mathbb{C}(X_r)} = r \,\mathrm{ord}_E$ by Lemma 2.6(1). By Lemma 2.6(2), $\mathrm{center}_{X_r} E = \{o_r\}$ and

$$a(E, X_r, -D_r) = \frac{a(F, \mathbb{A}^2, -C_r)}{r} = 1 - \frac{1}{r}.$$

Together with the lower bound, $\mathrm{mld}(X_r \ni o_r, -D_r) = 1 - \frac{1}{r} = 1 - \frac{1}{3n+1}$, which strictly increases with n. □

In Example 9.7, the boundary $-D_r$ has coefficient -1 along D_r, the image $\pi(C_r)$ of the curve C_r, whose multiplicity $r-3$ at the origin is unbounded. Since $-D_r$ has a negative coefficient, Example 9.7 is not a counterexample to the ACC for boundaries with coefficients in $[0, 1]$.

10. Double covers of threefold germs

In this section, we prove Proposition 10.5: for the double cover $(x^2 + h = 0)$ of $\mathbb{A}^3_{y,z,t}$ such that $((\mathbb{C}^*)^3, \frac{1}{2}\,\mathrm{div}(\mathrm{in}_u(h)))$ is lc for any $u \in \mathbb{R}^3_{>0}$, the quotient mld in $(0, 1)$ is a lattice Newton minimum, and the set of such values satisfies the ACC when (b_y, b_z, b_t) belongs to a fixed finite subset of $\mathbb{Q}^3_{\leq 0}$.

The following lemma is used to verify the hypothesis of Proposition 10.5 on leading forms.

Lemma 10.1.

(1) Let $(X \ni p)$ be a smooth surface germ and $C \geq 0$ a Weil divisor on X with $\mathrm{mult}_p C \leq 2$. Then $(X, \frac{1}{2}C)$ is lc near p.

(2) Let $\phi \in \mathbb{C}[u, v]$ be a non-zero polynomial of degree at most three which is not a constant multiple of the cube of a non-constant affine linear form. Then $(\mathbb{A}^2, \frac{1}{2}\,\mathrm{div}(\phi))$ is lc.

Proof. (1) Since the multiplicity of C at a closed point is upper semicontinuous, possibly replacing X with a neighborhood of p, we may assume that $\mathrm{mult}_q C \leq 2$ for any closed point $q \in X$. Let $0 < t < 1$. Then $\mathrm{mult}_q(\frac{t}{2}C) \leq t < 1$ for any closed point $q \in X$, so $(X, \frac{t}{2}C)$ is klt by [Laz04, Proposition 9.5.13]. For any prime divisor E over X, $a(E, X, \frac{t}{2}C)$ is an affine function of t which is positive for $0 < t < 1$. Letting $t \to 1$, we obtain $a(E, X, \frac{1}{2}C) \geq 0$. Thus $(X, \frac{1}{2}C)$ is lc.

(2) Let $q \in \mathbb{A}^2$ be a closed point. If $\mathrm{mult}_q \phi \leq 2$, then $(\mathbb{A}^2, \frac{1}{2}\,\mathrm{div}(\phi))$ is lc near q by (1). Suppose that $\mathrm{mult}_q \phi \geq 3$. Since $\deg \phi \leq 3$, we have $\mathrm{mult}_q \phi = \deg \phi = 3$. Write $q = (u_0, v_0)$. Since $\phi(u+u_0, v+v_0)$ is a non-zero polynomial of degree at most three which is not a constant multiple of the cube of a non-constant affine linear form, possibly replacing ϕ with $\phi(u+u_0, v+v_0)$, we may assume that $q = 0$. Then all terms of ϕ of degree at most two vanish, so ϕ is homogeneous of degree three, and

$$\phi = c\ell_1^{m_1} \cdots \ell_k^{m_k},$$

where $c \in \mathbb{C}^*$, $\ell_1, \ldots, \ell_k$ are pairwise non-proportional linear forms, and $m_1, \ldots, m_k$ are positive integers with $\sum_i m_i = 3$. By assumption $k \geq 2$, so $m_i \leq 2$ for any i. Let $f : Y \to \mathbb{A}^2$ be

the blow-up of 0 with exceptional curve E, and let L_i be the strict transform of $\operatorname{div}(\ell_i)$. Then $f^*\operatorname{div}(\phi) = \sum_i m_i L_i + 3E$ and $K_Y = f^*K_{\mathbb{A}^2} + E$, so

$$K_Y + \frac{1}{2}\sum_{i=1}^{k} m_i L_i + \frac{1}{2}E = f^*\Big(K_{\mathbb{A}^2} + \frac{1}{2}\operatorname{div}(\phi)\Big).$$

We have $(L_i \cdot E) = 1$ for any i, $L_i \cap L_j = \emptyset$ for $i \neq j$, and the points $L_1 \cap E, \dots, L_k \cap E$ are distinct. Thus $E + \sum_i L_i$ is an snc divisor, and the coefficients $\frac{1}{2}$ and $\frac{m_i}{2}$ are at most 1. Hence $(Y, \frac{1}{2}\sum_i m_i L_i + \frac{1}{2}E)$ is lc [KM98, Corollary 2.31(3)]. For any prime divisor F over $\mathbb{A}^2$, [KM98, Lemma 2.30] gives

$$a\Big(F, Y, \frac{1}{2}\sum_i m_i L_i + \frac{1}{2}E\Big) = a\Big(F, \mathbb{A}^2, \frac{1}{2}\operatorname{div}(\phi)\Big),$$

so $(\mathbb{A}^2, \frac{1}{2}\operatorname{div}(\phi))$ is lc near 0. □

The following lemma reduces the hypothesis of Proposition 10.5 on leading forms to integral weights and to the plane.

Lemma 10.2. *Let $P \in \mathbb{C}[[y,z,t]]$ be a non-zero power series, and for $u \in \mathbb{Z}^3_{>0}$ let $p_u(Y,Z) := \operatorname{in}_u(P)(Y,Z,1) \in \mathbb{C}[Y,Z]$, where in_u is taken with respect to y,z,t. If $((\mathbb{C}^*)^2, \frac{1}{2}\operatorname{div}(p_u))$ is lc for any $u \in \mathbb{Z}^3_{>0}$, then $((\mathbb{C}^*)^3, \frac{1}{2}\operatorname{div}(\operatorname{in}_u(P)))$ is lc for any $u \in \mathbb{R}^3_{>0}$.*

Proof. Let $V \subset \mathbb{Z}^3_{\geq 0}$ be the set of componentwise minimal exponents of P. By Definition-Lemma 7.3, V is finite, and for any $u \in \mathbb{R}^3_{>0}$ the exponents of the terms of $\operatorname{in}_u(P)$ belong to V. Let $u \in \mathbb{R}^3_{>0}$. Then the set of exponents of the terms of $\operatorname{in}_u(P)$ is

$$V_u := \{m \in V \mid \langle u, m\rangle = \operatorname{wt}_u(P)\}.$$

Let C_u be the set of $u' \in \mathbb{R}^3_{>0}$ such that $\langle u', m\rangle = \langle u', m'\rangle$ and $\langle u', m\rangle < \langle u', m''\rangle$ for any $m, m' \in V_u$ and $m'' \in V \backslash V_u$. For $u' \in C_u$, the elements of V_u have the least u'-weight among the elements of V, so the set of exponents of the terms of $\operatorname{in}_{u'}(P)$ is also V_u, and $\operatorname{in}_{u'}(P) = \operatorname{in}_u(P)$. The set C_u is defined by finitely many linear equations and strict linear inequalities with rational coefficients, and $u \in C_u$. Thus C_u is a non-empty open subset of a linear subspace of $\mathbb{R}^3$ defined over $\mathbb{Q}$, so C_u contains a vector $u' \in \mathbb{Q}^3_{>0}$. Since $\operatorname{in}_{ku'}(P) = \operatorname{in}_{u'}(P)$ for any positive integer k, we may assume that $u \in \mathbb{Z}^3_{>0}$.

Let $u = (u_y, u_z, u_t) \in \mathbb{Z}^3_{>0}$ and $e : (\mathbb{C}^*)^3 \to (\mathbb{C}^*)^3$ the homomorphism given by $y = s^{u_y}Y$, $z = s^{u_z}Z$, and $t = s^{u_t}$ in coordinates Y, Z, s on the domain of e. Since the determinant of the exponent matrix of e is $u_t \neq 0$, e is finite and étale. Since $\operatorname{in}_u(P)$ is a polynomial whose monomials have u-weight $\operatorname{wt}_u(P)$, we have

$$e^*\operatorname{in}_u(P) = s^{\operatorname{wt}_u(P)} p_u(Y,Z),$$

and s is a unit. Thus

$$e^*\frac{1}{2}\operatorname{div}(\operatorname{in}_u(P)) = \frac{1}{2}\operatorname{div}(p_u),$$

which is the pullback of $\frac{1}{2}\operatorname{div}(p_u)$ on $(\mathbb{C}^*)^2_{Y,Z}$ under the projection. The product of a log resolution of $((\mathbb{C}^*)^2, \frac{1}{2}\operatorname{div}(p_u))$ with $\mathbb{C}^*$ is a log resolution of $((\mathbb{C}^*)^3, \frac{1}{2}\operatorname{div}(p_u))$ in which, for each prime divisor E on the first resolution, $E \times \mathbb{C}^*$ has the discrepancy of E, so $((\mathbb{C}^*)^3, \frac{1}{2}\operatorname{div}(p_u))$ is lc [KM98, Lemma 2.30 and Corollary 2.31(3)], and log canonicity is preserved and reflected by the finite étale morphism e [KM98, Proposition 5.20]. Thus $((\mathbb{C}^*)^3, \frac{1}{2}\operatorname{div}(\operatorname{in}_u(P)))$ is lc. □

Set-up 10.3. Let b_y, b_z, b_t, h, X, p, G, B, and P be as follows.

(1) $b_y, b_z, b_t \in \mathbb{Q}_{\leq 0}$.
(2) $h(y,z,t)$ is a non-zero polynomial vanishing at 0 such that $X := (x^2 + h = 0) \subset \mathbb{A}^4$ is a normal klt threefold germ at 0.
(3) $p : X \to \mathbb{A}^3_{y,z,t}$ is the projection, through which $\mathbb{C}(y,z,t)$ is a subfield of $\mathbb{C}(X)$.

(4) G is a finite abelian group acting faithfully on $\mathbb{A}^4$ and diagonally in the coordinates x, y, z, t, such that the polynomial $x^2 + h$ is G-semi-invariant and the induced action of G on X is faithful.
(5) $B := b_y \operatorname{div}_X(y) + b_z \operatorname{div}_X(z) + b_t \operatorname{div}_X(t)$.
(6) $N_G \subset \mathbb{R}^4$ is taken with respect to the G-semi-invariant functions x, y, z, t, and $w \in \mathbb{R}^4$ is written $w = (w_x, w_y, w_z, w_t)$.
(7) For $w \in \mathbb{R}^4_{>0}$,

$$P(w) := w_x + (1 - b_y)w_y + (1 - b_z)w_z + (1 - b_t)w_t - \min\{2w_x, \operatorname{wt}_{(w_y,w_z,w_t)}(h)\}.$$

The following lemma is the main step in the proof of Proposition 10.5. Part (1) bounds the quotient mld from above by the values of P at lattice vectors, and part (2) proves the equality. In (2), the inequality between log discrepancies on $\mathbb{A}^3$ is assumed only at one prime divisor, which is the form used in Proposition 10.6.

Lemma 10.4. *Notation and conditions as in Set-up 10.3.*

(1) If $w \in N_G \cap \mathbb{R}^4_{>0}$ and $P(w) < 1$, then $\operatorname{mld}((X \ni 0, B)/G) \le P(w)$.
(2) Assume that $a := \operatorname{mld}((X \ni 0, B)/G) \in (0, 1)$ is attained by a prime divisor E over X/G, and that there exist a prime divisor F over X with $\operatorname{ord}_F|_{\mathbb{C}(X/G)} \in \mathbb{N}^+ \operatorname{ord}_E$, a positive integer ϱ, and a prime divisor ν over $\mathbb{A}^3_{y,z,t}$ with center 0 such that $\operatorname{ord}_F|_{\mathbb{C}(y,z,t)} = \varrho \operatorname{ord}_\nu$ and, for $v := (\operatorname{ord}_\nu y, \operatorname{ord}_\nu z, \operatorname{ord}_\nu t)$,

$$a(\nu, \mathbb{A}^3, 0) - \tfrac{1}{2} \operatorname{ord}_\nu h \ge v_y + v_z + v_t - \tfrac{1}{2} \operatorname{wt}_v(h).$$

Then $a = \min_{w \in N_G \cap \mathbb{R}^4_{>0}} P(w)$, attained at a primitive vector.

Proof. The proof is long, so we divide it into several steps. For $w \in \mathbb{R}^4_{\ge 0}$ let

$$\tau(w) := \operatorname{wt}_{(w_y,w_z,w_t)}(h) \quad \text{and} \quad R(w) := (1 - b_y)w_y + (1 - b_z)w_z + (1 - b_t)w_t - \frac{1}{2}\tau(w).$$

Both functions depend only on (w_y, w_z, w_t), and $\tau(sw) = s\tau(w)$, $R(sw) = sR(w)$ for any real number $s > 0$. Since h is non-zero and vanishes at 0, we have $\tau(w) > 0$ for $w \in \mathbb{R}^4_{>0}$. Considering the cases $2w_x \le \tau(w)$ and $2w_x \ge \tau(w)$ in the definition of P separately, we obtain

$$P(w) = R(w) + \left| w_x - \frac{1}{2}\tau(w) \right| \quad \text{for any } w \in \mathbb{R}^4_{>0}. \tag{10.1}$$

By adjunction [KM98, Proposition 5.73] applied to the closure of X in $\mathbb{P}^4$ (see also [Har77, Theorem III.7.11]), ω_X is generated near 0 by the image under the residue map of the local generator $dx \wedge dy \wedge dz \wedge dt/(x^2 + h)$ of $\omega_{\mathbb{A}^4}(X)$, which is

$$\Omega := \frac{dy \wedge dz \wedge dt}{\partial(x^2 + h)/\partial x} = \frac{dy \wedge dz \wedge dt}{2x}.$$

Thus for any prime divisor F' over X with $\operatorname{center}_X F' = \{0\}$,

$$a(F', X, B) = 1 + \operatorname{ord}_{F'} \Omega - b_y \operatorname{ord}_{F'} y - b_z \operatorname{ord}_{F'} z - b_t \operatorname{ord}_{F'} t, \tag{10.2}$$

where $\operatorname{ord}_{F'} \Omega$ is the order along F' of the pullback of Ω to a normal variety on which F' is a divisor.

Step 1. In this step, for any primitive vector $\alpha \in \mathbb{Z}^4_{>0}$ with $2\alpha_x = \tau(\alpha)$, we construct a prime divisor F_α over X such that $\operatorname{center}_X F_\alpha = \{0\}$, $\operatorname{ord}_{F_\alpha} x_i = \alpha_i$ for $x_i \in \{x, y, z, t\}$, and

$$a(F_\alpha, X, B) = R(\alpha). \tag{10.3}$$

We also show that $R(u) > 0$ for any $u \in \mathbb{Q}^4_{>0}$.

Write $f := x^2 + h$. Since $2\alpha_x = \tau(\alpha)$, both x^2 and the leading form $\operatorname{in}_{(\alpha_y,\alpha_z,\alpha_t)}(h)$ have α-weight $2\alpha_x$, so

$$\operatorname{wt}_\alpha(f) = 2\alpha_x \quad \text{and} \quad \operatorname{in}_\alpha(f) = x^2 + \operatorname{in}_{(\alpha_y,\alpha_z,\alpha_t)}(h).$$

Since α is primitive, there exist $\alpha^{(2)},\alpha^{(3)},\alpha^{(4)}\in\mathbb{Z}^4$ such that $\alpha,\alpha^{(2)},\alpha^{(3)},\alpha^{(4)}$ is a basis of $\mathbb{Z}^4$. Let $m_1,\dots,m_4$ be the dual basis with respect to the standard pairing $\langle\cdot,\cdot\rangle$. Let $\mu: Y\to\mathbb{A}^4$ be the weighted blow-up with weights α, that is, the toric morphism defined by the star subdivision of $\mathbb{R}^4_{\geq 0}$ along the ray $\mathbb{R}_{\geq 0}\alpha$, and let E_α be the exceptional divisor of μ. The affine open subset of Y corresponding to the ray $\mathbb{R}_{\geq 0}\alpha$ is

$$U_\alpha=\operatorname{Spec}\mathbb{C}[s,s_2^{\pm1},s_3^{\pm1},s_4^{\pm1}],\quad s:=x^{m_1},\ s_j:=x^{m_j},$$

where $x^\beta:=x^{\beta_x}y^{\beta_y}z^{\beta_z}t^{\beta_t}$ for $\beta\in\mathbb{Z}^4$, on which $E_\alpha\cap U_\alpha=(s=0)$ and, for any monomial x^β in x,y,z,t,

$$\mu^*x^\beta=s^{\langle\alpha,\beta\rangle}s_2^{\langle\alpha^{(2)},\beta\rangle}s_3^{\langle\alpha^{(3)},\beta\rangle}s_4^{\langle\alpha^{(4)},\beta\rangle}. \tag{10.4}$$

In particular, $\operatorname{ord}_{E_\alpha}x_i=\alpha_i$. By (10.4), $\mu^*f=s^{2\alpha_x}\tilde f$ for a regular function $\tilde f$ on U_α with $\tilde f(0,s_2,s_3,s_4)=\varphi(s_2,s_3,s_4)$, where

$$\varphi(s_2,s_3,s_4):=\operatorname{in}_\alpha(f)\big(\gamma(s_2,s_3,s_4)\big),\quad \gamma(s_2,s_3,s_4):=\big(s_2^{(\alpha^{(2)})_i}s_3^{(\alpha^{(3)})_i}s_4^{(\alpha^{(4)})_i}\big)_{i=x,y,z,t}.$$

The morphism $(\mathbb{C}^*)^4\to(\mathbb{C}^*)^4$, $(s,s_2,s_3,s_4)\to(s^{\alpha_i}\gamma(s_2,s_3,s_4)_i)_i$, is an isomorphism, and by weighted homogeneity

$$\operatorname{in}_\alpha(f)\big((s^{\alpha_i}\gamma(s_2,s_3,s_4)_i)_i\big)=s^{2\alpha_x}\varphi(s_2,s_3,s_4). \tag{10.5}$$

Since $\operatorname{in}_{(\alpha_y,\alpha_z,\alpha_t)}(h)$ is a non-zero polynomial, there exists $(y_0,z_0,t_0)\in(\mathbb{C}^*)^3$ with $\operatorname{in}_{(\alpha_y,\alpha_z,\alpha_t)}(h)(y_0,z_0,t_0)\neq 0$. Let x_0 be a square root of $-\operatorname{in}_{(\alpha_y,\alpha_z,\alpha_t)}(h)(y_0,z_0,t_0)$, and $q_0:=(x_0,y_0,z_0,t_0)\in(\mathbb{C}^*)^4$. Then $\operatorname{in}_\alpha(f)(q_0)=0$ and $\partial_x\operatorname{in}_\alpha(f)(q_0)=2x_0\neq 0$. Let (s^0,s_2^0,s_3^0,s_4^0) be the point corresponding to q_0 under the isomorphism $(s,s_2,s_3,s_4)\to(s^{\alpha_i}\gamma(s_2,s_3,s_4)_i)_i$ of $(\mathbb{C}^*)^4$. By (10.5), $\varphi(s_2^0,s_3^0,s_4^0)=0$, and the differential of $s^{2\alpha_x}\varphi$ at (s^0,s_2^0,s_3^0,s_4^0) equals $(s^0)^{2\alpha_x}d\varphi(s_2^0,s_3^0,s_4^0)$, which is non-zero. Hence $d\varphi(s_2^0,s_3^0,s_4^0)\neq 0$.

Let X' be the strict transform of X in Y. Since $\tilde f(0,s_2,s_3,s_4)=\varphi$ is non-zero, no irreducible component of $(\tilde f=0)$ is contained in E_α, so $X'\cap U_\alpha=(\tilde f=0)$. Let $q_1:=(0,s_2^0,s_3^0,s_4^0)\in E_\alpha\cap U_\alpha$. Then $\tilde f(q_1)=0$, and the restriction of $\tilde f$ to $E_\alpha\cap U_\alpha$ has non-zero differential $d\varphi$ at q_1. Thus X' is smooth at q_1, and $s|_{X'}$ is a regular parameter of $\mathcal{O}_{X',q_1}$ defining $E_\alpha\cap X'$ near q_1. Let F_α be the irreducible component of $E_\alpha\cap X'$ containing q_1. It is a prime divisor over X with $\operatorname{center}_X F_\alpha=\mu(F_\alpha)=\{0\}$. By (10.4), μ^*x_i is s^{α_i} times a unit on U_α, so $\operatorname{ord}_{F_\alpha}x_i=\alpha_i$.

A direct calculation shows that

$$\mu^*(dx\wedge dy\wedge dz\wedge dt)=\pm s^{|\alpha|-1}s_2^{c_2}s_3^{c_3}s_4^{c_4}ds\wedge ds_2\wedge ds_3\wedge ds_4$$

for some integers c_2,c_3,c_4, where $|\alpha|:=\alpha_x+\alpha_y+\alpha_z+\alpha_t$. Here the sign ±1 is the determinant of the matrix with rows $\alpha,\alpha^{(2)},\alpha^{(3)},\alpha^{(4)}$, obtained by taking $d\log$ of (10.4) for the four coordinates. Since Ω is the residue of $dx\wedge dy\wedge dz\wedge dt/f$ along X, and $\mu^*(dx\wedge dy\wedge dz\wedge dt/f)$ is $s^{|\alpha|-1-2\alpha_x}$ times a unit times $ds\wedge ds_2\wedge ds_3\wedge ds_4/\tilde f$ near q_1, taking residues along X' we obtain $\operatorname{ord}_{F_\alpha}\Omega=|\alpha|-1-2\alpha_x$. By (10.2),

$$a(F_\alpha,X,B)=\alpha_x+(1-b_y)\alpha_y+(1-b_z)\alpha_z+(1-b_t)\alpha_t-\operatorname{wt}_\alpha(f).$$

Since $\operatorname{wt}_\alpha(f)=2\alpha_x=\tau(\alpha)$, the right-hand side equals $R(\alpha)$, which proves (10.3).

Let $u\in\mathbb{Q}^4_{>0}$. Since u and $\tau(u)$ are rational, there exists an integer $k>0$ such that $k(u_y,u_z,u_t)\in\mathbb{Z}^3$ and $k\tau(u)\in 2\mathbb{Z}$. Let α be the primitive vector of $\mathbb{Z}^4$ on the ray through $(\frac12 k\tau(u),ku_y,ku_z,ku_t)\in\mathbb{Z}^4_{>0}$. Then $2\alpha_x=\tau(\alpha)$, and $\alpha=k'(\frac12\tau(u),u_y,u_z,u_t)$ for some rational number $k'>0$. By (10.3), $k'R(u)=R(\alpha)=a(F_\alpha,X,B)$, and $a(F_\alpha,X,B)\geq a(F_\alpha,X,0)>0$ since X is klt, $b_y,b_z,b_t\leq 0$, and y,z,t are regular on X. Thus $R(u)>0$.

Step 2. In this step we show that $\operatorname{mld}((X\ni 0,B)/G)\leq P(v')$ for any $v'\in N_G\cap\mathbb{R}^4_{>0}$ with $2v'_x=\tau(v')$.

Since the image of G in $\mathbb{T}^4$ is finite, $N_G\subset\mathbb{Q}^4$. Write $v'=\frac nr\alpha$, where $\alpha\in\mathbb{Z}^4_{>0}$ is primitive and n,r are coprime positive integers. Then $2\alpha_x=\tau(\alpha)$. Let F_α be the prime divisor of **Step 1**.

Since $v' \in N_G$, there exists $g \in G$ with $\chi_i(g) = \zeta^{n\alpha_i}$ for $x_i \in \{x, y, z, t\}$, where χ_i is the character of x_i and $\zeta := e^{2\pi\sqrt{-1}/r}$. Since G acts faithfully and diagonally on $\mathbb{A}^4$, for an integer k we have $g^k = 1$ if and only if $r \mid kn\alpha_i$ for any i. Since $\gcd(n, r) = 1$ and $\gcd_i(\alpha_i) = 1$, this holds if and only if $r \mid k$. Thus the order of g is r.

The automorphism g of $\mathbb{A}^4$ is the action of the element $(\zeta^{n\alpha_i})_{i=x,y,z,t}$ of the torus $(\mathbb{C}^*)^4$. Hence g lifts to Y, and on U_α the lift is given by $g^*s = \zeta^n s$ and $g^*s_j = s_j$ for $j = 2, 3, 4$, since $\langle m_1, \alpha\rangle = 1$ and $\langle m_j, \alpha\rangle = 0$. In particular, g fixes $E_\alpha \cap U_\alpha$ pointwise. Since g preserves X, it preserves X', so g maps F_α to itself and acts trivially on $F_\alpha \cap U_\alpha$. Thus $g \in \mathrm{Stab}_G(F_\alpha)$ and g acts trivially on the residue field of ord_{F_α}, that is, $\langle g\rangle \subset I_G(F_\alpha)$. Hence $|I_G(F_\alpha)| \geq r$. By Lemma 2.6, (10.3), and $R(\alpha) > 0$,

$$\mathrm{mld}((X \ni 0, B)/G) \leq \frac{a(F_\alpha, X, B)}{|I_G(F_\alpha)|} \leq \frac{R(\alpha)}{r} = \frac{R(v')}{n} \leq R(v').$$

Since $2v'_x = \tau(v')$, (10.1) implies $P(v') = R(v')$, which proves the claim of this step.

Step 3. In this step we prove (1).

Let $w \in N_G \cap \mathbb{R}^4_{>0}$ with $P(w) < 1$, and let $d := w_x - \frac{1}{2}\tau(w)$. Let $g \in G$ with $\chi_i(g) = e^{2\pi\sqrt{-1}w_i}$ for any i. Since the action of G is diagonal and $x^2 + h$ is semi-invariant, the characters of x^2 and of any monomial $y^{\beta_y}z^{\beta_z}t^{\beta_t}$ occurring in h are equal. Thus

$$2w_x - \beta_y w_y - \beta_z w_z - \beta_t w_t \in \mathbb{Z}$$

for any such monomial. Taking a monomial of h of least (w_y, w_z, w_t)-weight, we obtain $2d \in \mathbb{Z}$. By (10.1) and **Step 1**, $|d| = P(w) - R(w) < 1$. Thus $d \in \{0, \pm\frac{1}{2}\}$.

If $d = 0$, then $2w_x = \tau(w)$, and $\mathrm{mld}((X \ni 0, B)/G) \leq P(w)$ by **Step 2**.

Suppose that $|d| = \frac{1}{2}$. Let $v' := (\tau(w), 2w_y, 2w_z, 2w_t)$. Then $v' - 2w = (-2d, 0, 0, 0) \in \mathbb{Z}^4$ and $2w \in N_G$, so $v' \in N_G$. Moreover, $v' \in \mathbb{R}^4_{>0}$ and $2v'_x = 2\tau(w) = \tau(v')$. By **Step 2** and (10.1),

$$\mathrm{mld}((X \ni 0, B)/G) \leq P(v') = R(v') = 2R(w).$$

By (10.1), $P(w) = R(w) + \frac{1}{2} < 1$, so $R(w) < \frac{1}{2}$. Hence $2R(w) < R(w) + \frac{1}{2} = P(w)$, and $\mathrm{mld}((X \ni 0, B)/G) < P(w)$. This proves (1).

Step 4. In this step we start the proof of (2): we show that the vector

$$w := \frac{1}{e}(\mathrm{ord}_F x, \mathrm{ord}_F y, \mathrm{ord}_F z, \mathrm{ord}_F t)$$

belongs to $N_G \cap \mathbb{R}^4_{>0}$ and that $c := 2w_x - \tau(w)$ is an integer, where $e := |I_G(F)|$.

By Lemma 2.6, $\mathrm{ord}_F|_{\mathbb{C}(X/G)} = e\,\mathrm{ord}_E$, $a(F, X, B) = ea$, and $\pi(\mathrm{center}_X F) = \mathrm{center}_{X/G} E$, where $\pi : X \to X/G$ is the quotient morphism. Since E computes the mld at $\pi(0)$ and $\pi^{-1}(\pi(0)) = \{0\}$, we have $\mathrm{center}_X F = \{0\}$, so $w \in \mathbb{R}^4_{>0}$.

Let $I := I_G(F)$, let $\mathcal{O}_F$ be the valuation ring of ord_F, $\mathfrak{m}_F$ the maximal ideal of $\mathcal{O}_F$, and κ_F the residue field of $\mathcal{O}_F$. By Lemma 2.6(1), I is cyclic and acts faithfully on $\mathfrak{m}_F/\mathfrak{m}_F^2$. Since I acts trivially on κ_F and $\dim_{\kappa_F} \mathfrak{m}_F/\mathfrak{m}_F^2 = 1$, the group I acts on $\mathfrak{m}_F/\mathfrak{m}_F^2$ through a character $\lambda : I \to \mathbb{C}^*$, which is injective. Fix a generator g of I with $\lambda(g) = \exp(2\pi\sqrt{-1}/e)$. For a character χ of I and $u \in \mathcal{O}_F$ let

$$\Pi_\chi(u) := \frac{1}{e}\sum_{j=0}^{e-1} \chi(g)^{-j}(g^j)^*u.$$

Then $\Pi_\chi(u)$ is I-semi-invariant with character χ. Fix a uniformizer π_0 of $\mathcal{O}_F$ and elements $\xi_1^0, \xi_2^0 \in \mathcal{O}_F$ whose residue classes form a transcendence basis of κ_F over $\mathbb{C}$. Let $\varpi := \Pi_\lambda(\pi_0)$ and $\xi_k := \Pi_1(\xi_k^0)$ for $k = 1, 2$. Since $(g^j)^*\pi_0 \equiv \lambda(g)^j\pi_0$ modulo $\mathfrak{m}_F^2$, we have $\varpi \equiv \pi_0$ modulo $\mathfrak{m}_F^2$, so ϖ is a uniformizer with character λ. Since I acts trivially on κ_F, ξ_k is I-invariant and $\xi_k \equiv \xi_k^0$ modulo $\mathfrak{m}_F$.

We claim that for any I-semi-invariant rational function $u \neq 0$ on X with character χ,

$$\chi(g) = \lambda(g)^{\operatorname{ord}_F u}. \tag{10.6}$$

To prove (10.6), write $u = u_0 \varpi^k$ with $k = \operatorname{ord}_F u$. Then u_0 is a unit of $\mathcal{O}_F$ which is I-semi-invariant with character $\chi\lambda^{-k}$. The residue class $\bar{u}_0 \neq 0$ satisfies $\bar{u}_0 = g^*\bar{u}_0 = \chi(g)\lambda(g)^{-k}\bar{u}_0$, since I acts trivially on κ_F. Thus $\chi(g) = \lambda(g)^k$.

Applying (10.6) to the G-semi-invariant functions x, y, z, t, we obtain

$$\chi_i(g) = \exp(2\pi\sqrt{-1}\operatorname{ord}_F(x_i)/e) = \exp(2\pi\sqrt{-1}w_i) \quad \text{for any } i.$$

Thus $w \bmod \mathbb{Z}^4$ is the image of g in $\mathbb{T}^4$, and $w \in N_G$. The characters of x^2 and of any monomial $y^{\beta_y} z^{\beta_z} t^{\beta_t}$ occurring in h are equal, so by (10.6),

$$\beta_y \operatorname{ord}_F y + \beta_z \operatorname{ord}_F z + \beta_t \operatorname{ord}_F t \equiv 2 \operatorname{ord}_F x \pmod{e}.$$

The minimum of the left-hand side over the monomials of h is $e\tau(w)$. Hence

$$e\tau(w) \equiv 2 \operatorname{ord}_F x = 2ew_x \pmod{e},$$

that is, $c = 2w_x - \tau(w) \in \mathbb{Z}$.

Step 5. In this step we show that $D_0 := \frac{1}{e}\big(\varrho a(\nu, \mathbb{A}^3, 0) - \operatorname{ord}_F(yzt)\big)$ is an integer, that

$$a = R(w) + D_0 - \frac{c}{2}, \tag{10.7}$$

and that $D_0 - \frac{c}{2} \geq 0$, where e, w, and c are as in **Step 4**.

Let $S' \to \mathbb{A}^3_{y,z,t}$ be a proper birational morphism from a normal variety S' on which ν is a divisor, and X'' the normalization of S' in $\mathbb{C}(X)$. Since $X'' \to S'$ is finite and $\operatorname{ord}_F|_{\mathbb{C}(y,z,t)} = \varrho \operatorname{ord}_\nu$, the prime divisor F is a divisor on X'' with ramification index ϱ over ν. By the Hurwitz formula [Kol13, 2.41], $\operatorname{ord}_F \eta = \varrho \operatorname{ord}_\nu \eta + \varrho - 1$ for any rational 3-form $\eta \neq 0$ on $\mathbb{A}^3$. Since $\operatorname{ord}_\nu(dy \wedge dz \wedge dt) = a(\nu, \mathbb{A}^3, 0) - 1$, we obtain

$$\operatorname{ord}_F(dy \wedge dz \wedge dt) = \varrho a(\nu, \mathbb{A}^3, 0) - 1. \tag{10.8}$$

We show that $D_0 \in \mathbb{Z}$. On a smooth variety on which F is a divisor, ϖ defines F near a general point of F, and the restrictions of ξ_1, ξ_2 to F form a transcendence basis of $\kappa_F = \mathbb{C}(F)$. Thus ϖ, ξ_1, ξ_2 of **Step 4** are regular parameters at a general point of F, so $\operatorname{ord}_F(d\varpi \wedge d\xi_1 \wedge d\xi_2) = 0$. Moreover, $g^*(d\varpi \wedge d\xi_1 \wedge d\xi_2) = \lambda(g) d\varpi \wedge d\xi_1 \wedge d\xi_2$. Write $dy \wedge dz \wedge dt = u_0 d\varpi \wedge d\xi_1 \wedge d\xi_2$. The function u_0 is I-semi-invariant with character $\chi_y\chi_z\chi_t\lambda^{-1}$ and $\operatorname{ord}_F u_0 = \varrho a(\nu, \mathbb{A}^3, 0) - 1$ by (10.8). By (10.6) applied to u_0 and to yzt,

$$\lambda(g)^{\operatorname{ord}_F(yzt)} = \chi_y\chi_z\chi_t(g) = \lambda(g)^{\varrho a(\nu, \mathbb{A}^3, 0)}.$$

Since $\lambda(g)$ is a primitive e-th root of unity, e divides $\varrho a(\nu, \mathbb{A}^3, 0) - \operatorname{ord}_F(yzt)$. Thus $D_0 \in \mathbb{Z}$.

We show (10.7). By (10.2), (10.8), and $\Omega = dy \wedge dz \wedge dt/(2x)$, we have

$$ea = a(F, X, B) = \varrho a(\nu, \mathbb{A}^3, 0) - \operatorname{ord}_F x - b_y \operatorname{ord}_F y - b_z \operatorname{ord}_F z - b_t \operatorname{ord}_F t.$$

Substituting $\varrho a(\nu, \mathbb{A}^3, 0) = eD_0 + \operatorname{ord}_F(yzt)$ and dividing by e, we obtain

$$a = D_0 - w_x + (1 - b_y)w_y + (1 - b_z)w_z + (1 - b_t)w_t.$$

Since $w_x = \frac{1}{2}(\tau(w) + c)$, this is (10.7).

We show $D_0 - \frac{c}{2} \geq 0$. Since $\operatorname{ord}_F = \varrho \operatorname{ord}_\nu$ on $\mathbb{C}(y, z, t)$, we have $\varrho v = e(w_y, w_z, w_t)$, $\varrho \operatorname{wt}_v(h) = e\tau(w)$, and $\varrho \operatorname{ord}_\nu h = \operatorname{ord}_F h = \operatorname{ord}_F(x^2) = 2ew_x$. Multiplying the assumed inequality by ϱ, we obtain

$$\varrho a(\nu, \mathbb{A}^3, 0) - ew_x \geq \operatorname{ord}_F(yzt) - \frac{e}{2}\tau(w),$$

that is, $eD_0 \geq e(w_x - \frac{1}{2}\tau(w)) = \frac{ec}{2}$. Thus $D_0 - \frac{c}{2} \geq 0$.

Step 6. In this step we conclude the proof of (2).

Let $\delta := D_0 - \frac{c}{2}$, with D_0 and c as in **Steps 4 and 5**. By **Steps 4 and 5**, $\delta \in \frac{1}{2}\mathbb{Z}$ and $\delta \geq 0$. By (10.7) and **Step 1**, $\delta = a - R(w) < a < 1$. Thus $\delta \in \{0, \frac{1}{2}\}$, and $\delta = 0$ if and only if $c = 2D_0$ is even. Let w' be the vector with $w'_i = w_i$ for $i = y, z, t$ and

$$w'_x := \begin{cases} \frac{1}{2}\tau(w) & \text{if } c \text{ is even,} \\ \frac{1}{2}\tau(w) + \frac{1}{2} & \text{if } c \text{ is odd.} \end{cases}$$

Then $w'_x - w_x$ equals $-\frac{c}{2}$ or $\frac{1-c}{2}$, which is an integer, so $w' \in N_G$ by **Step 4**. Since $\tau(w) > 0$, $w' \in \mathbb{R}^4_{>0}$. By (10.1), (10.7), and $\tau(w') = \tau(w)$, we have

$$P(w') = R(w) + \left| w'_x - \frac{1}{2}\tau(w) \right| = R(w) + \delta = a.$$

Let $u \in N_G \cap \mathbb{R}^4_{>0}$. If $P(u) \geq 1$, then $P(u) > a$. If $P(u) < 1$, then $a \leq P(u)$ by (1). Thus $a = \min_{u \in N_G \cap \mathbb{R}^4_{>0}} P(u) = P(w')$. Suppose that $w' = kw''$ for some integer $k \geq 2$ and some $w'' \in N_G$. Then $w'' \in \mathbb{R}^4_{>0}$ and $P(w'') = \frac{1}{k}P(w') = \frac{a}{k} < a$ since $a > 0$, a contradiction. Thus w' is primitive in N_G. □

Proposition 10.5. *Notation and conditions as in Set-up 10.3. Assume that* $((\mathbb{C}^*)^3, \frac{1}{2}\operatorname{div}(\operatorname{in}_u(h)))$ *is lc for any* $u \in \mathbb{R}^3_{>0}$. *If* $a := \operatorname{mld}((X \ni 0, B)/G) \in (0, 1)$, *then* $a = \min_{w \in N_G \cap \mathbb{R}^4_{>0}} P(w)$, *attained at a primitive vector. In particular, for* (b_y, b_z, b_t) *in a fixed finite subset of* $\mathbb{Q}^3_{\leq 0}$, *the set of such numbers a satisfies the ACC.*

Proof. We first show that the inequality assumed in Lemma 10.4(2) holds for any prime divisor ν over $\mathbb{A}^3_{y,z,t}$ with center 0. We use Lemma 8.6(1), applied with $d = 3$, G trivial, $s = 1$, $f_1 = h$, and $b_1 = \frac{1}{2}$. In this case

$$\psi(v) = v_y + v_z + v_t - \frac{1}{2}\operatorname{wt}_v(h),$$

the weighted leading boundary B_w for $w \in \mathbb{R}^3_{>0}$ is $\frac{1}{2}\operatorname{div}(\operatorname{in}_w(h))$, Σ is a regular subdivision of the Newton fan of h, $\mu_\Sigma : V_\Sigma \to \mathbb{A}^3$ is the corresponding toric morphism, Δ_Σ is the sum of all toric prime divisors of V_Σ, among which are the strict transforms of the coordinate hyperplanes, and $B'_\Sigma = \frac{1}{2}H'$, where H' is the closure in V_Σ of $\operatorname{div}_{(\mathbb{C}^*)^3}(h)$, with multiplicities. In particular, H' contains no toric prime divisor. The hypothesis of Lemma 8.6(1) is the assumption of Proposition 10.5 on $\operatorname{in}_u(h)$, so $(V_\Sigma, \Delta_\Sigma + \frac{1}{2}H')$ is lc near $\mu_\Sigma^{-1}(0)$.

For a ray ρ of Σ, let n_ρ be its primitive generator in $\mathbb{Z}^3$ and D_ρ the corresponding toric prime divisor of V_Σ. By Lemma 8.5, applied with $d = 3$, G trivial, $s = 1$, $f_1 = h$, and $b_1 = \frac{1}{2}$,

$$K_{V_\Sigma} + \Delta_\Sigma + \frac{1}{2}H' = \mu_\Sigma^*\Big(K_{\mathbb{A}^3} + \frac{1}{2}\operatorname{div}(h)\Big) + \sum_\rho \psi(n_\rho)D_\rho,$$

and $\mu_\Sigma^* \operatorname{div}(x_i) = \sum_\rho (n_\rho)_i D_\rho$ for $x_i \in \{y, z, t\}$ [CLS11, Proposition 4.1.2]. Let ν be a prime divisor over $\mathbb{A}^3$ with center 0, and $v := (\operatorname{ord}_\nu y, \operatorname{ord}_\nu z, \operatorname{ord}_\nu t)$. Then $\operatorname{center}_{V_\Sigma} \nu \subset \mu_\Sigma^{-1}(0)$, and

$$a\Big(\nu, \mathbb{A}^3, \frac{1}{2}\operatorname{div}(h)\Big) = a\Big(\nu, V_\Sigma, \Delta_\Sigma + \frac{1}{2}H'\Big) + \sum_\rho \psi(n_\rho)\operatorname{mult}_\nu D_\rho \geq \sum_\rho \psi(n_\rho)\operatorname{mult}_\nu D_\rho.$$

Let $\sigma \in \Sigma$ be the cone whose orbit contains the generic point of $\operatorname{center}_{V_\Sigma} \nu$. Then $\operatorname{mult}_\nu D_\rho > 0$ only if $\rho \in \sigma(1)$, so $v = \sum_{\rho \in \sigma(1)} (\operatorname{mult}_\nu D_\rho) n_\rho$. Since Σ refines the Newton fan of h, the function ψ is linear on σ. Thus the right-hand side equals $\psi(v)$, while the left-hand side equals $a(\nu, \mathbb{A}^3, 0) - \frac{1}{2}\operatorname{ord}_\nu h$. This proves

$$a(\nu, \mathbb{A}^3, 0) - \frac{1}{2}\operatorname{ord}_\nu h \geq v_y + v_z + v_t - \frac{1}{2}\operatorname{wt}_v(h).$$

Next we show that a is attained. Let $\pi : X \to X/G$ be the quotient morphism and $\bar{0} := \pi(0)$. Since X is klt, $b_y, b_z, b_t \leq 0$, and y, z, t are regular on X, any prime divisor F' over X satisfies $a(F', X, B) \geq a(F', X, 0) > 0$, so by Lemma 2.6(2)(3) all log discrepancies of $(X, B)/G$ are

positive, that is, $(X,B)/G$ is sub-klt. Let $f : Y \to X/G$ be a log resolution of $(X,B)/G$ such that $f^{-1}(\bar{0})$ is a divisor and $f^{-1}(\bar{0}) \cup \operatorname{Supp} B_Y$ is snc, where $K_Y + B_Y = f^*(K_{X/G} + B_{X/G})$ and $f_*B_Y = B_{X/G}$. Since $(X,B)/G$ is sub-klt, $\operatorname{mult}_D B_Y < 1$ for any prime divisor D on Y. Let E' be a prime divisor over X/G with center $\bar{0}$, and η the generic point of $\operatorname{center}_Y E'$. Then $\eta \in f^{-1}(\bar{0})$. Take local equations of the components of $f^{-1}(\bar{0}) \cup \operatorname{Supp} B_Y$ through η as part of regular parameters of $\mathcal{O}_{Y,\eta}$, with coefficient 0 for the remaining parameters. By Lemma 2.13 and [KM98, Lemma 2.30],

$$a(E', X/G, B_{X/G}) = a(E', Y, B_Y) \geq \sum_D (1 - \operatorname{mult}_D B_Y) \operatorname{mult}_{E'} D \geq \min_D (1 - \operatorname{mult}_D B_Y),$$

where the sum runs over the components D of $f^{-1}(\bar{0}) \cup \operatorname{Supp} B_Y$ through η and the minimum over the finitely many components D of $f^{-1}(\bar{0})$. The second inequality holds since all terms are non-negative and some component D of $f^{-1}(\bar{0})$ passes through η with $\operatorname{mult}_{E'} D \geq 1$. Since $a(D, X/G, B_{X/G}) = 1 - \operatorname{mult}_D B_Y$ for any component D of $f^{-1}(\bar{0})$, the mld a is attained by a prime divisor E over X/G with center $\bar{0}$.

By Lemma 2.6(3), there exists a prime divisor F over X with $\operatorname{ord}_F|_{\mathbb{C}(X/G)} \in \mathbb{N}^+ \operatorname{ord}_E$, and $\operatorname{center}_X F = \{0\}$ by Lemma 2.6(2). The involution $\iota : x \to -x$ preserves X, and p is the quotient morphism of X by $\langle \iota \rangle$. By Lemma 2.6(1)(2) applied to $\langle \iota \rangle$, $\operatorname{ord}_F|_{\mathbb{C}(y,z,t)} = \varrho \operatorname{ord}_\nu$, where $\varrho = |I_{\langle \iota \rangle}(F)| \in \{1,2\}$ and ν is a prime divisor over $\mathbb{A}^3$ with $\operatorname{center}_{\mathbb{A}^3} \nu = p(\operatorname{center}_X F) = \{0\}$. By the first paragraph, ν satisfies the inequality assumed in Lemma 10.4(2). Thus $a = \min_{w \in N_G \cap \mathbb{R}^4_{>0}} P(w)$, attained at a primitive vector, by Lemma 10.4(2).

Finally, let $J \subset \mathbb{C}[x,y,z,t]$ be the monomial ideal generated by x^2 and the monomials occurring in h. Then $\operatorname{wt}_w(J) = \min\{2w_x, \operatorname{wt}_{(w_y,w_z,w_t)}(h)\}$, so

$$a = \inf\left\{ w_x + (1-b_y)w_y + (1-b_z)w_z + (1-b_t)w_t - \operatorname{wt}_w(J) \;\middle|\; w \in N_G \cap \mathbb{R}^4_{>0} \right\}.$$

If (b_y, b_z, b_t) belongs to a fixed finite subset of $\mathbb{Q}^3_{\leq 0}$, then $(1, 1-b_y, 1-b_z, 1-b_t)$ belongs to a fixed finite subset of $\mathbb{Q}^4_{>0}$. Since $N_G = N_H$ for the image H of G in $\mathbb{T}^4$, Corollary 4.6, applied with $d = 4$, $s = 1$, and $\Lambda = \{1\}$, implies that the set of such numbers a satisfies the ACC. □

In Theorem 8.14(ii), the leading forms of the equation of the threefold enter with coefficient one. In Proposition 10.5, only the leading forms of h enter, with coefficient $\frac{1}{2}$.

We now treat equations $x^2 + h$ with $h = y^3 + \alpha y + \beta$, $\deg_z \alpha \leq 5$, and $\deg_z \beta \leq 5$. For such an equation and a weight $u \in \mathbb{R}^3_{>0}$, the pair $((\mathbb{C}^*)^3, \frac{1}{2}\operatorname{div}(\operatorname{in}_u(h)))$, which appears in the hypothesis of Proposition 10.5, can fail to be lc only at a non-zero root of multiplicity at least three of the leading form of α or of β, and finitely many G-equivariant translations of z, adapted to a prime divisor computing the minimal log discrepancy, bring the equation to the situation of Lemma 10.4(2). The degree bound 5 covers the points of type cE_7 and cE_8 whose form α_3 (type cE_7) or β_5 (type cE_8) of the introduction of Section 12 has a single root in $\mathbb{P}^1$, treated in Subsection 12.6, since there $\deg_z \alpha \leq 3$ and $\deg_z \beta \leq 5$ by **Steps 4** and 5 of the proof of Theorem 12.1.

Proposition 10.6. *Let $\mathcal{B} \subset \mathbb{Q}^2_{\leq 0}$ be a finite set. Let $h(y,z,t) = y^3 + \alpha(z,t)y + \beta(z,t)$ be a polynomial with $h(0) = 0$, where $\alpha, \beta \in \mathbb{C}[z,t]$ are polynomials, possibly zero, of degree at most 5 in z, such that $X := (x^2 + h = 0) \subset \mathbb{A}^4$ is a normal klt threefold germ at 0. Let G be a finite abelian group acting faithfully on $\mathbb{A}^4$ and diagonally in the coordinates x,y,z,t, such that $x^2 + h$ is G-semi-invariant and the induced action of G on X is faithful, let $N_G \subset \mathbb{R}^4$ be taken with respect to x,y,z,t, let $(b_y, b_t) \in \mathcal{B}$, and let $B := b_y \operatorname{div}_X(y) + b_t \operatorname{div}_X(t)$. Assume that $a := \operatorname{mld}((X \ni 0, B)/G) \in (0,1)$. Then:*

(1) a is computed by a prime divisor over X/G, and for any prime divisor over X/G computing a, there exists a G-equivariant polynomial automorphism $z = z' + \varphi(t)$ of $\mathbb{A}^4$

with $\varphi \in t\mathbb{C}[t]$, fixing x, y, t, such that

$$a = \min_{w \in N_G \cap \mathbb{R}^4_{>0}} \Big(w_x + (1-b_y)w_y + w_{z'} + (1-b_t)w_t - \min\{2w_x, \mathrm{wt}_{(w_y, w_{z'}, w_t)}(h(y, z' + \varphi(t), t))\} \Big),$$

and the minimum is attained at a primitive vector of N_G;

(2) *the set of the numbers a, for all h, G, and $(b_y, b_t) \in \mathcal{B}$ satisfying the assumptions of Proposition 10.6, satisfies the ACC.*

Proof. For $u = (u_y, u_z, u_t) \in \mathbb{Z}^3_{>0}$, the monomials y^3, $yz^i t^j$, and $z^{i'} t^{j'}$ are pairwise distinct, so

$$\mathrm{in}_u(h) = \epsilon y^3 + \hat{\alpha}(z,t)y + \hat{\beta}(z,t),$$

where $\epsilon \in \{0,1\}$ and, for $\gamma \in \{\alpha, \beta\}$, either $\hat{\gamma} = 0$ or $\hat{\gamma} = \mathrm{in}_{(u_z,u_t)}(\gamma)$. Let y_1, z_1 be the coordinates of $(\mathbb{C}^*)^2$, and let

$$\bar{h}(y_1, z_1) := \mathrm{in}_u(h)(y_1, z_1, 1) = \epsilon y_1^3 + \bar{\alpha}(z_1)y_1 + \bar{\beta}(z_1), \quad \text{where } \bar{\gamma}(z_1) := \hat{\gamma}(z_1, 1) \text{ for } \gamma \in \{\alpha, \beta\}.$$

Since $u_t > 0$, two distinct monomials with equal u-weight have distinct exponents in (y, z). Thus $\bar{h} \neq 0$, $\bar{\gamma} \neq 0$ if and only if $\hat{\gamma} \neq 0$, and $\deg \bar{\gamma} \leq \deg_z \gamma \leq 5$.

Step 1. In this step we show that if $((\mathbb{C}^*)^2, \frac{1}{2}\operatorname{div}(\bar{h}))$ is not lc near a closed point ξ, then $\lambda := z_1(\xi) \in \mathbb{C}^*$ is a root of multiplicity at least three of $\bar{\alpha} \neq 0$ or of $\bar{\beta} \neq 0$.

Suppose that $\epsilon = 1$. Then $\partial^2 \bar{h}/\partial y_1^2 = 6y_1$ does not vanish at ξ, so $\mathrm{mult}_\xi \operatorname{div}(\bar{h}) \leq 2$, and $((\mathbb{C}^*)^2, \frac{1}{2}\operatorname{div}(\bar{h}))$ is lc near ξ by Lemma 10.1(1). Thus we may assume that $\epsilon = 0$. Then $\bar{h} = \bar{\alpha}(z_1)y_1 + \bar{\beta}(z_1)$, and $\bar{\alpha}$, $\bar{\beta}$ are not both zero. If $\bar{\alpha} = 0$ (resp. $\bar{\beta} = 0$), then, since y_1 is a unit on $(\mathbb{C}^*)^2$, near ξ we have $\operatorname{div}(\bar{h}) = m \operatorname{div}(z_1 - \lambda)$ with $m = \operatorname{ord}_\lambda \bar{\beta}$ (resp. $m = \operatorname{ord}_\lambda \bar{\alpha}$), and Lemma 10.1(1) shows that $m \geq 3$ if the pair is not lc near ξ.

Assume that $\bar{\alpha} \neq 0$ and $\bar{\beta} \neq 0$. Let $m := \min\{\operatorname{ord}_\lambda \bar{\alpha}, \operatorname{ord}_\lambda \bar{\beta}\}$, write $\bar{\alpha} = (z_1 - \lambda)^m \bar{\alpha}_1$ and $\bar{\beta} = (z_1 - \lambda)^m \bar{\beta}_1$, and let $q_1 := \bar{\alpha}_1(z_1)y_1 + \bar{\beta}_1(z_1)$, so that $\bar{h} = (z_1 - \lambda)^m q_1$. It suffices to show that the pair is lc near ξ when $m \leq 2$. Suppose that $m \leq 2$. If $\operatorname{ord}_\lambda \bar{\alpha} > m$, then $\bar{\beta}_1(\lambda) \neq 0$, so q_1 is a unit near ξ and $\mathrm{mult}_\xi \operatorname{div}(\bar{h}) = m \leq 2$. If $\operatorname{ord}_\lambda \bar{\alpha} = m$ and $q_1(\xi) \neq 0$, then again $\mathrm{mult}_\xi \operatorname{div}(\bar{h}) = m \leq 2$. In both cases Lemma 10.1(1) shows that the pair is lc near ξ. If $\operatorname{ord}_\lambda \bar{\alpha} = m$ and $q_1(\xi) = 0$, then $\frac{\partial q_1}{\partial y_1}(\xi) = \bar{\alpha}_1(\lambda) \neq 0$, so $z_1 - \lambda$ and q_1 are regular parameters at ξ, $\operatorname{div}(z_1 - \lambda) + \operatorname{div}(q_1)$ is snc near ξ, and near ξ

$$\frac{1}{2}\operatorname{div}(\bar{h}) = \frac{m}{2}\operatorname{div}(z_1 - \lambda) + \frac{1}{2}\operatorname{div}(q_1), \quad \frac{m}{2} \leq 1.$$

Since $\operatorname{div}(z_1 - \lambda) + \operatorname{div}(q_1)$ is snc near ξ and the coefficients $\frac{m}{2}$ and $\frac{1}{2}$ are at most 1, the pair is lc near ξ [KM98, Corollary 2.31(3)].

Step 2. In this step we fix $u \in \mathbb{Z}^3_{>0}$, $\gamma \in \{\alpha, \beta\}$ with $\bar{\gamma} \neq 0$, and a root $\lambda \in \mathbb{C}^*$ of $\bar{\gamma}$ of multiplicity at least three, and we show that $q := u_z/u_t$ is an integer, that λ is the only root of $\bar{\gamma}$ in $\mathbb{C}^*$ of multiplicity at least three, and that $z - \lambda t^q$ is G-semi-invariant with character equal to the character of z.

Write $u_z/u_t = k/n$ with coprime positive integers k, n, and let $\kappa := \mathrm{wt}_{(u_z,u_t)}(\gamma)$. Any monomial $z^i t^j$ of $\hat{\gamma}$ satisfies $u_z i + u_t j = \kappa$, hence $ki + nj = n\kappa/u_t$. Since k and n are coprime, the exponents i of the monomials of $\hat{\gamma}$ are congruent modulo n. Thus, if i_0 is one of them, then $\bar{\gamma}(\zeta z_1) = \zeta^{i_0}\bar{\gamma}(z_1)$ for any n-th root of unity ζ, so the n distinct numbers $\zeta\lambda$ are roots of $\bar{\gamma}$ of multiplicity at least three. Hence $3n \leq \deg \bar{\gamma} \leq 5$, so $n = 1$ and $q = k$ is an integer. Since each root of multiplicity at least three contributes at least 3 to $\deg \bar{\gamma}$, two distinct such roots would imply $\deg \bar{\gamma} \geq 6$, so λ is the only such root.

Let $\chi_x, \chi_y, \chi_z, \chi_t$ be the characters of x, y, z, t, and χ the character of $x^2 + h$ (Definition 2.14). Let $\sigma \in G$. Comparing the coefficients of y and 1 in $\sigma^*(x^2 + h) = \chi(\sigma)(x^2 + h)$, we obtain $c_\sigma \in \mathbb{C}^*$ with

$$\gamma(\chi_z(\sigma)z, \chi_t(\sigma)t) = c_\sigma \gamma(z, t).$$

Since the substitution of $\chi_z(\sigma)z$ for z and $\chi_t(\sigma)t$ for t multiplies any monomial by a constant, taking leading parts gives

$$\hat{\gamma}(\chi_z(\sigma)z, \chi_t(\sigma)t) = c_\sigma \hat{\gamma}(z,t).$$

Since q is an integer, any monomial $z^i t^j$ of $\hat{\gamma}$ equals $(z/t^q)^i t^{\kappa/u_t}$ with $\kappa/u_t = qi + j \in \mathbb{Z}$, hence $\hat{\gamma}(z,t) = t^{\kappa/u_t}\bar{\gamma}(z/t^q)$. Setting $t = 1$, we obtain

$$\bar{\gamma}\big(\chi_z(\sigma)\chi_t(\sigma)^{-q} z_1\big) = c_\sigma \chi_t(\sigma)^{-\kappa/u_t}\bar{\gamma}(z_1).$$

Thus $z_1 \to \chi_z(\sigma)\chi_t(\sigma)^{-q}z_1$ maps each root of $\bar{\gamma}$ to a root of $\bar{\gamma}$ of equal multiplicity. Since λ is the only non-zero root of multiplicity at least three, we have $\chi_z(\sigma) = \chi_t(\sigma)^q$, so

$$\sigma^*(z - \lambda t^q) = \chi_z(\sigma)(z - \lambda t^q).$$

Since $\sigma \in G$ is arbitrary, **Step 2** follows.

Steps 1 and 2 use only that $h = y^3 + \alpha y + \beta$ with $\deg_z \alpha \le 5$ and $\deg_z \beta \le 5$, and that G acts diagonally in the coordinates x, y, z, t with $x^2 + h$ G-semi-invariant.

Step 3. In this step we fix a prime divisor computing a and construct φ.

Since X is klt and $b_y, b_t \le 0$, (X, B) is sub-klt. By Lemma 2.6(2)(3), $(X, B)/G$ is sub-klt. Let $\pi : X \to X/G$ be the quotient morphism. We show that a is attained by a prime divisor. Let $\mu : W \to X/G$ be a log resolution of $(X/G, B_{X/G})$ such that $\mu^{-1}(\pi(0))$ is a divisor and $\operatorname{Exc}(\mu) \cup \mu^{-1}(\pi(0)) \cup \operatorname{Supp} \mu_*^{-1} B_{X/G}$ is snc, let $E_1, \dots, E_r$ be the components of $\mu^{-1}(\pi(0))$, and let $K_W + B_W := \mu^*(K_{X/G} + B_{X/G})$. The coefficients of B_W are less than 1. For a prime divisor P over X/G with $\operatorname{center}_{X/G} P = \{\pi(0)\}$, let E_j be a component containing $\operatorname{center}_W P$. By Lemma 2.13, applied at the generic point of $\operatorname{center}_W P$ with regular parameters defining the components of $\operatorname{Supp} B_W \cup \mu^{-1}(\pi(0))$ through it, we have

$$a(P, X/G, B_{X/G}) \ge a(E_j, X/G, B_{X/G}) \operatorname{mult}_P E_j \ge a(E_j, X/G, B_{X/G}).$$

Thus $a = \min_j a(E_j, X/G, B_{X/G})$.

Let $\bar{F}$ be a prime divisor over X/G computing a. By Lemma 2.6(3), there exists a prime divisor F over X with $\operatorname{ord}_F|_{\mathbb{C}(X/G)} \in \mathbb{N}^+ \operatorname{ord}_{\bar{F}}$, and $\pi(\operatorname{center}_X F) = \{\pi(0)\}$ by Lemma 2.6(2). Since 0 is fixed by G, $\pi^{-1}(\pi(0)) = \{0\}$, so $\operatorname{center}_X F = \{0\}$. Let $p : X \to \mathbb{A}^3_{y,z,t}$ be the projection. The involution $\iota : x \to -x$ preserves X, and p is the quotient morphism of X by the group $\langle \iota \rangle$. By Lemma 2.6(1)(2) for $\langle \iota \rangle$, we have

$$\operatorname{ord}_F|_{\mathbb{C}(y,z,t)} = \varrho \operatorname{ord}_\nu,$$

where $\varrho = |I_{\langle \iota \rangle}(F)| \in \{1, 2\}$ and ν is a prime divisor over $\mathbb{A}^3$ with $\operatorname{center}_{\mathbb{A}^3} \nu = p(\operatorname{center}_X F) = \{0\}$.

We construct φ by a finite procedure. At each stage we have $\varphi \in t\mathbb{C}[t]$ such that $z' := z - \varphi(t)$ is G-semi-invariant with character equal to the character of z. At the start, $\varphi = 0$. Let

$$h' := h(y, z' + \varphi(t), t) = y^3 + \alpha(z' + \varphi(t), t)y + \beta(z' + \varphi(t), t) \in \mathbb{C}[y, z', t].$$

The z'-degrees of $\alpha(z' + \varphi(t), t)$ and $\beta(z' + \varphi(t), t)$ are at most 5, G acts diagonally in the coordinates x, y, z', t and $x^2 + h'$ is G-semi-invariant, and $\mathbb{C}(y, z, t) = \mathbb{C}(y, z', t)$. Thus **Steps 1 and 2** apply to h' in the coordinates y, z', t, and we denote by $\bar{h}'$, $\bar{\alpha}'$, $\bar{\beta}'$ the polynomials defined there. Let

$$u = (u_y, u_z, u_t) := (\operatorname{ord}_\nu y, \operatorname{ord}_\nu z', \operatorname{ord}_\nu t) \in \mathbb{Z}^3_{>0},$$

write $u = c_0 \bar{u}$ with $c_0 \in \mathbb{N}^+$ and $\bar{u}$ primitive in $\mathbb{Z}^3$, let $g : Y \to \mathbb{A}^3$ be the toric morphism, with respect to the coordinates y, z', t, given by the star subdivision of $\mathbb{R}^3_{\ge 0}$ along the ray $\mathbb{R}_{\ge 0}\bar{u}$, let E be the exceptional divisor of g, let $Y^\circ \subset Y$ be the open toric subvariety given by the cone $\mathbb{R}_{\ge 0}\bar{u}$, let $O_E := E \cap Y^\circ$, and let D_Y be the strict transform of $\operatorname{div}(h)$. Then $Y^\circ \cong \mathbb{A}^1 \times (\mathbb{C}^*)^2$, O_E is the open torus orbit of E, and E is the only toric divisor of Y meeting Y° [CLS11, Example 1.2.21 and Theorem 3.2.6]. Moreover, E is the monomial prime divisor in y, z', t with weights $\bar{u}$ [CLS11, Proposition 4.1.1], so $\operatorname{ord}_E h = \operatorname{wt}_{\bar{u}}(h')$ (for a polynomial, ord_E is the minimum of ord_E over its monomials, since $Y^\circ \cong \mathbb{A}^1 \times (\mathbb{C}^*)^2$ has coordinates in which $E \cap Y^\circ$ is a coordinate hyperplane

and distinct monomials in y, z', t are distinct Laurent monomials, see [CLS11, Proposition 4.1.1]), and

$$g^* \operatorname{div}(h)|_{Y^\circ} = (D_Y + \operatorname{wt}_{\bar{u}}(h')E)|_{Y^\circ}.$$

For a Laurent monomial f in y, z', t with exponent vector θ, we have $\operatorname{ord}_\nu f = \langle u, \theta \rangle$. In particular, $\operatorname{ord}_\nu f \geq 0$ for any Laurent monomial f regular on Y°, and $\operatorname{ord}_\nu f = c_0 > 0$ for a Laurent monomial f defining E on Y°. Thus the generic point ξ_ν of $\operatorname{center}_Y \nu$ belongs to O_E, and $\operatorname{mult}_\nu E = c_0$. Let

$$\varpi : (\mathbb{C}^*)^2 \to O_E, \quad (y_1, z_1) \to \lim_{s \to 0} (s^{\bar{u}_y} y_1, s^{\bar{u}_z} z_1, s^{\bar{u}_t}),$$

where the limit is taken in Y° and exists by [CLS11, Proposition 3.2.2]. Then ϖ is a surjective homomorphism of tori with finite kernel $\{(s^{\bar{u}_y}, s^{\bar{u}_z}) \mid s^{\bar{u}_t} = 1\}$. Let f_0 be a Laurent monomial in y, z', t of $\bar{u}$-weight $\operatorname{wt}_{\bar{u}}(h')$, which exists since $\bar{u}$ is primitive. Then h'/f_0 is regular on Y° with divisor $D_Y|_{Y^\circ}$. For $(y_1, z_1) \in (\mathbb{C}^*)^2$ and $s \in \mathbb{C}^*$, a monomial of h' of $\bar{u}$-weight κ' takes at $(s^{\bar{u}_y} y_1, s^{\bar{u}_z} z_1, s^{\bar{u}_t})$ the value $s^{\kappa'}$ times its value at $(y_1, z_1, 1)$, and f_0 takes the value $s^{\operatorname{wt}_{\bar{u}}(h')} f_0(y_1, z_1, 1)$. Hence $(h'/f_0)(s^{\bar{u}_y} y_1, s^{\bar{u}_z} z_1, s^{\bar{u}_t})$ is $\operatorname{in}_{\bar{u}}(h')(y_1, z_1, 1)/f_0(y_1, z_1, 1)$ plus finitely many terms with positive powers of s, and letting $s \to 0$ and using $\operatorname{in}_{\bar{u}}(h') = \operatorname{in}_u(h')$, we obtain

$$\varpi^*\big((h'/f_0)|_{O_E}\big) = \frac{\bar{h}'(y_1, z_1)}{f_0(y_1, z_1, 1)}.$$

Hence $\varpi^*(D_Y|_{O_E}) = \operatorname{div}(\bar{h}')$. The morphism ϖ is the quotient morphism by $\operatorname{Ker} \varpi$, which acts freely by translations, and $\operatorname{div}(\bar{h}')$ is invariant under $\operatorname{Ker} \varpi$. Thus $(O_E, \frac{1}{2} D_Y|_{O_E})$ is the crepant quotient of $((\mathbb{C}^*)^2, \frac{1}{2} \operatorname{div}(\bar{h}'))$ by $\operatorname{Ker} \varpi$ (since ϖ is étale and $\varpi_* \varpi^*$ is multiplication by $\deg \varpi$, Lemma 2.6(0)), and, by Lemma 2.6(2)(3), for a closed point $\xi \in (\mathbb{C}^*)^2$, the pair $(O_E, \frac{1}{2} D_Y|_{O_E})$ is lc near $\varpi(\xi)$ if and only if $((\mathbb{C}^*)^2, \frac{1}{2} \operatorname{div}(\bar{h}'))$ is lc near ξ.

If $(O_E, \frac{1}{2} D_Y|_{O_E})$ is lc near ξ_ν, we stop. Otherwise, since the set of points near which a pair is not lc is closed (it is the image of the components of coefficient > 1 of the log pull-back on a log resolution, [KM98, Lemma 2.30 and Corollary 2.31(3)]), $(O_E, \frac{1}{2} D_Y|_{O_E})$ is not lc near any point of $\operatorname{center}_Y \nu \cap O_E$. By the previous paragraph and **Step 1**, for any closed point $\xi \in \varpi^{-1}(\operatorname{center}_Y \nu \cap O_E)$, $z_1(\xi)$ is a root of multiplicity at least three of $\bar{\alpha}' \neq 0$ or of $\bar{\beta}' \neq 0$. Let $\Sigma \subset \mathbb{C}^*$ be the finite set of such roots. Since $\varpi^{-1}(\operatorname{center}_Y \nu \cap O_E) \neq \emptyset$, **Step 2** implies that $q := u_z/u_t$ is an integer. The Laurent monomial z'/t^q has $\bar{u}$-weight 0, so it is regular on Y°, and $\varpi^*((z'/t^q)|_{O_E}) = z_1$. Thus z'/t^q takes values in Σ at the closed points of the irreducible set $\operatorname{center}_Y \nu \cap O_E$, so there exists $\lambda \in \Sigma$ with

$$\operatorname{center}_Y \nu \cap O_E \subset \{z'/t^q = \lambda\} \subset Y^\circ.$$

By **Step 2**, $z' - \lambda t^q$ is G-semi-invariant with character equal to the character of z', which is the character of z. Since the valuation ring of ν dominates $\mathcal{O}_{Y,\xi_\nu}$ and $z'/t^q - \lambda$ vanishes at ξ_ν, we have $\operatorname{ord}_\nu(z'/t^q - \lambda) > 0$. Hence

$$\operatorname{ord}_\nu(z' - \lambda t^q) > q \operatorname{ord}_\nu t = \operatorname{ord}_\nu z'.$$

We replace φ with $\varphi + \lambda t^q \in t\mathbb{C}[t]$, that is, z' with $z' - \lambda t^q$, and repeat.

The procedure stops: at any stage, Lemma 2.13, applied at 0 with the regular parameters y, z', t and all coefficients 0, implies

$$\operatorname{ord}_\nu y + \operatorname{ord}_\nu z' + \operatorname{ord}_\nu t \leq a(\nu, \mathbb{A}^3, 0).$$

The numbers $\operatorname{ord}_\nu y$, $\operatorname{ord}_\nu t$, and $a(\nu, \mathbb{A}^3, 0)$ do not depend on the stage, and the integer $\operatorname{ord}_\nu z'$ strictly increases at any stage at which we do not stop. Thus we stop after at most $a(\nu, \mathbb{A}^3, 0) - \operatorname{ord}_\nu y - \operatorname{ord}_\nu z - \operatorname{ord}_\nu t$ stages. From now on, φ, z', h', u, $\bar{u}$, c_0, $g : Y \to \mathbb{A}^3$, E, Y°, O_E, and D_Y are those of the last stage. In particular, $(O_E, \frac{1}{2} D_Y|_{O_E})$ is lc near ξ_ν.

Step 4. In this step we show that

$$a(\nu, \mathbb{A}^3, 0) - \tfrac{1}{2} \operatorname{ord}_\nu h \geq u_y + u_z + u_t - \tfrac{1}{2} \operatorname{wt}_u(h'). \tag{10.9}$$

Let $V \subset O_E$ be an open subset containing ξ_ν such that $(V, \frac{1}{2}D_Y|_V)$ is lc, and let $U := Y^\circ \backslash (O_E \backslash V)$. Then U is an open neighborhood of ξ_ν, both U and $E \cap U = V$ are smooth, and $E \cap U$ is not a component of D_Y. By adjunction, the different of $(U, (E + \frac{1}{2}D_Y)|_U)$ on $E \cap U$ is $\frac{1}{2}D_Y|_{E\cap U}$. Thus, by Theorem 2.23, $(U, (E + \frac{1}{2}D_Y)|_U)$ is lc near $E \cap U$. Since $\xi_\nu \in E \cap U$, we have

$$a(\nu, Y, E + \tfrac{1}{2}D_Y) \geq 0.$$

By [Amb06, Section 2, p. 360], $a(E, \mathbb{A}^3, 0) = \bar{u}_y + \bar{u}_z + \bar{u}_t$, so $K_Y = g^* K_{\mathbb{A}^3} + (\bar{u}_y + \bar{u}_z + \bar{u}_t - 1)E$, as E is the only exceptional divisor of g. Adding $E + \frac{1}{2}D_Y$ and using $g^* \operatorname{div}(h)|_{Y^\circ} = (D_Y + \operatorname{wt}_{\bar{u}}(h')E)|_{Y^\circ}$, we obtain

$$(K_Y + E + \tfrac{1}{2}D_Y)|_{Y^\circ} = \big(g^*(K_{\mathbb{A}^3} + \tfrac{1}{2}\operatorname{div}(h)) + (\bar{u}_y + \bar{u}_z + \bar{u}_t - \tfrac{1}{2}\operatorname{wt}_{\bar{u}}(h'))E\big)|_{Y^\circ}.$$

Since the generic point ξ_ν of $\operatorname{center}_Y \nu$ belongs to Y°, we obtain

$$a(\nu, \mathbb{A}^3, \tfrac{1}{2}\operatorname{div}(h)) = a(\nu, Y, E + \tfrac{1}{2}D_Y) + \big(\bar{u}_y + \bar{u}_z + \bar{u}_t - \tfrac{1}{2}\operatorname{wt}_{\bar{u}}(h')\big) \operatorname{mult}_\nu E.$$

Since $a(\nu, \mathbb{A}^3, \frac{1}{2}\operatorname{div}(h)) = a(\nu, \mathbb{A}^3, 0) - \frac{1}{2}\operatorname{ord}_\nu h$, $\operatorname{mult}_\nu E = c_0$, and $u = c_0\bar{u}$, we obtain (10.9).

Proof of (1). By **Step 3**, a is computed by a prime divisor over X/G, and **Steps 3** and 4 apply to any such prime divisor $\bar{F}$. The polynomial h' is non-zero and vanishes at 0, $X = (x^2 + h'(y, z', t) = 0)$, G acts faithfully on $\mathbb{A}^4$ and diagonally in the coordinates x, y, z', t, $x^2 + h'$ is G-semi-invariant, the induced action of G on X is faithful, and $B = b_y \operatorname{div}_X(y) + 0 \cdot \operatorname{div}_X(z') + b_t \operatorname{div}_X(t)$. Since z' and z have equal characters, N_G taken with respect to x, y, z', t equals N_G taken with respect to x, y, z, t. Possibly replacing z with z', and hence h with h', the data satisfy Set-up 10.3 with $b_z = 0$, and the function $P(w)$ of Set-up 10.3 is the function minimized in (1). Moreover, $\mathbb{C}(y, z, t) = \mathbb{C}(y, z', t)$, and $\operatorname{ord}_\nu h = \operatorname{ord}_\nu h'$ since $h'(y, z', t) = h(y, z, t)$. Thus $\bar{F}$, F, ϱ, ν from **Step 3** and (10.9) for $v = u$ satisfy the hypotheses of Lemma 10.4(2), with $\bar{F}$ in place of E. Hence, by Lemma 10.4(2),

$$a = \min_{w \in N_G \cap \mathbb{R}^4_{>0}} P(w),$$

and the minimum is attained at a primitive vector of N_G. Since $\varphi \in t\mathbb{C}[t]$ and $z' = z - \varphi(t)$ is G-semi-invariant with character equal to the character of z, the automorphism $z = z' + \varphi(t)$ fixing x, y, t is G-equivariant. This proves (1).

Proof of (2). Let a be as in the proposition. By (1), a is computed by a prime divisor over X/G. Let φ and z' be as in (1) for such a prime divisor, and let $J_1 \subset \mathbb{C}[x, y, z', t]$ be the monomial ideal generated by x^2 and the monomials of $h(y, z' + \varphi(t), t)$, so that

$$\operatorname{wt}_w(J_1) = \min\{2w_x, \operatorname{wt}_{(w_y, w_{z'}, w_t)}(h(y, z' + \varphi(t), t))\}$$

for $w \in \mathbb{R}^4_{>0}$. Let $H \subset \mathbb{T}^4$ be the image of G with respect to x, y, z', t. Since z' and z have equal characters, $N_H = N_G$. By (1),

$$a = \inf\Big\{w_x + (1 - b_y)w_y + w_{z'} + (1 - b_t)w_t - \operatorname{wt}_w(J_1) \;\Big|\; w \in N_H \cap \mathbb{R}^4_{>0}\Big\}.$$

Thus a belongs to the set of Corollary 4.6 with $d = 4$, $s = 1$, the coefficient 1, $\Lambda = \{1\}$, and the finite set

$$\mathcal{V} := \{(1, 1 - b_y, 1, 1 - b_t) \mid (b_y, b_t) \in \mathcal{B}\} \subset \mathbb{Q}^4_{>0}.$$

By Corollary 4.6, the numbers a satisfy the ACC. □

11. Quadratic rank one: type cD

In this section, we prove Theorem 11.2. Points of quadratic rank one are classified by their residual cubics, which the group action preserves up to a scalar. For type cD, the eight classes of residual cubics are treated in Propositions 11.8, 11.9, 11.11, 11.10, and 11.12, and, for three concurrent lines permuted cyclically by G, in the second part of Theorem 11.2.

Definition-Lemma 11.1. Let $(X \ni o)$ be a germ of a variety such that the completion of $\mathcal{O}_{X,o}$ is $\mathbb{C}[[x_1, \dots, x_d]]/(f)$ for an integer $d \geq 3$ and some $f \in (x_1, \dots, x_d)^2$. We regard homogeneous polynomials in $x_1, \dots, x_d$ as functions on $\mathbb{C}^d$, and the kernel of a quadratic form on $\mathbb{C}^d$ is the radical of the associated symmetric bilinear form. The *quadratic rank* of $(X \ni o)$ is the rank of the quadratic part f_2 of f. If the quadratic rank is one, the *residual cubic* of $(X \ni o)$ is the restriction of the cubic part f_3 of f to the kernel of f_2. If moreover $d = 4$, the residual cubic is a ternary cubic form, called the *residual ternary cubic*. The quadratic rank does not depend on the choice of $x_1, \dots, x_d$ and f, and the residual cubic depends on the choice of $x_1, \dots, x_d$ and f only up to linear coordinate changes and a non-zero scalar. An isolated cDV threefold singularity $(X \ni o)$ [KM98, Definition 5.32] is of *type* cA, cD, or cE if a general hyperplane section through o is a Du Val singularity of type A, D, or E, respectively, and of *type* cD_n or cE_n if a general hyperplane section through o is of type D_n or E_n [Ste05, Section 2].

Proof. We prove that the quadratic rank does not depend on the choice of $x_1, \dots, x_d$ and f, and that the residual cubic depends on the choice of $x_1, \dots, x_d$ and f only up to linear coordinate changes and a non-zero scalar. Let $\mathbb{C}[[x'_1, \dots, x'_d]]/(f')$, with $f' \in (x'_1, \dots, x'_d)^2$, be another presentation of the completion of $\mathcal{O}_{X,o}$ (the number of variables is d in both presentations since it is the embedding dimension $\dim_{\mathbb{C}} \mathfrak{m}/\mathfrak{m}^2$, where $\mathfrak{m}$ is the maximal ideal of the completion, as f and f' have order at least two). The isomorphism $\mathbb{C}[[x'_1, \dots, x'_d]]/(f') \cong \mathbb{C}[[x_1, \dots, x_d]]/(f)$ lifts to an isomorphism φ from $\mathbb{C}[[x'_1, \dots, x'_d]]$ to $\mathbb{C}[[x_1, \dots, x_d]]$ with $\varphi((f')) = (f)$: lift the images of the x'_i to $\mathbb{C}[[x_1, \dots, x_d]]$, and note that the induced homomorphism is an isomorphism on cotangent spaces because f and f' have order at least two. Thus $\varphi(f') = uf$ for a unit u. Write

$$\varphi(x'_i) = \sum_j A_{ij} x_j + \varphi_{2,i} + (\text{terms of degree at least } 3),$$

where $A = (A_{ij})$ is an invertible matrix and $\varphi_{2,1}, \dots, \varphi_{2,d}$ are quadratic forms, and let $\varphi_2 : \mathbb{C}^d \to \mathbb{C}^d$ be the map $(\varphi_{2,1}, \dots, \varphi_{2,d})$. Let B' be the symmetric bilinear form with $B'(v, v) = f'_2(v)$, and let u_1 be the linear part of u. Taking the parts of degree two and three of $\varphi(f') = uf$, we obtain

$$u(0) f_2(v) = f'_2(Av) \quad \text{and} \quad u(0) f_3(v) + u_1(v) f_2(v) = f'_3(Av) + 2B'(Av, \varphi_2(v))$$

for $v \in \mathbb{C}^d$. The first equality shows that $\operatorname{rank} f_2 = \operatorname{rank} f'_2$ and that $\operatorname{Ker} f_2 = A^{-1}(\operatorname{Ker} f'_2)$. For $v \in \operatorname{Ker} f_2$ we have $f_2(v) = 0$ and $B'(Av, \cdot) = 0$, so the second equality implies

$$u(0) f_3(v) = f'_3(Av).$$

Thus the restriction of f_3 to $\operatorname{Ker} f_2$ is $u(0)^{-1}$ times the composite of the linear isomorphism $A : \operatorname{Ker} f_2 \to \operatorname{Ker} f'_2$ with the restriction of f'_3 to $\operatorname{Ker} f'_2$. □

By [KM98, Corollary 5.38] (cf. [Rei83, Theorem (1.1)]), an isolated Gorenstein terminal threefold singularity is an isolated cDV singularity, so the type of such a singularity is defined. We use the residual cubic to classify points of quadratic rank one because a finite group acting on X and fixing o preserves the residual cubic up to a scalar. However, the group need not have bounded order: a residual cubic consisting of three non-concurrent lines is preserved by a two-dimensional torus.

Theorem 11.2. *Let $(X \ni o)$ be an isolated Gorenstein terminal threefold singularity of type cD_n for some $n \geq 4$, and let G be a finite abelian group acting faithfully on X and fixing o. Then the set of the numbers $\operatorname{mld}((X \ni o, 0)/G) \in (0, 1)$, for all such n, $(X \ni o)$, and G, satisfies the ACC.*

Moreover, assume that $n = 4$, the residual ternary cubic of $(X \ni o)$ is the product of three distinct concurrent lines, and the group of permutations of the three lines induced by G has order three. Then there exist a faithful diagonal action of G on $\mathbb{A}^4$ with coordinates x, y, z, t and polynomials $a_1, \dots, a_4 \in \mathbb{C}[t]$ such that, for

$$h_0 := y^3 + z^3 + a_1(t)yz + a_2(t)y + a_3(t)z + a_4(t),$$

the following hold.

(1) $x^2 + h_0$ *is* G*-semi-invariant.*
(2) $X_0 := (x^2 + h_0 = 0)$ *is an isolated Gorenstein terminal threefold germ at* 0*, and* $\operatorname{mld}((X_0 \ni 0, 0)/H) = \operatorname{mld}((X \ni o, 0)/H)$ *for any subgroup* $H \subset G$*.*
(3) If $\operatorname{mld}((X_0 \ni 0, 0)/G) < 1$*, then*

$$\operatorname{mld}((X_0 \ni 0, 0)/G) = \min_{w \in N_G \cap \mathbb{R}^4_{>0}} \left(w_x + w_y + w_z + w_t - \min\left\{ 2w_x, \operatorname{wt}_{(w_y, w_z, w_t)}(h_0) \right\} \right),$$

where $w = (w_x, w_y, w_z, w_t)$ *and* N_G *is taken with respect to the eigenparameters* x, y, z, t*.*

By Lemma 11.3 below, a singularity of type cD_n has quadratic rank one, so the residual ternary cubic in Theorem 11.2 is defined.

11.1. **Quadratic rank one.** In this subsection, we split off the square in the equation of a point of quadratic rank one equivariantly, find G-semi-invariant coordinates in which a binary cubic with three distinct linear factors has a normal form, and prepare the normal forms used in the rest of this section. For a finite abelian group G acting diagonally on a power series ring, we write χ_x for the character of a semi-invariant coordinate x and χ_F for the character of a semi-invariant power series F.

Lemma 11.3. *Let* $(X \ni o)$ *be an isolated cDV threefold singularity of quadratic rank one, and let* G *be a finite abelian group acting faithfully on* X *and fixing* o*. Then the following hold.*

(1) There exist a smooth fourfold germ $(Y \ni o)$ *with a faithful action of* G*, a* G*-equivariant closed embedding* $X \subset Y$ *with* $X = (f = 0)$ *for a regular* G*-semi-invariant function* f *on* Y*,* G*-semi-invariant regular parameters of* Y *at* o*, and* G*-semi-invariant elements* x, y, z, t *of the maximal ideal of the completion* $\widehat{\mathcal{O}}_{Y,o}$ *of* $\mathcal{O}_{Y,o}$ *such that* $\widehat{\mathcal{O}}_{Y,o} = \mathbb{C}[[x, y, z, t]]$*,* G *acts diagonally on* x, y, z, t*, and the completion of* $\mathcal{O}_{X,o}$ *is* $\mathbb{C}[[x, y, z, t]]/(x^2 + g(y, z, t))$ *for a* G*-semi-invariant power series* g *with* $\operatorname{ord} g \geq 3$*, whose cubic part* g_3 *is the residual ternary cubic, and* $x^2 + g$ *is the expansion of* f *in* $\widehat{\mathcal{O}}_{Y,o}$*. In particular,* $x^2 + g$ *is* G*-semi-invariant with character* χ_x^2*.*
(2) For any Y*,* x, y, z, t*, and* g *satisfying (1), the set of linear factors of* g_3*, counted with multiplicities, is* G*-stable, and any linear factor of* g_3 *whose multiplicity differs from the multiplicities of the other factors is* G*-semi-invariant.*
(3) For any Y*,* x, y, z, t*, and* g *satisfying (1),* $(X \ni o)$ *is of type cE if and only if* g_3 *is a cube of a linear form, and of type cD otherwise.*

Conversely, any isolated cDV threefold singularity of type cD or cE has quadratic rank one.

Proof. **Step 1.** In this step we prove (1).

By [KM98, Corollary 5.38], $(X \ni o)$ is terminal and K_X is Cartier. Since the completion of $\mathcal{O}_{X,o}$ has a presentation as in Definition-Lemma 11.1 and $\dim X = 3$, the embedding dimension of X at o is four, so X is singular at o. By Lemma 5.4, there exist a smooth fourfold germ $(Y \ni o)$ with a faithful action of G, a G-equivariant closed embedding $X \subset Y$ with $X = (f = 0)$ for a regular G-semi-invariant function f, and G-semi-invariant regular parameters $x_1, \dots, x_4$ of Y at o. Thus $\widehat{\mathcal{O}}_{Y,o} = \mathbb{C}[[x_1, \dots, x_4]]$, G acts diagonally on $x_1, \dots, x_4$, and the completion of $\mathcal{O}_{X,o}$ is $\mathbb{C}[[x_1, \dots, x_4]]/(F)$ for the expansion F of f, which is G-semi-invariant. Since X is singular at o, $F \in (x_1, \dots, x_4)^2$, and F_2 has rank one by Definition-Lemma 11.1.

Write $F_2 = l^2$ for a linear form l. For $\sigma \in G$, the automorphism σ^* preserves degrees, so $(\sigma^* l)^2 = \sigma^* F_2 = \chi_F(\sigma) l^2$ and $\sigma^* l \in \mathbb{C}^* l$. Thus $x := l$ is semi-invariant. Since G acts diagonally on the space of linear forms in $x_1, \dots, x_4$, there exist semi-invariant linear forms y', z', t' such that x, y', z', t' is a basis of the space of linear forms in $x_1, \dots, x_4$. Then $\mathbb{C}[[x_1, \dots, x_4]] = \mathbb{C}[[x, y', z', t']]$, G acts diagonally on x, y', z', t', the quadratic part of F is x^2, and $\chi_F = \chi_x^2$.

Since $\partial_x F = 2x + (\text{terms of degree at least } 2)$, the formal implicit function theorem yields a unique $\phi \in (y', z', t')\mathbb{C}[[y', z', t']]$ with $(\partial_x F)(\phi, y', z', t') = 0$, and $\phi \in (y', z', t')^2$ because

$(\partial_x F)(0, y', z', t') \in (y', z', t')^2$. For $\sigma \in G$, the series $\partial_x F$ is semi-invariant with character $\chi_F \chi_x^{-1}$, so

$$(\partial_x F)\big(\chi_x(\sigma)^{-1}\sigma^*\phi, y', z', t'\big) = \chi_F(\sigma)^{-1}\chi_x(\sigma)\sigma^*\big((\partial_x F)(\phi, y', z', t')\big) = 0.$$

By uniqueness, $\sigma^*\phi = \chi_x(\sigma)\phi$, that is, ϕ is zero or semi-invariant with character χ_x. Let $g := F(\phi, y', z', t') \in \mathbb{C}[[y', z', t']]$. Since $(\partial_x F)(\phi, y', z', t') = 0$, Taylor expansion in x at ϕ yields

$$F = g(y', z', t') + (x - \phi)^2 U$$

for some $U \in \mathbb{C}[[x, y', z', t']]$ with $U(0) = 1$. Here $U(0) = 1$ because the coefficient of x^2 in F is 1 and $\phi \in (y', z', t')^2$. The series g is non-zero, since otherwise $(F) = ((x - \phi)^2)$, whereas the completion of the reduced local ring $\mathcal{O}_{X,o}$ is reduced [Gro65, Scholie 7.8.3(ii),(iii),(vii)]. The series g is semi-invariant with character χ_F, and $x - \phi$ is semi-invariant with character χ_x, so U is G-invariant. Let $\sqrt{U}$ be the square root of U with constant term 1. For $\sigma \in G$, $\sigma^*\sqrt{U}$ is also a square root of U with constant term 1, so $\sqrt{U}$ is G-invariant. The element $(x - \phi)\sqrt{U}$ is semi-invariant with character χ_x and congruent to x modulo $(x, y', z', t')^2$. Replacing x with $(x - \phi)\sqrt{U}$ and writing y, z, t for y', z', t', we obtain $\widehat{\mathcal{O}}_{Y,o} = \mathbb{C}[[x, y, z, t]]$ with G acting diagonally on x, y, z, t and

$$F = x^2 + g(y, z, t).$$

Thus $x^2 + g$ is the expansion of f, and it is semi-invariant with character $\chi_F = \chi_x^2$. The quadratic part of F is x^2 in both coordinate systems, so $\operatorname{ord} g \geq 3$. In the presentation $\mathbb{C}[[x, y, z, t]]/(x^2 + g)$, the kernel of the quadratic part x^2 is $\{x = 0\}$ with coordinates y, z, t, and the restriction of the cubic part g_3 to $\{x = 0\}$ is g_3. Thus g_3 is the residual ternary cubic. This proves (1). See [GLS07, Theorem I.2.47] for the splitting lemma without group action.

Step 2. In this step we prove (2).

Let Y, x, y, z, t, and g satisfy (1). For $\sigma \in G$, the automorphism σ^* restricts to a linear automorphism of $\mathbb{C}[y, z, t]$ preserving degrees, and $\sigma^* g_3 = \chi_g(\sigma) g_3$. By unique factorization in $\mathbb{C}[y, z, t]$, for any linear factor L of g_3 of multiplicity m, the linear form $\sigma^* L$ is a factor of g_3 of multiplicity m. Thus the set of linear factors of g_3, counted with multiplicities, is G-stable. If no other linear factor of g_3 has multiplicity m, then $\sigma^* L \in \mathbb{C}^* L$ for any $\sigma \in G$, so L is G-semi-invariant.

Step 3. In this step we compute the quadratic rank of a general hyperplane section and prove that an isolated cDV threefold singularity is of type cD or cE if and only if it has quadratic rank one. In particular, $(X \ni o)$ is of type cD or cE, and the last sentence of the lemma holds.

Let $(X' \ni o')$ be an isolated cDV threefold singularity. A Du Val singularity is singular, so X' is singular at o'. By Lemma 5.4 applied to the trivial group and [KM98, Corollary 5.38], the completion of $\mathcal{O}_{X',o'}$ is $\mathbb{C}[[x_1, \dots, x_4]]/(f)$ for some $f \in (x_1, \dots, x_4)^2$. Let $H = (h = 0)$ be a general hyperplane section of X' through o', where h is a general element of the maximal ideal of $\mathcal{O}_{X',o'}$. The linear part of a lift $\hat{h} \in \mathbb{C}[[x_1, \dots, x_4]]$ of the expansion of h is a general linear form $\lambda_1 x_1 + \cdots + \lambda_4 x_4$. The linear form $\lambda_1 x_1 + \cdots + \lambda_4 x_4$ does not depend on the choice of $\hat{h}$ because $f \in (x_1, \dots, x_4)^2$. In particular $\lambda_4 \neq 0$, and by the formal implicit function theorem $(\hat{h}) = (x_4 - \tau)$ for some $\tau \in (x_1, x_2, x_3)\mathbb{C}[[x_1, x_2, x_3]]$ with linear part $\tau_1 = -\lambda_4^{-1}(\lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 x_3)$. Thus the completion of $\mathcal{O}_{H,o'}$ is $\mathbb{C}[[x_1, x_2, x_3]]/(q)$ for $q := f(x_1, x_2, x_3, \tau)$. Since $f \in (x_1, \dots, x_4)^2$, the quadratic part of q is

$$q_2 = f_2(x_1, x_2, x_3, \tau_1),$$

which is the restriction of f_2 to the general hyperplane $T := \{\lambda_1 x_1 + \cdots + \lambda_4 x_4 = 0\} \subset \mathbb{C}^4$, written in the coordinates x_1, x_2, x_3 of T.

Let $r := \operatorname{rank} f_2$ and $K := \operatorname{Ker} f_2$. If $K \neq 0$, then $K \not\subset T$ since T is general, so $T + K = \mathbb{C}^4$. As f_2 is the pullback of a non-degenerate quadratic form of rank r on $\mathbb{C}^4/K$ and $T \to \mathbb{C}^4/K$ is surjective, $\operatorname{rank}(f_2|_T) = r$. If $K = 0$, the kernel of $f_2|_T$ is $T \cap T^\perp$, where $T^\perp$ is the line orthogonal to T with respect to f_2, so $\operatorname{rank}(f_2|_T) \geq 2$. Thus the quadratic rank of H is one if and only if $r = 1$.

By [KM98, Theorem 4.20] and [Rei87, (1.2)], a Du Val singularity is analytically isomorphic to the germ at 0 of $\mathbb{C}[[u_1,u_2,u_3]]/(q')$, where q' is one of

$$\mathrm{A}_n:\ u_1^2+u_2^2+u_3^{n+1}\ (n\geq 1),\qquad \mathrm{D}_n:\ u_1^2+u_2^2u_3+u_3^{n-1}\ (n\geq 4),$$
$$\mathrm{E}_6:\ u_1^2+u_2^3+u_3^4,\qquad \mathrm{E}_7:\ u_1^2+u_2^3+u_2u_3^3,\qquad \mathrm{E}_8:\ u_1^2+u_2^3+u_3^5.$$

The quadratic part of q' has rank at least two for A_n and rank one for D_n and E_n. In the last two cases the kernel is $\{u_1=0\}$, and the residual cubic is $u_2^2u_3$ for D_n with $n\geq 5$, $u_2^2u_3+u_3^3=u_3(u_2+\sqrt{-1}u_3)(u_2-\sqrt{-1}u_3)$ for D_4, and u_2^3 for E_6, E_7, and E_8. By Definition-Lemma 11.1 with $d=3$, the quadratic rank and the property that the residual cubic is a cube of a linear form do not depend on the presentation. Thus a Du Val singularity is of type A if and only if the quadratic rank is at least two, of type E if and only if the quadratic rank is one and the residual cubic is a cube of a linear form, and of type D if and only if the quadratic rank is one and the residual cubic is non-zero and not a cube of a linear form.

Since H is Du Val, H is of type D or E if and only if the quadratic rank of H is one, that is, if and only if $r=1$. Thus $(X'\ni o')$ is of type cD or cE if and only if it has quadratic rank one.

Step 4. In this step we prove (3).

Let Y, x,y,z,t, and g satisfy (1). By **Step 3**, $(X\ni o)$ is of type cD or cE. Let $H=(h=0)$ be a general hyperplane section through o. As in **Step 3**, the linear part of a lift $\hat{h}\in\mathbb{C}[[x,y,z,t]]$ of the expansion of h is a general linear form $ax+by+cz+et$, so $e\neq 0$ and $(\hat{h})=(t-\tau)$ for some $\tau\in(x,y,z)\mathbb{C}[[x,y,z]]$ with linear part $\tau_1=-e^{-1}(ax+by+cz)$. Thus the completion of $\mathcal{O}_{H,o}$ is $\mathbb{C}[[x,y,z]]/(q)$ for $q:=x^2+g(y,z,\tau)$. Since $\operatorname{ord} g\geq 3$, we have $q_2=x^2$ and $q_3=g_3(y,z,\tau_1)$. The kernel of q_2 is $\{x=0\}$ with coordinates y,z, so the residual cubic of H is

$$q_3(0,y,z)=g_3\big(y,z,-e^{-1}(by+cz)\big),$$

which is the restriction of g_3 to the plane $\Pi:=\{t=-e^{-1}(by+cz)\}$ in $\mathbb{C}^3$ with coordinates y,z,t. Since (b,c,e) is general, Π is a general plane through 0. By **Step 3**, H is of type E if and only if $g_3|_\Pi$ is a cube of a linear form, and of type D otherwise. Moreover, $g_3|_\Pi\neq 0$, hence $g_3\neq 0$.

It remains to show that $g_3|_\Pi$ is a cube of a linear form for a general plane Π if and only if g_3 is a cube of a linear form. If $g_3=L^3$ for a linear form L, then $g_3|_\Pi=(L|_\Pi)^3$. Suppose that g_3 is not a cube of a linear form. Let $C:=(g_3=0)\subset\mathbb{P}^2$ and let $\ell\subset\mathbb{P}^2$ be the line defined by Π. If g_3 has no repeated factor, then C is a reduced curve with finitely many singular points. A general line avoids the singular points of C and is not tangent to C, since the lines tangent to C at smooth points form a family of dimension at most one. Thus $\ell\cap C$ consists of three distinct points, and $g_3|_\Pi$ has three distinct linear factors. If g_3 has a repeated factor, then $g_3=L^2M$ for linear forms L,M, since $\deg g_3=3$, and $M\notin\mathbb{C}L$ since g_3 is not a cube. For a general plane Π, the restrictions $L|_\Pi$ and $M|_\Pi$ are linearly independent, so $g_3|_\Pi$ has a double linear factor and a distinct simple linear factor. In both cases $g_3|_\Pi$ is not a cube of a linear form. This proves (3). □

Lemma 11.4. *Let G be a finite abelian group acting linearly on a two-dimensional complex vector space W, let $c=\ell_1\ell_2\ell_3\in\operatorname{Sym}^3 W$ be G-semi-invariant, where $\ell_1,\ell_2,\ell_3\in W$ are pairwise linearly independent, and let Σ be the group of permutations of $\{\mathbb{C}\ell_1,\mathbb{C}\ell_2,\mathbb{C}\ell_3\}$ induced by G. Then $|\Sigma|\in\{1,2,3\}$, and there exist G-semi-invariant elements y,z forming a basis of W such that c is a non-zero constant multiple of y^2z+z^3 if $|\Sigma|\leq 2$, and of y^3+z^3 if $|\Sigma|=3$.*

Proof. For $\sigma\in G$, the action of σ on $\operatorname{Sym} W$ is a graded ring automorphism, so

$$(\sigma\ell_1)(\sigma\ell_2)(\sigma\ell_3)=\sigma c\in\mathbb{C}^*c.$$

By unique factorization in the polynomial ring $\operatorname{Sym} W$, σ permutes $\mathbb{C}\ell_1,\mathbb{C}\ell_2,\mathbb{C}\ell_3$. Thus Σ is a quotient of G, hence abelian. An abelian subgroup of the symmetric group on three letters has order 1, 2, or 3, so $|\Sigma|\in\{1,2,3\}$.

Since G is finite abelian, W has a basis y,z of G-semi-invariant elements. Let $\psi:=\chi_y\chi_z^{-1}$. The monomials z^3,yz^2,y^2z,y^3 have characters $\chi_z^3,\chi_z^3\psi,\chi_z^3\psi^2,\chi_z^3\psi^3$, respectively, and since c

is semi-invariant, c is a linear combination of those monomials among z^3, yz^2, y^2z, y^3 whose character is the character of c. An element $\sigma \in G$ acts trivially on $\mathbb{P}(W)$ if and only if $\psi(\sigma) = 1$. A projective transformation of $\mathbb{P}(W) \cong \mathbb{P}^1$ fixing the three distinct points $\mathbb{C}\ell_1, \mathbb{C}\ell_2, \mathbb{C}\ell_3$ is the identity, so the image of G in $\mathrm{PGL}(W)$ is isomorphic to Σ. Thus $|\Sigma| = |G/\operatorname{Ker}\psi|$ is the order of ψ.

Suppose that $|\Sigma| = 1$. Then G acts on W by scalars, and any element of W is G-semi-invariant. Since $\mathrm{PGL}_2(\mathbb{C})$ acts transitively on ordered triples of distinct points of $\mathbb{P}^1$, there exists a basis y, z of W such that $\mathbb{C}\ell_1 = \mathbb{C}z$, $\mathbb{C}\ell_2 = \mathbb{C}(y + \sqrt{-1}z)$, and $\mathbb{C}\ell_3 = \mathbb{C}(y - \sqrt{-1}z)$. Thus c is a non-zero constant multiple of $z(y^2 + z^2) = y^2z + z^3$.

Suppose that $|\Sigma| = 2$. Then ψ has order two, z^3 and y^2z have character χ_z^3, and yz^2 and y^3 have character $\chi_z^3\psi \neq \chi_z^3$. Possibly exchanging y and z, we may assume that $c = \alpha z^3 + \beta y^2 z$ for some $\alpha, \beta \in \mathbb{C}$. If $\beta = 0$, then $c = \alpha z^3$, and if $\alpha = 0$, then $c = \beta y^2 z$. In both cases c does not have three pairwise independent linear factors. Thus $\alpha\beta \neq 0$, and replacing y with $(\beta/\alpha)^{1/2}y$, we obtain $c = \alpha(y^2z + z^3)$.

Suppose that $|\Sigma| = 3$. Then ψ has order three, so z^3 and y^3 have character χ_z^3, while yz^2 and y^2z have the characters $\chi_z^3\psi$ and $\chi_z^3\psi^2$, which are distinct and different from χ_z^3. A monomial does not have three pairwise independent linear factors, so $c = \alpha z^3 + \beta y^3$ for some $\alpha, \beta \in \mathbb{C}$, and $\alpha\beta \neq 0$ since otherwise c would be a monomial. Replacing y with $(\beta/\alpha)^{1/3}y$, we obtain $c = \alpha(y^3 + z^3)$.

In all cases the new y and z are G-semi-invariant, since they are constant multiples of semi-invariant elements. □

Lemma 11.5. *Let a finite abelian group G act diagonally on $\mathbb{C}[[x, y, z, t]]$, and let $h(y, z, t)$ be a power series such that $F := x^2 + y^2z + h(y, z, t)$ is G-semi-invariant, where any monomial occurring in h has degree at least three and the cubic part of h belongs to the linear span of yt^2 and t^3. Then there exist a G-equivariant automorphism φ of $\mathbb{C}[[x, y, z, t]]$ with $\varphi(x) = x$ and $\varphi(t) = t$, a number $\lambda \in \mathbb{C}$, an integer $\ell \geq 2$, and a power series $b(z, t) \in (z, t)^3$ which is zero or semi-invariant such that*

$$\varphi(F) = x^2 + y^2z + \lambda yt^\ell + b(z, t).$$

If moreover $h \in (y, z, t)^4$, then one can take $b \in (z, t)^4$, and $\ell \geq 3$ if $\lambda \neq 0$. If the ideal $(F) + (\partial_x F, \partial_y F, \partial_z F, \partial_t F)$ is primary to the maximal ideal of $\mathbb{C}[[x, y, z, t]]$, then $b \neq 0$.

Proof. We construct φ as a composite of G-equivariant automorphisms fixing x and t. Since x^2 and y^2z occur in F, we have $\chi_F = \chi_x^2 = \chi_y^2\chi_z$. Write $h_3 = \mu_1 yt^2 + \mu_2 t^3$ with $\mu_1, \mu_2 \in \mathbb{C}$.

Step 1. In this step we remove the monomials divisible by y^2 from h.

Let $N \geq 4$, and assume that $F = x^2 + y^2z + h(y, z, t)$, where h is zero or semi-invariant, any monomial of h has degree at least three, $h_3 = \mu_1 yt^2 + \mu_2 t^3$, and no monomial of h of degree in $[4, N-1]$ is divisible by y^2. Let q_N be the sum of the terms of h of degree N divisible by y^2. Then q_N/y^2 is zero or a homogeneous polynomial of degree $N - 2 \geq 2$ which is semi-invariant with character $\chi_F\chi_y^{-2} = \chi_z$. Let φ_N be the automorphism with $\varphi_N(z) = z - q_N/y^2$ fixing x, y, t. It is G-equivariant. We have $\varphi_N(x^2 + y^2z) = x^2 + y^2z - q_N$. The terms of h of degree three do not involve z, so φ_N fixes them. A monomial of h of degree $k \geq 4$ involving z is changed by φ_N only in degrees at least $k - 1 + (N - 2) \geq N + 1$. Thus

$$\varphi_N(F) = x^2 + y^2z + h'(y, z, t),$$

where h' is zero or semi-invariant, $h' - (h - q_N)$ has order at least $N + 1$, and h' satisfies the assumption above with $N + 1$ in place of N. Since $\varphi_N(z) - z \in (y, z, t)^{N-2}$, the composite $\varphi_N \circ \cdots \circ \varphi_5 \circ \varphi_4$ converges in the (x, y, z, t)-adic topology to a G-equivariant automorphism Φ fixing x, y, t. Possibly replacing F with $\Phi(F)$, we may assume that no monomial of h is divisible by y^2, that is,

$$F = x^2 + y^2z + a(z, t)y + b(z, t)$$

for some $a \in (z,t)^2$ and $b \in (z,t)^3$ with $a_2 = \mu_1 t^2$ and $b_3 = \mu_2 t^3$. If $h \in (y,z,t)^4$, then $\mu_1 = \mu_2 = 0$, so $a \in (z,t)^3$ and $b \in (z,t)^4$.

Step 2. In this step we remove the terms of ay divisible by yz.

Write $a(z,t) = a_0(t) + za_1(z,t)$. The monomials of ay have character χ_F, so a_1 is zero or semi-invariant with character $\chi_F\chi_y^{-1}\chi_z^{-1} = \chi_y$. Since $a_2 = \mu_1 t^2$, we have $a_1 \in (z,t)^2$. Let φ' be the automorphism with $\varphi'(y) = y - a_1/2$ fixing x, z, t. It is G-equivariant. A direct calculation shows

$$\varphi'(F) = x^2 + y^2 z + a_0(t)y + b_1(z,t), \qquad b_1 := b - \frac{a_0 a_1}{2} - \frac{z a_1^2}{4}.$$

Since $a_0 \in (t)^2$ and $a_1 \in (z,t)^2$, we have $b_1 - b \in (z,t)^4$. Thus $b_1 \in (z,t)^3$, and $b_1 \in (z,t)^4$ if $h \in (y,z,t)^4$. Possibly replacing F with $\varphi'(F)$, we may assume that $a = a_0(t)$ and $b = b_1$.

Step 3. In this step we normalize a_0.

If $a_0 = 0$, then we take $\lambda = 0$ and $\ell = 2$. Suppose that $a_0 \neq 0$. Write $a_0 = \lambda t^\ell U_0(t)$ with $\lambda \neq 0$ and $U_0(0) = 1$. Then $\ell \geq 2$, and $\ell \geq 3$ if $h \in (y,z,t)^4$. Each monomial of a_0 has character $\chi_y\chi_z$, so U_0 is G-invariant. Let φ'' be the automorphism with

$$\varphi''(y) = U_0(t)^{-1} y, \qquad \varphi''(z) = U_0(t)^2 z$$

fixing x, t. It is G-equivariant. Then

$$\varphi''(F) = x^2 + y^2 z + \lambda y t^\ell + b_1(U_0(t)^2 z, t).$$

The power series $b(z,t) := b_1(U_0(t)^2 z, t)$ is zero or semi-invariant, belongs to $(z,t)^3$, and belongs to $(z,t)^4$ if $h \in (y,z,t)^4$. This proves the first two sentences of the lemma.

Step 4. In this step we show that $b \neq 0$ under the last hypothesis.

Let $\mathfrak{J}(P) := (P) + (\partial_x P, \partial_y P, \partial_z P, \partial_t P)$ for $P \in \mathbb{C}[[x,y,z,t]]$. For an automorphism φ, the chain rule and the invertibility of the Jacobian matrix of φ imply $\mathfrak{J}(\varphi(F)) = \varphi(\mathfrak{J}(F))$. Thus $\mathfrak{J}(\varphi(F))$ is primary to the maximal ideal if $\mathfrak{J}(F)$ is. If $b = 0$, then $\varphi(F) = x^2 + y^2 z + \lambda y t^\ell$, and $\varphi(F)$ and the partial derivatives

$$2x, \quad 2yz + \lambda t^\ell, \quad y^2, \quad \lambda \ell y t^{\ell-1}$$

belong to the prime ideal (x,y,t), which is not primary to the maximal ideal. Thus $b \neq 0$. □

Lemma 11.6. *Let a finite abelian group G act diagonally on $\mathbb{C}[[x,y,z,t]]$, and let $c(y,z)$ be a G-semi-invariant binary cubic form with three distinct linear factors. Let $m_0 = 1, m_1, m_2, m_3$ be G-semi-invariant monomials of $\mathbb{C}[y,z]$ of degrees $0,1,1,2$ whose classes form a basis of $\mathbb{C}[y,z]/(\partial_y c, \partial_z c)$. Let $F = x^2 + c(y,z) + h(y,z,t)$ be G-semi-invariant with $h \in (y,z,t)^4$. Then there exist a G-equivariant automorphism φ of $\mathbb{C}[[x,y,z,t]]$ with $\varphi(x) = x$, $\varphi(t) = t$, and $\varphi(y) - y, \varphi(z) - z \in (y,z,t)^2$, and power series $r_0, \dots, r_3 \in \mathbb{C}[[t]]$, each zero or G-semi-invariant, with $\operatorname{ord} r_k \geq 4 - \deg m_k$, such that*

$$\varphi(F) = x^2 + c(y,z) + \sum_{k=0}^{3} r_k(t) m_k.$$

Proof. Let $J := (\partial_y c, \partial_z c) \subset \mathbb{C}[y,z]$. If $\partial_y c$ and $\partial_z c$ had a common zero $p \in \mathbb{P}^1$, then the Euler formula $3c = y\partial_y c + z\partial_z c$ would imply $c(p) = 0$, so p would be a multiple root of c, a contradiction. Thus the quadratic forms $\partial_y c$ and $\partial_z c$ have no common factor and form a regular sequence in $\mathbb{C}[y,z]$. Since J is generated by a regular sequence of two quadrics, the Koszul complex is a free resolution of $\mathbb{C}[y,z]/J$ [Eis95, Corollary 17.5]. Thus the Hilbert series of $\mathbb{C}[y,z]/J$ is $(1-s^2)^2/(1-s)^2 = (1+s)^2$, that is, the Hilbert function of $\mathbb{C}[y,z]/J$ is $(1,2,1)$. Hence any homogeneous polynomial of degree at least three belongs to J, and for $d \leq 2$ the space $\mathbb{C}[y,z]_d$ of homogeneous polynomials of degree d is the direct sum of $J_d := J \cap \mathbb{C}[y,z]_d$ and the span of the m_k with $\deg m_k = d$.

Since F is semi-invariant, c is semi-invariant with character χ_F, so $\partial_y c$ and $\partial_z c$ are semi-invariant with characters $\chi_F\chi_y^{-1}$ and $\chi_F\chi_z^{-1}$, and J is G-stable. For a character χ of G and a

power series P, let $P^{(\chi)} := |G|^{-1}\sum_{\sigma\in G}\chi(\sigma)^{-1}\sigma^*P$ be the χ-component of P. Let $p\in\mathbb{C}[y,z]_d$ be semi-invariant with character χ. If $d\geq 3$, write $p=\alpha\partial_y c+\beta\partial_z c$. Taking χ-components, we may replace α and β with their components of characters $\chi\chi_F^{-1}\chi_y$ and $\chi\chi_F^{-1}\chi_z$. If $d\leq 2$, write $p=\alpha\partial_y c+\beta\partial_z c+\sum_{\deg m_k=d}e_k m_k$ with $\alpha,\beta\in\mathbb{C}[y,z]_{d-2}$ and $e_k\in\mathbb{C}$. Since the decomposition $\mathbb{C}[y,z]_d=J_d\oplus\bigoplus_{\deg m_k=d}\mathbb{C}m_k$ is G-stable, taking χ-components shows that we may assume that α and β have characters $\chi\chi_F^{-1}\chi_y$ and $\chi\chi_F^{-1}\chi_z$, and that $e_k=0$ unless m_k has character χ.

We construct φ as a limit. Let $N\geq 4$, and assume that

$$F=x^2+c(y,z)+\sum_{k=0}^{3}r_k(t)m_k+h(y,z,t),$$

where $h\in(y,z,t)^N$ is zero or semi-invariant, and $r_k\in\mathbb{C}[t]$ is a polynomial such that r_km_k is zero or semi-invariant with character χ_F and any monomial of r_km_k has degree in $[4,N-1]$. For $N=4$, we take $r_k=0$. Write the degree-N part of h as $\sum_{j=0}^{N}t^jp_{N-j}(y,z)$ with $p_{N-j}\in\mathbb{C}[y,z]_{N-j}$ zero or semi-invariant with character $\chi_F\chi_t^{-j}$. By the previous paragraph,

$$p_{N-j}=\alpha_j\partial_y c+\beta_j\partial_z c+\sum_{\deg m_k=N-j}e_{k,j}m_k,$$

where $\alpha_j=\beta_j=0$ if $N-j\leq 1$, $e_{k,j}=0$ if $N-j\geq 3$, the polynomials $t^j\alpha_j$ and $t^j\beta_j$ are zero or semi-invariant with characters χ_y and χ_z, respectively, and $e_{k,j}t^jm_k$ is zero or semi-invariant with character χ_F. Let

$$A:=\sum_j t^j\alpha_j,\qquad B:=\sum_j t^j\beta_j,$$

which are homogeneous of degree $N-2\geq 2$, and let φ_N be the automorphism with $\varphi_N(y)=y-A$ and $\varphi_N(z)=z-B$ fixing x,t. It is G-equivariant. Then

$$\varphi_N(c)=c-A\partial_y c-B\partial_z c+(\text{terms of degree at least } 1+2(N-2)),$$

and $1+2(N-2)\geq N+1$. A monomial of degree $\delta\geq 4$ involving y or z is changed by φ_N only in degrees at least $\delta-1+(N-2)\geq N+1$, and φ_N fixes $r_0(t)m_0=r_0(t)$. Thus

$$\varphi_N(F)=x^2+c(y,z)+\sum_{k=0}^{3}\Big(r_k(t)+\sum_{j:N-j=\deg m_k}e_{k,j}t^j\Big)m_k+h'(y,z,t)$$

for some $h'\in(y,z,t)^{N+1}$ which is zero or semi-invariant. The new coefficient of m_k is $r_k+e_{k,j}t^j$ with $j=N-\deg m_k$, so F satisfies the assumption above with $N+1$ in place of N. Since $\varphi_N(y)-y,\varphi_N(z)-z\in(y,z,t)^{N-2}$, the composite $\varphi_N\circ\cdots\circ\varphi_5\circ\varphi_4$ converges in the (x,y,z,t)-adic topology to a G-equivariant automorphism φ with $\varphi(x)=x$, $\varphi(t)=t$, and $\varphi(y)-y,\varphi(z)-z\in(y,z,t)^2$, and the coefficients of m_k converge to power series $r_k\in\mathbb{C}[[t]]$. Each term added to the coefficient of m_k has t-degree $N-\deg m_k\geq 4-\deg m_k$, so $\operatorname{ord}r_k\geq 4-\deg m_k$, and r_k is zero or semi-invariant since r_km_k is zero or semi-invariant with character χ_F. Thus $\varphi(F)=x^2+c(y,z)+\sum_k r_k(t)m_k$. See [GLS07, Theorem I.2.23] for finite determinacy without group action. □

11.2. **The cD normal form.** In this subsection, we prove Theorem 11.7, which reduces the germs $(x^2+y^2z+\lambda yt^\ell+b(z,t)=0)$ with a diagonal group action to Proposition 9.5 by adjoining a square root of z to the function field. In the later subsections, we use Theorem 11.7 for the residual cubics consisting of a double line and a further line, the irreducible cuspidal cubic, the union of a smooth conic and a tangent line, and three concurrent lines whose induced permutation group has order at most two. After adjoining the square root and changing coordinates, the germ becomes a diagonal binary hypersurface, and the log discrepancy of a prime divisor divided by the order of the inertia group of the prime divisor does not change.

Theorem 11.7. *Let $b(z,t) \in (z,t)^3$ be a non-zero polynomial, $\ell \geq 2$ an integer, $\lambda \in \mathbb{C}$, and $X := (x^2 + y^2 z + \lambda y t^\ell + b(z,t) = 0) \subset \mathbb{A}^4$ a normal klt threefold germ at 0. Let G be a finite abelian group acting diagonally and faithfully on X and on $\mathbb{A}^4$ such that $x^2 + y^2 z + \lambda y t^\ell + b(z,t)$ is G-semi-invariant, and let N_G be the lattice of Definition 2.15 for the characters of x, y, z, t. For $w = (w_x, w_y, w_z, w_t) \in \mathbb{R}^4_{>0}$ let*

$$P(w) := w_x + w_y + w_z + w_t - \min\{2w_x, 2w_y + w_z, \ell w_t + w_y, \mathrm{wt}_{(w_z,w_t)}(b)\},$$

where the term $\ell w_t + w_y$ is omitted if $\lambda = 0$. Assume that $a := \mathrm{mld}((X \ni 0, 0)/G) \in (0,1)$. Then:

(1) $a = \min_{w \in N_G \cap \mathbb{R}^4_{>0}} P(w)$, and the minimum is attained at a primitive vector;

(2) the set of such numbers a satisfies the ACC.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we pass to a quadratic extension of $\mathbb{C}(X)$, which is the function field of a diagonal binary hypersurface X' with an action of a lifted group G', and we compare log discrepancies, inertia groups, and centers on X and on X'.

Let $f := x^2 + y^2 z + \lambda y t^\ell + b(z,t)$, and let $\chi_x, \chi_y, \chi_z, \chi_t, \chi_f$ be the characters of x, y, z, t, f. Since G acts diagonally and f is G-semi-invariant, any monomial occurring in f has character χ_f, that is,

$$\begin{aligned} &\chi_x^2 = \chi_y^2 \chi_z = \chi_f, \qquad \chi_y \chi_t^\ell = \chi_f \ \text{ if } \lambda \neq 0, \\ &\chi_z^i \chi_t^k = \chi_f \ \text{ for any monomial } z^i t^k \text{ occurring in } b. \end{aligned} \tag{11.1}$$

Let s be a square root of z in an algebraic closure of $\mathbb{C}(X)$. We show that $[\mathbb{C}(X)(s) : \mathbb{C}(X)] = 2$. The field $\mathbb{C}(X)$ is the quadratic extension of $\mathbb{C}(y,z,t)$ defined by $x^2 = -(y^2 z + \lambda y t^\ell + b)$. Suppose that $z = (\phi_0 + \phi_1 x)^2$ with $\phi_0, \phi_1 \in \mathbb{C}(y,z,t)$. Then $\phi_0 \phi_1 = 0$. If $\phi_1 = 0$, then $z = \phi_0^2$ would be a square in $\mathbb{C}(y,z,t)$. Hence $\phi_1 \neq 0$, and $\phi_0 = 0$. Thus

$$-\Big(y^2 + \frac{\lambda t^\ell}{z} y + \frac{b}{z}\Big) = -\frac{y^2 z + \lambda y t^\ell + b}{z} = \phi_1^{-2}$$

is a square in $\mathbb{C}(z,t)(y)$. The discriminant of $y^2 + (\lambda t^\ell/z) y + b/z \in \mathbb{C}(z,t)[y]$ is $(\lambda^2 t^{2\ell} - 4zb)/z^2$, which is non-zero: if $\lambda \neq 0$, then $\lambda^2 t^{2\ell}$ is not divisible by z, and if $\lambda = 0$, then $-4zb \neq 0$. Thus $y^2 + (\lambda t^\ell/z) y + b/z$ has two distinct roots in an algebraic closure of $\mathbb{C}(z,t)$, so $-(y^2 + (\lambda t^\ell/z) y + b/z)$ is not a square in $\mathbb{C}(z,t)(y)$, a contradiction.

Let

$$G' := \{(h, \epsilon) \mid h \in G,\ \epsilon \in \mathbb{C}^*,\ \epsilon^2 = \chi_z(h)\}.$$

Since $\mathbb{C}(X)(s) = \mathbb{C}(X)[S]/(S^2 - z)$, the element (h, ϵ) acts on $\mathbb{C}(X)(s)$ by h^* on $\mathbb{C}(X)$ and by $(h,\epsilon)^* s = \epsilon s$. The homomorphism $G' \to G$, $(h,\epsilon) \to h$, is surjective with kernel $\{(1,1), (1,-1)\}$, and $(1,-1)$ acts by the non-trivial element of $\mathrm{Gal}(\mathbb{C}(X)(s)/\mathbb{C}(X))$. Thus

$$\mathbb{C}(X)(s)^{G'} = \mathbb{C}(X)^G. \tag{11.2}$$

Let

$$x' := sx, \qquad y' := s^2 y + \frac{\lambda t^\ell}{2}, \qquad g(s,t) := s^2 b(s^2, t) - \frac{\lambda^2 t^{2\ell}}{4}.$$

A direct calculation shows that $x'^2 + y'^2 + g = zf = 0$ in $\mathbb{C}(X)(s)$, and

$$x = \frac{x'}{s}, \qquad y = \frac{1}{s^2}\Big(y' - \frac{\lambda t^\ell}{2}\Big), \qquad z = s^2. \tag{11.3}$$

Any monomial occurring in $s^2 b(s^2, t)$ is divisible by s^2, and $t^{2\ell}$ is not. Since $b \neq 0$, $b \in (z,t)^3$, and $\ell \geq 2$, we have $0 \neq g \in (s,t)^4$, and

$$\mathrm{wt}_{(c,d)}(g) = \min\{2c + \mathrm{wt}_{(2c,d)}(b), 2\ell d\} \quad \text{for any } (c,d) \in \mathbb{R}^2_{>0}, \tag{11.4}$$

where the term $2\ell d$ is omitted if $\lambda = 0$. Let

$$X' := (x'^2 + y'^2 + g(s,t) = 0) \subset \mathbb{A}^4_{x',y',s,t}$$

and let o' be the origin of $\mathbb{A}^4_{x',y',s,t}$. In the coordinates $x'+\sqrt{-1}y'$, $x'-\sqrt{-1}y'$, s, t, the equation of X' is $(x'+\sqrt{-1}y')(x'-\sqrt{-1}y')+g$, which has degree one in $x'+\sqrt{-1}y'$ with coprime coefficients $x'-\sqrt{-1}y'$ and g. Hence $x'^2+y'^2+g$ is irreducible. Since $\mathbb{C}(X)(s)$ has transcendence degree 3 and is generated by x',y',s,t by (11.3), we have $\mathbb{C}(X)(s)=\mathbb{C}(X')$.

By (11.1), x',y',s,t are G'-semi-invariant with characters

$$(h,\epsilon)\to\epsilon\chi_x(h),\qquad (h,\epsilon)\to\chi_y(h)\chi_z(h),\qquad (h,\epsilon)\to\epsilon,\qquad (h,\epsilon)\to\chi_t(h),$$

where for y' we use $\chi_t^\ell=\chi_y\chi_z$ when $\lambda\neq 0$. Moreover, x'^2, y'^2, and g have the character $(h,\epsilon)\to\chi_z(h)\chi_f(h)$. Thus G' acts diagonally on $\mathbb{A}^4_{x',y',s,t}$, the equation $x'^2+y'^2+g$ is G'-semi-invariant, and the induced action on $\mathbb{C}(X')$ is the action of G' on $\mathbb{C}(X)(s)$ above. If (h,ϵ) acts trivially on $\mathbb{A}^4_{x',y',s,t}$, then $\epsilon=1$, so $\chi_z(h)=\epsilon^2=1$ and $\chi_x(h)=\chi_y(h)=\chi_t(h)=1$, hence $h=1$ since G acts faithfully on $\mathbb{A}^4$. Thus G' acts faithfully on $\mathbb{A}^4_{x',y',s,t}$. Since the restrictions of x',y',s,t to X' are non-zero and generate $\mathbb{C}(X')$, G' also acts faithfully on X'.

By the Jacobian criterion, $\operatorname{Sing}X'\subset\{x'=y'=g=0\}$, which is a curve. Since X' is a hypersurface which is regular in codimension one, X' is normal [Har77, Proposition II.8.23]. Let

$$\Omega_{X'}:=\frac{dy'\wedge ds\wedge dt}{2x'},$$

which generates $\omega_{X'}$ near o' by [KM98, Proposition 5.73] applied to the closure of X' in $\mathbb{P}^4$. Since $x'dx'+y'dy'\equiv 0$ modulo ds,dt on X', we have

$$\Omega_{X'}=\frac{1}{2\sqrt{-1}}\cdot\frac{d(x'+\sqrt{-1}y')\wedge ds\wedge dt}{x'+\sqrt{-1}y'}.$$

The morphism from X' to the affine space $\mathbb{A}^3$ with coordinates $x'+\sqrt{-1}y',s,t$ is birational, because $x'-\sqrt{-1}y'=-g/(x'+\sqrt{-1}y')$ on X'. Let F' be a prime divisor over X' near o'. Since $x'+\sqrt{-1}y',s,t$ are regular on X', F' is also a prime divisor over $\mathbb{A}^3$, and

$$a(F',X',0)=a(F',\mathbb{A}^3,\operatorname{div}(x'+\sqrt{-1}y')),$$

because both sides are equal to $1+\operatorname{ord}_{F'}\Omega_{X'}$. If X' is smooth at the generic point of $\operatorname{center}_{X'}F'$, then $a(F',X',0)>0$ by Lemma 2.13 with all coefficients 0. Otherwise $\operatorname{center}_{X'}F'\subset\{x'=y'=0\}$, so $\operatorname{ord}_{F'}(x'+\sqrt{-1}y')>0$ and $\operatorname{ord}_{F'}g>0$. Let η be the generic point of $\operatorname{center}_{\mathbb{A}^3}F'$. Then η is a point of the plane $\{x'+\sqrt{-1}y'=0\}=\mathbb{A}^2_{s,t}$ at which g vanishes, so η is not the generic point of $\{x'+\sqrt{-1}y'=0\}$. Choose regular parameters of $\mathbb{A}^2_{s,t}$ at η: one parameter θ_1 if η is not a closed point, and two parameters θ_1,θ_2 if η is a closed point. Then $x'+\sqrt{-1}y'$ together with θ_1, or with θ_1,θ_2 if η is a closed point, form regular parameters of $\mathbb{A}^3$ at η, and Lemma 2.13, with coefficient 1 for $x'+\sqrt{-1}y'$ and 0 for the other parameters, implies that

$$a(F',X',0)=a(F',\mathbb{A}^3,\operatorname{div}(x'+\sqrt{-1}y'))\geq\operatorname{ord}_{F'}\theta_1>0.$$

Thus X' is a normal klt threefold germ at o'.

Let $\Omega_X:=\frac{dy\wedge dz\wedge dt}{2x}$, which generates ω_X near 0 by [KM98, Proposition 5.73] applied to the closure of X in $\mathbb{P}^4$. By (11.3), $dz=2sds$ and $dy\equiv s^{-2}dy'$ modulo ds,dt, so

$$\Omega_X=\frac{2s^{-1}dy'\wedge ds\wedge dt}{2s^{-1}x'}=2\Omega_{X'}\quad\text{in }\mathbb{C}(X'). \tag{11.5}$$

Let $\nu:Y\to X$ be the normalization of X in $\mathbb{C}(X')$, which is a finite morphism of degree 2 by the finiteness of integral closure [Har77, Theorem I.3.9A], and let B_Y be the divisor with $K_Y+B_Y=\nu^*K_X$. Then Y and X' are birational, and by (11.5), $a(F',Y,B_Y)=a(F',X',0)$ for any prime divisor F' over both Y and X', since both sides are equal to $1+\operatorname{ord}_{F'}\Omega_{X'}$. Let F' be a prime divisor over X' and F a prime divisor over X such that $\operatorname{ord}_{F'}|_{\mathbb{C}(X)}=r\operatorname{ord}_F$ for a positive integer r. Then F' is a prime divisor over Y, because $\mathcal{O}_Y$ is integral over $\mathcal{O}_X$. Since the ramification of ν along F' is tame in characteristic zero, [Kol13, 2.41 and (2.42.2)] implies that

$$a(F',X',0)=ra(F,X,0). \tag{11.6}$$

By (11.2), the function fields of X'/G' and X/G are both equal to $\mathbb{C}(X)^G$. Let E be the prime divisor over X/G such that $\operatorname{ord}_F|_{\mathbb{C}(X)^G} = |I_G(F)|\operatorname{ord}_E$ (Lemma 2.6(1) for (X,G)). Then $\operatorname{ord}_{F'}|_{\mathbb{C}(X)^G} = r|I_G(F)|\operatorname{ord}_E$, and Lemma 2.6(1) for (X',G') implies that

$$|I_{G'}(F')| = r|I_G(F)|. \tag{11.7}$$

By (11.6) and (11.7),

$$\frac{a(F',X',0)}{|I_{G'}(F')|} = \frac{a(F,X,0)}{|I_G(F)|}. \tag{11.8}$$

Finally, we compare centers. Let F be a prime divisor over X with $\operatorname{center}_X F = \{0\}$. By [Kol13, 2.42], there exists a divisorial valuation $\operatorname{ord}_{F'}$ of $\mathbb{C}(X')$ with $\operatorname{ord}_{F'}|_{\mathbb{C}(X)} = r\operatorname{ord}_F$ for a positive integer r. Since ord_F is positive on x,y,z,t, $\operatorname{ord}_{F'}$ is positive on s, $x' = sx$, $y' = s^2y + \frac{\lambda t^\ell}{2}$, and t, so F' is a prime divisor over X' with $\operatorname{center}_{X'} F' = \{o'\}$ [KM98, Lemma 2.45]. Conversely, let F' be a prime divisor over X' with $\operatorname{center}_{X'} F' = \{o'\}$. Let μ be the restriction of $\operatorname{ord}_{F'}$ to $\mathbb{C}(X)$. Since $\mathbb{C}(X')$ is finite over $\mathbb{C}(X)$, μ is non-trivial [Bou72, Chapter VI, §8, no. 1, Lemma 2], so the value group of μ is $r\mathbb{Z}$ for a positive integer r, and $\mu = rv$ for a discrete valuation v of $\mathbb{C}(X)$. The residue field of $\operatorname{ord}_{F'}$ is an extension of degree at most $[\mathbb{C}(X') : \mathbb{C}(X)] = 2$ of the residue field of v [Bou72, Chapter VI, §8, no. 1, Lemma 2], so the residue field of v has transcendence degree 2. The valuation v is positive on $z = s^2$ and t, and by (11.3), v is positive on x and y if and only if

$$\operatorname{ord}_{F'} x' > \operatorname{ord}_{F'} s \qquad \text{and} \qquad \operatorname{ord}_{F'}\left(y' - \frac{\lambda t^\ell}{2}\right) > 2\operatorname{ord}_{F'} s. \tag{11.9}$$

If (11.9) holds, then $v = \operatorname{ord}_F$ for a prime divisor F over X with $\operatorname{center}_X F = \{0\}$ [KM98, Lemma 2.45], and (11.8) holds for F and F'.

Step 2. In this step we check the hypotheses of Proposition 9.5 for (X',G').

We take the smooth surface germ of Proposition 9.5 to be $\mathbb{A}^2_{s,t}$ at the origin, with G'-eigenparameters s,t, and we take $b_z = b_t = 0$. Recall that $X' = (x'^2 + y'^2 + g(s,t) = 0)$. By **Step 1**, G' acts diagonally on x',y' and on $\mathbb{A}^2_{s,t}$, g is a non-zero G'-semi-invariant polynomial in $(s,t)^4$, $x'^2 + y'^2 + g$ is G'-semi-invariant, X' is a normal klt threefold germ at o', and G' acts faithfully on X'. The expansion of g is g itself. Let $N_{G'}$ be the lattice of Definition 2.15 for the characters of x',y',s,t, and for $(p,q,c,d) \in \mathbb{R}^4_{>0}$ let

$$P'(p,q,c,d) := p + q + c + d - \min\{2p, 2q, \operatorname{wt}_{(c,d)}(g)\}.$$

Thus Proposition 9.5(1) and (2) hold for (X',G') and P', with x',y',s,t in place of x,y,z,t. Note that $P(kw) = kP(w)$ and $P'(kw') = kP'(w')$ for any $k > 0$, since $\operatorname{wt}_{kw}(\cdot) = k\operatorname{wt}_w(\cdot)$.

Step 3. In this step we show that there exists $w \in N_G \cap \mathbb{R}^4_{>0}$ with $P(w) \le a$.

Let $\pi : X \to X/G$ be the quotient morphism. Since X is klt, Lemma 2.6(0)(2)(3) implies that $(X,0)/G$ is a klt pair. By [Amb99, Section 1, the second paragraph after Definition 1.1], there exists a prime divisor over X/G with center $\pi(0)$ computing a. By Lemma 2.6(2)(3) and since $\pi^{-1}(\pi(0)) = \{0\}$, there exists a prime divisor F over X with $\operatorname{center}_X F = \{0\}$ and $a(F,X,0)/|I_G(F)| = a$. By **Step 1**, there exists a prime divisor F' over X' with $\operatorname{center}_{X'} F' = \{o'\}$ and $\operatorname{ord}_{F'}|_{\mathbb{C}(X)} = r\operatorname{ord}_F$ for a positive integer r. Then F' satisfies (11.9), and by (11.8),

$$\frac{a(F',X',0)}{|I_{G'}(F')|} = a < 1.$$

Let

$$p := \operatorname{ord}_{F'} x', \quad q := \operatorname{ord}_{F'} y', \quad c := \operatorname{ord}_{F'} s, \quad d := \operatorname{ord}_{F'} t, \quad j := \operatorname{ord}_{F'}\left(y' - \frac{\lambda t^\ell}{2}\right),$$

which are positive integers. Let

$$w_F := \frac{1}{|I_G(F)|}(\operatorname{ord}_F x, \operatorname{ord}_F y, \operatorname{ord}_F z, \operatorname{ord}_F t).$$

Then $w_F \in N_G \cap \mathbb{R}^4_{>0}$ by Lemma 9.2(1)(3). By (11.3), the equality $\operatorname{ord}_{F'}|_{\mathbb{C}(X)} = r \operatorname{ord}_F$, and (11.7), we have

$$|I_{G'}(F')| w_F = (p - c, j - 2c, 2c, d).$$

By Proposition 9.5(1) for F', $P'(\frac{1}{|I_{G'}(F')|}(p, q, c, d)) = a$, that is, $P'(p, q, c, d) = a(F', X', 0)$. Thus it suffices to show that

$$P(p - c, j - 2c, 2c, d) \le a(F', X', 0), \tag{11.10}$$

since dividing (11.10) by $|I_{G'}(F')|$ implies that $P(w_F) \le a$.

Let $\beta := \operatorname{wt}_{(2c,d)}(b)$, and let $\alpha := \ell d$ if $\lambda \neq 0$. A direct calculation shows that

$$P(p - c, j - 2c, 2c, d) = p + j + c + d - \min\{2p, 2j, j + \alpha, 2c + \beta\},$$

and by (11.4), $\operatorname{wt}_{(c,d)}(g) = \min\{2c + \beta, 2\alpha\}$, where the terms involving α are omitted if $\lambda = 0$. If $\lambda = 0$, then $j = q$. If $\lambda \neq 0$, then $\operatorname{ord}_{F'}(\lambda t^\ell) = \alpha$, so $j \ge \min\{q, \alpha\}$, with equality if $q \neq \alpha$.

Case A: $\lambda = 0$, or $j = \min\{q, \alpha\}$. Then $j \le q$, and if $\lambda \neq 0$, then $j \le \alpha$, so $2j \le j + \alpha$ and $2j = \min\{2q, 2\alpha\}$. Thus

$$\min\{2p, 2j, j + \alpha, 2c + \beta\} = \min\{2p, 2q, 2\alpha, 2c + \beta\} = \min\{2p, 2q, \operatorname{wt}_{(c,d)}(g)\},$$

and

$$P(p - c, j - 2c, 2c, d) = P'(p, q, c, d) - (q - j) \le a(F', X', 0).$$

Case B: $\lambda \neq 0$ and $j > \min\{q, \alpha\}$. Then $q = \alpha < j$. We first show that $p > q$.

Suppose that $p < q$. By Proposition 9.5(1), the restriction of $\operatorname{ord}_{F'}$ to $\mathbb{C}(y', s, t)$ is a positive multiple of a monomial valuation in y', s, t. Since y' and t^ℓ are distinct monomials, $j = \min\{q, \alpha\}$, a contradiction.

Suppose that $p = q$. By Proposition 9.5(1), the restriction of $\operatorname{ord}_{F'}$ to $\mathbb{C}(x' + \sqrt{-1}y', s, t)$ is a positive multiple of a monomial valuation in $x' + \sqrt{-1}y', s, t$. Since $(x' + \sqrt{-1}y')(x' - \sqrt{-1}y') = -g$ and $2\sqrt{-1}y' = (x' + \sqrt{-1}y') - (x' - \sqrt{-1}y')$, we have

$$2\sqrt{-1}(x' + \sqrt{-1}y')\Big(y' - \frac{\lambda t^\ell}{2}\Big) = (x' + \sqrt{-1}y')^2 - \sqrt{-1}\lambda t^\ell (x' + \sqrt{-1}y') + g$$

on X'. The three terms on the right hand side have degrees 2, 1, and 0 in $x' + \sqrt{-1}y'$ and non-zero coefficients, and

$$\operatorname{ord}_{F'}(x' + \sqrt{-1}y') + \operatorname{ord}_{F'}(x' - \sqrt{-1}y') = \operatorname{ord}_{F'} g = \operatorname{wt}_{(c,d)}(g).$$

Thus

$$j = \min\{\operatorname{ord}_{F'}(x' + \sqrt{-1}y'),\ \alpha,\ \operatorname{ord}_{F'}(x' - \sqrt{-1}y')\}.$$

Since $\operatorname{ord}_{F'}(x' \pm \sqrt{-1}y') \ge \min\{p, q\} = q = \alpha$, we have $j = \alpha = q$, a contradiction.

Hence $p > q$. By Proposition 9.5(1), the restriction of $\operatorname{ord}_{F'}$ to $\mathbb{C}(x', s, t)$ is a positive multiple of a monomial valuation in x', s, t, and $a(F', X', 0) = p - q + c + d$. On X',

$$\Big(y' - \frac{\lambda t^\ell}{2}\Big)\Big(y' + \frac{\lambda t^\ell}{2}\Big) = -x'^2 - s^2 b(s^2, t). \tag{11.11}$$

Since $j > \alpha = \operatorname{ord}_{F'}(\lambda t^\ell)$, we have $\operatorname{ord}_{F'}(y' + \frac{\lambda t^\ell}{2}) = \alpha$. Since no monomial of x'^2 occurs in $s^2 b(s^2, t)$, the right hand side of (11.11) has value $m := \min\{2p, 2c + \beta\}$. Thus $j = m - \alpha$, and $m = j + \alpha > 2\alpha$. Then $2j = 2m - 2\alpha > m$, so $\min\{2p, 2j, j + \alpha, 2c + \beta\} = m$, and

$$P(p - c, j - 2c, 2c, d) = p + j + c + d - m = p - \alpha + c + d = p - q + c + d = a(F', X', 0).$$

Therefore (11.10) holds in both cases.

Step 4. In this step we show that for any $w \in N_G \cap \mathbb{R}^4_{>0}$ with $P(w) < 1$, there exists a prime divisor F over X with $\operatorname{center}_X F = \{0\}$ and $a(F, X, 0)/|I_G(F)| \le P(w)$. In particular, $a \le P(w)$.

Let $w = (w_x, w_y, w_z, w_t) \in N_G \cap \mathbb{R}^4_{>0}$ with $P(w) < 1$, let $\beta := \operatorname{wt}_{(w_z,w_t)}(b)$, and let $\alpha := \ell w_t$ if $\lambda \ne 0$. By Definition 2.15, there exists $h \in G$ with

$$\chi_u(h) = \exp(2\pi\sqrt{-1}w_u) \quad \text{for } u \in \{x, y, z, t\}.$$

Let $\epsilon := \exp(\pi\sqrt{-1}w_z)$. Then $(h, \epsilon) \in G'$, and by **Step 1** the characters of x', y', s, t take the values $\exp(2\pi\sqrt{-1}\gamma)$ at (h, ϵ) for $\gamma = w_x + \frac{w_z}{2}$, $w_y + w_z$, $\frac{w_z}{2}$, w_t respectively. If $\lambda \ne 0$, then $\chi_y\chi_z = \chi_t^\ell$ by (11.1), so the value for y' is also $\exp(2\pi\sqrt{-1}\alpha)$. Hence

$$\text{(11.12)} \quad \Big(w_x+\frac{w_z}{2},\ w_y+w_z,\ \frac{w_z}{2},\ w_t\Big) \in N_{G'}, \qquad \text{and} \qquad \Big(w_x+\frac{w_z}{2},\ \alpha,\ \frac{w_z}{2},\ w_t\Big) \in N_{G'} \ \text{ if } \lambda \ne 0.$$

By (11.4) with $(c, d) = (\frac{w_z}{2}, w_t)$, we have $\operatorname{wt}_{(c,d)}(g) = \min\{w_z + \beta, 2\alpha\}$, where 2α is omitted if $\lambda = 0$.

Case I: $\lambda = 0$, or $\alpha \ge w_y + w_z$. Let

$$(p, q, c, d) := \Big(w_x + \frac{w_z}{2},\ w_y + w_z,\ \frac{w_z}{2},\ w_t\Big) \in N_{G'}.$$

Since $2\alpha \ge 2w_y + 2w_z$, we have

$$\begin{aligned} P'(p, q, c, d) &= w_x + w_y + 2w_z + w_t - \min\{2w_x + w_z, 2w_y + 2w_z, w_z + \beta\} \\ &= w_x + w_y + w_z + w_t - \min\{2w_x, 2w_y + w_z, \beta\}. \end{aligned}$$

Since $\ell w_t + w_y = \alpha + w_y \ge 2w_y + w_z$, the term $\ell w_t + w_y$ does not change the minimum in $P(w)$, so $P'(p, q, c, d) = P(w) < 1$. By Proposition 9.5(2), there exist a positive rational number e and a prime divisor F' over X' with $\operatorname{center}_{X'} F' = \{o'\}$ such that $a(F', X', 0)/|I_{G'}(F')| \le P(w)$, $\operatorname{ord}_{F'} s = ec$, $\operatorname{ord}_{F'} t = ed$, $\operatorname{ord}_{F'} x' \ge ep$, and $\operatorname{ord}_{F'} y' \ge eq$, since in each of the three cases of Proposition 9.5(2), the lower bounds for $\operatorname{ord}_{F'} x'$ and $\operatorname{ord}_{F'} y'$ are at least ep and eq respectively. Thus

$$\operatorname{ord}_{F'} x' \ge e\Big(w_x + \frac{w_z}{2}\Big) > \frac{ew_z}{2} = \operatorname{ord}_{F'} s, \qquad \operatorname{ord}_{F'} y' \ge e(w_y + w_z) > ew_z = 2\operatorname{ord}_{F'} s,$$

and if $\lambda \ne 0$, then $\operatorname{ord}_{F'}(\lambda t^\ell) = e\alpha \ge e(w_y + w_z) > 2\operatorname{ord}_{F'} s$. Hence $\operatorname{ord}_{F'}(y' - \frac{\lambda t^\ell}{2}) > 2\operatorname{ord}_{F'} s$, and (11.9) holds. By **Step 1**, the restriction of $\operatorname{ord}_{F'}$ to $\mathbb{C}(X)$ is $r \operatorname{ord}_F$ for a positive integer r and a prime divisor F over X with $\operatorname{center}_X F = \{0\}$, and by (11.8), $a(F, X, 0)/|I_G(F)| \le P(w)$.

Case II: $\lambda \ne 0$ and $\alpha < w_y + w_z$. Let $n := w_y + w_z - \alpha$. Since

$$\exp(2\pi\sqrt{-1}(w_y + w_z)) = \chi_y(h)\chi_z(h) = \chi_t(h)^\ell = \exp(2\pi\sqrt{-1}\alpha)$$

by (11.1), n is a positive integer. Let $m_0 := \min\{2w_x, \beta\}$. By (11.1), $\exp(2\pi\sqrt{-1} \cdot 2w_x)$, $\exp(2\pi\sqrt{-1}\beta)$, and $\exp(2\pi\sqrt{-1}(w_y + \alpha))$ are all equal to $\chi_f(h)$, because β is the (w_z, w_t)-weight of a monomial occurring in b. Thus $m_0 - w_y - \alpha \in \mathbb{Z}$.

We show that $m_0 \ge w_y + \alpha$. Suppose that $m_0 < w_y + \alpha$, and let $n' := w_y + \alpha - m_0$, which is a positive integer. Since $2w_y + w_z = w_y + \alpha + n > m_0$, we have $P(w) = w_x + w_y + w_z + w_t - m_0$. Since $2w_x \ge m_0$ and $w_y + w_z - \alpha = n$,

$$P(w) \ge w_y + w_z + w_t - \frac{m_0}{2} = w_y + w_z + w_t - \frac{w_y + \alpha - n'}{2} = \frac{n + n' + w_z}{2} + w_t > 1,$$

a contradiction. Hence $m_0 \ge w_y + \alpha$. Since $2w_y + w_z = w_y + \alpha + n > w_y + \alpha$, the minimum in $P(w)$ is $w_y + \alpha$, and

$$\text{(11.13)} \qquad P(w) = w_x + w_z + w_t - \alpha.$$

Let

$$(p, q, c, d) := \Big(w_x + \frac{w_z}{2},\ \alpha,\ \frac{w_z}{2},\ w_t\Big) \in N_{G'}.$$

Since $m_0 \ge w_y + \alpha = 2\alpha + n - w_z$, we have

$$\text{(11.14)} \qquad 2p = 2w_x + w_z \ge m_0 + w_z \ge 2\alpha + n, \qquad w_z + \beta \ge w_z + m_0 \ge 2\alpha + n.$$

Thus $\mathrm{wt}_{(c,d)}(g) = 2\alpha$, $p > q$, and by (11.13),

$$P'(p,q,c,d) = p + \alpha + c + d - 2\alpha = w_x + w_z + w_t - \alpha = P(w) < 1.$$

In Proposition 9.5(2) for (p,q,c,d), we have $\eta = \alpha$, $\kappa = p - \alpha > 0$, and $\tau = 2\alpha = 2\eta$, so $\kappa = \frac{1}{2}$. By Proposition 9.5(2), there exist a positive rational number e and a prime divisor F' over X' with $\mathrm{center}_{X'} F' = \{o'\}$ such that $a(F', X', 0)/|I_{G'}(F')| \le P(w)$ and

$$(\mathrm{ord}_{F'} x', \mathrm{ord}_{F'} y') = (ep, e\alpha), \qquad \mathrm{ord}_{F'} s = \frac{ew_z}{2}, \qquad \mathrm{ord}_{F'} t = ew_t.$$

Recall the identity (11.11) on X'. By (11.14), the values of both factors on the left hand side of (11.11) have sum

$$\mathrm{ord}_{F'} \left(x'^2 + s^2 b(s^2, t)\right) \ge e \min\{2p, w_z + \beta\} \ge e(2\alpha + n).$$

Both factors have value at least $\min\{\mathrm{ord}_{F'} y', \mathrm{ord}_{F'}(\lambda t^\ell)\} = e\alpha$, and their difference λt^ℓ has value $e\alpha$. Hence one of the factors has value $e\alpha$, and the other has value at least $e(2\alpha + n) - e\alpha = e(w_y + w_z) > ew_z = 2\,\mathrm{ord}_{F'} s$.

Let $\iota : (x', y', s, t) \to (x', -y', s, t)$. Then ι is an automorphism of X' fixing o', and ι commutes with the diagonal action of G'. Let F'' be the prime divisor over X' with $\mathrm{ord}_{F''}(\phi) = \mathrm{ord}_{F'}(\iota^* \phi)$ for any $\phi \in \mathbb{C}(X')$. Then $a(F'', X', 0) = a(F', X', 0)$, $I_{G'}(F'') = I_{G'}(F')$, $\mathrm{center}_{X'} F'' = \{o'\}$, $\mathrm{ord}_{F''} \phi = \mathrm{ord}_{F'} \phi$ for $\phi \in \{x', s, t\}$, and

$$\mathrm{ord}_{F''} \left(y' - \frac{\lambda t^\ell}{2}\right) = \mathrm{ord}_{F'} \left(y' + \frac{\lambda t^\ell}{2}\right).$$

Possibly replacing F' with F'', we may assume that $\mathrm{ord}_{F'}(y' - \frac{\lambda t^\ell}{2}) > 2\,\mathrm{ord}_{F'} s$. Moreover, $\mathrm{ord}_{F'} x' - \mathrm{ord}_{F'} s = ep - \frac{ew_z}{2} = ew_x > 0$. Thus (11.9) holds. By **Step 1**, the restriction of $\mathrm{ord}_{F'}$ to $\mathbb{C}(X)$ is $r\,\mathrm{ord}_F$ for a positive integer r and a prime divisor F over X with $\mathrm{center}_X F = \{0\}$, and by (11.8), $a(F, X, 0)/|I_G(F)| \le P(w)$.

Finally, by Lemma 2.6, $a \le a(F, X, 0)/|I_G(F)| \le P(w)$ in both cases.

Step 5. In this step we prove (1) and (2).

By **Step 3**, there exists $w_F \in N_G \cap \mathbb{R}^4_{>0}$ with $P(w_F) \le a$. By **Step 4**, $P(w) \ge a$ for any $w \in N_G \cap \mathbb{R}^4_{>0}$ with $P(w) < 1$, and $P(w) \ge 1 > a$ for the other vectors $w \in N_G \cap \mathbb{R}^4_{>0}$. Thus $a = P(w_F) = \min_{w \in N_G \cap \mathbb{R}^4_{>0}} P(w)$. Suppose that $w \in N_G \cap \mathbb{R}^4_{>0}$ satisfies $P(w) = a$ and $w = kw_0$ for some $w_0 \in N_G$ and some integer $k \ge 2$. Then $w_0 \in \mathbb{R}^4_{>0}$ and $P(w_0) = \frac{a}{k} < a < 1$, which contradicts **Step 4**. Thus any vector at which the minimum is attained is primitive, and (1) holds.

Let H be the image of G in $\mathbb{T}^4$, so that $N_G = N_H$. The monomials occurring in f are x^2, $y^2 z$, yt^ℓ (if $\lambda \ne 0$), and the monomials occurring in b, which are pairwise distinct. Hence

$$\mathrm{wt}_w(f) = \min\{2w_x, 2w_y + w_z, \ell w_t + w_y, \mathrm{wt}_{(w_z, w_t)}(b)\} \quad \text{and} \quad P(w) = w_x + w_y + w_z + w_t - \mathrm{wt}_w(f),$$

where $\ell w_t + w_y$ is omitted if $\lambda = 0$. By (1),

$$a = \inf \left\{ w_x + w_y + w_z + w_t - \mathrm{wt}_w(f) \;\middle|\; w \in N_H \cap \mathbb{R}^4_{>0} \right\}.$$

By Theorem 4.5, with $d = 4$ and $m = 1$, the set of such numbers a satisfies the ACC. This proves (2). □

11.3. **Double lines, cusps, tangencies, and smooth or irreducible nodal cubics.** In this subsection, we treat isolated Gorenstein terminal threefold singularities of quadratic rank one whose residual ternary cubic is a non-zero multiple of $L^2 M$ for linearly independent linear forms L, M, defines an irreducible cuspidal cubic or the union of a smooth conic and a tangent line, or defines a smooth or an irreducible nodal plane cubic.

Proposition 11.8. *Let $(X \ni o)$ be an isolated Gorenstein terminal threefold singularity of quadratic rank one whose residual ternary cubic is cL^2M for a number $c \neq 0$ and linearly independent linear forms L, M, and let G be a finite abelian group acting faithfully on X and fixing o. Then there exist a faithful diagonal action of G on $\mathbb{A}^4$, a number $\lambda \in \mathbb{C}$, an integer $\ell \geq 2$ with $\ell \geq 3$ if $\lambda \neq 0$, and a non-zero polynomial $b \in (z,t)^4$ such that*

$$X_0 := (x^2 + y^2z + \lambda yt^\ell + b(z,t) = 0) \subset \mathbb{A}^4$$

is an isolated Gorenstein terminal threefold germ at 0 with G-semi-invariant equation and $\operatorname{mld}((X_0 \ni 0,0)/H) = \operatorname{mld}((X \ni o,0)/H)$ for any subgroup $H \subset G$. In particular, the set of such numbers $\operatorname{mld}((X \ni o,0)/G) \in (0,1)$ satisfies the ACC. For any singularity of type cD_n with $n \geq 5$, the quadratic rank is one and the residual ternary cubic is cL^2M for a number $c \neq 0$ and linearly independent linear forms L, M.

Proof. Since $(X \ni o)$ is singular, $(X \ni o)$ is an isolated cDV hypersurface singularity by Lemma 5.4. By Lemma 11.3, there exist a smooth fourfold germ $(Y \ni o)$ with a faithful action of G, a G-equivariant closed embedding $X \subset Y$ with $X = (f = 0)$ for a regular G-semi-invariant function f on Y, regular parameters $x_1, \dots, x_4$ of Y at o, and G-semi-invariant elements x, y, z, t of the maximal ideal of $\widehat{\mathcal{O}}_{Y,o} = \mathbb{C}[[x_1, \dots, x_4]]$ such that $\widehat{\mathcal{O}}_{Y,o} = \mathbb{C}[[x,y,z,t]]$, G acts diagonally on x, y, z, t, and the completion of $\mathcal{O}_{X,o}$ is $\mathbb{C}[[x,y,z,t]]/(x^2 + g(y,z,t))$ for a G-semi-invariant power series g with $\operatorname{ord} g \geq 3$ and cubic part $g_3 = cL^2M$. Moreover, $x^2 + g$ is the expansion $\hat{f}$ of f in $\mathbb{C}[[x_1, \dots, x_4]]$, and it is G-semi-invariant with character χ_x^2 (Lemma 11.3(1)).

Since L and M are linearly independent, their multiplicities as linear factors of g_3 are two and one, so L and M are G-semi-invariant by Lemma 11.3. Since G acts diagonally on the span of y, z, t, there exists a G-semi-invariant linear form t' in y, z, t such that L, M, t' are linearly independent. Possibly replacing (y, z, t) with (L, cM, t'), we may assume that

$$x^2 + g = x^2 + y^2z + h_4(y,z,t), \qquad h_4 \in (y,z,t)^4.$$

Let $F := x^2 + y^2z + h_4$, which is $x^2 + g$ in the new coordinates, hence G-semi-invariant. Since X is smooth near o outside o, the ideal $(\hat{f}) + (\partial\hat{f}/\partial x_1, \dots, \partial\hat{f}/\partial x_4)$ is primary to the maximal ideal of $\mathbb{C}[[x_1, \dots, x_4]]$ by the Jacobian criterion. Changing formal coordinates does not change the ideal generated by $\hat{f}$ and the first partial derivatives of $\hat{f}$, so $(F) + (\partial_x F, \partial_y F, \partial_z F, \partial_t F)$ is primary to the maximal ideal of $\mathbb{C}[[x,y,z,t]]$. By Lemma 11.5, there exist a G-equivariant automorphism φ of $\mathbb{C}[[x,y,z,t]]$, a number $\lambda \in \mathbb{C}$, an integer $\ell \geq 2$ with $\ell \geq 3$ if $\lambda \neq 0$, and a non-zero semi-invariant power series $\hat{b}(z,t) \in (z,t)^4$ such that

$$\varphi(F) = x^2 + y^2z + \lambda yt^\ell + \hat{b}(z,t). \tag{11.15}$$

Since F is a G-semi-invariant generator of the kernel of $\widehat{\mathcal{O}}_{Y,o} \to \widehat{\mathcal{O}}_{X,o}$, we use Corollary 7.5 for $(Y \ni o)$, the elements x, y, z, t, the generator F, the automorphism φ, and the unit 1, with e in the role of the truncation degree L there. We choose $e \gg 0$ with $e > \ell + 1$ and larger than the degree of some monomial occurring in $\hat{b}$, so that $b := \hat{b}_{<e} \neq 0$. Then

$$f_0 := (\varphi(F))_{<e} = x^2 + y^2z + \lambda yt^\ell + b(z,t)$$

with $b \in (z,t)^4$, and by Corollary 7.5, the action of G on $\mathbb{A}^4$ is faithful and diagonal, f_0 is G-semi-invariant, $X_0 := (f_0 = 0)$ is an isolated Gorenstein terminal threefold germ at 0, G acts faithfully on X_0, and $\operatorname{mld}((X_0 \ni 0,0)/H) = \operatorname{mld}((X \ni o,0)/H)$ for any subgroup $H \subset G$.

Assume that $a := \operatorname{mld}((X \ni o,0)/G) \in (0,1)$. The germ X_0 is normal and klt since it is terminal, $\ell \geq 2$, b is a non-zero polynomial in $(z,t)^3$, and G acts diagonally and faithfully on X_0 and on $\mathbb{A}^4$ with semi-invariant equation. Since $a = \operatorname{mld}((X_0 \ni 0,0)/G)$, Theorem 11.7(2) implies that the set of such numbers a satisfies the ACC.

We prove the last sentence. Let $(X \ni o)$ be of type cD_n with $n \geq 5$. By the last sentence of Lemma 11.3, $(X \ni o)$ has quadratic rank one. Let $S = X \cap (h = 0)$ be a general hyperplane section of X through o, where h is a general element of the maximal ideal of $\mathcal{O}_{X,o}$.

By Lemma 11.3 applied to the trivial group, the completion of $\mathcal{O}_{X,o}$ is

$$\widehat{\mathcal{O}}_{X,o} \cong \mathbb{C}[[x,y,z,t]]/(x^2+g(y,z,t))$$

for a power series g with $\operatorname{ord} g \geq 3$ whose cubic part g_3 is the residual ternary cubic. Let $\tilde{h} \in \mathbb{C}[[x,y,z,t]]$ be a lift of the image of h in the completion of $\mathcal{O}_{X,o}$. Since $x^2+g \in (x,y,z,t)^2$ and the class of h in $\mathfrak{m}_{X,o}/\mathfrak{m}_{X,o}^2$ is general, $\tilde{h} = \beta_1 x + \beta_2 y + \beta_3 z + \beta_4 t +$ (terms of degree ≥ 2) with $(\beta_1 : \beta_2 : \beta_3 : \beta_4)$ general. Then $\beta_4 \neq 0$, $t = \tau'(x,y,z,\tilde{h})$ for a power series τ', and the completion of $\mathcal{O}_{S,o}$ is $\mathbb{C}[[x,y,z]]/(x^2+g(y,z,\tau'(x,y,z,0)))$. Since $\operatorname{ord} g \geq 3$, the quadratic part of $x^2+g(y,z,\tau'(x,y,z,0))$ is x^2 and the cubic part is $g_3(y,z,-(\beta_1 x+\beta_2 y+\beta_3 z)/\beta_4)$. Thus the residual cubic of S is

$$g_3\Big(y,z,-\frac{\beta_2 y+\beta_3 z}{\beta_4}\Big),$$

the restriction of g_3 to the general line $(\beta_2 y+\beta_3 z+\beta_4 t=0)$ of the plane $\mathbb{P}^2$ with homogeneous coordinates $(y:z:t)$. Since $(X \ni o)$ is of type cD_n, the surface germ $(S \ni o)$ is a Du Val singularity of type D_n, so the completion of $\mathcal{O}_{S,o}$ is isomorphic to $\mathbb{C}[[x,y,z]]/(x^2+y^2z+z^{n-1})$ [KM98, Theorem 4.20], whose residual cubic is y^2z since $n-1 \geq 4$. By the uniqueness in Definition-Lemma 11.1, the restriction of g_3 to a general line is, up to a linear change of coordinates and a non-zero scalar, y^2z, that is, a non-zero binary cubic with a double root and a simple root different from the double root. Suppose that g_3 has no repeated irreducible factor. Then the curve $(g_3=0) \subset \mathbb{P}^2$ is reduced, so it has finitely many singular points, and the lines through one of them or tangent to the curve at a smooth point form a subset of dimension at most one of the dual plane. Thus a general line intersects $(g_3=0)$ at three distinct points, a contradiction. A repeated irreducible factor of the cubic form g_3 is linear, so $g_3 = L^2M$ for linear forms L, M. If M were proportional to L, then the restriction of g_3 to a general line would have a triple root. Thus $g_3 = cL^2M$ with $c \neq 0$ and L, M linearly independent. □

Proposition 11.9. *Let $(X \ni o)$ be an isolated Gorenstein terminal threefold singularity of quadratic rank one whose residual ternary cubic defines an irreducible cuspidal plane cubic or the union of a smooth conic and a tangent line, and let G be a finite abelian group acting faithfully on X and fixing o. Then there exist a faithful diagonal action of G on $\mathbb{A}^4$, a number $\lambda \in \mathbb{C}$, an integer $\ell \geq 2$, and a non-zero polynomial $b \in (z,t)^3$ such that*

$$X_0 := (x^2+y^2z+\lambda y t^\ell + b(z,t) = 0) \subset \mathbb{A}^4$$

is an isolated Gorenstein terminal threefold germ at 0 with G-semi-invariant equation and $\operatorname{mld}((X_0 \ni 0,0)/H) = \operatorname{mld}((X \ni o,0)/H)$ for any subgroup $H \subset G$. In particular, the set of such numbers $\operatorname{mld}((X \ni o,0)/G) \in (0,1)$ satisfies the ACC.

Proof. Since $(X \ni o)$ is singular, $(X \ni o)$ is an isolated cDV hypersurface singularity by Lemma 5.4. By Lemma 11.3, there exist a smooth fourfold germ $(Y \ni o)$ with a faithful action of G, a G-equivariant closed embedding $X \subset Y$ with $X=(f=0)$ for a regular G-semi-invariant function f on Y, regular parameters $x_1,\dots,x_4$ of Y at o, and G-semi-invariant elements x,y,z,t of the maximal ideal of $\widehat{\mathcal{O}}_{Y,o} = \mathbb{C}[[x_1,\dots,x_4]]$ such that $\widehat{\mathcal{O}}_{Y,o} = \mathbb{C}[[x,y,z,t]]$, G acts diagonally on x,y,z,t, and the completion of $\mathcal{O}_{X,o}$ is $\mathbb{C}[[x,y,z,t]]/(x^2+g(y,z,t))$ for a G-semi-invariant power series g with $\operatorname{ord} g \geq 3$ whose cubic part g_3 is the residual ternary cubic. Moreover, x^2+g is the expansion $\hat{f}$ of f in $\mathbb{C}[[x_1,\dots,x_4]]$, and it is G-semi-invariant with character χ_x^2 (Lemma 11.3(1)).

Let $C := (g_3=0) \subset \mathbb{P}^2$, where $\mathbb{P}^2$ has homogeneous coordinates $(y:z:t)$. Any $\gamma \in G$ multiplies each of y,z,t and g_3 by a non-zero number, so γ induces a projective automorphism $\bar{\gamma}$ of $\mathbb{P}^2$ preserving C. Let $\bar{G} := \{\bar{\gamma} \mid \gamma \in G\}$. A projective automorphism of $\mathbb{P}^2$ preserving C is determined by the restriction to a component of C of degree at least two, since such a component contains four points in general position. We find linearly independent linear forms y',z',t' in y,z,t such that any element of $\bar{G}$ is diagonal in the homogeneous coordinates $(y':z':t')$, and $g_3 = \kappa(y'^2z'+t'^3)$ or $g_3 = \kappa(y'^2z'+y't'^2)$ for a number $\kappa \neq 0$.

The cuspidal case. Choose homogeneous coordinates $(y_1 : z_1 : t_1)$, where y_1, z_1, t_1 are linear forms in y, z, t, in which the cusp of C is $(0 : 1 : 0)$. Since C has multiplicity two at $(0 : 1 : 0)$, $g_3 = z_1 q_2(y_1, t_1) + q_3(y_1, t_1)$ with binary forms q_2, q_3 of degree two and three, and q_2 is the square of a linear form since $(0 : 1 : 0)$ is a cusp. After a linear change of y_1, t_1, $q_2 = \kappa y_1^2$ with $\kappa \neq 0$, and the coefficient α of t_1^3 in q_3/κ is non-zero since C is irreducible. Write β for the coefficient of $y_1 t_1^2$ in q_3/κ. After replacing t_1 with $t_1 - \frac{\beta}{3\alpha} y_1$, we have $q_3/\kappa = \alpha t_1^3 + y_1^2 m(y_1, t_1)$ for a linear form m, so $g_3 = \kappa(y_1^2(z_1 + m(y_1, t_1)) + \alpha t_1^3)$. Taking $y' := y_1$, $z' := z_1 + m(y_1, t_1)$, and $t' := \alpha^{1/3} t_1$, we obtain $g_3 = \kappa(y'^2 z' + t'^3)$. The morphism

$$\nu : \mathbb{P}^1 \to C, \qquad \nu(s : r) = (s^3 : r^3 : -s^2 r)$$

is the normalization of C, the cusp is $\nu(0 : 1) = (0 : 1 : 0)$, and $p_0 := \nu(1 : 0) = (1 : 0 : 0)$. The Hessian of $y'^2 z' + t'^3$ is

$$\det \begin{pmatrix} 2z' & 2y' & 0 \\ 2y' & 0 & 0 \\ 0 & 0 & 6t' \end{pmatrix} = -24 y'^2 t',$$

and $(y'^2 t' = 0) \cap C = \{(0 : 1 : 0), p_0\}$. Thus p_0 is the unique flex of C at a smooth point. Any element of $\bar{G}$ fixes the cusp and p_0, so the restriction to C lifts to an automorphism of $\mathbb{P}^1$ fixing $(0 : 1)$ and $(1 : 0)$, that is, $(s : r) \to (cs : r)$ for some $c \in \mathbb{C}^*$. Since

$$\nu(cs : r) = (c^3 s^3 : r^3 : -c^2 s^2 r),$$

the element coincides on C with the diagonal automorphism $(y' : z' : t') \to (c^3 y' : z' : c^2 t')$, hence is equal to $(y' : z' : t') \to (c^3 y' : z' : c^2 t')$.

The case of a conic and a tangent line. Let $C = Q \cup \ell_0$, where Q is a smooth conic and ℓ_0 is tangent to Q at a point p. Since Q and ℓ_0 have different degrees, $\bar{G}$ preserves Q, ℓ_0, and $Q \cap \ell_0 = \{p\}$. Thus $\bar{G}$ acts faithfully on $Q \cong \mathbb{P}^1$ and fixes p, so $\bar{G}$ acts on $Q \backslash \{p\} \cong \mathbb{A}^1$ by affine transformations. The barycenter q of a $\bar{G}$-orbit in $Q \backslash \{p\}$ is fixed by $\bar{G}$. Any smooth conic is projectively equivalent to $(y'z' + t'^2 = 0)$, which is the image of the isomorphism

$$\iota : \mathbb{P}^1 \to (y'z' + t'^2 = 0), \qquad \iota(s : r) = (s^2 : r^2 : \sqrt{-1} sr),$$

and any automorphism of $\mathbb{P}^1$ is induced through ι by a projective automorphism of $\mathbb{P}^2$. Thus we may choose homogeneous coordinates $(y' : z' : t')$ with $Q = (y'z' + t'^2 = 0)$, $p = \iota(0 : 1) = (0 : 1 : 0)$, and $q = \iota(1 : 0) = (1 : 0 : 0)$. The tangent line to Q at $(0 : 1 : 0)$ is $(y' = 0)$, so $g_3 = \kappa y'(y'z' + t'^2) = \kappa(y'^2 z' + y' t'^2)$ for a number $\kappa \neq 0$. Any element of $\bar{G}$ fixes p and q, so the restriction to Q is $\iota \circ \tau \circ \iota^{-1}$ for $\tau(s : r) = (cs : r)$ with $c \in \mathbb{C}^*$. Since

$$\iota(cs : r) = (c^2 s^2 : r^2 : \sqrt{-1} csr),$$

the element coincides on Q with the diagonal automorphism $(y' : z' : t') \to (c^2 y' : z' : ct')$, hence is equal to $(y' : z' : t') \to (c^2 y' : z' : ct')$.

In both cases, y', z', t' are linear forms in the coordinates y, z, t fixed above, and any $\gamma \in G$ acts on the span of y, z, t by a linear map whose projectivization is diagonal in $(y' : z' : t')$, so y', z', t' are G-semi-invariant. Possibly replacing (y, z, t) with $(y', \kappa z', \kappa^{1/3} t')$ in the cuspidal case and with $(y', \kappa z', \kappa^{1/2} t')$ in the other case, we may assume that

$$x^2 + g = x^2 + y^2 z + h_3(y, z, t),$$

where any monomial occurring in h_3 has degree at least three and the cubic part of h_3 is t^3 or yt^2. Let $F := x^2 + y^2 z + h_3$, which is $x^2 + g$ in the new coordinates, hence G-semi-invariant. Since X is smooth near o outside o, the ideal $(\hat{f}) + (\partial \hat{f}/\partial x_1, \dots, \partial \hat{f}/\partial x_4)$ is primary to the maximal ideal by the Jacobian criterion, and the ideal generated by $\hat{f}$ and the first partial derivatives of $\hat{f}$ does not change under formal coordinate changes. Thus $(F) + (\partial_x F, \partial_y F, \partial_z F, \partial_t F)$ is primary to the maximal ideal of $\mathbb{C}[[x, y, z, t]]$. By Lemma 11.5, there exist a G-equivariant automorphism φ

of $\mathbb{C}[[x,y,z,t]]$, a number $\lambda \in \mathbb{C}$, an integer $\ell \geq 2$, and a non-zero semi-invariant power series $\hat{b}(z,t) \in (z,t)^3$ such that

$$\varphi(F) = x^2 + y^2 z + \lambda y t^\ell + \hat{b}(z,t). \tag{11.16}$$

Since F is a G-semi-invariant generator of the kernel of $\widehat{\mathcal{O}}_{Y,o} \to \widehat{\mathcal{O}}_{X,o}$, we use Corollary 7.5 for $(Y \ni o)$, the elements x,y,z,t, the generator F, the automorphism φ, and the unit 1, with e in the role of the truncation degree L there. We choose $e \gg 0$ with $e > \ell + 1$ and larger than the degree of some monomial occurring in $\hat{b}$, so that $b := \hat{b}_{<e} \neq 0$. Then

$$f_0 := (\varphi(F))_{<e} = x^2 + y^2 z + \lambda y t^\ell + b(z,t)$$

with $b \in (z,t)^3$, and by Corollary 7.5, the action of G on $\mathbb{A}^4$ is faithful and diagonal, f_0 is G-semi-invariant, $X_0 := (f_0 = 0)$ is an isolated Gorenstein terminal threefold germ at 0, G acts faithfully on X_0, and $\operatorname{mld}((X_0 \ni 0,0)/H) = \operatorname{mld}((X \ni o,0)/H)$ for any subgroup $H \subset G$.

Assume that $a := \operatorname{mld}((X \ni o,0)/G) \in (0,1)$. The germ X_0 is normal and klt since it is terminal, $\ell \geq 2$, b is a non-zero polynomial in $(z,t)^3$, and G acts diagonally and faithfully on X_0 and on $\mathbb{A}^4$ with semi-invariant equation. Since $a = \operatorname{mld}((X_0 \ni 0,0)/G)$, Theorem 11.7(2) implies that the set of such numbers a satisfies the ACC. □

For the smooth and the irreducible nodal residual cubics, we bound the order of the group instead of returning to the cD normal form. A weighted blow-up with weights $(3,2,2,2)$ extracts a prime divisor with log discrepancy three whose residue field is the function field of the projective plane, so the inertia group of the divisor is the kernel of the action of G on the plane, and the image of G preserves a smooth or a nodal plane cubic. For type cE, Propositions 12.6 and 12.7 bound the order of the group by a weighted blow-up in this way.

Proposition 11.10. *Let $(X \ni o)$ be an isolated Gorenstein terminal threefold singularity of quadratic rank one whose residual ternary cubic defines a smooth or an irreducible nodal plane cubic C, let G be a finite abelian group acting faithfully on X and fixing o, and let $a := \operatorname{mld}((X \ni o,0)/G) > 0$. Then there exists a prime divisor E over X with center o and $a(E,X,0) = 3$ whose inertia group is the kernel of the induced action of G on the projectivization of the three-dimensional kernel of the quadratic form. In particular, $|G| \leq \frac{162}{a}$ if C is smooth, $|G| \leq \frac{18}{a}$ if C is nodal, and for any $\epsilon > 0$ the set of such numbers $a \geq \epsilon$ is finite.*

Proof. **Step 1.** In this step, we construct E and show that $a(E,X,0) = 3$.

By Lemma 5.4, there exist a smooth fourfold germ $(Y \ni o)$ with a faithful action of G, a G-equivariant closed embedding $X \subset Y$ with $X = (f = 0)$ for a regular G-semi-invariant function f, and G-semi-invariant regular parameters $x_1, \dots, x_4$ of Y at o defining a morphism $Y \to \mathbb{A}^4$ étale at o. Let f_2 be the quadratic part of the expansion $\hat{f} \in \mathbb{C}[[x_1, \dots, x_4]]$ of f. Since the quadratic rank is one, $f_2 = \ell^2$ for a linear form ℓ in $x_1, \dots, x_4$. Since f is semi-invariant and G acts diagonally on $x_1, \dots, x_4$, $\gamma^* \ell^2$ is a multiple of ℓ^2 for any $\gamma \in G$, so $\gamma^* \ell$ is a multiple of ℓ. Thus $x' := \ell$ is semi-invariant, and there exist semi-invariant linear forms y', z', t' in $x_1, \dots, x_4$ such that x', y', z', t' form a basis of the linear forms in $x_1, \dots, x_4$. Possibly shrinking Y to a G-invariant neighborhood of o, we may assume that x', y', z', t' define an étale morphism $Y \to \mathbb{A}^4$ and that o is the only point of Y over 0. The morphism $Y \to \mathbb{A}^4$ is G-equivariant for the diagonal action of G on $\mathbb{A}^4$ by the characters of x', y', z', t'. Let $w := (3,2,2,2)$ and

$$g_3(y', z', t') := f_3(0, y', z', t'),$$

where f_3 is the cubic part of $\hat{f}$ in the coordinates x', y', z', t'. The kernel of f_2 on the tangent space of Y at o is $(x' = 0)$, so g_3 is the residual ternary cubic (Definition-Lemma 11.1), and $C = (g_3 = 0) \subset \mathbb{P}(\operatorname{Ker} f_2)$, where $(y' : z' : t')$ are homogeneous coordinates on $\mathbb{P}(\operatorname{Ker} f_2)$. A monomial of degree three divisible by x' has w-weight at least 7, and a monomial of degree at least four has w-weight at least 8, so

$$\operatorname{wt}_w(\hat{f}) = 6, \qquad \operatorname{wt}_w(\hat{f} - x'^2 - g_3) \geq 7.$$

Let $Y_0' \to \mathbb{A}^4$ be the weighted blow-up with weights w in the coordinates x', y', z', t', let $\pi : Y' := Y \times_{\mathbb{A}^4} Y_0' \to Y$, and let $E_Y := \pi^{-1}(o)$. Then $E_Y \cong \mathbb{P}(3,2,2,2)$. The exceptional divisor D_w of $Y_0' \to \mathbb{A}^4$ is the torus-invariant prime divisor of the ray spanned by w, whose valuation is the monomial valuation with weights w (Definition 2.16). Since $Y \to \mathbb{A}^4$ is étale and o is the only point of Y over 0, E_Y is the pullback of D_w and $\widehat{\mathcal{O}}_{Y,o} = \mathbb{C}[[x', y', z', t']]$, so ord_{E_Y} is the monomial valuation with weights w in x', y', z', t'. Moreover, G acts on Y' through the action of the diagonal torus on Y_0', and

$$a(E_Y, Y, 0) = 3 + 2 + 2 + 2 = 9$$

by [CLS11, Lemma 11.4.10]. Let X' be the strict transform of X. Since $\mathrm{ord}_{E_Y}(\hat{f}) = \mathrm{wt}_w(\hat{f}) = 6$, we have $\pi^* X = X' + 6E_Y$. Let

$$E_0 := (x'^2 + g_3(y', z', t') = 0) \subset \mathbb{P}(3,2,2,2) \cong E_Y.$$

Since $g_3 \neq 0$ has odd degree, $-g_3$ is not a square in $\mathbb{C}[y', z', t']$, so $x'^2 + g_3$ is irreducible and E_0 is reduced and irreducible. Since $(1:0:0:0) \notin E_0$ and $E_0 \cap (x' = 0) = (x' = g_3 = 0)$ has codimension one in E_0, we have $x' \neq 0$ and $(y', z', t') \neq (0,0,0)$ at a general point of E_0. The affine open subset $U \subset Y_0'$ where $x' \neq 0$ is the quotient of $\mathbb{A}^4$ with coordinates $\xi, \eta_1, \eta_2, \eta_3$, where $x' = \xi^3$ and $(y', z', t') = \xi^2(\eta_1, \eta_2, \eta_3)$, by μ_3 acting by scalar multiplication. There E_Y is the image of $(\xi = 0)$, and the quotient is smooth outside the image of the origin, which is the point $(1:0:0:0)$ of E_Y. Since $Y' \to Y_0'$ is étale, Y' is smooth at a general point of E_0. Since $\mathrm{wt}_w(\hat{f} - x'^2 - g_3) \geq 7$ and μ_3 acts on $\mathbb{A}^4_{\xi,\eta_1,\eta_2,\eta_3}$ freely outside the origin, on the pullback to Y' of $\mathbb{A}^4_{\xi,\eta} \to U$ we have $\pi^* f = \xi^6\big(1 + g_3(\eta_1, \eta_2, \eta_3) + \xi r\big)$ for a regular function r. Hence X' is there the image of $(1 + g_3(\eta) + \xi r = 0)$, whose intersection with $(\xi = 0)$ is the image of $(1 + g_3(\eta) = 0)$, that is, E_0 where $x' \neq 0$. Since $\sum_i \eta_i \partial_{\eta_i} g_3 = 3g_3 = -3$ on $(1 + g_3(\eta) = 0)$, the surface $(1 + g_3(\eta) = 0) \subset \mathbb{A}^3_\eta$ is smooth. Thus X' is smooth at a general point of E_0 and $X'|_{E_Y} = E_0$ near such a point. Let E be the prime divisor over X defined by $E_0 \subset X'$. The center of E on X is $\pi(E_0) = o$. Since

$$K_{Y'} + X' = \pi^*(K_Y + X) + (9 - 1 - 6)E_Y,$$

restricting to X' near a general point of E_0 and using the adjunction formula $K_X = (K_Y + X)|_X$, we obtain $K_{X'} = (\pi|_{X'})^* K_X + 2E_0$ near a general point of E_0. Thus $a(E, X, 0) = 3$.

Step 2. In this step, we compute the inertia group of E.

The residue field of ord_E is the function field of E_0, which consists of the fractions P/Q of weighted homogeneous polynomials P, Q in x', y', z', t' with $\mathrm{wt}_w(P) = \mathrm{wt}_w(Q)$. Multiplying P and Q by x' if $\mathrm{wt}_w(P)$ is odd, we may assume that $\mathrm{wt}_w(P)$ is even. A weighted homogeneous polynomial of even weight contains only even powers of x', and $x'^2 = -g_3$ on E_0, so the residue field of ord_E is $\mathbb{C}(y'/t', z'/t')$, the function field of $\mathbb{P}(\mathrm{Ker}\, f_2)$. The group G preserves E_Y and X', hence E_0, so $\mathrm{Stab}_G(E) = G$. Let $\chi_{y'}, \chi_{z'}, \chi_{t'}$ be the characters of y', z', t'. Since $\mathrm{ord}_E(y') = \mathrm{ord}_E(z') = \mathrm{ord}_E(t') = 2$, the residues of y'/t' and z'/t' are the functions y'/t' and z'/t' on E_0, and

$$\gamma^*\Big(\frac{y'}{t'}\Big) = \frac{\chi_{y'}(\gamma)}{\chi_{t'}(\gamma)} \cdot \frac{y'}{t'}, \qquad \gamma^*\Big(\frac{z'}{t'}\Big) = \frac{\chi_{z'}(\gamma)}{\chi_{t'}(\gamma)} \cdot \frac{z'}{t'}$$

for any $\gamma \in G$. Thus γ acts trivially on the residue field of ord_E if and only if $\chi_{y'}(\gamma) = \chi_{z'}(\gamma) = \chi_{t'}(\gamma)$, that is, if and only if γ acts trivially on $\mathbb{P}(\mathrm{Ker}\, f_2)$. Hence

$$I_G(E) = \mathrm{Ker}(G \to \mathrm{PGL}_3),$$

where PGL_3 is the group of projective automorphisms of $\mathbb{P}(\mathrm{Ker}\, f_2)$.

Step 3. In this step, we bound $|G|$ and prove the finiteness.

Let $\bar{G}$ be the image of G in PGL_3, so that $|G| = |I_G(E)| \cdot |\bar{G}|$. By Lemma 2.6, $a \leq a(E, X, 0)/|I_G(E)| = 3/|I_G(E)|$, so $|I_G(E)| \leq 3/a$. Since f is semi-invariant and G acts diagonally

on x', y', z', t', g_3 is semi-invariant, so $\bar{G}$ preserves C. If C is smooth, then $|\bar{G}| \leq 54$ since a smooth plane cubic has 18, 36, or 54 projective automorphisms [AF93, Section 3.6].

Assume that C is nodal, and let $\nu : \mathbb{P}^1 \to C$ be the normalization, with homogeneous coordinates $(s : r)$ on $\mathbb{P}^1$ such that the preimages of the node are $(0 : 1)$ and $(1 : 0)$. The pullbacks of the linear forms in y', z', t' form a three-dimensional subspace $V \subset H^0(\mathbb{P}^1, \mathcal{O}(3))$. The pullbacks of the forms vanishing at the node are the elements of V vanishing at $(0 : 1)$, that is, the elements $as^3 + bs^2r + csr^2 + dr^3$ of V with $d = 0$, and also the elements of V vanishing at $(1 : 0)$, that is, the elements with $a = 0$. Thus there exists $\kappa \neq 0$ with $V = \{as^3 + bs^2r + csr^2 + dr^3 \mid d = \kappa a\}$, and replacing s with $\kappa^{1/3}s$, we obtain

$$V = \operatorname{Span}(s^2r, sr^2, s^3 + r^3).$$

An element of $\bar{G}$ is determined by the restriction to C, since C contains four points in general position, and the restriction to C lifts to an automorphism τ of $\mathbb{P}^1$ preserving $\{(0 : 1), (1 : 0)\}$ with $\tau^*V = V$. Thus $\tau(s : r) = (cs : r)$ or $\tau(s : r) = (r : cs)$ for some $c \in \mathbb{C}^*$, and $\tau^*(s^3 + r^3) \in V$ implies $c^3 = 1$. Hence $|\bar{G}| \leq 6$.

Thus $|G| \leq \frac{3}{a} \cdot 54 = \frac{162}{a}$ if C is smooth, and $|G| \leq \frac{3}{a} \cdot 6 = \frac{18}{a}$ if C is nodal.

Since K_X is Cartier, $a(F, X, 0)$ is an integer for any prime divisor F over X, and $a(F, X, 0) \geq 1$ since X is terminal. By Lemma 2.6, $a = \inf_F a(F, X, 0)/|I_G(F)|$, where F runs over the prime divisors over X with center o, and

$$\frac{a(F, X, 0)}{|I_G(F)|} = \frac{a(F, X, 0) \cdot [G : I_G(F)]}{|G|} \in \frac{1}{|G|}\mathbb{N}^+.$$

Thus the infimum is attained and $a \in \frac{1}{|G|}\mathbb{Z}$. Moreover, $a \leq 3/|I_G(E)| \leq 3$. If $a \geq \epsilon$, then $|G| \leq 162/\epsilon$, so

$$a \in \bigcup_{1 \leq m \leq 162/\epsilon} \Big(\frac{1}{m}\mathbb{Z} \cap [\epsilon, 3]\Big),$$

which is a finite set. □

11.4. **Nodal residual cubics.** In this subsection, we prove Proposition 11.11. Proposition 11.10 does not cover the reducible residual cubics whose singular points are nodes, for instance three lines forming a triangle. Proposition 11.11 also covers smooth and irreducible nodal residual cubics, for which Proposition 11.10 also bounds the order of G. In the proof, we pass to a double cover Y of an affine piece of the weighted blow-up of X with weights $(3, 2, 2, 2)$. On Y the crepant boundary has coefficient -2 along one coordinate divisor and no other component, Lemma 3.1 shows that the minimal log discrepancy of a local quotient of Y by an inertia group equals $\operatorname{mld}((X \ni o, 0)/G)$, and the relevant points of Y are smooth points, generic points of curves, and singular points at which Corollary 9.6 can be used with the coefficients $(0, -2)$.

Proposition 11.11. *Let $(X \ni o)$ be an isolated Gorenstein terminal threefold singularity of quadratic rank one whose residual ternary cubic defines a reduced plane cubic whose singular points are nodes, that is, ordinary double points, and let G be a finite abelian group acting faithfully on X and fixing o. Then the set of such numbers $\operatorname{mld}((X \ni o, 0)/G) \in (0, 1)$ satisfies the ACC.*

Proof. The proof is long, so we divide it into several steps. Let $a := \operatorname{mld}((X \ni o, 0)/G)$ and assume that $a \in (0, 1)$.

Step 1. In this step we reduce to the case when X is a polynomial hypersurface $(x^2 + g = 0) \subset \mathbb{A}^4$.

By Lemma 11.3, there exist a smooth fourfold germ $(V \ni o)$ with a faithful action of G, a G-equivariant closed embedding $X \subset V$ with $X = (f = 0)$ for a regular G-semi-invariant function f on V, and G-semi-invariant elements x, y, z, t of the maximal ideal of $\mathfrak{R} := \widehat{\mathcal{O}}_{V,o}$ such that $\mathfrak{R} = \mathbb{C}[[x, y, z, t]]$, G acts diagonally on x, y, z, t, and the ideal of X in $\mathfrak{R}$ is generated by $x^2 + g(y, z, t)$ for a G-semi-invariant power series g with $\operatorname{ord} g \geq 3$ whose cubic part g_3 is the residual ternary cubic, and $x^2 + g$ is G-semi-invariant with character χ_x^2.

By Corollary 7.5, applied to $(V \ni o)$, the elements x, y, z, t, the generator $x^2 + g$ of the ideal of X in $\mathfrak{R}$ (the F of the corollary), $\psi = \mathrm{id}_{\mathfrak{R}}$, and $v = 1$, for an integer $L \geq 4$ large enough the polynomial $f_0 := x^2 + g_{<L}$ in the coordinates x, y, z, t of $\mathbb{A}^4$ is G-semi-invariant, G acts faithfully on $\mathbb{A}^4$ and on $X_0 := (f_0 = 0)$, X_0 is an isolated Gorenstein terminal threefold germ at 0, and $\mathrm{mld}((X_0 \ni 0, 0)/G) = a$. Since $L \geq 4$, the cubic part of $g_{<L}$ is g_3, so $(X_0 \ni 0)$ has quadratic rank one and residual ternary cubic g_3. Possibly replacing $(X \ni o)$ with $(X_0 \ni 0)$ and g with $g_{<L}$, we may assume that $X = (x^2 + g = 0) \subset \mathbb{A}^4$ for a polynomial $g \in \mathbb{C}[y, z, t]$ with $\mathrm{ord}\, g \geq 3$ and cubic part g_3, that $o = 0$, that G acts diagonally and faithfully on $\mathbb{A}^4$ and on X, and that $x^2 + g$ is G-semi-invariant. Shrinking X to a G-invariant affine open neighborhood of o, we may also assume that $X \backslash \{o\}$ is smooth. Let $\chi_x, \chi_y, \chi_z, \chi_t$ be the characters of x, y, z, t, and let $C := (g_3 = 0) \subset \mathbb{P}^2$.

Step 2. In this step we choose a prime divisor E over X/G computing a, and we construct a normal threefold Y over X and a prime divisor F over Y which is over E and has a center on Y.

Let $q : X \to X/G$ be the quotient morphism. Since X is terminal, $(X/G, B_{X/G})$ is klt [KM98, Proposition 5.20], so there exists a prime divisor E over X/G with $\mathrm{center}_{X/G} E = \{q(o)\}$ and $a(E, X/G, B_{X/G}) = a$ [Amb99, Section 1, the second paragraph after Definition 1.1]. Since q is finite and $q^{-1}(q(o)) = \{o\}$, there exists a prime divisor F_X over X with $\mathrm{ord}_{F_X}|_{\mathbb{C}(X/G)} \in \mathbb{N}^+ \mathrm{ord}_E$ and $\mathrm{center}_X F_X = \{o\}$. Let $\nu := \mathrm{ord}_{F_X}$. After permuting y, z, t, we may assume that $\nu(t) \leq \nu(y)$ and $\nu(t) \leq \nu(z)$. Since $x^2 = -g$ on X and any monomial of g has degree at least three in y, z, t,

$$2\nu(x) = \nu(g) \geq 3\nu(t) > 0. \tag{11.17}$$

For $d \geq 3$ let g_d be the homogeneous part of g of degree d, and let

$$\tilde{g}(y_1, z_1, \tau) := \sum_{d \geq 3} \tau^{2d-6} g_d(y_1, z_1, 1) \quad \text{and} \quad c(y_1, z_1) := \tilde{g}(y_1, z_1, 0) = g_3(y_1, z_1, 1).$$

Let $\bar{Y} := (x_1^2 + \tilde{g}(y_1, z_1, \tau) = 0) \subset \mathbb{A}^4_{x_1, y_1, z_1, \tau}$, and let $\phi : \bar{Y} \to \mathbb{A}^4$ be the morphism given by

$$x = \tau^3 x_1, \quad y = \tau^2 y_1, \quad z = \tau^2 z_1, \quad t = \tau^2.$$

Then $\phi^* g = \tau^6 \tilde{g}$ and $\phi^*(x^2 + g) = \tau^6(x_1^2 + \tilde{g})$, so $\phi(\bar{Y}) \subset (x^2 + g = 0)$. Let $Y := \phi^{-1}(X)$, an open subset of $\bar{Y}$.

We list the properties of Y used below. Since g_3 is reduced, t^2 does not divide g_3, so c is a reduced non-constant polynomial. In particular $\tilde{g}$ is not a square in $\mathbb{C}[y_1, z_1, \tau]$, so $x_1^2 + \tilde{g}$ is irreducible and $\bar{Y}$ is integral. The scheme $\bar{Y} \cap (\tau = 0)$ is the integral surface

$$D := (x_1^2 + c(y_1, z_1) = 0) \subset \mathbb{A}^3_{x_1, y_1, z_1},$$

and $D = \phi^{-1}(o) \subset Y$. Thus $\mathrm{div}(\tau) = D$ on Y. Let

$$N := \{(0, y_1, z_1, 0) \in D \mid (y_1 : z_1 : 1) \text{ is a singular point of } C\},$$

which is a finite set, and $(y_1 : z_1 : 1)$ is then a node of C. Since $\partial\tilde{g}/\partial\tau$ vanishes on $(\tau = 0)$, the Jacobian criterion shows that N is the set of the singular points of $\bar{Y}$ on D. By the Jacobian criterion, D is also smooth outside N. The morphism $(\tau \neq 0) \to (t \neq 0)$ of open subsets of $\mathbb{A}^4$ given by

$$(x_1, y_1, z_1, \tau) \to (\tau^3 x_1, \tau^2 y_1, \tau^2 z_1, \tau^2)$$

is étale, with Jacobian determinant $2\tau^8$, and $Y \cap (\tau \neq 0) = \phi^{-1}(X \cap (t \neq 0))$. Since $X \backslash \{o\}$ is smooth, Y is smooth outside N. Hence Y is normal, since Y is a hypersurface whose singular locus has codimension three [Har77, Proposition II.8.23].

Since $\mathbb{C}(Y) = \mathbb{C}(X)(\tau)$ and $\tau^2 = t$, the extension $\mathbb{C}(Y)/\mathbb{C}(X)$ is finite. Let F be a prime divisor over Y with $\mathrm{ord}_F|_{\mathbb{C}(X)} \in \mathbb{N}^+\nu$. Such F exists by Lemma 2.6(3) applied to the normalization of X in $\mathbb{C}(Y)$, which is Galois over X with group $\mathbb{Z}/2$ acting by $\tau \to -\tau$ (the involution $\iota : (x_1, y_1, z_1, \tau) \to (-x_1, y_1, z_1, -\tau)$ of $\bar{Y}$ preserves Y, satisfies $\phi \circ \iota = \phi$, and sends τ to $-\tau$, so $[\mathbb{C}(Y) : \mathbb{C}(X)] = 2$). By (11.17), ord_F is non-negative on $x_1 = x\tau^{-3}$, $y_1 = yt^{-1}$, and $z_1 = zt^{-1}$,

and positive on τ. Therefore F has a center $Z := \mathrm{center}_{\bar{Y}} F$ on $\bar{Y}$ with $Z \subset D \subset Y$. Thus Z is D, a curve, or a closed point.

Step 3. In this step we express a as the minimal log discrepancy, at the image of Z, of the quotient of $(Y, -2\,\mathrm{div}(\tau))$ by the inertia group of F in a group lifting the action of G to Y.

Let $\tilde{G} := \{(\sigma, \epsilon) \in G \times \mathbb{C}^* \mid \epsilon^2 = \chi_t(\sigma)\}$, and let $\tilde{G}$ act diagonally on $\mathbb{A}^4_{x_1,y_1,z_1,\tau}$ by

$$(\sigma,\epsilon)^*(x_1, y_1, z_1, \tau) = \big(\chi_x(\sigma)\epsilon^{-3}x_1,\ \chi_y(\sigma)\chi_t(\sigma)^{-1}y_1,\ \chi_z(\sigma)\chi_t(\sigma)^{-1}z_1,\ \epsilon\tau\big).$$

Then ϕ is equivariant with respect to the surjection $\tilde{G} \to G$, $(\sigma, \epsilon) \to \sigma$, and

$$(\sigma,\epsilon)^*(x_1^2 + \tilde{g}) = \chi_x(\sigma)^2\epsilon^{-6}(x_1^2 + \tilde{g}).$$

Thus $\tilde{G}$ acts on Y. If (σ, ϵ) acts trivially on Y, then σ acts trivially on X, so $\sigma = 1$ and $\epsilon = \pm 1$, and $\epsilon = 1$ since $(1, -1)^*\tau = -\tau$. Thus $\tilde{G}$ acts faithfully on Y. In particular $(1, -1)$ acts non-trivially on $\mathbb{C}(Y)$ and trivially on $\mathbb{C}(X)$, so $[\mathbb{C}(Y) : \mathbb{C}(X)] = 2$ and $[\mathbb{C}(Y) : \mathbb{C}(X/G)] = 2|G| = |\tilde{G}|$. Therefore $\mathbb{C}(Y)/\mathbb{C}(X/G)$ is a Galois extension with Galois group $\tilde{G}$. Let X' be the normalization of X/G in $\mathbb{C}(Y)$, let $\pi : X' \to X/G$ be the induced finite Galois morphism with Galois group $\tilde{G}$, and let B' be the $\mathbb{R}$-divisor with $K_{X'} + B' = \pi^*(K_{X/G} + B_{X/G})$. Since Y is normal with function field $\mathbb{C}(Y)$, the composite $q \circ \phi|_Y$ factors through a $\tilde{G}$-equivariant birational morphism $\rho : Y \to X'$.

Since X and $\bar{Y}$ are hypersurfaces, ω_X and ω_Y are generated by the residue forms

$$\Omega_X := \frac{dy \wedge dz \wedge dt}{2x} \quad \text{and} \quad \Omega_Y := \frac{dy_1 \wedge dz_1 \wedge d\tau}{2x_1},$$

respectively. Since $dy \wedge dz \wedge dt = 2\tau^5 dy_1 \wedge dz_1 \wedge d\tau$ and $2x = 2\tau^3 x_1$, we have $\phi^*\Omega_X = 2\tau^2\Omega_Y$. Since $K_X = \mathrm{div}(\Omega_X) = 0$ is the pullback of $K_{X/G} + B_{X/G}$ (Lemma 2.6(0)), representing K_Y by $\phi^*\Omega_X$ we get $K_Y = \mathrm{div}(2\tau^2\Omega_Y) = 2\,\mathrm{div}(\tau)$, and $K_Y + B_Y = \rho^*(K_{X'} + B')$ for $B_Y := -2\,\mathrm{div}(\tau) = -2D$. Let $J := I_{\tilde{G}}(F)$. We use Lemma 3.1, with (X, B), x, E, X', G, F, H, Y of the lemma taken to be $(X/G, B_{X/G})$, $q(o)$, E, X', $\tilde{G}$, F, J, Y. The hypotheses of Lemma 3.1 hold: Y is $\tilde{G}$-invariant, $\mathrm{ord}_F|_{\mathbb{C}(X/G)} \in \mathbb{N}^+\,\mathrm{ord}_E$, ρ is J-equivariant and birational, F has the center Z on Y, and $\mathrm{mld}(X/G \ni q(o), B_{X/G}) = a(E, X/G, B_{X/G}) = a > 0$. Thus

$$a = \mathrm{mld}(Y/J \ni \eta, B_{Y/J}),$$

where $(Y/J, B_{Y/J}) := (Y, B_Y)/J$ and η is the generic point of the image of Z in Y/J. Moreover, $J \subset \mathrm{Stab}_{\tilde{G}}(F)$ preserves Z, and J acts trivially on the function field of Z, which is contained in the residue field of ord_F.

Step 4. In this step we treat the cases when $Z = D$ and when Z is a closed point not in N.

Suppose that $Z = D$. Then η is the generic point of the image $\bar{D}$ of D in Y/J, and $a = a(\bar{D}, Y/J, B_{Y/J})$. Let r be the ramification index of $Y \to Y/J$ along D. Since $a(D, Y, B_Y) = 1 - (-2) = 3$ and $|I_J(D)| = r$ by Lemma 2.6(1), Lemma 2.6(2) implies that $3 = ra$. Thus a belongs to $\mathcal{S}_1 := \{\frac{3}{r} \mid r \in \mathbb{N}^+\}$, which satisfies the ACC.

Suppose that Z is a closed point $p \notin N$. Then J fixes p, and Y is smooth at p. Since J acts diagonally on $\mathbb{A}^4_{x_1,y_1,z_1,\tau}$ and fixes p, any coordinate which does not vanish at p has trivial character on J. Thus $x_1 - x_1(p)$, $y_1 - y_1(p)$, $z_1 - z_1(p)$, and τ are J-semi-invariant with the characters of x_1, y_1, z_1, τ. Since $p \notin N$, the differential of $x_1^2 + \tilde{g}$ at p is

$$2x_1(p)dx_1 + \frac{\partial c}{\partial y_1}(p)dy_1 + \frac{\partial c}{\partial z_1}(p)dz_1 \neq 0,$$

so τ and two of $x_1 - x_1(p)$, $y_1 - y_1(p)$, $z_1 - z_1(p)$ form J-eigenparameters τ, u_2, u_3 of $\mathcal{O}_{Y,p}$. The group J acts faithfully on Y and trivially on $\kappa(p) = \mathbb{C}$. By Lemma 3.4 with $k = \mathbb{C}$ and $B_Y = -2\,\mathrm{div}(\tau)$,

$$a = \mathrm{mld}((Y \ni p, B_Y)/J) = \min\{3w_1 + w_2 + w_3 \mid w \in N_J \cap \mathbb{R}^3_{>0}\},$$

where N_J is taken with respect to τ, u_2, u_3. Let

$$\mathcal{S}_2 := \left\{ \min\{3w_1 + w_2 + w_3 \mid w \in N_H \cap \mathbb{R}^3_{>0}\} \;\middle|\; H \subset \mathbb{T}^3 \text{ a finite subgroup} \right\}.$$

By Corollary 4.6 with $d = 3$, $s = 1$, $\mathcal{V} = \{(3,1,1)\}$, $\Lambda = \{0\}$, and the unit ideal as the monomial ideal, $\mathcal{S}_2$ satisfies the ACC, and $a \in \mathcal{S}_2$.

Step 5. In this step we treat the case when Z is a point $p \in N$.

The group J fixes p, $x_1(p) = \tau(p) = 0$, and $(y_1(p) : z_1(p) : 1)$ is a node of C. Let $y_1' := y_1 - y_1(p)$ and $z_1' := z_1 - z_1(p)$. Since J acts diagonally on $\mathbb{A}^4_{x_1,y_1,z_1,\tau}$ and fixes p, the functions y_1' and z_1' are J-semi-invariant with the characters of y_1 and z_1. Since $\tilde{g}$ is a polynomial in y_1, z_1, and τ^2, the quadratic part Q_p of $x_1^2 + \tilde{g}$ at p is

$$Q_p = x_1^2 + q_c(y_1', z_1') + g_4(y_1(p), z_1(p), 1)\tau^2,$$

where q_c is the quadratic part of c at $(y_1(p), z_1(p))$, which is non-degenerate since the singular point is a node. Since $\tilde{g}$ is a polynomial in y_1, z_1, τ^2 and x_1 occurs only in x_1^2, Q_p has no term $x_1\tau$, $y_1'\tau$, or $z_1'\tau$. Thus $\operatorname{rank} Q_p \geq 3$.

We show that Y is klt near p. Since p is an isolated singular point of Y, a general hyperplane section H_p of Y through p is smooth outside p near p [Har77, Corollary III.10.9, Remark III.10.9.2]. The quadratic part of the equation of H_p at p is the restriction of Q_p to a general hyperplane, of rank at least two. Thus H_p has a Du Val singularity of type A at p [KM98, Theorem 4.20 and Paragraph 4.25, Steps 2–3]. Since Y is normal and H_p is a normal Cartier divisor with $K_Y + H_p$ Cartier, (Y, H_p) is plt near p by [KM98, Theorem 5.50], so Y is klt near p. Since $B_Y \leq 0$, (Y, B_Y) is sub-klt near p.

We use Corollary 9.6 for the smooth fourfold germ $(\mathbb{A}^4 \ni p)$, the group J, the function $x_1^2 + \tilde{g}$, the J-eigenparameters (u_1, u_2, z, t) of the corollary taken to be (y_1', z_1', x_1, τ), and $(b_z, b_t) = (0, -2) \in \mathcal{B} := \{(0,-2)\}$. The hypotheses hold: $x_1^2 + \tilde{g}$ is J-semi-invariant and vanishes to order two at p, Y is a normal klt threefold germ at p on which J acts faithfully, the restriction of Q_p to the (y_1', z_1')-plane is q_c, $B_Y = 0 \cdot \operatorname{div}(x_1) - 2\operatorname{div}(\tau)$, and (Y, B_Y) is sub-klt near p. Since $a = \operatorname{mld}((Y \ni p, B_Y)/J) \in (0,1)$, a belongs to the set $\mathcal{S}_3$ of Corollary 9.6 for $\mathcal{B} = \{(0,-2)\}$, which satisfies the ACC.

Step 6. In this step we treat the case when Z is a curve $C' \subset D$.

Let ξ be the generic point of C'. Since N is finite, Y and D are smooth at ξ. Thus $\mathcal{O}_{Y,\xi}$ is a two-dimensional regular local ring, and τ is a regular parameter of $\mathcal{O}_{Y,\xi}$, since $\mathcal{O}_{Y,\xi}/(\tau) = \mathcal{O}_{D,\xi}$ is regular. Let u_0 be one of x_1, y_1, z_1 whose restriction to C' is not constant, and let

$$u_* := \frac{1}{|J|} \sum_{\sigma \in J} \sigma^* u_0.$$

Since J acts trivially on $\mathbb{C}(C')$, $u_*|_{C'} = u_0|_{C'}$ is not constant. Let $k := \mathbb{C}(u_*)$, and let Y_{u_*} be the generic fiber of $u_* : Y \to \mathbb{A}^1$, a normal quasi-projective variety over k on which J acts faithfully and k-linearly. Then ξ is a closed point of Y_{u_*} with $\mathcal{O}_{Y_{u_*},\xi} = \mathcal{O}_{Y,\xi}$ and $\kappa(\xi) = \mathbb{C}(C')$, J fixes ξ, and J acts trivially on $\kappa(\xi)$. The prime divisors over Y with center C' correspond bijectively to the prime divisors over Y_{u_*} with center ξ, with equal log discrepancies with respect to B_Y and $B_Y|_{Y_{u_*}}$, since du_* is part of a basis of Ω_Y at ξ, as $u_*|_{C'}$ is not constant. Since u_* is J-invariant, taking the quotient by J commutes with taking the generic fiber over k. Thus, by Remark 2.17,

$$a = \operatorname{mld}((Y_{u_*} \ni \xi, B_Y|_{Y_{u_*}})/J).$$

Since J is abelian and acts trivially on $\kappa(\xi) \supset \mathbb{C}$, the cotangent space $\mathfrak{m}_\xi/\mathfrak{m}_\xi^2$ of $\mathcal{O}_{Y_{u_*},\xi}$ is a direct sum of two J-eigenlines over $\kappa(\xi)$, one of which is spanned by the class of τ. Let $y_2 \in \mathfrak{m}_\xi$ be an element whose class spans the other eigenline, with character χ'. Replacing y_2 with $\frac{1}{|J|}\sum_{\sigma \in J} \chi'(\sigma)^{-1}\sigma^* y_2$, which is congruent to y_2 modulo $\mathfrak{m}_\xi^2$, we may assume that y_2

is J-semi-invariant. Then τ, y_2 are J-eigenparameters of $\mathcal{O}_{Y_{u_*},\xi}$. By Lemma 3.4 over k with $B_Y|_{Y_{u_*}} = -2\operatorname{div}(\tau)$,

$$a = \min\{3w_1 + w_2 \mid w \in N_J \cap \mathbb{R}^2_{>0}\},$$

where N_J is taken with respect to τ, y_2. Let

$$\mathcal{S}_4 := \left\{ \min\{3w_1 + w_2 \mid w \in N_H \cap \mathbb{R}^2_{>0}\} \;\middle|\; H \subset \mathbb{T}^2 \text{ a finite subgroup} \right\}.$$

By Corollary 4.6 with $d = 2$, $s = 1$, $\mathcal{V} = \{(3,1)\}$, $\Lambda = \{0\}$, and the unit ideal as the monomial ideal, $\mathcal{S}_4$ satisfies the ACC, and $a \in \mathcal{S}_4$.

Step 7. In this step we conclude the proof.

By **Step 2**, Z is D, a closed point not in N, a point of N, or a curve. By **Steps 4–6**, $a \in \mathcal{S}_1 \cup \mathcal{S}_2 \cup \mathcal{S}_3 \cup \mathcal{S}_4$. Since $\mathcal{S}_1, \dots, \mathcal{S}_4$ do not depend on $(X \ni o)$ and G and satisfy the ACC, the set of such numbers a satisfies the ACC. □

11.5. **Concurrent residual lines.** In this subsection, we treat residual cubics consisting of three distinct concurrent lines, and we prove Theorem 11.2. When the group of permutations of the three lines induced by G has order at most two, Lemmas 11.4 and 11.6 bring the equation into the family of Theorem 11.7. When the order is three, the equation is brought into the form of the Moreover part of Theorem 11.2, and Proposition 10.5 computes the minimal log discrepancy.

Proposition 11.12. *Let $(X \ni o)$ be an isolated Gorenstein terminal threefold singularity of quadratic rank one whose residual ternary cubic is the product of three distinct concurrent lines, and let G be a finite abelian group acting faithfully on X and fixing o such that the induced permutation group of the three lines has order at most two. Then the set of such numbers* $\operatorname{mld}((X \ni o, 0)/G) \in (0,1)$ *satisfies the ACC.*

Proof. Let $a := \operatorname{mld}((X \ni o, 0)/G)$ and assume that $a \in (0,1)$.

Step 1. In this step we bring the residual cubic into the form $y^2z + z^3$.

By Lemma 11.3, there exist a smooth fourfold germ $(Y \ni o)$ with a faithful action of G, a G-equivariant closed embedding $X \subset Y$ with $X = (f = 0)$ for a regular G-semi-invariant function f on Y, and G-semi-invariant elements x, y, z, t of the maximal ideal of $\mathfrak{R} := \widehat{\mathcal{O}}_{Y,o}$ such that $\mathfrak{R} = \mathbb{C}[[x, y, z, t]]$, G acts diagonally on x, y, z, t, and the ideal of X in $\mathfrak{R}$ is generated by $x^2 + g(y, z, t)$ for a G-semi-invariant power series g with $\operatorname{ord} g \geq 3$ whose cubic part g_3 is the residual ternary cubic. Write $g_3 = \ell_1\ell_2\ell_3$ for pairwise linearly independent linear forms ℓ_1, ℓ_2, ℓ_3 in y, z, t. Since the three lines are concurrent, ℓ_1, ℓ_2, ℓ_3 span a two-dimensional subspace W of the space of linear forms in y, z, t. By Lemma 11.3, the set of linear factors of g_3 is G-stable, so W is G-stable. Since G acts diagonally on the space of linear forms in y, z, t, there exists a G-semi-invariant linear form t' spanning a G-stable complement of W. By Lemma 11.4, since the induced permutation group has order at most two, there exist G-semi-invariant linear forms y', z' forming a basis of W such that $g_3 = \kappa(y'^2z' + z'^3)$ for a number $\kappa \neq 0$. Replacing y', z' with $\kappa^{1/3}y', \kappa^{1/3}z'$, we may assume that $\kappa = 1$. Replacing y, z, t with y', z', t', which is a G-equivariant linear change of coordinates fixing x, we may assume that the ideal of X in $\mathfrak{R}$ is generated by the G-semi-invariant power series

$$x^2 + y^2z + z^3 + h(y, z, t), \qquad h \in (y, z, t)^4.$$

Step 2. In this step we bring the equation into the family of Theorem 11.7.

Let $c := y^2z + z^3 = z(y + \sqrt{-1}z)(y - \sqrt{-1}z)$, which has three distinct linear factors. The ideal $(\partial_y c, \partial_z c) = (yz, y^2 + 3z^2)$ is generated by two linearly independent quadrics, and it contains any cubic monomial, since

$$3z^3 = z(y^2 + 3z^2) - y \cdot yz \quad \text{and} \quad y^3 = y(y^2 + 3z^2) - 3z \cdot yz.$$

Thus the classes of $1, y, z, z^2$ form a basis of $\mathbb{C}[y,z]/(\partial_y c, \partial_z c)$, and $1, y, z, z^2$ are G-semi-invariant since G acts diagonally. By Lemma 11.6 with $(m_0, m_1, m_2, m_3) = (1, y, z, z^2)$, there exist a G-equivariant automorphism φ of $\mathfrak{R}$ with $\varphi(x) = x$ and $\varphi(t) = t$, and power series $r_0, \dots, r_3 \in \mathbb{C}[[t]]$, each zero or G-semi-invariant, with $\operatorname{ord} r_0 \geq 4$, $\operatorname{ord} r_1, \operatorname{ord} r_2 \geq 3$, and $\operatorname{ord} r_3 \geq 2$, such that

$$\varphi(x^2 + y^2 z + z^3 + h) = x^2 + y^2 z + z^3 + r_0(t) + r_1(t) y + r_2(t) z + r_3(t) z^2.$$

If $r_1 = 0$, let $\lambda := 0$, $\ell := 2$, and $U := 1$. Otherwise write $r_1 = \lambda t^\ell U(t)$ with $\lambda \neq 0$, $\ell = \operatorname{ord} r_1 \geq 3$, and $U(0) = 1$. Since r_1 and t^ℓ are G-semi-invariant, so is the unit U, and U is G-invariant since $U(0) \neq 0$. Let θ be the G-equivariant automorphism of $\mathfrak{R}$ with $\theta(x) = x$, $\theta(y) = U^{-1} y$, $\theta(z) = U^2 z$, and $\theta(t) = t$. Then

$$\theta\varphi(x^2 + y^2 z + z^3 + h) = x^2 + y^2 z + \lambda y t^\ell + \hat{b}(z,t), \qquad \hat{b} := U^6 z^3 + r_3 U^4 z^2 + r_2 U^2 z + r_0,$$

where $\hat{b} \in (z,t)^3$ and the coefficient of z^3 in $\hat{b}$ is 1.

Step 3. In this step we pass to a polynomial model and conclude the proof.

By Corollary 7.5, applied to $(Y \ni o)$, the elements x, y, z, t of **Step 1**, the generator $F := x^2 + y^2 z + z^3 + h$ of the ideal of X in $\mathfrak{R}$, the automorphism $\theta\varphi$, and $v = 1$, for an integer $L > \ell + 1$ large enough the polynomial $f_0 := (\theta\varphi(F))_{<L} = x^2 + y^2 z + \lambda y t^\ell + b(z,t)$, where $b := \hat{b}_{<L}$, has the properties listed there. Then $b \in (z,t)^3$ is a non-zero polynomial, since the coefficient of z^3 in b is 1. By Corollary 7.5, $X_0 := (x^2 + y^2 z + \lambda y t^\ell + b(z,t) = 0) \subset \mathbb{A}^4$ is an isolated Gorenstein terminal threefold germ at 0, in particular a normal klt threefold germ, G acts diagonally and faithfully on X_0 and on $\mathbb{A}^4$ with G-semi-invariant equation, and $\operatorname{mld}((X_0 \ni 0, 0)/G) = a$. Thus Theorem 11.7(2), applied to b, $\ell \geq 2$, λ, X_0, and G, shows that a belongs to a set which satisfies the ACC and does not depend on $(X \ni o)$ and G. □

Proof of Theorem 11.2. The proof is long, so we divide it into several steps.

Step 1. In this step we show that $(X \ni o)$ has quadratic rank one.

By [KM98, Corollary 5.38], $(X \ni o)$ is an isolated cDV singularity. It is of type cD by assumption, so it has quadratic rank one by the last sentence of Lemma 11.3.

Step 2. In this step we assume that $n = 4$, that the residual ternary cubic is the product of three distinct concurrent lines, and that the group Σ of permutations of the three lines induced by G has order three, and we bring the residual cubic into the form $y^3 + z^3$.

By Lemma 11.3, there exist a smooth fourfold germ $(V \ni o)$ with a faithful action of G, a G-equivariant closed embedding $X \subset V$ with $X = (f = 0)$ for a regular G-semi-invariant function f on V, and G-semi-invariant elements x, y, z, t of the maximal ideal of $\mathfrak{R} := \widehat{\mathcal{O}}_{V,o}$ such that $\mathfrak{R} = \mathbb{C}[[x,y,z,t]]$, G acts diagonally on x, y, z, t, and the ideal of X in $\mathfrak{R}$ is generated by $x^2 + g(y,z,t)$ for a G-semi-invariant power series g with $\operatorname{ord} g \geq 3$ whose cubic part g_3 is the residual ternary cubic. Write $g_3 = \ell_1 \ell_2 \ell_3$ for pairwise linearly independent linear forms ℓ_1, ℓ_2, ℓ_3 in y, z, t. Since the three lines are concurrent, ℓ_1, ℓ_2, ℓ_3 span a two-dimensional subspace W of the space of linear forms in y, z, t, and W is G-stable since the set of linear factors of g_3 is G-stable (Lemma 11.3). Since G acts diagonally on the space of linear forms in y, z, t, there exists a G-semi-invariant linear form t' spanning a G-stable complement of W. By Lemma 11.4, since $|\Sigma| = 3$, there exist G-semi-invariant linear forms y', z' forming a basis of W such that $g_3 = \kappa(y'^3 + z'^3)$ for a number $\kappa \neq 0$. Replacing y', z' with $\kappa^{1/3} y', \kappa^{1/3} z'$ and then y, z, t with y', z', t', which is a G-equivariant linear change of coordinates fixing x, we may assume that the ideal of X in $\mathfrak{R}$ is generated by the G-semi-invariant power series

$$x^2 + y^3 + z^3 + h(y,z,t), \qquad h \in (y,z,t)^4.$$

Step 3. In this step we construct X_0 and h_0 of the Moreover part, keeping the notation and the assumptions of **Step 2**.

Let $c := y^3 + z^3$, which has three distinct linear factors. Since $(\partial_y c, \partial_z c) = (y^2, z^2)$, the classes of the G-semi-invariant monomials $1, y, z, yz$ form a basis of $\mathbb{C}[y,z]/(\partial_y c, \partial_z c)$. By Lemma 11.6

with $(m_0, m_1, m_2, m_3) = (1, y, z, yz)$, there exist a G-equivariant automorphism φ of $\mathfrak{R}$ with $\varphi(x) = x$ and $\varphi(t) = t$, and power series $r_0, \dots, r_3 \in \mathbb{C}[[t]]$, each zero or G-semi-invariant, with $\operatorname{ord} r_k \geq 4 - \deg m_k$, such that

$$\varphi(x^2 + y^3 + z^3 + h) = x^2 + y^3 + z^3 + r_3(t)yz + r_1(t)y + r_2(t)z + r_0(t).$$

By Corollary 7.5, applied to $(V \ni o)$, the elements x, y, z, t of **Step 2**, the generator $F := x^2 + y^3 + z^3 + h$ of the ideal of X in $\mathfrak{R}$, the automorphism φ, and $v = 1$, for an integer $L \geq 4$ large enough the polynomial $f_0 := (\varphi(F))_{<L}$ in the coordinates x, y, z, t of $\mathbb{A}^4$ has the properties listed there. Let

$$a_1 := (r_3)_{<L-2}, \quad a_2 := (r_1)_{<L-1}, \quad a_3 := (r_2)_{<L-1}, \quad a_4 := (r_0)_{<L}.$$

Then $a_1, \dots, a_4 \in \mathbb{C}[t]$ and $f_0 = x^2 + h_0$ for $h_0 := y^3 + z^3 + a_1(t)yz + a_2(t)y + a_3(t)z + a_4(t)$. By Corollary 7.5, $x^2 + h_0$ is G-semi-invariant, the action of G on $\mathbb{A}^4$ is faithful and diagonal, $X_0 := (x^2 + h_0 = 0)$ is an isolated Gorenstein terminal threefold germ at 0, and

$$\operatorname{mld}((X_0 \ni 0, 0)/H) = \operatorname{mld}((X \ni o, 0)/H) \quad \text{for any subgroup } H \subset G.$$

Step 4. In this step we prove the formula of the Moreover part, keeping the notation of **Step 3**.

We use Set-up 10.3 for $h = h_0$ and $b_y = b_z = b_t = 0$, so that $B = 0$. The conditions of Set-up 10.3 hold, since X_0 is a normal klt threefold germ at 0 and G acts diagonally and faithfully on X_0 and on $\mathbb{A}^4$ with G-semi-invariant equation. The function P of Set-up 10.3 is

$$P(w) = w_x + w_y + w_z + w_t - \min\{2w_x, \operatorname{wt}_{(w_y,w_z,w_t)}(h_0)\},$$

which is not the function P of Theorem 11.7. We verify the assumption of Proposition 10.5, that is, that $((\mathbb{C}^*)^3, \frac{1}{2}\operatorname{div}(\operatorname{in}_u(h_0)))$ is lc for any $u \in \mathbb{R}^3_{>0}$.

By Lemma 10.2, applied to the power series h_0, it suffices to show that $((\mathbb{C}^*)^2_{Y,Z}, \frac{1}{2}\operatorname{div}(p))$ is lc for any $u = (u_y, u_z, u_t) \in \mathbb{Z}^3_{>0}$, where $p(Y, Z) := \operatorname{in}_u(h_0)(Y, Z, 1)$. Fix such u. Each of $a_1(t)yz$, $a_2(t)y$, $a_3(t)z$, and $a_4(t)$ contributes at most one monomial to $\operatorname{in}_u(h_0)$, namely the term of least degree in t of the coefficient. Thus $\operatorname{in}_u(h_0)$ is a non-zero linear combination of $y^3, z^3, t^{k_1}yz, t^{k_2}y, t^{k_3}z, t^{k_4}$ for some integers $k_1, \dots, k_4 \geq 0$, and

$$p(Y, Z) = \alpha_1 Y^3 + \alpha_2 Z^3 + \alpha_3 YZ + \alpha_4 Y + \alpha_5 Z + \alpha_6$$

for numbers $\alpha_1, \dots, \alpha_6$, not all zero. The polynomial p has degree at most three.

We show that $((\mathbb{C}^*)^2_{Y,Z}, \frac{1}{2}\operatorname{div}(p))$ is lc. By Lemma 10.1(2), it suffices to treat the case when $p = \kappa M^3$ for a number $\kappa \neq 0$ and a non-constant affine linear form M. Then $\alpha_1 Y^3 + \alpha_2 Z^3$ is κ times the cube of the linear part of M. If $\alpha_1\alpha_2 \neq 0$, then $\alpha_1 Y^3 + \alpha_2 Z^3$ has three distinct linear factors, which is impossible. If $\alpha_1 \neq 0 = \alpha_2$, then $M = \mu Y + \mu'$ for numbers $\mu \neq 0$ and μ', and $\mu' = 0$ since p has no term Y^2. Thus $p = \alpha_1 Y^3$, which has no zero on $(\mathbb{C}^*)^2$. The case when $\alpha_2 \neq 0 = \alpha_1$ is symmetric, and p is not a cube of a non-constant affine linear form if $\alpha_1 = \alpha_2 = 0$, since then $\deg p \leq 2$. Thus $((\mathbb{C}^*)^2, \frac{1}{2}\operatorname{div}(p))$ is lc in any case.

Thus the assumption of Proposition 10.5 holds for X_0 and G. Since $(X_0/G, B_{X_0/G})$ is klt [KM98, Proposition 5.20], $\operatorname{mld}((X_0 \ni 0, 0)/G) > 0$. Hence, if $\operatorname{mld}((X_0 \ni 0, 0)/G) < 1$, then

$$\operatorname{mld}((X_0 \ni 0, 0)/G) = \min_{w \in N_G \cap \mathbb{R}^4_{>0}} \Big(w_x + w_y + w_z + w_t - \min\{2w_x, \operatorname{wt}_{(w_y,w_z,w_t)}(h_0)\}\Big).$$

Steps 3 and 4 prove the Moreover part.

Step 5. In this step we prove that the set of numbers $a := \operatorname{mld}((X \ni o, 0)/G) \in (0, 1)$ satisfies the ACC.

By **Step 1**, $(X \ni o)$ has quadratic rank one, so it has a residual ternary cubic g_3, and g_3 is not a cube of a linear form by Lemma 11.3, since $(X \ni o)$ is of type cD. Suppose that g_3 has a multiple linear factor. Then $g_3 = cL^2M$ for a number $c \neq 0$ and linearly independent linear forms L, M, and a belongs to the set of Proposition 11.8, which satisfies the ACC.

Suppose that g_3 is reduced. Then the plane cubic $(g_3 = 0)$ is one of the following: a smooth cubic, an irreducible nodal cubic, an irreducible cuspidal cubic, the union of a smooth conic and a secant line, the union of a smooth conic and a tangent line, three lines forming a triangle, or three distinct concurrent lines. The list is complete, since an irreducible plane cubic has arithmetic genus one, hence at most one singular point, a node or a cusp, and a reducible reduced cubic is a smooth conic with a secant or tangent line, or three lines, forming a triangle or concurrent.

- A smooth cubic, an irreducible nodal cubic, a smooth conic with a secant line, and a triangle are reduced plane cubics whose singular points are nodes. If $(g_3 = 0) \subset \mathbb{P}^2$ is one of them, then a belongs to the set of Proposition 11.11.
- For an irreducible cuspidal cubic and a smooth conic with a tangent line, a belongs to the set of Proposition 11.9.
- For three distinct concurrent lines, the group Σ of permutations of the three lines induced by G has order 1, 2, or 3 by Lemma 11.4. If $|\Sigma| \leq 2$, then a belongs to the set of Proposition 11.12. If $|\Sigma| = 3$, then $n = 4$: by the last sentence of Proposition 11.8, $n \geq 5$ would imply $g_3 = cL^2M$ for a number $c \neq 0$ and linearly independent linear forms L, M, which is not reduced, and $n \geq 4$ by assumption. Hence the assumptions of the Moreover part hold, and $a = \operatorname{mld}((X_0 \ni 0, 0)/G)$ by **Step 3**. Thus a belongs to the set of Proposition 10.5 for $(b_y, b_z, b_t) = (0, 0, 0)$, by **Step 4**.

Each of the sets of Propositions 11.8, 11.11, 11.9, 11.12, and 10.5 used above satisfies the ACC and does not depend on $(X \ni o)$ and G. Thus the set of such numbers a satisfies the ACC. □

12. Quadratic rank one: type cE

In this section, we treat the points of type cE. Together with Theorem 11.2 for the points of type cD, this completes the proof of the following theorem in Subsection 12.6.

Theorem 12.1. *Let $(X \ni o)$ be an isolated Gorenstein terminal threefold singularity of quadratic rank one, and G a finite abelian group acting faithfully on X and fixing o. Then the set of such numbers $\operatorname{mld}((X \ni o, 0)/G) \in (0, 1)$ satisfies the ACC.*

For a point of type cE, we study the following binary form under the conditions of Set-up 12.2 below: in type cE_6 the quartic part of $g(0, z, t)$, and in types cE_7 and cE_8, after writing g in the form of Lemma 12.4, the cubic part α_3 of α and the quintic part β_5 of β, respectively. In type cE_6 the quartic part of $g(0, z, t)$ has root type $(1, 1, 1, 1)$, $(2, 1, 1)$, $(2, 2)$, $(3, 1)$, or (4), and in types cE_7 and cE_8 the forms α_3 and β_5 have at least three, exactly two, or exactly one distinct roots. At least three distinct roots are treated in Propositions 12.6 and 12.7, a fourth power and a single root in Propositions 12.9 and 10.6, and two distinct roots in Propositions 12.10 and 12.13.

Set-up 12.2. Let $(X \ni o)$, G, x, y, z, t, and g be as follows.

(1) $(X \ni o)$ is an isolated Gorenstein terminal cDV threefold singularity of quadratic rank one, and G is a finite abelian group acting faithfully on X and fixing o.
(2) x, y, z, t are G-semi-invariant elements generating the maximal ideal of the completion of the local ring of an ambient smooth fourfold germ, on which G acts diagonally, such that the completion of $\mathcal{O}_{X,o}$ is $\mathbb{C}[[x, y, z, t]]/(x^2 + g(y, z, t))$ for a G-semi-invariant power series g with $\operatorname{ord} g \geq 3$ whose cubic part is y^3.

By Lemma 11.3, any $(X \ni o)$ and G as in Set-up 12.2(1) with $(X \ni o)$ of type cE admit x, y, z, t and g as in Set-up 12.2(2). Indeed, the residual ternary cubic is ℓ^3 for a G-semi-invariant linear form ℓ in the coordinates of Lemma 11.3, and after replacing one of the coordinates occurring in ℓ by ℓ, one may take $y = \ell$.

Lemma 12.3. *Notation and conditions as in Set-up 12.2. Let χ_x be the character of x, let $\mathfrak{R} := \mathbb{C}[[x, y, z, t]]$ with the diagonal action of G, and for a power series P and a positive integer L, let $P_{<L}$ be the sum of the terms of P of degree less than L. Then the following hold.*

(1) x^2+g *is G-semi-invariant with character χ_x^2. In particular, any monomial occurring in g has character χ_x^2.*

(2) *Let ψ be a G-equivariant automorphism of $\mathfrak{R}$ and v a G-invariant unit of $\mathfrak{R}$. Then there exists a positive integer L_1 such that for any integer $L \geq L_1$, the following holds for the polynomial $f_0 := (v\psi(x^2+g))_{<L}$ in the coordinates x,y,z,t of $\mathbb{A}^4$, on which G acts diagonally with the characters of x,y,z,t: the action of G on $\mathbb{A}^4$ is faithful, $X_0 := (f_0 = 0) \subset \mathbb{A}^4$ is an isolated Gorenstein terminal threefold germ at 0, G acts faithfully on X_0, and $\operatorname{mld}((X_0 \ni 0,0)/H) = \operatorname{mld}((X \ni o,0)/H)$ for any subgroup $H \subset G$. If moreover $L \geq 4$ and $v\psi(x^2+g) = x^2+g'$ for a power series $g' \in \mathbb{C}[[y,z,t]]$ with $\operatorname{ord} g' \geq 3$ whose cubic part is y^3, then $(X_0 \ni 0)$, G, the coordinates x,y,z,t, and $g'_{<L}$ satisfy the conditions of Set-up 12.2.*

(3) $\operatorname{mld}((X \ni o,0)/H) \in \frac{1}{|H|}\mathbb{N}^+$ *for any subgroup $H \subset G$.*

Proof. Let $(Y \ni o)$ be the ambient smooth fourfold germ in Set-up 12.2(2). Thus G acts on Y, X is a G-stable closed subvariety of Y, $\widehat{\mathcal{O}}_{Y,o} = \mathfrak{R}$, and the kernel of $\mathfrak{R} \to \widehat{\mathcal{O}}_{X,o}$ is the ideal (x^2+g).

(1) Let χ_g be the character of g and $\sigma \in G$. Since the ideal (x^2+g) is G-stable, there exists a unit $u_\sigma \in \mathfrak{R}$ such that

$$\chi_x(\sigma)^2 x^2 + \chi_g(\sigma) g = \sigma^*(x^2+g) = u_\sigma \cdot (x^2+g).$$

Since g does not involve x, $\operatorname{ord} g \geq 3$, and the cubic part of g is y^3, the coefficients of x^2 and of y^3 on the right-hand side are both equal to $u_\sigma(0)$. Thus $\chi_x(\sigma)^2 = u_\sigma(0) = \chi_g(\sigma)$. Since G acts diagonally, any monomial occurring in g has character $\chi_g = \chi_x^2$.

(2) By (1), x^2+g is a G-semi-invariant generator of the kernel of $\mathfrak{R} = \widehat{\mathcal{O}}_{Y,o} \to \widehat{\mathcal{O}}_{X,o}$, and $(X \ni o)$ is an isolated Gorenstein terminal threefold singularity by Set-up 12.2(1). By Corollary 7.5, applied to $(Y \ni o)$, the elements x,y,z,t, the generator x^2+g (the F of the corollary), ψ, and v, there exists L_1 such that for any $L \geq L_1$ the action of G on $\mathbb{A}^4$ is faithful, X_0 is an isolated Gorenstein terminal threefold germ at 0, G acts faithfully on X_0, and $\operatorname{mld}((X_0 \ni 0,0)/H) = \operatorname{mld}((X \ni o,0)/H)$ for any subgroup $H \subset G$.

Assume moreover that $L \geq 4$ and $v\psi(x^2+g) = x^2+g'$ as in (2). Then $f_0 = x^2 + g'_{<L}$. By [KM98, Corollary 5.38], $(X_0 \ni 0)$ is an isolated cDV singularity. Since $\operatorname{ord} g' \geq 3$ and $L \geq 4$, the quadratic part of f_0 is x^2, so $(X_0 \ni 0)$ has quadratic rank one (Definition-Lemma 11.1), and the cubic part of $g'_{<L}$ is y^3. By (1), $x^2+g' = v\psi(x^2+g)$ has character χ_x^2, so g' and $g'_{<L}$ are G-semi-invariant with character χ_x^2. Thus the conditions of Set-up 12.2 hold, with the ambient germ $(\mathbb{A}^4 \ni 0)$.

(3) Let F be a prime divisor over X with $\operatorname{center}_X F = \{o\}$. Since K_X is Cartier, $a(F,X,0) \in \mathbb{Z}$, and since X is terminal, $a(F,X,0) > 1$. Moreover, $|I_H(F)|$ divides $|H|$. Thus

$$\frac{a(F,X,0)}{|I_H(F)|} \in \frac{1}{|H|}\mathbb{N}^+.$$

By Lemma 2.6 applied to H, $\operatorname{mld}((X \ni o,0)/H)$ is the infimum of the numbers $a(F,X,0)/|I_H(F)|$, which is a minimum and belongs to $\frac{1}{|H|}\mathbb{N}^+$. □

Lemma 12.4. *Notation and conditions as in Set-up 12.2, and let χ_y be the character of y. Then there exist a G-semi-invariant power series $y' \in \mathbb{C}[[y,z,t]]$ with character χ_y and $y'-y \in (y,z,t)^2$, and power series $\alpha,\beta \in \mathbb{C}[[z,t]]$ with $\operatorname{ord}\alpha \geq 3$ and $\operatorname{ord}\beta \geq 4$, each of which is zero or G-semi-invariant, with characters χ_y^2 and χ_y^3, respectively, such that $g = y'^3 + \alpha(z,t)y' + \beta(z,t)$. In particular, condition (2) of Set-up 12.2 holds for x,y',z,t and $y'^3+\alpha y'+\beta$ in place of x,y,z,t and g.*

Proof. Let $\mathfrak{n}$ be the maximal ideal of $\mathbb{C}[[y,z,t]]$. Since g is G-semi-invariant and y^3 occurs in g, the character of g is χ_y^3. For any integer $N \geq 4$, we construct a G-semi-invariant power series $y_N \in \mathbb{C}[[y,z,t]]$ with character χ_y and $y_N - y \in \mathfrak{n}^2$, and power series $\alpha^{(N)}, \beta^{(N)} \in \mathbb{C}[[z,t]]$,

each zero or G-semi-invariant with character χ_y^2 and χ_y^3 respectively, such that $\operatorname{ord}\alpha^{(N)} \geq 3$, $\operatorname{ord}\beta^{(N)} \geq 4$, and

$$g - (y_N^3 + \alpha^{(N)} y_N + \beta^{(N)}) \in y_N^2 \cdot \mathfrak{n}^{N-2}.$$

Since $y_N - y \in \mathfrak{n}^2$, we have $\mathbb{C}[[y,z,t]] = \mathbb{C}[[y_N,z,t]]$, and since y_N, z, and t are G-semi-invariant, G acts diagonally on them.

For $N = 4$, let $y_4 := y$. Since $\operatorname{ord}(g - y^3) \geq 4$, expanding $g - y^3$ in y we may write $g - y^3 = y^2 r + \alpha^{(4)}(z,t)y + \beta^{(4)}(z,t)$ with $r \in \mathfrak{n}^2$, $\operatorname{ord}\alpha^{(4)} \geq 3$, and $\operatorname{ord}\beta^{(4)} \geq 4$. Since G acts diagonally and $g - y^3$ has character χ_y^3, the series $\alpha^{(4)}$ and $\beta^{(4)}$ are zero or semi-invariant with characters χ_y^2 and χ_y^3.

Assume that y_N, $\alpha^{(N)}$, and $\beta^{(N)}$ are constructed. Write

$$g - (y_N^3 + \alpha^{(N)} y_N + \beta^{(N)}) = y_N^2 P + R,$$

where P is a homogeneous polynomial of degree $N-2$ in y_N, z, t and $R \in y_N^2 \cdot \mathfrak{n}^{N-1}$. Since the left-hand side is zero or semi-invariant with character χ_y^3 and G acts diagonally on y_N, z, t, the polynomial P is zero or semi-invariant with character χ_y. Let $y_{N+1} := y_N + \frac{1}{3}P$. Then

$$y_N^3 + y_N^2 P = y_{N+1}^3 - \frac{1}{3} y_N P^2 - \frac{1}{27} P^3, \qquad \alpha^{(N)} y_N = \alpha^{(N)} y_{N+1} - \frac{1}{3}\alpha^{(N)} P,$$

and $y_N P^2$, P^3, $\alpha^{(N)} P$, and R belong to $\mathfrak{n}^{N+1}$, since $N \geq 4$. Thus

$$R' := g - (y_{N+1}^3 + \alpha^{(N)} y_{N+1} + \beta^{(N)}) \in \mathfrak{n}^{N+1},$$

and R' is zero or semi-invariant with character χ_y^3. Expanding R' in y_{N+1}, we write

$$R' = y_{N+1}^2 R'' + r_1(z,t) y_{N+1} + r_0(z,t),$$

where $y_{N+1}^2 R''$, $r_1 y_{N+1}$, and r_0 belong to $\mathfrak{n}^{N+1}$. Let $\alpha^{(N+1)} := \alpha^{(N)} + r_1$ and $\beta^{(N+1)} := \beta^{(N)} + r_0$. Then y_{N+1}, $\alpha^{(N+1)}$, and $\beta^{(N+1)}$ satisfy the conditions for $N+1$.

Since $y_{N+1} - y_N \in \mathfrak{n}^{N-2}$, $\alpha^{(N+1)} - \alpha^{(N)} \in \mathfrak{n}^N$, and $\beta^{(N+1)} - \beta^{(N)} \in \mathfrak{n}^{N+1}$, the sequences y_N, $\alpha^{(N)}$, and $\beta^{(N)}$ converge $\mathfrak{n}$-adically to power series y', α, and β. Then y' is G-semi-invariant with character χ_y and $y' - y \in \mathfrak{n}^2$. Each of α and β is zero or G-semi-invariant, with characters χ_y^2 and χ_y^3 respectively, $\operatorname{ord}\alpha \geq 3$, $\operatorname{ord}\beta \geq 4$, and $g - (y'^3 + \alpha y' + \beta) \in \bigcap_N \mathfrak{n}^N = 0$. Since $y' - y \in \mathfrak{n}^2$, the elements x, y', z, t generate the maximal ideal of $\mathbb{C}[[x,y,z,t]]$, and G acts diagonally on them. This implies the last part of the lemma. □

12.1. **Bounded groups.** In this subsection, we bound the order of G when the binary form of the section introduction has at least three distinct roots. The image of G in $\operatorname{Aut}\mathbb{P}^1$ permutes the roots, so for the weight vector $w = (6,4,3,3)$, $(9,6,4,4)$, or $(15,10,6,6)$ in type cE_6, cE_7, or cE_8, respectively, the subgroup G_w of the elements of G acting through the torus $\mathbf{T}_w$ has bounded index in G, and G_w is contained in the inertia group of the prime divisor $F = X' \cap E$ over X, where E is the exceptional divisor of the weighted blow-up of $\mathbb{A}^4$ with weights w and X' is the strict transform of X (Lemma 12.5).

Lemma 12.5. *Let G be a finite abelian group acting faithfully and diagonally on $\mathbb{A}^4$ with coordinates x,y,z,t, let $f = x^2 + g(y,z,t)$ be a G-semi-invariant polynomial, let $w = (w_x,w_y,w_z,w_t) \in \mathbb{Z}^4_{>0}$, and let $G_w \subset G$ be the subgroup of the elements of G acting on $\mathbb{A}^4$ through an element of $\mathbf{T}_w$. Assume that $X := (f = 0)$ is an isolated Gorenstein terminal threefold germ at 0 on which G acts faithfully, that $\gcd(w_x,w_y,w_z,w_t) = 1$, and that $\operatorname{in}_w(f) = x^2 + \phi(y,z,t)$ for a polynomial ϕ of degree three in y. Then*

$$|G_w| \cdot \operatorname{mld}((X \ni 0,0)/G) \leq w_x + w_y + w_z + w_t - \operatorname{wt}_w(f).$$

Proof. Let $\mu : Y' \to \mathbb{A}^4$ be the weighted blow-up with weights w in x,y,z,t, let $E \cong \mathbb{P}(w_x,w_y,w_z,w_t)$ be the exceptional divisor of μ, and let X' be the strict transform of X. Since ϕ has degree three in y, $-\phi$ is not a square in $\mathbb{C}[y,z,t]$, so $x^2 + \phi$ is irreducible. Let

$F := (x^2 + \phi = 0) \subset E$. Then F is a prime divisor on E which is not contained in any coordinate hyperplane of E. Thus the generic point of F is in the open torus orbit of E, along which Y' is smooth since w is primitive in $\mathbb{Z}^4$. A direct calculation shows that, near the generic point of F, $X' \cap E = F$ is defined in E by $\mathrm{in}_w(f)$, and since $\partial_x \mathrm{in}_w(f) = 2x$ does not vanish identically on F, X' is smooth and intersects E transversally along F. Since ord_E is the monomial valuation in x, y, z, t with weights w, we have $\mu^* X = X' + \mathrm{wt}_w(f)E$, and by [CLS11, Lemma 11.4.10],

$$K_{Y'} = \mu^* K_{\mathbb{A}^4} + (w_x + w_y + w_z + w_t - 1)E.$$

Thus

$$K_{Y'} + X' = \mu^*(K_{\mathbb{A}^4} + X) + (w_x + w_y + w_z + w_t - \mathrm{wt}_w(f) - 1)E,$$

and adjunction near the generic point of F implies

$$K_{X'} = (\mu|_{X'})^* K_X + (w_x + w_y + w_z + w_t - \mathrm{wt}_w(f) - 1)F.$$

Hence $a(F, X, 0) = w_x + w_y + w_z + w_t - \mathrm{wt}_w(f)$, and $\mathrm{center}_X F = \{0\}$ since $\mu(E) = \{0\}$.

Since G is a subgroup of the torus $(\mathbb{C}^*)^4$ of the toric variety Y', the group G acts on Y', the morphism μ is G-equivariant, and E, X', and F are G-stable. An element $\sigma \in G_w$ acts on $E = \mathbb{P}(w_x, w_y, w_z, w_t)$ through an element of $\mathbf{T}_w$, that is, trivially. Hence σ acts trivially on F and on $\mathbb{C}(F)$, which is the residue field of ord_F. Thus $G_w \subset I_G(F)$. By Lemma 2.6,

$$\mathrm{mld}((X \ni 0, 0)/G) \le \frac{a(F, X, 0)}{|I_G(F)|} \le \frac{w_x + w_y + w_z + w_t - \mathrm{wt}_w(f)}{|G_w|}.$$

□

Proposition 12.6. *Notation and conditions as in Set-up 12.2, and assume that the quartic part of $g(0, z, t)$ is a binary quartic with at least three distinct linear factors. Then $|G| \le \frac{192}{a}$, where $a := \mathrm{mld}((X \ni o, 0)/G) > 0$. In particular, for any $\epsilon > 0$ the set of such numbers $a \ge \epsilon$ is finite.*

Proof. Let $q(z, t)$ be the quartic part of $g(0, z, t)$, and let $\chi_x, \chi_y, \chi_z, \chi_t$ be the characters of x, y, z, t.

Step 1. In this step we reduce to the case where g is a polynomial. By Lemma 12.3(2) with $\psi = \mathrm{id}_{\mathfrak{R}}$ and $v = 1$, there exists an integer $L \ge 5$ such that $f_0 := x^2 + g_{<L}$ defines an isolated Gorenstein terminal threefold germ $X_0 \subset \mathbb{A}^4$ at 0, G acts faithfully on $\mathbb{A}^4$ and on X_0, $\mathrm{mld}((X_0 \ni 0, 0)/G) = a$, and $(X_0 \ni 0)$, G, x, y, z, t, and $g_{<L}$ satisfy the conditions of Set-up 12.2. Since $L \ge 5$, the quartic part of $g_{<L}(0, z, t)$ is q. Possibly replacing X with X_0 and g with $g_{<L}$, we may assume that x, y, z, t are coordinates of $\mathbb{A}^4$, $X = (x^2 + g = 0) \subset \mathbb{A}^4$, $o = 0$, and g is a polynomial.

Step 2. In this step we show that $|G_w| \cdot a \le 4$, where $w := (6, 4, 3, 3)$. A monomial $y^i z^j t^k$ of degree $d = i + j + k$ has w-weight $3d + i$. The monomials occurring in g other than y^3 have degree at least four, and those of degree four with $i = 0$ are the monomials of q. Hence any other monomial occurring in g has w-weight at least 13, and

$$\mathrm{in}_w(x^2 + g) = x^2 + y^3 + q(z, t), \qquad \mathrm{wt}_w(x^2 + g) = 12.$$

Since $\gcd(6, 4, 3, 3) = 1$, Lemma 12.5 implies

$$|G_w| \cdot a \le 6 + 4 + 3 + 3 - 12 = 4.$$

Step 3. In this step we show that $[G : G_w] \le 48$. By Lemma 12.3(1), the monomials x^2, y^3, and the monomials of q have character χ_x^2, so $\sigma^* q = \chi_x(\sigma)^2 q$ for any $\sigma \in G$. The group G acts on $\mathbb{P}^1$ with homogeneous coordinates z, t through the diagonal matrices with entries $\chi_z(\sigma), \chi_t(\sigma)$, which are elements of $\mathrm{PGL}(2, \mathbb{C}) = \mathrm{Aut}\,\mathbb{P}^1$ [Har77, Example II.7.1.1]. The zero divisor of q on $\mathbb{P}^1$ is G-stable, and the support of the zero divisor of q consists of k points with $3 \le k \le 4$. Since an element of $\mathrm{PGL}(2, \mathbb{C})$ fixing three distinct points of $\mathbb{P}^1$ is the identity, the image of G in

$\mathrm{PGL}(2,\mathbb{C})$ embeds into the symmetric group of the support of the zero divisor of q. Thus the kernel

$$G_0 := \{\sigma \in G \mid \chi_z(\sigma) = \chi_t(\sigma)\}$$

satisfies $[G : G_0] \le 4! = 24$. Let η be the restriction of χ_z to G_0, which is also the restriction of χ_t. Since q has a monomial $z^j t^{4-j}$, on G_0 we have

$$\chi_x^2 = \chi_y^3 = \eta^4.$$

Let $s := \chi_y \eta^{-1}$, a character of G_0. Then

$$s^3 = \chi_y^3 \eta^{-3} = \eta, \qquad s^4 = \chi_y^4 \eta^{-4} = \chi_y, \qquad (\chi_x s^{-6})^2 = \chi_x^2 s^{-12} = \chi_x^2 \eta^{-4} = 1.$$

Hence the kernel G_1 of the character $\chi_x s^{-6}$ of G_0 satisfies $[G_0 : G_1] \le 2$, and any $\sigma \in G_1$ multiplies x, y, z, t by $s(\sigma)^6, s(\sigma)^4, s(\sigma)^3, s(\sigma)^3$, that is, σ acts on $\mathbb{A}^4$ through an element of $\mathbf{T}_{(6,4,3,3)}$. Thus $G_1 \subset G_w$, and $[G : G_w] \le [G : G_1] \le 48$.

By **Steps 2 and 3**,

$$|G| \le 48|G_w| \le \frac{192}{a}.$$

Step 4. In this step we prove the finiteness. By **Step 2**, $a \le 4$, and by Lemma 12.3(3), $a \in \frac{1}{|G|}\mathbb{Z}$. If $a \ge \epsilon$, then $|G| \le \frac{192}{\epsilon}$, so

$$a \in [\epsilon, 4] \cap \bigcup_{1 \le n \le 192/\epsilon} \frac{1}{n}\mathbb{Z},$$

which is a finite set. □

Proposition 12.7. *Notation and conditions as in Set-up 12.2, and assume that* $g = y^3 + \alpha(z,t)y + \beta(z,t)$ *for power series* α, β *with* $\operatorname{ord}\alpha \ge 3$ *and* $\operatorname{ord}\beta \ge 4$, *and that one of the following holds:*

(E7) $\operatorname{ord}\beta \ge 5$ *and the cubic part* α_3 *of* α *has three distinct roots in* $\mathbb{P}^1$;

(E8) $\operatorname{ord}\alpha \ge 4$, $\operatorname{ord}\beta \ge 5$, *and the quintic part* β_5 *of* β *has at least three distinct roots in* $\mathbb{P}^1$.

Let $a := \operatorname{mld}((X \ni o, 0)/G) > 0$. *Then* $|G| \le \frac{30}{a}$ *in case (E7), and* $|G| \le \frac{840}{a}$ *in case (E8). In particular, for any* $\epsilon > 0$ *the set of such numbers* $a \ge \epsilon$ *is finite.*

Proof. Let $\chi_x, \chi_y, \chi_z, \chi_t$ be the characters of x, y, z, t.

Step 1. In this step we reduce to the case where α and β are polynomials. By Lemma 12.3(2) with $\psi = \mathrm{id}_{\mathfrak{R}}$ and $v = 1$, there exists an integer $L \ge 6$ such that $f_0 := x^2 + g_{<L}$ defines an isolated Gorenstein terminal threefold germ $X_0 \subset \mathbb{A}^4$ at 0, G acts faithfully on $\mathbb{A}^4$ and on X_0, $\operatorname{mld}((X_0 \ni 0, 0)/G) = a$, and $(X_0 \ni 0)$, G, x, y, z, t, and $g_{<L}$ satisfy the conditions of Set-up 12.2. We have

$$g_{<L} = y^3 + \alpha_{<L-1}y + \beta_{<L}.$$

Since $L \ge 6$, the parts of $\alpha_{<L-1}$ of degree at most four are those of α, and the parts of $\beta_{<L}$ of degree at most five are those of β. Hence $\alpha_{<L-1}$ and $\beta_{<L}$ satisfy the order conditions of the proposition, the part of $\alpha_{<L-1}$ of degree three is α_3, and the part of $\beta_{<L}$ of degree five is β_5. Possibly replacing X with X_0, α with $\alpha_{<L-1}$, and β with $\beta_{<L}$, we may assume that x, y, z, t are coordinates of $\mathbb{A}^4$, $X = (x^2 + g = 0) \subset \mathbb{A}^4$, $o = 0$, and α and β are polynomials.

Step 2. In this step we show that $|G_w| \cdot a \le 5$ in case (E7) and $|G_w| \cdot a \le 7$ in case (E8). In case (E7), let $w := (9, 6, 4, 4)$. The monomials x^2 and y^3 have w-weight 18, a monomial $yz^j t^k$ has w-weight $6 + 4(j + k)$, which is 18 if $j + k = 3$ and at least 22 if $j + k \ge 4$, and a monomial occurring in β has w-weight at least 20. Hence

$$\mathrm{in}_w(x^2 + g) = x^2 + y^3 + \alpha_3(z,t)y, \qquad \mathrm{wt}_w(x^2 + g) = 18.$$

In case (E8), let $w := (15, 10, 6, 6)$. The monomials x^2 and y^3 have w-weight 30, a monomial occurring in αy has w-weight at least $10 + 24 = 34$, and a monomial $z^j t^k$ occurring in β has w-weight $6(j + k)$, which is 30 if $j + k = 5$ and at least 36 otherwise. Hence

$$\mathrm{in}_w(x^2 + g) = x^2 + y^3 + \beta_5(z, t), \qquad \mathrm{wt}_w(x^2 + g) = 30.$$

Since $\gcd(9, 6, 4, 4) = \gcd(15, 10, 6, 6) = 1$, Lemma 12.5 implies

$$\begin{aligned} |G_w| \cdot a &\le 9 + 6 + 4 + 4 - 18 = 5 \quad \text{in case (E7)}, \\ |G_w| \cdot a &\le 15 + 10 + 6 + 6 - 30 = 7 \quad \text{in case (E8)}. \end{aligned}$$

Step 3. In this step we show that $[G : G_w] \le 6$ in case (E7) and $[G : G_w] \le 120$ in case (E8). Let $q := \alpha_3$ in case (E7) and $q := \beta_5$ in case (E8). By Lemma 12.3(1), the monomials occurring in $\alpha_3 y$ and in β_5 have character χ_x^2, so q is G-semi-invariant. The group G acts on $\mathbb{P}^1$ with homogeneous coordinates z, t through the diagonal matrices with entries $\chi_z(\sigma), \chi_t(\sigma)$ in $\mathrm{PGL}(2, \mathbb{C}) = \mathrm{Aut}\,\mathbb{P}^1$ [Har77, Example II.7.1.1], and the zero divisor of q is G-stable. The support of the zero divisor of q consists of k points, where $k = 3$ in case (E7) and $3 \le k \le 5$ in case (E8). Since an element of $\mathrm{PGL}(2, \mathbb{C})$ fixing three distinct points of $\mathbb{P}^1$ is the identity, the kernel

$$G_0 := \{\sigma \in G \mid \chi_z(\sigma) = \chi_t(\sigma)\}$$

satisfies $[G : G_0] \le 3! = 6$ in case (E7) and $[G : G_0] \le 5! = 120$ in case (E8). Let η be the restriction of χ_z to G_0, which is also the restriction of χ_t.

In case (E7), since $\alpha_3 y$ has a monomial $y z^j t^{3-j}$, on G_0 we have $\chi_x^2 = \chi_y^3 = \chi_y \eta^3$, so $\chi_y^2 = \eta^3$. Let $s := \eta^2 \chi_y^{-1}$ and $\rho := \chi_x s^{-4}$, characters of G_0. Then

$$s^2 = \eta^4 \chi_y^{-2} = \eta, \qquad s^3 = \eta^6 \chi_y^{-3} = \chi_y, \qquad \rho^2 = \chi_x^2 s^{-8} = \chi_y^3 s^{-8} = s,$$

hence

$$\rho^4 = \eta, \qquad \rho^6 = \chi_y, \qquad \rho^9 = \rho \cdot s^4 = \chi_x.$$

In case (E8), since β_5 has a monomial $z^j t^{5-j}$, on G_0 we have $\chi_x^2 = \chi_y^3 = \eta^5$. Let $\rho := \chi_x^{-1} \chi_y \eta$, a character of G_0. Then

$$\begin{aligned} \rho^6 &= \chi_x^{-6} \chi_y^6 \eta^6 = \chi_y^{-3} \eta^6 = \eta, \\ \rho^{10} &= \chi_x^{-10} \chi_y^{10} \eta^{10} = \chi_y^{10} \eta^{-15} = \chi_y, \\ \rho^{15} &= \chi_x^{-15} \chi_y^{15} \eta^{15} = \chi_x^{-15} \chi_x^{10} \chi_x^6 = \chi_x. \end{aligned}$$

In both cases, any $\sigma \in G_0$ multiplies x, y, z, t by $\rho(\sigma)^{w_x}, \rho(\sigma)^{w_y}, \rho(\sigma)^{w_z}, \rho(\sigma)^{w_t}$, that is, σ acts on $\mathbb{A}^4$ through an element of $\mathbf{T}_w$. Thus $G_0 \subset G_w$.

By **Steps 2 and 3**,

$$|G| \le 6|G_w| \le \frac{30}{a} \quad \text{in case (E7)}, \qquad |G| \le 120|G_w| \le \frac{840}{a} \quad \text{in case (E8)}.$$

Step 4. In this step we prove the finiteness. By **Step 2**, $a \le 7$, and by Lemma 12.3(3), $a \in \frac{1}{|G|}\mathbb{Z}$. If $a \ge \epsilon$, then $|G| \le \frac{840}{\epsilon}$, so

$$a \in [\epsilon, 7] \cap \bigcup_{1 \le n \le 840/\epsilon} \frac{1}{n}\mathbb{Z},$$

which is a finite set. □

12.2. **A fourth-power quartic.** In this subsection, we treat the points of type cE_6 whose quartic part is a fourth power. After an equivariant translation of z, the new equation x^2+h satisfies that the pair $((\mathbb{C}^*)^3, \frac{1}{2}\operatorname{div}(\mathrm{in}_u(h)))$ is lc for any $u \in \mathbb{R}^3_{>0}$ (Lemma 12.8), which is the hypothesis of Proposition 10.5 on leading forms, and Proposition 10.5 then implies the ACC.

Lemma 12.8. *Let a finite abelian group G act diagonally on $\mathbb{C}[[x,y,z,t]]$, let $\alpha_0,\alpha_1,\alpha_2,\beta_0,\beta_1,\beta_2 \in t\mathbb{C}[[t]]$, and let*

$$F = x^2+y^3+\alpha(z,t)y+\beta(z,t),$$
$$\alpha = \alpha_2(t)z^2+\alpha_1(t)z+\alpha_0(t), \qquad \beta = z^4+\beta_2(t)z^2+\beta_1(t)z+\beta_0(t).$$

Assume that F is G-semi-invariant. Then there exists $\theta \in t\mathbb{C}[[t]]$, either zero or G-semi-invariant with the character of z, such that for $z' := z-\theta(t)$ and

$$h(y,z',t) := y^3+\alpha(z'+\theta(t),t)y+\beta(z'+\theta(t),t),$$

so that $F = x^2+h(y,z',t)$, the pair $((\mathbb{C}^*)^3, \frac{1}{2}\operatorname{div}(\mathrm{in}_u(h)))$ is lc for any $u \in \mathbb{R}^3_{>0}$, where in_u is taken with respect to y,z',t.

Proof. Let χ_z,χ_t be the characters of z,t and χ the character of F. Since G acts diagonally, any monomial occurring in F has character χ. For a non-zero power series $P \in \mathbb{C}[[z,t]]$ and $v=(v_z,v_t) \in \mathbb{Z}^2_{>0}$, the leading form $\mathrm{in}_v(P)$ is a polynomial, and we let $P^v(Z) := \mathrm{in}_v(P)(Z,1) \in \mathbb{C}[Z]$. We say that v is *bad* for P if P^v has a non-zero root of multiplicity at least three.

Step 1. In this step we show: if $v=(v_z,v_t) \in \mathbb{Z}^2_{>0}$ is bad for β, then $n := v_z/v_t$ is a positive integer, and there exists $\lambda \in \mathbb{C}^*$ such that β_2,β_1,β_0 have orders $2n,3n,4n$ and leading coefficients $-6\lambda^2$, $8\lambda^3$, $-3\lambda^4$, respectively.

For $i \in \{0,1,2\}$ with $\beta_i \neq 0$, the term of $\beta_i z^i$ of least v-weight is $c_i t^{o_i} z^i$, where $o_i := \operatorname{ord}\beta_i$ and c_i is the leading coefficient of β_i. Thus $\mathrm{in}_v(\beta)$ is the sum of the terms of least v-weight among z^4 and the terms $c_i t^{o_i} z^i$, and β^v has no term in Z^3. If z^4 does not occur in $\mathrm{in}_v(\beta)$, then $\deg \beta^v \le 2$, and v is not bad. Assume that z^4 occurs in $\mathrm{in}_v(\beta)$, so that β^v is monic of degree four. A non-zero root λ of multiplicity four is impossible, since the coefficient of Z^3 in $(Z-\lambda)^4$ is $-4\lambda \neq 0$. If $\lambda \neq 0$ is a root of multiplicity three, then $\beta^v = (Z-\lambda)^3(Z-\lambda')$ with $-3\lambda-\lambda'=0$, so

$$\beta^v = (Z-\lambda)^3(Z+3\lambda) = Z^4-6\lambda^2Z^2+8\lambda^3Z-3\lambda^4.$$

Hence $\beta_2,\beta_1,\beta_0 \neq 0$, all the terms $c_i t^{o_i} z^i$ occur in $\mathrm{in}_v(\beta)$, $c_2=-6\lambda^2$, $c_1=8\lambda^3$, $c_0=-3\lambda^4$, and

$$4v_z = o_2v_t+2v_z = o_1v_t+v_z = o_0v_t.$$

Thus $o_2=2n$, $o_1=3n$, and $o_0=4n$, and $n=o_1-o_2$ is a positive integer.

Step 2. In this step we construct θ. If no $v \in \mathbb{Z}^2_{>0}$ is bad for β, let $\theta := 0$. Otherwise, fix a bad v, and let n and λ be as in **Step 1**. Write $\beta_2 = -6\lambda^2 t^{2n}\omega(t)$ with $\omega \in \mathbb{C}[[t]]$ and $\omega(0)=1$, let $\omega^{1/2} := \sum_{k\ge 0}\binom{1/2}{k}(\omega-1)^k$, and let

$$\theta := \lambda t^n \omega^{1/2}.$$

Then $\theta \in t\mathbb{C}[[t]]$ and $\theta^2 = -\frac{1}{6}\beta_2$. By **Step 1**, the monomials z^4, $t^{2n}z^2$, and $t^{3n}z$ occur in F, so $\chi_z^4 = \chi_t^{2n}\chi_z^2 = \chi_t^{3n}\chi_z$. Hence $\chi_z^2 = \chi_t^{2n}$ and $\chi_z^3 = \chi_t^{3n}$, so $\chi_z = \chi_t^n$. Any monomial $t^j z^2$ occurring in $\beta_2 z^2$ has character $\chi = \chi_t^{2n}\chi_z^2$, so $\chi_t^{j-2n}=1$. Thus ω is G-invariant, hence so is $\omega^{1/2}$, and θ is G-semi-invariant with character $\chi_t^n = \chi_z$.

Step 3. In this step we show that no $v \in \mathbb{Z}^2_{>0}$ is bad for $\beta' := \beta(z'+\theta(t),t) \in \mathbb{C}[[z',t]]$. If $\theta=0$, then $\beta'=\beta$ with $z'=z$, and the claim holds by the definition of θ. Assume that $\theta \neq 0$. Since $6\theta^2+\beta_2=0$, a direct calculation shows

$$\beta' = z'^4+4\theta z'^3+p_1z'+p_0, \qquad p_1 := \beta_1-8\theta^3, \qquad p_0 := \beta_0+\beta_1\theta-5\theta^4.$$

By **Step 1**, the terms of β_1 and β_0 of least order are $8\lambda^3t^{3n}$ and $-3\lambda^4t^{4n}$, and those of θ^3 and θ^4 are λ^3t^{3n} and λ^4t^{4n}. Since $8\lambda^3-8\lambda^3=0$ and $-3\lambda^4+8\lambda^4-5\lambda^4=0$, each of p_1 and p_0 is either zero or of order greater than $3n$ and $4n$, respectively. Let $v=(v_z,v_t) \in \mathbb{Z}^2_{>0}$ and $l := v_z/v_t$.

Divided by v_t, the v-weights of z'^4, of the term $4\lambda t^n z'^3$ of least order of $4\theta z'^3$, of the terms of $p_1 z'$, and of the terms of p_0 are

$$4l, \qquad n+3l, \qquad > 3n+l, \qquad > 4n,$$

respectively. If $l < n$, then $\mathrm{in}_v(\beta') = z'^4$, so $\beta'^v = Z^4$. If $l = n$, then $\beta'^v = Z^4 + 4\lambda Z^3 = Z^3(Z+4\lambda)$, whose only non-zero root -4λ is simple. If $l > n$, then $4l > n+3l$, so $\beta'^v = c_3 Z^3 + c_1' Z + c_0'$ for some $c_3 \in \{0, 4\lambda\}$ and $c_1', c_0' \in \mathbb{C}$. If $c_3 \neq 0$, a non-zero root ζ of multiplicity three would imply $\beta'^v = c_3(Z-\zeta)^3$, whose coefficient $-3c_3\zeta$ of Z^2 is non-zero, a contradiction. If $c_3 = 0$, then $\deg \beta'^v \leq 1$. Thus no v is bad for β'.

Step 4. In this step we conclude the proof. Let $\alpha' := \alpha(z' + \theta(t), t)$, a polynomial in z' of degree at most two with coefficients in $\mathbb{C}[[t]]$, so that $h = y^3 + \alpha' y + \beta'$. By Lemma 10.2, applied to $P = h$ in the coordinates y, z', t, it suffices to show that $((\mathbb{C}^*)^2, \frac{1}{2}\operatorname{div}(p))$ is lc for any $u = (u_y, u_z, u_t) \in \mathbb{Z}^3_{>0}$, where $p(Y, Z) := \mathrm{in}_u(h)(Y, Z, 1)$. Let $d := \mathrm{wt}_u(h)$. Then

$$\mathrm{in}_u(h) = \epsilon y^3 + yA + B,$$

where $\epsilon \in \{0, 1\}$, A is the sum of the terms of α' of u-weight $d - u_y$, and B is the sum of the terms of β' of u-weight d. Since $d \leq \mathrm{wt}_u(\beta')$, if $B \neq 0$, then $d = \mathrm{wt}_u(\beta')$ and $B = \mathrm{in}_{(u_z, u_t)}(\beta')$. Let $A_0(Z) := A(Z, 1)$ and $B_0(Z) := B(Z, 1)$. Then $\deg A_0 \leq 2$ if $A_0 \neq 0$, and by **Step 3**, any non-zero root of B_0 has multiplicity at most two if $B_0 \neq 0$.

Assume that $\epsilon = 1$. Then $p = Y^3 + A_0(Z)Y + B_0(Z)$. At any point $(Y_0, Z_0) \in (\mathbb{C}^*)^2$, we have $\partial_Y^2 p(Y_0, Z_0) = 6Y_0 \neq 0$, so the multiplicity of $\operatorname{div}(p)$ at (Y_0, Z_0) is at most two. By Lemma 10.1(1), $((\mathbb{C}^*)^2, \frac{1}{2}\operatorname{div}(p))$ is lc.

Assume that $\epsilon = 0$. Then $p = A_0(Z)Y + B_0(Z)$, and A_0 and B_0 are not both zero. Fix a point $(Y_0, \zeta) \in (\mathbb{C}^*)^2$, let $m_A := \mathrm{ord}_\zeta A_0$ and $m_B := \mathrm{ord}_\zeta B_0$, where the order of the zero polynomial is $+\infty$, and let $m := \min\{m_A, m_B\}$. If $A_0 \neq 0$, then $m_A \leq 2$. If $A_0 = 0$, then $B_0 \neq 0$, and $m_B \leq 2$ since $\zeta \neq 0$. Thus $m \leq 2$. Write $p = (Z-\zeta)^m Q$ with $Q = A_1(Z)Y + B_1(Z)$, where $A_1 := A_0/(Z-\zeta)^m$ and $B_1 := B_0/(Z-\zeta)^m$ are polynomials. If $m_A = m$, then $A_1(\zeta) \neq 0$, so near (Y_0, ζ), the divisor $\operatorname{div}(Q)$ is either zero or the smooth curve $Y = -B_1(Z)/A_1(Z)$, which is transverse to the line $(Z = \zeta)$. Thus near (Y_0, ζ), the $\mathbb{Q}$-divisor

$$\frac{1}{2}\operatorname{div}(p) = \frac{m}{2}(Z = \zeta) + \frac{1}{2}\operatorname{div}(Q)$$

has simple normal crossing support and coefficients at most one, so the pair is lc near (Y_0, ζ) [KM98, Corollary 2.31(3)]. If $m_A > m$, then $m_B = m$ and $Q(Y, \zeta) = B_1(\zeta) \neq 0$, so near (Y_0, ζ) we have $\frac{1}{2}\operatorname{div}(p) = \frac{m}{2}(Z = \zeta)$, and the pair is lc there since $\frac{m}{2} \leq 1$. Thus $((\mathbb{C}^*)^2, \frac{1}{2}\operatorname{div}(p))$ is lc. □

Proposition 12.9. *Notation and conditions as in Set-up 12.2, and assume that the quartic part of $g(0, z, t)$ is a non-zero fourth power of a linear form. Then there exist a faithful diagonal action of G on $\mathbb{A}^4$ and a polynomial $h_0(y, z, t)$ such that $X_0 := (x^2 + h_0 = 0)$ is an isolated Gorenstein terminal threefold germ at 0 with G-semi-invariant equation,* $\mathrm{mld}((X_0 \ni 0, 0)/H) = \mathrm{mld}((X \ni o, 0)/H)$ *for any subgroup $H \subset G$, and $((\mathbb{C}^*)^3, \frac{1}{2}\operatorname{div}(\mathrm{in}_u(h_0)))$ is lc for any $u \in \mathbb{R}^3_{>0}$. In particular, the set of such numbers* $\mathrm{mld}((X \ni o, 0)/G) \in (0, 1)$ *satisfies the ACC.*

Proof. Let $\chi_x, \chi_y, \chi_z, \chi_t$ be the characters of x, y, z, t, let $q(z, t)$ be the quartic part of $g(0, z, t)$, and write $q = c\ell^4$ with $c \in \mathbb{C}^*$ and $\ell = c_z z + c_t t$ a linear form.

Step 1. In this step we reduce to the case $q = z^4$. By Lemma 12.3(1), $\sigma^* q = \chi_x(\sigma)^2 q$ for any $\sigma \in G$, so $(\sigma^*\ell)^4 = \chi_x(\sigma)^2 \ell^4$. Since $\mathbb{C}[z, t]$ is factorial, $\sigma^*\ell \in \mathbb{C}^*\ell$, that is, ℓ is G-semi-invariant. If $c_z \neq 0$, comparing the coefficients of z in $\sigma^*\ell$ shows that ℓ has character χ_z, and we replace z with $c^{1/4}\ell$. If $c_z = 0$, then $\ell = c_t t$, and we replace z and t with $c^{1/4}\ell$ and z. In both cases x, y, and the new coordinates z, t satisfy condition (2) of Set-up 12.2, and the new quartic part of $g(0, z, t)$ is z^4. Possibly replacing z and t in this way, we may assume that $q = z^4$.

Step 2. In this step we construct a G-equivariant automorphism φ of $\mathbb{C}[[y,z,t]]$ with $\varphi(t)=t$ such that

$$\varphi(g)=y^3+a(z,t)y+b(z,t),\qquad a=a_2(t)z^2+a_1(t)z+a_0(t),\qquad b=z^4+b_2(t)z^2+b_1(t)z+b_0(t)$$

for power series $a_0,a_1,a_2,b_0,b_1,b_2\in t\mathbb{C}[[t]]$.

We use the w-weight of Definition 2.16 with $w=(4,3,3)$ in y,z,t: the $(4,3,3)$-weight of $y^iz^jt^k$ is $4i+3j+3k$. A monomial $y^iz^jt^k$ of degree d has $(4,3,3)$-weight $3d+i$. The monomials occurring in g other than y^3 have degree at least four, and those of degree four with $i=0$ are the monomials of $q=z^4$. Hence every monomial occurring in $g-y^3-z^4$ has $(4,3,3)$-weight at least 13. Let $\mathfrak{B}$ be the set of monomials $y^iz^jt^k$ with $i\le 1$ and $j\le 2$. Any monomial not in $\mathfrak{B}$ is divisible by y^2 or by z^3. By Lemma 12.3(1), any monomial occurring in g has character $\chi_x^2=\chi_y^3=\chi_z^4$.

Let $\varphi_{13}:=\mathrm{id}_{\mathbb{C}[[y,z,t]]}$. For an integer $N\ge 13$, assume that φ_N is a G-equivariant endomorphism of $\mathbb{C}[[y,z,t]]$ with $\varphi_N(t)=t$, $\varphi_N(y)-y$ of $(4,3,3)$-weight at least 5, and $\varphi_N(z)-z$ of $(4,3,3)$-weight at least 4, such that

$$\varphi_N(g)=y^3+z^4+b_N+r_N,$$

where b_N is a power series of $(4,3,3)$-weight at least 13 whose monomials belong to $\mathfrak{B}$, and r_N is a power series of $(4,3,3)$-weight at least N. Write the sum of the terms of r_N of $(4,3,3)$-weight N whose monomials are not in $\mathfrak{B}$ as y^2P+z^3Q, where P and Q are polynomials whose monomials have $(4,3,3)$-weights $N-8$ and $N-9$, respectively. Since φ_N is G-equivariant, any monomial occurring in $\varphi_N(g)$ has character χ_x^2, so P and Q are zero or G-semi-invariant with characters χ_y and χ_z, respectively. Let τ_N be the G-equivariant endomorphism of $\mathbb{C}[[y,z,t]]$ with

$$\tau_N(y)=y-\frac{1}{3}P,\qquad \tau_N(z)=z-\frac{1}{4}Q,\qquad \tau_N(t)=t.$$

Since P and Q have $(4,3,3)$-weights $N-8=4+(N-12)$ and $N-9=3+(N-12)$, for any monomial M of $(4,3,3)$-weight w_M, the power series $\tau_N(M)-M$ is of $(4,3,3)$-weight at least w_M+N-12. Moreover,

$$\tau_N(y^3)=y^3-y^2P+\frac{1}{3}yP^2-\frac{1}{27}P^3,\qquad \tau_N(z^4)=z^4-z^3Q+\frac{3}{8}z^2Q^2-\frac{1}{16}zQ^3+\frac{1}{256}Q^4,$$

and the terms after the second ones are of $(4,3,3)$-weight at least $2N-12\ge N+1$, since $N\ge 13$. Since any monomial occurring in b_N+r_N has $(4,3,3)$-weight at least 13, the series $\tau_N(b_N+r_N)-(b_N+r_N)$ is of $(4,3,3)$-weight at least $N+1$. Thus

$$\tau_N(\varphi_N(g))=y^3+z^4+b_{N+1}+r_{N+1},$$

where b_{N+1} is b_N plus the terms of r_N of $(4,3,3)$-weight N whose monomials belong to $\mathfrak{B}$, and r_{N+1} is of $(4,3,3)$-weight at least $N+1$. Let $\varphi_{N+1}:=\tau_N\circ\varphi_N$. Since $\varphi_N(y)$ and $\varphi_N(z)$ are of $(4,3,3)$-weight at least 4 and 3, the series $\varphi_{N+1}(y)-\varphi_N(y)$ and $\varphi_{N+1}(z)-\varphi_N(z)$ are of $(4,3,3)$-weight at least $N-8$ and $N-9$. Thus φ_{N+1} satisfies the conditions for $N+1$.

A power series of $(4,3,3)$-weight at least N belongs to $(y,z,t)^{\lceil N/4\rceil}$. Hence φ_N converges to a G-equivariant endomorphism φ of $\mathbb{C}[[y,z,t]]$ with $\varphi(t)=t$, and $\varphi(g)-y^3-z^4$ is a power series of $(4,3,3)$-weight at least 13 whose monomials belong to $\mathfrak{B}$. Since $\varphi(y)-y$ is of $(4,3,3)$-weight at least 5 and $\varphi(z)-z$ is of $(4,3,3)$-weight at least 4, we have $\varphi(y)-y\in(y,z,t)^2$ and $\varphi(z)-z-c'y\in(y,z,t)^2$ for some $c'\in\mathbb{C}$, where $c'\ne 0$ only if $\chi_y=\chi_z$. Thus the linear part of φ is invertible, and φ is an automorphism. Writing $\varphi(g)=y^3+ay+b$ with a and b as in the statement of **Step 2**, a monomial yz^jt^k of $(4,3,3)$-weight at least 13 with $j\le 2$ satisfies $k\ge 3-j\ge 1$, and a monomial z^jt^k of $(4,3,3)$-weight at least 13 with $j\le 2$ satisfies $k\ge 5-j\ge 3$. Thus $a_0,a_1,a_2,b_0,b_1,b_2\in t\mathbb{C}[[t]]$.

Step 3. In this step we construct, by Lemma 12.8, a G-equivariant translation ψ_1 of z. Let $h:=\varphi(g)=y^3+a(z,t)y+b(z,t)$, and extend φ to a G-equivariant automorphism of $\mathfrak{R}:=\mathbb{C}[[x,y,z,t]]$ with $\varphi(x)=x$. Then $x^2+h=\varphi(x^2+g)$ is G-semi-invariant with character

χ_x^2. By Lemma 12.8 applied to x^2+h, $\alpha=a$, and $\beta=b$, there exists $\theta\in t\mathbb{C}[[t]]$, either zero or G-semi-invariant with character χ_z, such that the power series

$$h'(y,z,t) := y^3 + a(z+\theta(t),t)y + b(z+\theta(t),t)$$

satisfies: $((\mathbb{C}^*)^3, \frac{1}{2}\operatorname{div}(\mathrm{in}_u(h')))$ is lc for any $u\in\mathbb{R}^3_{>0}$, where in_u is taken with respect to y,z,t. Let ψ_1 be the automorphism of $\mathfrak{R}$ with $\psi_1(z)=z+\theta(t)$ and $\psi_1(x)=x$, $\psi_1(y)=y$, $\psi_1(t)=t$. Then ψ_1 is G-equivariant and $\psi_1(x^2+h)=x^2+h'$.

Step 4. In this step we construct h_0. Let $\psi:=\psi_1\circ\varphi$ and $v:=1$, so that $v\psi(x^2+g)=x^2+h'$. By Lemma 12.3(2), there exists an integer $L\geq 4$, larger than the degrees of all componentwise minimal exponents of h', such that for $h_0:=h'_{<L}$ and $f_0:=x^2+h_0$, the following hold: G acts faithfully and diagonally on $\mathbb{A}^4$, $X_0=(f_0=0)$ is an isolated Gorenstein terminal threefold germ at 0, G acts faithfully on X_0, and $\mathrm{mld}((X_0\ni 0,0)/H)=\mathrm{mld}((X\ni o,0)/H)$ for any subgroup $H\subset G$. The equation f_0 is G-semi-invariant with character χ_x^2, since it is a truncation of x^2+h'. By Definition-Lemma 7.3, $\mathrm{in}_u(h_0)=\mathrm{in}_u(h')$ for any $u\in\mathbb{R}^3_{>0}$, and $((\mathbb{C}^*)^3,\frac{1}{2}\operatorname{div}(\mathrm{in}_u(h_0)))$ is lc for any $u\in\mathbb{R}^3_{>0}$. This proves the first part of the proposition.

Step 5. In this step we prove the ACC. We take $(b_y,b_z,b_t)=(0,0,0)$, $h=h_0$, $X=X_0$, and $B=0$ in Set-up 10.3. Since $\operatorname{ord} g\geq 3$ and ψ preserves orders, $\operatorname{ord} h'\geq 3$. Thus h_0 is a non-zero polynomial vanishing at 0, as y^3 occurs in h_0. The germ X_0 is normal and klt since it is terminal, and G acts diagonally and faithfully on X_0 and on $\mathbb{A}^4$ with semi-invariant equation. Thus the conditions of Set-up 10.3 hold, and by **Step 4** the assumption of Proposition 10.5 holds. By Proposition 10.5, applied with the finite set $\{(0,0,0)\}$, the set of numbers $\mathrm{mld}((X_0\ni 0,0)/G)\in(0,1)$ obtained in this way satisfies the ACC. Since $\mathrm{mld}((X\ni o,0)/G)=\mathrm{mld}((X_0\ni 0,0)/G)$ by **Step 4**, the set of such numbers $\mathrm{mld}((X\ni o,0)/G)\in(0,1)$ satisfies the ACC. □

12.3. **A quartic with two double roots.** In this subsection, we prove Proposition 12.10, which treats one of the cE cases of Theorem 12.1: the quartic part of the equation on the plane $(y=0)$ has two double roots. We extract the prime divisor of weights $(6,4,3,3)$, pass to root covers of the open subsets $(x_i\neq 0)$ of the weighted blow-up, and come back to X/G through Lemma 3.1.

Proposition 12.10. *Notation and conditions as in Set-up 12.2, and assume that the quartic part of $g(0,z,t)$ is $cL_1^2L_2^2$ for a number $c\neq 0$ and linearly independent linear forms L_1,L_2 in z,t. Then the set of such numbers $\mathrm{mld}((X\ni o,0)/G)\in(0,1)$ satisfies the ACC.*

Proof of Proposition 12.10. The proof is long, so we divide it into several steps. Let $a:=\mathrm{mld}((X\ni o,0)/G)$ and assume that $a\in(0,1)$.

Step 1. In this step we replace $(X\ni o)$ with a polynomial model and compute the leading form of the equation for the weights $d:=(6,4,3,3)$.

By Lemma 12.3(2) with $\psi=\mathrm{id}$ and $v=1$, there exists an integer $L\geq 5$ such that $f_0:=x^2+g_{<L}$ defines an isolated Gorenstein terminal threefold germ $X_0\subset\mathbb{A}^4$ at 0, G acts faithfully on $\mathbb{A}^4$ and on X_0, $\mathrm{mld}((X_0\ni 0,0)/G)=\mathrm{mld}((X\ni o,0)/G)$, and $(X_0\ni 0)$, G, x,y,z,t, and $g_{<L}$ satisfy the conditions of Set-up 12.2. Since $L\geq 5$, the terms of degree at most four of $g_{<L}$ are those of g. Possibly replacing $(X\ni o)$ with $(X_0\ni 0)$ and g with $g_{<L}$, we may assume that $X=(f=0)\subset\mathbb{A}^4$, where $f=x^2+g$ and g is a polynomial satisfying Set-up 12.2(2) and the assumption of Proposition 12.10.

Write $x_1,x_2,x_3,x_4:=x,y,z,t$, let χ_j be the character of x_j, and let g_4 be the quartic part of g. By Set-up 12.2(2), the cubic part of g is y^3. The d-weight of $y^az^bt^c$ is $4a+3(b+c)=3n+a$, where $n:=a+b+c$. Since $\operatorname{ord} g\geq 3$, the monomials of g of d-weight at most 12 are y^3 and the monomials of $g_4(0,z,t)$, all of d-weight 12. Moreover, a monomial of g of d-weight 13 has $n=4$ and $a=1$, so it is divisible by y. Thus

$$\mathrm{wt}_d(f)=12 \quad\text{and}\quad f_{12}:=\mathrm{in}_d(f)=x^2+y^3+cL_1^2L_2^2. \tag{12.1}$$

Step 2. In this step we construct the prime divisor E_0 of weights d, the root covers $\widetilde{X}_i$, and the boundary $-3\operatorname{div}(\tau)$ on them.

Let $h : \widehat{\mathbb{A}} \to \mathbb{A}^4$ be the weighted blow-up of the origin with weights d, D the exceptional divisor of h, and $\widehat{X}$ the strict transform of X. Then $D \cong \mathbb{P}(6,4,3,3)$ and $\widehat{X} \cap D = (f_{12} = 0) \subset D$. The polynomial f_{12} is irreducible, because $y^3 + cL_1^2L_2^2$ is not a square, and none of x, y, z, t divides f_{12}. Since $\gcd(6,4,3,3) = 1$, $\widehat{\mathbb{A}}$ is smooth at the points of D at which all coordinates are non-zero. Thus $\widehat{X} \cap D$ is a prime divisor E_0 on $\widehat{X}$, and $\widehat{X}$ is regular at the generic point of E_0, since D is Cartier there and $\widehat{X} \cap D$ is reduced. By [CLS11, Lemma 11.4.10], $a(D, \mathbb{A}^4, 0) = \sum_j d_j = 16$, and $\operatorname{ord}_D(f) = \operatorname{wt}_d(f) = 12$. A direct calculation with the adjunction formula near the generic point of E_0 shows that

$$a(E_0, X, 0) = 16 - 12 = 4.$$

Fix $i \in \{1,2,3,4\}$ and let $j_1 < j_2 < j_3$ be the other indices. Let $\varphi_i : \mathbb{A}^4_{\tau,\xi} \to \mathbb{A}^4$ be the morphism from the affine space with coordinates $\tau, \xi_{j_1}, \xi_{j_2}, \xi_{j_3}$ given by $x_i = \tau^{d_i}$ and $x_j = \tau^{d_j}\xi_j$ for $j \neq i$. Then φ_i factors through the quotient morphism q_i of $\mathbb{A}^4_{\tau,\xi}$ by $\mu_{d_i}(1, -d_{j_1}, -d_{j_2}, -d_{j_3})$ onto the open subset $U_i := (x_i \neq 0)$ of $\widehat{\mathbb{A}}$. Here, with $e_1, \dots, e_4$ the standard basis of $\mathbb{Z}^4$, U_i is the affine toric variety of the simplicial cone generated by d and e_j $(j \neq i)$, the sublattice of $\mathbb{Z}^4$ generated by d and e_j $(j \neq i)$ has index d_i, and the quotient of $\mathbb{Z}^4$ by it is generated by the class of $e_i = \frac{1}{d_i}(d - \sum_{j \neq i} d_j e_j)$. In the coordinates $\tau = x_i^{1/d_i}$ and $\xi_j = x_j\tau^{-d_j}$ dual to d, e_j, the class of e_i acts as $\mu_{d_i}(1, -d_{j_1}, -d_{j_2}, -d_{j_3})$ [CLS11, Proposition 1.3.18(c) and Example 1.3.20]. By (12.1), $\varphi_i^* f = \tau^{12}\tilde{f}_i$ for a polynomial $\tilde{f}_i$ not divisible by τ, and $\tilde{f}_i|_{\tau=0} = f_{12}(\xi_1, \dots, \xi_{i-1}, 1, \xi_{i+1}, \dots, \xi_4)$:

$$\tilde{f}_1|_{\tau=0} = 1 + \xi_2^3 + g_4(0, \xi_3, \xi_4), \qquad \tilde{f}_2|_{\tau=0} = \xi_1^2 + 1 + g_4(0, \xi_3, \xi_4),$$
$$\tilde{f}_3|_{\tau=0} = \xi_1^2 + \xi_2^3 + g_4(0, 1, \xi_4), \qquad \tilde{f}_4|_{\tau=0} = \xi_1^2 + \xi_2^3 + g_4(0, \xi_3, 1).$$

The polynomial $\tilde{f}_1|_{\tau=0}$ is irreducible, because it is a monic cubic in ξ_2 whose constant term has degree four, so it has no root in $\mathbb{C}[\xi_3, \xi_4]$. The other three are irreducible, because they have the form $\xi_1^2 + e$, where e is not a square, since either e has degree three in ξ_2, or $e - 1$ is a non-zero quartic form, in which case the terms of degree one and two of a square root of e would vanish, so the square root would be ± 1 and $e = 1$, a contradiction.

Let $\widetilde{X}_i := (\tilde{f}_i = 0) \subset \mathbb{A}^4_{\tau,\xi}$ and $S_i := \widetilde{X}_i \cap (\tau = 0)$, an integral surface. Since τ does not divide $\tilde{f}_i$, $\widetilde{X}_i = q_i^{-1}(\widehat{X} \cap U_i)$ and $S_i = q_i^{-1}(E_0 \cap U_i) \neq \emptyset$. We show that $\widetilde{X}_i$ is integral. Since $\widehat{X} \cap U_i$ is irreducible, μ_{d_i} permutes the irreducible components of $\widetilde{X}_i$ transitively. Since $(\tau = 0)$ is μ_{d_i}-stable and $S_i \neq \emptyset$, any irreducible component of $\widetilde{X}_i$ intersects $(\tau = 0)$ in a closed subset of S_i of pure dimension two, that is, in S_i. Since the Cartier divisor $S_i = (\tau = 0)$ of $\widetilde{X}_i$ is reduced, hence generically smooth, $\widetilde{X}_i$ is smooth at the generic point of S_i, so only one irreducible component of $\widetilde{X}_i$ contains S_i. Thus $\widetilde{X}_i$ is integral, μ_{d_i} acts faithfully on $\widetilde{X}_i$ with quotient $\widehat{X} \cap U_i$, and $\mathbb{C}(\widetilde{X}_i) = \mathbb{C}(X)(\tau)$ with $\tau^{d_i} = x_i$.

Let

$$\widetilde{G} := \{(\gamma, \zeta) \in G \times \mathbb{C}^* \mid \zeta^{d_i} = \chi_i(\gamma)\},$$

acting on $\mathbb{A}^4_{\tau,\xi}$ by $(\gamma,\zeta)^*\tau = \zeta\tau$ and $(\gamma,\zeta)^*\xi_j = \chi_j(\gamma)\zeta^{-d_j}\xi_j$. Then φ_i is equivariant with respect to the projection $\widetilde{G} \to G$, which is surjective with kernel μ_{d_i}, and $\tilde{f}_i$ is $\widetilde{G}$-semi-invariant. Thus the finite abelian group $\widetilde{G}$ acts on $\widetilde{X}_i$. The action is faithful, since G acts faithfully on $\mathbb{C}(X)$ and μ_{d_i} acts faithfully on $\widetilde{X}_i$. Moreover, $\mathbb{C}(\widetilde{X}_i)^{\widetilde{G}} = \mathbb{C}(X)^G = \mathbb{C}(X/G)$. Hence $\mathbb{C}(\widetilde{X}_i)$ is a Galois extension of $\mathbb{C}(X/G)$ with group $\widetilde{G}$.

Let Ω_X be the Poincaré residue of $dx_1 \wedge dx_2 \wedge dx_3 \wedge dx_4/f$ along X, and Ω_i that of $d\tau \wedge d\xi_{j_1} \wedge d\xi_{j_2} \wedge d\xi_{j_3}/\tilde{f}_i$ along $\widetilde{X}_i$. They generate ω_X and $\omega_{\widetilde{X}_i}$, by adjunction for hypersurfaces (cf. [KM98, Proposition 5.73]). Since

$$\varphi_i^*(dx_1 \wedge dx_2 \wedge dx_3 \wedge dx_4) = \pm d_i\tau^{15} d\tau \wedge d\xi_{j_1} \wedge d\xi_{j_2} \wedge d\xi_{j_3} \quad \text{and} \quad \varphi_i^* f = \tau^{12}\tilde{f}_i,$$

a direct calculation shows

$$\varphi_i^*\Omega_X = \pm d_i\tau^3\Omega_i. \tag{12.2}$$

Let $\pi : X \to X/G$ be the quotient morphism. By (12.2) and Lemma 2.6(0),

$$K_{\widetilde{X}_i} - 3\operatorname{div}(\tau) = \varphi_i^*K_X = (\pi\circ\varphi_i)^*(K_{X/G} + B_{X/G}). \tag{12.3}$$

Step 3. In this step we show that $\widetilde{X}_i$ is normal and klt near S_i, and we determine the closed points of S_i at which $\widetilde{X}_i$ is singular.

Let $\Sigma_i \subset S_i$ be the set of closed points at which all partial derivatives of $\tilde{f}_i|_{\tau=0}$ vanish. At a closed point $p \in S_i\backslash\Sigma_i$, S_i is smooth, so $\widetilde{X}_i$ is smooth at p and τ is a regular parameter of $\mathcal{O}_{\widetilde{X}_i,p}$.

- If $i \in \{1,2\}$, then $\Sigma_i = \emptyset$, because at a common zero of the partial derivatives, $\xi_2 = 0$ (resp. $\xi_1 = 0$) and, by the Euler formula, $g_4(0,\xi_3,\xi_4) = 0$, so $\tilde{f}_i|_{\tau=0}$ takes the value 1.
- If $i = 4$, let $b(\xi_3) := g_4(0,\xi_3,1) = cL_1(\xi_3,1)^2L_2(\xi_3,1)^2$. Since L_1, L_2 are linearly independent, any root of b has multiplicity two. Thus Σ_4 consists of the points with $\tau = \xi_1 = \xi_2 = 0$ at which ξ_3 is a root of b, and $|\Sigma_4| \le 2$. For $i = 3$, exchanging ξ_3 and ξ_4 and setting $b(\xi_4) := g_4(0,1,\xi_4)$, we obtain that Σ_3 consists of the points with $\tau = \xi_1 = \xi_2 = 0$ at which ξ_4 is a root of b, and $|\Sigma_3| \le 2$.

Let $p \in \Sigma_i$, so $i \in \{3,4\}$. Let f_{13} be the sum of the terms of f of d-weight 13. Then $\partial_\tau\tilde{f}_i|_{\tau=0} = f_{13}(\xi_1,\xi_2,\xi_3,\xi_4)|_{\xi_i=1}$, so it is divisible by ξ_2 by **Step 1**, and it vanishes at p. Hence $\tilde{f}_i$ vanishes to order at least two at p, and $\operatorname{Sing}(\widetilde{X}_i)\cap(\tau=0) = \Sigma_i$.

Only the germ of $\widetilde{X}_i$ along S_i is used below. Since Σ_i is finite, any irreducible component Z of $\operatorname{Sing}(\widetilde{X}_i)$ intersecting $(\tau = 0)$ satisfies $\dim Z \le 1$. Thus, possibly replacing $\widetilde{X}_i$ with the complement of the irreducible components of $\operatorname{Sing}(\widetilde{X}_i)$ disjoint from $(\tau = 0)$, which is a $\widetilde{G}$-stable open neighborhood of S_i, we may assume that $\dim Z \le 1$ for any irreducible component Z of $\operatorname{Sing}(\widetilde{X}_i)$, so that $\widetilde{X}_i$ is regular in codimension one. Since $\widetilde{X}_i$ is a hypersurface in $\mathbb{A}^4$ and regular in codimension one, $\widetilde{X}_i$ is normal [Har77, Proposition II.8.23(b)].

Let $p \in \Sigma_4$ and $s := \xi_3 - \xi_3(p)$. Then $b(\xi_3) = s^2e(s)$ for a polynomial e with $e(0) \ne 0$. Near p, the Cartier divisor $S_4 = (\tau = 0)$ of $\widetilde{X}_4$ is the surface $(\xi_1^2 + \xi_2^3 + s^2e(s) = 0) \subset \mathbb{A}^3$, which is analytically isomorphic to $(\xi_1^2 + \xi_2^3 + s^2 = 0)$, an A_2 singularity. Thus S_4 is Du Val, that is, canonical near p [KM98, Theorem 4.20]. Since S_4 and $K_{\widetilde{X}_4}$ are Cartier, $(\widetilde{X}_4, S_4)$ is plt near p by [KM98, Theorem 5.50(1)] with $B = 0$, so $\widetilde{X}_4$ is klt near p. For $p \in \Sigma_3$, exchanging ξ_3 and ξ_4 in the argument for $p \in \Sigma_4$ shows that $\widetilde{X}_3$ is klt near p. Since $\widetilde{X}_i$ is smooth at the points of $S_i\backslash\Sigma_i$, possibly shrinking $\widetilde{X}_i$ to a $\widetilde{G}$-stable open neighborhood of S_i, we may assume that $\widetilde{X}_i$ is klt. Since subtracting the Cartier divisor $3\operatorname{div}(\tau) \ge 0$ increases log discrepancies, $(\widetilde{X}_i, -3\operatorname{div}(\tau))$ is sub-klt.

Step 4. In this step we lift a prime divisor computing a to a root cover and show that a is the minimal log discrepancy of an inertia quotient of $(\widetilde{X}_i, -3\operatorname{div}(\tau))$, unless $a = 4/m$ for a positive integer m.

Let $\bar{o} := \pi(o)$. Since X is terminal, $(X/G, B_{X/G})$ is a klt pair near $\bar{o}$ by Lemma 2.6(0)(2)(3). Let $\mu : W \to X/G$ be a log resolution of $(X/G, B_{X/G})$ such that $\mu^{-1}(\bar{o})$ is a divisor and $\mu^{-1}(\bar{o}) \cup \operatorname{Exc}(\mu) \cup \operatorname{Supp}\mu_*^{-1}B_{X/G}$ is an snc divisor, and let $K_W + B_W = \mu^*(K_{X/G} + B_{X/G})$. Since $(X/G, B_{X/G})$ is klt, every coefficient of B_W is less than one, so a is the minimum of the log discrepancies of the components of $\mu^{-1}(\bar{o})$ [Amb99, Section 1, the second paragraph after Definition 1.1]. Thus there exists a prime divisor E over X/G with $\operatorname{center}_{X/G}E = \{\bar{o}\}$ and $a(E, X/G, B_{X/G}) = a$. By Lemma 2.6(3)(1)(2), there exists a prime divisor F over X with $\operatorname{ord}_F|_{\mathbb{C}(X/G)} = m\operatorname{ord}_E$, where $m = |I_G(F)|$, $\operatorname{center}_X F = \{o\}$, and $a(F, X, 0) = ma$.

Since h is proper, $V := \operatorname{center}_{\widehat{X}} F$ exists, and $h(V) = \{o\}$, so $V \subset \widehat{X} \cap D = E_0$. If $V = E_0$, then $F = E_0$ because $\widehat{X}$ is regular at the generic point of E_0, and Lemma 2.6(2) implies $4 = a(E_0, X, 0) = ma$, that is, $a = 4/m$.

Assume that $\dim V \le 1$, and choose i such that the generic point of V belongs to U_i. Lemma 2.6(3), applied to the normalization of X in $\mathbb{C}(\widetilde{X}_i)$ with the action of μ_{d_i}, implies that there exists a prime divisor $\widetilde{F}$ over $\widetilde{X}_i$ with $\operatorname{ord}_{\widetilde{F}}|_{\mathbb{C}(X)} = \varrho \operatorname{ord}_F$ for a positive integer ϱ. Since q_i is finite on the hypersurface $(\tilde{f}_i = 0) = q_i^{-1}(\widehat{X} \cap U_i)$ and the center of F on $\widehat{X} \cap U_i$ is $V \cap U_i$, the center of $\widetilde{F}$ on $(\tilde{f}_i = 0)$ exists, maps onto $V \cap U_i$, and is contained in $q_i^{-1}(E_0 \cap U_i) = S_i$, which is contained in $\widetilde{X}_i$. Thus $\widetilde{V} := \operatorname{center}_{\widetilde{X}_i} \widetilde{F}$ exists, $q_i(\widetilde{V}) = V \cap U_i$, $\widetilde{V} \subset S_i$, and $\dim \widetilde{V} = \dim V \le 1$. Let $I := I_{\widetilde{G}}(\widetilde{F})$. Since I preserves $\operatorname{ord}_{\widetilde{F}}$ and acts trivially on the residue field of $\operatorname{ord}_{\widetilde{F}}$, which contains the function field of $\widetilde{V}$, I fixes $\widetilde{V}$ pointwise.

In Lemma 3.1, take the pair $(X \ni x, B)$ to be $(X/G \ni \bar{o}, B_{X/G})$, the prime divisor E to be the prime divisor E over X/G with $a(E, X/G, B_{X/G}) = a$, the cover X' to be the normalization of X/G in $\mathbb{C}(\widetilde{X}_i)$ with Galois group $\widetilde{G}$ (**Step 2**), the prime divisor F to be $\widetilde{F}$, $H := I$, and $Y := \widetilde{X}_i$. Here $\widetilde{X}_i$ is normal (**Step 3**), and the $\widetilde{G}$-equivariant birational morphism $\widetilde{X}_i \to X'$ exists by the universal property of the normalization. By (12.3), the boundary B_Y of the lemma is $-3\operatorname{div}(\tau)$. Let $(\widetilde{X}_i/I, B_{\widetilde{X}_i/I}) := (\widetilde{X}_i, -3\operatorname{div}(\tau))/I$ and let η be the generic point of the image of $\widetilde{V}$ in $\widetilde{X}_i/I$. Lemma 3.1 implies

$$\operatorname{mld}(\widetilde{X}_i/I \ni \eta, B_{\widetilde{X}_i/I}) = a. \tag{12.4}$$

Step 5. In this step we compute the left-hand side of (12.4) in the three possible cases for the center $\widetilde{V}$ of $\widetilde{F}$ on the root cover $\widetilde{X}_i$ of **Step 2**, and we conclude.

Case 1: $\widetilde{V} = \{p\}$ for a closed point $p \in S_i \backslash \Sigma_i$. By **Step 3**, $\widetilde{X}_i$ is smooth at p and τ is a regular parameter of $\mathcal{O}_{\widetilde{X}_i,p}$. Since I is abelian, fixes p, and τ is I-semi-invariant, there exist $u_2, u_3 \in \mathcal{O}_{\widetilde{X}_i,p}$ such that τ, u_2, u_3 are I-eigenparameters at p: complete the class of τ to a basis of $\mathfrak{m}_p/\mathfrak{m}_p^2$ consisting of eigenvectors of I, and lift each of the other two basis vectors, with character χ, to the χ-component $\frac{1}{|I|}\sum_{\gamma \in I} \chi(\gamma)^{-1}\gamma^* u$ of any lift $u \in \mathfrak{m}_p$ of it. This χ-component belongs to $\mathfrak{m}_p$, and its class in $\mathfrak{m}_p/\mathfrak{m}_p^2$ is the class of u, since I fixes p and taking the χ-component commutes with $\mathfrak{m}_p \to \mathfrak{m}_p/\mathfrak{m}_p^2$. By Lemma 3.4 with $B = -3\operatorname{div}(\tau)$,

$$a = \operatorname{mld}((\widetilde{X}_i \ni p, -3\operatorname{div}(\tau))/I) = \min\{4w_1 + w_2 + w_3 \mid w \in N_I \cap \mathbb{R}^3_{>0}\}.$$

Case 2: $\widetilde{V} = \{p\}$ for a point $p \in \Sigma_i$. Then $i \in \{3, 4\}$, and we assume $i = 4$. For $i = 3$, exchange ξ_3 and ξ_4 in what follows. Let $s = \xi_3 - \xi_3(p)$ be as in **Step 3**. If $\xi_3(p) \ne 0$, then the character of ξ_3 is trivial on I, since I fixes p, so s is I-invariant. If $\xi_3(p) = 0$, then $s = \xi_3$. Thus ξ_1, s, ξ_2, τ are I-eigenparameters of $\mathbb{A}^4_{\tau,\xi}$ at p. By **Steps 2** and **3**, $\tilde{f}_4$ is I-semi-invariant and vanishes to order at least two at p, $\widetilde{X}_4$ is a normal klt germ at p on which I acts faithfully, and $(\widetilde{X}_4, -3\operatorname{div}(\tau))$ is sub-klt. The restriction of the quadratic part of $\tilde{f}_4$ at p to the (ξ_1, s)-plane is the quadratic part of $\xi_1^2 + s^2 e(s)$, that is, $\xi_1^2 + e(0)s^2$ with $e(0) \ne 0$, which is non-degenerate. Thus, by Corollary 9.6 for $(\mathbb{A}^4_{\tau,\xi} \ni p)$, the group I, the function $\tilde{f}_4$, the eigenparameters (u_1, u_2, z, t) of the corollary taken to be (ξ_1, s, ξ_2, τ), and $(b_z, b_t) = (0, -3)$, the number $a = \operatorname{mld}((\widetilde{X}_4 \ni p, -3\operatorname{div}(\tau))/I)$ belongs to the set of Corollary 9.6 for $\mathcal{B} := \{(0, -3)\}$.

Case 3: $\widetilde{V}$ is a curve C. Let η_C be the generic point of C. Since Σ_i is finite, **Step 3** implies that $\mathcal{O}_{\widetilde{X}_i,\eta_C}$ is a regular local ring of dimension two of which τ is a regular parameter. Let σ_0 be a regular function on $\widetilde{X}_i$ whose restriction to C is not constant, and let

$$\sigma := \frac{1}{|I|}\sum_{\gamma \in I} \gamma^* \sigma_0.$$

Then σ is I-invariant, and $\sigma|_C = \sigma_0|_C$ is not constant, since I fixes C pointwise. Let $k := \mathbb{C}(\sigma)$, let X_k be the generic fiber of σ over k, $c \in X_k$ the closed point corresponding to η_C, and $B_k := -3\operatorname{div}_{X_k}(\tau)$. By Lemma 3.6(3) with $(X, B) := (\widetilde{X}_i, -3\operatorname{div}(\tau))$, $t := \sigma$, and $H := I$, the group I acts on X_k fixing c, and by (12.4),

$$\operatorname{mld}((X_k \ni c, B_k)/I) = \operatorname{mld}(\widetilde{X}_i/I \ni \eta, B_{\widetilde{X}_i/I}) = a.$$

Since $\mathcal{O}_{X_k,c} = \mathcal{O}_{\widetilde{X}_i,\eta_C}$, X_k is smooth at c with $\dim \mathcal{O}_{X_k,c} = 2$. The action of I on X_k is faithful and k-linear, and it is trivial on $\kappa(c) = \mathbb{C}(C)$. Thus I acts $\kappa(c)$-linearly on $\mathfrak{m}_c/\mathfrak{m}_c^2$, and, as in Case 1, there exists $u \in \mathcal{O}_{X_k,c}$ such that τ, u are I-eigenparameters at c. By Lemma 3.4 over k (see Remark 2.17),

$$a = \min\{4w_1 + w_2 \mid w \in N_I \cap \mathbb{R}^2_{>0}\}.$$

Let $\Lambda_1 := \{4/m \mid m \in \mathbb{Z}_{\geq 1}\}$, let Λ_2 be the set of Corollary 9.6 for $\mathcal{B} = \{(0, -3)\}$, and let Λ_3 and Λ_4 be the sets of numbers $\min\{4w_1+w_2+w_3 \mid w \in N_H \cap \mathbb{R}^3_{>0}\}$ and $\min\{4w_1+w_2 \mid w \in N_H \cap \mathbb{R}^2_{>0}\}$, where H runs over the finite subgroups of $\mathbb{T}^3$ and of $\mathbb{T}^2$, respectively. By **Steps 4** and 5, $a \in \Lambda_1 \cup \Lambda_2 \cup \Lambda_3 \cup \Lambda_4$, and $\Lambda_1, \dots, \Lambda_4$ depend neither on $(X \ni o)$ nor on G. The set Λ_1 satisfies the ACC, because a strictly increasing sequence $4/m_1 < 4/m_2 < \cdots$ would yield a strictly decreasing sequence of positive integers $m_1 > m_2 > \cdots$. The set Λ_2 satisfies the ACC by Corollary 9.6. The sets Λ_3 and Λ_4 satisfy the ACC by Corollary 4.6, applied with $s = 1$, $\Lambda = \{0\}$, $\beta_1 = 0$, and $\mathcal{V} = \{(4, 1, 1)\}$, resp. $\mathcal{V} = \{(4, 1)\}$. Since a finite union of sets satisfying the ACC satisfies the ACC, the proof is complete. □

12.4. **Signed transverse D_4 and E_6 sections.** In this subsection, we prove Proposition 12.11, which treats threefold hypersurface germs whose section by an eigenparameter t is a Du Val singularity of type D_4 or E_6, with the boundary $b\operatorname{div}(t)$ for a fixed $b \leq 0$. Proposition 12.11 is used in Subsection 12.5 at singular points of fixed weighted blow-ups, where the coefficient of $\operatorname{div}(t)$ is fixed. Thus any change of coordinates below fixes t.

Proposition 12.11. *Fix a finite set $\mathcal{B} \subset \mathbb{Q}_{\leq 0}$. Let $(X \ni o)$ be a normal klt threefold hypersurface germ, smooth outside o, in a smooth fourfold germ $(Y \ni o)$ with a faithful action of a finite abelian group G fixing o and preserving X, and let t be a G-eigenparameter of Y at o such that $X \cap (t = 0)$ is a Du Val singularity of type D_4 or E_6. Let $b \in \mathcal{B}$ and $B := b\operatorname{div}_X(t)$. Then the set of such numbers $\operatorname{mld}((X \ni o, B)/G) \in (0, 1)$ satisfies the ACC. Moreover, there exist a faithful diagonal action of G on $\mathbb{A}^4$ with coordinates x, y, z, t and a polynomial $h(y, z, t)$ such that $((\mathbb{C}^*)^3, \frac{1}{2}\operatorname{div}(\operatorname{in}_u(h)))$ is lc for any $u \in \mathbb{R}^3_{>0}$, where in_u is taken with respect to y, z, t, and $X_0 := (x^2 + h = 0)$ satisfies*

$$\operatorname{mld}((X_0 \ni 0, b\operatorname{div}_{X_0}(t))/H) = \operatorname{mld}((X \ni o, B)/H)$$

for any subgroup $H \subset G$, and any such number in $(0, 1)$ equals $\min_{w \in N_G \cap \mathbb{R}^4_{>0}} P(w)$ for the data of Set-up 10.3 with X_0 in place of X, $b_y = b_z = 0$, and $b_t = b$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we show that G acts faithfully on X, and we fix an equation of X.

Since G is abelian and t is a G-eigenparameter, there exist G-semi-invariant regular functions x_1, x_2, x_3 near o such that x_1, x_2, x_3, t are G-eigenparameters of Y at o, since the class of t in $\mathfrak{m}_{Y,o}/\mathfrak{m}_{Y,o}^2$ extends to a basis of eigenvectors of G, and an eigenvector with character χ lifts to the χ-component of any lift in $\mathfrak{m}_{Y,o}$. Let

$$R := \widehat{\mathcal{O}}_{Y,o} = \mathbb{C}[[x_1, x_2, x_3, t]]$$

[Mat89, Theorem 29.7], on which G acts diagonally. For a G-semi-invariant element u of R, we denote by χ_u the character of u. Let f_1 be a generator of the ideal of X in $\mathcal{O}_{Y,o}$. The ideal (f_1) is G-stable, so G acts on the one-dimensional space $(f_1)/\mathfrak{m}_{Y,o}(f_1)$ by a character χ, and the χ-component f of f_1 generates the ideal by Nakayama's lemma. Shrinking Y, we have

$X = (f = 0)$ for the G-semi-invariant regular function f. Let $\hat{f} \in R$ be the image of f. Then $\widehat{\mathcal{O}}_{X,o} = R/(\hat{f})$, and the completion of the local ring of $X \cap (t = 0)$ at o is $R/(\hat{f}, t)$ [Mat89, Theorem 8.7].

Let $\sigma \in G$ act trivially on X, and suppose that $\sigma \neq 1$. Since G acts faithfully on Y, and hence on R, there exists $u \in \{x_1, x_2, x_3, t\}$ with $\chi_u(\sigma) \neq 1$. Since σ acts trivially on $R/(\hat{f})$,

$$(1 - \chi_u(\sigma))u = u - \sigma^* u \in (\hat{f}),$$

so $u \in (\hat{f})$. Since $\hat{f}$ is not a unit and $R/(u)$ is a domain, $(\hat{f}) = (u)$. If $u = t$, then $R/(\hat{f}, t) = R/(t)$ is of dimension three, which contradicts that $X \cap (t = 0)$ is a surface. If $u = x_i$, then $R/(\hat{f}, t) = R/(x_i, t)$ is regular, which contradicts that $X \cap (t = 0)$ is singular at o. Thus G acts faithfully on X, and so does any subgroup $H \subset G$. Since K_X and $\mathrm{div}_X(t)$ are Cartier and $\mathrm{div}_X(t)$ is G-invariant, Definition 2.5 defines $(X \ni o, B)/H$ for any subgroup $H \subset G$.

Step 2. In this step, we split off a square x^2 from $\hat{f}$ by a change of coordinates fixing t.

Let $\bar{f}$ be the image of $\hat{f}$ in $R/(t) = \mathbb{C}[[x_1, x_2, x_3]]$. Then $\bar{f}$ defines the completion of $X \cap (t = 0)$ at o. Since $X \cap (t = 0)$ is a Du Val singularity of type D_4 or E_6, $\bar{f}$ has order two, and the quadratic part of $\bar{f}$ is $\lambda \ell^2$ for some $\lambda \in \mathbb{C}^*$ and a linear form ℓ in x_1, x_2, x_3 [KM98, Theorem 4.20], because the rank of the quadratic part of an equation of order two depends neither on the equation nor on the coordinates, and the normal forms $x^2 + y^2 z + z^3$ and $x^2 + y^3 + z^4$ have quadratic part x^2. The quadratic part of $\bar{f}$ is G-semi-invariant, so the line $\mathbb{C}\ell$ is G-stable, and ℓ is a linear combination of the x_i with $\chi_{x_i} = \chi_\ell$. Possibly renumbering x_1, x_2, x_3, the coefficient of x_1 in ℓ is non-zero. Let $x := \lambda^{1/2}\ell$, $y := x_2$, and $z := x_3$. Then x, y, z, t are G-eigenparameters, $R = \mathbb{C}[[x, y, z, t]]$, $\bar{f} - x^2 \in (x, y, z)^3$, and the linear part of $\hat{f}$ is a multiple of t. Since x^2 occurs in $\hat{f}$, $\chi_x^2 = \chi_f$.

Thus $\partial_x \hat{f}(0) = 0$ and $\partial_x^2 \hat{f}(0) = 2$. By the implicit function theorem for formal power series, there exists a unique $\xi \in (y, z, t)\mathbb{C}[[y, z, t]]$ with $\partial_x \hat{f}(\xi, y, z, t) = 0$. For any $\sigma \in G$, $\chi_x(\sigma)^{-1}\sigma^*\xi$ is also such a solution, so ξ is zero or G-semi-invariant with $\chi_\xi = \chi_x$. By Taylor's formula in x,

$$\hat{f} = \hat{f}(\xi, y, z, t) + (x - \xi)^2 v$$

for some $v \in R$ with $v(0) = 1$. Since $\chi_x^2 = \chi_f$, v is G-invariant. Let $v^{1/2}$ be the square root of v with value 1 at 0, which is G-invariant by uniqueness, and let $x' := (x - \xi)v^{1/2}$. Then x', y, z, t are G-eigenparameters, $\chi_{x'} = \chi_x$, and $\hat{f} = x'^2 + \hat{f}(\xi, y, z, t)$. Since $\xi(y, z, 0) \in (y, z)^2$ and $\bar{f} - x^2 \in (x, y, z)^3$, the series $\hat{f}(\xi, y, z, 0)$ has order at least three, and the cubic part of $\hat{f}(\xi, y, z, 0)$ is the cubic part of $\bar{f}(0, y, z)$. Possibly replacing x with x', we may assume that

$$\hat{f} = x^2 + h(y, z, t) \tag{12.5}$$

for a G-semi-invariant $h \in \mathbb{C}[[y, z, t]]$ with $\chi_h = \chi_f$ such that $\bar{h} := h(y, z, 0)$ has order at least three. The parameter t has not been changed, so $\bar{f} = x^2 + \bar{h}$ defines the completion of $X \cap (t = 0)$ at o.

Step 3. In this step, we bring $\bar{h}$ to the normal form $y^2 z + z^3$ or $y^3 + z^3$ in the D_4 case, and $y^3 + z^4$ in the E_6 case, by G-equivariant changes of y, z alone.

Let c be the cubic part of $\bar{h}$. By the classification of simple singularities [AGV85, Section 16.2, Theorems 3, 4, 6_1, and 7_1], see also [KM98, Theorem 4.20], the following hold.

- If $X \cap (t = 0)$ is of type D_4, then c has three pairwise linearly independent linear factors.
- If $X \cap (t = 0)$ is of type E_6, then $c = \ell'^3$ for a linear form ℓ' in y, z, and if $\ell' = y$, then the coefficient of z^4 in $\bar{h}$ is non-zero.

Lemma 11.6 requires a cubic with three distinct linear factors, so it is used only in the D_4 case. We apply it to the series $x^2 + \bar{h}$, which does not involve t, and **Step 4** handles the dependence on t in both cases.

Assume that $X \cap (t = 0)$ is of type D_4. By Lemma 11.4, applied to c and the space $W := \mathbb{C}y \oplus \mathbb{C}z$ of linear forms in y, z, there exist G-semi-invariant $y', z' \in W$ forming a basis of W and $\lambda' \in \mathbb{C}^*$

such that $c = \lambda'(y'^2z' + z'^3)$ or $c = \lambda'(y'^3 + z'^3)$. Possibly replacing y, z with $\lambda'^{1/3}y', \lambda'^{1/3}z'$, we may assume that $c = c_0$, where $c_0 := y^2z + z^3$ or $c_0 := y^3 + z^3$. Let

$$(m_0, m_1, m_2, m_3) := \begin{cases} (1, y, z, z^2) & \text{if } c_0 = y^2z + z^3, \\ (1, y, z, yz) & \text{if } c_0 = y^3 + z^3. \end{cases}$$

A direct calculation shows that the classes of $m_0, \dots, m_3$ form a basis of $\mathbb{C}[y, z]/(\partial_y c_0, \partial_z c_0)$, and the m_k are G-semi-invariant since G acts diagonally. The series $x^2 + \bar{h} = x^2 + c_0 + (\bar{h} - c_0)$ is G-semi-invariant, and $\bar{h} - c_0 \in (y, z)^4$. By Lemma 11.6, there exist a G-equivariant automorphism φ of $\mathbb{C}[[x, y, z, t]]$ with $\varphi(x) = x$, $\varphi(t) = t$, and $\varphi(y) - y, \varphi(z) - z \in (y, z, t)^2$, and $r_0, \dots, r_3 \in \mathbb{C}[[t]]$ with $\operatorname{ord} r_k \geq 4 - \deg m_k \geq 2$, such that

$$\varphi(x^2 + \bar{h}) = x^2 + c_0 + \sum_{k=0}^{3} r_k(t) m_k.$$

Let $\epsilon : \mathbb{C}[[x, y, z, t]] \to \mathbb{C}[[y, z]]$ be the G-equivariant surjection with $\epsilon(x) = \epsilon(t) = 0$, and let ψ be the endomorphism of $\mathbb{C}[[y, z]]$ with $\psi(y) = \epsilon(\varphi(y))$ and $\psi(z) = \epsilon(\varphi(z))$. Then ψ is G-equivariant, and ψ is an automorphism since $\psi(y) - y, \psi(z) - z \in (y, z)^2$. Applying ϵ to the last equation and using $\varphi(x) = x$, $\varphi(t) = t$, and $r_k(0) = 0$, we obtain

$$\bar{h}(\psi(y), \psi(z)) = c_0(y, z), \quad \text{hence} \quad \bar{h}(y, z) = c_0(\psi^{-1}(y), \psi^{-1}(z)).$$

Extend ψ to R by $\psi(x) = x$ and $\psi(t) = t$. Then $x, \psi^{-1}(y), \psi^{-1}(z), t$ are G-eigenparameters with $\chi_{\psi^{-1}(y)} = \chi_y$ and $\chi_{\psi^{-1}(z)} = \chi_z$. Possibly replacing y, z with $\psi^{-1}(y), \psi^{-1}(z)$, we may assume that $\bar{h} = c_0$.

Assume that $X \cap (t = 0)$ is of type E_6. Since $c = \ell'^3$ is G-semi-invariant, the line $\mathbb{C}\ell'$ is G-stable, so ℓ' is G-semi-invariant. Possibly exchanging y and z, the coefficient of y in ℓ' is non-zero, and then $\chi_{\ell'} = \chi_y$. Possibly replacing y with ℓ', we may assume that $c = y^3$, so the coefficient of z^4 in $\bar{h}$ is non-zero. Replacing z with the product of z and a fourth root of the coefficient of z^4 in $\bar{h}$, we may assume that z^4 occurs in $\bar{h}$ with coefficient 1. Let y and z have weights 4 and 3. The monomials of degree three other than y^3 do not occur in $\bar{h}$, and the only monomial of degree at least four and weight at most 12 is z^4. Thus

$$\bar{h} = y^3 + z^4 + g,$$

where any monomial occurring in $g \in \mathbb{C}[[y, z]]$ has weight at least 13, and any monomial occurring in $\bar{h}$ has character $\chi_f = \chi_y^3 = \chi_z^4$.

Let $N \geq 13$, and assume that any monomial occurring in g has weight at least N. The monomials not in (y^2, z^3) are $1, z, z^2, y, yz, yz^2$, of weights $0, 3, 6, 4, 7, 10$. Thus any monomial of weight at least 11 is divisible by y^2 or by z^3, and dividing g monomial by monomial we can write

$$g = 3y^2a + 4z^3b'$$

with $a, b' \in \mathbb{C}[[y, z]]$ whose monomials have weights at least $N - 8$ and $N - 9$, respectively. Since any monomial of g has character $\chi_y^3 = \chi_z^4$, each of a and b' is zero or G-semi-invariant with $\chi_a = \chi_y$ and $\chi_{b'} = \chi_z$. Since $N - 9 \geq 4$, $a, b' \in (y, z)$. A direct calculation shows that

$$\bar{h}(y - a, z - b') = y^3 + z^4 + g', \qquad g' = 3ya^2 - a^3 + 6z^2b'^2 - 4zb'^3 + b'^4 + g(y - a, z - b') - g,$$

and any monomial occurring in g' has weight at least $2N - 12 \geq N + 1$. Iterating the substitutions $y \to y - a$, $z \to z - b'$, whose corrections have weights tending to infinity, we obtain a G-equivariant automorphism ψ of $\mathbb{C}[[y, z]]$ as the (y, z)-adic limit of the composites, with $\bar{h}(\psi(y), \psi(z)) = y^3 + z^4$. As in the D_4 case, possibly replacing y, z with $\psi^{-1}(y), \psi^{-1}(z)$, we may assume that $\bar{h} = y^3 + z^4$.

Let $h_c := c_0$ in the D_4 case and $h_c := y^3 + z^4$ in the E_6 case. Then

$$\hat{f} = x^2 + h(y, z, t), \qquad h(y, z, 0) = h_c. \tag{12.6}$$

Step 4. In this step, we remove the dependence of h on t modulo the Jacobian ideal of h_c by a t-adic division, keeping t fixed.

Let $M := \{m_0, m_1, m_2, m_3\}$ in the D_4 case and $M := \{1, z, z^2, y, yz, yz^2\}$ in the E_6 case, and let $\mathfrak{b} := (\partial_y h_c, \partial_z h_c) \subset \mathbb{C}[[y,z]]$, that is, $\mathfrak{b} = (yz, y^2+3z^2)$, (y^2, z^2), or (y^2, z^3). A direct calculation shows that $(y,z)^4 \subset \mathfrak{b}$ and that the classes of the monomials in M form a basis of $\mathbb{C}[[y,z]]/\mathfrak{b}$. Thus

$$\mathbb{C}[[y,z]] = \bigoplus_{m\in M} \mathbb{C}m \oplus \mathfrak{b},$$

and both summands are G-stable, since $\partial_y h_c$ and $\partial_z h_c$ are G-semi-invariant.

We construct, for $N \geq 1$, G-equivariant automorphisms ϕ_N of $\mathbb{C}[[y,z,t]]$ with $\phi_N(t) = t$ and $\phi_N(y) - y, \phi_N(z) - z \in t^N\mathbb{C}[[y,z,t]]$, and polynomials $r_{m,N} \in t\mathbb{C}[t]$ for $m \in M$ with $r_{m,N+1} - r_{m,N} \in t^N\mathbb{C}[t]$, each $r_{m,N}(t)m$ being zero or G-semi-invariant with character χ_f, such that $h_1 := h$ and $h_{N+1} := \phi_N(h_N)$ satisfy

$$h_N - h_c - \sum_{m\in M} r_{m,N}(t)m \in t^N\mathbb{C}[[y,z,t]]. \tag{12.7}$$

For $N = 1$, let $r_{m,1} := 0$. Then (12.7) holds by (12.6). Let $N \geq 1$, and assume that h_N and the $r_{m,N}$ are constructed. Write the left side of (12.7) as $t^N e + t^{N+1}e'$ with $e \in \mathbb{C}[[y,z]]$ and $e' \in \mathbb{C}[[y,z,t]]$, and write

$$e = \sum_{m\in M} e_m m + a\partial_y h_c + b'\partial_z h_c$$

with $e_m \in \mathbb{C}$ and $a, b' \in \mathbb{C}[[y,z]]$. The left side of (12.7) is zero or G-semi-invariant with character χ_f, so e is zero or G-semi-invariant with character $\chi_f\chi_t^{-N}$. Taking the $\chi_f\chi_t^{-N}$-components of both sides of the last equation, we may assume that $e_m = 0$ unless $\chi_m\chi_t^N = \chi_f$, and that $t^N a$ and $t^N b'$ are zero or G-semi-invariant with characters χ_y and χ_z. Let ϕ_N be the endomorphism of $\mathbb{C}[[y,z,t]]$ with

$$\phi_N(y) = y - t^N a, \qquad \phi_N(z) = z - t^N b', \qquad \phi_N(t) = t.$$

Then ϕ_N is a G-equivariant automorphism. Since $\phi_N(m) - m \in t^N\mathbb{C}[[y,z,t]]$, $r_{m,N} \in t\mathbb{C}[t]$, and $\phi_N(e) - e \in t^N\mathbb{C}[[y,z,t]]$, Taylor's formula yields

$$\begin{aligned}\phi_N(h_N) &\equiv h_c - t^N(a\partial_y h_c + b'\partial_z h_c) + \sum_{m\in M} r_{m,N}(t)m + t^N e \\ &\equiv h_c + \sum_{m\in M} \bigl(r_{m,N}(t) + e_m t^N\bigr)m \mod t^{N+1}.\end{aligned}$$

Thus $r_{m,N+1} := r_{m,N} + e_m t^N$ satisfy (12.7) for $N+1$.

Let $\Phi_N := \phi_N \circ \cdots \circ \phi_1$. Since ϕ_{N+1} is the identity modulo t^{N+1}, both $\Phi_{N+1}(y) - \Phi_N(y)$ and $\Phi_{N+1}(z) - \Phi_N(z)$ belong to $t^{N+1}\mathbb{C}[[y,z,t]]$. Thus the Φ_N converge t-adically, hence (y,z,t)-adically, to a G-equivariant endomorphism Φ with $\Phi(t) = t$ and $\Phi(y) - y, \Phi(z) - z \in t\mathbb{C}[[y,z,t]]$, which is an automorphism. Let $r_m := \lim_N r_{m,N} \in t\mathbb{C}[[t]]$. Then $\Phi(h) = \lim_N h_N = h_c + \sum_m r_m(t)m$. As in **Step 3**, possibly replacing y, z with $\Phi^{-1}(y), \Phi^{-1}(z)$, we may assume that

$$h = h_c + \sum_{m\in M} r_m(t)m, \qquad r_m \in t\mathbb{C}[[t]], \tag{12.8}$$

where each $r_m(t)m$ is zero or G-semi-invariant with character χ_f. No lower bound on $\operatorname{ord} r_m$ is claimed. In the D_4 case, we write $r_k := r_{m_k}$. In the E_6 case, we write $\alpha_0 := r_y$, $\alpha_1 := r_{yz}$, $\alpha_2 := r_{yz^2}$, $\beta_0 := r_1$, $\beta_1 := r_z$, and $\beta_2 := r_{z^2}$, so that

$$\begin{aligned}&h = y^3 + \alpha(z,t)y + \beta(z,t), \\ &\alpha = \alpha_2(t)z^2 + \alpha_1(t)z + \alpha_0(t), \qquad \beta = z^4 + \beta_2(t)z^2 + \beta_1(t)z + \beta_0(t).\end{aligned} \tag{12.9}$$

Step 5. In this step, we show that, after a G-equivariant translation of z by an element of $t\mathbb{C}[[t]]$,

$$\Bigl((\mathbb{C}^*)^3, \tfrac{1}{2}\operatorname{div}(\mathrm{in}_u(h))\Bigr) \text{ is lc for any } u \in \mathbb{R}^3_{>0}, \tag{12.10}$$

where weights are taken with respect to y, z, t.

Assume that $X \cap (t = 0)$ is of type E_6. Then h is as in (12.9) with $\alpha_k, \beta_k \in t\mathbb{C}[[t]]$. By Lemma 12.8, there exists $\theta \in t\mathbb{C}[[t]]$, zero or G-semi-invariant with $\chi_\theta = \chi_z$, such that h, written in y, $z' := z - \theta(t)$, and t, satisfies (12.10). Then x, y, z', t are G-eigenparameters, and possibly replacing z with z', (12.10) holds.

Assume that $X \cap (t = 0)$ is of type D_4. If $c_0 = y^2z + z^3$, then $h = y^2z + z^3 + r_3(t)z^2 + r_2(t)z + r_1(t)y + r_0(t)$. Since z^3 and $r_3(t)z^2$ occur in h with character χ_f, r_3 is zero or G-semi-invariant with $\chi_{r_3} = \chi_z$. Possibly replacing z with $z + r_3(t)/3$, a direct calculation shows that we may assume

$$h = y^2z + z^3 + \tilde{r}_3(t)y^2 + \tilde{r}_2(t)z + \tilde{r}_1(t)y + \tilde{r}_0(t), \tag{12.11}$$

where $\tilde{r}_3 = -r_3/3$, $\tilde{r}_2 = r_2 - r_3^2/3$, $\tilde{r}_1 = r_1$, and $\tilde{r}_0 = r_0 - r_2r_3/3 + 2r_3^3/27$ belong to $t\mathbb{C}[[t]]$. If $c_0 = y^3 + z^3$, then

$$h = y^3 + z^3 + r_3(t)yz + r_2(t)z + r_1(t)y + r_0(t). \tag{12.12}$$

By Lemma 10.2, applied to $P = h$, it suffices to show that $((\mathbb{C}^*)^2, \frac{1}{2}\operatorname{div}(p_1))$ is lc for any $u = (u_1, u_2, u_3) \in \mathbb{Z}^3_{>0}$, where $p_1(y_1, z_1) := \mathrm{in}_u(h)(y_1, z_1, 1)$. Fix such u, and let $p := \mathrm{in}_u(h)$.

Since $u_3 > 0$ and all monomials of p have u-weight $\mathrm{wt}_u(h)$, distinct monomials of p have distinct (y, z)-parts, so $p_1 \neq 0$. The monomials of h in (12.11) and (12.12) are of the form $\mathbf{m}t^k$ with $\mathbf{m} \in \{y^2z, z^3, y^2, z, y, 1\}$ and $\mathbf{m} \in \{y^3, z^3, yz, y, z, 1\}$, respectively. Thus there exist $a_1, \dots, a_6 \in \mathbb{C}$ such that

$$p_1 = a_1y_1^2z_1 + a_2z_1^3 + a_3y_1^2 + a_4z_1 + a_5y_1 + a_6 \quad \text{or} \quad p_1 = a_1y_1^3 + a_2z_1^3 + a_3y_1z_1 + a_4y_1 + a_5z_1 + a_6,$$

respectively. Suppose that $p_1 = \kappa(\mu_1y_1 + \mu_2z_1 + \mu_3)^3$ for some $\kappa \in \mathbb{C}^*$ and $\mu_1, \mu_2, \mu_3 \in \mathbb{C}$ with $(\mu_1, \mu_2) \neq (0, 0)$. In the first shape, the coefficient $\kappa\mu_1^3$ of y_1^3 vanishes, so $\mu_1 = 0$, and then the coefficient $3\kappa\mu_2^2\mu_3$ of z_1^2 vanishes, so $\mu_3 = 0$. In the second shape, the coefficients $3\kappa\mu_1^2\mu_2$ and $3\kappa\mu_1\mu_2^2$ of $y_1^2z_1$ and $y_1z_1^2$ vanish, so $\mu_1\mu_2 = 0$, and then the coefficient $3\kappa\mu_1^2\mu_3$ of y_1^2 or $3\kappa\mu_2^2\mu_3$ of z_1^2 vanishes, so $\mu_3 = 0$. In both shapes p_1 is a constant multiple of y_1^3 or z_1^3, and $\operatorname{div}(p_1) = 0$ on $(\mathbb{C}^*)^2$. Otherwise p_1 is a non-zero polynomial of degree at most three which is not a constant multiple of the cube of a non-constant affine linear form, and $(\mathbb{A}^2, \frac{1}{2}\operatorname{div}(p_1))$ is lc by Lemma 10.1(2). In either case $((\mathbb{C}^*)^2, \frac{1}{2}\operatorname{div}(p_1))$ is lc, so (12.10) holds.

After **Steps 2–5**, x, y, z, t are G-eigenparameters, $R = \mathbb{C}[[x, y, z, t]]$ with the diagonal action of G, and $\hat{f} = x^2 + h(y, z, t)$ with h satisfying (12.10). All changes of coordinates fixed t.

Step 6. In this step, we replace $\hat{f}$ with a polynomial by contact determinacy with derivatives only in x, y, z, which fixes t.

Let $\mathfrak{m}$ be the maximal ideal of R, and let

$$\mathfrak{j} := (\hat{f}, \partial_x\hat{f}, \partial_y\hat{f}, \partial_z\hat{f}) \quad \text{and} \quad \mathfrak{a} := (\hat{f}) + \mathfrak{m}(\partial_x\hat{f}, \partial_y\hat{f}, \partial_z\hat{f}).$$

We show that $\mathfrak{j}$ is $\mathfrak{m}$-primary. Suppose not. Then there exists a prime ideal $\mathfrak{p} \supset \mathfrak{j}$ of R with $\dim R/\mathfrak{p} = 1$. The normalization of the complete local domain $R/\mathfrak{p}$ is a complete discrete valuation ring with residue field $\mathbb{C}$, hence $\mathbb{C}[[s]]$, so there exists a local $\mathbb{C}$-algebra homomorphism $\gamma : R \to \mathbb{C}[[s]]$ with kernel $\mathfrak{p}$. If $\gamma(t) = 0$, then γ factors through $R/(t) = \mathbb{C}[[x, y, z]]$ and kills $\bar{f}, \partial_x\bar{f}, \partial_y\bar{f}, \partial_z\bar{f}$. The ideal $(\bar{f}, \partial_x\bar{f}, \partial_y\bar{f}, \partial_z\bar{f})$ of $\mathbb{C}[[x, y, z]]$ is primary to the maximal ideal, since the normal surface $X \cap (t = 0)$ is smooth outside o near o and $\bar{f}$ defines the completion at o (**Step 2**). This contradicts that the kernel of the induced map $\mathbb{C}[[x, y, z]] \to \mathbb{C}[[s]]$ is a prime ideal of dimension one. If $\gamma(t) \neq 0$, then $\gamma(t) \in s\mathbb{C}[[s]]$ is non-zero, so $d\gamma(t)/ds \neq 0$. Differentiating $\gamma(\hat{f}) = \hat{f}(\gamma(x), \gamma(y), \gamma(z), \gamma(t)) = 0$ with respect to s and using $\gamma(\partial_x\hat{f}) = \gamma(\partial_y\hat{f}) = \gamma(\partial_z\hat{f}) = 0$, we obtain

$$\gamma(\partial_t\hat{f}) \cdot \frac{d\gamma(t)}{ds} = 0,$$

so $\gamma(\partial_t\hat{f}) = 0$. Thus $\mathfrak{p}$ contains $\hat{f}$ and its partial derivatives $\partial_x\hat{f}$, $\partial_y\hat{f}$, $\partial_z\hat{f}$, $\partial_t\hat{f}$. By the chain rule, the ideal generated by $\hat{f}$ and its partial derivatives does not depend on the formal coordinates,

so it is the extension of the ideal of $\mathcal{O}_{Y,o}$ generated by f and the partial derivatives of f with respect to x_1, x_2, x_3, t, which is primary to $\mathfrak{m}_{Y,o}$ or equal to $\mathcal{O}_{Y,o}$ since X is smooth outside o. This contradicts $\dim R/\mathfrak{p} = 1$. Thus $\mathfrak{j}$ is $\mathfrak{m}$-primary.

Since $\mathfrak{m}\mathfrak{j} \subset \mathfrak{a} \subset \mathfrak{j}$, the ideal $\mathfrak{a}$ is $\mathfrak{m}$-primary, so there exists a positive integer q with $\mathfrak{m}^q \subset \mathfrak{a}$. Let d_h be the maximum of the degrees of the componentwise minimal exponents of h (Definition-Lemma 7.3), and fix an integer $L > \max\{2q, d_h\}$. By Lemma 7.1, applied to $F = \hat{f}$, the set S of the indices of x, y, z, and q in place of b, there exist a G-equivariant automorphism φ of R with $\varphi(t) = t$ and a G-invariant unit v of R such that

$$v\varphi(\hat{f}) = \hat{f}_{<L} = x^2 + h_{<L}, \tag{12.13}$$

where $h_{<L}$ is the sum of the terms of h of degree less than L, and the second equality holds since $L > 2$. Let $u \in \mathbb{R}^3_{>0}$. Since $L > d_h$, we have $\mathrm{wt}_u(h_{<L}) = \mathrm{wt}_u(h)$ and $\mathrm{in}_u(h_{<L}) = \mathrm{in}_u(h)$ by Definition-Lemma 7.3. By (12.10),

$$\left((\mathbb{C}^*)^3, \tfrac{1}{2}\,\mathrm{div}(\mathrm{in}_u(h_{<L}))\right) \text{ is lc for any } u \in \mathbb{R}^3_{>0}. \tag{12.14}$$

The polynomial $h_{<L}$ is non-zero, since y^3 or $y^2 z$ occurs in it, and $h_{<L}(0) = 0$.

Step 7. In this step, we compare the minimal log discrepancies of X and of the polynomial germ, and conclude.

Let G act diagonally on $\mathbb{A}^4$ with coordinates x, y, z, t by the characters $\chi_x, \chi_y, \chi_z, \chi_t$, and let $X_0 := (x^2 + h_{<L} = 0) \subset \mathbb{A}^4$, whose equation is G-semi-invariant with character χ_f. We identify $\widehat{\mathcal{O}}_{\mathbb{A}^4,0}$ with R G-equivariantly. Then $\widehat{\mathcal{O}}_{X_0,0} = R/(x^2 + h_{<L})$, and by (12.13), φ induces a G-equivariant isomorphism

$$\widehat{\mathcal{O}}_{X,o} = R/(\hat{f}) \xrightarrow{\sim} R/(x^2 + h_{<L}) = \widehat{\mathcal{O}}_{X_0,0}$$

mapping the class of t to the class of t. By **Step 1**, G acts faithfully on $\mathcal{O}_{X,o} \subset \widehat{\mathcal{O}}_{X,o}$, hence on $\widehat{\mathcal{O}}_{X_0,0}$, and hence on X_0 and on $\mathbb{A}^4$.

Since $\mathcal{O}_{X,o}$ is an excellent normal local ring, $\widehat{\mathcal{O}}_{X,o}$ is normal [Mat89, Theorem 32.2(i)]. Thus $\widehat{\mathcal{O}}_{X_0,0}$ is normal, and so is $\mathcal{O}_{X_0,0}$ [Mat89, Theorem 23.9, Corollary (i)]. By the chain rule and (12.13), the ideal of R generated by $x^2 + h_{<L}$ and the partial derivatives of $x^2 + h_{<L}$ is the image under φ of $(\hat{f}, \partial_x\hat{f}, \partial_y\hat{f}, \partial_z\hat{f}, \partial_t\hat{f})$, which is $\mathfrak{m}$-primary or equal to R by **Step 6**. Hence the corresponding ideal of $\mathcal{O}_{\mathbb{A}^4,0}$ has finite colength [Mat89, Theorem 8.7], and shrinking $\mathbb{A}^4$ around 0, X_0 is normal and smooth outside 0. By Lemma 3.3(2) with the zero boundary and the trivial group, applied to $(X \ni o)$ and to $(X_0 \ni 0)$, and by the isomorphism $\widehat{\mathcal{O}}_{X,o} \cong \widehat{\mathcal{O}}_{X_0,0}$ induced by φ, $\mathrm{mld}(X_0 \ni 0) = \mathrm{mld}(X \ni o) > 0$. Since K_{X_0} is Cartier and X_0 is smooth outside 0, X_0 is klt near 0. Thus the data

$$b_y = b_z = 0, \quad b_t = b, \quad h_{<L}, \quad X_0, \quad G, \quad B_0 := b\,\mathrm{div}_{X_0}(t)$$

satisfy the conditions of Set-up 10.3, with $h_{<L}$ in place of h and X_0 in place of X.

Let $H \subset G$ be a subgroup. The isomorphism $\widehat{\mathcal{O}}_{X,o} \cong \widehat{\mathcal{O}}_{X_0,0}$ induced by φ is H-equivariant and maps the pullback $\widehat{B}$ of $B = b\,\mathrm{div}_X(t)$ to $\widehat{\mathcal{O}}_{X,o}$ to the pullback $\widehat{B}_0$ of B_0 to $\widehat{\mathcal{O}}_{X_0,0}$, since it maps t to t. By Definition 3.2, the minimal log discrepancy of the formal quotient depends only on the complete local ring with the action of H and the boundary. Thus Lemma 3.3(2), applied to $(X \ni o, B)$ and to $(X_0 \ni 0, B_0)$, implies that

$$\mathrm{mld}((X_0 \ni 0, B_0)/H) = \mathrm{mld}((\widehat{X}_0 \ni \hat{0}, \widehat{B}_0)/H) = \mathrm{mld}((\widehat{X} \ni \hat{o}, \widehat{B})/H) = \mathrm{mld}((X \ni o, B)/H).$$

Thus X_0 and the polynomial $h_{<L}$ satisfy the equality in the statement, and by (12.14), $h_{<L}$ satisfies the lc condition in the statement.

Let $a := \mathrm{mld}((X \ni o, B)/G)$, and assume that $a \in (0, 1)$. Then $a = \mathrm{mld}((X_0 \ni 0, B_0)/G)$, and by (12.14), Proposition 10.5 implies that $a = \min_{w \in N_G \cap \mathbb{R}^4_{>0}} P(w)$, where N_G and P are taken for

X_0, $h_{<L}$, and $(b_y, b_z, b_t) = (0, 0, b)$. Since $(0, 0, b)$ belongs to the finite set $\{0\} \times \{0\} \times \mathcal{B} \subset \mathbb{Q}^3_{\leq 0}$, Proposition 10.5 also implies that the set of such numbers a satisfies the ACC. □

12.5. **Two distinct roots.** In this subsection, we prove Proposition 12.13, which completes type cE_6 and treats type cE_7 (resp. cE_8) when the cubic α_3 (resp. the quintic β_5) has at least two distinct roots. For each type we fix one weighted blow-up of $\mathbb{A}^4$ and pass to a finite cover of an affine open subset of it, on which a prime divisor computing the minimal log discrepancy of the quotient has a center. The possible centers are treated by Lemma 3.4, Corollary 9.6, and Proposition 12.11.

We first read off the type of $(X \ni o)$ from the homogeneous parts of low degree of the equation.

Lemma 12.12. *Notation and conditions as in Set-up 12.2, and assume that*

$$g = y^3 + \alpha(z,t)y + \beta(z,t)$$

for power series $\alpha, \beta \in \mathbb{C}[[z,t]]$ with $\operatorname{ord}\alpha \geq 3$ and $\operatorname{ord}\beta \geq 4$. Let α_3 be the cubic part of α, and let β_4 and β_5 be the quartic and quintic parts of β. Then α_3, β_4, and β_5 are not all zero. Moreover, $(X \ni o)$ is of type cE_6 if and only if $\beta_4 \neq 0$, of type cE_7 if and only if $\beta_4 = 0$ and $\alpha_3 \neq 0$, and of type cE_8 if and only if $\alpha_3 = \beta_4 = 0$ and $\beta_5 \neq 0$.

Proof. Let h be a general element of the maximal ideal of the local ring at o of the ambient smooth fourfold germ, and let $H := X \cap (h = 0)$. Since $(X \ni o)$ is an isolated cDV singularity, $(H \ni o)$ is a Du Val singularity [KM98, Definition 5.32]. Since X is smooth outside o, after shrinking, H is smooth outside o by Bertini's theorem, so H is normal [Har77, Proposition II.8.23].

The image $\hat{h}$ of h in $\mathbb{C}[[x,y,z,t]]$ is $c_x x + c_y y + c_z z + c_t t$ plus terms of degree at least 2, where $(c_x, c_y, c_z, c_t) \in \mathbb{C}^4$ is general since h is general. In particular, $c_t \neq 0$, so by the formal implicit function theorem, $(\hat{h}) = (t - \tau)$ for a power series

$$\tau = \lambda z + \mu x + \nu y + (\text{terms of degree at least } 2) \in \mathbb{C}[[x,y,z]], \qquad \lambda = -\frac{c_z}{c_t},$$

and λ is general. Thus $\widehat{\mathcal{O}}_{H,o} = \mathbb{C}[[x,y,z]]/(F_H)$, where

$$F_H := x^2 + y^3 + \alpha(z,\tau)y + \beta(z,\tau).$$

Assume that $\beta_4 \neq 0$ (resp. $\beta_4 = 0$ and $\alpha_3 \neq 0$, resp. $\alpha_3 = \beta_4 = 0$ and $\beta_5 \neq 0$). Give x, y, z the weights $(w_x, w_y, w_z) = (6, 4, 3)$ (resp. $(9, 6, 4)$, resp. $(15, 10, 6)$), and let $d := 2w_x = 12$ (resp. 18, resp. 30). Then every term of $\tau - \lambda z$ has weight at least $w_y > w_z$, so for a binary form φ of degree m, $\varphi(z,\tau) - \varphi(1,\lambda)z^m$ has weight at least $(m-1)w_z + w_y$. Since $\operatorname{ord}\alpha \geq 3$ and $\operatorname{ord}\beta \geq 4$, $\operatorname{ord}\beta \geq 5$ if $\beta_4 = 0$, and $\operatorname{ord}\alpha \geq 4$ if $\alpha_3 = 0$, a direct calculation shows that every term of F_H has weight at least d and that the terms of weight d add up to

$$x^2 + y^3 + \beta_4(1,\lambda)z^4, \qquad x^2 + y^3 + \alpha_3(1,\lambda)yz^3, \qquad x^2 + y^3 + \beta_5(1,\lambda)z^5,$$

respectively. Since λ is general, the coefficient $c := \beta_4(1,\lambda)$ (resp. $\alpha_3(1,\lambda)$, resp. $\beta_5(1,\lambda)$) is non-zero. After replacing z with $c^{-1/4}z$ (resp. $c^{-1/3}z$, resp. $c^{-1/5}z$), F_H is semiquasihomogeneous of degree d with quasihomogeneous part

$$f_0 := x^2 + y^3 + z^4, \qquad x^2 + y^3 + yz^3, \qquad x^2 + y^3 + z^5,$$

respectively, which has an isolated critical point at 0. The monomials

$$1, y, z, z^2, yz, yz^2; \qquad 1, y, yz, z, z^2, z^3, z^4; \qquad 1, y, yz, yz^2, yz^3, z, z^2, z^3$$

form a basis of the local algebra $\mathbb{C}[[x,y,z]]/(\partial_x f_0, \partial_y f_0, \partial_z f_0)$, and their weights are at most 10, 16, and 28, respectively, so all of them are less than d. Thus F_H is equivalent to f_0 [AGV85, Section 12.1, Theorem], and $(H \ni o)$ is of type E_6 (resp. E_7, resp. E_8) [KM98, Theorems 4.20 and 4.22] (the analytic type of the isolated singularity $(H \ni o)$ is determined by the completion $\widehat{\mathcal{O}}_{H,o}$, by Artin approximation [Art68, Corollary 1.6]).

Suppose that $\alpha_3 = \beta_4 = \beta_5 = 0$. Then $\operatorname{ord}\alpha \geq 4$ and $\operatorname{ord}\beta \geq 6$. Give x, y, z the weights $(3,2,1)$. Since every term of τ has weight at least 1, every term of F_H has weight at least 6. Consider the chart of the $(3,2,1)$-weighted blow-up of $\operatorname{Spec}\widehat{\mathcal{O}}_{H,o}$ given by the substitution

$$x = x_1 r^3, \qquad y = y_1 r^2, \qquad z = r.$$

A monomial $x^i y^j z^l$ becomes $r^{3i+2j+l} x_1^i y_1^j$, so $F_H(x_1 r^3, y_1 r^2, r) = r^6 \widetilde{F}$, where $\widetilde{F}$ is a power series in r with coefficients in $\mathbb{C}[x_1, y_1]$ and

$$\widetilde{F}|_{r=0} = x_1^2 + y_1^3 + c_1 y_1 + c_0$$

for some $c_0, c_1 \in \mathbb{C}$, the sum of the terms of F_H of weight 6 evaluated at $(x_1, y_1, 1)$. Since $y_1^3 + c_1 y_1 + c_0$ is not a square, $\widetilde{F}|_{r=0}$ is irreducible. Let E be the prime divisor $(\widetilde{F} = 0) \cap (r = 0)$. At a general point of E, $\partial_{x_1}\widetilde{F} = 2x_1 \neq 0$, so $(\widetilde{F} = 0)$ is smooth there, and r is a local parameter along E. The residue form $\Omega_H := dy \wedge dz / \partial_x F_H$ generates ω_H near o. Since $dy \wedge dz = r^2 dy_1 \wedge dr$ and $r^3(\partial_x F_H)(x_1 r^3, y_1 r^2, r) = \partial_{x_1}(r^6 \widetilde{F}) = r^6 \partial_{x_1}\widetilde{F}$, we obtain

$$\Omega_H = r^{-1} \cdot \frac{dy_1 \wedge dr}{\partial_{x_1}\widetilde{F}},$$

and $dy_1 \wedge dr / \partial_{x_1}\widetilde{F}$ generates the dualizing sheaf of $(\widetilde{F} = 0)$ at a general point of E. Thus the discrepancy of E with respect to $\operatorname{Spec}\widehat{\mathcal{O}}_{H,o}$ is -1. By Lemma 3.3(3), applied over $k = \mathbb{C}$ to $(H \ni o)$ with the zero boundary and the trivial group, the formal prime divisor E restricts to a prime divisor over H with center o and discrepancy -1, so H is not canonical at o, and $(H \ni o)$ is not Du Val [KM98, Theorem 4.20], a contradiction. Thus α_3, β_4, and β_5 are not all zero.

The following four cases are mutually exclusive and cover all possibilities: $\beta_4 \neq 0$, or $\beta_4 = 0$ and $\alpha_3 \neq 0$, or $\alpha_3 = \beta_4 = 0$ and $\beta_5 \neq 0$, or $\alpha_3 = \beta_4 = \beta_5 = 0$. The last case does not occur, and the Du Val singularities of types E_6, E_7, and E_8 are pairwise non-isomorphic, since the dual graphs of their minimal resolutions, which are unique, have 6, 7, and 8 vertices, respectively [KM98, Theorem 4.22]. By Definition-Lemma 11.1, the three equivalences follow. □

Proposition 12.13. *Notation and conditions as in Set-up 12.2. Assume that*

$$g = y^3 + \alpha(z,t)y + \beta(z,t)$$

for power series α, β with $\operatorname{ord}\alpha \geq 3$ and $\operatorname{ord}\beta \geq 4$, and that one of the following holds:

(1) $(X \ni o)$ is of type cE_6;
(2) $(X \ni o)$ is of type cE_7, and the cubic part α_3 of α has at least two distinct roots in $\mathbb{P}^1$;
(3) $(X \ni o)$ is of type cE_8, and the quintic part β_5 of β has at least two distinct roots in $\mathbb{P}^1$.

Then the set of such numbers $\operatorname{mld}((X \ni o, 0)/G) \in (0,1)$ satisfies the ACC.

Proof. The proof is long, so we divide it into several steps. Let α_3, β_4, and β_5 be the homogeneous parts of α and β of degrees 3, 4, and 5, and let $a := \operatorname{mld}((X \ni o, 0)/G)$. Assume that $a \in (0,1)$.

Step 1. In this step we reduce to four patterns of roots.

By Lemma 12.12, the type conditions in (1), (2), and (3) mean $\beta_4 \neq 0$, $\beta_4 = 0 \neq \alpha_3$, and $\alpha_3 = \beta_4 = 0 \neq \beta_5$, respectively.

Assume (1). Then $\beta_4 \neq 0$, and β_4 is the quartic part of $g(0,z,t) = \beta(z,t)$. If β_4 has at least three distinct linear factors, this case is treated in Proposition 12.6. If $\beta_4 = cL_1^2 L_2^2$ for a number $c \neq 0$ and linearly independent linear forms L_1, L_2, this case is treated in Proposition 12.10. If β_4 is a fourth power of a linear form, this case is treated in Proposition 12.9. Otherwise the roots of β_4 in $\mathbb{P}^1$ have multiplicities 3 and 1.

Assume (2). Then $\beta_4 = 0$, so $\operatorname{ord}\beta \geq 5$. If α_3 has three distinct roots, this case is treated in Proposition 12.7(E7). Otherwise the roots of α_3 have multiplicities 2 and 1.

Assume (3). Then $\alpha_3 = \beta_4 = 0$, so $\operatorname{ord}\alpha \geq 4$ and $\operatorname{ord}\beta \geq 5$. If β_5 has at least three distinct roots, this case is treated in Proposition 12.7(E8). Otherwise the roots of β_5 have multiplicities 3 and 2, or 4 and 1.

By Propositions 12.6 and 12.7, for any $\epsilon > 0$ only finitely many of the numbers a in the cases treated there satisfy $a \geq \epsilon$, so the numbers a in the cases of Propositions 12.6 and 12.7 satisfy the ACC. By Propositions 12.10 and 12.9, the numbers a in the cases treated there satisfy the ACC. By Lemma 2.10(4), it suffices to prove the ACC for the numbers a in the following cases:

(E6) $\beta_4 \neq 0$, and the roots of β_4 in $\mathbb{P}^1$ have multiplicities 3 and 1;
(E7) $\beta_4 = 0$, $\alpha_3 \neq 0$, and the roots of α_3 in $\mathbb{P}^1$ have multiplicities 2 and 1;
(E8) $\alpha_3 = \beta_4 = 0$, $\beta_5 \neq 0$, and the roots of β_5 in $\mathbb{P}^1$ have multiplicities 3 and 2, or 4 and 1.

Step 2. In this step we pass to a polynomial equation in $\mathbb{A}^4$ and fix the weights.

By Lemma 12.3(2) applied with $\psi = \mathrm{id}$ and $v = 1$, there exists an integer $L \geq 6$ such that $f_0 := (x^2 + g)_{<L}$, a polynomial in the coordinates x, y, z, t of $\mathbb{A}^4$, on which G acts diagonally and faithfully with the characters of x, y, z, t, defines an isolated Gorenstein terminal threefold germ $X_0 := (f_0 = 0)$ at 0, G acts faithfully on X_0, and $\mathrm{mld}((X_0 \ni 0, 0)/G) = a$. By Lemma 12.3(1), f_0 is G-semi-invariant. Since $L \geq 6$, we have

$$f_0 = x^2 + y^3 + \alpha_{<L-1}(z,t)y + \beta_{<L}(z,t),$$

and the homogeneous parts satisfy $(\alpha_{<L-1})_j = \alpha_j$ for $j \leq 4$ and $(\beta_{<L})_j = \beta_j$ for $j \leq 5$. Moreover, the singular points of $(f_0 = 0)$ other than 0 and the irreducible components of $(f_0 = 0)$ not containing 0 form a G-stable closed subset $Z \subset \mathbb{A}^4$ not containing 0, so there is a G-stable affine open neighborhood U of 0 in $\mathbb{A}^4$ disjoint from Z. Possibly replacing $(X \ni o)$ with $((f_0 = 0) \cap U \ni 0)$ and α, β with $\alpha_{<L-1}, \beta_{<L}$, we may assume that G acts faithfully and diagonally on $\mathbb{A}^4$ with coordinates x, y, z, t, that

$$X = (f = 0) \cap U, \qquad f = x^2 + y^3 + \alpha(z,t)y + \beta(z,t),$$

for polynomials α, β with $\operatorname{ord}\alpha \geq 3$ and $\operatorname{ord}\beta \geq 4$, that X is a normal variety smooth outside $o = 0$, and that one of (E6), (E7), and (E8) holds for α_3, β_4, and β_5.

Let $\chi_x, \chi_y, \chi_z, \chi_t$, and χ_f be the characters of x, y, z, t, and f. Since G acts diagonally, any monomial occurring in f has the character χ_f. Let

$$w = (w_x, w_y, w_z, w_t) := (6,4,3,3), \quad (9,6,4,4), \quad (15,10,6,6)$$

in cases (E6), (E7), and (E8), respectively, so that $2w_x = 3w_y$ and $w_z = w_t$, and let

$$k := w_y + w_z + w_t - w_x - 1,$$

so that $k = 3, 4, 6$, respectively. Since $\operatorname{ord}\alpha \geq 3$ and $\operatorname{ord}\beta \geq 4$, $\operatorname{ord}\beta \geq 5$ in case (E7), and $\operatorname{ord}\alpha \geq 4$ and $\operatorname{ord}\beta \geq 5$ in case (E8), a direct calculation shows that $\mathrm{wt}_w(f) = 2w_x$, that

$$\mathrm{in}_w(f) = x^2 + y^3 + \beta_4(z,t), \qquad x^2 + y^3 + \alpha_3(z,t)y, \qquad x^2 + y^3 + \beta_5(z,t)$$

in cases (E6), (E7), and (E8), respectively, and that the terms of f of w-weight $2w_x + 1$ add up to $\alpha_3(z,t)y$ in case (E6) and to 0 in cases (E7) and (E8).

Step 3. In this step we lift a prime divisor computing a to X and fix a coordinate $\ell \in \{y, t\}$.

Let $\pi_X : X \to X/G$ be the quotient morphism and $\bar{o} := \pi_X(o)$. Since X is terminal and $K_X = \pi_X^*(K_{X/G} + B_{X/G})$ by Lemma 2.6(0) with $B = 0$, $(X/G, B_{X/G})$ is klt [KM98, Proposition 5.20(4)], and a is attained by a prime divisor P over X/G with $\mathrm{center}_{X/G} P = \{\bar{o}\}$ [Amb99, Section 1, the second paragraph after Definition 1.1]. Since $\mathbb{C}(X)$ is a finite extension of $\mathbb{C}(X/G)$, the valuation ord_P extends to a discrete valuation of $\mathbb{C}(X)$ whose residue field is finite over that of ord_P, so by [KM98, Lemma 2.45] there exists a prime divisor F_0 over X such that $\mathrm{ord}_{F_0}|_{\mathbb{C}(X/G)}$ is a positive integer multiple of ord_P. Since $\pi_X^{-1}(\bar{o}) = \{o\}$, we have $\mathrm{center}_X F_0 = \{o\}$, so ord_{F_0} is positive on x, y, z, t. Let

$$\mu := \min\left\{\frac{\mathrm{ord}_{F_0}(x)}{w_x}, \frac{\mathrm{ord}_{F_0}(y)}{w_y}, \frac{\mathrm{ord}_{F_0}(z)}{w_z}, \frac{\mathrm{ord}_{F_0}(t)}{w_t}\right\} > 0.$$

Suppose that $\mathrm{ord}_{F_0}(z) > \mu w_z$ and $\mathrm{ord}_{F_0}(t) > \mu w_t$. Any monomial $x^{i_1}y^{i_2}z^{i_3}t^{i_4}$ occurring in $\alpha(z,t)y + \beta(z,t)$ satisfies $i_3 + i_4 \geq 1$ and has w-weight at least $2w_x$, so

$$\mathrm{ord}_{F_0}(x^{i_1}y^{i_2}z^{i_3}t^{i_4}) > \mu(i_1w_x + i_2w_y + i_3w_z + i_4w_t) \geq 2w_x\mu.$$

On the other hand, $\mathrm{ord}_{F_0}(x^2) \geq 2w_x\mu$ and $\mathrm{ord}_{F_0}(y^3) \geq 3w_y\mu = 2w_x\mu$, and equality holds for at least one of x^2 and y^3 since μ is attained at x or y. Since $f = 0$ on X, the least value of ord_{F_0} on the terms of f is attained at least twice. Thus $\mathrm{ord}_{F_0}(x^2) = \mathrm{ord}_{F_0}(y^3) = 2w_x\mu$, and in particular $\mathrm{ord}_{F_0}(y) = \mu w_y$. Exchanging z and t preserves the conditions (E6), (E7), and (E8), since they concern only the multiplicities of the roots of α_3, β_4, and β_5 in $\mathbb{P}^1$, and $w_z = w_t$. Possibly exchanging z and t, we may assume that either $\mathrm{ord}_{F_0}(t) = \mu w_t$, in which case we let $\ell := t$, or $\mathrm{ord}_{F_0}(z) > \mu w_z$, $\mathrm{ord}_{F_0}(t) > \mu w_t$, and $\mathrm{ord}_{F_0}(y) = \mu w_y$, in which case we let $\ell := y$. In both cases

$$\frac{\mathrm{ord}_{F_0}(\ell)}{w_\ell} = \mu.$$

Step 4. In this step we construct Y and determine the singularities of Y.

Let $\Phi : \mathbb{A}^5_{x_1,y_1,z_1,t_1,\tau} \to \mathbb{A}^4$ be the morphism with

$$\Phi^*x = \tau^{w_x}x_1, \quad \Phi^*y = \tau^{w_y}y_1, \quad \Phi^*z = \tau^{w_z}z_1, \quad \Phi^*t = \tau^{w_t}t_1,$$

and let $f_1 := \tau^{-2w_x}\Phi^*f$. By **Step 2**, f_1 is a polynomial and

$$f_1 = \mathrm{in}_w(f)(x_1, y_1, z_1, t_1) + \tau f_{1,1} + \tau^2 f_{1,2},$$

where $f_{1,1} = \alpha_3(z_1,t_1)y_1$ in case (E6), $f_{1,1} = 0$ in cases (E7) and (E8), and $f_{1,2}$ is a polynomial. Let $\ell_1 := t_1$ if $\ell = t$ and $\ell_1 := y_1$ if $\ell = y$, let $A_\ell := (\ell_1 = 1) \subset \mathbb{A}^5$, and let

$$Y := (f_1 = 0) \cap A_\ell \cap \Phi^{-1}(U), \qquad S := Y \cap (\tau = 0).$$

Since Φ maps $(\tau = 0)$ to $0 \in U$, S is the surface $(\mathrm{in}_w(f)(x_1, y_1, z_1, t_1) = 0)$ in the affine space $(\tau = 0) \cap A_\ell \cong \mathbb{A}^3$. On A_ℓ we have $\Phi^*\ell = \tau^{w_\ell}$. Thus the morphism $Y \backslash (\tau = 0) \to (X \backslash (\ell = 0)) \times \mathbb{G}_m$ given by Φ and τ is an isomorphism onto the closed subvariety $(\tau^{w_\ell} = \ell)$, with inverse given by $x_1 = x/\tau^{w_x}$, $y_1 = y/\tau^{w_y}$, $z_1 = z/\tau^{w_z}$, and $t_1 = t/\tau^{w_t}$. In particular, $Y \backslash (\tau = 0) \to X \backslash (\ell = 0)$ is finite étale of degree w_ℓ, and $Y \backslash (\tau = 0)$ is smooth since X is smooth outside $o \in (\ell = 0)$.

We determine the singular points of S.

- Let $\ell = y$. Then $S = (x_1^2 + 1 + \varphi(z_1, t_1) = 0)$, where φ is the binary form β_4, α_3, or β_5 of degree $m = 4$, 3, or 5 in case (E6), (E7), or (E8). At a singular point of S we have $x_1 = 0$ and $\partial\varphi/\partial z_1 = \partial\varphi/\partial t_1 = 0$, so $\varphi = 0$ by Euler's formula $m\varphi = z_1\partial\varphi/\partial z_1 + t_1\partial\varphi/\partial t_1$, which contradicts $x_1^2 + 1 + \varphi = 0$. Thus S is smooth.
- Let $\ell = t$ in case (E6) or (E8), and let $d = 4$ or 5, respectively. Then $S = (x_1^2 + y_1^3 + \beta_d(z_1, 1) = 0)$, and a singular point q of S has $x_1 = y_1 = 0$ and $z_1 = z_0$ for a root z_0 of $\beta_d(z_1, 1)$ of multiplicity $m \geq 2$. Write $\beta_d(z_1, 1) = (z_1 - z_0)^m e(z_1)$ with $e(z_0) \neq 0$. Then $\xi := (z_1 - z_0)e(z_1)^{1/m}$ is a coordinate of $\mathbb{C}[[x_1, y_1, z_1 - z_0]]$, and S is defined by $x_1^2 + y_1^3 + \xi^m$ near q.

  Here m is the multiplicity of the root $[z_0 : 1]$ of β_d in $\mathbb{P}^1$, so $m = 3$ in case (E6), and $m \in \{2, 3, 4\}$ in case (E8). If $m = 2$, then q is of type A_2, and the quadratic part $x_1^2 + e(z_0)(z_1 - z_0)^2$ is non-degenerate on the $(x_1, z_1 - z_0)$-plane. If $m = 3$, then $y_1^3 + \xi^3$ is a product of three pairwise non-proportional linear forms, so a linear change of the coordinates (y_1, ξ) takes it to $s_1^2 s_2 + s_2^3$ for new linear coordinates s_1, s_2, and q is of type D_4. If $m = 4$, then q is of type E_6. The types A_2, D_4, and E_6 are read off from the normal forms in [KM98, Theorems 4.20 and 4.22].
- Let $\ell = t$ in case (E7). Then $S = (x_1^2 + y_1^3 + \alpha_3(z_1, 1)y_1 = 0)$. At a singular point q of S we have $x_1 = 0$ and $y_1^3 + \alpha_3(z_1, 1)y_1 = 0 = 3y_1^2 + \alpha_3(z_1, 1)$. If $y_1 \neq 0$, then $y_1^2 = -\alpha_3(z_1, 1) = 3y_1^2$, a contradiction. Thus $y_1 = 0$ and $z_1 = z_0$ for a root z_0 of $\alpha_3(z_1, 1)$, and conversely any such point is singular.

If z_0 is a simple root, then $\xi := \alpha_3(z_1, 1) + y_1^2$ is a coordinate near q, S is defined by $x_1^2 + \xi y_1$, so q is of type A_1, and the quadratic part $\partial_z \alpha_3(z_0, 1)(z_1 - z_0)y_1$ of the equation on the $(y_1, z_1 - z_0)$-plane is non-degenerate. If z_0 is a double root, write $\alpha_3(z_1, 1) = (z_1 - z_0)^2 e(z_1)$ with $e(z_0) \neq 0$; then for $\xi := (z_1 - z_0)e(z_1)^{1/2}$, S is defined by $x_1^2 + y_1(y_1^2 + \xi^2)$ near q, and q is of type D_4 [KM98, Theorem 4.20]. By (E7), no other roots occur.

Thus S has finitely many singular points, and they are Du Val points of type D_4 or E_6, or of type A_2, where the quadratic part of the equation of S is non-degenerate on the $(x_1, z_1 - z_0)$-plane, or of type A_1, where the quadratic part of the equation of S is non-degenerate on the $(y_1, z_1 - z_0)$-plane. At a singular point q of S we have $y_1(q) = 0$, so $\partial_\tau f_1(q) = f_{1,1}(q) = 0$, and the derivatives of f_1 in the other coordinates of A_ℓ at q are those of the equation of S, which vanish. Thus f_1 vanishes to order at least two at any singular point q of S, and the quadratic part of f_1 at q restricts on $(\tau = 0)$ to the quadratic part of the equation of S. Since a smooth point of the Cartier divisor S is a smooth point of Y, the singular locus of Y is the finite set $\operatorname{Sing} S$. Hence Y and S are normal by [Har77, Proposition II.8.23]. Since S has only Du Val singularities, S is klt [KM98, Theorem 4.20], so (Y, S) is plt near S by [KM98, Theorem 5.50(1)]. Since $a(E, Y, 0) \geq a(E, Y, S)$ for any prime divisor E over Y, Y is klt near S. Thus Y is klt.

Step 5. In this step we lift the action of G to a component of Y.

Let

$$\widetilde{G} := \{(g, \zeta) \in G \times \mathbb{C}^* \mid \zeta^{w_\ell} = \chi_\ell(g)\},$$

a finite abelian group, act on $\mathbb{A}^5$ by

$$(g,\zeta)^* x_1 = \chi_x(g)\zeta^{-w_x} x_1, \quad (g,\zeta)^* y_1 = \chi_y(g)\zeta^{-w_y} y_1, \quad (g,\zeta)^* z_1 = \chi_z(g)\zeta^{-w_z} z_1,$$
$$(g,\zeta)^* t_1 = \chi_t(g)\zeta^{-w_t} t_1, \quad (g,\zeta)^* \tau = \zeta\tau.$$

Then Φ is equivariant with respect to the projection $\widetilde{G} \to G$, ℓ_1 is invariant, and $(g, \zeta)^* f_1 = \chi_f(g)\zeta^{-2w_x} f_1$, so Y is $\widetilde{G}$-stable. The projection $\widetilde{G} \to G$ is surjective with kernel $\{1\} \times \mu_{w_\ell}$, which acts simply transitively on the fibers of $Y\backslash(\tau = 0) \to X\backslash(\ell = 0)$ by **Step 4**. Since Y is normal, the irreducible components of Y are the connected components. Since $\dim S = 2$, any component of Y intersects $Y\backslash(\tau = 0)$, and $X\backslash(\ell = 0)$ is irreducible. Thus $\{1\} \times \mu_{w_\ell}$ permutes the components of Y transitively.

Let Y_1 be a component of Y and $G' \subset \widetilde{G}$ the stabilizer of Y_1. For $g \in G$ and $(g, \zeta) \in \widetilde{G}$, we have $(g, \zeta)Y_1 = (1, \zeta')Y_1$ for some $\zeta' \in \mu_{w_\ell}$, so $(g, \zeta/\zeta') \in G'$. Thus $G' \to G$ is surjective. The morphism $Y_1\backslash(\tau = 0) \to X\backslash(\ell = 0)$ is finite étale and Galois with group $G' \cap (\{1\} \times \mu_{w_\ell})$, hence

$$|G'| = |G| \cdot [\mathbb{C}(Y_1) : \mathbb{C}(X)] = [\mathbb{C}(Y_1) : \mathbb{C}(X)^G].$$

If $(g, \zeta) \in G'$ acts trivially on Y_1, then $\zeta = 1$ since $\tau \neq 0$ on Y_1, and g acts trivially on $\Phi^* x, \ldots, \Phi^* t$, that is, on $\mathbb{C}(X)$, so $g = 1$. Thus G' acts faithfully on $\mathbb{C}(Y_1)$ and fixes $\mathbb{C}(X)^G$, and by Artin's theorem $\mathbb{C}(Y_1)/\mathbb{C}(X)^G$ is Galois with group G'. Let $\pi : X' \to X/G$ be the normalization of X/G in $\mathbb{C}(Y_1)$. Then π is a finite Galois morphism with group G', and since Y_1 is normal, there exists a G'-equivariant birational morphism $\rho : Y_1 \to X'$ with $\pi \circ \rho = \pi_X \circ \Phi|_{Y_1}$.

Step 6. In this step we show that the local quotient of Y_1 at the center of a lift of P has minimal log discrepancy a.

Let $\Omega_X := dy \wedge dz \wedge dt/(2x)$ on X, and let $\Omega_Y := dy_1 \wedge dz_1 \wedge d\tau/(2x_1)$ if $\ell = t$ and $\Omega_Y := d\tau \wedge dz_1 \wedge dt_1/(2x_1)$ if $\ell = y$. Since $\partial f/\partial x = 2x$ and $\partial f_1/\partial x_1 = 2x_1$, the Poincaré residue shows that Ω_X and Ω_Y are nowhere vanishing sections of ω_X and ω_Y. On A_ℓ we have $d(\Phi^*\ell) = w_\ell \tau^{w_\ell - 1} d\tau$, so

$$\Phi^*(dy \wedge dz \wedge dt) = w_\ell \tau^{w_y + w_z + w_t - 1} \cdot 2x_1 \Omega_Y \quad \text{and} \quad \Phi^*(2x) = 2\tau^{w_x} x_1.$$

Hence

$$\Phi^*\Omega_X = w_\ell \tau^{w_y + w_z + w_t - w_x - 1} \Omega_Y = w_\ell \tau^k \Omega_Y, \tag{12.15}$$

so $K_{Y_1}+B_{Y_1}=\Phi^*K_X$ for $B_{Y_1}:=-k\,\mathrm{div}(\tau)|_{Y_1}$, where $\mathrm{div}(\tau)|_{Y_1}=S\cap Y_1$. Let B' be the $\mathbb{R}$-divisor on X' with $K_{X'}+B'=\pi^*(K_{X/G}+B_{X/G})$. Since $K_X=\pi_X^*(K_{X/G}+B_{X/G})$ and $\pi\circ\rho=\pi_X\circ\Phi|_{Y_1}$, we obtain

$$K_{Y_1}+B_{Y_1}=\rho^*(K_{X'}+B').$$

As in **Step 3**, by [KM98, Lemma 2.45] there exists a prime divisor F over Y_1 such that $\mathrm{ord}_F|_{\mathbb{C}(X)}$ is a positive integer multiple of ord_{F_0}, so $\mathrm{ord}_F|_{\mathbb{C}(X/G)}$ is a positive integer multiple of ord_P. On Y_1 we have $\ell=\tau^{w_\ell}$, so $\mathrm{ord}_F(\tau)=\mathrm{ord}_F(\ell)/w_\ell>0$, and by the choice of ℓ in **Step 3**,

$$\mathrm{ord}_F(x_1)=\mathrm{ord}_F(x)-w_x\frac{\mathrm{ord}_F(\ell)}{w_\ell}\geq 0;$$

likewise $\mathrm{ord}_F(y_1),\mathrm{ord}_F(z_1),\mathrm{ord}_F(t_1)\geq 0$. Thus F has a center on the closure of Y_1 in A_ℓ, contained in $(\tau=0)\subset\Phi^{-1}(U)$. Let $V:=\mathrm{center}_{Y_1}F\subset S$ and $H:=I_{G'}(F)$. Since H stabilizes F and acts trivially on the residue field of ord_F, which contains $\mathbb{C}(V)$, H fixes V pointwise.

By Lemma 3.1 applied to the pair $(X/G,B_{X/G})$, the point $\bar{o}$, the prime divisor P, the finite Galois morphism $\pi:X'\to X/G$ with group G', the $\mathbb{R}$-divisor B', the prime divisor F, the subgroup $H=I_{G'}(F)$, the morphism $\rho:Y_1\to X'$ in place of f, and B_{Y_1}, we obtain

$$\mathrm{mld}(Y_1/H\ni\eta,B_{Y_1/H})=a, \tag{12.16}$$

where $(Y_1/H,B_{Y_1/H}):=(Y_1,B_{Y_1})/H$ and η is the generic point of the image $\bar{V}$ of V in Y_1/H.

Step 7. In this step we show that a belongs to the union of five sets satisfying the ACC.

Assume first that V is a prime divisor, that is, a component of $S\cap Y_1$. Since H fixes V pointwise, the inertia group of V in H is H, so the ramification index of V over $\bar{V}$ is $|H|$ by Lemma 2.6(1). Since $\mathrm{mult}_V B_{Y_1}=-k$, the coefficient of $\bar{V}$ in $B_{Y_1/H}$ is $1-\frac{k+1}{|H|}$ by Lemma 2.6(4). Thus by (12.16),

$$a=\frac{k+1}{|H|}\in\mathcal{A}_1:=\left\{\frac{k+1}{e}\;\middle|\;k\in\{3,4,6\},\ e\in\mathbb{N}^+\right\},$$

and $\mathcal{A}_1$ satisfies the ACC since for any $\epsilon>0$ only finitely many elements of $\mathcal{A}_1$ are at least ϵ.

Assume next that $V=\{q\}$ for a closed point q. Then H fixes q, and $\mathrm{mld}((Y_1\ni q,B_{Y_1})/H)=a$ by (12.16).

- Let q be a smooth point of S. Then Y_1 is smooth at q and τ is a regular parameter at q. Since H is abelian and τ is H-semi-invariant, there exist H-eigenparameters τ,v_1,v_2 at q. By Lemma 3.4 with $\mathbb{C}$ as the base field, applied with $b_1=-k$ for τ and $b_2=b_3=0$,

  $$a=\min\left\{(k+1)u_1+u_2+u_3\;\middle|\;u\in N_H\cap\mathbb{R}^3_{>0}\right\},$$

  where N_H is the lattice of Definition 2.15 for the characters of τ,v_1,v_2. Let $\mathcal{A}_2$ be the set of such minima for $k\in\{3,4,6\}$ and all finite subgroups of $\mathbb{T}^3$. By Corollary 4.6 with $d=3$, $\mathcal{V}=\{(4,1,1),(5,1,1),(7,1,1)\}$, $s=1$, J_1 the unit ideal, and $\Lambda=\{0\}$, $\mathcal{A}_2$ satisfies the ACC.
- Let q be a singular point of S of type A_1 or A_2. By **Step 4**, $\ell=t$, $q=(0,0,z_0,0)$ in the coordinates x_1,y_1,z_1,τ of A_t, and q is of type A_1 only in case (E7) and of type A_2 only in case (E8); in case (E6), all singular points of S are of type D_4. The function z_1-z_0 is an H-eigenfunction: if $z_0\neq 0$ and $h\in H$, then evaluating $h^*z_1=\chi(h)z_1$ at the fixed point q shows that the character χ of z_1 is trivial on H. In case (E7), we use y_1,z_1-z_0,x_1,τ in place of u_1,u_2,z,t in Corollary 9.6, and in case (E8), we use x_1,z_1-z_0,y_1,τ in place of u_1,u_2,z,t.

  By **Step 4**, f_1 vanishes to order at least two at q, and the restriction of the quadratic part of f_1 at q to the (u_1,u_2)-plane is non-degenerate. Near q, $Y=Y_1$ since the irreducible components of Y are disjoint by **Step 5**. Thus near q, Y_1 is a normal klt hypersurface in A_t on which H acts faithfully by **Step 5**, and (Y_1,B_{Y_1}) is sub-klt since $B_{Y_1}\leq 0$. By Corollary 9.6 with $\mathcal{B}=\{(0,-3),(0,-4),(0,-6)\}$ and $(b_z,b_t)=(0,-k)$ (the coefficient

0 for x_1 in case (E7) and for y_1 in case (E8), and $-k$ for τ), a belongs to the set $\mathcal{A}_3$ of numbers in $(0,1)$ of that corollary, which satisfies the ACC.

- Let q be a singular point of S of type D_4 or E_6. Then $\ell = t$, and after shrinking, Y_1 is a normal klt hypersurface in $(A_t \ni q)$, smooth outside q, H acts faithfully on A_t, fixes q, and preserves Y_1, τ is an H-eigenparameter of A_t at q, and $Y_1 \cap (\tau = 0) = S$ near q, which is a Du Val singularity of type D_4 or E_6 at q by **Step 4**. By Proposition 12.11 with $\mathcal{B} = \{-3,-4,-6\}$, τ in place of t, and $b = -k$, a belongs to the set $\mathcal{A}_4$ of numbers in $(0,1)$ of that proposition, which satisfies the ACC.

Assume finally that V is a curve C with generic point η_C. Since $\operatorname{Sing} S$ is finite, S and Y_1 are regular at η_C, and τ is a regular parameter of $\mathcal{O}_{Y_1,\eta_C}$. Since $\tau|_C = 0$ and $\ell_1|_C = 1$, there exists $\sigma_0 \in \{x_1, y_1, z_1, t_1\}$ whose restriction to C is not constant. Let $\sigma := \sum_{h\in H} h^*\sigma_0$. Since H fixes C pointwise, $\sigma|_C = |H| \cdot \sigma_0|_C$ is not constant, and σ is H-invariant and regular at η_C. Moreover, H preserves B_{Y_1} since τ is semi-invariant. Let $Y_{1,\mathbb{C}(\sigma)}$ be the generic fiber of σ, c the closed point of $Y_{1,\mathbb{C}(\sigma)}$ corresponding to η_C, and $B_{\mathbb{C}(\sigma)} = -k \operatorname{div}(\tau)$ the restriction of B_{Y_1}. By Lemma 3.6(3) and (12.16),

$$\operatorname{mld}((Y_{1,\mathbb{C}(\sigma)} \ni c, B_{\mathbb{C}(\sigma)})/H) = \operatorname{mld}(Y_1/H \ni \eta, B_{Y_1/H}) = a.$$

The local ring of $Y_{1,\mathbb{C}(\sigma)}$ at c is $\mathcal{O}_{Y_1,\eta_C}$, which is regular of dimension two, hence $Y_{1,\mathbb{C}(\sigma)}$ is smooth over $\mathbb{C}(\sigma)$ at c, since $\operatorname{char}\mathbb{C}(\sigma) = 0$ [Gro67, Proposition 17.15.1]. Moreover, H acts $\mathbb{C}(\sigma)$-linearly and faithfully on $\mathcal{O}_{Y_1,\eta_C}$, and H acts trivially on the residue field $\kappa(c) = \mathbb{C}(C)$. Since H is abelian and τ is H-semi-invariant, there exists an H-eigenfunction v such that τ, v are H-eigenparameters at c. By Lemma 3.4 with $\mathbb{C}(\sigma)$ as the base field (Remark 2.17), applied with $b_1 = -k$ for τ and $b_2 = 0$,

$$a = \min\left\{(k+1)u_1 + u_2 \;\middle|\; u \in N_H \cap \mathbb{R}^2_{>0}\right\},$$

where N_H is the lattice of Definition 2.15 for the characters of τ, v. Let $\mathcal{A}_5$ be the set of such minima for $k \in \{3,4,6\}$ and all finite subgroups of $\mathbb{T}^2$. By Corollary 4.6 with $d = 2$, $\mathcal{V} = \{(4,1),(5,1),(7,1)\}$, $s = 1$, J_1 the unit ideal, and $\Lambda = \{0\}$, $\mathcal{A}_5$ satisfies the ACC.

Since $\operatorname{center}_{Y_1} F$ is a prime divisor, a curve, or a closed point, we have $a \in \mathcal{A}_1 \cup \cdots \cup \mathcal{A}_5$ in cases (E6), (E7), and (E8). The sets $\mathcal{A}_1, \dots, \mathcal{A}_5$ do not depend on $(X \ni o)$, G, x, y, z, t, and g, and satisfy the ACC. By Lemma 2.10(4), the numbers a in cases (E6), (E7), and (E8) satisfy the ACC, which completes the proof by **Step 1**. □

12.6. **The rank one theorem.** In this subsection, we prove Theorem 12.1. Theorem 11.2 and Proposition 12.13 leave the points of type cE_7 and cE_8 whose form α_3 (type cE_7) or β_5 (type cE_8) of the introduction of Section 12 is a power of a single linear form. At such a point we bring the equation to a form of bounded degree in one variable by two weighted eliminations, and then use Proposition 10.6.

Proof of Theorem 12.1. The proof is long, so we divide it into several steps. Let $\mathcal{A}_{\mathrm{D}}$ be the set of numbers in Theorem 11.2, let $\mathcal{A}_{\mathrm{E}}$ be the set of numbers in Proposition 12.13, and let $\mathcal{A}_{\mathrm{T}}$ be the set of numbers in Proposition 10.6(2) for $\mathcal{B} = \{(0,0)\}$. By Theorem 11.2, Proposition 12.13, and Proposition 10.6(2), the sets $\mathcal{A}_{\mathrm{D}}$, $\mathcal{A}_{\mathrm{E}}$, and $\mathcal{A}_{\mathrm{T}}$ satisfy the ACC. Let $(X \ni o)$ be an isolated Gorenstein terminal threefold singularity of quadratic rank one, which is an isolated cDV singularity by [KM98, Corollary 5.38], and G a finite abelian group acting faithfully on X and fixing o, such that

$$a := \operatorname{mld}((X \ni o, 0)/G) \in (0,1).$$

We show that

$$a \in \mathcal{A}_{\mathrm{D}} \cup \mathcal{A}_{\mathrm{E}} \cup \mathcal{A}_{\mathrm{T}}.$$

Step 1. In this step we reduce to the case when $(X \ni o)$ is of type cE and the equation of X has the form $x^2 + y^3 + \alpha(z,t)y + \beta(z,t)$.

By Lemma 11.3, $(X \ni o)$ is of type cD or cE. If $(X \ni o)$ is of type cD, then it is of type cD_n for some $n \geq 4$, and $a \in \mathcal{A}_{\mathrm{D}}$. Thus we may assume that $(X \ni o)$ is of type cE. By the

paragraph after Set-up 12.2, there exist x, y, z, t and g such that $(X \ni o)$, G, x, y, z, t, and g satisfy the conditions of Set-up 12.2. Let χ_y be the character of y. By Lemma 12.4, there exist a G-semi-invariant power series $y' \in \mathbb{C}[[y, z, t]]$ with the character of y, and power series $\alpha, \beta \in \mathbb{C}[[z, t]]$ with $\operatorname{ord}\alpha \geq 3$ and $\operatorname{ord}\beta \geq 4$, each of which is zero or G-semi-invariant, with characters χ_y^2 and χ_y^3, respectively, such that $g = y'^3 + \alpha(z,t)y' + \beta(z,t)$, and the conditions of Set-up 12.2 hold for x, y', z, t and g. Replacing y with y', we may assume that

$$g = y^3 + \alpha(z,t)y + \beta(z,t).$$

Step 2. In this step we read off the type of $(X \ni o)$ from α and β, and we reduce to two single-root cases.

Let α_3 be the cubic part of α, and β_4, β_5 the quartic and quintic parts of β. By Lemma 12.12, $(X \ni o)$ is of type cE_6 if and only if $\beta_4 \neq 0$, of type cE_7 if and only if $\beta_4 = 0$ and $\alpha_3 \neq 0$, and of type cE_8 if and only if $\alpha_3 = \beta_4 = 0$ and $\beta_5 \neq 0$. If $(X \ni o)$ is of type cE_6, then $a \in \mathcal{A}_{\mathrm{E}}$ by Proposition 12.13(1). If $(X \ni o)$ is of type cE_7 and α_3 has at least two distinct roots in $\mathbb{P}^1$, then $a \in \mathcal{A}_{\mathrm{E}}$ by Proposition 12.13(2). If $(X \ni o)$ is of type cE_8 and β_5 has at least two distinct roots in $\mathbb{P}^1$, then $a \in \mathcal{A}_{\mathrm{E}}$ by Proposition 12.13(3). Thus we may assume that one of the following holds for some $c \in \mathbb{C}\setminus\{0\}$ and a linear form L_0 in z, t:

(E7′) $\operatorname{ord}\beta \geq 5$ and $\alpha_3 = cL_0^3$;
(E8′) $\operatorname{ord}\alpha \geq 4$, $\operatorname{ord}\beta \geq 5$, and $\beta_5 = cL_0^5$.

Step 3. In this step we choose z and t adapted to L_0, and we write g as a weighted homogeneous polynomial plus terms of larger weight.

In case (E7′), $\alpha \neq 0$ since $\alpha_3 \neq 0$, and in case (E8′), $\beta \neq 0$ since $\beta_5 \neq 0$. Hence α (resp. β) is G-semi-invariant, and since the action of G on z, t is diagonal, the non-zero homogeneous part α_3 (resp. β_5) is G-semi-invariant with the character of α (resp. β). For any $\sigma \in G$, the form $\sigma^*\alpha_3 = c(\sigma^*L_0)^3$ (resp. $\sigma^*\beta_5 = c(\sigma^*L_0)^5$) is a multiple of cL_0^3 (resp. cL_0^5) in case (E7′) (resp. (E8′)). By unique factorization in $\mathbb{C}[z, t]$, σ^*L_0 is a multiple of L_0. Thus the line spanned by L_0 is G-stable, and L_0 is an eigenvector in the span of z and t. If the coefficient of z in L_0 is zero, then L_0 is a multiple of t, and we interchange z and t. Then L_0 and t are eigenvectors spanning the span of z and t. Possibly replacing z with $c^{1/3}L_0$ in case (E7′) and with $c^{1/5}L_0$ in case (E8′), which is a linear change of z, t by G-semi-invariant elements, we may assume that $\alpha_3 = z^3$ in case (E7′) and $\beta_5 = z^5$ in case (E8′). The orders of α and β do not change.

Let χ be the character of g, and χ_z the character of z. Then $\chi = \chi_y^3$, and $\chi = \chi_y\chi_z^3$ in case (E7′) and $\chi = \chi_z^5$ in case (E8′).

In case (E7′), let $w = (w_x, w_y, w_z, w_t) := (9, 6, 4, 4)$ and $f_7 := y^3 + yz^3$, so that f_7 is w-homogeneous of weight $2w_x = 3w_y = 18$. Then

$$g = f_7 + (\alpha - \alpha_3)y + \beta, \qquad \mathrm{wt}_w((\alpha - \alpha_3)y) \geq 4 \cdot 4 + 6 = 22, \qquad \mathrm{wt}_w(\beta) \geq 4 \cdot 5 = 20.$$

In case (E8′), let $w := (15, 10, 6, 6)$ and $f_8 := y^3 + z^5$, of weight $2w_x = 3w_y = 30$. Then

$$g = f_8 + \alpha y + (\beta - \beta_5), \qquad \mathrm{wt}_w(\alpha y) \geq 6 \cdot 4 + 10 = 34, \qquad \mathrm{wt}_w(\beta - \beta_5) \geq 6 \cdot 6 = 36.$$

In both cases, any monomial of $g - f_7$ (resp. $g - f_8$) has w-weight larger than $2w_x$.

Step 4. In this step, in case (E7′), we find a G-equivariant automorphism of $\mathbb{C}[[y, z, t]]$ fixing t which brings g to the form

$$y^3 + \big(z^3 + a_1(t)z + a_0(t)\big)y + \sum_{j=0}^{4} b_j(t)z^j \tag{12.17}$$

where $a_0, a_1, b_0, \ldots, b_4 \in t\mathbb{C}[[t]]$.

Let $J_7 := (\partial_y f_7, \partial_z f_7) = (3y^2 + z^3, 3yz^2) \subset \mathbb{C}[y, z]$. Modulo J_7 we have $y^2 \equiv -\frac{1}{3}z^3$ and $yz^2 \equiv 0$, hence $z^5 = z^2 \cdot z^3 \equiv -3y \cdot yz^2 \equiv 0$. Thus any monomial $y^i z^j$ is congruent modulo J_7 to 0 or to

cm for some $c \in \mathbb{C}$ and some element m of

$$\mathcal{M}_7 := \{1,\ z,\ z^2,\ z^3,\ z^4,\ y,\ yz\}$$

with $\mathrm{wt}_w(m) = 6i + 4j$, and $\mathcal{M}_7$ spans $\mathbb{C}[y,z]/J_7$. The elements of $\mathcal{M}_7$ are linearly independent modulo J_7, since J_7 is generated by w-homogeneous polynomials, the elements of $\mathcal{M}_7$ have the pairwise distinct weights $0, 4, 8, 12, 16, 6, 10$, the elements of J_7 of weight at most 16 are the multiples of $3y^2 + z^3$ by constants and by multiples of z (of weights 12 and 16) and the multiples of yz^2 by constants (of weight 14), and z^3, z^4 are not such multiples. Therefore $\mathcal{M}_7$ is a basis of $\mathbb{C}[y,z]/J_7$, which is one-dimensional in each of the weights $0, 4, 6, 8, 10, 12, 16$ and zero in the other weights. We call the elements of $\mathcal{M}_7$ the *basis monomials*. Their weights are at most $16 < 18$.

Since J_7 is generated by the w-homogeneous polynomials $\partial_y f_7$ and $\partial_z f_7$ of weights 12 and 14, any w-homogeneous polynomial $h \in \mathbb{C}[y,z,t]$ of weight $N > 18$ can be written as

$$h = A\partial_y f_7 + B\partial_z f_7 + b, \tag{12.18}$$

where $A, B \in \mathbb{C}[y,z,t]$ are w-homogeneous of weights $N-12$ and $N-14$, and b is a linear combination of the polynomials $t^k m$ with m a basis monomial and $4k + \mathrm{wt}_w(m) = N$, so that $k \geq 1$. If moreover h is G-semi-invariant with character χ, then we may take A, B, and b each zero or G-semi-invariant with characters χ_y, χ_z, and χ, respectively: indeed, $\partial_y f_7$ and $\partial_z f_7$ are semi-invariant with characters $\chi\chi_y^{-1}$ and $\chi\chi_z^{-1}$, and each $t^k m$ is semi-invariant, so it suffices to replace A, B, and b with their isotypic components for χ_y, χ_z, and χ.

We construct G-equivariant automorphisms Φ_N of $\mathbb{C}[[y,z,t]]$, for integers $N \geq 18$, fixing t, such that

$$\Phi_N(g) = f_7 + b_{\leq N} + (\text{terms of } w\text{-weight larger than } N), \tag{12.19}$$

where $b_{\leq N}$ is zero or a G-semi-invariant linear combination of polynomials $t^k m$ with $k \geq 1$, m a basis monomial, and $18 < 4k + \mathrm{wt}_w(m) \leq N$. Let $\Phi_{18} := \mathrm{id}_{\mathbb{C}[[y,z,t]]}$, which satisfies (12.19) by **Step 3**. Let $N > 18$, and let h_N be the part of weight N of $\Phi_{N-1}(g)$. Write $h_N = A_N\partial_y f_7 + B_N\partial_z f_7 + b_N$ as in (12.18), with the characters above. Let φ_N be the automorphism of $\mathbb{C}[[y,z,t]]$ with

$$\varphi_N(y) = y - A_N, \qquad \varphi_N(z) = z - B_N, \qquad \varphi_N(t) = t,$$

which is G-equivariant since A_N and B_N have the characters of y and z, and let $\Phi_N := \varphi_N \circ \Phi_{N-1}$. By Taylor's formula,

$$f_7(y - A_N, z - B_N) = f_7 - A_N\partial_y f_7 - B_N\partial_z f_7 + R_N,$$

where R_N is a sum of products of a partial derivative of f_7 of order $r \geq 2$ with r factors among A_N and B_N. A partial derivative of f_7 with respect to variables $v_1, \ldots, v_r \in \{y, z\}$ is w-homogeneous of weight $18 - \sum_i w_{v_i}$, and A_N, B_N have weights $(N-18) + w_y$, $(N-18) + w_z$. Thus the terms of R_N have weight $18 + r(N-18) \geq 2N - 18 > N$. Similarly, φ_N changes a monomial of weight $M > 18$ only by terms of weight at least $M + (N-18) > N$. Therefore $\Phi_N = \varphi_N \circ \Phi_{N-1}$ satisfies (12.19) with $b_{\leq N} := b_{\leq N-1} + b_N$.

Since A_N and B_N have weights $N-12$ and $N-14$, and the w-adic topology on $\mathbb{C}[[y,z,t]]$ equals the (y,z,t)-adic topology because the weights are positive, the sequence Φ_N converges to a G-equivariant endomorphism Φ of $\mathbb{C}[[y,z,t]]$ fixing t. Since $N - 12 > 6$, A_N has no linear term, and since $N - 14 > 4$, the linear term of B_N is a multiple of y. Thus the linear part of Φ is $y \to y$, $z \to z + c'y$, $t \to t$ for some $c' \in \mathbb{C}$, which is triangular with ones on the diagonal, and Φ is an automorphism. Letting $N \to +\infty$ in (12.19), $\Phi(g)$ equals f_7 plus a linear combination of the basis monomials with coefficients in $t\mathbb{C}[[t]]$, which is (12.17). The elements $\Phi^{-1}(y)$ and $\Phi^{-1}(z)$ are G-semi-invariant with the characters of y and z, and together with x and t they generate the maximal ideal of $\mathbb{C}[[x,y,z,t]]$. Possibly replacing y and z with $\Phi^{-1}(y)$ and $\Phi^{-1}(z)$, we may assume that g equals (12.17). In particular, $\deg_z \alpha \leq 3$ and $\deg_z \beta \leq 4$.

Step 5. In this step, in case (E8′), we find a G-equivariant automorphism of $\mathbb{C}[[y,z,t]]$ fixing t which brings g to the form

$$y^3+\big(a_3(t)z^3+a_2(t)z^2+a_1(t)z+a_0(t)\big)y+z^5+\sum_{j=0}^{3}b_j(t)z^j \tag{12.20}$$

where $a_0,\dots,a_3,b_0,\dots,b_3\in t\mathbb{C}[[t]]$.

Let $J_8:=(\partial_y f_8,\partial_z f_8)=(y^2,z^4)\subset\mathbb{C}[y,z]$. Since $J_8=(y^2,z^4)$ is generated by monomials, the monomials

$$1,\ z,\ z^2,\ z^3,\ y,\ yz,\ yz^2,\ yz^3$$

form a basis of $\mathbb{C}[y,z]/J_8$. For the weights $(w_y,w_z)=(10,6)$, the weights of the basis monomials are at most $28<30$. The argument of **Step 4** proves (12.20), with the following changes. The polynomial f_7, the weight 18, and the weights 12, 14 of $\partial_y f_7$, $\partial_z f_7$ are replaced with f_8, the weight 30, and the weights 20, 24 of $\partial_y f_8$, $\partial_z f_8$. For $N>30$, the coefficients A_N and B_N have weights $N-20=(N-30)+w_y$ and $N-24=(N-30)+w_z$, the terms of R_N have weight $30+r(N-30)>N$, a monomial of weight $M>30$ changes only by terms of weight at least $M+(N-30)>N$, and the polynomials $t^k m$ satisfy $6k+\mathrm{wt}_w(m)=N>28$, so $k\geq 1$. Since $N-20>10$, A_N has no linear term, and since $N-24>6$, the linear term of B_N is a multiple of y. Thus we may assume that g equals (12.20). In particular, $\deg_z\alpha\leq 3$ and $\deg_z\beta\leq 5$.

Step 6. In this step we replace X by a polynomial germ and conclude that $a\in\mathcal{A}_{\mathrm{T}}$.

By **Steps 4 and 5**, in both cases $g=y^3+\alpha y+\beta$ with $\alpha,\beta\in\mathbb{C}[[t]][z]$, $\deg_z\alpha\leq 3$, and $\deg_z\beta\leq 5$. The terms added in **Steps 4 and 5** have w-weight larger than $2w_x$, while any cubic monomial in y,z,t other than y^3 has w-weight smaller than $2w_x=3w_y$, so the cubic part of g is still y^3, and $(X\ni o)$, G, x,y,z,t, and g satisfy the conditions of Set-up 12.2. By Lemma 12.3(2) with $\psi=\mathrm{id}$ and $v=1$, there exists an integer $L\geq 6$ such that, for $f_0:=(x^2+g)_{<L}=x^2+h_0$ with $h_0:=y^3+\alpha_0(z,t)y+\beta_0(z,t)$, $\alpha_0:=\alpha_{<L-1}$, and $\beta_0:=\beta_{<L}$, the group G acts diagonally and faithfully on $\mathbb{A}^4$ and faithfully on $X_0:=(f_0=0)$, X_0 is an isolated Gorenstein terminal threefold germ at 0, hence a normal klt germ, and $\mathrm{mld}((X_0\ni 0,0)/G)=a$. Here α_0 and β_0 are polynomials with $\deg_z\alpha_0\leq 3$ and $\deg_z\beta_0\leq 5$, and f_0 is G-semi-invariant by Lemma 12.3(1). By Proposition 10.6(2) for $h=h_0$, the group G, and $(b_y,b_t)=(0,0)\in\mathcal{B}=\{(0,0)\}$, we obtain $a\in\mathcal{A}_{\mathrm{T}}$.

Step 7. In this step we conclude the proof.

By **Steps 1, 2**, and **6**, any number a in Theorem 12.1 belongs to $\mathcal{A}_{\mathrm{D}}\cup\mathcal{A}_{\mathrm{E}}\cup\mathcal{A}_{\mathrm{T}}$. By Lemma 2.10(4), $\mathcal{A}_{\mathrm{D}}\cup\mathcal{A}_{\mathrm{E}}\cup\mathcal{A}_{\mathrm{T}}$ satisfies the ACC. Thus the set of numbers $\mathrm{mld}((X\ni o,0)/G)\in(0,1)$ in Theorem 12.1 satisfies the ACC. □

13. Abelian quotients of Gorenstein threefolds

In this section, we prove Theorem 13.1. Theorem 13.1 and Construction-Theorem 5.3, applied with trivial G, imply that the minimal log discrepancies in $(0,1)$ at closed points of klt threefold pairs with standard coefficients satisfy the ACC. This is the part of the proof of Theorem 1.3 that concerns klt pairs and values below one. Subsection 13.1 contains Proposition 13.2, the ACC for finite abelian quotients admitting equivariant log smooth models on which the group preserves each component of the boundary, with coefficients in a fixed finite set, and Example 13.3. Neither of them is used in the proof of Theorem 1.3.

Unlike the proof of Theorem 1.5, which passes to Hausdorff limits of the groups (Theorem 16.1), here we compute an exact formula for each value. A crepant equivariant terminal extraction and the quotient by the inertia group of a computing divisor keep the minimal log discrepancy. Then we divide into three cases according to the dimension of the center of the computing divisor: a divisor yields the reciprocal of a positive integer, a curve yields a two-dimensional lattice minimum, and a point yields a three-dimensional lattice minimum if the terminal extraction is smooth there, and otherwise reduces to Sections 9–12.

Theorem 13.1. *Let $(X \ni x)$ be a klt threefold germ with K_X Cartier, and H a finite abelian group acting faithfully on X and fixing x. Then the set of such numbers* $\mathrm{mld}((X \ni x, 0)/H) \in (0,1)$ *satisfies the ACC.*

Proof. The proof is long, so we divide it into several steps. Let $\pi : X \to X/H$ be the quotient morphism, let $(X/H, B_{X/H}) := (X,0)/H$, and let

$$a := \mathrm{mld}((X \ni x, 0)/H).$$

Assume that $a \in (0,1)$.

Step 1. In this step we choose a prime divisor R over X computing a, and we define the group J and the integer e.

Since K_X is Cartier, for any resolution $f : X' \to X$ the divisor $K_{X'} - f^*K_X$ has integer coefficients. Thus $a(F, X, 0) \in \mathbb{Z}$ for any prime divisor F over X, and $a(F, X, 0) > 0$ because X is klt. By the In particular part of Lemma 2.6,

$$a = \inf_F \frac{a(F, X, 0)}{|I_H(F)|},$$

where F runs over the prime divisors over X with $\mathrm{center}_X F = \{x\}$. Since $a(F, X, 0)$ is a positive integer and $|I_H(F)|$ divides $|H|$ by Lagrange's theorem, each quotient in the infimum belongs to $\frac{1}{|H|}\mathbb{N}^+$. Thus the infimum is a minimum. Let R be a prime divisor over X with $\mathrm{center}_X R = \{x\}$ attaining it, and let

$$J := I_H(R), \qquad e := |J|.$$

Then

$$a(R, X, 0) = ea.$$

Let E be the prime divisor over X/H such that $\mathrm{ord}_R|_{\mathbb{C}(X/H)}$ is a positive multiple of ord_E. By Lemma 2.6(1) and (2), $\mathrm{ord}_R|_{\mathbb{C}(X/H)} = e\,\mathrm{ord}_E$, $\mathrm{center}_{X/H} E = \{\pi(x)\}$, and $a(E, X/H, B_{X/H}) = a$.

Step 2. In this step we construct a J-equivariant terminal extraction $g : Y \to X$ with $K_Y = g^*K_X$, and we show that the quotient of Y by J has minimal log discrepancy a at the generic point of the image of $\mathrm{center}_Y R$.

Let $\mathcal{S} := \mathbb{Z}_{\geq 0}$, that is, $N = 1$ and $\alpha_1 = 1$ in Construction-Theorem 5.1. Then

$$\Lambda_{\mathcal{S}} = \{0\}, \qquad \eta_{\mathcal{S}} = 1, \qquad R_{\mathcal{S}} = 1.$$

We apply Construction-Theorem 5.1 to $(X,0)$, x, and $\mathcal{S}$, with J in place of H. Hypothesis (i) holds since X is klt, (ii) holds since J is a subgroup of H, and (iii) holds by **Step 1**. We obtain a J-invariant open neighborhood U of x, the J-equivariant projective birational morphism $g : Y \to U$ of Construction-Theorem 5.1, and an $\mathbb{R}$-divisor $B_Y \geq 0$ on Y with $K_Y + B_Y = g^*K_U$. Since all $a(F, X, 0)$ are positive integers, the condition $a(F, X, 0) \leq 1$ in Construction-Theorem 5.1(1) means $a(F, X, 0) = 1$. Thus the g-exceptional prime divisors are exactly the exceptional prime divisors F over U with $a(F, X, 0) = 1$, and $\mathrm{mult}_F B_Y = 1 - a(F, X, 0) = 0$ for them. Since $g_*B_Y = 0$, we have $B_Y = 0$, that is,

$$K_Y = g^*K_U.$$

By Construction-Theorem 5.1(4), Y is terminal, and the Cartier index of K_Y at any closed point is at most $R_{\mathcal{S}} = 1$. Hence K_Y is Cartier.

Let $U' := \bigcap_{h \in H} hU$. Since x is fixed by H, U' is an H-invariant open neighborhood of x, and the restriction $g^{-1}(U') \to U'$ of g is a J-equivariant projective birational morphism with the properties above. Possibly replacing X with U', Y with $g^{-1}(U')$, and X/H with U'/H, we may assume that $U = X$. Then π is a finite Galois morphism with Galois group H, and $K_Y = g^*K_X$.

Let $Z := \mathrm{center}_Y R$ and let η_Z be the generic point of Z. Since $\mathrm{center}_X R = \{x\}$, we have $Z \subset g^{-1}(x)$. Since $J \subset \mathrm{Stab}_H(R)$, the subvariety Z is J-invariant. By Construction-Theorem 5.1(5), Z is a prime divisor, a complete curve, or a closed point.

Since $J \subset \mathrm{Stab}_H(R)$ acts trivially on the residue field $\kappa(R)$ of ord_R (Definition 2.5), which contains $\mathbb{C}(Z)$, J fixes Z pointwise.

Let $(Y/J, B_{Y/J}) := (Y,0)/J$, let Z^0 be the image of Z in Y/J, and let η be the generic point of Z^0. The hypotheses of Lemma 3.1, with X/H, X, H, R, J, g in place of X, X', G, F, H, f, and with the prime divisor E of **Step 1**, hold: $(X/H, B_{X/H})$ is a pair with $K_X = \pi^*(K_{X/H} + B_{X/H})$ by Lemma 2.6(0), $\operatorname{center}_{X/H} E = \{\pi(x)\}$, $\operatorname{ord}_R|_{\mathbb{C}(X/H)} = e \operatorname{ord}_E$, $J = I_H(R)$, Y is normal and quasi-projective, the crepant pullback of 0 to Y is 0 since $K_Y = g^*K_X$, and

$$\operatorname{mld}(X/H \ni \pi(x), B_{X/H}) = a = a(E, X/H, B_{X/H}) > 0$$

by **Step 1**. Thus, by Lemma 3.1,

$$\operatorname{mld}(Y/J \ni \eta, B_{Y/J}) = a. \tag{13.1}$$

Step 3. In this step we show that $a = \frac{1}{e}$ if Z is a divisor.

Suppose that Z is a divisor on Y. Then $\operatorname{ord}_R = \operatorname{ord}_Z$, and Z is g-exceptional since $g(Z) = \{x\}$. By **Step 2**, $a(R, X, 0) = 1$. By **Step 1**, $ea = 1$, that is,

$$a = \frac{1}{e}.$$

Step 4. In this step we show that, if Z is a curve, then a is a lattice minimum in dimension two.

Suppose that Z is a curve. Since Y is terminal, Y is smooth in codimension two [KM98, Corollary 5.18]. Hence $\mathcal{O}_{Y,\eta_Z}$ is a regular local ring of dimension two, with residue field $\mathbb{C}(Z)$.

Let $\bar{t}$ be a rational function on Y/J regular at η whose restriction to Z^0 is not constant, and let t be the pullback of $\bar{t}$ to Y. Then t is J-invariant and regular at η_Z. Since $Z \to Z^0$ is surjective, the restriction of t to Z is not constant. Let $k := \mathbb{C}(t)$, let Y_k be the generic fiber of t over k, and let $c \in Y_k$ be the closed point corresponding to η_Z. The hypotheses of Lemma 3.6(3), with $(Y,0)$, Z, J in place of (X,B), C, H, hold: J acts faithfully on Y, fixes Z pointwise by **Step 2**, and t is J-invariant. Thus, by Lemma 3.6(3), J acts on Y_k fixing c, and by (13.1),

$$\operatorname{mld}((Y_k \ni c, 0)/J) = \operatorname{mld}(Y/J \ni \eta, B_{Y/J}) = a.$$

Any non-zero polynomial in t with coefficients in $\mathbb{C}$ is a unit of $\mathcal{O}_{Y,\eta_Z}$, because the restriction of t to Z is not constant. Thus $\mathcal{O}_{Y_k,c} = \mathcal{O}_{Y,\eta_Z}$, which is regular of dimension two, and $\kappa(c) = \mathbb{C}(Z)$. Let $\mathfrak{m}_c$ be the maximal ideal of $\mathcal{O}_{Y_k,c}$. Since J acts trivially on $\kappa(c)$ by **Step 2**, J acts $\kappa(c)$-linearly on the two-dimensional $\kappa(c)$-vector space $\mathfrak{m}_c/\mathfrak{m}_c^2$. Since J is finite abelian and $\mathbb{C} \subset \kappa(c)$, the action of J on $\mathfrak{m}_c/\mathfrak{m}_c^2$ is diagonalizable, and the eigenvalues are roots of unity in $\mathbb{C}$. For an eigenvector with character $\chi : J \to \mathbb{C}^*$ and any lift $u \in \mathfrak{m}_c$ of the eigenvector,

$$\frac{1}{|J|} \sum_{h \in J} \chi(h)^{-1} h^* u$$

is a J-semi-invariant element of $\mathfrak{m}_c$ with character χ and whose image in $\mathfrak{m}_c/\mathfrak{m}_c^2$ equals the image of u. Applying the formula to the vectors of an eigenbasis of $\mathfrak{m}_c/\mathfrak{m}_c^2$, we obtain J-eigenparameters x_1, x_2 of Y_k at c.

Since $\mathcal{O}_{Y_k,c}$ is regular and $\operatorname{char} k = 0$, the variety Y_k is smooth over k at c [Gro67, Proposition 17.15.1], and J acts faithfully and k-linearly on Y_k, since J acts faithfully on $\mathbb{C}(Y)$ and fixes t. By Lemma 3.4, applied with k, Y_k, c, J, $d = 2$, and $B = 0$,

$$a = \min\{w_1 + w_2 \mid w \in N_J \cap \mathbb{R}^2_{>0}\},$$

where N_J is the lattice of Definition 2.15 for the characters of x_1, x_2. The right-hand side is a number of Corollary 4.6 for $d = 2$, $\mathcal{V} = \{(1,1)\}$, $s = 1$, the unit ideal, $\Lambda = \{0\}$, and the image of J in $\mathbb{T}^2$.

Step 5. In this step we treat the case when Z is a closed point.

Suppose that $Z = \{y\}$ is a closed point. By **Step 2**, y is fixed by J, and (13.1) reads

$$a = \operatorname{mld}((Y \ni y, 0)/J).$$

The germ $(Y \ni y)$ is terminal with K_Y Cartier, and J is a finite abelian group acting faithfully on Y and fixing y. By Lemma 5.4, with $(Y \ni y)$, J in place of $(X \ni x)$, G, either Y is smooth at y, or $(Y \ni y)$ is an isolated cDV hypersurface singularity, and there exist a smooth fourfold germ $(W \ni y)$ with a faithful action of J, a J-equivariant closed embedding $Y \subset W$ with $Y = (f = 0)$ for a regular J-semi-invariant function f, and J-semi-invariant regular parameters $x_1, \dots, x_4$ of W at y defining a morphism $W \to \mathbb{A}^4$ étale at y.

Suppose that Y is smooth at y. By Lemma 5.4, there exist J-semi-invariant regular parameters x_1, x_2, x_3 of Y at y, that is, J-eigenparameters of Y at y in the sense of Definition 2.15. Since y is a closed point, $\kappa(y) = \mathbb{C}$, and J acts trivially on it. By Lemma 3.4, applied with $k = \mathbb{C}$, $d = 3$, and $B = 0$,

$$a = \min\{w_1 + w_2 + w_3 \mid w \in N_J \cap \mathbb{R}^3_{>0}\},$$

where N_J is the lattice of Definition 2.15 for the characters of x_1, x_2, x_3. The right-hand side is a number of Corollary 4.6 for $d = 3$, $\mathcal{V} = \{(1,1,1)\}$, $s = 1$, the unit ideal, $\Lambda = \{0\}$, and the image of J in $\mathbb{T}^3$.

Suppose that Y is singular at y. Since $W \to \mathbb{A}^4$ is étale at y,

$$\widehat{\mathcal{O}}_{W,y} = \mathbb{C}[[x_1, \dots, x_4]], \qquad \widehat{\mathcal{O}}_{Y,y} = \mathbb{C}[[x_1, \dots, x_4]]/(f).$$

Thus the rank of the quadratic part of f is the quadratic rank of $(Y \ni y)$ in the sense of Definition-Lemma 11.1. Since Y is singular at y, f vanishes to order at least two at y. If the quadratic part of f is zero, then f vanishes to order at least three at y, so for any regular function u on W vanishing to order one at y, the hypersurface section $(u = 0) \cap Y$ of Y is defined by the restriction of f in the smooth threefold germ $(u = 0)$ and has multiplicity at least three at y. Such a surface singularity is not Du Val [KM98, Theorem 4.20], which contradicts the definition of cDV singularities [KM98, Definition 5.32]. Hence the quadratic part of f is non-zero, and the quadratic rank of $(Y \ni y)$ is at least one.

If the quadratic rank of $(Y \ni y)$ is at least two, then the hypotheses of Theorem 9.1, with $(W \ni y)$, J, f, Y in place of $(Y \ni o)$, G, f, X, hold: J acts faithfully on W and on Y, fixing y, f vanishes to order at least two at y, and Y is a normal klt threefold germ at y since Y is terminal. Thus a belongs to the set of Theorem 9.1(1).

If the quadratic rank of $(Y \ni y)$ is one, then $(Y \ni y)$ is an isolated Gorenstein terminal cDV threefold singularity of quadratic rank one, with a faithful action of J fixing y. Thus a belongs to the set of Theorem 12.1, applied with $(Y \ni y)$, J in place of $(X \ni o)$, G.

Step 6. In this step we conclude the proof.

By Construction-Theorem 5.1(5), **Steps 3–5** cover all cases. Thus a belongs to the union of the following sets:

(1) $\{\frac{1}{e} \mid e \in \mathbb{N}^+\}$;
(2) the set of numbers of Corollary 4.6 for $d = 2$, $\mathcal{V} = \{(1,1)\}$, $s = 1$, the unit ideal, and $\Lambda = \{0\}$, where the finite subgroup of $\mathbb{T}^2$ varies;
(3) the set of numbers of Corollary 4.6 for $d = 3$, $\mathcal{V} = \{(1,1,1)\}$, $s = 1$, the unit ideal, and $\Lambda = \{0\}$, where the finite subgroup of $\mathbb{T}^3$ varies;
(4) the set of numbers of Theorem 9.1(1);
(5) the set of numbers of Theorem 12.1.

The set (1) satisfies the ACC, since a strictly increasing sequence $\frac{1}{e_1} < \frac{1}{e_2} < \cdots$ would yield a strictly decreasing sequence of positive integers $e_1 > e_2 > \cdots$. The sets (2) and (3) satisfy the ACC by Corollary 4.6. The sets (4) and (5) satisfy the ACC by Theorems 9.1(1) and 12.1. By Lemma 2.10(4), the union of the sets (1)–(5) satisfies the ACC. Hence the numbers $\mathrm{mld}((X \ni x, 0)/H) \in (0,1)$ satisfy the ACC. □

13.1. **Equivariant log smooth models.** In this subsection, we prove an ACC criterion for crepant quotients of klt threefold germs by finite abelian groups whose actions lift to a smooth higher model on which the crepant pullback of the boundary has simple normal crossing support,

each of whose components is preserved by the group, and coefficients in a fixed finite set (Proposition 13.2). We also compute the normalized volumes of two families of diagonal quotients of $\mathbb{A}^3$ (Example 13.3). Neither result is used in the proofs of the main theorems of this paper. We include them in this section since they reduce minimal log discrepancies of abelian quotients to Lemma 3.4 and Theorem 4.5, which is also the reduction used for Theorem 13.1, there via Corollary 4.6.

Proposition 13.2. *Let $\mathcal{S}$ be a finite set of rational numbers less than* 1. *Consider data consisting of*

- *a klt threefold pair (X, B) with B a $\mathbb{Q}$-divisor and a closed point $x \in X$,*
- *a finite abelian group G acting faithfully on X, fixing x and preserving B, and*
- *a G-equivariant projective birational morphism $f : Y \to X$ from a smooth variety Y such that the $\mathbb{Q}$-divisor B_Y with $K_Y + B_Y = f^*(K_X + B)$ has simple normal crossing support, any non-zero coefficient of B_Y belongs to $\mathcal{S}$, and G preserves any component of* $\operatorname{Supp} B_Y$.

Then the set of numbers $\operatorname{mld}((X \ni x, B)/G)$ *for such data satisfies the ACC.*

Moreover, let (X, B) be a klt threefold pair with B a $\mathbb{Q}$-divisor, $\mathbf{T}$ an algebraic torus acting faithfully on X and preserving B, and $x \in X$ a $\mathbf{T}$-fixed closed point. Then the set

$$\{\operatorname{mld}((X \ni x, B)/G) \mid G \subset \mathbf{T} \text{ a finite subgroup}\}$$

satisfies the ACC, and so does the union of finitely many such sets.

Proof of Proposition 13.2. **Step 1.** In this step we pass to the quotient of Y by G and show that $\operatorname{mld}((X \ni x, B)/G)$ is attained by a prime divisor.

Since f is G-equivariant and G acts faithfully on X, G acts faithfully on Y and B_Y is G-invariant. Let $p : X \to X/G$ and $\pi : Y \to Z := Y/G$ be the quotient morphisms, let $\bar{x} := p(x)$, and write $(X, B)/G = (X/G, B_{X/G})$ and $(Y, B_Y)/G = (Z, C_Z)$. Since $p \circ f$ is G-invariant, $p \circ f = \bar{f} \circ \pi$ for a morphism $\bar{f} : Z \to X/G$. Since $p \circ f$ is proper and π is surjective, $\bar{f}$ is proper. Since $\deg \pi = \deg p = |G|$ and f is birational, $\bar{f}$ is birational. By Lemma 2.6(0),

$$\pi^*(K_Z + C_Z) = K_Y + B_Y = f^*(K_X + B) = f^*p^*(K_{X/G} + B_{X/G}) = \pi^*\bar{f}^*(K_{X/G} + B_{X/G}).$$

Since $\pi_*\pi^*D = |G|D$ for any $\mathbb{R}$-divisor D on Z, we obtain

$$K_Z + C_Z = \bar{f}^*(K_{X/G} + B_{X/G}). \tag{13.2}$$

Thus $a(F, Z, C_Z) = a(F, X/G, B_{X/G})$ for any prime divisor F over Z [KM98, Lemma 2.30]. By Lemma 2.6(0)(2)(3), $B_{X/G} \geq 0$, and any prime divisor E over X/G satisfies

$$a(E, X/G, B_{X/G}) = a(F, X, B)/|I_G(F)| > 0$$

for some prime divisor F over X. Hence $(X/G, B_{X/G})$ is klt and (Z, C_Z) is sub-klt.

Let $a := \operatorname{mld}((X \ni x, B)/G) = \operatorname{mld}(X/G \ni \bar{x}, B_{X/G})$. By [KM98, Theorem 0.2], there exists a log resolution $h : W \to X/G$ of $(X/G, B_{X/G})$ such that $h^{-1}(\bar{x})$ is a divisor and $\operatorname{Exc}(h) \cup h^{-1}(\bar{x}) \cup \operatorname{Supp} h_*^{-1}B_{X/G}$ is a simple normal crossing divisor. Write $K_W + B_W := h^*(K_{X/G} + B_{X/G})$ and let $E_1, \dots, E_l$ be the components of $h^{-1}(\bar{x})$. Since $\dim \bar{x} = 0$, any E_i is exceptional over X/G with $\operatorname{center}_{X/G} E_i = \{\bar{x}\}$, and $\operatorname{mult}_{E_i} B_W = 1 - a(E_i, X/G, B_{X/G})$. Since $(X/G, B_{X/G})$ is klt, $a = \min_i a(E_i, X/G, B_{X/G})$ [Amb99, Section 1, the second paragraph after Definition 1.1], and we let E be a component E_i with $a(E_i, X/G, B_{X/G}) = a$.

Step 2. In this step we express a as a minimal log discrepancy of (Z, C_Z) at a closed point.

Since $\bar{f}$ is proper, $\operatorname{center}_Z E$ exists. Let η be the generic point of $\operatorname{center}_Z E$ and $V := \operatorname{center}_Z E$. Then $\bar{f}(V) = \operatorname{center}_{X/G} E = \{\bar{x}\}$. Since $\bar{f}$ is birational, η is not the generic point of Z, so $\dim V \in \{0, 1, 2\}$. Any prime divisor F over Z with $\operatorname{center}_Z F = V$ satisfies $\operatorname{center}_{X/G} F = \bar{f}(V) = \{\bar{x}\}$, so by (13.2),

$$a(F, Z, C_Z) = a(F, X/G, B_{X/G}) \geq a = a(E, Z, C_Z).$$

Thus $a = \mathrm{mld}(Z \ni \eta, C_Z)$. Since (Z, C_Z) is sub-klt, by Lemma 2.4,

$$a = \mathrm{mld}(Z \ni z', C_Z) - \dim V \tag{13.3}$$

for a general closed point $z' \in V$.

Step 3. In this step we show that for any closed point $z' \in Z$, there exist a finite subgroup $H' \subset \mathbb{T}^3$ and $b_1, b_2, b_3 \in \mathcal{S} \cup \{0\}$ such that

$$\mathrm{mld}(Z \ni z', C_Z) = \min\Big\{ \sum_{j=1}^{3} (1 - b_j) w_j \;\Big|\; w \in N_{H'} \cap \mathbb{R}^3_{>0} \Big\}. \tag{13.4}$$

Let $y \in Y$ be a closed point with $\pi(y) = z'$ and let $H := \{\sigma \in G \mid \sigma(y) = y\}$. Let $P_1, \dots, P_r$ be the components of $\operatorname{Supp} B_Y$ passing through y. Then $r \le 3$, and P_j is H-invariant for any j since G preserves any component of $\operatorname{Supp} B_Y$. Let $I_j \subset \mathcal{O}_{Y,y}$ be the ideal of P_j, t_j a generator of I_j, and $\chi_j : H \to \mathbb{C}^*$ the character by which H acts on $I_j/\mathfrak{m}_y I_j$. Let

$$u_j := \frac{1}{|H|} \sum_{h \in H} \chi_j(h)^{-1} h^* t_j.$$

Then $u_j \in I_j$, $h^* u_j = \chi_j(h) u_j$ for any $h \in H$, and $u_j - t_j \in \mathfrak{m}_y I_j$, so u_j generates I_j by Nakayama's lemma. Since $\operatorname{Supp} B_Y$ is a simple normal crossing divisor, the images of $u_1, \dots, u_r$ in $\mathfrak{m}_y/\mathfrak{m}_y^2$ are linearly independent. Since H is finite abelian, $\mathfrak{m}_y/\mathfrak{m}_y^2$ has a basis of H-semi-invariant vectors containing the images of $u_1, \dots, u_r$. The formula defining u_j lifts the remaining basis vectors to H-semi-invariant functions $u_{r+1}, \dots, u_3 \in \mathfrak{m}_y$ with characters $\chi_{r+1}, \dots, \chi_3$. Thus u_1, u_2, u_3 are H-eigenparameters at y.

Let H act on $\mathbb{A}^3$ with coordinates x_1, x_2, x_3 by $h^* x_j = \chi_j(h) x_j$. The action of H on $\mathbb{A}^3$ is faithful: if $\chi_j(h) = 1$ for any j, then h acts trivially on $\widehat{\mathcal{O}}_{Y,y} = \mathbb{C}[[u_1, u_2, u_3]]$, hence on $\mathcal{O}_{Y,y} \subset \widehat{\mathcal{O}}_{Y,y}$ and on $\mathbb{C}(Y)$, so $h = \mathrm{id}_Y$. Let U be an H-invariant affine open neighborhood of y on which u_1, u_2, u_3 are regular, and let $\varphi := (u_1, u_2, u_3) : U \to \mathbb{A}^3$. Then φ is H-equivariant, $\varphi(y) = 0$, and φ is étale at y. Let $b_j := \mathrm{mult}_{P_j} B_Y \in \mathcal{S}$ for $j \le r$, $b_j := 0$ for $j > r$, and $B_{\mathbb{A}} := \sum_{j=1}^3 b_j \operatorname{div}(x_j)$. Since the étale locus of φ is open and u_j generates I_j for $j \le r$, possibly shrinking U to an H-invariant open neighborhood of y, we may assume that φ is étale, $(u_j = 0) = P_j \cap U$ for $j \le r$, and $\operatorname{Supp} B_Y \cap U = \bigcup_{j \le r} P_j \cap U$, so that $B_Y|_U = \varphi^* B_{\mathbb{A}}$.

We show directly that φ, which is compatible with $\operatorname{Supp} B_Y$, induces étale morphisms of quotients (cf. [Bar03, Proposition 3.4]). Let $q : U \to U/H$ and $q' : \mathbb{A}^3 \to \mathbb{A}^3/H$ be the quotient morphisms, $\bar{y} := q(y)$, $\bar{0} := q'(0)$, $\varphi_H : U/H \to \mathbb{A}^3/H$ the morphism induced by φ, and $g : U/H \to Z$ the morphism induced by $\pi|_U$. Let $\operatorname{Spec} A \subset Y$ be a G-invariant affine open neighborhood of the orbit $G \cdot y$. Since the invariant ring A^G is the kernel of the homomorphism $A \to \bigoplus_{\sigma \in G} A$, $s \to (\sigma^* s - s)_\sigma$, and since the completion of A^G at the maximal ideal of z' is flat over A^G, taking invariants commutes with completion at z'. Since the fiber $\pi^{-1}(z')$ is $G \cdot y$ and G permutes $G \cdot y$ transitively with stabilizer H of y, we obtain

$$\widehat{\mathcal{O}}_{Z,z'} \cong \Big(\prod_{y' \in G \cdot y} \widehat{\mathcal{O}}_{Y,y'} \Big)^G \cong (\widehat{\mathcal{O}}_{Y,y})^H.$$

Since H fixes y, the fiber $q^{-1}(\bar{y})$ is $\{y\}$. Replacing (Y, G, y) with (U, H, y) and with $(\mathbb{A}^3, H, 0)$, the argument above implies $\widehat{\mathcal{O}}_{U/H,\bar{y}} \cong (\widehat{\mathcal{O}}_{Y,y})^H$ and $\widehat{\mathcal{O}}_{\mathbb{A}^3/H,\bar{0}} \cong (\widehat{\mathcal{O}}_{\mathbb{A}^3,0})^H$. Hence g induces an isomorphism $\widehat{\mathcal{O}}_{Z,z'} \to \widehat{\mathcal{O}}_{U/H,\bar{y}}$, and taking H-invariants of the H-equivariant isomorphism $\widehat{\mathcal{O}}_{\mathbb{A}^3,0} \to \widehat{\mathcal{O}}_{U,y}$ induced by φ implies that φ_H induces an isomorphism $\widehat{\mathcal{O}}_{\mathbb{A}^3/H,\bar{0}} \to \widehat{\mathcal{O}}_{U/H,\bar{y}}$. Thus g and φ_H are étale at $\bar{y}$ [Har77, Exercise III.10.4] (cf. [Gro71, Exposé V, Proposition 2.2]). Since étale loci are open and $\bar{y}$ is an isolated point of $g^{-1}(z')$ and of $\varphi_H^{-1}(\bar{0})$, possibly shrinking U to an H-invariant open neighborhood of y, we may assume that g and φ_H are étale, $g^{-1}(z') = \{\bar{y}\}$, and $\varphi_H^{-1}(\bar{0}) = \{\bar{y}\}$.

Write $(U, B_Y|_U)/H = (U/H, C_U)$ and $(\mathbb{A}^3, B_{\mathbb{A}})/H = (\mathbb{A}^3/H, B_{\mathbb{A}^3/H})$. Since φ is étale, $K_U = \varphi^* K_{\mathbb{A}^3}$. By Lemma 2.6(0),

$$\begin{aligned} q^*(K_{U/H} + C_U) &= K_U + B_Y|_U = \varphi^*(K_{\mathbb{A}^3} + B_{\mathbb{A}}) \\ &= \varphi^* q'^*(K_{\mathbb{A}^3/H} + B_{\mathbb{A}^3/H}) = q^* \varphi_H^*(K_{\mathbb{A}^3/H} + B_{\mathbb{A}^3/H}) \end{aligned}$$

and

$$q^*(K_{U/H} + C_U) = (K_Y + B_Y)|_U = (\pi^*(K_Z + C_Z))|_U = q^* g^*(K_Z + C_Z).$$

Since $q_* q^* D = |H| D$ for any $\mathbb{R}$-divisor D on U/H, and since g and φ_H are étale, we obtain $C_U = g^* C_Z = \varphi_H^* B_{\mathbb{A}^3/H}$. By Lemma 9.3, applied to g and to φ_H, the prime divisors over Z with center z', those over U/H with center $\bar{y}$, and those over $\mathbb{A}^3/H$ with center $\bar{0}$ correspond bijectively to each other with equal log discrepancies. Hence

$$\operatorname{mld}(Z \ni z', C_Z) = \operatorname{mld}(U/H \ni \bar{y}, C_U) = \operatorname{mld}((\mathbb{A}^3 \ni 0, B_{\mathbb{A}})/H).$$

Let $H' \subset \mathbb{T}^3$ be the image of H under the characters χ_1, χ_2, χ_3 of x_1, x_2, x_3. Since $b_j < 1$ for any j, Lemma 3.4 implies (13.4).

Step 4. In this step we prove the first part of the proposition.

Let $M := \max(\{1\} \cup \{1 - b \mid b \in \mathcal{S}\})$ and

$$\Lambda := \left\{ 1 - \frac{1-b}{M} \;\middle|\; b \in \mathcal{S} \cup \{0\} \right\}.$$

Since $0 < 1 - b \le M$ for any $b \in \mathcal{S} \cup \{0\}$, Λ is a finite subset of $[0, 1)$. In particular, Λ is a DCC set. Let $\mathcal{A}$ be the set of finite numbers $\psi_{\min}(H', J, c)$, where $H' \subset \mathbb{T}^3$ is a finite subgroup, $J = (J_1, J_2, J_3)$ are non-zero monomial ideals of $\mathbb{C}[x_1, x_2, x_3]$, and $c \in \Lambda^3$, where

$$\psi_{\min}(H', J, c) = \inf\{\psi_{J,c}(w) \mid w \in N_{H'} \cap \mathbb{R}^3_{>0}\}, \quad \psi_{J,c}(w) = \sum_{i=1}^{3} w_i - \sum_{l=1}^{3} c_l \operatorname{wt}_w(J_l),$$

with $\operatorname{wt}_w(J_l) = \min\{\langle w, \alpha \rangle \mid x^\alpha \in J_l\}$ (Definition 2.16), and $N_{H'}$ is the lattice of Definition 2.15. By Theorem 4.5 with $d = s = 3$, $\mathcal{A}$ satisfies the ACC.

Let z', H', and b_1, b_2, b_3 be as in **Step 2** and **Step 3**, and let $b'_j := 1 - \frac{1-b_j}{M} \in \Lambda$. Since $\operatorname{wt}_w((x_j)) = w_j$ for any $w \in \mathbb{R}^3_{\ge 0}$,

$$\sum_{j=1}^{3} (1 - b_j) w_j = M\Big(\sum_{j=1}^{3} w_j - \sum_{j=1}^{3} b'_j \operatorname{wt}_w((x_j)) \Big) = M \psi_{J,b'}(w)$$

for $J = ((x_1), (x_2), (x_3))$ and $b' = (b'_1, b'_2, b'_3)$. By (13.4), $\operatorname{mld}(Z \ni z', C_Z) = M \psi_{\min}(H', J, b')$ and $\psi_{\min}(H', J, b') \in \mathcal{A}$. Since $\dim V = k \in \{0, 1, 2\}$ in (13.3),

$$\operatorname{mld}((X \ni x, B)/G) = a \in \bigcup_{k=0}^{2} \{Mt - k \mid t \in \mathcal{A}\}.$$

The right hand side depends only on $\mathcal{S}$. Since positive scaling and translations preserve the ACC, and a finite union of sets satisfying the ACC satisfies the ACC, the first part of Proposition 13.2 follows.

Step 5. In this step we prove the second part of the proposition.

Let $f_1 = \operatorname{BR}(X) : X_1 \to X$ be the functorial resolution of [Kol05, Theorem 36]. By [Kol05, Proposition 9.1], the action of $\mathbf{T}$ lifts to X_1, so f_1 is $\mathbf{T}$-equivariant. Let $E_1 := \operatorname{Exc}(f_1)$, which is a simple normal crossing divisor by [Kol05, Theorem 36(3)], and let I be the ideal of $f_1^{-1}(\operatorname{Supp} B) \cup E_1$. Let $\Pi : Y \to X_1$ be the principalization $\operatorname{BP}(X_1, I, E_1)$ of [Kol05, Theorem 35], and $f := f_1 \circ \Pi$. Let $\sigma, \operatorname{pr}_2 : \mathbf{T} \times X_1 \to X_1$ be the action morphism and the projection. Since $\mathbf{T}$ preserves B, we have $\sigma^* I = \operatorname{pr}_2^* I$ and $\sigma^* E_1 = \operatorname{pr}_2^* E_1$. Since σ and pr_2 are smooth, BP commutes with σ and pr_2 [Kol05, Theorem 35(4)], so the action of $\mathbf{T}$ lifts to Y (cf. [Kol05, Proposition 9.1],

which lifts the action to X_1 by the functoriality of BR), and f is $\mathbf{T}$-equivariant. The variety Y is smooth and f is projective birational. By [Kol05, Theorem 35(2)(3)], $\Pi^* I$ is the ideal of a simple normal crossing divisor whose support contains $\operatorname{Supp} B_Y$, where $K_Y + B_Y = f^*(K_X + B)$.

Since (X, B) is klt, $\operatorname{mult}_P B_Y = 1 - a(P, X, B) < 1$ for any prime divisor P on Y, and B_Y is a $\mathbb{Q}$-divisor since B is. Let $\mathcal{S}$ be the set of non-zero coefficients of B_Y. Then $\mathcal{S}$ is a finite set of rational numbers less than 1. Since f is $\mathbf{T}$-equivariant and $\mathbf{T}$ preserves B, $\mathbf{T}$ permutes the finitely many components of $\operatorname{Supp} B_Y$. The stabilizer in $\mathbf{T}$ of any component is a closed subgroup of finite index, hence equal to $\mathbf{T}$ since $\mathbf{T}$ is connected. Let $G \subset \mathbf{T}$ be a finite subgroup. Then G is abelian, acts faithfully on X, fixes x, and preserves B, f is G-equivariant, and G preserves any component of $\operatorname{Supp} B_Y$. By the first part of the proposition, the numbers $\operatorname{mld}((X \ni x, B)/G)$ belong to a set satisfying the ACC which depends only on $\mathcal{S}$. For finitely many such $(X, B, \mathbf{T}, x)$, the union of the corresponding sets is a finite union of sets satisfying the ACC, hence satisfies the ACC. □

The following example shows that the normalized volume of a finite abelian quotient singularity is not bounded from below by a positive number depending only on a positive lower bound of the minimal log discrepancy.

Example 13.3. For a klt germ $(X \ni x, B)$, let $\widehat{\operatorname{vol}}(X \ni x, B) := \inf_v a(v, X, B)^3 \operatorname{vol}(v)$ be the normalized volume [Blu18, Section 7.2], where v runs over the real valuations of $\mathbb{C}(X)$ centered at x, $a(v, X, B) \in \mathbb{R} \cup \{+\infty\}$ is the log discrepancy of v [Blu18, Section 7.1], and

$$\operatorname{vol}(v) := \limsup_{t \to +\infty} \frac{6\ell(\mathcal{O}_{X,x}/\{s \mid v(s) \geq t\})}{t^3}.$$

Let m be a positive integer, $B = \sum_j d_j(x_j = 0)$ with $d_j \in \{0, \frac{1}{m}, \dots, \frac{m-1}{m}\}$, and $G \subset (\mathbb{C}^*)^3$ a finite subgroup acting faithfully on $\mathbb{A}^3$. Then

$$\widehat{\operatorname{vol}}((\mathbb{A}^3 \ni 0, B)/G) = \frac{27}{|G|} \prod_{j=1}^{3} (1 - d_j),$$

and $B_{\mathbb{A}^3/G} \in \Psi(\{0, \frac{1}{m}, \dots, \frac{m-1}{m}\})$.

For $n \geq 1$ and $G_n := \mu_{3n+1}(1, n, n)$, we have

$$\operatorname{mld}((\mathbb{A}^3 \ni 0, 0)/G_n) = \frac{2n+1}{3n+1} \quad \text{and} \quad \widehat{\operatorname{vol}}((\mathbb{A}^3 \ni 0, 0)/G_n) = \frac{27}{3n+1}.$$

For $e \geq 4$ with $\gcd(e, 3) = 1$ and $G'_e := \mu_e \times \mu_3$ acting by $(\xi, \zeta) \cdot (x_1, x_2, x_3) = (\xi x_1, \zeta x_2, \zeta x_3)$ (the condition $\gcd(e, 3) = 1$ implies that $G'_e \cong \mu_e \times \mu_3$ is cyclic of order $3e$, like G_n, and $e \geq 4$ implies that the minimal log discrepancy $\frac{2}{3} + \frac{1}{e}$ belongs to $(\frac{2}{3}, 1)$), we have

$$\operatorname{mld}((\mathbb{A}^3 \ni 0, 0)/G'_e) = \frac{2}{3} + \frac{1}{e} \quad \text{and} \quad \widehat{\operatorname{vol}}((\mathbb{A}^3 \ni 0, 0)/G'_e) = \frac{9}{e}.$$

In particular, along both families the ratio of the normalized volume to the minimal log discrepancy tends to 0, while the minimal log discrepancies decrease to $\frac{2}{3}$.

For the first family, identify G_n with a subgroup of $\mathbb{T}^3$. Then G_n is the subgroup of elements of order dividing $3n + 1$ of the circle with weight vector $(-3, 1, 1)$, the G_n converge to the circle with weight vector $(-3, 1, 1)$ in the Hausdorff distance, and

$$\operatorname{Spec} \mathbb{C}[x_1, x_2, x_3]^{\mathbf{T}_{(-3,1,1)}} \cong \mathbb{A}^2/\mu_3(1, 1)$$

has minimal log discrepancy $\frac{2}{3}$ at the image of 0. Thus the limit $\frac{2}{3}$ of the minimal log discrepancies of the first family is the minimal log discrepancy at the image of 0 of the affine quotient by $\mathbf{T}_{(-3,1,1)}$, the complexification of the Hausdorff limit of the groups G_n, and the vectors $\frac{1}{3n+1}(1, n, n)$ computing $\operatorname{mld}((\mathbb{A}^3 \ni 0, 0)/G_n)$ converge to $(0, \frac{1}{3}, \frac{1}{3})$, on the boundary of $\mathbb{R}^3_{>0}$.

Proof of Example 13.3. A finite subgroup of $(\mathbb{C}^*)^3$ acts on $\mathbb{A}^3$ by

$$(g_1, g_2, g_3) \cdot (x_1, x_2, x_3) = (g_1 x_1, g_2 x_2, g_3 x_3),$$

so the character of x_j is $(g_1, g_2, g_3) \to g_j$, and we identify a finite subgroup of $(\mathbb{C}^*)^3$ with the image in $\mathbb{T}^3$ under the characters of x_1, x_2, x_3.

The normalized volume of $(\mathbb{A}^3 \ni 0, B)$. The pair $(\mathbb{A}^3, B)$ is klt [KM98, Corollary 2.31(3)], so $\widehat{\mathrm{vol}}(\mathbb{A}^3 \ni 0, B)$ is attained by a valuation v [Blu18, Main Theorem and Section 7.2]. The torus $(\mathbb{C}^*)^3$ acts on $(\mathbb{A}^3 \ni 0, B)$, so v is $(\mathbb{C}^*)^3$-invariant [XZ21, Corollary 1.2]. Let $s = \sum_\alpha c_\alpha x^\alpha$ be a non-zero polynomial and $\tau := v(s)$. Let L_τ be the subspace of the span of the monomials x^α with $c_\alpha \neq 0$ consisting of the polynomials s' with $v(s') \geq \tau$. Then L_τ is $(\mathbb{C}^*)^3$-invariant. Since distinct monomials have distinct characters, L_τ is spanned by monomials, and $s \in L_\tau$, so $v(x^\alpha) \geq \tau$ whenever $c_\alpha \neq 0$. Hence $v(s) = \min\{v(x^\alpha) \mid c_\alpha \neq 0\}$, that is, $v = v_w$ is the monomial valuation in x_1, x_2, x_3 with weights $w = (v(x_1), v(x_2), v(x_3)) \in \mathbb{R}^3_{>0}$. By [JM12, Proposition 5.1 and (5.1)], applied to the log smooth pair $(\mathbb{A}^3, \sum_j \mathrm{div}(x_j))$, and by [Blu18, Section 7.1],

$$a(v_w, \mathbb{A}^3, B) = \sum_{j=1}^3 w_j - v_w(B) = \sum_{j=1}^3 (1 - d_j) w_j.$$

The ideal $\{s \mid v_w(s) \geq t\}$ of $\mathcal{O}_{\mathbb{A}^3,0}$ is generated by the monomials x^α with $\langle w, \alpha \rangle \geq t$, so the colength of $\{s \mid v_w(s) \geq t\}$ is the number of $\alpha \in \mathbb{Z}^3_{\geq 0}$ with $\langle w, \alpha \rangle < t$, which is $\frac{t^3}{6 w_1 w_2 w_3} + O(t^2)$, since the number of points of $\mathbb{Z}^3$ in the simplex $\{u \in \mathbb{R}^3_{\geq 0} \mid \langle w, u \rangle < t\}$ differs from the volume $\frac{t^3}{6 w_1 w_2 w_3}$ of the simplex by $O(t^2)$. Thus $\mathrm{vol}(v_w) = \frac{1}{w_1 w_2 w_3}$. By the inequality of arithmetic and geometric means,

$$a(v_w, \mathbb{A}^3, B)^3 \, \mathrm{vol}(v_w) = \frac{\left(\sum_j (1 - d_j) w_j\right)^3}{w_1 w_2 w_3} \geq 27 \prod_{j=1}^3 (1 - d_j),$$

with equality for $w_j = \frac{1}{1 - d_j}$. Hence

$$\widehat{\mathrm{vol}}(\mathbb{A}^3 \ni 0, B) = 27 \prod_{j=1}^3 (1 - d_j). \tag{13.5}$$

The quotient. Let $p : \mathbb{A}^3 \to \mathbb{A}^3/G$ be the quotient morphism. Then p is a finite Galois morphism of degree $|G|$, $p^{-1}(p(0)) = \{0\}$, and $K_{\mathbb{A}^3} + B = p^*(K_{\mathbb{A}^3/G} + B_{\mathbb{A}^3/G})$ by Lemma 2.6(0). By Lemma 2.6(2)(3), $(\mathbb{A}^3/G, B_{\mathbb{A}^3/G})$ is klt. By [XZ21, Theorem 1.3] and (13.5),

$$\widehat{\mathrm{vol}}((\mathbb{A}^3 \ni 0, B)/G) = \frac{1}{|G|} \widehat{\mathrm{vol}}(\mathbb{A}^3 \ni 0, B) = \frac{27}{|G|} \prod_{j=1}^3 (1 - d_j).$$

Any prime divisor of $\mathbb{A}^3/G$ is $p(P)$ for a prime divisor P on $\mathbb{A}^3$. By Lemma 2.6(4), $\mathrm{mult}_{p(P)} B_{\mathbb{A}^3/G} = 1 - \frac{1 - \mathrm{mult}_P B}{e_P}$, where e_P is the ramification index of p along P. Since $\mathrm{mult}_P B \in \{0, \frac{1}{m}, \ldots, \frac{m-1}{m}\}$, $B_{\mathbb{A}^3/G} \in \Psi(\{0, \frac{1}{m}, \ldots, \frac{m-1}{m}\})$.

The first family. Let $r := 3n + 1$. Since $\gcd(n, r) = 1$, $G_n \subset \mathbb{T}^3$ is generated by $\frac{1}{r}(1, n, n) \bmod \mathbb{Z}^3$, and $N_{G_n} = \mathbb{Z}^3 + \mathbb{Z}\frac{1}{r}(1, n, n)$. Any coset of $\mathbb{Z}^3$ in N_{G_n} contains a unique vector in $(0, 1]^3$, and subtracting an integral vector with non-negative coordinates does not increase $w_1 + w_2 + w_3$. By Lemma 3.4 with $B = 0$,

$$\mathrm{mld}((\mathbb{A}^3 \ni 0, 0)/G_n) = \min_{0 \leq k \leq r-1} \sigma_k,$$

where $\sigma_0 = 3$ is the sum of the coordinates of $(1, 1, 1)$, and for $1 \leq k \leq r - 1$, $\sigma_k = \frac{k + 2 s_k}{r}$ is the sum of the coordinates of $\frac{1}{r}(k, s_k, s_k)$, with $s_k \in \{1, \ldots, r - 1\}$ the residue of kn modulo r. Since $3n \equiv -1 \bmod r$, the numbers s_k and $k + 2 s_k$ are given in Table 1.

k	s_k	$k+2s_k$
$3h+1,\ 0\le h\le n-1$	$n-h$	$2n+h+1$
$3h+2,\ 0\le h\le n-1$	$2n-h$	$4n+h+2$
$3h,\ 1\le h\le n$	$r-h$	$2r+h$

TABLE 1. The residues s_k of kn modulo $r=3n+1$ for $1\le k\le r-1$.

The smallest numerator in Table 1 is $2n+1$, attained at $k=1$, and $3r>2n+1$. Hence

$$\mathrm{mld}((\mathbb{A}^3\ni 0,0)/G_n)=\frac{2n+1}{3n+1}=\frac{2}{3}+\frac{1}{3(3n+1)}.$$

Since $|G_n|=3n+1$ and $d_j=0$, the formula for the normalized volume with $m=1$ implies $\widehat{\mathrm{vol}}((\mathbb{A}^3\ni 0,0)/G_n)=\frac{27}{3n+1}$. Thus

$$\frac{\widehat{\mathrm{vol}}((\mathbb{A}^3\ni 0,0)/G_n)}{\mathrm{mld}((\mathbb{A}^3\ni 0,0)/G_n)}=\frac{27}{2n+1}.$$

The second family. The group G'_e has order $3e$, and

$$N_{G'_e}=\frac{1}{e}\mathbb{Z}\times\left(\mathbb{Z}^2+\mathbb{Z}\left(\tfrac{1}{3},\tfrac{1}{3}\right)\right)\subset\mathbb{R}\times\mathbb{R}^2.$$

The least positive element of $\frac{1}{e}\mathbb{Z}$ is $\frac{1}{e}$. A vector $(w_2,w_3)\in\mathbb{Z}^2+\mathbb{Z}(\frac{1}{3},\frac{1}{3})$ with $w_2,w_3>0$ satisfies $w_2,w_3\in\mathbb{N}^+$, or $w_2,w_3\in\frac{1}{3}+\mathbb{Z}_{\ge 0}$, or $w_2,w_3\in\frac{2}{3}+\mathbb{Z}_{\ge 0}$, so $w_2+w_3\ge\frac{2}{3}$, with equality for $(w_2,w_3)=(\frac{1}{3},\frac{1}{3})$. By Lemma 3.4 with $B=0$,

$$\mathrm{mld}((\mathbb{A}^3\ni 0,0)/G'_e)=\frac{1}{e}+\frac{2}{3}.$$

The formula for the normalized volume with $m=1$ implies $\widehat{\mathrm{vol}}((\mathbb{A}^3\ni 0,0)/G'_e)=\frac{27}{3e}=\frac{9}{e}$. Thus

$$\frac{\widehat{\mathrm{vol}}((\mathbb{A}^3\ni 0,0)/G'_e)}{\mathrm{mld}((\mathbb{A}^3\ni 0,0)/G'_e)}=\frac{27}{2e+3}.$$

The circle. Let $\phi:\mathbb{R}/\mathbb{Z}\to\mathbb{T}^3$, $\phi(t):=\rho(t(-3,1,1))$. Since the second coordinate of $\phi(t)$ is t, ϕ is an injective homomorphism onto the circle $\rho(\mathbb{R}(-3,1,1))$, so the elements of order dividing r of the circle form $\phi(\frac{1}{r}\mathbb{Z}/\mathbb{Z})$. Since $-\frac{3n}{r}=\frac{1}{r}-1$, we have $\phi(\frac{n}{r})=\frac{1}{r}(1,n,n)\bmod\mathbb{Z}^3$. Since $\gcd(n,r)=1$, $\phi(\frac{n}{r})$ generates $\phi(\frac{1}{r}\mathbb{Z}/\mathbb{Z})$, so $G_n=\phi(\frac{1}{r}\mathbb{Z}/\mathbb{Z})$. The vectors $\frac{1}{r}(1,n,n)$, which compute $\mathrm{mld}((\mathbb{A}^3\ni 0,0)/G_n)$ by the first family, converge to $(0,\frac{1}{3},\frac{1}{3})$ as $n\to+\infty$. For any $t\in\mathbb{R}$ there exists $k\in\mathbb{Z}$ with $|t-\frac{k}{r}|\le\frac{1}{2r}$, so the distance from $\phi(t)$ to G_n is at most $\frac{\sqrt{11}}{2r}$ for the metric on $\mathbb{T}^3$ induced by the Euclidean metric. Since G_n is contained in the circle, the Hausdorff distance between G_n and the circle is at most $\frac{\sqrt{11}}{2(3n+1)}$, which tends to 0.

The torus $\mathbf{T}_{(-3,1,1)}$ acts diagonally, so $\mathbb{C}[x_1,x_2,x_3]^{\mathbf{T}_{(-3,1,1)}}$ is spanned by the monomials $x_1^{\alpha_1}x_2^{\alpha_2}x_3^{\alpha_3}$ of $\mathbf{T}_{(-3,1,1)}$-weight 0, that is, with $\alpha_2+\alpha_3=3\alpha_1$. The homomorphism sending $x_1^{\alpha_1}x_2^{\alpha_2}x_3^{\alpha_3}$ to $u^{\alpha_2}v^{\alpha_3}$ is injective on the monomials of $\mathbf{T}_{(-3,1,1)}$-weight 0, since $\alpha_1=\frac{\alpha_2+\alpha_3}{3}$, and the image is spanned by the monomials $u^{\alpha_2}v^{\alpha_3}$ with $3\mid\alpha_2+\alpha_3$, which span $\mathbb{C}[u,v]^{\mu_3(1,1)}$. Thus

$$\operatorname{Spec}\mathbb{C}[x_1,x_2,x_3]^{\mathbf{T}_{(-3,1,1)}}\cong\mathbb{A}^2/\mu_3(1,1),$$

and the image of 0 is the image of the origin of $\mathbb{A}^2$. Since no non-identity element of $\mu_3(1,1)$ fixes a divisor of $\mathbb{A}^2$, the crepant quotient boundary of $(\mathbb{A}^2,0)$ by $\mu_3(1,1)$ is zero. By Lemma 3.4 with $d=2$ and $B=0$, the minimal log discrepancy of $\mathbb{A}^2/\mu_3(1,1)$ at the image of the origin is the minimum of w_1+w_2 over the vectors $w\in(\mathbb{Z}^2+\mathbb{Z}(\frac{1}{3},\frac{1}{3}))\cap\mathbb{R}^2_{>0}$, which is $\frac{2}{3}$, attained at $(\frac{1}{3},\frac{1}{3})$. □

14. Mixed circles for smooth germs

In this section, we prove the ACC for minimal log discrepancies in $[1,+\infty)$ of quotients of smooth threefold germs by finite abelian groups (Theorem 14.2), and the ACC for quotients by groups which contain a large cyclic subgroup of a mixed circle with bounded index (Theorem 14.3). Theorem 14.3 is the core of Theorem 1.5. In both theorems, we compare the quotient with a quotient of the residual plane pair (T, B_T) of Set-up 14.1, and the ACC in dimension three is reduced to the ACC for surfaces. We say that a minimal log discrepancy is in the high range if it belongs to $[1,+\infty)$.

Set-up 14.1. Let X, u, $\mathbf{T}$, G, c_j, h_j, B, B^0, c_x, c_y, c_u, T, B_T, H, and b_H be as follows, where p,q are coprime positive integers, $P \geq \max\{p,q\}$ and e_0 are positive integers, and $\Gamma \subset [0,1]$ is a DCC set with $\Gamma\backslash\{0\} \neq \emptyset$ and $\gamma := \gamma(\Gamma)$.

(1) $X := \mathbb{A}^3_{x,y,t}$, $u := x^q y^p$, $\mathbf{T} := \mathbf{T}_{(p,-q,0)}$, and G is a finite diagonal abelian group acting faithfully on X and containing $\mu_r(p,-q,0)$ for a positive integer r.

(2) $c_1,\dots,c_s \in \Gamma$ are positive, and $h_j = h_j^0 + R_j$ are non-zero G-semi-invariant polynomials, where $h_j^0 = x^{a_j}y^{b_j}F_j(u,t)$ with non-negative integers a_j, b_j and non-zero polynomials F_j, h_j^0 has the character of h_j, and $R_j \in (x,y)^N$ for a positive integer N.

(3) $B := \sum_j c_j \operatorname{div}(h_j)$, $B^0 := \sum_j c_j \operatorname{div}(h_j^0)$, and (X,B) is klt near 0.

(4) $c_x := \sum_j c_j a_j$, $c_y := \sum_j c_j b_j$, $c_u := \max\{1-\frac{1-c_x}{q}, 1-\frac{1-c_y}{p}\}$, $T := \mathbb{A}^2_{u,t}$, and
$$B_T := \sum_j c_j \operatorname{div}_T(F_j) + c_u \operatorname{div}_T(u).$$
We call (T,B_T) the *residual plane pair*.

(5) When $G \cap \mathbf{T} = \mu_r(p,-q,0)$, we let $H \subset (\mathbb{C}^*)^2$ be the image of G under the characters of u and t, acting diagonally on T, and $b_H := \operatorname{mld}((T \ni 0, B_T)/H)$ whenever (T,B_T) is lc near 0.

The coefficient c_u is the larger of the coefficients of $\operatorname{div}_T(u)$ in the crepant quotients of (X,B^0) by μ_p on $(x \neq 0)$ and by μ_q on $(y \neq 0)$. Since c_u is the larger coefficient, (X,B^0) is lc near 0 if and only if (T,B_T) is lc near 0 (Lemma 14.5).

The main results of this section are the following two theorems.

Theorem 14.2. *Let $\Gamma \subset [0,1]$ be a DCC set. Then the following hold.*

(1) The set of numbers $\operatorname{mld}((\mathbb{A}^3 \ni 0, B)/G) \in [1,+\infty)$, where G is a finite diagonal abelian group acting faithfully on $\mathbb{A}^3$ and $B = \sum_j c_j \operatorname{div}(h_j)$ with $c_j \in \Gamma$ positive, h_j non-zero G-semi-invariant polynomials, and $(\mathbb{A}^3, B)$ klt near 0, satisfies the ACC.

(2) The set of numbers $\operatorname{mld}((X \ni x, B)/G) \in [1,+\infty)$, where $(X \ni x)$ is a smooth threefold germ, G is a finite abelian group acting faithfully on X and fixing x, and $B = \sum_j c_j \operatorname{div}(h_j)$ with $c_j \in \Gamma$ positive, h_j non-zero regular G-semi-invariant functions, and (X,B) klt near x, satisfies the ACC.

(3) Let $\epsilon > 0$. Assume that $\Gamma\backslash\{0\} \neq \emptyset$. Notation and conditions as in Set-up 14.1 for this Γ, and assume moreover that h_j^0 is the sum of all terms of h_j of $\mathbf{T}$-weight $pa_j - qb_j$ for any j, and that $G \cap \mathbf{T} = \mu_r(p,-q,0)$, $|H| \leq e_0$, and $N \geq \frac{r}{2P}$. If r and N are sufficiently large in terms of ϵ, P, e_0, and Γ, and $a := \operatorname{mld}((X \ni 0, B)/G) \geq 1+\epsilon$, then (T,B_T) is lc near 0 and $a \geq b_H$.

Theorem 14.3. *Let p,q be coprime positive integers, $e_0 \geq 1$ an integer, and $\Gamma \subset [0,1]$ a DCC set. Let $X = \mathbb{A}^3_{x,y,t}$ and $\mathbf{T} := \mathbf{T}_{(p,-q,0)}$. Then the set of positive numbers $\operatorname{mld}((X \ni 0, B)/G)$ satisfies the ACC, where G is a finite diagonal abelian group acting faithfully on X with $[G : G \cap \mathbf{T}] \leq e_0$, and $B = \sum_j c_j \operatorname{div}(h_j)$ with $c_j \in \Gamma$ positive, h_j non-zero G-semi-invariant polynomials, and (X,B) klt near 0.*

We prove Theorem 14.2 in Subsection 14.4 and Theorem 14.3 in Subsection 14.8.

14.1. **The mixed-circle normal form.** In this subsection, we compare log canonicity of the pairs in Set-up 14.1 with log canonicity of plane pairs (Lemma 14.5), and we show that strictly increasing sequences of minimal log discrepancies in the high range come from groups containing large cyclic subgroups of mixed circles (Proposition 14.6).

The following example shows the simplest case.

Example 14.4. Let $p = q = 1$, $r \geq 2$ an integer, $G = \mu_r(1,-1,0)$, $0 < c, d < 1$, and $B = d\operatorname{div}(x) + c\operatorname{div}(t)$ on $X = \mathbb{A}^3_{x,y,t}$. With $h_1 = x$, $h_2 = t$, $c_1 = d$, and $c_2 = c$ in Set-up 14.1, we have $c_x = d$, $c_y = 0$, $c_u = d$, H is trivial, and $(T, B_T) = (\mathbb{A}^2_{u,t}, d\operatorname{div}(u) + c\operatorname{div}(t))$, so $b_H = \operatorname{mld}(T \ni 0, B_T) = 2 - c - d$. Since x, y, t are G-eigenparameters at 0, Lemma 3.4 with $b_1 = d$, $b_2 = 0$, and $b_3 = c$ gives

$$\operatorname{mld}((X \ni 0, B)/G) = \min\{(1-d)w_1 + w_2 + (1-c)w_3 \mid w \in N_G \cap \mathbb{R}^3_{>0}\}.$$

For $w \in N_G \cap \mathbb{R}^3_{>0}$, the numbers $w_1 + w_2$ and w_3 are positive integers and $w_2 \geq \frac{1}{r}$. Thus

$$(1-d)w_1 + w_2 + (1-c)w_3 = (1-d)(w_1+w_2) + dw_2 + (1-c)w_3 \geq 2 - c - d + \frac{d}{r},$$

with equality for $w = (\frac{r-1}{r}, \frac{1}{r}, 1)$. Hence $\operatorname{mld}((X \ni 0, B)/G) = 2 - c - d + \frac{d}{r} = b_H + \frac{d}{r}$.

Lemma 14.5. *Notation and conditions as in Set-up 14.1. Let F be a prime divisor over X with center 0, and write $\operatorname{ord}_F|_{\mathbb{C}(u,t)} = \varrho \operatorname{ord}_\nu$ for a positive integer ϱ and a prime divisor ν over T. Let $c'_j \geq 0$, let F'_j be non-zero polynomials in u, t, let a'_j, b'_j be non-negative integers, and let $B' := \sum_j c'_j \operatorname{div}(x^{a'_j} y^{b'_j} F'_j(u,t))$, $c'_x := \sum_j c'_j a'_j$, $c'_y := \sum_j c'_j b'_j$, $c'_u := \max\{1 - \frac{1-c'_x}{q}, 1 - \frac{1-c'_y}{p}\}$, and $B'_T := \sum_j c'_j \operatorname{div}_T(F'_j) + c'_u \operatorname{div}_T(u)$.*

(1) $a(F, X, 0) \geq \varrho\big(a(\nu, T, 0) - \operatorname{ord}_\nu u\big) + \operatorname{ord}_F x + \operatorname{ord}_F y$.

(2) If (T, B'_T) is lc near 0, then

$$a(F, X, B') \geq \varrho a(\nu, T, B'_T) + \big(1 - c'_x - q(1-c'_u)\big)\operatorname{ord}_F x + \big(1 - c'_y - p(1-c'_u)\big)\operatorname{ord}_F y \geq 0,$$

the pair (X, B') is lc near 0, and (X, B') is klt near 0 if moreover (T, B'_T) is klt near 0.

(3) Conversely, if (X, B') is lc near 0, then (T, B'_T) is lc near 0.

Proof. Since $\gcd(p, q) = 1$, there exist integers l_1, l_2 with $ql_2 - pl_1 = 1$. Let $w := x^{l_1} y^{l_2}$. Then $x = u^{l_2} w^{-p}$ and $y = u^{-l_1} w^q$, so $\mathbb{C}(x, y, t) = \mathbb{C}(u, t)(w)$, and w is transcendental over $\mathbb{C}(u, t)$. Since $\frac{dx}{x} \wedge \frac{dy}{y} = (ql_2 - pl_1)\frac{du}{u} \wedge \frac{dw}{w} = \frac{du}{u} \wedge \frac{dw}{w}$, we have

$$dx \wedge dy \wedge dt = -\frac{xy}{u} \cdot \frac{dw}{w} \wedge du \wedge dt. \tag{14.1}$$

Step 1. In this step, we show that for any prime divisor E over X whose center contains 0 and such that $\operatorname{ord}_E|_{\mathbb{C}(u,t)}$ is non-trivial, we can write $\operatorname{ord}_E|_{\mathbb{C}(u,t)} = \varrho_E \operatorname{ord}_{\nu_E}$ for a positive integer ϱ_E and a prime divisor ν_E over T whose center contains 0.

Let η_E be the generic point of $\operatorname{center}_X E$ and ξ_E the image of η_E under the morphism $X \to T$ given by (u, t). Since $\mathcal{O}_{T,\xi_E} \to \mathcal{O}_{X,\eta_E}$ is a local homomorphism, ord_E is non-negative on $\mathcal{O}_{T,\xi_E}$ and positive on the maximal ideal of $\mathcal{O}_{T,\xi_E}$. By Lemma 2.12, applied to $\mathbb{C}(u,t) \subset \mathbb{C}(x,y,t)$, $\operatorname{ord}_E|_{\mathbb{C}(u,t)} = \varrho_E \operatorname{ord}_{\nu_E}$ for a positive integer ϱ_E and a prime divisor ν_E over T with $\operatorname{center}_T \nu_E = \overline{\{\xi_E\}}$. Since $0 \in \operatorname{center}_X E$ maps to $0 \in T$, the center of ν_E contains 0.

If $E = F$, then $\operatorname{ord}_F x, \operatorname{ord}_F y, \operatorname{ord}_F t > 0$, so $\operatorname{ord}_F u, \operatorname{ord}_F t > 0$ and $\operatorname{ord}_F|_{\mathbb{C}(u,t)}$ is non-trivial. In this case $\operatorname{ord}_\nu u, \operatorname{ord}_\nu t > 0$, and the center of ν on T is 0.

Step 2. In this step, for E, ϱ_E, and ν_E as in **Step 1**, we show

$$a(E, X, 0) \geq \varrho_E\big(a(\nu_E, T, 0) - \operatorname{ord}_{\nu_E} u\big) + \operatorname{ord}_E x + \operatorname{ord}_E y. \tag{14.2}$$

Recall that if E is a prime divisor over a normal variety Z, Z is smooth at the generic point z of $\operatorname{center}_Z E$, and a rational $(\dim Z)$-form σ on Z generates ω_Z at z, then $a(E,Z,0) = 1 + \operatorname{ord}_E \sigma$, where $a(E,Z,0)$ is computed on a smooth open neighborhood of z and $\operatorname{ord}_E \sigma$ is the order along E of σ on a variety birational to Z on which E is a divisor.

Let S be a normal surface with a birational morphism $S \to T$ such that ν_E is the valuation of a prime divisor C on S, and let η be the generic point of C. Let ϖ be a uniformizer of $\mathcal{O}_{S,\eta}$, and let $\tau \in \mathcal{O}_{S,\eta}$ be an element whose residue class is transcendental over $\mathbb{C}$. Then $d\varpi \wedge d\tau$ generates ω_S near η, and we write $du \wedge dt = g d\varpi \wedge d\tau$ with $g \in \mathbb{C}(S)^*$. Since $du \wedge dt$ generates ω_T, $\operatorname{ord}_{\nu_E} g = a(\nu_E, T, 0) - 1$. Fix an integer m such that $\operatorname{ord}_E(\varpi^m w) > 0$, and let $w' := \varpi^m w$. Let $V := S \times \mathbb{A}^1_{w'}$ and let ξ be the generic point of $C \times \{0\}$. Then $\mathbb{C}(V) = \mathbb{C}(S)(w') = \mathbb{C}(X)$. The valuation ring of ord_E contains $\mathcal{O}_{S,\eta}[w']$, and $\operatorname{ord}_E(\varpi) = \varrho_E > 0$ and $\operatorname{ord}_E(w') > 0$. Since (ϖ, w') is a maximal ideal of $\mathcal{O}_{S,\eta}[w']$, the center of E on V is $\overline{\{\xi\}}$.

Since $\frac{dw}{w} = \frac{dw'}{w'} - m\frac{d\varpi}{\varpi}$ and $d\varpi \wedge d\varpi = 0$, (14.1) implies

$$dx \wedge dy \wedge dt = -\frac{xyg}{uw'} dw' \wedge d\varpi \wedge d\tau.$$

The form $dx \wedge dy \wedge dt$ generates ω_X, and $dw' \wedge d\varpi \wedge d\tau$ generates ω_V near ξ. Thus

$$\begin{aligned} a(E,X,0) &= 1 + \operatorname{ord}_E(dx \wedge dy \wedge dt) \\ &= a(E,V,0) - \operatorname{ord}_E w' + \operatorname{ord}_E x + \operatorname{ord}_E y - \varrho_E \operatorname{ord}_{\nu_E} u + \varrho_E\big(a(\nu_E,T,0) - 1\big). \end{aligned}$$

Let $U \subset V$ be an open neighborhood of ξ such that U is smooth, ϖ and w' are regular on U, and $\operatorname{div}(\varpi) + \operatorname{div}(w')$ is an snc divisor on U. Since ϖ, w' are regular parameters of $\mathcal{O}_{V,\xi}$, Lemma 2.13 implies

$$a(E,V,0) - \operatorname{ord}_E w' = a(E,U,\operatorname{div}(w')) \geq \operatorname{ord}_E \varpi = \varrho_E.$$

Combining the last two formulas, we obtain (14.2).

Step 3. In this step, we prove (1) and (2).

Let E, ϱ_E, and ν_E be as in **Step 1**. For any j,

$$\operatorname{ord}_E\left(x^{a'_j} y^{b'_j} F'_j(u,t)\right) = a'_j \operatorname{ord}_E x + b'_j \operatorname{ord}_E y + \varrho_E \operatorname{ord}_{\nu_E} F'_j.$$

Moreover, $\varrho_E \operatorname{ord}_{\nu_E} u = \operatorname{ord}_E u = q \operatorname{ord}_E x + p \operatorname{ord}_E y$. By (14.2),

$$\begin{aligned} a(E,X,B') &= a(E,X,0) - c'_x \operatorname{ord}_E x - c'_y \operatorname{ord}_E y - \varrho_E \sum_j c'_j \operatorname{ord}_{\nu_E} F'_j \\ &\geq \varrho_E\Big(a(\nu_E,T,0) - \sum_j c'_j \operatorname{ord}_{\nu_E} F'_j - \operatorname{ord}_{\nu_E} u\Big) \\ &\quad + (1 - c'_x)\operatorname{ord}_E x + (1 - c'_y)\operatorname{ord}_E y \\ &= \varrho_E a(\nu_E, T, B'_T) - (1 - c'_u)\varrho_E \operatorname{ord}_{\nu_E} u + (1 - c'_x)\operatorname{ord}_E x + (1 - c'_y)\operatorname{ord}_E y \\ &= \varrho_E a(\nu_E, T, B'_T) + \big(1 - c'_x - q(1 - c'_u)\big)\operatorname{ord}_E x \\ &\quad + \big(1 - c'_y - p(1 - c'_u)\big)\operatorname{ord}_E y. \end{aligned} \tag{14.3}$$

Since $1 - c'_u = \min\{\frac{1-c'_x}{q}, \frac{1-c'_y}{p}\}$,

$$1 - c'_x - q(1 - c'_u) \geq 0 \quad \text{and} \quad 1 - c'_y - p(1 - c'_u) \geq 0. \tag{14.4}$$

Moreover, $\operatorname{ord}_E x, \operatorname{ord}_E y \geq 0$ because x, y are regular on X.

By **Step 1**, (14.2) for $E = F$ is (1), and (14.3) for $E = F$ is the first inequality of (2). Assume that (T, B'_T) is lc near 0. Since the center of ν on T is 0, $a(\nu, T, B'_T) \geq 0$, and (14.4) implies the second inequality of (2). Thus $a(F,X,B') \geq 0$ for any prime divisor F over X with center 0, that is, $\operatorname{mld}(X \ni 0, B') \geq 0$. If (X, B') were not lc near 0, then $\operatorname{mld}(X \ni 0, B') = -\infty$ (cf. [KM98, Corollary 2.31(1)]). Hence (X, B') is lc near 0.

Assume moreover that (T, B'_T) is klt near 0. Let E be a prime divisor over X whose center contains 0. If $\operatorname{ord}_E|_{\mathbb{C}(u,t)}$ is non-trivial, then the center of ν_E contains 0, so $a(\nu_E, T, B'_T) > 0$, and (14.3) and (14.4) imply $a(E, X, B') > 0$. If $\operatorname{ord}_E|_{\mathbb{C}(u,t)}$ is trivial, then $\operatorname{ord}_E u = \operatorname{ord}_E t = 0$. Since $q \operatorname{ord}_E x + p \operatorname{ord}_E y = \operatorname{ord}_E u = 0$ and $\operatorname{ord}_E x, \operatorname{ord}_E y \geq 0$, we have $\operatorname{ord}_E x = \operatorname{ord}_E y = 0$. Thus $\operatorname{ord}_E(x^{a'_j} y^{b'_j} F'_j(u,t)) = 0$ for any j, and $a(E, X, B') = a(E, X, 0) \geq 1$ because X is smooth. Therefore $a(E, X, B') > 0$ for any prime divisor E over X whose center contains 0. Let $f : X'' \to X$ be a log resolution of (X, B'), and let $X^\circ \subset X$ be the complement of the images of the f-exceptional prime divisors and of the components of $f_*^{-1}B'$ whose images do not contain 0. Then any f-exceptional prime divisor on $f^{-1}(X^\circ)$ and any component of $f_*^{-1}B'$ intersecting $f^{-1}(X^\circ)$ has positive log discrepancy with respect to (X, B'), so $(X^\circ, B'|_{X^\circ})$ is klt [KM98, Corollary 2.31(3)]. Hence (X, B') is klt near 0.

Step 4. In this step, we prove (3).

Let $X_x := X \backslash (x = 0)$ and let $\varphi : X_x \to T \times \mathbb{G}_m$ be the morphism given by (u, t, x). Since $\mathbb{C}[x^{\pm 1}, y, t]^{\mu_p} = \mathbb{C}[x^{\pm 1}, y^p, t] = \mathbb{C}[x^{\pm 1}, u, t]$ for the action of μ_p by $y \to \zeta y$, φ is the quotient morphism by the action of μ_p. On X_x, x is a unit, so

$$B'|_{X_x} = c'_y \operatorname{div}(y) + \sum_j c'_j \operatorname{div}(F'_j(u,t)),$$
$$K_{X_x} = \varphi^* K_{T \times \mathbb{G}_m} + (p-1)\operatorname{div}(y), \quad \text{and} \quad \varphi^* \operatorname{div}(u) = p \operatorname{div}(y).$$

Let

$$B'_{T,y} := \sum_j c'_j \operatorname{div}_T(F'_j) + \left(1 - \frac{1 - c'_y}{p}\right) \operatorname{div}_T(u)$$

and let $\operatorname{pr} : T \times \mathbb{G}_m \to T$ be the projection. Then

$$K_{X_x} + B'|_{X_x} = \varphi^*(K_{T \times \mathbb{G}_m} + \operatorname{pr}^* B'_{T,y}),$$

so $(T \times \mathbb{G}_m, \operatorname{pr}^* B'_{T,y})$ is the crepant quotient of $(X_x, B'|_{X_x})$ by μ_p (Lemma 2.6(0)(4)).

Let X° be a μ_p-invariant open neighborhood of 0 on which (X, B') is lc. Then X° contains the generic point of the line $(y = t = 0)$, and $X^\circ \cap X_x$ is a μ_p-invariant open subset containing the generic point of $(y = t = 0)$. By Lemma 2.6(2)(3), $(T \times \mathbb{G}_m, \operatorname{pr}^* B'_{T,y})$ is lc on $\varphi(X^\circ \cap X_x)$, which is an open neighborhood of the generic point of $\{0\} \times \mathbb{G}_m$. Let $g : T' \to T$ be a log resolution of $(T, B'_{T,y})$. Then $g \times \operatorname{id}_{\mathbb{G}_m} : T' \times \mathbb{G}_m \to T \times \mathbb{G}_m$ is a log resolution of $(T \times \mathbb{G}_m, \operatorname{pr}^* B'_{T,y})$, and for any prime divisor D on T',

$$a(D, T, B'_{T,y}) = a(D \times \mathbb{G}_m, T \times \mathbb{G}_m, \operatorname{pr}^* B'_{T,y}).$$

If the center of D on T contains 0, then the center of $D \times \mathbb{G}_m$ contains $\{0\} \times \mathbb{G}_m$, so $a(D, T, B'_{T,y}) \geq 0$. Removing from T the images of the g-exceptional prime divisors and of the components of $g_*^{-1}B'_{T,y}$ whose images do not contain 0, we deduce that $(T, B'_{T,y})$ is lc near 0 [KM98, Corollaries 2.31 and 2.32].

Interchanging x with y, p with q, and a'_j with b'_j does not change u, T, B', and $\sum_j c'_j \operatorname{div}_T(F'_j)$, interchanges c'_x and c'_y, and replaces X_x with $X \backslash (y = 0)$ and the action of μ_p by $y \to \zeta y$ with the action of μ_q by $x \to \zeta x$. After the interchange, the argument of the previous two paragraphs shows that $(T, B'_{T,x})$ is lc near 0, where

$$B'_{T,x} := \sum_j c'_j \operatorname{div}_T(F'_j) + \left(1 - \frac{1 - c'_x}{q}\right) \operatorname{div}_T(u).$$

Since c'_u is the larger of the coefficients of $\operatorname{div}_T(u)$ in $B'_{T,x}$ and $B'_{T,y}$, B'_T is equal to $B'_{T,x}$ or $B'_{T,y}$. Thus (T, B'_T) is lc near 0. □

The following proposition divides the groups of Theorem 4.8(2) into two classes: for groups whose images in $\mathbb{T}^3$ are contained in members of the list which are finite or whose identity components are coordinate circles, the quotient varieties belong to a finite list, and the groups in

a strictly increasing sequence of the remaining kind contain a large cyclic subgroup of a mixed circle with bounded index.

Proposition 14.6. *Fix $\epsilon > 0$ and a DCC set $\Gamma \subset [0,1]$, and let $\mathcal{M}$ be the set of numbers $\mathrm{mld}((\mathbb{A}^3 \ni 0, B)/G) \geq 1+\epsilon$, where G is a finite diagonal abelian group acting faithfully on $\mathbb{A}^3$ and $B = \sum_j c_j \operatorname{div}(h_j)$ with $c_j \in \Gamma$, h_j non-zero G-semi-invariant polynomials, and $(\mathbb{A}^3, B)$ klt near 0.*

(1) Let $H_1, \dots, H_m$ be the list of Theorem 4.8 for ϵ. The elements of $\mathcal{M}$ given by groups G whose image in $\mathbb{T}^3$ is contained in some H_a which is finite or whose identity component H_a^0 is a coordinate circle satisfy the ACC.

(2) Let $\alpha_i \in \mathcal{M}$ be a strictly increasing sequence, with $\alpha_i = \mathrm{mld}((\mathbb{A}^3 \ni 0, B_i)/G_i)$ and $B_i = \sum_j c_{ij} \operatorname{div}(h_{ij})$ as in the definition of $\mathcal{M}$. Then, after passing to a subsequence and permuting coordinates, there exist coprime positive integers $p, q \leq 1/\epsilon$ and a positive integer e_0 depending only on ϵ such that $G_i \cap \mathbf{T}_{(p,-q,0)} = \mu_{r_i}(p,-q,0)$, a subgroup of index at most e_0 in G_i, with $r_i \to +\infty$. If moreover $\Gamma \setminus \{0\} \neq \emptyset$, then, after deleting units and zero coefficients and passing to a further subsequence, the number s of equations is fixed with $s \leq \lceil 3/\gamma(\Gamma) \rceil$, the coefficients are nondecreasing, and there exist integers κ_j and $a_j, b_j \geq 0$ with $pa_j - qb_j = \kappa_j$ and ($a_j < q$ or $b_j < p$) such that $h_{ij} = x^{a_j} y^{b_j} F_{ij}(x^q y^p, t) + R_{ij}$ with non-zero polynomials F_{ij} and $R_{ij} \in (x,y)^{L_i}$, where $L_i := \max\{0, \lceil (r_i - C_0)/\max\{p,q\} \rceil\}$ and $C_0 := \max_j |\kappa_j|$, the summand $x^{a_j} y^{b_j} F_{ij}(x^q y^p, t)$ is the sum of all terms of h_{ij} of $\mathbf{T}_{(p,-q,0)}$-weight κ_j, and both summands, when non-zero, have the character of h_{ij}.

(3) The conclusions of (1) and (2) also hold for smooth threefold germs with faithful actions of finite abelian groups and boundaries $\sum_j c_j \operatorname{div}(h_j)$ given by regular semi-invariant functions, after replacing the germs by the polynomial models of Construction-Theorem 7.2.

Proof. We identify a finite diagonal subgroup $G \subset (\mathbb{C}^*)^3$ with the image of G in $\mathbb{T}^3$ under the characters of the coordinates, which determines G. If $|G| \leq n$, then G is a subgroup of the finite group $\mu_{n!}^3$ of triples of $n!$-th roots of unity. Hence there exist only finitely many finite diagonal subgroups of $(\mathbb{C}^*)^3$ of order at most n.

Step 1. In this step, we show that for finitely many normal threefolds $V_1, \dots, V_n$, the elements of $\mathcal{M}$ given by groups G such that $\mathbb{A}^3/G$ is isomorphic to some V_k satisfy the ACC.

Let $\alpha = \mathrm{mld}((\mathbb{A}^3 \ni 0, B)/G)$ be such an element, let $\pi : \mathbb{A}^3 \to \mathbb{A}^3/G$ be the quotient morphism, $o := \pi(0)$, and $(\mathbb{A}^3, B)/G = (\mathbb{A}^3/G, B_{\mathbb{A}^3/G})$. Since G preserves B and $(\mathbb{A}^3, B)$ is klt near 0, Lemma 2.10(5) implies that the coefficients of the components of $B_{\mathbb{A}^3/G}$ passing through o belong to the DCC set $\Psi(\Gamma_+ \cap [0,1])$. Let $B^\natural$ be the sum of the components of $B_{\mathbb{A}^3/G}$ passing through o, with their coefficients in $B_{\mathbb{A}^3/G}$. Since $\mathbb{A}^3/G$ is $\mathbb{Q}$-factorial [KM98, Lemma 5.16], $(\mathbb{A}^3/G, B^\natural)$ is a pair. Since $B^\natural = B_{\mathbb{A}^3/G}$ near o, $\alpha = \mathrm{mld}(\mathbb{A}^3/G \ni o, B^\natural)$. Thus α belongs to the union over k of the sets of $\mathrm{mld}(V_k \ni \eta, D)$, where $\eta \in V_k$ is a scheme-theoretic point and (V_k, D) is a pair with $D \in \Psi(\Gamma_+ \cap [0,1]) \setminus \{0\}$. By Theorem 2.19(1), each of the sets of $\mathrm{mld}(V_k \ni \eta, D)$ satisfies the ACC, and Lemma 2.10(4) implies the claim of this step.

Step 2. In this step, we prove (1).

Let H_a be a member of the list such that either H_a is finite or the identity component H_a^0 of H_a is a coordinate circle, and let G be a group as in (1) with image in $\mathbb{T}^3$ contained in H_a.

If H_a is finite, then $|G| \leq |H_a|$.

Assume that H_a^0 is a coordinate circle. After permuting coordinates, $H_a^0 = \rho(\mathbb{R}(1,0,0))$, so $G \cap H_a^0 = \mu_k(1,0,0)$ for $k := |G \cap H_a^0|$. The group $G \cap H_a^0$ is generated by pseudo-reflections, and $\mathbb{A}^3/(G \cap H_a^0) = \mathbb{A}^3_{x^k,y,t}$. The group $G/(G \cap H_a^0)$ acts diagonally and faithfully on the coordinates x^k, y, t, because an element of G acting trivially on x^k, y, and t is of the form $(\lambda, 1, 1)$ and so belongs to H_a^0. Since $G/(G \cap H_a^0)$ embeds into H_a/H_a^0, $|G/(G \cap H_a^0)| \leq |H_a/H_a^0|$. Moreover, $\mathbb{A}^3/G = \mathbb{A}^3_{x^k,y,t}/(G/(G \cap H_a^0))$.

In both cases, since the list $H_1, \dots, H_m$ is finite, the first paragraph of the proof implies that $\mathbb{A}^3/G$ is isomorphic to the quotient of $\mathbb{A}^3$ by one of finitely many finite diagonal groups. Thus **Step 1** implies (1).

Step 3. In this step, we prove the first part of (2).

For any i, $(\mathbb{A}^3, B_i)$ is klt on a G_i-invariant open neighborhood of 0, and x, y, t are G_i-eigenparameters at 0. By Theorem 4.8(2), the image of G_i in $\mathbb{T}^3$ is contained in a member H_{a_i} of the list. Passing to a subsequence, we may assume that $a_i = a$ for any i. Since α_i is strictly increasing, (1) implies that H_a is not finite and that H_a^0 is not a coordinate circle. By Theorem 4.8(1), after permuting coordinates, H_a^0 is the circle with weight vector $(p, -q, 0)$ for coprime positive integers $p, q \le 1/\epsilon$. Let $e_0 := \max_b |H_b/H_b^0|$, where b runs over $1, \dots, m$. Then e_0 depends only on ϵ.

An element $(\lambda^p, \lambda^{-q}, 1)$ of $\mathbf{T}_{(p,-q,0)}$ of finite order n satisfies $\lambda^{pn} = \lambda^{qn} = 1$, so $\lambda^n = 1$ because $\gcd(p, q) = 1$. Write $\lambda = e^{2\pi\sqrt{-1}\theta}$ with $\theta \in \mathbb{Q}$. Then the image of $(\lambda^p, \lambda^{-q}, 1)$ in $\mathbb{T}^3$ is $\rho(\theta(p, -q, 0)) \in H_a^0$. Conversely, an element of G_i whose image in $\mathbb{T}^3$ is $\rho(\theta(p, -q, 0))$ for some $\theta \in \mathbb{R}$ is $(\lambda^p, \lambda^{-q}, 1)$ with $\lambda = e^{2\pi\sqrt{-1}\theta}$. Thus $G_i \cap \mathbf{T}_{(p,-q,0)}$ corresponds to the intersection of the image of G_i in $\mathbb{T}^3$ with H_a^0, and $G_i/(G_i \cap \mathbf{T}_{(p,-q,0)})$ embeds into H_a/H_a^0. In particular, $[G_i : G_i \cap \mathbf{T}_{(p,-q,0)}] \le e_0$. Since $\gcd(p, q) = 1$, the homomorphism $\mathbb{C}^* \to \mathbf{T}_{(p,-q,0)}$, $\lambda \to (\lambda^p, \lambda^{-q}, 1)$, is an isomorphism. Hence the finite subgroup $G_i \cap \mathbf{T}_{(p,-q,0)}$ is $\mu_{r_i}(p, -q, 0)$ for $r_i := |G_i \cap \mathbf{T}_{(p,-q,0)}|$.

Suppose that $|G_i| \le n$ for infinitely many i and some n. By the first paragraph of the proof, the groups G_i with $|G_i| \le n$ belong to a finite set, so **Step 1** implies that the corresponding α_i form a set satisfying the ACC. This contradicts the assumption that α_i is strictly increasing. Thus $|G_i| \to +\infty$, and $r_i \ge |G_i|/e_0$ implies $r_i \to +\infty$.

Step 4. In this step, we prove the second part of (2). Assume that $\Gamma\backslash\{0\} \ne \emptyset$ and let $\gamma := \gamma(\Gamma)$. Then $\gamma > 0$ by Lemma 2.10(2).

Deleting the terms with $c_{ij} = 0$ does not change B_i. If $h_{ij}(0) \ne 0$, then deleting $c_{ij}\operatorname{div}(h_{ij})$ changes B_i only away from 0, and changes neither α_i nor the property that $(\mathbb{A}^3, B_i)$ is klt near 0. Thus we may assume that $c_{ij} \ge \gamma$ and $\operatorname{mult}_0 h_{ij} \ge 1$ for any i, j. Let E_0 be the exceptional divisor of the blow-up of $\mathbb{A}^3$ at 0. Since $(\mathbb{A}^3, B_i)$ is klt near 0,

$$0 < a(E_0, \mathbb{A}^3, B_i) = 3 - \sum_j c_{ij} \operatorname{mult}_0 h_{ij}$$

[KM98, Lemma 2.29]. Hence the number s_i of equations satisfies $\gamma s_i < 3$, so $s_i \le \lceil 3/\gamma \rceil$, and $\operatorname{mult}_0 h_{ij} < 3/\gamma$ for any j. Passing to a subsequence, we may assume that $s_i = s$ for any i. By Lemma 2.10(3), applied once for each j, we may assume that c_{ij} is nondecreasing in i for any j. Since $\operatorname{mult}_0 h_{ij} < 3/\gamma$, h_{ij} contains a monomial $x^{A_{ij}} y^{B_{ij}} t^{C_{ij}}$ of total degree less than $3/\gamma$. The set of such exponents is finite, so passing to a subsequence, we may assume that $(A_{ij}, B_{ij}, C_{ij}) = (A_j, B_j, C_j)$ does not depend on i. Let $\kappa_j := pA_j - qB_j$ and $C_0 := \max_j |\kappa_j|$.

Let $\zeta_i := e^{2\pi\sqrt{-1}/r_i}$ and $\sigma_i := (\zeta_i^p, \zeta_i^{-q}, 1)$, which generates $\mu_{r_i}(p, -q, 0)$. Then

$$\sigma_i^*(x^a y^b t^c) = \zeta_i^{pa-qb} x^a y^b t^c.$$

Since G_i acts diagonally and h_{ij} is G_i-semi-invariant with character χ_{ij}, any monomial occurring in h_{ij} is G_i-semi-invariant with character χ_{ij}. In particular, $\chi_{ij}(\sigma_i) = \zeta_i^{\kappa_j}$, and any monomial $x^a y^b t^c$ occurring in h_{ij} satisfies

$$pa - qb \equiv \kappa_j \pmod{r_i}. \tag{14.5}$$

Since $\gcd(p, q) = 1$, any two integer solutions of $pa - qb = \kappa_j$ differ by an integral multiple of (q, p). Subtracting (q, p) from (A_j, B_j) as long as both coordinates stay non-negative, we obtain a non-negative solution (a_j, b_j) with $a_j < q$ or $b_j < p$. Then the non-negative integer solutions of $pa - qb = \kappa_j$ are exactly $(a_j + ql, b_j + pl)$ with $l \in \mathbb{Z}_{\ge 0}$. Let $x^{a_j} y^{b_j} F_{ij}(x^q y^p, t)$ be the sum of

the terms of h_{ij} whose monomials $x^a y^b t^c$ satisfy $pa - qb = \kappa_j$, and let R_{ij} be the sum of the other terms. Since the $\mathbf{T}_{(p,-q,0)}$-weight of $x^a y^b t^c$ is $pa - qb$ (Definition 2.16), $x^{a_j} y^{b_j} F_{ij}(x^q y^p, t)$ is the sum of all terms of h_{ij} of $\mathbf{T}_{(p,-q,0)}$-weight κ_j. Then F_{ij} is a polynomial, and $F_{ij} \neq 0$ because $x^{A_j} y^{B_j} t^{C_j}$ occurs in h_{ij}. By (14.5), any monomial $x^a y^b t^c$ occurring in R_{ij} satisfies $pa - qb = \kappa_j + nr_i$ for a non-zero integer n. Thus

$$\max\{p, q\}(a + b) \geq |pa - qb| \geq r_i - |\kappa_j| \geq r_i - C_0.$$

Since $a + b$ is a non-negative integer, $a + b \geq L_i$. Hence $R_{ij} \in (x, y)^{L_i}$. Since any monomial occurring in h_{ij} has the character χ_{ij}, both $x^{a_j} y^{b_j} F_{ij}(x^q y^p, t)$ and R_{ij}, when non-zero, have the character of h_{ij}.

Step 5. In this step, we prove (3).

Let $(X \ni x)$ be a smooth threefold germ, G a finite abelian group acting faithfully on X and fixing x, h_j regular G-semi-invariant functions, and $c_j \in \Gamma$ such that (X, B) is klt near x and $\operatorname{mld}((X \ni x, B)/G) \geq 1 + \epsilon$, where $B = \sum_j c_j \operatorname{div}(h_j)$. Deleting the terms with $c_j = 0$, we may assume that $c_j > 0$ for any j. Since G is finite abelian, the action of G on $\mathfrak{m}_x/\mathfrak{m}_x^2$ is diagonalizable, and the eigenvectors of the action lift to G-semi-invariant elements of $\mathfrak{m}_x$ by the projections onto the isotypic components. Thus there exist G-eigenparameters x_1, x_2, x_3 at x. By Construction-Theorem 7.2 with $n = 3$, $Y = X$, $H = G$, $\beta_j = c_j$, and no vectors $\beta^{(k)}$ and no prime divisors P_k, there exist a faithful diagonal action of G on $\mathbb{A}^3$ in which the coordinates have the characters of x_1, x_2, x_3 and semi-invariant polynomials $h_{j,0}$ with the characters of h_j such that, for $B_0 := \sum_j c_j \operatorname{div}(h_{j,0})$, $(\mathbb{A}^3, B_0)$ is klt near 0 and

$$\operatorname{mld}((\mathbb{A}^3 \ni 0, B_0)/G) = \operatorname{mld}((X \ni x, B)/G)$$

by Construction-Theorem 7.2(1) applied to the subgroup G. Hence $\operatorname{mld}((X \ni x, B)/G) \in \mathcal{M}$, and the image of G in $\mathbb{T}^3$ under the characters of the coordinates of $\mathbb{A}^3$ is equal to the image of G under the characters of x_1, x_2, x_3. Thus (1) for smooth threefold germs follows from (1). Applying (2) to the polynomial models $(\mathbb{A}^3, B_0)$ with the action of G, we obtain (2) for smooth threefold germs, where the decompositions in (2) are those of the polynomials $h_{j,0}$. □

14.2. **The residual plane pair.** In this subsection, we bound $\operatorname{mld}((X \ni 0, B)/G)$ in Set-up 14.1 from above by the closed-point mld of the residual plane pair (T, B_T) and by the closed-point mld of the crepant quotient $(T, B_T)/H$.

Construction-Proposition 14.7. Let $\Gamma \subset [0, 1]$ be a DCC set with $\Gamma\backslash\{0\} \neq \emptyset$, and let

$$\Gamma' := \Gamma \cup \Psi(\Gamma_+ \cap [0, 1]), \qquad \Xi := \Gamma'_+ \cap [0, 1].$$

Then Γ', Ξ, and $\Psi(\Xi)$ are DCC subsets of $[0, 1]$, and $\Xi \subset \Psi(\Xi)$. We fix a positive integer ℓ given by Theorem 2.19(2) for the threefold $\mathbb{A}^3$ and the DCC set $\Gamma\backslash\{0\} \subset \mathbb{R}_{>0}$, and a positive integer L given by Theorem 2.18(2) for the DCC set $\Psi(\Xi)$. In particular, ℓ and L depend only on Γ.

Proof. By Lemma 2.10(1), applied to Γ, to $\Gamma_+ \cap [0, 1]$, to Γ', and to Ξ, and since a union of two DCC sets satisfies the DCC, the sets Γ', Ξ, and $\Psi(\Xi)$ are DCC subsets of $[0, 1]$. Moreover, $\Xi \subset \Psi(\Xi)$, since $c = 1 - \frac{1-c}{1}$ for any c. The integer ℓ depends only on $\mathbb{A}^3$ and $\Gamma\backslash\{0\}$, and the integer L depends only on $\Psi(\Xi)$, so ℓ and L depend only on Γ. □

Proposition 14.8. *Let $P \geq 1$ be an integer, $\Gamma \subset [0, 1]$ a DCC set with $\Gamma\backslash\{0\} \neq \emptyset$, and ℓ and L the integers of Construction-Proposition 14.7 for Γ. Notation and conditions as in Set-up 14.1 for P and Γ, and assume that $N \geq \ell$ and $N > 3/\gamma$. Then:*

(1) $a_j + b_j \leq \operatorname{mult}_0 h_j^0 = \operatorname{mult}_0 h_j < \frac{3}{\gamma}$ for any j, (X, B^0) is lc near 0, and $\operatorname{mld}(X \ni 0, B^0) = \operatorname{mld}(X \ni 0, B)$;

(2) (T, B_T) is lc near 0, $c_x, c_y \in \Gamma_+ \cap [0, 1]$, $c_u \in \Psi(\Gamma_+ \cap [0, 1])$, the coefficients of B_T belong to a DCC set depending only on Γ and P, and $b := \operatorname{mld}(T \ni 0, B_T)$ is attained by a prime divisor ν over T with $\operatorname{center}_T \nu = \{0\}$ and $a(\nu, T, 0) \leq L$.

Proof. (1) Let E_0 be the exceptional divisor of the blow-up of X at 0. Since (X, B) is klt near 0, by [KM98, Lemma 2.29],

$$a(E_0, X, B) = 3 - \sum_j c_j \operatorname{mult}_0 h_j > 0.$$

As $c_j \geq \gamma$ for any j, we obtain $\operatorname{mult}_0 h_j < 3/\gamma < N \leq \operatorname{mult}_0 R_j$, so $\operatorname{mult}_0 h_j^0 = \operatorname{mult}_0 h_j$ (if $h_j(0) \neq 0$, both sides are 0). Since $h_j^0 = x^{a_j} y^{b_j} F_j(u, t)$, we have $a_j + b_j \leq \operatorname{mult}_0 h_j^0$.

Let $\mathfrak{m}_0 := (x, y, t)$. Since $N \geq \ell$, for any j we have

$$h_j - h_j^0 = R_j \in (x, y)^N \subset \mathfrak{m}_0^\ell, \quad \text{hence} \quad (h_j) + \mathfrak{m}_0^\ell = (h_j^0) + \mathfrak{m}_0^\ell.$$

For any prime divisor E over X,

$$a\Big(E, X, 0, \prod_j (h_j)^{c_j}\Big) = a(E, X, B) \quad \text{and} \quad a\Big(E, X, 0, \prod_j (h_j^0)^{c_j}\Big) = a(E, X, B^0).$$

Since $c_j \in \Gamma \backslash \{0\}$ for any j, by the choice of ℓ in Construction-Proposition 14.7, Theorem 2.19(2), applied to the pair $(X, 0)$, the point 0, and the $\mathbb{R}$-ideals $\prod_j (h_j)^{c_j}$ and $\prod_j (h_j^0)^{c_j}$, implies that

$$\operatorname{mld}(X \ni 0, B^0) = \operatorname{mld}(X \ni 0, B).$$

By [Amb99, Section 1, the second paragraph after Definition 1.1], for an $\mathbb{R}$-divisor $D \geq 0$ on X, $\operatorname{mld}(X \ni 0, D)$ is the minimum of $a(E_i, X, D)$ over finitely many prime divisors E_i over X with center 0 if (X, D) is lc near 0, and $\operatorname{mld}(X \ni 0, D) = -\infty$ otherwise. Since (X, B) is klt near 0, $\operatorname{mld}(X \ni 0, B) > 0$. Thus $\operatorname{mld}(X \ni 0, B^0) > 0$, and (X, B^0) is lc near 0.

(2) The $\mathbb{R}$-divisor B^0 is the $\mathbb{R}$-divisor B' of Lemma 14.5 for $c_j' = c_j$, $a_j' = a_j$, $b_j' = b_j$, and $F_j' = F_j$, and then $B_T' = B_T$. By (1) and Lemma 14.5(3), (T, B_T) is lc near 0.

The prime divisor $\{x = 0\}$ of X passes through 0. Since $\operatorname{mult}_{\{x=0\}} \operatorname{div}(F_j(u, t)) \geq 0$ for any j, we have $\operatorname{mult}_{\{x=0\}} B^0 \geq c_x$. Since (X, B^0) is lc near 0, $c_x \leq 1$. Likewise, the prime divisor $\{y = 0\}$ of X passes through 0 and $\operatorname{mult}_{\{y=0\}} B^0 \geq c_y$, so $c_y \leq 1$. As c_x and c_y are finite sums of elements of Γ, we obtain

$$c_x, c_y \in \Gamma_+ \cap [0, 1], \quad c_u \in \Psi(\Gamma_+ \cap [0, 1]), \quad \text{and} \quad 0 \leq c_u \leq 1.$$

In particular, $B_T \geq 0$. Let Γ' and Ξ be as in Construction-Proposition 14.7. Then $B_T = \sum_j c_j \operatorname{div}_T(F_j) + c_u \operatorname{div}_T(u)$ with $c_j, c_u \in \Gamma'$. Since F_j and u are regular functions on T, the coefficient of B_T along any prime divisor of T is a combination of the c_j and c_u with non-negative integer coefficients, so it belongs to Γ'_+. The set $\Gamma' \subset [0, 1]$ is a DCC set by Construction-Proposition 14.7, and Γ'_+ satisfies the DCC by Lemma 2.10(1) applied to Γ'. Thus the coefficients of B_T belong to the DCC set Γ'_+, which depends only on Γ.

By Lemma 2.10(5), applied to T, the point 0, the trivial group, and the DCC set Γ', there exists an affine open neighborhood T' of 0 such that $(T', B_T|_{T'})$ is lc and the coefficients of $B_T|_{T'}$ belong to Ξ, as Theorem 2.18(2) requires. The prime divisors over T with center 0 are the prime divisors over T' with center 0, and their log discrepancies with respect to (T, B_T) and $(T', B_T|_{T'})$ coincide, so $b = \operatorname{mld}(T' \ni 0, B_T|_{T'})$. Since $\Xi \subset \Psi(\Xi)$, by the choice of L in Construction-Proposition 14.7, Theorem 2.18(2), applied to the lc surface germ $(T' \ni 0, B_T|_{T'})$, implies that there exists a prime divisor ν over T with $\operatorname{center}_T \nu = \{0\}$, $a(\nu, T, B_T) = b$, and $a(\nu, T, 0) \leq L$. □

We bound $\operatorname{mld}((X \ni 0, B)/G)$ from above by $a(F, X, B)/|I_G(F)|$ for a prime divisor F over X given by a Gauss extension (Definition-Proposition 2.11) of a multiple of a prime divisor ν over T such that the prime divisor over T/H induced by ν computes b_H and has log discrepancy at most L with respect to T/H (Theorem 2.18(2)).

Proposition 14.9. *In Set-up 14.1, assume that $G \cap \mathbf{T} = \mu_r(p, -q, 0)$, $|H| \leq e_0$, $N \geq \ell$, and $N > 3/\gamma$, where ℓ and L are the integers of Construction-Proposition 14.7 for Γ. Then the kernel*

of the homomorphism $G \to H$ is $G \cap \mathbf{T}$, (T, B_T) is lc near 0, and for any η with $0 < \eta \le \frac{1}{8Pe_0}$, $r \ge \frac{2P}{\eta}$, and $N > \frac{4L}{\gamma\eta}$,

$$\operatorname{mld}((X \ni 0, B)/G) \le b_H + 2\eta.$$

In particular, for any $\epsilon > 0$ there exist r_0, N_0 depending only on ϵ, P, e_0, and Γ such that if $r \ge r_0$, $N \ge N_0$, and $\operatorname{mld}((X \ni 0, B)/G) \ge 1 + \epsilon$, then $b_H > 1 + \frac{\epsilon}{2}$, T/H is smooth, the exceptional divisor of the blow-up of T/H at the image $\bar{0}$ of 0 is the unique prime divisor over T/H computing b_H, $\operatorname{mult}_{\bar{0}} B_{T/H} = 2 - b_H$, $(T/H, B_{T/H})$ is klt near $\bar{0}$, the pairs (T, B_T) and (X, B^0) are klt near 0, and H is a cyclic group of order at most e_0 acting on u or on t by a character and trivially on the other coordinate. Moreover, for fixed Γ, P, and e_0, the set of the numbers b_H for all data as in Set-up 14.1 with $G \cap \mathbf{T} = \mu_r(p, -q, 0)$, $|H| \le e_0$, $N \ge \ell$, and $N > 3/\gamma$ satisfies the ACC.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove the first two conclusions, and we bound the coefficients of $(T, B_T)/H$.

We identify $g \in G$ with $(g_1, g_2, g_3) \in (\mathbb{C}^*)^3$, where $g^*x = g_1x$, $g^*y = g_2y$, and $g^*t = g_3t$. Then $g^*u = g_1^q g_2^p u$, and the homomorphism $G \to H$ is $g \to (g_1^q g_2^p, g_3)$. For $g = (s^p, s^{-q}, 1) \in \mathbf{T}$ we have $g_1^q g_2^p = s^{pq-pq} = 1$, so $G \cap \mathbf{T}$ is contained in the kernel. Fix integers l_1, l_2 with $ql_2 - pl_1 = 1$. If $g_1^q g_2^p = 1$ and $g_3 = 1$, let $s := g_1^{-l_1} g_2^{-l_2}$. Using $g_2^p = g_1^{-q}$ and $g_1^q = g_2^{-p}$, we obtain

$$s^p = g_1^{-pl_1} g_1^{ql_2} = g_1 \quad \text{and} \quad s^{-q} = g_2^{-pl_1} g_2^{ql_2} = g_2,$$

so $g \in \mathbf{T}$. Thus the kernel of $G \to H$ is $G \cap \mathbf{T} = \mu_r(p, -q, 0)$. By Proposition 14.8(2), (T, B_T) is lc near 0.

By Proposition 14.8(2), $c_x, c_y \in \Gamma_+ \cap [0, 1]$ and $c_u \in \Psi(\Gamma_+ \cap [0, 1])$. Let Γ' and Ξ be as in Construction-Proposition 14.7. Then $c_j, c_u \in \Gamma'$ for any j.

Since G is diagonal and h_j^0 is G-semi-invariant, all the monomials $x^{a_j+qi} y^{b_j+pi} t^{i'}$ occurring in $h_j^0 = x^{a_j} y^{b_j} F_j(u, t)$ have the character of h_j^0. Thus F_j is H-semi-invariant, u and t are H-semi-invariant, and B_T is H-invariant. By Lemma 2.10(5), applied to T, the point 0, the group H, and the DCC set Γ', there exists an H-invariant affine open neighborhood T' of 0 such that $(T', B_T|_{T'})$ is lc and the coefficients of $B_{T'/H}$ belong to $\Psi(\Xi)$, where $(T'/H, B_{T'/H}) := (T', B_T|_{T'})/H$. Let $\pi : T \to T/H$ be the quotient morphism and $\bar{0} := \pi(0)$. Then $(T'/H, B_{T'/H})$ is an open subset of $(T/H, B_{T/H})$ containing $\bar{0}$, it is lc by Lemma 2.6(2)(3), and $b_H = \operatorname{mld}(T'/H \ni \bar{0}, B_{T'/H})$.

By Proposition 14.8(1), $a_j + b_j < \frac{3}{\gamma}$ for any j.

Step 2. In this step we choose a prime divisor ν over T related to b_H, and we introduce a subgroup J of G, two characters α, β of J, and the lattice N_J.

The surface T/H is $\mathbb{Q}$-factorial [KM98, Lemma 5.16], so $K_{T/H}$ is $\mathbb{Q}$-Cartier and $a(E', T/H, 0)$ is defined for any prime divisor E' over T/H. By Theorem 2.18(2), applied to the lc surface germ $(T'/H \ni \bar{0}, B_{T'/H})$, there exists a prime divisor E over T/H with $\operatorname{center}_{T/H} E = \{\bar{0}\}$, $a(E, T/H, B_{T/H}) = b_H$, and $a(E, T/H, 0) \le L$. By Lemma 2.6(3)(1), there exists a prime divisor ν over T with $\operatorname{ord}_\nu|_{\mathbb{C}(T/H)} = e_1 \operatorname{ord}_E$, where $J_0 := I_H(\nu)$ and $e_1 := |J_0| \le |H| \le e_0$. Since H fixes 0, $\pi^{-1}(\bar{0}) = \{0\}$, so $\operatorname{center}_T \nu = \{0\}$ by Lemma 2.6(2). Write $(T, 0)/H = (T/H, \Delta_H)$, where $\Delta_H \ge 0$ by Lemma 2.6(0). By Lemma 2.6(2),

$$a(\nu, T, B_T) = e_1 b_H \quad \text{and} \quad a(\nu, T, 0) = e_1 a(E, T/H, \Delta_H) \le e_1 a(E, T/H, 0) \le e_1 L,$$

where $a(E, T/H, \Delta_H) \le a(E, T/H, 0)$ because $\Delta_H \ge 0$ [KM98, Lemma 2.27]. Let $m := \operatorname{ord}_\nu u$. Since $\operatorname{center}_T \nu = \{0\}$, we have $m \ge 1$ and $\operatorname{ord}_\nu t \ge 1$. Since T is smooth with regular parameters u, t at 0 and $\operatorname{center}_T \nu = \{0\}$, Lemma 2.13, applied with $b_1 = b_2 = 0$, implies that $a(\nu, T, 0) \ge m + \operatorname{ord}_\nu t$. Since $B_T \ge 0$ and $a(\nu, T, B_T) \ge 0$, we have $c_j \operatorname{ord}_\nu F_j \le a(\nu, T, 0)$ for

any j. As $c_j \geq \gamma$, we obtain

$$1 \leq m \leq e_1 L \quad \text{and} \quad \operatorname{ord}_\nu F_j \leq \frac{e_1 L}{\gamma} \quad \text{for any } j. \tag{14.6}$$

Let J be the preimage of J_0 under $G \to H$. Let $\zeta := \exp(2\pi\sqrt{-1}/r)$ and $g_0 := (\zeta^p, \zeta^{-q}, 1)$, a generator of $\mu_r(p,-q,0)$. Let $w := x^{l_1}y^{l_2}$, and let $\beta : J \to \mathbb{C}^*$ be the character with $g^*w = \beta(g)w$, that is, $\beta(g) = g_1^{l_1}g_2^{l_2}$. Then $\beta(g_0) = \zeta^{pl_1 - ql_2} = \zeta^{-1}$. Thus g_0 has order r, and β maps $\mu_r(p,-q,0)$ isomorphically onto μ_r. Since $G \to H$ is surjective with kernel $\mu_r(p,-q,0)$ (**Step 1**), $|J| = re_1$.

Let $\mathfrak{m}_\nu$ be the maximal ideal of the valuation ring of ord_ν and $\kappa(\nu)$ the residue field. By Lemma 2.6(1), J_0 is cyclic and acts faithfully on $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$. Since J_0 acts trivially on $\kappa(\nu)$ and $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$ is a $\kappa(\nu)$-vector space of dimension one, any $h \in J_0$ acts on $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$ by multiplication by a root of unity $\alpha_0(h) \in \mathbb{C}^*$, since h has finite order and the roots of unity of $\kappa(\nu)$ belong to $\mathbb{C}$. The map $\alpha_0 : J_0 \to \mathbb{C}^*$ is injective. Let $\alpha : J \to \mathbb{C}^*$ be the composition of $J \to J_0$ and α_0. Then the kernel of α is $\mu_r(p,-q,0)$, $\alpha(J) = \mu_{e_1}$, and for a uniformizer z of ord_ν and $g \in J$,

$$g^*z - \alpha(g)z \in \mathfrak{m}_\nu^2.$$

Since $\operatorname{Ker}\alpha = \mu_r(p,-q,0)$ and β is injective on $\mu_r(p,-q,0)$, the homomorphism $(\alpha,\beta) : J \to (\mathbb{C}^*)^2$ is injective.

Let $\Lambda \subset \mathbb{T}^2$ be the subgroup of the vectors $(\vartheta_1, \vartheta_2) \bmod \mathbb{Z}^2$ for which there exists $g \in J$ with $\alpha(g) = \exp(2\pi\sqrt{-1}\vartheta_1)$ and $\beta(g) = \exp(2\pi\sqrt{-1}\vartheta_2)$, and let $N_J := \rho^{-1}(\Lambda)$ (Definition 2.15). Since Λ is finite, $N_J \subset \mathbb{Q}^2$. Since $\alpha(J) = \mu_{e_1}$, the first coordinates of the vectors of N_J form $\frac{1}{e_1}\mathbb{Z}$. A vector $(0,\vartheta_2)$ belongs to N_J if and only if $\vartheta_2 \in \frac{1}{r}\mathbb{Z}$, because $\operatorname{Ker}\alpha = \mu_r(p,-q,0)$ and $\beta(\mu_r(p,-q,0)) = \mu_r$. Thus there exists $\sigma_0 \in \mathbb{Q}$ such that

$$N_J \cap \left(\left\{\frac{1}{e_1}\right\} \times \mathbb{R}\right) = \left\{\left(\frac{1}{e_1}, \sigma_0 + \frac{i}{r}\right) \middle| i \in \mathbb{Z}\right\}. \tag{14.7}$$

Step 3. In this step, for coprime integers $\varrho \geq 1$ and k, we compute the log discrepancy and the inertia group in J of the Gauss extension of $\varrho \operatorname{ord}_\nu$ with w-value k.

Since $ql_2 - pl_1 = 1$, we have $x = u^{l_2}w^{-p}$, $y = u^{-l_1}w^q$, and $\mathbb{C}(X) = \mathbb{C}(u,t)(w)$. A direct calculation shows that $\frac{dx}{x} \wedge \frac{dy}{y} = \frac{du}{u} \wedge \frac{dw}{w}$, hence

$$dx \wedge dy \wedge dt = -\frac{xy}{u}\frac{dw}{w} \wedge du \wedge dt. \tag{14.8}$$

Let $\varrho \geq 1$ and k be coprime integers such that $\varrho l_2 m - pk > 0$ and $qk - \varrho l_1 m > 0$. Let v_k be the Gauss extension of $\varrho \operatorname{ord}_\nu$ to $\mathbb{C}(u,t)(w)$ with w-value k (Definition-Proposition 2.11). Since $\varrho \operatorname{ord}_\nu(\mathbb{C}(u,t)^*) = \varrho\mathbb{Z}$ and $\gcd(\varrho, k) = 1$, $v_k(\mathbb{C}(X)^*) = \mathbb{Z}$. Moreover,

$$v_k(x) = \varrho l_2 m - pk > 0, \quad v_k(y) = qk - \varrho l_1 m > 0, \quad \text{and} \quad v_k(t) = \varrho \operatorname{ord}_\nu t > 0.$$

Let $T_1 \to T$ be a projective birational morphism from a smooth surface T_1 on which ν is a prime divisor, and let $z, s \in \mathcal{O}_{T_1,\nu}$ be elements such that z is a uniformizer of ord_ν and $dz \wedge ds$ generates $\Omega^2_{T_1}$ at the generic point of ν. Write $du \wedge dt = \varphi dz \wedge ds$, so that $\operatorname{ord}_\nu \varphi = a(\nu, T, 0) - 1$. Let $w_1 := w$ if $k \geq 0$ and $w_1 := w^{-1}$ if $k < 0$, let $Y := T_1 \times \mathbb{A}^1_{w_1}$, and let $\xi \in Y$ be the generic point of $\nu \times \{0\}$. Then z, w_1 are regular parameters of $\mathcal{O}_{Y,\xi}$, and v_k is the monomial valuation in z, w_1 with weights $(\varrho, |k|)$: for $a_i \in \mathcal{O}_{T_1,\nu}$, $v_k(\sum_i a_i w_1^i) = \min_i(\varrho \operatorname{ord}_\nu a_i + i|k|)$. Thus the residue field of v_k is $\kappa(\nu)(\theta)$, where θ is the residue of $w^\varrho z^{-k}$ and is transcendental over $\kappa(\nu)$. This is the standard description of the residue field of a monomial valuation with coprime weights, given by [Bou72, Chapter VI, §10, no. 1, Proposition 2] applied to the restriction of v_k to $\mathbb{C}(u,t)(w^\varrho z^{-k})$, over which $\mathbb{C}(X)$ has degree ϱ and v_k has ramification index ϱ [Bou72, Chapter VI, §8, no. 1, Lemma 2]. Since v_k is non-negative on $\mathbb{C}[x,y,t]$ and the transcendence degree of $\kappa(\nu)(\theta)$ over $\mathbb{C}$ is 2, $v_k = \operatorname{ord}_F$ for a prime divisor F over X [KM98, Lemma 2.45], and $\operatorname{center}_X F = \{0\}$. We claim that

$$a(F, X, 0) = \varrho\big(a(\nu, T, 0) - m\big) + \operatorname{ord}_F x + \operatorname{ord}_F y, \tag{14.9}$$

that is, equality holds in Lemma 14.5(1) for F.

We first compute $a(F,Y,0)$. Let $T_1^\circ \subset T_1$ be an affine open neighborhood of the generic point of ν such that z and s are regular on T_1° and $dz \wedge ds$ generates $\Omega^2_{T_1^\circ}$, and let $U := T_1^\circ \times \mathbb{A}^1_{w_1}$, an open neighborhood of ξ in Y. Then dz, ds, dw_1 form a basis of Ω^1_U, so $\phi := (z,s,w_1) : U \to \mathbb{A}^3$ is étale. Let z', s', w' be the coordinates of $\mathbb{A}^3$. Since $dz \wedge ds$ does not vanish at the generic point of ν, $s|_\nu$ is not constant, so $\phi(\xi)$ is the generic point of the line $\{z' = w' = 0\}$, and $g(s)$ is a unit at the generic point of ν for any non-zero $g \in \mathbb{C}[s']$. Let F' be the toric prime divisor over $\mathbb{A}^3$ given by the primitive vector $(\varrho, 0, |k|)$, so that $\mathrm{ord}_{F'}$ is the monomial valuation in z', s', w' with weights $(\varrho, 0, |k|)$. For a non-zero polynomial $f = \sum_{a,c} g_{a,c}(s') z'^a w'^c$ with $g_{a,c} \in \mathbb{C}[s']$, the formula for v_k above gives

$$v_k(\phi^* f) = \min\{\varrho a + |k|c \mid g_{a,c} \neq 0\} = \mathrm{ord}_{F'} f.$$

Thus v_k restricts to $\mathrm{ord}_{F'}$ on $\mathbb{C}(\mathbb{A}^3)$.

Let $W \to \mathbb{A}^3$ be a proper birational morphism from a normal variety W on which F' is a divisor. The base change along ϕ of $W \to \mathbb{A}^3$ is a proper birational morphism $W_U \to U$, and $W_U \to W$ is étale. The center of v_k on W_U maps to the generic point of F', the center of $\mathrm{ord}_{F'}$ on W. As $W_U \to W$ is étale, the center of v_k on W_U is the generic point of a prime divisor D of W_U. Since v_k is non-negative on the discrete valuation ring $\mathcal{O}_{W_U,D}$ and positive on its maximal ideal, and both v_k and ord_D take the value 1 at the pullback of a uniformizer of $\mathcal{O}_{W,F'}$, we have $v_k = \mathrm{ord}_D$, that is, the prime divisor F over U corresponding to F' is a divisor on W_U. Since $W_U \to W$ is étale, the relative canonical divisors correspond, so $a(F,U,0) = a(F',\mathbb{A}^3,0)$. By [CLS11, Lemma 11.4.10], $a(F',\mathbb{A}^3,0) = \varrho + |k|$. Hence

$$a(F,Y,0) = a(F,U,0) = a(F',\mathbb{A}^3,0) = \varrho + |k| = \mathrm{ord}_F(zw_1).$$

Since $\frac{dw}{w} = \pm\frac{dw_1}{w_1}$, (14.8) implies that

$$dx \wedge dy \wedge dt = \pm\frac{xy\varphi}{uw_1} dz \wedge ds \wedge dw_1.$$

As $dx \wedge dy \wedge dt$ and $dz \wedge ds \wedge dw_1$ generate the canonical sheaves of X and Y near the centers 0 and ξ of F, we obtain

$$a(F,X,0) = a(F,Y,0) + \mathrm{ord}_F \frac{xy\varphi}{uw_1} = \varrho + |k| + \mathrm{ord}_F\, x + \mathrm{ord}_F\, y + \varrho\big(a(\nu,T,0) - 1\big) - \varrho m - |k|,$$

which is (14.9).

Let $\psi := \alpha^{-k}\beta^\varrho : J \to \mathbb{C}^*$. We show that

$$I_J(F) = \mathrm{Ker}\,\psi. \tag{14.10}$$

Let $g \in J$. For $a_i \in \mathbb{C}(u,t)$, $g^*(\sum_i a_i w^i) = \sum_i (g^* a_i)\beta(g)^i w^i$ and $\mathrm{ord}_\nu(g^* a_i) = \mathrm{ord}_\nu a_i$, since the image of g in H belongs to J_0. Thus g preserves v_k by Definition-Proposition 2.11, that is, $J \subset \mathrm{Stab}_G(F)$. The element g acts trivially on $\kappa(\nu)$, and

$$g^*(w^\varrho z^{-k}) = \psi(g) w^\varrho z^{-k} \Big(\frac{g^* z}{\alpha(g) z}\Big)^{-k},$$

where $\mathrm{ord}_F\left(\frac{g^* z}{\alpha(g)z} - 1\right) = \varrho\,\mathrm{ord}_\nu\left(\frac{g^* z - \alpha(g) z}{\alpha(g) z}\right) > 0$. Hence g maps θ to $\psi(g)\theta$, and g acts trivially on $\kappa(\nu)(\theta)$ if and only if $\psi(g) = 1$. This proves (14.10). Let $e := |I_J(F)|$. Then $e = |J|/|\psi(J)| = |J^*/\langle \alpha^{-k}\beta^\varrho\rangle|$, where J^* is the character group of J.

Finally, we show that

$$N_J \cap \mathbb{Q}(\varrho, k) = \frac{1}{e}\mathbb{Z}(\varrho, k). \tag{14.11}$$

Since (α, β) is injective, for any $\vartheta \in N_J$ there exists a unique $g_\vartheta \in J$ with

$$(\alpha(g_\vartheta), \beta(g_\vartheta)) = \big(\exp(2\pi\sqrt{-1}\vartheta_1), \exp(2\pi\sqrt{-1}\vartheta_2)\big),$$

and $\Phi : \vartheta \to g_\vartheta$ is a homomorphism $N_J \to J$ with kernel $\mathbb{Z}^2$. For $\vartheta = c(\varrho, k) \in N_J$, $\psi(g_\vartheta) = \exp(2\pi\sqrt{-1}c(\varrho k - k\varrho)) = 1$, so $\Phi(N_J \cap \mathbb{Q}(\varrho, k)) \subset \operatorname{Ker}\psi$. Conversely, let $g \in \operatorname{Ker}\psi$, and let $\vartheta \in N_J$ with $g_\vartheta = g$. Then $\varrho\vartheta_2 - k\vartheta_1 \in \mathbb{Z}$. Since $\gcd(\varrho, k) = 1$, there exists $(a, a') \in \mathbb{Z}^2$ with $\varrho a' - ka = \varrho\vartheta_2 - k\vartheta_1$, and then $\vartheta - (a, a') \in N_J \cap \mathbb{Q}(\varrho, k)$ and $g_{\vartheta-(a,a')} = g$. Thus Φ maps $N_J \cap \mathbb{Q}(\varrho, k)$ onto $\operatorname{Ker}\psi$ with kernel $\mathbb{Z}^2 \cap \mathbb{Q}(\varrho, k) = \mathbb{Z}(\varrho, k)$. Hence $N_J \cap \mathbb{Q}(\varrho, k)$ is a subgroup of $\mathbb{Q}(\varrho, k)$ containing $\mathbb{Z}(\varrho, k)$ with index $|\operatorname{Ker}\psi| = e$, which is (14.11).

Step 4. In this step we choose a lattice point of N_J and prove $\operatorname{mld}((X \ni 0, B)/G) \le b_H + 2\eta$.

For $\vartheta = (\vartheta_1, \vartheta_2) \in \mathbb{R}^2$ let

$$\sigma_x(\vartheta) := l_2 m\vartheta_1 - p\vartheta_2 \quad \text{and} \quad \sigma_y(\vartheta) := -l_1 m\vartheta_1 + q\vartheta_2,$$

so that $q\sigma_x(\vartheta) + p\sigma_y(\vartheta) = m\vartheta_1$. Since $1 - c_u = \min\{\frac{1-c_x}{q}, \frac{1-c_y}{p}\}$, one of the following two cases holds.

Case A. $c_u = 1 - \frac{1-c_y}{p}$. Let $\kappa_0 := 1 - c_x - q(1 - c_u)$. Then $1 - c_y - p(1 - c_u) = 0$ and $0 \le \kappa_0 \le 1$, since $q(1 - c_u) \le 1 - c_x \le 1$. Along the points $(\frac{1}{e_1}, \sigma_0 + \frac{i}{r})$ of (14.7), σ_x changes by steps of $\frac{p}{r} \le \frac{P}{r} \le \frac{\eta}{2}$. Thus there exists $\vartheta \in N_J$ with $\vartheta_1 = \frac{1}{e_1}$ and $\sigma_x := \sigma_x(\vartheta) \in [\eta, 2\eta]$. Let $\sigma_y := \sigma_y(\vartheta)$. Since $m \ge 1$, $e_1 \le e_0$, $p, q \le P$, and $8Pe_0\eta \le 1$,

$$\sigma_y = \frac{m/e_1 - q\sigma_x}{p} \ge \frac{1/e_0 - 2P\eta}{P} \ge \frac{3}{4Pe_0} \ge \eta.$$

Case B. $c_u = 1 - \frac{1-c_x}{q}$. Let $\kappa_0 := 1 - c_y - p(1 - c_u)$. Then $1 - c_x - q(1 - c_u) = 0$ and $0 \le \kappa_0 \le 1$. Along the points of (14.7), σ_y changes by steps of $\frac{q}{r} \le \frac{\eta}{2}$, so there exists $\vartheta \in N_J$ with $\vartheta_1 = \frac{1}{e_1}$ and $\sigma_y := \sigma_y(\vartheta) \in [\eta, 2\eta]$. Let $\sigma_x := \sigma_x(\vartheta)$. Since $m \ge 1$, $e_1 \le e_0$, $p, q \le P$, and $8Pe_0\eta \le 1$,

$$\sigma_x = \frac{m/e_1 - p\sigma_y}{q} \ge \frac{1/e_0 - 2P\eta}{P} \ge \frac{3}{4Pe_0} \ge \eta.$$

In both cases $\sigma_x, \sigma_y \ge \eta$. Since $q\sigma_x + p\sigma_y = m/e_1$, (14.6) implies that $\sigma_x, \sigma_y \le m/e_1 \le L$. Let $\varrho \ge 1$ and k be the coprime integers with $\vartheta \in \mathbb{Q}_{>0}(\varrho, k)$. By (14.11), $\vartheta = \frac{d}{e}(\varrho, k)$ for a positive integer d, where $e = |\operatorname{Ker}\psi|$ and $\psi = \alpha^{-k}\beta^\varrho$ for (ϱ, k). Thus $\varrho = \frac{e}{de_1}$, and

$$\varrho l_2 m - pk = \frac{e}{d}\sigma_x > 0 \quad \text{and} \quad qk - \varrho l_1 m = \frac{e}{d}\sigma_y > 0.$$

Let F be the prime divisor of **Step 3** for (ϱ, k). Then $\operatorname{ord}_F x = \frac{e}{d}\sigma_x$, $\operatorname{ord}_F y = \frac{e}{d}\sigma_y$, and $\operatorname{center}_X F = \{0\}$.

We show that $\operatorname{ord}_F h_j = \operatorname{ord}_F h_j^0$ for any j. By Proposition 14.8(1) and (14.6),

$$\operatorname{ord}_F h_j^0 = a_j \operatorname{ord}_F x + b_j \operatorname{ord}_F y + \varrho \operatorname{ord}_\nu F_j = \frac{e}{d}\Big(a_j\sigma_x + b_j\sigma_y + \frac{\operatorname{ord}_\nu F_j}{e_1}\Big) < \frac{e}{d}\Big(\frac{3L}{\gamma} + \frac{L}{\gamma}\Big) = \frac{e}{d} \cdot \frac{4L}{\gamma}.$$

Since $R_j \in (x, y)^N$ and $N > \frac{4L}{\gamma\eta}$,

$$\operatorname{ord}_F R_j \ge N \min\{\operatorname{ord}_F x, \operatorname{ord}_F y\} \ge \frac{e}{d}N\eta > \frac{e}{d} \cdot \frac{4L}{\gamma} > \operatorname{ord}_F h_j^0.$$

Thus $\operatorname{ord}_F h_j = \operatorname{ord}_F h_j^0$.

Since $\operatorname{ord}_F h_j = a_j \operatorname{ord}_F x + b_j \operatorname{ord}_F y + \varrho \operatorname{ord}_\nu F_j$, $q \operatorname{ord}_F x + p \operatorname{ord}_F y = \operatorname{ord}_F u = \varrho m$, and $a(\nu, T, B_T) = a(\nu, T, 0) - \sum_j c_j \operatorname{ord}_\nu F_j - c_u m$, (14.9) implies that

$$a(F, X, B) = \varrho a(\nu, T, B_T) + \big(1 - c_x - q(1 - c_u)\big) \operatorname{ord}_F x + \big(1 - c_y - p(1 - c_u)\big) \operatorname{ord}_F y. \tag{14.12}$$

Since $\varrho a(\nu, T, B_T) = \varrho e_1 b_H = \frac{e}{d}b_H$, (14.12) reads $a(F, X, B) = \frac{e}{d}(b_H + \kappa_0\sigma_x)$ in Case A and $a(F, X, B) = \frac{e}{d}(b_H + \kappa_0\sigma_y)$ in Case B. We treat Case A. For Case B, replace σ_x by σ_y in the rest of this step. Since (X, B) is klt near 0, $a(F, X, B) > 0$. Since $J \subset G$, $I_J(F) \subset I_G(F)$, so

$|I_G(F)| \geq e$ by (14.10). By the formula for $\mathrm{mld}((X \ni x, B)/G)$ at the end of Lemma 2.6, using $d \geq 1$, $\kappa_0 \leq 1$, and $\sigma_x \leq 2\eta$,

$$\mathrm{mld}((X \ni 0, B)/G) \leq \frac{a(F, X, B)}{|I_G(F)|} \leq \frac{a(F, X, B)}{e} = \frac{b_H + \kappa_0 \sigma_x}{d} \leq b_H + 2\eta.$$

Step 5. In this step we prove the conclusions in the case $\mathrm{mld}((X \ni 0, B)/G) \geq 1 + \epsilon$.

Let $\eta := \min\{\frac{\epsilon}{8}, \frac{1}{8Pe_0}\}$, let r_0 be an integer with $r_0 \geq \frac{2P}{\eta}$, and let N_0 be an integer with $N_0 \geq \ell$, $N_0 > 3/\gamma$, and $N_0 > \frac{4L}{\gamma\eta}$. Then r_0 and N_0 depend only on ϵ, P, e_0, and Γ. Assume that $r \geq r_0$, $N \geq N_0$, and $\mathrm{mld}((X \ni 0, B)/G) \geq 1 + \epsilon$. By **Step 4**,

$$1 + \epsilon \leq b_H + 2\eta \leq b_H + \frac{\epsilon}{4},$$

so $b_H \geq 1 + \frac{3\epsilon}{4} > 1 + \frac{\epsilon}{2}$.

Since $B_{T/H} \geq 0$, $\mathrm{mld}(T/H \ni \bar{0}) \geq b_H > 1$ [KM98, Lemma 2.27], so the germ $T/H \ni \bar{0}$ is terminal (the singular points of T/H are isolated). By [KM98, Theorem 4.5(1)], applied with $\Delta = 0$, T/H is smooth near $\bar{0}$. By [HL23, Lemma 2.17], applied to the lc surface germ $(T/H \ni \bar{0}, B_{T/H})$, we have $b_H = 2 - \mathrm{mult}_{\bar{0}} B_{T/H}$, and exactly one prime divisor over $T/H \ni \bar{0}$ computes b_H. The exceptional divisor $E_{\bar{0}}$ of the blow-up of T/H at $\bar{0}$ satisfies $a(E_{\bar{0}}, T/H, B_{T/H}) = 2 - \mathrm{mult}_{\bar{0}} B_{T/H} = b_H$ [KM98, Lemma 2.29]. Thus $E_{\bar{0}}$ is the unique prime divisor over T/H computing b_H, and $\mathrm{mult}_{\bar{0}} B_{T/H} = 2 - b_H$.

Since $\mathrm{mult}_{\bar{0}} B_{T/H} = 2 - b_H < 1$, we may replace the coefficient of each component of $B_{T/H}$ through $\bar{0}$ by a rational number not smaller than it so that the multiplicity at $\bar{0}$ stays less than 1. On a neighborhood of $\bar{0}$ meeting no other component of $B_{T/H}$, we obtain a $\mathbb{Q}$-divisor $B^* \geq B_{T/H}$ with $\mathrm{mult}_{\bar{0}} B^* < 1$. By [KM98, Theorem 4.5(1)], the germ $(T/H \ni \bar{0}, B^*)$ is terminal, and so is $(T/H \ni \bar{0}, B_{T/H})$, since $B^* \geq B_{T/H}$ [KM98, Lemma 2.27]. Moreover, any component of $B_{T/H}$ through $\bar{0}$ has coefficient less than 1. Thus $(T/H, B_{T/H})$ is klt near $\bar{0}$. By Lemma 2.6(2)(3), (T, B_T) is klt near 0. Since B^0 and B_T are the $\mathbb{R}$-divisors B' and B'_T of Lemma 14.5 for $c'_j = c_j$, $a'_j = a_j$, $b'_j = b_j$, and $F'_j = F_j$, Lemma 14.5(2) implies that (X, B^0) is klt near 0.

We identify H with the image of H in $\mathbb{T}^2$ under the characters of u and t. By [CLS11, Proposition 1.3.18], T/H is the affine toric variety of the cone $\mathbb{R}^2_{\geq 0}$ with respect to the lattice N_H (Definition 2.15). Since $\mathbb{Z}^2 \subset N_H \subset \mathbb{Q}^2$, the primitive vectors of N_H on the rays $\mathbb{R}_{\geq 0}(1, 0)$ and $\mathbb{R}_{\geq 0}(0, 1)$ are $(\frac{1}{n_1}, 0)$ and $(0, \frac{1}{n_2})$ for positive integers n_1, n_2. Since T/H is smooth near $\bar{0}$ and $\bar{0}$ belongs to the closure of any torus orbit of T/H [CLS11, Theorem 3.2.6(d)], T/H is smooth, so $(\frac{1}{n_1}, 0)$ and $(0, \frac{1}{n_2})$ form a basis of N_H [CLS11, Theorem 1.3.12]. Hence

$$N_H = \mathbb{Z}\Big(\frac{1}{n_1}, 0\Big) + \mathbb{Z}\Big(0, \frac{1}{n_2}\Big),$$

that is, $H = \mu_{n_1} \times \mu_{n_2}$, where μ_{n_1} acts on u and μ_{n_2} acts on t. In particular, $T/H = \mathbb{A}^2_{u^{n_1}, t^{n_2}}$, and π has ramification index n_1 along $\{u = 0\}$ and n_2 along $\{t = 0\}$. By Lemma 2.6(4) and $B_T \geq 0$, the coefficients of $B_{T/H}$ along the coordinate axes $\{u^{n_1} = 0\}$ and $\{t^{n_2} = 0\}$ of T/H are at least $1 - \frac{1}{n_1}$ and $1 - \frac{1}{n_2}$. Thus

$$b_H = 2 - \mathrm{mult}_{\bar{0}} B_{T/H} \leq 2 - \Big(1 - \frac{1}{n_1}\Big) - \Big(1 - \frac{1}{n_2}\Big) = \frac{1}{n_1} + \frac{1}{n_2}.$$

If $n_1 \geq 2$ and $n_2 \geq 2$, then $b_H \leq 1$, a contradiction. Thus $\min\{n_1, n_2\} = 1$, and H is a cyclic group of order $n_1 n_2 = |H| \leq e_0$ acting on u or on t by a character and trivially on the other coordinate.

Step 6. In this step we prove the ACC for the numbers b_H.

By **Step 1**, $b_H = \mathrm{mld}(T'/H \ni \bar{0}, B_{T'/H})$, where $(T'/H, B_{T'/H})$ is an lc surface pair whose coefficients belong to $\Psi(\Xi)$, and $\Psi(\Xi)$ is a DCC set depending only on Γ. Thus the ACC for the numbers b_H follows from Theorem 2.18(1), applied to the DCC set $\Psi(\Xi)$. □

Corollary 14.10. *Let $P \geq 1$ be an integer, $\Gamma \subset [0,1]$ a DCC set with $\Gamma\backslash\{0\} \neq \emptyset$, and ℓ and L the integers of Construction-Proposition 14.7 for Γ. Notation and conditions as in Set-up 14.1 for P and Γ, assume that $N \geq \ell$ and $N > 3/\gamma$, and let $b := \mathrm{mld}(T \ni 0, B_T)$. Then:*

(1) for any η with $0 < \eta \leq \frac{1}{8P}$, $r \geq \frac{2P}{\eta}$ and $N > \frac{4L}{\gamma\eta}$, we have $\mathrm{mld}((X \ni 0, B)/G) \leq b + 2\eta$;

(2) for any $\epsilon > 0$ there exist r_0, N_0 depending only on ϵ, P, and Γ such that if $r \geq r_0$, $N \geq N_0$, and $\mathrm{mld}((X \ni 0, B)/G) \geq 1 + \epsilon$, then (X, B^0) and (T, B_T) are klt near 0.

Proof. Let $G_r := \mu_r(p, -q, 0) \subset G$. For any prime divisor F over X with $\mathrm{center}_X F = \{0\}$, $I_{G_r}(F) \subset I_G(F)$ and $a(F, X, B) > 0$, since (X, B) is klt near 0. Thus $a(F, X, B)/|I_G(F)| \leq a(F, X, B)/|I_{G_r}(F)|$, and by the formula for $\mathrm{mld}((X \ni x, B)/G)$ at the end of Lemma 2.6,

$$\mathrm{mld}((X \ni 0, B)/G) \leq \mathrm{mld}((X \ni 0, B)/G_r). \tag{14.13}$$

The group G_r is a finite diagonal abelian group acting faithfully on X. Since $G_r \subset G$, h_j and h_j^0 are G_r-semi-invariant, and the character of h_j equals that of h_j^0. Thus Set-up 14.1 holds with G replaced by G_r. Since $G_r \subset \mathbf{T}$, $G_r \cap \mathbf{T} = G_r = \mu_r(p, -q, 0)$. Since u and t are G_r-invariant, the image of G_r under the characters of u and t is trivial. Thus, for G_r in place of G, H is the trivial group, $(T, B_T)/H = (T, B_T)$, $b_H = b$, and $|H| \leq 1$.

(1) By Proposition 14.9, applied to G_r in place of G and with $e_0 = 1$, $\mathrm{mld}((X \ni 0, B)/G_r) \leq b + 2\eta$. By (14.13), $\mathrm{mld}((X \ni 0, B)/G) \leq b + 2\eta$.

(2) Let r_0 and N_0 be the integers of Proposition 14.9 for ϵ, P, $e_0 = 1$, and Γ. Assume that $r \geq r_0$, $N \geq N_0$, and $\mathrm{mld}((X \ni 0, B)/G) \geq 1 + \epsilon$. By (14.13), $\mathrm{mld}((X \ni 0, B)/G_r) \geq 1 + \epsilon$. By Proposition 14.9, applied to G_r in place of G and with $e_0 = 1$, for which H is trivial and $b_H = b$, the pairs (X, B^0) and (T, B_T) are klt near 0. □

The following elementary lemma is used several times below.

Lemma 14.11. *Let $(\alpha_i)_{i\geq 1}$ be a strictly increasing sequence of real numbers and $(b_i)_{i\geq 1}$ a sequence of real numbers such that $b_i \leq \alpha_i$ for any i and $\lim_i(\alpha_i - b_i) = 0$. Then the set $\{b_i \mid i \geq 1\}$ does not satisfy the ACC.*

Proof. Suppose that $\{b_i \mid i \geq 1\}$ satisfies the ACC. Any sequence of real numbers has a monotone subsequence, and a non-decreasing sequence in a set satisfying the ACC is eventually constant, so, passing to a subsequence, we may assume that (b_i) is non-increasing. Then $\alpha_i \leq b_1 + (\alpha_i - b_i)$ is bounded, so the strictly increasing sequence (α_i) converges to some A with $\alpha_i < A$ for any i. Since $\lim_i(\alpha_i - b_i) = 0$, b_i converges to A, and since (b_i) is non-increasing, $b_i \geq A > \alpha_i$ for any i, contradicting $b_i \leq \alpha_i$. □

14.3. **Uniform margins.** In this subsection, we show that high values of $\mathrm{mld}((X \ni 0, B)/G)$ in Set-up 14.1 bound the prime divisors F over X with $a(F, X, B)/|I_G(F)| \leq 3$, and that the lifts of the prime divisors computing the terms of an increasing sequence of values $\mathrm{mld}((X \ni 0, B)/G) \geq 1 + \epsilon$ change the order of some equation h_j.

Proposition 14.12(3) is the key to the lower bound in Theorem 14.2. For minimal log discrepancies below one, Proposition 14.12(1) has no uniform analogue, and Theorem 14.19 uses generic limits instead.

Proposition 14.12. *In Set-up 14.1, assume that $G \cap \mathbf{T} = \mu_r(p, -q, 0)$, $|H| \leq e_0$, and $a := \mathrm{mld}((X \ni 0, B)/G) \geq 1 + \epsilon$ for some $\epsilon > 0$. Let $\epsilon_0 := \min\{\epsilon, 1\}$, $\lambda := 1 + \frac{\epsilon_0}{8}$, $Q := \frac{3\lambda}{\lambda - 1}$, $C_1 := (P + 1)e_0Q$, and $C_2 := \frac{Q(3+C_1)}{\gamma}$. Then there exist r_0 and N_0 depending only on ϵ, P, e_0, and Γ such that (1)–(3) hold whenever $r \geq r_0$ and $N \geq N_0$, and (4) holds:*

(1) $(X, \lambda B^0)$ is klt near 0 and $(X, \lambda B)$ is lc near 0;

(2) for any prime divisor F over X with center 0 and $a(F, X, B)/|I_G(F)| \leq 3$, with $e := |I_G(F)|$ and $\mathrm{ord}_F|_{\mathbb{C}(u,t)} = \varrho\,\mathrm{ord}_\nu$ for a positive integer ϱ and a prime divisor ν over T, we have $\frac{a(F,X,0)}{e} \leq Q$, $\frac{\mathrm{ord}_F h_j}{e} \leq \frac{3}{\gamma(\lambda-1)}$, $\frac{1}{e_0} \leq \frac{\varrho}{e} \leq Q$, $\mathrm{ord}_\nu t \leq e_0Q$, $\mathrm{ord}_\nu u \leq Pe_0Q$, and $a(\nu, T, 0) \leq C_1$, the number $\sigma_F := a(F, X, 0) - \varrho(a(\nu, T, 0) - \mathrm{ord}_\nu u) - \mathrm{ord}_F x - \mathrm{ord}_F y$

satisfies $\frac{\sigma_F}{e} \in \mathbb{Z} \cap [0, Q]$, *and* ν *has a generating sequence* [dFGPM23, Definition 2.9] *of at most* $\lfloor e_0 Q \rfloor + 2$ *elements which are* $I_G(F)$*-semi-invariant polynomials;*

(3) *for* F *as in (2),* $\frac{\operatorname{ord}_F h_j^0}{e} \le C_2$ *for any* j*, and if* $\operatorname{ord}_F h_j \ne \operatorname{ord}_F h_j^0$ *for some* j*, then* $\min\{\operatorname{ord}_F x, \operatorname{ord}_F y\} \le \frac{eC_2}{N} \le \frac{e_0 r C_2}{N}$*;*

(4) *if* α_i *is a strictly increasing sequence of numbers a for data in Set-up 14.1 with* $G \cap \mathbf{T} = \mu_r(p, -q, 0)$*,* $|H| \le e_0$*, and* $a \ge 1 + \epsilon$*, with fixed* P*,* e_0*,* Γ*, and* ϵ*, with* $r = r_i$ *and* $N = N_i$*, where* $r_i, N_i \to +\infty$*, then, after deleting finitely many terms, any lift* F *to* X *of any prime divisor computing* α_i *satisfies* $\operatorname{ord}_F h_j \ne \operatorname{ord}_F h_j^0$ *for some* j*.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we choose r_0 and N_0, and we prove (1).

Let ℓ and L be the integers of Construction-Proposition 14.7 for Γ, and let $\eta := \min\{\frac{\epsilon_0}{8}, \frac{1}{8P}\}$. Let ℓ_λ be the integer of Theorem 2.19(2) for the threefold $\mathbb{A}^3$ and the DCC set $\lambda(\Gamma\backslash\{0\}) \subset \mathbb{R}_{>0}$. Since λ depends only on ϵ, the integer ℓ_λ depends only on ϵ and Γ. Let $r_0 := \lceil \frac{2P}{\eta} \rceil$, and let N_0 be the least integer such that $N_0 \ge \max\{\ell, \ell_\lambda\}$, $N_0 > \frac{3}{\gamma}$, and $N_0 > \frac{4L}{\gamma\eta}$. Then r_0 and N_0 depend only on ϵ, P, and Γ. In **Steps 1–4** we assume that $r \ge r_0$ and $N \ge N_0$.

By Proposition 14.8(2) and Corollary 14.10(1), (T, B_T) is lc near 0, and $b := \operatorname{mld}(T \ni 0, B_T)$ satisfies

$$b \ge a - 2\eta \ge 1 + \epsilon_0 - \frac{\epsilon_0}{4} > 1 + \frac{\epsilon_0}{2}.$$

Let E_T be the exceptional divisor of the blow-up of T at 0, and let $s_0 := \sum_j c_j \operatorname{mult}_0 F_j$. Since $\operatorname{mult}_0 \operatorname{div}_T(u) = 1$, we have $a(E_T, T, B_T) = 2 - c_u - s_0$ [KM98, Lemma 2.29]. Thus $2 - c_u - s_0 \ge b > 1 + \frac{\epsilon_0}{2}$, that is, $c_u + s_0 < 1 - \frac{\epsilon_0}{2}$. Since $c_u \ge 1 - \frac{1 - c_x}{q}$ and $c_u \ge 1 - \frac{1 - c_y}{p}$, we obtain

$$c_x + q s_0 < 1 - \frac{q\epsilon_0}{2} \le 1 - \frac{\epsilon_0}{2} \quad \text{and} \quad c_y + p s_0 < 1 - \frac{p\epsilon_0}{2} \le 1 - \frac{\epsilon_0}{2}. \tag{14.14}$$

In Lemma 14.5, take $c'_j := \lambda c_j$, $a'_j := a_j$, $b'_j := b_j$, and $F'_j := F_j$. Then $B' = \lambda B^0$, $c'_x = \lambda c_x$, $c'_y = \lambda c_y$, and

$$B'_T = \lambda \sum_j c_j \operatorname{div}_T(F_j) + c'_u \operatorname{div}_T(u), \qquad c'_u = \max\left\{1 - \frac{1 - \lambda c_x}{q}, 1 - \frac{1 - \lambda c_y}{p}\right\}.$$

Since $\lambda c_x \ge 0$, we have $c'_u \ge 1 - \frac{1}{q} \ge 0$, so $B'_T \ge 0$. Since $\lambda(1 - \frac{\epsilon_0}{2}) \le 1 - \frac{\epsilon_0}{4}$, (14.14) implies that $\lambda(c_x + q s_0) < 1 - \frac{\epsilon_0}{4}$ and $\lambda(c_y + p s_0) < 1 - \frac{\epsilon_0}{4}$. Hence

$$\operatorname{mult}_0 B'_T = c'_u + \lambda s_0 = \max\left\{1 - \frac{1 - \lambda(c_x + q s_0)}{q}, 1 - \frac{1 - \lambda(c_y + p s_0)}{p}\right\} < 1 - \frac{\epsilon_0}{4P} < 1.$$

Since multiplicity is upper semicontinuous, $\operatorname{mult}_z B'_T < 1$ for any closed point $z \in T$ near 0. Replacing the coefficients of the components of B'_T through 0 by rational numbers not smaller than them and keeping the multiplicity at 0 less than one, we obtain, on a neighborhood of 0, a $\mathbb{Q}$-divisor $B^* \ge B'_T$ with $\operatorname{mult}_z B^* < 1$ for any closed point z near 0. Then (T, B^*) is terminal near 0 [KM98, Theorem 4.5(1)], and so is (T, B'_T) since $B'_T \le B^*$ [KM98, Lemma 2.27]. Moreover, the coefficients of B'_T are less than one, so (T, B'_T) is klt near 0. By Lemma 14.5(2), $(X, \lambda B^0)$ is klt near 0.

Let $\mathfrak{m}_0$ be the maximal ideal of $0 \in X$, and let $\mathfrak{a} := \prod_j (h_j)^{\lambda c_j}$ and $\mathfrak{b} := \prod_j (h_j^0)^{\lambda c_j}$. The exponents λc_j belong to $\lambda(\Gamma\backslash\{0\})$. Since $h_j - h_j^0 = R_j \in (x, y)^N \subset \mathfrak{m}_0^{\ell_\lambda}$, we have $(h_j) + \mathfrak{m}_0^{\ell_\lambda} = (h_j^0) + \mathfrak{m}_0^{\ell_\lambda}$ for any j. By Theorem 2.19(2), applied with the zero boundary,

$$\operatorname{mld}(X \ni 0, \lambda B) = \operatorname{mld}(X \ni 0, 0, \mathfrak{a}) = \operatorname{mld}(X \ni 0, 0, \mathfrak{b}) = \operatorname{mld}(X \ni 0, \lambda B^0) > 0,$$

where the last inequality holds because $(X, \lambda B^0)$ is klt near 0: on a log resolution of $(X, \lambda B^0)$ near 0 the crepant boundary is snc with coefficients less than one, so by [KM98, Lemma 2.30 and

Corollary 2.31(3)] the log discrepancies of the prime divisors over X with center 0 are bounded below by the minimum of finitely many positive numbers.

We show that $(X, \lambda B)$ is lc near 0. Let $f : Y \to X$ be a log resolution of $(X, \lambda B)$ such that $f^{-1}(0)$ is a divisor and $f^{-1}(0) \cup \mathrm{Exc}(f) \cup \mathrm{Supp}\, f_*^{-1}B$ is snc, which exists since we may first blow up 0 and then take a log resolution (cf. [KM98, Notation 0.4(10)]), and let $K_Y + B_Y := f^*(K_X + \lambda B)$. Suppose that $(X, \lambda B)$ is not lc near 0. Since (Y, B_Y) is log smooth, there exists a prime divisor E on Y with $0 \in f(E)$ and $\mathrm{mult}_E B_Y > 1$ [KM98, Corollary 2.31(3)], that is, $a(E, X, \lambda B) < 0$. If $f(E) = \{0\}$, then $a(E, X, \lambda B) < 0$ contradicts $\mathrm{mld}(X \ni 0, \lambda B) > 0$. Otherwise, E is not a component of $f^{-1}(0)$, and there exists a component D of $f^{-1}(0)$ with $E \cap D \neq \emptyset$. Let $Y_1 \to Y$ be the blow-up along a component Z_1 of $E \cap D$, with exceptional divisor E_1. For $k \geq 2$, let $Y_k \to Y_{k-1}$ be the blow-up along a component Z_k of the intersection of E_{k-1} with the strict transform of E, with exceptional divisor E_k. At the generic point of Z_k, the support of the boundary consists of the strict transform of E and of the strict transform of D (if $k = 1$) or E_{k-1} (if $k \geq 2$), and the strict transform of E intersects the strict transform of D (resp. E_{k-1}) transversally there. Hence [KM98, Lemma 2.29] implies that $a(E_1, X, \lambda B) = a(D, X, \lambda B) + a(E, X, \lambda B)$ and $a(E_k, X, \lambda B) = a(E_{k-1}, X, \lambda B) + a(E, X, \lambda B)$, so

$$a(E_k, X, \lambda B) = a(D, X, \lambda B) + k a(E, X, \lambda B).$$

Since $Z_1 \subset D$ and $Z_k \subset E_{k-1}$, the center of E_k on X is $\{0\}$ for any k. As $a(E, X, \lambda B) < 0$, we have $a(E_k, X, \lambda B) < 0$ for $k \gg 0$, which contradicts $\mathrm{mld}(X \ni 0, \lambda B) > 0$. Thus $(X, \lambda B)$ is lc near 0 (cf. [KM98, Corollary 2.31(1)]). This proves (1).

Step 2. In this step we restrict ord_F to $\mathbb{C}(u, t)$, compare the inertia group of F with that of the prime divisor over T defined by the restriction, and prove the first two inequalities of (2).

Let F be a prime divisor over X with center 0, $I := I_G(F)$, and $e := |I|$. Let l_1 and l_2 be integers with $ql_2 - pl_1 = 1$, and let $w := x^{l_1} y^{l_2}$. Then x and y are Laurent monomials in u and w, so $\mathbb{C}(X) = \mathbb{C}(u, t)(w)$. Since the center of F is 0, we have $\mathrm{ord}_F u > 0$ and $\mathrm{ord}_F t > 0$. Hence ord_F is non-negative on $\mathcal{O}_{T,0}$ and positive on the maximal ideal of $\mathcal{O}_{T,0}$, and Lemma 2.12, applied to $\mathbb{C}(u, t) \subset \mathbb{C}(X)$ and the point $0 \in T$, implies that $\mathrm{ord}_F|_{\mathbb{C}(u,t)} = \varrho\, \mathrm{ord}_\nu$ for a positive integer ϱ and a prime divisor ν over T with $\mathrm{center}_T \nu = \{0\}$.

Let $\mathcal{O}_F$ be the valuation ring of ord_F, with maximal ideal $\mathfrak{m}_F$ and residue field $\kappa(F)$. By Definition 2.5, I acts on $\mathcal{O}_F$ and trivially on $\kappa(F)$. Hence I acts $\kappa(F)$-linearly on the one-dimensional $\kappa(F)$-vector space $\mathfrak{m}_F/\mathfrak{m}_F^2$, by a character $\chi_F : I \to \mathbb{C}^*$. By Lemma 2.6(1), I is cyclic and χ_F is injective, so the order of χ_F is e. Applying $\frac{1}{e}\sum_{g \in I} \chi_F(g)^{-1} g^*$ to an element of $\mathfrak{m}_F \backslash \mathfrak{m}_F^2$, we obtain an I-semi-invariant element $\pi_F \in \mathfrak{m}_F \backslash \mathfrak{m}_F^2$ with character χ_F. Let ϕ be an I-semi-invariant rational function on X. Then $\phi = v \pi_F^{\mathrm{ord}_F \phi}$ for a unit $v \in \mathcal{O}_F$, and v is I-semi-invariant. If χ is the character of v, then $g^* v = \chi(g) v$ for $g \in I$, and the residue class of v is non-zero and I-invariant, so χ is trivial. Hence

$$\text{the character of } \phi \text{ on } I \text{ is } \chi_F^{\mathrm{ord}_F \phi}. \tag{14.15}$$

For $g \in G$, the pullback g^* on $\mathbb{C}(u, t)$ is the pullback by the image of g in H. Let J be the image of I in H. We regard characters of J as characters of I via the surjection $I \to J$. Since I fixes ord_F and acts trivially on $\kappa(F)$, the group J fixes ord_ν and acts trivially on the residue field $\kappa(\nu) \subset \kappa(F)$ of ord_ν, so $J \subset I_H(\nu)$. Since H acts faithfully on T, Lemma 2.6(1) implies that $I_H(\nu)$ acts faithfully on $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$, where $\mathfrak{m}_\nu$ is the maximal ideal of the valuation ring of ord_ν. Let χ_ν be the character by which J acts on $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$. Then χ_ν is injective on J. Applying $\frac{1}{|J|}\sum_{g \in J} \chi_\nu(g)^{-1} g^*$ to an element of $\mathfrak{m}_\nu \backslash \mathfrak{m}_\nu^2$, we obtain a J-semi-invariant element $\pi_\nu \in \mathbb{C}(u, t)$ with $\mathrm{ord}_\nu \pi_\nu = 1$ and character χ_ν. Since $\mathrm{ord}_F \pi_\nu = \varrho$, (14.15) implies that $\chi_\nu = \chi_F^\varrho$ as characters of I. The order of χ_F^ϱ is $e/\gcd(e, \varrho)$, and the order of χ_ν is $|J|$. Thus $|J| = e/\gcd(e, \varrho)$. Since $|J| \leq |H| \leq e_0$, we obtain

$$\frac{e}{|J|} = \gcd(e, \varrho) \text{ divides } \varrho \quad \text{and} \quad \frac{\varrho}{e} \geq \frac{1}{|J|} \geq \frac{1}{e_0}. \tag{14.16}$$

Now assume moreover that $a(F,X,B)/e \le 3$. By (1),

$$a(F,X,\lambda B) = a(F,X,0) - \lambda \sum_j c_j \operatorname{ord}_F h_j \ge 0.$$

Hence

$$a(F,X,B) = a(F,X,0) - \sum_j c_j \operatorname{ord}_F h_j \ge \Big(1 - \frac{1}{\lambda}\Big) a(F,X,0) = \frac{\lambda-1}{\lambda} a(F,X,0)$$

and

$$a(F,X,B) = a(F,X,\lambda B) + (\lambda - 1) \sum_j c_j \operatorname{ord}_F h_j \ge (\lambda-1)\gamma \operatorname{ord}_F h_j$$

for any j, as $c_j \ge \gamma$. Since $a(F,X,B) \le 3e$, we obtain $\frac{a(F,X,0)}{e} \le Q$ and $\frac{\operatorname{ord}_F h_j}{e} \le \frac{3}{\gamma(\lambda-1)}$.

Step 3. In this step we prove the remaining parts of (2).

Let F be as in (2), and recall from **Step 2** the group J, the characters χ_F and χ_ν, and $w = x^{l_1} y^{l_2}$. By Lemma 14.5(1), $\sigma_F \ge 0$. Since T is smooth at 0 with regular parameters u, t and the center of ν is 0, Lemma 2.13 implies that $a(\nu,T,0) \ge \operatorname{ord}_\nu u + \operatorname{ord}_\nu t$. Hence $a(\nu,T,0) - \operatorname{ord}_\nu u \ge \operatorname{ord}_\nu t \ge 1$. By the definition of σ_F and $\sigma_F \ge 0$,

$$\varrho\big(a(\nu,T,0) - \operatorname{ord}_\nu u\big) + \operatorname{ord}_F x + \operatorname{ord}_F y \le a(F,X,0) \le eQ.$$

Thus $\frac{\varrho}{e} \le Q$ and $\frac{\operatorname{ord}_F x + \operatorname{ord}_F y}{e} \le Q$. By (14.16), $a(\nu,T,0) - \operatorname{ord}_\nu u \le \frac{eQ}{\varrho} \le e_0 Q$, so $\operatorname{ord}_\nu t \le e_0 Q$. Since $u = x^q y^p$,

$$\varrho \operatorname{ord}_\nu u = \operatorname{ord}_F u = q \operatorname{ord}_F x + p \operatorname{ord}_F y \le P(\operatorname{ord}_F x + \operatorname{ord}_F y) \le PeQ,$$

so (14.16) implies that $\operatorname{ord}_\nu u \le Pe_0 Q$. Hence $a(\nu,T,0) \le e_0 Q + Pe_0 Q = C_1$. Moreover, $0 \le \sigma_F \le a(F,X,0) \le eQ$.

We show that e divides σ_F. Since G is diagonal, x, y, u, w, $dx \wedge dy \wedge dt$, and $du \wedge dt$ are G-semi-invariant. Let $\Omega_F := \Omega^3_{\mathcal{O}_F/\mathbb{C}}$. Since $\mathcal{O}_F$ is the local ring of a smooth variety at the generic point of a divisor, Ω_F is a free $\mathcal{O}_F$-module of rank one, and I acts on Ω_F by pullback. Since the extension $\kappa(F)/\mathbb{C}$ is separable, the second fundamental exact sequence

$$0 \to \mathfrak{m}_F/\mathfrak{m}_F^2 \to \Omega^1_{\mathcal{O}_F/\mathbb{C}} \otimes \kappa(F) \to \Omega^1_{\kappa(F)/\mathbb{C}} \to 0$$

is exact [Mat89, Theorems 25.2 and 26.9]. Taking third exterior powers,

$$\Omega_F \otimes \kappa(F) \cong \Omega^2_{\kappa(F)/\mathbb{C}} \otimes_{\kappa(F)} \mathfrak{m}_F/\mathfrak{m}_F^2.$$

Since I acts trivially on $\kappa(F)$, it acts on $\Omega_F \otimes \kappa(F)$ by χ_F. Applying $\frac{1}{e}\sum_{g\in I} \chi_F(g)^{-1} g^*$ to a generator of Ω_F, we obtain an I-semi-invariant generator ω_F of Ω_F with character χ_F. By the definition of $a(F,X,0)$, $dx \wedge dy \wedge dt = v\pi_F^{a(F,X,0)-1}\omega_F$ for a unit $v \in \mathcal{O}_F$, and as in (14.15) the character of v is trivial. Hence the character of $dx \wedge dy \wedge dt$ on I is $\chi_F^{a(F,X,0)}$.

Next we compute the character of $du \wedge dt$. Let $\mathcal{O}_\nu$ be the valuation ring of ord_ν, and let $\Omega_\nu := \Omega^2_{\mathcal{O}_\nu/\mathbb{C}}$. Since $\mathcal{O}_\nu$ is the local ring of a smooth surface at the generic point of a curve, Ω_ν is a free $\mathcal{O}_\nu$-module of rank one, and J acts on Ω_ν by pullback. Similarly [Mat89, Theorems 25.2 and 26.9], taking second exterior powers,

$$\Omega_\nu \otimes \kappa(\nu) \cong \Omega^1_{\kappa(\nu)/\mathbb{C}} \otimes_{\kappa(\nu)} \mathfrak{m}_\nu/\mathfrak{m}_\nu^2.$$

Since J acts trivially on $\kappa(\nu)$, it acts on $\Omega_\nu \otimes \kappa(\nu)$ by χ_ν. Applying $\frac{1}{|J|}\sum_{g\in J} \chi_\nu(g)^{-1} g^*$ to a generator of Ω_ν, we obtain a J-semi-invariant generator ω_ν of Ω_ν with character χ_ν. By the definition of $a(\nu,T,0)$, $du \wedge dt = v'\pi_\nu^{a(\nu,T,0)-1}\omega_\nu$ for a unit $v' \in \mathcal{O}_\nu$. Since $du \wedge dt$, π_ν, and ω_ν are J-semi-invariant, so is v'. If χ' is the character of v', then $g^* v' = \chi'(g) v'$ for $g \in J$, and the residue class of v' is non-zero and J-invariant, so χ' is trivial. Hence the character of $du \wedge dt$ on

J is $\chi_\nu^{a(\nu,T,0)}$, which is $\chi_F^{\varrho a(\nu,T,0)}$ on I. Since $\frac{du}{u} = q\frac{dx}{x} + p\frac{dy}{y}$, $\frac{dw}{w} = l_1\frac{dx}{x} + l_2\frac{dy}{y}$, and $ql_2 - pl_1 = 1$, we have

$$dx \wedge dy \wedge dt = -\frac{xy}{u} \cdot \frac{dw}{w} \wedge du \wedge dt.$$

The form $\frac{dw}{w}$ is G-invariant. By (14.15) and $\operatorname{ord}_F u = \varrho \operatorname{ord}_\nu u$, the character of the right hand side on I is

$$\chi_F^{\operatorname{ord}_F x + \operatorname{ord}_F y - \varrho \operatorname{ord}_\nu u + \varrho a(\nu,T,0)}.$$

Thus $\chi_F^{\sigma_F} = 1$. Since the order of χ_F is e, we obtain $\frac{\sigma_F}{e} \in \mathbb{Z} \cap [0, Q]$.

We prove the claim on generating sequences. Let C be a curvetta of ν [dFGPM23, Definition 2.5] whose strict transform intersects the divisor of ν at a point outside the strict transform of $\{t = 0\}$. Then $(C \cdot \{t = 0\})_0 = \operatorname{ord}_\nu t$ [dFGPM23, Definition 2.5 and Remark 2.6]. Since the local intersection number is at least the product of the multiplicities,

$$\operatorname{mult}_0 C \le (C \cdot \{t = 0\})_0 = \operatorname{ord}_\nu t \le \lfloor e_0 Q \rfloor.$$

Let $g(C)$ be the number of characteristic exponents of C. Let $\bar\beta_0$ be the least positive element of the semigroup of C [dFGPM23, Definition 1.15]. Then $2^{g(C)} \le \bar\beta_0$ [dFGPM23, Notation 2.2 and Proposition 2.3(1)], and $\bar\beta_0 \le (L_0 \cdot C)_0 = \operatorname{mult}_0 C$ for a general line L_0 through 0 [Har77, Exercise I.5.4(b)]. Hence $g(C) \le \operatorname{mult}_0 C - 1$, and ν has a generating sequence $\xi_1, \dots, \xi_k \in \widehat{\mathcal{O}} := \mathbb{C}[[u,t]]$ with $k \le g(C) + 2 \le \lfloor e_0 Q \rfloor + 1 < \lfloor e_0 Q \rfloor + 2$ [dFGPM23, Propositions 2.12 and 2.14]. Let $\widehat{\mathfrak{m}}$ be the maximal ideal of $\widehat{\mathcal{O}}$.

For $\theta \ge 0$ let $\mathcal{P}_\theta := \{\phi \in \widehat{\mathcal{O}} \mid \operatorname{ord}_\nu \phi \ge \theta\}$, as in [dFGPM23, Definition 2.9], and let $d_i := \operatorname{ord}_\nu \xi_i$. Let ξ_i'' be the sum of the terms of ξ_i of degree at most d_i. Since $\operatorname{ord}_\nu u \ge 1$ and $\operatorname{ord}_\nu t \ge 1$, we have $\operatorname{ord}_\nu \phi \ge \operatorname{mult}_0 \phi$ for $\phi \in \widehat{\mathcal{O}}$, so $\operatorname{ord}_\nu(\xi_i - \xi_i'') > d_i$. For a character χ of J let $\operatorname{pr}_\chi := \frac{1}{|J|}\sum_{g \in J} \chi(g)^{-1} g^*$. As J acts diagonally on u, t, the element $\xi_i' := \operatorname{pr}_{\chi_\nu^{d_i}}(\xi_i'')$ is a J-semi-invariant polynomial in u and t, hence an I-semi-invariant polynomial. Since J fixes ord_ν, any pr_χ preserves any $\mathcal{P}_\theta$. Since J acts trivially on $\kappa(\nu)$, $g^*\phi - \chi_\nu(g)^d \phi \in \mathcal{P}_{d+1}$ for any $\phi \in \mathcal{P}_d$ and $g \in J$, so $\operatorname{pr}_\chi(\mathcal{P}_d) \subset \mathcal{P}_{d+1}$ for $\chi \ne \chi_\nu^d$. Here we may assume that ϕ is a polynomial, since for $\phi \in \widehat{\mathcal{O}}$ and an integer $M > d + 1$, the sum of the terms of ϕ of degree at least M belongs to $\widehat{\mathfrak{m}}^M \subset \mathcal{P}_{d+1}$. Thus $\operatorname{ord}_\nu(\xi_i'' - \xi_i') > d_i$, and therefore $\operatorname{ord}_\nu(\xi_i' - \xi_i) > d_i = \operatorname{ord}_\nu \xi_i'$. Writing $\xi^\beta := \prod_i \xi_i^{\beta_i}$, $\xi'^\beta := \prod_i \xi_i'^{\beta_i}$, and $\langle d, \beta \rangle := \sum_i d_i \beta_i$ for $\beta \in \mathbb{Z}^k_{\ge 0}$, we obtain $\operatorname{ord}_\nu(\xi'^\beta - \xi^\beta) > \langle d, \beta \rangle$.

Let $\mathfrak{a}_\theta \subset \widehat{\mathcal{O}}$ be the ideal generated by the ξ'^β with $\langle d, \beta \rangle \ge \theta$, so that $\mathfrak{a}_\theta \subset \mathcal{P}_\theta$. For $\theta' \ge \theta$ and $\phi \in \mathcal{P}_{\theta'}$, write $\phi = \sum_\beta \phi_\beta \xi^\beta$ with $\phi_\beta \in \widehat{\mathcal{O}}$ and $\langle d, \beta \rangle \ge \theta'$, which is possible because $\xi_1, \dots, \xi_k$ is a generating sequence. Then $\phi - \sum_\beta \phi_\beta \xi'^\beta \in \mathcal{P}_{\theta'+1}$, so $\mathcal{P}_{\theta'} \subset \mathfrak{a}_\theta + \mathcal{P}_{\theta'+1}$. By induction, $\mathcal{P}_\theta \subset \mathfrak{a}_\theta + \mathcal{P}_{\theta+n'}$ for any positive integer n'. The divisor of ν is reached by finitely many point blow-ups [Abh56, Proposition 3], and the multiplicities of strict transforms do not increase under point blow-ups, so induction on the number of blow-ups implies that $\operatorname{ord}_\nu \phi \le c_\nu \operatorname{mult}_0 \phi$ for any $\phi \in \widehat{\mathcal{O}}$, where c_ν depends only on ν. Hence $\mathcal{P}_{\theta+n'} \subset \widehat{\mathfrak{m}}^{\lceil (\theta+n')/c_\nu \rceil}$. By the Krull intersection theorem,

$$\mathcal{P}_\theta \subset \bigcap_{n'} (\mathfrak{a}_\theta + \widehat{\mathfrak{m}}^{n'}) = \mathfrak{a}_\theta.$$

Thus $\mathcal{P}_\theta = \mathfrak{a}_\theta$ for any θ, and $\xi_1', \dots, \xi_k'$ is a generating sequence of ν of at most $\lfloor e_0 Q \rfloor + 2$ elements consisting of $I_G(F)$-semi-invariant polynomials in u and t. This proves (2).

Step 4. In this step we prove (3).

Let F be as in (2). By Proposition 14.8(1), $a_j + b_j < \frac{3}{\gamma}$. By Proposition 14.8(2), (T, B_T) is lc near 0. Since $c_u \ge 0$ and $a(\nu, T, B_T) \ge 0$, we have $c_j \operatorname{ord}_\nu F_j \le a(\nu, T, 0) \le C_1$, so $\operatorname{ord}_\nu F_j \le \frac{C_1}{\gamma}$. By **Step 3**, $\frac{\operatorname{ord}_F x + \operatorname{ord}_F y}{e} \le Q$ and $\frac{\varrho}{e} \le Q$. Hence

$$\frac{\operatorname{ord}_F h_j^0}{e} = \frac{a_j \operatorname{ord}_F x + b_j \operatorname{ord}_F y}{e} + \frac{\varrho}{e} \operatorname{ord}_\nu F_j \le \frac{3Q}{\gamma} + \frac{QC_1}{\gamma} = C_2.$$

Suppose that $\operatorname{ord}_F h_j \neq \operatorname{ord}_F h_j^0$ for some j. Since $h_j = h_j^0 + R_j$, we have $\operatorname{ord}_F R_j \leq \operatorname{ord}_F h_j^0$. Since $R_j \in (x,y)^N$,

$$N \min\{\operatorname{ord}_F x, \operatorname{ord}_F y\} \leq \operatorname{ord}_F R_j \leq \operatorname{ord}_F h_j^0 \leq eC_2.$$

By Proposition 14.9, the kernel of $G \to H$ is $G \cap \mathbf{T} = \mu_r(p, -q, 0)$, which has order r as p and q are coprime. Thus $e \leq |G| = r|H| \leq e_0 r$, and we obtain $\min\{\operatorname{ord}_F x, \operatorname{ord}_F y\} \leq \frac{eC_2}{N} \leq \frac{e_0 r C_2}{N}$. This proves (3).

Step 5. In this step we bound log discrepancies with respect to (X, B^0) from below by b_H.

Assume that $N \geq \ell$ and $N > \frac{3}{\gamma}$. Let F be a prime divisor over X with center 0, and let e, ϱ, ν, and J be as in **Step 2**. We show that

$$\frac{a(F, X, B^0)}{e} \geq b_H. \tag{14.17}$$

By Proposition 14.9, (T, B_T) is lc near 0. By Lemma 14.5(2), applied with $B' = B^0$,

$$a(F, X, B^0) \geq \varrho a(\nu, T, B_T) + \big(1 - c_x - q(1 - c_u)\big) \operatorname{ord}_F x + \big(1 - c_y - p(1 - c_u)\big) \operatorname{ord}_F y.$$

Since $c_u \geq 1 - \frac{1-c_x}{q}$ and $c_u \geq 1 - \frac{1-c_y}{p}$, we have $1 - c_x - q(1 - c_u) \geq 0$ and $1 - c_y - p(1 - c_u) \geq 0$, so $a(F, X, B^0) \geq \varrho a(\nu, T, B_T)$. Since h_j^0 and $x^{a_j} y^{b_j}$ are G-semi-invariant, each F_j is H-semi-invariant, so B_T is H-invariant. By **Step 2**, the center of ν is 0 and $J \subset I_H(\nu)$. By Lemma 2.6 and $a(\nu, T, B_T) \geq 0$,

$$\frac{a(\nu, T, B_T)}{|J|} \geq \frac{a(\nu, T, B_T)}{|I_H(\nu)|} \geq b_H \geq 0.$$

By (14.16), $\varrho/(e/|J|)$ is a positive integer. Hence

$$\frac{a(F, X, B^0)}{e} \geq \frac{\varrho}{e} a(\nu, T, B_T) = \frac{\varrho}{e/|J|} \cdot \frac{a(\nu, T, B_T)}{|J|} \geq b_H.$$

Step 6. In this step we prove (4).

Let α_i, r_i, N_i be as in (4), and let G_i, H_i, $h_{i,j}$, $h_{i,j}^0$, B_i, and B_i^0 be the corresponding data. Suppose that for infinitely many i there exists a lift F_i to X of a prime divisor computing α_i such that $\operatorname{ord}_{F_i} h_{i,j} = \operatorname{ord}_{F_i} h_{i,j}^0$ for any j. Possibly passing to a subsequence, we may assume that F_i exists for any i. Since 0 is fixed by G_i, Lemma 2.6(2) implies that the center of F_i is 0 and $\alpha_i = a(F_i, X, B_i)/|I_{G_i}(F_i)|$. Since $\operatorname{ord}_{F_i} h_{i,j} = \operatorname{ord}_{F_i} h_{i,j}^0$ for any j, we have $a(F_i, X, B_i) = a(F_i, X, B_i^0)$. Since $N_i \to +\infty$, (14.17) implies that $\alpha_i \geq b_{H_i}$ for $i \gg 0$.

Let $\eta_i := \max\{\frac{2P}{r_i}, \frac{8L}{\gamma N_i}\}$. Then $\eta_i \to 0$, $r_i \geq \frac{2P}{\eta_i}$, and $N_i \geq \frac{8L}{\gamma \eta_i} > \frac{4L}{\gamma \eta_i}$. For $i \gg 0$, we have $\eta_i \leq \frac{1}{8Pe_0}$, $N_i \geq \ell$, and $N_i > \frac{3}{\gamma}$. Hence, for $i \gg 0$, Proposition 14.9 implies that

$$0 \leq \alpha_i - b_{H_i} \leq 2\eta_i \to 0,$$

and that the numbers b_{H_i} belong to a set satisfying the ACC. Since $\alpha_i \geq b_{H_i}$ for $i \gg 0$, after deleting finitely many terms this contradicts Lemma 14.11. Therefore only finitely many i admit such a lift F_i. After deleting finitely many terms, any lift F to X of any prime divisor computing α_i satisfies $\operatorname{ord}_F h_{i,j} \neq \operatorname{ord}_F h_{i,j}^0$ for some j. This proves (4). □

14.4. **The high range for smooth germs.** In this subsection, we prove Theorem 14.2. The lower bound in (3) is the key new step of the main line of the proof of the main theorems.

Proof of Theorem 14.2. The proof is long, so we divide it into several steps.

Step 1. In this step we fix the thresholds in (3), and we choose a prime divisor F over X with center 0 such that $a = a(F, X, B)/|I_G(F)|$.

Assume that ϵ and the data of Set-up 14.1 satisfy the hypotheses of (3). Let ℓ and L be the integers of Construction-Proposition 14.7 for Γ, and let λ, Q, C_1, and C_2 be the constants of Proposition 14.12. They depend only on ϵ, P, e_0, and Γ. Let r_0 be the maximum of the constants r_0 in the In particular part of Proposition 14.9 and in Proposition 14.12, and let N_0

be the maximum of the corresponding constants N_0. We prove the conclusion of (3) under the conditions

$$r \geq \max\Big\{r_0,\ 8P^3e_0^2C_2^2,\ \frac{P}{\gamma}+\frac{6P^2e_0}{\gamma(\lambda-1)}\Big\} \quad \text{and} \quad N \geq \max\Big\{N_0,\ \ell,\ \frac{3}{\gamma}+1\Big\}.$$

By Proposition 14.9, (T, B_T) is lc near 0, which is the first conclusion of (3). Since $a \geq 1+\epsilon$, the In particular part of Proposition 14.9 implies that $(T/H, B_{T/H})$ is klt near $\bar{0}$, where $\bar{0}$ is the image of 0 in T/H, and (X, B^0) is klt near 0. It remains to prove $a \geq b_H$.

Since $\operatorname{div}(F_j(u,t)) \geq 0$ for any j, we have $B^0 \geq c_x \operatorname{div}(x) + c_y \operatorname{div}(y)$. Thus $c_x < 1$ and $c_y < 1$. Since $c_j \geq \gamma$ for any j, we obtain $a_j \leq c_x/c_j < 1/\gamma$ and $b_j \leq c_y/c_j < 1/\gamma$. Hence, for $\kappa_j := pa_j - qb_j$,

$$|\kappa_j| \leq \max\{pa_j, qb_j\} < \frac{P}{\gamma} \quad \text{for any } j. \tag{14.18}$$

Let $\pi : X \to X/G$ be the quotient morphism. Since (X, B) is klt near 0, Lemma 2.6(2)(3) implies that $(X/G, B_{X/G})$ is klt near $\pi(0)$. Hence a is attained by a prime divisor E over X/G with $\operatorname{center}_{X/G} E = \{\pi(0)\}$ [Amb99, Section 1, the second paragraph after Definition 1.1]. By Lemma 2.6(1)–(3), there exists a prime divisor F over X with $\operatorname{ord}_F|_{\mathbb{C}(X/G)} = e \operatorname{ord}_E$, where $e := |I_G(F)|$, such that $\pi(\operatorname{center}_X F) = \{\pi(0)\}$ and

$$a = a(E, X/G, B_{X/G}) = \frac{a(F, X, B)}{e}. \tag{14.19}$$

Since 0 is fixed by G, $\pi^{-1}(\pi(0)) = \{0\}$, hence $\operatorname{center}_X F = \{0\}$. Let E_0 be the exceptional divisor of the blow-up of X at 0. Then $a(E_0, X, B) = 3 - \sum_j c_j \operatorname{mult}_0 h_j \leq 3$ [KM98, Lemma 2.29], and the In particular part of Lemma 2.6 implies $a \leq a(E_0, X, B)/|I_G(E_0)| \leq 3$. Thus $a(F, X, B)/e \leq 3$, and Proposition 14.12(2)(3) hold for F.

Step 2. In this step we replace each h_j by a polynomial h_j^+ with $\operatorname{ord}_F h_j^+ = \operatorname{ord}_F h_j$ whose monomials have $\mathbf{T}$-weight at least κ_j.

We distinguish two cases.

- (**Case 1**) $\operatorname{ord}_F h_j = \operatorname{ord}_F h_j^0$ for any j.
- (**Case 2**) $\operatorname{ord}_F h_j \neq \operatorname{ord}_F h_j^0$ for some j.

Interchanging x with y and p with q preserves u, $\mathbf{T}$, $\mu_r(p,-q,0)$, c_u, B_T, H, and the conditions of Set-up 14.1, and replaces each κ_j by $-\kappa_j$, so (14.18) still holds. Possibly interchanging x with y and p with q, we may assume that $m := \operatorname{ord}_F x \leq \operatorname{ord}_F y$ in Case 2.

Since u and t have $\mathbf{T}$-weight 0, any monomial occurring in h_j^0 has $\mathbf{T}$-weight κ_j. Since G is diagonal, any monomial occurring in a G-semi-invariant polynomial has the character of that polynomial. The generator $\sigma := (\zeta^p, \zeta^{-q}, 1)$ of $\mu_r(p,-q,0) \subset G$, $\zeta = e^{2\pi\sqrt{-1}/r}$, satisfies $\sigma^*(x^{i_1}y^{i_2}t^{i_3}) = \zeta^{pi_1-qi_2}x^{i_1}y^{i_2}t^{i_3}$, and h_j^0 has the character of h_j. Thus any monomial $x^{i_1}y^{i_2}t^{i_3}$ occurring in h_j or in R_j satisfies

$$pi_1 - qi_2 = \kappa_j + rl \quad \text{for an integer } l. \tag{14.20}$$

By the assumption of (3) that h_j^0 is the sum of all terms of h_j of $\mathbf{T}$-weight κ_j, any monomial occurring in R_j satisfies (14.20) with $l \neq 0$.

In Case 1, let $h_j^+ := h_j^0$ for any j.

In Case 2, Proposition 14.12(3) implies $m \leq eC_2/N \leq e_0rC_2/N$. Since $m \geq 1$ and $N \geq \frac{r}{2P}$, we obtain

$$1 \leq m \leq 2Pe_0C_2 \quad \text{and} \quad \frac{1}{e} \leq \frac{C_2}{N} \leq \frac{2PC_2}{r}. \tag{14.21}$$

Since $\operatorname{center}_X F = \{0\}$, by Lemma 14.5, $\operatorname{ord}_F|_{\mathbb{C}(u,t)} = \varrho \operatorname{ord}_\nu$ for a positive integer ϱ and a prime divisor ν over T. Then $\varrho \operatorname{ord}_\nu u = \operatorname{ord}_F u = qm + p \operatorname{ord}_F y > 0$. Since ν is a prime divisor over T

and u is a regular function on T, $\operatorname{ord}_\nu u$ is a positive integer, so $\operatorname{ord}_\nu u \geq 1$. Since $\varrho/e \geq 1/e_0$ by Proposition 14.12(2), (14.21) and $r \geq 8P^3e_0^2C_2^2$ imply

$$\frac{\operatorname{ord}_F y}{e} \geq \frac{1}{p}\Big(\frac{\varrho}{e} - \frac{qm}{e}\Big) \geq \frac{1}{P}\Big(\frac{1}{e_0} - \frac{4P^3e_0C_2^2}{r}\Big) \geq \frac{1}{2Pe_0}. \tag{14.22}$$

Let h_j^+ be the sum of h_j^0 and of the terms of R_j with $l > 0$ in (14.20), and let h_j^- be the sum of the terms of R_j with $l < 0$, so that $h_j = h_j^+ + h_j^-$. Let $x^{i_1}y^{i_2}t^{i_3}$ be a monomial occurring in h_j^-. Then $l < 0$, so $qi_2 = pi_1 - \kappa_j - rl \geq r - \kappa_j > r - \frac{P}{\gamma}$ by (14.18), and (14.22) implies

$$\frac{\operatorname{ord}_F(x^{i_1}y^{i_2}t^{i_3})}{e} \geq \frac{i_2 \operatorname{ord}_F y}{e} > \frac{r - P/\gamma}{2P^2e_0} \geq \frac{3}{\gamma(\lambda-1)}.$$

Since $\operatorname{ord}_F h_j/e \leq \frac{3}{\gamma(\lambda-1)}$ by Proposition 14.12(2), we obtain $\operatorname{ord}_F h_j^- > \operatorname{ord}_F h_j$ whenever $h_j^- \neq 0$, hence $\operatorname{ord}_F h_j^+ = \operatorname{ord}_F h_j$. Moreover, $h_j^+ \neq 0$, because the terms of R_j with $l > 0$ have $\mathbf{T}$-weights different from κ_j.

In both cases, for any j, the following hold.

(a) h_j^+ is a non-zero G-semi-invariant polynomial with the character of h_j.

(b) $\operatorname{ord}_F h_j^+ = \operatorname{ord}_F h_j$.

(c) Any monomial occurring in h_j^+ satisfies (14.20) with $l \geq 0$, and the sum of the terms of h_j^+ with $l = 0$ is h_j^0.

Property (a) holds because any monomial occurring in h_j has the character of h_j. By (b) and (14.19),

$$ea = a(F, X, B) = a(F, X, 0) - \sum_j c_j \operatorname{ord}_F h_j^+. \tag{14.23}$$

Step 3. In this step we pass to the degree-p extension $x = z^p$, $y = z^{-q}y'$ of $\mathbb{C}(x, y, t)$, in which $z^{-\kappa_j}h_j^+$ becomes a polynomial $\tilde{h}_j$ in $Z := z^r$, y', and t.

Let $Y_0 := \mathbb{A}^3_{z,y',t}$, and regard $\mathbb{C}(x, y, t)$ as a subfield of $\mathbb{C}(z, y', t)$ via $x = z^p$ and $y = z^{-q}y'$. Then $\mathbb{C}(z, y', t) = \mathbb{C}(x, y, t)(z)$ because $y' = z^q y$, and z is a root of $\tau^p - x \in \mathbb{C}(x, y, t)[\tau]$, which is irreducible by the Eisenstein criterion for the prime element x of $\mathbb{C}[x, y, t]$. Thus $[\mathbb{C}(z, y', t) : \mathbb{C}(x, y, t)] = p$.

Let $\psi : (\mathbb{C}^*)^3 \to (\mathbb{C}^*)^3$ be the surjective homomorphism $(\xi_1, \xi_2, \xi_3) \to (\xi_1^p, \xi_1^{-q}\xi_2, \xi_3)$, and let $\tilde{G} := \psi^{-1}(G)$, a finite group acting diagonally on Y_0. For $\sigma \in \tilde{G}$, the restriction of σ^* to $\mathbb{C}(x, y, t)$ is $\psi(\sigma)^*$. The kernel of ψ is $\mu_p(1, q, 0)$, so $|\tilde{G}| = p|G|$. Since $\mu_p(1, q, 0)$ fixes x and y and has order p, $\mathbb{C}(Y_0)^{\mu_p(1,q,0)} = \mathbb{C}(x, y, t)$. Hence $\mathbb{C}(Y_0)^{\tilde{G}} = \mathbb{C}(x, y, t)^G$.

Since $\psi(\mu_r(1, 0, 0)) = \mu_r(p, -q, 0) \subset G$, we have $\mu_r(1, 0, 0) \subset \tilde{G}$. Let $Y := \mathbb{A}^3_{Z,y',t}$ with $Z := z^r$, so that $Y_0 \to Y$, $(z, y', t) \to (z^r, y', t)$, is the quotient morphism by $\mu_r(1, 0, 0)$. Let $\bar{G}$ be the image of $\tilde{G}$ under the homomorphism $(\xi_1, \xi_2, \xi_3) \to (\xi_1^r, \xi_2, \xi_3)$. The kernel of $(\xi_1, \xi_2, \xi_3) \to (\xi_1^r, \xi_2, \xi_3)$ is $\mu_r(1, 0, 0)$, so $\bar{G} = \tilde{G}/\mu_r(1, 0, 0)$ is a finite diagonal group acting faithfully on Y, and

$$\mathbb{C}(Y)^{\bar{G}} = \mathbb{C}(Y_0)^{\tilde{G}} = \mathbb{C}(x, y, t)^G = \mathbb{C}(X/G). \tag{14.24}$$

Let $S' := (Z = 0) = \mathbb{A}^2_{y',t} \subset Y$. We show that $\bar{G}$ acts faithfully on S'. Let $(\xi_1, \xi_2, \xi_3) \in \tilde{G}$ be an element such that the image $(\xi_1^r, \xi_2, \xi_3) \in \bar{G}$ acts trivially on S'. Then $\xi_2 = \xi_3 = 1$, and $\psi(\xi_1, 1, 1) = (\xi_1^p, \xi_1^{-q}, 1) \in G \cap \mathbf{T} = \mu_r(p, -q, 0)$. Thus $(\xi_1^p, \xi_1^{-q}) = (\zeta^{kp}, \zeta^{-kq})$ for an integer k, so $(\xi_1\zeta^{-k})^p = (\xi_1\zeta^{-k})^q = 1$. Since p and q are coprime, $\xi_1 = \zeta^k$ and $\xi_1^r = 1$.

By (14.20) and (c), any monomial $x^{i_1}y^{i_2}t^{i_3}$ occurring in h_j^+ satisfies $z^{-\kappa_j}x^{i_1}y^{i_2}t^{i_3} = z^{pi_1-qi_2-\kappa_j}y'^{i_2}t^{i_3} = Z^l y'^{i_2}t^{i_3}$ with $l \geq 0$. Hence $\tilde{h}_j := z^{-\kappa_j}h_j^+$ is a polynomial in Z, y', and t, and, since $u = x^q y^p = y'^p$, (c) implies

$$\tilde{h}_j(0, y', t) = z^{-\kappa_j}h_j^0 = z^{-\kappa_j}z^{pa_j-qb_j}y'^{b_j}F_j(y'^p, t) = y'^{b_j}F_j(y'^p, t) \neq 0. \tag{14.25}$$

By (a), h_j^+ and z are $\tilde{G}$-semi-invariant, hence so is $\tilde{h}_j$. Since $\tilde{h}_j$ is a polynomial in Z, y', and t, it is $\mu_r(1,0,0)$-invariant, so $\tilde{h}_j$ is a $\bar{G}$-semi-invariant polynomial on Y.

Step 4. In this step we express a as a log discrepancy over Y, and we show that the boundary $\hat{B}$ below does not increase it.

Let

$$\beta := p - q - \sum_j c_j \kappa_j = p - q - pc_x + qc_y, \qquad \beta_+ := \max\left\{0, -\frac{\beta}{q}\right\},$$

and let

$$B_\beta := \left(1 - \frac{\beta}{r}\right) \operatorname{div}(Z) + \sum_j c_j \operatorname{div}(\tilde{h}_j), \qquad \hat{B} := \operatorname{div}(Z) + \sum_j c_j \operatorname{div}(\tilde{h}_j) + \beta_+ \operatorname{div}(y')$$

on Y. The $\mathbb{R}$-divisor B_β may have negative coefficients, while $\hat{B} \geq 0$. Both are $\bar{G}$-invariant by **Step 3**.

Recall from **Step 1** that E is a prime divisor over X/G computing a and that $\operatorname{ord}_F|_{\mathbb{C}(X/G)} = e \operatorname{ord}_E$. The coordinate ring $\mathbb{C}[Y_0]^{\mu_p(1,q,0)}$ of $Y_0/\mu_p(1,q,0)$ is contained in $\mathbb{C}[x,y,t]$, because an invariant monomial $z^{i_1} y'^{i_2} t^{i_3}$ satisfies $p \mid i_1 + qi_2$ and equals $x^{(i_1+qi_2)/p} y^{i_2} t^{i_3}$. Thus $X \to Y_0/\mu_p(1,q,0)$ is a birational morphism. Since ord_F is non-negative on $\mathbb{C}[Y_0]^{\mu_p(1,q,0)} \subset \mathbb{C}[x,y,t]$, ord_F has a center on $Y_0/\mu_p(1,q,0)$, so F is a prime divisor over $Y_0/\mu_p(1,q,0)$ [KM98, Lemma 2.45]. By Lemma 2.6(3), there exist a prime divisor F' over Y_0 and a positive integer θ with $\operatorname{ord}_{F'}|_{\mathbb{C}(x,y,t)} = \theta \operatorname{ord}_F$. Let F_Y be the prime divisor over Y with $\operatorname{ord}_{F'}|_{\mathbb{C}(Y)} = \theta_r \operatorname{ord}_{F_Y}$ for a positive integer θ_r (Lemma 2.6 for the action of $\mu_r(1,0,0)$ on Y_0). Since $\operatorname{ord}_F x > 0$, $\operatorname{ord}_F y > 0$, and $\operatorname{ord}_F t > 0$,

$$\operatorname{ord}_{F'} z = \frac{\theta}{p} \operatorname{ord}_F x > 0, \qquad \operatorname{ord}_{F'} y' = q \operatorname{ord}_{F'} z + \theta \operatorname{ord}_F y > 0, \qquad \operatorname{ord}_{F'} t = \theta \operatorname{ord}_F t > 0.$$

Thus $\operatorname{ord}_{F_Y} Z > 0$, $\operatorname{ord}_{F_Y} y' > 0$, and $\operatorname{ord}_{F_Y} t > 0$, so $\operatorname{center}_Y F_Y$ is the origin of Y. By (14.24), the restrictions of ord_F and of ord_{F_Y} to $\mathbb{C}(Y)^{\bar{G}} = \mathbb{C}(X/G)$ are positive multiples of the restriction of $\operatorname{ord}_{F'}$. Hence $\operatorname{ord}_{F_Y}|_{\mathbb{C}(Y)^{\bar{G}}} = e_Y \operatorname{ord}_E$ for a positive integer e_Y, and restricting $\operatorname{ord}_{F'}$ to $\mathbb{C}(X/G)$ through $\mathbb{C}(x,y,t)$ and through $\mathbb{C}(Y)$ implies

$$\theta e = \theta_r e_Y. \tag{14.26}$$

By Lemma 2.6(1)(2) for the action of $\bar{G}$ on Y, $e_Y = |I_{\bar{G}}(F_Y)|$, and the center of E on $Y/\bar{G}$ is the image of the origin.

Let $\omega := dx \wedge dy \wedge dt$ and $\omega_0 := dz \wedge dy' \wedge dt$. A direct calculation shows $\omega = pz^{p-q-1}\omega_0$. Let $\operatorname{ord}_F \omega$ be the order of ω along F on a higher model of X on which F is a divisor, and similarly for F'. Since ω and ω_0 are nowhere vanishing on X and on Y_0, we have $a(F,X,0) = 1 + \operatorname{ord}_F \omega$ and $a(F',Y_0,0) = 1 + \operatorname{ord}_{F'} \omega_0$.

We show that $1 + \operatorname{ord}_{F'} \omega = \theta(1 + \operatorname{ord}_F \omega)$. Let $\pi_0 : Y_0 \to W := Y_0/\mu_p(1,q,0)$ be the quotient morphism. Since $\gcd(p,q) = 1$, a non-trivial element $(\zeta_p^k, \zeta_p^{kq}, 1)$ of $\mu_p(1,q,0)$, where $\zeta_p := e^{2\pi\sqrt{-1}/p}$, fixes only the line $(z = y' = 0)$, so π_0 is quasi-étale. Let $B_\omega := -(p-q-1)\operatorname{div}(z)$, a $\mu_p(1,q,0)$-invariant $\mathbb{R}$-divisor on Y_0, and let $(W, B_W) := (Y_0, B_\omega)/\mu_p(1,q,0)$. Since $\operatorname{div}_{Y_0}(\omega) = (p-q-1)\operatorname{div}(z) = -B_\omega$, we have $a(F', Y_0, B_\omega) = 1 + \operatorname{ord}_{F'} \omega$. Since $\mathbb{C}(W) = \mathbb{C}(x,y,t)$, ω is also a rational differential form on W. Since π_0 is quasi-étale, $\operatorname{div}_{Y_0}(\omega) = \pi_0^* \operatorname{div}_W(\omega)$, and Lemma 2.6(0) implies $B_\omega = \pi_0^* B_W$. Thus $\pi_0^*(\operatorname{div}_W(\omega) + B_W) = 0$, hence $\operatorname{div}_W(\omega) + B_W = 0$, and $a(F, W, B_W) = 1 + \operatorname{ord}_F \omega$. Since $\operatorname{ord}_{F'}|_{\mathbb{C}(x,y,t)} = \theta \operatorname{ord}_F$, Lemma 2.6(1)(2) for the action of $\mu_p(1,q,0)$ on Y_0 implies $|I_{\mu_p(1,q,0)}(F')| = \theta$ and

$$1 + \operatorname{ord}_{F'} \omega = a(F', Y_0, B_\omega) = \theta a(F, W, B_W) = \theta(1 + \operatorname{ord}_F \omega).$$

Since $\omega = pz^{p-q-1}\omega_0$, we obtain

$$\theta a(F, X, 0) = a(F', Y_0, 0) + (p - q - 1) \operatorname{ord}_{F'} z.$$

By Lemma 2.6(4), the crepant quotient of $(Y_0, 0)$ by $\mu_r(1,0,0)$ is $(Y, (1-\frac{1}{r})\operatorname{div}(Z))$, so Lemma 2.6(2) implies $a(F', Y_0, 0) = \theta_r a(F_Y, Y, (1-\frac{1}{r})\operatorname{div}(Z))$. Moreover,

$$\operatorname{ord}_{F'} h_j^+ = \kappa_j \operatorname{ord}_{F'} z + \operatorname{ord}_{F'} \tilde{h}_j, \quad \operatorname{ord}_{F'} z = \frac{\theta_r}{r} \operatorname{ord}_{F_Y} Z, \quad \operatorname{ord}_{F'} \tilde{h}_j = \theta_r \operatorname{ord}_{F_Y} \tilde{h}_j.$$

Thus (14.23) implies

$$\begin{aligned} \theta ea &= \theta a(F, X, 0) - \sum_j c_j \operatorname{ord}_{F'} h_j^+ = a(F', Y_0, 0) - (1-\beta)\operatorname{ord}_{F'} z - \sum_j c_j \operatorname{ord}_{F'} \tilde{h}_j \\ &= \theta_r \Big(a(F_Y, Y, 0) - \Big(1 - \frac{1}{r}\Big) \operatorname{ord}_{F_Y} Z - \frac{1-\beta}{r} \operatorname{ord}_{F_Y} Z - \sum_j c_j \operatorname{ord}_{F_Y} \tilde{h}_j \Big) = \theta_r a(F_Y, Y, B_\beta). \end{aligned}$$

By (14.26), we obtain

$$a = \frac{a(F_Y, Y, B_\beta)}{e_Y}. \tag{14.27}$$

By the definitions of B_β and $\hat{B}$,

$$a(F_Y, Y, B_\beta) - a(F_Y, Y, \hat{B}) = \frac{\beta}{r} \operatorname{ord}_{F_Y} Z + \beta_+ \operatorname{ord}_{F_Y} y'. \tag{14.28}$$

If $\beta \geq 0$, then $\beta_+ = 0$ and the right-hand side of (14.28) is $\frac{\beta}{r}\operatorname{ord}_{F_Y} Z \geq 0$. If $\beta < 0$, then $\beta_+ = -\frac{\beta}{q}$, and $\operatorname{ord}_{F_Y} Z = \frac{r}{\theta_r}\operatorname{ord}_{F'} z$ and $\operatorname{ord}_{F_Y} y' = \frac{1}{\theta_r}(q \operatorname{ord}_{F'} z + \theta \operatorname{ord}_F y)$ imply

$$\frac{\beta}{r} \operatorname{ord}_{F_Y} Z + \beta_+ \operatorname{ord}_{F_Y} y' = \frac{-\beta}{\theta_r} \Big(\frac{q \operatorname{ord}_{F'} z + \theta \operatorname{ord}_F y}{q} - \operatorname{ord}_{F'} z \Big) = \frac{-\beta\theta}{q\theta_r} \operatorname{ord}_F y > 0.$$

Let $(Y/\bar{G}, \hat{B}_{Y/\bar{G}}) := (Y, \hat{B})/\bar{G}$. By (14.27), (14.28), $e_Y = |I_{\bar{G}}(F_Y)|$, and Lemma 2.6(2),

$$a \geq \frac{a(F_Y, Y, \hat{B})}{e_Y} = a(E, Y/\bar{G}, \hat{B}_{Y/\bar{G}}). \tag{14.29}$$

Step 5. In this step we show that $(T, B_T)/H$ is the crepant quotient of a pair on S' by $\bar{G}$, and we finish the proof of (3) by Theorem 8.3.

Let

$$C' := \sum_j c_j \operatorname{div}(F_j(y'^p, t)) + (c_y + \beta_+) \operatorname{div}(y')$$

on S'. By (14.25),

$$C' = \sum_j c_j \operatorname{div}(\tilde{h}_j|_{S'}) + \beta_+ \operatorname{div}(y'|_{S'}).$$

Since $c_y - \frac{\beta}{q} = 1 - \frac{p(1-c_x)}{q}$, we have

$$pc_u - (p-1) = \max\Big\{1 - \frac{p(1-c_x)}{q},\ c_y\Big\} = c_y + \beta_+. \tag{14.30}$$

Let $f : S' \to T$ be the morphism $(y', t) \to (y'^p, t)$. The ramification divisor of f is $(p-1)\operatorname{div}(y')$, $f^* \operatorname{div}(u) = p \operatorname{div}(y')$, and $f^* \operatorname{div}(F_j(u,t)) = \operatorname{div}(F_j(y'^p, t))$. Thus (14.30) implies

$$K_{S'} + C' = f^*(K_T + B_T).$$

For $(\xi_1, \xi_2, \xi_3) \in \tilde{G}$, the image $(\xi_1^r, \xi_2, \xi_3) \in \bar{G}$ multiplies $u = y'^p$ by ξ_2^p and t by ξ_3, and $\psi(\xi_1, \xi_2, \xi_3) \in G$ multiplies $u = x^q y^p$ by $\xi_1^{pq}(\xi_1^{-q}\xi_2)^p = \xi_2^p$ and t by ξ_3. Since $\psi(\tilde{G}) = G$, f is equivariant for a surjective homomorphism $\bar{G} \to H$. Let $\bar{G}_0$ be the kernel of $\bar{G} \to H$. Then $\bar{G}_0$ fixes u and t, so $\bar{G}_0$ multiplies y' by p-th roots of unity, and $\bar{G}_0$ acts faithfully on S' by **Step 3**. Moreover, $\bar{G}_0$ contains the image $(\zeta_p^{kr}, \zeta_p^{kq}, 1)$ of any element $(\zeta_p^k, \zeta_p^{kq}, 1)$ of $\mu_p(1, q, 0)$, and ζ_p^{kq} runs over all p-th roots of unity as k varies since p and q are coprime. Hence $S'/\bar{G}_0 = T$ with quotient morphism f, and the composite $\pi_{S'} : S' \to T \to T/H$ is the quotient morphism

by $\bar{G}$. By Lemma 2.6(0) for the action of H on (T, B_T), $K_{S'} + C' = \pi_{S'}^*(K_{T/H} + B_{T/H})$. By Lemma 2.6(0) for the action of $\bar{G}$ on (S', C') and the injectivity of $\pi_{S'}^*$ on $\mathbb{R}$-divisors,

$$(S', C')/\bar{G} = (T, B_T)/H, \quad \text{hence} \quad b_H = \mathrm{mld}((S' \ni 0, C')/\bar{G}). \tag{14.31}$$

By **Step 1**, $(T/H, B_{T/H})$ is klt near $\bar{0}$. By (14.31) and Lemma 2.6(2) for the action of $\bar{G}$ on (S', C'), (S', C') is klt near 0.

We take $\bar{G}$ in place of both G and H in Theorem 8.3, and the following data: $n = 3$, $c = 1$, the diagonal action of $\bar{G}$ on $Y = \mathbb{A}^3_{Z,y',t}$, $f_1 = Z$, the polynomials $\tilde{h}_j$ with the coefficients c_j, and the polynomial y' with the coefficient β_+ (omitted if $\beta_+ = 0$). The common zero scheme of f_1 is the smooth plane S', $\bar{G}$ acts faithfully on S' by **Step 3**, the restrictions of $\tilde{h}_j$ and y' to S' are non-zero by (14.25), the boundary on S' is C', which is klt near 0, and the boundary on Y is $\hat{B}$. The In particular part of Theorem 8.3 and (14.31) imply that any prime divisor over $Y/\bar{G}$ with center the image of 0 has log discrepancy at least b_H with respect to $(Y, \hat{B})/\bar{G}$. Since the center of E on $Y/\bar{G}$ is the image of 0 by **Step 4**, (14.29) implies

$$a \geq a(E, Y/\bar{G}, \hat{B}_{Y/\bar{G}}) \geq b_H.$$

This proves (3).

Step 6. In this step we prove (1).

Since $\Gamma \cup \{1\}$ is a DCC subset of $[0, 1]$ containing Γ, the set in (1) for Γ is contained in the set in (1) for $\Gamma \cup \{1\}$. Possibly replacing Γ with $\Gamma \cup \{1\}$, we may assume that $\Gamma \backslash \{0\} \neq \emptyset$. Suppose that α_i is a strictly increasing sequence of numbers in the set in (1). At most one term of the sequence equals 1. Removing that term, let $\epsilon := \alpha_1 - 1 > 0$. Then $\alpha_i \geq 1 + \epsilon$ for any i, so α_i belongs to the set $\mathcal{M}$ of Proposition 14.6 for ϵ and Γ.

By Proposition 14.6(2), after passing to a subsequence and permuting coordinates, we obtain coprime positive integers $p, q \leq 1/\epsilon$, a positive integer e_0, integers $r_i \to +\infty$ with $G_i \cap \mathbf{T} = \mu_{r_i}(p, -q, 0)$ of index at most e_0 in G_i, and the decompositions $h_{ij} = x^{a_j} y^{b_j} F_{ij}(u, t) + R_{ij}$ with the integers L_i and C_0 as there. Let $P := \max\{p, q\}$, $N_i := L_i$, and $h^0_{ij} := x^{a_j} y^{b_j} F_{ij}(u, t)$. Since $r_i \to +\infty$, after deleting finitely many terms we have $r_i \geq 2C_0$, hence $N_i \geq \frac{r_i - C_0}{P} \geq \frac{r_i}{2P} > 0$. Thus the data of α_i satisfy the conditions of Set-up 14.1 with $r = r_i$ and $N = N_i$, and h^0_{ij} is the sum of all terms of h_{ij} of $\mathbf{T}$-weight $pa_j - qb_j$ by Proposition 14.6(2). Since p and q are coprime, any element of $(\mathbb{C}^*)^3$ multiplying u and t by 1 belongs to $\mathbf{T}$. Hence the kernel of $G_i \to H_i$ is $G_i \cap \mathbf{T}$, and $|H_i| \leq e_0$. Since $r_i, N_i \to +\infty$ and $\alpha_i \geq 1 + \epsilon$, (3) implies $\alpha_i \geq b_{H_i}$ for any i after deleting finitely many terms.

Let ℓ and L be the integers of Construction-Proposition 14.7 for Γ, and let

$$\eta_i := \max\left\{\frac{2P}{r_i}, \frac{8L}{\gamma N_i}\right\}.$$

Then $\eta_i \to 0$, $r_i \geq \frac{2P}{\eta_i}$, and $N_i \geq \frac{8L}{\gamma \eta_i} > \frac{4L}{\gamma \eta_i}$. After deleting finitely many terms, $\eta_i \leq \frac{1}{8Pe_0}$, $N_i \geq \ell$, and $N_i > \frac{3}{\gamma}$ for any i. Thus Proposition 14.9 implies $\alpha_i \leq b_{H_i} + 2\eta_i$, hence

$$0 \leq \alpha_i - b_{H_i} \leq 2\eta_i \quad \text{for any } i. \tag{14.32}$$

By the Moreover part of Proposition 14.9, the numbers b_{H_i} belong to an ACC set depending only on Γ, P, and e_0. By (14.32), this contradicts Lemma 14.11. This proves (1).

Step 7. In this step we prove (2).

Let $(X' \ni x')$ be a smooth threefold germ, G' a finite abelian group acting faithfully on X' and fixing x', and $B' = \sum_j c_j \operatorname{div}(h_j)$, where $c_j \in \Gamma$ satisfy $c_j > 0$, h_j are regular G'-semi-invariant functions, and (X', B') is klt near x'. Let $\mathfrak{m}_{x'}$ be the maximal ideal of $\mathcal{O}_{X',x'}$. Since G' fixes x', $\sigma^* \mathfrak{m}_{x'} = \mathfrak{m}_{x'}$ for any $\sigma \in G'$. Since G' is a finite abelian group, there exist $f_1, f_2, f_3 \in \mathfrak{m}_{x'}$ whose

images in $\mathfrak{m}_{x'}/\mathfrak{m}_{x'}^2$ form a basis of G'-semi-invariant vectors with characters χ_1, χ_2, χ_3. Let

$$x_i := \frac{1}{|G'|} \sum_{\sigma \in G'} \chi_i(\sigma)^{-1} \sigma^* f_i \quad \text{for } i = 1, 2, 3.$$

Since G' is abelian, $x_i \in \mathfrak{m}_{x'}$ is G'-semi-invariant with character χ_i, and $x_i - f_i \in \mathfrak{m}_{x'}^2$. By Nakayama's lemma, x_1, x_2, x_3 are G'-semi-invariant regular parameters of $\mathcal{O}_{X',x'}$. By Construction-Theorem 7.2(1) with $n = 3$ and X' in place of Y, the group G', the functions h_j with the coefficients c_j, no vectors in (iv), and no prime divisors in (v), there exist a faithful diagonal action of G' on $\mathbb{A}^3$ and non-zero G'-semi-invariant polynomials $h_{j,0}$ such that, for $B_0 := \sum_j c_j \operatorname{div}(h_{j,0})$, $(\mathbb{A}^3, B_0)$ is klt near 0 and

$$\operatorname{mld}((\mathbb{A}^3 \ni 0, B_0)/G') = \operatorname{mld}((X' \ni x', B')/G').$$

Hence any number in the set in (2) belongs to the set in (1). Thus (2) follows from (1). □

14.5. **Jets of residual pairs.** In this subsection, we show that on a fixed quotient surface, the minimal log discrepancy of a crepant quotient is determined by a bounded jet of the equations and is constant along generic limits of the equations.

Construction 14.13. Let H be a finite diagonal subgroup of $(\mathbb{C}^*)^2$ acting on $T = \mathbb{A}^2_{u,t}$, let N and s be positive integers, and for $i \geq 1$ let $(f_{i,j})_{1 \leq j \leq s}$ be tuples of H-semi-invariant polynomials in $\mathbb{C}[u,t]$ with $\operatorname{ord}_0 f_{i,j} \leq N$ such that the character of $f_{i,j}$ depends only on j. After passing to a subsequence and multiplying each $f_{i,j}$ by a non-zero constant, which does not change $\operatorname{div}(f_{i,j})$, we construct an algebraically closed field $k \supset \mathbb{C}$ and non-zero H-semi-invariant power series $f_{\infty,j} \in k[[u,t]]$, $f_{\infty,j}$ having the character of $f_{i,j}$, with the following property. For any positive integer d, let $\mathcal{J}_d$ be the affine space of s-tuples of polynomials of degree less than d in u, t. Then the tuple of the parts of degree less than d of the $f_{\infty,j}$, a k-point of $\mathcal{J}_d$, maps to the generic point of an irreducible closed subset $Z_d \subset \mathcal{J}_d$, and any non-empty open subset of Z_d contains the tuple of the parts of degree less than d of the $f_{i,j}$ for all but finitely many i. We call $(f_{\infty,j})_j$ a *generic limit* of the tuples $(f_{i,j})_j$ [Kaw24, Section 4] (cf. [dFM09], [dFEM11, Section 3]).

Proof. We first normalize the polynomials. Since $\operatorname{ord}_0 f_{i,j} \leq N$, some monomial of degree at most N occurs in $f_{i,j}$, and there exist only finitely many monomials of degree at most N. Hence, possibly passing to a subsequence, we may assume that for any j there exists a monomial m_j of degree at most N which occurs in $f_{i,j}$ for any i. Multiplying $f_{i,j}$ by a non-zero constant, we may assume that m_j occurs in $f_{i,j}$ with coefficient 1. Let χ_j be the character of $f_{i,j}$.

For any $d \geq 1$ and $i \geq 1$, let $\phi_{i,d} \in \mathcal{J}_d(\mathbb{C})$ be the tuple of the parts of degree less than d of the $f_{i,j}$, and let $p_d : \mathcal{J}_{d+1} \to \mathcal{J}_d$ be the linear projection taking parts of degree less than d, so that $p_d(\phi_{i,d+1}) = \phi_{i,d}$. Let $L_d \subset \mathcal{J}_d$ be the closed affine-linear subspace defined by the following conditions: for any j, any monomial whose character is not χ_j has coefficient 0 in the j-th polynomial, and, if $d > N$, the monomial m_j has coefficient 1 in the j-th polynomial. Then $\phi_{i,d} \in L_d$ for any i and d.

Set $\Sigma_0 := \mathbb{Z}_{\geq 1}$. By induction on $d \geq 1$, we construct an infinite subset $\Sigma_d \subset \Sigma_{d-1}$ and an irreducible closed subset $Z_d \subset \mathcal{J}_d$. Since closed subsets of $\mathcal{J}_d$ satisfy the descending chain condition and $\mathcal{J}_d$ contains $\phi_{i,d}$ for any $i \in \Sigma_{d-1}$, there exists an irreducible closed subset $Z_d \subset \mathcal{J}_d$ which is minimal among the irreducible closed subsets of $\mathcal{J}_d$ containing $\phi_{i,d}$ for infinitely many $i \in \Sigma_{d-1}$. Let Σ_d be the set of $i \in \Sigma_{d-1}$ such that $\phi_{i,d} \in Z_d$.

We show that

(14.33) $\quad Z_d \subset V \quad$ for any closed subset $V \subset \mathcal{J}_d$ containing $\phi_{i,d}$ for infinitely many $i \in \Sigma_d$.

In fact, for such V, $V \cap Z_d$ has finitely many irreducible components, and one of them contains $\phi_{i,d}$ for infinitely many $i \in \Sigma_d \subset \Sigma_{d-1}$. By minimality, it is equal to Z_d. We obtain the following.

(a) For any non-empty open subset $U \subset Z_d$, $\phi_{i,d} \in U$ for all but finitely many $i \in \Sigma_d$. Otherwise $Z_d \subset Z_d \backslash U$ by (14.33) with $V = Z_d \backslash U$, which is impossible.

(b) $Z_d \subset L_d$, by (14.33) with $V = L_d$.
(c) $p_d(Z_{d+1})$ is a dense subset of Z_d. In fact, $p_d^{-1}(Z_d)$ contains $\phi_{i,d+1}$ for any $i \in \Sigma_{d+1}$, so $Z_{d+1} \subset p_d^{-1}(Z_d)$ by (14.33) with $d+1$ in place of d and $V = p_d^{-1}(Z_d)$. Moreover, the closure of $p_d(Z_{d+1})$ contains $\phi_{i,d}$ for any $i \in \Sigma_{d+1}$, so it contains Z_d.

By (c), p_d induces an inclusion $\mathbb{C}(Z_d) \subset \mathbb{C}(Z_{d+1})$ of function fields.

Choose integers $i_1 < i_2 < \cdots$ with $i_d \in \Sigma_d$, and replace the sequence of tuples by $(f_{i_d,j})_j$, $d \geq 1$. Since $i_{d'} \in \Sigma_{d'} \subset \Sigma_d$ for any $d' \geq d$, (a) implies that any non-empty open subset of Z_d contains $\phi_{i_{d'},d}$ for all but finitely many d'.

Let k be an algebraic closure of the field $\bigcup_d \mathbb{C}(Z_d)$. For any j and any monomial μ of degree less than d, let $x_{j,\mu,d}$ be the coordinate function on $\mathcal{J}_d$ given by the coefficient of μ in the j-th polynomial. Then $p_d^* x_{j,\mu,d} = x_{j,\mu,d+1}$, so the restrictions $x_{j,\mu,d}|_{Z_d} \in \mathbb{C}(Z_d)$ for $d > \deg \mu$ define one element $\xi_{j,\mu} \in k$. Set

$$f_{\infty,j} := \sum_{\mu} \xi_{j,\mu} \mu \in k[[u,t]].$$

For any d, the tuple of the parts of degree less than d of the $f_{\infty,j}$ is the k-point of $\mathcal{J}_d$ given by the homomorphism $\mathbb{C}[\mathcal{J}_d] \to \mathbb{C}[Z_d] \subset \mathbb{C}(Z_d) \subset k$, whose kernel is the ideal of Z_d. Thus it maps to the generic point of Z_d. By (b), $\xi_{j,m_j} = 1$, and $\xi_{j,\mu} = 0$ for any monomial μ whose character is not χ_j. Hence $f_{\infty,j}$ is a non-zero H-semi-invariant power series with character χ_j. □

In Theorem 14.19, we take a generic limit of residual plane pairs, and we need the quotient minimal log discrepancy to be preserved in the limit. Lemma 14.14 below proves the preservation of the quotient minimal log discrepancy for power series over algebraically closed extension fields of $\mathbb{C}$, using Theorem 2.19(2) on the product of the quotient surface with an affine line and Lemma 8.1.

Lemma 14.14. *Let H be a finite diagonal subgroup of $(\mathbb{C}^*)^2$ acting on $T = \mathbb{A}^2_{u,t}$, and $\Lambda \subset \mathbb{R}_{>0}$ a finite set.*

(1) There exists a positive integer ℓ such that, for any algebraically closed field $k \supset \mathbb{C}$ and any two finite tuples (f_j), (f'_j) of non-zero H-semi-invariant power series in $k[[u,t]]$ with $f_j - f'_j \in (u,t)^\ell$, and any $c_j \in \Lambda$, the crepant quotients of $(\operatorname{Spec} k[[u,t]], \sum_j c_j \operatorname{div}(f_j))$ and $(\operatorname{Spec} k[[u,t]], \sum_j c_j \operatorname{div}(f'_j))$ by H have equal minimal log discrepancies at the closed point, where the minimal log discrepancies are defined with formal prime divisors as in Definition 3.2.
(2) Let $(f_{i,j})_j$, $i \geq 1$, k, and $(f_{\infty,j})_j$ be as in Construction 14.13 for this H, and fix $c_1, \ldots, c_s \in \Lambda$. Then, on an infinite subsequence, the minimal log discrepancies at the closed point of the crepant quotients by H of $(T, \sum_j c_j \operatorname{div}(f_{i,j}))$ are equal to that of $(\operatorname{Spec} k[[u,t]], \sum_j c_j \operatorname{div}(f_{\infty,j}))$.

Proof. The proof is long, so we divide it into several steps.

Let $d := |H|$, $Q := T/H = \operatorname{Spec} \mathbb{C}[u,t]^H$, and $\Lambda' := \{c/d \mid c \in \Lambda\}$. For a field $k \supset \mathbb{C}$, let $T_k := \mathbb{A}^2_k$, $Q_k := T_k/H = \operatorname{Spec} k[u,t]^H$, let π denote the quotient morphism $T_k \to Q_k$, denote the image of 0 in Q_k again by 0, and let $\mathfrak{n}_k \subset k[u,t]^H$ be the ideal of 0. We write $\mathfrak{n} := \mathfrak{n}_{\mathbb{C}}$. Since H acts diagonally, a power series is H-invariant if and only if any monomial occurring in it is H-invariant. In particular, $\mathfrak{n}_k$ is spanned by the non-constant H-invariant monomials, and $R[u,t]^H = R \otimes_{\mathbb{C}} \mathbb{C}[u,t]^H$ for any $\mathbb{C}$-algebra R. For a power series f and a positive integer n, let $f^{<n}$ be the part of f of degree less than n. If f is H-semi-invariant, then so is $f^{<n}$, with the character of f when $f^{<n} \neq 0$.

Let $e_u := |H \cap (\mathbb{C}^* \times \{1\})|$ and $e_t := |H \cap (\{1\} \times \mathbb{C}^*)|$. The inertia group of a prime divisor on T_k is the subgroup of H fixing it pointwise, so it is $H \cap (\mathbb{C}^* \times \{1\})$ for $(u = 0)$, $H \cap (\{1\} \times \mathbb{C}^*)$ for $(t = 0)$, and trivial for any other prime divisor. By Lemma 2.6(1)(4) and Remark 2.17,

$(T_k, 0)/H = (Q_k, \Delta_H)$, where

$$\Delta_H := \Big(1 - \frac{1}{e_u}\Big)\pi(\{u = 0\}) + \Big(1 - \frac{1}{e_t}\Big)\pi(\{t = 0\}).$$

By Lemma 2.6(0), $K_{Q_k} + \Delta_H$ is $\mathbb{R}$-Cartier. For non-zero ideals $\mathfrak{a}_j \subset k[u,t]^H$ and real numbers $r_j > 0$, we use the $\mathbb{R}$-ideal $\mathfrak{a} = \prod_j \mathfrak{a}_j^{r_j}$ on Q_k, the minimal log discrepancy $\mathrm{mld}(Q_k \ni 0, \Delta_H, \mathfrak{a})$ of Definition 2.2, and $\pi^*\mathfrak{a} := \prod_j (\mathfrak{a}_j \mathcal{O}_{T_k})^{r_j}$.

Step 1. In this step we show that, for any field $k \supset \mathbb{C}$ and any positive integer m,

$$(u,t)^{dm} \cap k[[u,t]]^H \subset \mathfrak{n}_k^m k[[u,t]]^H \quad \text{and} \quad (u,t)^{dm} \cap k[u,t]^H \subset \mathfrak{n}_k^m, \tag{14.34}$$

and that for H-semi-invariant power series $f, f' \in k[[u,t]]$ with $f - f' \in (u,t)^{dm}$, we have $f^d, f'^d \in k[[u,t]]^H$ and $f^d - f'^d \in \mathfrak{n}_k^m k[[u,t]]^H$, where $f^d - f'^d \in \mathfrak{n}_k^m$ if f and f' are polynomials.

The group $\mathrm{Hom}(H, \mathbb{C}^*)$ of characters of H has order d. Let $\psi_1, \dots, \psi_q$ be characters of H with $q \geq d$. The $d+1$ products $\psi_1 \cdots \psi_a$, $0 \leq a \leq d$, take at most d values, so $\psi_1 \cdots \psi_a = \psi_1 \cdots \psi_b$ for some $0 \leq a < b \leq d$. Thus the block $\psi_{a+1}, \dots, \psi_b$ has length between 1 and d and trivial product.

Let μ be an H-invariant monomial of degree at least dm, and write $\mu = y_1 \cdots y_q$ with $y_l \in \{u, t\}$ and $q = \deg \mu$. Removing a block of length at most d with trivial product deletes at most d factors, and after $m' < m$ removals at least $q - m'd \geq d$ factors remain. Thus we may remove m such blocks, and we obtain

$$\mu = \mu_1 \cdots \mu_m \nu,$$

where $\mu_1, \dots, \mu_m$ are non-constant H-invariant monomials of degree at most d and ν is a monomial. Since μ and the μ_l are H-invariant, so is ν. Let $g_1, \dots, g_w$ be the non-constant H-invariant monomials of degree at most d. Then $g_1, \dots, g_w \in \mathfrak{n}_k$. For $f \in (u,t)^{dm} \cap k[[u,t]]^H$, any monomial occurring in f is H-invariant of degree at least dm. Writing each such monomial as $\mu_1 \cdots \mu_m \nu$ and grouping the terms according to the product $\mu_1 \cdots \mu_m = g_{l_1} \cdots g_{l_m}$, we obtain

$$f = \sum_{1 \leq l_1 \leq \cdots \leq l_m \leq w} g_{l_1} \cdots g_{l_m} \varphi_{l_1, \dots, l_m}$$

with $\varphi_{l_1,\dots,l_m} \in k[[u,t]]^H$, which are polynomials if f is a polynomial. This proves (14.34).

Let f and f' be H-semi-invariant with $f - f' \in (u,t)^{dm}$. Since the characters of f and f' have order dividing d, the power series f^d and f'^d are H-invariant. Moreover,

$$f^d - f'^d = (f - f') \sum_{a=0}^{d-1} f^a f'^{d-1-a} \in (u,t)^{dm} \cap k[[u,t]]^H,$$

so the last claim follows from (14.34).

Step 2. In this step we express quotient minimal log discrepancies on T_k by $\mathbb{R}$-ideals on Q_k. Let $k \supset \mathbb{C}$ be a field and $\mathfrak{b} = \prod_j \mathfrak{b}_j^{r_j}$ with non-zero ideals $\mathfrak{b}_j \subset k[u,t]^H$ and $r_j > 0$. Let F be a prime divisor over T_k with $\mathrm{center}_{T_k} F = \{0\}$, and E the prime divisor over Q_k such that ord_F restricted to the function field of Q_k is a multiple of ord_E. By Lemma 2.6(2) and Remark 2.17, $\mathrm{center}_{Q_k} E = \{0\}$ and $a(F, T_k, 0, \pi^*\mathfrak{b}) = |I_H(F)| \cdot a(E, Q_k, \Delta_H, \mathfrak{b})$. Since any prime divisor over Q_k arises in this way (Lemma 2.6(3)), we obtain

$$\mathrm{mld}(Q_k \ni 0, \Delta_H, \mathfrak{b}) = \inf_F \frac{a(F, T_k, 0, \pi^*\mathfrak{b})}{|I_H(F)|}, \tag{14.35}$$

where F runs over the prime divisors over T_k with $\mathrm{center}_{T_k} F = \{0\}$. Let $f_1, \dots, f_s \in k[u,t]$ be non-zero H-semi-invariant polynomials and $c_1, \dots, c_s > 0$. Then $f_j^d \in k[u,t]^H$ by **Step 1**, and for

$\mathfrak{b} = \prod_j (f_j^d)^{c_j/d}$ we have $\operatorname{ord}_F(\pi^*\mathfrak{b}) = \sum_j c_j \operatorname{ord}_F f_j$, so $a(F, T_k, 0, \pi^*\mathfrak{b}) = a(F, T_k, \sum_j c_j \operatorname{div}(f_j))$. By (14.35) and the last sentence of Lemma 2.6,

$$\operatorname{mld}\Big(\Big(T_k \ni 0, \sum_j c_j \operatorname{div}(f_j)\Big)/H\Big) = \operatorname{mld}\Big(Q_k \ni 0, \Delta_H, \prod_j (f_j^d)^{c_j/d}\Big). \tag{14.36}$$

Step 3. In this step we compute closed-point minimal log discrepancies on Q_k by a log resolution (cf. [KM98, Corollaries 2.31 and 2.32]). Let $k \supset \mathbb{C}$ be an algebraically closed field, $\mathfrak{a} = \prod_{j=1}^s \mathfrak{a}_j^{r_j}$ with non-zero ideals $\mathfrak{a}_j \subset k[u,t]^H$ and $r_j > 0$, and $g : Y \to Q_k$ a log resolution of Δ_H, $\mathfrak{a}_1, \dots, \mathfrak{a}_s$, and $\mathfrak{n}_k$ [KM98, Notation 0.4(10)]. Thus Y is a smooth surface, $\mathfrak{a}_j\mathcal{O}_Y = \mathcal{O}_Y(-A_j)$ for $1 \le j \le s$ and $\mathfrak{n}_k\mathcal{O}_Y = \mathcal{O}_Y(-A_0)$ for divisors $A_0, \dots, A_s \ge 0$, and $\operatorname{Exc}(g) \cup \operatorname{Supp} g_*^{-1}\Delta_H \cup \bigcup_{j=0}^s \operatorname{Supp} A_j$ is an snc divisor with irreducible components $E_1, \dots, E_m$. Let $a_i := a(E_i, Q_k, \Delta_H, \mathfrak{a})$,

$$I_0 := \{i \mid g(E_i) = \{0\}\}, \quad \text{and} \quad I_1 := \{i \mid E_i \cap g^{-1}(0) \ne \emptyset\}.$$

We show that

$$\operatorname{mld}(Q_k \ni 0, \Delta_H, \mathfrak{a}) = \begin{cases} \min_{i \in I_0} a_i & \text{if } a_i \ge 0 \text{ for any } i \in I_1, \\ -\infty & \text{otherwise.} \end{cases} \tag{14.37}$$

By the definition of the a_i,

$$K_Y + \sum_{i=1}^m (1 - a_i)E_i = g^*(K_{Q_k} + \Delta_H) + \sum_{j=1}^s r_j A_j.$$

Hence

$$a(F, Q_k, \Delta_H, \mathfrak{a}) = a\Big(F, Y, \sum_i (1 - a_i)E_i\Big)$$

for any prime divisor F over Y [KM98, Lemma 2.30]. Since $\mathfrak{n}_k\mathcal{O}_Y = \mathcal{O}_Y(-A_0)$ is invertible and not equal to $\mathcal{O}_Y$, we have $g^{-1}(0) = \operatorname{Supp} A_0 = \bigcup_{i \in I_0} E_i$ and $I_0 \ne \emptyset$.

Assume that $a_i \ge 0$ for any $i \in I_1$. Let F be a prime divisor over Q_k with $\operatorname{center}_{Q_k} F = \{0\}$. Then $\operatorname{center}_Y F \subset g^{-1}(0)$. If $\operatorname{center}_Y F$ is a curve, then $F = E_i$ for some $i \in I_0$. Otherwise $\operatorname{center}_Y F$ is a closed point $p \in g^{-1}(0)$. Then $p \in E_{i_0}$ for some $i_0 \in I_0$, and at most one other component $E_{i'}$ passes through p, in which case $i' \in I_1$. Let x_1, x_2 be regular parameters at p such that $E_{i_0} = (x_1 = 0)$ near p and, if $E_{i'}$ exists, $E_{i'} = (x_2 = 0)$ near p. Let $b := a_{i'}$ if $E_{i'}$ exists and $b := 1$ otherwise. Lemma 2.13 and Remark 2.17, applied on a neighborhood of p on which $\sum_i (1 - a_i)E_i = (1 - a_{i_0}) \operatorname{div}(x_1) + (1 - b) \operatorname{div}(x_2)$, imply

$$a(F, Q_k, \Delta_H, \mathfrak{a}) \ge a_{i_0} \operatorname{ord}_F x_1 + b \operatorname{ord}_F x_2 \ge a_{i_0},$$

since $\operatorname{ord}_F x_1 \ge 1$ and $a_{i_0}, b \ge 0$. Thus $\operatorname{mld}(Q_k \ni 0, \Delta_H, \mathfrak{a}) \ge \min_{i \in I_0} a_i$. The reverse inequality holds since E_i is a prime divisor over Q_k with center 0 for any $i \in I_0$.

Assume that $a_{i_1} < 0$ for some $i_1 \in I_1$. Pick $p_0 \in E_{i_1} \cap g^{-1}(0)$. Let G_1 be the exceptional curve of the blow-up of Y at p_0, and for $n \ge 1$ let G_{n+1} be the exceptional curve of the blow-up at the intersection point p_n of G_n and the strict transform of E_{i_1}. No other component of the strict transform of $\sum_i E_i + \sum_{n' < n} G_{n'}$ passes through p_n, so [KM98, Lemmas 2.29 and 2.30] imply

$$a(G_{n+1}, Q_k, \Delta_H, \mathfrak{a}) = a(G_n, Q_k, \Delta_H, \mathfrak{a}) + a_{i_1}$$

for any $n \ge 1$. Since the center of any G_n on Q_k is 0, we obtain $\operatorname{mld}(Q_k \ni 0, \Delta_H, \mathfrak{a}) = -\infty$. This proves (14.37).

Step 4. In this step we find the constant ℓ' for ideals on Q. Let Γ be the finite set consisting of 1, the positive coefficients of Δ_H, and the elements of Λ'. Let $X := Q \times \mathbb{A}^1_\tau$, $\eta := (0,0) \in X$, and $B^+ := \operatorname{pr}_1^* \Delta_H + (\tau = 0)$. Then X is a normal threefold, and (X, B^+) is a pair with $B^+ \in \Gamma$, since $K_Q + \Delta_H$ is $\mathbb{R}$-Cartier. Let ℓ' be the positive integer l of Theorem 2.19(2) for X and Γ. We show that, for any non-zero ideals $\mathfrak{a}_j, \mathfrak{b}_j \subset \mathbb{C}[u,t]^H$ with

$$\mathfrak{a}_j + \mathfrak{n}^{\ell'} = \mathfrak{b}_j + \mathfrak{n}^{\ell'} \quad (1 \le j \le s)$$

and any $r_1, \dots, r_s \in \Lambda'$, the $\mathbb{R}$-ideals $\mathfrak{a} := \prod_j \mathfrak{a}_j^{r_j}$ and $\mathfrak{b} := \prod_j \mathfrak{b}_j^{r_j}$ satisfy

$$\operatorname{mld}(Q \ni 0, \Delta_H, \mathfrak{a}) = \operatorname{mld}(Q \ni 0, \Delta_H, \mathfrak{b}). \tag{14.38}$$

We first prove (14.40) below, where $\operatorname{pr}_1^* \mathfrak{a} := \prod_j (\mathfrak{a}_j \mathcal{O}_X)^{r_j}$. We use Lemma 8.1 for (Q, Δ_H) with the trivial group, where τ is the coordinate of $\mathbb{A}^1$. Let F be a prime divisor over X with $\operatorname{center}_X F = \{\eta\}$. By Lemma 8.1(1), there exist a positive integer ϱ and a prime divisor E over Q with $\operatorname{center}_Q E = \{0\}$ such that $\operatorname{ord}_F|_{\mathbb{C}(Q)} = \varrho \operatorname{ord}_E$ and $a(F, X, B^+) \geq \varrho a(E, Q, \Delta_H)$. Since $\mathfrak{a}_j$ is generated by regular functions on Q, $\operatorname{mult}_F(\mathfrak{a}_j \mathcal{O}_X) = \varrho \operatorname{mult}_E(\mathfrak{a}_j)$. Hence

$$a(F, X, B^+, \operatorname{pr}_1^* \mathfrak{a}) \geq \varrho a(E, Q, \Delta_H, \mathfrak{a}). \tag{14.39}$$

Conversely, let E be a prime divisor over Q with $\operatorname{center}_Q E = \{0\}$. By Lemma 8.1(2), where $q = 1$ since the group is trivial, the Gauss extension of ord_E with τ-value 1 is ord_F for a prime divisor F over X with $\operatorname{center}_X F = \{\eta\}$ and $a(F, X, B^+) = a(E, Q, \Delta_H)$. Since $\operatorname{ord}_F|_{\mathbb{C}(Q)} = \operatorname{ord}_E$, we obtain $a(F, X, B^+, \operatorname{pr}_1^* \mathfrak{a}) = a(E, Q, \Delta_H, \mathfrak{a})$. Thus $\operatorname{mld}(X \ni \eta, B^+, \operatorname{pr}_1^* \mathfrak{a}) \leq \operatorname{mld}(Q \ni 0, \Delta_H, \mathfrak{a})$. By **Step 3** with $k = \mathbb{C}$, $\operatorname{mld}(Q \ni 0, \Delta_H, \mathfrak{a})$ is either non-negative or $-\infty$. In the first case, $a(E, Q, \Delta_H, \mathfrak{a}) \geq 0$ for any E as in (14.39), so

$$\varrho a(E, Q, \Delta_H, \mathfrak{a}) \geq a(E, Q, \Delta_H, \mathfrak{a}) \geq \operatorname{mld}(Q \ni 0, \Delta_H, \mathfrak{a}),$$

and (14.39) implies the reverse inequality. In the second case, both sides are $-\infty$. Thus

$$\operatorname{mld}(X \ni \eta, B^+, \operatorname{pr}_1^* \mathfrak{a}) = \operatorname{mld}(Q \ni 0, \Delta_H, \mathfrak{a}), \tag{14.40}$$

and (14.40) also holds with $\mathfrak{b}$ in place of $\mathfrak{a}$.

Since $\mathfrak{n}\mathcal{O}_{X,\eta} \subset \mathfrak{m}_\eta$, we have $\mathfrak{a}_j \mathcal{O}_{X,\eta} + \mathfrak{m}_\eta^{\ell'} = \mathfrak{b}_j \mathcal{O}_{X,\eta} + \mathfrak{m}_\eta^{\ell'}$ for any j. As $B^+ \in \Gamma$ and $r_j \in \Gamma$, Theorem 2.19(2) implies

$$\operatorname{mld}(X \ni \eta, B^+, \operatorname{pr}_1^* \mathfrak{a}) = \operatorname{mld}(X \ni \eta, B^+, \operatorname{pr}_1^* \mathfrak{b}).$$

Together with (14.40) for $\mathfrak{a}$ and $\mathfrak{b}$, this proves (14.38).

Step 5. In this step we compare minimal log discrepancies over an algebraically closed field $k \supset \mathbb{C}$ with those of specializations to closed points. Let $R \subset k$ be a finitely generated $\mathbb{C}$-subalgebra, $V := \operatorname{Spec} R$, $r_1, \dots, r_s$ positive real numbers, and $\mathfrak{a}_1, \dots, \mathfrak{a}_s$ ideals of $R[u,t]^H$ such that $\mathfrak{a}_{j,k} := \mathfrak{a}_j k[u,t]^H \neq 0$ for any j. For a closed point $z \in V$, let $\mathfrak{a}_j(z) \subset \mathbb{C}[u,t]^H$ be the image of $\mathfrak{a}_j$ under the homomorphism $R[u,t]^H \to \mathbb{C}[u,t]^H$ induced by z. We show that there exists a non-empty open subset $U \subset V$ such that, for any closed point $z \in U$, we have $\mathfrak{a}_j(z) \neq 0$ for any j and

$$\operatorname{mld}\Big(Q \ni 0, \Delta_H, \prod_j \mathfrak{a}_j(z)^{r_j}\Big) = \operatorname{mld}\Big(Q_k \ni 0, \Delta_H, \prod_j \mathfrak{a}_{j,k}^{r_j}\Big). \tag{14.41}$$

Let $g : Y \to Q_k$, $A_0, \dots, A_s$, $E_1, \dots, E_m$, $a_1, \dots, a_m$, I_0, and I_1 be as in **Step 3** for the ideals $\mathfrak{a}_{j,k}$ and the exponents r_j. By [Gro66, Théorème 8.8.2, Proposition 8.9.1], there exist a finitely generated $\mathbb{C}$-subalgebra $R' \subset k$ containing R, a projective birational morphism $\mathcal{G} : \mathcal{Y} \to Q \times V'$ with $V' := \operatorname{Spec} R'$, and divisors $\mathcal{A}_0, \dots, \mathcal{A}_s$ and $\mathcal{E}_1, \dots, \mathcal{E}_m$ on $\mathcal{Y}$, whose base changes by $\operatorname{Spec} k \to V'$ are g, the A_j, and the E_i, such that $\mathfrak{a}_j \mathcal{O}_{\mathcal{Y}} = \mathcal{O}_{\mathcal{Y}}(-\mathcal{A}_j)$ for $j \geq 1$, $\mathfrak{n}\mathcal{O}_{\mathcal{Y}} = \mathcal{O}_{\mathcal{Y}}(-\mathcal{A}_0)$, and

$$K_{\mathcal{Y}/V'} + \sum_{i=1}^m (1 - a_i)\mathcal{E}_i = \mathcal{G}^*\big(K_{(Q \times V')/V'} + \Delta_H \times V'\big) + \sum_{j=1}^s r_j \mathcal{A}_j.$$

By [Gro66, Théorème 8.10.5] and generic smoothness [Har77, Corollary III.10.7], after replacing V' by a non-empty open subset, we may assume the following: $\mathcal{Y}$, any $\mathcal{E}_i$, and any intersection of two distinct $\mathcal{E}_i$ are smooth over V', any intersection of three distinct $\mathcal{E}_i$ is empty, $\mathcal{G}$ is an isomorphism over the complement of $\mathcal{G}(\bigcup_i \mathcal{E}_i)$, $\mathcal{E}_i \subset \mathcal{G}^{-1}(0 \times V')$ for $i \in I_0$, $\mathcal{E}_i \cap \mathcal{G}^{-1}(0 \times V')$ is empty for $i \notin I_1$, and $\mathcal{E}_i \cap \mathcal{G}^{-1}(0 \times V')$ maps surjectively onto V' for $i \in I_1$.

Let $z' \in V'$ be a closed point and $g_{z'} : Y_{z'} \to Q$ the fiber of $\mathcal{G}$ over z'. Since the divisors, ideals, and relative canonical divisors above commute with restriction to the fiber over z', $g_{z'}$ is

a log resolution of Δ_H, $\mathfrak{a}_1(z'),\dots,\mathfrak{a}_s(z')$, and $\mathfrak{n}$, where $\mathfrak{a}_j(z')\subset\mathbb{C}[u,t]^H$ is the image of $\mathfrak{a}_j$ under the homomorphism $R[u,t]^H\to\mathbb{C}[u,t]^H$ induced by $R\subset R'$ and z'. Moreover, any irreducible component of $\mathcal{E}_i|_{Y_{z'}}$ has log discrepancy a_i with respect to $(Q,\Delta_H,\prod_j\mathfrak{a}_j(z')^{r_j})$, the irreducible components of $g_{z'}^{-1}(0)$ are the irreducible components of $\mathcal{E}_i|_{Y_{z'}}$ for $i\in I_0$, and for any $i\in I_1$, at least one irreducible component of $\mathcal{E}_i|_{Y_{z'}}$ and no irreducible component of $\mathcal{E}_{i''}|_{Y_{z'}}$ with $i''\notin I_1$ intersects $g_{z'}^{-1}(0)$. In particular, $\mathfrak{a}_j(z')\mathcal{O}_{Y_{z'}}$ is invertible, so $\mathfrak{a}_j(z')\neq 0$. Applying (14.37) to g and to $g_{z'}$, we obtain (14.41) with z' in place of z.

By Chevalley's theorem, the image of $V'\to V$ is constructible. It contains the generic point of V, so it contains a non-empty open subset $U\subset V$. For a closed point $z\in U$, the fiber of $V'\to V$ over z is a non-empty scheme of finite type over $\mathbb{C}$, so it contains a closed point z' of V', and $\mathfrak{a}_j(z')=\mathfrak{a}_j(z)$ for any j. Thus U satisfies the claim.

Step 6. In this step we show that ℓ' works over any algebraically closed field $k\supset\mathbb{C}$. Let $\mathfrak{a}_j,\mathfrak{b}_j\subset k[u,t]^H$, $1\le j\le s$, be non-zero ideals with $\mathfrak{a}_j+\mathfrak{n}_k^{\ell'}=\mathfrak{b}_j+\mathfrak{n}_k^{\ell'}$, and $r_1,\dots,r_s\in\Lambda'$. We show that

$$\operatorname{mld}\Big(Q_k\ni 0,\Delta_H,\prod_j\mathfrak{a}_j^{r_j}\Big)=\operatorname{mld}\Big(Q_k\ni 0,\Delta_H,\prod_j\mathfrak{b}_j^{r_j}\Big). \tag{14.42}$$

Fix finitely many generators of the $\mathfrak{a}_j$ and the $\mathfrak{b}_j$. Since $\mathfrak{n}_k^{\ell'}$ is generated by the products of ℓ' non-constant H-invariant monomials, any generator of $\mathfrak{a}_j$ is a $k[u,t]^H$-linear combination of the generators of $\mathfrak{b}_j$ and of finitely many such products, and conversely. Let $R\subset k$ be the $\mathbb{C}$-subalgebra generated by the coefficients of the generators and of the polynomials appearing in the linear combinations above, $V:=\operatorname{Spec}R$, and $\mathfrak{a}_{j,R},\mathfrak{b}_{j,R}\subset R[u,t]^H$ the ideals generated by the generators of $\mathfrak{a}_j$ and $\mathfrak{b}_j$. Then $\mathfrak{a}_{j,R}k[u,t]^H=\mathfrak{a}_j$, $\mathfrak{b}_{j,R}k[u,t]^H=\mathfrak{b}_j$, and for any closed point $z\in V$, the linear combinations specialize, so

$$\mathfrak{a}_{j,R}(z)+\mathfrak{n}^{\ell'}=\mathfrak{b}_{j,R}(z)+\mathfrak{n}^{\ell'}.$$

Since R is a domain, V is irreducible. Hence **Step 5**, applied to the $\mathfrak{a}_{j,R}$ and to the $\mathfrak{b}_{j,R}$, implies that there exists a closed point $z\in V$ such that $\mathfrak{a}_{j,R}(z)$ and $\mathfrak{b}_{j,R}(z)$ are non-zero and (14.41) holds for both. Combining (14.41) with (14.38) for the ideals $\mathfrak{a}_{j,R}(z)$ and $\mathfrak{b}_{j,R}(z)$, we obtain (14.42).

Step 7. In this step we treat power series and prove (1) with $\ell:=d\ell'$. Let $k\supset\mathbb{C}$ be an algebraically closed field, $\widehat{T}_k:=\operatorname{Spec}k[[u,t]]$ with closed point $\hat{0}$, $f_1,\dots,f_s\in k[[u,t]]$ non-zero H-semi-invariant power series, and $c_1,\dots,c_s\in\Lambda$. Let

$$\hat{\mu}(f):=\operatorname{mld}\Big(\Big(\widehat{T}_k\ni\hat{0},\sum_j c_j\operatorname{div}(f_j)\Big)/H\Big)$$

be as in Definition 3.2, and for $n\ge 1$ let

$$\mathfrak{b}_n(f):=\prod_j\big(((f_j^{<dn})^d)+\mathfrak{n}_k^n\big)^{c_j/d}\quad\text{and}\quad m_n(f):=\operatorname{mld}(Q_k\ni 0,\Delta_H,\mathfrak{b}_n(f)).$$

Here $(f_j^{<dn})^d\in k[u,t]^H$ by **Step 1**, and the ideals in $\mathfrak{b}_n(f)$ are non-zero since they contain $\mathfrak{n}_k^n$.

Let $f_1',\dots,f_s'\in k[[u,t]]$ be non-zero H-semi-invariant power series with $f_j-f_j'\in(u,t)^{d\ell'}$, and $n,n'\ge\ell'$. We show that $m_n(f)=m_{n'}(f')$. The parts of degree less than $d\ell'$ of f_j and f_j' coincide, so $f_j^{<dn}-f_j'^{<dn'}\in(u,t)^{d\ell'}$. By **Step 1**, $(f_j^{<dn})^d-(f_j'^{<dn'})^d\in\mathfrak{n}_k^{\ell'}$. Since $n,n'\ge\ell'$, we obtain

$$((f_j^{<dn})^d)+\mathfrak{n}_k^n+\mathfrak{n}_k^{\ell'}=((f_j^{<dn})^d)+\mathfrak{n}_k^{\ell'}=((f_j'^{<dn'})^d)+\mathfrak{n}_k^{\ell'}=((f_j'^{<dn'})^d)+\mathfrak{n}_k^{n'}+\mathfrak{n}_k^{\ell'}.$$

As $c_j/d\in\Lambda'$, **Step 6** implies $m_n(f)=m_{n'}(f')$. In particular, $m_n(f)=m_{\ell'}(f)$ for any $n\ge\ell'$.

We show that $\hat{\mu}(f)=m_{\ell'}(f)$. By Lemma 3.3(3), applied to the germ $(T_k\ni 0)$ over k with $B=0$ and the group H, the restriction of valuations from the fraction field of $k[[u,t]]$ to $k(u,t)$

sends any formal prime divisor P over $\widehat{T}_k$ with center $\hat{0}$ to ord_{E_P} for a prime divisor E_P over T_k with $\mathrm{center}_{T_k} E_P = \{0\}$, such that the residue fields of ord_P and ord_{E_P} are isomorphic, $a(P, \widehat{T}_k, 0) = a(E_P, T_k, 0)$, $\mathrm{Stab}_H(P) = \mathrm{Stab}_H(E_P)$, and $I_H(P) = I_H(E_P)$. Moreover, by the last part of Lemma 3.3(3), any prime divisor over T_k with center 0 is E_P for some P (cf. [dFEM11, Lemma 2.21 and Appendix A]). Let P be such a formal prime divisor, and let $\mathrm{mult}_{E_P}(\mathfrak{n}_k) \geq 1$ denote the order of $\mathfrak{n}_k \mathcal{O}_{T_k}$ along E_P. By **Step 1**, $f_j^d - (f_j^{<dn})^d \in \mathfrak{n}_k^n k[[u,t]]^H$, so $\mathrm{ord}_P(f_j^d - (f_j^{<dn})^d) \geq n \,\mathrm{mult}_{E_P}(\mathfrak{n}_k)$. Therefore

$$\mathrm{mult}_{E_P}\left(((f_j^{<dn})^d) + \mathfrak{n}_k^n\right) = \min\{d \,\mathrm{ord}_P f_j,\ n \,\mathrm{mult}_{E_P}(\mathfrak{n}_k)\} \leq d \,\mathrm{ord}_P f_j,$$

with equality if $n \geq d \,\mathrm{ord}_P f_j$. Since

$$a\Big(P, \widehat{T}_k, \sum_j c_j \,\mathrm{div}(f_j)\Big) = a(P, \widehat{T}_k, 0) - \sum_j c_j \,\mathrm{ord}_P f_j,$$

we obtain

$$a\Big(P, \widehat{T}_k, \sum_j c_j \,\mathrm{div}(f_j)\Big) \leq a(E_P, T_k, 0, \pi^* \mathfrak{b}_n(f))$$

for any n, with equality if $n \geq d \max_j \mathrm{ord}_P f_j$. Dividing by $|I_H(P)| = |I_H(E_P)|$ and taking the infimum over P, (14.35) implies $\hat{\mu}(f) \leq m_n(f)$ for any n. On the other hand, for any P and any $n \geq \max\{\ell', d \max_j \mathrm{ord}_P f_j\}$, (14.35) implies

$$\frac{a(P, \widehat{T}_k, \sum_j c_j \,\mathrm{div}(f_j))}{|I_H(P)|} = \frac{a(E_P, T_k, 0, \pi^* \mathfrak{b}_n(f))}{|I_H(E_P)|} \geq m_n(f) = m_{\ell'}(f).$$

Thus $\hat{\mu}(f) = m_{\ell'}(f)$.

Applying $\hat{\mu}(f) = m_{\ell'}(f)$ to f and to f', and $m_{\ell'}(f) = m_{\ell'}(f')$, we obtain $\hat{\mu}(f) = \hat{\mu}(f')$ whenever $f_j - f_j' \in (u,t)^{d\ell'}$ for any j. This proves (1) with $\ell = d\ell'$.

Finally, assume that $f_1, \ldots, f_s$ are polynomials, and let $n \geq \ell'$ with $dn > \deg f_j$ for any j. Then $f_j^{<dn} = f_j$, and $((f_j^d) + \mathfrak{n}_k^n) + \mathfrak{n}_k^{\ell'} = (f_j^d) + \mathfrak{n}_k^{\ell'}$. By **Step 6** and (14.36),

$$\hat{\mu}(f) = m_n(f) = \mathrm{mld}\Big(Q_k \ni 0, \Delta_H, \prod_j (f_j^d)^{c_j/d}\Big) = \mathrm{mld}\Big(\Big(T_k \ni 0, \sum_j c_j \,\mathrm{div}(f_j)\Big)/H\Big). \tag{14.43}$$

Step 8. In this step we prove (2). Let N be the integer of Construction 14.13, $\ell = d\ell'$ as in **Step 7**, and $D := \max\{\ell, N+1\}$. For any i, let $\phi_i \in \mathcal{J}_D(\mathbb{C})$ be the tuple $(f_{i,j}^{<D})_j$, and let $\zeta \in \mathcal{J}_D(k)$ be the tuple $(f_{\infty,j}^{<D})_j$. By Construction 14.13, ζ maps to the generic point of Z_D, and any non-empty open subset of Z_D contains ϕ_i for all but finitely many i. Since $\mathrm{ord}_0 f_{i,j} \leq N < D$, the j-th component of ϕ_i is non-zero. Hence Z_D is not contained in the closed subset of $\mathcal{J}_D$ where the j-th component vanishes, and $f_{\infty,j}^{<D} \neq 0$.

Since $f_{\infty,j} - f_{\infty,j}^{<D} \in (u,t)^\ell$ and $f_{\infty,j}^{<D}$ is a non-zero H-semi-invariant polynomial, (1) and (14.43) imply

$$\mathrm{mld}\Big(\Big(\mathrm{Spec}\, k[[u,t]] \ni \hat{0}, \sum_j c_j \,\mathrm{div}(f_{\infty,j})\Big)/H\Big) = \mathrm{mld}\Big(\Big(T_k \ni 0, \sum_j c_j \,\mathrm{div}(f_{\infty,j}^{<D})\Big)/H\Big).$$

Similarly, (1) with $k = \mathbb{C}$ and (14.43) applied to $(f_{i,j})_j$ and $(f_{i,j}^{<D})_j$ imply

$$\mathrm{mld}\Big(\Big(T \ni 0, \sum_j c_j \,\mathrm{div}(f_{i,j})\Big)/H\Big) = \mathrm{mld}\Big(\Big(T \ni 0, \sum_j c_j \,\mathrm{div}(f_{i,j}^{<D})\Big)/H\Big)$$

for any i.

Let $R := \mathbb{C}[Z_D]$. Since ζ maps to the generic point of Z_D, the homomorphism $\mathbb{C}[\mathcal{J}_D] \to k$ given by ζ induces an injection $R \subset k$. Let $\mathbf{F}_j \in R[u,t]$ be the polynomial whose coefficients are the restrictions to Z_D of the coordinate functions of the j-th component, so that $\mathbf{F}_j$ maps to $f_{\infty,j}^{<D}$ under $R[u,t] \subset k[u,t]$, and $\mathbf{F}_j$ evaluated at a closed point $z \in Z_D$ is the j-th component

of z. Since $R[u,t] \subset k[u,t]$ and $f_{\infty,j}^{<D}$ is H-semi-invariant, so is $\mathbf{F}_j$, and $\mathbf{F}_j^d \in R[u,t]^H$. Applying **Step 5** to the ideals $(\mathbf{F}_j^d) \subset R[u,t]^H$, whose extensions $((f_{\infty,j}^{<D})^d)$ to $k[u,t]^H$ are non-zero, and to the exponents c_j/d, we obtain a non-empty open subset $U \subset Z_D$ such that, for any closed point $z \in U$, the j-th component $\mathbf{F}_j(z)$ of z is non-zero for any j and

$$\operatorname{mld}\Big(Q \ni 0, \Delta_H, \prod_j (\mathbf{F}_j(z)^d)^{c_j/d}\Big) = \operatorname{mld}\Big(Q_k \ni 0, \Delta_H, \prod_j ((f_{\infty,j}^{<D})^d)^{c_j/d}\Big).$$

By (14.36) for $\mathbb{C}$ and for k, the left-hand side and the right-hand side are

$$\operatorname{mld}\Big(\Big(T \ni 0, \sum_j c_j \operatorname{div}(\mathbf{F}_j(z))\Big)/H\Big) \quad \text{and} \quad \operatorname{mld}\Big(\Big(T_k \ni 0, \sum_j c_j \operatorname{div}(f_{\infty,j}^{<D})\Big)/H\Big),$$

respectively. Since $\phi_i \in U$ for all but finitely many i, combining the three equalities above with $z = \phi_i$, we obtain

$$\operatorname{mld}\Big(\Big(T \ni 0, \sum_j c_j \operatorname{div}(f_{i,j})\Big)/H\Big) = \operatorname{mld}\Big(\Big(\operatorname{Spec} k[[u,t]] \ni \hat{0}, \sum_j c_j \operatorname{div}(f_{\infty,j})\Big)/H\Big)$$

for all but finitely many i. In particular, (2) holds. □

14.6. **A lower bound under a uniform margin.** In this subsection, we prove Proposition 14.15. It is the lower bound argument for Theorem 14.2, which passes to a field extension of degree p, takes finite quotients, and uses inversion of adjunction, carried out under a uniform lc margin λ. We use it in the proof of Theorem 14.19 when the generic limit of the residual plane pairs is klt.

Proposition 14.15. *Let P and e_0 be positive integers, $\Gamma \subset [0,1]$ a DCC set with $\Gamma\backslash\{0\} \neq \emptyset$, and $\lambda > 1$. Then there exist r_0 and N_0, depending only on λ, P, e_0, and Γ, such that the following holds. Notation as in Set-up 14.1 for these P, e_0, and Γ. Assume that h_j^0 is the sum of all terms of h_j of $\mathbf{T}$-weight $pa_j - qb_j$ for any j, that $G \cap \mathbf{T} = \mu_r(p,-q,0)$, $|H| \le e_0$, $r \ge r_0$, and $N \ge N_0$, and that $(X, \lambda B^0)$, $(X, \lambda B)$, and $(T, \lambda B_T)$ are lc near 0. Then $\operatorname{mld}((X \ni 0, B)/G) \ge b_H$.*

Proof. The proof is long, so we divide it into several steps.

Let $C_1 := \frac{3\lambda}{\lambda - 1}$ and $C_2 := \frac{C_1}{\lambda\gamma}$, where $\gamma = \gamma(\Gamma)$, let ℓ be the integer of Construction-Proposition 14.7 for Γ, and let

$$r_0 := \Big\lceil 2P^2 e_0 C_2 + \frac{3P}{\gamma} \Big\rceil \quad \text{and} \quad N_0 := \max\Big\{\ell, \Big\lfloor \frac{3}{\gamma} \Big\rfloor + 1, \lceil 2Pe_0C_2 \rceil\Big\}.$$

Then r_0 and N_0 depend only on λ, P, e_0, and Γ. Assume that the data of Set-up 14.1 satisfy the assumptions of the proposition. Since $G \cap \mathbf{T} = \mu_r(p,-q,0)$, $|H| \le e_0$, $N \ge N_0 \ge \ell$, and $N \ge N_0 > 3/\gamma$, the assumptions of Proposition 14.9 hold. By Proposition 14.9, the kernel of $G \to H$ is $G \cap \mathbf{T}$. Let $\pi : X \to X/G$ be the quotient morphism, and write $(X,B)/G = (X/G, B_{X/G})$.

Since $c_x \ge 0$ and $q \ge 1$, we have $c_u \ge 0$, so $B_T \ge 0$. Let $V \ni 0$ be an open subset of T such that $(V, \lambda B_T|_V)$ is lc. For any prime divisor ν over T whose center intersects V, we have $a(\nu, T, 0) \ge 1$ as T is smooth, so

$$a(\nu, T, B_T) = \frac{\lambda - 1}{\lambda} a(\nu, T, 0) + \frac{1}{\lambda} a(\nu, T, \lambda B_T) \ge \frac{\lambda - 1}{\lambda} > 0.$$

Thus (T, B_T) is klt near 0. In particular, $b_H = \operatorname{mld}((T \ni 0, B_T)/H)$ is defined.

Step 1. In this step we fix a prime divisor E over X/G with $\operatorname{center}_{X/G} E = \{\pi(0)\}$ and $a(E, X/G, B_{X/G}) \le 3$, and we bound the orders of h_j along a prime divisor over X corresponding to E.

By Lemma 2.6(1)–(3), there exists a prime divisor F over X such that $\operatorname{ord}_F|_{\mathbb{C}(X/G)} = e \operatorname{ord}_E$ with $e = |I_G(F)|$, $\pi(\operatorname{center}_X F) = \{\pi(0)\}$, and $a(F, X, B) = ea(E, X/G, B_{X/G}) \le 3e$. Since G fixes 0, $\pi^{-1}(\pi(0)) = \{0\}$, so $\operatorname{center}_X F = \{0\}$. Since $(X, \lambda B)$ is lc near 0,

$$0 \le a(F, X, \lambda B) = \lambda a(F, X, B) - (\lambda - 1) a(F, X, 0) \le 3\lambda e - (\lambda - 1) a(F, X, 0).$$

Moreover, $a(F,X,\lambda B) = a(F,X,0) - \lambda\sum_j c_j \operatorname{ord}_F h_j \geq 0$, $\operatorname{ord}_F h_j \geq 0$ for any j, and $c_j \geq \gamma$. Hence

$$a(F,X,0) \leq C_1 e \quad \text{and} \quad \operatorname{ord}_F h_j \leq \frac{a(F,X,0)}{\lambda c_j} \leq C_2 e \quad \text{for any } j. \tag{14.44}$$

By Proposition 14.8(1), $a_j + b_j \leq \operatorname{mult}_0 h_j^0 = \operatorname{mult}_0 h_j < 3/\gamma$ for any j. Let $\kappa_j := pa_j - qb_j$. Then κ_j is the $\mathbf{T}$-weight of h_j^0 (Definition 2.16), and since $P \geq \max\{p,q\}$,

$$|\kappa_j| \leq P(a_j + b_j) < \frac{3P}{\gamma}. \tag{14.45}$$

The generator $(\zeta^p, \zeta^{-q}, 1)$ of $\mu_r(p,-q,0) \subset G$, $\zeta = e^{2\pi\sqrt{-1}/r}$, multiplies a monomial $x^k y^l t^m$ by ζ^{pk-ql}. Since h_j is G-semi-invariant with the character of h_j^0, any monomial $x^k y^l t^m$ occurring in h_j satisfies

$$pk - ql = \kappa_j + nr \quad \text{for some } n \in \mathbb{Z}, \tag{14.46}$$

and by assumption, the terms of h_j with $n = 0$ are exactly the terms of h_j^0.

Step 2. In this step we construct G-semi-invariant polynomials h_j^+ with $\operatorname{ord}_F h_j^+ = \operatorname{ord}_F h_j$ such that any monomial occurring in h_j^+ satisfies (14.46) with $n \geq 0$.

Since $\operatorname{center}_X F = \{0\}$ and u and t vanish at 0, the restriction of ord_F to $\mathbb{C}(u,t)$ is non-trivial, non-negative on $\mathcal{O}_{T,0}$, and positive on the maximal ideal of $\mathcal{O}_{T,0}$. As $\mathbb{C}(X)$ has transcendence degree one over $\mathbb{C}(u,t)$, by Lemma 2.12 we may write $\operatorname{ord}_F|_{\mathbb{C}(u,t)} = \varrho \operatorname{ord}_\nu$ for a positive integer ϱ and a prime divisor ν over T with $\operatorname{center}_T \nu = \{0\}$.

We show that

$$\frac{\varrho}{e} \geq \frac{1}{e_0}. \tag{14.47}$$

Let $I_0 := I_G(F) \cap G \cap \mathbf{T}$. By Lemma 2.6(1), $I_G(F)$ is cyclic and acts faithfully on $\mathfrak{m}_F/\mathfrak{m}_F^2$. Since $I_G(F)$ acts trivially on the residue field of ord_F, it acts on the one-dimensional space $\mathfrak{m}_F/\mathfrak{m}_F^2$ through an injective character χ, and on $\mathfrak{m}_F^\varrho/\mathfrak{m}_F^{\varrho+1}$ through χ^ϱ. Pick $w_0 \in \mathbb{C}(u,t)$ with $\operatorname{ord}_\nu w_0 = 1$. Then $\operatorname{ord}_F w_0 = \varrho$, and w_0 is fixed by I_0 because $G \cap \mathbf{T}$ acts trivially on u and t. Thus $\chi(\sigma)^\varrho = 1$, that is, $\sigma^\varrho = 1$, for any $\sigma \in I_0$. Since I_0 is cyclic, $|I_0|$ divides ϱ. Since the kernel of $G \to H$ is $G \cap \mathbf{T}$, $I_G(F)/I_0$ is isomorphic to a subgroup of H. Therefore $e = |I_G(F)| \leq \varrho|H| \leq \varrho e_0$, which is (14.47).

Case 1: $\operatorname{ord}_F h_j^0 = \operatorname{ord}_F h_j$ *for any* j. Let $h_j^+ := h_j^0$.

Case 2: $\operatorname{ord}_F h_{j_0}^0 \neq \operatorname{ord}_F h_{j_0}$ *for some* j_0. Then

$$\operatorname{ord}_F R_{j_0} = \min\{\operatorname{ord}_F h_{j_0}, \operatorname{ord}_F h_{j_0}^0\} \leq C_2 e$$

by (14.44). Since $R_{j_0} \in (x,y)^N$, we have $\operatorname{ord}_F R_{j_0} \geq N \min\{\operatorname{ord}_F x, \operatorname{ord}_F y\}$, so $\min\{\operatorname{ord}_F x, \operatorname{ord}_F y\} \leq C_2 e/N$. Possibly interchanging (x,p,a_j) and (y,q,b_j), which preserves u, c_u, B_T, $\mathbf{T}$, and $G \cap \mathbf{T} = \mu_r(p,-q,0)$ and changes the sign of the $\mathbf{T}$-weights, we may assume that $\operatorname{ord}_F x \leq C_2 e/N$. Whether we interchange depends on F. From now on F is fixed, and the constructions below, namely the terms of h_j that we delete and the coordinate whose p-th root we adjoin in **Step 3**, are made for the fixed F. Since $u = x^q y^p$, $\operatorname{ord}_\nu u \geq 1$, $q \leq P$, and $N \geq 2Pe_0C_2$, (14.47) implies

$$p \operatorname{ord}_F y = \varrho \operatorname{ord}_\nu u - q \operatorname{ord}_F x \geq \frac{e}{e_0} - \frac{PC_2 e}{N} \geq \frac{e}{2e_0},$$

so $\operatorname{ord}_F y \geq \frac{e}{2Pe_0}$. Let h_j^- be the sum of the terms of h_j with $n < 0$ in (14.46), and let $h_j^+ := h_j - h_j^-$. If $x^k y^l t^m$ occurs in h_j^-, then $pk - ql \leq \kappa_j - r$, so by (14.45) and $r \geq r_0$,

$$l \geq \frac{r - \kappa_j}{q} > \frac{r - 3P/\gamma}{P} \geq 2Pe_0C_2.$$

Hence $\operatorname{ord}_F(x^k y^l t^m) \geq l \operatorname{ord}_F y > C_2 e \geq \operatorname{ord}_F h_j$ by (14.44). Thus $\operatorname{ord}_F h_j^- > \operatorname{ord}_F h_j$, and $\operatorname{ord}_F h_j^+ = \operatorname{ord}_F h_j$.

In both cases, h_j^+ is a non-zero polynomial whose terms are terms of h_j, so h_j^+ is G-semi-invariant with the character of h_j. Moreover, $\operatorname{ord}_F h_j^+ = \operatorname{ord}_F h_j$, any monomial occurring in h_j^+ satisfies (14.46) with $n \geq 0$, and the terms of h_j^+ with $n = 0$ are exactly the terms of h_j^0. Let $B^+ := \sum_j c_j \operatorname{div}(h_j^+)$. Then B^+ is G-invariant and

$$a(F, X, B^+) = a(F, X, B). \tag{14.48}$$

Step 3. In this step we pass to a field extension of degree p and construct the group $\tilde{H}$, the polynomials g_j, and the plane pair $(T', B_{T'})$.

Let $X'' := \mathbb{A}^3_{z,y_1,t}$, and embed $\mathbb{C}(X)$ into $\mathbb{C}(X'')$ by $x = z^p$, $y = z^{-q} y_1$. Then $u = y_1^p$, $\mathbb{C}(X'') = \mathbb{C}(X)(z)$, and $[\mathbb{C}(X'') : \mathbb{C}(X)] = p$. Let

$$\tau : (\mathbb{C}^*)^3 \to (\mathbb{C}^*)^3, \quad (\xi_1, \xi_2, \xi_3) \to (\xi_1^p, \xi_1^{-q} \xi_2, \xi_3).$$

For the diagonal actions on X'' and X, the embedding $\mathbb{C}(X) \subset \mathbb{C}(X'')$ is τ-equivariant. The homomorphism τ is surjective with kernel $\mu_p(1, q, 0)$. Let $\tilde{G} := \tau^{-1}(G)$. Then $\tilde{G}$ is a finite diagonal abelian group of order $p|G|$ acting faithfully on X''. Since $\mu_p(1, q, 0)$ fixes x, y, and t and acts faithfully on $\mathbb{C}(X'')$, we have $\mathbb{C}(X'')^{\mu_p(1,q,0)} = \mathbb{C}(X)$, so $\mathbb{C}(X'')^{\tilde{G}} = \mathbb{C}(X)^G$.

For $\xi^r = 1$, $\tau(\xi, 1, 1) = (\xi^p, \xi^{-q}, 1) \in \mu_r(p, -q, 0)$, and $\xi^p = \xi^{-q} = 1$ implies $\xi = 1$ as $\gcd(p, q) = 1$. Thus $\mu_r(1, 0, 0) \subset \tilde{G}$, and τ maps $\mu_r(1, 0, 0)$ isomorphically onto $\mu_r(p, -q, 0) = G \cap \mathbf{T}$. Let $Z := z^r$, $Y := \mathbb{A}^3_{Z,y_1,t}$, $\varpi : X'' \to Y$ the quotient morphism by $\mu_r(1, 0, 0)$, and $\tilde{H} := \tilde{G}/\mu_r(1, 0, 0)$. Then $\tilde{H}$ acts diagonally and faithfully on Y, and $\mathbb{C}(Y)^{\tilde{H}} = \mathbb{C}(X'')^{\tilde{G}} = \mathbb{C}(X)^G$. Let $T' := \{Z = 0\} = \mathbb{A}^2_{y_1,t} \subset Y$.

We show that $\tilde{H}$ acts faithfully on T'. Let $\sigma = (\xi_1, \xi_2, \xi_3) \in \tilde{G}$ with $\xi_2 = \xi_3 = 1$. Then $\tau(\sigma)$ acts trivially on $u = y_1^p$ and on t, so $\tau(\sigma) \in G \cap \mathbf{T}$ as the kernel of $G \to H$ is $G \cap \mathbf{T}$. Let $\sigma' \in \mu_r(1, 0, 0)$ with $\tau(\sigma') = \tau(\sigma)$. Then $\sigma\sigma'^{-1} = (\xi, \xi^q, 1)$ with $\xi^p = 1$, and $\xi^q = 1$ because $\sigma\sigma'^{-1}$ fixes y_1. Hence $\xi = 1$ and $\sigma \in \mu_r(1, 0, 0)$.

By (14.46), a monomial $x^k y^l t^m$ occurring in h_j^+ equals $z^{\kappa_j} Z^n y_1^l t^m$ with $n \geq 0$, and $x^{a_j} y^{b_j} u^i t^m = z^{\kappa_j} y_1^{b_j + pi} t^m$. Hence

$$h_j^+ = z^{\kappa_j} g_j(Z, y_1, t), \qquad g_j(0, y_1, t) = y_1^{b_j} F_j(y_1^p, t) \neq 0, \tag{14.49}$$

for a polynomial $g_j \in \mathbb{C}[Z, y_1, t]$. Since h_j^+ and z are $\tilde{G}$-semi-invariant, so is g_j, and $\mu_r(1, 0, 0)$ fixes g_j. Thus g_j is a regular $\tilde{H}$-semi-invariant function on Y. Let

$$\beta := p - q - pc_x + qc_y \quad \text{and} \quad \beta_+ := \max\left\{0, -\frac{\beta}{q}\right\},$$

and define on Y and on T'

$$B_\beta := \left(1 - \frac{\beta}{r}\right) \operatorname{div}(Z) + \sum_j c_j \operatorname{div}(g_j), \qquad \hat{B} := \operatorname{div}(Z) + \sum_j c_j \operatorname{div}(g_j) + \beta_+ \operatorname{div}(y_1),$$

$$B_{T'} := \sum_j c_j \operatorname{div}_{T'}\left(F_j(y_1^p, t)\right) + (c_y + \beta_+) \operatorname{div}_{T'}(y_1).$$

The $\mathbb{R}$-divisors B_β, $\hat{B}$, and $B_{T'}$ are $\tilde{H}$-invariant, and $B_{T'} \geq 0$. By the definition of c_u in Set-up 14.1(4) and since $c_y + \beta_+ = \max\{c_y, 1 - \frac{p(1-c_x)}{q}\}$, we have

$$pc_u - (p - 1) = \max\left\{1 - \frac{p(1 - c_x)}{q}, c_y\right\} = c_y + \beta_+. \tag{14.50}$$

We show that $(T', B_{T'})/\tilde{H} = (T, B_T)/H$. Let $H_1 \subset \tilde{H}$ be the image of $\mu_p(1, q, 0)$. Since $\mu_p(1, q, 0) \cap \mu_r(1, 0, 0) = \{1\}$, $H_1 \cong \mu_p$, and H_1 acts on T' by $y_1 \to \xi^q y_1$, $t \to t$ with $\xi^p = 1$. As

$\gcd(p,q)=1$, the morphism $\varphi : T' \to T$, $(y_1,t) \to (y_1^p,t)$, is the quotient morphism by H_1, and $R(\varphi) = (p-1)\operatorname{div}(y_1)$. Since $B_{T'} = \varphi^*(B_T - c_u \operatorname{div}(u)) + (c_y+\beta_+)\operatorname{div}(y_1)$, (14.50) implies

$$\frac{1}{p}\varphi_*(B_{T'}+R(\varphi)) = B_T - c_u\operatorname{div}(u) + \frac{c_y+\beta_++p-1}{p}\operatorname{div}(u) = B_T,$$

that is, $(T',B_{T'})/H_1 = (T,B_T)$. The image of $\sigma \in \tilde{G}$ acts on u and t as $\tau(\sigma)$ does, so $\tilde{H}$ acts on $T = T'/H_1$ through H. An element of $\tilde{H}$ acting trivially on T multiplies y_1 by a p-th root of unity and fixes t, so it belongs to H_1 because $\tilde{H}$ acts faithfully on T'. Thus $\tilde{H}/H_1 \cong H$ as groups acting on T, $|\tilde{H}| = p|H|$, and the quotient morphism $T' \to T'/\tilde{H}$ is $\pi_H \circ \varphi$, where $\pi_H : T \to T/H$ is the quotient morphism. By the Hurwitz formula [Kol13, 2.41], $R(\pi_H\circ\varphi) = R(\varphi) + \varphi^*R(\pi_H)$. Since $\varphi_*\varphi^*R(\pi_H) = pR(\pi_H)$, Definition 2.5 implies

$$\begin{aligned}\frac{1}{p|H|}(\pi_H\circ\varphi)_*\big(B_{T'}+R(\pi_H\circ\varphi)\big) &= \frac{1}{|H|}\pi_{H*}\Big(\frac{1}{p}\varphi_*(B_{T'}+R(\varphi)) + R(\pi_H)\Big)\\ &= \frac{1}{|H|}\pi_{H*}(B_T+R(\pi_H)).\end{aligned}$$

Hence $(T',B_{T'})/\tilde{H} = (T,B_T)/H$, and

$$\operatorname{mld}((T' \ni 0, B_{T'})/\tilde{H}) = b_H. \tag{14.51}$$

Since $(T',B_{T'})/H_1 = (T,B_T)$ and (T,B_T) is klt near 0, Lemma 2.6(2)(3) implies that $(T',B_{T'})$ is klt near 0.

Step 4. In this step we express $a(E,X/G,B_{X/G})$ by a log discrepancy over Y with respect to B_β and compare it with the one with respect to $\hat{B}$.

Let $X' := \mathbb{A}^3_{z,y,t}$, let $f : X' \to X$, $(z,y,t) \to (z^p,y,t)$, and let $\theta : X' \to X''$, $(z,y,t) \to (z,z^qy,t)$. Then f is the quotient morphism by $\mu_p(1,q,0)$, which acts on X' as $\mu_p(1,0,0)$, and θ is a birational morphism. Both are compatible with $\mathbb{C}(X) \subset \mathbb{C}(X') = \mathbb{C}(X'')$. Let F' be a prime divisor over X' such that $\operatorname{ord}_{F'}|_{\mathbb{C}(X)} = d_1 \operatorname{ord}_F$ for a positive integer d_1 (Lemma 2.6(3)). Let F_Y be the prime divisor over Y such that $\operatorname{ord}_{F'}|_{\mathbb{C}(Y)} = d_2 \operatorname{ord}_{F_Y}$ for a positive integer d_2, let E' be the prime divisor over $Y/\tilde{H}$ such that $\operatorname{ord}_{F_Y}|_{\mathbb{C}(Y)^{\tilde{H}}} = e_{\tilde{H}} \operatorname{ord}_{E'}$, and $e_{\tilde{H}} = |I_{\tilde{H}}(F_Y)|$ (Lemma 2.6(1)).

Claim 14.16. $\operatorname{center}_Y F_Y = \{0\}$, $\operatorname{center}_{Y/\tilde{H}} E'$ *is the image of* 0, $d_1e = d_2e_{\tilde{H}}$, *and*

$$a(E,X/G,B_{X/G}) = \frac{a(F_Y,Y,B_\beta)}{e_{\tilde{H}}}. \tag{14.52}$$

Proof. The restriction of $\operatorname{ord}_{F'}$ to $\mathbb{C}(X)^G = \mathbb{C}(Y)^{\tilde{H}}$ equals both $d_1e\operatorname{ord}_E$ and $d_2e_{\tilde{H}}\operatorname{ord}_{E'}$. Since ord_E and $\operatorname{ord}_{E'}$ are discrete valuations with value group $\mathbb{Z}$, $\operatorname{ord}_E = \operatorname{ord}_{E'}$ and $d_1e = d_2e_{\tilde{H}}$.

(i) Let $B_{X'} := f^*B^+ - (p-1)\operatorname{div}(z)$, a $\mu_p(1,0,0)$-invariant sub-boundary. Since $R(f) = (p-1)\operatorname{div}(z)$ and $\frac{1}{p}f_*f^*B^+ = B^+$, Definition 2.5 implies $(X',B_{X'})/\mu_p(1,0,0) = (X,B^+)$. By Lemma 2.6(2), $d_1 = |I_{\mu_p(1,0,0)}(F')|$, $f(\operatorname{center}_{X'}F') = \{0\}$, so $\operatorname{center}_{X'}F' = \{0\}$, and $a(F',X',B_{X'}) = d_1a(F,X,B^+)$.

(ii) Let

$$B_{X''} := (1-\beta)\operatorname{div}(z) + \sum_j c_j \operatorname{div}\big(g_j(z^r,y_1,t)\big)$$

on X''. Since $\operatorname{center}_{X''}F' = \theta(\operatorname{center}_{X'}F') = \{0\}$, both $a(F',X',B_{X'})$ and $a(F',X'',B_{X''})$ can be computed on a normal variety W with a birational morphism $W \to X'$ on which F' is a prime divisor. The 3-forms $\eta := dz\wedge dy\wedge dt$ and $\omega := dz\wedge dy_1\wedge dt$ have no zeros and poles on X' and on X'', respectively, so

$$a(F',X',B_{X'}) = 1 + \operatorname{ord}_{F'}\eta - \operatorname{mult}_{F'}B_{X'} \quad\text{and}\quad a(F',X'',B_{X''}) = 1 + \operatorname{ord}_{F'}\omega - \operatorname{mult}_{F'}B_{X''},$$

where $\operatorname{ord}_{F'}$ of a 3-form is the order along F' of the pullback of the 3-form to W (cf. [KM98, Lemma 2.30]). Since $dy_1 = z^q dy + qz^{q-1}ydz$, we have $\omega = z^q\eta$. By (14.49), $\operatorname{ord}_{F'} h_j^+ = \kappa_j \operatorname{ord}_{F'} z + \operatorname{ord}_{F'} g_j(z^r, y_1, t)$, and $\sum_j c_j\kappa_j = pc_x - qc_y$. Hence

$$a(F', X'', B_{X''}) - a(F', X', B_{X'}) = \big(q - (1-\beta) + pc_x - qc_y - (p-1)\big)\operatorname{ord}_{F'} z = 0.$$

(iii) The $\mathbb{R}$-divisor $B_{X''}$ is $\mu_r(1,0,0)$-invariant, $R(\varpi) = (r-1)\operatorname{div}(z)$, $\varpi_*\operatorname{div}(z) = \operatorname{div}(Z)$, and $\varpi_*\operatorname{div}(g_j(z^r, y_1, t)) = r\operatorname{div}(g_j)$. Thus

$$\frac{1}{r}\varpi_*\big(B_{X''} + R(\varpi)\big) = \frac{r-\beta}{r}\operatorname{div}(Z) + \sum_j c_j \operatorname{div}(g_j) = B_\beta,$$

so $(X'', B_{X''})/\mu_r(1,0,0) = (Y, B_\beta)$ by Definition 2.5. By Lemma 2.6(2), $d_2 = |I_{\mu_r(1,0,0)}(F')|$, $\operatorname{center}_Y F_Y = \varpi(\operatorname{center}_{X''} F') = \{0\}$, and $a(F', X'', B_{X''}) = d_2 a(F_Y, Y, B_\beta)$. By Lemma 2.6(2) for the action of $\tilde{H}$ on Y, $\operatorname{center}_{Y/\tilde{H}} E'$ is the image of 0.

By Lemma 2.6(2), (14.48), (i), (ii), and (iii),

$$\begin{aligned} a(E, X/G, B_{X/G}) &= \frac{a(F, X, B^+)}{e} = \frac{a(F', X', B_{X'})}{d_1 e} = \frac{a(F', X'', B_{X''})}{d_1 e} \\ &= \frac{d_2 a(F_Y, Y, B_\beta)}{d_1 e} = \frac{a(F_Y, Y, B_\beta)}{e_{\tilde{H}}}. \end{aligned}$$

□

Proof of Proposition 14.15 continued. We show that

$$a(F_Y, Y, B_\beta) - a(F_Y, Y, \hat{B}) = \frac{\beta}{r}\operatorname{ord}_{F_Y} Z + \beta_+ \operatorname{ord}_{F_Y} y_1 \geq 0. \tag{14.53}$$

The equality holds as $\hat{B} - B_\beta = \frac{\beta}{r}\operatorname{div}(Z) + \beta_+\operatorname{div}(y_1)$. If $\beta \geq 0$, then $\beta_+ = 0$, and the inequality holds as $\operatorname{ord}_{F_Y} Z > 0$. Suppose that $\beta < 0$. Then $\beta_+ = -\beta/q$. Since $y_1 = z^q y$, $Z = z^r$, $\operatorname{ord}_{F'}|_{\mathbb{C}(Y)} = d_2 \operatorname{ord}_{F_Y}$, and $\operatorname{ord}_{F'}|_{\mathbb{C}(X)} = d_1 \operatorname{ord}_F$, we have

$$\frac{\beta}{r}\operatorname{ord}_{F_Y} Z + \beta_+ \operatorname{ord}_{F_Y} y_1 = \frac{-\beta}{d_2}\Big(\frac{\operatorname{ord}_{F'} y_1}{q} - \operatorname{ord}_{F'} z\Big) = \frac{-\beta d_1}{q d_2}\operatorname{ord}_F y \geq 0.$$

Step 5. In this step we conclude the proof using Theorem 8.3 on Y.

Recall from **Step 3** that $\tilde{H}$ is a finite abelian group acting diagonally on $Y = \mathbb{A}^3_{Z,y_1,t}$, that Z is $\tilde{H}$-semi-invariant with zero scheme $T' \cong \mathbb{A}^2$, that $\tilde{H}$ acts faithfully on T', and that the g_j are $\tilde{H}$-semi-invariant polynomials with non-zero restrictions to T' by (14.49). By (14.49),

$$\sum_j c_j \operatorname{div}(g_j|_{T'}) + \beta_+ \operatorname{div}(y_1|_{T'}) = B_{T'},$$

and $(T', B_{T'})$ is klt near 0. Take $n = 3$, $c = 1$, $f_1 = Z$, the polynomials g_j with coefficients c_j, and y_1 with coefficient β_+ if $\beta_+ > 0$ in Theorem 8.3. Then $B_{\mathbb{A}^n} = \hat{B}$. By Theorem 8.3 and (14.51), any prime divisor over $Y/\tilde{H}$ with center the image of 0 has log discrepancy at least b_H with respect to $(Y, \hat{B})/\tilde{H}$. Recall from **Step 4** that F_Y is a prime divisor over Y with $\operatorname{ord}_{F_Y}|_{\mathbb{C}(Y)^{\tilde{H}}} = e_{\tilde{H}} \operatorname{ord}_{E'}$. By Claim 14.16, $\operatorname{center}_{Y/\tilde{H}} E'$ is the image of 0, so Lemma 2.6(2) implies

$$a(F_Y, Y, \hat{B}) = e_{\tilde{H}} a(E', Y/\tilde{H}, \hat{B}_{Y/\tilde{H}}) \geq e_{\tilde{H}} b_H,$$

where $(Y, \hat{B})/\tilde{H} = (Y/\tilde{H}, \hat{B}_{Y/\tilde{H}})$. By (14.52) and (14.53),

$$a(E, X/G, B_{X/G}) = \frac{a(F_Y, Y, B_\beta)}{e_{\tilde{H}}} \geq \frac{a(F_Y, Y, \hat{B})}{e_{\tilde{H}}} \geq b_H.$$

It remains to consider the prime divisors E over X/G with $\operatorname{center}_{X/G} E = \{\pi(0)\}$ and $a(E, X/G, B_{X/G}) > 3$. Let E_0 be the exceptional divisor of the blow-up of T at 0. Since H acts diagonally on T, E_0 is H-invariant. Since $B_T \geq 0$, $a(E_0, T, B_T) = 2 - \operatorname{mult}_0 B_T \leq 2$ by

[KM98, Lemma 2.29]. By Lemma 2.6(2), the prime divisor over T/H under E_0 has center the image of 0 and log discrepancy $a(E_0, T, B_T)/|I_H(E_0)| \leq 2$ with respect to $(T, B_T)/H$. Thus $b_H \leq 2 < a(E, X/G, B_{X/G})$ for such E.

Since E is an arbitrary prime divisor over X/G with $\operatorname{center}_{X/G} E = \{\pi(0)\}$, we conclude that $\operatorname{mld}((X \ni 0, B)/G) \geq b_H$. □

14.7. **Generic limits.** In this subsection, we prove Theorem 14.19.

In the proof of Theorem 14.19, Proposition 14.9, including the ACC in it, reduces the problem to bounding the quotient minimal log discrepancies from below by the limit of the minimal log discrepancies of the crepant quotients of the residual plane pairs. We obtain the lower bound from a generic limit of the residual plane pairs. If the generic limit is klt, then it implies a uniform lc margin, and Proposition 14.15 implies the lower bound. If the generic limit is lc but not klt, then it has exactly one curve with coefficient one, the curve is smooth, and a fixed multiple of one coordinate function of T can be added to the boundaries without destroying log canonicity. The log canonicity of the enlarged pairs implies the lower bound directly.

Definition-Lemma 14.17. For an $\mathbb{R}$-divisor $D \geq 0$ on $T = \mathbb{A}^2_{u,t}$, let $\operatorname{lct}_0(T; D) := \sup\{c \geq 0 \mid (T, cD) \text{ is lc near } 0\}$. For an algebraically closed field $k \supset \mathbb{C}$ and an $\mathbb{R}$-divisor $D \geq 0$ on $\operatorname{Spec} k[[u,t]]$, we say that $(\operatorname{Spec} k[[u,t]], D)$ is lc (resp. klt) if $a(P, \operatorname{Spec} k[[u,t]], D) \geq 0$ (resp. > 0) for any formal prime divisor P over $\operatorname{Spec} k[[u,t]]$ (Definition 3.2), and we let $\operatorname{lct}_0(\operatorname{Spec} k[[u,t]]; D) := \sup\{c \geq 0 \mid (\operatorname{Spec} k[[u,t]], cD) \text{ is lc}\}$. By [dFEM11, Section 2], whether $(\operatorname{Spec} k[[u,t]], D)$ is lc or klt can be checked on one log resolution, which exists by [dFEM11, Theorems 2.3 and 2.4], so $\operatorname{lct}_0(\operatorname{Spec} k[[u,t]]; D)$ is the log canonical threshold of [dFEM11, Section 2]. Let H, N, s, the tuples $(f_{i,j})_{1 \leq j \leq s}$, k, and $(f_{\infty,j})_{1 \leq j \leq s}$ be as in Construction 14.13, and assume that $f_{i,j}(0) = 0$ for any i and j. Then:

(1) for any subset $S \subset \{1, \dots, s\}$, the tuple $(f_{\infty,j})_{j \in S}$ is a generic limit of the tuples $(f_{i,j})_{j \in S}$ in the sense of Construction 14.13;

(2) for any real numbers $\theta_1, \dots, \theta_s \geq 0$, after passing to a subsequence,

$$\operatorname{lct}_0\Big(T; \sum_j \theta_j \operatorname{div}(f_{i,j})\Big) \to \operatorname{lct}_0\Big(\operatorname{Spec} k[[u,t]]; \sum_j \theta_j \operatorname{div}(f_{\infty,j})\Big).$$

Proof. For $d \geq 1$, let Z_d^S be the closure of the image of Z_d under the projection of $\mathcal{J}_d$ to the space of the subtuples indexed by S. Then Z_d^S is irreducible, the projection maps the generic point of Z_d to the generic point of Z_d^S, and the inverse image in Z_d of a non-empty open subset of Z_d^S is a non-empty open subset of Z_d. Thus (1) holds.

We prove (2). If all θ_j vanish, then both sides are $+\infty$. Otherwise, dropping the indices j with $\theta_j = 0$, which changes neither side, we may assume that $\theta_j > 0$ for any j by (1). Let $D_i := \sum_j \theta_j \operatorname{div}(f_{i,j})$ and $D_\infty := \sum_j \theta_j \operatorname{div}(f_{\infty,j})$. We use the generic limit construction of [dFEM11, Section 3] in the variables u, t, with the zero ideals as the ideals of the ambient spaces and with the sequences of the ideals of $\mathbb{C}[[u,t]]$ generated by $f_{i,1}, \dots, f_{i,s}$. For $d \geq 1$, sending a tuple $(f_j)_j \in \mathcal{J}_d$ to the tuple of ideals $((f_j) + (u,t)^d)_j$ of $\mathbb{C}[u,t]/(u,t)^d$ is a morphism on a non-empty open subset Z_d° of Z_d, by generic flatness. Let Z'_d be the closure of the image of Z_d° in the product of the Hilbert schemes of ideals of $\mathbb{C}[u,t]/(u,t)^d$. The truncation $Z_{d+1} \to Z_d$ maps the generic point to the generic point, so the induced rational map $Z'_{d+1} \dashrightarrow Z'_d$ is dominant, which is condition $(\star)$ of [dFEM11, Section 3]. The inverse image in Z_d° of a non-empty open subset of Z'_d is a non-empty open subset of Z_d, which contains the tuple of the parts of degree less than d of the $f_{i,j}$ for all but finitely many i. Thus condition $(\star\star)$ of [dFEM11, Section 3] holds.

Let $K := \bigcup_d \mathbb{C}(Z'_d)$, and let $\mathfrak{a}_j \subset K[[u,t]]$ be the generic limit ideal at the position j, which is proper by [dFEM11, Lemma 3.1(ii)]. The dominant morphisms $Z_d^\circ \to Z'_d$ and the k-points of the Z_d given by $(f_{\infty,j})_j$ induce an embedding $K \subset k$, and $\mathfrak{a}_j k[[u,t]] + (u,t)^d = (f_{\infty,j}) + (u,t)^d$ for any d. Hence $\mathfrak{a}_j k[[u,t]] = (f_{\infty,j})$ by the Krull intersection theorem. The ambient spaces $\operatorname{Spec} \mathbb{C}[[u,t]]$

are smooth, hence klt, and the ideals $(f_{i,j})$ are proper since $f_{i,j}(0) = 0$. Moreover $\mathrm{lct}_0(T; D_i)$ is the log canonical threshold on $\operatorname{Spec}\mathbb{C}[[u,t]]$ of the $\mathbb{R}$-ideal $\prod_j (f_{i,j})^{\theta_j}$ [dFEM11, Remark 2.10 and Proposition 2.11(ii)]. By [dFEM11, Proposition 2.19(ii)], the log canonical threshold on $\operatorname{Spec} K[[u,t]]$ of $\prod_j \mathfrak{a}_j^{\theta_j}$ and $\mathrm{lct}_0(\operatorname{Spec} k[[u,t]]; D_\infty)$ are the limits as $d \to +\infty$ of the log canonical thresholds of $\prod_j (\mathfrak{a}_j + (u,t)^d)^{\theta_j}$ and of $\prod_j ((f_{\infty,j}) + (u,t)^d)^{\theta_j}$, respectively. The ideals $\mathfrak{a}_j + (u,t)^d$ are generated by polynomials over K, so by [dFEM11, Remark 2.10 and Proposition 2.11(ii)] the log canonical thresholds of $\prod_j (\mathfrak{a}_j + (u,t)^d)^{\theta_j}$ and of $\prod_j ((f_{\infty,j}) + (u,t)^d)^{\theta_j}$ are the log canonical thresholds at 0 of the corresponding $\mathbb{R}$-ideals on $\mathbb{A}^2_K$ and on $\mathbb{A}^2_k$. A log resolution over K of the $\mathbb{R}$-ideal on $\mathbb{A}^2_K$ base changes to a log resolution over k of the $\mathbb{R}$-ideal on $\mathbb{A}^2_k$, and each component of the base change of a prime divisor F on the resolution has discrepancy and orders of the ideals equal to those of F. Thus the log canonical threshold on $\operatorname{Spec} K[[u,t]]$ of $\prod_j \mathfrak{a}_j^{\theta_j}$ is equal to $\mathrm{lct}_0(\operatorname{Spec} k[[u,t]]; D_\infty)$.

Hence [dFEM11, Proposition 3.3(ii) and Corollary 3.4] imply that, on an infinite subsequence, $\mathrm{lct}_0(T; D_i) \to \mathrm{lct}_0(\operatorname{Spec} k[[u,t]]; D_\infty)$. □

Remark 14.18. Let $k \supset \mathbb{C}$ be an algebraically closed field. Theorem 2.23 holds over k, since [Kaw07] works over any algebraically closed field of characteristic zero. Theorem 8.3 also holds over k, that is, with the varieties, polynomials, divisors, and group actions in the statement of Theorem 8.3 defined over k instead of $\mathbb{C}$. Given such data over k, let $k_1 \subset k$ be the algebraic closure in k of the field generated over $\mathbb{C}$ by the coefficients of finitely many polynomials defining the data. Then $|k_1| = |\mathbb{C}|$, so k_1 and $\mathbb{C}$ are algebraically closed fields of characteristic zero whose transcendence degrees over $\mathbb{Q}$ are both equal to $|\mathbb{C}|$, and there exists a field isomorphism $\varphi : k_1 \to \mathbb{C}$ (cf. [Lan02, Theorem VIII.1.1 and Corollary V.2.9]). Applying φ to the coefficients carries the varieties, the polynomials, the finite diagonal groups, whose entries are roots of unity, the pairs, their crepant quotients, log discrepancies, log canonicity, and the klt property to the corresponding objects over $\mathbb{C}$. Thus Theorem 8.3 holds over k_1. Moreover, log canonicity and the klt property near a closed point and the minimal log discrepancy at a closed point do not change under the base change from k_1 to k. To see this, let (X', B') be a pair over k_1, let $x' \in X'$ be a closed point, let $h : Y \to X'$ be a projective birational morphism over k_1 from a smooth variety Y such that $\mathfrak{m}_{x'}\mathcal{O}_Y$ is invertible and $h^{-1}(x') \cup \operatorname{Exc}(h) \cup \operatorname{Supp} h_*^{-1}B'$ is an snc divisor, and write $K_Y + \sum_j d_j D_j = h^*(K_{X'} + B')$ with distinct prime divisors D_j. The base change of h to k is again such a log resolution, each D_j stays a prime divisor since k_1 is algebraically closed, and the coefficients d_j do not change. The pair is lc (resp. klt) near x' if and only if $d_j \le 1$ (resp. $d_j < 1$) for every D_j meeting $h^{-1}(x')$ [KM98, Corollary 2.31(3)], and if the pair is lc near x', then the minimal log discrepancy of the pair at x' is the minimum of $1 - d_j$ over the components D_j of $h^{-1}(x')$ (Lemma 3.5), while otherwise it is $-\infty$ if $\dim X' \ge 2$, and $1 - \mathrm{mult}_{x'} B'$ if $\dim X' = 1$ [Amb99, Section 1, the second paragraph after Definition 1.1]. For the data of Theorem 8.3, normality near 0 descends from k to k_1 [Mat89, Theorem 23.9, Corollary], hence so does integrality near 0, and the codimension and the faithfulness of the group action near 0 do not change under the base change from k_1 to k. Moreover, the crepant quotient of the base change by a finite group acting over k_1 is the base change of the crepant quotient, since taking invariants of a finite group and forming the ramification divisor commute with the flat base change from k_1 to k. Applying the description by the coefficients d_j to the pairs in the statement of Theorem 8.3 and to their crepant quotients, and using Lemma 2.6(2)(3) to pass between a pair on $\mathbb{A}^n$ and its crepant quotient, we obtain Theorem 8.3 over k.

Theorem 14.19. *Fix coprime positive integers p, q, an integer $e_0 \ge 1$, and a DCC set $\Gamma \subset [0,1]$ with $\Gamma\backslash\{0\} \ne \emptyset$. For any $i \ge 1$, let G_i, $h_{i,j} = h^0_{i,j} + R_{i,j}$, $c_{i,j}$, B_i, r_i, and N_i be data as in Set-up 14.1 for $P := \max\{p,q\}$, e_0, and Γ, with $G_i \cap \mathbf{T} = \mu_{r_i}(p, -q, 0)$ and $|G_i/(G_i \cap \mathbf{T})| \le e_0$, such that $h^0_{i,j}$ is the sum of all terms of $h_{i,j}$ of $\mathbf{T}$-weight $pa_{i,j} - qb_{i,j}$ for any j, and assume that $r_i \to +\infty$ and $N_i \to +\infty$. Then the numbers $\mathrm{mld}((X \ni 0, B_i)/G_i)$ do not form a strictly increasing sequence.*

Proof. The proof is long, so we divide it into several steps.

Let $\gamma := \gamma(\Gamma)$. Then $\gamma > 0$ (Lemma 2.10(2)). For the i-th data we use the notation of Set-up 14.1 with an index i: B_i^0, $c_{x,i}$, $c_{y,i}$, $c_{u,i}$, $B_{T,i}$, H_i, and b_{H_i}, and we write $h_{i,j}^0 = x^{a_{i,j}} y^{b_{i,j}} F_{i,j}(u,t)$. Let $\alpha_i := \mathrm{mld}((X \ni 0, B_i)/G_i)$. For a pair (X', B') with a faithful action of a finite group G' preserving B' and a prime divisor E over X'/G', we write $a(E, (X', B')/G')$ for the log discrepancy of E with respect to the crepant quotient $(X', B')/G'$. If F is a prime divisor over X' with $\mathrm{ord}_F|_{\mathbb{C}(X'/G')} = e\,\mathrm{ord}_E$, then

$$a(F, X', B') = ea(E, (X', B')/G') \tag{14.54}$$

by Lemma 2.6(1)(2). Suppose for contradiction that the sequence α_i is strictly increasing. We pass to subsequences several times, and the sequence α_i stays strictly increasing.

Step 1. In this step we show that b_{H_i} decreases to a limit $b_\infty \geq \lim_i \alpha_i$, and we fix the group H_i.

Since (X, B_i) is klt near 0, the crepant quotient $(X, B_i)/G_i$ is klt near the image of 0 [KM98, Proposition 5.20], so $\alpha_i > 0$. If $h_{i,j}(0) \neq 0$, then $h_{i,j}^0(0) \neq 0$ because $R_{i,j} \in (x,y)^{N_i}$, so $a_{i,j} = b_{i,j} = 0$ and $F_{i,j}(0,0) \neq 0$. Removing the labels j with $h_{i,j}(0) \neq 0$ changes neither B_i nor $B_{T,i}$ near 0, nor the numbers α_i and b_{H_i}. Thus we may assume that $h_{i,j}(0) = 0$ for any i, j. Let E_0 be the exceptional divisor of the blow-up of the origin of X. Then

$$0 < a(E_0, X, B_i) = 3 - \sum_j c_{i,j} \,\mathrm{mult}_0\, h_{i,j} \leq 3$$

[KM98, Lemma 2.29]. Since $c_{i,j} \geq \gamma$, the number of labels is less than $3/\gamma$. Discarding finitely many indices, we may assume that $N_i > 3/\gamma$ and $N_i \geq \ell$, where ℓ and L are the integers of Construction-Proposition 14.7 for Γ. By Proposition 14.8(1),

$$\mathrm{mult}_0\, h_{i,j}^0 = \mathrm{mult}_0\, h_{i,j} < \frac{3}{\gamma}, \qquad a_{i,j} + b_{i,j} < \frac{3}{\gamma}.$$

Passing to a subsequence, we may assume that the number s of labels, the exponents $a_{i,j}$ and $b_{i,j}$, and the set J_1 of the labels j with $F_{i,j}(0,0) = 0$ do not depend on i. By Lemma 2.10(3), passing to a subsequence, we may assume that $c_{i,j}$ is nondecreasing in i for any j, with limit $c_j^* \in [\gamma, 1]$. Then $c_{x,i}$ and $c_{y,i}$ are nondecreasing, so $c_{u,i}$ is nondecreasing, with limit c_u^*. Since $c_{x,i} \geq 0$, we have $c_{u,i} \geq 1 - \frac{1}{q} \geq 0$. In particular, if $c_u^* = 0$, then $c_{u,i} = 0$ for any i.

The kernel of $G_i \to H_i$ contains $G_i \cap \mathbf{T}$, since $\mathbf{T}$ fixes u and t. Thus $|H_i| \leq |G_i/(G_i \cap \mathbf{T})| \leq e_0$, so the assumptions of Proposition 14.9 hold. Hence the kernel of $G_i \to H_i$ is $G_i \cap \mathbf{T}$, and $(T, B_{T,i})$ is lc near 0. Let

$$\eta_i := \max\left\{\frac{2P}{r_i}, \frac{8L}{\gamma N_i}\right\}. \tag{14.55}$$

Since $r_i, N_i \to +\infty$, discarding finitely many indices, we may assume that $\eta_i \leq \frac{1}{8Pe_0}$. Moreover $r_i \geq \frac{2P}{\eta_i}$ and $N_i \geq \frac{8L}{\gamma\eta_i} > \frac{4L}{\gamma\eta_i}$. Thus Proposition 14.9 implies that $\alpha_i \leq b_{H_i} + 2\eta_i$, where $\eta_i \to 0$, and that the set of the numbers b_{H_i} satisfies the ACC. Since any sequence in a set satisfying the ACC has a nonincreasing subsequence, passing to a subsequence, we may assume that b_{H_i} is nonincreasing. The divisor E_0 is G_i-invariant, and by (14.54) the image of E_0 over X/G_i has log discrepancy at most $a(E_0, X, B_i) \leq 3$. Hence α_i increases to a limit $\alpha_\infty \in (0, 3]$. Since $b_{H_i} \geq \alpha_i - 2\eta_i \geq \alpha_1 - 2\eta_i$, the sequence b_{H_i} converges to a limit b_∞, and $b_\infty \geq \alpha_\infty$. Thus

$$b_{H_i} \downarrow b_\infty, \qquad \alpha_i < \alpha_\infty \leq b_\infty, \qquad b_\infty > 0. \tag{14.56}$$

It remains to find an index i with $\alpha_i \geq b_\infty$.

Any element of a diagonal subgroup of $(\mathbb{C}^*)^2$ of order at most e_0 has order dividing $e_0!$, so there exist only finitely many such subgroups. Passing to a subsequence, we may assume that $H_i = H$ for any i. Since $h_{i,j}^0$ and $x^{a_{i,j}} y^{b_{i,j}}$ are G_i-semi-invariant, the polynomial $F_{i,j}(x^q y^p, t)$ is

G_i-semi-invariant. Since G_i acts on $\mathbb{C}[u,t]$ through H, the polynomial $F_{i,j} \in \mathbb{C}[u,t]$ is H-semi-invariant. Passing to a subsequence, we may assume that the character of $F_{i,j}$ depends only on j.

Step 2. In this step we replace the coefficients of $B_{T,i}$ with their limits, and we show that the numbers b_{H_i} change by an amount tending to 0.

Let

$$B^*_{T,i} := \sum_j c^*_j \operatorname{div}_T(F_{i,j}) + c^*_u \operatorname{div}_T(u).$$

Near 0, only the labels $j \in J_1$ contribute to $B_{T,i}$ and $B^*_{T,i}$, and $B_{T,i} \le B^*_{T,i}$. Let τ_i be the minimum of the numbers $c_{i,j}/c^*_j$ and, if $c^*_u > 0$, of $c_{u,i}/c^*_u$. Discarding finitely many indices, we may assume that $\tau_i > 0$. Then $\tau_i \to 1$ and $\tau_i B^*_{T,i} \le B_{T,i}$. Since $(T, B_{T,i})$ is lc near 0, $\operatorname{lct}_0(T; B^*_{T,i}) \ge \tau_i$. By [dFEM11, Theorem 1.1], the numbers $\operatorname{lct}_0(T; \sum_{j\in J_1} c^*_j \operatorname{div}(f_j) + c^*_u \operatorname{div}(u))$, where the f_j are non-zero polynomials vanishing at 0, satisfy the ACC, since T is fixed and the exponents belong to a finite set. If infinitely many of the numbers $\operatorname{lct}_0(T; B^*_{T,i})$ were less than 1, then, since $\tau_i \to 1$, they would contain a strictly increasing subsequence. Thus, discarding finitely many indices, we may assume that $(T, B^*_{T,i})$ is lc near 0. Let $b^*_{H_i} := \operatorname{mld}((T \ni 0, B^*_{T,i})/H)$. We show that

$$0 \le b_{H_i} - b^*_{H_i} \le l\Big[\sum_{j\in J_1} \frac{c^*_j - c_{i,j}}{c^*_j} + \frac{c^*_u - c_{u,i}}{c^*_u}\Big] \to 0, \tag{14.57}$$

where l is defined below and the last term is omitted if $c^*_u = 0$.

The first inequality follows from $B_{T,i} \le B^*_{T,i}$. Near 0, the coefficients of $B^*_{T,i}$ belong to the set Λ_0 of the numbers $\sum_j n_j c^*_j + n c^*_u \le 1$ with non-negative integers n_j, n, which is finite because $c^*_j \ge \gamma$. Hence, near the image of 0, by Definition 2.5, the coefficients of the boundary of $(T, B^*_{T,i})/H$ belong to the finite set

$$\Lambda_1 := \Big\{1 - \frac{1-v}{e} \;\Big|\; v \in \Lambda_0 \cup \{0\},\ 1 \le e \le |H|\Big\}.$$

Let l be the integer of Theorem 2.18(2) for Λ_1. Since $(T, B^*_{T,i})/H$ is lc near the image of 0 [KM98, Proposition 5.20], there exists a prime divisor E_i over T/H with center the image of 0 such that $a(E_i, (T, B^*_{T,i})/H) = b^*_{H_i}$ and $a(E_i, T/H, 0) \le l$. Let E'_i be a prime divisor over T over E_i, with ramification index e_i. Then the center of E'_i on T is 0. Since the boundary of the crepant quotient $(T, 0)/H$ is effective (Definition 2.5), (14.54) implies that

$$a(E'_i, T, 0) = e_i a(E_i, (T, 0)/H) \le e_i a(E_i, T/H, 0) \le e_i l.$$

Since $(T, B^*_{T,i})$ is lc near 0,

$$0 \le a(E'_i, T, B^*_{T,i}) = a(E'_i, T, 0) - \sum_{j\in J_1} c^*_j \operatorname{ord}_{E'_i} F_{i,j} - c^*_u \operatorname{ord}_{E'_i} u.$$

Thus $\operatorname{ord}_{E'_i} F_{i,j} \le e_i l/c^*_j$ for any $j \in J_1$, and $\operatorname{ord}_{E'_i} u \le e_i l/c^*_u$ if $c^*_u > 0$. By (14.54),

$$b_{H_i} \le \frac{a(E'_i, T, B_{T,i})}{e_i} = b^*_{H_i} + \sum_{j\in J_1} (c^*_j - c_{i,j}) \frac{\operatorname{ord}_{E'_i} F_{i,j}}{e_i} + (c^*_u - c_{u,i}) \frac{\operatorname{ord}_{E'_i} u}{e_i},$$

which implies (14.57). In particular, $b^*_{H_i} \to b_\infty$.

Step 3. In this step we construct a generic limit $B_{T,\infty}$ of the pairs $(T, B^*_{T,i})$, and we show that $(\operatorname{Spec} k[[u,t]], B_{T,\infty})$ is lc with quotient minimal log discrepancy b_∞.

Since $(T, B_{T,i})$ is lc near 0, the exceptional divisor of the blow-up of the origin of T gives $2 - \operatorname{mult}_0 B_{T,i} \ge 0$ [KM98, Lemma 2.29], so $\operatorname{mult}_0 F_{i,j} \le 2/c_{i,j} \le 2/\gamma$ for any $j \in J_1$.

By **Step 1**, the tuples $((F_{i,j})_{j\in J_1}, u, t)$ consist of H-semi-invariant polynomials of order at most $\lfloor 2/\gamma \rfloor$ at 0 whose characters depend only on the position in the tuple. We apply Construction 14.13

to the tuples $((F_{i,j})_{j\in J_1}, u, t)$: after passing to a subsequence and multiplying the entries by non-zero constants, which does not change their divisors, we obtain an algebraically closed field $k \supset \mathbb{C}$ and a generic limit $((F_j^*)_{j\in J_1}, u', t')$ of non-zero H-semi-invariant power series in $k[[u,t]]$. Since $F_{i,j}(0,0) = 0$ for $j \in J_1$, all entries vanish at 0, which is the assumption of Lemma 14.17. The entries at the position of u are $\kappa_i u$ with non-zero constants κ_i. By Lemma 14.17(1), u' is a generic limit of the one-entry tuples $(\kappa_i u)$. Thus the closed subsets Z_d of Construction 14.13 for the tuples $(\kappa_i u)$ are contained in the line spanned by u, and u' is a non-zero constant multiple of u. Likewise t' is a non-zero constant multiple of t. We replace u' and t' with u and t, which does not change their divisors.

Let

$$B_{T,\infty} := \sum_{j\in J_1} c_j^* \operatorname{div}(F_j^*) + c_u^* \operatorname{div}(u) \tag{14.58}$$

on $\operatorname{Spec} k[[u,t]]$. Since $\operatorname{lct}_0(T; B_{T,i}^*) \geq 1$ for any i by **Step 2**, Lemma 14.17(2), applied with the coefficients c_j^* on the entries $F_{i,j}$, c_u^* on u, and 0 on t, implies that $(\operatorname{Spec} k[[u,t]], B_{T,\infty})$ is lc. Let Λ be the set of the numbers c_j^*, $j \in J_1$, together with c_u^* if $c_u^* > 0$. By Lemma 14.17(1) and Lemma 14.14(2) for the subtuples consisting of the entries $F_{i,j}$, $j \in J_1$, and of u if $c_u^* > 0$, with the coefficients in Λ, on an infinite subsequence $b_{H_i}^*$ is equal to the minimal log discrepancy at the closed point of $(\operatorname{Spec} k[[u,t]], B_{T,\infty})/H$. Since $b_{H_i}^* \to b_\infty$,

$$\operatorname{mld}((\operatorname{Spec} k[[u,t]] \ni 0, B_{T,\infty})/H) = b_\infty > 0. \tag{14.59}$$

Step 4. In this step we show that $(\operatorname{Spec} k[[u,t]], B_{T,\infty})$ is not klt.

Suppose that $(\operatorname{Spec} k[[u,t]], B_{T,\infty})$ is klt. Since a log resolution of $(\operatorname{Spec} k[[u,t]], B_{T,\infty})$ has finitely many prime divisors in the support of the boundary and the exceptional locus, $\operatorname{lct}_0(\operatorname{Spec} k[[u,t]]; B_{T,\infty}) > 1$ by Lemma 14.17 and [dFEM11, Theorems 2.3, 2.4 and formula (5)]. Let λ be a real number with $1 < \lambda < \operatorname{lct}_0(\operatorname{Spec} k[[u,t]]; B_{T,\infty})$. By Lemma 14.17(2), applied with the coefficients of **Step 3**, passing to a subsequence and discarding finitely many indices, we may assume that $\operatorname{lct}_0(T; B_{T,i}^*) > \lambda$. Thus $(T, \lambda B_{T,i}^*)$ is lc near 0, and so is $(T, \lambda B_{T,i})$.

We show that $(X, \lambda B_i^0)$ is lc near 0. In the notation of Lemma 14.5, let $c_j' := \lambda c_{i,j}$, $F_j' := F_{i,j}$, $a_j' := a_{i,j}$, and $b_j' := b_{i,j}$. Then $B' = \lambda B_i^0$, $c_x' = \lambda c_{x,i}$, and $c_y' = \lambda c_{y,i}$. For $c \geq 0$ and $\lambda \geq 1$,

$$1 - \frac{1-\lambda c}{q} = \lambda\Big(1 - \frac{1-c}{q}\Big) - (\lambda - 1)\Big(1 - \frac{1}{q}\Big) \leq \lambda\Big(1 - \frac{1-c}{q}\Big),$$

and likewise $1 - \frac{1-\lambda c}{p} \leq \lambda\Big(1 - \frac{1-c}{p}\Big)$. Thus $0 \leq c_u' \leq \lambda c_{u,i}$, and $0 \leq B_T' \leq \lambda B_{T,i}$. Hence (T, B_T') is lc near 0, and Lemma 14.5(2) implies that $(X, \lambda B_i^0)$ is lc near 0.

We show that $(X, \lambda B_i)$ is lc near 0. Let l_1 be the integer of Theorem 2.19(2) for the threefold X and the DCC set $\lambda(\Gamma\backslash\{0\})$. Discarding finitely many indices, we may assume that $N_i \geq l_1$. Since $h_{i,j} - h_{i,j}^0 \in (x,y)^{N_i} \subset \mathfrak{m}_0^{l_1}$, where $\mathfrak{m}_0$ is the maximal ideal of $0 \in X$, Theorem 2.19(2), applied at the closed point 0 to the $\mathbb{R}$-ideals $\prod_j (h_{i,j})^{\lambda c_{i,j}}$ and $\prod_j (h_{i,j}^0)^{\lambda c_{i,j}}$, implies that

$$\operatorname{mld}(X \ni 0, \lambda B_i) = \operatorname{mld}(X \ni 0, \lambda B_i^0) \geq 0.$$

Thus $(X, \lambda B_i)$ is lc near 0, since otherwise $\operatorname{mld}(X \ni 0, \lambda B_i) = -\infty$ [Amb99, Section 1, the second paragraph after Definition 1.1].

Let r_0 and N_0 be the numbers of Proposition 14.15 for λ, P, e_0, and Γ. Discarding finitely many indices, we may assume that $r_i \geq r_0$ and $N_i \geq N_0$. By assumption, $h_{i,j}^0$ is the sum of all terms of $h_{i,j}$ of $\mathbf{T}$-weight $pa_{i,j} - qb_{i,j}$, $G_i \cap \mathbf{T} = \mu_{r_i}(p, -q, 0)$, and $|H_i| \leq e_0$ by **Step 1**. We have shown that $(X, \lambda B_i^0)$, $(X, \lambda B_i)$, and $(T, \lambda B_{T,i})$ are lc near 0. Thus Proposition 14.15 and (14.56) imply that $\alpha_i \geq b_{H_i} \geq b_\infty$, which contradicts (14.56). Hence $(\operatorname{Spec} k[[u,t]], B_{T,\infty})$ is not klt.

Step 5. In this step we show that $B_{T,\infty} = C + \Delta$ for a smooth curve C, and we choose a coordinate $z \in \{u, t\}$ which is a parameter of C.

By **Steps 3** and 4, $(\operatorname{Spec} k[[u,t]], B_{T,\infty})$ is lc but not klt. No formal prime divisor over $\operatorname{Spec} k[[u,t]]$ with center 0 has log discrepancy 0 with respect to $B_{T,\infty}$: otherwise such a formal prime divisor P gives $\operatorname{mld}((\operatorname{Spec} k[[u,t]] \ni 0, B_{T,\infty})/H) \le a(P, \operatorname{Spec} k[[u,t]], B_{T,\infty})/|I_H(P)| = 0$ by Definition 3.2, which contradicts (14.59). A prime divisor over $\operatorname{Spec} k[[u,t]]$ is either a prime divisor on $\operatorname{Spec} k[[u,t]]$ or has center 0. Thus some prime divisor on $\operatorname{Spec} k[[u,t]]$ has coefficient 1 in $B_{T,\infty}$. If the sum C_1 of the prime divisors with coefficient 1 in $B_{T,\infty}$ had $\operatorname{mult}_0 C_1 \ge 2$, then the exceptional divisor E of the blow-up of 0 would satisfy $a(E, \operatorname{Spec} k[[u,t]], B_{T,\infty}) = 2 - \operatorname{mult}_0 B_{T,\infty} \le 0$ by Lemma 3.3(3) and [KM98, Lemma 2.29], hence $= 0$ since the pair is lc, a contradiction. Thus exactly one prime divisor C has coefficient 1 in $B_{T,\infty}$, and $\operatorname{mult}_0 C = 1$, so C is smooth. Write

$$B_{T,\infty} = C + \Delta,$$

where $\Delta \ge 0$ does not contain C. Since F_j^* and u are H-semi-invariant, H preserves $B_{T,\infty}$, hence C.

Let g be a generator of the ideal of C. For $\rho \in H$, write $\rho^* g = w_\rho g$ with a unit w_ρ, and let $\chi(\rho) := w_\rho(0)$. Then χ is a character of H. Let

$$f := \frac{1}{|H|} \sum_{\rho \in H} \chi(\rho)^{-1} \rho^* g.$$

Then f is H-semi-invariant with character χ, and $f = g \cdot \frac{1}{|H|} \sum_{\rho \in H} \chi(\rho)^{-1} w_\rho$, where the second factor is a unit with value 1 at 0. Thus f is an H-semi-invariant generator of the ideal of C. Since C is smooth, $f \notin (u,t)^2$. If $\frac{\partial f}{\partial u}(0) \ne 0$, let $z := t$. Otherwise $\frac{\partial f}{\partial t}(0) \ne 0$, and we let $z := u$. Then f, z are regular parameters of $k[[u,t]]$, and $z|_C$ is a parameter of $C \cong \operatorname{Spec} k[[z]]$.

Let χ_u, χ_t, χ_f, and χ_z be the characters of u, t, f, and z. Since all monomials of an H-semi-invariant power series have the character of the series and f has a non-zero linear term in the coordinate other than z, we have $\chi_f = \chi_u$ if $z = t$ and $\chi_f = \chi_t$ if $z = u$. Let $I := |\chi_z(H)|$ and $H_0 := \operatorname{Ker}(\chi_z) \subset H$. We distinguish two cases.

(i) C is neither $(u = 0)$ nor $(t = 0)$. Suppose that $z = t$. Then $f(0,t) \ne 0$, since otherwise u divides f and $C = (u = 0)$. Let $m \ge 1$ be the order of $f(0,t)$. Then $\chi_u = \chi_f = \chi_t^m$. Hence any element of H_0 fixes t and u, so it is the identity, as H acts faithfully on T. If $z = u$, then $f(u,0) \ne 0$, since otherwise t divides f and $C = (t = 0)$, and with $m \ge 1$ the order of $f(u,0)$ we have $\chi_t = \chi_f = \chi_u^m$, so again any element of H_0 fixes u and t and is the identity. Thus $H_0 = \{1\}$ and $I = |H|$.

(ii) C is $(u = 0)$ or $(t = 0)$. Then we take $f := u$ or $f := t$, respectively, and z is the other coordinate. Since H acts faithfully on T and H_0 fixes z, the restriction of χ_f to H_0 is injective. Thus H_0 is cyclic, $k[[f,z]]^{H_0} = k[[f^{m_0}, z]]$ for $m_0 := |H_0|$, and H/H_0 acts diagonally on f^{m_0} and z, faithfully on the line $(f^{m_0} = 0)$, with $|H/H_0| = I$.

Dividing the entries F_j^*, $j \in J_1$, and u by the highest powers of f dividing them, we write $\Delta = \sum_{j=1}^{s'} \delta_j \operatorname{div}(g_j)$, where $\delta_j > 0$ and the $g_j \in k[[u,t]]$ are H-semi-invariant and not divisible by f. Let

$$\beta_0 := \sum_{j=1}^{s'} \delta_j \operatorname{ord}_0(g_j|_C), \tag{14.60}$$

where $g_j|_C \in k[[z]]$ is the restriction. In case (ii), the power series $g_j(0,z)$ in the coordinates (f,z) is non-zero, so g_j has a monomial z^a, and the character of g_j is χ_z^a, which is trivial on H_0. Hence $g_j \in k[[f^{m_0}, z]]$ in case (ii).

Step 6. In this step we show that $\beta_0 < 1$ and

$$b_\infty = \frac{1 - \beta_0}{I}. \tag{14.61}$$

Let $\sigma \in (0,1)$ with $\sigma\beta_0 < 1$. We show that

$$\operatorname{mld}((\operatorname{Spec} k[[u,t]] \ni 0, C + \sigma\Delta)/H) = \frac{1-\sigma\beta_0}{I}. \tag{14.62}$$

Let ℓ_σ be the integer of Lemma 14.14(1) for H and the finite set $\{1\} \cup \{\sigma\delta_j\}_j$, and let $n_1 > \ell_\sigma$ be an integer, depending on σ, with $n_1 > \operatorname{ord}_0(g_j|_C)$ for any j. Let g'_j be the part of degree less than n_1 of g_j as a power series in f and z. Since H acts diagonally on f and z, g'_j is an H-semi-invariant polynomial in f and z. Moreover $g_j - g'_j \in (f,z)^{n_1} = (u,t)^{n_1}$, and $\operatorname{ord}_0 g'_j(0,z) = \operatorname{ord}_0(g_j|_C)$ since $n_1 > \operatorname{ord}_0(g_j|_C)$. Let H act on $\mathbb{A}^2_{f,z}$ by the characters χ_f and χ_z. Then the identification of the completion of $\mathbb{A}^2_{f,z}$ at 0 with $k[[u,t]]$ is H-equivariant. By Lemma 14.14(1) and Lemma 3.3(3), the left side of (14.62) is equal to

$$\operatorname{mld}\Big(\Big(\mathbb{A}^2_{f,z} \ni 0, \operatorname{div}(f) + \sigma \sum_j \delta_j \operatorname{div}(g'_j)\Big)/H\Big).$$

In case (i), H acts faithfully on the line $(f=0) = \mathbb{A}^1_z$, and near 0 the restriction of $\sigma\sum_j \delta_j \operatorname{div}(g'_j)$ to $\mathbb{A}^1_z$ is $\sigma\beta_0 \cdot [0]$, which is klt near 0 because $\sigma\beta_0 < 1$. Thus Theorem 8.3 (Remark 14.18) with $n=2$, $c=1$, $f_1 = f$, the polynomials g'_j, and the coefficients $\sigma\delta_j$ implies that the last minimal log discrepancy is equal to $\operatorname{mld}((\mathbb{A}^1_z \ni 0, \sigma\beta_0 \cdot [0])/H)$. The group H acts on $\mathbb{A}^1_z$ through χ_z, and its image $\chi_z(H)$ is a cyclic group of order I acting faithfully, so the quotient morphism $\mathbb{A}^1_z \to \mathbb{A}^1_z/H \cong \mathbb{A}^1$ has ramification index I at 0. By Lemma 2.6(4) with $b = \sigma\beta_0$ and $e_P = I$, the crepant quotient of $(\mathbb{A}^1_z, \sigma\beta_0 \cdot [0])$ is $(\mathbb{A}^1, (1 - \frac{1-\sigma\beta_0}{I}) \cdot [0])$ near the image of 0, which has minimal log discrepancy $\frac{1-\sigma\beta_0}{I}$ at 0.

In case (ii), the g'_j are polynomials in $f' := f^{m_0}$ and z by **Step 5**. The quotient morphism $\mathbb{A}^2_{f,z} \to \mathbb{A}^2_{f,z}/H_0 = \mathbb{A}^2_{f',z}$ is ramified only along $(f=0)$, with index m_0, so the crepant quotient of $(\mathbb{A}^2_{f,z}, \operatorname{div}(f) + \sigma\sum_j \delta_j \operatorname{div}(g'_j))$ by H_0 is $(\mathbb{A}^2_{f',z}, \operatorname{div}(f') + \sigma\sum_j \delta_j \operatorname{div}(g'_j))$ (Definition 2.5). Since the crepant identity $K_{X'} + B' = \pi^*(K_{X'/G'} + B'_{X'/G'})$ of Lemma 2.6(0) determines $B'_{X'/G'}$, the crepant quotient by H is the crepant quotient by H/H_0 of $(\mathbb{A}^2_{f',z}, \operatorname{div}(f') + \sigma\sum_j \delta_j \operatorname{div}(g'_j))$. The group H/H_0 acts faithfully on $(f'=0) = \mathbb{A}^1_z$ through χ_z, and the restriction of $\sigma\sum_j \delta_j \operatorname{div}(g'_j)$ to $\mathbb{A}^1_z$ is again $\sigma\beta_0 \cdot [0]$ near 0. Thus the argument of case (i) for $\mathbb{A}^2_{f',z}$ and H/H_0 implies (14.62).

Suppose that $\beta_0 \geq 1$. For $\sigma \in (0, 1/\beta_0)$, we have $C + \sigma\Delta \leq B_{T,\infty}$, so (14.59) and (14.62) imply that $b_\infty \leq \frac{1-\sigma\beta_0}{I}$. Letting $\sigma \to 1/\beta_0$, we obtain $b_\infty \leq 0$, which contradicts (14.59). Thus $\beta_0 < 1$.

For any $\sigma \in (0,1)$, we have $\sigma\beta_0 < 1$ and $C + \sigma\Delta \leq B_{T,\infty}$, so (14.59) and (14.62) imply that $b_\infty \leq \frac{1-\sigma\beta_0}{I}$, and letting $\sigma \to 1$ we obtain $b_\infty \leq \frac{1-\beta_0}{I}$. Conversely, let P be a formal prime divisor over $\operatorname{Spec} k[[u,t]]$ with center the closed point, and let $\operatorname{mult}_P \Delta := \sum_j \delta_j \operatorname{ord}_P g_j$. Then

$$a(P, \operatorname{Spec} k[[u,t]], C + \sigma\Delta) = a(P, \operatorname{Spec} k[[u,t]], C) - \sigma \operatorname{mult}_P \Delta$$

is an affine function of σ. For $\sigma \in (0,1)$, (14.62) and Definition 3.2 imply that $a(P, \operatorname{Spec} k[[u,t]], C + \sigma\Delta)/|I_H(P)| \geq \frac{1-\sigma\beta_0}{I}$. Letting $\sigma \to 1$, we obtain $a(P, \operatorname{Spec} k[[u,t]], B_{T,\infty})/|I_H(P)| \geq \frac{1-\beta_0}{I}$. Taking the infimum over P, we obtain $b_\infty \geq \frac{1-\beta_0}{I}$ by Definition 3.2 and (14.59), and (14.61) holds.

Step 7. In this step we show that the coordinate z with a fixed coefficient can be added to $B_{T,\infty}$ keeping the pair lc.

Let

$$c_z := I b_\infty = 1 - \beta_0 \in (0,1], \tag{14.63}$$

where the equality is (14.61) and $c_z > 0$ by (14.59). We show that

$$(\operatorname{Spec} k[[u,t]], B_{T,\infty} + c_z \operatorname{div}(z)) \text{ is lc.} \tag{14.64}$$

In the coordinates (f, z),

$$B_{T,\infty} + c_z \operatorname{div}(z) = \operatorname{div}(f) + \sum_j \delta_j \operatorname{div}(g_j) + c_z \operatorname{div}(z),$$

and $(z = 0) \neq C$. Fix a formal prime divisor ν over $\operatorname{Spec} k[[u, t]]$ with center 0. Let n_2 be an integer, depending on ν, with $n_2 > \operatorname{ord}_0(g_j|_C)$ and $n_2 \operatorname{ord}_\nu(u, t) > \operatorname{ord}_\nu g_j$ for any j, and let g_j'' be the part of degree less than n_2 of g_j as a power series in f and z. Then $\operatorname{ord}_\nu(g_j - g_j'') \geq n_2 \operatorname{ord}_\nu(u, t) > \operatorname{ord}_\nu g_j$, so $\operatorname{ord}_\nu g_j'' = \operatorname{ord}_\nu g_j$. Consider the pair $(\mathbb{A}^2_{f,z}, \operatorname{div}(f) + \sum_j \delta_j \operatorname{div}(g_j'') + c_z \operatorname{div}(z))$ over k. Near 0, the restriction of $\sum_j \delta_j \operatorname{div}(g_j'') + c_z \operatorname{div}(z)$ to the smooth line $(f = 0) = \mathbb{A}^1_z$ is $(\beta_0 + c_z) \cdot [0] = [0]$ by (14.60) and (14.63), and $(\mathbb{A}^1_z, [0])$ is lc near 0. Since $(f = 0)$ is a smooth Cartier divisor on the smooth surface $\mathbb{A}^2_{f,z}$, the different is the restriction $[0]$, and Theorem 2.23 (Remark 14.18), applied on an open neighborhood U of 0 in $\mathbb{A}^2_{f,z}$ such that the restriction to $(f = 0) \cap U$ is $[0]$, implies that the pair on $\mathbb{A}^2_{f,z}$ is lc near 0. By Lemma 3.3(3), ν restricts to a prime divisor ν_0 over $\mathbb{A}^2_{f,z}$ with center 0, and the log discrepancy of ν with respect to the completion of the pair on $\mathbb{A}^2_{f,z}$ is equal to the log discrepancy of ν_0 with respect to the pair on $\mathbb{A}^2_{f,z}$, which is non-negative. Since $\operatorname{ord}_\nu g_j'' = \operatorname{ord}_\nu g_j$ for any j, the log discrepancy of ν with respect to $B_{T,\infty} + c_z \operatorname{div}(z)$ is equal to the log discrepancy of ν with respect to the completion of the pair on $\mathbb{A}^2_{f,z}$, so it is non-negative. Thus any formal prime divisor with center 0 has non-negative log discrepancy with respect to $B_{T,\infty} + c_z \operatorname{div}(z)$. For the exceptional divisor E of the blow-up of 0, this implies that $2 - \operatorname{mult}_0(B_{T,\infty} + c_z \operatorname{div}(z)) \geq 0$ by Lemma 3.3(3) and [KM98, Lemma 2.29]. Since C has coefficient 1, any prime divisor on $\operatorname{Spec} k[[u, t]]$ other than C has coefficient at most 1 in $B_{T,\infty} + c_z \operatorname{div}(z)$. As a prime divisor over $\operatorname{Spec} k[[u, t]]$ is either a prime divisor on $\operatorname{Spec} k[[u, t]]$ or has center 0, (14.64) holds.

Step 8. In this step we show that $\alpha_i \geq b_\infty$ for some i, which contradicts (14.56).

Recall that $z \in \{u, t\}$ and $c_z = Ib_\infty$ are fixed in **Steps 5** and 7. Let $D_i := B^*_{T,i} + c_z \operatorname{div}_T(z)$ and $D_\infty := B_{T,\infty} + c_z \operatorname{div}(z)$. Then D_i and D_∞ coincide near 0 with the divisors in Lemma 14.17(2) for the tuples of **Step 3** and their generic limit, with the coefficients c_j^* on the entries $F_{i,j}$, the coefficient c_u^* on u, and the coefficient c_z added on the entry z. By Lemma 14.17(2) and (14.64), on an infinite subsequence,

$$\operatorname{lct}_0(T; D_i) \to \operatorname{lct}_0(\operatorname{Spec} k[[u, t]]; D_\infty) \geq 1.$$

By [dFEM11, Theorem 1.1], regarding $\operatorname{div}(z)$ as a separate factor, the numbers $\operatorname{lct}_0(T; D_i)$ on the fixed smooth surface T satisfy the ACC, since the exponents belong to the finite set $\{c_j^*\}_j \cup \{c_u^*, c_z\}$. If infinitely many of them were less than 1, then, since the limit is at least 1, they would contain a strictly increasing subsequence. Thus, discarding finitely many indices, (T, D_i) is lc near 0, and so is

$$(T, B_{T,i} + c_z \operatorname{div}_T(z)). \tag{14.65}$$

We regard z as the polynomial $x^q y^p$ or t on X. In the notation of Lemma 14.5, consider the coefficients $c_{i,j}$, the polynomials $F_{i,j}$, and the exponents $a_{i,j}, b_{i,j}$, together with one more term with coefficient c_z, polynomial z, and exponents $0, 0$. Then $c'_x = c_{x,i}$, $c'_y = c_{y,i}$, $c'_u = c_{u,i}$, $B'_T = B_{T,i} + c_z \operatorname{div}_T(z)$, and $B' = B^0_i + c_z \operatorname{div}(z)$. By (14.65) and Lemma 14.5(2), $(X, B^0_i + c_z \operatorname{div}(z))$ is lc near 0. Let l_2 be the integer of Theorem 2.19(2) for X and the DCC set $(\Gamma \backslash \{0\}) \cup \{c_z\}$. Discarding finitely many indices, we may assume that $N_i \geq l_2$. The $\mathbb{R}$-ideals $\prod_j (h_{i,j})^{c_{i,j}} \cdot (z)^{c_z}$ and $\prod_j (h^0_{i,j})^{c_{i,j}} \cdot (z)^{c_z}$ have corresponding factors congruent modulo $\mathfrak{m}_0^{l_2}$, since $h_{i,j} - h^0_{i,j} \in (x, y)^{N_i}$ and the factors (z) are equal. Thus Theorem 2.19(2) and [Amb99, Section 1, the second paragraph after Definition 1.1], as in **Step 4**, imply that

$$(X, B_i + c_z \operatorname{div}(z)) \text{ is lc near } 0. \tag{14.66}$$

Since $\mathbf{T}$ fixes u and t, the character of z on G_i is trivial on $G_i \cap \mathbf{T}$, and it factors through H with image of order I, where H is the group fixed in **Step 1** and $I = |\chi_z(H)|$ is as in **Step 5**. Thus z^I is a G_i-invariant regular function on X vanishing at 0, that is, a regular function on X/G_i vanishing at the image of 0. Let E be any prime divisor over X/G_i with center the image of 0, and let F be a prime divisor over X over E, with ramification index e. Since G_i fixes 0, the center of F on X is 0. Then $\operatorname{ord}_E(z^I) \geq 1$ and $I \operatorname{ord}_F(z) = e \operatorname{ord}_E(z^I) \geq e$. By (14.54) and (14.66),

$$\begin{aligned} a(E, (X, B_i)/G_i) &= \frac{a(F, X, B_i)}{e} \\ &= \frac{a(F, X, B_i + c_z \operatorname{div}(z)) + c_z \operatorname{ord}_F(z)}{e} \geq \frac{c_z \operatorname{ord}_F(z)}{e} \geq \frac{c_z}{I} = b_\infty. \end{aligned}$$

Since E is arbitrary, $\alpha_i \geq b_\infty$, which contradicts (14.56).

By **Steps 4** and 8, both the case that $(\operatorname{Spec} k[[u,t]], B_{T,\infty})$ is klt and the case that it is not klt contradict $\alpha_i < \alpha_\infty \leq b_\infty$ in (14.56). Thus the sequence α_i is not strictly increasing. □

14.8. **The mixed-circle class.** In this subsection, we prove Theorem 14.3.

Proof of Theorem 14.3. Since $\Gamma \cup \{1\} \subset [0,1]$ is a DCC set containing Γ, possibly replacing Γ with $\Gamma \cup \{1\}$, we may assume that $\Gamma \backslash \{0\} \neq \emptyset$. Let $\gamma := \gamma(\Gamma)$. By Lemma 2.10(2), $\gamma > 0$. Let $P := \max\{p, q\}$.

Suppose that the theorem does not hold. Then for any $i \geq 1$, there exist a finite diagonal abelian group G_i acting faithfully on X with $[G_i : G_i \cap \mathbf{T}] \leq e_0$, positive real numbers $c_{i,1}, \dots, c_{i,s_i} \in \Gamma$, and non-zero G_i-semi-invariant polynomials $h_{i,1}, \dots, h_{i,s_i}$, such that (X, B_i) is klt near 0, where $B_i := \sum_j c_{i,j} \operatorname{div}(h_{i,j})$, and the numbers $\operatorname{mld}((X \ni 0, B_i)/G_i)$ form a strictly increasing sequence. Let $r_i := |G_i \cap \mathbf{T}|$. Possibly passing to a subsequence, we may assume that either $r_i \to +\infty$, or there exists a positive integer m such that $r_i \leq m$ for any i.

Step 1. In this step we show that the data with $|G \cap \mathbf{T}| \geq 6P/\gamma$ are as in Set-up 14.1.

Let G be a finite diagonal abelian group acting faithfully on X with $[G : G \cap \mathbf{T}] \leq e_0$, let $c_1, \dots, c_s \in \Gamma$ be positive real numbers, let $h_1, \dots, h_s$ be non-zero G-semi-invariant polynomials with characters $\chi_1, \dots, \chi_s$, and let $B := \sum_j c_j \operatorname{div}(h_j)$. Assume that (X, B) is klt near 0, and let $r := |G \cap \mathbf{T}|$.

Since p and q are coprime, $\lambda^p = \lambda^q = 1$ implies $\lambda = 1$ for any $\lambda \in \mathbb{C}^*$. Thus the homomorphism $\mathbb{C}^* \to \mathbf{T}$ given by $\lambda \to (\lambda^p, \lambda^{-q}, 1)$ is an isomorphism. Since $G \cap \mathbf{T}$ is a subgroup of $\mathbf{T}$ of order r, it is the image of the group of r-th roots of unity, that is,

$$G \cap \mathbf{T} = \mu_r(p, -q, 0).$$

Let E be the exceptional divisor of the blow-up of X at 0. By [KM98, Lemma 2.29],

$$a(E, X, B) = 3 - \sum_j c_j \operatorname{ord}_0(h_j).$$

Since (X, B) is klt near 0 and $\operatorname{center}_X E = \{0\}$, we have $a(E, X, B) > 0$, hence $\sum_j c_j \operatorname{ord}_0(h_j) < 3$. Since $c_j \geq \gamma$ for any j, we obtain

$$\operatorname{ord}_0(h_j) < \frac{3}{\gamma} \quad \text{for any } j. \tag{14.67}$$

Fix an index j. Pick a monomial $x^{k_j} y^{l_j} t^{d_j}$ occurring in h_j with $k_j + l_j + d_j = \operatorname{ord}_0(h_j)$, let $\kappa_j := pk_j - ql_j$, and let h_j^0 be the sum of all terms of h_j of $\mathbf{T}$-weight κ_j. Then $h_j^0 \neq 0$, and by (14.67),

$$|\kappa_j| \leq P(k_j + l_j) < \frac{3P}{\gamma}. \tag{14.68}$$

For any $\sigma = (\sigma_1, \sigma_2, \sigma_3) \in G$ and any monomial $x^k y^l t^d$, we have

$$\sigma^*(x^k y^l t^d) = \sigma_1^k \sigma_2^l \sigma_3^d x^k y^l t^d.$$

Comparing the coefficients of $\sigma^* h_j$ and $\chi_j(\sigma) h_j$ for any $\sigma \in G$, we see that any monomial occurring in h_j is G-semi-invariant with character χ_j. Thus h_j^0 is G-semi-invariant with character χ_j. Let $R_j := h_j - h_j^0$. Then $h_j = h_j^0 + R_j$, and R_j is either zero or G-semi-invariant with character χ_j.

We describe h_j^0. Let $x^k y^l t^d$ and $x^{k'} y^{l'} t^{d'}$ be two monomials occurring in h_j^0. Then $pk - ql = pk' - ql' = \kappa_j$, that is, $p(k - k') = q(l - l')$. Since p and q are coprime, $(k - k', l - l') = n(q, p)$ for some $n \in \mathbb{Z}$. Let $x^{a_j} y^{b_j} t^d$ be a monomial occurring in h_j^0 such that a_j is minimal among the exponents of x in the monomials occurring in h_j^0. Then any monomial occurring in h_j^0 is of the form

$$x^{a_j + nq} y^{b_j + np} t^{d'} = x^{a_j} y^{b_j} u^n t^{d'}$$

for some non-negative integers n and d', where $u = x^q y^p$. Thus

$$h_j^0 = x^{a_j} y^{b_j} F_j(u, t)$$

for a non-zero polynomial F_j. Moreover, $pa_j - qb_j = \kappa_j$, so h_j^0 is the sum of all terms of h_j of $\mathbf{T}$-weight $pa_j - qb_j$.

We bound the tail R_j. Assume that $r \geq 6P/\gamma$, and let $x^k y^l t^d$ be a monomial occurring in R_j. Let $\zeta := e^{2\pi\sqrt{-1}/r}$ and $\sigma := (\zeta^p, \zeta^{-q}, 1)$. Then $\sigma \in G$, and

$$\sigma^*(x^k y^l t^d) = \zeta^{pk-ql} x^k y^l t^d \quad \text{and} \quad \sigma^*(x^{k_j} y^{l_j} t^{d_j}) = \zeta^{\kappa_j} x^{k_j} y^{l_j} t^{d_j}.$$

Since both monomials have character χ_j, we have $\zeta^{pk-ql} = \zeta^{\kappa_j}$, that is, $pk - ql - \kappa_j \in r\mathbb{Z}$. Since $x^k y^l t^d$ does not occur in h_j^0, $pk - ql \neq \kappa_j$, hence $|pk - ql - \kappa_j| \geq r$. By (14.68),

$$P(k + l) \geq |pk - ql| \geq r - |\kappa_j| > r - \frac{3P}{\gamma} \geq \frac{r}{2}.$$

Let $N := \lceil \frac{r}{2P} \rceil$. Since $k + l$ is an integer and $k + l > \frac{r}{2P}$, we have $k + l \geq N$. Thus $R_j \in (x, y)^N$.

Thus, if $r \geq 6P/\gamma$, then G, $h_j = h_j^0 + R_j$, c_j, B, r, and N are data as in Set-up 14.1 with $G \cap \mathbf{T} = \mu_r(p, -q, 0)$ and $|G/(G \cap \mathbf{T})| \leq e_0$, and h_j^0 is the sum of all terms of h_j of $\mathbf{T}$-weight $pa_j - qb_j$ for any j.

Step 2. In this step we show that $r_i \to +\infty$ is impossible.

Assume that $r_i \to +\infty$. Possibly passing to a subsequence, we may assume that $r_i \geq 6P/\gamma$ for any i. By **Step 1**, for any i, there exist decompositions $h_{i,j} = h_{i,j}^0 + R_{i,j}$ such that, with $N_i := \lceil \frac{r_i}{2P} \rceil$, the i-th data satisfy all hypotheses of Theorem 14.19 except possibly $r_i \to +\infty$ and $N_i \to +\infty$. Moreover, $r_i \to +\infty$ and $N_i \to +\infty$. By Theorem 14.19, the numbers $\mathrm{mld}((X \ni 0, B_i)/G_i)$ do not form a strictly increasing sequence, a contradiction.

Step 3. In this step we show that $r_i \leq m$ for any i is impossible, which finishes the proof.

Assume that $r_i \leq m$ for any i. Then

$$|G_i| = |G_i \cap \mathbf{T}| \cdot [G_i : G_i \cap \mathbf{T}] \leq me_0.$$

For any $g = (g_1, g_2, g_3) \in G_i$, the order of g divides $|G_i|$, which divides $(me_0)!$. Thus g_1, g_2, g_3 are roots of unity of order dividing $(me_0)!$, and

$$G_i \subset \mu_{(me_0)!}^3,$$

where $\mu_{(me_0)!} \subset \mathbb{C}^*$ is the group of $(me_0)!$-th roots of unity. Since $\mu_{(me_0)!}^3$ is a finite group, it has only finitely many subgroups. Thus there exist only finitely many quotients X/G_i.

For any i, let $\pi_i : X \to X/G_i$ be the quotient morphism, write $(X, B_i)/G_i = (X/G_i, (B_i)_{X/G_i})$, and let B_i' be the sum of the terms bS of $(B_i)_{X/G_i}$, where S is a prime divisor passing through

$\pi_i(0)$ and b is the coefficient of S in $(B_i)_{X/G_i}$. Then $B_i' = (B_i)_{X/G_i}$ near $\pi_i(0)$. Since the minimal log discrepancy at a closed point depends only on a neighborhood of the point,

$$\mathrm{mld}((X \ni 0, B_i)/G_i) = \mathrm{mld}(X/G_i \ni \pi_i(0), B_i').$$

By [KM98, Lemma 5.16], X/G_i is $\mathbb{Q}$-factorial, hence $K_{X/G_i} + B_i'$ is $\mathbb{R}$-Cartier, and $(X/G_i, B_i')$ is a pair.

We show that the coefficients of B_i' belong to

$$\Lambda := \Psi(\Gamma_+ \cap [0,1]) \backslash \{0\}.$$

Since $h_{i,j}$ is G_i-semi-invariant, $\sigma^* \operatorname{div}(h_{i,j}) = \operatorname{div}(h_{i,j})$ for any $\sigma \in G_i$, hence G_i preserves B_i. Since (X, B_i) is klt near 0, by Lemma 2.10(5), there exists a G_i-invariant affine open neighborhood X' of 0 such that the coefficients of $(B_i|_{X'})_{X'/G_i}$ belong to $\Psi(\Gamma_+ \cap [0,1])$. Since X' is G_i-invariant, $X' = \pi_i^{-1}(X'/G_i)$, and X'/G_i is an open neighborhood of $\pi_i(0)$ in X/G_i on which $(B_i)_{X/G_i}$ restricts to $(B_i|_{X'})_{X'/G_i}$, because the ramification divisor of $\pi_i|_{X'}$ is the restriction of the ramification divisor of π_i. Any prime divisor S passing through $\pi_i(0)$ intersects X'/G_i, so the coefficient of S in B_i' is a coefficient of $(B_i|_{X'})_{X'/G_i}$. Thus the coefficients of B_i' belong to Λ. By Lemma 2.10(1), $\Gamma_+ \cap [0,1]$ is a DCC subset of $[0,1]$, hence $\Psi(\Gamma_+ \cap [0,1])$ satisfies the DCC, again by Lemma 2.10(1). Thus Λ is a DCC subset of $\mathbb{R}_{>0}$.

For any subgroup $G \subset \mu^3_{(me_0)!}$ acting faithfully on X, the quotient X/G is a normal threefold. By Theorem 2.19(1), the set of $\mathrm{mld}(X/G \ni \eta, B')$, where $\eta \in X/G$ is a scheme-theoretic point and $(X/G, B')$ is a pair with $B' \in \Lambda$, satisfies the ACC. By Lemma 2.10(4), the union $\mathcal{A}$ of the sets above over the finitely many such G satisfies the ACC. Since $\pi_i(0)$ is a closed point of X/G_i and $B_i' \in \Lambda$, the set $\mathcal{A}$ contains $\mathrm{mld}((X \ni 0, B_i)/G_i)$ for any i. Thus the numbers $\mathrm{mld}((X \ni 0, B_i)/G_i)$ do not form a strictly increasing sequence, a contradiction.

By **Steps 2 and 3**, the theorem holds. □

15. Singular points above one

In this section, we prove Proposition 15.1, the ACC for minimal log discrepancies in $[1, +\infty)$ of abelian quotients of isolated Gorenstein terminal cDV threefold singularities. We first use the lower bound on the minimal log discrepancy to bound the group up to a subgroup $\mu_r(1,-1,0,0)$ (Theorem 15.5). Bounded groups are treated in Proposition 15.6. In the remaining case, the equation is of type cA and contains the monomial xy, and we compare the quotient with a quotient of the plane $\mathbb{A}^2_{z,t}$ with a boundary determined by the equation and the boundary functions (Theorem 15.9).

Proposition 15.1. *Let $\Gamma \subset [0,1]$ be a DCC set. Then the set of numbers $\mathrm{mld}((X \ni o, B)/G)$ that belong to $[1, +\infty)$ satisfies the ACC, where $(X \ni o)$ is an isolated Gorenstein terminal cDV threefold singularity, G is a finite abelian group acting faithfully on X and fixing o, $X = (f = 0)$ for a G-semi-invariant $f \in \mathfrak{m}^2_{Y,o}$ under a G-equivariant closed embedding of X into a smooth fourfold germ $(Y \ni o)$, and $B = \sum_j c_j \operatorname{div}_X(h_j)$ with $c_j \in \Gamma$ positive, h_j non-zero regular G-semi-invariant functions, and (X, B) klt near o.*

We prove Proposition 15.1 in Subsection 15.5.

15.1. **The group dichotomy.** In this subsection, we prove Theorem 15.5: we bound the group up to a subgroup $\mu_r(1,-1,0,0)$ by passing to a limit of the groups in $\mathbb{T}^4$.

The following lemma bounds the weights of the equations along a convergent sequence of finite groups.

Lemma 15.2. *Let $(H_i)_{i\geq 1}$ be finite subgroups of $\mathbb{T}^d$ converging in the Hausdorff distance to a closed subgroup H, and let $\mathrm{Lie}(H^0) \subset \mathbb{R}^d$ be the Lie algebra of the identity component H^0 of H. Let $\alpha_0 \in \mathbb{Z}^d_{\geq 0}$, and for any i let f_i be a non-zero polynomial in $x_1, \ldots, x_d$ such that $\langle \alpha - \alpha_0, u\rangle \in \mathbb{Z}$ for any $u \in N_{H_i}$ and any exponent α of a monomial occurring in f_i. Let $\mathcal{A}$ be the set of $\alpha \in \mathbb{Z}^d_{\geq 0}$*

such that x^α occurs in f_i for infinitely many i and $\langle \alpha - \alpha_0, v\rangle = 0$ for any $v \in \mathrm{Lie}(H^0)$. Let $w_i \in \mathbb{R}^d_{>0}$ be vectors converging to $w \in \mathbb{R}^d_{>0}$, and L a real number with $\langle \alpha, w\rangle \geq L$ for any $\alpha \in \mathcal{A}$. Then $\liminf_i \mathrm{wt}_{w_i}(f_i) \geq L$.

Proof. Since $\mathrm{wt}_{w_i}(f_i) \geq 0$, we may assume that $L > 0$. By Lemma 4.3(1), $H^0 = \rho(\mathrm{Lie}(H^0))$. Choose $0 < \mu < \min_k w_k/2$. For $i \gg 0$, any coordinate of w_i is at least μ, so a monomial x^α with $\langle \alpha, w_i\rangle < L + 1$ satisfies $\sum_k \alpha_k < (L+1)/\mu$. Let S be the finite set of $\alpha \in \mathbb{Z}^d_{\geq 0}$ with $\sum_k \alpha_k < (L+1)/\mu$.

Let $\alpha \in S$ be an exponent such that x^α occurs in f_i for infinitely many i. Then the continuous homomorphism $\mathbb{T}^d \to \mathbb{T}^1$, $\rho(u) \to \langle \alpha - \alpha_0, u\rangle \bmod \mathbb{Z}$, vanishes on H_{i_l} for a subsequence $(H_{i_l})_{l\geq 1}$. Since the H_{i_l} converge to H in the Hausdorff distance, any point of H is a limit of points of H_{i_l}, and the homomorphism vanishes on H. In particular, $\langle \alpha - \alpha_0, v\rangle \in \mathbb{Z}$ for any $v \in \mathrm{Lie}(H^0)$, and by linearity $\langle \alpha - \alpha_0, v\rangle = 0$. Thus $\alpha \in \mathcal{A}$.

Hence, for $i \gg 0$, any monomial x^α occurring in f_i satisfies either $\alpha \notin S$, in which case $\langle \alpha, w_i\rangle \geq L + 1$, or $\alpha \in S \cap \mathcal{A}$. Therefore

$$\mathrm{wt}_{w_i}(f_i) \geq \min\Big\{L + 1,\ \min_{\alpha \in S\cap\mathcal{A}} \langle \alpha, w_i\rangle\Big\} \quad \text{for } i \gg 0.$$

Since $S \cap \mathcal{A}$ is finite, we obtain $\liminf_i \mathrm{wt}_{w_i}(f_i) \geq L$ from $\langle \alpha, w_i\rangle \to \langle \alpha, w\rangle \geq L$ for any $\alpha \in S \cap \mathcal{A}$. □

Lemma 15.3. *Let $X = (f = 0) \subset \mathbb{A}^4$ be a normal klt hypersurface germ at 0, G a finite diagonal abelian group acting faithfully on X and preserving f up to a character, and $B \geq 0$ a G-invariant $\mathbb{R}$-divisor on X such that (X, B) is klt near 0. Let N_G be the lattice of Definition 2.15 for the characters of the coordinates $x_1, \ldots, x_4$. Then*

$$\sum_{j=1}^4 w_j - \mathrm{wt}_w(f) \geq \mathrm{mld}((X \ni 0, B)/G) \quad \textit{for any } w \in N_G \cap \mathbb{R}^4_{>0}.$$

Proof. Let $\pi : X \to X/G$ be the quotient morphism. Since $B \geq 0$, the boundary of $(X, B)/G$ is the boundary of $(X, 0)/G$ plus $\frac{1}{|G|}\pi_* B \geq 0$ (Definition 2.5). Hence

$$\mathrm{mld}((X \ni 0, 0)/G) \geq \mathrm{mld}((X \ni 0, B)/G).$$

Let $(\mathbb{A}^4/G, \Delta) := (\mathbb{A}^4, \mathrm{div}(f))/G$. We take $n = 4$, $c = 1$, $f_1 = f$, $s = 1$, $h_1 = 1$, and $b_1 = 1$ in Theorem 8.3. Here 1 is a G-semi-invariant polynomial with trivial character and $\mathrm{div}(1) = 0$, X is integral near 0 since X is normal, and $(X, 0)$ is klt near 0 since (X, B) is klt near 0 and $B \geq 0$. Thus the boundaries of Theorem 8.3 are 0 on X and $\mathrm{div}(f)$ on $\mathbb{A}^4$, and any prime divisor over $\mathbb{A}^4/G$ with center the image of 0 has log discrepancy at least $\mathrm{mld}((X \ni 0, 0)/G)$ with respect to $(\mathbb{A}^4/G, \Delta)$.

Let $v \in N_G \cap \mathbb{R}^4_{>0}$ be primitive. By Lemma 8.5 (with $d = 4$, $s = 1$, $f_1 = f$, and $b_1 = 1$), v defines a toric prime divisor E_v over $\mathbb{A}^4/G$ with

$$a(E_v, \mathbb{A}^4/G, \Delta) = \sum_{j=1}^4 v_j - \mathrm{wt}_v(f).$$

Since all coordinates of v are positive, the center of E_v on $\mathbb{A}^4/G$ is the image of 0. Hence $\sum_j v_j - \mathrm{wt}_v(f) \geq \mathrm{mld}((X \ni 0, 0)/G)$. Since $(X, 0)$ is klt near 0, $(X, 0)/G$ is klt near the image of 0 [KM98, Proposition 5.20], so $\mathrm{mld}((X \ni 0, 0)/G) > 0$ (by [KM98, Corollary 2.31(3)] on a log resolution of $(X, 0)/G$ near the image of 0, together with [KM98, Lemma 2.30]). Since G is finite, N_G is a lattice containing $\mathbb{Z}^4$, so any $w \in N_G \cap \mathbb{R}^4_{>0}$ is of the form kv for a primitive vector $v \in N_G \cap \mathbb{R}^4_{>0}$ and an integer $k \geq 1$. Since $\mathrm{wt}_{kv}(f) = k\,\mathrm{wt}_v(f)$, we obtain

$$\sum_j w_j - \mathrm{wt}_w(f) = k\Big(\sum_j v_j - \mathrm{wt}_v(f)\Big) \geq k\,\mathrm{mld}((X \ni 0, 0)/G) \geq \mathrm{mld}((X \ni 0, 0)/G) \geq \mathrm{mld}((X \ni 0, B)/G).$$

□

Lemma 15.4. *For any $\epsilon > 0$ there exists a positive integer $N_4(\epsilon)$ with the following property. Let $X = (f = 0) \subset \mathbb{A}^4$ be a normal klt hypersurface germ at 0 with $\operatorname{mult}_0 f = 2$, G a finite diagonal abelian group acting faithfully on X and preserving f up to a character, and $B \geq 0$ a G-invariant $\mathbb{R}$-divisor such that (X, B) is klt and $\operatorname{mld}((X \ni 0, B)/G) \geq 1 + \epsilon$. Then one of the characters of the four coordinates has order at most $N_4(\epsilon)$.*

Proof. The proof is long, so we divide it into several steps. Let $x_1, \dots, x_4$ be the coordinates of $\mathbb{A}^4$.

Step 1. By Lemma 15.3, since $\operatorname{mld}((X \ni 0, B)/G) \geq 1 + \epsilon$, we have

$$\sum_{j=1}^{4} w_j - \operatorname{wt}_w(f) \geq 1 + \epsilon \quad \text{for any } w \in N_G \cap \mathbb{R}^4_{>0}, \tag{15.1}$$

where N_G is the lattice of Definition 2.15 for the characters of $x_1, \dots, x_4$.

Step 2. In this step we assume that $N_4(\epsilon)$ does not exist, pass to a limit subgroup of $\mathbb{T}^4$, and find a vector of the Lie algebra of the identity component of the limit with four non-zero coordinates.

Suppose that $N_4(\epsilon)$ does not exist. Then for any positive integer i there exist $X_i = (f_i = 0)$, G_i, and B_i satisfying the assumptions of the lemma such that the characters of $x_1, \dots, x_4$ on G_i have orders greater than i. Since G_i acts faithfully on X_i, G_i acts faithfully on $\mathbb{A}^4$, and we identify G_i with the image $H_i \subset \mathbb{T}^4$ of G_i under the characters of $x_1, \dots, x_4$. Let $p_j : \mathbb{T}^4 \to \mathbb{T}^1$ be the j-th projection. Then the order of the character of x_j on G_i is $|p_j(H_i)|$.

Since $\operatorname{mult}_0 f_i = 2$, some monomial of degree two occurs in f_i. There exist ten monomials of degree two in four variables, so possibly passing to a subsequence, there exists $\alpha_0 \in \mathbb{Z}^4_{\geq 0}$ with $\sum_j (\alpha_0)_j = 2$ such that x^{α_0} occurs in f_i for any i. Since G_i acts diagonally and f_i is G_i-semi-invariant, any monomial occurring in f_i has the character of x^{α_0}, that is,

$$\langle \alpha - \alpha_0, u \rangle \in \mathbb{Z} \quad \text{for any } u \in N_{H_i} \text{ and any exponent } \alpha \text{ of } f_i. \tag{15.2}$$

By Lemma 4.4, possibly passing to a subsequence, the H_i converge in the Hausdorff distance to a closed subgroup $H \subset \mathbb{T}^4$, and $H_i \subset H$ for $i \gg 0$. By Lemma 4.3(1), H/H^0 is finite and $H^0 = \rho(\operatorname{Lie}(H^0))$.

We show that the linear form $v \to v_j$ does not vanish on $\operatorname{Lie}(H^0)$ for any j. Suppose that $v_j = 0$ for any $v \in \operatorname{Lie}(H^0)$. Then $p_j(H^0) = 0$, so $p_j(H)$ is a quotient of H/H^0 and $|p_j(H)| \leq |H/H^0|$. Since $H_i \subset H$ for $i \gg 0$, we obtain $|p_j(H_i)| \leq |H/H^0|$ for $i \gg 0$, which contradicts $|p_j(H_i)| > i$. Since a real vector space is not a union of four proper linear subspaces, there exists $v \in \operatorname{Lie}(H^0)$ with $v_j \neq 0$ for any j.

For $0 < s < 1/\max_j |v_j|$, let $w(s) \in (0,1)^4$ be the vector with

$$w(s)_j = sv_j \ \text{ if } v_j > 0, \qquad w(s)_j = 1 + sv_j \ \text{ if } v_j < 0.$$

Then $\rho(w(s)) = \rho(sv) \in H^0$. Since the H_i converge to H in the Hausdorff distance and ρ is a local isometry, there exist $w_i \in N_{H_i}$ with $w_i \to w(s)$. For $i \gg 0$, $w_i \in \mathbb{R}^4_{>0}$, so (15.1) for X_i implies

$$\sum_j (w_i)_j - \operatorname{wt}_{w_i}(f_i) \geq 1 + \epsilon \quad \text{for } i \gg 0. \tag{15.3}$$

Step 3. In this step we choose the sign of v and the number s so that (15.3) fails, which proves the lemma. Since all coordinates of v are non-zero, one of the following cases holds.

Case 1. v or $-v$ has at most one negative coordinate. Possibly replacing v with $-v$, we may assume that v has at most one negative coordinate. Then $\sum_j w(s)_j$ is $s\sum_j v_j$ or $1 + s\sum_j v_j$, so for s small enough, $\sum_j w(s)_j < 1 + \epsilon$. Since $w_i \to w(s)$, we obtain $\sum_j (w_i)_j < 1 + \epsilon$ for $i \gg 0$. Since $\operatorname{wt}_{w_i}(f_i) \geq 0$, this contradicts (15.3).

Case 2. v has exactly two negative coordinates. Possibly replacing v with $-v$, we may assume that

$$\langle \alpha_0, v\rangle \le 0.$$

Let $Q := \{j \mid v_j < 0\}$, so that $|Q| = 2$. Fix ϵ_1 with $0 < \epsilon_1 < \epsilon/3$. Since $\sum_j w(s)_j = 2 + s\sum_j v_j$ and $w(s)_j = 1 + sv_j$ for $j \in Q$, we may choose s small enough so that

$$\sum_j w(s)_j < 2 + \epsilon_1 \quad \text{and} \quad w(s)_j > 1 - \epsilon_1 \ \text{ for any } j \in Q.$$

By (15.2), Lemma 15.2 is available for (H_i), α_0, and (f_i). Let $\mathcal{A}$ be the set of Lemma 15.2, and let $\alpha \in \mathcal{A}$. Since $f_i(0) = 0$, $\alpha \ne 0$. Moreover, $\langle \alpha, v\rangle = \langle \alpha_0, v\rangle \le 0$. If $\alpha_j = 0$ for any $j \in Q$, then $\langle \alpha, v\rangle = \sum_{j\notin Q} \alpha_j v_j > 0$, a contradiction. Thus $\alpha_j \ge 1$ for some $j \in Q$, and

$$\langle \alpha, w(s)\rangle \ge w(s)_j > 1 - \epsilon_1.$$

By Lemma 15.2 with $L = 1 - \epsilon_1$, $\liminf_i \mathrm{wt}_{w_i}(f_i) \ge 1 - \epsilon_1$. Hence, for $i \gg 0$, $\mathrm{wt}_{w_i}(f_i) > 1 - 2\epsilon_1$ and $\sum_j (w_i)_j < 2 + \epsilon_1$, so

$$\sum_j (w_i)_j - \mathrm{wt}_{w_i}(f_i) < 1 + 3\epsilon_1 < 1 + \epsilon,$$

which contradicts (15.3).

Thus $N_4(\epsilon)$ exists. □

Theorem 15.5. *For any $\epsilon > 0$ there exists an integer $e(\epsilon) \ge 3$ with the following property. Let $X = (f = 0) \subset \mathbb{A}^4$ be a normal integral isolated klt hypersurface germ at 0 with $\mathrm{mult}_0 f = 2$, and G a finite diagonal abelian group acting faithfully on X and preserving f up to a character, such that $\mathrm{mld}((X \ni 0, B)/G) \ge 1 + \epsilon$ for some G-invariant $\mathbb{R}$-divisor $B \ge 0$ with (X, B) klt. Then one of the following holds:*

(1) $|G| \le e(\epsilon)$;

(2) after a permutation of the coordinates x, y, z, t, the group $R := G \cap \mathbf{T}_{(1,-1,0,0)}$ is $\mu_r(1, -1, 0, 0)$ for an integer $r > 2$, $[G : R] \le e(\epsilon)$, the character of f is trivial on R, and the quadratic part of f is $cxy + q_2(z, t)$ with $c \ne 0$.

In particular, if the quadratic part of f has rank one, then $|G| \le e(\epsilon)$. The conclusion also holds if X is replaced by an isolated Gorenstein terminal cDV threefold singularity $(X \ni o)$ with a faithful action of a finite abelian group G fixing o and a G-equivariant closed embedding into a smooth fourfold germ $(Y \ni o)$ such that $X = (f = 0)$ for a G-semi-invariant $f \in \mathfrak{m}^2_{Y,o}$, and the coordinates x, y, z, t are replaced by G-eigenparameters of Y at o.

Proof of Theorem 15.5. The proof is long, so we divide it into several steps. We write x_1, x_2, x_3, x_4 for x, y, z, t, and $e_1, \dots, e_4$ for the standard basis of $\mathbb{R}^4$.

Step 1. By Lemma 15.3, since $\mathrm{mld}((X \ni 0, B)/G) \ge 1 + \epsilon$, we have

$$\sum_{j=1}^4 w_j - \mathrm{wt}_w(f) \ge 1 + \epsilon \quad \text{for any } w \in N_G \cap \mathbb{R}^4_{>0}, \tag{15.4}$$

where N_G is the lattice of Definition 2.15 for the characters of $x_1, \dots, x_4$.

Step 2. In this step we assume that $e(\epsilon)$ does not exist, pass to a limit subgroup H of $\mathbb{T}^4$, and derive an inequality (15.7) which we use in the remaining steps.

Suppose that no integer $e(\epsilon) \ge 3$ as in the theorem exists. Then for any integer $i \ge 3$ there exist $X_i = (f_i = 0)$, G_i, and B_i satisfying the assumptions of the theorem such that $|G_i| > i$ and (2) fails with $e(\epsilon)$ replaced by i for any permutation of the coordinates. Since G_i acts faithfully on X_i, G_i acts faithfully on $\mathbb{A}^4$, and we identify G_i with the image $H_i \subset \mathbb{T}^4$ of G_i under the characters of $x_1, \dots, x_4$. Let $p_j : \mathbb{T}^4 \to \mathbb{T}^1$ be the j-th projection.

Since $\mathrm{mult}_0 f_i = 2$ and there exist ten monomials of degree two in four variables, possibly passing to a subsequence, there exists $\alpha_0 \in \mathbb{Z}^4_{\geq 0}$ with $\sum_j (\alpha_0)_j = 2$ such that x^{α_0} occurs in f_i for any i. Since G_i acts diagonally and f_i is G_i-semi-invariant, the character of f_i is the character of x^{α_0}, and

$$\langle \alpha - \alpha_0, u \rangle \in \mathbb{Z} \quad \text{for any } u \in N_{H_i} \text{ and any exponent } \alpha \text{ of } f_i. \tag{15.5}$$

By Lemma 4.4, possibly passing to a subsequence, the H_i converge in the Hausdorff distance to a closed subgroup $H \subset \mathbb{T}^4$, and $H_i \subset H$ for $i \gg 0$. By Lemma 4.3(1), H/H^0 is finite, $H^0 = \rho(\mathrm{Lie}(H^0))$, and $\mathrm{Lie}(H^0)$ is spanned by $\mathrm{Lie}(H^0) \cap \mathbb{Z}^4$. If $H^0 = \{0\}$, then H is finite and $|H_i| \leq |H|$ for $i \gg 0$, which contradicts $|H_i| > i$. Thus $\mathrm{Lie}(H^0) \neq 0$.

Let $R_i := H_i \cap H^0$. For $i \gg 0$, the homomorphism $H_i/R_i \to H/H^0$ is injective, so

$$[H_i : R_i] \leq |H/H^0|,$$

and $|R_i| \to +\infty$ since $|H_i| > i$. We show that the R_i converge to H^0 in the Hausdorff distance. The connected components of H are finitely many disjoint compact sets, so there exists $d_0 > 0$ such that the distance between two distinct components is at least d_0. Let $i \gg 0$ be an integer such that $H_i \subset H$ and the Hausdorff distance d_i between H_i and H is less than d_0. For any $h \in H^0$ there exists $g \in H_i$ with distance at most d_i from h. Since $g \in H$, $g \in H^0$, so $g \in R_i$. Since $R_i \subset H^0$, the Hausdorff distance between R_i and H^0 is at most $d_i \to 0$. Hence, for $\beta \in \mathbb{Z}^4$,

$$\text{if } \langle \beta, u \rangle \in \mathbb{Z} \text{ for any } u \in N_{R_i} \text{ for infinitely many } i, \text{ then } \langle \beta, v \rangle = 0 \text{ for any } v \in \mathrm{Lie}(H^0). \tag{15.6}$$

Under the hypothesis of (15.6), the continuous homomorphism $\mathbb{T}^4 \to \mathbb{T}^1$, $\rho(u) \to \langle \beta, u \rangle \bmod \mathbb{Z}$, vanishes on H^0, so $\langle \beta, v \rangle \in \mathbb{Z}$ for any $v \in \mathrm{Lie}(H^0)$, and $\langle \beta, v \rangle = 0$ by linearity.

Let I be the set of indices j such that $v_j \neq 0$ for some $v \in \mathrm{Lie}(H^0)$, and let $k := |I|$. Then $k \geq 1$. For $j \notin I$, $p_j(H^0) = 0$, so R_i acts trivially on x_j. For $j \in I$, the order of the character of x_j on R_i tends to $+\infty$. Otherwise, there exist an integer M and a subsequence along which $p_j(R_i)$ is contained in the set F_M of elements of $\mathbb{T}^1$ of order at most M. Since F_M is finite and the R_i converge to H^0 in the Hausdorff distance, $p_j(H^0) \subset F_M$, which contradicts $p_j(H^0) = \rho(\{v_j \mid v \in \mathrm{Lie}(H^0)\}) = \mathbb{T}^1$.

Let λ be the linear form $v \to \langle \alpha_0, v \rangle$ on $\mathrm{Lie}(H^0)$, and let

$$\mathcal{A}_0 := \Big\{ \alpha \in \mathbb{Z}^4_{\geq 0} \;\Big|\; \sum_j \alpha_j \geq 2 \text{ and } \langle \alpha, v \rangle = \lambda(v) \text{ for any } v \in \mathrm{Lie}(H^0) \Big\}.$$

We show the following. Let $(f_{i_l})_{l \geq 1}$ be a subsequence of (f_i), and let $\mathcal{A}'$ be the set of $\alpha \in \mathcal{A}_0$ such that x^α occurs in f_{i_l} for infinitely many l. Let $w \in N_H \cap \mathbb{R}^4_{>0}$ and let L be a real number with $\langle \alpha, w \rangle \geq L$ for any $\alpha \in \mathcal{A}'$. Then

$$\sum_{j=1}^{4} w_j - L \geq 1 + \epsilon. \tag{15.7}$$

Since the H_{i_l} converge to H in the Hausdorff distance, $\rho(w) \in H$, and ρ is a local isometry, there exist $w_l \in N_{H_{i_l}}$ with $w_l \to w$, and $w_l \in \mathbb{R}^4_{>0}$ for $l \gg 0$. Since $\mathrm{mult}_0 f_i = 2$, any exponent α of f_i satisfies $\sum_j \alpha_j \geq 2$. Thus, by (15.5), the set $\mathcal{A}$ of Lemma 15.2 for (H_{i_l}), α_0, and (f_{i_l}) is $\mathcal{A}'$, and Lemma 15.2 implies $\liminf_l \mathrm{wt}_{w_l}(f_{i_l}) \geq L$. By (15.4) for X_{i_l}, $\sum_j (w_l)_j - \mathrm{wt}_{w_l}(f_{i_l}) \geq 1 + \epsilon$ for $l \gg 0$. Taking the limit superior as $l \to +\infty$, we obtain (15.7).

Since $\mathcal{A}' \subset \mathcal{A}_0$, taking $(f_{i_l}) = (f_i)$, we obtain (15.7) for any $w \in N_H \cap \mathbb{R}^4_{>0}$ and any real number L with $\langle \alpha, w \rangle \geq L$ for any $\alpha \in \mathcal{A}_0$.

Step 3. In this step we show that $k \neq 4$. Suppose that $k = 4$. By **Step 2**, the orders of the characters of $x_1, \ldots, x_4$ on R_i tend to $+\infty$, and so do the orders of the characters on $G_i \supset R_i$. Since X_i, G_i, and B_i satisfy the assumptions of Lemma 15.4, one of the characters of $x_1, \ldots, x_4$ on G_i has order at most $N_4(\epsilon)$, a contradiction for $i \gg 0$. Thus $1 \leq k \leq 3$.

Step 4. In this step we use that X_i has an isolated singularity at 0 to restrict λ. Let χ_f and χ_j be the characters of f_i and x_j restricted to R_i, and let

$$V := \{x_j = 0 \text{ for any } j \in I\} \subset \mathbb{A}^4,$$

so that $V \cong \mathbb{A}^{4-k}$ and R_i fixes V pointwise by **Step 2**. Since R_i fixes V pointwise, any R_i-semi-invariant polynomial whose character on R_i is non-trivial vanishes on V.

Suppose that χ_f is non-trivial. Then $f_i|_V = 0$. For any j, the derivative $\partial f_i/\partial x_j$ is semi-invariant with character $\chi_f\chi_j^{-1}$ on R_i. Since χ_j is trivial for $j \notin I$, $\partial f_i/\partial x_j$ vanishes on V unless $j \in I$ and $\chi_j = \chi_f$. Let I_f be the set of $j \in I$ with $\chi_j = \chi_f$, and $m := |I_f|$. For $j \in I_f$, $(\partial f_i/\partial x_j)|_V$ vanishes at 0 since $\mathrm{mult}_0 f_i = 2$. If $m < 4-k$, then the common zero locus $Z_i \subset V$ of the polynomials $(\partial f_i/\partial x_j)|_V$, $j \in I_f$, satisfies $\dim_0 Z_i \geq 4-k-m \geq 1$, since the maximal ideal of the local ring $\mathcal{O}_{V,0}$ has height $4-k$ and, by [Mat89, Theorem 13.6(ii)], its image modulo the m germs $(\partial f_i/\partial x_j)|_V$, $j \in I_f$, has height at least $4-k-m$. Since f_i and all the derivatives $\partial f_i/\partial x_j$ vanish on Z_i, Z_i is contained in the singular locus of X_i, which contradicts that X_i has an isolated singularity at 0. Hence $m \geq 4-k$.

Possibly passing to a subsequence, we may assume that whether χ_f is trivial, and the set of $j \in I$ with $\chi_j = \chi_f$, do not depend on i. If χ_f is trivial, then $\langle \alpha_0, u\rangle \in \mathbb{Z}$ for any $u \in N_{R_i}$, and (15.6) with $\beta = \alpha_0$ implies $\lambda = 0$. If $\chi_j = \chi_f$, then (15.6) with $\beta = \alpha_0 - e_j$ implies $\lambda(v) = v_j$ for any $v \in \mathrm{Lie}(H^0)$. Since $I_f \subset I$, we have $m \leq k$. Thus, if χ_f is non-trivial, then $k \neq 1$ since $4-k \leq m \leq k$, $I_f = I$ if $k = 2$ since $2 \leq m \leq 2$, and $I_f \neq \emptyset$ if $k = 3$ since $m \geq 1$. Therefore:

(a) if $k = 1$, then $\lambda = 0$;
(b) if $k = 2$, then $\lambda = 0$, or $\lambda(v) = v_j$ for any $j \in I$ and any $v \in \mathrm{Lie}(H^0)$;
(c) if $k = 3$, then $\lambda = 0$, or there exists $j \in I$ with $\lambda(v) = v_j$ for any $v \in \mathrm{Lie}(H^0)$.

Step 5. In this step we show that $k \neq 1$. Suppose that $k = 1$. After a permutation of the coordinates, $I = \{1\}$, so $\mathrm{Lie}(H^0) = \mathbb{R}e_1$, and $\lambda = 0$ by **Step 4**. Thus $\mathcal{A}_0$ is the set of α with $\alpha_1 = 0$ and $\sum_j \alpha_j \geq 2$. Let $0 < s < \min\{1, \epsilon\}$ and

$$w := (s, 1, 1, 1) = se_1 + (0, 1, 1, 1) \in N_H \cap \mathbb{R}^4_{>0}.$$

For any $\alpha \in \mathcal{A}_0$, $\langle \alpha, w\rangle = \alpha_2 + \alpha_3 + \alpha_4 \geq 2$. By (15.7) with $L = 2$, $1 + s = \sum_j w_j - 2 \geq 1 + \epsilon$, a contradiction.

Step 6. In this step we show that $k \neq 3$. Suppose that $k = 3$. After a permutation of the coordinates, $I = \{1, 2, 3\}$. Since $\mathrm{Lie}(H^0)$ is not a union of three proper linear subspaces, there exists $v \in \mathrm{Lie}(H^0)$ with $v_1, v_2, v_3 \neq 0$, and $v_4 = 0$. By **Step 4**, $\lambda(v) \in \{0, v_1, v_2, v_3\}$. The relation $\lambda(v) \in \{0, v_1, v_2, v_3\}$ still holds if v is replaced by $-v$ or x_1, x_2, x_3 are permuted.

Case 1. v_1, v_2, v_3 are all positive or all negative. Possibly replacing v with $-v$, we may assume that $v_1, v_2, v_3 > 0$. For $s > 0$ small enough, $w := sv + e_4 = (sv_1, sv_2, sv_3, 1) \in N_H \cap \mathbb{R}^4_{>0}$ and $\sum_j w_j = 1 + s(v_1 + v_2 + v_3) < 1 + \epsilon$. Since $\langle \alpha, w\rangle \geq 0$ for any α, (15.7) with $L = 0$ implies $\sum_j w_j \geq 1 + \epsilon$, a contradiction.

Case 2. v_1, v_2, v_3 are neither all positive nor all negative. Possibly replacing v with $-v$ and permuting x_1, x_2, x_3, we may assume that $v = (a, b, c, 0)$ with $0 < a \leq b$ and $c < 0$. Let $\ell := \lambda(v) \in \{0, a, b, c\}$. Any $\alpha \in \mathcal{A}_0$ satisfies

$$a\alpha_1 + b\alpha_2 + c\alpha_3 = \ell. \tag{15.8}$$

For $s > 0$ small enough, let

$$w^+(s) := sv + e_3 + e_4 = (sa, sb, 1 + sc, 1) \in N_H \cap \mathbb{R}^4_{>0}.$$

By (15.8), $\langle \alpha, w^+(s)\rangle = \alpha_3 + \alpha_4 + s\ell$ for any $\alpha \in \mathcal{A}_0$.

Suppose first that $\alpha_3 + \alpha_4 \geq 1$ for any $\alpha \in \mathcal{A}_0$. By (15.7) with $L = 1 + s\ell$,

$$1 + \epsilon \leq \sum_j w^+(s)_j - 1 - s\ell = 1 + s(a + b + c - \ell),$$

which fails for s small enough.

Suppose next that there exists $\alpha \in \mathcal{A}_0$ with $\alpha_3 = \alpha_4 = 0$. Then $\ell = a\alpha_1 + b\alpha_2$ with $\alpha_1 + \alpha_2 \geq 2$, so $\ell \geq 2a > a > 0$. Since $\ell \in \{0, a, b, c\}$, we obtain $\ell = b$, and $b \geq 2a > a$. We show that

$$\alpha_1 + \alpha_2 + \alpha_4 \geq 2 \quad \text{for any } \alpha \in \mathcal{A}_0.$$

Let $\alpha \in \mathcal{A}_0$. If $\alpha_1 + \alpha_2 = 0$, then (15.8) reads $c\alpha_3 = b$, which is impossible since $c\alpha_3 \leq 0 < b$. Thus $\alpha_1 + \alpha_2 \geq 1$. If $\alpha_1 + \alpha_2 + \alpha_4 = 1$, then $\alpha_4 = 0$ and (α_1, α_2) is $(1, 0)$ or $(0, 1)$. In the first case, (15.8) reads $a + c\alpha_3 = b$, which is impossible since $a + c\alpha_3 \leq a < b$. In the second case, (15.8) reads $b + c\alpha_3 = b$, so $\alpha_3 = 0$ and $\alpha = e_2$, which contradicts $\sum_j \alpha_j \geq 2$. Hence $\alpha_1 + \alpha_2 + \alpha_4 \geq 2$. For $s > 0$ small enough, let

$$w^-(s) := -sv + e_1 + e_2 + e_4 = (1 - sa, 1 - sb, -sc, 1) \in N_H \cap \mathbb{R}^4_{>0}.$$

By (15.8), for any $\alpha \in \mathcal{A}_0$,

$$\langle \alpha, w^-(s) \rangle = \alpha_1 + \alpha_2 + \alpha_4 - s\ell \geq 2 - sb.$$

By (15.7) with $L = 2 - sb$,

$$1 + \epsilon \leq \sum_j w^-(s)_j - 2 + sb = 1 - s(a + c),$$

which fails for s small enough. Thus $k \neq 3$.

Step 7. In this step we show that, after a permutation of the coordinates, $I = \{1, 2\}$, $\operatorname{Lie}(H^0) = \mathbb{R}(1, -1, 0, 0)$, and $\lambda = 0$. By **Steps 3, 5**, and **6**, $k = 2$, and after a permutation of the coordinates, $I = \{1, 2\}$. Thus $\operatorname{Lie}(H^0) \subset \mathbb{R}e_1 + \mathbb{R}e_2$.

Case 1. $\lambda \neq 0$. By **Step 4**, $\lambda(v) = v_1 = v_2$ for any $v \in \operatorname{Lie}(H^0)$, so $\operatorname{Lie}(H^0) = \mathbb{R}(1, 1, 0, 0)$, and $\mathcal{A}_0$ is the set of α with $\alpha_1 + \alpha_2 = 1$ and $\sum_j \alpha_j \geq 2$. Let $0 < s < \min\{1, \epsilon\}$ and $w := (s, s, 1, 1) \in N_H \cap \mathbb{R}^4_{>0}$. For any $\alpha \in \mathcal{A}_0$,

$$\langle \alpha, w \rangle = s + \alpha_3 + \alpha_4 = s + \sum_j \alpha_j - 1 \geq 1 + s.$$

By (15.7) with $L = 1 + s$, $1 + s = \sum_j w_j - 1 - s \geq 1 + \epsilon$, a contradiction.

Case 2. $\lambda = 0$. Suppose that $\operatorname{Lie}(H^0)$ contains a vector $v = (a, b, 0, 0)$ with $a, b > 0$. Then any $\alpha \in \mathcal{A}_0$ satisfies $a\alpha_1 + b\alpha_2 = 0$, so $\alpha_1 = \alpha_2 = 0$ and $\alpha_3 + \alpha_4 \geq 2$. For $s > 0$ small enough, $w := sv + e_3 + e_4 = (sa, sb, 1, 1) \in N_H \cap \mathbb{R}^4_{>0}$ and $\langle \alpha, w \rangle \geq 2$ for any $\alpha \in \mathcal{A}_0$. By (15.7) with $L = 2$, $s(a + b) = \sum_j w_j - 2 \geq 1 + \epsilon$, which fails for s small enough. If $\dim \operatorname{Lie}(H^0) = 2$, then $\operatorname{Lie}(H^0) = \mathbb{R}e_1 + \mathbb{R}e_2$ contains $(1, 1, 0, 0)$, which was just excluded. Hence $\dim \operatorname{Lie}(H^0) = 1$, and $\operatorname{Lie}(H^0)$ is spanned by a vector $(a, b, 0, 0)$ with $a, b \neq 0$ since $I = \{1, 2\}$. Here a and b have opposite signs, since otherwise $(a, b, 0, 0)$ or $(-a, -b, 0, 0)$ has two positive coordinates, which was just excluded. Since $\operatorname{Lie}(H^0)$ is spanned by $\operatorname{Lie}(H^0) \cap \mathbb{Z}^4$, we obtain

$$\operatorname{Lie}(H^0) = \mathbb{R}(p, -q, 0, 0) \quad \text{for coprime integers } p, q > 0.$$

Then $\mathcal{A}_0$ is the set of $\alpha = (jq, jp, \alpha_3, \alpha_4)$ with integers $j, \alpha_3, \alpha_4 \geq 0$ and $j(p + q) + \alpha_3 + \alpha_4 \geq 2$. In other words, $x^\alpha = (x_1^q x_2^p)^j x_3^{\alpha_3} x_4^{\alpha_4}$.

Suppose that $p \neq q$. Possibly interchanging x_1 and x_2, we may assume that $p < q$, and then $q \geq 2$.

If $p \geq 2$, let $0 < s < 1/q$ and

$$w := s(p, -q, 0, 0) + (0, 1, 1, 1) = (ps, 1 - qs, 1, 1) \in N_H \cap \mathbb{R}^4_{>0}.$$

The w-weight of $x_1^q x_2^p$ is $qps + p(1 - qs) = p \geq 2$. Thus any $\alpha \in \mathcal{A}_0$ satisfies $\langle \alpha, w \rangle \geq 2$: if $j \geq 1$, then $\langle \alpha, w \rangle \geq jp \geq 2$, and if $j = 0$, then $\langle \alpha, w \rangle = \alpha_3 + \alpha_4 \geq 2$. By (15.7) with $L = 2$, $1 + (p - q)s = \sum_j w_j - 2 \geq 1 + \epsilon$, a contradiction since $p < q$.

If $p=1$, choose $\tau>0$ with $q\tau<1$ and $(q-1)\tau<\epsilon$, and let

$$w:=-\tau(1,-q,0,0)+(1,0,1,1)=(1-\tau,q\tau,1,1)\in N_H\cap\mathbb{R}^4_{>0}.$$

The w-weight of $x_1^q x_2$ is $q(1-\tau)+q\tau=q\geq 2$, so any $\alpha=(jq,j,\alpha_3,\alpha_4)\in\mathcal{A}_0$ satisfies $\langle\alpha,w\rangle=jq+\alpha_3+\alpha_4\geq 2$: if $j\geq 1$, then $jq\geq 2$, and if $j=0$, then $\alpha_3+\alpha_4\geq 2$. By (15.7) with $L=2$, $1+(q-1)\tau=\sum_j w_j-2\geq 1+\epsilon$, a contradiction.

Hence $p=q=1$, $\mathrm{Lie}(H^0)=\mathbb{R}(1,-1,0,0)$, and $H^0=\rho(\mathbb{R}(1,-1,0,0))$.

Step 8. In this step we show that the monomial x_1x_2 occurs in f_i for $i\gg 0$, and we deduce alternative (2) with bound i, which contradicts the choice of the data and proves the theorem for hypersurfaces in $\mathbb{A}^4$. By **Step 7**, $\mathcal{A}_0$ is the set of α with $\alpha_1=\alpha_2$ and $\sum_j\alpha_j\geq 2$, that is, $x^\alpha=(x_1x_2)^j x_3^{\alpha_3}x_4^{\alpha_4}$.

Suppose that there exists a subsequence (f_{i_l}) such that x_1x_2 does not occur in f_{i_l} for any l. Then the set $\mathcal{A}'$ of **Step 2** for (f_{i_l}) is contained in $\mathcal{A}_0\backslash\{(1,1,0,0)\}$. Let

$$w:=(\tfrac12,\tfrac12,1,1)=\tfrac12(1,-1,0,0)+(0,1,1,1)\in N_H\cap\mathbb{R}^4_{>0}.$$

For $\alpha=(j,j,\alpha_3,\alpha_4)\in\mathcal{A}_0\backslash\{(1,1,0,0)\}$, $\langle\alpha,w\rangle=j+\alpha_3+\alpha_4\geq 2$: if $j=0$, then $\alpha_3+\alpha_4\geq 2$, if $j=1$, then $\alpha_3+\alpha_4\geq 1$ since $\alpha\neq(1,1,0,0)$, and if $j\geq 2$, this is clear. By (15.7) for (f_{i_l}) with $L=2$, $1=\sum_j w_j-2\geq 1+\epsilon$, a contradiction. Thus x_1x_2 occurs in f_i for $i\gg 0$.

Let $i\gg 0$. By Lemma 4.3(2) for the weight vector $(1,-1,0,0)$, the image of $G_i\cap\mathbf{T}_{(1,-1,0,0)}$ in $\mathbb{T}^4$ is $H_i\cap\rho(\mathbb{R}(1,-1,0,0))=H_i\cap H^0=R_i$. Since $G_i\cap\mathbf{T}_{(1,-1,0,0)}$ is a finite subgroup of $\mathbf{T}_{(1,-1,0,0)}\cong\mathbb{C}^*$, it is $\mu_{r_i}(1,-1,0,0)$ with $r_i=|R_i|$. By **Step 2**, $[G_i:G_i\cap\mathbf{T}_{(1,-1,0,0)}]=[H_i:R_i]\leq|H/H^0|$ and $r_i=|R_i|\to+\infty$, so $r_i>2$. Since x_1x_2 occurs in f_i, the character of f_i is the character of x_1x_2, which is trivial on $\mathbf{T}_{(1,-1,0,0)}$. Thus any monomial x^α occurring in f_i satisfies $r_i\mid\alpha_1-\alpha_2$. For a monomial of degree two, $|\alpha_1-\alpha_2|\leq 2<r_i$, so $\alpha_1=\alpha_2$. Hence the quadratic part of f_i is $c_ix_1x_2+q_{i,2}(x_3,x_4)$ with $c_i\neq 0$, since x_1x_2 occurs in f_i. For $i>|H/H^0|$, this is alternative (2) with $e(\epsilon)$ replaced by i, which contradicts the choice of X_i and G_i. Therefore $e(\epsilon)$ exists.

If the quadratic part of f has rank one, alternative (2) does not hold, since $cxy+q_2(z,t)$ with $c\neq 0$ has rank at least two. Thus $|G|\leq e(\epsilon)$.

Step 9. In this step we treat isolated Gorenstein terminal cDV threefold singularities. Let $(X\ni o)$, G, $(Y\ni o)$, $f\in\mathfrak{m}^2_{Y,o}$, and G-eigenparameters x,y,z,t of Y at o be as in the last sentence of the theorem, and let $B\geq 0$ be a G-invariant $\mathbb{R}$-divisor with (X,B) klt and $\mathrm{mld}((X\ni o,B)/G)\geq 1+\epsilon$. The homomorphism $G\to(\mathbb{C}^*)^4$ given by the characters of x,y,z,t is injective, since an element acting trivially on x,y,z,t acts trivially on $\widehat{\mathcal{O}}_{Y,o}\supset\mathcal{O}_{Y,o}$, hence on X (alternatively, use the faithful diagonal action of Construction-Theorem 7.2 below), and we identify G with the image of G under the homomorphism.

By [KM98, Definition 5.32], a general hyperplane section of X through o is a Du Val singularity, and the equation of a Du Val singularity has multiplicity two [KM98, Theorem 4.20]. Since the lowest-degree homogeneous part of f does not vanish on a general hyperplane, $\mathrm{mult}_o f$ is the multiplicity of the equation of a general hyperplane section. Hence $\mathrm{mult}_o f=2$. Since $B\geq 0$, the boundary of $(X,B)/G$ is the boundary of $(X,0)/G$ plus $\frac{1}{|G|}\pi_*B\geq 0$ for the quotient morphism $\pi:X\to X/G$ (Definition 2.5), so $\mathrm{mld}((X\ni o,0)/G)\geq\mathrm{mld}((X\ni o,B)/G)\geq 1+\epsilon$.

We apply Construction-Theorem 7.2 with $H=G$, $n=4$, $s=1$, $h_1=1$, $\beta_1=1$, and no $\beta^{(j)}$ or P_i. Here $(X,0)$ is klt near o since X is terminal. By Construction-Theorem 7.2(4) with $\psi=\mathrm{id}_{\mathfrak{R}}$ and $v=1$, we may take $h_{1,0}=1$, so the boundary B_0 is 0. Thus we obtain a faithful diagonal action of G on $\mathbb{A}^4$ in which the coordinates x_0,y_0,z_0,t_0 have the characters of x,y,z,t, and an isolated Gorenstein terminal germ $X_0=(f_0=0)\subset\mathbb{A}^4$ at 0 for a semi-invariant polynomial f_0 with the character of f, such that

$$\mathrm{mld}((X_0\ni 0,0)/G)=\mathrm{mld}((X\ni o,0)/G)\geq 1+\epsilon,$$

and a G-equivariant isomorphism Φ between the completions $\widehat{\mathcal{O}}_{X,o}$ and $\widehat{\mathcal{O}}_{X_0,0}$. Since X_0 is an isolated Gorenstein terminal germ, X_0 is normal and klt, integral near 0 since the normal local ring $\mathcal{O}_{X_0,0}$ is a domain, and has an isolated singularity at 0. By Construction-Theorem 7.2(1), G acts faithfully on X_0. Since $f \in \mathfrak{m}_{Y,o}^2$, $\widehat{\mathcal{O}}_{X,o}$ is not regular, so X_0 is singular at 0. Since X_0 is an isolated Gorenstein terminal threefold singularity, X_0 is a cDV singularity [KM98, Corollary 5.38], and the previous paragraph for X_0 instead of X implies $\operatorname{mult}_0 f_0 = 2$. By the case of hypersurfaces in $\mathbb{A}^4$ proved in **Steps 1–8**, applied to X_0, G, and $B = 0$, either $|G| \le e(\epsilon)$, or after a permutation of x_0, y_0, z_0, t_0, the group $R := G \cap \mathbf{T}_{(1,-1,0,0)}$ is $\mu_r(1,-1,0,0)$ for an integer $r > 2$, $[G : R] \le e(\epsilon)$, the character of f_0 is trivial on R, and the quadratic part of f_0 is $c_0 x_0 y_0 + q_{0,2}(z_0, t_0)$ with $c_0 \neq 0$.

In the first case we are done. In the second case, we apply to x, y, z, t the permutation applied to x_0, y_0, z_0, t_0. Since x_0, y_0, z_0, t_0 have the characters of x, y, z, t, the group $R = G \cap \mathbf{T}_{(1,-1,0,0)}$ defined by the coordinates x, y, z, t coincides with the group R defined by x_0, y_0, z_0, t_0, so $R = \mu_r(1,-1,0,0)$ with $r > 2$ and $[G : R] \le e(\epsilon)$, and the character of f, which is the character of f_0, is trivial on R. It remains to show that the quadratic part of f is $cxy + q_2(z,t)$ with $c \neq 0$.

Let $\mathfrak{m}_X$ and $\mathfrak{m}_{X_0}$ be the maximal ideals of $\widehat{\mathcal{O}}_{X,o}$ and $\widehat{\mathcal{O}}_{X_0,0}$. Since $f \in \mathfrak{m}_{Y,o}^2$, $\mathfrak{m}_X/\mathfrak{m}_X^2 = \mathfrak{m}_{Y,o}/\mathfrak{m}_{Y,o}^2$ has basis $\bar{x}, \bar{y}, \bar{z}, \bar{t}$, and the kernel of the multiplication map $\operatorname{Sym}^2(\mathfrak{m}_X/\mathfrak{m}_X^2) \to \mathfrak{m}_X^2/\mathfrak{m}_X^3$ is spanned by the class $f_2 \neq 0$ of f in $\mathfrak{m}_{Y,o}^2/\mathfrak{m}_{Y,o}^3 = \operatorname{Sym}^2(\mathfrak{m}_{Y,o}/\mathfrak{m}_{Y,o}^2)$, since $f\mathfrak{m}_{Y,o} \subset \mathfrak{m}_{Y,o}^3$. Similarly, the corresponding kernel for X_0 is spanned by the quadratic part $f_{0,2}$ of f_0. The isomorphism Φ induces a G-equivariant linear isomorphism $\varphi : \mathfrak{m}_X/\mathfrak{m}_X^2 \to \mathfrak{m}_{X_0}/\mathfrak{m}_{X_0}^2$ compatible with the multiplication maps, so $\operatorname{Sym}^2 \varphi$ maps one kernel onto the other, and

$$\operatorname{Sym}^2 \varphi(f_2) = \kappa f_{0,2} \quad \text{for some } \kappa \neq 0.$$

Let $\zeta := e^{2\pi\sqrt{-1}/r}$. On R, the characters of x, y, z, and t take the values ζ, ζ^{-1}, 1, and 1 at the generator $(\zeta, \zeta^{-1}, 1, 1)$, and similarly for x_0, y_0, z_0, t_0. Since $r > 2$, the numbers ζ, ζ^{-1}, and 1 are pairwise distinct, so the eigenspaces of R on $\mathfrak{m}_X/\mathfrak{m}_X^2$ are $\mathbb{C}\bar{x}$, $\mathbb{C}\bar{y}$, and $\mathbb{C}\bar{z} + \mathbb{C}\bar{t}$, and similarly for X_0. Since φ is R-equivariant, $\varphi(\bar{x}) = a\bar{x}_0$ and $\varphi(\bar{y}) = b\bar{y}_0$ for some $a, b \neq 0$, and $\varphi(\mathbb{C}\bar{z} + \mathbb{C}\bar{t}) = \mathbb{C}\bar{z}_0 + \mathbb{C}\bar{t}_0$. Thus $\operatorname{Sym}^2 \varphi(\bar{x}\bar{y}) = ab\bar{x}_0\bar{y}_0$, and for any other monomial of degree two in $\bar{x}, \bar{y}, \bar{z}, \bar{t}$, the coefficient of $\bar{x}_0\bar{y}_0$ in the image under $\operatorname{Sym}^2 \varphi$ is 0. Comparing the coefficients of $\bar{x}_0\bar{y}_0$, the coefficient of $\bar{x}\bar{y}$ in f_2 is $\kappa c_0/(ab) \neq 0$. Since the character of f is trivial on R, possibly replacing f_i with f and r_i with r, the argument of **Step 8** gives $\alpha_1 = \alpha_2$ for any monomial $\bar{x}^{\alpha_1}\bar{y}^{\alpha_2}\bar{z}^{\alpha_3}\bar{t}^{\alpha_4}$ of degree two occurring in f_2. Hence the quadratic part of f is $cxy + q_2(z,t)$ with $c \neq 0$. The statement on quadratic rank one follows as in **Step 8**. □

15.2. **Bounded groups.** In this subsection, we prove that the minimal log discrepancies in $[1, +\infty)$ of crepant quotients of Gorenstein klt threefold germs by groups of bounded order satisfy the ACC. We also prove the ACC for minimal log discrepancies in $[1, +\infty)$ of abelian quotients of isolated cDV points of quadratic rank one, without a bound on the order of the group (Corollary 15.7).

The proof passes to a terminal model of the quotient whose boundary coefficients belong to a DCC set depending only on the bound on the order of the group, and treats separately the cases where a divisor computing the minimal log discrepancy has a closed point or a curve as center on the terminal model. The first case follows from [HLL25, Theorem 4.1], and this is the only place in the paper where we use [HLL25, Theorem 4.1] directly.

Proposition 15.6. *Fix an integer $L \ge 1$ and a DCC set $\Gamma \subset [0,1]$. Let $(X \ni x)$ be a normal Gorenstein klt threefold germ, G a finite group of order at most L acting faithfully on X and fixing x, $h_1, \dots, h_s$ non-zero regular G-semi-invariant functions, $c_j \in \Gamma$ positive, and $B := \sum_j c_j \operatorname{div}_X(h_j)$ with (X, B) klt near x. Then the set of such numbers $\operatorname{mld}((X \ni x, B)/G) \in [1, +\infty)$ satisfies the ACC. More precisely, let $M := L!$, $\gamma := \gamma(\Gamma)$ if $\Gamma \backslash \{0\} \neq \emptyset$ and $\gamma := 1$*

otherwise, and

$$\Lambda := \Big\{1 - \frac{k}{M} \;\Big|\; 1 \le k \le M\Big\}, \quad N' := \lceil M/\gamma \rceil,$$
$$\Sigma := \Big(\Lambda + \Big\{\frac{1}{M}\sum_{i=1}^{n}\gamma_i \;\Big|\; 0 \le n \le N',\ \gamma_i \in \Gamma\backslash\{0\}\Big\}\Big) \cap [0,1).$$

Then Σ is a DCC set depending only on L and Γ. For any $a := \mathrm{mld}((X \ni x, B)/G) > 1$ and any prime divisor E computing a, there exist a projective birational morphism $Y \to X/G$ and an $\mathbb{R}$-divisor $C_Y \ge 0$ such that Y is $\mathbb{Q}$-factorial and terminal, $C_Y \in \Sigma$, (Y, C_Y) is klt, the center of E on Y is a closed point or a curve in the fiber over the image of x with generic point η, and $\mathrm{mld}(Y \ni \eta, C_Y) = a$. If the center is a curve, then Y is smooth at η, $\mathrm{mult}_\eta C_Y < 1$, and $a = 2 - \mathrm{mult}_\eta C_Y$, where $\mathrm{mult}_\eta C_Y$ is the sum of the coefficients of the components of C_Y through η multiplied by their multiplicities at η.

Proof. The proof is long, so we divide it into several steps. Let $\pi : X \to X/G$ be the quotient morphism and $q := \pi(x)$.

Step 1. In this step we write $B_{X/G}$ as the boundary of $(X,0)/G$ plus Cartier divisors, and we show that the log discrepancies of $(X,0)/G$ belong to $\frac{1}{M}\mathbb{N}^+$.

Let V be an affine open neighborhood of x such that V is Gorenstein, $h_1, \dots, h_s$ are regular on V, and $(V, B|_V)$ is klt. Since X is separated and G fixes x, $\bigcap_{\sigma\in G}\sigma(V)$ is a G-invariant affine open neighborhood of x contained in V. Since $\mathrm{mld}((X \ni x, B)/G)$ depends only on a neighborhood of q, possibly replacing X with $\bigcap_{\sigma\in G}\sigma(V)$, we may assume that X is Gorenstein, that $h_1, \dots, h_s$ are regular on X, and that (X, B) is klt. In particular, X is klt. Let $(X/G, \Delta_0) := (X,0)/G$. By Lemma 2.6(0), $K_{X/G} + \Delta_0$ is $\mathbb{R}$-Cartier and $K_X = \pi^*(K_{X/G} + \Delta_0)$. Since $\Delta_0 \ge 0$, $(X/G, \Delta_0)$ is a pair.

Let E be a prime divisor over X/G. By Lemma 2.6(3), there exists a prime divisor F over X with $\mathrm{ord}_F|_{\mathbb{C}(X/G)} = e\,\mathrm{ord}_E$ for some positive integer e. Since K_X is Cartier and X is klt, $a(F, X, 0)$ is a positive integer. By Lemma 2.6(1)(2),

$$a(F, X, 0) = |I_G(F)| \cdot a(E, X/G, \Delta_0).$$

Since $|I_G(F)|$ divides $|G|$ and $|G| \le L$, $|I_G(F)|$ divides M. Thus

$$a(E, X/G, \Delta_0) \in \tfrac{1}{M}\mathbb{N}^+ \quad \text{for any prime divisor } E \text{ over } X/G. \tag{15.9}$$

In particular, $(X/G, \Delta_0)$ is klt.

The order of any character of G divides $|G|$, hence divides M. Therefore h_j^M is G-invariant, and $h_j^M = \pi^*\phi_j$ for a regular function ϕ_j on X/G. Since $\pi^*\,\mathrm{div}(\phi_j) = M\,\mathrm{div}(h_j)$, we have $B = \pi^*\big(\sum_j \frac{c_j}{M}\,\mathrm{div}(\phi_j)\big)$. As $\pi_*\pi^*D = |G|D$ for any $\mathbb{R}$-divisor D on X/G, Definition 2.5 implies

$$B_{X/G} = \Delta_0 + \sum_j \frac{c_j}{M}\,\mathrm{div}(\phi_j). \tag{15.10}$$

By Lemma 2.6(2)(3) applied to (X, B), the pair $(X/G, B_{X/G})$ is klt.

Step 2. In this step we construct Y and C_Y.

Let $\mathcal{S} := \frac{1}{M}\mathbb{Z}_{\ge 0}$, the semigroup generated by $\frac{1}{M}$. Then $1 \in \mathcal{S}$ and $\Lambda_{\mathcal{S}} = \Lambda$. By (15.9), we may apply Construction-Theorem 5.1 to $(X/G, \Delta_0)$, the point q, and the trivial group. Possibly shrinking X/G to a neighborhood of q and X to the preimage, we obtain a projective birational morphism $g : Y \to X/G$ and an $\mathbb{R}$-divisor $C_0 \ge 0$ on Y such that $K_Y + C_0 = g^*(K_{X/G} + \Delta_0)$, $C_0 \in \Lambda$, and Y is $\mathbb{Q}$-factorial and terminal (Construction-Theorem 5.1(1)–(4)). Let

$$C_Y := C_0 + \sum_j \frac{c_j}{M} g^*\,\mathrm{div}(\phi_j).$$

By (15.10), $K_Y + C_Y = g^*(K_{X/G} + B_{X/G})$, so $a(P, Y, C_Y) = a(P, X/G, B_{X/G})$ for any prime divisor P over Y [KM98, Lemma 2.30]. Since $(X/G, B_{X/G})$ is klt and $C_Y \geq 0$, (Y, C_Y) is klt.

Step 3. In this step we show that $C_Y \in \Sigma$ and that Σ is a DCC set.

Let T be a prime divisor on Y, $d_T := \operatorname{mult}_T C_0$, and $n_j := \operatorname{mult}_T g^* \operatorname{div}(\phi_j) \in \mathbb{Z}_{\geq 0}$. Since $0 = 1 - \frac{M}{M} \in \Lambda$, we have $d_T \in \Lambda$. Since (Y, C_Y) is klt,

$$\operatorname{mult}_T C_Y = d_T + \frac{1}{M} \sum_j c_j n_j < 1.$$

As $d_T \geq 0$ and $c_j \geq \gamma$, we get $\gamma \sum_j n_j < M$, so $n := \sum_j n_j \leq N'$. Writing $\sum_j c_j n_j$ as a sum of n elements of $\Gamma \backslash \{0\}$ in which c_j occurs n_j times, we obtain $\operatorname{mult}_T C_Y \in \Sigma$. Thus $C_Y \in \Sigma$.

By Lemma 2.10(3), the set of sums of at most N' elements of $\Gamma \backslash \{0\}$ satisfies the DCC: given a sequence of such sums, add zeros so that each sum has exactly N' summands in $\Gamma \cup \{0\}$, and pass N' times to a subsequence on which the i-th summands are nondecreasing. Multiplying by $\frac{1}{M}$, adding the finite set Λ, and intersecting with $[0, 1)$ preserve the DCC. Hence Σ is a DCC set, and by definition Σ depends only on L and Γ.

Step 4. In this step we find the center of E on Y and show that $\operatorname{mld}(Y \ni \eta, C_Y) = a$.

Assume that $a > 1$ and that E computes a, that is, $\operatorname{center}_{X/G} E = \{q\}$ and $a(E, X/G, B_{X/G}) = a$. By (15.10), $B_{X/G} - \Delta_0$ is a non-negative real linear combination of Cartier divisors, so

$$a(E, X/G, \Delta_0) \geq a(E, X/G, B_{X/G}) = a > 1$$

by [KM98, Lemma 2.27]. By Construction-Theorem 5.1(5), $Z := \operatorname{center}_Y E$ is a closed point or a complete curve in $g^{-1}(q)$. Let η be the generic point of Z. If P is a prime divisor over Y with $\operatorname{center}_Y P = Z$, then $\operatorname{center}_{X/G} P = g(Z) = \{q\}$, so $a(P, Y, C_Y) = a(P, X/G, B_{X/G}) \geq a$ by **Step 2**. Since $a(E, Y, C_Y) = a$, we obtain

$$\operatorname{mld}(Y \ni \eta, C_Y) = a. \tag{15.11}$$

Step 5. In this step we assume that Z is a curve, and we show that Y is smooth at η, $\operatorname{mult}_\eta C_Y < 1$, and $a = 2 - \operatorname{mult}_\eta C_Y$.

Since a terminal threefold is smooth in codimension two [KM98, Corollary 5.18], Y is smooth at η, and $\mathcal{O}_{Y,\eta}$ is a two-dimensional regular local ring with residue field $\mathbb{C}(Z)$. Let $m := \operatorname{mult}_\eta C_Y$. All prime divisors considered below have centers on Y equal to Z, and their log discrepancies are computed at the generic points of their centers. Hence we may replace Y by a neighborhood of η on which Y and Z are smooth, and a blow-up of the closed point of a two-dimensional regular local ring dominating $\mathcal{O}_{Y,\eta}$ corresponds to the blow-up of a curve on a threefold over such a neighborhood. By [KM98, Lemma 2.29], the exceptional divisor E_1 of the blow-up of Y along Z satisfies $a(E_1, Y, C_Y) = 2 - m$. Since $\operatorname{center}_Y E_1 = Z$, (15.11) implies $a \leq 2 - m$, so

$$m \leq 2 - a < 1.$$

Let P be a prime divisor over Y with $\operatorname{center}_Y P = Z$. Then ord_P is centered at the maximal ideal of $\mathcal{O}_{Y,\eta}$, and the residue field of ord_P has transcendence degree one over $\mathbb{C}(Z)$. By [Abh56, Proposition 3], the quadratic sequence along ord_P starting from $\mathcal{O}_{Y,\eta}$ is finite. Thus there exist regular two-dimensional schemes $S_0 := \operatorname{Spec} \mathcal{O}_{Y,\eta}, S_1, \ldots, S_k$ and closed points $p_i \in S_i$ for $0 \leq i \leq k-1$ such that S_{i+1} is the blow-up of S_i at p_i, p_{i+1} is the center of ord_P on S_{i+1}, and $\operatorname{ord}_P = \operatorname{ord}_{E_k}$, where $E_i \subset S_i$ is the exceptional curve of $S_i \to S_{i-1}$. Let $A_i := a(E_i, Y, C_Y)$, and for $j \geq i$ let $E_i^{(j)} \subset S_j$ be the strict transform of E_i. On S_i the curves $E_1^{(i)}, \ldots, E_i^{(i)}$ form a simple normal crossing divisor (we omit the induction on i, which uses that the exceptional curve of the blow-up of a two-dimensional regular scheme at a closed point p is $\mathbb{P}^1_{\kappa(p)}$, which is regular and intersects the strict transforms of the earlier exceptional curves transversally), so p_i belongs to at most two of them.

Let $D_1, \dots, D_n$ be the components of C_Y through η, with coefficients $d_1, \dots, d_n$ and multiplicities $m_1, \dots, m_n$ at η, so that $m = \sum_l d_l m_l$. For a closed point $p \in S_i$ over p_0 let $m_p := \sum_l d_l \operatorname{mult}_p D_l^{(i)}$, where $D_l^{(i)}$ is the strict transform of D_l on S_i. Then

$$m_p \le m,$$

since the multiplicity of a strict transform at a closed point of the exceptional curve of one blow-up is at most the multiplicity at the blown-up point: if u, v are regular parameters at the blown-up point, φ is a local equation of multiplicity μ with leading form $\varphi_\mu(u, v)$, and $v = uv'$ in one chart, then $\varphi = u^\mu \varphi'$ with $\varphi'|_{(u=0)} = \varphi_\mu(1, v')$, a non-zero polynomial of degree at most μ, so φ' has order at most μ at any closed point of $(u = 0)$. The remaining point of the exceptional curve belongs to the other chart, where the roles of u and v are exchanged.

The crepant pullback of $K_Y + C_Y$ to S_i has boundary the strict transform of C_Y plus $\sum_{j \le i}(1 - A_j)E_j^{(i)}$. Hence [KM98, Lemma 2.29] implies $A_1 = 2 - m$, and for $1 \le i \le k - 1$,

$$A_{i+1} = \begin{cases} 1 + A_j - m_{p_i} & \text{if } E_j^{(i)} \text{ is the only one of } E_1^{(i)}, \dots, E_i^{(i)} \text{ through } p_i, \\ A_j + A_{j'} - m_{p_i} & \text{if } p_i \in E_j^{(i)} \cap E_{j'}^{(i)},\ j \ne j'. \end{cases}$$

We show $A_i \ge 2 - m$ by induction on i. If $A_j, A_{j'} \ge 2 - m$, then $m_{p_i} \le m < 1$ implies

$$1 + A_j - m_{p_i} \ge A_j + (1 - m) > 2 - m \quad \text{and} \quad A_j + A_{j'} - m_{p_i} \ge (2 - m) + 2(1 - m) > 2 - m.$$

Thus $a(P, Y, C_Y) = A_k \ge 2 - m$. Since P is an arbitrary prime divisor over Y with center Z, (15.11) implies $a \ge 2 - m$. Therefore $a = 2 - m$.

By **Steps 2–5**, the conclusions on Σ, Y, C_Y, and E in the statement hold.

Step 6. In this step we prove the ACC for groups of order at most L.

Let $a_1 < a_2 < \cdots$ be a strictly increasing sequence, where $a_i = \operatorname{mld}((X_i \ni x_i, B_i)/G_i) \in [1, +\infty)$ for data $(X_i \ni x_i)$, G_i, B_i as in the statement. Possibly removing a_1, we may assume that $a_i > 1$ for any i. Since $(X_i/G_i, (B_i)_{X_i/G_i})$ is klt by **Step 1**, there exists a prime divisor E_i computing a_i [Amb06, Section 2, p. 360]. **Steps 2–5** applied to E_i imply that there exist a $\mathbb{Q}$-factorial terminal threefold Y_i, an $\mathbb{R}$-divisor $C_{Y_i} \in \Sigma$ on Y_i, and a point $\eta_i \in Y_i$ which is a closed point or the generic point of a curve, such that (Y_i, C_{Y_i}) is klt and $a_i = \operatorname{mld}(Y_i \ni \eta_i, C_{Y_i})$. Possibly passing to a subsequence, either η_i is a closed point for any i, or η_i is the generic point of a curve for any i.

In the first case, $a_i \ge 1$, Y_i is terminal near η_i, and $C_{Y_i} \in \Sigma$, where $\Sigma \subset [0, 1)$ is a DCC set by **Step 3**. This contradicts [HLL25, Theorem 4.1]. In the second case, $a_i = 2 - m_i$ with $m_i := \operatorname{mult}_{\eta_i} C_{Y_i} < 1$ by **Step 5**. Since m_i is a sum of elements of Σ with positive integer multiplicities, $m_i \in \Sigma_+ \cap [0, 1]$, which satisfies the DCC by Lemma 2.10(1). This contradicts the fact that (m_i) is strictly decreasing. Hence the set of numbers $\operatorname{mld}((X \ni x, B)/G) \in [1, +\infty)$ with $|G| \le L$ satisfies the ACC. □

Corollary 15.7. *Fix a DCC set $\Gamma \subset [0, 1]$. Let $(X \ni x)$ be an isolated Gorenstein terminal cDV threefold singularity of quadratic rank one, G a finite abelian group acting faithfully on X and fixing x, $h_1, \dots, h_s$ non-zero regular G-semi-invariant functions, $c_j \in \Gamma$ positive, and $B := \sum_j c_j \operatorname{div}_X(h_j)$ with (X, B) klt near x. Then the set of such numbers $\operatorname{mld}((X \ni x, B)/G)$ that belong to $[1, +\infty)$ satisfies the ACC.*

Proof. Let $a_1 < a_2 < \cdots$ be a strictly increasing sequence, where $a_i = \operatorname{mld}((X_i \ni x_i, B_i)/G_i) \in [1, +\infty)$ for data $(X_i \ni x_i)$, G_i, and $B_i = \sum_j c_{ij} \operatorname{div}_{X_i}(h_{ij})$ as in the statement. Let $\epsilon := a_2 - 1 > 0$. Then $a_i \ge 1 + \epsilon$ for any $i \ge 2$.

Since X_i is singular at x_i, by Lemma 5.4(2) there exist a smooth fourfold germ $(Y_i \ni x_i)$ with a faithful action of G_i, a G_i-equivariant closed embedding $X_i \subset Y_i$ with $X_i = (f_i = 0)$ for a regular G_i-semi-invariant function f_i, and G_i-eigenparameters $x_{i,1}, \dots, x_{i,4}$ of Y_i at x_i. Hence $f_i \in \mathfrak{m}_{Y_i, x_i}^2$. By [Mat89, Theorems 29.7 and 8.7], the completion of $\mathcal{O}_{X_i, x_i}$ is

$$\mathbb{C}[[x_{i,1}, \dots, x_{i,4}]]/(f_i),$$

so by Definition-Lemma 11.1 the quadratic part of f_i has rank one. The $\mathbb{R}$-divisor B_i is G_i-invariant since the h_{ij} are G_i-semi-invariant. By the last sentence of Theorem 15.5, applied to $X_i \subset Y_i$, f_i, B_i, and the eigenparameters $x_{i,1}, \dots, x_{i,4}$, together with the sentence of Theorem 15.5 on quadratic parts of rank one, $|G_i| \leq e(\epsilon)$ for any $i \geq 2$. Since X_i is a normal Gorenstein klt threefold, Proposition 15.6 with $L = e(\epsilon)$ implies that $(a_i)_{i\geq 2}$ is not strictly increasing, a contradiction. □

15.3. **Split polynomial models.** In this subsection, we replace a germ in alternative (2) of Theorem 15.5, where the order of G is not bounded and Proposition 15.6 does not apply, together with a boundary given by semi-invariant functions, by a polynomial hypersurface $xy = g_0(z,t)$ whose coordinates x, y, z, t and equation $xy - g_0$ have the characters of the eigenparameters and the equation of the germ, without changing the minimal log discrepancies of the quotients by subgroups (Construction-Theorem 15.8). It is used in Case 2 of the proof of Proposition 15.1 in Subsection 15.5.

Construction-Theorem 15.8. Let $(X \ni o)$ be an isolated Gorenstein terminal singular threefold germ, $X = (f = 0)$ in a smooth fourfold germ $(Y \ni o)$, G a finite abelian group acting on Y, fixing o, preserving X, and acting faithfully on X, and x, y, z, t G-eigenparameters of Y at o such that f is semi-invariant with quadratic part $cxy + q_2(z,t)$, $c \neq 0$, and a subgroup $R \subset G$ of order $r > 2$ acts as $\mu_r(1, -1, 0, 0)$. Let $h_1, \dots, h_s$ be non-zero regular G-semi-invariant functions on X, $c_j > 0$, and $B := \sum_j c_j \operatorname{div}_X(h_j)$ with (X, B) klt near o. Then there exist a non-zero polynomial $g_0 \in (z,t)^2$ and semi-invariant polynomials $h_{1,0}, \dots, h_{s,0}$ whose restrictions to $X_0 := (xy - g_0(z,t) = 0) \subset \mathbb{A}^4$ are non-zero, such that the following hold.

(1) X_0 is an isolated Gorenstein terminal threefold germ at 0, G acts diagonally and faithfully on X_0 with the coordinates x, y, z, t having the characters of the eigenparameters x, y, z, t, the equation $xy - g_0$ has the character of f, and R acts as $\mu_r(1, -1, 0, 0)$. If $R = G \cap \mathbf{T}_{(1,-1,0,0)}$ for the eigenparameters on Y, then $R = G \cap \mathbf{T}_{(1,-1,0,0)}$ on $\mathbb{A}^4$.
(2) $h_{j,0}$ has the character of h_j, and (X_0, B_0) is klt near 0 for $B_0 := \sum_j c_j \operatorname{div}_{X_0}(h_{j,0})$.
(3) $\operatorname{mld}((X \ni o, B)/H) = \operatorname{mld}((X_0 \ni 0, B_0)/H)$ for any subgroup $H \subset G$.
(4) There exists a G-equivariant isomorphism between the completions of $\mathcal{O}_{X,o}$ and $\mathcal{O}_{X_0,0}$.

Proof. The proof is long, so we divide it into several steps. Let $\chi_x, \chi_y, \chi_z, \chi_t$, and χ_f be the characters of x, y, z, t, and f (Definition 2.14). Let $\mathfrak{R} := \mathbb{C}[[x, y, z, t]]$ be the completion of $\mathcal{O}_{Y,o}$ [Mat89, Theorem 29.7], with the induced diagonal action of G, let $\mathfrak{m}_{\mathfrak{R}}$ be the maximal ideal of $\mathfrak{R}$, and let $\hat{f} \in \mathfrak{R}$ be the expansion of f. Then the completion $\widehat{\mathcal{O}}_{X,o}$ of $\mathcal{O}_{X,o}$ is $\mathfrak{R}/(\hat{f})$ [Mat89, Theorem 8.7]. Let $\mathfrak{m}$ be the maximal ideal of $\widehat{\mathcal{O}}_{X,o}$. For a character χ of G and an element w of a ring on which G acts, let

$$e_\chi(w) := \frac{1}{|G|} \sum_{\sigma \in G} \chi(\sigma)^{-1} \sigma^* w.$$

Then $e_\chi(w)$ is zero or G-semi-invariant with character χ, e_χ commutes with G-equivariant homomorphisms, and $e_\chi(w) = w$ if w is G-semi-invariant with character χ. Since f is semi-invariant and xy occurs in $\hat{f}$ with coefficient $c \neq 0$, we have $\chi_f = \chi_x \chi_y$. In particular, χ_f is trivial on R, while $\chi_x|_R$ and $\chi_y|_R$ have order $r > 2$.

Step 1. In this step we show that there exist G-semi-invariant elements $\hat{x}, \hat{y}, \hat{z}, \hat{t} \in \mathfrak{m}$ with characters $\chi_x, \chi_y, \chi_z, \chi_t$ and a non-zero semi-invariant $\hat{g} \in (z,t)^2 \subset \mathbb{C}[[z,t]]$ such that the continuous homomorphism

$$\rho : \mathfrak{R} \to \widehat{\mathcal{O}}_{X,o}, \quad x \to \hat{x},\ y \to \hat{y},\ z \to \hat{z},\ t \to \hat{t}$$

is G-equivariant and surjective with kernel $(xy - \hat{g}(z,t))$, that is,

$$\widehat{\mathcal{O}}_{X,o} \cong \mathbb{C}[[\hat{x}, \hat{y}, \hat{z}, \hat{t}]]/(\hat{x}\hat{y} - \hat{g}(\hat{z}, \hat{t})).$$

Since X is singular at o, $f \in \mathfrak{m}_{Y,o}^2$. The quadratic part of f has rank at least two, and the restriction of the quadratic part to the (x,y)-plane is cxy, which is non-degenerate. Since X is terminal, $(X,0)$ is klt. Thus we may use Theorem 9.1(3) in the situation of Theorem 9.1(2) with $u_1 = x$, $u_2 = y$, and $b_z = b_t = 0$. Let $(V \ni o_V)$, Q_2, g, and $(X' \ni o')$ be as there, where we write o_V, x', y', X_1, o_1 for v, x, y, X_0, o_0 there, since X_0 and x, y already denote the polynomial model and the eigenparameters on Y. Then z, t are G-eigenparameters of V at o_V, G acts diagonally on $\mathbb{A}^2$ with coordinates x', y', $Q_2 \in \{x'y', x'^2 + y'^2\}$, $X_1 = (Q_2(x', y') + g = 0) \subset \mathbb{A}^2 \times V$, with G acting on $\mathbb{A}^2 \times V$ by the product of the diagonal action on $\mathbb{A}^2$ and the action on V, the morphisms $X' \to X$ and $X' \to X_1$ are G-equivariant and étale at o', the pullbacks to X' of z and t on X coincide with those of z and t on X_1, and g is a non-zero regular G-semi-invariant function on V vanishing to order at least two at o_V. Since the residue fields at o, o', and o_1 are $\mathbb{C}$, the morphisms $X' \to X$ and $X' \to X_1$ induce isomorphisms of completions [Har77, Exercise III.10.4]. Let $\hat{g}_1 \in \mathbb{C}[[z,t]]$ be the expansion of g. Then $\hat{g}_1$ is non-zero, belongs to $(z,t)^2$, and is semi-invariant with the character χ_g of g. By [Mat89, Theorems 29.7 and 8.7], we obtain a G-equivariant isomorphism

$$\Phi : \widehat{\mathcal{O}}_{X,o} \to \mathbb{C}[[x', y', z, t]]/(Q_2 + \hat{g}_1)$$

with $\Phi(z) = z$ and $\Phi(t) = t$.

Since $f \in \mathfrak{m}_{\mathfrak{R}}^2$ and the quadratic part $f_2 := cxy + q_2(z,t)$ of $\hat{f}$ is non-zero, the classes of x, y, z, t form a basis of $\mathfrak{m}/\mathfrak{m}^2$, and the kernel of the multiplication map $\mathrm{Sym}^2(\mathfrak{m}/\mathfrak{m}^2) \to \mathfrak{m}^2/\mathfrak{m}^3$ is the line spanned by f_2, on which G acts by χ_f. Likewise, the classes of x', y', z, t form a basis of the cotangent space of $\mathbb{C}[[x', y', z, t]]/(Q_2 + \hat{g}_1)$, and the kernel of the corresponding multiplication map is the line spanned by $Q_2 + g_2$, where g_2 is the quadratic part of $\hat{g}_1$. The isomorphism Φ induces a G-equivariant linear isomorphism from $\mathfrak{m}/\mathfrak{m}^2$ to the cotangent space of $\mathbb{C}[[x', y', z, t]]/(Q_2 + \hat{g}_1)$ which fixes the classes of z, t and maps the first kernel line onto the second. Hence the characters of x', y' are χ_x, χ_y up to order, and G acts on the line spanned by $Q_2 + g_2$ by χ_f. If $Q_2 = x'^2 + y'^2$, then the monomial x'^2 occurs in $Q_2 + g_2$, so χ_f is the square of the character of x'. Since χ_f is trivial on R and the character of x' restricted to R has order $r > 2$, this is impossible. Thus $Q_2 = x'y'$, and possibly swapping x' and y', we may assume that x' and y' have characters χ_x and χ_y.

Let $\hat{g} := -\hat{g}_1$, $\hat{x} := \Phi^{-1}(x')$, $\hat{y} := \Phi^{-1}(y')$, and let $\hat{z}, \hat{t}$ be the images of z, t in $\widehat{\mathcal{O}}_{X,o}$. Then ρ is the composition of the isomorphism $\mathfrak{R} \to \mathbb{C}[[x', y', z, t]]$ sending x, y, z, t to x', y', z, t, the quotient map, and Φ^{-1}. Hence ρ is surjective with kernel $(xy - \hat{g})$, and ρ is G-equivariant since $\hat{x}, \hat{y}, \hat{z}, \hat{t}$ have characters $\chi_x, \chi_y, \chi_z, \chi_t$.

Step 2. In this step we construct a G-equivariant automorphism ψ of $\mathfrak{R}$ and a G-invariant unit v of $\mathfrak{R}$ such that

$$v\psi(\hat{f}) = xy - \hat{g}.$$

Let $p : \mathfrak{R} \to \widehat{\mathcal{O}}_{X,o} = \mathfrak{R}/(\hat{f})$ be the quotient map. For $w \in \{x, y, z, t\}$ fix $a_w \in \mathfrak{R}$ with $p(a_w) = \rho(w)$. Since p is G-equivariant and $\rho(w)$ is semi-invariant with character χ_w, we have $p(e_{\chi_w}(a_w)) = \rho(w)$, so possibly replacing a_w with $e_{\chi_w}(a_w)$, we may assume that a_w is semi-invariant with character χ_w. Let $\varphi : \mathfrak{R} \to \mathfrak{R}$ be the continuous homomorphism with $\varphi(w) = a_w$ for $w \in \{x, y, z, t\}$. Then φ is G-equivariant and $p \circ \varphi = \rho$. Since p and ρ are surjective with kernels contained in $\mathfrak{m}_{\mathfrak{R}}^2$, both induce isomorphisms $\mathfrak{m}_{\mathfrak{R}}/\mathfrak{m}_{\mathfrak{R}}^2 \to \mathfrak{m}/\mathfrak{m}^2$. Hence φ induces an isomorphism of $\mathfrak{m}_{\mathfrak{R}}/\mathfrak{m}_{\mathfrak{R}}^2$. Since $\mathfrak{m}_{\mathfrak{R}}^k/\mathfrak{m}_{\mathfrak{R}}^{k+1}$ is the k-th symmetric power of $\mathfrak{m}_{\mathfrak{R}}/\mathfrak{m}_{\mathfrak{R}}^2$, φ induces an isomorphism of each $\mathfrak{m}_{\mathfrak{R}}^k/\mathfrak{m}_{\mathfrak{R}}^{k+1}$, hence a bijection of each $\mathfrak{R}/\mathfrak{m}_{\mathfrak{R}}^k$, and taking the inverse limit, φ is an automorphism of the complete local ring $\mathfrak{R}$. Let $\psi := \varphi^{-1}$. Since $p \circ \varphi = \rho$, we have $\varphi^{-1}((\hat{f})) = (xy - \hat{g})$, so $\psi(\hat{f}) = w_0(xy - \hat{g})$ for a unit w_0 of $\mathfrak{R}$.

We show that $xy - \hat{g}$ is semi-invariant with character χ_f. The ideal $(xy - \hat{g}) = \mathrm{Ker}\,\rho$ is G-stable. For $\sigma \in G$, since $\chi_x\chi_y = \chi_f$,

$$\sigma^*(xy - \hat{g}) - \chi_f(\sigma)(xy - \hat{g}) = (\chi_f(\sigma) - \chi_g(\sigma))\hat{g}.$$

If $\chi_g(\sigma) \neq \chi_f(\sigma)$ for some σ, then $\hat{g} \in (xy - \hat{g})$, hence $xy \in (xy - \hat{g})$, and $\widehat{\mathcal{O}}_{X,o} \cong \mathfrak{R}/(xy - \hat{g})$ is a quotient of $\mathfrak{R}/(xy, \hat{g})$. This contradicts $\dim \mathfrak{R}/(xy, \hat{g}) = 2 < 3 = \dim \widehat{\mathcal{O}}_{X,o}$. Thus $\chi_g = \chi_f$, and $xy - \hat{g}$ is semi-invariant with character χ_f. Since ψ is G-equivariant, $\psi(\hat{f})$ is also semi-invariant with character χ_f, so w_0 is G-invariant. Let $v := w_0^{-1}$. Then $v\psi(\hat{f}) = xy - \hat{g}$.

Step 3. In this step we construct g_0 and $h_{1,0}, \dots, h_{s,0}$ and prove (1)–(4).

Since the statement depends only on a neighborhood of o, possibly shrinking Y to a G-invariant affine neighborhood of o, we may assume that Y is affine. For each j, fix a regular function h'_j on Y with $h'_j|_X = h_j$. Since X is G-stable and h_j is semi-invariant, $e_\chi(h'_j)|_X = e_\chi(h_j) = h_j$ for the character χ of h_j. Hence, possibly replacing h_j with $e_\chi(h'_j)$, we may assume that h_j is a regular G-semi-invariant function on Y with character χ, non-zero on X. Let $\hat{h}_j \in \mathfrak{R}$ be the expansion of h_j.

We apply Construction-Theorem 7.2 with $H = G$, $n = 4$, the parameters x, y, z, t, the functions h_j with $\beta_j = c_j$, no vectors in (iv), no prime divisors in (v), and, in (4), the automorphism ψ and the unit v of **Step 2**. Fix $N \geq N_1$ and then an integer $L \geq \max\{L_1, \operatorname{ord} \hat{g} + 1\}$, where N_1 and L_1 are the integers in Construction-Theorem 7.2(4). Let $h_{j,0} := \psi(\hat{h}_j)_{<N}$ and $g_0 := \hat{g}_{<L}$. Since $\operatorname{ord} \hat{g} \geq 2$, we have $L \geq 3$, so

$$f_0 := (v\psi(\hat{f}))_{<L} = (xy - \hat{g})_{<L} = xy - g_0.$$

Since $\hat{g} \in (z,t)^2$ and $L > \operatorname{ord} \hat{g}$, g_0 is a non-zero polynomial in $(z,t)^2$. By Construction-Theorem 7.2, G acts faithfully and diagonally on $\mathbb{A}^4$ with the coordinates x, y, z, t having characters $\chi_x, \chi_y, \chi_z, \chi_t$, $X_0 = (xy - g_0 = 0)$ is an isolated Gorenstein terminal germ at 0, $xy - g_0$ has character χ_f, and $h_{j,0}$ is a semi-invariant polynomial with the character of h_j whose restriction to X_0 is non-zero. Moreover, Construction-Theorem 7.2(1) implies (2) and (3), and Construction-Theorem 7.2(2) implies (4).

It remains to prove the rest of (1). Let $\sigma \in G$ with $\sigma \neq 1$. Since G acts faithfully on X, σ acts non-trivially on $\mathbb{C}(X)$, hence on $\mathcal{O}_{X,o}$ and on $\widehat{\mathcal{O}}_{X,o} \supset \mathcal{O}_{X,o}$. By the G-equivariant isomorphism in (4), σ acts non-trivially on the completion of $\mathcal{O}_{X_0,0}$, hence on X_0. Thus G acts faithfully on X_0. Whether an element of G belongs to R and acts as a given element of $\mu_r(1, -1, 0, 0)$, and whether it belongs to $\mathbf{T}_{(1,-1,0,0)}$, depends only on the characters of the four coordinates. Since the coordinates x, y, z, t of $\mathbb{A}^4$ have the characters of the eigenparameters x, y, z, t of Y, R acts on $\mathbb{A}^4$ as $\mu_r(1, -1, 0, 0)$, and $R = G \cap \mathbf{T}_{(1,-1,0,0)}$ on $\mathbb{A}^4$ if $R = G \cap \mathbf{T}_{(1,-1,0,0)}$ for the eigenparameters on Y. This proves (1). □

15.4. **Split hypersurfaces with a mixed circle.** In this subsection, we prove Theorem 15.9. The quotient mld $\operatorname{mld}((X \ni 0, B)/G)$ is bounded above by means of a prime divisor defined by a Gauss extension (Definition-Proposition 2.11) and below by comparison with log discrepancies over T and by adjunction along $(x^r = 0)$, in terms of the quotient mld of the plane pair (T, B_T) of Theorem 15.9(2) by H.

Theorem 15.9. *Fix an integer $e_0 \geq 1$ and a DCC set $\Gamma \subset [0,1]$. Let $X := (xy - g(z,t) = 0) \subset \mathbb{A}^4$ with $0 \neq g \in (z,t)^2$, let G be a finite diagonal abelian group preserving $xy - g$ up to a character and acting faithfully on X, and let $R := G \cap \mathbf{T}_{(1,-1,0,0)} = \mu_r(1, -1, 0, 0)$ and $H := G/R$. Let $h_1, \dots, h_s$ be G-semi-invariant polynomials with non-zero restrictions to X, $c_j \in \Gamma$ positive, and $B := \sum_j c_j \operatorname{div}_X(h_j)$. Assume that X is a normal klt threefold germ at 0, $|H| \leq e_0$, and (X, B) is klt near 0. Then the set of such numbers $\operatorname{mld}((X \ni 0, B)/G) \in [1, +\infty)$ satisfies the ACC. More precisely, write each h_j modulo $xy - g$ uniquely as $A_{j,0} + \sum_{k \geq 1} x^k A_{j,k} + \sum_{l \geq 1} y^l A_{j,-l}$ with $A_{j,k} \in \mathbb{C}[z,t]$ (the classes of 1, x^k, y^l with $k, l \geq 1$ form a basis of $\mathbb{C}[x,y,z,t]/(xy - g)$ over $\mathbb{C}[z,t]$). For any $\epsilon > 0$, if r is sufficiently large in terms of ϵ, e_0, and Γ, and $a := \operatorname{mld}((X \ni 0, B)/G) \geq 1 + \epsilon$, then the following hold.*

(1) For each j there exist non-negative integers a_j, b_j with $\min\{a_j, b_j\} = 0$ and $|a_j - b_j|$ bounded in terms of Γ such that $F_j := A_{j,a_j - b_j} \neq 0$, and any integer $k \neq a_j - b_j$ with

$A_{j,k} \neq 0$ *satisfies* $k \equiv a_j - b_j \pmod r$ *and* $|k| \geq r - C$ *for a constant* C *depending only on* Γ. *Let* $h_j^0 := x^{a_j} y^{b_j} F_j(z,t)$.

(2) *Let* $c_x := \sum_j c_j a_j$, $c_y := \sum_j c_j b_j$, $c_g := \max\{c_x, c_y\}$, $T := \mathbb{A}^2_{z,t}$, *and*

$$B_T := \sum_j c_j \operatorname{div}_T(F_j) + c_g \operatorname{div}_T(g).$$

Then H *acts faithfully on* T, (T, B_T) *is klt near* 0, *and the coefficients of* B_T *and* $B_{T/H}$ *belong to a DCC set depending only on* Γ.

(3) *The number* $b_H := \operatorname{mld}((T \ni 0, B_T)/H)$ *satisfies* $b_H \leq a \leq b_H + \frac{2}{\sqrt{r}}$.

Proof of Theorem 15.9. The proof is long, so we divide it into several steps.

Step 1. In this step, we write the h_j in normal form, bound a from above by weight vectors, and prove (1).

Let $\gamma := \gamma(\Gamma)$ if $\Gamma \backslash \{0\} \neq \emptyset$ and $\gamma := 1$ otherwise. Then $\gamma > 0$ (Lemma 2.10(2)) and $c_j \geq \gamma$ for any j. Let $C_0 := \lceil 1/\gamma \rceil + 1$ and fix $\epsilon > 0$. Each lower bound for r below depends only on ϵ, e_0, and Γ.

Since G is diagonal and $f := xy - g$ is G-semi-invariant, the character of any monomial of g equals the character of xy. Let $\tilde{h}_j := A_{j,0} + \sum_{k\geq 1} x^k A_{j,k} + \sum_{l \geq 1} y^l A_{j,-l}$. Replacing xy by g in the monomials of h_j, repeatedly, turns h_j into $\tilde{h}_j$. Thus $\tilde{h}_j$ is G-semi-invariant, the character of $\tilde{h}_j$ equals the character of h_j, and $\tilde{h}_j|_X = h_j|_X$. Possibly replacing h_j with $\tilde{h}_j$, we may assume that

$$h_j = \sum_{k \in \mathbb{Z}} h_{j,k}, \qquad h_{j,k} := x^k A_{j,k} \ \text{ for } k \geq 0, \qquad h_{j,k} := y^{-k} A_{j,k} \ \text{ for } k < 0,$$

where $A_{j,k} = 0$ for all but finitely many k.

Let $B_{\mathbb{A}^4} := \operatorname{div}(f) + \sum_j c_j \operatorname{div}(h_j)$, let $N_G \subset \mathbb{R}^4$ be the lattice of G with respect to the eigenparameters x, y, z, t of $\mathbb{A}^4$ at 0 (Definition 2.15), and let $J := (J(f), J(h_1), \dots, J(h_s))$ and $c := (1, c_1, \dots, c_s)$, so that $\psi_{J,c}(w) = \sum_i w_i - \operatorname{wt}_w(f) - \sum_j c_j \operatorname{wt}_w(h_j)$ (Definition 2.16). We show that

$$a \leq \psi_{J,c}(w) \quad \text{for any } w \in N_G \cap \mathbb{R}^4_{>0}. \tag{15.12}$$

If $h_j(0) \neq 0$, then $\operatorname{div}(h_j) = 0$ near 0 and $\operatorname{wt}_w(h_j) = 0$, so in the proof of (15.12) we may assume that $h_j(0) = 0$ for any j. By Theorem 8.3, applied to the hypersurface $X = (f = 0) \subset \mathbb{A}^4$ and the subgroup G, $a = \operatorname{mld}((\mathbb{A}^4 \ni 0, B_{\mathbb{A}^4})/G) > 0$, and any prime divisor over $\mathbb{A}^4/G$ with center the image of 0 has log discrepancy at least a with respect to $(\mathbb{A}^4, B_{\mathbb{A}^4})/G$. Let $w \in N_G \cap \mathbb{R}^4_{>0}$ be primitive. By Lemma 8.5, applied with $d = 4$ to the polynomial f with coefficient one and the polynomials h_j with coefficients c_j, w defines a toric prime divisor E_w over $\mathbb{A}^4/G$ with $a(E_w, \mathbb{A}^4/G, (B_{\mathbb{A}^4})_{\mathbb{A}^4/G}) = \psi_{J,c}(w)$, and the center of E_w is the image of 0, since the smallest face of $\mathbb{R}^4_{\geq 0}$ containing w is $\mathbb{R}^4_{\geq 0}$ itself [CLS11, Theorem 3.2.6 and Lemma 3.3.21(b)]. Thus $a \leq \psi_{J,c}(w)$. Any $w \in N_G \cap \mathbb{R}^4_{>0}$ is of the form $k'w'$ for an integer $k' \geq 1$ and a primitive $w' \in N_G \cap \mathbb{R}^4_{>0}$, and $\psi_{J,c}(w) = k'\psi_{J,c}(w') \geq k'a \geq a$ since $a > 0$. This proves (15.12).

Since $g \in (z,t)^2$, $\operatorname{wt}_{(1,1,1,1)}(f) = 2$. As $(1,1,1,1) \in \mathbb{Z}^4 \subset N_G$, (15.12) implies

$$a \leq 2 - \sum_j c_j \operatorname{mult}_0 h_j. \tag{15.13}$$

From now on until the end of **Step 8**, we assume that $a \geq 1 + \epsilon$ and $r > 2C_0$, and we let $n_r := r - C_0$. By (15.13), $\sum_j c_j \operatorname{mult}_0 h_j \leq 1 - \epsilon$, so $\operatorname{mult}_0 h_j < 1/\gamma < C_0$ for any j. A monomial occurring in $h_{j,k}$ has degree at least $|k|$, so there exists an integer κ_j with $|\kappa_j| < C_0$ such that h_{j,κ_j} contains a monomial of degree $\operatorname{mult}_0 h_j$. The element $(\zeta, \zeta^{-1}, 1, 1)$ of R, where $\zeta = e^{2\pi\sqrt{-1}/r}$, multiplies $h_{j,k}$ by ζ^k under pullback (Definition 2.14). Since h_j is G-semi-invariant, $h_{j,k} \neq 0$ implies $k \equiv \kappa_j \pmod r$, and then $|k| \geq r - |\kappa_j| > n_r$ if moreover $k \neq \kappa_j$. Let $a_j := \max\{\kappa_j, 0\}$

and $b_j := \max\{-\kappa_j, 0\}$. Then $\min\{a_j, b_j\} = 0$, $|a_j - b_j| = |\kappa_j| < C_0$, $F_j = A_{j,\kappa_j} \neq 0$, and $h_j^0 = h_{j,\kappa_j}$, so (1) holds with $C := C_0$. Moreover,

$$h_j - h_j^0 = \sum_{k \neq \kappa_j} h_{j,k} \in (x, y)^{n_r} \subset \mathbb{C}[x, y, z, t]. \tag{15.14}$$

Since $n_r > C_0 > \operatorname{mult}_0 h_j$, (15.14) implies $\operatorname{mult}_0 h_j^0 = \operatorname{mult}_0 h_j$. Let $\sigma_0 := \sum_j c_j \operatorname{mult}_0 F_j$. As $\operatorname{mult}_0 h_j^0 = a_j + b_j + \operatorname{mult}_0 F_j$, (15.13) implies

$$c_x + c_y + \sigma_0 \leq 1 - \epsilon. \tag{15.15}$$

Step 2. In this step, we prove (2).

Let $m := \operatorname{mult}_0 g \geq 2$, assume $r \geq 4$, and choose $\theta \in \frac{1}{r}\mathbb{Z}$ with $r^{-1/2} \leq \theta < r^{-1/2} + \frac{1}{r}$, so that $\theta < 2r^{-1/2} \leq 1$. Let

$$w_+ := (m - \theta, \theta, 1, 1), \qquad w_- := (\theta, m - \theta, 1, 1).$$

The differences $w_+ - (m, 0, 1, 1)$ and $w_- - (0, m, 1, 1)$ are integer multiples of $(-\frac{1}{r}, \frac{1}{r}, 0, 0)$ and $(\frac{1}{r}, -\frac{1}{r}, 0, 0)$ respectively, which belong to $N_R \subset N_G$, so $w_\pm \in N_G \cap \mathbb{R}^4_{>0}$. The sum of the coordinates of $w_\pm$ is $m + 2$, and $\operatorname{wt}_{w_\pm}(f) = m$. By (15.12) at $w_\pm$,

$$\sum_j c_j \operatorname{wt}_{w_\pm}(h_j) \leq 1 - \epsilon,$$

so $\operatorname{wt}_{w_\pm}(h_j) < 1/\gamma$ for any j. A monomial occurring in $h_{j,k}$ with $k \neq \kappa_j$ has $w_\pm$-weight at least $|k|\theta > n_r r^{-1/2}$. Assuming $n_r r^{-1/2} > 1/\gamma$, (15.14) implies $\operatorname{wt}_{w_\pm}(h_j) = \operatorname{wt}_{w_\pm}(h_j^0)$. If $c_x \geq c_y$, then the inequality at w_+ is

$$c_x(m - \theta) + c_y\theta + \sigma_0 = c_g m + \sigma_0 - |c_x - c_y|\theta \leq 1 - \epsilon,$$

and if $c_x < c_y$, then the inequality at w_- is $c_x\theta + c_y(m - \theta) + \sigma_0 \leq 1 - \epsilon$, whose left-hand side is again $c_g m + \sigma_0 - |c_x - c_y|\theta$. By (15.15), $|c_x - c_y| \leq 1$. Assuming $r \geq 16/\epsilon^2$, so that $\theta < 2r^{-1/2} \leq \epsilon/2$, we obtain

$$\operatorname{mult}_0 B_T = c_g m + \sigma_0 \leq 1 - \frac{\epsilon}{2}. \tag{15.16}$$

Let $\lambda' := 1 + \frac{\epsilon}{4}$ and $\lambda := 1 + \frac{\epsilon}{8}$. By (15.16), $\lambda' \operatorname{mult}_0 B_T \leq (1 + \frac{\epsilon}{4})(1 - \frac{\epsilon}{2}) < 1$. Let $B' \geq \lambda' B_T$ be a $\mathbb{Q}$-divisor with $\operatorname{Supp} B' = \operatorname{Supp} B_T$ and $\operatorname{mult}_0 B' < 1$. By [Laz04, Proposition 9.5.13], (T, B') is klt near 0, hence

$$(T, \lambda' B_T) \text{ is klt near } 0. \tag{15.17}$$

In particular, (T, B_T) is klt near 0.

An element of G acting trivially on T multiplies x and y by some $\mu_x, \mu_y \in \mathbb{C}^*$ and fixes $g \neq 0$. Since f is G-semi-invariant, $\mu_x\mu_y = 1$, so the element belongs to $G \cap \mathbf{T}_{(1,-1,0,0)} = R$. As R acts trivially on T, H acts faithfully on T.

If $c_g = c_x$, then $B_T = \sum_j c_j \operatorname{div}_T(F_j g^{a_j})$, and if $c_g = c_y$, then $B_T = \sum_j c_j \operatorname{div}_T(F_j g^{b_j})$. The polynomials $F_j = h_j^0/(x^{a_j} y^{b_j})$ and g are G-semi-invariant functions of z, t, hence H-semi-invariant functions on T, and H preserves B_T. We first prove that the coefficients of B_T and $B_{T/H}$ belong to a DCC set depending only on Γ. Every coefficient of B_T is a finite sum of elements of Γ, since $c_g \in \{c_x, c_y\} \subset \Gamma_+$. Thus it belongs to Γ_+, and Γ_+ satisfies the DCC by Lemma 2.10(1). Let $\pi_H : T \to T/H$ be the quotient morphism. Since H acts diagonally and faithfully on T, a non-identity element of H fixes pointwise at most one of the curves $(z = 0)$ and $(t = 0)$, and no other curve. Thus the inertia group of a prime divisor P of T other than $(z = 0)$ and $(t = 0)$ is trivial, and $\operatorname{mult}_{\pi_H(P)} B_{T/H} = \operatorname{mult}_P B_T \in \Gamma_+$ by Lemma 2.6(1) and (4). The curves $(z = 0)$ and $(t = 0)$ pass through 0, so $\operatorname{mult}_{(z=0)} B_T < 1$ and $\operatorname{mult}_{(t=0)} B_T < 1$ because (T, B_T) is klt near 0. Thus $\operatorname{mult}_{(z=0)} B_T, \operatorname{mult}_{(t=0)} B_T \in \Gamma_+ \cap [0, 1]$, and by Lemma 2.6(4) the coefficients of

their images in $B_{T/H}$ belong to $\Psi(\Gamma_+ \cap [0,1])$. Hence all coefficients of B_T and $B_{T/H}$ belong to $\Gamma_+ \cup \Psi(\Gamma_+ \cap [0,1])$, which is a DCC set depending only on Γ (Lemma 2.10(1)). This proves (2).

Moreover, by Lemma 2.10(5), applied to (T, B_T), H, and Γ, near 0 the coefficients of B_T belong to $\Gamma_+ \cap [0,1]$ and the coefficients of $B_{T/H}$ belong to $\Psi(\Gamma_+ \cap [0,1])$, which is used in **Steps 5 and 9**.

Step 3. In this step, we bound log discrepancies of prime divisors over X from below by log discrepancies of prime divisors over T.

Let $B^0 := \sum_j c_j \operatorname{div}_X(h_j^0)$. Here x, y, and F_j do not vanish identically on X. Since $y = g/x$ on X, $\mathbb{C}(X) = \mathbb{C}(T)(x)$. The canonical sheaf ω_X is generated by

$$\omega := \frac{dx}{x} \wedge dz \wedge dt,$$

the residue of $\frac{dx \wedge dy \wedge dz \wedge dt}{f}$ along X. Let E be a prime divisor over X. As x and y are regular on X, $\operatorname{ord}_E(x), \operatorname{ord}_E(y) \geq 0$.

Suppose that $\operatorname{ord}_E|_{\mathbb{C}(T)}$ is trivial. Then $\operatorname{ord}_E(x) + \operatorname{ord}_E(y) = \operatorname{ord}_E(g) = 0$, so $\operatorname{ord}_E(x) = \operatorname{ord}_E(y) = 0$ and $\operatorname{ord}_E(h_j^0) = 0$ for any j.

Suppose that $\operatorname{ord}_E|_{\mathbb{C}(T)}$ is non-trivial. Let η_E be the generic point of $\operatorname{center}_X E$ and ξ its image under the projection $X \to T$ to the coordinates z, t. The local homomorphism $\mathcal{O}_{T,\xi} \to \mathcal{O}_{X,\eta_E}$ maps the maximal ideal of $\mathcal{O}_{T,\xi}$ into that of $\mathcal{O}_{X,\eta_E}$, so Lemma 2.12 implies that $\operatorname{ord}_E|_{\mathbb{C}(T)} = \rho_E \operatorname{ord}_{\nu_E}$ for a positive integer ρ_E and a prime divisor ν_E over T. For a prime divisor ν over T, let $\mathcal{O}_\nu$ be the valuation ring of ord_ν, $\mathfrak{m}_\nu$ the maximal ideal of $\mathcal{O}_\nu$, and $\mathbb{C}(\nu)$ the residue field. Let $T' \to T$ be a proper birational morphism from a smooth surface on which ν_E is a prime divisor, let $u \in \mathcal{O}_{\nu_E}$ be a uniformizer, and let $u' \in \mathcal{O}_{\nu_E}$ be an element such that $du \wedge du'$ generates $\Omega^2_{T'}$ at the generic point of ν_E. As $dz \wedge dt$ generates ω_T,

$$dz \wedge dt \in u^{a(\nu_E,T,0)-1} \mathcal{O}^*_{\nu_E} du \wedge du'.$$

Let $V := T' \times \mathbb{A}^1_x$, so that $\mathbb{C}(V) = \mathbb{C}(X)$. Since $\operatorname{ord}_E \geq 0$ on $\mathcal{O}_{\nu_E}$ and $\operatorname{ord}_E(x) \geq 0$, E has a center on V with generic point η, and η maps to the generic point of ν_E. Thus $dx \wedge du \wedge du'$ generates ω_V near η, and comparing $a(E, X, 0) = 1 + \operatorname{ord}_E(\omega)$ with $a(E, V, \operatorname{div}(x)) = 1 + \operatorname{ord}_E(dx \wedge du \wedge du') - \operatorname{ord}_E(x)$, we obtain

$$a(E, X, 0) = a(E, V, \operatorname{div}(x)) + \rho_E\big(a(\nu_E, T, 0) - 1\big). \tag{15.18}$$

Either x does not vanish at η, or η is the generic point of $\nu_E \times \{0\}$ and u, x is a regular system of parameters of $\mathcal{O}_{V,\eta}$. In both cases, by Lemma 2.13, applied near η to a regular system of parameters containing u, with coefficient one for x in the second case and coefficient zero for the other parameters, $a(E, V, \operatorname{div}(x)) \geq \operatorname{ord}_E(u) = \rho_E$. By (15.18),

$$a(E, X, 0) \geq \rho_E a(\nu_E, T, 0). \tag{15.19}$$

As $xy = g$ on X, $\operatorname{ord}_E(x) + \operatorname{ord}_E(y) = \rho_E \operatorname{ord}_{\nu_E}(g)$. Since $c_x, c_y \leq c_g$ and $\operatorname{ord}_E(x), \operatorname{ord}_E(y) \geq 0$, we obtain $c_x \operatorname{ord}_E(x) + c_y \operatorname{ord}_E(y) \leq c_g \rho_E \operatorname{ord}_{\nu_E}(g)$, hence

$$\sum_j c_j \operatorname{ord}_E(h_j^0) = c_x \operatorname{ord}_E(x) + c_y \operatorname{ord}_E(y) + \rho_E \sum_j c_j \operatorname{ord}_{\nu_E}(F_j) \leq \rho_E \operatorname{mult}_{\nu_E} B_T,$$

and by (15.19), for any real number $\mu \geq 0$,

$$a(E, X, \mu B^0) \geq \rho_E a(\nu_E, T, \mu B_T). \tag{15.20}$$

Let $T^\circ \subset T$ be an open neighborhood of 0 on which $(T, \lambda' B_T)$ is klt (15.17), and let $X^\circ \subset X$ be an open neighborhood of 0 which is klt and contained in the preimage of T°. Let E be a prime divisor over X whose center on X intersects X°. If $\operatorname{ord}_E|_{\mathbb{C}(T)}$ is trivial, then $a(E, X, \lambda' B^0) = a(E, X, 0) > 0$. Otherwise, the generic point of the center of E on X maps to

the generic point of the center of ν_E on T, so the center of ν_E intersects T°, and $a(E, X, \lambda' B^0) > 0$ by (15.20). Thus

$$(X, \lambda' B^0) \text{ is klt near } 0. \tag{15.21}$$

Step 4. In this step, we prove that $(X, \lambda B)$ is lc near 0.

Recall that a pair (X', B') is lc near a closed point x' if and only if $\mathrm{mld}(X' \ni x', B') \geq 0$ (cf. [KM98, Corollary 2.31(1)]). Choose an integer q, depending only on ϵ and Γ, with $\frac{1}{q} < (\lambda' - \lambda)\gamma$, and let $p_j := \lceil q\lambda c_j \rceil$. As $c_j \geq \gamma$,

$$\lambda c_j \leq \frac{p_j}{q} < \lambda c_j + \frac{1}{q} < \lambda' c_j.$$

By (15.21) and Theorem 8.3, applied to $X \subset \mathbb{A}^4$, the polynomials h_j^0 with coefficients $\lambda' c_j$, and the trivial subgroup, $\mathrm{mld}(\mathbb{A}^4 \ni 0, \mathrm{div}(f) + \sum_j \lambda' c_j \mathrm{div}(h_j^0)) > 0$. Let $\Phi_0 := f^q \prod_j (h_j^0)^{p_j}$. Since $\frac{1}{q}\mathrm{div}(\Phi_0) \leq \mathrm{div}(f) + \sum_j \lambda' c_j \mathrm{div}(h_j^0)$, the pair $(\mathbb{A}^4, \frac{1}{q}\mathrm{div}(\Phi_0))$ is lc near 0.

For a non-zero polynomial Φ' with $\Phi'(0) = 0$, let $\mathrm{lct}_0(\Phi')$ be the largest real number c' such that $(\mathbb{A}^4, c'\mathrm{div}(\Phi'))$ is lc near 0. Then $\mathrm{lct}_0(\Phi_0) \geq \frac{1}{q}$. Let $\Phi := f^q \prod_j h_j^{p_j}$. By (15.14), $\Phi - \Phi_0 \in \mathfrak{m}^{n_r}$, where $\mathfrak{m} := (x, y, z, t)$. By [dFM09, Corollary 2.10], applied to $\mathfrak{a} = (\Phi)$ and $\mathfrak{b} = (\Phi_0) \subset (\Phi) + \mathfrak{m}^{n_r}$, as $\dim \mathcal{O}_{\mathbb{A}^4,0} = 4$,

$$\mathrm{lct}_0(\Phi) \geq \mathrm{lct}_0(\Phi_0) - \frac{4}{n_r} \geq \frac{1}{q} - \frac{4}{n_r}.$$

Since $(\mathbb{A}^4, \mathrm{lct}_0(\Phi)\mathrm{div}(\Phi))$ is lc on an open neighborhood U of 0, $\mathrm{lct}_0(\Phi)$ is the log canonical threshold of $\Phi|_U$ on the smooth fourfold U. By [dFEM10, Theorems 1.1 and 5.1], the log canonical thresholds on smooth fourfolds satisfy the ACC, so there exists $\omega_q > 0$, depending only on q, such that none of them belongs to $(\frac{1}{q} - \omega_q, \frac{1}{q})$. Assuming $n_r > 4/\omega_q$, we obtain $\mathrm{lct}_0(\Phi) \geq \frac{1}{q}$, that is, $(\mathbb{A}^4, \mathrm{div}(f) + \sum_j \frac{p_j}{q}\mathrm{div}(h_j))$ is lc near 0. Theorem 8.3 requires the pair on X to be klt, which is not known here. Instead, as X is normal and klt near 0, Theorem 2.22, applied with the trivial group, $n = 4$, $c = 1$, $f_1 = f$, and the $\mathbb{R}$-ideal $\prod_j (h_j)^{p_j/q}$, implies

$$\mathrm{mld}\Big(X \ni 0, \sum_j \frac{p_j}{q}\mathrm{div}_X(h_j)\Big) = \mathrm{mld}\Big(\mathbb{A}^4 \ni 0, \mathrm{div}(f) + \sum_j \frac{p_j}{q}\mathrm{div}(h_j)\Big) \geq 0.$$

As $\lambda c_j \leq \frac{p_j}{q}$, we conclude that

$$(X, \lambda B) \text{ is lc near } 0. \tag{15.22}$$

Step 5. In this step, we prove that $a \leq b_H + \frac{2}{\sqrt{r}}$.

For a prime divisor ν over T with center 0, let $e_\nu := |I_H(\nu)| \leq e_0$ and let $G_\nu \subset G$ be the preimage of $I_H(\nu)$. Since $I_H(\nu)$ acts trivially on $\mathbb{C}(\nu)$, it acts on the one-dimensional $\mathbb{C}(\nu)$-vector space $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$ by a character, which is faithful by Lemma 2.6(1). The projection of a uniformizer of $\mathcal{O}_\nu$ to the isotypic component of $\mathbb{C}(T)$ for the character of $I_H(\nu)$ on $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$ is an $I_H(\nu)$-semi-invariant uniformizer u_ν of $\mathcal{O}_\nu$. As G_ν acts on $\mathbb{C}(T)$ through $I_H(\nu)$, u_ν is G_ν-semi-invariant with a character α_ν of G_ν, the kernel of α_ν is R, and $\alpha_\nu(G_\nu)$ is the group of e_ν-th roots of unity. Let χ_x be the character of the G-semi-invariant function x. Then $\chi_x(R)$ is the group of r-th roots of unity.

By Lemma 2.6(2)(3) and part (2) of the theorem, $(T/H, B_{T/H})$ is klt near the image of 0, so $b_H > 0$. By **Step 2** and Theorem 2.18(2), applied to the DCC set $\Psi(\Gamma_+ \cap [0, 1])$, there exist a positive integer l depending only on Γ and a prime divisor E_T over T/H with center the image of 0 such that $a(E_T, T/H, B_{T/H}) = b_H$ and $a(E_T, T/H, 0) \leq l$. By Lemma 2.6(1)–(3), there exists a prime divisor ν over T with $\mathrm{ord}_\nu|_{\mathbb{C}(T/H)} = e_\nu \mathrm{ord}_{E_T}$, and the center of ν is 0, since

$\pi_H^{-1}(\pi_H(0)) = \{0\}$. Let $(T,0)/H = (T/H, B^{\mathrm{br}})$, so that $B^{\mathrm{br}} \geq 0$ (Lemma 2.6(0)). As T/H is $\mathbb{Q}$-factorial [KM98, Lemma 5.16], Lemma 2.6(2) and [KM98, Lemma 2.27] imply

$$\frac{a(\nu,T,0)}{e_\nu} = a(E_T, T/H, B^{\mathrm{br}}) \leq a(E_T, T/H, 0) \leq l, \qquad \frac{a(\nu,T,B_T)}{e_\nu} = b_H. \tag{15.23}$$

Let $u := u_\nu$ and $\alpha := \alpha_\nu$. Choose $\sigma_1 \in G_\nu$ with $\alpha(\sigma_1) = e^{2\pi\sqrt{-1}/e_\nu}$ and $\xi_1 \in \mathbb{Q}$ with $\chi_x(\sigma_1) = e^{2\pi\sqrt{-1}\xi_1}$. Replacing σ_1 with $\sigma_1\sigma_2$ for $\sigma_2 \in R$, we see that for any $\xi \in \xi_1 + \frac{1}{r}\mathbb{Z}$ there exists $\sigma \in G_\nu$ with

$$\alpha(\sigma) = e^{2\pi\sqrt{-1}/e_\nu} \quad \text{and} \quad \chi_x(\sigma) = e^{2\pi\sqrt{-1}\xi}. \tag{15.24}$$

As ν has center 0 and $g \in (z,t)^2$, $\operatorname{ord}_\nu(g) \geq 2$, so $\operatorname{ord}_\nu(g)/e_\nu \geq 2/e_0$. For $\xi \in \mathbb{Q}$ let $\xi_y := \operatorname{ord}_\nu(g)/e_\nu - \xi$. If $c_x \geq c_y$, choose $\xi \in \xi_1 + \frac{1}{r}\mathbb{Z}$ with $r^{-1/2} \leq \xi_y < r^{-1/2} + \frac{1}{r}$, and let $\tau := \xi_y$. If $c_x < c_y$, choose $\xi \in \xi_1 + \frac{1}{r}\mathbb{Z}$ with $r^{-1/2} \leq \xi < r^{-1/2} + \frac{1}{r}$, and let $\tau := \xi$. Assuming $r \geq 4e_0^2$, so that $\frac{2}{e_0} - 2r^{-1/2} \geq 2r^{-1/2}$, in both cases $\xi, \xi_y > 0$, $\tau = \min\{\xi, \xi_y\} < 2r^{-1/2}$, and

$$c_x\xi + c_y\xi_y = c_g\frac{\operatorname{ord}_\nu(g)}{e_\nu} - |c_x - c_y|\tau. \tag{15.25}$$

Write $(\frac{1}{e_\nu}, \xi) = \frac{1}{m_0}(m_1, m_2)$ with coprime positive integers m_1, m_2. Then $m_0 = m_1 e_\nu$ is a positive integer. Let v' be the Gauss extension of $m_1 \operatorname{ord}_\nu$ to $\mathbb{C}(X) = \mathbb{C}(T)(x)$ with x-value m_2 (Definition-Proposition 2.11). As $m_1 \operatorname{ord}_\nu(\mathbb{C}(T)^*) = m_1\mathbb{Z}$ and $\gcd(m_1, m_2) = 1$, $v'(\mathbb{C}(X)^*) = \mathbb{Z}$. For $b_0, \dots, b_n \in \mathcal{O}_\nu$ not all in $\mathfrak{m}_\nu$, the definition of v' implies

$$v'\Big(\sum_k b_k\big(x^{m_1}u^{-m_2}\big)^k\Big) = \min_k\big(m_1\operatorname{ord}_\nu(b_k) - km_1m_2 + km_1m_2\big) = 0.$$

Thus the residue class of $x^{m_1}u^{-m_2}$ is transcendental over $\mathbb{C}(\nu)$, the residue field of v' has transcendence degree two over $\mathbb{C}$, and $v' = \operatorname{ord}_{E'}$ for a prime divisor E' over X [KM98, Lemma 2.45]. Since $\operatorname{ord}_{E'}(x) = m_2 > 0$, $\operatorname{ord}_{E'}(y) = m_1\operatorname{ord}_\nu(g) - m_2 = m_0\xi_y > 0$, and $\operatorname{ord}_{E'}(z), \operatorname{ord}_{E'}(t) \geq m_1 > 0$, the center of E' on X is $\{0\}$.

We compute $a(E', X, B)$. Let T' and $V = T' \times \mathbb{A}^1_x$ be as in **Step 3** for ν, and let η be the generic point of $\nu \times \{0\} \subset V$. Then u, x is a regular system of parameters of $\mathcal{O}_{V,\eta}$. Let val be the monomial valuation at η in u, x with weights (m_1, m_2) [JM12, Proposition 3.1]. Both val and $\operatorname{ord}_{E'}$ are valuations of $\mathbb{C}(X)$, the fraction field of $\mathcal{O}_\nu[x]$, so it suffices to compare them on $\mathcal{O}_\nu[x]$. Let $\phi = \sum_k b_k x^k$ with $b_k \in \mathcal{O}_\nu$. Expanding each b_k in $\widehat{\mathcal{O}_\nu} \cong \mathbb{C}(\nu)[[u]]$, by [JM12, Proposition 3.1] we obtain

$$\operatorname{val}(\phi) = \min_k\big(m_1\operatorname{ord}_\nu(b_k) + km_2\big) = \operatorname{ord}_{E'}(\phi).$$

By [JM12, Proposition 5.1 and (5.1)], applied to the log smooth pair $(V, \operatorname{div}(u) + \operatorname{div}(x))$ near η, $a(E', V, 0) = m_1 + m_2$, so $a(E', V, \operatorname{div}(x)) = m_1$. By (15.18) for E', where $\nu_{E'} = \nu$ and $\rho_{E'} = m_1$,

$$a(E', X, 0) = m_1 a(\nu, T, 0) = \frac{m_0}{e_\nu}a(\nu, T, 0).$$

In $\mathbb{C}(T)[x, x^{-1}]$, $h_{j,k} = A_{j,k}x^k$ for $k \geq 0$ and $h_{j,k} = A_{j,k}g^{-k}x^k$ for $k < 0$. By the definition of v', $\operatorname{ord}_{E'}(h_j) = \min_k \operatorname{ord}_{E'}(h_{j,k})$, where

$$\frac{\operatorname{ord}_{E'}(h_{j,k})}{m_0} = \frac{\operatorname{ord}_\nu(A_{j,k})}{e_\nu} + k\xi \quad \text{for } k \geq 0, \qquad \frac{\operatorname{ord}_{E'}(h_{j,k})}{m_0} = \frac{\operatorname{ord}_\nu(A_{j,k})}{e_\nu} - k\xi_y \quad \text{for } k < 0.$$

Thus $\operatorname{ord}_{E'}(h_{j,k})/m_0 \geq |k|\tau > n_r r^{-1/2}$ if $k \neq \kappa_j$ and $h_{j,k} \neq 0$. On the other hand, as $c_x\xi + c_y\xi_y \leq c_g(\xi + \xi_y)$, (15.23) implies

$$\sum_j c_j\frac{\operatorname{ord}_{E'}(h_j^0)}{m_0} = \sum_j c_j\frac{\operatorname{ord}_\nu(F_j)}{e_\nu} + c_x\xi + c_y\xi_y \leq \frac{\operatorname{mult}_\nu B_T}{e_\nu} < \frac{a(\nu,T,0)}{e_\nu} \leq l,$$

so $\operatorname{ord}_{E'}(h_j^0)/m_0 < l/\gamma$. Assuming $n_r r^{-1/2} > l/\gamma$, we obtain $\operatorname{ord}_{E'}(h_j) = \operatorname{ord}_{E'}(h_j^0)$ for any j. By (15.25) and (15.23),

$$\frac{a(E',X,B)}{m_0} = \frac{a(\nu,T,0)}{e_\nu} - \sum_j c_j \frac{\operatorname{ord}_\nu(F_j)}{e_\nu} - c_x\xi - c_y\xi_y = \frac{a(\nu,T,B_T)}{e_\nu} + |c_x - c_y|\tau = b_H + |c_x - c_y|\tau.$$

As $|c_x - c_y| \le 1$ and $\tau < 2r^{-1/2}$, we obtain $0 < a(E',X,B)/m_0 < b_H + \frac{2}{\sqrt{r}}$.

We show that $|I_G(E')| \in m_0\mathbb{Z}$. Let $\sigma' \in I_G(E')$. Then σ' preserves $\operatorname{ord}_{E'}|_{\mathbb{C}(T)} = m_1 \operatorname{ord}_\nu$ and acts trivially on the residue field of $\operatorname{ord}_{E'}$, which contains $\mathbb{C}(\nu)$. Thus the image of σ' in H belongs to $I_H(\nu)$, that is, $\sigma' \in G_\nu$. Hence $I_G(E') = I_{G_\nu}(E')$, and by Lemma 2.6(1) applied to G_ν, the value group of $\operatorname{ord}_{E'}|_{\mathbb{C}(X)^{G_\nu}}$ is $|I_G(E')|\mathbb{Z}$. Since G_ν preserves $\mathbb{C}(T)[x,x^{-1}]$, $\mathbb{C}(X)^{G_\nu}$ is the fraction field of $\mathbb{C}(T)[x,x^{-1}]^{G_\nu}$ (multiply the numerator and the denominator by the product of the other conjugates of the denominator). As x is G_ν-semi-invariant, any element of $\mathbb{C}(T)[x,x^{-1}]^{G_\nu}$ is a sum of G_ν-invariant elements $s'x^i$ with $s' \in \mathbb{C}(T)$ and $i \in \mathbb{Z}$, and $\operatorname{ord}_{E'}$ of the sum is the minimum of the $\operatorname{ord}_{E'}(s'x^i)$. Let $s'x^i \ne 0$ be G_ν-invariant and $k' := \operatorname{ord}_\nu(s')$. Then s' is G_ν-semi-invariant, and $w' := s'u^{-k'}$ is a G_ν-semi-invariant unit of $\mathcal{O}_\nu$. Since G_ν acts trivially on $\mathbb{C}(\nu)$ and the residue class of w' is non-zero, w' is G_ν-invariant. Thus $\alpha^{k'}\chi_x^i$ is trivial on G_ν, and evaluating at σ as in (15.24), $\frac{k'}{e_\nu} + i\xi \in \mathbb{Z}$. Therefore

$$\operatorname{ord}_{E'}(s'x^i) = m_1k' + m_2i = m_0\Big(\frac{k'}{e_\nu} + i\xi\Big) \in m_0\mathbb{Z},$$

so $|I_G(E')| \in m_0\mathbb{N}^+$. By Lemma 2.6,

$$a \le \frac{a(E',X,B)}{|I_G(E')|} \le \frac{a(E',X,B)}{m_0} < b_H + \frac{2}{\sqrt{r}}. \tag{15.26}$$

Step 6. In this step, we fix a prime divisor E computing a, bound orders along E, and prove that $a \ge b_H$ if $\operatorname{ord}_E(h_j) = \operatorname{ord}_E(h_j^0)$ for any j.

By Lemma 2.6(2)(3), $(X/G, B_{X/G})$ is klt near the image of 0, so some prime divisor over X/G with center the image of 0 computes a [Amb06, Section 2]. By Lemma 2.6(2)(3), there exists a prime divisor E over X with center $\{0\}$ such that $a(E,X,B) = ea$, where $e := |I_G(E)|$. By (15.13), $a \le 2$. Let

$$M_3 := \frac{2\lambda}{\lambda - 1}, \qquad M_2 := \frac{2}{\gamma(\lambda-1)}, \qquad M_1 := \frac{M_3}{\lambda'\gamma}.$$

By (15.22), $a(E,X,\lambda B) \ge 0$, so

$$(\lambda-1)\sum_j c_j \operatorname{ord}_E(h_j) = a(E,X,B) - a(E,X,\lambda B) \le a(E,X,B) \le 2e.$$

Thus $a(E,X,0) = a(E,X,B) + \sum_j c_j \operatorname{ord}_E(h_j) \le 2e + \frac{2e}{\lambda-1}$, and

$$a(E,X,0) \le eM_3, \qquad \operatorname{ord}_E(h_j) \le eM_2 \quad \text{for any } j. \tag{15.27}$$

As the center of E is $\{0\}$, $\operatorname{ord}_E(z), \operatorname{ord}_E(t) > 0$, so $\operatorname{ord}_E|_{\mathbb{C}(T)}$ is non-trivial. Let $\rho := \rho_E$ and ν_E be as in **Step 3**. Then ν_E has center 0. By (15.21), $a(E,X,\lambda'B^0) > 0$, so $\lambda'\gamma \operatorname{ord}_E(h_j^0) \le \lambda' \sum_{j'} c_{j'} \operatorname{ord}_E(h_{j'}^0) < a(E,X,0) \le eM_3$, that is,

$$\operatorname{ord}_E(h_j^0) < eM_1 \quad \text{for any } j. \tag{15.28}$$

Since $I_G(E)$ preserves $\operatorname{ord}_E|_{\mathbb{C}(T)} = \rho \operatorname{ord}_{\nu_E}$ and acts trivially on the residue field of ord_E, which contains $\mathbb{C}(\nu_E)$, the image of $I_G(E)$ in H is contained in $I_H(\nu_E)$, so $I_G(E) \subset G_{\nu_E}$. By Lemma 2.6(1), $I_G(E)$ is cyclic of order e, and it acts on $\mathfrak{m}_E/\mathfrak{m}_E^2$ by a faithful character χ_E, where $\mathfrak{m}_E$ is the maximal ideal of the valuation ring of ord_E. As $\operatorname{ord}_E(u_{\nu_E}) = \rho$ and $I_G(E)$ acts on $\mathfrak{m}_E^\rho/\mathfrak{m}_E^{\rho+1}$ by χ_E^ρ, we have $\alpha_{\nu_E}|_{I_G(E)} = \chi_E^\rho$. Since the kernel of α_{ν_E} is R, the kernel of χ_E^ρ is

$I_G(E) \cap R$. Let $e_R := |I_G(E) \cap R|$ and $e_H := e/e_R$, the order of the image of $I_G(E)$ in H. As χ_E is faithful, $e_R = \gcd(e, \rho)$. Thus ρ/e_R is a positive integer, $e_H \le |I_H(\nu_E)| \le e_0$, and

$$\frac{\rho}{e} \ge \frac{e_R}{e} = \frac{1}{e_H} \ge \frac{1}{e_0}. \tag{15.29}$$

Suppose that $\operatorname{ord}_E(h_j) = \operatorname{ord}_E(h_j^0)$ for any j. Then $a(E, X, B) = a(E, X, B^0)$. By (15.20) with $\mu = 1$, (15.29), $a(\nu_E, T, B_T) > 0$, and Lemma 2.6 applied to H acting on T,

$$a = \frac{a(E, X, B^0)}{e} \ge \frac{\rho}{e} a(\nu_E, T, B_T) = \frac{\rho}{e_R} \cdot \frac{a(\nu_E, T, B_T)}{e_H} \ge \frac{a(\nu_E, T, B_T)}{|I_H(\nu_E)|} \ge b_H.$$

Step 7. In this step, we assume that $\operatorname{ord}_E(h_{j_0}) \ne \operatorname{ord}_E(h_{j_0}^0)$ for some j_0, and we show that the terms $h_{j,k}$ with $k < \kappa_j$ do not contribute to $\operatorname{ord}_E(h_j)$.

Since $\operatorname{ord}_E(h_{j_0}) \ne \operatorname{ord}_E(h_{j_0}^0)$, we have $\operatorname{ord}_E(h_{j_0} - h_{j_0}^0) \le \operatorname{ord}_E(h_{j_0}^0)$. By (15.14) and (15.28),

$$n_r \min\{\operatorname{ord}_E(x), \operatorname{ord}_E(y)\} \le \operatorname{ord}_E(h_{j_0} - h_{j_0}^0) < eM_1.$$

Interchanging x and y interchanges a_j and b_j, c_x and c_y, and $h_{j,k}$ and $h_{j,-k}$, and replaces κ_j with $-\kappa_j$. It preserves R, H, B_T, b_H, and the conclusions of **Steps 1**–**6**. Thus we may assume that $\operatorname{ord}_E(x) \le \operatorname{ord}_E(y)$. As ν_E has center 0, $\operatorname{ord}_{\nu_E}(g) \ge 2$, so by (15.29),

$$\frac{\operatorname{ord}_E(x) + \operatorname{ord}_E(y)}{e} = \frac{\rho}{e} \operatorname{ord}_{\nu_E}(g) \ge \frac{2}{e_0}.$$

Assuming $n_r \ge e_0 M_1$, we obtain

$$\frac{\operatorname{ord}_E(y)}{e} \ge \frac{2}{e_0} - \frac{M_1}{n_r} \ge \frac{1}{e_0}. \tag{15.30}$$

Let $h_j^- := \sum_{k<\kappa_j} h_{j,k}$ and $h_j^+ := h_j - h_j^- = \sum_{k \ge \kappa_j} h_{j,k}$. If $k < \kappa_j$ and $h_{j,k} \ne 0$, then $k \le \kappa_j - r < 0$ by **Step 1**, so $h_{j,k} = y^{-k} A_{j,k}$ with $-k \ge r - \kappa_j > n_r$. Thus $\operatorname{ord}_E(h_j^-) \ge n_r \operatorname{ord}_E(y) \ge n_r e/e_0$ by (15.30). Assuming $n_r > e_0 M_2$, (15.27) implies $\operatorname{ord}_E(h_j^-) > \operatorname{ord}_E(h_j)$. Hence

$$\operatorname{ord}_E(h_j^+) = \operatorname{ord}_E(h_j) \quad \text{for any } j. \tag{15.31}$$

Step 8. In this step, we prove that $a \ge b_H$ under the assumption of **Step 7**, by adjunction along $(\bar{x} = 0)$, where $\bar{x} := x^r$.

Let $Y := \mathbb{A}^3_{\bar{x},z,t}$. Since $\mathbb{C}(X) = \mathbb{C}(x, z, t)$ and R multiplies x by r-th roots of unity and fixes z, t, $\mathbb{C}(X)^R = \mathbb{C}(\bar{x}, z, t) = \mathbb{C}(Y)$. Let $h_{j,k} \ne 0$ with $k \ge \kappa_j$. Then $k = \kappa_j + rl'$ for an integer $l' \ge 0$. If $k \ge 0$, then $h_{j,k} = x^{\kappa_j} \bar{x}^{l'} A_{j,k}$. If $k < 0$, then $l' = 0$ as $\kappa_j + r > 0$, so $k = \kappa_j < 0$ and $h_{j,k} = y^{b_j} F_j = x^{\kappa_j} g^{b_j} F_j$ on X. Thus, on X,

$$h_j^+ = x^{\kappa_j} \phi_j(\bar{x}, z, t), \qquad \phi_j \in \mathbb{C}[\bar{x}, z, t], \qquad \phi_j(0, z, t) = g^{b_j} F_j. \tag{15.32}$$

The element $(\zeta, \zeta^{-1}, 1, 1)$ of R multiplies h_j^+ and x^{κ_j} by ζ^{κ_j}, so ϕ_j is R-invariant and G-semi-invariant, that is, ϕ_j is an H-semi-invariant polynomial on Y. The group H acts diagonally on Y, and faithfully on $T = (\bar{x} = 0)$ by **Step 2**.

Let $\beta := c_x - c_y = \sum_j c_j \kappa_j$, and let

$$B_Y := \Big(1 + \frac{\beta}{r}\Big) \operatorname{div}(\bar{x}) + \sum_j c_j \operatorname{div}(\phi_j), \qquad B'_Y := \operatorname{div}(\bar{x}) + \sum_j c_j \operatorname{div}(\phi_j) + \max\{\beta, 0\} \operatorname{div}(g)$$

on Y. Since $c_y + \max\{\beta, 0\} = c_g$, (15.32) implies $\sum_j c_j \operatorname{div}_T(\phi_j|_T) + \max\{\beta, 0\} \operatorname{div}_T(g) = B_T$.

As $I_R(E) = I_G(E) \cap R$, Lemma 2.6(1) applied to R implies that $\operatorname{ord}_E|_{\mathbb{C}(Y)} = e_R \operatorname{ord}_{E_Y}$ for a prime divisor E_Y over Y. The center of E_Y on Y is 0, as $\operatorname{ord}_{E_Y}(\bar{x}) = r \operatorname{ord}_E(x)/e_R > 0$ and $\operatorname{ord}_{E_Y}(z), \operatorname{ord}_{E_Y}(t) > 0$. By Lemma 2.6(1) applied to G acting on X and to H acting on Y, $|I_H(E_Y)| = e/e_R = e_H$.

A non-identity element of R fixes exactly the points of X with $x = y = 0$, which form the curve $(x = y = g = 0)$. Thus the quotient morphism $X \to X/R$ is quasi-étale, and $(X, 0)/R = (X/R, 0)$. Since R is finite, $\mathbb{C}[X]^R$ is the image of $\mathbb{C}[x, y, z, t]^R$, which is generated by x^r, y^r, xy, z, t. As $xy = g$ on X, $\mathbb{C}[X]^R$ is generated by $\bar{x}$, $y' := y^r$, z, t, which satisfy $\bar{x}y' = g^r$. The polynomial $\bar{x}y' - g^r$ is irreducible since $g \neq 0$, and both $\mathbb{C}[\bar{x}, y', z, t]/(\bar{x}y' - g^r)$ and $\mathbb{C}[X]^R$ are domains of dimension 3, so $X/R = (\bar{x}y' - g^r = 0) \subset \mathbb{A}^4_{\bar{x},y',z,t}$. The projection $X/R \to Y$ is birational, and the residue form $\frac{d\bar{x}}{\bar{x}} \wedge dz \wedge dt$ generates both $\omega_{X/R}$ and $\omega_Y(\operatorname{div}(\bar{x}))$, so $(X/R, 0)$ and $(Y, \operatorname{div}(\bar{x}))$ are crepant birational. By Lemma 2.6(2) applied to R,

$$a(E, X, 0) = e_R a(E_Y, X/R, 0) = e_R a(E_Y, Y, \operatorname{div}(\bar{x})).$$

By (15.31) and (15.32), and as $\operatorname{ord}_E(x) = e_R \operatorname{ord}_{E_Y}(\bar{x})/r$,

$$\sum_j c_j \operatorname{ord}_E(h_j) = \sum_j c_j\big(\kappa_j \operatorname{ord}_E(x) + \operatorname{ord}_E(\phi_j)\big) = e_R\Big(\frac{\beta}{r}\operatorname{ord}_{E_Y}(\bar{x}) + \sum_j c_j \operatorname{ord}_{E_Y}(\phi_j)\Big).$$

Hence $a(E, X, B) = e_R a(E_Y, Y, B_Y)$, and

$$a = \frac{a(E, X, B)}{e} = \frac{a(E_Y, Y, B_Y)}{e_H}. \tag{15.33}$$

By Theorem 8.3, applied with $n = 3$ to the hypersurface $T = (\bar{x} = 0) \subset Y$, the group H, the polynomials ϕ_j with coefficients c_j, and, if $\beta > 0$, the polynomial g with coefficient β, any prime divisor over Y/H with center the image of 0 has log discrepancy at least $\operatorname{mld}((T \ni 0, B_T)/H) = b_H$ with respect to $(Y, B'_Y)/H$. Here $\phi_j|_T \neq 0$ by (15.32), and (T, B_T) is klt near 0 by (2). By Lemma 2.6(2) applied to H acting on Y,

$$\frac{a(E_Y, Y, B'_Y)}{e_H} \geq b_H. \tag{15.34}$$

Finally,

$$a(E_Y, Y, B_Y) - a(E_Y, Y, B'_Y) = -\frac{\beta}{r}\operatorname{ord}_{E_Y}(\bar{x}) + \max\{\beta, 0\}\operatorname{ord}_{E_Y}(g).$$

If $\beta \leq 0$, the right-hand side is $\frac{-\beta}{r}\operatorname{ord}_{E_Y}(\bar{x}) \geq 0$. If $\beta > 0$, since $g = xy$ on X, the right-hand side is

$$\beta\Big(\operatorname{ord}_{E_Y}(g) - \frac{\operatorname{ord}_{E_Y}(\bar{x})}{r}\Big) = \frac{\beta}{e_R}\big(\operatorname{ord}_E(g) - \operatorname{ord}_E(x)\big) = \frac{\beta}{e_R}\operatorname{ord}_E(y) \geq 0.$$

By (15.33) and (15.34), $a \geq b_H$. Together with **Step 6** and (15.26), this proves (3).

Step 9. In this step, we prove the ACC.

Suppose that $a^{(1)} < a^{(2)} < \cdots$ is a strictly increasing sequence of numbers

$$a^{(i)} = \operatorname{mld}((X^{(i)} \ni 0, B^{(i)})/G^{(i)}) \in [1, +\infty)$$

as in the theorem, with groups $R^{(i)} = \mu_{r^{(i)}}(1, -1, 0, 0)$ and $H^{(i)}$. Let $\epsilon := a^{(2)} - 1 > 0$, so that $a^{(i)} \geq 1 + \epsilon$ for any $i \geq 2$. If $r^{(i)}$ is bounded on an infinite subsequence, then $|G^{(i)}| = r^{(i)}|H^{(i)}| \leq r^{(i)}e_0$ is bounded on the subsequence, and since the hypersurface $X^{(i)}$ is Gorenstein, Proposition 15.6 contradicts the strict increase of $a^{(i)}$. Otherwise, passing to a subsequence, we may assume that $r^{(i)} \to +\infty$. Then for $i \gg 0$, (3) implies

$$b^{(i)} \leq a^{(i)} \leq b^{(i)} + \frac{2}{\sqrt{r^{(i)}}},$$

where $b^{(i)} = \operatorname{mld}((T \ni 0, B_T^{(i)})/H^{(i)})$. By **Step 2** applied to each $i \gg 0$, $b^{(i)}$ is the mld of an lc surface germ whose boundary has coefficients in $\Psi(\Gamma_+ \cap [0, 1])$ near the point, so by Theorem 2.18(1), the set of the $b^{(i)}$ with $i \gg 0$ satisfies the ACC. Since $b^{(i)} \leq a^{(i)}$ for $i \gg 0$ and $a^{(i)} - b^{(i)} \to 0$, Lemma 14.11, applied to the sequences $(a^{(i)})_{i \geq i_0}$ and $(b^{(i)})_{i \geq i_0}$ for some i_0, gives a contradiction. □

15.5. **The singular high range.** In this subsection, we prove Proposition 15.1.

Proof of Proposition 15.1. Suppose that the proposition does not hold. Then there exists a strictly increasing sequence $a_1 < a_2 < \cdots$ of numbers in the set of Proposition 15.1. Let $\epsilon := a_2 - 1$. Since $a_2 > a_1 \geq 1$, we have $\epsilon > 0$ and $a_i \geq 1 + \epsilon$ for any $i \geq 2$. Let $e(\epsilon)$ be the integer in Theorem 15.5, let $\mathcal{A}_1$ be the set of numbers in Proposition 15.6 for $L := e(\epsilon)$ and Γ, and let $\mathcal{A}_2$ be the set of numbers in Theorem 15.9 for $e_0 := e(\epsilon)$ and Γ. By Proposition 15.6 and Theorem 15.9, $\mathcal{A}_1$ and $\mathcal{A}_2$ satisfy the ACC.

We show that any number $a \geq 1 + \epsilon$ in the set of Proposition 15.1 belongs to $\mathcal{A}_1 \cup \mathcal{A}_2$. Write $a = \mathrm{mld}((X \ni o, B)/G)$, where $(X \ni o)$, G, $(Y \ni o)$, f, and $B = \sum_j c_j \operatorname{div}_X(h_j)$ satisfy the assumptions of Proposition 15.1.

Since G is abelian and fixes o, the induced action of G on $\mathfrak{m}_{Y,o}/\mathfrak{m}_{Y,o}^2$ is diagonalizable. For a character χ of G, the map

$$p_\chi(u) := \frac{1}{|G|} \sum_{\sigma \in G} \chi(\sigma)^{-1} \sigma^* u$$

sends $\mathfrak{m}_{Y,o}$ to the χ-semi-invariant elements of $\mathfrak{m}_{Y,o}$ and induces on $\mathfrak{m}_{Y,o}/\mathfrak{m}_{Y,o}^2$ the projection onto the χ-eigenspace. Thus applying the maps p_χ to lifts of a basis of $\mathfrak{m}_{Y,o}/\mathfrak{m}_{Y,o}^2$ consisting of eigenvectors, we obtain G-eigenparameters x, y, z, t of Y at o.

Since $c_j > 0$ and each h_j is a regular G-semi-invariant function, $B \geq 0$ is G-invariant. Moreover, (X, B) is klt near o and $\mathrm{mld}((X \ni o, B)/G) = a \geq 1 + \epsilon$. Thus Theorem 15.5, in the form for isolated Gorenstein terminal cDV threefold singularities with the eigenparameters x, y, z, t, implies that one of the following two cases occurs.

Case 1. $|G| \leq e(\epsilon)$. Since X is terminal, X is normal and klt. Moreover, G acts faithfully on X and fixes o, each h_j is a non-zero regular G-semi-invariant function, $c_j \in \Gamma$ and $c_j > 0$ for each j, and (X, B) is klt near o. Thus $a \in \mathcal{A}_1$.

Case 2. After a permutation of x, y, z, t, the group $R := G \cap \mathbf{T}_{(1,-1,0,0)}$ is $\mu_r(1, -1, 0, 0)$ for an integer $r > 2$, $[G : R] \leq e(\epsilon)$, and the quadratic part of f is $\lambda xy + q_2(z, t)$ with $\lambda \neq 0$. Since $f \in \mathfrak{m}_{Y,o}^2$, X is singular at o. We apply Construction-Theorem 15.8 to $(X \ni o)$, $(Y \ni o)$, G, x, y, z, t, R, and the functions h_j. Thus there exist a non-zero polynomial $g_0 \in (z, t)^2$ and G-semi-invariant polynomials $h_{j,0}$ whose restrictions to

$$X_0 := (xy - g_0(z, t) = 0) \subset \mathbb{A}^4$$

are non-zero, such that (1), (2), and (3) of Construction-Theorem 15.8 hold. Let

$$B_0 := \sum_j c_j \operatorname{div}_{X_0}(h_{j,0}).$$

We show that X_0, G, the polynomials $h_{j,0}$, and the numbers c_j satisfy the assumptions of Theorem 15.9 for $e_0 = e(\epsilon)$ and Γ. By Construction-Theorem 15.8(1), X_0 is an isolated Gorenstein terminal threefold germ at 0, hence a normal klt threefold germ at 0. By Construction-Theorem 15.8(1), G is a finite diagonal abelian group acting faithfully on X_0, and $xy - g_0$ has the character of f, so G preserves $xy - g_0$ up to a character. Since $R = G \cap \mathbf{T}_{(1,-1,0,0)}$ for the eigenparameters x, y, z, t of Y, the last sentence of Construction-Theorem 15.8(1) implies that $G \cap \mathbf{T}_{(1,-1,0,0)} = R = \mu_r(1, -1, 0, 0)$ on $\mathbb{A}^4$, and

$$|G/R| = [G : R] \leq e(\epsilon) = e_0.$$

By Construction-Theorem 15.8(2), each $h_{j,0}$ has the character of h_j, and (X_0, B_0) is klt near 0. Finally, $c_j \in \Gamma$ and $c_j > 0$ for each j. By Construction-Theorem 15.8(3) for the subgroup G of G,

$$a = \mathrm{mld}((X \ni o, B)/G) = \mathrm{mld}((X_0 \ni 0, B_0)/G).$$

Since $a \geq 1 + \epsilon$, the number $\mathrm{mld}((X_0 \ni 0, B_0)/G)$ belongs to $[1, +\infty)$, hence to $\mathcal{A}_2$. Thus $a \in \mathcal{A}_2$.

Therefore $a_i \in \mathcal{A}_1 \cup \mathcal{A}_2$ for any $i \geq 2$. Since $\mathcal{A}_1$ or $\mathcal{A}_2$ contains a_i for infinitely many i, possibly passing to a subsequence, we may assume that $a_i \in \mathcal{A}_1$ for any $i \geq 2$, or $a_i \in \mathcal{A}_2$ for any $i \geq 2$. In either case, the strictly increasing sequence $(a_i)_{i\geq 2}$ contradicts the ACC for $\mathcal{A}_1$ or for $\mathcal{A}_2$. □

16. Limits of diagonal groups

In this section, we prove Theorem 16.1. Theorem 16.1 reduces the case of smooth diagonal quotients in Theorem 1.5 to a fixed mixed two-coordinate circle, and this case is treated in Section 14. Recall that $\rho : \mathbb{R}^3 \to \mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$ is the quotient map of Definition 2.15.

Theorem 16.1. *Let $\Gamma \subset [0,1]$ be a DCC set. For any $i \geq 1$, let G_i be a finite diagonal group acting faithfully on $X := \mathbb{A}^3_{x,y,z}$, and $B_i := \sum_j c_{i,j} \operatorname{div}(h_{i,j})$ with $c_{i,j} \in \Gamma$ and non-zero G_i-semi-invariant polynomials $h_{i,j}$, such that (X, B_i) is klt near 0 and $a_i := \operatorname{mld}((X \ni 0, B_i)/G_i) > 0$ is strictly increasing. Then, after passing to a subsequence and permuting the coordinates, there exist coprime positive integers p, q and a positive integer d such that, for $\mathbf{T} := \mathbf{T}_{(p,-q,0)}$ and $R_i := G_i \cap \mathbf{T}$, we have $R_i = \mu_{n_i}(p, -q, 0)$ with $n_i \to +\infty$ and $[G_i : R_i] = d$. More precisely, the G_i, identified with their images in $\mathbb{T}^3$, converge in the Hausdorff distance to a closed subgroup H of $\mathbb{T}^3$ with $G_i \subset H$, the identity component H^0 of H is $\rho(\mathbb{R}(p, -q, 0))$, and $d = |H/H^0|$.*

We prove Theorem 16.1 in Subsection 16.11.

16.1. **Weighted degenerations.** In this subsection, we prove Proposition 16.3. When G is trivial and the boundary is replaced by a single ideal of $\mathbb{C}[[x_1, x_2, x_3]]$ with exponent one, the inequality of Proposition 16.3 is [Shi21, Proposition 3.3], and **Steps 1 and 2** follow the method of [Shi21]. We use the formula of [NS26b, Theorem 5.3, Remark 5.4] for the minimal log discrepancies of $\mathbb{R}$-ideals on quotients of the affine space, in the case of the setting of [NS26b, Section 3] in which the number c of equations is zero. In this case the Jacobian ideal is the unit ideal, and the correction terms of [NS26b, Theorem 5.3] coming from the equations and from the Jacobian ideal vanish. We only state the case of diagonal groups acting on $\mathbb{A}^3$.

Definition-Theorem 16.2 ([NS26b, Theorem 5.3, Remark 5.4])**.** Let $X = \mathbb{A}^3$ with coordinates x_1, x_2, x_3, let G be a finite diagonal subgroup of $(\mathbb{C}^*)^3$ which acts faithfully on X and contains no pseudo-reflection, let $\pi : X \to X/G$ be the quotient morphism, let $\mathfrak{a}_1, \dots, \mathfrak{a}_s$ be non-zero ideals of $\mathbb{C}[x_1, x_2, x_3]^G$, and let $\tau_1, \dots, \tau_s$ be non-negative real numbers. For $i = 1, 2, 3$, let χ_i be the character of x_i.

(1) For $\gamma \in G$, let $\epsilon_1, \epsilon_2, \epsilon_3 \in [0, 1)$ be the real numbers such that $\chi_i(\gamma) = e^{2\pi\sqrt{-1}\epsilon_i}$ for any i, and let $\operatorname{age}(\gamma) := \epsilon_1 + \epsilon_2 + \epsilon_3$.

(2) Let $\mathcal{A}$ be the set of families $u = (u_{i,l})_{1\leq i\leq 3, l\geq 0}$ of complex numbers, and let $u_i(t) := \sum_{l\geq 0} u_{i,l} t^l$. For $\gamma \in G$, any G-invariant monomial x^α satisfies $\sum_i \epsilon_i \alpha_i \in \mathbb{Z}_{\geq 0}$. Thus, for any $h \in \mathbb{C}[x_1, x_2, x_3]^G$ and $u \in \mathcal{A}$,

$$h_\gamma(u) := h\big(t^{\epsilon_1} u_1(t), t^{\epsilon_2} u_2(t), t^{\epsilon_3} u_3(t)\big) \in \mathbb{C}[[t]],$$

and the coefficient of t^m in $h_\gamma(u)$ is a polynomial in the $u_{i,l}$ with $l \leq m$. For an ideal $\mathfrak{a}$ of $\mathbb{C}[x_1, x_2, x_3]^G$, let $\operatorname{ord}_t \mathfrak{a}_\gamma(u) := \min\{\operatorname{ord}_t h_\gamma(u) \mid h \in \mathfrak{a}\}$.

(3) For $\gamma \in G$ and $k = (k_1, \dots, k_s) \in \mathbb{Z}^s_{\geq 0}$, let $C_{\gamma,k} \subset \mathcal{A}$ be the set of the $u \in \mathcal{A}$ such that $\operatorname{ord}_t h_\gamma(u) \geq 1$ for any $h \in \mathbb{C}[x_1, x_2, x_3]^G$ with $h(0) = 0$, that is, the arc of X/G defined by u is centered at $\pi(0)$, and $\operatorname{ord}_t(\mathfrak{a}_j)_\gamma(u) \geq k_j$ for any j. For any integer $n \geq \max_j k_j$, $C_{\gamma,k}$ is the preimage of a closed subset $C_{\gamma,k,n}$ of the space $\mathbb{C}^{3(n+1)}$ of the $(u_{i,l})_{l\leq n}$, and we let

$$\operatorname{codim} C_{\gamma,k} := \dim \mathbb{C}^{3(n+1)} - \dim C_{\gamma,k,n},$$

which does not depend on n, with $\operatorname{codim} \emptyset := +\infty$.

Then

$$\operatorname{mld}\big(X/G \ni \pi(0), \mathfrak{a}_1^{\tau_1} \cdots \mathfrak{a}_s^{\tau_s}\big) = \inf\Big\{ \operatorname{codim} C_{\gamma,k} + \operatorname{age}(\gamma) - \sum_{j=1}^{s} \tau_j k_j \;\Big|\; \gamma \in G,\ k \in \mathbb{Z}^s_{\geq 0} \Big\}.$$

A factor with $\tau_j = 0$ may be omitted, since $C_{\gamma,k}$ decreases in k_j and the infimum over k_j is attained at $k_j = 0$.

Proposition 16.3 is used in the proof of Theorem 16.4 to bound quotient mlds from below by those of the leading forms.

Proposition 16.3. *Let $X = \mathbb{A}^3$ with coordinates x_1, x_2, x_3, let G be a finite diagonal subgroup of $(\mathbb{C}^*)^3$ acting faithfully on X, let $f_1, \dots, f_s$ be non-zero G-semi-invariant polynomials, $c_j \geq 0$, $w \in \mathbb{Z}^3_{>0}$, $f_j^0 := \operatorname{in}_w(f_j)$, $B := \sum_j c_j \operatorname{div}(f_j)$, and $B^0 := \sum_j c_j \operatorname{div}(f_j^0)$, and assume that (X, B) and (X, B^0) are klt near 0. Then*

$$\operatorname{mld}((X \ni 0, B)/G) \geq \operatorname{mld}((X \ni 0, B^0)/G).$$

Proof. The proof is long, so we divide it into several steps. Let $\pi : X \to X/G$ be the quotient morphism and $r := |G|$. For any j, since G is diagonal, any monomial occurring in f_j is G-semi-invariant with character equal to the character of f_j. Hence f_j^0 is G-semi-invariant with character equal to the character of f_j. Let N be a positive integer divisible by r, and let $g_j := f_j^N$ and $g_j^0 := (f_j^0)^N$. Then g_j and g_j^0 are G-invariant, and $g_j^0 = \operatorname{in}_w(g_j)$, since the leading form of a product of polynomials is the product of the leading forms. Moreover,

$$B = \sum_j \frac{c_j}{N} \operatorname{div}(g_j) \quad \text{and} \quad B^0 = \sum_j \frac{c_j}{N} \operatorname{div}(g_j^0).$$

We regard G-invariant polynomials as regular functions on X/G.

Step 1. In this step, we assume that G contains no pseudo-reflection, and we reduce the inequality to an inequality between the heights of two ideals of a polynomial ring.

Since G contains no pseudo-reflection, π is étale in codimension one, so $R(\pi) = 0$. Since $\operatorname{div}_X(g_j) = \pi^* \operatorname{div}_{X/G}(g_j)$, we have $\pi_* \operatorname{div}_X(g_j) = r \operatorname{div}_{X/G}(g_j)$, hence

$$B_{X/G} = \frac{1}{r}\pi_* B = \sum_j \frac{c_j}{N} \operatorname{div}_{X/G}(g_j) \quad \text{and} \quad B^0_{X/G} = \sum_j \frac{c_j}{N} \operatorname{div}_{X/G}(g_j^0).$$

Thus

$$\operatorname{mld}((X \ni 0, B)/G) = \operatorname{mld}\Big(X/G \ni \pi(0), \prod_j (g_j)^{\frac{c_j}{N}}\Big) \quad \text{and}$$

$$\operatorname{mld}((X \ni 0, B^0)/G) = \operatorname{mld}\Big(X/G \ni \pi(0), \prod_j (g_j^0)^{\frac{c_j}{N}}\Big).$$

For $\gamma \in G$ and $k \in \mathbb{Z}^s_{\geq 0}$, let $C_{\gamma,k}$ and $C^0_{\gamma,k}$ be the sets of Definition-Theorem 16.2 for the ideals $(g_1), \dots, (g_s)$ and for the ideals $(g_1^0), \dots, (g_s^0)$, respectively. The numbers $\operatorname{age}(\gamma)$ and $\sum_j \frac{c_j}{N} k_j$ do not change when the ideals (g_j) are replaced by the ideals (g_j^0). By Definition-Theorem 16.2, it suffices to show that

$$\operatorname{codim} C_{\gamma,k} \geq \operatorname{codim} C^0_{\gamma,k} \tag{16.1}$$

for any $\gamma \in G$ and $k \in \mathbb{Z}^s_{\geq 0}$.

Fix γ and k, and let $\epsilon_1, \epsilon_2, \epsilon_3$ be as in Definition-Theorem 16.2(1). We show that, for $u \in \mathcal{A}$, the arc of X/G defined by u is centered at $\pi(0)$ if and only if

$$u_{i,0} = 0 \text{ for any } i \text{ with } \epsilon_i = 0. \tag{16.2}$$

If $\epsilon_i = 0$, then x_i^r is G-invariant, vanishes at 0, and the constant coefficient of $(x_i^r)_\gamma(u)$ is $u_{i,0}^r$, so if the arc is centered at $\pi(0)$, then $u_{i,0} = 0$. Conversely, if (16.2) holds, then $\operatorname{ord}_t\big(t^{\epsilon_i} u_i(t)\big) > 0$ for

any i, so $\operatorname{ord}_t(x^\alpha)_\gamma(u)$ is at least 1 for any G-invariant monomial $x^\alpha \neq 1$. Since G is diagonal, any $h \in \mathbb{C}[x_1, x_2, x_3]^G$ with $h(0) = 0$ is a linear combination of such monomials, hence $\operatorname{ord}_t h_\gamma(u) \geq 1$.

Let $n \geq \max_j k_j$ be an integer, let z_γ be the number of i with $\epsilon_i = 0$, and let R_n be the polynomial ring in the variables $u_{i,l}$ with $0 \leq l \leq n$, except the $u_{i,0}$ with $\epsilon_i = 0$. For $1 \leq j \leq s$ and $0 \leq m < k_j$, let $P_{j,m} \in R_n$ and $P^0_{j,m} \in R_n$ be the coefficients of t^m in $(g_j)_\gamma(u)$ and in $(g_j^0)_\gamma(u)$, respectively, after substituting $u_{i,0} = 0$ for any i with $\epsilon_i = 0$. Let $\mathfrak{b} \subset R_n$ be the ideal generated by the $P_{j,m}$, and $\mathfrak{b}^0 \subset R_n$ the ideal generated by the $P^0_{j,m}$. By (16.2), $C_{\gamma,k,n}$ is, as a set, the zero locus of $\mathfrak{b}$ in the linear subspace $\operatorname{Spec} R_n = \{u_{i,0} = 0 \text{ for any } i \text{ with } \epsilon_i = 0\}$ of $\mathbb{C}^{3(n+1)}$, and likewise for $C^0_{\gamma,k,n}$ and $\mathfrak{b}^0$. For a proper ideal $\mathfrak{c}$ of R_n, $\dim R_n - \dim R_n/\mathfrak{c} = \operatorname{ht} \mathfrak{c}$ [Eis95, Corollary 13.4]. With the convention $\operatorname{ht} R_n := +\infty$, which corresponds to $C_{\gamma,k,n} = \emptyset$ and $\operatorname{codim} \emptyset = +\infty$, we obtain

$$\operatorname{codim} C_{\gamma,k} = z_\gamma + \operatorname{ht} \mathfrak{b} \quad \text{and} \quad \operatorname{codim} C^0_{\gamma,k} = z_\gamma + \operatorname{ht} \mathfrak{b}^0.$$

Therefore, (16.1) follows from

$$\operatorname{ht} \mathfrak{b} \geq \operatorname{ht} \mathfrak{b}^0. \tag{16.3}$$

Step 2. In this step, we prove (16.3), which concludes the proof of the inequality when G contains no pseudo-reflection.

If $\mathfrak{b} = R_n$, then $\operatorname{ht} \mathfrak{b} = +\infty$ and there is nothing to prove, so we may assume that $\mathfrak{b} \neq R_n$. We assign to each variable $u_{i,l}$ of R_n two degrees: the *arc degree* $r(l + \epsilon_i)$ and the *degeneration degree* w_i. Both are positive integers: $r\epsilon_i \in \mathbb{Z}$ since $\chi_i(\gamma)^r = 1$, $l + \epsilon_i > 0$ since $u_{i,0}$ is not a variable of R_n when $\epsilon_i = 0$, and $w_i > 0$. Any monomial $\prod_\nu u_{i_\nu,l_\nu}$ occurring in the coefficient of t^m of $(g_j)_\gamma(u)$ satisfies $\sum_\nu (l_\nu + \epsilon_{i_\nu}) = m$. Thus $P_{j,m}$ is homogeneous of arc degree rm, and $\mathfrak{b}$ is homogeneous for the arc degree. For a non-zero $P \in R_n$, let $\operatorname{in}_w(P)$ be the sum of the terms of P of least degeneration degree, and let $\operatorname{in}_w(\mathfrak{b})$ be the ideal generated by the leading forms $\operatorname{in}_w(P)$ of the non-zero elements $P \in \mathfrak{b}$.

We show that $R_n/\mathfrak{b}$ and $R_n/\operatorname{in}_w(\mathfrak{b})$ have equal Hilbert functions for the arc degree. For integers $\delta \geq 0$ and μ, let $\mathfrak{b}_\delta$ be the space of elements of $\mathfrak{b}$ that are homogeneous of arc degree δ, let $F^\mu \mathfrak{b}_\delta \subset \mathfrak{b}_\delta$ be the subspace of the elements all of whose terms have degeneration degree at least μ, and let $\operatorname{in}_w(\mathfrak{b})_{\delta,\mu}$ be the space of elements of $\operatorname{in}_w(\mathfrak{b})$ that are homogeneous of arc degree δ and of degeneration degree μ. Since $\mathfrak{b}$ is homogeneous for the arc degree, the leading form of any non-zero element of $\mathfrak{b}$ is a sum of leading forms of non-zero elements of the $\mathfrak{b}_\delta$. Moreover, $Q \operatorname{in}_w(P) = \operatorname{in}_w(QP)$ for any monomial Q, and for $P, P' \in \mathfrak{b}_\delta$ with $\operatorname{in}_w(P)$ and $\operatorname{in}_w(P')$ of degeneration degree μ, the sum $\operatorname{in}_w(P) + \operatorname{in}_w(P')$ is either 0 or $\operatorname{in}_w(P + P')$. Thus $\operatorname{in}_w(\mathfrak{b})_{\delta,\mu}$ consists of 0 and of the leading forms of degeneration degree μ of the elements of $\mathfrak{b}_\delta$, and taking the part of degeneration degree μ induces an isomorphism

$$F^\mu \mathfrak{b}_\delta / F^{\mu+1} \mathfrak{b}_\delta \cong \operatorname{in}_w(\mathfrak{b})_{\delta,\mu}.$$

Since any variable of R_n has positive arc degree, the space of elements of R_n homogeneous of arc degree δ has finite dimension, so $F^\mu \mathfrak{b}_\delta \neq F^{\mu+1} \mathfrak{b}_\delta$ for only finitely many μ. Summing over μ, we obtain

$$\dim_{\mathbb{C}} \mathfrak{b}_\delta = \sum_\mu \dim_{\mathbb{C}} \operatorname{in}_w(\mathfrak{b})_{\delta,\mu}$$

for any δ. This proves the claim.

In particular, since $\mathfrak{b}$ contains no non-zero constant, neither does $\operatorname{in}_w(\mathfrak{b})$, so $\operatorname{in}_w(\mathfrak{b}) \neq R_n$. The rings $R_n/\mathfrak{b}$ and $R_n/\operatorname{in}_w(\mathfrak{b})$ are finitely generated positively graded $\mathbb{C}$-algebras with equal Hilbert functions, hence $\dim R_n/\mathfrak{b} = \dim R_n/\operatorname{in}_w(\mathfrak{b})$ [BH98, Proposition 4.4.1]. By [Eis95, Corollary 13.4],

$$\operatorname{ht} \mathfrak{b} = \operatorname{ht} \operatorname{in}_w(\mathfrak{b}).$$

Next we show that $\mathfrak{b}^0 \subset \operatorname{in}_w(\mathfrak{b})$. Let $\lambda_j := \operatorname{wt}_w(g_j)$. For a monomial x^α, the coefficient of t^m in $(x^\alpha)_\gamma(u)$ is homogeneous of degeneration degree $\langle w, \alpha \rangle$, since each of the α_i factors coming

from x_i contributes a variable $u_{i,l}$ of degeneration degree w_i. The monomials occurring in g_j^0 are the monomials occurring in g_j of w-weight λ_j, and the other monomials occurring in g_j have w-weight greater than λ_j. Since substituting $u_{i,0} = 0$ preserves the degeneration degree, $P_{j,m}^0$ is homogeneous of degeneration degree λ_j, and $P_{j,m} - P_{j,m}^0$ is a sum of terms of degeneration degree greater than λ_j. Thus, if $P_{j,m}^0 \neq 0$, then $P_{j,m}^0 = \mathrm{in}_w(P_{j,m}) \in \mathrm{in}_w(\mathfrak{b})$. Hence $\mathfrak{b}^0 \subset \mathrm{in}_w(\mathfrak{b})$, and

$$\mathrm{ht}\,\mathfrak{b} = \mathrm{ht}\,\mathrm{in}_w(\mathfrak{b}) \geq \mathrm{ht}\,\mathfrak{b}^0.$$

This proves (16.3). By **Step 1**, the inequality $\mathrm{mld}((X \ni 0, B)/G) \geq \mathrm{mld}((X \ni 0, B^0)/G)$ holds when G contains no pseudo-reflection.

Step 3. In this step, we prove the inequality for an arbitrary finite diagonal group G by first taking the quotient by the subgroup generated by pseudo-reflections.

For $i = 1, 2, 3$, let $G^{(i)} \subset G$ be the subgroup of the elements acting trivially on $x_{i'}$ for any $i' \neq i$. Since G acts faithfully on X, χ_i embeds $G^{(i)}$ into $\mathbb{C}^*$, so $G^{(i)}$ is cyclic of some order r_i and $\chi_i(G^{(i)})$ is the group of r_i-th roots of unity. An element $\sigma \in G$ is a pseudo-reflection if and only if exactly one of $\chi_1(\sigma), \chi_2(\sigma), \chi_3(\sigma)$ is not 1, so the subgroup of G generated by pseudo-reflections is $G_{\mathrm{ref}} = G^{(1)}G^{(2)}G^{(3)}$. Let $y_i := x_i^{r_i}$. Then $\mathbb{C}[x_1, x_2, x_3]^{G_{\mathrm{ref}}} = \mathbb{C}[y_1, y_2, y_3]$, so $X' := X/G_{\mathrm{ref}}$ is $\mathbb{A}^3$ with coordinates y_1, y_2, y_3. Let $\pi' : X \to X'$ be the quotient morphism. The group G/G_{ref} acts diagonally on X', and $X/G = X'/(G/G_{\mathrm{ref}})$.

We show that G/G_{ref} acts faithfully on X' and contains no pseudo-reflection. Let $g \in G$ and $i \in \{1, 2, 3\}$, and assume that g acts trivially on $y_{i'}$ for any $i' \neq i$. Then $\chi_{i'}(g)^{r_{i'}} = 1$, so there exists $h_{i'} \in G^{(i')}$ with $\chi_{i'}(h_{i'}) = \chi_{i'}(g)$ for any $i' \neq i$. Thus $g\prod_{i' \neq i} h_{i'}^{-1}$ acts trivially on $x_{i'}$ for any $i' \neq i$, so $g\prod_{i' \neq i} h_{i'}^{-1} \in G^{(i)}$, and $g \in G_{\mathrm{ref}}$. This proves the claim.

Since g_j and g_j^0 are G_{ref}-invariant, there exist polynomials $\bar{g}_j$ and $\bar{g}_j^0$ in y_1, y_2, y_3 with $g_j = \bar{g}_j(y_1, y_2, y_3)$ and $g_j^0 = \bar{g}_j^0(y_1, y_2, y_3)$. Let $(X', B') := (X, B)/G_{\mathrm{ref}}$ and $(X', B'^0) := (X, B^0)/G_{\mathrm{ref}}$. The ramification index of π' along $(x_i = 0)$ is r_i, π' is unramified along any other prime divisor of X, and $\mathrm{ord}_{(x_i=0)}(g_j) = r_i\,\mathrm{ord}_{(y_i=0)}(\bar{g}_j)$. Thus Lemma 2.6(4) implies that

$$B' = \sum_{i=1}^{3}\left(1 - \frac{1}{r_i}\right)\mathrm{div}(y_i) + \sum_j \frac{c_j}{N}\,\mathrm{div}(\bar{g}_j) \quad \text{and}$$

$$B'^0 = \sum_{i=1}^{3}\left(1 - \frac{1}{r_i}\right)\mathrm{div}(y_i) + \sum_j \frac{c_j}{N}\,\mathrm{div}(\bar{g}_j^0).$$

In particular, $B' \geq 0$ and $B'^0 \geq 0$, and (X', B') and (X', B'^0) are klt near the origin by Lemma 2.6(2)(3). Let $w' := (r_1w_1, r_2w_2, r_3w_3) \in \mathbb{Z}^3_{>0}$. The w'-weight of a monomial y^β is $\sum_i r_iw_i\beta_i$, which is the w-weight of $y^\beta = \prod_i x_i^{r_i\beta_i}$ as a monomial in x_1, x_2, x_3. Hence $\mathrm{in}_{w'}(\bar{g}_j) = \bar{g}_j^0$ for any j, and $\mathrm{in}_{w'}(y_i) = y_i$ for any i.

By Lemma 2.6(5), applied to the normal subgroup G_{ref} of the abelian group G, $(X', B')/(G/G_{\mathrm{ref}}) = (X, B)/G$ and $(X', B'^0)/(G/G_{\mathrm{ref}}) = (X, B^0)/G$. **Steps 1 and 2**, applied to X', the group G/G_{ref}, the semi-invariant polynomials $y_1, y_2, y_3, \bar{g}_1, \ldots, \bar{g}_s$ with the coefficients $1 - \frac{1}{r_1}, 1 - \frac{1}{r_2}, 1 - \frac{1}{r_3}, \frac{c_1}{N}, \ldots, \frac{c_s}{N}$, and the weight w', imply that

$$\begin{aligned}\mathrm{mld}((X \ni 0, B)/G) &= \mathrm{mld}((X' \ni 0, B')/(G/G_{\mathrm{ref}})) \\ &\geq \mathrm{mld}((X' \ni 0, B'^0)/(G/G_{\mathrm{ref}})) = \mathrm{mld}((X \ni 0, B^0)/G).\end{aligned}$$

□

16.2. **Limits of dimension at least two.** In this subsection, we prove Theorem 16.4, which excludes, in the proof of Theorem 16.1, Hausdorff limits H with $\dim H^0 \geq 2$.

Theorem 16.4. *Let $\Gamma \subset [0, 1]$ be a DCC set and $\epsilon > 0$. For any $i \geq 1$, let G_i be a finite diagonal subgroup of $(\mathbb{C}^*)^3$ acting faithfully on $\mathbb{A}^3$, identified with the image of G_i in $\mathbb{T}^3$ as*

in Definition 2.15, and $B_i := \sum_j c_{i,j} \operatorname{div}(h_{i,j})$ *with* $c_{i,j} \in \Gamma$ *and non-zero* G_i*-semi-invariant polynomials* $h_{i,j}$*, such that* $(\mathbb{A}^3, B_i)$ *is klt near* 0 *and* $a_i := \operatorname{mld}((\mathbb{A}^3 \ni 0, B_i)/G_i) \geq \epsilon$*. Assume that the* G_i *converge in the Hausdorff distance to a closed subgroup* $H \subset \mathbb{T}^3$*.*

(1) If $\dim H^0 = 3$*, then no such sequence exists.*

(2) If $\dim H^0 = 2$*, let* $m \in \mathbb{Z}^3$ *be a primitive generator of the lattice of characters of* $\mathbb{T}^3$ *vanishing on* H^0*, and* $d := |\chi_m(H)|$ *for the character* $\chi_m : \mathbb{T}^3 \to \mathbb{T}^1$ *defined by* m*. Then* m *can be taken with* $m_k \geq 0$ *for any* k*,* $d \max_k m_k \leq \frac{1}{\epsilon}$*, and, after deleting the terms of* B_i *with* $c_{i,j} = 0$ *or* $h_{i,j}(0) \neq 0$ *and passing to a subsequence, there exist fixed* $\alpha_j \in \mathbb{Z}^3_{\geq 0}$*, polynomials* $U_{i,j}$ *in one variable* u*, and decompositions* $h_{i,j} = x^{\alpha_j} U_{i,j}(x^m) + R_{i,j}$ *with* $U_{i,j}(0) \neq 0$*,* $U_{i,j} \in \mathbb{C}[u^d]$*,* $R_{i,j} \in (x_1, x_2, x_3)^{\nu_i}$*, and* $\nu_i \to +\infty$*. Let* $J := \{k \mid m_k > 0\}$*,* $\beta_{i,k} := \sum_j c_{i,j}\alpha_{j,k}$*,* $\varrho_i := \min_{k \in J} \frac{1-\beta_{i,k}}{m_k}$*, and* $b_i := \varrho_i/d$*. Then* $b_i = \operatorname{mld}((\mathbb{A}^1 \ni 0, (1-\varrho_i) \cdot 0)/\mu_d)$*, the numbers* $1 - \frac{\varrho_i}{d}$ *belong to a DCC set depending only on* Γ*,* m*,* d*, and the* α_j*, the sequence* b_i *is nonincreasing, and there exist* $\eta_i \geq 0$ *with* $\eta_i \to 0$ *and* $b_i \leq a_i \leq b_i + \eta_i$*.*

In particular, if the sequence a_i *is strictly increasing, then* $\dim H^0 \leq 1$*.*

Proof. The proof is long, so we divide it into several steps. For $a \in \mathbb{Z}^3_{\geq 0}$ we write $|a| := a_1 + a_2 + a_3$, and for $w \in \mathbb{R}^3$ we write $\|w\|_1 := \sum_k |w_k|$. We write $\mathbb{T}^1 = \mathbb{R}/\mathbb{Z}$ additively, so that $\chi_m(g) = \langle m, \xi \rangle + \mathbb{Z}$ for $g \in \mathbb{T}^3$ represented by $\xi \in \mathbb{R}^3$.

Step 1. In this step we bound the log discrepancies of toric prime divisors over $\mathbb{A}^3/G_i$ from below and from above, and we fix the approximation of points of H by elements of G_i.

For any i and j, the divisor $\operatorname{div}(h_{i,j})$ is G_i-invariant, and it does not contain 0 if $h_{i,j}(0) \neq 0$. Thus deleting the terms of B_i with $c_{i,j} = 0$ or $h_{i,j}(0) \neq 0$ changes neither B_i on a G_i-invariant open neighborhood of 0, nor the hypotheses on B_i, nor a_i, so we may assume that $c_{i,j} > 0$ and $h_{i,j}(0) = 0$ for any i and j. Let s_i be the number of the polynomials $h_{i,j}$ for fixed i.

For any i let

$$\psi_i(v) := \sum_{k=1}^{3} v_k - \sum_j c_{i,j} \operatorname{wt}_v(h_{i,j}), \qquad v \in \mathbb{R}^3_{\geq 0},$$

which is the function ψ of Set-up 8.4 for $\mathbb{A}^3$, the group G_i, the polynomials $h_{i,j}$, and the coefficients $c_{i,j}$. By Lemma 8.5, any primitive vector v of N_{G_i} with $v \in \mathbb{R}^3_{>0}$ defines a toric prime divisor E_v over $\mathbb{A}^3/G_i$ with

$$a(E_v, \mathbb{A}^3/G_i, (B_i)_{\mathbb{A}^3/G_i}) = \psi_i(v).$$

Since $\mathbb{A}^3/G_i$ is the affine toric variety of the cone $\mathbb{R}^3_{\geq 0}$ in the lattice N_{G_i} [CLS11, Proposition 1.3.18] and v is in the interior of $\mathbb{R}^3_{\geq 0}$, so that $\mathbb{R}^3_{\geq 0}$ is the minimal cone containing v, the center of E_v on $\mathbb{A}^3/G_i$ is the torus-fixed point $O(\mathbb{R}^3_{\geq 0})$ [CLS11, Lemma 3.3.21(b), Theorem 3.2.6(b)], that is, the image $\bar{0}$ of 0. Hence $\psi_i(v) \geq a_i$. Any $v \in N_{G_i} \cap \mathbb{R}^3_{>0}$ can be written as $v = lv'$, where l is a positive integer and v' is a primitive vector of N_{G_i}, because N_{G_i} is a lattice. Since $\operatorname{wt}_{lv'} = l \operatorname{wt}_{v'}$, we have $\psi_i(v) = l\psi_i(v') \geq la_i \geq a_i$. Since $c_{i,j} > 0$ and $\operatorname{wt}_v(h_{i,j}) \geq 0$, we obtain

$$\epsilon \leq a_i \leq \psi_i(v) \leq \sum_{k=1}^{3} v_k \quad \text{for any } i \text{ and any } v \in N_{G_i} \cap \mathbb{R}^3_{>0}. \tag{16.4}$$

Let θ_i be the Hausdorff distance between G_i and H. Then $\theta_i \to 0$. Since G_i is finite and the metric of $\mathbb{T}^3$ is induced by the Euclidean metric of $\mathbb{R}^3$, for any $\xi \in \mathbb{R}^3$ with $\xi + \mathbb{Z}^3 \in H$, the distance from $\xi + \mathbb{Z}^3$ to G_i is at most θ_i, and it equals the Euclidean distance from ξ to some $v \in N_{G_i}$. Thus, for any i:

$$\text{if } \xi \in \mathbb{R}^3 \text{ and } \xi + \mathbb{Z}^3 \in H, \text{ then some } v \in N_{G_i} \text{ has } |v_k - \xi_k| \leq \theta_i \text{ for any } k. \tag{16.5}$$

Step 2. In this step we prove (1).

Assume that $\dim H^0 = 3$. Then H^0 is open in $\mathbb{T}^3$, so $H = \mathbb{T}^3$. Since G_i is finite, $\theta_i > 0$. Applying (16.5) to $\xi = \sqrt{\theta_i}(1,1,1)$, we obtain $v_i \in N_{G_i}$ with

$$v_{i,k} \geq \sqrt{\theta_i} - \theta_i \quad \text{and} \quad \sum_{k=1}^{3} v_{i,k} \leq 3(\sqrt{\theta_i} + \theta_i).$$

For $i \gg 0$ we have $\theta_i < 1$, so $v_i \in \mathbb{R}^3_{>0}$, and $3(\sqrt{\theta_i} + \theta_i) < \epsilon$. This contradicts (16.4). Thus (1) holds.

Step 3. In this step we assume that $\dim H^0 = 2$, and we prove that $H = \chi_m^{-1}(\frac{1}{d}\mathbb{Z}/\mathbb{Z})$, and that $G_i \subset H$ and $\chi_m(G_i) = \frac{1}{d}\mathbb{Z}/\mathbb{Z}$ for $i \gg 0$.

Since H is infinite and G_i is finite, $\theta_i > 0$ for any i. Since m is primitive, m is the first row of some $Q \in \mathrm{GL}_3(\mathbb{Z})$. The automorphism of $\mathbb{T}^3$ induced by Q maps $\operatorname{Ker}\chi_m$ onto $\{\xi + \mathbb{Z}^3 \mid \xi_1 \in \mathbb{Z}\}$, which is a two-dimensional torus. Thus $\operatorname{Ker}\chi_m$ is a connected closed subgroup of $\mathbb{T}^3$ with $\dim \operatorname{Ker}\chi_m = 2$. Since χ_m vanishes on H^0, we have $H^0 \subset \operatorname{Ker}\chi_m$, and H^0 is open in $\operatorname{Ker}\chi_m$ as $\dim H^0 = 2$. Hence $H^0 = \operatorname{Ker}\chi_m$. Since H is compact, H/H^0 is finite, and $\chi_m(H)$ is the image of H/H^0. Thus $\chi_m(H)$ is a subgroup of $\mathbb{T}^1$ of order d, that is, $\chi_m(H) = \frac{1}{d}\mathbb{Z}/\mathbb{Z}$. Since $\operatorname{Ker}\chi_m = H^0 \subset H$, we obtain

$$H = \chi_m^{-1}\big(\tfrac{1}{d}\mathbb{Z}/\mathbb{Z}\big).$$

Let $g \in G_i$. Choose $h \in H$ whose distance to g is at most θ_i, and $\xi, \xi' \in \mathbb{R}^3$ representing g and h respectively with $|\xi_k - \xi'_k| \leq \theta_i$ for any k. Since $d\langle m, \xi'\rangle \in \mathbb{Z}$, the element $d\chi_m(g) \in \mathbb{T}^1$ is represented by $d\langle m, \xi - \xi'\rangle$, and

$$|d\langle m, \xi - \xi'\rangle| \leq d\|m\|_1\theta_i.$$

Hence, if $d\|m\|_1\theta_i < \frac{1}{4}$, then the subgroup $\{d\chi_m(g) \mid g \in G_i\}$ of $\mathbb{T}^1$ is contained in the image of $(-\frac{1}{4}, \frac{1}{4})$ in $\mathbb{T}^1$, so it is trivial by Lemma 4.4. Thus $\chi_m(G_i) \subset \frac{1}{d}\mathbb{Z}/\mathbb{Z}$, that is, $G_i \subset H$, for $i \gg 0$.

Since m is primitive, there exists $a \in \mathbb{Z}^3$ with $\langle m, a\rangle = 1$. Then $\chi_m(\frac{1}{d}a + \mathbb{Z}^3) = \frac{1}{d} + \mathbb{Z}$, so $\frac{1}{d}a + \mathbb{Z}^3 \in H$. By (16.5), there exists $v \in N_{G_i}$ with $|v_k - \frac{1}{d}a_k| \leq \theta_i$ for any k. If $G_i \subset H$, then $\langle m, v\rangle \in \frac{1}{d}\mathbb{Z}$, and $|\langle m, v\rangle - \frac{1}{d}| \leq \|m\|_1\theta_i$. Hence, if moreover $\|m\|_1\theta_i < \frac{1}{d}$, then $\langle m, v\rangle = \frac{1}{d}$ and $\frac{1}{d} + \mathbb{Z} \in \chi_m(G_i)$. Possibly removing finitely many terms of the sequence, we may assume that for any i,

$$G_i \subset H, \quad \chi_m(G_i) = \tfrac{1}{d}\mathbb{Z}/\mathbb{Z}, \quad \text{and} \quad \langle m, v\rangle \in \tfrac{1}{d}\mathbb{Z} \text{ for any } v \in N_{G_i}. \tag{16.6}$$

Step 4. In this step we prove that m can be taken with $m_k \geq 0$ for any k.

Suppose that m has a positive coordinate and a negative coordinate. Let

$$\lambda := \frac{\sum_{m_k<0}(-m_k)}{\sum_{m_k>0} m_k} > 0,$$

and let $w \in \mathbb{R}^3_{>0}$ be the vector with $w_k = \lambda$ if $m_k > 0$ and $w_k = 1$ if $m_k \leq 0$. Then $\langle m, w\rangle = 0$. Thus $\sqrt{\theta_i}w + \mathbb{Z}^3 \in \operatorname{Ker}\chi_m = H^0 \subset H$. By (16.5), there exists $v_i \in N_{G_i}$ with

$$v_{i,k} \geq \sqrt{\theta_i}w_k - \theta_i \quad \text{and} \quad \sum_{k=1}^{3} v_{i,k} \leq \sqrt{\theta_i}\sum_{k=1}^{3} w_k + 3\theta_i.$$

For $i \gg 0$ we have $\sqrt{\theta_i} < \min_k w_k$, so $v_i \in \mathbb{R}^3_{>0}$, and $\sum_k v_{i,k} < \epsilon$. This contradicts (16.4). Hence the non-zero coordinates of m are either all positive or all negative. Since replacing m with $-m$ does not change $\operatorname{Ker}\chi_m$ or d, we may assume that $m_k \geq 0$ for any k. Let $M := m_1 + m_2 + m_3$. Then $M \geq 1$, and $J = \{k \mid m_k > 0\}$ is non-empty.

Step 5. In this step we pass to a subsequence and construct the decompositions of the $h_{i,j}$ in Theorem 16.4(2).

If $s_i = 0$ for infinitely many i, then we pass to the subsequence of such i, let $s := 0$, and the remaining steps hold with all sums over j empty and $\beta_{i,k} = 0$ for any i and k. Otherwise,

possibly removing finitely many terms, we may assume that $s_i \geq 1$ for any i. Then $\Gamma\backslash\{0\} \neq \emptyset$, and $\gamma := \gamma(\Gamma) > 0$ by Lemma 2.10(2). Let E_0 be the exceptional divisor of the blow-up of $\mathbb{A}^3$ at 0. Since $(\mathbb{A}^3, B_i)$ is klt near 0,

$$0 < a(E_0, \mathbb{A}^3, B_i) = 3 - \sum_j c_{i,j} \operatorname{mult}_0 h_{i,j}$$

[KM98, Lemma 2.29]. Since $c_{i,j} \geq \gamma$ and $\operatorname{mult}_0 h_{i,j} \geq 1$, we have $s_i < \frac{3}{\gamma}$ and $\operatorname{mult}_0 h_{i,j} < \frac{3}{\gamma}$ for any i and j. Thus, possibly passing to a subsequence, we may assume that $s_i = s$ for any i. By Lemma 2.10(3) applied to the sequences $(c_{i,j})_i$ for $j = 1, \dots, s$ in turn, possibly passing to a subsequence, we may assume that $c_{i,j}$ is nondecreasing in i for any j. For any i and j, choose a monomial $x^{\alpha_{i,j}}$ of $h_{i,j}$ with $|\alpha_{i,j}| = \operatorname{mult}_0 h_{i,j}$. Since $|\alpha_{i,j}| < \frac{3}{\gamma}$, there exist finitely many possibilities for $\alpha_{i,j}$. Thus, possibly passing to a subsequence, we may assume that $\alpha_{i,j} = \alpha_j$ does not depend on i. In particular, x^{α_j} is a monomial of $h_{i,j}$ of least total degree, and

$$\beta_{i,k} \geq 0, \quad \sum_{k=1}^{3} \beta_{i,k} = \sum_j c_{i,j}|\alpha_j| < 3, \quad \text{and } \beta_{i,k} \text{ is nondecreasing in } i \text{ for any } k. \tag{16.7}$$

Let $g \in G_i$ be represented by $\xi \in N_{G_i}$. Then $g^* x^a = e^{2\pi\sqrt{-1}\langle a,\xi\rangle} x^a$ for any $a \in \mathbb{Z}^3_{\geq 0}$. Since $h_{i,j}$ is G_i-semi-invariant and distinct monomials are linearly independent, the G_i-character of any monomial x^a of $h_{i,j}$ equals the G_i-character of $h_{i,j}$, that is,

$$\chi_{a-\alpha_j}(g) = 0 \quad \text{for any } g \in G_i, \tag{16.8}$$

where $\chi_{a-\alpha_j} : \mathbb{T}^3 \to \mathbb{T}^1$ is the character defined by $a - \alpha_j$.

Fix j and $a \in \mathbb{Z}^3_{\geq 0}$ with $a - \alpha_j \notin \mathbb{Z}m$. Suppose that x^a is a monomial of $h_{i,j}$ for infinitely many i. Any $h \in H$ is the limit of a sequence of elements of the groups G_i for such i, so (16.8) and the continuity of $\chi_{a-\alpha_j}$ imply that $\chi_{a-\alpha_j}$ vanishes on H, hence on H^0. Thus $a - \alpha_j \in \mathbb{Z}m$ by the definition of m, a contradiction. Hence x^a is a monomial of $h_{i,j}$ for only finitely many i. Since there exist finitely many $a \in \mathbb{Z}^3_{\geq 0}$ with $|a| \leq n$ for any n, the number

$$\nu'_i := \min\{|a| \mid x^a \text{ is a monomial of some } h_{i,j} \text{ with } a - \alpha_j \notin \mathbb{Z}m\},$$

with $\nu'_i := +\infty$ if there is no such monomial, satisfies $\nu'_i \to +\infty$. Let $\nu_i := \min\{\nu'_i, i\}$. Then $\nu_i \to +\infty$.

Let $x^{\alpha_j + lm}$, $l \in \mathbb{Z}$, be a monomial of $h_{i,j}$. Since $|\alpha_j + lm| = |\alpha_j| + lM$ and x^{α_j} has least total degree, $l \geq 0$. By (16.8), $l\chi_m = \chi_{lm}$ vanishes on G_i, so $\frac{l}{d} \in \mathbb{Z}$ by (16.6). Let $U_{i,j}(u) := \sum_l q_l u^l$, where q_l is the coefficient of $x^{\alpha_j + lm}$ in $h_{i,j}$, and let $R_{i,j} := h_{i,j} - x^{\alpha_j} U_{i,j}(x^m)$. Then $U_{i,j} \in \mathbb{C}[u^d]$, $U_{i,j}(0) = q_0 \neq 0$, and any monomial x^a of $R_{i,j}$ satisfies $a - \alpha_j \notin \mathbb{Z}m$, so $|a| \geq \nu'_i \geq \nu_i$, that is, $R_{i,j} \in (x_1, x_2, x_3)^{\nu_i}$. Moreover, any monomial of $x^{\alpha_j} U_{i,j}(x^m)$ other than x^{α_j} has total degree $|\alpha_j| + lM > |\alpha_j|$. Since $\nu_i \to +\infty$, possibly removing finitely many terms, we may assume that $\nu_i > |\alpha_j|$ for any j. Thus

$$\operatorname{in}_{(1,1,1)}(h_{i,j}) = U_{i,j}(0) x^{\alpha_j} \quad \text{for any } i \text{ and } j. \tag{16.9}$$

Step 6. In this step we construct vectors $v_{i,k} \in N_{G_i}$ close to $\frac{1}{dm_k} e_k$ for $k \in J$, and we prove that $\beta_{i,k} \leq 1 - \frac{dm_k \epsilon}{2}$ for $k \in J$ and $i \gg 0$, and that $d \max_k m_k \leq \frac{1}{\epsilon}$.

Let $\sigma_i := \max\{\sqrt{\theta_i}, \nu_i^{-1/2}\}$. Then $\sigma_i \to 0$, $\frac{\theta_i}{\sigma_i} \leq \sqrt{\theta_i} \to 0$, and $\nu_i \sigma_i \geq \sqrt{\nu_i} \to +\infty$. For $k \in J$ let $p_{i,k} \in \mathbb{R}^3$ be the vector with

$$p_{i,k,l} := \sigma_i \ (l \neq k) \quad \text{and} \quad p_{i,k,k} := \frac{1}{m_k}\Big(\frac{1}{d} - \sigma_i \sum_{l \neq k} m_l\Big).$$

Then $\langle m, p_{i,k}\rangle = \frac{1}{d}$, so $p_{i,k} + \mathbb{Z}^3 \in \chi_m^{-1}(\frac{1}{d} + \mathbb{Z}) \subset H$. By (16.5), there exists $v_{i,k} \in N_{G_i}$ with $|v_{i,k,l} - p_{i,k,l}| \le \theta_i$ for any l. Let $C := M + 5$. Since $m_k \ge 1$ and $\theta_i \le \sigma_i$ for $i \gg 0$,

$$\Big\|v_{i,k} - \frac{1}{dm_k} e_k\Big\|_1 \le \Big(2 + \frac{1}{m_k}\sum_{l\ne k} m_l\Big)\sigma_i + 3\theta_i \le C\sigma_i. \tag{16.10}$$

Since $\theta_i \le \frac{\sigma_i}{2}$ and $\frac{1}{dm_k} - C\sigma_i \ge \frac{\sigma_i}{2}$ for $i \gg 0$, we obtain $\min_l v_{i,k,l} \ge \frac{\sigma_i}{2}$ for $i \gg 0$. In particular, $v_{i,k} \in N_{G_i} \cap \mathbb{R}^3_{>0}$.

We compute $\mathrm{wt}_{v_{i,k}}(h_{i,j})$ for $i \gg 0$. Since $\langle m, v_{i,k}\rangle > 0$, any monomial $x^{\alpha_j + lm}$ of $x^{\alpha_j} U_{i,j}(x^m)$ has $v_{i,k}$-weight $\langle \alpha_j, v_{i,k}\rangle + l\langle m, v_{i,k}\rangle \ge \langle \alpha_j, v_{i,k}\rangle$, with equality for $l = 0$. A monomial x^a of $R_{i,j}$ has $v_{i,k}$-weight

$$\langle a, v_{i,k}\rangle \ge |a| \min_l v_{i,k,l} \ge \frac{\nu_i \sigma_i}{2} \to +\infty,$$

while $\langle \alpha_j, v_{i,k}\rangle \le |\alpha_j| \max_l v_{i,k,l}$ is bounded by (16.10). Thus $\mathrm{wt}_{v_{i,k}}(h_{i,j}) = \langle \alpha_j, v_{i,k}\rangle$ for $i \gg 0$. Let

$$L_i(v) := \sum_{l=1}^{3} (1 - \beta_{i,l}) v_l, \qquad v \in \mathbb{R}^3.$$

Then $\psi_i(v_{i,k}) = L_i(v_{i,k})$ for $i \gg 0$. By (16.7), $|1 - \beta_{i,l}| \le 2$ for any l. Hence (16.10) and (16.4) imply that, for $i \gg 0$,

$$\epsilon \le a_i \le L_i(v_{i,k}) \le \frac{1 - \beta_{i,k}}{dm_k} + 2C\sigma_i \quad \text{for any } k \in J. \tag{16.11}$$

Since $2C\sigma_i \le \frac{\epsilon}{2}$ for $i \gg 0$, possibly removing finitely many terms, we may assume that

$$\beta_{i,k} \le 1 - \frac{dm_k \epsilon}{2} < 1 \quad \text{for any } i \text{ and any } k \in J. \tag{16.12}$$

Moreover, (16.4) and (16.10) imply that $\epsilon \le \sum_l v_{i,k,l} \to \frac{1}{dm_k}$ for any $k \in J$. Since $m_k = 0$ for $k \notin J$, we obtain $d \max_k m_k \le \frac{1}{\epsilon}$.

Step 7. In this step we prove that $\beta_{i,k} \le 1$ for any i and any $k \notin J$.

Suppose that $\beta_{i_0,k} > 1$ for some i_0 and some $k \notin J$. Let $w \in \mathbb{R}^3_{>0}$ be the vector with $w_l := \frac{1}{dM}$ for $l \in J$, $w_l := 1$ for $l \notin J \cup \{k\}$, and $w_k := T$, where $T > 0$ is so large that

$$L_{i_0}(w) = \sum_{l\ne k} (1 - \beta_{i_0,l}) w_l + (1 - \beta_{i_0,k}) T < 0.$$

Then $\langle m, w\rangle = \sum_{l\in J} \frac{m_l}{dM} = \frac{1}{d}$, so $w + \mathbb{Z}^3 \in H$. By (16.5), there exists $w_i \in N_{G_i}$ with $|w_{i,l} - w_l| \le \theta_i$ for any l. For $i \gg 0$ we have $\frac{1}{2}\min_l w_l \le w_{i,l} \le 2\max_l w_l$ for any l, so $w_i \in N_{G_i} \cap \mathbb{R}^3_{>0}$. Since $\langle m, w_i\rangle > 0$, any monomial $x^{\alpha_j + lm}$ of $x^{\alpha_j} U_{i,j}(x^m)$ has w_i-weight at least $\langle \alpha_j, w_i\rangle$, with equality for $l = 0$, and $\langle \alpha_j, w_i\rangle \le 2|\alpha_j| \max_l w_l$. Any monomial x^a of $R_{i,j}$ has w_i-weight at least $\frac{\nu_i}{2}\min_l w_l \to +\infty$. Thus $\psi_i(w_i) = L_i(w_i)$ for $i \gg 0$. By (16.7), $\beta_{i,l} \ge \beta_{i_0,l}$ for any $i \ge i_0$ and any l, so

$$\psi_i(w_i) = L_i(w_i) \le L_{i_0}(w_i) \to L_{i_0}(w) < 0.$$

Hence $\psi_i(w_i) < 0$ for $i \gg 0$, which contradicts (16.4). Thus $\beta_{i,k} \le 1$ for any i and any $k \notin J$.

Let $B_i^0 := \sum_{k=1}^3 \beta_{i,k} \operatorname{div}(x_k)$. By (16.9), $B_i^0 = \sum_j c_{i,j} \operatorname{div}(\mathrm{in}_{(1,1,1)}(h_{i,j}))$. By (16.7), (16.12), and the last paragraph, $0 \le \beta_{i,k} \le 1$ for any i and k. Since $\sum_k \operatorname{div}(x_k)$ is an snc divisor, $(\mathbb{A}^3, \sum_k \operatorname{div}(x_k))$ is lc and $(\mathbb{A}^3, t'\sum_k \operatorname{div}(x_k))$ is klt for any rational number $0 \le t' < 1$ [KM98, Corollary 2.31(3)]. For any prime divisor E over $\mathbb{A}^3$, $a(E, \mathbb{A}^3, \sum_k b_k \operatorname{div}(x_k)) = a(E, \mathbb{A}^3, 0) - \sum_k b_k \operatorname{ord}_E(x_k)$ is nonincreasing in each $b_k \ge 0$. Since $0 \le \beta_{i,k} \le 1$, $(\mathbb{A}^3, B_i^0)$ is lc. For $0 \le t < 1$, choosing a rational $t' \in (t, 1)$, we have $t\beta_{i,k} < t'$, so $(\mathbb{A}^3, tB_i^0)$ is klt.

Step 8. In this step we prove that $a_i \ge \mathrm{mld}((\mathbb{A}^3 \ni 0, B_i^0)/G_i)$ for any i.

Proposition 16.3 requires both $(\mathbb{A}^3, B_i)$ and $(\mathbb{A}^3, B_i^0)$ to be klt near 0, while **Step 7** only shows that $(\mathbb{A}^3, B_i^0)$ is lc, so we first multiply the boundaries by some $t < 1$ and then let $t \to 1^-$. Fix i

and $0 < t < 1$. Since $B_i \geq 0$, we have $a(F, \mathbb{A}^3, tB_i) \geq a(F, \mathbb{A}^3, B_i)$ for any prime divisor F over $\mathbb{A}^3$, so $(\mathbb{A}^3, tB_i)$ is klt near 0. By the end of **Step 7**, $(\mathbb{A}^3, tB_i^0)$ is klt. By (16.9), Proposition 16.3 applied to $f_j := h_{i,j}$, the coefficients $tc_{i,j}$, and $w := (1,1,1)$ implies that

$$\operatorname{mld}((\mathbb{A}^3 \ni 0, tB_i)/G_i) \geq \operatorname{mld}((\mathbb{A}^3 \ni 0, tB_i^0)/G_i). \tag{16.13}$$

Let D be B_i or B_i^0. For a prime divisor F over $\mathbb{A}^3$ with $\operatorname{center}_{\mathbb{A}^3} F = \{0\}$ and $0 < t \leq 1$, let

$$A_t(F) := \frac{a(F, \mathbb{A}^3, tD)}{|I_{G_i}(F)|} = \frac{a(F, \mathbb{A}^3, 0) - t \operatorname{mult}_F D}{|I_{G_i}(F)|}.$$

Since $D \geq 0$, $A_t(F)$ is a nonincreasing affine function of t. Let $\phi(t) := \operatorname{mld}((\mathbb{A}^3 \ni 0, tD)/G_i)$. Since tD is G_i-invariant, Lemma 2.6 implies that $\phi(t) = \inf_F A_t(F)$, where F runs over the prime divisors over $\mathbb{A}^3$ with $\operatorname{center}_{\mathbb{A}^3} F = \{0\}$. In particular, ϕ is nonincreasing, and

$$\lim_{t \to 1-0} \phi(t) = \inf_{0<t<1} \phi(t) = \inf_{0<t<1} \inf_F A_t(F) = \inf_F \inf_{0<t<1} A_t(F) = \inf_F A_1(F) = \phi(1),$$

where the fourth equality holds since $A_t(F)$ is nonincreasing and continuous in t. Applying the last equality to $D = B_i$ and $D = B_i^0$ and letting $t \to 1-0$ in (16.13), we obtain

$$a_i \geq \operatorname{mld}((\mathbb{A}^3 \ni 0, B_i^0)/G_i). \tag{16.14}$$

Step 9. In this step we prove the conclusions of (2).

Fix i. Consider Set-up 8.4 for $\mathbb{A}^3$ and the group G_i, with the polynomials x_k for the indices k with $\beta_{i,k} > 0$ and the coefficients $\beta_{i,k}$, so that the boundary is B_i^0. If $\beta_{i,k} = 0$ for any k, then there is no polynomial and $B_i^0 = 0$. The polynomials x_k are reduced and pairwise coprime, $\beta_{i,k} \leq 1$ by **Step 7**, and $(\mathbb{A}^3, B_i^0)$ is lc. Condition (a) of Proposition 8.8 holds: for any $w \in \mathbb{R}^3_{>0}$, $\operatorname{in}_w(x_k) = x_k$ has no zero in $(\mathbb{C}^*)^3$, so the common zero set in $(\mathbb{C}^*)^3$ of any non-empty set of the polynomials x_k is empty. Since $\operatorname{wt}_v(x_k) = v_k$, the function ψ of Set-up 8.4 for $\mathbb{A}^3$, G_i, and B_i^0 is $\sum_k v_k - \sum_k \beta_{i,k} v_k = L_i(v)$. Thus Proposition 8.8 implies that

$$\operatorname{mld}((\mathbb{A}^3 \ni 0, B_i^0)/G_i) = \min_{v \in N_{G_i} \cap \mathbb{R}^3_{>0}} L_i(v).$$

By (16.12), $\varrho_i \geq \frac{d\epsilon}{2} > 0$. By the definition of ϱ_i, $1 - \beta_{i,k} \geq \varrho_i m_k$ for $k \in J$, and $1 - \beta_{i,k} \geq 0 = \varrho_i m_k$ for $k \notin J$ by **Step 7**. For $v \in N_{G_i} \cap \mathbb{R}^3_{>0}$ we have $\langle m, v \rangle > 0$, so $\langle m, v \rangle \geq \frac{1}{d}$ by (16.6). Hence

$$L_i(v) \geq \varrho_i \langle m, v \rangle \geq \frac{\varrho_i}{d} = b_i,$$

and (16.14) implies that $a_i \geq b_i$.

Conversely, choose $k \in J$ with $\varrho_i = \frac{1-\beta_{i,k}}{m_k}$, and recall that $v_{i,k} \in N_{G_i} \cap \mathbb{R}^3_{>0}$ is the vector of **Step 6** close to $\frac{1}{dm_k} e_k$. By (16.11), $a_i \leq b_i + 2C\sigma_i$ for $i \gg 0$. Let $\eta_i := a_i - b_i$. Then $\eta_i \geq 0$ and $\eta_i \leq 2C\sigma_i$ for $i \gg 0$, so $\eta_i \to 0$ and $b_i \leq a_i \leq b_i + \eta_i$.

Since $\beta_{i,k} \geq 0$ and $m_k \geq 1$ for $k \in J$, we have $\varrho_i \leq 1$. Thus $1 - \varrho_i \in [0, 1)$. Let $\zeta := e^{2\pi\sqrt{-1}/d}$, and let μ_d act on $\mathbb{A}^1$ with coordinate z by $z \to \zeta z$. The quotient $\mathbb{A}^1/\mu_d$ is the affine line with coordinate z^d, and the ramification index of 0 over the image $\bar{0}$ of 0 is d. By Lemma 2.6(4), the coefficient of $\bar{0}$ in the crepant quotient boundary of $(1 - \varrho_i) \cdot 0$ is

$$1 - \frac{1 - (1 - \varrho_i)}{d} = 1 - \frac{\varrho_i}{d}.$$

Since the only prime divisor over a smooth curve with center a closed point is that point, $\operatorname{mld}((\mathbb{A}^1 \ni 0, (1 - \varrho_i) \cdot 0)/\mu_d) = \frac{\varrho_i}{d} = b_i$.

Let Ξ be the set of the numbers

$$1 - \frac{1}{d} \min_{k \in J} \frac{1 - \sum_{j=1}^s \alpha_{j,k} c'_j}{m_k}, \qquad (c'_1, \dots, c'_s) \in \Gamma^s,$$

which depends only on Γ, m, d, and the α_j, and contains $1 - \frac{\varrho_i}{d}$ for any i. If a sequence in Ξ were strictly decreasing, then, by Lemma 2.10(3) applied to the coordinates of representing tuples, we

could pass to a subsequence along which any coordinate is nondecreasing, so any $\sum_j \alpha_{j,k} c'_j$ is nondecreasing and the number in Ξ is nondecreasing, a contradiction. Thus Ξ satisfies the DCC.

Finally, along the subsequence of **Step 5**, along which $c_{i,j}$ is nondecreasing in i for any j, $\beta_{i,k}$ is nondecreasing in i for any k by (16.7), so ϱ_i and b_i are nonincreasing. Thus (2) holds.

Step 10. In this step we prove that if a_i is strictly increasing, then $\dim H^0 \le 1$.

Suppose that a_i is strictly increasing and $\dim H^0 \ge 2$. By (1), $\dim H^0 = 2$. Passing to the subsequence of (2), a_i is still strictly increasing, $b_i \le a_i$ for any i, and $a_i - b_i \to 0$. Since b_i is nonincreasing, the set $\{b_i \mid i \ge 1\}$ satisfies the ACC, which contradicts Lemma 14.11. □

16.3. **Positive circles.** In this subsection, we prove the ACC for quotient minimal log discrepancies at the origin of $\mathbb{A}^3$ when the groups have bounded index over their intersections with the one-parameter subgroup $\mathbf{T}_{(p,q,0)}$, where p and q are coprime positive integers.

For a finite diagonal group G, the proof extracts the divisor of the vector $\frac{1}{n}(p,q,0)$, where $n = |G \cap \mathbf{T}_{(p,q,0)}|$, by a star subdivision and uses Theorem 2.19 on finitely many fixed affine toric threefolds.

Theorem 16.5. *Fix coprime positive integers p, q, an integer $e_0 \ge 1$, and a DCC set $\Gamma \subset [0,1]$. Let $\mathbf{T} := \mathbf{T}_{(p,q,0)}$ act on $X := \mathbb{A}^3_{x,y,z}$. Then the set of numbers $\operatorname{mld}((X \ni 0, B)/G)$ satisfies the ACC, where G is a finite diagonal group acting faithfully on X with $[G : G \cap \mathbf{T}] \le e_0$, and $B = \sum_j c_j \operatorname{div}(h_j)$ with $c_j \in \Gamma$, h_j non-zero G-semi-invariant polynomials, and (X, B) klt near 0.*

Moreover, let $n := |G \cap \mathbf{T}|$, $d_j := \min\{pa+qb \mid x^a y^b z^c$ occurs in $h_j\}$, and $A := p+q-\sum_j c_j d_j$. If $n > p+q$, then $A > 0$, the vector $v := \frac{1}{n}(p,q,0)$ is primitive in N_G, and the star subdivision at v yields a projective birational morphism $f : Y_G \to X/G$ extracting one divisor E_G over the image of the z-axis. Let B_{Y_G} be the $\mathbb{R}$-divisor on Y_G with $K_{Y_G} + B_{Y_G} = f^*(K_{X/G} + B_{X/G})$ and $f_* B_{Y_G} = B_{X/G}$. Then $B_{Y_G} \ge 0$, (Y_G, B_{Y_G}) is klt near the fiber over the image of 0, $\operatorname{mult}_{E_G} B_{Y_G} = 1 - \frac{A}{n}$, and Y_G is the union of two affine open subsets isomorphic to $\mathbb{A}^3/G_x$ and $\mathbb{A}^3/G_y$, where G_x and G_y are finite diagonal groups of orders $p[G : G \cap \mathbf{T}]$ and $q[G : G \cap \mathbf{T}]$.

Proof. The proof is long, so we divide it into several steps. Let e_x, e_y, e_z be the standard basis of $\mathbb{Z}^3 \subset \mathbb{R}^3$. For G and $B = \sum_{j=1}^s c_j \operatorname{div}(h_j)$ as in the statement, let $\pi : X \to X/G$ be the quotient morphism and $\bar{0} := \pi(0)$, and identify G with the image of G in $\mathbb{T}^3$ under the characters of x, y, z. Since G acts faithfully, $[N_G : \mathbb{Z}^3] = |G|$.

Step 1. In this step we assume that $n > p+q$, and we prove that v is primitive in N_G, that f and E_G exist, and the statements about B_{Y_G}.

Let $\iota : \mathbb{R}/\mathbb{Z} \to \mathbf{T}$ be the homomorphism

$$\iota(s \bmod \mathbb{Z}) := (e^{2\pi\sqrt{-1}sp}, e^{2\pi\sqrt{-1}sq}, 1).$$

For $s \in \mathbb{R}$, the class $\rho(s(p,q,0))$ is the image of $\iota(s \bmod \mathbb{Z})$ in $\mathbb{T}^3$, so $s(p,q,0) \in N_G$ if and only if $\iota(s \bmod \mathbb{Z}) \in G \cap \mathbf{T}$. The homomorphism ι is injective: if $sp, sq \in \mathbb{Z}$, then $s = \alpha sp + \beta sq \in \mathbb{Z}$ for integers α, β with $\alpha p + \beta q = 1$. The image of ι contains $G \cap \mathbf{T}$, because an element $(\lambda^p, \lambda^q, 1)$ of finite order has $|\lambda| = 1$. Thus the classes $s \bmod \mathbb{Z}$ with $s(p,q,0) \in N_G$ form the subgroup of $\mathbb{R}/\mathbb{Z}$ of order n, that is, $\frac{1}{n}\mathbb{Z}/\mathbb{Z}$, and

$$N_G \cap \mathbb{R}(p,q,0) = \mathbb{Z} \cdot \tfrac{1}{n}(p,q,0) = \mathbb{Z}v.$$

If $\frac{1}{k}v \in N_G$ for an integer $k \ge 2$, then $\frac{1}{k} \in \mathbb{Z}$, which is impossible. Thus v is primitive in N_G.

By [CLS11, Proposition 1.3.18], X/G is the affine toric variety of the cone $\sigma := \operatorname{Cone}(e_x, e_y, e_z)$ with respect to the lattice N_G. The vector v belongs to the relative interior of the face $\operatorname{Cone}(e_x, e_y)$ of σ. Thus the star subdivision Σ of σ at v has the maximal cones

$$\sigma_x := \operatorname{Cone}(v, e_y, e_z) \quad \text{and} \quad \sigma_y := \operatorname{Cone}(e_x, v, e_z),$$

and any face of σ not containing $\operatorname{Cone}(e_x, e_y)$ is a cone of Σ. Let $f : Y_G \to X/G$ be the toric morphism defined by Σ. Then f is projective and birational, the toric prime divisor E_G of

Y_G corresponding to the ray $\mathbb{R}_{\geq 0}v$ is the unique exceptional prime divisor of f, and f is an isomorphism over the complement of the closure of the orbit of $\mathrm{Cone}(e_x, e_y)$, which is the image of the z-axis [CLS11, Lemma 11.1.3, Proposition 11.1.6, and Theorem 3.2.6]. In particular, $f(E_G)$ is the image of the z-axis. By Lemma 2.6(0), $K_{X/G} + B_{X/G}$ is $\mathbb{R}$-Cartier and $B_{X/G} \geq 0$, so B_{Y_G} is defined and

$$B_{Y_G} = f_*^{-1} B_{X/G} + (\mathrm{mult}_{E_G} B_{Y_G}) E_G.$$

We compute $\mathrm{mult}_{E_G} B_{Y_G}$. A term with $c_j = 0$ does not contribute to B or to A. An h_j with $h_j(0) \neq 0$ has $d_j = 0$, because the constant monomial occurs in h_j, and $\mathrm{div}(h_j)$ does not contain the z-axis. Thus removing the terms with $c_j = 0$ or $h_j(0) \neq 0$ changes neither A nor B near the generic point of the z-axis, hence changes neither A nor $a(E_G, X/G, B_{X/G})$. By the sentence of Lemma 8.5 beginning "Moreover", with $\mathbb{A}^3$ in place of $\mathbb{A}^d$, and with the remaining h_j and c_j in place of the f_j and b_j of Set-up 8.4, E_G is the toric prime divisor E_v and

$$a(E_G, X/G, B_{X/G}) = \psi(v) = v_1 + v_2 + v_3 - \sum_j c_j \,\mathrm{wt}_v(h_j).$$

By Definition 2.16,

$$\mathrm{wt}_v(h_j) = \frac{1}{n} \min\{pa + qb \mid x^a y^b z^c \text{ occurs in } h_j\} = \frac{d_j}{n}.$$

By [KM98, Lemma 2.30], $a(E_G, Y_G, B_{Y_G}) = a(E_G, X/G, B_{X/G})$. Thus

$$a(E_G, X/G, B_{X/G}) = \frac{p+q}{n} - \sum_j c_j \frac{d_j}{n} = \frac{A}{n} \quad \text{and} \quad \mathrm{mult}_{E_G} B_{Y_G} = 1 - a(E_G, Y_G, B_{Y_G}) = 1 - \frac{A}{n}.$$

Since (X, B) is klt near 0, Lemma 2.6(2)(3) implies that $(X/G, B_{X/G})$ is klt on an open neighborhood V of $\bar{0}$. Since $\bar{0} \in f(E_G)$, we have $\frac{A}{n} = a(E_G, X/G, B_{X/G}) > 0$, hence $A > 0$. Since $c_j \geq 0$ and $d_j \geq 0$, we have $A \leq p + q < n$, hence

$$0 < 1 - \frac{A}{n} < 1.$$

Since $B_{X/G} \geq 0$, the formula for B_{Y_G} implies $B_{Y_G} \geq 0$. Let P be a prime divisor over Y_G whose center on Y_G intersects $f^{-1}(V)$. Then $\mathrm{center}_{X/G} P = f(\mathrm{center}_{Y_G} P)$ intersects V, so

$$a(P, Y_G, B_{Y_G}) = a(P, X/G, B_{X/G}) > 0$$

by [KM98, Lemma 2.30]. Thus (Y_G, B_{Y_G}) is klt on $f^{-1}(V)$, in particular near $f^{-1}(\bar{0})$.

Step 2. In this step we assume that $n > p + q$, we show that Y_G is the union of affine open subsets $U_x \cong \mathbb{A}^3/G_x$ and $U_y \cong \mathbb{A}^3/G_y$ with $|G_x| = p[G : G \cap \mathbf{T}]$ and $|G_y| = q[G : G \cap \mathbf{T}]$, and we show that $\mathbb{A}^3/G_x$ and $\mathbb{A}^3/G_y$ belong to a finite list depending only on p, q, and e_0.

Let

$$\Lambda_x := \mathbb{Z}v + \mathbb{Z}e_y + \mathbb{Z}e_z \quad \text{and} \quad \Lambda_y := \mathbb{Z}e_x + \mathbb{Z}v + \mathbb{Z}e_z.$$

Since $v \in N_G$ and $\mathbb{Z}^3 \subset N_G$, we have $\Lambda_x, \Lambda_y \subset N_G$. We have

$$|\det(v, e_y, e_z)| = \frac{p}{n}, \qquad |\det(e_x, v, e_z)| = \frac{q}{n},$$

while the covolume of N_G is $\frac{1}{[N_G : \mathbb{Z}^3]} = \frac{1}{|G|}$. Thus

$$[N_G : \Lambda_x] = \frac{p|G|}{n} = p[G : G \cap \mathbf{T}] \quad \text{and} \quad [N_G : \Lambda_y] = \frac{q|G|}{n} = q[G : G \cap \mathbf{T}].$$

The cone σ_x is generated by the basis v, e_y, e_z of Λ_x, so the affine toric variety of σ_x with respect to Λ_x is $\mathbb{A}^3$, with coordinates the characters dual to v, e_y, e_z. Let $U_x \subset Y_G$ be the affine toric open subset corresponding to σ_x. Then $U_x \cong \mathbb{A}^3/G_x$, where $G_x := N_G/\Lambda_x$ acts faithfully on $\mathbb{A}^3$ through the characters dual to v, e_y, e_z [CLS11, Proposition 1.3.18]. Thus G_x is a finite diagonal group of order $p[G : G \cap \mathbf{T}]$. By the argument for U_x with σ_y and Λ_y in place of σ_x and Λ_x, the

affine toric open subset $U_y \subset Y_G$ corresponding to σ_y is isomorphic to $\mathbb{A}^3/G_y$ for a finite diagonal group G_y of order $q[G : G \cap \mathbf{T}]$. Since σ_x and σ_y are the maximal cones of Σ, $Y_G = U_x \cup U_y$.

For any positive integer M, let $l := \operatorname{lcm}(1, \dots, M)$. The order of any element of a finite diagonal group $H \subset (\mathbb{C}^*)^3$ with $|H| \leq M$ divides $|H|$, hence divides l. Thus H is a subgroup of the finite group $\{(\zeta_1, \zeta_2, \zeta_3) \mid \zeta_1^l = \zeta_2^l = \zeta_3^l = 1\}$. Therefore there exist only finitely many finite diagonal groups $H \subset (\mathbb{C}^*)^3$ with $|H| \leq M$, and hence only finitely many varieties $\mathbb{A}^3/H$. Taking $M = e_0 \max\{p, q\}$, so that $|G_x| \leq M$ and $|G_y| \leq M$, we see that $\mathbb{A}^3/G_x$ and $\mathbb{A}^3/G_y$ belong to a finite list depending only on p, q, and e_0.

Step 3. In this step we show that the coefficients of $B_{X/G}$ at the prime divisors containing $\bar{0}$ belong to the set

$$\Xi_0 := \Psi(\Gamma_+ \cap [0, 1]),$$

which satisfies the DCC and depends only on Γ, and we determine the components of B_{Y_G} whose supports intersect $f^{-1}(\bar{0})$ when $n > p + q$.

Since (X, B) is klt near 0, it is lc near 0, and G preserves B because the h_j are G-semi-invariant. By Lemma 2.10(5), applied to X, 0, G, the h_j, and the c_j, there exists a G-invariant affine open neighborhood X' of 0 such that the coefficients of $B_{X'/G}$, where $(X', B|_{X'})/G = (X'/G, B_{X'/G})$, belong to $\Psi(\Gamma_+ \cap [0, 1]) = \Xi_0$, and Ξ_0 satisfies the DCC. Let D be a prime divisor on X/G with $\bar{0} \in D$. Then D intersects X'/G, and the restriction of $B_{X/G}$ to X'/G is $B_{X'/G}$, so $\operatorname{mult}_D B_{X/G} \in \Xi_0$.

Now assume that $n > p + q$, and let D' be a component of B_{Y_G} whose support intersects $f^{-1}(\bar{0})$. If $D' \neq E_G$, then $D' = f_*^{-1}D$ for a component D of $B_{X/G}$ by the formula for B_{Y_G} in **Step 1**, and $\bar{0} \in f(D') = D$. Thus the coefficients of the components of B_{Y_G} whose supports intersect $f^{-1}(\bar{0})$ belong to $\Xi_0 \cup \{1 - \frac{A}{n}\}$.

Step 4. In this step we begin the proof of the ACC and treat the case when n is bounded.

Suppose that the ACC fails. Then there exist finite diagonal groups G_i acting faithfully on X with $[G_i : G_i \cap \mathbf{T}] \leq e_0$, non-zero G_i-semi-invariant polynomials $h_{i,j}$, and $c_{i,j} \in \Gamma$, such that, for $B_i := \sum_j c_{i,j} \operatorname{div}(h_{i,j})$, the pair (X, B_i) is klt near 0 and the numbers

$$a_i := \operatorname{mld}((X \ni 0, B_i)/G_i)$$

form a strictly increasing sequence. Let $n_i := |G_i \cap \mathbf{T}|$ and $A_i := p + q - \sum_j c_{i,j} d_{i,j}$, where $d_{i,j}$ is defined from $h_{i,j}$ as d_j is defined from h_j. We write $\bar{0}$ for the image of 0 in X/G_i.

Claim 16.6. *Let U be a normal $\mathbb{Q}$-factorial variety, $B_U \geq 0$ an $\mathbb{R}$-divisor on U, $\eta \in U$ a scheme-theoretic point, and B'' an $\mathbb{R}$-divisor with $0 \leq B'' \leq B_U$ whose support does not intersect $\overline{\{\eta\}}$. Then*

$$\operatorname{mld}(U \ni \eta, B_U - B'') = \operatorname{mld}(U \ni \eta, B_U).$$

Proof. Since U is $\mathbb{Q}$-factorial, B'' is $\mathbb{R}$-Cartier, and $\operatorname{mult}_P B'' = 0$ for any prime divisor P over U with $\operatorname{center}_U P = \overline{\{\eta\}}$. Thus $a(P, U, B_U - B'') = a(P, U, B_U)$ for any such P. □

Assume that n_i is bounded on an infinite subsequence. Then $|G_i| = n_i[G_i : G_i \cap \mathbf{T}] \leq e_0 n_i$ is bounded on that subsequence. By the finiteness of the finite diagonal groups of order at most M in **Step 2**, applied with M an upper bound of $e_0 n_i$ on that subsequence, there exist only finitely many groups G_i on that subsequence. Thus, possibly passing to a subsequence, we may assume that $G_i = G$ for any i. Let B_i' be the sum of the components of $(B_i)_{X/G}$ containing $\bar{0}$, with their coefficients in $(B_i)_{X/G}$. The variety X/G is $\mathbb{Q}$-factorial [KM98, Lemma 5.16], so $(X/G, B_i')$ is a pair, and Claim 16.6 implies $\operatorname{mld}(X/G \ni \bar{0}, B_i') = a_i$. By **Step 3**, the coefficients of B_i' belong to the DCC set $\Xi_0 \backslash \{0\} \subset \mathbb{R}_{>0}$. Thus Theorem 2.19(1), applied to the normal threefold X/G and $\Xi_0 \backslash \{0\}$, implies that $\{a_i\}$ satisfies the ACC, a contradiction.

Step 5. In this step we treat the case when n is unbounded: we construct a DCC set Ξ containing the relevant coefficients, and we realize a_i as a minimal log discrepancy at a scheme-theoretic point of Y_{G_i}.

By **Step 4**, n_i is unbounded on any infinite subsequence. Possibly passing to a subsequence, we may assume that n_i is strictly increasing and $n_i > p+q$ for any i. Let $f_i : Y_i := Y_{G_i} \to X/G_i$, $E_i := E_{G_i}$, and $B_{Y_i} := B_{Y_{G_i}}$ be as in the statement for G_i and B_i, and let

$$\theta_i := \operatorname{mult}_{E_i} B_{Y_i} = 1 - \frac{A_i}{n_i}.$$

By **Step 1**, $0 < A_i \le p+q$, hence

$$1 - \frac{p+q}{n_i} \le \theta_i < 1.$$

The set $\Theta := \{\theta_i \mid i \ge 1\}$ satisfies the DCC: if $\theta_{i_1} > \theta_{i_2} > \cdots$, then the indices i_k are pairwise distinct, so $n_{i_k} \to +\infty$ and $\theta_{i_k} \to 1$, contradicting $\theta_{i_k} < \theta_{i_1} < 1$ for any $k \ge 2$. Thus

$$\Xi := \Xi_0 \cup \Theta$$

satisfies the DCC, as a union of two DCC sets. By **Step 3**, the coefficients of the components of B_{Y_i} whose supports intersect $f_i^{-1}(\bar{0})$ belong to $\Xi\backslash\{0\}$.

By **Step 1**, $(X/G_i, (B_i)_{X/G_i})$ is klt near $\bar{0}$, so a_i is attained [Amb99, Section 1, the second paragraph after Definition 1.1]: there exists a prime divisor P_i over X/G_i with $\operatorname{center}_{X/G_i} P_i = \{\bar{0}\}$ and $a(P_i, X/G_i, (B_i)_{X/G_i}) = a_i$. Since f_i is proper, P_i has a center Z_i on Y_i, and $f_i(Z_i) = \{\bar{0}\}$. Let η_i be the generic point of Z_i. For any prime divisor P over Y_i with $\operatorname{center}_{Y_i} P = Z_i$, we have $\operatorname{center}_{X/G_i} P = f_i(Z_i) = \{\bar{0}\}$, so

$$a(P, Y_i, B_{Y_i}) = a(P, X/G_i, (B_i)_{X/G_i}) \ge a_i$$

by [KM98, Lemma 2.30], with equality for $P = P_i$. Thus

$$\operatorname{mld}(Y_i \ni \eta_i, B_{Y_i}) = a_i.$$

Step 6. In this step we conclude the proof by Theorem 2.19(1) on a fixed affine toric threefold.

By **Step 2**, η_i belongs to an affine open subset $U_i \subset Y_i$ isomorphic to $\mathbb{A}^3/H_i$ for a finite diagonal group H_i of order at most $e_0 \max\{p,q\}$, where U_i is the affine toric open subset corresponding to $\operatorname{Cone}(v_i, e_y, e_z)$ or $\operatorname{Cone}(e_x, v_i, e_z)$ and $v_i := \frac{1}{n_i}(p,q,0)$. By the finiteness in **Step 2**, applied with $M = e_0 \max\{p,q\}$, possibly passing to a subsequence, we may assume that there exist a finite diagonal group H and isomorphisms $\phi_i : U_i \to U$ for any i, where $U := \mathbb{A}^3/H$. Recall that $Z_i = \overline{\{\eta_i\}} \subset Y_i$ is the center of P_i, so $Z_i \subset f_i^{-1}(\bar{0})$.

Let B_i' be the sum of the components of B_{Y_i} whose supports intersect $f_i^{-1}(\bar{0})$, with their coefficients in B_{Y_i}. By **Step 5**, $B_i' \ge 0$ has coefficients in $\Xi\backslash\{0\}$, and the support of $B_{Y_i} - B_i'$ does not intersect Z_i. The variety U is $\mathbb{Q}$-factorial [KM98, Lemma 5.16], so $K_U + \phi_{i*}(B_i'|_{U_i})$ is $\mathbb{R}$-Cartier. The prime divisors over U_i with center $Z_i \cap U_i$ are the prime divisors over Y_i with center Z_i, and $a(P, U_i, B_{Y_i}|_{U_i}) = a(P, Y_i, B_{Y_i})$ for any such P. Thus **Step 5** and Claim 16.6 imply

$$\operatorname{mld}(U \ni \phi_i(\eta_i), \phi_{i*}(B_i'|_{U_i})) = \operatorname{mld}(U_i \ni \eta_i, B_{Y_i}|_{U_i}) = \operatorname{mld}(Y_i \ni \eta_i, B_{Y_i}) = a_i.$$

Since U is affine and $\mathbb{Q}$-factorial, $(U, \phi_{i*}(B_i'|_{U_i}))$ is a pair (Definition 2.1). Its coefficients belong to the DCC set $\Xi\backslash\{0\} \subset \mathbb{R}_{>0}$ for any i, and Theorem 2.19(1), applied to the normal threefold U and $\Xi\backslash\{0\}$, implies that $\{a_i\}$ satisfies the ACC. This contradicts the strict increase of a_i. Thus the set of numbers $\operatorname{mld}((X \ni 0, B)/G)$ in the statement satisfies the ACC. □

16.4. **Gauss extensions along a circle.** In this subsection, we express the quotient mld of a boundary defined by $\mathbf{T}_{(p,q,-r_0)}$-semi-invariant equations by Gauss extensions over the invariant surface. This is used in the proofs of Construction-Theorem 16.9 and Proposition 16.11.

Proposition 16.7. *Let p, q, r_0 be positive integers with $\gcd(p,q,r_0) = 1$, $\mathbf{b} = (b_1, b_2, b_3) := (p, q, -r_0)$, and $\mathbf{T} := \mathbf{T}_{\mathbf{b}}$ acting on $X := \mathbb{A}^3_{x_1,x_2,x_3}$. Let G be a finite diagonal group acting faithfully on X, h_j non-zero polynomials which are semi-invariant for G and $\mathbf{T}$, with $\mathbf{T}$-weights κ_j, $c_j \ge 0$, and $B := \sum_j c_j \operatorname{div}(h_j)$ with (X, B) lc near 0. Let $S := \operatorname{Spec} \mathbb{C}[x_1, x_2, x_3]^{\mathbf{T}}$, $\bar{0} \in S$*

the image of 0, B_S *the reduced toric boundary of* S, *and* $s := x^\eta$ *for some* $\eta \in \mathbb{Z}^3$ *with* $\langle \mathbf{b}, \eta\rangle = 1$, *so that* $\mathbb{C}(X) = \mathbb{C}(S)(s)$. *Let* $f_l := x_l s^{-b_l}$ *and* $\phi_j := h_j s^{-\kappa_j}$ *in* $\mathbb{C}(S)$, *and* β *the character of* s. *For a prime divisor* ν *over* S *with center* $\bar{0}$, *let* $\mathfrak{m}_\nu$ *be the maximal ideal of the valuation ring of* ord_ν, $J_\nu := I_G(\nu)$ *for the action of* G *on* S, *which need not be faithful,* α_ν *the character of* J_ν *on* $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$, *and for coprime integers* $m \geq 1$ *and* k *with* $n_l := m \operatorname{ord}_\nu f_l + b_l k > 0$ *for* $l = 1, 2, 3$, *let*

$$e_\nu(m,k) := \left|\operatorname{Ker}\left(\beta^m \alpha_\nu^{-k} : J_\nu \to \mathbb{C}^*\right)\right|$$

and

$$R_\nu(m,k) := ma(\nu, S, B_S) + \sum_l n_l - \sum_j c_j\left(m \operatorname{ord}_\nu \phi_j + \kappa_j k\right).$$

Then:

(1) $\operatorname{mld}((X \ni 0, B)/G) = \min_{\nu,m,k} R_\nu(m,k)/e_\nu(m,k)$, *the minimum is attained, and each term is* $a(F, X, B)/|I_G(F)|$ *for the Gauss extension* F *of* $m \operatorname{ord}_\nu$ *to* $\mathbb{C}(S)(s)$ *with* s*-value* k, *for which* $I_G(F) = \operatorname{Ker}(\beta^m \alpha_\nu^{-k})$ *and* $\operatorname{Stab}_G(F) = \operatorname{Stab}_G(\nu)$.

(2) *Assume that* $\operatorname{mld}((X \ni 0, B)/G) > 0$. *Let* E *be a prime divisor over* X *with center* 0 *such that* $\operatorname{ord}_E|_{\mathbb{C}(X/G)}$ *is a positive multiple of* ord_P *for a prime divisor* P *over* X/G *computing* $\operatorname{mld}((X \ni 0, B)/G)$. *Then there is a Gauss extension* F *as in (1) with* $I_G(F) = I_G(E)$, $\operatorname{ord}_F x_l = \operatorname{ord}_E x_l$ *for* $l = 1, 2, 3$, $\operatorname{ord}_F h_j = \operatorname{ord}_E h_j$ *for every* j, *and* $a(F, X, B) = a(E, X, B)$.

Proof. The proof is long, so we divide it into several steps. For a prime divisor P over a variety, $\mathcal{O}_P$, $\mathfrak{m}_P$, and $\mathbb{C}(P)$ denote the valuation ring, the maximal ideal, and the residue field of ord_P. The order of a rational differential form along P is taken on a smooth variety, projective and birational over the given one, on which P is a prime divisor.

Step 1. In this step we show that $\mathbb{C}(X) = \mathbb{C}(S)(s)$ and $f_l, \phi_j \in \mathbb{C}(S)$, and we relate the canonical forms of X and of (S, B_S).

Let e_1, e_2, e_3 be the standard basis of $\mathbb{Z}^3$, $L := \{a \in \mathbb{Z}^3 \mid \langle \mathbf{b}, a\rangle = 0\}$, and a_1, a_2 a basis of L. Any $a \in \mathbb{Z}^3$ is the sum of $a - \langle \mathbf{b}, a\rangle\eta \in L$ and $\langle \mathbf{b}, a\rangle\eta$, so (a_1, a_2, η) is a basis of $\mathbb{Z}^3$. Let $U := x^{a_1}$ and $V := x^{a_2}$. Since the monomial x^a has $\mathbf{T}$-weight $\langle \mathbf{b}, a\rangle$, the ring $\mathbb{C}[x_1, x_2, x_3]^{\mathbf{T}}$ is spanned by the monomials x^a with $a \in L \cap \mathbb{Z}^3_{\geq 0}$. The vector $w_0 := (r_0, r_0, p+q)$ belongs to $L \cap \mathbb{Z}^3_{>0}$, and $a + Nw_0 \in \mathbb{Z}^3_{\geq 0}$ for any $a \in L$ and any integer $N \gg 0$. Thus $L \cap \mathbb{Z}^3_{\geq 0}$ generates the group L, and

$$\mathbb{C}(S) = \mathbb{C}(U, V) \quad \text{and} \quad \mathbb{C}(X) = \mathbb{C}(U, V, s) = \mathbb{C}(S)(s).$$

An element of $\mathbb{C}(U, V)$ which is integral over $\mathbb{C}[x_1, x_2, x_3]^{\mathbf{T}}$ is a $\mathbf{T}$-invariant element of the normal ring $\mathbb{C}[x_1, x_2, x_3]$. Thus S is the normal affine toric surface of the semigroup $L \cap \mathbb{Z}^3_{\geq 0}$, with torus $\operatorname{Spec} \mathbb{C}[L]$. The ideal of $\bar{0}$ is spanned by the non-constant monomials of $\mathbb{C}[x_1, x_2, x_3]^{\mathbf{T}}$. Since $\langle \mathbf{b}, e_l - b_l\eta\rangle = 0$, and $\langle \mathbf{b}, \alpha - \kappa_j\eta\rangle = 0$ for any monomial x^α occurring in h_j, both $f_l = x^{e_l - b_l\eta}$ and ϕ_j are Laurent polynomials in U and V. Any $\sigma \in G$ commutes with $\mathbf{T}$, so σ^* preserves $\mathbb{C}[x_1, x_2, x_3]^{\mathbf{T}}$ and the ideal of $\bar{0}$, and $\sigma^* s = \beta(\sigma)s$. Thus G acts on S and fixes $\bar{0}$.

Let $\omega_S := \frac{dU}{U} \wedge \frac{dV}{V}$. Then ω_S is regular and nowhere vanishing on $\operatorname{Spec} \mathbb{C}[L]$, and $\operatorname{div}(\omega_S) = -B_S$ [CLS11, Theorem 8.2.3 and §8.2, pp. 369–371]. Thus $K_S + B_S$ is Cartier, and

$$a(\nu, S, B_S) = 1 + \operatorname{ord}_\nu \omega_S$$

for any prime divisor ν over S. Since $\det(a_1, a_2, \eta) = \pm 1$, we have

$$\frac{dU}{U} \wedge \frac{dV}{V} \wedge \frac{ds}{s} = \pm \frac{dx_1}{x_1} \wedge \frac{dx_2}{x_2} \wedge \frac{dx_3}{x_3},$$

that is,

$$dx_1 \wedge dx_2 \wedge dx_3 = \pm x_1 x_2 x_3 \omega_S \wedge \frac{ds}{s}. \tag{16.15}$$

Step 2. In this step we show that any prime divisor E over X with center 0 restricts to a multiple of ord_ν for a prime divisor ν over S with center $\bar{0}$, and we bound $a(E,X,B)$ from below in terms of ν.

Let E be a prime divisor over X with $\mathrm{center}_X E = \{0\}$. The monomial $x_1^{r_0}x_3^p$ belongs to $\mathbb{C}[x_1,x_2,x_3]^{\mathbf{T}}$ and $\mathrm{ord}_E(x_1^{r_0}x_3^p) > 0$, so the restriction of ord_E to $\mathbb{C}(S)$ is non-trivial, non-negative on $\mathbb{C}[x_1,x_2,x_3]^{\mathbf{T}}$, and positive on the ideal of $\bar{0}$. By Lemma 2.12, applied to $\mathbb{C}(S) \subset \mathbb{C}(X)$ and the point $\bar{0} \in S$, $\mathrm{ord}_E|_{\mathbb{C}(S)} = m\,\mathrm{ord}_\nu$ for a positive integer m and a prime divisor ν over S with $\mathrm{center}_S \nu = \bar{0}$.

Now let E be any prime divisor over X such that $\mathrm{ord}_E|_{\mathbb{C}(S)} = m\,\mathrm{ord}_\nu$ for a prime divisor ν over S and an integer $m \geq 1$, and let $k := \mathrm{ord}_E(s)$, which may be negative or zero. Since $x_l = f_l s^{b_l}$ and $h_j = \phi_j s^{\kappa_j}$,

$$\mathrm{ord}_E x_l = m\,\mathrm{ord}_\nu f_l + b_l k \quad \text{and} \quad \mathrm{ord}_E h_j = m\,\mathrm{ord}_\nu \phi_j + \kappa_j k. \tag{16.16}$$

Let $t \in \mathbb{C}(S)$ with $\mathrm{ord}_\nu t = 1$, let $y \in \mathcal{O}_\nu$ be an element whose residue in $\mathbb{C}(\nu)$ is transcendental over $\mathbb{C}$, and let $\theta := s^m t^{-k}$. Then $\mathrm{ord}_E \theta = mk - km = 0$. Let $\bar{\theta}, \bar{y} \in \mathbb{C}(E)$ be the residues of θ and y. We show that

$$a(E,X,0) \geq m a(\nu,S,B_S) + \sum_l \mathrm{ord}_E x_l, \tag{16.17}$$

with equality if $d\bar{\theta} \wedge d\bar{y} \neq 0$ in $\Omega^2_{\mathbb{C}(E)/\mathbb{C}}$.

On a smooth surface, projective and birational over S, on which ν is a prime divisor, t and $y - y(z)$ are local coordinates at a general closed point z of ν, since the residue of y is transcendental over $\mathbb{C}$ and the characteristic is zero. By **Step 1**,

$$\omega_S = u t^{a(\nu,S,B_S)-1} dt \wedge dy$$

for some $u \in \mathcal{O}_\nu^*$. Let w be a uniformizer of ord_E, and write $t = w^m u_1$ and $s = w^k u_2$ with $u_1, u_2 \in \mathcal{O}_E^*$, so that $\theta = u_2^m u_1^{-k}$. A direct calculation shows that

$$dt \wedge \frac{ds}{s} = w^{m-1}\Big(u_1 dw \wedge \frac{d\theta}{\theta} + w du_1 \wedge \frac{du_2}{u_2}\Big).$$

Since $\mathrm{ord}_E u = m\,\mathrm{ord}_\nu u = 0$, $\mathrm{ord}_E t = m$, and $y \in \mathcal{O}_E$, we obtain

$$\mathrm{ord}_E\Big(\omega_S \wedge \frac{ds}{s}\Big) \geq m\big(a(\nu,S,B_S) - 1\big) + m - 1,$$

with equality if $dw \wedge d\theta \wedge dy$ generates the canonical sheaf at the generic point of E. The restriction of $d\theta \wedge dy$ to E is $d\bar{\theta} \wedge d\bar{y}$, so $dw \wedge d\theta \wedge dy$ is such a generator if $d\bar{\theta} \wedge d\bar{y} \neq 0$. Since $a(E,X,0) = 1 + \mathrm{ord}_E(dx_1 \wedge dx_2 \wedge dx_3)$, (16.15) implies (16.17). By (16.17),

$$\begin{aligned} a(E,X,B) &= a(E,X,0) - \sum_j c_j \,\mathrm{ord}_E h_j \\ &\geq m a(\nu,S,B_S) + \sum_l \mathrm{ord}_E x_l - \sum_j c_j \,\mathrm{ord}_E h_j, \end{aligned} \tag{16.18}$$

with equality if $d\bar{\theta} \wedge d\bar{y} \neq 0$.

Step 3. In this step we show that each Gauss extension in (1) is a prime divisor over X with center 0 and log discrepancy $R_\nu(m,k)$, and we compute the residue field.

Let ν be a prime divisor over S with center $\bar{0}$, and $m \geq 1$ and k coprime integers with $n_l > 0$ for $l = 1,2,3$. Let $v := m\,\mathrm{ord}_\nu$, F the Gauss extension of v to $\mathbb{C}(S)(s)$ with s-value k, $t \in \mathbb{C}(S)$ with $\mathrm{ord}_\nu t = 1$, $y \in \mathcal{O}_\nu$ an element whose residue in $\mathbb{C}(\nu)$ is transcendental over $\mathbb{C}$, as in **Step 2**, and $\theta := s^m t^{-k}$. By Definition-Proposition 2.11, $F(\mathbb{C}(X)^*) = \mathbb{Z}$, and $F(\theta) = mk - km = 0$. We show that the residue $\bar{\theta}$ of θ is transcendental over $\mathbb{C}(\nu)$ and that $\mathbb{C}(F) = \mathbb{C}(\nu)(\bar{\theta})$.

Let $Q = \sum_i a_i s^i \in \mathbb{C}(S)[s]$ be non-zero, $\gamma := F(Q)$, and $I_Q := \{i \mid v(a_i) + ik = \gamma\}$. For $i, i' \in I_Q$, $(i-i')k = v(a_{i'}) - v(a_i) \in m\mathbb{Z}$, so m divides $i - i'$. Fix $i_0 \in I_Q$, and for $i = i_0 + mj \in I_Q$ let $\epsilon_i := a_i a_{i_0}^{-1} t^{kj}$. Then $v(\epsilon_i) = -mjk + mkj = 0$ and $a_i s^i = a_{i_0} s^{i_0} \epsilon_i \theta^j$, so

$$Q = a_{i_0} s^{i_0} \Big(\sum_{i_0 + mj \in I_Q} \epsilon_{i_0+mj} \theta^j \Big) + Q' \quad \text{with} \quad F(Q') > \gamma. \tag{16.19}$$

For $\delta_j \in \mathcal{O}_v$, the definition of F implies $F(\sum_j \delta_j \theta^j) = F(\sum_j \delta_j t^{-kj} s^{mj}) = \min_j v(\delta_j)$. Thus $\sum_j \bar{\delta}_j \bar{\theta}^j \neq 0$ whenever some residue $\bar{\delta}_j$ is non-zero, that is, $\bar{\theta}$ is transcendental over $\mathbb{C}(\nu)$. Any element of $\mathcal{O}_F^*$ is Q_1/Q_2 with $Q_1, Q_2 \in \mathbb{C}(S)[s]$ and $F(Q_1) = F(Q_2)$. Writing Q_1 and Q_2 in the form (16.19) with indices i_1 and i_2 in place of i_0, we have $i_1 \equiv i_2 \bmod m$ and $a_{i_1} s^{i_1} / (a'_{i_2} s^{i_2}) = \delta \theta^{(i_1 - i_2)/m}$ with $\delta \in \mathcal{O}_v^*$, where a_{i_1} and a'_{i_2} are the corresponding coefficients of Q_1 and Q_2. Hence the residue of Q_1/Q_2 belongs to $\mathbb{C}(\nu)(\bar{\theta})$, and $\mathbb{C}(F) = \mathbb{C}(\nu)(\bar{\theta})$.

Thus F is a discrete valuation of $\mathbb{C}(X)$ with value group $\mathbb{Z}$, trivial on $\mathbb{C}$, whose residue field has transcendence degree two over $\mathbb{C}$. Moreover, $F(x_l) = v(f_l) + b_l k = n_l > 0$, so F is non-negative on $\mathbb{C}[x_1, x_2, x_3]$ and positive on x_1, x_2, x_3. By [KM98, Lemma 2.45], F is the valuation of a prime divisor over X, which we also denote by F, and $\operatorname{center}_X F = \{0\}$. Since $F|_{\mathbb{C}(S)} = m \operatorname{ord}_\nu$ and $\operatorname{ord}_F(s) = k$, (16.16) and (16.18) hold for $E = F$. The residue $\bar{y}$ of y is transcendental over $\mathbb{C}$ and belongs to $\mathbb{C}(\nu)$, so $\bar{\theta}$ and $\bar{y}$ are algebraically independent over $\mathbb{C}$, and $d\bar{\theta} \wedge d\bar{y} \neq 0$ in characteristic zero. Hence equality holds in (16.18), and by (16.16),

$$\operatorname{ord}_F x_l = n_l, \quad \operatorname{ord}_F h_j = m \operatorname{ord}_\nu \phi_j + \kappa_j k, \quad \text{and} \quad a(F, X, B) = R_\nu(m, k). \tag{16.20}$$

Since (X, B) is lc near 0, $R_\nu(m, k) \geq 0$.

Step 4. In this step we compute $\operatorname{Stab}_G(F)$ and $I_G(F)$ for the Gauss extension F of **Step 3**.

Let $\sigma \in G$. Then σ^* preserves $\mathbb{C}(S)$ and $\sigma^* s = \beta(\sigma) s$, so for $a_i \in \mathbb{C}(S)$,

$$F\Big(\sigma^* \sum_i a_i s^i\Big) = F\Big(\sum_i \sigma^*(a_i) \beta(\sigma)^i s^i\Big) = \min_i \big(v(\sigma^* a_i) + ik\big).$$

Thus $F \circ \sigma^*$ is the Gauss extension of $v \circ \sigma^*$ with s-value k. Since a Gauss extension restricts to the given valuation on $\mathbb{C}(S)$, $F \circ \sigma^* = F$ if and only if $v \circ \sigma^* = v$. Hence $\operatorname{Stab}_G(F) = \operatorname{Stab}_G(\nu)$.

The kernel of the action of G on S acts trivially on $\mathbb{C}(S)$, so it fixes ord_ν and acts trivially on $\mathbb{C}(\nu)$. Thus the kernel is contained in J_ν. For $\sigma \in J_\nu$, σ^* acts $\mathbb{C}(\nu)$-linearly on the one-dimensional $\mathbb{C}(\nu)$-vector space $\mathfrak{m}_\nu / \mathfrak{m}_\nu^2$, so $\alpha_\nu(\sigma)$ is the residue of $\sigma^* t / t$. It is a root of unity in $\mathbb{C}(\nu)$, hence an element of $\mathbb{C}^*$.

Since $\mathbb{C}(\nu) \subset \mathbb{C}(F)$, any element of $I_G(F)$ belongs to $\operatorname{Stab}_G(\nu)$ and acts trivially on $\mathbb{C}(\nu)$, that is, $I_G(F) \subset J_\nu$. For $\sigma \in J_\nu$,

$$\sigma^* \theta = \beta(\sigma)^m s^m (\sigma^* t)^{-k} = \beta(\sigma)^m \Big(\frac{\sigma^* t}{t}\Big)^{-k} \theta, \quad \text{so} \quad \sigma^* \bar{\theta} = \beta(\sigma)^m \alpha_\nu(\sigma)^{-k} \bar{\theta}.$$

Since $\mathbb{C}(F) = \mathbb{C}(\nu)(\bar{\theta})$ with $\bar{\theta} \neq 0$ (**Step 3**) and σ acts trivially on $\mathbb{C}(\nu)$, σ acts trivially on $\mathbb{C}(F)$ if and only if $\beta(\sigma)^m \alpha_\nu(\sigma)^{-k} = 1$. Thus

$$I_G(F) = \operatorname{Ker}\big(\beta^m \alpha_\nu^{-k} : J_\nu \to \mathbb{C}^*\big) \quad \text{and} \quad |I_G(F)| = e_\nu(m, k). \tag{16.21}$$

Claim 16.8. *Let E be a prime divisor over X with $\operatorname{center}_X E = \{0\}$, let $m \operatorname{ord}_\nu$ be the restriction of ord_E to $\mathbb{C}(S)$ (**Step 2**), $k := \operatorname{ord}_E(s)$, $g := \gcd(m, k)$, where $\gcd(m, 0) = m$, and E_0 the Gauss extension of $\frac{m}{g} \operatorname{ord}_\nu$ to $\mathbb{C}(S)(s)$ with s-value $\frac{k}{g}$. Then $\frac{m}{g} \operatorname{ord}_\nu f_l + b_l \frac{k}{g} > 0$ for $l = 1, 2, 3$, $\operatorname{ord}_E x_l = g \operatorname{ord}_{E_0} x_l$, $\operatorname{ord}_E h_j = g \operatorname{ord}_{E_0} h_j$, $I_G(E) \subset I_G(E_0)$, and*

$$\frac{a(E_0, X, B)}{|I_G(E_0)|} \leq \frac{a(E_0, X, B)}{|I_G(E)|} \leq \frac{g a(E_0, X, B)}{|I_G(E)|} \leq \frac{a(E, X, B)}{|I_G(E)|}.$$

Proof. By (16.16), $\frac{m}{g}\operatorname{ord}_\nu f_l + b_l\frac{k}{g} = \frac{1}{g}\operatorname{ord}_E x_l > 0$. Thus $(\nu, \frac{m}{g}, \frac{k}{g})$ is a triple in the minimum of (1), and (16.16) and (16.20) imply $\operatorname{ord}_E x_l = g\operatorname{ord}_{E_0} x_l$ and $\operatorname{ord}_E h_j = g\operatorname{ord}_{E_0} h_j$. By (16.18) and (16.20),

$$a(E,X,B) \geq ma(\nu,S,B_S) + \sum_l \operatorname{ord}_E x_l - \sum_j c_j \operatorname{ord}_E h_j = ga(E_0,X,B).$$

Let $\sigma \in I_G(E)$. Then σ fixes ord_E, hence fixes $\operatorname{ord}_E|_{\mathbb{C}(S)} = m\operatorname{ord}_\nu$. Since $\mathcal{O}_\nu = \mathcal{O}_E \cap \mathbb{C}(S)$, we have $\mathbb{C}(\nu) \subset \mathbb{C}(E)$, so σ acts trivially on $\mathbb{C}(\nu)$ and $\sigma \in J_\nu$. Let w be a uniformizer of ord_E, $t \in \mathbb{C}(S)$ with $\operatorname{ord}_\nu t = 1$, and write $t = w^m u_1$ and $s = w^k u_2$ with $u_1, u_2 \in \mathcal{O}_E^*$. Since σ acts trivially on $\mathbb{C}(E)$, the residues of $\sigma^* u_1/u_1$ and $\sigma^* u_2/u_2$ are 1. Let $\lambda \in \mathbb{C}(E)^*$ be the residue of $\sigma^* w/w$. Then

$$\alpha_\nu(\sigma) = \lambda^m \quad \text{and} \quad \beta(\sigma) = \lambda^k, \quad \text{so} \quad \beta(\sigma)^{m/g}\alpha_\nu(\sigma)^{-k/g} = \lambda^{mk/g - km/g} = 1.$$

By (16.21) applied to E_0, $\sigma \in I_G(E_0)$. Hence $I_G(E) \subset I_G(E_0)$.

Since $\operatorname{center}_X E_0 = \{0\}$ and (X,B) is lc near 0, $a(E_0,X,B) \geq 0$. Since $|I_G(E)| \leq |I_G(E_0)|$ and $g \geq 1$, the claimed inequalities follow. □

Proof of Proposition 16.7 continued.

Step 5. In this step we prove (1) and (2).

Since the h_j are G-semi-invariant, B is G-invariant, so the hypotheses of Lemma 2.6 hold for (X,B). Let (ν,m,k) be a triple in the minimum of (1) and F the corresponding Gauss extension. By **Steps 3 and 4**, $\operatorname{center}_X F = \{0\}$ and $a(F,X,B)/|I_G(F)| = R_\nu(m,k)/e_\nu(m,k)$, so $R_\nu(m,k)/e_\nu(m,k) \geq \operatorname{mld}((X \ni 0, B)/G)$ by Lemma 2.6.

Let X° be a G-invariant open neighborhood of 0 on which (X,B) is lc. By Lemma 2.6(2) and (3), applied to $(X^\circ, B|_{X^\circ})$, any prime divisor over X°/G has non-negative log discrepancy with respect to $(X,B)/G$. Thus $(X,B)/G$ is lc near the image of 0 in X/G, and $\operatorname{mld}((X \ni 0, B)/G) = a(P, X/G, B_{X/G})$ for a prime divisor P over X/G with center the image of 0 [Amb99, Section 1, the second paragraph after Definition 1.1]. By Lemma 2.6(3), there exists a prime divisor E over X such that $\operatorname{ord}_E|_{\mathbb{C}(X/G)}$ is a positive multiple of ord_P. Let $\pi : X \to X/G$ be the quotient morphism. By Lemma 2.6(2), $\pi(\operatorname{center}_X E)$ is the image of 0, and $\pi^{-1}(\pi(0)) = \{0\}$ since G fixes 0. Thus $\operatorname{center}_X E = \{0\}$ and, by Lemma 2.6(1) and (2),

$$\frac{a(E,X,B)}{|I_G(E)|} = \operatorname{mld}((X \ni 0, B)/G).$$

Let $m\operatorname{ord}_\nu := \operatorname{ord}_E|_{\mathbb{C}(S)}$, $k := \operatorname{ord}_E s$, $g := \gcd(m,k)$, and let E_0 be as in the Claim. The Claim implies that E_0 satisfies

$$\frac{a(E_0,X,B)}{|I_G(E_0)|} \leq \operatorname{mld}((X \ni 0, B)/G).$$

This proves (1).

For (2), let E be a prime divisor over X with center 0 such that $\operatorname{ord}_E|_{\mathbb{C}(X/G)}$ is a positive multiple of ord_P for a prime divisor P over X/G with $a(P, X/G, B_{X/G}) = \operatorname{mld}((X \ni 0, B)/G) > 0$. By Lemma 2.6(1) and (2), $a(E,X,B)/|I_G(E)| = \operatorname{mld}((X \ni 0, B)/G)$. Let g and E_0 be attached to E by the Claim. By (1) and the Claim,

$$\begin{aligned}\operatorname{mld}((X \ni 0, B)/G) &\leq \frac{a(E_0,X,B)}{|I_G(E_0)|} \leq \frac{a(E_0,X,B)}{|I_G(E)|} \\ &\leq \frac{ga(E_0,X,B)}{|I_G(E)|} \leq \frac{a(E,X,B)}{|I_G(E)|} = \operatorname{mld}((X \ni 0, B)/G).\end{aligned}$$

Thus equality holds throughout, and $a(E_0,X,B) > 0$. Hence $g = 1$ and $|I_G(E)| = |I_G(E_0)|$, so $I_G(E) = I_G(E_0)$ and $a(E,X,B) = a(E_0,X,B)$. By the Claim with $g = 1$, $\operatorname{ord}_E x_l = \operatorname{ord}_{E_0} x_l$ and $\operatorname{ord}_E h_j = \operatorname{ord}_{E_0} h_j$. Hence $F := E_0$ satisfies (2). □

16.5. **Circle-homogeneous equations.** In this subsection, we bound from both sides the quotient minimal log discrepancy of a boundary whose equations are homogeneous for a mixed circle (Definition 4.1) with three non-zero weights.

Construction-Theorem 16.9 passes to a cyclic cover on which the equations become polynomials in two variables, and bounds the quotient minimal log discrepancy from below by a minimal log discrepancy of a surface pair with coefficients in a fixed DCC set, and from above by the sum of the lower bound and an error which tends to 0 as the order of the intersection of the group with the circle tends to infinity. If the number τ of Construction-Theorem 16.9 is negative, the surface pair is the quotient by a finite group of a sub-pair on a weighted blow-up of a neighborhood of the origin of the plane, and Example 16.10 shows that the corresponding quotient of the pair on the plane does not then give a lower bound.

Construction-Theorem 16.9. Let p, q, r be positive integers with $\gcd(p,q,r) = 1$, $\mathbf{T} := \mathbf{T}_{(p,q,-r)}$ acting on $X := \mathbb{A}^3_{x,y,z}$, G a finite diagonal group acting faithfully on X, $h_1, \dots, h_s$ non-zero polynomials semi-invariant for G and $\mathbf{T}$ with $\mathbf{T}$-weights κ_j, $c_j \geq 0$, and $B := \sum_j c_j \operatorname{div}(h_j)$ with (X, B) lc near 0. Let $a := \operatorname{mld}((X \ni 0, B)/G)$, $n := |G \cap \mathbf{T}|$, $H := G/(G \cap \mathbf{T})$, and $\tau := r - p - q + \sum_j c_j \kappa_j$. Consider the cover $z = t^r$ with $\mathbb{C}(X) \subset \mathbb{C}(x,y,t)$, let $u := xt^p$ and $v := yt^q$, write $h_j(x,y,t^r) = t^{-\kappa_j} P_j(u,v)$ with $P_j \in \mathbb{C}[u,v]$, and let $C := \sum_j c_j \operatorname{div}(P_j)$ on $\mathbb{A}^2_{u,v}$. Let $\tilde{G} := \{(g,\alpha) \mid g = (g_x, g_y, g_z) \in G,\ \alpha^r = g_z\}$ act on (x,y,t) by (g_x, g_y, α), and let $\tilde{H}$ be the image of $\tilde{G}$ in $\operatorname{Aut}(\mathbb{A}^2_{u,v})$. Then $\tilde{H}$ has order $r|H|$, the image of $\{(1,\alpha) \mid \alpha^r = 1\}$ in $\tilde{H}$ is $\mu_r(p,q)$, $\tilde{H}/\mu_r(p,q) \cong H$, $\mathbb{C}(x,y,t)^{\tilde{G}} = \mathbb{C}(X)^G$, and $(\mathbb{A}^2_{u,v}, C)$ is lc near 0. Let V be an $\tilde{H}$-invariant affine open neighborhood of 0 on which $(V, C|_V)$ is lc.

(1) If $\tau \geq 0$, let $(Z, B_Z) := (V, C|_V)/\tilde{H}$ and $b := \operatorname{mld}(Z \ni \bar{0}, B_Z)$ for the image $\bar{0}$ of 0.
(2) If $\tau < 0$, let
   (i) $g_0 := \gcd(p,q)$,
   (ii) $f_V : V' \to V$ the weighted blow-up with weights $(p/g_0, q/g_0)$ and exceptional curve E,
   (iii) $\beta_z := \sum_j c_j \operatorname{mult}_{(z=0)} \operatorname{div}(h_j) \in [0,1]$, and
   (iv) $B_{V'} := (f_V)^{-1}_*(C|_V) + \big(1 - \frac{r(1-\beta_z)}{g_0}\big)E$.

   Then $(V', B_{V'})$ is sub-lc over an $\tilde{H}$-invariant affine open neighborhood of 0 in V, and we replace V by such a neighborhood. Let $(Z, B_Z) := (V', B_{V'})/\tilde{H}$ and let b be the minimum of $a(F, Z, B_Z)$ over the prime divisors F over Z whose center on Z is contained in the image $\bar{E}$ of E.

Then $B_Z \geq 0$, (Z, B_Z) is lc, $b \geq 0$ is attained, and $b \leq a$. If moreover $n > r|H| \max\{p,q\}$, then

$$a \leq b + \frac{|\tau|}{n}.$$

If $\tau < 0$, then r divides the order e_E of the inertia group of E in $\tilde{H}$, $e_E = re_{\mathrm{res}}$ with $1 \leq e_{\mathrm{res}} \leq |H|$, and $\operatorname{mult}_{\bar{E}} B_Z = 1 - \frac{1-\beta_z}{g_0 e_{\mathrm{res}}}$. In particular, for fixed p, q, r, a fixed integer $e_0 \geq 1$ and a DCC set $\Gamma \subset [0,1]$, if $c_j \in \Gamma$ and $|H| \leq e_0$, then the coefficients of B_Z belong to a DCC set depending only on Γ, p, q, r, e_0, the numbers b satisfy the ACC, and if $(a_i)_i$ is a strictly increasing sequence of such numbers a, with corresponding numbers n_i, then $(n_i)_i$ is bounded. Moreover, let $\bar{0} \in V/\tilde{H}$ be the image of 0, $\tau^- := \max\{-\tau, 0\}$, and $\mathfrak{b}$ the ideal sheaf on $V/\tilde{H}$ generated by $u^{qr|H|}$ and $v^{pr|H|}$, which are $\tilde{H}$-invariant since $\tilde{H}$ is a diagonal group of order $r|H|$. Here $\mathfrak{b}^0$ means $\mathcal{O}_{V/\tilde{H}}$. Then $\big(V/\tilde{H}, (C|_V)_{V/\tilde{H}}, \mathfrak{b}^{\tau^-/(pqr|H|)}\big)$ is lc near $\bar{0}$ and

$$b = \operatorname{mld}\big(V/\tilde{H} \ni \bar{0}, (C|_V)_{V/\tilde{H}}, \mathfrak{b}^{\tau^-/(pqr|H|)}\big).$$

Proof. The proof is long, so we divide it into several steps. Let $\tilde{X} := \mathbb{A}^3_{x,y,t}$ and let $\pi : \tilde{X} \to X$ be the morphism $(x,y,t) \to (x,y,t^r)$.

Step 1. In this step we prove the properties of $\tilde{H}$ and of the invariant fields, and we express a by prime divisors over $\tilde{X}$.

Let $x^{i_1}y^{i_2}z^{i_3}$ be a monomial occurring in h_j. Then $pi_1+qi_2-ri_3=\kappa_j$, so $x^{i_1}y^{i_2}t^{ri_3}=u^{i_1}v^{i_2}t^{-\kappa_j}$. Thus $h_j(x,y,t^r)=t^{-\kappa_j}P_j(u,v)$ for a polynomial P_j, and $P_j\neq 0$ since i_3 is determined by (i_1,i_2). Moreover, if a monomial $u^{i_1}v^{i_2}$ occurs in P_j, then the monomial $x^{i_1}y^{i_2}z^{i_3}$ occurs in h_j for an integer $i_3\geq 0$ with

$$pi_1+qi_2=\kappa_j+ri_3. \tag{16.22}$$

An element $(g,\alpha)\in\tilde{G}$ multiplies u by $g_x\alpha^p$ and v by $g_y\alpha^q$, so $\tilde{H}$ is a finite diagonal group. The group $\tilde{G}$ acts faithfully on $\tilde{X}$, since $g_x=g_y=\alpha=1$ implies $g_z=\alpha^r=1$. Since $h_j(x,y,t^r)$ and t are $\tilde{G}$-semi-invariant, so is P_j. Hence C is $\tilde{H}$-invariant.

Let $\tilde{G}_0$ be the kernel of the action of $\tilde{G}$ on $\mathbb{A}^2_{u,v}$, and let β_t be the character of t, so that $\beta_t((g,\alpha))=\alpha$. An element $(g,\alpha)\in\tilde{G}$ belongs to $\tilde{G}_0$ if and only if $g_x=\alpha^{-p}$ and $g_y=\alpha^{-q}$. Since $g_z=\alpha^r$, this holds if and only if $g=(\alpha^{-p},\alpha^{-q},\alpha^r)$. Thus $(g,\alpha)\to g$ maps $\tilde{G}_0$ onto $G\cap\mathbf{T}$: for $g=(\gamma^p,\gamma^q,\gamma^{-r})\in G\cap\mathbf{T}$ with $\gamma\in\mathbb{C}^*$, we have $(g,\gamma^{-1})\in\tilde{G}_0$. If $(g,\alpha)\in\tilde{G}_0$ and $g=1$ or $\alpha=1$, then $\alpha^p=\alpha^q=\alpha^r=1$ and $g=(\alpha^{-p},\alpha^{-q},\alpha^r)$, so $\alpha=1$ and $g=1$ by $\gcd(p,q,r)=1$. Hence $\tilde{G}_0\cong G\cap\mathbf{T}$, $|\tilde{G}_0|=n$, β_t is injective on $\tilde{G}_0$, and

$$|\tilde{H}|=\frac{|\tilde{G}|}{n}=\frac{r|G|}{n}=r|H|.$$

Let $\zeta:=e^{2\pi\sqrt{-1}/r}$. The image of $\{(1,\alpha)\mid\alpha^r=1\}$ in $\tilde{H}$ is generated by (ζ^p,ζ^q), so it is $\mu_r(p,q)$. Since $\gcd(r,\gcd(p,q))=1$, we have $(\zeta^{pl},\zeta^{ql})=(1,1)$ if and only if $r\mid l$, so $|\mu_r(p,q)|=r$. The homomorphism $\tilde{G}\to G$, $(g,\alpha)\to g$, is surjective with kernel $\{(1,\alpha)\mid\alpha^r=1\}$. Thus the kernel of the induced surjection $\tilde{G}\to H$ is the preimage of $G\cap\mathbf{T}$, which is $\tilde{G}_0\cdot\{(1,\alpha)\mid\alpha^r=1\}$ by the previous paragraph. Hence $\tilde{G}\to H$ induces an isomorphism

$$\tilde{H}/\mu_r(p,q)=\tilde{G}/\big(\tilde{G}_0\cdot\{(1,\alpha)\mid\alpha^r=1\}\big)\cong H.$$

Since $\mathbb{C}[x,y,t]^{\{(1,\alpha)|\alpha^r=1\}}=\mathbb{C}[x,y,t^r]$, the morphism π is the quotient morphism of $\tilde{X}$ by $\{(1,\alpha)\mid\alpha^r=1\}$, and $\mathbb{C}[x,y,t]^{\tilde{G}}=\mathbb{C}[x,y,z]^G$. Thus $\tilde{X}/\tilde{G}=X/G$ and $\mathbb{C}(x,y,t)^{\tilde{G}}=\mathbb{C}(X)^G$.

Let

$$\tilde{D}:=\sum_j c_j\operatorname{div}(h_j(x,y,t^r))-(r-1)(t=0),$$

a $\tilde{G}$-invariant $\mathbb{R}$-divisor on $\tilde{X}$. By the Hurwitz formula, $K_{\tilde{X}}=\pi^*K_X+(r-1)(t=0)$, so $K_{\tilde{X}}+\tilde{D}=\pi^*(K_X+B)$. Let $\rho:\tilde{X}\to X/G$ be the quotient morphism by $\tilde{G}$, and write $(\tilde{X},\tilde{D})/\tilde{G}=(X/G,\tilde{D}_{X/G})$. By Lemma 2.6(0), applied to (X,B) and G and to $(\tilde{X},\tilde{D})$ and $\tilde{G}$,

$$\rho^*(K_{X/G}+B_{X/G})=K_{\tilde{X}}+\tilde{D}=\rho^*(K_{X/G}+\tilde{D}_{X/G}).$$

Since ρ is finite, $\tilde{D}_{X/G}=B_{X/G}$, that is, $(\tilde{X},\tilde{D})/\tilde{G}=(X,B)/G$. By Lemma 2.6,

$$a=\inf_F\frac{a(F,\tilde{X},\tilde{D})}{|I_{\tilde{G}}(F)|}, \tag{16.23}$$

where F runs over the prime divisors over $\tilde{X}$ with $\operatorname{center}_{\tilde{X}}F=\{0\}$. Similarly, $K_{\tilde{X}}+\tilde{D}=\pi^*(K_X+B)$ and Lemma 2.6(0) imply $(\tilde{X},\tilde{D})/\{(1,\alpha)\mid\alpha^r=1\}=(X,B)$. Let W_0 be an open neighborhood of $0\in X$ on which (X,B) is lc. By Lemma 2.6(2), applied to $\pi^{-1}(W_0)$, we have $a(F,\tilde{X},\tilde{D})\geq 0$ for any prime divisor F over $\pi^{-1}(W_0)$.

Step 2. In this step we show that $(\mathbb{A}^2_{u,v},C)$ is lc near 0 and that V exists.

The morphism $\tilde{X}\backslash(t=0)\to\mathbb{A}^2_{u,v}\times\mathbb{G}_m$, $(x,y,t)\to(xt^p,yt^q,t)$, is an isomorphism with inverse $(u,v,t)\to(ut^{-p},vt^{-q},t)$. Since t is a unit on $\tilde{X}\backslash(t=0)$, the restriction of $\tilde{D}$ to $\tilde{X}\backslash(t=0)$ is pr^*C, where $\operatorname{pr}:\mathbb{A}^2_{u,v}\times\mathbb{G}_m\to\mathbb{A}^2_{u,v}$ is the projection. Let F_0 be a prime divisor over $\mathbb{A}^2_{u,v}$ with

$0 \in \operatorname{center}_{\mathbb{A}^2_{u,v}} F_0$, realized as a prime divisor on a normal surface Y_0 with a proper birational morphism $Y_0 \to \mathbb{A}^2_{u,v}$. Then $F_0 \times \mathbb{G}_m$ is a prime divisor on $Y_0 \times \mathbb{G}_m$, and

$$a(F_0 \times \mathbb{G}_m, \mathbb{A}^2_{u,v} \times \mathbb{G}_m, \operatorname{pr}^* C) = a(F_0, \mathbb{A}^2_{u,v}, C),$$

since $K_{Y_0 \times \mathbb{G}_m}$ and the pullback of $K_{\mathbb{A}^2_{u,v} \times \mathbb{G}_m} + \operatorname{pr}^* C$ are pullbacks of the corresponding divisors on Y_0 and $\mathbb{A}^2_{u,v}$. The center of $F_0 \times \mathbb{G}_m$ on $\tilde{X}$ contains the closure of $\{0\} \times \mathbb{G}_m$, which is the t-axis $\{x = y = 0\}$ and contains 0. Hence $F_0 \times \mathbb{G}_m$ is a prime divisor over $\pi^{-1}(W_0)$, and **Step 1** implies $a(F_0, \mathbb{A}^2_{u,v}, C) \geq 0$. Since $C \geq 0$, it follows that $(\mathbb{A}^2_{u,v}, C)$ is lc near 0: on a log resolution of $(\mathbb{A}^2_{u,v}, C)$, the images of the prime divisors with negative log discrepancy form a closed subset not containing 0, and the pair is lc on the complement [KM98, Corollary 2.31(3)].

Let V_1 be an affine open neighborhood of 0 such that $(V_1, C|_{V_1})$ is lc. Then $\bigcap_{\sigma \in \tilde{H}} \sigma(V_1)$ is an $\tilde{H}$-invariant affine open neighborhood of 0 on which C restricts to an lc pair ($\mathbb{A}^2_{u,v}$ is separated, so the intersection is affine, and each σ fixes 0 and preserves C). In particular, the coefficients of $C|_V$ belong to $[0, 1]$.

Step 3. In this step we compare $a(F, \tilde{X}, \tilde{D})$ with log discrepancies over $\mathbb{A}^2_{u,v}$.

Let F be a prime divisor over $\tilde{X}$ with $\operatorname{center}_{\tilde{X}} F = \{0\}$, and let $k := \operatorname{ord}_F t$. Then $k > 0$, $\operatorname{ord}_F u = \operatorname{ord}_F x + pk > 0$, $\operatorname{ord}_F v = \operatorname{ord}_F y + qk > 0$, and $\operatorname{ord}_F \geq 0$ on $\mathbb{C}[u, v]$. Thus the restriction of ord_F to $\mathbb{C}(u, v)$ is a non-trivial valuation with center 0 on $\mathbb{A}^2_{u,v}$.

By Lemma 2.12, applied to $\mathbb{C}(u, v) \subset \mathbb{C}(x, y, t)$ and the point $0 \in \mathbb{A}^2_{u,v}$, the restriction is $m \operatorname{ord}_\nu$ for a positive integer m and a prime divisor ν over $\mathbb{A}^2_{u,v}$ with $\operatorname{center}_{\mathbb{A}^2_{u,v}} \nu = \{0\}$. Since $x = ut^{-p}$ and $y = vt^{-q}$,

$$\operatorname{ord}_F x = m \operatorname{ord}_\nu u - pk > 0, \qquad \operatorname{ord}_F y = m \operatorname{ord}_\nu v - qk > 0, \qquad k > 0. \tag{16.24}$$

We show that

$$a(F, \tilde{X}, \tilde{D}) \geq m a(\nu, \mathbb{A}^2_{u,v}, C) + \tau k. \tag{16.25}$$

Let $Y \to \mathbb{A}^2_{u,v}$ be a proper birational morphism from a smooth surface such that ν is a prime divisor P on Y, let ξ be the generic point of P, and let $\varpi_\nu \in \mathcal{O}_{Y,\xi}$ be a local equation of P. Then $du \wedge dv = \epsilon \varpi_\nu^{a(\nu, \mathbb{A}^2_{u,v}, 0) - 1} \omega$ for a generator ω of Ω^2_Y at ξ and a unit $\epsilon \in \mathcal{O}_{Y,\xi}$. The valuation ord_F is non-negative on $\mathcal{O}_{Y,\xi}[t]$ and positive on ϖ_ν and t, so ord_F has center the generic point of the curve $P \times \{0\} \subset Y \times \mathbb{A}^1_t$, at which ϖ_ν and t are regular parameters. By [KM98, Corollary 2.31(3)], applied to the snc pair $(Y \times \mathbb{A}^1_t, P \times \mathbb{A}^1_t + Y \times \{0\})$ near the generic point of $P \times \{0\}$,

$$a(F, Y \times \mathbb{A}^1_t, 0) \geq \operatorname{ord}_F \varpi_\nu + \operatorname{ord}_F t = m + k.$$

For a smooth variety W with $\mathbb{C}(W) = \mathbb{C}(x, y, t)$ on which ord_F has a center, $a(F, W, 0) = 1 + \operatorname{ord}_F \omega_W$ for any generator ω_W of Ω^3_W at the generic point of $\operatorname{center}_W F$. Since

$$dx \wedge dy \wedge dt = t^{-p-q} du \wedge dv \wedge dt = t^{-p-q} \epsilon \varpi_\nu^{a(\nu, \mathbb{A}^2_{u,v}, 0) - 1} \omega \wedge dt$$

and $\operatorname{ord}_F \epsilon = 0$, we obtain

$$a(F, \tilde{X}, 0) = a(F, Y \times \mathbb{A}^1_t, 0) + m\big(a(\nu, \mathbb{A}^2_{u,v}, 0) - 1\big) - (p + q)k \geq m a(\nu, \mathbb{A}^2_{u,v}, 0) + (1 - p - q)k.$$

Since $\operatorname{ord}_F h_j(x, y, t^r) = m \operatorname{ord}_\nu P_j - \kappa_j k$ and $\operatorname{ord}_F t = k$,

$$a(F, \tilde{X}, \tilde{D}) = a(F, \tilde{X}, 0) - \sum_j c_j (m \operatorname{ord}_\nu P_j - \kappa_j k) + (r - 1)k.$$

Thus (16.25) follows from $a(\nu, \mathbb{A}^2_{u,v}, C) = a(\nu, \mathbb{A}^2_{u,v}, 0) - \sum_j c_j \operatorname{ord}_\nu P_j$ and $\tau = r - p - q + \sum_j c_j \kappa_j$.

Conversely, let ν be a prime divisor over $\mathbb{A}^2_{u,v}$ with $\operatorname{center}_{\mathbb{A}^2_{u,v}} \nu = \{0\}$, and let $m \geq 1$ and k be coprime integers satisfying (16.24). Let $F_\nu(m, k)$ be the prime divisor over $\tilde{X}$ whose valuation is the Gauss extension of $m \operatorname{ord}_\nu$ to $\mathbb{C}(u, v)(t)$ with t-value k (Definition-Proposition 2.11). It is also the Gauss extension of $m \operatorname{ord}_\nu$ to $\mathbb{C}(u, v)(t^{-1})$ with t^{-1}-value $-k$. By (16.24), $\operatorname{center}_{\tilde{X}} F_\nu(m, k) =$

$\{0\}$. Let $J_\nu := I_{\tilde G}(\nu)$ for the action of $\tilde G$ on $\mathbb{A}^2_{u,v}$, and let α_ν be the character of J_ν on $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$. We use Proposition 16.7(1) for $(\tilde X, 0)$, the group $\tilde G$, the weights $(p,q,-1)$, and $\eta = (0,0,-1)$. In this case, the invariant surface is $\mathbb{A}^2_{u,v}$ with reduced toric boundary $\operatorname{div}(u)+\operatorname{div}(v)$, the functions f_1, f_2, f_3 are u, v, 1, the Gauss variable is t^{-1}, and the positivity conditions are (16.24). Since

$$a\big(\nu, \mathbb{A}^2_{u,v}, \operatorname{div}(u)+\operatorname{div}(v)\big) = a(\nu, \mathbb{A}^2_{u,v}, 0) - \operatorname{ord}_\nu u - \operatorname{ord}_\nu v,$$

Proposition 16.7(1) implies $a(F_\nu(m,k), \tilde X, 0) = ma(\nu, \mathbb{A}^2_{u,v}, 0) + (1-p-q)k$ and $I_{\tilde G}(F_\nu(m,k)) = \operatorname{Ker}(\beta_t^{-m}\alpha_\nu^k)$, because the character of t^{-1} is β_t^{-1}. We have $\operatorname{Ker}(\beta_t^{-m}\alpha_\nu^k) = \operatorname{Ker}(\beta_t^m\alpha_\nu^{-k})$, since $\beta_t^{-m}\alpha_\nu^k$ is the inverse of $\beta_t^m\alpha_\nu^{-k}$. Since $\operatorname{ord}_{F_\nu(m,k)} h_j(x,y,t^r) = m\operatorname{ord}_\nu P_j - \kappa_j k$ and $\operatorname{ord}_{F_\nu(m,k)} t = k$, the computation of $a(F, \tilde X, \tilde D)$ from $a(F, \tilde X, 0)$ in the proof of (16.25) holds with $F_\nu(m,k)$ in place of F, and we obtain

$$\begin{aligned} a(F_\nu(m,k), \tilde X, \tilde D) &= ma(\nu, \mathbb{A}^2_{u,v}, C) + \tau k, \\ I_{\tilde G}(F_\nu(m,k)) &= \operatorname{Ker}(\beta_t^m\alpha_\nu^{-k} : J_\nu \to \mathbb{C}^*). \end{aligned} \tag{16.26}$$

Step 4. In this step we assume $\tau < 0$, prove a formula for log discrepancies over V', and shrink V as in (2).

Let $w_E := (\frac{p}{g_0}, \frac{q}{g_0})$. The variety V' is an open subset of the toric surface whose fan has rays spanned by $(1,0)$, w_E, and $(0,1)$. Two-dimensional fans are simplicial, so V' is $\mathbb{Q}$-factorial [CLS11, Proposition 4.2.7] and $K_{V'} + B_{V'}$ is $\mathbb{R}$-Cartier. The diagonal group $\tilde H$ acts on V' and preserves E. Since (X,B) is lc near 0 and $\beta_z = \operatorname{mult}_{(z=0)} B$, we have $\beta_z \in [0,1]$.

The valuation ord_E is the monomial valuation in u, v with weights w_E [CLS11, Proposition 4.1.1], and $a(E,V,0) = \frac{p+q}{g_0}$ [CLS11, Lemma 11.4.10]. Let $\beta_j := \operatorname{mult}_{(z=0)} \operatorname{div}(h_j)$, the least exponent of z in a monomial of h_j, so that $\beta_z = \sum_j c_j\beta_j$. By (16.22), $\operatorname{ord}_E P_j = \frac{\kappa_j + r\beta_j}{g_0}$. Thus

$$a(E, V, C|_V) = \frac{p+q-\sum_j c_j\kappa_j - r\beta_z}{g_0} = \frac{r(1-\beta_z)}{g_0} + \frac{|\tau|}{g_0}.$$

Comparing the coefficients of E, we obtain

$$K_{V'} + B_{V'} = f_V^*(K_V + C|_V) + \frac{|\tau|}{g_0}E.$$

On the affine toric open subset of the cone spanned by $(1,0)$ and w_E, the monomial $u^q v^{-p}$ is regular since $\langle (q,-p), (1,0)\rangle = q$ and $\langle (q,-p), w_E\rangle = 0$, and $\operatorname{div}(v^p) = \frac{pq}{g_0}E$ [CLS11, Propositions 4.1.1 and 4.1.2]. The affine toric open subset of the cone spanned by w_E and $(0,1)$ is symmetric. Thus $(u^q, v^p)\mathcal{O}_{V'} = \mathcal{O}_{V'}(-\frac{pq}{g_0}E)$, so $\min\{q\operatorname{ord}_F u, p\operatorname{ord}_F v\} = \frac{pq}{g_0}\operatorname{mult}_F E$ for any prime divisor F over V', where $\operatorname{mult}_F E$ is the coefficient of F in the pullback of the $\mathbb{Q}$-Cartier divisor E. Hence, for any prime divisor F over V',

$$a(F, V', B_{V'}) = a(F, V, C|_V) - \frac{|\tau|}{pq}\min\{q\operatorname{ord}_F u, p\operatorname{ord}_F v\}. \tag{16.27}$$

For a prime divisor ν over V with $\operatorname{center}_V \nu = \{0\}$, let $\lambda_\nu := \min\{\frac{\operatorname{ord}_\nu u}{p}, \frac{\operatorname{ord}_\nu v}{q}\}$. Then (16.27) reads $a(\nu, V', B_{V'}) = a(\nu, V, C|_V) - |\tau|\lambda_\nu$.

We show that $a(\nu, V', B_{V'}) \ge 0$ for such ν. Let $(k_i/m_i)_i$ be a sequence of positive rational numbers in lowest terms with $k_i/m_i < \lambda_\nu$ and $k_i/m_i \to \lambda_\nu$. Then (m_i, k_i) satisfies (16.24). By **Step 1** and (16.26),

$$0 \le a(F_\nu(m_i,k_i), \tilde X, \tilde D) = m_i a(\nu, V, C|_V) + \tau k_i.$$

Dividing by m_i and letting $i \to +\infty$ implies $a(\nu, V, C|_V) - |\tau|\lambda_\nu \ge 0$.

Since $f_V^{-1}(0) = E$, a prime divisor over V' whose center is contained in E is such a ν. A prime divisor over V' whose center intersects E and is not contained in E is a prime divisor $Q \ne E$ on V', and $a(Q, V', B_{V'}) = 1 - \operatorname{mult}_Q B_{V'} \ge 0$, since $\operatorname{mult}_Q B_{V'}$ is a coefficient of $C|_V$ or 0, and the

coefficients of $C|_V$ belong to $[0,1]$. Let $N_0 \subset V'$ be the union of the images of the prime divisors with negative log discrepancy with respect to $(V', B_{V'})$ on a log resolution of $(V', B_{V'})$. Then N_0 is closed, $N_0 \cap E = \emptyset$, and $(V'\backslash N_0, B_{V'}|_{V'\backslash N_0})$ is sub-lc [KM98, Corollary 2.31(3)]. Since f_V is proper, $f_V(N_0)$ is closed and $0 \notin f_V(N_0)$. Let V_2 be an affine open neighborhood of 0 in $V\backslash f_V(N_0)$. Then $\bigcap_{\sigma\in\tilde{H}} \sigma(V_2)$ is an $\tilde{H}$-invariant affine open neighborhood of 0 over which $(V', B_{V'})$ is sub-lc (V is separated, so the intersection is affine, and each σ fixes 0 and preserves $B_{V'}$). From now on, V denotes such a neighborhood, and V' and $B_{V'}$ are restricted over V. Then $(V', B_{V'})$ is sub-lc, and (16.27) still holds.

Step 5. In this step we show that $B_Z \geq 0$ and that (Z, B_Z) is lc, compute $\operatorname{mult}_{\bar{E}} B_Z$, and prove the formula for b in terms of $\mathfrak{b}$.

If $\tau \geq 0$, then $B_Z \geq 0$ by Lemma 2.6(0) since $C|_V \geq 0$, and (Z, B_Z) is lc by Lemma 2.6(2) and (3).

Assume $\tau < 0$. By **Step 4** and Lemma 2.6(2) and (3), (Z, B_Z) is sub-lc. We show that $\mu_r(p,q) \subset I_{\tilde{H}}(E)$. Since $(p,q) = g_0 w_E$, the element (ζ^p, ζ^q) multiplies any monomial $u^{i_1}v^{i_2}$ by $\zeta^{g_0\langle w_E,(i_1,i_2)\rangle}$. Thus (ζ^p,ζ^q) fixes $\operatorname{in}_{w_E}(f)/\operatorname{in}_{w_E}(f')$ for any non-zero $f, f' \in \mathbb{C}[u,v]$ with $\operatorname{wt}_{w_E}(f) = \operatorname{wt}_{w_E}(f')$. The residue field of ord_E is generated by the residues of such quotients f/f', and the residue of f/f' equals the residue of $\operatorname{in}_{w_E}(f)/\operatorname{in}_{w_E}(f')$, because

$$\operatorname{ord}_E\big(f\operatorname{in}_{w_E}(f') - f'\operatorname{in}_{w_E}(f)\big) > \operatorname{wt}_{w_E}(f) + \operatorname{wt}_{w_E}(f').$$

Since diagonal elements preserve ord_E, we obtain $\mu_r(p,q) \subset I_{\tilde{H}}(E)$. Hence $r = |\mu_r(p,q)|$ divides e_E, and e_E divides $|\tilde{H}| = r|H|$. Thus $e_{\mathrm{res}} := e_E/r$ is an integer with $1 \leq e_{\mathrm{res}} \leq |H|$.

Let Q be a prime divisor on V'. If $Q \neq E$, then $\operatorname{mult}_Q B_{V'}$ is a coefficient of $C|_V$ or 0, so $\operatorname{mult}_Q B_{V'} \in [0,1]$, and Lemma 2.6(4) implies that the coefficient of B_Z along the image of Q belongs to $[0,1]$. By Lemma 2.6(1) and (4),

$$\operatorname{mult}_{\bar{E}} B_Z = 1 - \frac{r(1-\beta_z)}{g_0 e_E} = 1 - \frac{1-\beta_z}{g_0 e_{\mathrm{res}}} \in [0,1],$$

since $\beta_z \in [0,1]$. Thus $B_Z \geq 0$ and (Z, B_Z) is lc.

We prove the formula for b. If $\tau \geq 0$, then $\mathfrak{b}^0 = \mathcal{O}_{V/\tilde{H}}$, and the formula is the definition of b in (1), with (Z, B_Z) lc. Assume $\tau < 0$, so $\tau^- = |\tau|$, and let $\rho_V : V \to V/\tilde{H}$ and $\rho_{V'} : V' \to Z$ be the quotient morphisms. For any prime divisor F over V, which is also a prime divisor over V', we have $\operatorname{ord}_F(\rho_V^*\mathfrak{b}) = r|H|\min\{q\operatorname{ord}_F u, p\operatorname{ord}_F v\}$. Thus (16.27) implies

$$a\big(F, V, C|_V, (\rho_V^*\mathfrak{b})^{|\tau|/(pqr|H|)}\big) = a(F, V', B_{V'}).$$

Since f_V is $\tilde{H}$-equivariant, it induces a proper birational morphism $Z \to V/\tilde{H}$, so the prime divisors over Z are exactly the prime divisors over $V/\tilde{H}$. Let $\bar{F}$ be the prime divisor over $V/\tilde{H}$ under F. By Lemma 2.6(2), applied to $(V, C|_V)$ with the $\mathbb{R}$-ideal $\mathfrak{b}^{|\tau|/(pqr|H|)}$ and to $(V', B_{V'})$,

$$a\big(\bar{F}, V/\tilde{H}, (C|_V)_{V/\tilde{H}}, \mathfrak{b}^{|\tau|/(pqr|H|)}\big) = a(\bar{F}, Z, B_Z) \geq 0.$$

By Lemma 2.6(3), this holds for any prime divisor $\bar{F}$ over $V/\tilde{H}$, so the triple is lc. Moreover, $\operatorname{center}_{V/\tilde{H}} \bar{F} = \{\bar{0}\}$ if and only if $\operatorname{center}_V F = \{0\}$, if and only if $\operatorname{center}_{V'} F \subset E$, if and only if $\operatorname{center}_Z \bar{F} \subset \bar{E}$, because $\rho_V^{-1}(\bar{0}) = \{0\}$, $f_V^{-1}(0) = E$, and $\rho_{V'}^{-1}(\bar{E}) = E$. Taking the infimum over the prime divisors $\bar{F}$ over $V/\tilde{H}$ with $\operatorname{center}_{V/\tilde{H}} \bar{F} = \{\bar{0}\}$ implies the formula for b.

Step 6. In this step we show that $b \geq 0$ and that b is attained.

If $\tau \geq 0$, then b is the minimal log discrepancy of the lc surface pair (Z, B_Z) at a closed point, so $b \geq 0$ and b is attained [KM98, Lemma 2.29 and Corollary 2.31(3)]. Assume $\tau < 0$ and let $c_E := \operatorname{mult}_{\bar{E}} B_Z$. A prime divisor over Z whose center is contained in the irreducible curve $\bar{E}$ is either $\bar{E}$, with $a(\bar{E}, Z, B_Z) = 1 - c_E$, or has center a closed point of $\bar{E}$. Let $S_0 \subset \bar{E}$ be the set of closed points at which Z or $\bar{E}$ is singular, together with the intersection points of $\bar{E}$ with the components of $\operatorname{Supp} B_Z$ other than $\bar{E}$. The set S_0 is finite. Let $y \in \bar{E}\backslash S_0$ and let $\bar{E}$ and $(w_y = 0)$

be defined by regular parameters of $\mathcal{O}_{Z,y}$. For any prime divisor F over Z with $\operatorname{center}_Z F = \{y\}$, [KM98, Corollary 2.31(3)] implies

$$a(F, Z, B_Z) \geq (1 - c_E) \operatorname{ord}_F \bar{E} + \operatorname{ord}_F w_y \geq 1 - c_E.$$

Thus $b = \min\big(\{1 - c_E\} \cup \{\operatorname{mld}(Z \ni y, B_Z) \mid y \in S_0\}\big)$. Each of the numbers $1 - c_E$ and $\operatorname{mld}(Z \ni y, B_Z)$, $y \in S_0$, is non-negative and attained, as in the case $\tau \geq 0$, since (Z, B_Z) is lc. Hence $b \geq 0$ and b is attained.

Step 7. In this step we prove $b \leq a$.

Let F be a prime divisor over $\tilde{X}$ with $\operatorname{center}_{\tilde{X}} F = \{0\}$, and let m, ν, and k be as in **Step 3**. Let $\bar{\nu}$ be the prime divisor over Z under ν, where ν is regarded as a prime divisor over V if $\tau \geq 0$ and over V' if $\tau < 0$. Then $\operatorname{center}_Z \bar{\nu} = \{\bar{0}\}$ if $\tau \geq 0$, and $\operatorname{center}_Z \bar{\nu} \subset \bar{E}$ if $\tau < 0$. If $\tau \geq 0$, then $\tau k \geq 0$, and (16.25) and Lemma 2.6(2) imply

$$a(F, \tilde{X}, \tilde{D}) \geq m a(\nu, V, C|_V) = m|I_{\tilde{H}}(\nu)| a(\bar{\nu}, Z, B_Z) \geq m|I_{\tilde{H}}(\nu)| b.$$

If $\tau < 0$, then $k < m\lambda_\nu$ by (16.24), and (16.25), (16.27), and Lemma 2.6(2) imply

$$a(F, \tilde{X}, \tilde{D}) \geq m\big(a(\nu, V, C|_V) - |\tau|\lambda_\nu\big) = m a(\nu, V', B_{V'}) = m|I_{\tilde{H}}(\nu)| a(\bar{\nu}, Z, B_Z) \geq m|I_{\tilde{H}}(\nu)| b.$$

We bound $|I_{\tilde{G}}(F)|$. An element of $I_{\tilde{G}}(F)$ preserves ord_F and $\mathbb{C}(u, v)$, hence preserves ord_ν, and acts trivially on the residue field of ord_ν, which is contained in the residue field of ord_F. Thus $I_{\tilde{G}}(F) \subset J_\nu$, and the image of $I_{\tilde{G}}(F)$ in $\tilde{H}$ is contained in $I_{\tilde{H}}(\nu)$. Let $I_0 := I_{\tilde{G}}(F) \cap \tilde{G}_0$, and let χ_F be the character of $I_{\tilde{G}}(F)$ on $\mathfrak{m}_F/\mathfrak{m}_F^2$, which is injective by Lemma 2.6(1). Let $\varpi_\nu \in \mathbb{C}(u, v)$ with $\operatorname{ord}_\nu \varpi_\nu = 1$, and write $\varpi_\nu = w_F^m \epsilon_F$ for a uniformizer w_F and a unit ϵ_F of the valuation ring of ord_F. For $\sigma \in I_{\tilde{G}}(F)$, we have $\sigma^* w_F \equiv \chi_F(\sigma) w_F$ modulo $\mathfrak{m}_F^2$ and $\sigma^* \epsilon_F \equiv \epsilon_F$ modulo $\mathfrak{m}_F$, so $\sigma^* \varpi_\nu \equiv \chi_F(\sigma)^m \varpi_\nu$ modulo $\mathfrak{m}_F^{m+1}$. For $\sigma \in I_0$, we have $\sigma^* \varpi_\nu = \varpi_\nu$, so $\chi_F(\sigma)^m = 1$. Since χ_F is injective on I_0, $|I_0|$ divides m. Hence

$$|I_{\tilde{G}}(F)| = |I_0| \cdot |I_{\tilde{G}}(F)/I_0| \leq m|I_{\tilde{H}}(\nu)|.$$

Since $b \geq 0$ by **Step 6**, we obtain $a(F, \tilde{X}, \tilde{D})/|I_{\tilde{G}}(F)| \geq b$. Since F is arbitrary, (16.23) implies $a \geq b$.

Step 8. In this step we assume $n > r|H| \max\{p, q\}$ and prove $a \leq b + \frac{|\tau|}{n}$.

By **Steps 5** and 6, b is attained by a prime divisor over Z with center $\{\bar{0}\}$ or center contained in $\bar{E}$. If $\tau < 0$, then f_V induces a proper birational morphism $Z \to V/\tilde{H}$, since f_V is $\tilde{H}$-equivariant, so the prime divisors over Z are exactly the prime divisors over $V/\tilde{H}$. Since $\rho_V^{-1}(\bar{0}) = \{0\}$, $\rho_{V'}^{-1}(\bar{E}) = E$, and $f_V(E) = \{0\}$, Lemma 2.6(2) and (3) imply that there exists a prime divisor ν over V with $\operatorname{center}_V \nu = \{0\}$ such that $a(\nu, V, C|_V) = |I_{\tilde{H}}(\nu)| b$ if $\tau \geq 0$, and $a(\nu, V', B_{V'}) = |I_{\tilde{H}}(\nu)| b$ if $\tau < 0$.

Recall that $J_\nu = I_{\tilde{G}}(\nu)$, with characters α_ν on $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$ and β_t on t. The image of J_ν in $\tilde{H}$ is $I_{\tilde{H}}(\nu)$, and $\tilde{G}_0 \subset J_\nu$. Since $I_{\tilde{H}}(\nu)$ acts faithfully on $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$ (Lemma 2.6(1)), $\alpha_\nu(J_\nu)$ is cyclic of order $|I_{\tilde{H}}(\nu)|$ and $\operatorname{Ker} \alpha_\nu = \tilde{G}_0$. By **Step 1**, β_t is injective on $\tilde{G}_0$, and $\beta_t(\tilde{G}_0)$ is the group of n-th roots of unity. Thus $(\alpha_\nu, \beta_t) : J_\nu \to (\mathbb{C}^*)^2$ is injective. Let N_{J_ν} be the set of $w' \in \mathbb{R}^2$ such that $(e^{2\pi\sqrt{-1}w'_1}, e^{2\pi\sqrt{-1}w'_2}) = (\alpha_\nu(\sigma), \beta_t(\sigma))$ for some $\sigma \in J_\nu$. Then $\mathbb{Z}^2 \subset N_{J_\nu}$, $J_\nu \cong N_{J_\nu}/\mathbb{Z}^2$, the projection of N_{J_ν} to the first coordinate is $\frac{1}{|I_{\tilde{H}}(\nu)|}\mathbb{Z}$, and $N_{J_\nu} \cap (\mathbb{Z} \times \mathbb{R}) = \mathbb{Z} \times \frac{1}{n}\mathbb{Z}$. Hence $\{w_2 \mid (\frac{1}{|I_{\tilde{H}}(\nu)|}, w_2) \in N_{J_\nu}\}$ is a coset of $\frac{1}{n}\mathbb{Z}$ in $\mathbb{R}$, and there exists $w = (\frac{1}{|I_{\tilde{H}}(\nu)|}, w_2) \in N_{J_\nu}$ with

$$0 < w_2 \leq \frac{1}{n} \ \text{ if } \tau \geq 0, \qquad \frac{\lambda_\nu}{|I_{\tilde{H}}(\nu)|} - \frac{1}{n} \leq w_2 < \frac{\lambda_\nu}{|I_{\tilde{H}}(\nu)|} \ \text{ if } \tau < 0, \tag{16.28}$$

where $\lambda_\nu = \min\{\frac{\operatorname{ord}_\nu u}{p}, \frac{\operatorname{ord}_\nu v}{q}\}$ as in **Step 4**. Since $\operatorname{ord}_\nu u, \operatorname{ord}_\nu v \geq 1$ and $|I_{\tilde{H}}(\nu)| \leq |\tilde{H}| = r|H|$,

$$\frac{\lambda_\nu}{|I_{\tilde{H}}(\nu)|} \geq \frac{1}{r|H| \max\{p, q\}} > \frac{1}{n}.$$

Thus $0 < w_2 < \frac{\lambda_\nu}{|I_{\tilde{H}}(\nu)|}$ in both cases.

No vector $\frac{1}{l}w$ with an integer $l \geq 2$ belongs to N_{J_ν}, since $\frac{1}{l|I_{\tilde{H}}(\nu)|} \notin \frac{1}{|I_{\tilde{H}}(\nu)|}\mathbb{Z}$. The set of $\theta \in \mathbb{R}$ with $\theta w \in N_{J_\nu}$ is a discrete subgroup of $\mathbb{R}$ containing 1, hence equal to $\frac{1}{l}\mathbb{Z}$ for an integer $l \geq 1$, and $l = 1$. Thus $\mathbb{R}w \cap N_{J_\nu} = \mathbb{Z}w$. Let e be the least positive integer with $ew \in \mathbb{Z}^2$ and let $(m,k) := ew$. If an integer $l \geq 2$ divides m and k, then $\frac{1}{l}(m,k) \in \mathbb{Z}^2 \cap \mathbb{R}w \subset \mathbb{Z}w$, so $\frac{1}{l}(m,k) = jw$ for an integer $0 < j < e$ with $jw \in \mathbb{Z}^2$, a contradiction. Hence $\gcd(m,k) = 1$. Since $\frac{k}{m} = |I_{\tilde{H}}(\nu)|w_2 < \lambda_\nu$ and $k > 0$, the pair (m,k) satisfies (16.24). Let $F := F_\nu(m,k)$.

We compute $|I_{\tilde{G}}(F)|$. Let $\sigma \in J_\nu$ correspond to $w' \in N_{J_\nu}$. Then

$$(\beta_t^m \alpha_\nu^{-k})(\sigma) = e^{2\pi\sqrt{-1}\det((m,k),w')}, \qquad \det((m,k),w') = mw_2' - kw_1'.$$

By (16.26), $\sigma \in I_{\tilde{G}}(F)$ if and only if $\det((m,k),w') \in \mathbb{Z}$. Since $\gcd(m,k) = 1$, the map $\det((m,k),\cdot)$ sends $\mathbb{Z}^2$ onto $\mathbb{Z}$. Thus for $w' \in N_{J_\nu}$ with $\det((m,k),w') \in \mathbb{Z}$, there exists $z' \in \mathbb{Z}^2$ with $\det((m,k),w'-z') = 0$, so $w'-z' \in \mathbb{R}w \cap N_{J_\nu} = \mathbb{Z}w$. Conversely, $\det((m,k),w) = 0$. Hence $I_{\tilde{G}}(F)$ corresponds to $(\mathbb{Z}^2 + \mathbb{Z}w)/\mathbb{Z}^2$, a cyclic group of order e, and $|I_{\tilde{G}}(F)| = e$.

Since $\frac{m}{e} = \frac{1}{|I_{\tilde{H}}(\nu)|}$ and $\frac{k}{e} = w_2$, (16.26) implies

$$\frac{a(F,\tilde{X},\tilde{D})}{|I_{\tilde{G}}(F)|} = \frac{ma(\nu,V,C|_V)+\tau k}{e} = \frac{a(\nu,V,C|_V)}{|I_{\tilde{H}}(\nu)|} + \tau w_2.$$

If $\tau \geq 0$, then the right side is $b + \tau w_2 \leq b + \frac{\tau}{n}$. If $\tau < 0$, then $a(\nu,V,C|_V) = a(\nu,V',B_{V'}) + |\tau|\lambda_\nu$ by (16.27), so the right side is

$$b + |\tau|\Big(\frac{\lambda_\nu}{|I_{\tilde{H}}(\nu)|} - w_2\Big) \leq b + \frac{|\tau|}{n}$$

by (16.28). Since $\operatorname{center}_{\tilde{X}} F = \{0\}$, (16.23) implies $a \leq b + \frac{|\tau|}{n}$.

Step 9. In this step we prove the statements on DCC and ACC sets.

Fix p,q,r,e_0,Γ, assume that $c_j \in \Gamma$ and $|H| \leq e_0$, and let $\Gamma' := \Gamma_+ \cap [0,1]$. Since $(V, C|_V)$ is lc, any coefficient of $C|_V$ is at most 1 and is a finite sum of numbers c_j, so it belongs to Γ'. Similarly $\beta_z = \sum_j c_j\beta_j \in \Gamma'$. By Lemma 2.6(4) and **Step 5**, any coefficient of B_Z is $1 - \frac{1-c}{e'}$ with $c \in \Gamma'$ and a positive integer e': for the image of a prime divisor $Q \neq E$, the number c is a coefficient of $C|_V$ or 0, and for $\bar{E}$, we have $c = \beta_z$ and $e' = g_0 e_{\mathrm{res}}$. Thus the coefficients of B_Z belong to $\Psi(\Gamma')$, which is a DCC set depending only on Γ by Lemma 2.10(1), applied to Γ and to Γ'.

If $\tau \geq 0$, then b is the minimal log discrepancy at a closed point of an lc surface pair with coefficients in $\Psi(\Gamma')$. If $\tau < 0$, then by **Step 6**, b is either $1 - \operatorname{mult}_{\bar{E}} B_Z \in \{1-c \mid c \in \Psi(\Gamma')\}$ or such a minimal log discrepancy. The set $\{1-c \mid c \in \Psi(\Gamma')\}$ satisfies the ACC since $\Psi(\Gamma')$ satisfies the DCC, and the set of minimal log discrepancies at closed points of lc surface pairs with coefficients in $\Psi(\Gamma')$ satisfies the ACC by Theorem 2.18(1). By Lemma 2.10(4), the numbers b belong to an ACC set depending only on Γ.

Let E_0 be the exceptional divisor of the blow-up of $0 \in X$. Since (X,B) is lc near 0, [KM98, Lemma 2.29] implies $0 \leq a(E_0,X,B) = 3 - \sum_j c_j \operatorname{mult}_0 h_j$, since the discrepancy of [KM98] is our log discrepancy minus 1. A monomial $x^{i_1}y^{i_2}z^{i_3}$ of h_j with $i_1+i_2+i_3 = \operatorname{mult}_0 h_j$ has $\mathbf{T}$-weight $\kappa_j = pi_1 + qi_2 - ri_3$, so $|\kappa_j| \leq \max\{p,q,r\}\operatorname{mult}_0 h_j$. Hence

$$|\tau| \leq |r-p-q| + 3\max\{p,q,r\}. \tag{16.29}$$

Suppose that $(a_i)_i$ is a strictly increasing sequence with $a_i = \operatorname{mld}((X \ni 0, B_i)/G_i)$ for data satisfying the hypotheses above with the fixed p,q,r,e_0,Γ, and that the corresponding numbers n_i are unbounded. Passing to a subsequence, we may assume that $n_i \to +\infty$ and $n_i > re_0\max\{p,q\}$ for any i. Let b_i be the corresponding numbers b, which belong to the ACC set above. By **Steps 7** and 8 and (16.29),

$$b_i \leq a_i \leq b_i + \frac{|r-p-q| + 3\max\{p,q,r\}}{n_i},$$

so $a_i - b_i \to 0$, which contradicts Lemma 14.11. □

Example 16.10. For an integer $n \geq 2$, let $G_n := \mu_n(1,1,-1)$ act on $\mathbb{A}^3_{x,y,z}$, and let $B := c\,\mathrm{div}(x-y) + d\,\mathrm{div}(z)$ with real numbers $0 \leq c, d < 1$. Then $\mathrm{mld}((\mathbb{A}^3 \ni 0, B)/G_n) = 1-d+\frac{1+d-c}{n}$, attained by the prime divisor under the monomial valuation with weights $(1,1,n-1)$, whose inertia group is G_n. In Construction-Theorem 16.9 with $(p,q,r) = (1,1,1)$ applied to the equations $x-y$ and z, we have $\tau = c-d-1 < 0$ and $b = 1-d$, so the upper bound $\mathrm{mld}((\mathbb{A}^3 \ni 0, B)/G_n) \leq b + \frac{|\tau|}{n}$ of Construction-Theorem 16.9 is an equality, while the minimal log discrepancy at the origin of $(\mathbb{A}^2_{u,v}, c\,\mathrm{div}(u-v))$ is $2-c > \mathrm{mld}((\mathbb{A}^3 \ni 0, B)/G_n)$.

Proof. We check the hypotheses of Proposition 8.8(a) for Set-up 8.4 in dimension 3 with $G = G_n$, $f_1 = x-y$, $f_2 = z$, $b_1 = c$, and $b_2 = d$, where a term with coefficient 0 is omitted. The polynomials $x-y$ and z are reduced and coprime, and $(\mathbb{A}^3, B)$ is lc near 0 by [KM98, Corollary 2.31(3)], since $x-y$ and z are part of a system of coordinates and $c, d < 1$. Condition (a) holds: for $w \in \mathbb{R}^3_{>0}$, the polynomial $\mathrm{in}_w(z) = z$ has no zero in $(\mathbb{C}^*)^3$, and $\mathrm{in}_w(x-y)$ is x, $-y$, or $x-y$, which defines the empty set or a smooth hypersurface of $(\mathbb{C}^*)^3$. The function of Set-up 8.4 is

$$\psi(v) = v_1 + v_2 + v_3 - c\min\{v_1, v_2\} - dv_3,$$

which is nondecreasing in each coordinate because $c, d < 1$. The lattice N_{G_n} is $\mathbb{Z}^3 + \mathbb{Z}\frac{(1,1,-1)}{n}$. For $1 \leq k \leq n-1$, the coordinatewise least element of $(\frac{(k,k,-k)}{n} + \mathbb{Z}^3) \cap \mathbb{R}^3_{>0}$ is $\frac{(k,k,n-k)}{n}$, with ψ-value $1 - d + \frac{(1+d-c)k}{n}$. The least element of $\mathbb{Z}^3 \cap \mathbb{R}^3_{>0}$ is $(1,1,1)$, with ψ-value $3-c-d$. Since $1+d-c > 0$ and

$$3-c-d-\left(1-d+\frac{(1+d-c)k}{n}\right) > 2-c-(1+d-c) = 1-d > 0$$

for $k < n$, the minimum of ψ on $N_{G_n} \cap \mathbb{R}^3_{>0}$ is $1-d+\frac{1+d-c}{n}$, attained at the primitive vector $v := \frac{(1,1,n-1)}{n}$. By Proposition 8.8, $\mathrm{mld}((\mathbb{A}^3 \ni 0, B)/G_n) = 1-d+\frac{1+d-c}{n}$. The primitive vector of $\mathbb{Z}^3$ on the ray of v is $nv = (1,1,n-1)$, so the monomial prime divisor F over $\mathbb{A}^3$ with weights $(1,1,n-1)$ satisfies $|I_{G_n}(F)| = n$, that is, $I_{G_n}(F) = G_n$, and $a(F, \mathbb{A}^3, B) = n\psi(v)$. By Lemma 2.6(2), the prime divisor over $\mathbb{A}^3/G_n$ under F has log discrepancy $\psi(v)$.

In Construction-Theorem 16.9 with $(p,q,r) = (1,1,1)$, we have $G_n \subset \mathbf{T}_{(1,1,-1)}$, so $n = |G_n|$ and H is trivial. The polynomial $x-y$ has $\mathbf{T}$-weight 1 and z has $\mathbf{T}$-weight -1, so $\tau = 1-1-1+c-d = c-d-1 < 0$. Since $u = xt$, $v = yt$, $x-y = t^{-1}(u-v)$, and $z = t$, we have $C = c\,\mathrm{div}(u-v)$. Since $u-v$ and u are coordinates, any prime divisor F' over $\mathbb{A}^2_{u,v}$ with center 0 satisfies

$$a(F', \mathbb{A}^2_{u,v}, c\,\mathrm{div}(u-v)) \geq (1-c)\,\mathrm{ord}_{F'}(u-v) + \mathrm{ord}_{F'}\,u \geq 2-c$$

by [KM98, Corollary 2.31(3)], with equality for the blow-up of 0. Thus the minimal log discrepancy at the origin of $(\mathbb{A}^2_{u,v}, c\,\mathrm{div}(u-v))$ is $2-c$, and

$$2-c-\left(1-d+\frac{1+d-c}{n}\right) = (1+d-c)\left(1-\frac{1}{n}\right) > 0.$$

In particular, $2-c$ is not a lower bound for $\mathrm{mld}((\mathbb{A}^3 \ni 0, B)/G_n)$. In Construction-Theorem 16.9 with $V = \mathbb{A}^2_{u,v}$, the group $\tilde{H}$ is trivial, $g_0 = 1$, $\beta_z = d$, V' is the blow-up of 0, and $B_{V'}$ is dE plus c times the strict transform of $(u-v=0)$. Here $a(E, V', B_{V'}) = 1-d$, and any prime divisor F' over V' whose center is a closed point of E satisfies $a(F', V', B_{V'}) \geq (1-d)+(1-c) > 1-d$ by [KM98, Corollary 2.31(3)]. Thus $b = 1-d$, and the upper bound $b + \frac{|\tau|}{n} = 1-d+\frac{1+d-c}{n}$ is attained. □

16.6. **Upper bounds.** In this subsection, we prove Proposition 16.11: the number b of Construction-Theorem 16.9 bounds the minimal log discrepancies of diagonal quotients of $\mathbb{A}^3$ from above also for equations which are not semi-invariant for $\mathbf{T}$.

By Theorem 2.18(2), the number b is computed by a prime divisor with bounded log discrepancy. The P_j have bounded orders along a lift of such a prime divisor to $\mathbb{A}^2_{u,v}$, and we take a Gauss

extension of the lift along which x, y, and z have order at least η times the order of the inertia group, so that terms of large degree do not change the orders of the equations. For arbitrary semi-invariant equations, the characters of $G \cap \mathbf{T}$ force the terms whose $\mathbf{T}$-weight differs from the $\mathbf{T}$-weight of a monomial of least degree to have large degree, and [Kaw24, Theorem 1.3] shows that the sum of the other terms defines an lc pair.

Proposition 16.11. *Fix positive integers p, q, r with $\gcd(p,q,r) = 1$, an integer $e_0 \geq 1$, and a DCC set $\Gamma \subset [0,1]$ with $\Gamma \backslash \{0\} \neq \emptyset$, and let $\gamma := \gamma(\Gamma)$, $P := \max\{p,q\}$, $P_{\max} := \max\{p,q,r\}$, $T_0 := r+p+q+3P_{\max}$, $\eta_0 := \frac{1}{4re_0P}$, and $\mathbf{T} := \mathbf{T}_{(p,q,-r)}$. There exists a positive integer U_0 depending only on p, q, r, e_0, Γ such that the following hold for finite diagonal groups G acting faithfully on $X := \mathbb{A}^3_{x,y,z}$ with $|G/(G \cap \mathbf{T})| \leq e_0$, $n := |G \cap \mathbf{T}|$, and positive $c_j \in \Gamma$.*

(1) Let h_j^0 be non-zero polynomials semi-invariant for G and $\mathbf{T}$ such that (X, B^0) is lc near 0 for $B^0 := \sum_j c_j \operatorname{div}(h_j^0)$, let N be a positive integer, let $h_j = h_j^0 + R_j$ have the character of h_j^0 with $R_j \in (x,y,z)^N$, and let $B := \sum_j c_j \operatorname{div}(h_j)$. Let b be the number of Construction-Theorem 16.9 for the equations h_j^0. Then for any η with $0 < \eta \leq \eta_0$, $n\eta \geq 1$, and $N\eta > U_0$, there exists a prime divisor E over X/G with center the image of 0 such that $a(E, X/G, B_{X/G}) \leq b + 2T_0\eta$ and any prime divisor F over X lying over E satisfies $\operatorname{ord}_F h_j = \operatorname{ord}_F h_j^0$ for any j. In particular,

$$\operatorname{mld}((X \ni 0, B)/G) \leq b + 2T_0\eta.$$

(2) Let h_j be arbitrary non-zero G-semi-invariant polynomials with (X, B) klt near 0 for $B := \sum_j c_j \operatorname{div}(h_j)$, and discard those h_j with $h_j(0) \neq 0$. For any j, choose a monomial m_j of least degree occurring in h_j, let κ_j be the $\mathbf{T}$-weight of m_j, and let h_j^0 be the sum of the terms of h_j of $\mathbf{T}$-weight κ_j. Let $C_1 := 3P_{\max}/\gamma$, $N_n := \max\{0, \lceil (n - C_1)/P_{\max} \rceil\}$, and $C_0 := 4T_0 \max\{1, 2U_0P_{\max}\}$. Then there exists a positive integer n_0 depending only on p, q, r, e_0, Γ such that, if $n \geq n_0$, then $(X, \sum_j c_j \operatorname{div}(h_j^0))$ is lc near 0, $h_j - h_j^0 \in (x,y,z)^{N_n}$, and $\operatorname{mld}((X \ni 0, B)/G) \leq b + \frac{C_0}{n}$ for the number b of Construction-Theorem 16.9 for the h_j^0. In particular, if moreover $\operatorname{mld}((X \ni 0, B)/G) \geq \epsilon > 0$ and $n \geq \max\{n_0, 2C_0/\epsilon\}$, then $b \geq \frac{\epsilon}{2}$.

Proof. The proof is long, so we divide it into several steps. In **Steps 1–4** we prove (1), and in **Step 5** we prove (2). For (1), we use the notation of Construction-Theorem 16.9 for G, the equations h_j^0, and the numbers c_j. In particular, κ_j is the $\mathbf{T}$-weight of h_j^0, $h_j^0(x,y,t^r) = t^{-\kappa_j} P_j(u,v)$ with $P_j \in \mathbb{C}[u,v]$, $C = \sum_j c_j \operatorname{div}(P_j)$, $H = G/(G \cap \mathbf{T})$, and $|\tilde{H}| = r|H| \leq re_0$. Let $g_0 := \gcd(p,q)$. When $\tau < 0$, we write E_V for the exceptional curve E of f_V in Construction-Theorem 16.9(2), and $\bar{E}_V$ for the image $\bar{E}$ of E_V on Z.

Step 1. In this step, we show that

$$|\kappa_j| \leq \frac{3P_{\max}}{\gamma} \text{ for any } j, \qquad \sum_j c_j|\kappa_j| \leq 3P_{\max}, \qquad \text{and} \qquad |\tau| \leq T_0. \tag{16.30}$$

Let E_0 be the exceptional divisor of the blow-up of X at 0. Since (X, B^0) is lc near 0, [KM98, Lemma 2.29] implies

$$0 \leq a(E_0, X, B^0) = 3 - \sum_j c_j \operatorname{ord}_0 h_j^0.$$

A monomial $x^{a_1}y^{a_2}z^{a_3}$ has $\mathbf{T}$-weight $pa_1 + qa_2 - ra_3$, and $|pa_1 + qa_2 - ra_3| \leq P_{\max}(a_1 + a_2 + a_3)$. For a monomial of degree $\operatorname{ord}_0 h_j^0$ occurring in h_j^0, we obtain $|\kappa_j| \leq P_{\max} \operatorname{ord}_0 h_j^0$. Thus $\sum_j c_j|\kappa_j| \leq 3P_{\max}$, and $|\kappa_j| \leq 3P_{\max}/\gamma$ since $c_j \geq \gamma$. Hence

$$|\tau| \leq |r - p - q| + \sum_j c_j|\kappa_j| \leq r + p + q + 3P_{\max} = T_0.$$

Step 2. In this step, we choose a prime divisor S' over $\mathbb{A}^2_{u,v}$ with center 0 lying over a prime divisor computing b, we bound the orders of u, v, and the P_j along S', and we define U_0.

Let $W := V$ and $B_W := C|_V$ if $\tau \geq 0$, and let $W := V'$ and $B_W := B_{V'}$ if $\tau < 0$. Then $(Z, B_Z) = (W, B_W)/\tilde{H}$. By Construction-Theorem 16.9, $B_Z \geq 0$, (Z, B_Z) is lc, and the coefficients of B_Z belong to a DCC set depending only on Γ, p, q, r, e_0. Let l_1 be the integer l of Theorem 2.18(2) for the DCC set of Construction-Theorem 16.9. The integer l_1 depends only on p, q, r, e_0, Γ. The surface W is an open subset of $\mathbb{A}^2_{u,v}$ or of a weighted blow-up of $\mathbb{A}^2_{u,v}$, so W is $\mathbb{Q}$-factorial [CLS11, Proposition 4.2.7], and Z is $\mathbb{Q}$-factorial by [KM98, Lemma 5.16].

There exists a prime divisor S over Z such that $a(S, Z, B_Z) = b$, $a(S, Z, 0) \leq l_1$, and $\operatorname{center}_Z S$ is $\{\bar{0}\}$ if $\tau \geq 0$ and is contained in $\bar{E}_V$ if $\tau < 0$. If $\tau \geq 0$, then $b = \operatorname{mld}(Z \ni \bar{0}, B_Z)$, and Theorem 2.18(2) implies the existence of S. Assume that $\tau < 0$. If $b = a(\bar{E}_V, Z, B_Z)$, then we let $S := \bar{E}_V$, and $a(S, Z, 0) = 1 \leq l_1$. Otherwise, by Construction-Theorem 16.9, $b = a(S_1, Z, B_Z)$ for a prime divisor S_1 over Z whose center is a closed point $z_1 \in \bar{E}_V$. By the definition of b, any prime divisor over Z with center $\{z_1\}$ has log discrepancy at least b, so $\operatorname{mld}(Z \ni z_1, B_Z) = b$, and Theorem 2.18(2) at z_1 implies the existence of S.

By Lemma 2.6(3), there exists a prime divisor S' over W such that $\operatorname{ord}_{S'}|_{\mathbb{C}(Z)} = e_{S'} \operatorname{ord}_S$, where $e_{S'} := |I_{\tilde{H}}(S')|$ by Lemma 2.6(1). Then $1 \leq e_{S'} \leq |\tilde{H}| \leq re_0$. By Lemma 2.6(2), $\operatorname{center}_Z S$ is the image of $\operatorname{center}_W S'$. If $\tau \geq 0$, then $\operatorname{center}_V S' = \{0\}$, since 0 is the only point of V over $\bar{0}$. If $\tau < 0$, then $E_V = f_V^{-1}(0)$ is $\tilde{H}$-invariant, so E_V is the preimage of $\bar{E}_V$ in V', $\operatorname{center}_{V'} S' \subset E_V$, and $\operatorname{center}_V S' = \{0\}$. Thus in both cases S' is a prime divisor over $\mathbb{A}^2_{u,v}$ with center 0, $\operatorname{ord}_{S'} u \geq 1$, $\operatorname{ord}_{S'} v \geq 1$, and Lemma 2.6(2) implies

$$a(S', W, B_W) = e_{S'} a(S, Z, B_Z) = e_{S'} b. \tag{16.31}$$

Write $(W, 0)/\tilde{H} = (Z, \Delta_Z)$. Then $\Delta_Z \geq 0$ by Lemma 2.6(0). By Lemma 2.6(2) and [KM98, Lemma 2.27],

$$a(S', W, 0) = e_{S'} a(S, Z, \Delta_Z) \leq e_{S'} a(S, Z, 0) \leq e_{S'} l_1.$$

If $\tau \geq 0$, then $a(S', \mathbb{A}^2_{u,v}, 0) = a(S', W, 0) \leq e_{S'} l_1$. Assume that $\tau < 0$. The curve E_V is the toric divisor of the primitive vector $(p/g_0, q/g_0)$, so $a(E_V, V, 0) = (p+q)/g_0$ [CLS11, Lemma 11.4.10], that is,

$$K_{V'} = f_V^* K_V + \Big(\frac{p+q}{g_0} - 1\Big) E_V.$$

The surface V' is an open subset of the toric surface of the fan with rays $(1, 0)$, $(p/g_0, q/g_0)$, $(0, 1)$, which is $\mathbb{Q}$-factorial [CLS11, Proposition 4.2.7], and E_V is the torus-invariant prime divisor of the ray $(p/g_0, q/g_0)$. Since lc is a local property, the pair (V', E_V) is lc [CLS11, Proposition 11.4.24(a)]. Thus $a(S', V', 0) - \operatorname{mult}_{S'} E_V = a(S', V', E_V) \geq 0$, and

$$a(S', V, 0) = a(S', V', 0) + \Big(\frac{p+q}{g_0} - 1\Big) \operatorname{mult}_{S'} E_V \leq \frac{p+q}{g_0} a(S', V', 0) \leq \frac{p+q}{g_0} e_{S'} l_1.$$

Let $l_2 := \frac{p+q}{g_0} l_1$. Then $a(S', \mathbb{A}^2_{u,v}, 0) \leq e_{S'} l_2$ in both cases. Since $(\mathbb{A}^2_{u,v}, (u = 0) + (v = 0))$ is lc [KM98, Corollary 2.31(3)], $\operatorname{ord}_{S'} u + \operatorname{ord}_{S'} v \leq a(S', \mathbb{A}^2_{u,v}, 0)$. Together with $\operatorname{ord}_{S'} u, \operatorname{ord}_{S'} v \geq 1$ and $e_{S'} \leq re_0$, we obtain

$$\frac{1}{re_0} \leq \frac{\operatorname{ord}_{S'} u}{e_{S'}}, \ \frac{\operatorname{ord}_{S'} v}{e_{S'}} \leq \frac{a(S', \mathbb{A}^2_{u,v}, 0)}{e_{S'}} \leq l_2. \tag{16.32}$$

Since $(V, C|_V)$ is lc and $0 \in V$, $\sum_j c_j \operatorname{ord}_{S'} P_j \leq a(S', \mathbb{A}^2_{u,v}, 0)$. As $c_j \geq \gamma$, we obtain

$$\frac{\operatorname{ord}_{S'} P_j}{e_{S'}} \leq \frac{l_2}{\gamma} \quad \text{for any } j. \tag{16.33}$$

Let

$$U_0 := \Big\lceil \max\Big\{1, \frac{l_2(1 + 3P_{\max})}{\gamma}\Big\} \Big\rceil.$$

Then U_0 is a positive integer depending only on p, q, r, e_0, Γ. Since (X, B^0) is lc near 0, [KM98, Lemma 2.29] implies $\sum_i c_i \operatorname{ord}_0 h_i^0 \le 3$, so $\operatorname{ord}_0 h_j^0 \le 3/\gamma < U_0$ for any j, since $l_2 = \frac{p+q}{g_0} l_1 \ge 2$ (as $l_1 \ge 1$ and p/g_0, q/g_0 are positive integers). If $0 < \eta \le \eta_0$ and $N\eta > U_0$, then $N > U_0 > \operatorname{ord}_0 h_j^0$, hence $h_j = h_j^0 + R_j \notin (x, y, z)^N$. In particular, $h_j \ne 0$ and B is defined.

Step 3. In this step, for η as in (1), we construct a prime divisor $\tilde{F}$ over $X' := \mathbb{A}^3_{x,y,t}$ along which x, y, and $z = t^r$ have order at least $\eta |I_{\tilde{G}}(\tilde{F})|$.

Let

$$M_0 := \min\left\{ \frac{\operatorname{ord}_{S'} u}{p e_{S'}}, \frac{\operatorname{ord}_{S'} v}{q e_{S'}} \right\}.$$

By (16.32), $\frac{1}{re_0 P} \le M_0 \le l_2$. Since $\eta \le \eta_0 = \frac{1}{4re_0 P}$, we have $2\eta \le M_0/2$.

The group $\tilde{G}$ acts on $\mathbb{C}(u, v)$ through $\tilde{H}$, so the inertia group $I_{\tilde{G}}(S')$ for the action of $\tilde{G}$ on $\mathbb{A}^2_{u,v}$ is the preimage $\tilde{J}$ of $I_{\tilde{H}}(S')$ in $\tilde{G}$. Let χ_1 be the character of $\tilde{J}$ on $\mathfrak{m}_{S'}/\mathfrak{m}_{S'}^2$, where $\mathfrak{m}_{S'}$ is the maximal ideal of the valuation ring of $\operatorname{ord}_{S'}$, and let χ_2 be the character of t, that is, $\chi_2((g, \alpha)) = \alpha$. The kernel $\operatorname{Ker}(\tilde{G} \to \tilde{H})$ consists of the elements (g, α) with $g_x \alpha^p = g_y \alpha^q = 1$, it is contained in $\tilde{J}$, and

$$|\operatorname{Ker}(\tilde{G} \to \tilde{H})| = \frac{|\tilde{G}|}{|\tilde{H}|} = \frac{r|G|}{r|H|} = n.$$

Since $I_{\tilde{H}}(S')$ is cyclic of order $e_{S'}$ and acts faithfully on $\mathfrak{m}_{S'}/\mathfrak{m}_{S'}^2$ (Lemma 2.6(1)), $\chi_1(\tilde{J})$ is the group of $e_{S'}$-th roots of unity and $\operatorname{Ker}\chi_1 = \operatorname{Ker}(\tilde{G} \to \tilde{H})$. If $(g, \alpha) \in \operatorname{Ker}(\tilde{G} \to \tilde{H})$ and $\alpha = 1$, then $g_x = g_y = 1$ and $g_z = \alpha^r = 1$, so χ_2 is injective on $\operatorname{Ker}(\tilde{G} \to \tilde{H})$, and $\chi_2(\operatorname{Ker}(\tilde{G} \to \tilde{H}))$ is the group of n-th roots of unity. In particular, $(\chi_1, \chi_2) : \tilde{J} \to (\mathbb{C}^*)^2$ is injective.

Let $N_{\tilde{J}} \subset \mathbb{R}^2$ be the set of vectors (w_1, w_2) for which there exists $g \in \tilde{J}$ with $\chi_1(g) = e^{2\pi\sqrt{-1}w_1}$ and $\chi_2(g) = e^{2\pi\sqrt{-1}w_2}$. Then $N_{\tilde{J}}$ is a subgroup of $\frac{1}{|\tilde{J}|}\mathbb{Z}^2$ containing $\mathbb{Z}^2$, and $\tilde{J} \cong N_{\tilde{J}}/\mathbb{Z}^2$ since (χ_1, χ_2) is injective. The projection of $N_{\tilde{J}}$ to the first coordinate is $\frac{1}{e_{S'}}\mathbb{Z}$. If $g_1 \in \tilde{J}$ satisfies $\chi_1(g_1) = e^{2\pi\sqrt{-1}/e_{S'}}$, then the elements $g \in \tilde{J}$ with $\chi_1(g) = \chi_1(g_1)$ are the products $g_1 g'$ with $g' \in \operatorname{Ker}(\tilde{G} \to \tilde{H})$. Hence the set of second coordinates of the points of $N_{\tilde{J}}$ with first coordinate $\frac{1}{e_{S'}}$ is $\sigma_1 + \frac{1}{n}\mathbb{Z}$ for some $\sigma_1 \in \mathbb{R}$. Since $\frac{1}{n} \le \eta$, the set $\sigma_1 + \frac{1}{n}\mathbb{Z}$ intersects any closed interval of length η. We choose $w := (\frac{1}{e_{S'}}, \sigma) \in N_{\tilde{J}}$ with

$$\sigma \in [\eta, 2\eta] \ \text{ if } \tau \ge 0, \qquad \sigma \in [M_0 - 2\eta, M_0 - \eta] \ \text{ if } \tau < 0. \tag{16.34}$$

Since $2\eta \le M_0/2$, we have $0 < \sigma < M_0$ in both cases, and $\sigma \ge M_0/2$ if $\tau < 0$. The vector w is primitive in $N_{\tilde{J}}$, since for any integer $k \ge 2$ the first coordinate of $\frac{1}{k}w$ is not in $\frac{1}{e_{S'}}\mathbb{Z}$.

Since $w \in \mathbb{Q}^2$, there exist a positive integer e and a primitive vector $(k_1, k_2) \in \mathbb{Z}^2$ with $(k_1, k_2) = ew$. Then $k_1 = e/e_{S'}$ is a positive integer and $k_2 = e\sigma > 0$. Since $u = xt^p$ and $v = yt^q$, we have $\mathbb{C}(x, y, t) = \mathbb{C}(u, v)(t)$. Let $\tilde{F}$ be the prime divisor over X' such that $\operatorname{ord}_{\tilde{F}}$ is the Gauss extension of $k_1 \operatorname{ord}_{S'}$ to $\mathbb{C}(u, v)(t)$ with t-value k_2 (Definition-Proposition 2.11). Since $k_1 \operatorname{ord}_{S'}(\mathbb{C}(u, v)^*) = k_1\mathbb{Z}$ and $\gcd(k_1, k_2) = 1$, we have $\operatorname{ord}_{\tilde{F}}(\mathbb{C}(x, y, t)^*) = \mathbb{Z}$.

The data for Proposition 16.7(1) are X', the weights $(p, q, -1)$, the group $\tilde{G}$, the empty boundary, and the vector $(0, 0, -1) \in \mathbb{Z}^3$. The group $\tilde{G}$ is a finite diagonal group acting faithfully on X': if $(g_x, g_y, \alpha) = (1, 1, 1)$, then $g_z = \alpha^r = 1$. The surface of Proposition 16.7 is $\operatorname{Spec}\mathbb{C}[x, y, t]^{\mathbf{T}_{(p,q,-1)}} = \mathbb{A}^2_{u,v}$ with reduced toric boundary $(u = 0) + (v = 0)$, the Gauss variable is t^{-1}, and the functions f_1, f_2, f_3 of Proposition 16.7 are $u, v, 1$. We take $\nu := S'$, $m := k_1$, and $k := -k_2$. Then the Gauss extension of $k_1 \operatorname{ord}_{S'}$ with t^{-1}-value $-k_2$ is $\operatorname{ord}_{\tilde{F}}$, and

$$n_1 = k_1 \operatorname{ord}_{S'} u - pk_2 = e\Big(\frac{\operatorname{ord}_{S'} u}{e_{S'}} - p\sigma\Big) > 0, \quad n_2 = e\Big(\frac{\operatorname{ord}_{S'} v}{e_{S'}} - q\sigma\Big) > 0, \quad n_3 = k_2 > 0,$$

because $\sigma < M_0$. The character of t^{-1} on $\tilde{J}$ is χ_2^{-1}. Thus Proposition 16.7(1) implies

$$I_{\tilde{G}}(\tilde{F}) = \mathrm{Ker}\left(\chi_2^{k_1}\chi_1^{-k_2} : \tilde{J} \to \mathbb{C}^*\right)$$

and

$$\begin{aligned} a(\tilde{F}, X', 0) &= k_1 a\big(S', \mathbb{A}^2_{u,v}, (u=0)+(v=0)\big) + n_1 + n_2 + n_3 \\ &= k_1 a(S', \mathbb{A}^2_{u,v}, 0) + (1-p-q)k_2. \end{aligned} \tag{16.35}$$

Claim 16.12. $|I_{\tilde{G}}(\tilde{F})| = e$.

Proof. Under the isomorphism $\tilde{J} \cong N_{\tilde{J}}/\mathbb{Z}^2$, the element g with $\chi_i(g) = e^{2\pi\sqrt{-1}w_i}$ satisfies

$$\chi_2^{k_1}\chi_1^{-k_2}(g) = e^{2\pi\sqrt{-1}(k_1w_2 - k_2w_1)}.$$

Hence $I_{\tilde{G}}(\tilde{F}) \cong L/\mathbb{Z}^2$, where

$$L := \{(w_1, w_2) \in N_{\tilde{J}} \mid k_1w_2 - k_2w_1 \in \mathbb{Z}\}.$$

Since (k_1, k_2) is primitive in $\mathbb{Z}^2$, there exists $(a_1, a_2) \in \mathbb{Z}^2$ with $k_1a_2 - k_2a_1 = 1$, and any $w' \in \mathbb{R}^2$ is $w' = \lambda(k_1, k_2) + \mu(a_1, a_2)$ with $\mu = k_1w'_2 - k_2w'_1$. Thus $L = (N_{\tilde{J}} \cap \mathbb{R}w) + \mathbb{Z}(a_1, a_2)$, and since $\mathbb{Z}^2 \cap \mathbb{R}w = \mathbb{Z}(k_1, k_2)$,

$$L/\mathbb{Z}^2 \cong (N_{\tilde{J}} \cap \mathbb{R}w)/\mathbb{Z}(k_1, k_2).$$

The group $N_{\tilde{J}} \cap \mathbb{R}w$ is a discrete subgroup of $\mathbb{R}w$, so it is $\mathbb{Z}w''$ for some w'', and $w = mw''$ for an integer $m \neq 0$. Since w is primitive in $N_{\tilde{J}}$, $|m| = 1$, so $N_{\tilde{J}} \cap \mathbb{R}w = \mathbb{Z}w$. Hence $|L/\mathbb{Z}^2| = |\mathbb{Z}w/\mathbb{Z}ew| = e$. □

Proof of Proposition 16.11 continued. Since $x = ut^{-p}$, $y = vt^{-q}$, and $z = t^r$, we have $\mathrm{ord}_{\tilde{F}} x = k_1 \mathrm{ord}_{S'} u - pk_2$, $\mathrm{ord}_{\tilde{F}} y = k_1 \mathrm{ord}_{S'} v - qk_2$, and $\mathrm{ord}_{\tilde{F}} z = rk_2$, that is,

$$\frac{\mathrm{ord}_{\tilde{F}} x}{e} = \frac{\mathrm{ord}_{S'} u}{e_{S'}} - p\sigma, \qquad \frac{\mathrm{ord}_{\tilde{F}} y}{e} = \frac{\mathrm{ord}_{S'} v}{e_{S'}} - q\sigma, \qquad \frac{\mathrm{ord}_{\tilde{F}} z}{e} = r\sigma. \tag{16.36}$$

We show that the three numbers in (16.36) are at least η. If $\tau \geq 0$, then by (16.32), (16.34), and $\eta \leq \eta_0$,

$$\frac{\mathrm{ord}_{S'} u}{e_{S'}} - p\sigma \geq \frac{1}{re_0} - 2P\eta \geq \frac{1}{2re_0} \geq \eta,$$

replacing u, p by v, q, we obtain $\frac{\mathrm{ord}_{S'} v}{e_{S'}} - q\sigma \geq \eta$, and $r\sigma \geq \eta$. If $\tau < 0$, then by the definition of M_0 and (16.34),

$$\frac{\mathrm{ord}_{S'} u}{e_{S'}} - p\sigma \geq p(M_0 - \sigma) \geq p\eta \geq \eta,$$

replacing u, p by v, q, we obtain $\frac{\mathrm{ord}_{S'} v}{e_{S'}} - q\sigma \geq q\eta \geq \eta$, and $r\sigma \geq M_0/2 \geq \frac{1}{2re_0P} \geq \eta$. Thus, for any non-negative integers a_1, a_2, a_3,

$$\mathrm{ord}_{\tilde{F}}(x^{a_1}y^{a_2}z^{a_3}) \geq (a_1 + a_2 + a_3)e\eta. \tag{16.37}$$

In particular, $\mathrm{center}_{X'} \tilde{F} = \{0\}$.

Step 4. In this step, we show that the orders of the h_j and the h_j^0 along $\tilde{F}$ coincide, compute the log discrepancy, and conclude the proof of (1).

Since $h_j^0(x, y, t^r) = t^{-\kappa_j}P_j(u, v)$ and $P_j \in \mathbb{C}(u, v)$,

$$\mathrm{ord}_{\tilde{F}} h_j^0(x, y, t^r) = k_1 \mathrm{ord}_{S'} P_j - \kappa_j k_2.$$

By (16.30), (16.33), and $0 < \sigma < M_0 \leq l_2$,

$$\frac{\mathrm{ord}_{\tilde{F}} h_j^0(x, y, t^r)}{e} = \frac{\mathrm{ord}_{S'} P_j}{e_{S'}} - \kappa_j\sigma \leq \frac{l_2}{\gamma} + \frac{3P_{\max}}{\gamma} l_2 \leq U_0.$$

Any monomial occurring in R_j has degree at least N, so (16.37) implies $\operatorname{ord}_{\tilde F} R_j(x,y,t^r) \geq Ne\eta > eU_0$ if $R_j \neq 0$. Thus

$$\operatorname{ord}_{\tilde F} h_j(x,y,t^r) = \operatorname{ord}_{\tilde F} h_j^0(x,y,t^r) \quad \text{for any } j. \tag{16.38}$$

Let $f : X' \to X$ be the morphism $(x,y,t) \to (x,y,t^r)$, and let

$$B' := \sum_j c_j \operatorname{div}\big(h_j(x,y,t^r)\big) - (r-1)\operatorname{div}(t).$$

By the Hurwitz formula, $K_{X'} + B' = f^*(K_X + B)$. The subgroup $\mu_r := \{(1,\alpha) \mid \alpha^r = 1\}$ of $\tilde G$ has invariant ring $\mathbb{C}[x,y,t^r] = \mathbb{C}[x,y,z]$, and $\tilde G/\mu_r \cong G$ acts on $X = X'/\mu_r$ as G. Hence $X'/\tilde G = X/G$, and the quotient morphism of X' by $\tilde G$ is $\pi \circ f$, where $\pi : X \to X/G$ is the quotient morphism. The $\mathbb{R}$-divisor B' is $\tilde G$-invariant, since $h_j(x,y,t^r)$ and t are $\tilde G$-semi-invariant. Write $(X',B')/\tilde G = (X/G, B'_{X/G})$. By Lemma 2.6(0),

$$(\pi \circ f)^*(K_{X/G} + B'_{X/G}) = K_{X'} + B' = f^*\pi^*(K_{X/G} + B_{X/G}),$$

so $B'_{X/G} = B_{X/G}$, that is, $(X',B')/\tilde G = (X,B)/G$. By (16.35), (16.38), and $\operatorname{ord}_{\tilde F} t = k_2$, we have

$$\begin{aligned} a(\tilde F, X', B') &= a(\tilde F, X', 0) - \sum_j c_j\big(k_1 \operatorname{ord}_{S'} P_j - \kappa_j k_2\big) + (r-1)k_2 \\ &= k_1 a(S', \mathbb{A}^2_{u,v}, C) + \Big(r - p - q + \sum_j c_j\kappa_j\Big)k_2 = k_1 a(S', \mathbb{A}^2_{u,v}, C) + \tau k_2. \end{aligned}$$

Let E be the prime divisor over X/G such that $\operatorname{ord}_{\tilde F}|_{\mathbb{C}(X/G)} = |I_{\tilde G}(\tilde F)|\operatorname{ord}_E$. By Claim 16.12 and Lemma 2.6(2), $\operatorname{center}_{X/G} E$ is the image of 0, and

$$a(E, X/G, B_{X/G}) = \frac{a(\tilde F, X', B')}{e} = \frac{a(S', \mathbb{A}^2_{u,v}, C)}{e_{S'}} + \tau\sigma. \tag{16.39}$$

If $\tau \geq 0$, then $a(S', \mathbb{A}^2_{u,v}, C) = a(S', V, C|_V) = e_{S'}b$ by (16.31). By (16.30) and (16.34), (16.39) implies

$$a(E, X/G, B_{X/G}) = b + \tau\sigma \leq b + 2T_0\eta.$$

Assume that $\tau < 0$, and let D_u, D_v be the strict transforms of $(u=0)$, $(v=0)$ on V'. Then

$$f_V^* \operatorname{div}(u) = \frac{p}{g_0}E_V + D_u, \qquad f_V^* \operatorname{div}(v) = \frac{q}{g_0}E_V + D_v,$$

and $D_u \cap D_v = \emptyset$, since the rays of $(1,0)$ and $(0,1)$ do not span a cone of the fan of V'. Thus $\operatorname{center}_{V'} S'$ is not contained in both D_u and D_v, and

$$\operatorname{mult}_{S'} E_V = \min\left\{\frac{g_0 \operatorname{ord}_{S'} u}{p}, \frac{g_0 \operatorname{ord}_{S'} v}{q}\right\} = g_0 e_{S'} M_0.$$

The valuation ord_{E_V} is the monomial valuation in u, v with weights $(p/g_0, q/g_0)$. Since

$$x^{a_1}y^{a_2}(t^r)^{a_3} = t^{ra_3 - pa_1 - qa_2}u^{a_1}v^{a_2}$$

and $h_j^0(x,y,t^r) = t^{-\kappa_j}P_j(u,v)$, the map $x^{a_1}y^{a_2}z^{a_3} \to u^{a_1}v^{a_2}$ is a bijection from the monomials of h_j^0 to the monomials of P_j, and $pa_1 + qa_2 = \kappa_j + ra_3$ for any monomial $x^{a_1}y^{a_2}z^{a_3}$ of h_j^0, so

$$g_0 \operatorname{ord}_{E_V} P_j = \kappa_j + r \operatorname{mult}_{(z=0)} \operatorname{div}(h_j^0).$$

Recall that $\beta_z = \sum_j c_j \operatorname{mult}_{(z=0)} \operatorname{div}(h_j^0)$. Hence

$$a(E_V, V, C|_V) = \frac{p+q}{g_0} - \sum_j c_j \operatorname{ord}_{E_V} P_j = \frac{p + q - \sum_j c_j\kappa_j - r\beta_z}{g_0} = \frac{r(1-\beta_z) + |\tau|}{g_0}.$$

Let C' be the $\mathbb{R}$-divisor on V' with $K_{V'}+C'=f_V^*(K_V+C|_V)$ and $(f_V)_*C'=C|_V$. Then

$$C'=(f_V)_*^{-1}(C|_V)+\bigl(1-a(E_V,V,C|_V)\bigr)E_V=B_{V'}-\frac{|\tau|}{g_0}E_V.$$

By [KM98, Lemma 2.30], $a(S',V',C')=a(S',V,C|_V)$, so

$$a(S',V',B_{V'})=a(S',V,C|_V)-\frac{|\tau|}{g_0}\operatorname{mult}_{S'}E_V=a(S',\mathbb{A}^2_{u,v},C)-|\tau|e_{S'}M_0.$$

Together with (16.31), $a(S',\mathbb{A}^2_{u,v},C)/e_{S'}=b+|\tau|M_0$. By (16.30), (16.34), and (16.39),

$$a(E,X/G,B_{X/G})=b+|\tau|(M_0-\sigma)\le b+2T_0\eta.$$

It remains to show the equality of orders. Let F_0 be the prime divisor over $X=X'/\mu_r$ under $\tilde{F}$ for the faithful action of μ_r on X' (Definition 2.7). By Lemma 2.6(1), $\operatorname{ord}_{\tilde{F}}|_{\mathbb{C}(X)}=d\operatorname{ord}_{F_0}$ for $d:=|I_{\mu_r}(\tilde{F})|$. By (16.38), $\operatorname{ord}_{F_0}h_j=\operatorname{ord}_{F_0}h_j^0$ for any j. The restriction of ord_{F_0} to $\mathbb{C}(X/G)$ is a positive multiple of ord_E, so F_0 is a prime divisor over X lying over E. Let F be a prime divisor over X lying over E. Since $\mathbb{C}(X)/\mathbb{C}(X)^G$ is a Galois extension with group G, the valuation rings of $\mathbb{C}(X)$ over the valuation ring of ord_E form one G-orbit [Ser79, Chapter I, §7, Proposition 19], so there exists $g\in G$ with $\operatorname{ord}_F(\varphi)=\operatorname{ord}_{F_0}(g^*\varphi)$ for any $\varphi\in\mathbb{C}(X)^*$. Since h_j and h_j^0 are G-semi-invariant, g^*h_j and $g^*h_j^0$ are non-zero constant multiples of h_j and h_j^0, and hence

$$\operatorname{ord}_F h_j=\operatorname{ord}_{F_0}h_j=\operatorname{ord}_{F_0}h_j^0=\operatorname{ord}_F h_j^0.$$

Since $\operatorname{center}_{X/G}E$ is the image of 0, $\operatorname{mld}((X\ni 0,B)/G)\le a(E,X/G,B_{X/G})\le b+2T_0\eta$. This proves (1).

Step 5. In this step, we prove (2).

Since (X,B) is klt near 0, [KM98, Lemma 2.29] implies $0<a(E_0,X,B)=3-\sum_j c_j\operatorname{ord}_0 h_j$ for the exceptional divisor E_0 of the blow-up of X at 0. Hence $\operatorname{ord}_0 h_j<3/\gamma$, and the inequality $|pa_1+qa_2-ra_3|\le P_{\max}(a_1+a_2+a_3)$ of **Step 1** applied to m_j implies $|\kappa_j|<3P_{\max}/\gamma=C_1$.

Since $\gcd(p,q,r)=1$, the homomorphism $s\to(s^p,s^q,s^{-r})$ from $\mathbb{C}^*$ to $\mathbf{T}$ is injective, so $G\cap\mathbf{T}$ is the image of the group of n-th roots of unity, and the element with parameter s multiplies a monomial of $\mathbf{T}$-weight κ by s^κ. Since G is diagonal and h_j is G-semi-invariant, any monomial occurring in h_j has the character of h_j. Hence the $\mathbf{T}$-weight of any monomial occurring in h_j is κ_j+ln for some integer l. If $l\ne 0$, then $|\kappa_j+ln|\ge n-|\kappa_j|>n-C_1$, so the degree of the monomial is greater than $(n-C_1)/P_{\max}$, hence at least N_n. Thus $h_j-h_j^0\in(x,y,z)^{N_n}$. The polynomial h_j^0 contains m_j, so $h_j^0\ne 0$, and h_j^0 is semi-invariant for $\mathbf{T}$ with $\mathbf{T}$-weight κ_j and for G with the character of h_j.

Let ℓ be the integer l of Theorem 2.19(2) for the threefold X and the DCC set $\Gamma\backslash\{0\}$. The integer ℓ depends only on Γ. Let $\mathfrak{a}:=\prod_j(h_j)^{c_j}$ and $\mathfrak{a}^0:=\prod_j(h_j^0)^{c_j}$. For any prime divisor F over X, $a(F,X,0,\mathfrak{a})=a(F,X,B)$ and $a(F,X,0,\mathfrak{a}^0)=a(F,X,B^0)$, where $B^0:=\sum_j c_j\operatorname{div}(h_j^0)$. Assume that $N_n\ge\ell$. Then $(h_j)+\mathfrak{m}_0^\ell=(h_j^0)+\mathfrak{m}_0^\ell$ for any j, where $\mathfrak{m}_0$ is the maximal ideal of 0, and Theorem 2.19(2), applied to the closed point $0\in X$ and the boundary 0, implies

$$\operatorname{mld}(X\ni 0,B^0)=\operatorname{mld}(X\ni 0,0,\mathfrak{a}^0)=\operatorname{mld}(X\ni 0,0,\mathfrak{a})=\operatorname{mld}(X\ni 0,B)\ge 0,$$

where the last inequality holds since (X,B) is klt near 0. Hence (X,B^0) is lc near 0, since otherwise $\operatorname{mld}(X\ni 0,B^0)=-\infty$ [Amb99, Section 1, the second paragraph after Definition 1.1].

Let n_0 be the least integer such that $n_0\ge 2C_1$, $(n_0-C_1)/P_{\max}\ge\ell$, and $2\max\{1,2U_0P_{\max}\}/n_0\le\eta_0$. The integer n_0 depends only on p,q,r,e_0,Γ. Assume that $n\ge n_0$. Then $N_n\ge\ell\ge 1$, $N_n\ge(n-C_1)/P_{\max}\ge n/(2P_{\max})$, and the conclusions above hold. Let

$$\eta:=\max\left\{\frac{2}{n},\frac{2U_0}{N_n}\right\}.$$

Then

$$\eta \le \max\left\{\frac{2}{n}, \frac{4U_0P_{\max}}{n}\right\} = \frac{2\max\{1, 2U_0P_{\max}\}}{n} \le \eta_0,$$

$n\eta \ge 2$, and $N_n\eta \ge 2U_0 > U_0$. By (1) for the polynomials h_j^0, the integer N_n, and $R_j := h_j - h_j^0$,

$$\operatorname{mld}((X \ni 0, B)/G) \le b + 2T_0\eta \le b + \frac{4T_0\max\{1, 2U_0P_{\max}\}}{n} = b + \frac{C_0}{n}.$$

Finally, if $\operatorname{mld}((X \ni 0, B)/G) \ge \epsilon > 0$, then $b \ge \epsilon - \frac{C_0}{n} \ge \frac{\epsilon}{2}$ for $n \ge \max\{n_0, 2C_0/\epsilon\}$. □

16.7. **Thickenings along a marked curve.** In this subsection, we prove that on the product of a smooth surface germ with $\mathbb{A}^1$, perturbations divisible by a high power of a local equation of a curve do not change the mld (Theorem 16.13), and we extend Theorem 16.13 to quotients by finite diagonal groups of bounded order (Proposition 16.15).

In the proof of Theorem 16.21, the terms of higher order are divisible by a high power of a local equation d of the marked curve, and we need that such perturbations do not change the mld of the product with $\mathbb{A}^1$, with the exponent depending only on a DCC set containing the other coefficients and on a lower bound for the coefficients along the curve.

Theorem 16.13. *Let $\Gamma \subset [0,1]$ be a DCC set with $\Gamma\backslash\{0\} \ne \emptyset$, $\gamma := \gamma(\Gamma)$, $\delta := \delta(\Gamma)$, and $H_M := 2 + \frac{1+M}{\delta}$ for real numbers $M \ge 0$. Let $(S \ni p)$ be a smooth surface germ, D a non-zero reduced curve through p with local equation d, C an $\mathbb{R}$-divisor supported on D with coefficients in $[-M, 1]$, and $B_0 := \sum_l b_l \operatorname{div}(f_l)$ with $b_l \in \Gamma$ positive and f_l non-zero regular functions near p, such that $(S, C + B_0)$ is sub-lc near p and $\mu := \operatorname{mld}(S \ni p, C + B_0) \ge 0$. Let $X := S \times \mathbb{A}^1_t$, $o := (p, 0)$, and $C_X := \operatorname{pr}_S^* C$. Then:*

(1) μ is attained by a prime divisor E over S with center p such that $a(E, S, 0) \le H_M \operatorname{ord}_E d$ and $\operatorname{ord}_E f_l < N \operatorname{ord}_E d$ for any l and any integer N with $\gamma N > H_M + M$;

(2) for any integer N with $\gamma N > H_M + M$ and any non-zero regular functions h_l near o with $h_l - f_l \in (d^N)\mathcal{O}_X$,

$$\operatorname{mld}\Big(X \ni o, C_X + \sum_l b_l \operatorname{div}(h_l)\Big) = \mu + 1,$$

and the Gauss extension of ord_E to $\mathbb{C}(S)(t)$ with t-value 1 computes both $\operatorname{mld}(X \ni o, C_X + \sum_l b_l \operatorname{div}(h_l))$ and $\operatorname{mld}(X \ni o, C_X + \operatorname{pr}_S^* B_0)$ and has equal values on f_l and on h_l for any l.

Proof. The proof is long, so we divide it into several steps. By Lemma 2.10(2), $\gamma > 0$ and $\delta > 0$.

Step 1. In this step we fix a prime divisor E' over S with center p, introduce the blow-ups extracting E', and bound A_i/d_i as long as the multiplicity m_i of the strict transform of B_0 is larger than one.

Let E' be a prime divisor over S with center p. Blowing up the centers of $\operatorname{ord}_{E'}$ one after another, we obtain blow-ups

$$S_r \to S_{r-1} \to \cdots \to S_1 \to S_0 = S$$

such that E' is the exceptional curve of $S_r \to S_{r-1}$ [KM98, Lemma 2.45]. Here $S_i \to S_{i-1}$ is the blow-up of a closed point $p_{i-1} \in S_{i-1}$ with exceptional curve E_i, and $p_i \in E_i$ for $1 \le i < r$. Let $g_i : S_i \to S$ be the composite morphism. Let $D_i := (g_i)_*^{-1} D$ and $B_i := (g_i)_*^{-1} B_0$, and for $k < i$ let E_k also stand for the strict transform of E_k on S_i. The divisor $E_1 + \cdots + E_i$ on S_i is snc, since blowing up a closed point of a smooth surface keeps the exceptional configuration snc (cf. [Har77, Proposition V.3.6]). We say that p_i is *free* if $p_i \notin E_k$ for any $k < i$, and *satellite* otherwise. Since $E_1 + \cdots + E_i$ is snc and $p_i \in E_i$, a satellite p_i belongs to exactly one E_k with $k < i$. For $1 \le i \le r$ let

$$A_i := a(E_i, S, 0), \quad d_i := \operatorname{ord}_{E_i} d, \quad a_i := a(E_i, S, C + B_0), \quad m_i := \operatorname{mult}_{p_{i-1}} B_{i-1}.$$

Since d vanishes at p, $d_i \ge 1$. Since $\operatorname{center}_S E_i = \{p\}$, $a_i \ge \mu \ge 0$.

Since $K_{S_i} = g_i^* K_S + \sum_{k\le i}(A_k - 1)E_k$ and $g_i^* \operatorname{div}(d) = D_i + \sum_{k\le i} d_k E_k$, [KM98, Lemma 2.29] implies that for $1 \le i < r$,

$$\begin{aligned} &A_{i+1} = A_i + 1, \quad d_{i+1} = d_i + \operatorname{mult}_{p_i} D_i && \text{if } p_i \text{ is free,} \\ &A_{i+1} = A_i + A_k, \quad d_{i+1} = d_i + d_k + \operatorname{mult}_{p_i} D_i && \text{if } p_i \in E_k \text{ with } k < i. \end{aligned} \tag{16.40}$$

By [Har77, Corollary V.3.7] (the computation is local, and both sides are linear in B_i), $(B_i \cdot E_i) = m_i$ for $1 \le i \le r$. Since E_i is smooth, [Har77, Exercise I.5.4(a)] (the argument is local, and it extends to $\mathbb{R}$-divisors by linearity) implies that for $1 \le i < r$,

$$m_{i+1} = \operatorname{mult}_{p_i} B_i \le (B_i \cdot E_i)_{p_i} \le (B_i \cdot E_i) = m_i.$$

Thus $m_1 \ge m_2 \ge \cdots \ge m_r$. Each m_i is a finite sum of elements of $\Gamma\backslash\{0\}$, namely of the coefficients b_l counted with multiplicities, so $m_i > 1$ implies $m_i \ge 1 + \delta$.

Let $1 \le i \le r$ with $m_i > 1$. Then $m_k \ge 1 + \delta$ for any $k \le i$. For $k \le i$ let r_k be the order along E_i of the maximal ideal of $p_{k-1} \in S_{k-1}$, which is a positive integer. Since the maximal ideal of p_{k-1} generates $\mathcal{O}_{S_k}(-E_k)$, K_{S_k} is the pullback of $K_{S_{k-1}}$ plus E_k, and the pullback of B_{k-1} to S_k is $B_k + m_k E_k$ [Har77, Proposition V.3.6], we have

$$A_i = 1 + \sum_{k\le i} r_k \quad \text{and} \quad \operatorname{mult}_{E_i} B_0 = \sum_{k\le i} r_k m_k.$$

Since $C \ge -MD$, $-\operatorname{mult}_{E_i} C \le M d_i$. Hence

$$0 \le a_i = A_i - \operatorname{mult}_{E_i} B_0 - \operatorname{mult}_{E_i} C \le 1 + \sum_{k\le i} r_k(1 - m_k) + M d_i \le 1 - \delta \sum_{k\le i} r_k + M d_i,$$

so $A_i \le 1 + \frac{1+Md_i}{\delta}$. Since $d_i \ge 1$, we obtain

$$\frac{A_i}{d_i} \le 1 + \frac{1+M}{\delta} = H_M - 1 \quad \text{if } m_i > 1. \tag{16.41}$$

Step 2. In this step we define a stopping index $j \le r$ and prove that

$$A_i \le H_M d_i \quad \text{for any } 1 \le i \le j. \tag{16.42}$$

Let j be the least integer $1 \le j < r$ such that $m_j \le 1$, p_j is free, and $p_j \notin D_j$. If no such integer exists, let $j := r$.

We prove (16.42) by induction on i. For $i = 1$, $A_1 = 2$ and $d_1 = \operatorname{mult}_p D \ge 1$, so $A_1/d_1 \le 2 < H_M$. Let $2 \le i \le j$. If $m_i > 1$, then (16.42) for i follows from (16.41). Suppose that $m_i \le 1$. If $p_{i-1} \in E_k$ for some $k < i - 1$, then by (16.40) and induction,

$$\frac{A_i}{d_i} \le \frac{A_{i-1} + A_k}{d_{i-1} + d_k} \le \max\left\{\frac{A_{i-1}}{d_{i-1}}, \frac{A_k}{d_k}\right\} \le H_M.$$

Suppose that p_{i-1} is free. If $m_{i-1} > 1$, then by (16.40) and (16.41),

$$\frac{A_i}{d_i} \le \frac{A_{i-1} + 1}{d_{i-1}} \le \frac{A_{i-1}}{d_{i-1}} + 1 \le H_M.$$

If $m_{i-1} \le 1$, then $p_{i-1} \in D_{i-1}$ by the minimality of j, since $i - 1 < j$. Thus $d_i \ge d_{i-1} + 1$ by (16.40), and

$$\frac{A_i}{d_i} \le \frac{A_{i-1} + 1}{d_{i-1} + 1} \le \max\left\{\frac{A_{i-1}}{d_{i-1}}, 1\right\} \le H_M.$$

Step 3. In this step we prove that for any integer N with $\gamma N > H_M + M$,

$$\operatorname{ord}_{E_i} f_l < N d_i \quad \text{for any } l \text{ and any } 1 \le i \le j. \tag{16.43}$$

Let $1 \le i \le j$. Since $b_l \operatorname{div}(f_l) \ge 0$ for any l, $a_i \ge 0$, $-\operatorname{mult}_{E_i} C \le M d_i$, and by (16.42),

$$b_l \operatorname{ord}_{E_i} f_l \le \operatorname{mult}_{E_i} B_0 \le A_i - \operatorname{mult}_{E_i} C \le A_i + M d_i \le (H_M + M) d_i < \gamma N d_i.$$

Since $b_l \ge \gamma$, (16.43) follows.

Step 4. In this step we prove (1).

Since $(S, C + B_0)$ is sub-lc near p, μ is the minimum of the log discrepancies of finitely many prime divisors over S with center p [Amb99, Section 1, the second paragraph after Definition 1.1]. Thus there exists a prime divisor E' over S with center p and $a(E', S, C + B_0) = \mu$. Let $S_r \to \cdots \to S_0$, E_i, p_i, a_i, m_i, and j be the data of **Steps 1** and 2 for E'. If $j = r$, then $E := E_r = E'$ satisfies (1) by (16.42) and (16.43).

Suppose that $j < r$. Since p_j is free and $p_j \notin D_j$, near p_j the curve E_j is the only exceptional curve of g_j and the strict transform of C is zero. Hence near p_j,

$$K_{S_j} + (1 - a_j)E_j + B_j = g_j^*(K_S + C + B_0).$$

Since $(B_j \cdot E_j) = m_j \le 1$, the coefficient ℓ of p_j in $B_j|_{E_j}$ satisfies $0 \le \ell \le 1$. Since S_j and E_j are smooth, the different of $(S_j, E_j + B_j)$ on E_j is $B_j|_{E_j}$, which is ℓp_j near p_j. Thus Theorem 2.23, applied on a neighborhood of p_j, implies that $(S_j, E_j + B_j)$ is lc near p_j. Since $\operatorname{center}_{S_j} E' = \{p_j\}$, $\operatorname{mult}_{E'} E_j \ge 1$, and $a_j \ge 0$, [KM98, Lemma 2.30] implies

$$\mu = a(E', S_j, (1 - a_j)E_j + B_j) = a(E', S_j, E_j + B_j) + a_j \operatorname{mult}_{E'} E_j \ge a_j \operatorname{mult}_{E'} E_j \ge a_j \ge \mu.$$

Thus $a_j = \mu$, and $E := E_j$ satisfies (1) by (16.42) and (16.43).

Step 5. In this step we fix an integer N with $\gamma N > H_M + M$ and functions h_l as in (2), let $B^h := C_X + \sum_l b_l \operatorname{div}(h_l)$ near o, and prove that $a(F, X, B^h) \ge \mu + 1$ for any prime divisor F over X with center o.

Let F be a prime divisor over X with center o. Then ord_F is non-negative on $\mathcal{O}_{S,p}$ and positive on the maximal ideal of p, so the restriction of ord_F to $\mathbb{C}(S)$ is a non-trivial valuation with center p. By Lemma 2.12, applied to $\mathbb{C}(S) \subset \mathbb{C}(X)$ and the point p, the restriction of ord_F to $\mathbb{C}(S)$ equals $e \operatorname{ord}_{E'}$ for an integer $e \ge 1$ and a prime divisor E' over S with center p. Let $S_r \to \cdots \to S_0$, E_i, p_i, a_i, m_i, and j be the data of **Steps 1** and 2 for E'. Let $X_i := S_i \times \mathbb{A}^1_t$. Write $h_l = f_l + d^N \phi_l$ with ϕ_l regular near o.

For $i \le j$, (16.43) and $\operatorname{ord}_{E_i \times \mathbb{A}^1} d^N = N d_i$ imply $\operatorname{ord}_{E_i \times \mathbb{A}^1} h_l = \operatorname{ord}_{E_i} f_l$. Since $K_{X_i} - (g_i \times \mathrm{id})^* K_X$ is the pullback of $K_{S_i} - g_i^* K_S$ and $\operatorname{mult}_{E_i \times \mathbb{A}^1} C_X = \operatorname{mult}_{E_i} C$, we obtain $a(E_i \times \mathbb{A}^1, X, B^h) = a_i$ for $i \le j$.

Suppose that $j < r$. The image in S_j of the center of F on X_j is the center p_j of $\operatorname{ord}_{E'}$ on S_j, and $\operatorname{ord}_F t > 0$. Hence the center of F on X_j is $(p_j, 0)$. Let u be a local equation of E_j at p_j. Since near p_j the curve E_j is the only exceptional curve of g_j and $D_j = 0$, we may write near p_j

$$g_j^* d = u^{d_j} w \quad \text{and} \quad g_j^* f_l = u^{c_l} \tilde{f}_l,$$

where w is a unit, $c_l := \operatorname{ord}_{E_j} f_l$, and $\tilde{f}_l$ is a local equation of the strict transform of $\operatorname{div}(f_l)$. Then

$$(g_j \times \mathrm{id})^* h_l = u^{c_l} \tilde{h}_l, \qquad \tilde{h}_l := \tilde{f}_l + u^{N d_j - c_l} w^N (g_j \times \mathrm{id})^* \phi_l,$$

and $N d_j - c_l \ge 1$ by (16.43). Thus $\tilde{h}_l|_{E_j \times \mathbb{A}^1}$ is the pullback of $\tilde{f}_l|_{E_j} \ne 0$ under the projection $E_j \times \mathbb{A}^1 \to E_j$. Let $B_j^h := \sum_l b_l \operatorname{div}(\tilde{h}_l)$ near $(p_j, 0)$. Then $B_j^h \ge 0$, near $(p_j, 0)$

$$K_{X_j} + (1 - a_j)(E_j \times \mathbb{A}^1) + B_j^h = (g_j \times \mathrm{id})^*(K_X + B^h),$$

and $B_j^h|_{E_j \times \mathbb{A}^1} = \ell(\{p_j\} \times \mathbb{A}^1)$ near $(p_j, 0)$, where ℓ is the coefficient of p_j in $B_j|_{E_j}$. Since $(B_j \cdot E_j) = m_j \le 1$, $0 \le \ell \le 1$. Since X_j and $E_j \times \mathbb{A}^1$ are smooth, the different of $(X_j, E_j \times \mathbb{A}^1 + B_j^h + \operatorname{div}(t))$ on $E_j \times \mathbb{A}^1$ is $\ell(\{p_j\} \times \mathbb{A}^1) + E_j \times \{0\}$ near $(p_j, 0)$, which is an snc divisor with coefficients at most one, hence lc [KM98, Corollary 2.31(3)]. By Theorem 2.23, applied on a neighborhood of $(p_j, 0)$, the pair $(X_j, E_j \times \mathbb{A}^1 + B_j^h + \operatorname{div}(t))$ is lc near $(p_j, 0)$. Since $\operatorname{mult}_F(E_j \times \mathbb{A}^1) \ge 1$, $\operatorname{ord}_F t \ge 1$, and $a_j \ge \mu \ge 0$, [KM98, Lemma 2.30] implies

$$\begin{aligned} a(F, X, B^h) &= a(F, X_j, E_j \times \mathbb{A}^1 + B_j^h + \operatorname{div}(t)) + a_j \operatorname{mult}_F(E_j \times \mathbb{A}^1) + \operatorname{ord}_F t \\ &\ge a_j + 1 \ge \mu + 1. \end{aligned} \tag{16.44}$$

Suppose that $j = r$. The image in S_r of the center of F on X_r is the generic point of E_r, and $\operatorname{ord}_F t > 0$. Hence the center of F on X_r is $E_r \times \{0\}$. Let η be the generic point of $E_r \times \{0\}$ and u a local equation of E_r at the generic point of E_r. Then u, t are regular parameters of $\mathcal{O}_{X_r,\eta}$. The curves E_k with $k < r$, D_r, and the strict transforms of $\operatorname{div}(f_l)$ do not contain E_r, so at the generic point of E_r we may write $g_r^* d = u^{d_r} w$ and $g_r^* f_l = u^{c_l} \tilde{f}_l$ with w, $\tilde{f}_l$ units and $c_l := \operatorname{ord}_{E_r} f_l$. Then

$$(g_r \times \mathrm{id})^* h_l = u^{c_l} \big(\tilde{f}_l + u^{N d_r - c_l} w^N (g_r \times \mathrm{id})^* \phi_l \big),$$

where $N d_r - c_l \geq 1$ by (16.43), so the second factor is a unit at η. Thus there exists a smooth open neighborhood U of η in X_r on which u and t are regular, $\operatorname{div}(u) + \operatorname{div}(t)$ is snc, and $K_U + (1 - a_r) \operatorname{div}(u)$ is the restriction of $(g_r \times \mathrm{id})^*(K_X + B^h)$. By [KM98, Lemma 2.30] and Lemma 2.13,

$$a(F, X, B^h) = a(F, U, (1 - a_r) \operatorname{div}(u)) \geq a_r \operatorname{ord}_F u + \operatorname{ord}_F t \geq \mu + 1. \tag{16.45}$$

By (16.44) and (16.45), $a(F, X, B^h) \geq \mu + 1$.

Step 6. In this step we prove (2).

Let N, h_l, and B^h be as in **Step 5**, and let E be the prime divisor of **Step 4**. Let $S' \to S$ be the composite of the blow-ups extracting E (**Step 1**), $X' := S' \times \mathbb{A}^1_t$, η' the generic point of $E \times \{0\} \subset X'$, and u a local equation of E at the generic point of E. Then u, t are regular parameters of $\mathcal{O}_{X',\eta'}$. Let P be the exceptional divisor of the blow-up of a neighborhood of η' along the closure of η'. For $a_s \in \mathbb{C}(S)$, write $a_s = u^{\operatorname{ord}_E a_s} w_s$ with w_s a unit at the generic point of E. Since u, t are regular parameters of $\mathcal{O}_{X',\eta'}$, ord_P is the order of vanishing along the maximal ideal of $\mathcal{O}_{X',\eta'}$, whose associated graded ring is the polynomial ring in the classes of u and t. Since each w_s is a unit at η', after multiplying by a power of u we obtain

$$\operatorname{ord}_P \Big(\sum_s a_s t^s \Big) = \min_s (\operatorname{ord}_E a_s + s).$$

Thus ord_P is the Gauss extension of ord_E to $\mathbb{C}(S)(t)$ with t-value 1 (Definition-Proposition 2.11), and $\operatorname{center}_X P = \{o\}$. Since $a(P, X', 0) = 2$ and $K_{X'} - (\text{pullback of } K_X)$ is the pullback of $K_{S'} - (\text{pullback of } K_S)$, which has coefficient $a(E, S, 0) - 1$ along E, we obtain $a(P, X, 0) = a(E, S, 0) + 1$. By (1),

$$\operatorname{ord}_P (h_l - f_l) \geq N \operatorname{ord}_P d = N \operatorname{ord}_E d > \operatorname{ord}_E f_l = \operatorname{ord}_P f_l,$$

so $\operatorname{ord}_P h_l = \operatorname{ord}_P f_l = \operatorname{ord}_E f_l$. Since moreover $\operatorname{mult}_P C_X = \operatorname{mult}_E C$,

$$a(P, X, B^h) = a(E, S, 0) + 1 - \operatorname{mult}_E C - \sum_l b_l \operatorname{ord}_E f_l = a(E, S, C + B_0) + 1 = \mu + 1.$$

The computation of $a(P, X, B^h)$ with $h_l = f_l$ implies $a(P, X, C_X + \operatorname{pr}_S^* B_0) = \mu + 1$. Since **Step 5** holds both for the functions h_l and for the functions f_l, P computes $\operatorname{mld}(X \ni o, B^h) = \mu + 1$ and $\operatorname{mld}(X \ni o, C_X + \operatorname{pr}_S^* B_0) = \mu + 1$. This proves (2). □

We next descend Theorem 16.13 to quotients by finite diagonal groups of bounded order with an invariant parameter. The constants are attached to the following set-up, which is also used in Proposition 16.16.

Set-up 16.14. Let e_0, Γ, Γ_*, γ_*, and δ_* be as follows.

(1) $e_0 \geq 1$ is an integer and $\Gamma \subset [0, 1]$ is a DCC set.
(2) $\gamma_* := \gamma(\Gamma_*)$ and $\delta_* := \delta(\Gamma_*)$, where

$$\Gamma_* := \bigcup_{q=1}^{e_0} \frac{1}{q} \Gamma \cup \Big\{ \frac{k}{q} \;\Big|\; 1 \leq k \leq q \leq e_0 \Big\}.$$

Proposition 16.15. *Notation as in Set-up 16.14. Fix a real number $M \geq 0$, and let $H_M := 2 + \frac{1+M}{\delta_*}$. Let J be a faithful finite diagonal group of order $q \leq e_0$ on $S := \mathbb{A}^2_{x,y}$, $X := S \times \mathbb{A}^1_t$ with t invariant, $C := c \operatorname{div}_S(x)$ with $-M \leq c \leq 1$, f_i non-zero regular J-semi-invariant functions near 0, $b_i \in \Gamma$ positive, such that $(S, C + \sum_i b_i \operatorname{div}(f_i))$ is sub-lc near 0, and let $\mu := \operatorname{mld}((S \ni 0, C + \sum_i b_i \operatorname{div}(f_i))/J) \geq 0$. Let N be an integer with $\gamma_* \lfloor N/e_0 \rfloor > H_M + M$. Then for any non-zero regular J-semi-invariant functions h_i near $(0,0)$ with the characters of f_i and $h_i - f_i \in (x^N)$,*

$$\operatorname{mld}\Big((X \ni (0,0), \operatorname{pr}_S^* C + \sum_i b_i \operatorname{div}(h_i))/J\Big) = \mu + 1.$$

Moreover, there exists a prime divisor E over S/J computing μ such that the Gauss extension of ord_E to $\mathbb{C}(S/J)(t)$ with t-value 1 computes both

$$\operatorname{mld}\Big((X \ni (0,0), \operatorname{pr}_S^* C + \sum_i b_i \operatorname{div}(h_i))/J\Big) \quad \text{and}$$

$$\operatorname{mld}\Big((X \ni (0,0), \operatorname{pr}_S^* C + \sum_i b_i \operatorname{pr}_S^* \operatorname{div}(f_i))/J\Big),$$

and has equal values on h_i^q and f_i^q for any i.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we compute the crepant quotients and descend the divisibility $h_i - f_i \in (x^N)$.

By Lemma 2.10(1), $\bigcup_{q'=1}^{e_0} \frac{1}{q'}\Gamma$ is a DCC set, so $\Gamma_* \subset [0,1]$ is a DCC set, as the union with a finite set. Since $1 \in \Gamma_*$, Lemma 2.10(2) implies $\gamma_* > 0$ and $\delta_* > 0$. The constants $\gamma(\Gamma_*) = \gamma_*$, $\delta(\Gamma_*) = \delta_*$, and H_M are the constants of Theorem 16.13 for the DCC set Γ_*.

Let $\pi : S \to S/J$ be the quotient morphism and $\bar{0} := \pi(0)$. Since J acts trivially on t, we have $X/J = (S/J) \times \mathbb{A}^1_t$, and $\pi \times \mathrm{id}$ is the quotient morphism of X. Since J fixes 0, any neighborhood of 0 contains a J-invariant affine one. Possibly shrinking S and X to J-invariant affine neighborhoods of 0 and $(0,0)$, we may assume that the functions f_i and h_i are regular and that (S, B) is sub-lc, where $B := C + \sum_i b_i \operatorname{div}(f_i)$. Let $B^h := \operatorname{pr}_S^* C + \sum_i b_i \operatorname{div}(h_i)$ on X, $(S,B)/J = (S/J, B_{S/J})$, and $(X, B^h)/J = ((S/J) \times \mathbb{A}^1_t, B^h_{X/J})$. Then $(X, \operatorname{pr}_S^* B)/J = ((S/J) \times \mathbb{A}^1_t, \operatorname{pr}^* B_{S/J})$ and $\mu = \operatorname{mld}(S/J \ni \bar{0}, B_{S/J})$.

The functions x, y are J-semi-invariant. Let e_x (resp. e_y) be the order of the subgroup of J consisting of the elements σ with $\sigma^* y = y$ (resp. $\sigma^* x = x$). A non-trivial element $\sigma \in J$ with $\sigma^* x \neq x$ and $\sigma^* y \neq y$ fixes only the origin, so the prime divisors of S with non-trivial inertia group are among $(x = 0)$ and $(y = 0)$, with inertia groups of orders e_x and e_y. By the Hurwitz formula,

$$R(\pi) = (e_x - 1) \operatorname{div}_S(x) + (e_y - 1) \operatorname{div}_S(y) \quad \text{and} \quad R(\pi \times \mathrm{id}) = \operatorname{pr}_S^* R(\pi).$$

Let $d_x := x^q$ and $d_y := y^q$. Since the characters of J have orders dividing q, the functions d_x, d_y, f_i^q, and h_i^q are J-invariant, hence regular on S/J or on $(S/J) \times \mathbb{A}^1_t$. Since π has degree q, $\pi_* \pi^*$ is multiplication by q on divisors, and $\pi^* \operatorname{div}(f_i^q) = q \operatorname{div}_S(f_i)$ implies $\pi_* \operatorname{div}_S(f_i) = \operatorname{div}(f_i^q)$. Likewise $\pi_* \operatorname{div}_S(x) = \operatorname{div}(d_x)$, $\pi_* \operatorname{div}_S(y) = \operatorname{div}(d_y)$, and, since $\pi \times \mathrm{id}$ also has degree q,

$$(\pi \times \mathrm{id})_* \operatorname{pr}_S^* \operatorname{div}_S(x) = \operatorname{pr}^* \operatorname{div}(d_x), \quad (\pi \times \mathrm{id})_* \operatorname{pr}_S^* \operatorname{div}_S(y) = \operatorname{pr}^* \operatorname{div}(d_y),$$
$$(\pi \times \mathrm{id})_* \operatorname{div}(h_i) = \operatorname{div}(h_i^q).$$

By Definition 2.5,

$$\begin{aligned} B_{S/J} &= \frac{c + e_x - 1}{q} \operatorname{div}(d_x) + \frac{e_y - 1}{q} \operatorname{div}(d_y) + \sum_i \frac{b_i}{q} \operatorname{div}(f_i^q), \\ B^h_{X/J} &= \operatorname{pr}^* \Big(\frac{c + e_x - 1}{q} \operatorname{div}(d_x) + \frac{e_y - 1}{q} \operatorname{div}(d_y) \Big) + \sum_i \frac{b_i}{q} \operatorname{div}(h_i^q). \end{aligned} \tag{16.46}$$

The numbers $\frac{e_x-1}{q}$ and $\frac{e_y-1}{q}$, when non-zero, and $\frac{b_i}{q}$ belong to $\Gamma_*\backslash\{0\}$, since $e_x, e_y \le q \le e_0$.

Let $L := \lfloor N/q \rfloor$. Since $q \le e_0$, $\gamma_* L \ge \gamma_* \lfloor N/e_0 \rfloor > H_M + M$. Since

$$h_i^q - f_i^q = (h_i - f_i) \sum_{a=0}^{q-1} h_i^{q-1-a} f_i^a$$

and x^{qL} divides x^N, the function $\phi_i := (h_i^q - f_i^q)/d_x^L$ is regular near $(0,0)$. Since ϕ_i is J-invariant, ϕ_i is regular on $(S/J) \times \mathbb{A}^1_t$ near $(\bar{0}, 0)$. Thus

$$h_i^q - f_i^q \in (d_x^L)\mathcal{O}_{(S/J)\times\mathbb{A}^1_t,(\bar{0},0)}. \tag{16.47}$$

Step 2. In this step we pull back to the minimal resolution of S/J and check the hypotheses of Theorem 16.13 at the points over $\bar{0}$.

Let $g : Y \to S/J$ be the minimal resolution, with $g = \mathrm{id}_{S/J}$ if S/J is smooth. By [CLS11, Proposition 1.3.18], S/J is the affine toric surface of the cone $\mathbb{R}^2_{\ge 0}$ in the lattice N_J of Definition 2.15, formed with the characters of x and y. A vector $(v_x, 0)$ belongs to N_J if and only if $e^{2\pi\sqrt{-1}v_x}$ is the character of x at an element $\sigma \in J$ with $\sigma^* y = y$. Thus $N_J \cap \mathbb{R}_{\ge 0}(1,0)$ and $N_J \cap \mathbb{R}_{\ge 0}(0,1)$ are generated by $(\frac{1}{e_x}, 0)$ and $(0, \frac{1}{e_y})$. For a primitive $v = (v_x, v_y) \in N_J \cap \mathbb{R}^2_{\ge 0}$, let E_v be the corresponding toric prime divisor over S/J. The linear function $e_x v_x + e_y v_y$ takes the value 1 at $(\frac{1}{e_x}, 0)$ and $(0, \frac{1}{e_y})$, so $a(E_v, S/J, 0) = e_x v_x + e_y v_y$ by [CLS11, Lemma 11.4.10], and, since d_x and d_y are the characters of $(q,0)$ and $(0,q)$, $\mathrm{ord}_{E_v} d_x = q v_x$ and $\mathrm{ord}_{E_v} d_y = q v_y$ [CLS11, Proposition 4.1.2]. The minimal resolution g is toric: the toric resolution of S/J of [Ful93, §2.6, pp. 45–47] has exceptional curves E' with $(E'^2) \le -2$ [Ful93, §2.6, p. 47, Exercise (c)], so it is the minimal resolution, and g satisfies the hypothesis of [KM98, Corollary 4.3] that K_Y is g-nef. Moreover, the exceptional curves of g are the curves E_v for finitely many primitive $v \in N_J \cap \mathbb{R}^2_{\ge 0}$. By [KM98, Corollary 4.3 and Proposition 5.20], $0 < a(E_v, S/J, 0) \le 1$ for any exceptional curve E_v of g, so $0 < v_x \le \frac{1}{e_x} \le 1$.

Let D_Y be the reduced divisor with support $\mathrm{Supp}\,\mathrm{div}_Y(g^* d_x)$. The components of D_Y are the strict transform on Y of the image of $(x=0)$ and the exceptional curves of g, so $g^{-1}(\bar{0}) \subset \mathrm{Supp}\, D_Y$. Let

$$C_Y := \sum_E (1 - a(E, S/J, 0))E + \frac{c}{q}\,\mathrm{div}_Y(g^* d_x),$$

where E runs over the exceptional curves of g, let

$$B_Y := \frac{e_x - 1}{q}\,\mathrm{div}_Y(g^* d_x) + \frac{e_y - 1}{q}\,\mathrm{div}_Y(g^* d_y) + \sum_i \frac{b_i}{q}\,\mathrm{div}_Y(g^* f_i^q),$$

and let

$$B_Y^h := \frac{e_x - 1}{q}\,\mathrm{pr}^*\,\mathrm{div}_Y(g^* d_x) + \frac{e_y - 1}{q}\,\mathrm{pr}^*\,\mathrm{div}_Y(g^* d_y) + \sum_i \frac{b_i}{q}\,\mathrm{div}((g\times\mathrm{id})^* h_i^q)$$

on $Y \times \mathbb{A}^1_t$. Since $K_Y = g^* K_{S/J} + \sum_E (a(E, S/J, 0) - 1)E$, (16.46) implies

$$\begin{aligned} K_Y + C_Y + B_Y &= g^*(K_{S/J} + B_{S/J}), \\ K_{Y\times\mathbb{A}^1_t} + \mathrm{pr}^* C_Y + B_Y^h &= (g\times\mathrm{id})^*(K_{(S/J)\times\mathbb{A}^1_t} + B^h_{X/J}). \end{aligned} \tag{16.48}$$

We have $B_Y \ge 0$, and the non-zero numbers among $\frac{e_x-1}{q}$, $\frac{e_y-1}{q}$, and $\frac{b_i}{q}$ in the definition of B_Y belong to $\Gamma_*\backslash\{0\}$ by **Step 1**.

The $\mathbb{R}$-divisor C_Y is supported on D_Y. The coefficient of an exceptional curve E_v in C_Y is $1 - e_x v_x - e_y v_y + c v_x$. Since $e_x v_x + e_y v_y \le 1$ and $0 < v_x \le 1$, $1 - e_x v_x - e_y v_y + c v_x \ge \min\{0, c\} \ge -M$. Since π^* of the image of $(x=0)$ is e_x times $(x=0)$, the coefficient of the strict transform of the image of $(x=0)$ in C_Y is $\frac{c}{e_x} \ge -M$. By Lemma 2.6(2) and (3), $(S/J, B_{S/J})$ is sub-lc near $\bar{0}$. Hence $(Y, C_Y + B_Y)$ is sub-lc near $g^{-1}(\bar{0})$, so the coefficients of $C_Y + B_Y$ near $g^{-1}(\bar{0})$ are at most one. Since $B_Y \ge 0$, the coefficients of C_Y near $g^{-1}(\bar{0})$ belong to $[-M, 1]$.

Let $y_0 \in g^{-1}(\bar{0})$ be a closed point and d a local equation of D_Y at y_0. Since the coefficients of $\operatorname{div}_Y(g^*d_x)$ are positive integers, $g^*d_x \in (d)\mathcal{O}_{Y,y_0}$, and (16.47) implies

$$(g \times \mathrm{id})^* h_i^q - g^* f_i^q \in (d^L)\mathcal{O}_{Y\times\mathbb{A}^1_t,(y_0,0)}. \tag{16.49}$$

Any prime divisor over Y with center y_0 has center $\bar{0}$ on S/J, so by (16.48) and [KM98, Lemma 2.30], $\operatorname{mld}(Y \ni y_0, C_Y + B_Y) \geq \mu \geq 0$.

Step 3. In this step we prove that $a(P, (S/J) \times \mathbb{A}^1_t, B^h_{X/J}) \geq \mu + 1$ for any prime divisor P over $(S/J) \times \mathbb{A}^1_t$ with center $(\bar{0}, 0)$ whose center on $Y \times \mathbb{A}^1_t$ is a closed point.

The center of P on $Y \times \mathbb{A}^1_t$ is a closed point $(y_0, 0)$ with $y_0 \in g^{-1}(\bar{0})$. By **Steps 1** and 2 and (16.49), the hypotheses of Theorem 16.13 hold for the DCC set Γ_*, the germ $(Y \ni y_0)$, the curve D_Y with local equation d, the $\mathbb{R}$-divisor C_Y, the $\mathbb{R}$-divisor B_Y written as in **Step 2**, the integer L, and the functions g^*d_x, g^*d_y, $(g \times \mathrm{id})^* h_i^q$ in place of g^*d_x, g^*d_y, $g^*f_i^q$. By Theorem 16.13(2), (16.48), and [KM98, Lemma 2.30],

$$a(P, (S/J) \times \mathbb{A}^1_t, B^h_{X/J}) = a(P, Y \times \mathbb{A}^1_t, \operatorname{pr}^* C_Y + B^h_Y) \geq \operatorname{mld}(Y \ni y_0, C_Y + B_Y) + 1 \geq \mu + 1.$$

Step 4. In this step we prove that $a(P, (S/J) \times \mathbb{A}^1_t, B^h_{X/J}) \geq \mu + 1$ for any prime divisor P over $(S/J) \times \mathbb{A}^1_t$ whose center on $Y \times \mathbb{A}^1_t$ is $E \times \{0\}$ for an exceptional curve E of g, and that $E \times \mathbb{A}^1_t$ is the only component of $\operatorname{pr}^* C_Y + B^h_Y$ through the generic point of $E \times \{0\}$.

Let E be an exceptional curve of g, c_E the coefficient of E in C_Y, and $\alpha_E := a(E, S/J, B_{S/J})$. Since $\operatorname{center}_{S/J} E = \{\bar{0}\}$, $\alpha_E \geq \mu \geq 0$. By (16.48), $\operatorname{mult}_E B_Y = 1 - \alpha_E - c_E \leq 1 + M$. Since $\frac{b_i}{q} \geq \gamma_*$ and $\gamma_* L > H_M + M > 1 + M$,

$$\operatorname{ord}_E g^* f_i^q \leq \frac{1+M}{\gamma_*} < L \leq L \operatorname{ord}_E g^* d_x. \tag{16.50}$$

Let η be the generic point of $E \times \{0\}$ and z a local equation of E at the generic point of E. Then z, t are regular parameters of $\mathcal{O}_{Y\times\mathbb{A}^1_t,\eta}$. By (16.47) and (16.50), $(g \times \mathrm{id})^* h_i^q = z^{k_i} w_i$ with $k_i := \operatorname{ord}_E g^* f_i^q$ and w_i a unit at η. The other components of C_Y and of $\operatorname{div}_Y(g^*d_x)$, $\operatorname{div}_Y(g^*d_y)$ do not contain E. Hence $E \times \mathbb{A}^1_t$ is the only component of $\operatorname{pr}^* C_Y + B^h_Y$ through η, and the coefficient of $E \times \mathbb{A}^1_t$ in $\operatorname{pr}^* C_Y + B^h_Y$ equals the coefficient $1 - \alpha_E$ of E in $C_Y + B_Y$. Let U be a smooth open neighborhood of η on which z and t are regular, $\operatorname{div}(z) + \operatorname{div}(t)$ is snc, and $\operatorname{pr}^* C_Y + B^h_Y$ restricts to $(1 - \alpha_E)\operatorname{div}(z)$. For a prime divisor P with $\operatorname{center}_{Y\times\mathbb{A}^1_t} P = E \times \{0\}$, (16.48), [KM98, Lemma 2.30], and Lemma 2.13 imply

$$\begin{aligned} a(P, (S/J) \times \mathbb{A}^1_t, B^h_{X/J}) &= a(P, U, (1 - \alpha_E)\operatorname{div}(z)) \\ &\geq \alpha_E \operatorname{ord}_P z + \operatorname{ord}_P t \geq \alpha_E + 1 \geq \mu + 1. \end{aligned} \tag{16.51}$$

Step 5. In this step we prove the equality and construct E.

Let P be a prime divisor over $(S/J)\times\mathbb{A}^1_t$ with center $(\bar{0}, 0)$. Since $Y\times\mathbb{A}^1_t \to (S/J)\times\mathbb{A}^1_t$ is proper, the center of P on $Y \times \mathbb{A}^1_t$ is contained in $g^{-1}(\bar{0}) \times \{0\}$, and $\dim g^{-1}(\bar{0}) \leq 1$. Thus the center of P on $Y \times \mathbb{A}^1_t$ is a closed point or $E \times \{0\}$ for an exceptional curve E of g, and **Steps 3** and 4 imply $a(P, (S/J) \times \mathbb{A}^1_t, B^h_{X/J}) \geq \mu + 1$. The case $h_i = f_i$ implies $a(P, (S/J) \times \mathbb{A}^1_t, \operatorname{pr}^* B_{S/J}) \geq \mu + 1$.

Since $(S/J, B_{S/J})$ is sub-lc near $\bar{0}$ (**Step 2**), μ is the minimum of the log discrepancies of finitely many prime divisors over S/J with center $\bar{0}$ [Amb99, Section 1, the second paragraph after Definition 1.1]. Thus there exists a prime divisor E_0 over S/J with $a(E_0, S/J, B_{S/J}) = \mu$.

Suppose that the center of E_0 on Y is a closed point y_0. Then by **Step 2**,

$$\mu \leq \operatorname{mld}(Y \ni y_0, C_Y + B_Y) \leq a(E_0, Y, C_Y + B_Y) = \mu,$$

so $\operatorname{mld}(Y \ni y_0, C_Y + B_Y) = \mu$. By Theorem 16.13(1) and (2), applied as in **Step 3**, there exists a prime divisor E over Y with center y_0 and $a(E, Y, C_Y + B_Y) = \mu$ such that the Gauss extension of ord_E to $\mathbb{C}(Y)(t) = \mathbb{C}(S/J)(t)$ with t-value 1 has log discrepancy $\mu + 1$ with respect to both $(Y \times \mathbb{A}^1_t, \operatorname{pr}^* C_Y + B^h_Y)$ and $(Y \times \mathbb{A}^1_t, \operatorname{pr}^*(C_Y + B_Y))$, and has equal values on $(g \times \mathrm{id})^* h_i^q$ and

$g^* f_i^q$. By (16.48) and [KM98, Lemma 2.30], E is a prime divisor over S/J computing μ, and the Gauss extension has center $(\bar{0}, 0)$ and log discrepancy $\mu + 1$ with respect to $((S/J) \times \mathbb{A}^1_t, B^h_{X/J})$ and $((S/J) \times \mathbb{A}^1_t, \mathrm{pr}^* B_{S/J})$.

Suppose that the center of E_0 on Y is the generic point of an exceptional curve E of g. Then $\mathrm{ord}_{E_0} = \mathrm{ord}_E$ and $\alpha_E = \mu$. Let η and z be as in **Step 4**. Let U_h and U_f be the neighborhoods U of η in **Step 4** for the functions h_i and for the functions f_i in place of h_i, and let $U := U_h \cap U_f$. Then U is a smooth open neighborhood of η on which z and t are regular, and $\mathrm{pr}^* C_Y + B^h_Y$ and $\mathrm{pr}^*(C_Y + B_Y)$ both restrict to $(1 - \mu)\,\mathrm{div}(z)$ on U. Let P' be the exceptional divisor of the blow-up of U along the closure of η. For $a_s \in \mathbb{C}(S/J)$, write $a_s = z^{\mathrm{ord}_E a_s} w'_s$ with w'_s a unit at the generic point of E. Since z, t are regular parameters of $\mathcal{O}_{Y \times \mathbb{A}^1_t, \eta}$, $\mathrm{ord}_{P'}$ is the order of vanishing along the maximal ideal of $\mathcal{O}_{Y \times \mathbb{A}^1_t, \eta}$, whose associated graded ring is the polynomial ring in the classes of z and t. Since each w'_s is a unit at η, after multiplying by a power of z we obtain

$$\mathrm{ord}_{P'} \Big(\sum_s a_s t^s \Big) = \min_s (\mathrm{ord}_E a_s + s).$$

Hence $\mathrm{ord}_{P'}$ is the Gauss extension of ord_E with t-value 1 (Definition-Proposition 2.11), and $\mathrm{center}_{(S/J) \times \mathbb{A}^1_t} P' = (\bar{0}, 0)$. Thus by (16.48) and [KM98, Lemmas 2.29 and 2.30], P' has log discrepancy $2 - (1 - \mu) = \mu + 1$ with respect to $((S/J) \times \mathbb{A}^1_t, B^h_{X/J})$ and $((S/J) \times \mathbb{A}^1_t, \mathrm{pr}^* B_{S/J})$. By (16.47) and (16.50),

$$\mathrm{ord}_{P'} \big((g \times \mathrm{id})^* h_i^q - g^* f_i^q \big) \geq L \,\mathrm{ord}_E\, g^* d_x > \mathrm{ord}_E\, g^* f_i^q = \mathrm{ord}_{P'}\, g^* f_i^q,$$

so $\mathrm{ord}_{P'}$ has equal values on h_i^q and f_i^q.

In both cases, together with the lower bound above, the Gauss extension computes both mlds, which are equal to $\mu + 1$. Since $(X, B^h)/J = ((S/J) \times \mathbb{A}^1_t, B^h_{X/J})$ and $(X, \mathrm{pr}_S^* B)/J = ((S/J) \times \mathbb{A}^1_t, \mathrm{pr}^* B_{S/J})$, Proposition 16.15 follows. □

16.8. **Perturbations at interior nodes.** In this subsection, we show that replacing the functions $f_j(uv, z)$ on $\mathbb{A}^3_{u,v,z}$ by functions congruent to them modulo $(uv)^N$ does not change the minimal log discrepancies of the quotient by a bounded diagonal group at the points of the image of $(uv = 0)$.

Proposition 16.16. *Notation as in Set-up 16.14. Fix a real number $\kappa \geq 1$, and let $N \geq 1$ be an integer with*

$$\gamma_* \Big(\frac{N}{e_0} - 1 \Big) > 1 + \kappa \Big(1 + \frac{1}{\delta_*} \Big).$$

Let J be a faithful finite diagonal group of order at most e_0 acting on $V := \mathbb{A}^3_{u,v,z}$, let $0 \leq \lambda_0, \lambda_1 \leq \kappa$ be real numbers, and for $1 \leq j \leq s$ let $c_j \in \Gamma$ be positive, $f_j(w, z)$ non-zero polynomials with $f_j(0, z) \neq 0$, and $H_j(u, v, z)$ non-zero polynomials such that H_j and $f_j(uv, z)$ are J-semi-invariant with equal characters and $H_j - f_j(uv, z) \in ((uv)^N)$. Let

$$B_0 := (1 - \lambda_0)(u = 0) + (1 - \lambda_1)(v = 0) + \sum_j c_j \,\mathrm{div}(f_j(uv, z)),$$

$$B_H := (1 - \lambda_0)(u = 0) + (1 - \lambda_1)(v = 0) + \sum_j c_j \,\mathrm{div}(H_j),$$

and let ξ be a scheme-theoretic point of the image Z of $(uv = 0)$ in V/J such that (V, B_0) is sub-lc near the inverse image of ξ. Then

$$\mathrm{mld}(V/J \ni \xi, (B_H)_{V/J}) = \mathrm{mld}(V/J \ni \xi, (B_0)_{V/J}).$$

More precisely, if ξ is a closed point, then for any prime divisor E over V/J with center $\overline{\{\xi\}}$ there exists a prime divisor E' over V/J with center $\overline{\{\xi\}}$ such that

$$a(E', V/J, (B_H)_{V/J}) = a(E', V/J, (B_0)_{V/J}) \leq \min\{a(E, V/J, (B_H)_{V/J}), a(E, V/J, (B_0)_{V/J})\}.$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we fix the notation, and in Claim 16.17 we reduce the case when ξ is a closed point to finding one prime divisor $E^\sharp$ over the quotient U/I by the inertia group I.

Let $\pi : V \to V/J$ be the quotient morphism. Since H_j and $f_j(uv, z)$ are J-semi-invariant with equal characters, $\Phi_j := H_j/f_j(uv, z)$ belongs to $\mathbb{C}(V)^J$, and

$$B_H - B_0 = \sum_j c_j \operatorname{div}_V(\Phi_j).$$

We use the following consequence of the last equality. Let G be a finite group acting faithfully on a normal quasi-projective variety X, and let B and B' be G-invariant $\mathbb{R}$-divisors on X such that $K_X + B$ is $\mathbb{R}$-Cartier and $B' - B = \sum_j c_j \operatorname{div}_X(\Psi_j)$ for some $\Psi_j \in \mathbb{C}(X)^G$. Since the pullback of $\mathbb{R}$-divisors under $X \to X/G$ is injective and sends $\operatorname{div}_{X/G}(\Psi_j)$ to $\operatorname{div}_X(\Psi_j)$, we have $B'_{X/G} = B_{X/G} + \sum_j c_j \operatorname{div}_{X/G}(\Psi_j)$, hence

$$a(D, X/G, B'_{X/G}) = a(D, X/G, B_{X/G}) - \sum_j c_j \operatorname{ord}_D \Psi_j \tag{16.52}$$

for any prime divisor D over X/G.

For any integer $m \geq 1$, let $M_m := \kappa m - 1 \geq 0$. Since $\lfloor \frac{mN}{e_0} \rfloor > \frac{mN}{e_0} - 1 \geq m(\frac{N}{e_0} - 1)$, the hypothesis on N implies

$$\gamma_* \Big\lfloor \frac{mN}{e_0} \Big\rfloor > m + \kappa m\Big(1 + \frac{1}{\delta_*}\Big) \geq 1 + \kappa m + \frac{\kappa m}{\delta_*} = 2 + \frac{1 + M_m}{\delta_*} + M_m. \tag{16.53}$$

Until the end of **Step 5**, we assume that ξ is a closed point, and we fix a prime divisor E over V/J with $\operatorname{center}_{V/J} E = \{\xi\}$. For $\star \in \{0, H\}$ and any prime divisor D over V/J, let $a_\star(D) := a(D, V/J, (B_\star)_{V/J})$. By Lemma 2.6(3), there exists a prime divisor F over V such that $\operatorname{ord}_F|_{\mathbb{C}(V/J)} = e \operatorname{ord}_E$ for some integer e. Let $I := I_J(F)$. By Lemma 2.6(1)(2), $e = |I|$, $a(F, V, B_\star) = |I| a_\star(E)$, and $\pi(\operatorname{center}_V F) = \{\xi\}$, so $\operatorname{center}_V F = \{x\}$ for a closed point $x = (u_0, v_0, z_0)$. Since $\pi^{-1}(\xi)$ is contained in the J-invariant divisor $(uv = 0)$, we have $u_0 v_0 = 0$. Since I fixes F, it fixes x. For $s' \in \{u, v, z\}$ and $g \in J$, write $g^* s' = \chi_{s'}(g) s'$. If $s'(x) \neq 0$, then $\chi_{s'}(g) s'(x) = s'(g \cdot x) = s'(x)$ for any $g \in I$, so $\chi_{s'}|_I = 1$. In particular, $z - z_0$ is I-semi-invariant.

Let Y be a normal variety with an action of I and an I-equivariant proper birational morphism $Y \to V$, let $U \subset Y$ be an I-invariant quasi-projective open subset such that F has a center on U (in **Steps 2** and **3**, U is affine), and let $\varphi : U \to V$ be the induced morphism. For $\star \in \{0, H\}$, define $B_{U,\star}$ by $K_U + B_{U,\star} = \varphi^*(K_V + B_\star)$, let $\pi_U : U \to U/I$ be the quotient morphism, and let $a_{U,\star}(D) := a(D, U/I, (B_{U,\star})_{U/I})$ for any prime divisor D over U/I. Let E_I be the prime divisor over U/I such that $\operatorname{ord}_F|_{\mathbb{C}(U/I)} = e_I \operatorname{ord}_{E_I}$ for some integer e_I. Since $I_I(F) = I$, Lemma 2.6(1)(2) and [KM98, Lemma 2.30] imply that $e_I = |I|$ and

$$a_{U,\star}(E_I) = \frac{a(F, U, B_{U,\star})}{|I|} = \frac{a(F, V, B_\star)}{|I|} = a_\star(E). \tag{16.54}$$

Claim 16.17. *Let U and $\varphi : U \to V$ be as defined before (16.54), and let $E^\sharp$ be a prime divisor over U/I such that $\operatorname{center}_{U/I} E^\sharp \subset \pi_U(\varphi^{-1}(x))$ and*

$$a_{U,H}(E^\sharp) = a_{U,0}(E^\sharp) \leq \min\{a_{U,H}(E_I), a_{U,0}(E_I)\}.$$

Then there exists a prime divisor E' over V/J with $\operatorname{center}_{V/J} E' = \{\xi\}$ and

$$a_H(E') = a_0(E') \leq \min\{a_H(E), a_0(E)\}.$$

Proof. By Lemma 2.6(1)–(3) applied to the action of I on U, there exists a prime divisor $F^\sharp$ over U such that $\operatorname{ord}_{F^\sharp}|_{\mathbb{C}(U/I)} = |I_I(F^\sharp)| \operatorname{ord}_{E^\sharp}$ and $\pi_U(\operatorname{center}_U F^\sharp) = \operatorname{center}_{U/I} E^\sharp$. Since I fixes x, $\varphi^{-1}(x)$ is I-invariant, so $\operatorname{center}_U F^\sharp \subset \varphi^{-1}(x)$, and $F^\sharp$ is a prime divisor over V with

center x. Let E' be the prime divisor over V/J such that $\operatorname{ord}_{F^\sharp}|_{\mathbb{C}(V/J)} = |I_J(F^\sharp)|\operatorname{ord}_{E'}$. Then $\operatorname{center}_{V/J} E' = \{\pi(x)\} = \{\xi\}$, and Lemma 2.6(1)(2) and [KM98, Lemma 2.30] imply

$$|I_J(F^\sharp)|a_\star(E') = a(F^\sharp, V, B_\star) = a(F^\sharp, U, B_{U,\star}) = |I_I(F^\sharp)|a_{U,\star}(E^\sharp).$$

Since $I_I(F^\sharp) = I \cap I_J(F^\sharp)$ is a subgroup of $I_J(F^\sharp)$, the number $d := |I_J(F^\sharp)|/|I_I(F^\sharp)|$ is a positive integer, and $a_\star(E') = a_{U,\star}(E^\sharp)/d$ for both $\star$. Thus $a_H(E') = a_0(E')$. Since (V, B_0) is sub-lc near x, $a(F^\sharp, V, B_0) \geq 0$, so $a_{U,0}(E^\sharp) \geq 0$. Therefore, by (16.54),

$$a_H(E') = a_0(E') \leq a_{U,0}(E^\sharp) \leq \min\{a_{U,H}(E_I), a_{U,0}(E_I)\} = \min\{a_H(E), a_0(E)\}.$$

□

Proof of Proposition 16.16 continued. In **Steps 2–5**, we construct an open subset U and a prime divisor $E^\sharp$ satisfying the hypotheses of Claim 16.17, according to the position of x and of $\operatorname{center}_U F$.

Step 2. In this step, we treat the case when $f_j(0, z_0) \neq 0$ for all j and the case when x is not a node, that is, $(u_0, v_0) \neq (0, 0)$.

Assume that $f_j(0, z_0) \neq 0$ for all j. Since uv vanishes at x and $H_j - f_j(uv, z) \in ((uv)^N)$, we have $H_j(x) = f_j(0, z_0) \neq 0$. Thus $\operatorname{ord}_F \Phi_j = 0$, so $\operatorname{ord}_E \Phi_j = 0$ for all j, and (16.52) applied to J acting on V with $B = B_0$ and $B' = B_H$ implies $a_H(E) = a_0(E)$. Hence $E' := E$ satisfies the conclusion.

Assume that x is not a node. Possibly interchanging u and v, and λ_0 and λ_1, we may assume that $u_0 = 0$ and $v_0 \neq 0$. By **Step 1**, $\chi_v|_I = 1$. Let $Y := V$, $U := (v \neq 0) \subset V$, φ the inclusion, $w := uv$, and $S := \mathbb{A}^2_{w,z}$. The functions w, z, v induce an I-equivariant isomorphism $U \cong S \times \mathbb{G}_{m,v}$, where I acts on S diagonally through χ_u and χ_z, and trivially on $\mathbb{G}_{m,v}$. The action of I on S is faithful: an element of I acting trivially on w and z acts trivially on $u = w/v$, v, and z, hence is the identity since J acts faithfully on V. Write $H_j = f_j(uv, z) + (uv)^N R_j$ with $R_j \in \mathbb{C}[u, v, z]$. On U, we have $(u = 0) = (w = 0)$ and $(v = 0) \cap U = \emptyset$, so

$$B_{U,0} = \operatorname{pr}_S^* B_S, \quad B_S := (1 - \lambda_0)(w = 0) + \sum_j c_j \operatorname{div}_S(f_j(w, z)),$$

and $B_{U,H} = (1 - \lambda_0)(w = 0) + \sum_j c_j \operatorname{div}_U(H_j)$, where $H_j|_U = f_j(w, z) + w^N R_j(w/v, v, z)$.

For any prime divisor D over S, the prime divisor $D \times \mathbb{G}_m$ over U satisfies

$$a(D \times \mathbb{G}_m, U, B_{U,0}) = a(D, S, B_S),$$

as one sees on $S' \times \mathbb{G}_m$ for a higher model $S' \to S$ on which D is a divisor. Since (V, B_0) is sub-lc near x, (S, B_S) is sub-lc near $o := (0, z_0)$, and $\mu := \operatorname{mld}((S \ni o, B_S)/I) \geq 0$ by Lemma 2.6.

The group I acting on S with eigenparameters w and $z - z_0$, the variety $X := S \times \mathbb{A}^1_t$ with $t := v - v_0$, the numbers $c := 1 - \lambda_0$ and $M := M_1 = \kappa - 1$, and the functions $f_j(w, z)$ with coefficients c_j satisfy the hypotheses of Proposition 16.15. Here $-M \leq c \leq 1$, $|I| \leq |J| \leq e_0$, and the numerical hypothesis on N holds by (16.53) with $m = 1$. Near x, U is an I-invariant open subset of X, and $B_{U,H}$ and $B_{U,0}$ are the boundaries of Proposition 16.15 for $h_j := H_j|_U$ and $h_j := f_j(w, z)$ respectively. Both choices satisfy $h_j - f_j(w, z) \in (w^N)$ and have the characters of the $f_j(w, z)$. Thus Proposition 16.15 implies that

$$\operatorname{mld}(U/I \ni \pi_U(x), (B_{U,\star})_{U/I}) = \mu + 1$$

for both $\star$, and that there exists a prime divisor E_1 over U/I computing both minimal log discrepancies. Hence $\operatorname{center}_{U/I} E_1 = \{\pi_U(x)\}$ and $a_{U,H}(E_1) = a_{U,0}(E_1) = \mu + 1$. Since $\operatorname{center}_{U/I} E_I = \{\pi_U(x)\}$, we have $a_{U,\star}(E_I) \geq \mu + 1$ for both $\star$. Thus $E^\sharp := E_1$ satisfies the conditions of Claim 16.17, and Claim 16.17 implies the existence of E'.

Step 3. In this step, we assume that x is a node, that is, $u_0 = v_0 = 0$, and we construct U, an étale cover W of U, and prime divisors computing the minimal log discrepancies at closed points of W/I.

Since $u(x) = v(x) = 0$, we have $\operatorname{ord}_F u > 0$ and $\operatorname{ord}_F v > 0$. Let $n := \gcd(\operatorname{ord}_F u, \operatorname{ord}_F v)$, $a := \operatorname{ord}_F u/n$, $b := \operatorname{ord}_F v/n$, and $m := a + b \geq 2$, and fix integers m_1, m_2 with $am_1 + bm_2 = 1$. Let

$$\sigma := u^{m_1} v^{m_2}, \quad y_1 := u^{-b} v^a, \quad \text{so that} \quad u = \sigma^a y_1^{-m_2}, \quad v = \sigma^b y_1^{m_1}, \quad uv = \sigma^m y_1^{m_1 - m_2}.$$

Let $Y_2 \to \mathbb{A}^2_{u,v}$ be the toric morphism given by the subdivision of the positive quadrant along the ray $\mathbb{R}_{\geq 0}(a, b)$, and let $Y := Y_2 \times \mathbb{A}^1_z$. Since (m_1, m_2) and $\pm(-b, a)$ generate the monoid of lattice points pairing non-negatively with (a, b), the affine open subset of Y_2 corresponding to $\mathbb{R}_{\geq 0}(a, b)$ is $\operatorname{Spec} \mathbb{C}[\sigma, y_1^{\pm 1}]$. Let $U := \operatorname{Spec} \mathbb{C}[\sigma, y_1^{\pm 1}, z] \subset Y$ and let $\varphi : U \to V$ be the induced morphism. Since I acts diagonally on u, v, and z, it acts on Y through the torus, U is I-invariant, φ is I-equivariant, and σ and y_1 are eigenfunctions. We have

$$\operatorname{ord}_F \sigma = m_1 \operatorname{ord}_F u + m_2 \operatorname{ord}_F v = n > 0, \quad \operatorname{ord}_F y_1 = -b \operatorname{ord}_F u + a \operatorname{ord}_F v = 0,$$

and $\operatorname{ord}_F(z - z_0) > 0$, so ord_F is non-negative on $\mathbb{C}[\sigma, y_1^{\pm 1}, z]$, F has a center on U, and

$$\operatorname{center}_U F \subset C := (\sigma = 0, z = z_0) \subset \varphi^{-1}(x).$$

The group I acts trivially on the residue field of ord_F, and y_1 has a non-zero residue there since $\operatorname{ord}_F y_1 = 0$. Comparing residues in $g^* y_1 = \chi_{y_1}(g) y_1$ for $g \in I$, we obtain $\chi_{y_1}|_I = 1$. Hence I acts trivially on $C \cong \mathbb{G}_{m, y_1}$, and $C_I := \pi_U(C)$ is a curve isomorphic to C.

A direct calculation shows that

$$\varphi^*(du \wedge dv \wedge dz) = \sigma^{m-1} y_1^{m_1 - m_2 - 1} d\sigma \wedge dy_1 \wedge dz,$$

$\varphi^*(u = 0) = a(\sigma = 0)$, and $\varphi^*(v = 0) = b(\sigma = 0)$. Thus

$$B_{U,0} = (1 - \alpha)(\sigma = 0) + \sum_j c_j \operatorname{div}_U(f_j(uv, z)), \quad B_{U,H} = (1 - \alpha)(\sigma = 0) + \sum_j c_j \operatorname{div}_U(H_j),$$

where $\alpha := a\lambda_0 + b\lambda_1$. Since $0 \leq \alpha \leq \kappa m$, we have $-M_m \leq 1 - \alpha \leq 1$.

Let $W := \operatorname{Spec} \mathbb{C}[\sigma', \varsigma^{\pm 1}, z]$, and let $\psi : W \to U$ be the morphism given by

$$y_1 = \varsigma^m, \quad \sigma = \sigma' \varsigma^{m_2 - m_1}.$$

Then $\psi^*(uv) = \sigma'^m$, and $\psi^* u$ and $\psi^* v$ are σ'^a and σ'^b times integral powers of ς. Let the group $G := \mu_m(m_1 - m_2, 1, 0)$ act on W through the coordinates (σ', ς, z). Then $\mathbb{C}[\sigma', \varsigma^{\pm 1}, z]^G = \mathbb{C}[\sigma, y_1^{\pm 1}, z]$, and G acts freely on W since it acts freely on the coordinate ς. Hence ψ is the quotient morphism by G, and ψ is étale. Let I act on W by $g^* \sigma' = \chi_\sigma(g) \sigma'$, $g^* \varsigma = \varsigma$, and $g^* z = \chi_z(g) z$. Since $\chi_{y_1}|_I = 1$, ψ is I-equivariant, and the actions of I and G on W commute. From now on, the letters S, B_S, o, μ, X, t, c, M, and h_j denote the new objects defined below. Their meanings in **Step 2** are not used in **Steps 3–5**. Let $S := \mathbb{A}^2_{\sigma', z}$, so that $W = S \times \mathbb{G}_{m, \varsigma}$ and I acts on S diagonally with eigenparameters σ' and $z - z_0$. The action of I on S is faithful: an element of I acting trivially on σ' and z acts trivially on W, hence on U and on V, as ψ and φ are dominant. Write $H_j = f_j(uv, z) + (uv)^N R_j$ with $R_j \in \mathbb{C}[u, v, z]$, and let

$$h_j := \psi^* \varphi^* H_j = f_j(\sigma'^m, z) + \sigma'^{mN} R'_j, \quad R'_j := \psi^* \varphi^* R_j \in \mathbb{C}[\sigma', \varsigma^{\pm 1}, z].$$

For $\star \in \{0, H\}$, let $B_{W,\star} := \psi^* B_{U,\star}$, so that $K_W + B_{W,\star} = \psi^*(K_U + B_{U,\star})$, and

$$B_S := (1 - \alpha)(\sigma' = 0) + \sum_j c_j \operatorname{div}_S(f_j(\sigma'^m, z)),$$

$$B_{W,0} = \operatorname{pr}_S^* B_S,$$

$$B_{W,H} = (1 - \alpha)(\sigma' = 0) + \sum_j c_j \operatorname{div}_W(h_j).$$

We have $W/I = (S/I) \times \mathbb{G}_{m, \varsigma}$ since I acts trivially on ς, and $(B_{W,0})_{W/I} = \operatorname{pr}_1^*(B_S)_{S/I}$. The action of G descends to a free action on W/I, and the induced morphism $\psi_I : W/I \to U/I$ is the quotient morphism by G, hence étale. For any prime divisor D over W/I, let

$a_{W,\star}(D) := a(D, W/I, (B_{W,\star})_{W/I})$. Since the pullback of $\mathbb{R}$-divisors under $W \to W/I$ is injective, $K_{W/I}+(B_{W,\star})_{W/I} = \psi_I^*(K_{U/I}+(B_{U,\star})_{U/I})$, so $(B_{W,\star})_{W/I} = \psi_I^*(B_{U,\star})_{U/I}$ and the crepant quotient of $(W/I, (B_{W,\star})_{W/I})$ by G is $(U/I, (B_{U,\star})_{U/I})$. The inertia group in G of a prime divisor D over W/I acts trivially on the residue field of ord_D, so it fixes $\mathrm{center}_{W/I} D$ pointwise, and it is trivial since G acts freely. Thus Lemma 2.6(1)–(3) applied to G acting on W/I implies the following: for any prime divisor D over W/I, $\mathrm{ord}_D|_{\mathbb{C}(U/I)} = \mathrm{ord}_{D_U}$ for a prime divisor D_U over U/I with

$$a_{U,\star}(D_U) = a_{W,\star}(D) \quad \text{and} \quad \mathrm{center}_{U/I} D_U = \psi_I(\mathrm{center}_{W/I} D),$$

and any prime divisor over U/I is D_U for some prime divisor D over W/I.

Since (V, B_0) is sub-lc near x, [KM98, Lemma 2.30] implies that $(U, B_{U,0})$ is sub-lc near C, and Lemma 2.6(2)(3) applied to G acting freely on W implies that $(W, B_{W,0})$ is sub-lc near $\psi^{-1}(C) = \widetilde{C} := (\sigma' = 0, z = z_0)$. Since $B_{W,0} = \mathrm{pr}_S^* B_S$, for any prime divisor D over S we have $a(D \times \mathbb{G}_m, W, B_{W,0}) = a(D, S, B_S)$, as one sees on $S' \times \mathbb{G}_m$ for a higher model $S' \to S$ on which D is a divisor. Since $\widetilde{C} = \{o\} \times \mathbb{G}_m$ for $o := (0, z_0)$, it follows that (S, B_S) is sub-lc near o. Let $\mu := \mathrm{mld}((S \ni o, B_S)/I)$. Then $\mu \geq 0$ by Lemma 2.6.

Let $p' \in W/I$ be the image of a closed point $(0, \varsigma_0, z_0) \in \widetilde{C}$. The group I acting on S with eigenparameters σ' and $z - z_0$, the variety $X := S \times \mathbb{A}^1_t$ with $t := \varsigma - \varsigma_0$, the numbers $c := 1-\alpha$ and $M := M_m$, the functions $f_j(\sigma'^m, z)$ with coefficients c_j, and the integer mN in place of N satisfy the hypotheses of Proposition 16.15. In particular, the numerical hypothesis holds by (16.53). Near $(0, \varsigma_0, z_0)$, W is an I-invariant open subset of X, and $B_{W,H}$ and $B_{W,0}$ are the boundaries of Proposition 16.15 for the functions h_j and $f_j(\sigma'^m, z)$ respectively, which are congruent to $f_j(\sigma'^m, z)$ modulo σ'^{mN} and are I-semi-invariant with the characters of the $f_j(\sigma'^m, z)$. Thus Proposition 16.15 implies that

$$\mathrm{mld}(W/I \ni p', (B_{W,\star})_{W/I}) = \mu + 1 \tag{16.55}$$

for both $\star$, and that there exists a prime divisor $E_{p'}$ over W/I with $\mathrm{center}_{W/I} E_{p'} = \{p'\}$ and $a_{W,H}(E_{p'}) = a_{W,0}(E_{p'}) = \mu + 1$. Let $y \in C_I$ be a closed point. The prime divisors over U/I with center $\{y\}$ are the D_U for the prime divisors D over W/I whose center is a closed point of $\psi_I^{-1}(y)$, and any such closed point is the image of a closed point of $\widetilde{C}$. Hence (16.55) implies

$$\mathrm{mld}(U/I \ni y, (B_{U,\star})_{U/I}) = \mu + 1 \quad \text{for any closed point } y \in C_I \text{ and both } \star. \tag{16.56}$$

Step 4. In this step, we treat the case when x is a node and $\mathrm{center}_U F$ is a closed point $p \in C$.

In this case, $\mathrm{center}_{U/I} E_I = \{\pi_U(p)\}$. By **Step 3**, there exists a prime divisor $\widetilde{E}$ over W/I with $\widetilde{E}_U = E_I$, so $\mathrm{center}_{W/I} \widetilde{E}$ is a closed point $p' \in \psi_I^{-1}(\pi_U(p))$, p' is the image of a closed point of $\widetilde{C}$, and $a_{W,\star}(\widetilde{E}) = a_{U,\star}(E_I)$. By (16.55), $a_{W,\star}(\widetilde{E}) \geq \mu + 1$ for both $\star$. Let $E^\sharp := (E_{p'})_U$. By **Step 3**,

$$a_{U,H}(E^\sharp) = a_{U,0}(E^\sharp) = \mu + 1 \leq \min\{a_{U,H}(E_I), a_{U,0}(E_I)\},$$

and $\mathrm{center}_{U/I} E^\sharp = \{\psi_I(p')\} = \{\pi_U(p)\} \subset \pi_U(\varphi^{-1}(x))$. Thus Claim 16.17 implies the existence of E'.

Step 5. In this step, we treat the case when x is a node and $\mathrm{center}_U F = C$.

In this case, $\mathrm{center}_{U/I} E_I = C_I$. We first show that $a_{U,\star}(E_I) \geq \mu$ for both $\star$. Let η be the generic point of C_I. Since $(U, B_{U,0})$ is sub-lc near C, $(U/I, (B_{U,0})_{U/I})$ is sub-lc near η by Lemma 2.6(2)(3). The sub-pair $(U/I, (B_{U,H})_{U/I})$ is also sub-lc near η: otherwise, since the non-sub-lc locus of a sub-pair is closed, being the image of the components with coefficient > 1 of the crepant pull-back to a log resolution (cf. [Amb99, Section 1, the second paragraph after Definition 1.1]), the center of some prime divisor over U/I with negative log discrepancy with respect to $(B_{U,H})_{U/I}$ contains η, hence contains a closed point $y \in C_I$, and then $\mathrm{mld}(U/I \ni y, (B_{U,H})_{U/I}) = -\infty$ by [Amb99, Section 1, the second paragraph after Definition 1.1], which contradicts (16.56). By Lemma 2.4 and (16.56),

$$\mathrm{mld}(U/I \ni \eta, (B_{U,\star})_{U/I}) = (\mu + 1) - 1 = \mu$$

for both $\star$. Hence $a_{U,\star}(E_I) \geq \mu$.

Next we construct a prime divisor over W/I with center $\widetilde{C}_I$, the image of $\widetilde{C}$ in W/I, at which both log discrepancies are equal to μ. Let o_I be the image of o in S/I, and let $\mathcal{E}$ be the set of prime divisors D over S/I such that $\operatorname{center}_{S/I} D = \{o_I\}$ and $a(D, S/I, (B_S)_{S/I}) = \mu$.

Claim 16.18. *The set $\mathcal{E}$ is countable.*

Proof. Possibly replacing S/I with a neighborhood of o_I, we may assume that $(S/I, (B_S)_{S/I})$ is sub-lc. By [KM98, Theorem 0.2], there exists a log resolution $g_0 : S_0 \to S/I$ of $(S/I, (B_S)_{S/I})$ such that $g_0^{-1}(o_I)$ is a divisor and $\operatorname{Supp} g_{0*}^{-1}(B_S)_{S/I} \cup \operatorname{Exc}(g_0) \cup g_0^{-1}(o_I)$ is snc. Consider a finite sequence of blow-ups $S_k \to S_{k-1} \to \cdots \to S_0$, where $S_{i+1} \to S_i$ is the blow-up of a closed point p_i over o_I. Define B_{S_i} by letting $K_{S_i} + B_{S_i}$ be the pullback of $K_{S/I} + (B_S)_{S/I}$, and let Λ_i be the union of $\operatorname{Supp} B_{S_i}$ and the inverse image of o_I in S_i. Then Λ_i is snc, and all coefficients of B_{S_i} are at most 1.

Let $D \in \mathcal{E}$ be a prime divisor such that $\operatorname{center}_{S_i} D$ is a closed point p. We show that p is contained in two irreducible components of Λ_i. Since p is over o_I, it is contained in a component P of the inverse image of o_I, and $a(P, S_i, B_{S_i}) \geq \mu$. Suppose that P is the only component of Λ_i containing p. Then near p we have $B_{S_i} = bP$ for some $b \leq 1$, and

$$a(D, S_i, B_{S_i}) = a(D, S_i, P) + (1-b)\operatorname{mult}_D P \geq 1 + (1-b) > 1 - b = a(P, S_i, B_{S_i}) \geq \mu,$$

where $a(D, S_i, P) \geq 1$ by [KM98, Corollary 2.31(3)] and $\operatorname{mult}_D P \geq 1$. This contradicts $D \in \mathcal{E}$.

By [KM98, Lemma 2.45], any $D \in \mathcal{E}$ is either a component of $g_0^{-1}(o_I)$ or the exceptional divisor of $S_k \to S_{k-1}$ for a finite sequence of blow-ups $S_k \to \cdots \to S_0$ of closed points $p_i \in S_i$ over o_I with $p_i = \operatorname{center}_{S_i} D$ for any i. By the previous paragraph, each p_i is then a point contained in two components of Λ_i. Since each Λ_i has finitely many such points, for any k there exist finitely many such sequences of length k. Hence $\mathcal{E}$ is countable. □

Proof of Proposition 16.16 continued. Let $q := |I|$. For $D \in \mathcal{E}$ and $1 \leq j \leq s$, write $h_j^q = \sum_{l \in \mathbb{Z}} g_{j,l} \varsigma^l$ with $g_{j,l} \in \mathbb{C}[S]^I$, which is possible since h_j^q is I-invariant and I acts trivially on ς. Let $r_{D,j} := \min_l \operatorname{ord}_D g_{j,l}$, and let L be the set of l with $\operatorname{ord}_D g_{j,l} = r_{D,j}$. For $\varsigma_0 \in \mathbb{C}^*$, we have $\operatorname{ord}_D(\sum_l g_{j,l}\varsigma_0^l) > r_{D,j}$ if and only if $\sum_{l \in L} \bar{g}_{j,l}\varsigma_0^l = 0$, where $\bar{g}_{j,l} \neq 0$ is the class of $g_{j,l}$ in the $\mathbb{C}$-vector space

$$\{g \in \mathbb{C}[S]^I \mid \operatorname{ord}_D g \geq r_{D,j}\}/\{g \in \mathbb{C}[S]^I \mid \operatorname{ord}_D g > r_{D,j}\}.$$

Hence $\operatorname{ord}_D(h_j^q(\sigma', \varsigma_0, z)) > r_{D,j}$ for at most finitely many $\varsigma_0 \in \mathbb{C}^*$. Also, $h_j(\sigma', \varsigma_0, z) = 0$ for at most finitely many ς_0 since $h_j \neq 0$. Thus each j and each $D \in \mathcal{E}$ exclude only finitely many $\varsigma_0 \in \mathbb{C}^*$. Since $\mathcal{E}$ is countable by Claim 16.18 and $\mathbb{C}^*$ is uncountable, there exists $\varsigma_0 \in \mathbb{C}^*$ such that, for any j and any $D \in \mathcal{E}$,

$$\tilde{h}_j := h_j(\sigma', \varsigma_0, z) \neq 0 \quad \text{and} \quad \operatorname{ord}_D(\tilde{h}_j^q) = r_{D,j}.$$

Consider the data of Proposition 16.15 fixed in **Step 3** at the point $(0, \varsigma_0, z_0)$, now with the functions $\tilde{h}_j$, which do not depend on $t = \varsigma - \varsigma_0$. They are non-zero, I-semi-invariant with the characters of the $f_j(\sigma'^m, z)$, and $\tilde{h}_j - f_j(\sigma'^m, z) = \sigma'^{mN} R'_j(\sigma', \varsigma_0, z) \in (\sigma'^{mN})$. Thus, by Proposition 16.15, there exists $D_0 \in \mathcal{E}$ whose Gauss extension with t-value 1 has equal values on $\tilde{h}_j^q$ and $f_j(\sigma'^m, z)^q$. As $\tilde{h}_j^q$ and $f_j(\sigma'^m, z)^q$ belong to $\mathbb{C}(S/I)$, on which the Gauss extension restricts to ord_{D_0}, we obtain $\operatorname{ord}_{D_0}(\tilde{h}_j^q) = \operatorname{ord}_{D_0}(f_j(\sigma'^m, z)^q)$. By the choice of ς_0,

$$r_{D_0,j} = \operatorname{ord}_{D_0}(f_j(\sigma'^m, z)^q) \quad \text{for any } j.$$

Let E_0 be the prime divisor over $W/I = (S/I) \times \mathbb{G}_{m,\varsigma}$ given by $D_0 \times \mathbb{G}_m$. Then $\operatorname{ord}_{E_0}(\sum_l g_l \varsigma^l) = \min_l \operatorname{ord}_{D_0} g_l$ for $g_l \in \mathbb{C}(S/I)$, so

$$\operatorname{ord}_{E_0}(h_j^q) = r_{D_0,j} = \operatorname{ord}_{E_0}(f_j(\sigma'^m, z)^q).$$

Since $\psi^*\varphi^*\Phi_j = h_j / f_j(\sigma'^m, z)$, we obtain $\operatorname{ord}_{E_0}(\psi^*\varphi^*\Phi_j) = 0$ for any j. Since $(B_{W,0})_{W/I} = \operatorname{pr}_1^*(B_S)_{S/I}$, we have $a_{W,0}(E_0) = a(D_0, S/I, (B_S)_{S/I}) = \mu$, as one sees on $S'' \times \mathbb{G}_m$ for a higher

model $S'' \to S/I$ on which D_0 is a divisor. Since $B_{W,H} - B_{W,0} = \sum_j c_j \operatorname{div}_W(\psi^*\varphi^*\Phi_j)$, (16.52) applied to I acting on W implies $a_{W,H}(E_0) = a_{W,0}(E_0) = \mu$. Moreover, $\operatorname{center}_{W/I} E_0 = \{o_I\} \times \mathbb{G}_m = \widetilde{C}_I$. Let $E^\sharp := (E_0)_U$. By **Step 3**,

$$a_{U,H}(E^\sharp) = a_{U,0}(E^\sharp) = \mu \le \min\{a_{U,H}(E_I), a_{U,0}(E_I)\},$$

and $\operatorname{center}_{U/I} E^\sharp = \psi_I(\widetilde{C}_I) = C_I \subset \pi_U(\varphi^{-1}(x))$. Thus Claim 16.17 implies the existence of E'.

Step 6. In this step, we conclude the proof.

Assume that ξ is a closed point. **Steps 2, 4**, and **5** cover all positions of x and of $\operatorname{center}_U F$, so for any prime divisor E over V/J with center $\{\xi\}$ there exists E' as in the statement. Then

$$\operatorname{mld}(V/J \ni \xi, (B_0)_{V/J}) \le a_0(E') = a_H(E') \le a_H(E),$$
$$\operatorname{mld}(V/J \ni \xi, (B_H)_{V/J}) \le a_H(E') = a_0(E') \le a_0(E).$$

Taking the infimum over E, we obtain $\operatorname{mld}(V/J \ni \xi, (B_0)_{V/J}) \le \operatorname{mld}(V/J \ni \xi, (B_H)_{V/J})$ and $\operatorname{mld}(V/J \ni \xi, (B_H)_{V/J}) \le \operatorname{mld}(V/J \ni \xi, (B_0)_{V/J})$.

Assume that ξ is not a closed point, and let $\Xi := \overline{\{\xi\}} \subset Z$. The locus where (V, B_0) is not sub-lc is closed, being the image of the components with coefficient > 1 of the crepant pull-back to a log resolution (cf. [Amb99, Section 1, the second paragraph after Definition 1.1]), and it is disjoint from $\pi^{-1}(\xi)$, and π is finite, so (V, B_0) is sub-lc near $\pi^{-1}(y)$ for a general closed point $y \in \Xi$. For such y, the closed case implies

$$\operatorname{mld}(V/J \ni y, (B_H)_{V/J}) = \operatorname{mld}(V/J \ni y, (B_0)_{V/J}) \ge 0,$$

where the inequality holds since $(V/J, (B_0)_{V/J})$ is sub-lc near y by Lemma 2.6(2)(3). In particular, $(V/J, (B_0)_{V/J})$ is sub-lc near ξ. The sub-pair $(V/J, (B_H)_{V/J})$ is sub-lc near y, since otherwise the center of some prime divisor over V/J with negative log discrepancy with respect to $(B_H)_{V/J}$ contains y, and then $\operatorname{mld}(V/J \ni y, (B_H)_{V/J}) = -\infty$ by [Amb99, Section 1, the second paragraph after Definition 1.1]. Hence $(V/J, (B_H)_{V/J})$ is sub-lc near ξ. By Lemma 2.4 applied to both sub-pairs, for a general closed point $y \in \Xi$,

$$\begin{aligned}\operatorname{mld}(V/J \ni \xi, (B_H)_{V/J}) &= \operatorname{mld}(V/J \ni y, (B_H)_{V/J}) - \dim \Xi \\ &= \operatorname{mld}(V/J \ni y, (B_0)_{V/J}) - \dim \Xi \\ &= \operatorname{mld}(V/J \ni \xi, (B_0)_{V/J}).\end{aligned}$$

□

16.9. **Adjunction along a section.** In this subsection, we bound log discrepancies of quotients of $\mathbb{A}^1_U \times S$ from below by minimal log discrepancies of quotients of a surface S, when $(U = 0)$ has coefficient one.

In the proof of Theorem 16.21, lower bounds near the hyperplane $(U = 0)$ follow from quotient adjunction to $(U = 0)$ with all branch divisors of the quotient included in the boundary. The branch divisor may intersect $(U = 0)$ in codimension one, so we add two coordinates with the characters of the surface coordinates, which removes pseudoreflections, so that the hypotheses of Theorem 2.22 hold.

Lemma 16.19. *Let H_0 be a finite diagonal group acting on $\mathbb{A}^1_U \times \mathbb{A}^2_{z_1,z_2}$, $S \subset \mathbb{A}^2_{z_1,z_2}$ an H_0-invariant affine open subset on which H_0 acts faithfully, $V_0 := \mathbb{A}^1_U \times S$, $\pi : S \to S/H_0$ the quotient morphism, and $\operatorname{pr} : V_0/H_0 \to S/H_0$ the morphism induced by the projection. Let $K_j(U, z_1, z_2)$ be regular H_0-semi-invariant functions on V_0 such that the functions $P_j := K_j(0, z_1, z_2)$ on S are non-zero. Let $c_j \ge 0$ be real numbers, $B_S := \sum_j c_j \operatorname{div}(P_j)$, $B_{V_0} := (U = 0) + \sum_j c_j \operatorname{div}(K_j)$, and $\mathfrak{a}$ an $\mathbb{R}$-ideal on S/H_0.*

(1) Let $s \in S$ be a closed point such that $(S, B_S, \pi^*\mathfrak{a})$ is lc near s, and let

$$b_s := \operatorname{mld}\big(S/H_0 \ni \pi(s), (B_S)_{S/H_0}, \mathfrak{a}\big).$$

Then any prime divisor E over V_0/H_0 with center the image of $(0,s)$ satisfies $a\big(E, V_0/H_0, (B_{V_0})_{V_0/H_0}, \mathrm{pr}^*\,\mathfrak{a}\big) \geq b_s$.

(2) *Let $C \subset S$ be an integral curve such that $(S, B_S, \pi^*\mathfrak{a})$ is lc near the generic point of C. Let b_C be the minimal log discrepancy of $\big(S/H_0, (B_S)_{S/H_0}, \mathfrak{a}\big)$ at the generic point of $\pi(C)$. Then any prime divisor E over V_0/H_0 with center the image of $\{0\} \times C$ satisfies $a\big(E, V_0/H_0, (B_{V_0})_{V_0/H_0}, \mathrm{pr}^*\,\mathfrak{a}\big) \geq b_C$.*

Proof. Proof of (1). The proof is long, so we divide it into several steps.

Step 1. In this step we reduce to the case when s is the origin, H_0 fixes s, $S = \mathbb{A}^2_{z_1,z_2}$, and the K_j are polynomials.

Let $H_s := \mathrm{Stab}_{H_0}(s)$. Since H_0 acts on U by a character, $\mathrm{Stab}_{H_0}((0,s)) = H_s$. Let $f : V_0/H_s \to V_0/H_0$ and $f_S : S/H_s \to S/H_0$ be the induced morphisms, and let x' and s' be the images of $(0,s)$ and s in V_0/H_s and S/H_s respectively. By [Gro71, Exp. V, Proposition 2.2], applied to the quotient morphisms $V_0 \to V_0/H_0$ and $S \to S/H_0$ with the subgroup H_s, which is the stabilizer of $(0,s)$ and of s in H_0, the morphism f is étale at x' and f_S is étale at s'. Since f and f_S are finite, after shrinking V_0/H_s and S/H_s to open neighborhoods of x' and s', we may assume that f and f_S are étale, $f^{-1}(f(x')) = \{x'\}$, and $f_S^{-1}(f_S(s')) = \{s'\}$. By Lemma 2.6(0)(5), applied to the normal subgroup H_s of the abelian group H_0, and since f and f_S are étale, $(B_{V_0})_{V_0/H_s} = f^*(B_{V_0})_{V_0/H_0}$ and $(B_S)_{S/H_s} = f_S^*(B_S)_{S/H_0}$.

By Lemma 9.3, for any prime divisor E over V_0/H_0 with center $f(x')$ there exists a prime divisor E' over V_0/H_s with center x' such that $\mathrm{ord}_{E'}|_{\mathbb{C}(V_0/H_0)} = \mathrm{ord}_E$ and

$$a\big(E', V_0/H_s, (B_{V_0})_{V_0/H_s}\big) = a\big(E, V_0/H_0, (B_{V_0})_{V_0/H_0}\big).$$

The ideals of $f^*\,\mathrm{pr}^*\,\mathfrak{a}$ are generated by the pullbacks of generators of the ideals of $\mathrm{pr}^*\,\mathfrak{a}$, so $\mathrm{ord}_{E'}|_{\mathbb{C}(V_0/H_0)} = \mathrm{ord}_E$ implies that

$$a\big(E', V_0/H_s, (B_{V_0})_{V_0/H_s}, f^*\,\mathrm{pr}^*\,\mathfrak{a}\big) = a\big(E, V_0/H_0, (B_{V_0})_{V_0/H_0}, \mathrm{pr}^*\,\mathfrak{a}\big).$$

Moreover, $f^*\,\mathrm{pr}^*\,\mathfrak{a} = \mathrm{pr}_s^*\, f_S^*\mathfrak{a}$, where $\mathrm{pr}_s : V_0/H_s \to S/H_s$ is induced by the projection, and the pullback of $f_S^*\mathfrak{a}$ to S is $\pi^*\mathfrak{a}$. By Lemma 9.3 applied to f_S, and since the ideals of $f_S^*\mathfrak{a}$ are generated by the pullbacks of generators of the ideals of $\mathfrak{a}$,

$$b_s = \mathrm{mld}\big(S/H_s \ni s', (B_S)_{S/H_s}, f_S^*\mathfrak{a}\big).$$

Thus, possibly replacing H_0 with H_s and $\mathfrak{a}$ with $f_S^*\mathfrak{a}$, we may assume that H_0 fixes s.

For $i = 1, 2$, let χ_i be the character of H_0 such that $\sigma^* z_i = \chi_i(\sigma) z_i$ for any $\sigma \in H_0$. Write $s = (s_1, s_2)$. For $i = 1, 2$, the function $z_i - s_i$ is H_0-semi-invariant with character χ_i: this is clear if $s_i = 0$, and if $s_i \neq 0$, then χ_i is trivial since H_0 fixes s. Thus, possibly replacing z_i with $z_i - s_i$ for $i = 1, 2$, we may assume that $s = 0$.

The set $\mathbb{A}^2_{z_1,z_2} \backslash S$ is closed and H_0-invariant, so the image of $\mathbb{A}^2_{z_1,z_2} \backslash S$ in $\mathbb{A}^2_{z_1,z_2}/H_0 = \mathrm{Spec}\,\mathbb{C}[z_1, z_2]^{H_0}$ is closed and does not contain the image of 0. Hence there exists $\varphi \in \mathbb{C}[z_1, z_2]^{H_0}$ such that $\varphi(0) \neq 0$ and the principal open subset $D(\varphi)$ is contained in S. The coordinate ring of $\mathbb{A}^1_U \times D(\varphi)$ is $\mathbb{C}[U, z_1, z_2]_\varphi$, so there exists an integer $m \geq 0$ such that $k_j := \varphi^m K_j$ are polynomials for all j. Each k_j is H_0-semi-invariant with the character of K_j, $\mathrm{div}(k_j) = \mathrm{div}(K_j)$ on $\mathbb{A}^1_U \times D(\varphi)$, and $k_j(0, z_1, z_2) = \varphi^m P_j \neq 0$. Write $\mathfrak{a} = \prod_l \mathfrak{a}_l^{r_l}$. The restriction of each $\mathfrak{a}_l$ to $D(\varphi)/H_0 = \mathrm{Spec}\,\mathbb{C}[z_1, z_2]_\varphi^{H_0}$ is generated by an ideal $\mathfrak{a}'_l$ of $\mathbb{C}[z_1, z_2]^{H_0}$. The log discrepancies in (1), b_s, and the condition that $(S, B_S, \pi^*\mathfrak{a})$ is lc near 0 only depend on the H_0-invariant open neighborhood $D(\varphi)$ of 0. Thus, possibly replacing S with $\mathbb{A}^2_{z_1,z_2}$, K_j with k_j, and $\mathfrak{a}$ with $\prod_l (\mathfrak{a}'_l)^{r_l}$, we may assume that $S = \mathbb{A}^2_{z_1,z_2}$ and the K_j are polynomials.

In the rest of the proof of (1), we denote by $\bar{0}$ the image of the origin in any quotient by H_0, let $r := |H_0|$, let $\pi_{V_0} : V_0 \to V_0/H_0$ be the quotient morphism, and let $\mathfrak{a}_{V_0} := \pi_{V_0}^*\,\mathrm{pr}^*\,\mathfrak{a}$. The ideals of $\mathfrak{a}_{V_0}$ are generated by the pullbacks under the projection $V_0 \to S$ of generators of the ideals of $\pi^*\mathfrak{a}$.

Step 2. In this step we add two coordinates so that the hypotheses of Theorem 2.22 hold.

Let H_0 act on $\mathbb{A}^5 := V_0 \times \mathbb{A}^2_{y_1,y_2}$ by the given action on V_0 and $\sigma^* y_i = \chi_i(\sigma) y_i$ for $i = 1, 2$, and let $X := (U = 0) \subset \mathbb{A}^5$, so $X = S \times \mathbb{A}^2_{y_1,y_2}$. Let $\sigma \in H_0$ be a non-identity element. Since H_0 acts faithfully on S, there exists $i \in \{1, 2\}$ such that $\chi_i(\sigma) \neq 1$, so the fixed locus of σ in $\mathbb{A}^5$ is contained in $(z_i = y_i = 0)$. Hence H_0 contains no pseudoreflection, and the quotient morphisms $\mathbb{A}^5 \to \mathbb{A}^5/H_0$ and $X \to X/H_0$ have no ramification divisor. Moreover, $\mathbb{A}^5 \to \mathbb{A}^5/H_0$ is étale at any point with trivial stabilizer [Gro71, Exp. V, Proposition 2.2(ii)], so the locus where $\mathbb{A}^5 \to \mathbb{A}^5/H_0$ is not étale is contained in

$$(z_1 = y_1 = 0) \cup (z_2 = y_2 = 0),$$

whose intersection with X has codimension two in X. Since $X \to X/H_0$ is finite, the image in $\mathbb{A}^5/H_0$ of the locus where $\mathbb{A}^5 \to \mathbb{A}^5/H_0$ is not étale intersects X/H_0 in codimension at least two.

Since $\chi^r = 1$ for any character χ of H_0, the functions U^r, y_1^r, y_2^r, and K_j^r are H_0-invariant. Let $\mathfrak{a}_{\mathbb{A}^5/H_0}$ be the pullback of $\mathfrak{a}$ under the morphism $\mathbb{A}^5/H_0 \to S/H_0$ induced by the projection, and let

$$\mathfrak{b} := (y_1^r)^{\frac{1}{r}} (y_2^r)^{\frac{1}{r}} \prod_{c_j > 0} (K_j^r)^{\frac{c_j}{r}} \cdot \mathfrak{a}_{\mathbb{A}^5/H_0},$$

an $\mathbb{R}$-ideal on $\mathbb{A}^5/H_0$. Since $K_j|_X = P_j \neq 0$, the restriction of $\mathfrak{b}$ to X/H_0 is non-zero. Since $X \cong \mathbb{A}^4$ is smooth and defined by the H_0-semi-invariant polynomial U, Theorem 2.22 implies that

$$\operatorname{mld}\big(X/H_0 \ni \bar{0}, \mathfrak{b}|_{X/H_0}\big) = \operatorname{mld}\big(\mathbb{A}^5/H_0 \ni \bar{0}, (U^r)^{\frac{1}{r}} \mathfrak{b}\big). \tag{16.57}$$

Let $B_{\mathbb{A}^5}$ and B_X be the pullbacks of B_{V_0} and B_S to $\mathbb{A}^5$ and X respectively, plus $(y_1 = 0) + (y_2 = 0)$, and let $\mathfrak{a}_{\mathbb{A}^5}$ and $\mathfrak{a}_X$ be the pullbacks of $\mathfrak{a}_{V_0}$ and $\pi^*\mathfrak{a}$ to $\mathbb{A}^5$ and X respectively. For any prime divisor F over $\mathbb{A}^5$, the pullback of the ideal (K_j^r) has multiplicity $r \operatorname{mult}_F(K_j = 0)$ along F, and the pullbacks of (U^r) and (y_i^r) have multiplicities $r \operatorname{mult}_F(U = 0)$ and $r \operatorname{mult}_F(y_i = 0)$ along F. Hence the log discrepancy of F with respect to $\mathbb{A}^5$ and the pullback of $(U^r)^{\frac{1}{r}} \mathfrak{b}$ is $a(F, \mathbb{A}^5, B_{\mathbb{A}^5}, \mathfrak{a}_{\mathbb{A}^5})$, and for any prime divisor F' over X, the log discrepancy of F' with respect to X and the pullback of $\mathfrak{b}|_{X/H_0}$ is $a(F', X, B_X, \mathfrak{a}_X)$. The crepant quotients of $(\mathbb{A}^5, 0)$ and $(X, 0)$ by H_0 have zero boundary, and the origin is the only point over $\bar{0}$. Thus Lemma 2.6(2)(3) implies that

$$\operatorname{mld}\big(\mathbb{A}^5/H_0 \ni \bar{0}, (U^r)^{\frac{1}{r}} \mathfrak{b}\big) = \inf_F \frac{a(F, \mathbb{A}^5, B_{\mathbb{A}^5}, \mathfrak{a}_{\mathbb{A}^5})}{|I_{H_0}(F)|} \tag{16.58}$$

and

$$\operatorname{mld}\big(X/H_0 \ni \bar{0}, \mathfrak{b}|_{X/H_0}\big) = \inf_{F'} \frac{a(F', X, B_X, \mathfrak{a}_X)}{|I_{H_0}(F')|}, \tag{16.59}$$

where F (resp. F') runs over the prime divisors over $\mathbb{A}^5$ (resp. X) with center the origin.

Step 3. In this step we show that the right-hand side of (16.59) is at least b_s.

Let F' be a prime divisor over X with center the origin. Lemma 8.1(1) applied to $(S \times \mathbb{A}^1_{y_1}, \operatorname{pr}_1^* B_S + (y_1 = 0))$ and the character χ_2, and then to (S, B_S) and the character χ_1, implies that there exist a positive integer ϱ and a prime divisor E over S with center the origin such that $\operatorname{ord}_{F'}|_{\mathbb{C}(S)} = \varrho \operatorname{ord}_E$, $I_{H_0}(F') \subset I_{H_0}(E)$, and

$$a(F', X, B_X) \geq \varrho a(E, S, B_S).$$

The ideals of $\mathfrak{a}_X$ are generated by functions on S, so $\operatorname{ord}_{F'}|_{\mathbb{C}(S)} = \varrho \operatorname{ord}_E$ implies that $\operatorname{mult}_{F'}$ of any ideal of $\mathfrak{a}_X$ is ϱ times mult_E of the corresponding ideal of $\pi^*\mathfrak{a}$. Hence

$$a(F', X, B_X, \mathfrak{a}_X) \geq \varrho a(E, S, B_S, \pi^*\mathfrak{a}).$$

Since $(S, B_S, \pi^*\mathfrak{a})$ is lc near 0, $a(E, S, B_S, \pi^*\mathfrak{a}) \geq 0$. Let $\bar{E}$ be the prime divisor over S/H_0 such that $\operatorname{ord}_E|_{\mathbb{C}(S/H_0)} = |I_{H_0}(E)| \operatorname{ord}_{\bar{E}}$. Then $\operatorname{center}_{S/H_0} \bar{E} = \{\bar{0}\}$, and Lemma 2.6(1)(2) implies that

$$\frac{a(F', X, B_X, \mathfrak{a}_X)}{|I_{H_0}(F')|} \geq \frac{\varrho a(E, S, B_S, \pi^*\mathfrak{a})}{|I_{H_0}(F')|} \geq \frac{a(E, S, B_S, \pi^*\mathfrak{a})}{|I_{H_0}(E)|} = a\big(\bar{E}, S/H_0, (B_S)_{S/H_0}, \mathfrak{a}\big) \geq b_s.$$

Step 4. In this step we conclude the proof of (1).

Let E be a prime divisor over V_0/H_0 with center $\bar{0}$. By Lemma 2.6(3), there exists a prime divisor F_0 over V_0 such that $\operatorname{ord}_{F_0}|_{\mathbb{C}(V_0/H_0)} = |I_{H_0}(F_0)| \operatorname{ord}_E$, and $\operatorname{center}_{V_0} F_0$ is the origin since the origin is the only point of V_0 over $\bar{0}$. By Lemma 2.6(2),

$$a\big(E, V_0/H_0, (B_{V_0})_{V_0/H_0}, \operatorname{pr}^*\mathfrak{a}\big) = \frac{a(F_0, V_0, B_{V_0}, \mathfrak{a}_{V_0})}{|I_{H_0}(F_0)|}.$$

Lemma 8.1(2) applied to (V_0, B_{V_0}) and the character χ_1, and then to $(V_0 \times \mathbb{A}^1_{y_1}, \operatorname{pr}_1^* B_{V_0} + (y_1 = 0))$ and the character χ_2, implies that there exists a prime divisor F over $\mathbb{A}^5$ with center the origin such that $\operatorname{ord}_F|_{\mathbb{C}(V_0)} = \operatorname{ord}_{F_0}$, $I_{H_0}(F) = I_{H_0}(F_0)$, and $a(F, \mathbb{A}^5, B_{\mathbb{A}^5}) = a(F_0, V_0, B_{V_0})$. The equality $\operatorname{ord}_F|_{\mathbb{C}(V_0)} = \operatorname{ord}_{F_0}$ holds because a Gauss extension of ord_{F_0} restricts to ord_{F_0} on $\mathbb{C}(V_0)$. The ideals of $\mathfrak{a}_{\mathbb{A}^5}$ are generated by functions on V_0, so $a(F, \mathbb{A}^5, B_{\mathbb{A}^5}, \mathfrak{a}_{\mathbb{A}^5}) = a(F_0, V_0, B_{V_0}, \mathfrak{a}_{V_0})$. By (16.58), (16.57), (16.59), and **Step 3**,

$$\begin{aligned} a\big(E, V_0/H_0, (B_{V_0})_{V_0/H_0}, \operatorname{pr}^*\mathfrak{a}\big) &= \frac{a(F, \mathbb{A}^5, B_{\mathbb{A}^5}, \mathfrak{a}_{\mathbb{A}^5})}{|I_{H_0}(F)|} \\ &\geq \operatorname{mld}\big(\mathbb{A}^5/H_0 \ni \bar{0}, (U^r)^{\frac{1}{r}}\mathfrak{b}\big) = \operatorname{mld}\big(X/H_0 \ni \bar{0}, \mathfrak{b}|_{X/H_0}\big) \geq b_s. \end{aligned}$$

This proves (1).

Proof of (2). Let $\Delta := (B_{V_0})_{V_0/H_0}$, $\Delta_S := (B_S)_{S/H_0}$, and $\bar{C} \subset V_0/H_0$ the image of $\{0\} \times C$. Let E be a prime divisor over V_0/H_0 with $\operatorname{center}_{V_0/H_0} E = \bar{C}$, and let $a_E := a(E, V_0/H_0, \Delta, \operatorname{pr}^*\mathfrak{a})$.

First we show that there exists a dense open subset $\bar{C}^\circ \subset \bar{C}$ such that

$$\operatorname{mld}\big(V_0/H_0 \ni y, \Delta, \operatorname{pr}^*\mathfrak{a}\big) \leq a_E + 1 \tag{16.60}$$

for any closed point $y \in \bar{C}^\circ$. Write $\operatorname{pr}^*\mathfrak{a} = \prod_l \mathfrak{b}_l^{r_l}$. By [KM98, Lemma 2.45], there exists a projective birational morphism $Y_1 \to V_0/H_0$ such that Y_1 is normal and E is a divisor on Y_1. By [KM98, Theorem 0.2] for Y_1 and the ideals $\mathfrak{b}_l\mathcal{O}_{Y_1}$, there exists a log resolution $h : Y \to V_0/H_0$ of $(V_0/H_0, \Delta)$ factoring through Y_1 such that $\mathfrak{b}_l\mathcal{O}_Y = \mathcal{O}_Y(-M_l)$ for some divisors $M_l \geq 0$, and $\operatorname{Exc}(h) \cup \operatorname{Supp} h_*^{-1}\Delta \cup \operatorname{Supp} \sum_l M_l$ is a simple normal crossing divisor. Since Y_1 is normal at the generic point of E and the inverse of the birational proper morphism $Y \to Y_1$ is defined in codimension one, E is a divisor on Y. Let

$$K_Y + B_Y := h^*(K_{V_0/H_0} + \Delta) + \sum_l r_l M_l.$$

Since the $\mathfrak{b}_l\mathcal{O}_Y$ are invertible, $\operatorname{mult}_G \mathfrak{b}_l = \operatorname{mult}_G M_l$ for any prime divisor G over Y, so $a(G, V_0/H_0, \Delta, \operatorname{pr}^*\mathfrak{a}) = a(G, Y, B_Y)$. Since $h(E) = \bar{C}$ and $\dim E = 2$, for a general closed point $y \in \bar{C}$ the fiber $E \cap h^{-1}(y)$ is a non-empty curve. For any component $D \neq E$ of $\operatorname{Exc}(h) \cup \operatorname{Supp} B_Y$, $\dim(D \cap E) \leq 1$, so $D \cap E$ contains components of $E \cap h^{-1}(y)$ for only finitely many y. Hence there exists a dense open subset $\bar{C}^\circ \subset \bar{C}$ such that for any closed point $y \in \bar{C}^\circ$, $E \cap h^{-1}(y)$ is a non-empty curve, and no component of $E \cap h^{-1}(y)$ is contained in a component $D \neq E$ of $\operatorname{Exc}(h) \cup \operatorname{Supp} B_Y$. Let $y \in \bar{C}^\circ$ be a closed point, Γ a component of $E \cap h^{-1}(y)$, and $Y^\circ \subset Y$ an open subset such that $\Gamma \cap Y^\circ$ is a non-empty smooth curve and

$$B_Y|_{Y^\circ} = (1 - a_E)E|_{Y^\circ}.$$

Let G be the exceptional divisor of the blow-up of Y° along $\Gamma \cap Y^\circ$. By [KM98, Lemma 2.29],

$$a(G, Y, B_Y) = 2 - (1 - a_E) = a_E + 1.$$

Since $\operatorname{center}_{V_0/H_0} G = h(\Gamma) = \{y\}$, (16.60) holds.

Next we show that there exists a dense open subset $\pi(C)^\circ \subset \pi(C)$ such that

$$\operatorname{mld}\big(S/H_0 \ni y, \Delta_S, \mathfrak{a}\big) = b_C + 1 \tag{16.61}$$

for any closed point $y \in \pi(C)^\circ$. Since B_S and $\pi^*\mathfrak{a}$ are H_0-invariant, $(S, B_S, \pi^*\mathfrak{a})$ is lc on an H_0-invariant open neighborhood of the generic point of C. Thus Lemma 2.6(2)(3) implies that $(S/H_0, \Delta_S, \mathfrak{a})$ is lc near the generic point η of $\pi(C)$. Since S/H_0 is normal, $\mathcal{O}_{S/H_0,\eta}$ is a discrete valuation ring. Write $\mathfrak{a} = \prod_l \mathfrak{a}_l^{r_l}$. Two coherent ideal sheaves with equal stalks at η coincide near η, so there exist an open neighborhood W of η and integers $m_l \geq 0$ such that W is smooth, $\pi(C) \cap W$ is a smooth prime divisor on W, and $\mathfrak{a}_l|_W = \mathcal{O}_W(-m_l(\pi(C) \cap W))$ for any l. Let

$$\Delta_W := \Delta_S|_W + \sum_l r_l m_l (\pi(C) \cap W).$$

Then $a(\bar{E}, S/H_0, \Delta_S, \mathfrak{a}) = a(\bar{E}, W, \Delta_W)$ for any prime divisor $\bar{E}$ over S/H_0 whose center intersects W. Hence (W, Δ_W) is lc near η, $b_C = \operatorname{mld}(W \ni \eta, \Delta_W)$, and

$$\operatorname{mld}(S/H_0 \ni y, \Delta_S, \mathfrak{a}) = \operatorname{mld}(W \ni y, \Delta_W)$$

for any closed point $y \in \pi(C) \cap W$. Thus Lemma 2.4 implies (16.61).

Finally, the morphism $C \to \pi(C)$ and the morphism $C \to \bar{C}$ sending s to the image of $(0, s)$ are finite and surjective, and $(S, B_S, \pi^*\mathfrak{a})$ is lc on an open neighborhood of the generic point of C. Hence there exists a closed point $s \in C$ such that $(S, B_S, \pi^*\mathfrak{a})$ is lc near s, $\pi(s) \in \pi(C)^\circ$, and the image y of $(0, s)$ is contained in $\bar{C}^\circ$. By (1), any prime divisor over V_0/H_0 with center y has log discrepancy at least b_s with respect to $(V_0/H_0, \Delta, \operatorname{pr}^*\mathfrak{a})$, so $\operatorname{mld}(V_0/H_0 \ni y, \Delta, \operatorname{pr}^*\mathfrak{a}) \geq b_s$. By (16.60) and (16.61),

$$a_E \geq \operatorname{mld}\big(V_0/H_0 \ni y, \Delta, \operatorname{pr}^*\mathfrak{a}\big) - 1 \geq b_s - 1 = b_C.$$

This proves (2). □

16.10. **Mixed circles with three non-zero weights.** In this subsection, we prove Theorem 16.21: along a mixed circle with three non-zero weights, the number b of Construction-Theorem 16.9 bounds the quotient mld from both sides, and the quotient mlds satisfy the ACC. We first prove a uniform jet cutoff on finitely many fixed quotients of $\mathbb{A}^3$.

Lemma 16.20. *Let M be a positive integer and $\Lambda \subset \mathbb{R}_{>0}$ a DCC set. Then there exists a positive integer l with the following property. Let J be a finite diagonal group of order at most M acting faithfully on $\mathbb{A}^3$ with coordinates x_1, x_2, x_3, $\pi : \mathbb{A}^3 \to \mathbb{A}^3/J$ the quotient morphism, $B \geq 0$ an $\mathbb{R}$-divisor on $\mathbb{A}^3/J$ with $B \in \Lambda$, $c_1, \dots, c_s \in \Lambda$, $R \geq Ml$ an integer, and F_j, F_j' non-zero J-semi-invariant polynomials with $F_j - F_j' \in (x_1^R)$ for $1 \leq j \leq s$. Let $\mathfrak{a} := \prod_j (F_j^{|J|})^{c_j/|J|}$ and $\mathfrak{a}' := \prod_j \left((F_j')^{|J|}\right)^{c_j/|J|}$ be the corresponding $\mathbb{R}$-ideals on $\mathbb{A}^3/J$. Then*

$$\operatorname{mld}(\mathbb{A}^3/J \ni \eta, B, \mathfrak{a}) = \operatorname{mld}(\mathbb{A}^3/J \ni \eta, B, \mathfrak{a}')$$

for any scheme-theoretic point η of $\pi((x_1 = 0))$.

Proof. Any element of a diagonal group of order at most M has order at most M, so the eigenvalues of the element belong to $\mu_{M!}$. Thus there exist only finitely many finite diagonal groups of order at most M acting faithfully on $\mathbb{A}^3$, namely subgroups of $\mu_{M!}^3$. Let $\Lambda' := \Lambda \cup \bigcup_{d=1}^{M} \frac{1}{d}\Lambda$, a finite union of DCC sets, hence a DCC set. For any such group J, the variety $\mathbb{A}^3/J$ is $\mathbb{Q}$-factorial [KM98, Lemma 5.16], so $(\mathbb{A}^3/J, B)$ is a pair for any $\mathbb{R}$-divisor $B \geq 0$. Theorem 2.19(2), applied to $\mathbb{A}^3/J$ and Λ', gives a positive integer l_J, and we let l be the maximum of the l_J.

Let J, B, c_j, F_j, F_j', R be as in the statement and $m := \lfloor R/|J| \rfloor$, so that $m \geq l$ because $R \geq Ml \geq |J|l$. The polynomials $F_j^{|J|}$ and $(F_j')^{|J|}$ are J-invariant, and

$$F_j^{|J|} - (F_j')^{|J|} = (F_j - F_j') \sum_{i=0}^{|J|-1} F_j^i (F_j')^{|J|-1-i} \in (x_1^R).$$

Since $|J|m \le R$, the quotient $\left(F_j^{|J|} - (F_j')^{|J|}\right)/(x_1^{|J|})^m$ is a polynomial, and it is J-invariant because $x_1^{|J|}$ is J-invariant. Hence $F_j^{|J|} - (F_j')^{|J|}$ belongs to the ideal of $\mathbb{C}[x_1, x_2, x_3]^J$ generated by $(x_1^{|J|})^m$. The invariant function $x_1^{|J|}$ vanishes on $\pi((x_1 = 0))$, so it belongs to the maximal ideal $\mathfrak{m}_\eta$ of $\mathcal{O}_{\mathbb{A}^3/J,\eta}$. Thus

$$F_j^{|J|}\mathcal{O}_{\mathbb{A}^3/J,\eta} + \mathfrak{m}_\eta^l = (F_j')^{|J|}\mathcal{O}_{\mathbb{A}^3/J,\eta} + \mathfrak{m}_\eta^l$$

for any j, because $m \ge l$. The exponents $c_j/|J|$ belong to Λ' and $B \in \Lambda \subset \Lambda'$. Thus $\operatorname{mld}(\mathbb{A}^3/J \ni \eta, B, \mathfrak{a}) = \operatorname{mld}(\mathbb{A}^3/J \ni \eta, B, \mathfrak{a}')$ by Theorem 2.19(2). □

To prove Theorem 16.21, we divide the prime divisors over the quotient into three classes according to their orders along two coordinates of a finite cover. For the first two classes, Lemma 16.20 deletes the terms of high order and Lemma 16.19 bounds the log discrepancies from below by b. For the third class, Proposition 16.16 compares the equations with the homogeneous model.

Theorem 16.21. *Fix positive integers p, q, r with $\gcd(p, q, r) = 1$, an integer $e_0 \ge 1$, and a DCC set $\Gamma \subset [0, 1]$, and let $\mathbf{T} := \mathbf{T}_{(p,q,-r)}$ act on $X := \mathbb{A}^3_{x,y,z}$. Let G be a finite diagonal group acting faithfully on X with $|G/(G \cap \mathbf{T})| \le e_0$, let $n := |G \cap \mathbf{T}|$, and let $B = \sum_j c_j \operatorname{div}(h_j)$ with $c_j \in \Gamma$ and h_j non-zero G-semi-invariant polynomials such that (X, B) is klt near 0. Delete units and zero coefficients, for any j choose a monomial m_j of least degree occurring in h_j, let κ_j be the $\mathbf{T}$-weight of m_j, and let h_j^0 be the sum of the terms of h_j of $\mathbf{T}$-weight κ_j. Then:*

(1) the set of positive numbers $\operatorname{mld}((X \ni 0, B)/G)$, for all such G and B, satisfies the ACC;

(2) there exist $n_0 \ge 1$ and $C_0 \ge 0$ depending only on p, q, r, e_0, Γ such that if $n \ge n_0$, then $(X, \sum_j c_j \operatorname{div}(h_j^0))$ is lc near 0, and the number b of Construction-Theorem 16.9 for the h_j^0 satisfies

$$b \le \operatorname{mld}((X \ni 0, B)/G) \le b + \frac{C_0}{n}.$$

Proof. The proof is long, so we divide it into several steps. Let $H := G/(G \cap \mathbf{T})$, so that $|H| \le e_0$.

Step 1. In this step we reduce (2) to the lower bound $b \le a(E, X/G, B_{X/G})$ and introduce the integers A_j, β_j and the numbers a_s, a_t.

If no h_j remains after the deletion, then $B = 0$ and $\tau = r-p-q$, and Construction-Theorem 16.9 implies $b \le \operatorname{mld}((X \ni 0, 0)/G) \le b + \frac{|r-p-q|}{n}$ for $n > re_0 \max\{p, q\}$. Thus (2) holds in this case for any $n_0 > re_0 \max\{p, q\}$ and $C_0 \ge |r - p - q|$. In the rest of the proof we assume that some h_j remains, so that $\Gamma\backslash\{0\} \ne \emptyset$. Let $\gamma := \gamma(\Gamma) > 0$ (Lemma 2.10(2)) and $P_{\max} := \max\{p, q, r\}$. For the exceptional divisor E_0 of the blow-up of the origin of X, the klt condition implies

$$a(E_0, X, B) = 3 - \sum_j c_j \operatorname{mult}_0 h_j > 0$$

[KM98, Lemma 2.29]. Thus the number of the h_j is less than $3/\gamma$, $\deg m_j = \operatorname{mult}_0 h_j < 3/\gamma$, and

$$|\kappa_j| \le P_{\max} \deg m_j < \frac{3P_{\max}}{\gamma}.$$

Let n_{up} be the integer n_0 of Proposition 16.11(2), which depends only on p, q, r, e_0, Γ. By Proposition 16.11(2), if $n \ge n_{\mathrm{up}}$, then (X, B^0) is lc near 0 for $B^0 := \sum_j c_j \operatorname{div}(h_j^0)$, and $\operatorname{mld}((X \ni 0, B)/G) \le b + \frac{C_0}{n}$ for the constant C_0 of Proposition 16.11(2), which depends only on p, q, r, e_0, Γ. Thus it remains to prove $b \le a(E, X/G, B_{X/G})$ for any prime divisor E over X/G with center the image of 0 (equivalently $b \le \operatorname{mld}((X \ni 0, B)/G)$, by Definition 2.2), once n is larger than a constant depending only on p, q, r, e_0, Γ. We assume that $n \ge n_{\mathrm{up}}$.

Let $\beta_j := \operatorname{ord}_z h_j^0$ be the largest integer such that z^{β_j} divides h_j^0, and $A_j := \kappa_j + r\beta_j$. Any monomial $x^{\alpha_1}y^{\alpha_2}z^{\alpha_3}$ of h_j^0 satisfies $p\alpha_1 + q\alpha_2 = \kappa_j + r\alpha_3 \ge A_j$, with equality if $\alpha_3 = \beta_j$. Thus A_j

is the least $\mathbf{T}_{(p,q,0)}$-weight of a monomial of h_j^0, and $A_j \geq 0$. Let $\beta_z := \sum_j c_j\beta_j$, the coefficient of $(z=0)$ in B^0. If $\tau < 0$, then β_z is the number of Construction-Theorem 16.9(2) for the h_j^0. Since $(z=0)$ passes through 0 and (X, B^0) is lc near 0, we have $0 \leq \beta_z \leq 1$, that is, $\sum_j c_j r\beta_j \leq r$. Let E_1 be the toric prime divisor over X of the primitive vector $\frac{1}{\gcd(p,q)}(p,q,0)$. Since A_j is the least $\mathbf{T}_{(p,q,0)}$-weight of a monomial of h_j^0, we have $\operatorname{ord}_{E_1} h_j^0 = \frac{A_j}{\gcd(p,q)}$. Thus

$$a(E_1, X, B^0) = a(E_1, X, 0) - \sum_j c_j \operatorname{ord}_{E_1} h_j^0 = \frac{1}{\gcd(p,q)}\Big(p+q-\sum_j c_j A_j\Big)$$

[CLS11, (11.4.7)], which is non-negative because (X, B^0) is lc near 0. Hence $\sum_j c_j A_j \leq p+q$. In particular $\beta_j \leq \frac{1}{\gamma}$ and $A_j \leq \frac{p+q}{\gamma}$ for any j. Let

$$a_s := p+q-\sum_j c_j A_j, \qquad a_t := r - \sum_j c_j r\beta_j = r(1-\beta_z), \qquad a_{\min} := \min\{a_s, a_t\}.$$

Then $0 \leq a_s \leq p+q$, $0 \leq a_t \leq r$, and $a_t - a_s = r-p-q+\sum_j c_j\kappa_j = \tau$ for the number τ of Construction-Theorem 16.9. Let $\tau^- := \max\{-\tau, 0\}$ as in Construction-Theorem 16.9 and $\tau^+ := \max\{\tau, 0\}$, so that $\tau = \tau^+ - \tau^-$. Moreover $a_s - \tau^- = a_{\min}$ and $a_t - \tau^+ = a_{\min}$.

The group $G \cap \mathbf{T}$ is the image of μ_n under $\alpha \to (\alpha^p, \alpha^q, \alpha^{-r})$, which is injective because $\gcd(p,q,r) = 1$, and α acts on a monomial of $\mathbf{T}$-weight κ by α^κ. Since G is diagonal, any monomial occurring in a G-semi-invariant polynomial h is G-semi-invariant with the character of h. Thus any monomial occurring in h_j has $\mathbf{T}$-weight $\kappa_j + kn$ for an integer k, and h_j^0, together with any sum of terms of h_j, has the character of h_j.

Step 2. In this step we fix the integers N, L_t, L_0, M, l, R and the DCC set Λ, all depending only on p, q, r, e_0, Γ.

Let γ_* and δ_* be the numbers of Set-up 16.14 for $(re_0\max\{p,q\}, \Gamma)$ in place of (e_0, Γ), and let $N \geq 1$ be an integer with

$$\gamma_*\Big(\frac{N}{re_0\max\{p,q\}} - 1\Big) > 1 + \max\{p+q, r\}\Big(1+\frac{1}{\delta_*}\Big).$$

Let $L_t := \lfloor 1/\gamma \rfloor + N + 3$ and $L_0 := \lfloor (p+q)/\gamma \rfloor + N + 3$. By **Step 1**,

$$L_t > \max_j \beta_j, \quad rL_t \geq \max_j r\beta_j + N + 3, \quad L_0 \geq \max_j A_j + N + 3.$$

Let L_s be the unique integer with $L_0 \leq L_s \leq L_0 + r - 1$ and $L_s \equiv n \bmod r$, and let $M := \max\{(L_0 + r)re_0,\ L_t e_0 \max\{p,q\}\}$. Since Γ_+ satisfies the DCC (Lemma 2.10(1)), so do $\Gamma_+ \cap [0, p+q]$ and $\Gamma_+ \cap [0, r]$. Hence

$$\Lambda := \Big(\Big\{1 - \frac{p+q-x}{e},\ 1 - \frac{r-y}{e},\ 1 - \frac{1}{e} \;\Big|\; x \in \Gamma_+ \cap [0, p+q],\ y \in \Gamma_+ \cap [0, r],\ 1 \leq e \leq rM\Big\} \cap \mathbb{R}_{>0}\Big) \cup (\Gamma \setminus \{0\})$$

is a DCC set, as a finite union of DCC sets. Let l be the integer of Lemma 16.20 for (M, Λ), and $R := Ml$. We assume that

$$n \geq L_0 + r + \frac{r}{\gamma} + R \quad \text{and} \quad n \geq r\Big(R + L_t + \frac{1}{\gamma}\Big) + \frac{3P_{\max}}{\gamma}.$$

In particular $n \geq L_s + \max_j r\beta_j + R$, $n \geq rL_t$, and $\frac{n}{r} - L_t - \frac{|\kappa_j|}{r} - \beta_j \geq R$ for any j.

Since $\sum_j c_j A_j \in \Gamma_+ \cap [0, p+q]$ and $\sum_j c_j r\beta_j \in \Gamma_+ \cap [0, r]$, any positive number of the form $1 - \frac{a_{\min}}{e}$ or $1 - \frac{1}{e}$ with $1 \leq e \leq rM$ belongs to Λ, and $c_j \in \Lambda$ for any j.

Step 3. In this step we prove $a(E, X/G, B_{X/G}) \geq b$ for the prime divisors E whose center on a finite quotient of $\mathbb{A}^3_{u_1,v_1,w}$ defined below is contained in the image of $(w=0)$.

As in Construction-Theorem 16.9, consider the cover $z = t^r$, $u := xt^p$, and $v := yt^q$, and let $\tilde{G}$ act on (x,y,t). An element $(g,\alpha) \in \tilde{G}$ acts on (u,v,t) by $(g_x\alpha^p, g_y\alpha^q, \alpha)$. It acts trivially on

(u,v) if and only if $g=(\alpha^{-p},\alpha^{-q},\alpha^r)$, which means $g\in G\cap\mathbf{T}$ and $\alpha\in\mu_n$. Thus the kernel of $\tilde{G}\to\tilde{H}$ acts on t through all of μ_n. Let $U:=t^n$. Then $\tilde{H}$ acts diagonally and faithfully on $\mathbb{A}^3_{u,v,U}$, with characters χ_u,χ_v,χ_U, and

$$\mathbb{C}(u,v,U)^{\tilde{H}}=\mathbb{C}(x,y,t)^{\tilde{G}}=\mathbb{C}(X)^G.$$

A monomial $x^{\alpha_1}y^{\alpha_2}z^{\alpha_3}$ of $\mathbf{T}$-weight κ_j+kn becomes $t^{-\kappa_j}U^{-k}u^{\alpha_1}v^{\alpha_2}$. Hence

$$h_j(x,y,t^r)=t^{-\kappa_j}\sum_{k\in\mathbb{Z}}U^{-k}P_{j,k}(u,v)$$

with $P_{j,0}=P_j$, and any monomial $u^{\alpha_1}v^{\alpha_2}$ of $P_{j,k}$ satisfies $p\alpha_1+q\alpha_2-\kappa_j-kn=r\alpha_3\geq 0$.

Let $V_s:=\mathbb{A}^3_{u_1,v_1,w}$ with $U=w^{L_s}$, $u=u_1w^p$, and $v=v_1w^q$, and let

$$J_s:=\{(h,\beta)\mid h\in\tilde{H},\ \beta^{L_s}=\chi_U(h)\}$$

act on V_s by $(u_1,v_1,w)\to(\chi_u(h)\beta^{-p}u_1,\chi_v(h)\beta^{-q}v_1,\beta w)$. The subgroup $\{(1,\beta)\mid\beta^{L_s}=1\}$ has invariant field $\mathbb{C}(u,v,U)$, so $\mathbb{C}(V_s)^{J_s}=\mathbb{C}(X)^G$. The group J_s acts faithfully, because $\tilde{H}$ acts faithfully on (u,v), and $|J_s|=L_sr|H|\leq M$. Let

$$\phi_j:=w^{-A_j}t^{\kappa_j}h_j=\sum_{k\in\mathbb{Z}}w^{-A_j-kL_s}P_{j,k}(u_1w^p,v_1w^q).$$

A monomial $u^{\alpha_1}v^{\alpha_2}$ of $P_{j,k}$ contributes the exponent $p\alpha_1+q\alpha_2-A_j-kL_s$ of w. For $k=0$ it is non-negative, because the monomials $u^{\alpha_1}v^{\alpha_2}$ of P_j correspond to the monomials $x^{\alpha_1}y^{\alpha_2}z^{\alpha_3}$ of h_j^0, with $p\alpha_1+q\alpha_2=\kappa_j+r\alpha_3$, so A_j is the least (p,q)-weight of a monomial of P_j (**Step 1**). The terms with exponent 0 correspond to the monomials of h_j^0 whose exponent of z is β_j, and their sum is the non-zero polynomial $\mathrm{in}_{(p,q)}(P_j)(u_1,v_1)$ in u_1,v_1. For $k<0$ it is at least $L_s-A_j>0$. For $k>0$ it is at least

$$\kappa_j+kn-A_j-kL_s=k(n-L_s)-r\beta_j\geq n-L_s-r\beta_j\geq R.$$

Thus ϕ_j is a polynomial with $\phi_j|_{(w=0)}\neq 0$. It is J_s-semi-invariant, and so is the truncation ϕ_j^- of ϕ_j consisting of the terms with $k\leq 0$, with the character of ϕ_j (**Step 1**).

Since $dx\wedge dy\wedge dz$ is a non-zero constant times $t^{r-1-p-q}dt\wedge du\wedge dv$ and $t^n=U=w^{L_s}$, a direct calculation shows that $K_{V_s}+B_s$ is the pullback of $K_{X/G}+B_{X/G}$ for

$$B_s:=\Big(1-a_s-\frac{L_s\tau}{n}\Big)(w=0)+\sum_j c_j\,\mathrm{div}(\phi_j):$$

the form contributes $1-p-q-\frac{L_s(r-p-q)}{n}$ and $\sum_j c_j\,\mathrm{div}(h_j)$ contributes $\sum_j c_j\big(A_j-\frac{L_s\kappa_j}{n}\big)$ to the coefficient of $(w=0)$, because $t^n=w^{L_s}$ and $h_j=w^{A_j}t^{-\kappa_j}\phi_j$. Since G is diagonal, $\mathbb{C}[x,y,z]^G$ is generated by G-invariant monomials. A G-invariant monomial $x^{\alpha_1}y^{\alpha_2}z^{\alpha_3}$ has $\mathbf{T}$-weight kn for an integer k, and it pulls back to $u_1^{\alpha_1}v_1^{\alpha_2}w^{k(n-L_s)+r\alpha_3}$, where $k(n-L_s)+r\alpha_3\geq 0$ because $r\alpha_3\geq\max\{0,-kn\}$ and $n\geq L_s$. Thus the G-invariant monomials pull back to regular functions on V_s, and $V_s/J_s\to X/G$ is a birational morphism, crepant for $(V_s,B_s)/J_s$ and $(X/G,B_{X/G})$ (Lemma 2.6(0)). Hence

$$a(E,X/G,B_{X/G})=a(E,V_s/J_s,(B_s)_{V_s/J_s})$$

for any prime divisor E over X/G with a center on V_s/J_s [KM98, 2.23 and 2.25]. Since $n\geq L_s$,

$$a_s+\frac{L_s\tau}{n}=\Big(1-\frac{L_s}{n}\Big)a_s+\frac{L_s}{n}a_t\geq a_{\min}.$$

Let

$$B_s^+:=(1-a_{\min})(w=0)+\sum_j c_j\,\mathrm{div}(\phi_j)\geq B_s.$$

Let ζ be a primitive r-th root of unity and $h_\zeta\in\tilde{H}$ the image of $(1,\zeta)\in\tilde{G}$. Then $\chi_U(h_\zeta)=\zeta^n=\zeta^{L_s}$, so $(h_\zeta,\zeta)\in J_s$, and it acts by $(u_1,v_1,w)\to(u_1,v_1,\zeta w)$ because h_ζ acts on (u,v) by (ζ^p,ζ^q).

Thus the inertia group of $(w=0)$ in J_s has order e_w divisible by r, and $a_{\min} \le a_t \le r \le e_w$. Let $B_{\mathrm{bg}} := \big((1-a_{\min})(w=0)\big)_{V_s/J_s}$. The coefficients of B_{bg} are $1-\frac{a_{\min}}{e_w} \ge 0$ along the image of $(w=0)$ and $1-\frac{1}{e'}$ along the other branch divisors, so $B_{\mathrm{bg}} \ge 0$ and $B_{\mathrm{bg}} \in \Lambda$ (**Step 2**). Let

$$\mathfrak{a}_s := \prod_j \big(\phi_j^{|J_s|}\big)^{c_j/|J_s|}, \qquad \mathfrak{a}_s^- := \prod_j \big((\phi_j^-)^{|J_s|}\big)^{c_j/|J_s|},$$

so that

$$(B_s^+)_{V_s/J_s} = B_{\mathrm{bg}} + \sum_j \frac{c_j}{|J_s|}\operatorname{div}(\phi_j^{|J_s|}).$$

Since $\phi_j - \phi_j^- \in (w^R)$, Lemma 16.20 implies

$$\operatorname{mld}(V_s/J_s \ni \eta, B_{\mathrm{bg}}, \mathfrak{a}_s) = \operatorname{mld}(V_s/J_s \ni \eta, B_{\mathrm{bg}}, \mathfrak{a}_s^-)$$

for any scheme-theoretic point η of the image of $(w=0)$.

Let $\mathfrak{b}_s$ be the ideal on V_s/J_s generated by $u_1^{q|J_s|}$ and $v_1^{p|J_s|}$. By Lemma 2.6(2), the pullback of $\mathfrak{b}_s^{\tau^-/(pq|J_s|)}$ to V_s has the orders of $(u_1^q, v_1^p)^{\tau^-/(pq)}$ along any prime divisor. Let

$$K_j := \sum_{k\le 0} U^{-k} P_{j,k}(u,v) \in \mathbb{C}[U,u,v],$$

a $\tilde{H}$-semi-invariant polynomial with $K_j(0,u,v) = P_j$. The ideal (u^q, v^p) pulls back to $w^{pq}(u_1^q, v_1^p)$, and a direct calculation shows that the triple

$$\Big(V_s, (1-a_{\min})(w=0) + \sum_j c_j \operatorname{div}(\phi_j^-), (u_1^q, v_1^p)^{\tau^-/(pq)}\Big)$$

is the pullback of

$$\Big(\mathbb{A}^3_{U,u,v}, (U=0) + \sum_j c_j \operatorname{div}(K_j), (u^q, v^p)^{\tau^-/(pq)}\Big),$$

where the factor w^{τ^-} of the pullback of the $\mathbb{R}$-ideal $(u^q, v^p)^{\tau^-/(pq)}$ is moved into the coefficient of $(w=0)$: the divisor $(U=0)$ and the K_j contribute $1-a_s$ to the coefficient of $(w=0)$, and the ideal contributes τ^-, while $a_s - \tau^- = a_{\min}$.

Let $V \subset \mathbb{A}^2_{u,v}$ be the $\tilde{H}$-invariant affine open neighborhood of 0 of Construction-Theorem 16.9, $\bar{0} \in V/\tilde{H}$ the image of 0, and $\mathfrak{b}$ the ideal on $V/\tilde{H}$ generated by $u^{qr|H|}$ and $v^{pr|H|}$. By Construction-Theorem 16.9, $\big(V/\tilde{H}, (C|_V)_{V/\tilde{H}}, \mathfrak{b}^{\tau^-/(pqr|H|)}\big)$ is lc near $\bar{0}$ with mld b at $\bar{0}$, and the pullback of $\big((C|_V)_{V/\tilde{H}}, \mathfrak{b}^{\tau^-/(pqr|H|)}\big)$ to V has the orders of $(C|_V, (u^q, v^p)^{\tau^-/(pq)})$ along any prime divisor (Lemma 2.6(2)). Let F be a prime divisor over V_s/J_s with center $\overline{\{\eta\}}$ contained in the image of $(w=0)$. Then $\operatorname{ord}_F(w) > 0$ and $\operatorname{ord}_F(u_1), \operatorname{ord}_F(v_1) \ge 0$, so $U = w^{L_s}$, $u = u_1 w^p$, and $v = v_1 w^q$ have positive orders along F, and the center of F on $(\mathbb{A}^1_U \times V)/\tilde{H}$ is the image of $(0,0)$. Here $B_S = C|_V$, the triple $\big(V, C|_V, \pi^*\mathfrak{b}^{\tau^-/(pqr|H|)}\big)$ is lc near 0 by Lemma 2.6(2) for the quotient morphism $\pi : V \to V/\tilde{H}$, and by the last part of Construction-Theorem 16.9, the number b_s of Lemma 16.19(1) at the point $0 \in V$ equals b. Thus Lemma 16.19(1), with $H_0 := \tilde{H}$, $S := V$, the closed point $0 \in V$ in place of s, the functions K_j, and $\mathfrak{a} := \mathfrak{b}^{\tau^-/(pqr|H|)}$, implies

$$a\big(F, V_s/J_s, B_{\mathrm{bg}}, \mathfrak{a}_s^- \mathfrak{b}_s^{\tau^-/(pq|J_s|)}\big) \ge b$$

(both quotient triples have function field $\mathbb{C}(X)^G$ and pull back to triples on V_s with equal log discrepancies along any prime divisor over V_s, so their log discrepancies are equal by Lemma 2.6(2) and the ramification formula for log discrepancies under the generically finite morphism $V_s \to \mathbb{A}^3_{U,u,v}$, cf. [KM98, Proposition 5.20]).

Let E be a prime divisor over X/G whose center on V_s/J_s is $\overline{\{\eta\}}$ for a point η of the image of $(w=0)$. Then

$$\begin{aligned} a(E,X/G,B_{X/G}) &\geq a(E,V_s/J_s,(B_s^+)_{V_s/J_s}) \\ &\geq \mathrm{mld}(V_s/J_s \ni \eta, B_{\mathrm{bg}}, \mathfrak{a}_s) = \mathrm{mld}(V_s/J_s \ni \eta, B_{\mathrm{bg}}, \mathfrak{a}_s^-), \end{aligned}$$

and the last mld is at least $\mathrm{mld}\big(V_s/J_s \ni \eta, B_{\mathrm{bg}}, \mathfrak{a}_s^- \mathfrak{b}_s^{\tau^-/(pq|J_s|)}\big) \geq b$. The exponent $\tau^-/(pq|J_s|)$ was not used in Lemma 16.20, so it need not belong to a DCC set.

Step 4. In this step we prove $a(E,X/G,B_{X/G}) \geq b$ for the prime divisors E whose center on a finite quotient of $\mathbb{A}^3_{y_1,y_2,\xi}$ defined below is contained in the image of $(y_1=0)$.

Let $g_0 := \gcd(p,q)$, and consider the morphism $\mathbb{A}^3_{s,\xi,z} \to X$, $(s,\xi,z) \to (s^p, s^q\xi, z)$, a finite cover of the complement of the strict transform of $(x=0)$ in the weighted blow-up of X with weights $(p/g_0, q/g_0, 0)$, and let $U' := s^n$ and $y_3 := s^r z$. The group $\{(g,\alpha) \mid g \in G,\ \alpha^p = g_x\}$ of order $p|G|$ acts on (s,ξ,z) by $(\alpha, g_y\alpha^{-q}, g_z)$, hence on (y_3,ξ) by the characters $\alpha^r g_z$ and $g_y\alpha^{-q}$. An element acts trivially on (y_3,ξ) if and only if $g = (\alpha^p, \alpha^q, \alpha^{-r})$, which means $g \in G \cap \mathbf{T}$ and $\alpha \in \mu_n$. Thus the residual group J_x of order $p|H|$ acts diagonally and faithfully on $\mathbb{A}^3_{U',y_3,\xi}$ with characters $\chi_{U'}, \chi_{y_3}, \chi_\xi$, and $\mathbb{C}(U',y_3,\xi)^{J_x} = \mathbb{C}(X)^G$. A monomial $x^{\alpha_1}y^{\alpha_2}z^{\alpha_3}$ of $\mathbf{T}$-weight $\kappa_j + kn$ becomes $s^{\kappa_j}U'^k y_3^{\alpha_3}\xi^{\alpha_2}$. Hence

$$h_j(s^p, s^q\xi, z) = s^{\kappa_j}\tilde{h}_j(U', y_3, \xi), \qquad \tilde{h}_j = \sum_{k\in\mathbb{Z}} U'^k Q_{j,k}(y_3,\xi).$$

Let f_j be the polynomial in two variables such that $h_j^0(s^p, s^q\xi, t^r) = s^{A_j}t^{r\beta_j}f_j(st,\xi)$ in the polynomial ring $\mathbb{C}[s,t,\xi]$, where t is an indeterminate ($z = t^r$ as in **Step 3**). It exists because a monomial of h_j^0 becomes $s^{A_j}t^{r\beta_j}(st)^{r(\alpha_3-\beta_j)}\xi^{\alpha_2}$, and $f_j(st,\xi)$ is a polynomial in $(st)^r$ and ξ, with $f_j(0,\xi) \neq 0$. Let $\bar{f}_j$ be the polynomial with $f_j(st,\xi) = \bar{f}_j((st)^r,\xi)$. Then $Q_{j,0}(y_3,\xi) = y_3^{\beta_j}\bar{f}_j(y_3,\xi)$.

Let $V_t := \mathbb{A}^3_{y_1,y_2,\xi}$ with $U' = y_1^{L_t}$ and $y_3 = y_1y_2$, and let

$$J_t := \{(h,\alpha) \mid h \in J_x,\ \alpha^{L_t} = \chi_{U'}(h)\}$$

act on V_t by $(y_1,y_2,\xi) \to (\alpha y_1, \chi_{y_3}(h)\alpha^{-1}y_2, \chi_\xi(h)\xi)$. As in **Step 3**, J_t acts faithfully, $|J_t| = L_t p|H| \leq M$, and $\mathbb{C}(V_t)^{J_t} = \mathbb{C}(X)^G$. Let $\phi_j := y_1^{-\beta_j}\tilde{h}_j(y_1^{L_t}, y_1y_2, \xi)$. A monomial $U'^k y_3^{\alpha_3}\xi^{\alpha_2}$ contributes the exponent $kL_t + \alpha_3 - \beta_j$ of y_1. For $k=0$ it is non-negative, and the terms with exponent 0 have the non-zero sum $y_2^{\beta_j}\bar{f}_j(0,\xi)$. For $k \geq 1$ it is at least $L_t - \beta_j > 0$. For $k = -m < 0$, the regularity of h_j implies $r\alpha_3 = p\alpha_1 + q\alpha_2 + mn - \kappa_j \geq mn - \kappa_j$, so the exponent is at least

$$m\Big(\frac{n}{r} - L_t\Big) - \frac{\kappa_j}{r} - \beta_j \geq \frac{n}{r} - L_t - \frac{|\kappa_j|}{r} - \beta_j \geq R.$$

Thus ϕ_j is a J_t-semi-invariant polynomial with $\phi_j|_{(y_1=0)} \neq 0$, and so is the truncation ϕ_j^+ of ϕ_j consisting of the terms with $k \geq 0$, with the character of ϕ_j.

Since $dx \wedge dy \wedge dz$ is a non-zero constant times $s^{p+q-r-1}ds \wedge d\xi \wedge dy_3$ and $dU' \wedge dy_3 = L_t y_1^{L_t} dy_1 \wedge dy_2$, a direct calculation shows that $K_{V_t} + B_t$ is the pullback of $K_{X/G} + B_{X/G}$ for

$$B_t := \Big(\beta_z + \frac{L_t\tau}{n}\Big)(y_1=0) + \sum_j c_j \operatorname{div}(\phi_j) :$$

the form contributes $\frac{L_t(r-p-q)}{n}$ and $\sum_j c_j \operatorname{div}(h_j)$ contributes $\sum_j c_j\big(\beta_j + \frac{L_t\kappa_j}{n}\big)$ to the coefficient of $(y_1=0)$, because $s^n = y_1^{L_t}$ and $h_j = s^{\kappa_j}y_1^{\beta_j}\phi_j$. A G-invariant monomial $x^{\alpha_1}y^{\alpha_2}z^{\alpha_3}$ of $\mathbf{T}$-weight kn pulls back to $y_1^{kL_t+\alpha_3}y_2^{\alpha_3}\xi^{\alpha_2}$, where $kL_t + \alpha_3 \geq 0$ because $r\alpha_3 \geq -kn$ and $n \geq rL_t$. Thus, as in **Step 3**, the G-invariant monomials pull back to regular functions on V_t, $V_t/J_t \to X/G$ is a crepant birational morphism, and $a(E,X/G,B_{X/G}) = a(E,V_t/J_t,(B_t)_{V_t/J_t})$ for any prime

divisor E over X/G with a center on V_t/J_t. Let $\beta_* := \beta_z + \frac{\tau^+}{r} = 1 - \frac{a_{\min}}{r} \in [0,1]$. Since $n \geq rL_t$, we have $\frac{L_t\tau}{n} \leq \frac{\tau^+}{r}$, and

$$B_t^+ := \beta_*(y_1 = 0) + \sum_j c_j \operatorname{div}(\phi_j) \geq B_t.$$

Let $B'_{\mathrm{bg}} := \big(\beta_*(y_1 = 0)\big)_{V_t/J_t}$. The coefficients of B'_{bg} are $1 - \frac{a_{\min}}{re}$ along the image of $(y_1 = 0)$, where $e \leq M$ is the order of the inertia group, and $1 - \frac{1}{e'}$ along the other branch divisors. Thus $B'_{\mathrm{bg}} \geq 0$ and $B'_{\mathrm{bg}} \in \Lambda$. With $\mathfrak{a}_t$ and $\mathfrak{a}_t^+$ defined from ϕ_j and ϕ_j^+ as in **Step 3**, Lemma 16.20 implies

$$\operatorname{mld}(V_t/J_t \ni \eta, B'_{\mathrm{bg}}, \mathfrak{a}_t) = \operatorname{mld}(V_t/J_t \ni \eta, B'_{\mathrm{bg}}, \mathfrak{a}_t^+)$$

for any scheme-theoretic point η of the image of $(y_1 = 0)$.

As in **Step 3**, we add the $\mathbb{R}$-divisor $\frac{\tau^+}{r}(y_2 = 0)$ only now. Let

$$K'_j := \sum_{k \geq 0} U'^k Q_{j,k}(y_3, \xi) \in \mathbb{C}[U', y_3, \xi],$$

a J_x-semi-invariant polynomial with $K'_j(0, y_3, \xi) = Q_{j,0}$. A direct calculation shows that

$$\Big(V_t, \beta_*(y_1 = 0) + \frac{\tau^+}{r}(y_2 = 0) + \sum_j c_j \operatorname{div}(\phi_j^+)\Big)$$

is the pullback of

$$\Big(\mathbb{A}^3_{U',y_3,\xi}, (U' = 0) + \frac{\tau^+}{r}(y_3 = 0) + \sum_j c_j \operatorname{div}(K'_j)\Big) :$$

the divisor $(U' = 0)$ and the K'_j contribute β_z to the coefficient of $(y_1 = 0)$, and $\frac{\tau^+}{r}(y_3 = 0)$ contributes $\frac{\tau^+}{r}$ to the coefficients of $(y_1 = 0)$ and $(y_2 = 0)$. The restriction of $\frac{\tau^+}{r}(y_3 = 0) + \sum_j c_j \operatorname{div}(K'_j)$ to $(U' = 0) \cong \mathbb{A}^2_{y_3,\xi}$ is

$$B_{y_3,\xi} := \frac{\tau^+}{r}(y_3 = 0) + \sum_j c_j \operatorname{div}(Q_{j,0}) = \Big(1 - \frac{a_{\min}}{r}\Big)(y_3 = 0) + \sum_j c_j \operatorname{div}(\bar{f}_j).$$

Consider the finite cover $\mathbb{A}^2_{w,\xi}$, given by $u = w^p$ and $v = w^q\xi$, of the complement of the strict transform of $(u = 0)$ in the weighted blow-up of $\mathbb{A}^2_{u,v}$ with weights $(p/g_0, q/g_0)$. Let S' be the inverse image of V in $\mathbb{A}^2_{w,\xi}$, and let $S \subset \mathbb{A}^2_{y_3,\xi}$ be the image of S' under $y_3 = w^r$. Then S is a J_x-invariant affine open subset containing $(y_3 = 0)$.

Claim 16.22. *The pair $(S, B_{y_3,\xi}|_S)$ is lc near $(y_3 = 0)$, and $a(F, S/J_x, (B_{y_3,\xi}|_S)_{S/J_x}) \geq b$ for any prime divisor F over S/J_x whose center is contained in the image of $(y_3 = 0)$.*

Proof. If $\tau < 0$, let $f_V : V' \to V$, $B_{V'}$, and (Z, B_Z) be as in Construction-Theorem 16.9(2), E_V the exceptional curve of f_V, and $\bar{E}_V$ the image of E_V in Z. The group $J_w := \{(g, \alpha, \beta) \mid g \in G,\ \alpha^p = g_x,\ \beta^r = g_z\}$ acts on (s, t, ξ) by $(\alpha, \beta, g_y\alpha^{-q})$, hence on $(w, \xi) = (st, \xi)$ by the characters $\alpha\beta$ and $g_y\alpha^{-q}$. It acts on $u = w^p$ by $g_x\beta^p$ and on $v = w^q\xi$ by $g_y\beta^q$, which is the action of $(g, \beta) \in \tilde{G}$. The elements $(1, \alpha, 1)$ with $\alpha^p = 1$ act by $(w, \xi) \to (\alpha w, \alpha^{-q}\xi)$, and their invariant field in $\mathbb{C}(w, \xi)$ is $\mathbb{C}(u, v)$. Thus the image of J_w in $\operatorname{Aut}(\mathbb{A}^2_{w,\xi})$ has invariant field $\mathbb{C}(u, v)^{\tilde{H}}$. The elements $(1, 1, \beta)$ with $\beta^r = 1$ act by $(w, \xi) \to (\beta w, \xi)$, and the quotient by them is $y_3 = w^r$. The remaining action on (y_3, ξ) has the characters $\alpha^r g_z$ and $g_y\alpha^{-q}$, which is the action of J_x. Hence S/J_x is a quotient of S' by the image of J_w, and it is birational to $V/\tilde{H}$.

Since $du \wedge dv = pw^{p+q-1} dw \wedge d\xi$ and $P_j(w^p, w^q\xi) = w^{A_j} f_j(w, \xi)$, the pullback of $C|_V$ to S' is $(1 - a_s)(w = 0) + \sum_j c_j \operatorname{div}(f_j)$, and the ideal (u^q, v^p) pulls back to (w^{pq}). Hence the pullback to

S' of $\big((C|_V)_{V/\tilde{H}}, \mathfrak{b}^{\tau^-/(pqr|H|)}\big)$ has the orders of

$$(1-a_{\min})(w=0)+\sum_j c_j \operatorname{div}(f_j) \tag{16.62}$$

along any prime divisor, because $a_s-\tau^-=a_{\min}$. If $\tau<0$, the $\mathbb{R}$-divisor in (16.62) is the pullback of $B_{V'}$, because $a_{\min}=a_t$ and $(w=0)$ maps to E_V with ramification index g_0. The quotient by $(1,1,\beta)$ is ramified with index r along $(w=0)$, and $f_j(w,\xi)=\bar{f}_j(w^r,\xi)$, so the crepant quotient by the elements $(1,1,\beta)$ is $(S,B_{y_3,\xi}|_S)$. Thus $(S,B_{y_3,\xi}|_S)/J_x$ is crepant birational to $(V,C|_V)/\tilde{H}$ if $\tau\geq 0$ (then $\tau^-=0$, so $\mathfrak{b}^{\tau^-/(pqr|H|)}=\mathcal{O}_{V/\tilde{H}}$), and to (Z,B_Z) if $\tau<0$.

If $\tau\geq 0$, the image of $(y_3=0)$ maps to $\bar{0}$, so any F as in the claim has center $\bar{0}$ on $V/\tilde{H}$, and $a(F,S/J_x,(B_{y_3,\xi}|_S)_{S/J_x})\geq b$ by Construction-Theorem 16.9. Moreover, $(V,C|_V)/\tilde{H}$ is lc near $\bar{0}$ by Construction-Theorem 16.9, and the inverse image in S/J_x of an lc neighborhood of $\bar{0}$ is an open subset containing the image of $(y_3=0)$, so $(S,B_{y_3,\xi}|_S)$ is lc near $(y_3=0)$ (Lemma 2.6(2)). If $\tau<0$, the image of $(y_3=0)$ is an open subset of $\bar{E}_V$, so $a(F,S/J_x,(B_{y_3,\xi}|_S)_{S/J_x})\geq b$ by the definition of b, and $(S,B_{y_3,\xi}|_S)$ is lc near $(y_3=0)$ because (Z,B_Z) is lc. □

Proof of Theorem 16.21 continued. In Lemma 16.19, take $H_0:=J_x$, U' in place of U, $(z_1,z_2):=(y_3,\xi)$, the surface S, the functions K'_j with coefficients c_j together with the function y_3 with coefficient $\frac{\tau^+}{r}$, and $\mathfrak{a}:=\mathcal{O}_{S/J_x}$. Then $B_{V_0}=(U'=0)+\frac{\tau^+}{r}(y_3=0)+\sum_j c_j\operatorname{div}(K'_j)$ and $B_S=B_{y_3,\xi}|_S$. Let $\Sigma:=(y_3=0)\subset S$. By Claim 16.22, for any closed point o_S of Σ, the number b_s of Lemma 16.19(1) with o_S in place of s is at least b, and the number b_Σ of Lemma 16.19(2) with Σ in place of C satisfies $b_\Sigma\geq b$, because the image of Σ is a prime divisor over S/J_x with center in the image of $(y_3=0)$. Let F be a prime divisor over V_t/J_t with center contained in the image of $(y_1=0)$. Then $U'=y_1^{L_t}$ and $y_3=y_1y_2$ have positive orders along F, so the center of F on $(\mathbb{A}^1_{U'}\times S)/J_x$ is the image of $(0,o_S)$ for a closed point $o_S\in\Sigma$ or of $\{0\}\times\Sigma$. Let $B''_{\mathrm{bg}}:=\big(\beta_*(y_1=0)+\frac{\tau^+}{r}(y_2=0)\big)_{V_t/J_t}\geq B'_{\mathrm{bg}}$. Thus Lemma 16.19 implies $a(F,V_t/J_t,B''_{\mathrm{bg}},\mathfrak{a}_t^+)\geq b$ (the quotient triples on V_t/J_t and on $(\mathbb{A}^1_{U'}\times S)/J_x$ have equal log discrepancies as in **Step 3**), and hence $\operatorname{mld}(V_t/J_t\ni\eta,B'_{\mathrm{bg}},\mathfrak{a}_t^+)\geq b$ for any point η of the image of $(y_1=0)$. As in **Step 3**, for any prime divisor E over X/G whose center on V_t/J_t is $\overline{\{\eta\}}$ for a point η of the image of $(y_1=0)$,

$$a(E,X/G,B_{X/G})\geq \operatorname{mld}(V_t/J_t\ni\eta,B'_{\mathrm{bg}},\mathfrak{a}_t)=\operatorname{mld}(V_t/J_t\ni\eta,B'_{\mathrm{bg}},\mathfrak{a}_t^+)\geq b.$$

With the roles of x and y exchanged, namely for the morphism $(s,\xi,z)\to(s^p\xi,s^q,z)$, the argument of **Step 4** implies $a(E,X/G,B_{X/G})\geq b$ for the prime divisors E whose center on the corresponding quotient V_t/J_t is contained in the image of $(y_1=0)$, with $|J_t|=L_tq|H|\leq M$.

Step 5. In this step we prove $a(E,X/G,B_{X/G})\geq b$ for the prime divisors E whose center on a finite quotient of the affine toric open subset U_{σ_i}, $\sigma_i:=\operatorname{Cone}(\nu_i,\nu_{i+1})$, defined below is contained in the image of $(u_0v_0=0)$, for any i with $\min\{i-\max_j r\beta_j,n-i-1-\max_j A_j\}\geq N$.

Consider the morphism $\mathbb{A}^3_{s,t,\xi}\to X$, $(s,t,\xi)\to(s^p,s^q\xi,t^r)$. The group

$$\{(g,\alpha,\beta)\mid g\in G,\ \alpha^p=g_x,\ \beta^r=g_z\}$$

of order $pr|G|$ acts on $\mathbb{A}^3_{s,t,\xi}$ by $(\alpha,\beta,g_y\alpha^{-q})$ with invariant field $\mathbb{C}(X)^G$. The elements (g,α,β) acting trivially on (st,ξ) are those with $g=(\alpha^p,\alpha^q,\alpha^{-r})$ and $\beta=\alpha^{-1}$, which means $\alpha\in\mu_n$. They form a subgroup isomorphic to μ_n acting by $(s,t,\xi)\to(\alpha s,\alpha^{-1}t,\xi)$, and the remaining group has order $pr|H|\leq re_0\max\{p,q\}$. Since $dx\wedge dy\wedge dz$ is a non-zero constant times $s^{p+q-1}t^{r-1}ds\wedge dt\wedge d\xi$, the pullback of (X,B) is

$$\Big(\mathbb{A}^3_{s,t,\xi},(1-p-q)(s=0)+(1-r)(t=0)+\sum_j c_j\operatorname{div}\big(h_j(s^p,s^q\xi,t^r)\big)\Big).$$

The quotient $\mathbb{A}^2_{s,t}/\mu_n$ is the affine toric surface of the cone $\mathbb{R}^2_{\geq 0}$ in the lattice $\mathbb{Z}^2+\mathbb{Z}(\frac{1}{n},\frac{n-1}{n})$, and the rays of

$$\nu_i := \Big(\frac{i}{n},\frac{n-i}{n}\Big), \qquad 0\leq i\leq n,$$

define the minimal resolution of $\mathbb{A}^2_{s,t}/\mu_n$ [Ful93, §2.6, pp. 45–47]: consecutive ν_i,ν_{i+1} form a lattice basis. The affine toric open subset $U_{\sigma_i}\times\mathbb{A}^1_\xi = \mathbb{A}^3_{u_0,v_0,\xi}$ has coordinates with

$$st = u_0v_0, \qquad s^n = u_0^i v_0^{i+1}, \qquad t^n = u_0^{n-i}v_0^{n-i-1},$$

where $(u_0=0)$ and $(v_0=0)$ are the divisors of ν_i and ν_{i+1}. The remaining group commutes with the torus, so it acts diagonally on $\mathbb{A}^3_{u_0,v_0,\xi}$. Let J_i be the image of the remaining group in $\mathrm{Aut}(\mathbb{A}^3_{u_0,v_0,\xi})$, of order at most $re_0\max\{p,q\}$, so that $\mathbb{C}(u_0,v_0,\xi)^{J_i}=\mathbb{C}(X)^G$.

The pullback of h_j^0 is $s^{A_j}t^{r\beta_j}f_j(st,\xi)$, and $s^{A_j}t^{r\beta_j}$ is semi-invariant for μ_n with the character of the pullback of h_j. A monomial $s^{\alpha_1}t^{\alpha_2}\xi^{\alpha_3}$ of the pullback of h_j satisfies $\alpha_1-\alpha_2=\kappa_j+kn$, and the quotient of the monomial by $s^{A_j}t^{r\beta_j}$ is

$$(u_0v_0)^{\alpha_2-r\beta_j}s^{kn}\xi^{\alpha_3} = u_0^{\alpha_2-r\beta_j+ik}v_0^{\alpha_2-r\beta_j+(i+1)k}\xi^{\alpha_3}.$$

For $k>0$ both exponents are at least $i-r\beta_j$. For $k<0$, the inequality $\alpha_1\geq 0$ implies $\alpha_2\geq -\kappa_j-kn$, and both exponents are at least $-A_j+|k|(n-i-1)\geq n-i-1-A_j$. Thus the μ_n-invariant function $H_j := s^{-A_j}t^{-r\beta_j}h_j(s^p,s^q\xi,t^r)$ is a polynomial in u_0,v_0,ξ and satisfies

$$H_j - f_j(u_0v_0,\xi)\in((u_0v_0)^N).$$

The polynomials H_j and $f_j(u_0v_0,\xi)$ are J_i-semi-invariant, and $f_j(u_0v_0,\xi)$ has the character of H_j, because h_j^0 has the character of h_j (**Step 1**) and both h_j and h_j^0 are divided by the monomial $s^{A_j}t^{r\beta_j}$. Since $f_j(0,\xi)\neq 0$, the order of $f_j(st,\xi)$ along the divisor of ν_i is 0. Thus, by [CLS11, Lemma 11.4.10 and Proposition 4.1.1], the divisor of ν_i has log discrepancy

$$\frac{i}{n}(p+q)+\frac{n-i}{n}r-\sum_j c_j\Big(\frac{i}{n}A_j+\frac{n-i}{n}r\beta_j\Big)=\omega_i := \frac{i}{n}a_s+\Big(1-\frac{i}{n}\Big)a_t\in[0,\max\{p+q,r\}]$$

with respect to the pullback of (X,B^0). Hence the pullbacks of (X,B) and (X,B^0) to $\mathbb{A}^3_{u_0,v_0,\xi}$ are

$$B_{H,i} := (1-\omega_i)(u_0=0)+(1-\omega_{i+1})(v_0=0)+\sum_j c_j\,\mathrm{div}(H_j),$$

$$B_{0,i} := (1-\omega_i)(u_0=0)+(1-\omega_{i+1})(v_0=0)+\sum_j c_j\,\mathrm{div}\big(f_j(u_0v_0,\xi)\big).$$

Since $1\leq i<i+1\leq n-1$, the non-constant G-invariant monomials, which are regular on $\mathbb{A}^3_{u_0,v_0,\xi}$, vanish along $(u_0v_0=0)$, and $(u_0v_0=0)$ maps to the image of 0 in X/G. Thus $\mathbb{A}^3_{u_0,v_0,\xi}/J_i\to X/G$ is a crepant birational morphism for $(B_{H,i})_{\mathbb{A}^3/J_i}$ and $B_{X/G}$, and $a(E,X/G,B_{X/G})=a(E,\mathbb{A}^3_{u_0,v_0,\xi}/J_i,(B_{H,i})_{\mathbb{A}^3/J_i})$ for any prime divisor E over X/G with a center on $\mathbb{A}^3_{u_0,v_0,\xi}/J_i$. Similarly $a(E,X/G,B^0_{X/G})=a(E,\mathbb{A}^3_{u_0,v_0,\xi}/J_i,(B_{0,i})_{\mathbb{A}^3/J_i})$ for any such E, where $(X,B^0)/G=(X/G,B^0_{X/G})$. Since (X,B^0) is lc near 0, $(\mathbb{A}^3_{u_0,v_0,\xi},B_{0,i})$ is sub-lc near $(u_0v_0=0)$. By Proposition 16.16 with Set-up 16.14 for $(re_0\max\{p,q\},\Gamma)$ in place of (e_0,Γ), $\kappa=\max\{p+q,r\}$, $(\lambda_0,\lambda_1)=(\omega_i,\omega_{i+1})$, $J=J_i$, the coordinates u_0,v_0,ξ in place of u,v,z, the $\mathbb{R}$-divisors $B_{H,i},B_{0,i}$ in place of B_H,B_0, and the point η in place of the point ξ of the proposition, for any scheme-theoretic point η of the image of $(u_0v_0=0)$ in $\mathbb{A}^3_{u_0,v_0,\xi}/J_i$,

$$\mathrm{mld}\big(\mathbb{A}^3_{u_0,v_0,\xi}/J_i\ni\eta,(B_{H,i})_{\mathbb{A}^3/J_i}\big)=\mathrm{mld}\big(\mathbb{A}^3_{u_0,v_0,\xi}/J_i\ni\eta,(B_{0,i})_{\mathbb{A}^3/J_i}\big).$$

Any prime divisor over $\mathbb{A}^3_{u_0,v_0,\xi}/J_i$ with center $\overline{\{\eta\}}$ has center the image of 0 on X/G, so the right-hand side is at least $\mathrm{mld}((X\ni 0,B^0)/G)\geq b$ by Construction-Theorem 16.9. Thus $a(E,X/G,B_{X/G})\geq b$ for any prime divisor E over X/G whose center on $\mathbb{A}^3_{u_0,v_0,\xi}/J_i$ is contained in the image of $(u_0v_0=0)$. With the roles of x and y exchanged, the argument of **Step 5** implies

$a(E, X/G, B_{X/G}) \geq b$ for the prime divisors E whose center on the corresponding quotient is contained in the image of $(u_0 v_0 = 0)$.

Step 6. In this step we prove that any prime divisor E over X/G with center the image of 0 is treated in **Steps 3–5**. This proves (2).

Let K be a finite Galois extension of $\mathbb{C}(X)^G$ containing $\mathbb{C}(s,t,\xi)$ for both morphisms $(s,t,\xi) \to (s^p, s^q\xi, t^r)$ and $(s,t,\xi) \to (s^p\xi, s^q, t^r)$, the function field of V_s, and the function fields of the varieties V_t of **Step 4** for both roles of x and y. Let Y be the normalization of X/G in K, on which $\mathrm{Gal}(K/\mathbb{C}(X)^G)$ acts with quotient X/G. By Lemma 2.6(3), there exists a prime divisor E' over Y such that $\mathrm{ord}_{E'}|_{\mathbb{C}(X)^G}$ is a positive multiple of ord_E. The functions x, y, z have positive orders along E', because $x^{|G|}, y^{|G|}, z^{|G|}$ are G-invariant regular functions vanishing at 0. Since $\mathrm{ord}_{E'}(y)/q \geq \mathrm{ord}_{E'}(x)/p$ or the reverse inequality holds, possibly exchanging the roles of x and y we may assume that $\mathrm{ord}_{E'}(x)/p \leq \mathrm{ord}_{E'}(y)/q$. We now use the morphism $(s,t,\xi) \to (s^p, s^q\xi, t^r)$ of **Steps 4** and **5**, for which

$$\mathrm{ord}_{E'}(\xi) = \mathrm{ord}_{E'}(y) - \frac{q}{p}\mathrm{ord}_{E'}(x) \geq 0.$$

Thus $\mathrm{ord}_{E'}$ is non-negative on s, t, ξ, and by Lemma 2.6 applied to the normalization of $\mathbb{A}^3_{s,t,\xi}$ in K, the restriction of $\mathrm{ord}_{E'}$ to $\mathbb{C}(s,t,\xi)$ is a positive multiple of $\mathrm{ord}_{E''}$ for a prime divisor E'' over $\mathbb{A}^3_{s,t,\xi}$. Let

$$\theta := \frac{\mathrm{ord}_{E'}(s)}{\mathrm{ord}_{E'}(s) + \mathrm{ord}_{E'}(t)} \in (0,1).$$

Then $\mathrm{ord}_{E'}(u)/p = \mathrm{ord}_{E'}(s) + \mathrm{ord}_{E'}(t) \leq \mathrm{ord}_{E'}(v)/q$ and $\mathrm{ord}_{E'}(U) = n\,\mathrm{ord}_{E'}(t)$. The valuation of E' has a center on V_s if and only if $\mathrm{ord}_{E'}(U) \leq L_s \min\{\mathrm{ord}_{E'}(u)/p, \mathrm{ord}_{E'}(v)/q\}$, that is,

$$n(1-\theta) \leq L_s,$$

and then $\mathrm{ord}_{E'}(w) = \mathrm{ord}_{E'}(U)/L_s > 0$. Similarly $\mathrm{ord}_{E'}(U') = n\,\mathrm{ord}_{E'}(s)$ and $\mathrm{ord}_{E'}(y_3) = r(\mathrm{ord}_{E'}(s) + \mathrm{ord}_{E'}(t))$, so the valuation of E' has a center on V_t if and only if $\mathrm{ord}_{E'}(U') \leq L_t\,\mathrm{ord}_{E'}(y_3)$, that is,

$$n\theta \leq rL_t,$$

and then $\mathrm{ord}_{E'}(y_1) = \mathrm{ord}_{E'}(U')/L_t > 0$. If $n(1-\theta) \leq L_s$ or $n\theta \leq rL_t$, then **Step 3** or **Step 4** implies $a(E, X/G, B_{X/G}) \geq b$.

Suppose that $n(1-\theta) > L_s$ and $n\theta > rL_t$. Let i be an integer with $i \leq n\theta \leq i+1$. Then $(\mathrm{ord}_{E'}(s), \mathrm{ord}_{E'}(t)) \in \sigma_i$, so the valuation of E' has a center on $\mathbb{A}^3_{u_0,v_0,\xi}$, and $\mathrm{ord}_{E'}(u_0v_0) = \mathrm{ord}_{E'}(st) > 0$. Moreover

$$i - \max_j r\beta_j > rL_t - 1 - \max_j r\beta_j \geq N+2, \qquad n - i - 1 - \max_j A_j > L_s - 1 - \max_j A_j \geq N+2,$$

so **Step 5** implies $a(E, X/G, B_{X/G}) \geq b$. Taking $n_0 > re_0 \max\{p,q\}$ and, if $\Gamma\backslash\{0\} \neq \emptyset$, also larger than the bounds of **Step 1** and **Step 2**, and taking for C_0 the constant of Proposition 16.11(2) if $\Gamma\backslash\{0\} \neq \emptyset$, which is at least $4(r+p+q) > |r-p-q|$ by definition and so also serves in the case $B = 0$ of **Step 1**, and $C_0 := |r-p-q|$ if $\Gamma\backslash\{0\} = \emptyset$, we obtain (2).

Step 7. In this step we prove (1).

Suppose that there exists a strictly increasing sequence $a_i = \mathrm{mld}((X \ni 0, B_i)/G_i)$ of positive numbers as in (1), with $n_i := |G_i \cap \mathbf{T}|$. If the n_i are bounded, then $|G_i| \leq e_0 n_i$ is bounded, so there exist only finitely many groups G_i, and possibly passing to a subsequence we may assume $G_i = G$ for any i. The variety X/G is $\mathbb{Q}$-factorial [KM98, Lemma 5.16]. Let o be the image of 0. Since G preserves B_i and (X, B_i) is klt near 0, Lemma 2.10(5) implies that the coefficients of the components of $(B_i)_{X/G}$ passing through o belong to the DCC set $\Psi(\Gamma_+ \cap [0,1])$. Let $B_i^\natural$ be the sum of the components of $(B_i)_{X/G}$ passing through o, with their coefficients in $(B_i)_{X/G}$. Then $(X/G, B_i^\natural)$ is a pair, and $a_i = \mathrm{mld}(X/G \ni o, B_i^\natural)$ because $B_i^\natural = (B_i)_{X/G}$ near o.

Thus Theorem 2.19(1), applied to the fixed threefold X/G and the DCC set $\Psi(\Gamma_+ \cap [0,1])\backslash\{0\}$, contradicts the strict increase of the a_i.

Otherwise, possibly passing to a subsequence we may assume that $n_i \geq n_0$ for any i and $n_i \to +\infty$. By (2), $b_i \leq a_i \leq b_i + \frac{C_0}{n_i}$ for the numbers b_i of Construction-Theorem 16.9, which satisfy the ACC. Since $a_i - b_i \to 0$, this contradicts Lemma 14.11. □

16.11. **Reduction to mixed two-coordinate circles.** In this subsection, we prove Theorem 16.1.

Proof of Theorem 16.1. The proof is long, so we divide it into several steps. As in Definition 2.15, we identify G_i with the image of G_i in $\mathbb{T}^3$ under the characters of x, y, z. Since the coordinates of the elements of the finite group G_i are roots of unity, the map $G_i \to \mathbb{T}^3$ of Definition 2.15 is defined on all of G_i, and it is injective since an element $g \in G_i$ is determined by (g_x, g_y, g_z).

Step 1. In this step, we pass to a subsequence such that the G_i converge to a closed subgroup $H \subset \mathbb{T}^3$, and we show that $[G_i : G_i \cap H^0] = |H/H^0|$.

By Lemma 4.4, after passing to a subsequence, there exists a closed subgroup $H \subset \mathbb{T}^3$ such that the G_i converge to H in the Hausdorff distance and, for any $i \gg 0$, $G_i \subset H$ and G_i intersects any connected component of H. Removing finitely many terms of the sequence, we may assume that $G_i \subset H$ and that G_i intersects any connected component of H for any i. Let H^0 be the connected component of H containing 0. By Lemma 4.3(1), H/H^0 is finite and $H^0 = \rho(V)$ for a linear subspace $V \subset \mathbb{R}^3$ spanned by $V \cap \mathbb{Z}^3$ (for the structure of closed subgroups of $\mathbb{R}^3$ and of $\mathbb{T}^3$, see [Bou98, Chapter VII, §1, no. 2, Theorem 2] and [HR79, (24.10) Theorem]). Since G_i intersects any connected component of H, the composition $G_i \to H \to H/H^0$ is surjective with kernel $G_i \cap H^0$. Thus

$$[G_i : G_i \cap H^0] = |H/H^0| \quad \text{for any } i.$$

Step 2. In this step, we show that H^0 is a circle, we define m, R_i, n_i, and d, and we show that $n_i \to +\infty$.

Since the a_i are strictly increasing, $a_i \geq a_1 > 0$ for any i. If $\dim H^0 \geq 2$, then Theorem 16.4 with $\epsilon := a_1$ implies that the a_i are not strictly increasing, a contradiction. Suppose that $\dim H^0 = 0$. Then H is finite and $|G_i| \leq |H|$. Let $\mathbf{T}' := \mathbf{T}_{(1,1,0)}$. Then

$$[G_i : G_i \cap \mathbf{T}'] \leq |G_i| \leq |H|.$$

By Theorem 16.5 with $(p,q) = (1,1)$, $e_0 = |H|$, and the DCC set Γ, the set $\{a_i\}_{i\geq 1}$ satisfies the ACC, which contradicts the strict increase of the a_i. Hence $\dim H^0 = 1$. By Lemma 4.3(1), H^0 is a circle. Let $m = (m_1, m_2, m_3) \in \mathbb{Z}^3$ be a weight vector of H^0 (Definition 4.1). Then m is primitive, $H^0 = \rho(\mathbb{R}m)$, and m is determined up to sign.

Since m is primitive, the morphism $\mathbb{C}^* \to \mathbf{T}_m$, $s \to (s^{m_1}, s^{m_2}, s^{m_3})$, is an isomorphism. Let $R_i := G_i \cap \mathbf{T}_m$ and $n_i := |R_i|$. Since R_i is a finite subgroup of $\mathbf{T}_m \cong \mathbb{C}^*$,

$$R_i = \mu_{n_i}(m_1, m_2, m_3).$$

By Lemma 4.3(2), the image of R_i in $\mathbb{T}^3$ is $G_i \cap \rho(\mathbb{R}m) = G_i \cap H^0$. Let $d := |H/H^0|$. By **Step 1**,

$$[G_i : R_i] = d \quad \text{for any } i.$$

We show that $n_i \to +\infty$. Otherwise, there exist an integer $M \geq 1$ and infinitely many i with $|G_i| = dn_i \leq M$. Since H^0 is a circle, there exist points $w_0, \ldots, w_M \in H^0$ and a real number $\eta > 0$ such that the distance between w_k and w_l is at least 3η for any $k \neq l$. Since the G_i converge to H in the Hausdorff distance, for any $i \gg 0$ and any k there exists $g_k \in G_i$ whose distance to w_k is less than η. By the triangle inequality, $g_0, \ldots, g_M$ are pairwise distinct, so $|G_i| \geq M + 1$ for any $i \gg 0$, a contradiction. Thus $n_i \to +\infty$.

Step 3. In this step, we show that H^0 is not a coordinate circle.

Suppose that exactly one coordinate of m is non-zero. Since $\rho(\mathbb{R}m) = \rho(\mathbb{R}(-m))$ and $\mathbf{T}_m = \mathbf{T}_{-m}$, possibly permuting the coordinates and replacing m with $-m$, we may assume that $m = (1,0,0)$. Then $R_i = \mu_{n_i}(1,0,0)$ is generated by $(\zeta_i, 1, 1)$, where $\zeta_i := e^{2\pi\sqrt{-1}/n_i}$, and

$$\mathbb{C}[x,y,z]^{R_i} = \mathbb{C}[x',y,z], \quad x' := x^{n_i}.$$

Let $X' := \mathbb{A}^3_{x',y,z} = X/R_i$, $\pi_1 : X \to X'$ the quotient morphism, and $J_i := G_i/R_i$. Then $|J_i| = d$.

For any j, let $a_{i,j}$ be the least exponent of x in the monomials occurring in $h_{i,j}$. The element $(\zeta_i, 1, 1)$ multiplies a monomial $x^a y^b z^c$ by ζ_i^a. Since $h_{i,j}$ is R_i-semi-invariant, the exponents of x in the monomials occurring in $h_{i,j}$ are congruent to $a_{i,j}$ modulo n_i. Thus

$$h_{i,j} = x^{a_{i,j}} F_{i,j}(x',y,z)$$

for a polynomial $F_{i,j}$ not divisible by x'. Let

$$\delta_i := \sum_j c_{i,j} a_{i,j} = \operatorname{mult}_{(x=0)} B_i.$$

Since (X, B_i) is klt near 0 and $0 \in (x = 0)$, we have $\delta_i \in \Gamma_+ \cap [0,1)$.

The non-identity elements of R_i fix pointwise only the hyperplane $(x = 0)$. Thus π_1 is ramified only along $(x = 0)$, with ramification index n_i over $(x' = 0)$, and the coefficient in B_i of a prime divisor on X other than $(x = 0)$ is the coefficient of the image prime divisor in $\sum_j c_{i,j} \operatorname{div}(F_{i,j})$. By Lemma 2.6(4), $(X, B_i)/R_i = (X', B'_i)$, where

$$B'_i := \Big(1 - \frac{1-\delta_i}{n_i}\Big) \operatorname{div}(x') + \sum_j c_{i,j} \operatorname{div}(F_{i,j}).$$

Since π_1 is finite and $\pi_1^{-1}(0) = \{0\}$, there exists a neighborhood U of $0 \in X'$ such that (X, B_i) is klt on $\pi_1^{-1}(U)$. By Lemma 2.6(1)(2)(3), for any prime divisor E over X', there exists a prime divisor F over X with $\operatorname{ord}_F|_{\mathbb{C}(X')} = |I_{R_i}(F)| \operatorname{ord}_E$, and

$$a(E, X', B'_i) = \frac{a(F, X, B_i)}{|I_{R_i}(F)|}, \qquad \operatorname{center}_{X'} E = \pi_1(\operatorname{center}_X F).$$

Hence (X', B'_i) is klt on U.

An element $g = (g_x, g_y, g_z) \in G_i$ acts on X' by $(g_x^{n_i}, g_y, g_z)$, so J_i acts diagonally on X'. If $(g_x^{n_i}, g_y, g_z) = (1,1,1)$, then $g \in G_i \cap \mathbf{T}_{(1,0,0)} = R_i$. Thus J_i acts faithfully on X'. Let χ_x and $\chi_{i,j}$ be the characters of x and $h_{i,j}$. Then $F_{i,j} = x^{-a_{i,j}} h_{i,j}$ is G_i-semi-invariant with character $\chi_{i,j}\chi_x^{-a_{i,j}}$. Since a monomial with exponent $a_{i,j}$ of x occurs in $h_{i,j}$, the restrictions of $\chi_{i,j}$ and $\chi_x^{a_{i,j}}$ to R_i coincide. Hence $F_{i,j}$ is J_i-semi-invariant, and so is x'.

Since $\mathbb{C}[x',y,z]^{J_i} = \mathbb{C}[x,y,z]^{G_i}$, we have $X'/J_i = X/G_i$. By Lemma 2.6(5), applied to the normal subgroup R_i of G_i, with $X' = X/R_i$, $(X', B'_i) = (X, B_i)/R_i$, and $J_i = G_i/R_i$, we have $(X, B_i)/G_i = (X', B'_i)/J_i$. Thus

$$a_i = \operatorname{mld}((X' \ni 0, B'_i)/J_i).$$

Let $\Xi := \Psi(\Gamma_+ \cap [0,1])$. By Lemma 2.10(1) applied to Γ, the set $\Gamma_+ \cap [0,1]$ satisfies the DCC. By Lemma 2.10(1) applied to $\Gamma_+ \cap [0,1]$, the set Ξ satisfies the DCC, and $\Xi \subset [0,1]$. Moreover, $\Gamma \subset \Xi$ (take $e = 1$), and $1 - \frac{1-\delta_i}{n_i} \in \Xi$ since $\delta_i \in \Gamma_+ \cap [0,1]$. Thus the coefficients of B'_i belong to Ξ. For $\mathbf{T}' := \mathbf{T}_{(1,1,0)}$ acting on X', we have $[J_i : J_i \cap \mathbf{T}'] \le |J_i| = d$. By Theorem 16.5 with $(p,q) = (1,1)$, $e_0 = d$, and the DCC set Ξ, the set $\{a_i\}_{i \ge 1}$ satisfies the ACC, which contradicts the strict increase of the a_i. Thus at least two coordinates of m are non-zero.

Step 4. In this step, we exclude the remaining sign patterns of m other than a mixed two-coordinate circle, and we conclude the proof.

Case 1. Suppose that exactly two coordinates of m are non-zero and that they are both positive or both negative. Possibly permuting the coordinates and replacing m with $-m$, we may assume that $m = (p,q,0)$ for coprime positive integers p, q (p, q are coprime since m is primitive). By

Step 2, $[G_i : G_i \cap \mathbf{T}_{(p,q,0)}] = d$. By Theorem 16.5 with $e_0 = d$, the set $\{a_i\}_{i\geq 1}$ satisfies the ACC, a contradiction.

Case 2. Suppose that the three coordinates of m are non-zero and are all positive or all negative. Possibly replacing m with $-m$, we may assume that $m_1, m_2, m_3 > 0$. Let F be the monomial prime divisor in x, y, z with weights m (Definition 2.16). Then $\operatorname{center}_X F = \{0\}$. Since m is primitive, F is the toric prime divisor over X defined by m, so $a(F, X, 0) = m_1 + m_2 + m_3$ [Amb06, Section 2, p. 360]. Since G_i is diagonal, G_i fixes ord_F. Let $g = (s^{m_1}, s^{m_2}, s^{m_3}) \in R_i$. For any non-zero polynomial f whose monomials all have m-weight k, we have $g^* f = s^k f$. If f_1, f_2 are non-zero polynomials with $\operatorname{ord}_F f_1 = \operatorname{ord}_F f_2$, then the residue class of f_1/f_2 is the residue class of $\operatorname{in}_m(f_1)/\operatorname{in}_m(f_2)$ (their difference has positive order along F), which is fixed by g^*. Since any element of the valuation ring of ord_F with value zero is of the form f_1/f_2 with non-zero polynomials f_1, f_2 and $\operatorname{ord}_F f_1 = \operatorname{ord}_F f_2$, g acts trivially on the residue field of ord_F. Thus $R_i \subset I_{G_i}(F)$ and $|I_{G_i}(F)| \geq n_i$. Let E_i be the prime divisor over X/G_i with $\operatorname{ord}_F|_{\mathbb{C}(X/G_i)} = |I_{G_i}(F)| \operatorname{ord}_{E_i}$. By Lemma 2.6(1)(2), the center of E_i on X/G_i is the image of 0, and

$$a(E_i, X/G_i, B_{X/G_i}) = \frac{a(F, X, B_i)}{|I_{G_i}(F)|} = \frac{m_1 + m_2 + m_3 - \sum_j c_{i,j} \operatorname{ord}_F h_{i,j}}{|I_{G_i}(F)|} \leq \frac{m_1 + m_2 + m_3}{n_i}.$$

Thus $a_1 \leq a_i \leq \frac{m_1+m_2+m_3}{n_i}$ for any i, which contradicts $n_i \to +\infty$ (**Step 2**).

Case 3. Suppose that the three coordinates of m are non-zero and are neither all positive nor all negative. Possibly permuting the coordinates and replacing m with $-m$, we may assume that $m = (p, q, -r)$ for positive integers p, q, r with $\gcd(p, q, r) = 1$ since m is primitive. By **Step 2**, $|G_i/(G_i \cap \mathbf{T}_m)| = d$. By Theorem 16.21(1) with $e_0 = d$, the set of the positive numbers a_i satisfies the ACC, a contradiction.

Conclusion. By **Step 3** and Cases 1–3, exactly two coordinates of m are non-zero and they have opposite signs. Possibly permuting the coordinates and replacing m with $-m$, we may assume that $m = (p, -q, 0)$ for coprime positive integers p, q (p, q are coprime since m is primitive). Then $\mathbf{T} = \mathbf{T}_m$ and, by **Step 2**,

$$R_i = G_i \cap \mathbf{T} = \mu_{n_i}(p, -q, 0), \qquad n_i \to +\infty, \qquad [G_i : R_i] = d.$$

By **Step 1** and **Step 2**, the G_i converge in the Hausdorff distance to H, $G_i \subset H$, and $d = |H/H^0|$, where $H^0 = \rho(\mathbb{R}(p, -q, 0))$. This proves Theorem 16.1. □

16.12. **A lower bound by the parts of largest weight.** In this subsection, we prove Proposition 16.23, which bounds from below the log discrepancies of the prime divisors over X/G with center the image of 0 by the number b of Construction-Theorem 16.9 for the parts h_j^0 of largest $\mathbf{T}$-weight of the h_j. Proposition 16.23 is not used in the proofs of the main theorems of this paper.

Proposition 16.23. *Let p, q, r be positive integers with $\gcd(p, q, r) = 1$, $\mathbf{T} := \mathbf{T}_{(p,q,-r)}$ acting on $X := \mathbb{A}^3_{x,y,z}$, and G a finite diagonal group acting faithfully on X. Let $h_1, \dots, h_s$ be non-zero G-semi-invariant polynomials and $c_1, \dots, c_s \geq 0$ real numbers. For any j, let κ_j be the largest $\mathbf{T}$-weight of a monomial occurring in h_j and h_j^0 the sum of the terms of h_j of $\mathbf{T}$-weight κ_j. Assume that (X, B) is klt near 0 for $B := \sum_j c_j \operatorname{div}(h_j)$ and that (X, B^0) is lc near 0 for $B^0 := \sum_j c_j \operatorname{div}(h_j^0)$, and let b be the number defined in Construction-Theorem 16.9(1) if $\tau \geq 0$ and in Construction-Theorem 16.9(2) if $\tau < 0$, for $h_1^0, \dots, h_s^0$ and $c_1, \dots, c_s$, where $\tau := r - p - q + \sum_j c_j \kappa_j$. Then any prime divisor E over X/G with center the image of 0 satisfies $a(E, X/G, B_{X/G}) \geq b$. In particular, $b \leq \operatorname{mld}((X \ni 0, B)/G)$.*

Proof. The proof is long, so we divide it into several steps. Since G is diagonal, all monomials occurring in h_j have the G-character of h_j, so h_j^0 is a non-zero polynomial which is semi-invariant for G and $\mathbf{T}$, with $\mathbf{T}$-weight κ_j. We use the notation of Construction-Theorem 16.9 for $h_1^0, \dots, h_s^0$,

with τ as in the statement: $n = |G \cap \mathbf{T}|$, $H = G/(G \cap \mathbf{T})$, the cover $z = t^r$, $u = xt^p$, $v = yt^q$, the polynomials P_j with $h_j^0(x, y, t^r) = t^{-\kappa_j} P_j(u, v)$, $C = \sum_j c_j \operatorname{div}(P_j)$, and the groups $\tilde{G}$ and $\tilde{H}$. Let $R := G \cap \mathbf{T}$, let $S := \mathbb{A}^2_{u,v}$, and let $\pi : S \to S/\tilde{H}$ be the quotient morphism. For $\sigma = (g, \alpha) \in \tilde{G}$, we have $\sigma^* x = g_x x$, $\sigma^* y = g_y y$, and $\sigma^* t = \alpha t$.

Step 1. In this step, we write h_j on the cover $z = t^r$ as $t^{-\kappa_j} Q_j(t^n, u, v)$ for a polynomial Q_j, and we describe the action of $\tilde{H}$ on $\mathbb{A}^3_{U,u,v}$, where $U := t^n$.

Since $\gcd(p, q, r) = 1$, the morphism $\mathbb{C}^* \to \mathbf{T}$, $\lambda \to (\lambda^p, \lambda^q, \lambda^{-r})$, is an isomorphism, so $R = \{(\lambda^p, \lambda^q, \lambda^{-r}) \mid \lambda^n = 1\}$. The element $(\lambda^p, \lambda^q, \lambda^{-r})$ multiplies a monomial of $\mathbf{T}$-weight k by λ^k. Since all monomials occurring in h_j have the R-character of h_j, their $\mathbf{T}$-weights are congruent modulo n. Since κ_j is the largest one, the $\mathbf{T}$-weights of the monomials occurring in h_j are of the form $\kappa_j - ln$ with $l \in \mathbb{Z}_{\geq 0}$. A monomial $x^a y^b z^c$ of $\mathbf{T}$-weight $\kappa_j - ln$ satisfies

$$x^a y^b t^{rc} = u^a v^b t^{rc-pa-qb} = t^{-\kappa_j} u^a v^b t^{ln}.$$

Thus

$$h_j(x, y, t^r) = t^{-\kappa_j} Q_j(U, u, v), \qquad Q_j(U, u, v) = \sum_{l \geq 0} U^l P_{j,l}(u, v), \qquad Q_j(0, u, v) = P_j(u, v),$$

where $P_{j,l} \in \mathbb{C}[u, v]$. In particular, $Q_j \in \mathbb{C}[U, u, v]$. Let $V_0 := \mathbb{A}^1_U \times S = \mathbb{A}^3_{U,u,v}$.

We show four properties of the groups.

(a) For any $g \in G$, there exist exactly r numbers α with $\alpha^r = g_z$. Thus $|\tilde{G}| = r|G| = rn|H|$. Since $|\tilde{H}| = r|H|$ (Construction-Theorem 16.9), the kernel $\tilde{R}$ of $\tilde{G} \to \tilde{H}$ satisfies $|\tilde{R}| = n$.

(b) Let $(g, \alpha) \in \tilde{R}$. Then $g_x \alpha^p = g_y \alpha^q = 1$ and $g_z = \alpha^r$. For $\lambda := \alpha^{-1}$, we have $g = (\lambda^p, \lambda^q, \lambda^{-r}) \in G \cap \mathbf{T} = R$, so $\lambda^n = 1$. If $\alpha = 1$, then $g = \mathrm{id}_X$. Thus $\tilde{R}$ acts faithfully on t and trivially on u, v. Since $\mathbb{C}(x, y, t) = \mathbb{C}(u, v, t)$ and $|\tilde{R}| = n$, and since $\mathbb{C}(x, y, t)^{\tilde{G}} = \mathbb{C}(X)^G$ (Construction-Theorem 16.9) and $\tilde{H} = \tilde{G}/\tilde{R}$, we have

$$\mathbb{C}(x, y, t)^{\tilde{R}} = \mathbb{C}(U, u, v),$$
$$\mathbb{C}(U, u, v)^{\tilde{H}} = \mathbb{C}(X)^G.$$

(c) For $\sigma = (g, \alpha) \in \tilde{G}$, we have $\sigma^* U = \alpha^n U$, $\sigma^* u = g_x \alpha^p u$, and $\sigma^* v = g_y \alpha^q v$. By (b), $\alpha^n = 1$ for $(g, \alpha) \in \tilde{R}$. Thus $\tilde{H}$ acts diagonally on V_0 by the characters of U, u, v, and $\tilde{H}$ acts faithfully on S by the definition of $\tilde{H}$.

(d) Let χ_j be the character of h_j. Then $Q_j = t^{\kappa_j} h_j(x, y, t^r)$ is $\tilde{G}$-semi-invariant with character $(g, \alpha) \to \chi_j(g) \alpha^{\kappa_j}$. For $(g, \alpha) \in \tilde{R}$ and $\lambda = \alpha^{-1}$, the element $g = (\lambda^p, \lambda^q, \lambda^{-r})$ multiplies any monomial occurring in h_j by $\lambda^{\kappa_j - ln} = \lambda^{\kappa_j}$, so $\chi_j(g) \alpha^{\kappa_j} = \lambda^{\kappa_j} \lambda^{-\kappa_j} = 1$. Hence Q_j is a regular $\tilde{H}$-semi-invariant function on V_0.

Step 2. In this step, we express the log discrepancy of a prime divisor E over X/G with center the image of 0 as a log discrepancy over $V_0/\tilde{H}$ with respect to a sub-boundary Δ on V_0.

Let $Y := \mathbb{A}^3_{x,y,t}$, $\psi : Y \to X$ the morphism $(x, y, t) \to (x, y, t^r)$, and $Y' := \mathbb{A}^3_{u,v,t}$. The group $\tilde{G}$ acts faithfully on Y: if $(g_x, g_y, \alpha) = (1, 1, 1)$, then $g_z = \alpha^r = 1$ and $g = \mathrm{id}_X$. The kernel of $\tilde{G} \to G$, $(g, \alpha) \to g$, is $\{(\mathrm{id}_X, \alpha) \mid \alpha^r = 1\}$, and the invariant ring of $\mathbb{C}[x, y, t]$ under $\{(\mathrm{id}_X, \alpha) \mid \alpha^r = 1\}$ is $\mathbb{C}[x, y, t^r] = \mathbb{C}[x, y, z]$. Thus $\mathbb{C}[x, y, t]^{\tilde{G}} = \mathbb{C}[x, y, z]^G$, $Y/\tilde{G} = X/G$, and the quotient morphism $\pi_Y : Y \to X/G$ is $\pi_X \circ \psi$, where $\pi_X : X \to X/G$ is the quotient morphism. Let

$$B_Y := \psi^* B - (r - 1) \operatorname{div}(t).$$

Since ψ is ramified only along $(t = 0)$, with ramification index r, the Hurwitz formula implies $K_Y + B_Y = \psi^*(K_X + B)$. By Lemma 2.6(0) applied to G and $\tilde{G}$,

$$\pi_Y^* \big(K_{X/G} + (B_Y)_{X/G}\big) = K_Y + B_Y = \psi^* \pi_X^* \big(K_{X/G} + B_{X/G}\big).$$

Since the pullback of $\mathbb{R}$-divisors under a finite surjective morphism is injective, we obtain

$$(Y, B_Y)/\tilde{G} = (X, B)/G.$$

The group $\tilde{R}$ acts on Y' through t, and $\mathbb{C}[u, v, t]^{\tilde{R}} = \mathbb{C}[U, u, v]$ by **Step 1**(b), so $Y'/\tilde{R} = V_0$. The non-identity elements of $\tilde{R}$ fix pointwise only $(t = 0)$, so $Y' \to V_0$ is ramified only along $(t = 0)$, with ramification index n over $(U = 0)$. Since $Q_j(0, u, v) = P_j \neq 0$, the polynomial $Q_j(t^n, u, v)$ is not divisible by t. Let

$$\Delta' := (1 - \tau)\operatorname{div}(t) + \sum_j c_j \operatorname{div}\big(Q_j(t^n, u, v)\big) \quad \text{and} \quad \Delta := \Big(1 - \frac{\tau}{n}\Big)\operatorname{div}(U) + \sum_j c_j \operatorname{div}(Q_j).$$

By Lemma 2.6(4), $(Y', \Delta')/\tilde{R} = (V_0, \Delta)$. The coefficient $1 - \frac{\tau}{n}$ may be negative, and (V_0, Δ) is a sub-pair. By **Step 1**(c)(d), Δ is $\tilde{H}$-invariant. Let $\pi_0 : V_0 \to V_0/\tilde{H}$ be the quotient morphism.

Let E be a prime divisor over X/G with center $\pi_X(0)$. By Lemma 2.6(3), there exist a prime divisor F over Y and a positive integer $e_{\tilde{G}}$ with $\operatorname{ord}_F|_{\mathbb{C}(X/G)} = e_{\tilde{G}} \operatorname{ord}_E$, and $e_{\tilde{G}} = |I_{\tilde{G}}(F)|$ by Lemma 2.6(1). By Lemma 2.6(2), $\pi_Y(\operatorname{center}_Y F) = \{\pi_X(0)\}$. Since $\tilde{G}$ is diagonal, $\pi_Y^{-1}(\pi_X(0)) = \{0\}$, so $\operatorname{center}_Y F = \{0\}$. Thus $\operatorname{ord}_F(x), \operatorname{ord}_F(y), \operatorname{ord}_F(t) > 0$, and

$$\begin{aligned}
\operatorname{ord}_F(u) &= \operatorname{ord}_F(x) + p \operatorname{ord}_F(t) > p \operatorname{ord}_F(t),\\
\operatorname{ord}_F(v) &= \operatorname{ord}_F(y) + q \operatorname{ord}_F(t) > q \operatorname{ord}_F(t),\\
\operatorname{ord}_F(U) &= n \operatorname{ord}_F(t) > 0.
\end{aligned}$$

In particular, F is a prime divisor over Y' with center the origin. Since

$$dx \wedge dy \wedge dt = t^{-p-q} du \wedge dv \wedge dt,$$

we have $a(F, Y, 0) = a(F, Y', 0) - (p + q)\operatorname{ord}_F(t)$. Since $h_j(x, y, t^r) = t^{-\kappa_j} Q_j(t^n, u, v)$, we obtain

$$\begin{aligned}
a(F, Y, B_Y) &= a(F, Y', 0) - (p + q)\operatorname{ord}_F(t) + (r - 1)\operatorname{ord}_F(t)\\
&\quad - \sum_j c_j \big(\operatorname{ord}_F Q_j(t^n, u, v) - \kappa_j \operatorname{ord}_F(t)\big)\\
&= a(F, Y', \Delta').
\end{aligned}$$

Let F_0 be the prime divisor over V_0 with $\operatorname{ord}_F|_{\mathbb{C}(V_0)} = e_{\tilde{R}} \operatorname{ord}_{F_0}$, where $e_{\tilde{R}} := |I_{\tilde{R}}(F)|$, and let E_0 be the prime divisor over $V_0/\tilde{H}$ with $\operatorname{ord}_{F_0}|_{\mathbb{C}(V_0/\tilde{H})} = e_{\tilde{H}} \operatorname{ord}_{E_0}$, where $e_{\tilde{H}} := |I_{\tilde{H}}(F_0)|$. By Lemma 2.6(2), the center of F_0 on V_0 is the origin, the center of E_0 on $V_0/\tilde{H}$ is $\pi_0(0)$, and $a(F, Y', \Delta') = e_{\tilde{R}} a(F_0, V_0, \Delta)$. By **Step 1**(b), $\mathbb{C}(V_0/\tilde{H}) = \mathbb{C}(X/G)$. Restricting ord_F to $\mathbb{C}(X/G)$, we obtain $e_{\tilde{G}} \operatorname{ord}_E = e_{\tilde{R}} e_{\tilde{H}} \operatorname{ord}_{E_0}$. Since the value groups of ord_E and ord_{E_0} are $\mathbb{Z}$, we have $\operatorname{ord}_E = \operatorname{ord}_{E_0}$ and $e_{\tilde{G}} = e_{\tilde{R}} e_{\tilde{H}}$. Hence

$$a(E, X/G, B_{X/G}) = \frac{a(F, Y, B_Y)}{e_{\tilde{G}}} = \frac{a(F, Y', \Delta')}{e_{\tilde{R}} e_{\tilde{H}}} = \frac{a(F_0, V_0, \Delta)}{e_{\tilde{H}}}. \tag{16.63}$$

Dividing the inequalities for $\operatorname{ord}_F(u)$, $\operatorname{ord}_F(v)$, and $\operatorname{ord}_F(U)$ by $e_{\tilde{R}}$, we obtain

$$\frac{\operatorname{ord}_{F_0}(u)}{p} > \frac{\operatorname{ord}_{F_0}(U)}{n} \quad \text{and} \quad \frac{\operatorname{ord}_{F_0}(v)}{q} > \frac{\operatorname{ord}_{F_0}(U)}{n}. \tag{16.64}$$

Step 3. In this step, we bound $a(F_0, V_0, B_{V_0}, \pi_0^* \mathfrak{c})$ from below by $e_{\tilde{H}} b$, where $\mathfrak{c}$ is an $\mathbb{R}$-ideal on $V_0/\tilde{H}$ defined below and

$$B_{V_0} := \operatorname{div}(U) + \sum_j c_j \operatorname{div}(Q_j).$$

Let $\mathfrak{b}_S$ be the ideal sheaf on $S/\tilde{H}$ generated by the $\tilde{H}$-invariant functions $u^{qr|H|}$ and $v^{pr|H|}$, and let $\mathfrak{a} := \mathfrak{b}_S^{\tau^-/(pqr|H|)}$, where $\mathfrak{b}_S^0$ means $\mathcal{O}_{S/\tilde{H}}$. Let $V \subset S$ and $\bar{0} \in V/\tilde{H}$ be as in Construction-Theorem 16.9. Then $V/\tilde{H}$ is an open neighborhood of $\bar{0} = \pi(0)$ in $S/\tilde{H}$, the restriction of $C_{S/\tilde{H}}$

to $V/\tilde{H}$ is $(C|_V)_{V/\tilde{H}}$, and the restriction of $\mathfrak{b}_S$ to $V/\tilde{H}$ is the ideal sheaf $\mathfrak{b}$ of Construction-Theorem 16.9. Since minimal log discrepancies at $\pi(0)$ only depend on a neighborhood of $\pi(0)$, the Moreover part of Construction-Theorem 16.9 implies that $(S/\tilde{H}, C_{S/\tilde{H}}, \mathfrak{a})$ is lc near $\pi(0)$ and

$$b = \operatorname{mld}\big(S/\tilde{H} \ni \pi(0), C_{S/\tilde{H}}, \mathfrak{a}\big).$$

By Lemma 2.6(2)(3), $(S, C, \pi^*\mathfrak{a})$ is lc near 0.

We check the hypotheses of Lemma 16.19(1) for $H_0 = \tilde{H}$, $(z_1, z_2) = (u, v)$, S, $V_0 = \mathbb{A}^1_U \times S$, $K_j = Q_j$, the closed point $0 \in S$ in place of the point s of Lemma 16.19(1), and $\mathfrak{a}$. By **Step 1**(c)(d), $\tilde{H}$ acts diagonally on V_0 and faithfully on S, and the Q_j are regular $\tilde{H}$-semi-invariant functions on V_0 with $Q_j(0, u, v) = P_j \neq 0$. Moreover, $B_S = C$, and the number b_s of Lemma 16.19(1) for the point $0 \in S$ equals b by the formula for b above. Let $\operatorname{pr} : V_0/\tilde{H} \to S/\tilde{H}$ be the morphism induced by the projection, and $\mathfrak{c} := \operatorname{pr}^* \mathfrak{a}$. Since the center of E_0 on $V_0/\tilde{H}$ is $\pi_0(0)$, Lemma 16.19(1) implies

$$a\big(E_0, V_0/\tilde{H}, (B_{V_0})_{V_0/\tilde{H}}, \mathfrak{c}\big) \geq b.$$

The $\mathbb{R}$-divisor B_{V_0} is $\tilde{H}$-invariant, and $\pi_0^*\mathfrak{c} = \mathfrak{b}_0^{\tau^-/(pqr|H|)}$, where $\mathfrak{b}_0$ is the ideal of V_0 generated by $u^{qr|H|}$ and $v^{pr|H|}$. By Lemma 2.6(2),

$$a\big(F_0, V_0, B_{V_0}, \mathfrak{b}_0^{\tau^-/(pqr|H|)}\big) = e_{\tilde{H}} a\big(E_0, V_0/\tilde{H}, (B_{V_0})_{V_0/\tilde{H}}, \mathfrak{c}\big) \geq e_{\tilde{H}} b.$$

Step 4. In this step, we compare $a(F_0, V_0, \Delta)$ with $a(F_0, V_0, B_{V_0}, \mathfrak{b}_0^{\tau^-/(pqr|H|)})$ and conclude the proof.

We have $B_{V_0} - \Delta = \frac{\tau}{n} \operatorname{div}(U)$ and

$$\operatorname{mult}_{F_0} \mathfrak{b}_0 = \min\big\{qr|H| \operatorname{ord}_{F_0}(u), pr|H| \operatorname{ord}_{F_0}(v)\big\} = pqr|H| \min\left\{\frac{\operatorname{ord}_{F_0}(u)}{p}, \frac{\operatorname{ord}_{F_0}(v)}{q}\right\}.$$

Thus

$$a(F_0, V_0, \Delta) - a\big(F_0, V_0, B_{V_0}, \mathfrak{b}_0^{\tau^-/(pqr|H|)}\big) = \frac{\tau}{n} \operatorname{ord}_{F_0}(U) + \tau^- \min\left\{\frac{\operatorname{ord}_{F_0}(u)}{p}, \frac{\operatorname{ord}_{F_0}(v)}{q}\right\}.$$

If $\tau \geq 0$, then $\tau^- = 0$ and the right-hand side is $\frac{\tau}{n} \operatorname{ord}_{F_0}(U) \geq 0$. If $\tau < 0$, then $\tau^- = -\tau$ and the right-hand side is

$$\tau^-\left(\min\left\{\frac{\operatorname{ord}_{F_0}(u)}{p}, \frac{\operatorname{ord}_{F_0}(v)}{q}\right\} - \frac{\operatorname{ord}_{F_0}(U)}{n}\right) > 0$$

by (16.64). In both cases, **Step 3** implies $a(F_0, V_0, \Delta) \geq e_{\tilde{H}} b$. By (16.63),

$$a(E, X/G, B_{X/G}) = \frac{a(F_0, V_0, \Delta)}{e_{\tilde{H}}} \geq b.$$

Since E is an arbitrary prime divisor over X/G with center the image of 0, we have

$$b \leq \operatorname{mld}((X \ni 0, B)/G).$$

□

17. Finite real and rational coefficients

In this section, we reduce strictly increasing sequences of minimal log discrepancies at closed points of threefold pairs with coefficients in a fixed finite set to cyclic quotients of terminal points carrying pairs with rational proxy boundaries (Construction-Theorem 17.6). The results of this section are not used in the proofs of the main theorems of this paper. Some of them are used in the sequels to this paper.

17.1. **Rational proxy boundaries.** In this subsection, we express the boundary B of Construction-Theorem 5.2 as a convex combination of klt $\mathbb{Q}$-boundaries such that, on the index one cover, a log discrepancy with respect to the pullback of each of these $\mathbb{Q}$-boundaries is less than, equal to, or greater than 1 exactly when the log discrepancy with respect to the pullback of B is.

Construction-Proposition 17.1. Let $\Gamma \subset [0,1]$ be a finite set, and let s, q, and $\lambda_1, \dots, \lambda_s$ be as in Construction-Theorem 5.2. Then there exist a positive integer L, positive real numbers $\mu_1, \dots, \mu_s$ with $\sum_k \mu_k = 1$, and an invertible $s \times s$ matrix $\mathbf{R} = (r_{kj})$ with positive rational entries and row sums 1, all depending only on Γ, such that $\lambda_j = \sum_k \mu_k r_{kj}$ for any j and the matrix $\mathbf{A} = (\mathbf{A}_{kj}) := L\mathbf{R}$ has integer entries.

Moreover, let (X, B), x, $B^{(j)}$, $\pi : X' \to X$, x', G, B', $B'^{(j)}$, and χ_j be as in Construction-Theorem 5.2, and define

$$C^{(k)} := \sum_j r_{kj} B^{(j)} \text{ on } X, \qquad C'^{(k)} := \pi^* C^{(k)} \text{ on } X'$$

for $1 \le k \le s$. Then, after shrinking X near x, the following hold.

(1) $B = \sum_k \mu_k C^{(k)}$, and $(X, C^{(k)})$ and $(X', C'^{(k)})$ are klt for any k.
(2) For any prime divisor F over X', the following hold.
  (a) For any k, the signs of $a(F, X', C'^{(k)}) - 1$ and $a(F, X', B') - 1$ are equal.
  (b) If $a(F, X', B') = 1$, then $a(F, X', B'^{(j)}) = 1$ for any j.
(3) For a subgroup $H \subset G$ and an H-equivariant proper birational morphism $f : Y \to X'$ from a normal quasi-projective variety, with B_Y and $C_Y^{(k)}$ defined by $K_Y + B_Y = f^*(K_{X'} + B')$ and $K_Y + C_Y^{(k)} = f^*(K_{X'} + C'^{(k)})$, the following hold.
  (a) If $B_Y \ge 0$, then $C_Y^{(k)} \ge 0$, $(Y, C_Y^{(k)})$ is klt, and $C_Y^{(k)} \in \{0, \frac{1}{qL}, \dots, \frac{qL-1}{qL}\}$ for any k.
  (b) If $B_Y \ge 0$ and (Y, B_Y) is terminal, then $(Y, C_Y^{(k)})$ is terminal for any k.
(4) With $\nu_k := \prod_j \chi_j^{\mathbf{A}_{kj}}$ for $1 \le k \le s$, the following hold.
  (a) For any k, the sheaf

$$\mathcal{O}_{X'}(qL(K_{X'} + C'^{(k)})) = \bigotimes_j \mathcal{O}_{X'}(q(K_{X'} + B'^{(j)}))^{\otimes \mathbf{A}_{kj}}$$

  is invertible, with character ν_k at x'.
  (b) For any subgroup $G_1 \subset G$, the order of $\bigcap_k \operatorname{Ker}(\nu_k|_{G_1})$ divides $|\det \mathbf{A}|$.
  (c) In the setting of (3), for any k,

$$\mathcal{O}_Y(qL(K_Y + C_Y^{(k)})) = f^* \mathcal{O}_{X'}(qL(K_{X'} + C'^{(k)})),$$

  and the character of $\mathcal{O}_Y(qL(K_Y + C_Y^{(k)}))$ at any H-fixed point $y \in Y$ over x' is $\nu_k|_H$.
(5) Assume, in the setting of (3), that F is a prime divisor over X' as in Construction-Theorem 5.2(2), that $H = I := I_G(F)$, that $B_Y \ge 0$, and that $\operatorname{center}_Y F$ is a closed point y, let $h : Y' \to Y$, y', J, $\hat{I}$, and ω be as in Construction-Theorem 5.2(5), and let $C_{Y'}^{(k)} := h^* C_Y^{(k)}$. Then the following hold.
  (a) For any k, $C_{Y'}^{(k)} \ge 0$ near y', and $qLC_{Y'}^{(k)}$ is Cartier near y'.
  (b) Near y', there exist regular semi-invariant functions $\varphi_1, \dots, \varphi_s$ with $\operatorname{div}(\varphi_k) = qLC_{Y'}^{(k)}$ such that the character of φ_k is the character of $\omega^{\otimes qL}$ divided by the pullback of $\nu_k|_I$ to $\hat{I}$.
  (c) For $\varphi_1, \dots, \varphi_s$ as in (b), the joint kernel of the characters of $\omega^{\otimes qL}/\varphi_1, \dots, \omega^{\otimes qL}/\varphi_s$ has order dividing $|J| \cdot |\det \mathbf{A}|$.

Proof. By Construction-Theorem 5.2, $\lambda_1, \dots, \lambda_s$ are linearly independent over $\mathbb{Q}$. Since $\sum_j \lambda_j = 1$, we have $\sum_j \lambda_j n_j - q = \sum_j \lambda_j (n_j - q)$ for any $n = (n_1, \dots, n_s) \in \mathbb{Z}^s$. Thus, for any $n \in \mathbb{Z}^s_{\geq 0}$,

$$\sum_j \lambda_j n_j = q \quad \text{if and only if} \quad n_1 = \cdots = n_s = q. \tag{17.1}$$

We construct L, μ_k, and $\mathbf{R}$ such that

$$\operatorname{sign}\Big(\sum_j r_{kj} n_j - q\Big) = \operatorname{sign}\Big(\sum_j \lambda_j n_j - q\Big) \quad \text{for any } k \text{ and any } n \in \mathbb{Z}^s_{\geq 0}. \tag{17.2}$$

If $s = 1$, then $\lambda_1 = 1$, and we let $L := 1$, $\mu_1 := 1$, and $\mathbf{R} := (1)$. Then (17.2) holds since $r_{11} = \lambda_1$. Assume that $s \geq 2$. Let

$$\mathcal{P} := \Big\{ w \in \mathbb{R}^s \;\Big|\; \sum_j w_j = 1 \Big\}, \qquad \mathcal{N} := \Big\{ n \in \mathbb{Z}^s_{\geq 0} \;\Big|\; \sum_j \lambda_j n_j \leq 2q \Big\}.$$

The set $\mathcal{N}$ is finite since $n_j \leq 2q/\lambda_j$ for any $n \in \mathcal{N}$ and any j. Let U be the set of $w \in \mathcal{P}$ such that $w_j > \frac{\lambda_j}{2}$ for any j and

$$\operatorname{sign}\Big(\sum_j w_j n_j - q\Big) = \operatorname{sign}\Big(\sum_j \lambda_j n_j - q\Big)$$

for any $n \in \mathcal{N}$ with $n \neq (q, \dots, q)$. By (17.1), $\sum_j \lambda_j n_j - q \neq 0$ for any such n, so U is defined in $\mathcal{P}$ by finitely many strict inequalities which hold at $\lambda := (\lambda_1, \dots, \lambda_s)$. Thus U is an open neighborhood of λ in $\mathcal{P}$. We show that

$$\operatorname{sign}\Big(\sum_j w_j n_j - q\Big) = \operatorname{sign}\Big(\sum_j \lambda_j n_j - q\Big) \quad \text{for any } w \in U \text{ and any } n \in \mathbb{Z}^s_{\geq 0}. \tag{17.3}$$

If $n = (q, \dots, q)$, then both sides are 0 since $\sum_j w_j = \sum_j \lambda_j = 1$. If $n \in \mathcal{N}$ and $n \neq (q, \dots, q)$, then (17.3) is the definition of U. If $n \notin \mathcal{N}$, then $\sum_j \lambda_j n_j > 2q$, and $w_j > \frac{\lambda_j}{2}$ implies

$$\sum_j w_j n_j \geq \frac{1}{2} \sum_j \lambda_j n_j > q.$$

Since $\mathcal{P}$ is an affine hyperplane defined over $\mathbb{Q}$, the rational points of $\mathcal{P}$ are dense in $\mathcal{P}$. Thus there exist affinely independent points $r_1, \dots, r_s \in U \cap \mathbb{Q}^s$ such that λ is in the interior of the simplex $\operatorname{Conv}(r_1, \dots, r_s) \subset \mathcal{P}$: take an $(s-1)$-simplex in U whose interior in $\mathcal{P}$ contains λ, and replace the vertices of the simplex with sufficiently close rational points of $\mathcal{P}$. Write $\lambda = \sum_k \mu_k r_k$ with $\mu_k > 0$ and $\sum_k \mu_k = 1$, and let $\mathbf{R}$ be the matrix with rows $r_k = (r_{k1}, \dots, r_{ks})$. Then $r_{kj} > \frac{\lambda_j}{2} > 0$, $\sum_j r_{kj} = 1$, and $\lambda_j = \sum_k \mu_k r_{kj}$ for any k and j. If $c \in \mathbb{R}^s$ and $\sum_k c_k r_k = 0$, then the sum of the coordinates of $\sum_k c_k r_k$ is $\sum_k c_k = 0$, and the affine independence of $r_1, \dots, r_s$ implies $c = 0$. Thus $\mathbf{R}$ is invertible. Let L be a positive integer such that $L\mathbf{R}$ has integer entries. Since s, q, and λ depend only on Γ, so do L, μ_k, and $\mathbf{R}$. Applying (17.3) to $w = r_k$, we obtain (17.2). Since $r_{kj} > 0$, $\sum_j r_{kj} = 1$, and $\mathbf{R}$ is invertible, the matrix $\mathbf{A} = L\mathbf{R}$ is an integer matrix with positive entries, row sums L, and $\det \mathbf{A} \neq 0$.

We prove (1). By Construction-Theorem 5.2, $B = \sum_j \lambda_j B^{(j)}$, so

$$B = \sum_j \sum_k \mu_k r_{kj} B^{(j)} = \sum_k \mu_k C^{(k)}.$$

Since $B^{(j)} \geq 0$ and $r_{kj} > 0$, $C^{(k)} \geq 0$, and $K_X + C^{(k)} = \sum_j r_{kj}(K_X + B^{(j)})$ is $\mathbb{Q}$-Cartier. Since $K_X + B = \sum_j \lambda_j (K_X + B^{(j)})$, for any prime divisor E over X we have

$$a(E, X, B) = \sum_j \lambda_j a(E, X, B^{(j)}), \qquad a(E, X, C^{(k)}) = \sum_j r_{kj} a(E, X, B^{(j)}).$$

Since $(X, B^{(j)})$ is lc, $a(E, X, B^{(j)}) \geq 0$ for any j. Since (X, B) is klt, $a(E, X, B) > 0$, so $a(E, X, B^{(j)}) > 0$ for some j. Since $r_{kj} > 0$ for any j, $a(E, X, C^{(k)}) > 0$. Thus $(X, C^{(k)})$ is klt. By Construction-Theorem 5.2(1), $K_{X'} + B'^{(j)} = \pi^*(K_X + B^{(j)})$ for any j, so

$$K_{X'} + C'^{(k)} = \pi^*(K_X + C^{(k)}),$$

and $(X', C'^{(k)})$ is klt [KM98, Proposition 5.20].

We prove (2). By Construction-Theorem 5.2(1), the sheaf $\mathcal{O}_{X'}(q(K_{X'} + B'^{(j)}))$ is invertible near x' for any j. Since π is finite and $\pi^{-1}(x) = \{x'\}$, after shrinking X near x, $q(K_{X'} + B'^{(j)})$ is Cartier on X' for any j. For a prime divisor F over X', let $n_j(F) := qa(F, X', B'^{(j)})$. Since $q(K_{X'} + B'^{(j)})$ is Cartier, $n_j(F) \in \mathbb{Z}$, and $n_j(F) \geq 0$ since $(X', B'^{(j)})$ is lc (Construction-Theorem 5.2(1), [KM98, Proposition 5.20]). Since $K_{X'} + B' = \sum_j \lambda_j (K_{X'} + B'^{(j)})$ and $K_{X'} + C'^{(k)} = \sum_j r_{kj}(K_{X'} + B'^{(j)})$, we have

$$a(F, X', B') = \frac{1}{q} \sum_j \lambda_j n_j(F), \qquad a(F, X', C'^{(k)}) = \frac{1}{q} \sum_j r_{kj} n_j(F). \tag{17.4}$$

By (17.2) applied to $n = (n_1(F), \dots, n_s(F))$, the signs of $a(F, X', C'^{(k)}) - 1$ and $a(F, X', B') - 1$ are equal. If $a(F, X', B') = 1$, then $\sum_j \lambda_j n_j(F) = q$, so $n_j(F) = q$ for any j by (17.1), that is, $a(F, X', B'^{(j)}) = 1$.

We prove (3). Assume that $B_Y \geq 0$, and let P be a prime divisor on Y. If P is not f-exceptional, then $\operatorname{mult}_P C_Y^{(k)} = \operatorname{mult}_{f_*P} C'^{(k)} \geq 0$. If P is f-exceptional, then $a(P, X', B') = 1 - \operatorname{mult}_P B_Y \leq 1$, so $a(P, X', C'^{(k)}) \leq 1$ by (2), and $\operatorname{mult}_P C_Y^{(k)} = 1 - a(P, X', C'^{(k)}) \geq 0$. Thus $C_Y^{(k)} \geq 0$. For any prime divisor P' over Y, $a(P', Y, C_Y^{(k)}) = a(P', X', C'^{(k)}) > 0$ by (1), so $(Y, C_Y^{(k)})$ is klt. By (17.4) and $\mathbf{A} = L\mathbf{R}$, for any prime divisor P on Y,

$$\operatorname{mult}_P C_Y^{(k)} = 1 - a(P, X', C'^{(k)}) = \frac{qL - \sum_j \mathbf{A}_{kj} n_j(P)}{qL} \in \frac{1}{qL}\mathbb{Z},$$

and $0 \leq \operatorname{mult}_P C_Y^{(k)} < 1$. Thus $C_Y^{(k)} \in \{0, \frac{1}{qL}, \dots, \frac{qL-1}{qL}\}$. If (Y, B_Y) is terminal and P' is a prime divisor exceptional over Y, then $a(P', X', B') = a(P', Y, B_Y) > 1$, so

$$a(P', Y, C_Y^{(k)}) = a(P', X', C'^{(k)}) > 1$$

by (2). Thus $(Y, C_Y^{(k)})$ is terminal.

We prove (4). Since $\sum_j \mathbf{A}_{kj} = L$ and $C'^{(k)} = \sum_j \frac{\mathbf{A}_{kj}}{L} B'^{(j)}$, we have

$$qL(K_{X'} + C'^{(k)}) = \sum_j \mathbf{A}_{kj} q(K_{X'} + B'^{(j)}).$$

Thus

$$\mathcal{O}_{X'}(qL(K_{X'} + C'^{(k)})) = \bigotimes_j \mathcal{O}_{X'}(q(K_{X'} + B'^{(j)}))^{\otimes \mathbf{A}_{kj}}$$

as G-linearized invertible sheaves, and the character of a tensor product at x' is the product of the characters, so the character at x' is $\nu_k = \prod_j \chi_j^{\mathbf{A}_{kj}}$. Let $G_1 \subset G$ be a subgroup. Since $\bigcap_j \operatorname{Ker} \chi_j = \{1\}$ (Construction-Theorem 5.2(1)), the homomorphism

$$\iota : G_1 \to (\mathbb{C}^*)^s, \qquad \sigma \to (\chi_1(\sigma), \dots, \chi_s(\sigma))$$

is injective. For an integer $s \times s$ matrix $\mathbf{M}$, let $\Phi_{\mathbf{M}} : (\mathbb{C}^*)^s \to (\mathbb{C}^*)^s$ be the homomorphism with $\Phi_{\mathbf{M}}(z_1, \dots, z_s)_k = \prod_j z_j^{\mathbf{M}_{kj}}$. Then $\nu_k(\sigma) = \Phi_{\mathbf{A}}(\iota(\sigma))_k$, so ι maps $\bigcap_k \operatorname{Ker}(\nu_k|_{G_1})$ injectively into $\operatorname{Ker} \Phi_{\mathbf{A}}$. By the Smith normal form [Jac85, Theorem 3.8], there exist $\mathbf{U}, \mathbf{V} \in \mathrm{GL}_s(\mathbb{Z})$ and positive integers $d_1, \dots, d_s$ with $\prod_j d_j = |\det \mathbf{A}|$ such that $\mathbf{A} = \mathbf{U} \operatorname{diag}(d_1, \dots, d_s)\mathbf{V}$. For integer $s \times s$ matrices $\mathbf{M}$ and $\mathbf{M}'$, we have

$$\Phi_{\mathbf{M}\mathbf{M}'} = \Phi_{\mathbf{M}} \circ \Phi_{\mathbf{M}'},$$

so the homomorphisms $\Phi_{\mathbf{U}}$ and $\Phi_{\mathbf{V}}$ are automorphisms, and

$$\operatorname{Ker}\Phi_{\mathbf{A}} = \Phi_{\mathbf{V}}^{-1}\big(\operatorname{Ker}\Phi_{\operatorname{diag}(d_1,\dots,d_s)}\big) \cong \mu_{d_1}\times\cdots\times\mu_{d_s}$$

has order $|\det\mathbf{A}|$. Thus the order of $\bigcap_k \operatorname{Ker}(\nu_k|_{G_1})$ divides $|\det\mathbf{A}|$. In the setting of (3),

$$qL(K_Y + C_Y^{(k)}) = f^*(qL(K_{X'} + C'^{(k)})),$$
$$\mathcal{O}_Y(qL(K_Y + C_Y^{(k)})) = f^*\mathcal{O}_{X'}(qL(K_{X'} + C'^{(k)}))$$

as H-linearized invertible sheaves. The pullback of an H-linearized invertible sheaf along the H-equivariant morphism f has, at an H-fixed point y over x', the character at x' restricted to H, which is $\nu_k|_H$.

We prove (5). By (3), $C_Y^{(k)} \geq 0$. Define $B_Y^{(j)}$ by $K_Y + B_Y^{(j)} = f^*(K_{X'} + B'^{(j)})$ as in Construction-Theorem 5.2(4). Since $\sum_j r_{kj} = 1$,

$$K_Y + C_Y^{(k)} = \sum_j r_{kj} f^*(K_{X'} + B'^{(j)}) = K_Y + \sum_j r_{kj} B_Y^{(j)},$$

so $C_Y^{(k)} = \sum_j r_{kj} B_Y^{(j)}$. By Construction-Theorem 5.2(5), near y' there exist semi-invariant rational functions $g_1,\dots,g_s$ with $\operatorname{div}(g_j) = qh^*B_Y^{(j)}$ such that the character of $\omega^{\otimes q}/g_j$ is the pullback of $\chi_j|_I$ to $\hat{I}$. Let $\varphi_k := \prod_j g_j^{\mathbf{A}_{kj}}$, a semi-invariant rational function. Then

$$\operatorname{div}(\varphi_k) = \sum_j \mathbf{A}_{kj} qh^* B_Y^{(j)} = qLh^* C_Y^{(k)} = qLC_{Y'}^{(k)},$$

and, since $\sum_j \mathbf{A}_{kj} = L$,

$$\omega^{\otimes qL}/\varphi_k = \prod_j (\omega^{\otimes q}/g_j)^{\mathbf{A}_{kj}}.$$

Thus the character of $\omega^{\otimes qL}/\varphi_k$ is the pullback of $\prod_j \chi_j^{\mathbf{A}_{kj}}|_I = \nu_k|_I$ to $\hat{I}$. Since $C_Y^{(k)} \geq 0$ and h is quasi-étale, $C_{Y'}^{(k)} \geq 0$. Since Y' is normal (Definition 2.5) and $\operatorname{div}(\varphi_k) \geq 0$ near y', φ_k is regular near y', and $qLC_{Y'}^{(k)} = \operatorname{div}(\varphi_k)$ is Cartier near y'. Let $p : \hat{I} \to I$ be the surjection with kernel J. The joint kernel of the characters of $\omega^{\otimes qL}/\varphi_1,\dots,\omega^{\otimes qL}/\varphi_s$ is $p^{-1}\big(\bigcap_k \operatorname{Ker}(\nu_k|_I)\big)$, whose order is $|J|\cdot|\bigcap_k \operatorname{Ker}(\nu_k|_I)|$. By (4), the order of $\bigcap_k \operatorname{Ker}(\nu_k|_I)$ divides $|\det\mathbf{A}|$, which implies (5). □

17.2. **Cyclic covers trivializing a multiple of the log canonical divisor.** In this subsection, we construct, for a klt threefold pair (X,B) with $\mathbb{Q}$-boundary and a positive integer m such that mB is a Weil divisor, the cyclic cover on which $m(K_X+B)$ becomes principal, and an equivariant terminal model over the cover whose boundary coefficients and Cartier indices are bounded in terms of m. The integer m is fixed in advance: it does not depend on the denominators of log discrepancies of (X,B).

Construction-Theorem 17.2. Let (X,B) be a threefold pair, $x \in X$ a closed point, and m a positive integer. Assume that (X,B) is klt near x, B is a $\mathbb{Q}$-divisor, and mB is a Weil divisor. Let r be the Cartier index of K_X+B at x and $s := r/\gcd(r,m)$. After shrinking X near x, let h be a rational function with $\operatorname{div}(h) = sm(K_X+B)$, and let $\pi : X' \to X$ be the normalization of X in $\mathbb{C}(X)(z)$, where $z^s = h$. Then the following hold.

(1) With $B' := \pi^*B$, $\mathcal{M}_{X'} := \mathcal{O}_{X'}(m(K_{X'}+B'))$, and a rational section ω_0 of $\omega_X^{\otimes m}$ with $\operatorname{div}(\omega_0) = mK_X$, the following hold.
   (a) The morphism π is a quasi-étale Galois morphism of degree s with cyclic Galois group $G = \mu_s$, where $\zeta \in \mu_s$ acts by $\zeta^* z = \zeta z$.
   (b) The fiber $\pi^{-1}(x)$ is a single point x'.
   (c) For any prime divisor P' on X', $\operatorname{mult}_{P'} B' = \operatorname{mult}_{\pi(P')} B$.
   (d) We have $K_{X'} + B' = \pi^*(K_X+B)$ and $m(K_{X'}+B') = \operatorname{div}(z)$.
   (e) The pair (X',B') is klt.

(f) The sheaf $\mathcal{M}_{X'}$ is invertible with generator $z^{-1}\pi^*\omega_0$ and faithful character $\chi(\zeta)=\zeta^{-1}$ at x'.
(g) For any prime divisor F over X', $a(F,X',B')\in\frac{1}{m}\mathbb{N}^+$.

(2) Let E be a prime divisor over X with $\operatorname{center}_X E=\{x\}$ and $a(E,X,B)=\frac{p}{q}$ for coprime positive integers p and q, let F be a prime divisor over X' lying over E, let $\kappa(E)$ and $\kappa(F)$ be the residue fields of ord_E and ord_F, let $\mathfrak{m}_F$ be the maximal ideal of the valuation ring of ord_F, let $H:=\operatorname{Stab}_G(F)$, $I:=I_G(F)$, $e:=|I|$, $f:=[\kappa(F):\kappa(E)]$, and $t:=[G:H]$, and let ρ_F be the character of I on $\mathfrak{m}_F/\mathfrak{m}_F^2$. Then the following hold.
(a) We have $\operatorname{ord}_F|_{\mathbb{C}(X)}=e\operatorname{ord}_E$ and $e=\frac{q}{\gcd(q,m)}$.
(b) We have $a(F,X',B')=\frac{p}{\gcd(q,m)}$.
(c) We have $s=eft$.
(d) We have $\operatorname{center}_{X'}F=\{x'\}$.
(e) The extension $\kappa(F)/\kappa(E)$ is a cyclic Galois extension of degree $f=|H/I|$.
(f) The character ρ_F is faithful, and $\chi|_I=\rho_F^{ma(F,X',B')}$.
(g) The character of any non-zero rational G-semi-invariant function u restricts to $\rho_F^{\operatorname{ord}_F(u)}$ on I.

(3) For F and H as in (2), after shrinking X near x, there exist an H-equivariant projective birational morphism $g:Y\to X'$ and an $\mathbb{R}$-divisor $B_Y\geq 0$ on Y with $K_Y+B_Y=g^*(K_{X'}+B')$ such that the following hold.
(a) The variety Y is H-$\mathbb{Q}$-factorial.
(b) We have $B_Y\in\{0,\frac{1}{m},\dots,\frac{m-1}{m}\}$, and (Y,B_Y) is terminal.
(c) The g-exceptional prime divisors are exactly the exceptional prime divisors over X' with log discrepancy at most 1 with respect to (X',B').
(d) We have $\mathcal{O}_Y(m(K_Y+B_Y))=g^*\mathcal{M}_{X'}$ as H-linearized invertible sheaves.
(e) The variety Y is terminal, and the Cartier index of K_Y at any closed point of Y is at most m.
(f) Both $\operatorname{lcm}(1,\dots,m)K_Y$ and $\operatorname{lcm}(1,\dots,m)B_Y$ are Cartier.
(g) If $a(F,X',B')\leq 1$, then F is a prime divisor on Y with $\operatorname{mult}_F B_Y=1-a(F,X',B')$.
(h) If $a(F,X',B')>1$, then $\operatorname{center}_Y F$ is an H-invariant closed point or an H-invariant complete curve in $g^{-1}(x')$.

Proof. We prove (1). Since K_X+B is an $\mathbb{R}$-Cartier $\mathbb{Q}$-divisor, K_X+B is $\mathbb{Q}$-Cartier near x: near x, K_X+B is a real linear combination of Cartier divisors, and the coefficients of such a combination are the solutions of a system of linear equations with rational coefficients, so they can be taken rational. Thus r is well defined. Let $D:=m(K_X+B)$, a Weil divisor since mB is a Weil divisor. For an integer j, jD is Cartier at x if and only if $r\mid jm$, that is, if and only if $s\mid j$. Thus s is the Cartier index of D at x, and $\operatorname{div}(h)=sD$.

Since $\mu_s\subset\mathbb{C}(X)$, the extension $\mathbb{C}(X)(z)/\mathbb{C}(X)$ is Galois, and the Galois group is a subgroup $G\subset\mu_s$ acting by $\zeta^*z=\zeta z$. Let $j:=|G|$, and suppose that $j<s$. Then $j\mid s$, z^j is G-invariant, so $z^j\in\mathbb{C}(X)$, and $(z^j)^{s/j}=h$ implies $\operatorname{div}(z^j)=\frac{j}{s}\operatorname{div}(h)=jD$ near x. Thus jD is Cartier at x with $0<j<s$, a contradiction. Hence $G=\mu_s$ and $\deg\pi=s$.

Let P be a prime divisor on X, v_P a local parameter of $\mathcal{O}_{X,P}$, and $d_P:=\operatorname{mult}_P D$. Then $w_P:=zv_P^{-d_P}$ satisfies $w_P^s=hv_P^{-sd_P}\in\mathcal{O}_{X,P}^*$, so $\mathcal{O}_{X,P}[w_P]=\bigoplus_{i=0}^{s-1}\mathcal{O}_{X,P}w_P^i$ is finite étale over $\mathcal{O}_{X,P}$, hence it is the integral closure of $\mathcal{O}_{X,P}$ in $\mathbb{C}(X')$. Thus π is quasi-étale, and $vz^i=vv_P^{id_P}w_P^i$ with $v\in\mathbb{C}(X)$ and $0\leq i<s$ is integral over $\mathcal{O}_{X,P}$ if and only if $\operatorname{ord}_P(v)+id_P\geq 0$. Since $\pi_*\mathcal{O}_{X'}$ is G-stable and X' is normal, we obtain

$$\pi_*\mathcal{O}_{X'}=\bigoplus_{i=0}^{s-1}\mathcal{O}_X(iD)z^i, \tag{17.5}$$

that is, π is the cyclic cover associated to D and the nowhere vanishing section h^{-1} of $\mathcal{O}_X(sD)$ as in [KM98, Definition 5.19].

We show that $\pi^{-1}(x)$ is a single point. Otherwise, since G is abelian and acts transitively on $\pi^{-1}(x)$, there exists a non-trivial character ψ of G, trivial on the stabilizer of a point $y_1 \in \pi^{-1}(x)$, with $\psi(\zeta) = \zeta^i$ for some integer $0 < i < s$. Let u_0 be a regular function near $\pi^{-1}(x)$ with $u_0(\sigma(y_1)) = \psi(\sigma)$ for any $\sigma \in G$. The projection $u := \frac{1}{s}\sum_{\sigma\in G}\psi(\sigma)^{-1}\sigma^* u_0$ of u_0 onto the character ψ satisfies $\sigma^* u = \psi(\sigma)u$ and $u(\sigma(y_1)) = \psi(\sigma) \neq 0$ for any $\sigma \in G$. By (17.5), $u = vz^i$ with $v \in \mathcal{O}_X(iD)_x$, and $u^s = v^s h^i$ is a G-invariant unit near $\pi^{-1}(x)$, so $v^s h^i \in \mathcal{O}^*_{X,x}$. Thus $iD = -\operatorname{div}(v)$ is Cartier at x with $0 < i < s$, a contradiction. Hence $\pi^{-1}(x)$ is a single point x'.

Since π is étale over the generic point of any prime divisor on X, $\operatorname{mult}_{P'} B' = \operatorname{mult}_{\pi(P')} B$ for any prime divisor P' on X' and $K_{X'} = \pi^* K_X$, so $K_{X'} + B' = \pi^*(K_X + B)$. Since $z^s = h$, we have

$$m(K_{X'} + B') = \pi^* D = \frac{1}{s}\pi^* \operatorname{div}(h) = \operatorname{div}(z).$$

The pair (X', B') is klt by [KM98, Proposition 5.20]. Since π is quasi-étale, $\pi^*\omega_0$ is a rational section of $\omega_{X'}^{\otimes m}$ with $\operatorname{div}(\pi^*\omega_0) = \pi^*(mK_X) = mK_{X'}$, so $\operatorname{div}(z^{-1}\pi^*\omega_0) + mB' = 0$. Thus

$$\mathcal{M}_{X'} = \mathcal{O}_{X'} \cdot z^{-1}\pi^*\omega_0,$$

and the sheaf $\mathcal{M}_{X'}$ is invertible with generator $z^{-1}\pi^*\omega_0$. Since $\pi^*\omega_0$ is G-invariant, $\zeta^*(z^{-1}\pi^*\omega_0) = \zeta^{-1}z^{-1}\pi^*\omega_0$. Since x' is fixed by G, the character of $\mathcal{M}_{X'}$ at x' is $\chi(\zeta) = \zeta^{-1}$ (Definition 2.14), which is faithful. Let F be a prime divisor over X', and let W be a higher model of X' on which F is a divisor. Since $m(K_{X'} + B') = \operatorname{div}(z)$,

$$ma(F, X', B') = m + m \operatorname{mult}_F K_W - \operatorname{ord}_F(z) \in \mathbb{Z},$$

and $a(F, X', B') > 0$ since (X', B') is klt. Thus $a(F, X', B') \in \frac{1}{m}\mathbb{N}^+$.

We prove (2). By (1), π is quasi-étale, $X = X'/G$, and $\pi_* B' = sB$, so $(X', B')/G = (X, B)$ (Definition 2.5). By Lemma 2.6(1)(2), $\operatorname{ord}_F|_{\mathbb{C}(X)} = e \operatorname{ord}_E$, ρ_F is faithful,

$$a(F, X', B') = ea(E, X, B) = \frac{ep}{q},$$

and $\pi(\operatorname{center}_{X'} F) = \operatorname{center}_X E = \{x\}$, so $\operatorname{center}_{X'} F = \{x'\}$. The integral closure of the valuation ring of ord_E in $\mathbb{C}(X')$ is a semilocal Dedekind domain whose localizations at maximal ideals are the valuation rings of the prime divisors over X' lying over E (cf. [Ser79, Chapter I, §4, Propositions 8–10] and [KM98, Lemma 2.45]). By [Ser79, Chapter I, §7, Propositions 19–21 and the Corollaries to Propositions 19 and 21], G permutes the prime divisors over X' lying over E transitively, so the number of such prime divisors is t, $\kappa(F)/\kappa(E)$ is a Galois extension with group H/I (the residue fields have characteristic 0), and $s = |G| = eft$. Since H/I is a quotient of a subgroup of the cyclic group G, $\kappa(F)/\kappa(E)$ is cyclic of degree $f = |H/I|$. The last statement of (2) is Lemma 9.2(1), where the character λ there is ρ_F.

We compute $\chi|_I$. Since z^{-1} is G-semi-invariant with character χ, Lemma 9.2(1) implies $\chi|_I = \rho_F^{-\operatorname{ord}_F(z)}$. Let W_E be a higher model of X on which E is a divisor, and let $c := \operatorname{mult}_E K_{W_E} \in \mathbb{Z}$ and $N := \operatorname{ord}_E(h)$. Since the pullback of $K_X + B$ to W_E is $\frac{1}{sm}\operatorname{div}(h)$, we have $a(E, X, B) = 1 + c - \frac{N}{sm}$. Since $z^s = h$ and $\operatorname{ord}_F|_{\mathbb{C}(X)} = e \operatorname{ord}_E$, we have $\operatorname{ord}_F(z) = \frac{eN}{s}$. Thus

$$ma(F, X', B') = ema(E, X, B) = em(1 + c) - \operatorname{ord}_F(z). \tag{17.6}$$

Since ρ_F has order e, (17.6) implies $\chi|_I = \rho_F^{-\operatorname{ord}_F(z)} = \rho_F^{ma(F,X',B')}$.

We show that $e = \frac{q}{\gcd(q,m)}$. Let $e' := \frac{q}{\gcd(q,m)}$ and $m' := \frac{m}{\gcd(q,m)}$, so that $\gcd(e', m'p) = 1$. By (1),

$$ma(F, X', B') = \frac{mep}{q} = \frac{em'p}{e'} \in \mathbb{Z},$$

so $e' \mid e$. Since χ is faithful on G, the character $\chi|_I = \rho_F^{ma(F,X',B')}$ has order $e = |I|$, so $ma(F, X', B') = \frac{e}{e'}m'p$ is coprime to e. Since $\frac{e}{e'}$ divides both $ma(F, X', B')$ and e, we have $e = e'$. Thus $a(F, X', B') = \frac{ep}{q} = \frac{p}{\gcd(q,m)}$.

We prove (3). We use Construction-Theorem 5.1 for (X', B'), the point x', the group H, $N = 1$, and $\alpha_1 = \frac{1}{m}$. Then $\mathcal{S} = \frac{1}{m}\mathbb{Z}_{\geq 0}$ and $1 = m \cdot \frac{1}{m} \in \mathcal{S}$. We check the assumptions. By (1), (X', B') is klt near x', and any log discrepancy of (X', B') belongs to $\frac{1}{m}\mathbb{N}^+ \subset \mathcal{S}$. The group $H \subset G$ acts faithfully on X', fixes x' since $\pi^{-1}(x) = \{x'\}$, and preserves $B' = \pi^* B$. Moreover,

$$\Lambda_{\mathcal{S}} = \left\{0, \frac{1}{m}, \dots, \frac{m-1}{m}\right\}, \quad \eta_{\mathcal{S}} = \frac{1}{m}, \quad R_{\mathcal{S}} = m, \quad M_{\mathcal{S}} = \operatorname{lcm}(1, \dots, m).$$

Since π is finite and $\pi^{-1}(x) = \{x'\}$, any H-invariant neighborhood of x' contains the preimage of a neighborhood of x, so shrinking X' to an H-invariant neighborhood of x' may be done by shrinking X near x. By Construction-Theorem 5.1, after shrinking X near x, there exist an H-equivariant projective birational morphism $g : Y \to X'$ and an $\mathbb{R}$-divisor $B_Y \geq 0$ on Y satisfying the following. By Construction-Theorem 5.1(1), $K_Y + B_Y = g^*(K_{X'} + B')$, and the g-exceptional prime divisors are exactly the exceptional prime divisors E' over X' with $a(E', X', B') \leq 1$, with $\operatorname{mult}_{E'} B_Y = 1 - a(E', X', B')$. By Construction-Theorem 5.1(2), $B_Y \in \Lambda_{\mathcal{S}} = \{0, \frac{1}{m}, \dots, \frac{m-1}{m}\}$ and (Y, B_Y) is terminal. By Construction-Theorem 5.1(3), Y is H-$\mathbb{Q}$-factorial. By Construction-Theorem 5.1(4), Y is terminal, the Cartier index of K_Y at any closed point of Y is at most $R_{\mathcal{S}} = m$, and $\operatorname{lcm}(1, \dots, m)K_Y$ is Cartier. By (2), $\operatorname{center}_{X'} F = \{x'\}$, so F is exceptional over X'. If $a(F, X', B') \leq 1$, then F is a prime divisor on Y by Construction-Theorem 5.1(5), and $\operatorname{mult}_F B_Y = 1 - a(F, X', B')$ by Construction-Theorem 5.1(1). If $a(F, X', B') > 1$, then $\operatorname{center}_Y F$ is a closed point or a complete curve in $g^{-1}(x')$ by Construction-Theorem 5.1(5), which is H-invariant since $H = \operatorname{Stab}_G(F)$.

We use the Moreover part of Construction-Theorem 5.1 with one boundary $B^{(1)} := B'$, with the integer q there equal to m, and with $\lambda_1 := 1$: the $\mathbb{Q}$-divisor $B' \geq 0$ is H-invariant, (X', B') is lc, and $m(K_{X'} + B') = \operatorname{div}(z)$ is Cartier by (1). Thus

$$\mathcal{O}_Y(m(K_Y + B_Y)) \cong g^* \mathcal{O}_{X'}(m(K_{X'} + B')) = g^* \mathcal{M}_{X'}$$

as H-linearized invertible sheaves. Finally, $m(K_Y + B_Y) = g^* \operatorname{div}(z)$ and $\operatorname{lcm}(1, \dots, m)K_Y$ are Cartier and $m \mid \operatorname{lcm}(1, \dots, m)$, so

$$\operatorname{lcm}(1, \dots, m)B_Y = \operatorname{lcm}(1, \dots, m)(K_Y + B_Y) - \operatorname{lcm}(1, \dots, m)K_Y$$

is Cartier. □

17.3. **Proxy boundaries on canonical covers.** In this subsection, we compute the characters of the proxy boundaries of Construction-Proposition 17.1 on the canonical cover of a terminal point of an equivariant higher model of the index one cover, and we write the pullbacks of the proxy boundaries by regular equations on a smooth germ.

Corollary 17.3. *Let $\Gamma \subset [0, 1]$ be a finite set, let s, q, μ_k, L, and $\mathbf{A}$ be as in Construction-Proposition 17.1, and let $m := qL$. Let (X, B), x, $\pi : X' \to X$, G, B', and χ_j be as in Construction-Theorem 5.2, and let $C'^{(k)}$ and ν_k be as in Construction-Proposition 17.1. Let E be a prime divisor computing $a := \operatorname{mld}(X \ni x, B) > 0$, let F be a prime divisor over X' lying over E, and let $I := I_G(F)$. Let $g : T \to X'$ be an I-equivariant proper birational morphism from a normal quasi-projective variety T, and define B_T and $C_T^{(k)}$ by $K_T + B_T = g^*(K_{X'} + B')$ and $K_T + C_T^{(k)} = g^*(K_{X'} + C'^{(k)})$. Assume that $B_T \geq 0$, T is terminal, and $\operatorname{center}_T F$ is a closed point c. Let r be the Cartier index of K_T at c. Then the conclusions of Construction-Theorem 5.5 hold for $(T \ni c, B_T)$, $H = I$, and F, with $h : T' \to T$, $\hat{I}$, and F' as in (1) and (2) there and $H' := I_{\hat{I}}(F')$, and moreover, where a character of G is restricted to I and inflated to $\hat{I}$ through the surjection $\hat{I} \to I$ of Construction-Theorem 5.5(1):*

(1) the characters χ_j, inflated to $\hat{I}$, have joint kernel μ_r, and together with the character κ of a generator of $\omega_{T'}$ they have trivial joint kernel, and the characters ν_k, inflated to $\hat{I}$, have joint kernel of order dividing $r|\det \mathbf{A}|$;

(2) *on H', the inflated characters χ_j have joint kernel of order u, where u is the denominator of $a(F,T,0)$, and the inflated characters ν_k have joint kernel of order dividing $u|\det \mathbf{A}|$, and the joint kernel of κ and the ν_k has order dividing $|\det \mathbf{A}|$;*

(3) *there exist regular $\hat{I}$-semi-invariant functions $h_1,\dots,h_s$ on the smooth germ Y of Construction-Theorem 5.5(3), non-zero on T', with $\operatorname{div}(h_k|_{T'}) = mh^*C_T^{(k)}$,*

$$h^*B_T = \sum_k \frac{\mu_k}{m}\operatorname{div}(h_k|_{T'}),$$

and $\nu_k = \kappa^m/d_k$ on $\hat{I}$, where d_k is the character of h_k.

Proof. We first check the hypotheses of Construction-Theorem 5.5 for $(T \ni c, B_T)$, $H = I$, and F. Since X is a threefold and T is terminal, $(T \ni c)$ is a terminal threefold germ. By Lemma 2.6(1), applied to the faithful action of G on X', the group I is cyclic. Since g is I-equivariant and birational, I acts faithfully on T, and since $I \subset \operatorname{Stab}_G(F)$, I preserves $\operatorname{center}_T F = \{c\}$. Since $B' = \pi^*B$ is G-invariant and g is I-equivariant, B_T is I-invariant. The log discrepancies of (T, B_T) are those of (X', B'), and (X', B') is klt by Construction-Theorem 5.2(1). Since $B_T \geq 0$, (T, B_T) is klt. Since E computes $\operatorname{mld}(X \ni x, B)$, $\operatorname{center}_X E = \{x\}$. By Lemma 3.1, applied with $X' \to X$, G, F, $H = I$, $Y = T$, $f = g$, and η the image of c in T/I, we have

$$\operatorname{mld}((T \ni c, B_T)/I) = a > 0,$$

and $\operatorname{mld}((T \ni c, B_T)/I)$ is attained by the prime divisor E_T over T/I under F. Since $I_I(F) = I$, Lemma 2.6(2), applied to (T, B_T) and the group I, implies

$$a(F, T, B_T) = |I| \cdot a(E_T, T/I, B_{T/I}) = |I|a.$$

Thus the hypotheses of Construction-Theorem 5.5 hold, and after shrinking I-invariant neighborhoods of c, the conclusions of Construction-Theorem 5.5 hold for $(T \ni c, B_T)$, $H = I$, and F. Let c' be the point of T' over c.

Since $\operatorname{center}_{X'} F \subset \pi^{-1}(x) = \{x'\}$, we have $g(c) = x'$. By Construction-Theorem 5.5(1), h is the quotient morphism of T' by μ_r, hence a finite quasi-étale Galois morphism with group μ_r, the variety T' is Gorenstein, $\hat{I}$ fixes c', and the exact sequence $1 \to \mu_r \to \hat{I} \to I \to 1$ is compatible with h. Thus Construction-Theorem 5.2(4)(5) and Construction-Proposition 17.1(3)(5) can be applied with the letters Y, f, y, Y', y', J there taken to be T, g, c, T', c', μ_r, and with the group $\hat{I}$ of Construction-Theorem 5.5(1). As in the statement, we denote the inflated characters of χ_j and ν_k again by χ_j and ν_k. Let

$$I_\nu := \bigcap_k \operatorname{Ker}(\nu_k|_I) \subset I.$$

By Construction-Proposition 17.1(4), $|I_\nu|$ divides $|\det \mathbf{A}|$.

We prove (1). By Construction-Theorem 5.2(1), $\bigcap_j \operatorname{Ker}\chi_j = \{1\}$ in G, and hence the restrictions $\chi_j|_I$ have trivial joint kernel. Since $\hat{I} \to I$ is surjective with kernel μ_r, the joint kernel of the χ_j on $\hat{I}$ is μ_r. By Construction-Theorem 5.5(1), κ is faithful on μ_r. Thus the joint kernel of κ and the χ_j on $\hat{I}$ is $\mu_r \cap \operatorname{Ker}\kappa = \{1\}$. The joint kernel of the ν_k on $\hat{I}$ is the inverse image of I_ν under $\hat{I} \to I$, of order $r|I_\nu|$, which divides $r|\det \mathbf{A}|$.

We prove (2). By Construction-Theorem 5.5(2), the homomorphism $H' \to I$ is surjective and $|H' \cap \mu_r| = u$, where u is the denominator of $a(F,T,0)$. By (1), the joint kernel of the χ_j on H' is $H' \cap \mu_r$, of order u. The joint kernel of the ν_k on H' is the inverse image of I_ν under $H' \to I$, of order $u|I_\nu|$, which divides $u|\det \mathbf{A}|$. The homomorphism $H' \to I$ maps $\operatorname{Ker}(\kappa|_{H'}) \cap \bigcap_k \operatorname{Ker}(\nu_k|_{H'})$ into I_ν, and the kernel of the induced homomorphism is contained in $\mu_r \cap \operatorname{Ker}\kappa = \{1\}$. Thus

$$\left|\operatorname{Ker}(\kappa|_{H'}) \cap \bigcap_k \operatorname{Ker}(\nu_k|_{H'})\right| \text{ divides } |I_\nu|,$$

and hence divides $|\det \mathbf{A}|$.

We prove (3). The generator ω of $\omega_{T'}$ near c' in Construction-Theorem 5.2(5) has character κ (Definition 2.14), and $m = qL$. Thus Construction-Proposition 17.1(5) (with the substitutions above) yields regular semi-invariant functions on T' with divisor $mh^*C_T^{(k)}$ and character κ^m/ν_k. By Construction-Theorem 5.5(3), each lifts to a regular semi-invariant function h_k on the smooth germ Y of Construction-Theorem 5.5(3) with character κ^m/ν_k. Thus $h_k|_{T'} \neq 0$, $\operatorname{div}(h_k|_{T'}) = mh^*C_T^{(k)}$, and $d_k = \kappa^m/\nu_k$, that is, $\nu_k = \kappa^m/d_k$ on $\hat{I}$. By Construction-Proposition 17.1(1), $B = \sum_k \mu_k C^{(k)}$, and hence $B' = \sum_k \mu_k C'^{(k)}$. Since $\sum_k \mu_k = 1$,

$$K_T + \sum_k \mu_k C_T^{(k)} = \sum_k \mu_k g^*(K_{X'} + C'^{(k)}) = g^*(K_{X'} + B') = K_T + B_T.$$

Thus $B_T = \sum_k \mu_k C_T^{(k)}$, and

$$h^*B_T = \sum_k \mu_k h^* C_T^{(k)} = \sum_k \frac{\mu_k}{m} \operatorname{div}(h_k|_{T'}).$$

□

17.4. **Formal presentations.** In this subsection, we write the completed local ring of the canonical cover, the pullback of the boundary, and the characters of Corollary 17.3 in power series of eigenparameters, for the morphism of Construction-Theorem 5.1.

Construction-Proposition 17.4. Let $\Gamma \subset [0,1]$ be a finite set, let $s, q, \lambda_j, \mu_k, L$ be as in Construction-Proposition 17.1, $m := qL$,

$$\mathcal{S} := \Big\{ \sum_j \frac{\lambda_j}{q} n_j \;\Big|\; n_j \in \mathbb{Z}_{\geq 0} \Big\},$$

and $R_{\mathcal{S}}$ as in Construction-Theorem 5.1. Let (X, B), x, $\pi : X' \to X$, and G be as in Construction-Theorem 5.2, $B' := \pi^*B$, E a prime divisor computing $a := \operatorname{mld}(X \ni x, B) > 0$, F a prime divisor over X' lying over E, and $I := I_G(F)$. Let $g : T \to X'$ be an I-equivariant projective birational morphism as in Construction-Theorem 5.1 for (X', B'), $H = I$, and $\mathcal{S}$, and B_T the crepant pullback of B'. Assume that $\operatorname{center}_T F$ is a closed point c. Then the Cartier index r of K_T at c is at most $R_{\mathcal{S}}$. Let $h : T' \to T$, $\hat{I}$, and F' be as in Construction-Theorem 5.5(1)(2) for $(T \ni c, B_T)$, $H = I$, and F, and $H' := I_{\hat{I}}(F')$, let c' be the point of T' over c, $\widehat{\mathcal{O}}$ the completion of $\mathcal{O}_{T',c'}$, and ν_k the characters of Construction-Proposition 17.1(4), inflated to $\hat{I}$. Then the following hold.

(1) (a) $\widehat{\mathcal{O}} \cong \mathbb{C}[[x_1, x_2, x_3]]$ if T' is smooth at c', and $\widehat{\mathcal{O}} \cong \mathbb{C}[[x_1, \dots, x_4]]/(\hat{f})$ with $\hat{f}$ semi-invariant defining an isolated cDV singularity otherwise, where $\hat{I}$ acts diagonally on the x_i.
   (b) There exist semi-invariant power series $\hat{h}_1, \dots, \hat{h}_s$ in the x_i whose images in $\widehat{\mathcal{O}}$ are non-zero, such that the pullback of h^*B_T to $\operatorname{Spec} \widehat{\mathcal{O}}$ is $\sum_k \frac{\mu_k}{m} \operatorname{div}(\hat{h}_k)$.
   (c) For $H'' \in \{\hat{I}, H'\}$, the completion of the local ring of T'/H'' at the image of c' is $\widehat{\mathcal{O}}^{H''}$.

(2) For $H'' \in \{\hat{I}, H'\}$, $a = \inf_P a(P, \operatorname{Spec} \widehat{\mathcal{O}}, \widehat{B})/|I_{H''}(P)|$, where P runs over the formal prime divisors over $\operatorname{Spec} \widehat{\mathcal{O}}$ with center the closed point and $\widehat{B}$ is the pullback of h^*B_T.

(3) There exists a formal prime divisor $\hat{F}'$ over $\operatorname{Spec} \widehat{\mathcal{O}}$ restricting to F', with isomorphic residue field and equal decomposition and inertia groups in $\hat{I}$. In particular $I_{\hat{I}}(\hat{F}') = H'$.

(4) If c_i, d_f, and d_k are the characters of x_i, $\hat{f}$ ($d_f := 1$ if T' is smooth at c') and $\hat{h}_k$, and κ is the character of a generator of $\omega_{T'}$ at c', then $\kappa = \prod_i c_i/d_f$ and $\nu_k = \kappa^m/d_k$ on $\hat{I}$.

Proof. Since $\sum_j \lambda_j = 1$, taking $n_j = q$ for any j shows that $1 \in \mathcal{S}$. By Construction-Theorem 5.2(1), (X', B') is klt, $\pi^{-1}(x) = \{x'\}$, and $a(P, X', B') \in \mathcal{S}$ for any prime divisor P over X'. Since G acts faithfully on X' and B' is G-invariant, I acts faithfully on X', fixes x',

and preserves B'. Thus the hypotheses of Construction-Theorem 5.1 hold for (X', B'), x', $H = I$, and $\mathcal{S}$. By Construction-Theorem 5.1(1)(2)(4), $B_T \geq 0$, T is terminal, and the Cartier index of K_T at any closed point is at most $R_{\mathcal{S}}$. In particular, $r \leq R_{\mathcal{S}}$. Since X' is quasi-projective and g is projective, T is a normal quasi-projective variety. Define $C_T^{(k)}$ by $K_T + C_T^{(k)} = g^*(K_{X'} + C'^{(k)})$, where $C'^{(k)}$ is as in Construction-Proposition 17.1. By Corollary 17.3, the conclusions of Construction-Theorem 5.5 hold for $(T \ni c, B_T)$, $H = I$, and F, and so do (1), (2), and (3) of Corollary 17.3.

Let Y, n, the $\hat{I}$-eigenparameters $x_1, \ldots, x_n$ of Y at c', and, if $n = 4$, the $\hat{I}$-semi-invariant function f be as in Construction-Theorem 5.5(3) for $(T \ni c, B_T)$ and $H = I$. By Corollary 17.3(3), there exist regular $\hat{I}$-semi-invariant functions $h_1, \ldots, h_s$ on Y, non-zero on T', such that the character of h_k is κ^m/ν_k for any k and

$$h^* B_T = \sum_k \frac{\mu_k}{m} \operatorname{div}(h_k|_{T'}).$$

Since B' is G-invariant and g is I-equivariant, B_T is I-invariant, and hence $h^* B_T$ is $\hat{I}$-invariant. Since T' is Gorenstein and $h^* B_T = \sum_k \frac{\mu_k}{m} \operatorname{div}(h_k|_{T'})$ is an $\mathbb{R}$-linear combination of principal divisors, $K_{T'} + h^* B_T$ is $\mathbb{R}$-Cartier near c'. By Construction-Theorem 5.5(1), $\hat{I}$ acts faithfully on T' and fixes c', and hence so does any subgroup of $\hat{I}$. Let $\widehat{X}$ and $\hat{x}$ be as in Definition 3.2 for $(T' \ni c')$, and let $H'' \in \{\hat{I}, H'\}$.

We prove (1). Let $\widehat{\mathcal{O}}_{Y,c'}$ be the completion of $\mathcal{O}_{Y,c'}$. Since Y is smooth at c' and $x_1, \ldots, x_n$ are regular parameters of $\mathcal{O}_{Y,c'}$, we have $\widehat{\mathcal{O}}_{Y,c'} = \mathbb{C}[[x_1, \ldots, x_n]]$ by [Mat89, Theorem 29.7]. Since $\sigma^* x_i = c_i(\sigma) x_i$ for any $\sigma \in \hat{I}$, where c_i is the character of x_i, the group $\hat{I}$ acts diagonally on the x_i. If $n = 3$, then $T' = Y$ and $\widehat{\mathcal{O}} = \mathbb{C}[[x_1, x_2, x_3]]$. Assume that $n = 4$, and let $\hat{f}$ be the image of f in $\widehat{\mathcal{O}}_{Y,c'}$. Here and below, the hat on $\hat{f}$ and on $\hat{h}_k$ means the image in $\widehat{\mathcal{O}}_{Y,c'}$, and is unrelated to the notation $\hat{I}$. Since $\mathcal{O}_{T',c'} = \mathcal{O}_{Y,c'}/(f)$, we have $\widehat{\mathcal{O}} = \widehat{\mathcal{O}}_{Y,c'}/(\hat{f})$ [Mat89, Theorem 8.7]. The power series $\hat{f}$ is semi-invariant with the character of f, and $\hat{f}$ defines the completion of the isolated cDV singularity $(T' \ni c')$ of Construction-Theorem 5.5(3).

In both cases, let $\hat{h}_k$ be the image of h_k in $\widehat{\mathcal{O}}_{Y,c'}$. Then $\hat{h}_k$ is semi-invariant with the character of h_k. The class of $\hat{h}_k$ in $\widehat{\mathcal{O}}$ is the image of $h_k|_{T'}$. Since T' is a variety and $h_k|_{T'} \neq 0$, the germ of $h_k|_{T'}$ in $\mathcal{O}_{T',c'}$ is non-zero. Since $\widehat{\mathcal{O}}$ is faithfully flat over $\mathcal{O}_{T',c'}$ [Mat89, Theorem 8.14], the homomorphism $\mathcal{O}_{T',c'} \to \widehat{\mathcal{O}}$ is injective, and hence the image of $\hat{h}_k$ in $\widehat{\mathcal{O}}$ is non-zero. Since $\widehat{X} \to T'$ is flat, the pullback of $\operatorname{div}(h_k|_{T'})$ to $\widehat{X}$ is $\operatorname{div}(\hat{h}_k)$. Thus the pullback of $h^* B_T$ to $\widehat{X}$ is

$$\widehat{B} = \sum_k \frac{\mu_k}{m} \operatorname{div}(\hat{h}_k).$$

By Lemma 3.3(1), applied to $(T' \ni c', h^* B_T)$ and the group H'', the completion of the local ring of T'/H'' at the image of c' is $\widehat{\mathcal{O}}^{H''}$.

We prove (2). By Lemma 3.3(3), applied to $(T' \ni c', h^* B_T)$ and the group H'', restriction maps the formal prime divisors over $\widehat{X}$ with center $\hat{x}$ onto the prime divisors over T' with center c', and preserves log discrepancies with respect to $\widehat{B}$ and $h^* B_T$ and inertia groups in H''. Thus, by the formula for $\operatorname{mld}((X \ni x, B)/G)$ in the last sentence of Lemma 2.6, applied to the faithful action of H'' on T' and the closed point c',

$$\inf_P \frac{a(P, \widehat{X}, \widehat{B})}{|I_{H''}(P)|} = \operatorname{mld}((T' \ni c', h^* B_T)/H''),$$

where P runs over the formal prime divisors over $\widehat{X}$ with center $\hat{x}$. By Lemma 3.1, applied with $X' \to X$, G, F, $H = I$, $Y = T$, $f = g$, and η the image of c in T/I, we have

$$\operatorname{mld}((T \ni c, B_T)/I) = a.$$

If $H'' = H'$, then Construction-Theorem 5.5(2) implies

$$\mathrm{mld}((T' \ni c', h^*B_T)/H') = \mathrm{mld}((T \ni c, B_T)/I) = a.$$

Assume that $H'' = \hat{I}$. By Construction-Theorem 5.5(1), $T'/\hat{I} = T/I$, and the quotient morphism $T' \to T/I$ is $p \circ h$, where $p : T \to T/I$ is the quotient morphism. Since $\mathrm{Ker}(\hat{I} \to I)$ is the group μ_r of Construction-Theorem 5.5(1), $T = T'/\mu_r$ and h is the quotient morphism. Since h is quasi-étale, Lemma 2.6(4) implies that $(T', h^*B_T)/\mu_r = (T, B_T)$, and Lemma 2.6(5), applied to the normal subgroup μ_r of $\hat{I}$, implies that $(T', h^*B_T)/\hat{I} = (T, B_T)/I$. Thus $\mathrm{mld}((T' \ni c', h^*B_T)/\hat{I}) = \mathrm{mld}((T \ni c, B_T)/I) = a$.

We prove (3). Since $\mathrm{center}_T F = \{c\}$ and $h^{-1}(c) = \{c'\}$, we have $\mathrm{center}_{T'} F' = \{c'\}$. By Lemma 3.3(3), applied to $(T' \ni c', h^*B_T)$ and the group $\hat{I}$, there exists a formal prime divisor $\hat{F}'$ over $\widehat{X}$ with center $\hat{x}$ which restricts to F', with isomorphic residue field and equal decomposition and inertia groups in $\hat{I}$. In particular, $I_{\hat{I}}(\hat{F}') = I_{\hat{I}}(F') = H'$.

We prove (4). Since $\mathcal{O}_{Y,c'} \to \widehat{\mathcal{O}}_{Y,c'}$ is $\hat{I}$-equivariant, the characters of x_i, $\hat{f}$, and $\hat{h}_k$ in $\widehat{\mathcal{O}}_{Y,c'}$ are the characters of x_i, f, and h_k in $\mathcal{O}_{Y,c'}$. By Construction-Theorem 5.5(3), $\kappa = \prod_i c_i/d_f$. Since the character of h_k is κ^m/ν_k, we have $\nu_k = \kappa^m/d_k$ on $\hat{I}$. □

17.5. **Curve centers for rational coefficients.** In this subsection, we prove Proposition 17.5. For the cover $\pi : X' \to X$ and the variety Y of Construction-Theorem 17.2, it shows that along a strictly increasing sequence of minimal log discrepancies, the center on Y of a prime divisor lying over a divisor computing the minimal log discrepancy is a closed point for all but finitely many terms. We use Proposition 17.5 only in Construction-Theorem 17.6(6).

Proposition 17.5. *Let (X, B), x, m, $\pi : X' \to X$, G, and B' be as in Construction-Theorem 17.2, and let $\Psi_m := \Psi(\{0, \frac{1}{m}, \dots, \frac{m-1}{m}\})$. Let E be a prime divisor computing $a := \mathrm{mld}(X \ni x, B) > 0$, F a prime divisor over X' lying over E, $H := \mathrm{Stab}_G(F)$, $I := I_G(F)$, and $g : Y \to X'$ and B_Y as in Construction-Theorem 17.2(3). Then the following hold.*

(1) Assume that $C := \mathrm{center}_Y F$ is a curve, and let J be the kernel of the action of H on the function field of C. Then $I \subset J$, and the coefficients of $B_{Y/J}$ belong to Ψ_m, where $(Y/J, B_{Y/J}) := (Y, B_Y)/J$. Moreover, for the generic point $\bar{\eta}$ of the image of C in Y/J and the prime divisor P_0 over Y/J with $\mathrm{ord}_F|_{\mathbb{C}(Y/J)} \in \mathbb{N}^+ \mathrm{ord}_{P_0}$, we have $\mathrm{center}_{Y/J} P_0 = \overline{\{\bar{\eta}\}}$ and $\mathrm{mld}(Y/J \ni \bar{\eta}, B_{Y/J}) = a(P_0, Y/J, B_{Y/J}) = a$.

(2) Let $\Gamma \subset \mathbb{Q} \cap [0, 1]$ be a finite set with $m\Gamma \subset \mathbb{Z}$, and $0 < \epsilon < A_0$. Let $a_i = \mathrm{mld}(X_i \ni x_i, B_i)$ be a strictly increasing sequence in $[\epsilon, A_0]$, where (X_i, B_i) is a threefold pair, klt near the closed point x_i, with $B_i \in \Gamma$. Let $\pi_i : X_i' \to X_i$ and B_i' be the objects π and B' of Construction-Theorem 17.2 for (X_i, B_i), x_i, and m, and let E_i, F_i, and Y_i be the objects E, F, and Y defined before (1), with (X_i, B_i) and x_i in place of (X, B) and x. Then $a(F_i, X_i', B_i') > 1$ and $\mathrm{center}_{Y_i} F_i$ is a closed point for any $i \gg 0$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we prove (1).

Assume that $C := \mathrm{center}_Y F$ is a curve, and let J be the kernel of the action of H on $\mathbb{C}(C)$. Let η be the generic point of C, let $h : Y \to Y/J$ be the quotient morphism, and let $\bar{C} := h(C)$, so that $\bar{\eta}$ is the generic point of $\bar{C}$. Since g is H-equivariant and H fixes ord_F, H stabilizes C and acts on $\mathbb{C}(C)$. Since G acts faithfully on X' and g is H-equivariant and birational, J acts faithfully on Y. Since ord_F is centered at η, the valuation ring of ord_F dominates $\mathcal{O}_{Y,\eta}$, so $\mathbb{C}(C)$ embeds H-equivariantly into the residue field of ord_F. By definition, I acts trivially on the residue field of ord_F, hence on $\mathbb{C}(C)$. Thus $I \subset J$.

By Construction-Theorem 17.2(1)–(3), π is a finite Galois morphism with Galois group G, $K_{X'} + B' = \pi^*(K_X + B)$, $\mathrm{ord}_F|_{\mathbb{C}(X)} = |I| \mathrm{ord}_E$, Y is a normal quasi-projective variety, and $K_Y + B_Y = g^*(K_{X'} + B')$. Moreover, $\mathrm{center}_X E = \{x\}$ and $a(E, X, B) = a > 0$, since E computes

a. Since $I_G(F) = I \subset J$ and g is J-equivariant, Lemma 3.1, applied with $H := J$ and $f := g$, implies that

$$\mathrm{mld}(Y/J \ni \bar{\eta}, B_{Y/J}) = \mathrm{mld}(X \ni x, B) = a,$$

and that, for the prime divisor P_0 over Y/J with $\mathrm{ord}_F|_{\mathbb{C}(Y/J)} \in \mathbb{N}^+ \mathrm{ord}_{P_0}$, we have $\mathrm{center}_{Y/J} P_0 = \overline{\{\bar{\eta}\}}$ and $a(P_0, Y/J, B_{Y/J}) = a$.

Let P be a prime divisor on Y/J, P_Y a prime divisor on Y with $h(P_Y) = P$, b the coefficient of P_Y in B_Y, and u the ramification index of h along P_Y. By Lemma 2.6(4), the coefficient of P in $B_{Y/J}$ is

$$1 - \frac{1-b}{u}.$$

Since $b \in \{0, \frac{1}{m}, \dots, \frac{m-1}{m}\}$ by Construction-Theorem 17.2(3), the coefficient of P in $B_{Y/J}$ belongs to Ψ_m. This proves (1).

Step 2. In this step, we show that, in the setting of (2), the numbers a_i are rational, $a(F_i, X_i', B_i') > 1$, and $\mathrm{center}_{Y_i} F_i$ is a closed point or a curve for $i \gg 0$.

Since (X_i, B_i) is klt near x_i, $\mathrm{mld}(X_i \ni x_i, B_i)$ is attained by a prime divisor E_i over X_i [Amb99, Section 1, the second paragraph after Definition 1.1]. Since B_i is a $\mathbb{Q}$-divisor and $K_{X_i} + B_i$ is $\mathbb{R}$-Cartier, $K_{X_i} + B_i$ is $\mathbb{Q}$-Cartier (being $\mathbb{R}$-Cartier is a system of linear equations with rational coefficients, which has a rational solution if it has a real one), so $a_i = a(E_i, X_i, B_i) \in \mathbb{Q}$. Since $m\Gamma \subset \mathbb{Z}$, mB_i is a Weil divisor, so (X_i, B_i), x_i, and m satisfy the hypotheses of Construction-Theorem 17.2. Let G_i, $g_i : Y_i \to X_i'$, and B_{Y_i} be the corresponding objects, and let $H_i := \mathrm{Stab}_{G_i}(F_i)$.

Write $a_i = \frac{p_i}{q_i}$ in lowest terms. For any positive integer N, the interval $[\epsilon, A_0]$ contains only finitely many rational numbers with denominator at most N, and the a_i are pairwise distinct. Thus $q_i > N$ for $i \gg 0$, that is, $\lim_{i \to +\infty} q_i = +\infty$. By Construction-Theorem 17.2(2),

$$a(F_i, X_i', B_i') = \frac{p_i}{\gcd(q_i, m)} = \frac{a_i q_i}{\gcd(q_i, m)} \geq \frac{\epsilon q_i}{m}.$$

Hence $a(F_i, X_i', B_i') > 1$ for $i \gg 0$, and Construction-Theorem 17.2(3) implies that $\mathrm{center}_{Y_i} F_i$ is a closed point or a curve for $i \gg 0$.

Step 3. In this step, we conclude the proof of (2).

Suppose that $C_i := \mathrm{center}_{Y_i} F_i$ is a curve for infinitely many i. Passing to a subsequence, which is again strictly increasing in $[\epsilon, A_0]$, we may assume that C_i is a curve for any i. Let J_i be the kernel of the action of H_i on the function field of C_i, let $(Y_i/J_i, B_{Y_i/J_i}) := (Y_i, B_{Y_i})/J_i$, and let $\bar{\eta}_i$ be the generic point of the image of C_i in Y_i/J_i. By (1), applied to (X_i, B_i), the coefficients of B_{Y_i/J_i} belong to Ψ_m and $\mathrm{mld}(Y_i/J_i \ni \bar{\eta}_i, B_{Y_i/J_i}) = a_i > 0$. Since $\Psi_m \subset [0, 1)$, we have $B_{Y_i/J_i} \geq 0$, so $(Y_i/J_i, B_{Y_i/J_i})$ is a threefold pair by Lemma 2.6(0). Since $\{0, \frac{1}{m}, \dots, \frac{m-1}{m}\}$ is finite, Ψ_m satisfies the DCC by Lemma 2.10(1). Thus Lemma 3.7, applied with $\Theta := \Psi_m$, implies that the set $\{a_i\}$ satisfies the ACC, which contradicts the assumption that (a_i) is strictly increasing. Hence $\mathrm{center}_{Y_i} F_i$ is a curve for only finitely many i, and by **Step 2**, $\mathrm{center}_{Y_i} F_i$ is a closed point for any $i \gg 0$. This proves (2). □

17.6. **Finite real coefficients.** In this subsection, we prove Construction-Theorem 17.6. It realizes a strictly increasing sequence of minimal log discrepancies of klt threefold pairs with coefficients in a finite set as minimal log discrepancies at closed points of quotients of terminal threefold pairs by cyclic groups.

Construction-Theorem 17.6. Let Γ, ϵ, A_0, and, for each i, (X_i, B_i), x_i, and a_i be as follows.

(i) $\Gamma \subset [0, 1]$ is a finite set.
(ii) $0 < \epsilon < A_0$.
(iii) For each i, (X_i, B_i) is a threefold pair klt near a closed point x_i with $B_i \in \Gamma$, and the numbers $a_i := \mathrm{mld}(X_i \ni x_i, B_i)$ form a strictly increasing sequence in $[\epsilon, A_0]$.

Let s, q, and $\lambda_1, \dots, \lambda_s$ be as in Construction-Theorem 5.2, and let μ_k, $\mathbf{R} = (r_{kj})$, L, and $\mathbf{A}$ be as in Construction-Proposition 17.1. Let $\mathcal{S} := \{\sum_j \frac{\lambda_j}{q} n_j \mid n_j \in \mathbb{Z}_{\geq 0}\}$, let $\Lambda_{\mathcal{S}}$, $\eta_{\mathcal{S}}$, $R_{\mathcal{S}}$, and $M_{\mathcal{S}}$ be as in Construction-Theorem 5.1, and let

$$\Xi_j := \left\{ 1 - \frac{n}{q} \;\middle|\; n \in \mathbb{Z},\ 0 \leq n \leq \lfloor q/\lambda_j \rfloor \right\}.$$

Then there exist prime divisors over X_i computing a_i, and, after deleting finitely many terms, for any prime divisors E_i computing a_i the following hold.

(1) There exist an affine neighborhood U_i of x_i and a small $\mathbb{Q}$-factorialization $f_i : X_i'' \to U_i$ such that the following hold for the strict transform B_i'' of B_i.
  (a) $(U_i, B_i|_{U_i})$ is klt.
  (b) $K_{X_i''} + B_i'' = f_i^*((K_{X_i} + B_i)|_{U_i})$.
  (c) $B_i'' \in \Gamma$.
  (d) (X_i'', B_i'') is klt.
  (e) $\operatorname{center}_{X_i''} E_i$ is a closed point x_i''.
  (f) $\operatorname{mld}(X_i'' \ni x_i'', B_i'') = a_i$.

  If X_i is $\mathbb{Q}$-factorial near x_i, then f_i can be taken to be the identity.

(2) Let $\rho_i : X_i' \to X_i''$ be the cover of Construction-Theorem 5.2 at x_i'', with group G_i and point o_i over x_i'', let $B_i''^{(j)}$ be the $\mathbb{Q}$-divisors $B^{(j)}$ of Construction-Theorem 5.2 for (X_i'', B_i'') at x_i'', and let $\chi_{i,j}$ be the characters of $\mathcal{O}_{X_i'}(q(K_{X_i'} + \rho_i^* B_i''^{(j)}))$ at o_i. Let F_i be a prime divisor over X_i' with $\operatorname{ord}_{F_i}|_{\mathbb{C}(X_i)} \in \mathbb{N}^+ \operatorname{ord}_{E_i}$, and let $I_i := I_{G_i}(F_i)$ and $e_i := |I_i|$. Then the following hold.
  (a) $\bigcap_j \operatorname{Ker} \chi_{i,j} = \{1\}$.
  (b) $a(F, X_i', \rho_i^* B_i'') \in \mathcal{S}$ for any prime divisor F over X_i'.
  (c) I_i is cyclic.
  (d) $\operatorname{ord}_{F_i}|_{\mathbb{C}(X_i)} = e_i \operatorname{ord}_{E_i}$.
  (e) $e_i \to +\infty$.
  (f) $a(F_i, X_i', \rho_i^* B_i'') = e_i a_i > 1$.

(3) There exists an I_i-equivariant projective birational morphism $g_i : Y_i \to X_i'$ such that the following hold for the $\mathbb{R}$-divisor B_{Y_i} defined by $K_{Y_i} + B_{Y_i} = g_i^*(K_{X_i'} + \rho_i^* B_i'')$.
  (a) $B_{Y_i} \geq 0$.
  (b) (Y_i, B_{Y_i}) is terminal.
  (c) $B_{Y_i} \in \Lambda_{\mathcal{S}}$.
  (d) Y_i is terminal.
  (e) The Cartier index of K_{Y_i} at any closed point is at most $\min\{R_{\mathcal{S}}, qL\}$.
  (f) $M_{\mathcal{S}} K_{Y_i}$ is Cartier.
  (g) $\operatorname{center}_{Y_i} F_i$ is a closed point y_i fixed by I_i.

(4) Let $(Z_i, \Delta_i) := (Y_i, B_{Y_i})/I_i$, and let z_i be the image of y_i. Then the following hold.
  (a) $\Delta_i \in \Psi(\Lambda_{\mathcal{S}})$.
  (b) (Z_i, Δ_i) is klt.
  (c) $\operatorname{mld}(Z_i \ni z_i, \Delta_i) = a_i$, attained by the prime divisor under F_i.

(5) Let $C_i''^{(k)} := \sum_j r_{kj} B_i''^{(j)}$, define $B_{Y_i}^{(j)}$ and $C_{Y_i}^{(k)}$ by $K_{Y_i} + B_{Y_i}^{(j)} = g_i^*(K_{X_i'} + \rho_i^* B_i''^{(j)})$ and $K_{Y_i} + C_{Y_i}^{(k)} = g_i^*(K_{X_i'} + \rho_i^* C_i''^{(k)})$, and let $(Z_i, \Delta_i^{(k)}) := (Y_i, C_{Y_i}^{(k)})/I_i$. Then the following hold.
  (a) $B_{Y_i} = \sum_j \lambda_j B_{Y_i}^{(j)}$.
  (b) $B_{Y_i}^{(j)} \in \Xi_j$.
  (c) The characters of $\mathcal{O}_{Y_i}(q(K_{Y_i} + B_{Y_i}^{(j)}))$ at y_i have trivial joint kernel on I_i.
  (d) $B_{Y_i} = \sum_k \mu_k C_{Y_i}^{(k)}$.
  (e) $C_{Y_i}^{(k)} \in \{0, \frac{1}{qL}, \dots, \frac{qL-1}{qL}\}$.

(f) $(Y_i, C_{Y_i}^{(k)})$ is terminal.
(g) $qL(K_{Y_i} + C_{Y_i}^{(k)})$ is Cartier.
(h) The joint kernel on I_i of the characters of $\mathcal{O}_{Y_i}(qL(K_{Y_i} + C_{Y_i}^{(k)}))$ at y_i has order dividing $|\det \mathbf{A}|$.
(i) For any prime divisor F over X_i' and any k, the signs of $a(F, X_i', \rho_i^* C_i''^{(k)}) - 1$ and $a(F, X_i', \rho_i^* B_i'') - 1$ are equal.
(j) $\Delta_i = \sum_k \mu_k \Delta_i^{(k)}$.
(k) $\Delta_i^{(k)} \geq 0$.
(l) $\Delta_i^{(k)} \in \Psi(\{0, \frac{1}{qL}, \dots, \frac{qL-1}{qL}\})$.

(6) Assume moreover that $\Gamma \subset \mathbb{Q}$, and let m be a positive integer with $m\Gamma \subset \mathbb{Z}$. The objects X_i', G_i, o_i, F_i, g_i, Y_i, B_{Y_i}, Z_i, Δ_i, y_i, and z_i in (6) are defined anew below and are independent of the identically named objects in (2)–(5). Let $\pi_i : X_i' \to X_i$ be the cover of Construction-Theorem 17.2 at x_i for m, with group G_i, point o_i over x_i, and $B_i' := \pi_i^* B_i$, let F_i be a prime divisor over X_i' with $\operatorname{ord}_{F_i}|_{\mathbb{C}(X_i)} \in \mathbb{N}^+ \operatorname{ord}_{E_i}$, let $H_i := \operatorname{Stab}_{G_i}(F_i)$, let $g_i : Y_i \to X_i'$ and B_{Y_i} be as in Construction-Theorem 17.2(3) for F_i and H_i, and let $(Z_i, \Delta_i) := (Y_i, B_{Y_i})/H_i$. Then, after deleting finitely many terms, the following hold.
(a) $a(F_i, X_i', B_i') > 1$.
(b) $B_{Y_i} \in \{0, \frac{1}{m}, \dots, \frac{m-1}{m}\}$.
(c) (Y_i, B_{Y_i}) is terminal.
(d) Y_i is terminal.
(e) The Cartier index of K_{Y_i} at any closed point is at most m.
(f) $\operatorname{lcm}(1, \dots, m) K_{Y_i}$ is Cartier.
(g) The character of $\mathcal{M}_{X_i'}$ is faithful on H_i.
(h) $\operatorname{center}_{Y_i} F_i$ is a closed point y_i fixed by H_i.
(i) $\Delta_i \in \Psi(\{0, \frac{1}{m}, \dots, \frac{m-1}{m}\})$.
(j) (Z_i, Δ_i) is klt.
(k) For the image z_i of y_i, $\operatorname{mld}(Z_i \ni z_i, \Delta_i) = a_i$, attained by the prime divisor under F_i.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we show that, for any DCC set $\Theta \subset [0, 1]$, only finitely many indices i have the following property: there exist a threefold pair (W, D) with $D \in \Theta$ and an integral curve $C \subset W$ with generic point ξ such that $\operatorname{mld}(W \ni \xi, D) = a_i$.

By Lemma 3.7, the set $\mathcal{T}$ of the numbers $\operatorname{mld}(W \ni \xi, D) > 0$, where (W, D) is a threefold pair with $D \in \Theta$ and ξ is the generic point of an integral curve on W, satisfies the ACC. Since (a_i) is a strictly increasing sequence of positive numbers, only finitely many a_i belong to $\mathcal{T}$.

Step 2. In this step we show that a_i is attained, and we reduce the assertions on arbitrary choices of prime divisors and of the morphisms g_i in (6) to fixed choices.

Fix i. Shrinking X_i near x_i changes neither the prime divisors over X_i with center x_i nor their log discrepancies, so we may assume that (X_i, B_i) is klt. By [KM98, Theorem 0.2], there exists a log resolution $h : W \to X_i$ of (X_i, B_i) such that $\mathfrak{m}_{x_i}\mathcal{O}_W$ is invertible and $\operatorname{Exc}(h) \cup \operatorname{Supp} h_*^{-1} B_i \cup h^{-1}(x_i)$ is an snc divisor. Write $h^{-1}(x_i) = P_1 \cup \cdots \cup P_n$. Since (X_i, B_i) is klt, $a_i = \min_l a(P_l, X_i, B_i)$ [Amb99, Section 1, the second paragraph after Definition 1.1], and a_i is attained by some P_l.

Any infinite subsequence of (a_i) is a strictly increasing sequence in $[\epsilon, A_0]$. Thus, to prove that a property of the prime divisors E_i, and of prime divisors F_i over them, holds for any choices after deleting finitely many terms, it suffices to prove that, for any fixed choices, the property holds for $i \gg 0$. If the property failed for some choices for infinitely many i, then fixing such choices along the corresponding subsequence would contradict the case of fixed choices. The choices of the morphisms g_i in (6) reduce to fixed choices by the argument of the previous two sentences.

Step 3. In this step we prove (1).

Choose an affine open neighborhood U_i of x_i such that $(U_i, B_i|_{U_i})$ is klt and, if X_i is $\mathbb{Q}$-factorial near x_i, such that U_i is $\mathbb{Q}$-factorial. The prime divisors over U_i with center x_i are the prime divisors over X_i with center x_i, and each of them has equal log discrepancies with respect to $(U_i, B_i|_{U_i})$ and (X_i, B_i). By [BCHM10, Corollary 1.4.3], there exists a small $\mathbb{Q}$-factorialization $f_i : X_i'' \to U_i$, and we take $f_i = \mathrm{id}_{U_i}$ if U_i is $\mathbb{Q}$-factorial. Let B_i'' be the strict transform of B_i. Then $B_i'' \in \Gamma$. Since X_i'' is $\mathbb{Q}$-factorial, $K_{X_i''} + B_i''$ is $\mathbb{R}$-Cartier, and the push-forward to U_i of the $\mathbb{R}$-divisor

$$f_i^*((K_{X_i} + B_i)|_{U_i}) - (K_{X_i''} + B_i'')$$

is zero. Thus the support of $f_i^*((K_{X_i} + B_i)|_{U_i}) - (K_{X_i''} + B_i'')$ consists of f_i-exceptional prime divisors, and there exist none because f_i is small. Hence $K_{X_i''} + B_i'' = f_i^*((K_{X_i} + B_i)|_{U_i})$, so $a(F, X_i'', B_i'') = a(F, U_i, B_i|_{U_i})$ for any prime divisor F over X_i'', and (X_i'', B_i'') is klt.

The center $\mathrm{center}_{X_i''} E_i$ is contained in $f_i^{-1}(x_i)$, which contains no prime divisor because f_i is small. Thus $\mathrm{center}_{X_i''} E_i$ is a closed point or a curve. Assume that $\mathrm{center}_{X_i''} E_i$ is a curve C_i with generic point ξ_i. Any prime divisor F over X_i'' with $\mathrm{center}_{X_i''} F = C_i$ satisfies $\mathrm{center}_{U_i} F = \{x_i\}$, hence $a(F, X_i'', B_i'') = a(F, X_i, B_i) \geq a_i$, with equality for $F = E_i$. Thus $\mathrm{mld}(X_i'' \ni \xi_i, B_i'') = a_i$. Since Γ is finite, it is a DCC set, and **Step 1** for $\Theta = \Gamma$ shows that $\mathrm{center}_{X_i''} E_i$ is a curve only for finitely many i, for any choices of E_i. Deleting finitely many terms, $\mathrm{center}_{X_i''} E_i$ is a closed point x_i''. Any prime divisor F over X_i'' with $\mathrm{center}_{X_i''} F = \{x_i''\}$ satisfies $\mathrm{center}_{U_i} F = \{x_i\}$, hence $a(F, X_i'', B_i'') \geq a_i$, with equality for $F = E_i$. Thus $\mathrm{mld}(X_i'' \ni x_i'', B_i'') = a_i$.

Step 4. In this step we prove (2).

By **Step 3**, (X_i'', B_i'') is a $\mathbb{Q}$-factorial klt threefold pair with $B_i'' \in \Gamma$, so the conditions of Construction-Theorem 5.2 hold at x_i''. By Construction-Theorem 5.2(1), ρ_i is a quasi-étale Galois morphism with group G_i, $\rho_i^{-1}(x_i'') = \{o_i\}$, $K_{X_i'} + \rho_i^* B_i'' = \rho_i^*(K_{X_i''} + B_i'')$, the pair $(X_i', \rho_i^* B_i'')$ is klt, $\bigcap_j \mathrm{Ker}\, \chi_{i,j} = \{1\}$, and $a(F, X_i', \rho_i^* B_i'') \in \mathcal{S}$ for any prime divisor F over X_i'. Since ρ_i is quasi-étale, $R(\rho_i) = 0$, and as $\rho_{i*}\rho_i^* B_i'' = |G_i| B_i''$, we have $(X_i', \rho_i^* B_i'')/G_i = (X_i'', B_i'')$. Since $\mathbb{C}(X_i'') = \mathbb{C}(X_i)$, **Step 3** shows that E_i is a prime divisor over X_i'' with $\mathrm{center}_{X_i''} E_i = \{x_i''\}$ and $a(E_i, X_i'', B_i'') = a_i$. By Lemma 2.6(1)(2), I_i is cyclic, $\mathrm{ord}_{F_i}|_{\mathbb{C}(X_i)} = e_i \,\mathrm{ord}_{E_i}$,

$$a(F_i, X_i', \rho_i^* B_i'') = e_i a(E_i, X_i'', B_i'') = e_i a_i,$$

and $\rho_i(\mathrm{center}_{X_i'} F_i) = \{x_i''\}$, so $\mathrm{center}_{X_i'} F_i = \{o_i\}$. By Construction-Theorem 5.2(3), applied to the strictly increasing sequence $a_i = \mathrm{mld}(X_i'' \ni x_i'', B_i'')$ in $[\epsilon, A_0]$, we have $e_i \to +\infty$ and $e_i a_i > 1$ for $i \gg 0$. By **Step 2**, after deleting finitely many terms, $e_i a_i > 1$ for any choices of E_i and F_i, and $e_i \to +\infty$ uniformly in the choices.

Step 5. In this step we prove (3), except for the bound qL on the Cartier index, and (4).

Let $\alpha_j := \lambda_j/q$ for $1 \leq j \leq s$. Then $\mathcal{S} = \{\sum_j \alpha_j n_j \mid n_j \in \mathbb{Z}_{\geq 0}\}$, and $1 \in \mathcal{S}$ by taking $n_j = q$ for any j, as $\sum_j \lambda_j = 1$. By **Step 4**, $(X_i', \rho_i^* B_i'')$ is klt near o_i, the group $I_i \subset G_i$ acts faithfully on X_i', fixing o_i and preserving $\rho_i^* B_i''$, and any log discrepancy of $(X_i', \rho_i^* B_i'')$ belongs to $\mathcal{S}$. By Construction-Theorem 5.1 with $H = I_i$, the set $\Lambda_{\mathcal{S}} \subset [0, 1)$ is finite, and, after shrinking X_i' to an I_i-invariant neighborhood of o_i, there exists an I_i-equivariant projective birational morphism $g_i : Y_i \to X_i'$ with $B_{Y_i} \geq 0$ such that (Y_i, B_{Y_i}) is terminal, $B_{Y_i} \in \Lambda_{\mathcal{S}}$, Y_i is terminal, K_{Y_i} is $\mathbb{Q}$-Cartier with Cartier index at most $R_{\mathcal{S}}$ at any closed point, and $M_{\mathcal{S}} K_{Y_i}$ is Cartier. Since ρ_i is finite and $\rho_i^{-1}(x_i'') = \{o_i\}$, any neighborhood of o_i contains $\rho_i^{-1}(V)$ for a neighborhood V of x_i''. Thus, possibly shrinking X_i'' near x_i'' and replacing X_i' and Y_i with the preimages, we may assume that $X_i' = \rho_i^{-1}(X_i'')$, so that $\rho_i : X_i' \to X_i''$ is a Galois morphism with group G_i. Since $a(F_i, X_i', \rho_i^* B_i'') > 1$ and $\mathrm{center}_{X_i'} F_i = \{o_i\}$ by **Step 4**, and since I_i stabilizes F_i, Construction-Theorem 5.1(5) implies that $C_i := \mathrm{center}_{Y_i} F_i$ is an I_i-invariant closed point or complete curve in $g_i^{-1}(o_i)$.

Since g_i is birational and I_i-equivariant, I_i acts faithfully on Y_i. Let ζ_i be the generic point of the image of C_i in Z_i. Since $1 \in \mathcal{S}$ (see the first paragraph of this step) and $\Lambda_{\mathcal{S}} = \{1 - s \mid s \in \mathcal{S},\ 0 < s \le 1\}$, we have $0 \in \Lambda_{\mathcal{S}}$. Hence $\operatorname{mult}_P B_{Y_i} \in \Lambda_{\mathcal{S}}$ for any prime divisor P on Y_i, whether or not P is a component of B_{Y_i}. As any prime divisor on Z_i is the image of a prime divisor on Y_i, Lemma 2.6(4) implies $\Delta_i \in \Psi(\Lambda_{\mathcal{S}})$. By Lemma 2.6(0), $\Delta_i \ge 0$ and $K_{Z_i} + \Delta_i$ is $\mathbb{R}$-Cartier. Let P be a prime divisor over Z_i. By Lemma 2.6(2)(3), there exists a prime divisor F' over Y_i with

$$a(P, Z_i, \Delta_i) = \frac{a(F', Y_i, B_{Y_i})}{|I_{I_i}(F')|} > 0,$$

where the inequality holds because (Y_i, B_{Y_i}) is terminal and $B_{Y_i} \in \Lambda_{\mathcal{S}} \subset [0, 1)$. Thus (Z_i, Δ_i) is klt. By **Step 3**, $\operatorname{mld}(X_i'' \ni x_i'', B_i'') = a(E_i, X_i'', B_i'') = a_i > 0$. Thus Lemma 3.1, with (X_i'', B_i''), x_i'', E_i, ρ_i, G_i, $\rho_i^* B_i''$, F_i, I_i, and g_i in place of (X, B), x, E, π, G, B', F, H, and f, implies

$$\operatorname{mld}(Z_i \ni \zeta_i, \Delta_i) = a_i,$$

attained by the prime divisor over Z_i under F_i.

Assume that C_i is a curve. Then the image of C_i in Z_i is a curve with generic point ζ_i. Since $\Lambda_{\mathcal{S}}$ is finite, $\Psi(\Lambda_{\mathcal{S}}) \subset [0, 1)$ is a DCC set by Lemma 2.10(1). Thus **Step 1** for $\Theta = \Psi(\Lambda_{\mathcal{S}})$ shows that C_i is a curve only for finitely many i, for any choices of E_i and F_i. Deleting finitely many terms, C_i is a closed point y_i, which is fixed by I_i because C_i is I_i-invariant, and $\zeta_i = z_i$. This proves (3), except for the bound qL, and (4).

Step 6. In this step we prove (5) and the bound qL in (3).

By Construction-Theorem 5.2, $B_i'' = \sum_j \lambda_j B_i''^{(j)}$, where the $B_i''^{(j)} \ge 0$ are $\mathbb{Q}$-divisors. The $\mathbb{Q}$-divisors $\rho_i^* B_i''^{(j)}$ are G_i-invariant, and by Construction-Theorem 5.2(1), after shrinking near x_i'', the pair $(X_i', \rho_i^* B_i''^{(j)})$ is lc and $q(K_{X_i'} + \rho_i^* B_i''^{(j)})$ is Cartier for any j. By the Moreover part of Construction-Theorem 5.1, with $H = I_i$ and $\rho_i^* B_i''^{(j)}$ in place of $B^{(j)}$, we have $B_{Y_i} = \sum_j \lambda_j B_{Y_i}^{(j)}$, $B_{Y_i}^{(j)} \in \Xi_j$, and the character of $\mathcal{O}_{Y_i}(q(K_{Y_i} + B_{Y_i}^{(j)}))$ at y_i is $\chi_{i,j}|_{I_i}$. By (2)(a), the joint kernel of the $\chi_{i,j}|_{I_i}$ is $I_i \cap \bigcap_j \operatorname{Ker} \chi_{i,j} = \{1\}$.

Since $\sum_j r_{kj} = 1$ and $\mathbf{A} = L\mathbf{R}$,

$$qL(K_{X_i'} + \rho_i^* C_i''^{(k)}) = \sum_j \mathbf{A}_{kj} q(K_{X_i'} + \rho_i^* B_i''^{(j)}),$$

which is Cartier. Thus $C_{Y_i}^{(k)}$ is defined, and $qL(K_{Y_i} + C_{Y_i}^{(k)}) = g_i^*(qL(K_{X_i'} + \rho_i^* C_i''^{(k)}))$ is Cartier. By Construction-Proposition 17.1(1), $B_i'' = \sum_k \mu_k C_i''^{(k)}$. Since $\sum_k \mu_k = 1$,

$$K_{Y_i} + B_{Y_i} = g_i^*\Big(\sum_k \mu_k (K_{X_i'} + \rho_i^* C_i''^{(k)})\Big) = K_{Y_i} + \sum_k \mu_k C_{Y_i}^{(k)},$$

so $B_{Y_i} = \sum_k \mu_k C_{Y_i}^{(k)}$. The $\mathbb{Q}$-divisor $\rho_i^* C_i''^{(k)}$ is the $\mathbb{Q}$-divisor $C'^{(k)}$ of Construction-Proposition 17.1 for (X_i'', B_i'') at x_i''. Since $B_{Y_i} \ge 0$ and (Y_i, B_{Y_i}) is terminal, Construction-Proposition 17.1(3), with $H = I_i$ and $f = g_i$, implies that $C_{Y_i}^{(k)} \ge 0$, $C_{Y_i}^{(k)} \in \{0, \frac{1}{qL}, \dots, \frac{qL-1}{qL}\}$, and $(Y_i, C_{Y_i}^{(k)})$ is terminal. Possibly shrinking near x_i'', Construction-Proposition 17.1(2) implies that the signs of $a(F, X_i', \rho_i^* C_i''^{(k)}) - 1$ and $a(F, X_i', \rho_i^* B_i'') - 1$ are equal for any prime divisor F over X_i' and any k. By Construction-Proposition 17.1(4), the character of $\mathcal{O}_{X_i'}(qL(K_{X_i'} + \rho_i^* C_i''^{(k)}))$ at o_i is $\nu_{i,k} := \prod_j \chi_{i,j}^{\mathbf{A}_{kj}}$, the character of $\mathcal{O}_{Y_i}(qL(K_{Y_i} + C_{Y_i}^{(k)}))$ at y_i is $\nu_{i,k}|_{I_i}$, and the joint kernel of the $\nu_{i,k}|_{I_i}$ has order dividing $|\det \mathbf{A}|$.

Let $h_i : Y_i \to Z_i$ be the quotient morphism. Since $\sum_k \mu_k = 1$ and $B_{Y_i} = \sum_k \mu_k C^{(k)}_{Y_i}$, Definition 2.5 implies

$$\Delta_i = \frac{1}{|I_i|} h_{i*}(B_{Y_i} + R(h_i)) = \sum_k \mu_k \frac{1}{|I_i|} h_{i*}(C^{(k)}_{Y_i} + R(h_i)) = \sum_k \mu_k \Delta^{(k)}_i.$$

Since $C^{(k)}_{Y_i} \geq 0$ and 0 belongs to $\{0, \frac{1}{qL}, \dots, \frac{qL-1}{qL}\}$, Lemma 2.6(0)(4) implies $\Delta^{(k)}_i \geq 0$ and $\Delta^{(k)}_i \in \Psi(\{0, \frac{1}{qL}, \dots, \frac{qL-1}{qL}\})$.

Let $y \in Y_i$ be a closed point and r the Cartier index of K_{Y_i} at y. Since K_{Y_i} is $\mathbb{Q}$-Cartier and $qL(K_{Y_i} + C^{(1)}_{Y_i})$ is Cartier, $C^{(1)}_{Y_i}$ is $\mathbb{Q}$-Cartier. Let P be a prime divisor over Y_i with $\operatorname{center}_{Y_i} P = \{y\}$. As $qL(K_{Y_i} + C^{(1)}_{Y_i})$ is Cartier, $qL \cdot a(P, Y_i, C^{(1)}_{Y_i}) \in \mathbb{Z}$. As $(Y_i, C^{(1)}_{Y_i})$ is terminal and P is exceptional over Y_i, $a(P, Y_i, C^{(1)}_{Y_i}) > 1$. Since $C^{(1)}_{Y_i} \geq 0$ is $\mathbb{Q}$-Cartier, [KM98, Lemma 2.27] implies

$$a(P, Y_i, 0) \geq a(P, Y_i, C^{(1)}_{Y_i}) \geq 1 + \frac{1}{qL}.$$

If $r > 1$, then by [Kaw93, Theorem] there exists a prime divisor P over Y_i with $\operatorname{center}_{Y_i} P = \{y\}$ and $a(P, Y_i, 0) = 1 + \frac{1}{r}$, so $r \leq qL$. Together with **Step 5**, $r \leq \min\{R_{\mathcal{S}}, qL\}$. This proves (3) and (5).

Step 7. In this step we prove (6). In this step the notation of (6) replaces the notation of (2)–(5).

Since mB_i is a Weil divisor and (X_i, B_i) is klt near x_i, the conditions of Construction-Theorem 17.2 hold for (X_i, B_i), x_i, and m. By Construction-Theorem 17.2(1), π_i is a Galois morphism with group G_i, $\pi_i^{-1}(x_i) = \{o_i\}$, $K_{X'_i} + B'_i = \pi_i^*(K_{X_i} + B_i)$, and the character of $\mathcal{M}_{X'_i}$ is faithful on G_i, hence on H_i. Since mB_i is a Weil divisor, B_i is a $\mathbb{Q}$-divisor, and $K_{X_i} + B_i$ is $\mathbb{Q}$-Cartier near x_i (being $\mathbb{R}$-Cartier is a system of linear equations with rational coefficients, which has a rational solution if it has a real one). Thus $a_i = a(E_i, X_i, B_i)$ is a positive rational number, and Construction-Theorem 17.2(2) implies $\operatorname{center}_{X'_i} F_i = \{o_i\}$. By Proposition 17.5(2) and **Step 2**, after deleting finitely many terms, $a(F_i, X'_i, B'_i) > 1$ for any choices of E_i and F_i.

By Construction-Theorem 17.2(3), $B_{Y_i} \in \{0, \frac{1}{m}, \dots, \frac{m-1}{m}\}$, the pair (Y_i, B_{Y_i}) is terminal, Y_i is terminal with Cartier index of K_{Y_i} at most m at any closed point, $\operatorname{lcm}(1, \dots, m)K_{Y_i}$ is Cartier, and, as $a(F_i, X'_i, B'_i) > 1$, $\operatorname{center}_{Y_i} F_i$ is an H_i-invariant closed point or complete curve in $g_i^{-1}(o_i)$. By Proposition 17.5(2) and **Step 2**, after deleting finitely many terms, $\operatorname{center}_{Y_i} F_i$ is a closed point y_i for any choices of E_i, F_i, and g_i, and y_i is fixed by H_i.

Since g_i is birational and H_i-equivariant, H_i acts faithfully on Y_i. Replacing I_i and $\Lambda_{\mathcal{S}}$ with H_i and $\{0, \frac{1}{m}, \dots, \frac{m-1}{m}\}$, the argument of **Step 5** shows that $\Delta_i \in \Psi(\{0, \frac{1}{m}, \dots, \frac{m-1}{m}\})$ and that (Z_i, Δ_i) is klt. Since $I_{G_i}(F_i) \subset \operatorname{Stab}_{G_i}(F_i) = H_i$ and $\operatorname{mld}(X_i \ni x_i, B_i) = a(E_i, X_i, B_i) = a_i > 0$, Lemma 3.1, with (X_i, B_i), x_i, E_i, π_i, G_i, B'_i, F_i, H_i, and g_i in place of (X, B), x, E, π, G, B', F, H, and f, implies

$$\operatorname{mld}(Z_i \ni z_i, \Delta_i) = a_i,$$

attained by the prime divisor over Z_i under F_i. □

18. Gauss valuations on split hypersurfaces

In this section, we study prime divisors computing minimal log discrepancies of finite quotients of split hypersurfaces $(xy + g(z, t) = 0) \subset \mathbb{A}^4$ and of complete intersections, by means of Gauss extensions and monomial valuations. The results of this section are not used in the proofs of the main theorems of this paper. Some of them are used in the sequels to this paper.

18.1. **Log discrepancies of Gauss extensions.** In this subsection, we compute log discrepancies and inertia groups of Gauss extensions, and we show that on a split hypersurface with a boundary pulled back from the plane, any prime divisor with quotient log discrepancy less than one is a Gauss extension of a plane valuation.

Lemma 18.1. *Let X be a normal variety, B an $\mathbb{R}$-divisor on X such that $K_X + B$ is $\mathbb{R}$-Cartier, E a prime divisor over X, and ϱ, c coprime positive integers. Then the Gauss extension of $\varrho \operatorname{ord}_E$ to $\mathbb{C}(X)(t) = \mathbb{C}(X \times \mathbb{A}^1_t)$ with t-value c (Definition-Proposition 2.11) is ord_F for a prime divisor F over $X \times \mathbb{A}^1_t$ with $\operatorname{center}_{X\times\mathbb{A}^1_t} F = \operatorname{center}_X E \times \{0\}$, and*

$$a(F, X \times \mathbb{A}^1_t, \operatorname{pr}_X^* B) = \varrho a(E, X, B) + c.$$

Moreover, for any $u \in \mathbb{C}(X)$ with $\operatorname{ord}_E u = 1$, the residue R of t^ϱ/u^c is transcendental over the residue field $\kappa(E)$ of ord_E, and the residue field of ord_F is $\kappa(E)(R)$.

Proof. Let v be the Gauss extension of $\varrho \operatorname{ord}_E$ to $\mathbb{C}(X)(t)$ with t-value c. Let $f : Y \to X$ be a proper birational morphism from a normal variety Y such that E is a prime divisor on Y, and let B_Y be the $\mathbb{R}$-divisor on Y such that $K_Y + B_Y = f^*(K_X + B)$ and $f_* B_Y = B$. Then $\operatorname{mult}_E B_Y = 1 - a(E, X, B)$. Let η be the generic point of E, τ a uniformizer of $\mathcal{O}_{Y,\eta}$, and ζ the generic point of $E \times \{0\} \subset Y \times \mathbb{A}^1_t$. Then τ, t are regular parameters of $\mathcal{O}_{Y\times\mathbb{A}^1,\zeta}$, and

$$\widehat{\mathcal{O}}_{Y\times\mathbb{A}^1,\zeta} = \widehat{\mathcal{O}}_{Y,\eta}[[t]] \cong \kappa(E)[[\tau, t]],$$

where the expansion of any non-zero $u \in \mathcal{O}_{Y,\eta}$ in $\kappa(E)[[\tau]]$ has least τ-exponent $\operatorname{ord}_E u$. Let v_0 be the monomial valuation at ζ in τ, t with weights (ϱ, c) [JM12, Proposition 3.1] (on a regular open neighborhood of ζ). For $P = \sum_i u_i t^i$ with $u_i \in \mathcal{O}_{Y,\eta}$, the monomials of the expansion of $u_i t^i$ have t-exponent i, so

$$v_0(P) = \min_i (\varrho \operatorname{ord}_E u_i + ic) = v(P).$$

Thus v and v_0 coincide on $\mathcal{O}_{Y,\eta}[t]$, hence on the fraction field $\mathbb{C}(X)(t)$ of $\mathcal{O}_{Y,\eta}[t]$.

Since $\operatorname{ord}_E(\mathbb{C}(X)^*) = \mathbb{Z}$ and $\gcd(\varrho, c) = 1$, $v(\mathbb{C}(X)(t)^*) = \mathbb{Z}$ (Definition-Proposition 2.11). Let $\kappa(v)$ be the residue field of v. Since $v|_{\mathbb{C}(X)} = \varrho \operatorname{ord}_E$, $\kappa(E)$ is a subfield of $\kappa(v)$. Since $v(t^\varrho/\tau^c) = \varrho c - \varrho c = 0$, the residue $R \in \kappa(v)$ of t^ϱ/τ^c is defined. Let $0 \neq P = \sum_i u_i t^i$ with $u_i \in \mathbb{C}(X)$, let S be the set of indices i with $u_i \neq 0$ and $\varrho \operatorname{ord}_E u_i + ic = v(P)$, and let $i_0 := \min S$. For $i \in S$,

$$\varrho(\operatorname{ord}_E u_i - \operatorname{ord}_E u_{i_0}) = c(i_0 - i),$$

so $\gcd(\varrho, c) = 1$ implies that $i = i_0 + \varrho e$ for an integer $e \geq 0$ and $\operatorname{ord}_E u_i = \operatorname{ord}_E u_{i_0} - ce$. Thus

$$\frac{P}{u_{i_0} t^{i_0}} = \sum_{e\geq 0,\ i_0+\varrho e\in S} b_e \Big(\frac{t^\varrho}{\tau^c}\Big)^e + \sum_{i\notin S} \frac{u_i t^i}{u_{i_0} t^{i_0}}, \qquad b_e := \frac{u_{i_0+\varrho e}\tau^{ce}}{u_{i_0}},$$

where each b_e is a unit of $\mathcal{O}_{Y,\eta}$, and the last sum has positive v-value by the formula of Definition-Proposition 2.11. Hence the residue of $P/(u_{i_0} t^{i_0})$ is $\sum_e \bar{b}_e R^e$, a polynomial in R whose coefficients $\bar{b}_e \in \kappa(E)$ are all non-zero. Every element of $\mathbb{C}(X)(t)$ with v-value 0 is P/Q for some non-zero $P, Q \in \mathbb{C}(X)[t]$ with $v(P) = v(Q)$. Write $Q = \sum_j u'_j t^j$ with $u'_j \in \mathbb{C}(X)$, and let j_0 be the least index j with $u'_j \neq 0$ and $\varrho \operatorname{ord}_E u'_j + jc = v(Q)$. Since $v(u_{i_0} t^{i_0}) = v(P) = v(Q) = v(u'_{j_0} t^{j_0})$, we have $\varrho(\operatorname{ord}_E u_{i_0} - \operatorname{ord}_E u'_{j_0}) = c(j_0 - i_0)$, so $\gcd(\varrho, c) = 1$ implies that $i_0 - j_0 = \varrho e$ and $\operatorname{ord}_E u_{i_0} - \operatorname{ord}_E u'_{j_0} = -ce$ for an integer e. Thus

$$\frac{u_{i_0} t^{i_0}}{u'_{j_0} t^{j_0}} = b' \Big(\frac{t^\varrho}{\tau^c}\Big)^e, \qquad b' := \frac{u_{i_0}\tau^{ce}}{u'_{j_0}},$$

where b' is a unit of $\mathcal{O}_{Y,\eta}$. Since P/Q is the product of $P/(u_{i_0} t^{i_0})$, $u'_{j_0} t^{j_0}/Q$, and $b'(t^\varrho/\tau^c)^e$, the computation of residues above, applied to P and to Q, shows that the residue of P/Q belongs to $\kappa(E)(R)$. Since $\kappa(E) \subset \kappa(v)$ and $R \in \kappa(v)$, we obtain $\kappa(v) = \kappa(E)(R)$.

We show that R is transcendental over $\kappa(E)$. Let $b_0, \dots, b_N \in \mathcal{O}_{Y,\eta}$ be elements whose residues $\bar{b}_0, \dots, \bar{b}_N \in \kappa(E)$ are not all zero. By the formula of Definition-Proposition 2.11,

$$v\Big(\tau^{-cN} \sum_{e=0}^{N} b_e \tau^{c(N-e)} t^{\varrho e}\Big) = -\varrho cN + \min_e \big(\varrho \operatorname{ord}_E b_e + \varrho c(N-e) + \varrho ce\big) = 0,$$

and the element on the left is $\sum_e b_e (t^\varrho/\tau^c)^e$. Thus its residue $\sum_e \bar{b}_e R^e$ is non-zero, and R is transcendental over $\kappa(E)$. If $u \in \mathbb{C}(X)$ satisfies $\operatorname{ord}_E u = 1$, then u/τ is a unit of $\mathcal{O}_{Y,\eta}$ with residue $\bar{w} \in \kappa(E)^*$, so the residue of t^ϱ/u^c is $\bar{w}^{-c}R$, which is transcendental over $\kappa(E)$ and satisfies $\kappa(E)(\bar{w}^{-c}R) = \kappa(E)(R)$.

Since $\kappa(E)$ has transcendence degree $\dim X - 1$ over $\mathbb{C}$, the residue field $\kappa(v) = \kappa(E)(R)$ has transcendence degree $\dim X = \dim(X \times \mathbb{A}^1) - 1$. Thus $v = \operatorname{ord}_F$ for a prime divisor F over $X \times \mathbb{A}^1_t$ [KM98, Lemma 2.45]. Since v_0 is non-negative on $\mathcal{O}_{Y\times\mathbb{A}^1,\zeta}$ and positive on τ and t, $\operatorname{center}_{Y\times\mathbb{A}^1} F = E \times \{0\}$, and $\operatorname{center}_{X\times\mathbb{A}^1_t} F = f(E) \times \{0\} = \operatorname{center}_X E \times \{0\}$.

Let $U \subset Y \times \mathbb{A}^1$ be an open neighborhood of ζ such that U is smooth, τ is regular on U, $\operatorname{div}(\tau) + \operatorname{div}(t)$ is an snc divisor on U, and $(\operatorname{pr}_Y^* B_Y)|_U = (1 - a(E,X,B))\operatorname{div}(\tau)$. A direct calculation on the weighted blow-up of U along the closure of ζ with weights (ϱ, c) in τ, t shows that $a(F, U, 0) = \varrho + c$ (see also [JM12, Proposition 5.1 and (5.1)]). Thus

$$a(F, U, (1 - a(E,X,B))\operatorname{div}(\tau)) = \varrho + c - \varrho(1 - a(E,X,B)) = \varrho a(E,X,B) + c.$$

Since

$$K_{Y\times\mathbb{A}^1} + \operatorname{pr}_Y^* B_Y = (f \times \mathrm{id}_{\mathbb{A}^1})^*(K_{X\times\mathbb{A}^1} + \operatorname{pr}_X^* B),$$

the left-hand side is $a(F, X \times \mathbb{A}^1_t, \operatorname{pr}_X^* B)$ [KM98, Lemma 2.30]. □

The next lemma collects the parts of the analysis of a split hypersurface that do not involve a boundary. It is used for Proposition 18.3 and in Subsection 18.5.

Lemma 18.2. *Let $T := \mathbb{A}^2_{z,t}$, $0 \neq g \in (z,t)^2$, and $X := (xy + g(z,t) = 0) \subset \mathbb{A}^4_{x,y,z,t}$, normal near 0. Let G be a finite diagonal abelian group preserving X and acting faithfully on X, and β the character of x. For a prime divisor ν over T with center 0, let $J_\nu := I_G(\nu)$ for the action of G on T, which need not be faithful, and let α_ν be the character of J_ν on $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$. Then the following hold.*

(1) Let ν be a prime divisor over T with center 0, and ϱ, k coprime positive integers with $k < \varrho \operatorname{ord}_\nu g$. Then the Gauss extension of $\varrho\operatorname{ord}_\nu$ to $\mathbb{C}(X) = \mathbb{C}(T)(x)$ with x-value k (Definition-Proposition 2.11) is ord_F for a prime divisor F over X with $\operatorname{center}_X F = \{0\}$, and

$$\operatorname{ord}_F y = \varrho \operatorname{ord}_\nu g - k, \qquad a(F,X,0) = \varrho a(\nu, T, 0), \qquad I_G(F) = \operatorname{Ker}(\beta^\varrho \alpha_\nu^{-k} : J_\nu \to \mathbb{C}^*).$$

(2) Let F be a prime divisor over X with $\operatorname{center}_X F = \{0\}$. Then $\operatorname{ord}_F|_{\mathbb{C}(T)} = \varrho\operatorname{ord}_\nu$ for a positive integer ϱ and a prime divisor ν over T with center 0, $I_G(F) \subset J_\nu$, and $0 < \operatorname{ord}_F x < \varrho \operatorname{ord}_\nu g$. Moreover, $d_F := a(F,X,0) - \varrho a(\nu,T,0)$ is a non-negative integer, and $d_F = 0$ if and only if ord_F is the Gauss extension of $\varrho\operatorname{ord}_\nu$ to $\mathbb{C}(X) = \mathbb{C}(T)(x)$ with x-value $\operatorname{ord}_F x$. In this case, ϱ and $\operatorname{ord}_F x$ are coprime.

Proof. Since $y = -g/x$ on X, $\mathbb{C}(X) = \mathbb{C}(T)(x) = \mathbb{C}(\mathbb{A}^3_{z,t,x})$, and x is transcendental over $\mathbb{C}(T)$. Since X is a hypersurface, ω_X is generated near 0 by the residue form $\Omega := \frac{dx}{x} \wedge dz \wedge dt$ by adjunction, and $\omega_{\mathbb{A}^3_{z,t,x}}$ is generated by $dx \wedge dz \wedge dt = x\Omega$. Let F be a prime divisor over X with $\operatorname{center}_X F = \{0\}$. Then $\operatorname{ord}_F z > 0$, $\operatorname{ord}_F t > 0$, and $\operatorname{ord}_F x > 0$, so the center of F on $\mathbb{A}^3_{z,t,x}$ is 0. The orders of Ω and $x\Omega$ along F are $a(F,X,0) - 1$ and $a(F, \mathbb{A}^3_{z,t,x}, 0) - 1$, so

$$a(F,X,0) = a(F, \mathbb{A}^3_{z,t,x}, 0) - \operatorname{ord}_F x. \tag{18.1}$$

We first show the following: $(*)$ if F is a prime divisor over X with center 0 such that $\operatorname{ord}_F|_{\mathbb{C}(T)} = \varrho\operatorname{ord}_\nu$ for a positive integer ϱ and a prime divisor ν over T with center 0, then

$I_G(F) \subset J_\nu$. Let $\sigma \in I_G(F)$. Since G acts diagonally, σ^* preserves $\mathbb{C}(T) = \mathbb{C}(z,t)$, so σ^* preserves $\operatorname{ord}_F|_{\mathbb{C}(T)} = \varrho \operatorname{ord}_\nu$, and σ acts trivially on the residue field $\kappa(\nu)$ of ord_ν, which is a subfield of the residue field $\kappa(F)$ of ord_F. Thus $\sigma \in J_\nu$. Here the elements of G acting trivially on T belong to J_ν: they act trivially on $\mathbb{C}(T)$, so they fix ord_ν and act trivially on $\kappa(\nu)$.

(1) By Lemma 18.1, applied to $(T,0)$, the prime divisor ν, and the coordinate x of $T \times \mathbb{A}^1_x = \mathbb{A}^3_{z,t,x}$ with x-value k, the Gauss extension is ord_F for a prime divisor F over $\mathbb{A}^3_{z,t,x}$ with

$$a(F, \mathbb{A}^3_{z,t,x}, 0) = \varrho a(\nu, T, 0) + k.$$

Since $\mathbb{C}(X) = \mathbb{C}(\mathbb{A}^3_{z,t,x})$, F is a prime divisor over X. We have $\operatorname{ord}_F x = k > 0$, $\operatorname{ord}_F z = \varrho \operatorname{ord}_\nu z > 0$, $\operatorname{ord}_F t = \varrho \operatorname{ord}_\nu t > 0$, and

$$\operatorname{ord}_F y = \operatorname{ord}_F g - \operatorname{ord}_F x = \varrho \operatorname{ord}_\nu g - k > 0.$$

Since x, y, z, t generate the coordinate ring of X, $\operatorname{center}_X F = \{0\}$. By (18.1),

$$a(F, X, 0) = \varrho a(\nu, T, 0).$$

Let $\tau \in \mathbb{C}(T)$ with $\operatorname{ord}_\nu \tau = 1$. By Lemma 18.1, applied to $(T,0)$, the prime divisor ν, and the coordinate x with $u := \tau$, we have $\kappa(F) = \kappa(\nu)(R)$, where R is the residue of x^ϱ/τ^k. Let $\sigma \in J_\nu$. Since σ^* preserves ord_ν on $\mathbb{C}(T)$ and $\sigma^* x = \beta(\sigma) x$, the formula of Definition-Proposition 2.11 implies that σ^* preserves ord_F. Since $\sigma^*\tau/\tau$ has residue $\alpha_\nu(\sigma)$ in $\kappa(\nu)$, $\sigma^*(x^\varrho/\tau^k)$ has residue $\beta(\sigma)^\varrho \alpha_\nu(\sigma)^{-k} R$. Since σ acts trivially on $\kappa(\nu)$, $\sigma \in I_G(F)$ if and only if $\beta^\varrho \alpha_\nu^{-k}(\sigma) = 1$. By $(*)$, $I_G(F) \subset J_\nu$. Therefore $I_G(F) = \operatorname{Ker}(\beta^\varrho \alpha_\nu^{-k} : J_\nu \to \mathbb{C}^*)$.

(2) Since z, t vanish at 0 and ord_F is non-negative on $\mathcal{O}_{X,0}$, the restriction $\operatorname{ord}_F|_{\mathbb{C}(T)}$ is non-negative on $\mathcal{O}_{T,0}$, positive on the maximal ideal of $\mathcal{O}_{T,0}$, and non-trivial. By Lemma 2.12, applied to $\mathbb{C}(T) \subset \mathbb{C}(X)$ and the point $0 \in T$, $\operatorname{ord}_F|_{\mathbb{C}(T)} = \varrho \operatorname{ord}_\nu$ for a positive integer ϱ and a prime divisor ν over T with $\operatorname{center}_T \nu = \{0\}$. By $(*)$, $I_G(F) \subset J_\nu$.

Since x and y vanish at 0, $\operatorname{ord}_F x > 0$ and $\operatorname{ord}_F y > 0$. Since

$$\operatorname{ord}_F x + \operatorname{ord}_F y = \operatorname{ord}_F(-g) = \varrho \operatorname{ord}_\nu g,$$

we have $0 < \operatorname{ord}_F x < \varrho \operatorname{ord}_\nu g$. Let $k := \operatorname{ord}_F x$.

Let $h : T' \to T$ be a proper birational morphism from a smooth surface T' such that ν is a prime divisor E_ν on T'. Let η be the generic point of E_ν, τ a uniformizer of $\mathcal{O}_{T',\eta}$, and ζ the generic point of $E_\nu \times \{0\} \subset T' \times \mathbb{A}^1_x$. Since $\operatorname{ord}_F|_{\mathbb{C}(T)} = \varrho \operatorname{ord}_\nu$ and $k > 0$, the generic point of the center of F on $T' \times \mathbb{A}^1_x$ maps to η, and x vanishes at it, so $\operatorname{center}_{T' \times \mathbb{A}^1_x} F = E_\nu \times \{0\}$. Since E_ν is the only h-exceptional prime divisor on T' through η, near ζ we have

$$K_{T' \times \mathbb{A}^1} = (h \times \mathrm{id}_{\mathbb{A}^1})^* K_{\mathbb{A}^3_{z,t,x}} + (a(\nu, T, 0) - 1)(E_\nu \times \mathbb{A}^1). \tag{18.2}$$

Let $U \subset T' \times \mathbb{A}^1_x$ be an open neighborhood of ζ such that U is smooth and connected, τ is regular on U, $\operatorname{div}(\tau) = (E_\nu \times \mathbb{A}^1) \cap U$, $D_U := \operatorname{div}(\tau) + \operatorname{div}(x)$ is an snc divisor on U, and (18.2) holds on U. By [KM98, Lemma 2.30] and Lemma 2.13, applied with the regular parameters τ, x of $\mathcal{O}_{T' \times \mathbb{A}^1, \zeta}$,

$$a(F, \mathbb{A}^3_{z,t,x}, 0) = a\big(F, U, (1 - a(\nu, T, 0)) \operatorname{div}(\tau)\big) \geq \varrho a(\nu, T, 0) + k.$$

By (18.1), $d_F \geq 0$. Since X is Gorenstein near 0, T is smooth, and $\varrho \in \mathbb{Z}$, d_F is an integer.

Assume that $d_F = 0$. Since $\operatorname{ord}_F \tau = \varrho$, we have $a\big(F, U, (1 - a(\nu, T, 0)) \operatorname{div}(\tau)\big) = a(F, U, 0) - \varrho(1 - a(\nu, T, 0))$, so by (18.1),

$$0 = d_F = a(F, U, 0) - \varrho(1 - a(\nu, T, 0)) - k - \varrho a(\nu, T, 0) = a(F, U, 0) - \varrho - k.$$

Let v_0 be the monomial valuation at ζ in τ, x with weights (ϱ, k) [JM12, Proposition 3.1], which is the image of ord_F under the retraction r_{U,D_U} onto the set $\mathrm{QM}(U, D_U)$ of quasi-monomial valuations of (U, D_U) [JM12, §4.3]. Since U is regular and connected, D_U is a reduced snc divisor, and ord_F has center $E_\nu \times \{0\}$ on U, with generic point ζ, the setting of [JM12, §4.3] holds with $X = Y = U$, and by definition, $r_{U,D_U}(\operatorname{ord}_F)$ is the valuation in $\mathrm{QM}(U, D_U)$ with values $\operatorname{ord}_F \tau = \varrho$ and $\operatorname{ord}_F x = k$ on $\operatorname{div}(\tau)$ and $\operatorname{div}(x)$, namely v_0. By [JM12, Proposition 5.1 and

Corollary 5.4], applied to the log smooth pair (U, D_U), the log discrepancy $a(F, U, 0)$ is at least the log discrepancy $\varrho + k$ of v_0, with equality if and only if $\operatorname{ord}_F = v_0$. Thus $\operatorname{ord}_F = v_0$. Since

$$\widehat{\mathcal{O}}_{U,\zeta} = \widehat{\mathcal{O}}_{T',\eta}[[x]] \quad \text{and} \quad \widehat{\mathcal{O}}_{T',\eta} \cong \kappa(\nu)[[\tau]],$$

we have $v_0(\sum_i u_i x^i) = \min_i(\varrho \operatorname{ord}_\nu u_i + ik)$ for $u_i \in \mathcal{O}_{T',\eta}$. Thus ord_F coincides with the Gauss extension of $\varrho \operatorname{ord}_\nu$ with x-value k on $\mathcal{O}_{T',\eta}[x]$, hence on $\mathbb{C}(T)(x) = \mathbb{C}(X)$. Since $\operatorname{ord}_F(\mathbb{C}(X)^*) = \mathbb{Z}$ and, by the formula of Definition-Proposition 2.11, $\operatorname{ord}_F(\mathbb{C}(X)^*) \subset \varrho\mathbb{Z}+k\mathbb{Z}$, we have $\gcd(\varrho, k) = 1$.

Conversely, assume that ord_F is the Gauss extension of $\varrho \operatorname{ord}_\nu$ with x-value k. Then ϱ and k are coprime, since $\operatorname{ord}_F(\mathbb{C}(X)^*) = \mathbb{Z}$ and $\operatorname{ord}_F(\mathbb{C}(X)^*) \subset \varrho\mathbb{Z} + k\mathbb{Z}$ by the formula of Definition-Proposition 2.11. Since $0 < k < \varrho \operatorname{ord}_\nu g$, (1) implies $a(F, X, 0) = \varrho a(\nu, T, 0)$, that is, $d_F = 0$. □

Proposition 18.3. *Let $T := \mathbb{A}^2_{z,t}$, $0 \neq g \in (z,t)^2$, and $X := (xy + g(z,t) = 0) \subset \mathbb{A}^4_{x,y,z,t}$ a normal klt threefold germ at 0. Let G be a finite diagonal abelian group preserving X and acting faithfully on X. Let $h_1, \dots, h_s$ be non-zero G-semi-invariant polynomials in z, t, $c_1, \dots, c_s \geq 0$ real numbers, $B_T := \sum_j c_j \operatorname{div}_T(h_j)$, and $B := \sum_j c_j \operatorname{div}_X(h_j)$. Assume that (X, B) is klt and $a := \operatorname{mld}((X \ni 0, B)/G) \in (0, 1)$. For a prime divisor ν over T with center 0, let $l_\nu := \operatorname{ord}_\nu g$, $b_\nu := a(\nu, T, B_T)$, $J_\nu := I_G(\nu)$ for the action of G on T, which need not be faithful, α_ν the character of J_ν on $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$, and β the character of x. For coprime positive integers ϱ, k with $k < \varrho l_\nu$, let*

$$e(\nu; \varrho, k) := |\operatorname{Ker}(\beta^\varrho \alpha_\nu^{-k} : J_\nu \to \mathbb{C}^*)|.$$

Then $b_\nu > 0$ and $l_\nu \geq 2$ for any ν, and

$$a = \min_{\nu,\varrho,k} \frac{\varrho b_\nu}{e(\nu; \varrho, k)},$$

the minimum being attained. Each term is $a(F, X, B)/|I_G(F)|$ for the prime divisor F over X given by the Gauss extension of $\varrho \operatorname{ord}_\nu$ to $\mathbb{C}(X) = \mathbb{C}(T)(x)$ with x-value k, for which $\operatorname{ord}_F y = \varrho l_\nu - k$ and $I_G(F) = \operatorname{Ker}(\beta^\varrho \alpha_\nu^{-k})$. Moreover, any prime divisor F over X with center 0 and $a(F, X, B)/|I_G(F)| < 1$ is given by such a Gauss extension. If (X, B) is terminal, then $b_\nu > 1$ for any ν.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we show that $l_\nu \geq 2$ and $b_\nu > 0$, that $b_\nu > 1$ if (X, B) is terminal, and that each term of the minimum is $a(F, X, B)/|I_G(F)|$ for the Gauss extension F.

Let ν be a prime divisor over T with center 0. Since $\operatorname{ord}_\nu z \geq 1$, $\operatorname{ord}_\nu t \geq 1$, and $g \in (z,t)^2$, $l_\nu = \operatorname{ord}_\nu g \geq 2$. Let F be a prime divisor over X with center 0 such that $\operatorname{ord}_F|_{\mathbb{C}(T)} = \varrho \operatorname{ord}_\nu$ for a positive integer ϱ. Since $h_j \in \mathbb{C}(T)$,

$$a(F, X, B) = a(F, X, 0) - \varrho \sum_j c_j \operatorname{ord}_\nu h_j \quad \text{and} \quad b_\nu = a(\nu, T, 0) - \sum_j c_j \operatorname{ord}_\nu h_j. \tag{18.3}$$

Let ϱ, k be coprime positive integers with $k < \varrho l_\nu$, and let F be the prime divisor over X given by the Gauss extension of $\varrho \operatorname{ord}_\nu$ with x-value k. By Lemma 18.2(1), $\operatorname{center}_X F = \{0\}$, $\operatorname{ord}_F y = \varrho l_\nu - k$, $I_G(F) = \operatorname{Ker}(\beta^\varrho \alpha_\nu^{-k})$, and $a(F, X, 0) = \varrho a(\nu, T, 0)$. By (18.3), $a(F, X, B) = \varrho b_\nu$, so

$$\frac{\varrho b_\nu}{e(\nu; \varrho, k)} = \frac{a(F, X, B)}{|I_G(F)|}.$$

Since $l_\nu \geq 2$, we may take $\varrho = k = 1$, and then $a(F, X, B) = b_\nu$. Since (X, B) is klt, $b_\nu > 0$. If (X, B) is terminal, then $b_\nu > 1$, because F is exceptional over X, as $\operatorname{center}_X F$ is a closed point of the threefold X.

Step 2. In this step, we show that for any prime divisor F over X with center 0, writing $\operatorname{ord}_F|_{\mathbb{C}(T)} = \varrho \operatorname{ord}_\nu$ with ϱ a positive integer and ν a prime divisor over T with center 0 (Lemma 18.2(2)), the integer d_F of Lemma 18.2(2) satisfies $a(F, X, B) = \varrho b_\nu + d_F$ and is divisible by $|I_G(F)|$.

By Lemma 18.2(2), d_F is a non-negative integer, and by (18.3), $a(F,X,B) = \varrho b_\nu + d_F$. Let $I := I_G(F)$ and λ the character of I on $\mathfrak{m}_F/\mathfrak{m}_F^2$. By Lemma 2.6(1), I is cyclic and λ is faithful. Let β_y, χ_z, and χ_t be the characters of y, z, and t. For $\sigma \in G$, $\sigma^*(xy+g)$ is a polynomial whose monomials are exactly the monomials of $xy+g$ and vanishes on X, so it is a constant multiple of $xy+g$. Since xy occurs in $xy+g$, the polynomial $xy+g$ is G-semi-invariant with character $\beta\beta_y$. By Lemma 9.2(2), applied to $\mathbb{A}^4$ with the G-eigenparameters x,y,z,t at 0 and to $xy+g$,

$$(\chi_z\chi_t)|_I = \lambda^{a(F,X,0)}. \tag{18.4}$$

By Lemma 18.2(2), $I \subset J_\nu$. Let $\sigma \in I$, and let $\kappa(\nu)$ and $\kappa(F)$ be the residue fields of ord_ν and ord_F. Let $\mathcal{O}_\nu \subset \mathbb{C}(T)$ be the valuation ring of ord_ν, τ a uniformizer, and $s \in \mathcal{O}_\nu$ an element whose residue is transcendental over $\mathbb{C}$. Since $\mathcal{O}_\nu$ is the local ring of a smooth surface at the generic point of a prime divisor, $d\tau \wedge ds$ generates the free $\mathcal{O}_\nu$-module $\Omega^2_{\mathcal{O}_\nu/\mathbb{C}}$, and

$$dz \wedge dt = u\tau^{a(\nu,T,0)-1} d\tau \wedge ds$$

for a unit $u \in \mathcal{O}_\nu^*$. Write $\sigma^*\tau = \tau w$ with $w \in \mathcal{O}_\nu^*$. Then w has residue $\alpha_\nu(\sigma)$. Since σ acts trivially on $\kappa(\nu)$, $\sigma^*s - s \in \tau\mathcal{O}_\nu$ and $\sigma^*u - u \in \tau\mathcal{O}_\nu$. Write $\sigma^*s = s + \tau s_1$ with $s_1 \in \mathcal{O}_\nu$. Then

$$\sigma^*(d\tau\wedge ds) = (wd\tau+\tau dw)\wedge(ds+s_1d\tau+\tau ds_1) = wd\tau\wedge ds+\tau\big(wd\tau\wedge ds_1+dw\wedge ds+s_1dw\wedge d\tau+\tau dw\wedge ds_1\big),$$

and since $d\tau, ds$ form a basis of $\Omega^1_{\mathcal{O}_\nu/\mathbb{C}}$, we have $dw, ds_1 \in \mathcal{O}_\nu d\tau + \mathcal{O}_\nu ds$, so the bracket belongs to $\mathcal{O}_\nu d\tau \wedge ds$. Hence $\sigma^*(d\tau \wedge ds) \in (w + \tau\mathcal{O}_\nu) d\tau \wedge ds$. Comparing residues of the coefficients in $\sigma^*(dz \wedge dt) = \chi_z(\sigma)\chi_t(\sigma) dz \wedge dt$, we obtain $\chi_z(\sigma)\chi_t(\sigma) = \alpha_\nu(\sigma)^{a(\nu,T,0)}$. On the other hand, write $\tau = \pi^\varrho u'$, where π is a uniformizer of ord_F and u' is a unit of the valuation ring of ord_F. Since σ acts trivially on $\kappa(F)$, the residue of $\sigma^*\tau/\tau = (\sigma^*\pi/\pi)^\varrho \cdot \sigma^*u'/u'$ in $\kappa(F)$ is $\lambda(\sigma)^\varrho$. Thus $\alpha_\nu(\sigma) = \lambda(\sigma)^\varrho$. Since $\sigma \in I$ is arbitrary,

$$(\chi_z\chi_t)|_I = \lambda^{\varrho a(\nu,T,0)}. \tag{18.5}$$

By (18.4) and (18.5), $\lambda^{d_F} = 1$. Since λ is a faithful character of the cyclic group I, $|I|$ divides d_F.

As a consequence,

$$\frac{a(F,X,B)}{|I_G(F)|} = \frac{\varrho b_\nu}{|I_G(F)|} + \frac{d_F}{|I_G(F)|},$$

where the first summand is positive by **Step 1** and the second summand is a non-negative integer. Thus

$$\frac{a(F,X,B)}{|I_G(F)|} < 1 \implies d_F = 0. \tag{18.6}$$

Step 3. In this step, we show that a is the minimum of the terms and that the minimum is attained.

Let $\pi : X \to X/G$ be the quotient morphism. Since the h_j are G-semi-invariant, B is G-invariant, and $B \geq 0$. By Lemma 2.6(0), $(X,B)/G = (X/G, B_{X/G})$ is a pair and $K_X + B = \pi^*(K_{X/G} + B_{X/G})$. Since (X,B) is klt, $(X/G, B_{X/G})$ is klt [KM98, Proposition 5.20]. By [Amb99, Section 1, the second paragraph after Definition 1.1], there exists a prime divisor E over X/G with $\mathrm{center}_{X/G} E = \{\pi(0)\}$ and $a(E, X/G, B_{X/G}) = a$. By Lemma 2.6(3), there exists a prime divisor F over X such that $\mathrm{ord}_F|_{\mathbb{C}(X/G)}$ is a positive integer multiple of ord_E, and by Lemma 2.6(2), $a(F,X,B) = |I_G(F)| \cdot a$ and $\pi(\mathrm{center}_X F) = \{\pi(0)\}$. Since 0 is fixed by G, $\pi^{-1}(\pi(0)) = \{0\}$, so $\mathrm{center}_X F = \{0\}$. As $a < 1$, we have $d_F = 0$ by (18.6), so Lemma 18.2(2) implies that ord_F is the Gauss extension of $\varrho\,\mathrm{ord}_\nu$ with x-value $k := \mathrm{ord}_F x$ for the data ϱ and ν of Lemma 18.2(2), where ϱ and k are coprime and $0 < k < \varrho l_\nu$. By **Step 1**,

$$a = \frac{a(F,X,B)}{|I_G(F)|} = \frac{\varrho b_\nu}{e(\nu;\varrho,k)}.$$

Conversely, by **Step 1**, each term $\varrho b_\nu / e(\nu;\varrho,k)$ is $a(F',X,B)/|I_G(F')|$ for a prime divisor F' over X with $\mathrm{center}_X F' = \{0\}$, which is at least a by the in-particular part of Lemma 2.6, since

$\operatorname{center}_X F' = \{0\}$ and 0 is fixed by G. Thus a is the minimum of the terms, and the minimum is attained.

Finally, let F be a prime divisor over X with center 0 and $a(F,X,B)/|I_G(F)| < 1$. By (18.6), $d_F = 0$, and by Lemma 18.2(2), ord_F is the Gauss extension of $\varrho \operatorname{ord}_\nu$ with x-value $k := \operatorname{ord}_F x$, where $\gcd(\varrho, k) = 1$ and $0 < k < \varrho l_\nu$. □

18.2. **Toric computing valuations.** In this subsection, we show that if the minimal log discrepancy of a complete intersection quotient is computed by a prime divisor which is monomial in the coordinates, then the minimal log discrepancy is a lattice Newton minimum with an auxiliary coordinate of coefficient one.

Set-up 18.4. Let n, r, s, H, $x_1, \dots, x_{n+1}$, X, $f_1, \dots, f_r$, $h_1, \dots, h_s$, $c_1, \dots, c_s$, B, χ, N_H, and $\psi_\chi(X,B)$ be as follows.

(1) $n \geq 2$ and $1 \leq r < n$ are integers, H is a finite cyclic group acting diagonally on $\mathbb{A}^n$ with respect to coordinates $x_1, \dots, x_n$, and $X \subset \mathbb{A}^n$ is the common zero scheme of H-semi-invariant polynomials $f_1, \dots, f_r$ vanishing at 0, which is, near 0, a normal integral complete intersection of codimension r with klt singularities, on which H acts faithfully.

(2) $s \geq 0$ is an integer, $h_1, \dots, h_s$ are H-semi-invariant polynomials in $x_1, \dots, x_n$ with non-zero restrictions to X, $c_1, \dots, c_s > 0$ are real numbers, and $B := \sum_j c_j \operatorname{div}(h_j|_X)$ with (X,B) klt near 0.

(3) χ is a character of H, H acts on $\mathbb{A}^{n+1}$ with coordinates $x_1, \dots, x_{n+1}$ through the action on $\mathbb{A}^n$ and by χ on x_{n+1}, $N_H \subset \mathbb{R}^{n+1}$ is the lattice of Definition 2.15 for the action of H on $\mathbb{A}^{n+1}$, and

$$\psi_\chi(X,B) := \inf_{w \in N_H \cap \mathbb{R}^{n+1}_{>0}} \Big(\sum_{i=1}^{n} w_i - \sum_{l=1}^{r} \operatorname{wt}_w(f_l) - \sum_{j=1}^{s} c_j \operatorname{wt}_w(h_j) \Big).$$

The number $\psi_\chi(X,B)$ depends on the coordinates, the polynomials f_l and h_j, and the coefficients c_j, and not only on the pair (X,B). Its value does not depend on χ, since w_{n+1} does not occur in the expression in (3), $\mathbb{Z}^{n+1} \subset N_H$, and the projection of N_H to the first n coordinates does not depend on χ. The character χ only determines which vectors of N_H attain the infimum.

The next corollary is used for the ACC statements later in this section.

Corollary 18.5. *Let n_0, r_0, and s_0 be positive integers and $\Lambda \subset (0, +\infty)$ a DCC set. Then the set of finite numbers $\psi_\chi(X,B)$, for data as in Set-up 18.4 with $n \leq n_0$, $r \leq r_0$, $s \leq s_0$, and $c_1, \dots, c_s \in \Lambda$, satisfies the ACC.*

Proof. Fix data as in Set-up 18.4. Let $H' \subset \mathbb{T}^{n+1}$ be the image of H, so that $N_{H'} = N_H$, and let

$$J := (J(f_1), \dots, J(f_r), J(h_1), \dots, J(h_s), (x_{n+1})) \quad \text{and} \quad c' := (1, \dots, 1, c_1, \dots, c_s, 1),$$

an $(r+s+1)$-tuple of non-zero monomial ideals of $\mathbb{C}[x_1, \dots, x_{n+1}]$ and an $(r+s+1)$-tuple of real numbers. Since $\operatorname{wt}_w((x_{n+1})) = w_{n+1}$, the number $\psi_{J,c'}(w)$ of Definition 2.16 is the expression in Set-up 18.4(3) whose infimum is $\psi_\chi(X,B)$. Thus $\psi_\chi(X,B) = \psi_{\min}(H', J, c')$. For fixed n, r, and s, Theorem 4.5, applied with $d = n+1$, $r+s+1$ monomial ideals, and the DCC set $\Lambda \cup \{1\}$, implies that the finite numbers $\psi_\chi(X,B)$ satisfy the ACC. Since there are finitely many triples (n,r,s) with $n \leq n_0$, $r \leq r_0$, and $0 \leq s \leq s_0$, Corollary 18.5 follows from Lemma 2.10(4). □

Lemma 18.6. *Notation and conditions as in Set-up 18.4. Let F be a prime divisor over X with center 0 such that $I_H(F) = H$ and $a(F,X,B)/|H| = \operatorname{mld}((X \ni 0, B)/H)$, and assume that χ is the character of H on $\mathfrak{m}_F/\mathfrak{m}_F^2$. Assume that there exists $\tilde{w} \in \mathbb{Z}^n_{>0}$ with $\gcd(\tilde{w}_1, \dots, \tilde{w}_n) = 1$, $\operatorname{ord}_F x_i = \tilde{w}_i$ for $1 \leq i \leq n$, $\operatorname{ord}_F h_j = \operatorname{wt}_{\tilde{w}}(h_j)$ for any j, and $a(F,X,0) = \sum_i \tilde{w}_i - \sum_l \operatorname{wt}_{\tilde{w}}(f_l)$. Then $\operatorname{mld}((X \ni 0, B)/H) = \psi_\chi(X,B)$, and the infimum is attained at $\frac{1}{|H|}(\tilde{w}, 1)$.*

Proof. Let $m := \mathrm{mld}((X \ni 0, B)/H)$ and $B_{\mathbb{A}^n} := \sum_l \mathrm{div}(f_l) + \sum_j c_j \,\mathrm{div}(h_j)$. For $w \in \mathbb{R}^{n+1}_{\geq 0}$, let

$$\psi(w) := \sum_{i=1}^{n} w_i - \sum_{l=1}^{r} \mathrm{wt}_w(f_l) - \sum_{j=1}^{s} c_j \,\mathrm{wt}_w(h_j),$$

so that $\psi_\chi(X, B) = \inf\{\psi(w) \mid w \in N_H \cap \mathbb{R}^{n+1}_{>0}\}$, and $\psi(qw) = q\psi(w)$ for any real number $q > 0$.

By Theorem 8.3, applied with $G = H$ and the subgroup H,

$$m = \mathrm{mld}((\mathbb{A}^n \ni 0, B_{\mathbb{A}^n})/H) > 0.$$

Since H acts faithfully on X, H acts faithfully on $\mathbb{A}^n$. Let $B^+ := \mathrm{pr}_1^* B_{\mathbb{A}^n} + (x_{n+1} = 0)$ on $\mathbb{A}^{n+1} = \mathbb{A}^n \times \mathbb{A}^1_{x_{n+1}}$. By the in-particular part of Lemma 8.1, applied to $(\mathbb{A}^n, B_{\mathbb{A}^n})$, the point 0, the group H, and the character χ, with the coordinate t of Lemma 8.1 taken to be x_{n+1},

$$\mathrm{mld}((\mathbb{A}^{n+1} \ni 0, B^+)/H) = m. \tag{18.7}$$

We show that $\psi(w) \geq m$ for any $w \in N_H \cap \mathbb{R}^{n+1}_{>0}$. Let M_H be the dual lattice of N_H, so that

$$M_H = \{\alpha \in \mathbb{Z}^{n+1} \mid \langle u, \alpha\rangle \in \mathbb{Z} \text{ for any } u \in N_H\},$$
$$N_H = \{u \in \mathbb{R}^{n+1} \mid \langle u, \alpha\rangle \in \mathbb{Z} \text{ for any } \alpha \in M_H\}.$$

By Definition 2.15, a monomial x^α with $\alpha \in \mathbb{Z}^{n+1}_{\geq 0}$ is H-invariant if and only if $\alpha \in M_H$. Since $|H|u \in \mathbb{Z}^{n+1}$ for any $u \in N_H$, $|H|\mathbb{Z}^{n+1} \subset M_H$, so M_H is generated as a group by $M_H \cap \mathbb{Z}^{n+1}_{\geq 0}$.

First, let $w \in N_H \cap \mathbb{R}^{n+1}_{>0}$ be primitive in N_H. Then $N_H \cap \mathbb{R}w = \mathbb{Z}w$, and $\langle w, M_H\rangle = \mathbb{Z}$, since $\langle w, M_H\rangle = p\mathbb{Z}$ for a positive integer p implies $\frac{1}{p}w \in N_H$. Let v be the primitive vector of $\mathbb{Z}^{n+1}$ on the ray $\mathbb{R}_{>0}w$. Since $\mathbb{Z}^{n+1} \subset N_H$, $v = qw$ for a positive integer q. Let F_v be the monomial prime divisor over $\mathbb{A}^{n+1}$ in $x_1, \dots, x_{n+1}$ with weights v (Definition 2.16), namely the exceptional divisor of the weighted blow-up of $\mathbb{A}^{n+1}$ at 0 with weights v. Since $v \in \mathbb{Z}^{n+1}_{>0}$, $\mathrm{center}_{\mathbb{A}^{n+1}} F_v = \{0\}$. By [JM12, Proposition 5.1 and (5.1)], $a(F_v, \mathbb{A}^{n+1}, 0) = \sum_{i=1}^{n+1} v_i$, and by Definition 2.16, $\mathrm{ord}_{F_v} P = \mathrm{wt}_v(P)$ for any non-zero polynomial P. Since f_l and h_j do not involve x_{n+1},

$$a(F_v, \mathbb{A}^{n+1}, B^+) = \sum_{i=1}^{n+1} v_i - \sum_l \mathrm{wt}_v(f_l) - \sum_j c_j \,\mathrm{wt}_v(h_j) - v_{n+1} = \psi(v) = q\psi(w).$$

The field $\mathbb{C}(\mathbb{A}^{n+1})^H$ is the fraction field of $\mathbb{C}[x_1, \dots, x_{n+1}]^H$, which is spanned by the monomials x^α with $\alpha \in M_H \cap \mathbb{Z}^{n+1}_{\geq 0}$. Since ord_{F_v} of a polynomial is the least v-weight of the monomials occurring in it, and $\langle v, \alpha\rangle = q\langle w, \alpha\rangle$, the value group of $\mathrm{ord}_{F_v}|_{\mathbb{C}(\mathbb{A}^{n+1})^H}$ is $q\langle w, M_H\rangle = q\mathbb{Z}$. Thus $\mathrm{ord}_{F_v}|_{\mathbb{C}(\mathbb{A}^{n+1}/H)} = q\,\mathrm{ord}_E$ for a prime divisor E over $\mathbb{A}^{n+1}/H$, and by Lemma 2.6(1)(2), $q = |I_H(F_v)|$, E has center the image of 0, and

$$a(E, (\mathbb{A}^{n+1}, B^+)/H) = \frac{a(F_v, \mathbb{A}^{n+1}, B^+)}{q} = \psi(w)$$

(cf. [Amb06, Section 2]). By (18.7), $\psi(w) \geq m$.

Next, let $w \in N_H \cap \mathbb{R}^{n+1}_{>0}$ be arbitrary. Then $w = pw_0$ for a positive integer p and a primitive vector w_0 of N_H on $\mathbb{R}_{>0}w$, so $\psi(w) = p\psi(w_0) \geq pm \geq m$, since $m > 0$. Thus $\psi_\chi(X, B) \geq m$.

Finally, we show that $\psi(\frac{1}{|H|}(\tilde{w}, 1)) = m$ and $\frac{1}{|H|}(\tilde{w}, 1) \in N_H$. Let λ be the character of $H = I_H(F)$ on $\mathfrak{m}_F/\mathfrak{m}_F^2$. By Lemma 2.6(1), λ is faithful, and by assumption, $\chi = \lambda$. Let χ_i be the character of x_i for $1 \leq i \leq n$. Since $\mathrm{ord}_F x_i = \tilde{w}_i$ is finite, $x_i|_X \neq 0$, and by Lemma 9.2(1), $\chi_i = \lambda^{\tilde{w}_i}$. Since H is cyclic and λ is faithful, there exists a generator σ of H with $\lambda(\sigma) = e^{2\pi\sqrt{-1}/|H|}$. Then $\chi_i(\sigma) = e^{2\pi\sqrt{-1}\tilde{w}_i/|H|}$ for $1 \leq i \leq n$ and $\chi(\sigma) = e^{2\pi\sqrt{-1}/|H|}$, so

$\frac{1}{|H|}(\tilde{w},1) \in N_H \cap \mathbb{R}^{n+1}_{>0}$ by Definition 2.15. Since f_l and h_j do not involve x_{n+1},

$$\psi\Big(\frac{1}{|H|}(\tilde{w},1)\Big) = \frac{1}{|H|}\Big(\sum_i \tilde{w}_i - \sum_l \operatorname{wt}_{\tilde{w}}(f_l) - \sum_j c_j \operatorname{wt}_{\tilde{w}}(h_j)\Big)$$
$$= \frac{a(F,X,0) - \sum_j c_j \operatorname{ord}_F h_j}{|H|} = \frac{a(F,X,B)}{|H|} = m.$$

Since $\psi_\chi(X,B) \geq m$, we obtain $m = \psi_\chi(X,B)$, and the infimum is attained at $\frac{1}{|H|}(\tilde{w},1)$. □

18.3. **Generating sequences and toric embeddings.** In this subsection, we prove Proposition 18.7: a plane prime divisor with a minimal generating sequence of $m+1$ elements becomes the monomial prime divisor of a primitive weight vector on an equivariant complete intersection in $\mathbb{A}^{m+3}$. Generating sequences are taken in the sense of [dFGPM23, Definition 2.9].

Proposition 18.7. *Let $T := \mathbb{A}^2_{z,t}$, let H be a finite cyclic group acting diagonally on T, not necessarily faithfully, let ν be an H-invariant prime divisor over T with center 0, let m be a non-negative integer, and let $N := m+3$. Assume that H acts trivially on the residue field of ord_ν and that ν has a minimal generating sequence* [dFGPM23, Definition 2.9] *with $m+1$ elements. Then there exist H-semi-invariant polynomials $q_0,\dots,q_m \in \mathbb{C}[z,t]$, the closed embedding $\iota = (z,t,q_0,\dots,q_m) : T \to \mathbb{A}^N$, where H acts diagonally on $\mathbb{A}^N$ so that ι is H-equivariant, H-semi-invariant polynomials $f_1,\dots,f_{N-2}$ on $\mathbb{A}^N$, and a vector $v \in \mathbb{Z}^N_{>0}$ with $\gcd(v_1,\dots,v_N)=1$, such that:*

(1) $f_1,\dots,f_{N-2}$ generate the ideal of $\iota(T)$ near 0, and $\operatorname{in}_v(f_1),\dots,\operatorname{in}_v(f_{N-2})$ form a regular sequence generating a prime ideal whose zero locus is smooth at a point of $(\mathbb{C}^*)^N$;
(2) for any non-zero polynomial $h \in \mathbb{C}[z,t]$ there exists a polynomial $\tilde{h}$ on $\mathbb{A}^N$ with $\tilde{h}\circ\iota = h$ and $\operatorname{wt}_v(\tilde{h}) = \operatorname{ord}_\nu h$, and $\tilde{h}$ can be taken H-semi-invariant with the character of h if h is H-semi-invariant;
(3) $a(\nu,T,0) = \sum_{i=1}^N v_i - \sum_{l=1}^{N-2} \operatorname{wt}_v(f_l)$;
(4) $v = (\operatorname{ord}_\nu z, \operatorname{ord}_\nu t, \operatorname{ord}_\nu q_0,\dots,\operatorname{ord}_\nu q_m)$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we compare ord_ν with the order of vanishing at 0, introduce the graded ring of ord_ν, and compute the action of H on it.

Let $R := \mathbb{C}[z,t]$, $\mathfrak{m} := (z,t) \subset R$, $\widehat{R} := \mathbb{C}[[z,t]]$, and $e_\nu := \min\{\operatorname{ord}_\nu z, \operatorname{ord}_\nu t\}$. Since $\operatorname{center}_T \nu = \{0\}$, $e_\nu \geq 1$. For a non-zero $h \in \widehat{R}$ let $\operatorname{ord}_0 h := \max\{L \mid h \in \mathfrak{m}^L \widehat{R}\}$. We show that there exists a positive integer r_ν such that for any non-zero $h \in R$,

$$e_\nu \operatorname{ord}_0 h \leq \operatorname{ord}_\nu h \leq 2^{r_\nu - 1} \operatorname{ord}_0 h. \tag{18.8}$$

The left inequality holds because h is an R-linear combination of the monomials $z^a t^b$ with $a+b = \operatorname{ord}_0 h$. For the right inequality, let $T_0 := T$ and $p_0 := 0$. For $k \geq 0$, if the center of ν on T_k is a closed point p_k, let $T_{k+1} \to T_k$ be the blow-up of p_k, with exceptional curve E_{k+1}. By [KM98, Lemma 2.45], there exists $r_\nu \geq 1$ such that the center of ν on T_{r_ν} is E_{r_ν}, that is, $\operatorname{ord}_\nu = \operatorname{ord}_{E_{r_\nu}}$. Let D_k be the pullback of $\operatorname{div}(h)$ to T_k and $\mu_k := \operatorname{mult}_{p_k} D_k$. Then $D_{k+1} = \mu_k E_{k+1} + D'$ with $E_{k+1} \not\subset \operatorname{Supp} D'$, and since D_{k+1} is a pullback from T_k,

$$(D' \cdot E_{k+1}) = (D_{k+1} \cdot E_{k+1}) - \mu_k (E_{k+1}^2) = \mu_k.$$

Restricting a local equation of D' to the smooth curve E_{k+1} shows

$$\operatorname{mult}_{p_{k+1}} D' \leq (D' \cdot E_{k+1})_{p_{k+1}} \leq \mu_k.$$

Thus $\mu_{k+1} = \mu_k + \operatorname{mult}_{p_{k+1}} D' \leq 2\mu_k$, and $\operatorname{ord}_\nu h = \mu_{r_\nu - 1} \leq 2^{r_\nu - 1}\mu_0 = 2^{r_\nu - 1} \operatorname{ord}_0 h$.

Let $h \in \widehat{R}$ be non-zero and let L be an integer with $e_\nu L > 2^{r_\nu - 1} \operatorname{ord}_0 h$. If $h', h'' \in R$ satisfy $h - h', h - h'' \in \mathfrak{m}^L \widehat{R}$, then $\operatorname{ord}_0 h' = \operatorname{ord}_0 h$, because $e_\nu \leq \operatorname{ord}_\nu z \leq 2^{r_\nu - 1}$ by (18.8) and so $L > \operatorname{ord}_0 h$. By (18.8), either $h' = h''$, or $\operatorname{ord}_\nu(h' - h'') \geq e_\nu L$ and $e_\nu L > \operatorname{ord}_\nu h'$. Thus $\operatorname{ord}_\nu h :=$

$\operatorname{ord}_\nu h'$ does not depend on h'. Computing with such truncations, $\operatorname{ord}_\nu(ab) = \operatorname{ord}_\nu a + \operatorname{ord}_\nu b$ and $\operatorname{ord}_\nu(a+b) \geq \min\{\operatorname{ord}_\nu a, \operatorname{ord}_\nu b\}$ for non-zero $a, b, a+b \in \widehat{R}$, and $\operatorname{ord}_\nu h \geq e_\nu \operatorname{ord}_0 h$ for any non-zero $h \in \widehat{R}$.

For an integer $d \geq 0$ let $\mathfrak{a}_d := \{h \in \widehat{R}\backslash\{0\} \mid \operatorname{ord}_\nu h \geq d\} \cup \{0\}$, $\mathcal{G}_d := \mathfrak{a}_d/\mathfrak{a}_{d+1}$, and let $\mathcal{G} := \bigoplus_{d\geq 0} \mathcal{G}_d$ with the multiplication induced by $\widehat{R}$. For a non-zero $h \in \widehat{R}$ with $\operatorname{ord}_\nu h = d$, let $\operatorname{in}_\nu(h) \in \mathcal{G}_d$ be the class of h. Since $\operatorname{in}_\nu(a)\operatorname{in}_\nu(b) = \operatorname{in}_\nu(ab) \neq 0$, $\mathcal{G}$ is a domain. Since $\mathfrak{m}^L \widehat{R} \subset \mathfrak{a}_{d+1}$ when $e_\nu L \geq d+1$, any non-zero element of $\mathcal{G}_d$ is $\operatorname{in}_\nu(h)$ for some $h \in R$.

Let $\mathcal{O}_\nu \subset \mathbb{C}(T)$ be the valuation ring of ord_ν, with maximal ideal $\mathfrak{m}_\nu$. Since ν is H-invariant, H preserves $\mathcal{O}_\nu$. Let $\lambda : H \to \mathbb{C}^*$ be the character by which H acts on the line $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$ through pullback. We claim that for any non-zero $h \in R$ and any $\sigma \in H$, with $d := \operatorname{ord}_\nu h$,

$$\sigma^* h = \lambda(\sigma)^d h \quad \text{or} \quad \operatorname{ord}_\nu\left(\sigma^* h - \lambda(\sigma)^d h\right) > d. \tag{18.9}$$

To prove (18.9), let u be a uniformizer of $\mathcal{O}_\nu$ and write $h = u^d \epsilon$ with $\epsilon \in \mathcal{O}_\nu^*$. Then $\sigma^* u = \lambda(\sigma) u(1+\epsilon_1)$ for some $\epsilon_1 \in \mathfrak{m}_\nu$, and $\sigma^*\epsilon - \epsilon \in \mathfrak{m}_\nu$ because H acts trivially on the residue field of ord_ν. Thus

$$\sigma^* h - \lambda(\sigma)^d h = \lambda(\sigma)^d u^d\big((1+\epsilon_1)^d \sigma^*\epsilon - \epsilon\big) \in u^d \mathfrak{m}_\nu.$$

Moreover $H = I_H(\nu)$ (Definition 2.5), so by Lemma 9.2(1), the characters of z and t are $\lambda^{\operatorname{ord}_\nu z}$ and $\lambda^{\operatorname{ord}_\nu t}$.

Step 2. In this step we choose $q_0, \ldots, q_m$, define ι and v, and describe $\mathcal{G}$ by generators and relations.

Let $y_0, \ldots, y_m \in \mathfrak{m}\widehat{R}$ be a minimal generating sequence of ord_ν such that (y_0, y_1) is a system of local coordinates of $\widehat{R}$ and $y_2, \ldots, y_m \in \mathbb{C}[[y_0]][y_1]$ are monic polynomials in y_1 with $\deg_{y_1} y_j = \dim_{\mathbb{C}} \widehat{R}/(y_0, y_j)$, that is, the Weierstrass polynomials of the branches $(y_j = 0)$, which exists by [dFGPM23, Remark 2.1] and the first paragraph of [dFGPM23, §2.1.1]. Moreover $\deg_{y_1} y_j = n_1 \cdots n_{j-1}$ for $2 \leq j \leq m$ [dFGPM23, §2.1.1] (see also [dFGPM23, Proposition 2.3(3) and Notation 2.15]), where $n_j = e_{j-1}/e_j$ is the integer of [dFGPM23, Notation 2.2] for a curvetta A of E_{r_ν} (see the paragraph after (ii)), which is the n_j of (i) below. Let $\bar{\beta}_i := \operatorname{ord}_\nu y_i$. For $I = (i_0, \ldots, i_m) \in \mathbb{Z}_{\geq 0}^{m+1}$ let $y^I := y_0^{i_0} \cdots y_m^{i_m}$, and let $\bar{\beta} := (\bar{\beta}_0, \ldots, \bar{\beta}_m)$. By [dFGPM23, Definition 2.9], any $h \in \mathfrak{a}_d$ is a finite sum $\sum_k a_k y^{I_k}$ with $a_k \in \widehat{R}$ and $\langle \bar{\beta}, I_k\rangle \geq d$. Since $(a_k - a_k(0))y^{I_k} \in \mathfrak{a}_{d+1}$, if $\operatorname{ord}_\nu h = d$ then $\operatorname{in}_\nu(h) = \sum_{\langle \bar{\beta}, I_k\rangle = d} a_k(0)\operatorname{in}_\nu(y^{I_k})$. Thus

$$\mathcal{G} \text{ is generated by } \operatorname{in}_\nu(y_0), \ldots, \operatorname{in}_\nu(y_m) \text{ as a } \mathbb{C}\text{-algebra.} \tag{18.10}$$

In particular $m \geq 1$, for if $m = 0$, then by (18.10), for any non-zero $a, b \in R$ with $\operatorname{ord}_\nu a = \operatorname{ord}_\nu b$ there exists $c \in \mathbb{C}^*$ with $\operatorname{ord}_\nu(a - cb) > \operatorname{ord}_\nu b$, so the residue field of ord_ν is $\mathbb{C}$, while it is the function field of the curve E_{r_ν} of **Step 1**.

Let $\mathcal{S} := \{I \in \mathbb{Z}_{\geq 0}^{m+1} \mid i_j < n_j \text{ for } 1 \leq j \leq m-1\}$, where the integers n_j are those of (i) below. The following hold.

(i) For $1 \leq j \leq m-1$ there exist a positive integer n_j, $\theta_j \in \mathbb{C}^*$, and non-negative integers $b_0^j, \ldots, b_{j-1}^j$ such that $n_j \bar{\beta}_j = \sum_{i=0}^{j-1} b_i^j \bar{\beta}_i$ and

$$\operatorname{in}_\nu(y_j)^{n_j} = \theta_j \prod_{i=0}^{j-1} \operatorname{in}_\nu(y_i)^{b_i^j}.$$

(ii) The elements $\operatorname{in}_\nu(y^I)$, $I \in \mathcal{S}$, of $\mathcal{G}$ are linearly independent over $\mathbb{C}$.

Here $\bar{\beta}_i$ is the $\bar{\beta}_i$ of [dFGPM23, Proposition 2.14 and Notation 2.15]. Here A is a curvetta of E_{r_ν} [dFGPM23, Definition 2.5], $g(A)+1$ is the number of minimal generators of the semigroup of A [dFGPM23, Notation 2.2], and $\ell_\nu \geq 0$ is the number of blow-ups by which $T_{r_\nu} \to T$ of **Step 1** exceeds the minimal embedded resolution of A (the integer ℓ_E of [dFGPM23, Proposition 2.14] for $E = E_{r_\nu}$), so that $m = g(A)$ if $\ell_\nu = 0$ and $m = g(A)+1$ if $\ell_\nu > 0$ [dFGPM23, Proposition 2.14]. For $1 \leq j \leq m-1$ we have $\bar{\beta}_{j+1} > n_j \bar{\beta}_j$. For $j \leq g(A) - 1$ this is (2.3) in [dFGPM23,

Proposition 2.3(2)]. When $\ell_\nu > 0$, the remaining case $j = m-1$ follows from the formula $\bar\beta_m = n_{m-1}\bar\beta_{m-1} + \ell_\nu$ of [dFGPM23, Notation 2.15].

Proof of (i). By [dFGPM23, Proposition 2.22], $y_{j+1} = y_j^{n_j} - \theta_j y_0^{b_0^j}\cdots y_{j-1}^{b_{j-1}^j} + p_j$, where $\theta_j \in \mathbb{C}^*$, $n_j\bar\beta_j = \sum_{i=0}^{j-1} b_i^j\bar\beta_i$, and any monomial of p_j has $\bar\beta$-weight larger than $n_j\bar\beta_j$. Thus $\operatorname{ord}_\nu p_j > n_j\bar\beta_j$. Since moreover $\operatorname{ord}_\nu y_{j+1} = \bar\beta_{j+1} > n_j\bar\beta_j$, (i) follows by taking classes in $\mathcal{G}_{n_j\bar\beta_j}$.

Proof of (ii). Since $\deg_{y_1} y_j = n_1\cdots n_{j-1}$ for $2 \le j \le m$ and $n_j > 1$ for $1 \le j \le m-1$ [dFGPM23, Proposition 2.3(1)], [dFGPM23, Proposition 2.20] shows that any $h \in \mathbb{C}[[y_0]][y_1]$ can be written uniquely as $h = \sum \gamma_I(y_0) y_1^{i_1}\cdots y_m^{i_m}$ with $\gamma_I \in \mathbb{C}[[y_0]]$, $0 \le i_j < n_j$ for $1 \le j \le m-1$, and $0 \le i_m \le \deg_{y_1} h/\deg_{y_1} y_m$. This is the $(y_0,\dots,y_m)$-adic expansion of h. Let $h = \sum_{I\in\mathcal{S}} a_I y^I$ be a finite sum with $a_I \in \mathbb{C}$. The y_1-degree of y^I is $D(I) := i_1 + i_2 n_1 + \cdots + i_m n_1\cdots n_{m-1}$, and $D(I) \ne D(I')$ for $I, I' \in \mathcal{S}$ with $(i_1,\dots,i_m) \ne (i_1',\dots,i_m')$. Thus $\deg_{y_1} h = \max\{D(I) \mid a_I \ne 0\}$, any I with $a_I \ne 0$ satisfies $i_m \le D(I)/\deg_{y_1} y_m \le \deg_{y_1} h/\deg_{y_1} y_m$, and grouping the terms of h by $(i_1,\dots,i_m)$ yields the $(y_0,\dots,y_m)$-adic expansion of h. Let $\mathcal{S}_d := \{I \in \mathcal{S} \mid \langle\bar\beta, I\rangle = d\}$. Suppose that $\sum_{I\in\mathcal{S}_d} a_I \operatorname{in}_\nu(y^I) = 0$ with not all a_I zero, and let $h := \sum_{I\in\mathcal{S}_d} a_I y^I$. Then $\operatorname{ord}_\nu h > d$, which contradicts the equality $\operatorname{ord}_\nu h = d$ given by [dFGPM23, Remark 2.23]. Since homogeneous elements of distinct degrees are linearly independent, (ii) follows.

For $0 \le i \le m$ fix $y_i' \in R$ with $y_i - y_i' \in \mathfrak{m}^L\widehat{R}$, where $e_\nu L > \bar\beta_i$, and let

$$q_i := \frac{1}{|H|}\sum_{\sigma\in H} \lambda(\sigma)^{-\bar\beta_i}\sigma^* y_i'.$$

Since H acts diagonally on z, t, we have $q_i \in R$ and $\sigma^* q_i = \lambda(\sigma)^{\bar\beta_i} q_i$ for any $\sigma \in H$. Since $\operatorname{ord}_\nu(y_i - y_i') \ge e_\nu L > \bar\beta_i$ and, by (18.9), $\operatorname{ord}_\nu(q_i - y_i') > \bar\beta_i$, we have $\operatorname{in}_\nu(q_i) = \operatorname{in}_\nu(y_i)$. In particular $q_i(0) = 0$.

Let $x_1,\dots,x_N$ be the coordinates of $\mathbb{A}^N$, so that $\iota^* x_1 = z$, $\iota^* x_2 = t$, and $\iota^* x_{i+3} = q_i$ for $0 \le i \le m$. Then ι is the graph of $(q_0,\dots,q_m)$, hence a closed embedding with $\iota(0) = 0$. Let

$$v := (\operatorname{ord}_\nu z, \operatorname{ord}_\nu t, \bar\beta_0,\dots,\bar\beta_m) \in \mathbb{Z}^N_{>0},$$

so $v_i = \operatorname{ord}_\nu(\iota^* x_i)$ for any i, which is (4). Let H act on $\mathbb{A}^N$ by $\sigma^* x_i := \lambda(\sigma)^{v_i} x_i$. By **Step 1** and the choice of q_i, ι is H-equivariant. For $a \in \mathbb{Z}^N_{\ge 0}$ let $x^a := \prod_i x_i^{a_i}$. Then $\sigma^* x^a = \lambda(\sigma)^{\langle v,a\rangle} x^a$ and $\operatorname{ord}_\nu(\iota^* x^a) = \langle v, a\rangle$. We call a polynomial on $\mathbb{A}^N$ *v-homogeneous of weight d* if any monomial occurring in it has v-weight d. Hence any v-homogeneous polynomial of weight d is H-semi-invariant with character λ^d, and for any polynomial G on $\mathbb{A}^N$ with $\iota^* G \ne 0$,

$$\operatorname{wt}_v(G) \le \operatorname{ord}_\nu(\iota^* G) \quad\text{and}\quad \iota^*\big(G - \operatorname{in}_v(G)\big) \in \mathfrak{a}_{\operatorname{wt}_v(G)+1}. \tag{18.11}$$

Let $\Phi : \mathbb{C}[x_1,\dots,x_N] \to \mathcal{G}$ be the $\mathbb{C}$-linear map sending a v-homogeneous polynomial P of weight d to the class of $\iota^* P \in \mathfrak{a}_d$ in $\mathcal{G}_d$. Then Φ is a homomorphism of $\mathbb{C}$-algebras, graded by $\deg x_i = v_i$, and it is surjective by (18.10), because $\Phi(x_{i+3}) = \operatorname{in}_\nu(q_i) = \operatorname{in}_\nu(y_i)$. Let $\mathfrak{q} := \operatorname{Ker}\Phi$. Since $\mathcal{G}$ is a domain, $\mathfrak{q}$ is a prime ideal generated by v-homogeneous polynomials. By (18.11), for any non-zero polynomial G on $\mathbb{A}^N$ with $d := \operatorname{wt}_v(G)$,

$$\operatorname{in}_v(G) \in \mathfrak{q} \iff \iota^* G \in \mathfrak{a}_{d+1}. \tag{18.12}$$

Step 3. In this step we prove (2) and that $v_1,\dots,v_N$ have greatest common divisor 1.

Let $h \in R$ be non-zero and $d := \operatorname{ord}_\nu h$. Let $h_0 := h$. For $k \ge 0$, if $h_k \ne 0$, the surjectivity of Φ yields a v-homogeneous polynomial P_k of weight $\operatorname{ord}_\nu h_k$ with $\Phi(P_k) = \operatorname{in}_\nu(h_k)$, and we let $h_{k+1} := h_k - \iota^* P_k \in R$, so that $h_{k+1} = 0$ or $\operatorname{ord}_\nu h_{k+1} > \operatorname{ord}_\nu h_k$. If $h_k = 0$, let $P_k := 0$ and $h_{k+1} := 0$. Thus $h_k = 0$ or $\operatorname{ord}_\nu h_k \ge d + k$ for any k. Fix an integer $K \ge 2^{r_\nu - 1} d/e_\nu$ and let

$$\tilde h := \sum_{k=0}^{K-1} P_k + h_K(x_1, x_2).$$

Then $\iota^*\tilde{h} = h$. Any non-zero P_k has v-weight at least d. If $h_K \neq 0$, then by (18.8), $\operatorname{ord}_0 h_K \geq (d+K)/2^{r_\nu - 1} \geq d/e_\nu$, so $\operatorname{wt}_v(h_K(x_1, x_2)) \geq e_\nu \operatorname{ord}_0 h_K \geq d$. Thus $\operatorname{wt}_v(\tilde{h}) \geq d$, and $\operatorname{wt}_v(\tilde{h}) \leq d$ by (18.11).

If h is H-semi-invariant with character χ, let

$$\tilde{h}' := \frac{1}{|H|} \sum_{\sigma \in H} \chi(\sigma)^{-1} \sigma^* \tilde{h}.$$

Since H acts diagonally on $\mathbb{A}^N$, $\tilde{h}'$ is the sum of the terms of $\tilde{h}$ whose monomials have character χ, so $\operatorname{wt}_v(\tilde{h}') \geq d$. Since ι is H-equivariant, $\iota^*\tilde{h}' = h$, and $\operatorname{wt}_v(\tilde{h}') \leq d$ by (18.11). Thus $\tilde{h}'$ is H-semi-invariant with character χ, $\iota^*\tilde{h}' = h$, and $\operatorname{wt}_v(\tilde{h}') = \operatorname{ord}_\nu h$. This proves (2).

By (2), $\operatorname{ord}_\nu h \in \sum_i \mathbb{Z}_{\geq 0} v_i$ for any non-zero $h \in R$. Since $\operatorname{ord}_\nu(\mathbb{C}(T)^*) = \mathbb{Z}$, the integers $v_1, \ldots, v_N$ have greatest common divisor 1.

Step 4. In this step we show that $\mathfrak{q}$ is generated by a regular sequence of $N-2$ v-homogeneous polynomials and that $V(\mathfrak{q})$ is smooth at a point of $(\mathbb{C}^*)^N$.

Let $S := \mathbb{C}[x_3, \ldots, x_N]$. By (18.10), $\Phi(S) = \mathcal{G}$, so there exist v-homogeneous polynomials $Z_1, T_1 \in S$ of weights v_1, v_2 with $\Phi(Z_1) = \Phi(x_1)$ and $\Phi(T_1) = \Phi(x_2)$. Let

$$g_1 := x_1 - Z_1, \quad g_2 := x_2 - T_1, \quad g_{j+2} := x_{j+3}^{n_j} - \theta_j \prod_{i=0}^{j-1} x_{i+3}^{b_i^j} \quad (1 \leq j \leq m-1).$$

Since $m \geq 1$, the list $g_1, \ldots, g_{N-2}$ consists of $N-2 = m+1$ polynomials. Each g_l is v-homogeneous, and we let d_l be the weight of g_l, so $d_1 = v_1$, $d_2 = v_2$, and $d_{j+2} = n_j \bar{\beta}_j$. By the choice of Z_1, T_1 and by (i) in **Step 2**, $g_l \in \mathfrak{q}$ for any l.

For $0 \leq j \leq m-1$ let $S_j := S/(g_3, \ldots, g_{j+2})$ and $S_j' := \mathbb{C}[x_3, \ldots, x_{j+3}]/(g_3, \ldots, g_{j+2})$, so $S_j = S_j'[x_{j+4}, \ldots, x_N]$. For $j \leq m-2$, g_{j+3} is a monic polynomial of degree n_{j+1} in x_{j+4} whose other coefficients belong to S_j'. Since division by a monic polynomial works over any commutative ring, g_{j+3} is a nonzerodivisor on S_j, and S_{j+1} is a free $S_j'[x_{j+5}, \ldots, x_N]$-module with basis $1, x_{j+4}, \ldots, x_{j+4}^{n_{j+1}-1}$. By induction on j, $g_3, \ldots, g_{N-2}$ is a regular sequence in S, and S_{m-1} is spanned over $\mathbb{C}$ by the classes of the monomials $x_3^{i_0} x_4^{i_1} \cdots x_N^{i_m}$ with $(i_0, \ldots, i_m) \in \mathcal{S}$. Since $g_l \in \mathfrak{q} = \operatorname{Ker} \Phi$ for any l, $\Phi|_S$ factors through S_{m-1}, and it sends the class of $x_3^{i_0} \cdots x_N^{i_m}$ to $\operatorname{in}_\nu(q_0)^{i_0} \cdots \operatorname{in}_\nu(q_m)^{i_m} = \operatorname{in}_\nu(y^I)$, $I = (i_0, \ldots, i_m)$. By (ii) in **Step 2**, the induced map $S_{m-1} \to \mathcal{G}$ is injective, that is,

$$\mathfrak{q} \cap S = (g_3, \ldots, g_{N-2})S.$$

For any $P \in \mathfrak{q}$, $P - P(Z_1, T_1, x_3, \ldots, x_N) \in (g_1, g_2)$ and $P(Z_1, T_1, x_3, \ldots, x_N) \in \mathfrak{q} \cap S$. Thus $\mathfrak{q} = (g_1, \ldots, g_{N-2})$. Moreover $g_1 \neq 0$, the image of g_2 in $\mathbb{C}[x_1, \ldots, x_N]/(g_1) \cong \mathbb{C}[x_2, \ldots, x_N]$ is $x_2 - T_1 \neq 0$, and $\mathbb{C}[x_1, \ldots, x_N]/(g_1, g_2) \cong S$ via $x_1 \to Z_1$, $x_2 \to T_1$. Hence $g_1, \ldots, g_{N-2}$ is a regular sequence.

Since $\mathfrak{q}$ is prime and generated by a regular sequence of length $N-2$, $V(\mathfrak{q}) \subset \mathbb{A}^N$ is an integral surface whose ideal is $\mathfrak{q}$, and the smooth locus of $V(\mathfrak{q})$ is a dense open subset. Since $\Phi(x_i) = \operatorname{in}_\nu(\iota^* x_i) \neq 0$, no x_i belongs to $\mathfrak{q}$, so $V(\mathfrak{q}) \cap (\mathbb{C}^*)^N$ is a dense open subset of $V(\mathfrak{q})$. Thus there exists a point of $V(\mathfrak{q}) \cap (\mathbb{C}^*)^N$ at which $V(\mathfrak{q})$ is smooth, that is, at which the Jacobian matrix of $g_1, \ldots, g_{N-2}$ has rank $N-2$.

Step 5. In this step we construct $f_1, \ldots, f_{N-2}$ and prove (1).

Fix $1 \leq l \leq N-2$. Since $g_l \in \mathfrak{q}$, (18.12) implies $\iota^* g_l = 0$ or $\operatorname{ord}_\nu(\iota^* g_l) > d_l$. In the second case, **Step 3** yields a polynomial G_l with $\iota^* G_l = \iota^* g_l$ and $\operatorname{wt}_v(G_l) = \operatorname{ord}_\nu(\iota^* g_l) > d_l$. In the first case let $G_l := 0$. Let

$$f_l := \frac{1}{|H|} \sum_{\sigma \in H} \lambda(\sigma)^{-d_l} \sigma^*(g_l - G_l),$$

which is the sum of the terms of $g_l - G_l$ whose monomials have character λ^{d_l}. Since g_l is v-homogeneous of weight d_l and ι is H-equivariant, f_l is an H-semi-invariant polynomial with

$$\iota^* f_l = 0, \qquad \mathrm{in}_v(f_l) = g_l, \qquad \mathrm{wt}_v(f_l) = d_l.$$

In particular $f_l(0) = 0$, and by **Step 4**, $\mathrm{in}_v(f_1), \dots, \mathrm{in}_v(f_{N-2})$ form a regular sequence generating the prime ideal $\mathfrak{q}$, whose zero locus is smooth at a point of $(\mathbb{C}^*)^N$.

Let $\mathfrak{p} \subset \mathbb{C}[x_1, \dots, x_N]$ be the ideal of $\iota(T)$. It is generated by $\rho_i := x_{i+3} - q_i(x_1, x_2)$, $0 \le i \le m$, and it contains $f_1, \dots, f_{N-2}$. Let $\widehat{\mathcal{O}} := \mathbb{C}[[x_1, \dots, x_N]]$. Since $\iota^* x_i \in \mathfrak{m}$, ι^* extends to a continuous homomorphism $\hat{\iota}^* : \widehat{\mathcal{O}} \to \widehat{R}$ vanishing on $\mathfrak{p}\widehat{\mathcal{O}}$. We show that $\mathfrak{p}\widehat{\mathcal{O}} = (f_1, \dots, f_{N-2})\widehat{\mathcal{O}}$. Let $Q \in \mathfrak{p}\widehat{\mathcal{O}}$ be non-zero and $d := \mathrm{wt}_v(Q)$. Fix L with $e_\nu L \ge d+1$ and write $Q - \mathrm{in}_v(Q) = Q' + Q''$, where Q' is a polynomial whose monomials have v-weight larger than d and $Q'' \in (x_1, \dots, x_N)^L \widehat{\mathcal{O}}$. Then

$$\iota^* \mathrm{in}_v(Q) = -\iota^* Q' - \hat{\iota}^* Q'',$$

with $\iota^* Q' \in \mathfrak{a}_{d+1}$ by (18.11) and $\hat{\iota}^* Q'' \in \mathfrak{m}^L \widehat{R} \subset \mathfrak{a}_{d+1}$. Thus $\Phi(\mathrm{in}_v(Q)) = 0$, that is, $\mathrm{in}_v(Q) \in \mathfrak{q}$. Since $\mathfrak{q} = (g_1, \dots, g_{N-2})$, there exist v-homogeneous polynomials a_l of weight $d - d_l$ (with $a_l = 0$ if $d < d_l$) such that $\mathrm{in}_v(Q) = \sum_l a_l g_l$. Since $\mathrm{in}_v(f_l) = g_l$, $Q_1 := Q - \sum_l a_l f_l \in \mathfrak{p}\widehat{\mathcal{O}}$ satisfies $Q_1 = 0$ or $\mathrm{wt}_v(Q_1) > d$. Repeating with Q_1 in place of Q, we obtain v-homogeneous polynomials $a_{l,k}$ of weight at least $d + k - d_l$ (or zero) such that, for any K, $Q - \sum_l \sum_{k<K} a_{l,k} f_l$ is zero or has v-weight at least $d + K$. Since $v \in \mathbb{Z}^N_{>0}$, the total degree of $a_{l,k}$ tends to infinity with k, so $A_l := \sum_k a_{l,k} \in \widehat{\mathcal{O}}$. For any K,

$$Q - \sum_l A_l f_l = \Big(Q - \sum_l \sum_{k<K} a_{l,k} f_l\Big) - \sum_l \Big(\sum_{k \ge K} a_{l,k}\Big) f_l$$

is zero or has v-weight at least $d + K$. Thus $Q = \sum_l A_l f_l$.

For $Q \in \widehat{\mathcal{O}}$ with $Q(0) = 0$ let $\ell(Q) \in \bigoplus_i \mathbb{C} x_i$ be the linear part of Q. By the previous paragraph, $\rho_i = \sum_l A_{il} f_l$ for some $A_{il} \in \widehat{\mathcal{O}}$, and since $f_l \in \mathfrak{p}$, $f_l = \sum_i B_{li} \rho_i$ for some polynomials B_{li}. Taking linear parts,

$$\ell(\rho_i) = \sum_l A_{il}(0) \ell(f_l) = \sum_{l,i'} A_{il}(0) B_{li'}(0) \ell(\rho_{i'}).$$

Since $\ell(\rho_i) = x_{i+3} - \ell(q_i(x_1, x_2))$, $0 \le i \le m$, are linearly independent, the product of the square matrices $(A_{il}(0))$ and $(B_{li}(0))$ is the identity matrix. Thus $\det(B_{li})$ is a unit in $\mathcal{O}_{\mathbb{A}^N,0}$, and $\rho_0, \dots, \rho_m$ belong to $(f_1, \dots, f_{N-2})\mathcal{O}_{\mathbb{A}^N,0}$. Hence $f_1, \dots, f_{N-2}$ generate $\mathfrak{p}\mathcal{O}_{\mathbb{A}^N,0}$, so there exists an open neighborhood U of 0 in $\mathbb{A}^N$ such that $f_1, \dots, f_{N-2}$ generate the ideal of $\iota(T) \cap U$ in $\mathcal{O}_U$. This proves (1). Moreover $\ell(f_1), \dots, \ell(f_{N-2})$ are linearly independent.

Step 6. In this step we prove (3).

Let $X := \iota(T)$, which we identify with T through ι. Fix $P = (P_{ik}) \in \mathrm{GL}_N(\mathbb{Z})$ whose first column is v, which exists since $\gcd(v_1, \dots, v_N) = 1$. Let $Y := \mathbb{A}^1_s \times (\mathbb{C}^*)^{N-1}$ with coordinates $s, u_2, \dots, u_N$, let $E := (s = 0) \subset Y$, and let $\phi : Y \to \mathbb{A}^N$ be the morphism with

$$\phi^* x_i = s^{v_i} \prod_{k=2}^{N} u_k^{P_{ik}}, \qquad 1 \le i \le N,$$

which is a morphism because $v \in \mathbb{Z}^N_{>0}$. Then $\phi(E) = \{0\}$, and $\phi|_{Y \setminus E} : Y \setminus E \to (\mathbb{C}^*)^N$ is an isomorphism since $P \in \mathrm{GL}_N(\mathbb{Z})$. Let $\phi_1 : (\mathbb{C}^*)^{N-1} \to (\mathbb{C}^*)^N$, $u \to \phi(1, u)$. Then $\phi(s, u) = (s^{v_1}, \dots, s^{v_N}) \cdot \phi_1(u)$, so for any non-zero polynomial G on $\mathbb{A}^N$ with $d := \mathrm{wt}_v(G)$,

$$\phi^* G = s^d G^\phi \quad \text{with} \quad G^\phi \in \mathcal{O}(Y) \quad \text{and} \quad G^\phi(0, u) = \mathrm{in}_v(G)(\phi_1(u)). \tag{18.13}$$

A direct calculation shows

$$\bigwedge_i \phi^*(dx_i/x_i) = \det(P) ds/s \wedge du_2/u_2 \wedge \cdots \wedge du_N/u_N,$$

hence

$$\phi^*(dx_1 \wedge \cdots \wedge dx_N) = \epsilon s^{\sum_i v_i - 1} ds \wedge du_2 \wedge \cdots \wedge du_N, \tag{18.14}$$

where $\epsilon := \det(P) \prod_{k=2}^N u_k^{\sum_i P_{ik} - 1} \in \mathcal{O}(Y)^*$. In other words, the monomial prime divisor in $x_1, \ldots, x_N$ with weights v has log discrepancy $\sum_i v_i$ over $\mathbb{A}^N$ [CLS11, Lemma 11.4.10].

Let $\Gamma := \{u \in (\mathbb{C}^*)^{N-1} \mid g_l(\phi_1(u)) = 0 \text{ for } 1 \le l \le N-2\}$. Since $g_l(\phi(s,u)) = s^{d_l} g_l(\phi_1(u))$, the isomorphism $\phi|_{Y \setminus E}$ maps $\mathbb{C}^* \times \Gamma$ onto $V(\mathfrak{q}) \cap (\mathbb{C}^*)^N$. Thus Γ is irreducible, and by **Step 4** there exists $(s_0, \xi) \in \mathbb{C}^* \times \Gamma$ such that the Jacobian matrix of the functions $s^{d_l} g_l(\phi_1(u))$ with respect to $s, u_2, \ldots, u_N$ has rank $N-2$ at (s_0, ξ). Since $g_l(\phi_1(\xi)) = 0$, the derivatives with respect to s vanish at (s_0, ξ), so the Jacobian matrix of $g_1 \circ \phi_1, \ldots, g_{N-2} \circ \phi_1$ with respect to $u_2, \ldots, u_N$ has rank $N-2$ at ξ.

By **Step 5** and (18.13), $\phi^* f_l = s^{d_l} f_l^\phi$ with $f_l^\phi(0,u) = g_l(\phi_1(u))$. Hence the Jacobian matrix of $s, f_1^\phi, \ldots, f_{N-2}^\phi$ has rank $N-1$ at $(0,\xi)$. Fix an open neighborhood $V \subset Y$ of $(0,\xi)$ with $\phi(V) \subset U$, where U is as in **Step 5**, such that the Jacobian matrix of $s, f_1^\phi, \ldots, f_{N-2}^\phi$ has rank $N-1$ at any point of V and $X^\phi := V \cap V(f_1^\phi, \ldots, f_{N-2}^\phi)$ is irreducible. Then X^ϕ is a smooth surface, $F := E \cap X^\phi = \{(0,u) \in V \mid u \in \Gamma\}$ is a smooth irreducible curve, and $s|_{X^\phi}$ is a local equation of F on X^ϕ. On $V \setminus E$ we have $f_l^\phi = s^{-d_l} \phi^* f_l$, so $X^\phi \setminus E = \phi^{-1}(X \cap U) \cap (V \setminus E)$, which is a non-empty open subset of $\phi^{-1}(X \cap (\mathbb{C}^*)^N) \cong X \cap (\mathbb{C}^*)^N$. Thus ϕ restricts to a birational morphism $\phi_X : X^\phi \to X$, and F is a prime divisor over X with center 0.

We show that $\mathrm{ord}_F = \mathrm{ord}_\nu$ on $\mathbb{C}(X) = \mathbb{C}(T)$. Let $h \in R$ be non-zero, $d := \mathrm{ord}_\nu h$, and $\tilde{h}$ as in (2). By (18.12), $\mathrm{in}_v(\tilde{h}) \notin \mathfrak{q}$. Since $\mathfrak{q}$ is the ideal of $V(\mathfrak{q})$ and $V(\mathfrak{q}) \cap (\mathbb{C}^*)^N$ is dense in $V(\mathfrak{q})$, $\mathrm{in}_v(\tilde{h})$ does not vanish identically on $V(\mathfrak{q}) \cap (\mathbb{C}^*)^N$. Since $\mathrm{in}_v(\tilde{h})(\phi(s,u)) = s^d \,\mathrm{in}_v(\tilde{h})(\phi_1(u))$, the function $\mathrm{in}_v(\tilde{h}) \circ \phi_1$ does not vanish identically on Γ. As Γ is irreducible and F is isomorphic to an open subset of Γ, it does not vanish identically on F. By (18.13), $\phi_X^* h = (\phi^* \tilde{h})|_{X^\phi} = s^d \tilde{h}^\phi|_{X^\phi}$ with $\tilde{h}^\phi|_F = \mathrm{in}_v(\tilde{h}) \circ \phi_1|_F \not\equiv 0$. Thus $\mathrm{ord}_F(\phi_X^* h) = d$. Hence ord_F and ord_ν coincide on R, so on $\mathbb{C}(T)$, and $a(\nu, T, 0) = a(F, X, 0)$.

It remains to compute $a(F, X, 0)$. By **Step 5**, $df_1(0), \ldots, df_{N-2}(0)$ are linearly independent. Fix $1 \le a < b \le N$ such that $df_1(0), \ldots, df_{N-2}(0), dx_a(0), dx_b(0)$ is a basis of the cotangent space of $\mathbb{A}^N$ at 0. Shrinking U and V, there exists $\epsilon_0 \in \mathcal{O}(U)^*$ with

$$dx_1 \wedge \cdots \wedge dx_N = \epsilon_0 dx_a \wedge dx_b \wedge df_1 \wedge \cdots \wedge df_{N-2} \quad \text{on } U.$$

Let $\widetilde{\Omega} := \epsilon_0 dx_a \wedge dx_b$ and $\Omega_X := \widetilde{\Omega}|_X$. Since $x_a|_X, x_b|_X$ are local coordinates of X at 0, Ω_X generates ω_X near 0. Since $ds, df_1^\phi, \ldots, df_{N-2}^\phi$ are linearly independent at any point of V, after shrinking V there exists a 2-form $\widetilde{\Omega}'$ on V with

$$\widetilde{\Omega}' \wedge df_1^\phi \wedge \cdots \wedge df_{N-2}^\phi = ds \wedge du_2 \wedge \cdots \wedge du_N, \tag{18.15}$$

and $\Omega_{X^\phi} := \widetilde{\Omega}'|_{X^\phi}$ generates ω_{X^ϕ}. By (18.14), on V,

$$\phi^* \widetilde{\Omega} \wedge \bigwedge_{l=1}^{N-2} \left(s^{d_l} df_l^\phi + d_l s^{d_l - 1} f_l^\phi ds \right) = \phi^*(dx_1 \wedge \cdots \wedge dx_N) = \epsilon s^{\sum_i v_i - 1} ds \wedge du_2 \wedge \cdots \wedge du_N. \tag{18.16}$$

Let

$$\beta := \epsilon^{-1} s^{\sum_l d_l - \sum_i v_i + 1} \phi^* \widetilde{\Omega} - \widetilde{\Omega}',$$

a 2-form on $V \setminus E$. At any point $y \in X^\phi \setminus E$, all $f_l^\phi(y)$ vanish, so (18.15) and (18.16) imply $\beta \wedge df_1^\phi \wedge \cdots \wedge df_{N-2}^\phi = 0$ at y. Since $df_1^\phi(y), \ldots, df_{N-2}^\phi(y)$ are linearly independent and $\dim Y = N$, $\beta(y)$ belongs to the ideal of the exterior algebra generated by $df_1^\phi(y), \ldots, df_{N-2}^\phi(y)$, so β restricts to zero on the tangent space of X^ϕ at y. Thus

$$\phi_X^* \Omega_X = \epsilon|_{X^\phi} s^{\sum_i v_i - \sum_l d_l - 1} \Omega_{X^\phi}$$

on $X^\phi\backslash E$, hence as rational 2-forms on X^ϕ. Since ϵ is a unit and $s|_{X^\phi}$ is a local equation of F, the coefficient of F in $K_{X^\phi} - \phi_X^* K_X$ is $\sum_i v_i - \sum_l d_l - 1$, that is, $a(F, X, 0) = \sum_i v_i - \sum_l d_l$. Since $a(\nu, T, 0) = a(F, X, 0)$ and $d_l = \mathrm{wt}_v(f_l)$, this proves (3). □

18.4. **Split hypersurfaces with plane boundary.** In this subsection, we prove the ACC for the minimal log discrepancies in $(0, 1)$ of finite quotients of split cA hypersurfaces $xy + g(z, t) = 0$ with boundary defined by polynomials in z, t (Theorem 18.8).

Theorem 18.8. *Let $s \geq 0$ be an integer and $\Gamma \subset [0, 1]$ a DCC set, let $\epsilon := \gamma(\Gamma)$ if $\Gamma\backslash\{0\} \neq \emptyset$ and $\epsilon := 1$ otherwise, and let $M := 2 + \lceil 2/\epsilon \rceil$. Let $X := (xy + g(z, t) = 0) \subset \mathbb{A}^4$ with $0 \neq g \in (z, t)^2$, let G be a finite diagonal abelian group preserving the equation up to a character and acting faithfully on X, let $h_1, \dots, h_s$ be non-zero G-semi-invariant polynomials in z, t, let $c_1, \dots, c_s \in \Gamma$, and let $B := \sum_j c_j \operatorname{div}_X(h_j)$. Assume that X is a normal klt threefold germ at 0 and that (X, B) is klt. Let $\tau : X \to X/G$ be the quotient morphism and $T := \mathbb{A}^2_{z,t}$. Then:*

(1) for fixed s and Γ, the set of the numbers $\mathrm{mld}((X \ni 0, B)/G) \in (0, 1)$, where X, G, $h_1, \dots, h_s$, and $c_1, \dots, c_s$ vary, satisfies the ACC;

(2) if $a := \mathrm{mld}((X \ni 0, B)/G) \in (0, 1)$, $\bar{F}$ is a prime divisor over X/G with $\mathrm{center}_{X/G} \bar{F} = \{\tau(0)\}$ and $a(\bar{F}, X/G, B_{X/G}) = a$, and F is a prime divisor over X with $\mathrm{ord}_F|_{\mathbb{C}(X/G)} \in \mathbb{N}^+ \mathrm{ord}_{\bar{F}}$, then $\mathrm{mld}((X \ni 0, B)/I_G(F)) = a$;

(3) for a and F as in (2), the prime divisor ν over T with $\mathrm{ord}_F|_{\mathbb{C}(T)} \in \mathbb{N}^+ \mathrm{ord}_\nu$ (Lemma 18.2(2)) has a minimal generating sequence with at most $M + 4$ elements;

(4) for a and F as in (2), $a = \psi_\chi(X', B')$ for some data n, H, X', B', and χ as in Set-up 18.4 such that $n + 1 \leq M + 9$, X' is defined by at most $M + 5$ equations, and B' is defined by at most s polynomials with coefficients in Γ.

Proof of Theorem 18.8. The proof is long, so we divide it into several steps.

Step 1. In this step we fix $\bar{F}$ and F as in (2), prove (2), and bound the multiplicity at 0 of the plane boundary.

Recall $T = \mathbb{A}^2_{z,t}$, and let $B_T := \sum_j c_j \operatorname{div}_T(h_j)$. Since G is diagonal, G acts on T through the coordinates z, t, and the projection $X \to T$, $(x, y, z, t) \to (z, t)$, is G-equivariant. For a non-zero rational function or rational differential form u on X or on T which is semi-invariant under a subgroup of G, we denote by χ_u the character of u (Definition 2.14). Since G is diagonal and preserves $xy + g$ up to a character, g is G-semi-invariant and $\chi_g = \chi_x\chi_y = \chi_{xy+g}$.

Assume that $a := \mathrm{mld}((X \ni 0, B)/G) \in (0, 1)$. By Lemma 2.6(0), $K_X + B = \tau^*(K_{X/G} + B_{X/G})$, and $(X/G, B_{X/G})$ is klt since (X, B) is klt [KM98, Proposition 5.20]. The infimum defining a is attained, since $(X/G, B_{X/G})$ is klt near $\tau(0)$ and a can be computed on a log resolution of $(X/G, B_{X/G})$ on which the fiber over $\tau(0)$ is a divisor ([KM98, Theorem 0.2] and Lemma 2.13). Thus, by [Amb99, Section 1, the second paragraph after Definition 1.1], there exists a prime divisor $\bar{F}$ over X/G with $\mathrm{center}_{X/G} \bar{F} = \{\tau(0)\}$ and $a(\bar{F}, X/G, B_{X/G}) = a$, and by Lemma 2.6(3) there exists a prime divisor F over X with $\mathrm{ord}_F|_{\mathbb{C}(X/G)} \in \mathbb{N}^+ \mathrm{ord}_{\bar{F}}$. Fix such $\bar{F}$ and F. Since G fixes 0, we have $\tau^{-1}(\tau(0)) = \{0\}$, so $\mathrm{center}_X F = \{0\}$ by Lemma 2.6(2). Let $I := I_G(F)$ and $e := |I|$. By Lemma 2.6(1)(2), I is cyclic, the character λ of I on $\mathfrak{m}_F/\mathfrak{m}_F^2$ is faithful, and $a = a(F, X, B)/e < 1$. Since G preserves $xy + g$ up to a character, G preserves X, so Proposition 18.3 and Lemma 18.2 apply to X, G, and B. By Proposition 18.3, there exist a prime divisor ν over T with center 0 and coprime positive integers ϱ, k with $k < \varrho \mathrm{ord}_\nu g$ such that ord_F is the Gauss extension of $\varrho \mathrm{ord}_\nu$ to $\mathbb{C}(X) = \mathbb{C}(T)(x)$ with x-value k (Definition-Proposition 2.11), $\mathrm{ord}_F y = \varrho \mathrm{ord}_\nu g - k$, and

$$a(F, X, B) = \varrho b_\nu, \qquad b_\nu := a(\nu, T, B_T) > 0, \qquad a = \frac{\varrho b_\nu}{e}. \tag{18.17}$$

Since $\operatorname{ord}_F|_{\mathbb{C}(T)} = \varrho \operatorname{ord}_\nu$, we have $\operatorname{ord}_F h_j = \varrho \operatorname{ord}_\nu h_j$ for any j, and hence

$$\begin{aligned} a(F,X,0) &= a(F,X,B) + \sum_j c_j \operatorname{ord}_F h_j \\ &= \varrho\Big(a(\nu,T,B_T) + \sum_j c_j \operatorname{ord}_\nu h_j\Big) = \varrho a(\nu,T,0). \end{aligned} \tag{18.18}$$

Since $\operatorname{ord}_F|_{\mathbb{C}(T)} = \varrho \operatorname{ord}_\nu$, the prime divisor ν is the one in (3). By Lemma 18.2(2), $I \subset J_\nu$ for the group J_ν of Lemma 18.2, that is, I fixes ord_ν and acts trivially on the residue field of ord_ν.

We prove (2). We have $\operatorname{mld}(X/G \ni \tau(0), B_{X/G}) = a(\bar{F}, X/G, B_{X/G}) = a > 0$. By Lemma 3.1, applied to the pair $(X/G, B_{X/G})$, the point $\tau(0)$, the prime divisor $\bar{F}$, the Galois morphism τ with group G (the crepant pullback of $K_{X/G} + B_{X/G}$ is $K_X + B$ by Lemma 2.6(0)), the prime divisor F, the subgroup $H := I$, and the identity morphism of X (after shrinking X to an I-invariant affine neighborhood of 0), we obtain $\operatorname{mld}((X \ni 0, B)/I) = a$. Thus (2) holds.

Next we show that $\operatorname{mult}_P B_T < 1$ for any prime divisor P on T with $0 \in P$. Let ϕ be an irreducible polynomial with $P = (\phi = 0)$. If $\phi \nmid g$, then $X \to T$ is smooth over the generic point of P, where the fiber is the irreducible conic $xy = -g$, and the fibers over the finitely many points of P with $g = 0$ are curves. Hence $(\phi = 0) \subset X$ is a prime divisor P_X with $0 \in P_X$ and $\operatorname{ord}_{P_X} \phi = 1$. If $\phi \mid g$, let $P_X := (x = \phi = 0) \subset X$. Near the generic point of P_X, the function y is a unit and X is the graph of $x = -g/y$ over $\mathbb{A}^3_{y,z,t}$, so ϕ is a regular parameter of $\mathcal{O}_{X,P_X}$. In both cases $\operatorname{ord}_{P_X}|_{\mathbb{C}(T)} = \operatorname{ord}_P$, so $\operatorname{mult}_{P_X} B = \operatorname{mult}_P B_T$. Since (X,B) is klt near 0 and $0 \in P_X$, we obtain $\operatorname{mult}_P B_T < 1$.

Finally, let E_0 be the exceptional curve of the blow-up of $0 \in T$. By [KM98, Lemma 2.29], $a(E_0, T, B_T) = 2 - \sum_j c_j \operatorname{mult}_0 h_j$, which is positive by Proposition 18.3. Let $C \subset T$ be the reduced curve $(zt \prod_{j:c_j>0} h_j = 0)$. Since G is diagonal and $h_1, \dots, h_s$ are G-semi-invariant, C is G-invariant. If $c_j > 0$, then $\Gamma\backslash\{0\} \neq \emptyset$, and $c_j \geq \epsilon > 0$ by Lemma 2.10(2). Thus

$$\operatorname{mult}_0 C \leq 2 + \sum_{j:c_j>0} \operatorname{mult}_0 h_j \leq 2 + \frac{1}{\epsilon}\sum_j c_j \operatorname{mult}_0 h_j < 2 + \frac{2}{\epsilon} \leq M. \tag{18.19}$$

Step 2. In this step we resolve C equivariantly and show that, if the center of ν on the resolution is a closed point p, then ν is the exceptional curve of a sequence of blow-ups starting with the blow-up of p, each centered at an intersection point of two boundary curves.

Let $\mathcal{K}$ be the set of all proper or infinitely near singular points of the germ $(C,0)$ in the sense of [Cas00, Section 3.8, p. 93], that is, of the points on C that are multiple on C, are satellite points, or precede a satellite point on C. By the definition in [Cas00, Section 3.8, p. 93] and [Cas00, Corollary 3.5.4], any point that precedes a point of $\mathcal{K}$ belongs to $\mathcal{K}$, so $\mathcal{K}$ is a cluster. Let $\pi : T' \to T$ be the composition of the blow-ups of the points of $\mathcal{K}$. By [Cas00, Section 4.3, p. 134], π is an embedded resolution of $(C,0)$, that is, the reduced total transform $\pi^{-1}(C)$ is an snc divisor near $\pi^{-1}(0)$, and π does not depend on the order of the blow-ups. The set $\mathcal{K}$ is determined by C, and C is I-invariant. Thus the action of I on T lifts to T' and π is I-equivariant. Since $(zt = 0) \subset C$, the point 0 is a singular point of C, so $\pi^{-1}(0)$ is a divisor. Any point of $\mathcal{K}$ is a point of C, that is, it is contained in the strict transform of C on the surface where it is blown up [Cas00, Section 3.8, p. 93]. For $\mathfrak{q} \in \mathcal{K}$, let $E_\mathfrak{q} \subset T'$ be the strict transform of the exceptional curve of the blow-up of $\mathfrak{q}$. If $E_\mathfrak{q} \cap E_{\mathfrak{q}'} \neq \emptyset$ for $\mathfrak{q} \neq \mathfrak{q}'$, then one of $\mathfrak{q}, \mathfrak{q}'$ is infinitely near to the other, since otherwise $E_\mathfrak{q}$ and $E_{\mathfrak{q}'}$ are mapped to distinct points of some intermediate surface.

By **Step 1**, I fixes ord_ν, so the center of ν on T' is I-invariant, and it is contained in $\pi^{-1}(0)$. Thus either

(a) the center of ν on T' is a curve, and $\nu = E_\mathfrak{q}$ for some $\mathfrak{q} \in \mathcal{K}$, or
(b) the center of ν on T' is a closed point $p \in \pi^{-1}(0)$ fixed by I.

Assume that we are in case (b). Since $\pi^{-1}(C)$ is snc near p and $p \in \pi^{-1}(0)$, at most two components of $\pi^{-1}(C)$ pass through p, and at least one of them is exceptional. Any $\sigma \in I$ maps

a component through p to a component through p, preserves the exceptional locus of π, and maps $E_{\mathfrak{q}}$ to $E_{\sigma(\mathfrak{q})}$, where $\sigma(\mathfrak{q}) \in \mathcal{K}$ is infinitely near to exactly as many points of $\mathcal{K}$ as $\mathfrak{q}$. If $E_{\mathfrak{q}}$ and $E_{\mathfrak{q}'}$ both pass through p, then one of $\mathfrak{q}, \mathfrak{q}'$ is infinitely near to the other, so the numbers of points of $\mathcal{K}$ to which $\mathfrak{q}$ and $\mathfrak{q}'$ are infinitely near differ, and σ does not exchange $E_{\mathfrak{q}}$ and $E_{\mathfrak{q}'}$. Hence I preserves each component of $\pi^{-1}(C)$ through p. Since I is a finite abelian group fixing p, there exist I-eigenparameters u_1, u_2 at p such that $(u_1 = 0)$ is exceptional and any component of $\pi^{-1}(C)$ through p is $(u_1 = 0)$ or $(u_2 = 0)$, since the Reynolds projection of a local equation of an I-invariant smooth curve onto the character of I on the conormal line of the curve is again a local equation, and, for an I-eigenvector $\bar{u}$ of $\mathfrak{m}_p/\mathfrak{m}_p^2$ complementary to the conormal line, the Reynolds projection of a lift of $\bar{u}$ onto the character of $\bar{u}$ is an I-semi-invariant regular parameter.

Let $B_{T'}$ be the $\mathbb{R}$-divisor with $K_{T'} + B_{T'} = \pi^*(K_T + B_T)$. Near p, the support of $B_{T'}$ and the support of $K_{T'} - \pi^* K_T$ are contained in $\pi^{-1}(C)$, hence in $(u_1 u_2 = 0)$. Thus there exist integers $r_1, r_2 \geq 0$ and a regular function U near p with $U(p) \neq 0$ such that

$$\pi^*(dz \wedge dt) = U u_1^{r_1} u_2^{r_2} du_1 \wedge du_2, \qquad B_{T'} = (1 - b_1)\operatorname{div}(u_1) + (1 - b_2)\operatorname{div}(u_2) \text{ near } p,$$

where $b_i := 1 - \operatorname{mult}_{(u_i=0)} B_{T'}$. We show that $b_1, b_2 > 0$. If $(u_i = 0)$ is exceptional over T, then the center of $(u_i = 0)$ on T is 0 and $b_i = a((u_i = 0), T, B_T) > 0$ by Proposition 18.3. Otherwise $(u_i = 0)$ is the strict transform of a curve P through 0, and $b_i = 1 - \operatorname{mult}_P B_T > 0$ by **Step 1**. Let $d_i := \operatorname{ord}_\nu u_i \geq 1$. Since $a(\nu, T, B_T) = a(\nu, T', B_{T'})$ and $K_{T'} - \pi^* K_T = r_1 \operatorname{div}(u_1) + r_2 \operatorname{div}(u_2)$ near p,

$$b_\nu = b_1 d_1 + b_2 d_2 + \big(a(\nu, T', 0) - d_1 - d_2\big), \qquad a(\nu, T, 0) = a(\nu, T', 0) + r_1 d_1 + r_2 d_2. \tag{18.20}$$

The integer $a(\nu, T', 0) - d_1 - d_2$ equals $a(\nu, T', \operatorname{div}(u_1) + \operatorname{div}(u_2))$, which is non-negative by Lemma 2.13.

We show that e divides $\varrho(a(\nu, T', 0) - d_1 - d_2)$. By Lemma 9.2(1), applied to the group I acting on X, any non-zero I-semi-invariant $h \in \mathbb{C}(T) \subset \mathbb{C}(X)$ satisfies $\chi_h = \lambda^{\operatorname{ord}_F h} = \lambda^{\varrho \operatorname{ord}_\nu h}$. In particular $\chi_{u_i} = \lambda^{\varrho d_i}$, and $\chi_{du_1 \wedge du_2} = \chi_{u_1}\chi_{u_2}$. Since π is I-equivariant and $dz \wedge dt$, u_1, and u_2 are I-semi-invariant, the forms $\pi^*(dz \wedge dt)$ and $u_1^{r_1} u_2^{r_2} du_1 \wedge du_2$ are I-semi-invariant, and so is their ratio U. The rational function U is regular and non-zero at the fixed point p, so $\chi_U(\sigma)U(p) = (\sigma^* U)(p) = U(\sigma(p)) = U(p) \neq 0$ for any $\sigma \in I$, and χ_U is trivial. Since π is I-equivariant,

$$\chi_z \chi_t|_I = \chi_{dz \wedge dt}|_I = \chi_U \chi_{u_1}^{r_1+1} \chi_{u_2}^{r_2+1} = \lambda^{\varrho((r_1+1)d_1 + (r_2+1)d_2)}.$$

On the other hand, $\chi_{xy+g} = \chi_x \chi_y$, so Lemma 9.2(2), applied to the group I, the eigenparameters x, y, z, t of $\mathbb{A}^4$ at 0, and $xy + g$, implies that $\chi_z \chi_t|_I = \lambda^{a(F,X,0)}$. Since λ is a faithful character of the cyclic group I of order e, we deduce from (18.18) and (18.20) that e divides

$$\begin{aligned} a(F, X, 0) - \varrho\big((r_1 + 1)d_1 + (r_2 + 1)d_2\big) &= \varrho\big(a(\nu, T, 0) - (r_1 + 1)d_1 - (r_2 + 1)d_2\big) \\ &= \varrho\big(a(\nu, T', 0) - d_1 - d_2\big). \end{aligned}$$

By (18.17) and (18.20),

$$a = \frac{\varrho(b_1 d_1 + b_2 d_2)}{e} + \frac{\varrho(a(\nu, T', 0) - d_1 - d_2)}{e},$$

where the first summand is positive and the second summand is a non-negative integer. Since $a < 1$, the second summand is zero. Thus

$$a\big(\nu, T', \operatorname{div}(u_1) + \operatorname{div}(u_2)\big) = 0. \tag{18.21}$$

Let $T'_0 \subset T'$ be a neighborhood of p on which u_1, u_2 are regular and $D_0 := \operatorname{div}(u_1) + \operatorname{div}(u_2)$ is snc, and let $p_0 := p$. Inductively, as long as the center of ν on T'_i is a closed point p_i, let $T'_{i+1} \to T'_i$ be the blow-up of p_i, let E'_{i+1} be the exceptional curve, and let D_{i+1} be the sum of the strict transform of D_i and E'_{i+1}. By [KM98, Lemma 2.45], there exists $\kappa \geq 0$ such that the center of ν on T'_κ is a curve. We show by induction that, for any $0 \leq i < \kappa$,

$$p_i \text{ is the intersection point of two components of } D_i, \qquad a(\nu, T'_i, D_i) = 0. \tag{18.22}$$

For $i = 0$, the first part of (18.22) holds since $u_1(p) = u_2(p) = 0$, and the second part is (18.21). Assume (18.22) for some $i < \kappa - 1$. Since $\operatorname{mult}_{p_i} D_i = 2$, the divisor $K_{T'_{i+1}} + D_{i+1}$ is the pullback of $K_{T'_i} + D_i$, so $a(\nu, T'_{i+1}, D_{i+1}) = 0$. The point p_{i+1} is contained in E'_{i+1}. If p_{i+1} were contained in no other component of D_{i+1}, then, for a local equation ξ_1 of E'_{i+1} and a regular parameter ξ_2 at p_{i+1} with (ξ_1, ξ_2) a regular system of parameters, Lemma 2.13 would imply that $a(\nu, T'_{i+1}, D_{i+1}) \geq \operatorname{ord}_\nu \xi_2 \geq 1$, a contradiction. Thus (18.22) holds for $i + 1$. In particular, $\nu = E'_\kappa$ is the exceptional curve of the blow-up of $p_{\kappa-1}$, and each p_i with $0 \leq i < \kappa$ is the intersection point of two components of D_i. Since the exceptional curves over T through p are among the components of D_0, any exceptional curve over T through p_i is one of the components of D_i through p_i.

Step 3. In this step we prove (3). More precisely, we show that ν has a minimal generating sequence with at most $M + 2$ elements.

For a prime divisor E over T with center 0, let θ_E be the minimal resolution of ord_E [dFGPM23, Definition 2.8]: the composition of the blow-ups of the centers of ord_E, starting with the blow-up of 0, until the center of ord_E is a curve. Any blow-up in θ_E after the first is centered at a point of the exceptional curve created by the previous blow-up.

We call such a blow-up *free* if the center is contained in exactly one exceptional curve, and *satellite* if the center is the intersection point of two exceptional curves. Let $L(E) := 2$ if θ_E is the blow-up of 0, and otherwise let $L(E)$ be the number of vertices of valence one in the dual graph of the exceptional curves of θ_E. We use the following four facts.

(i) The valuation ord_E has a minimal generating sequence with $L(E)$ elements. If θ_E is not the blow-up of 0, then by [dFGPM23, Definition 2.11 and Proposition 2.12] a minimal generating sequence of ord_E consists of the equations of one curvetta [dFGPM23, Definition 2.5] for each vertex of valence one of the dual graph of θ_E. If θ_E is the blow-up of 0, then $\operatorname{ord}_E = \operatorname{ord}_0$, and z, t form a minimal generating sequence [dFGPM23, Example 2.13].

(ii) Let $\tilde{E}$ be a prime divisor over T such that $\theta_{\tilde{E}}$ is θ_E followed by one blow-up at a point of E. A free blow-up adds a vertex adjacent only to E, and a satellite blow-up at the intersection point of E and another exceptional curve $\hat{E}$ replaces the edge between E and $\hat{E}$ by a new vertex adjacent to both. Thus $L(\tilde{E}) = L(E) + 1$ if the blow-up is free and E has valence at least two, and $L(\tilde{E}) = L(E)$ otherwise. In particular, L does not decrease along a minimal resolution, and a sequence of consecutive free blow-ups increases L by at most one, since after the first of them the blow-up is centered on a vertex of valence one.

(iii) Let A be a singular branch at 0, and let E be a prime divisor over T such that A is a curvetta of E at θ_E and θ_E is the minimal embedded resolution of A followed by $\ell_E \geq 0$ blow-ups at the points of the strict transforms of A. By [dFGPM23, Proposition 2.14] and (i), $L(E) = g(A) + 1$ if $\ell_E = 0$, and $L(E) = g(A) + 2$ if $\ell_E > 0$, where $g(A) + 1$ is the number of minimal generators of the semigroup of A [dFGPM23, Notation 2.2].

(iv) For any branch A of C at 0, $g(A) \leq M - 1$. Let $\bar{\beta}_0 < \cdots < \bar{\beta}_{g(A)}$ be the minimal generators of the semigroup of A, $e_0 := \bar{\beta}_0$, $e_j := \gcd(e_{j-1}, \bar{\beta}_j)$, and $n_j := e_{j-1}/e_j$. By [dFGPM23, Notation 2.2 and Proposition 2.3(1)], $n_j > 1$ for any j, so $2^{g(A)} \leq n_1 \cdots n_{g(A)} = e_0/e_{g(A)} \leq \bar{\beta}_0$. For a general linear form ℓ in z, t, [Cas00, Proposition 2.6.7] and (18.19) imply, since $\bar{\beta}_0 = \operatorname{mult}_0 A$ is the least positive element of the semigroup of A [Cas00, Remark 5.8.4], that

$$\bar{\beta}_0 \leq (Z(\ell) \cdot A)_0 \leq (Z(\ell) \cdot C)_0 = \operatorname{mult}_0 C \leq M.$$

Thus $g(A) + 1 \leq 2^{g(A)} \leq M$.

In case (a), let $\mathfrak{q} \in \mathcal{K}$ with $\nu = E_\mathfrak{q}$. In case (b), let $\mathfrak{q} \in \mathcal{K}$ be a point such that $E_\mathfrak{q}$ passes through p, and such that $\mathfrak{q}$ is infinitely near to $\mathfrak{q}'$ if a second exceptional curve $E_{\mathfrak{q}'}$ passes through p. In both cases let $E := E_\mathfrak{q}$. The centers of the blow-ups in θ_E are points $\mathfrak{q}_0 = 0, \mathfrak{q}_1, \ldots, \mathfrak{q}_j = \mathfrak{q}$

of $\mathcal{K}$, where $\mathfrak{q}_0, \dots, \mathfrak{q}_{j-1}$ are the points to which $\mathfrak{q}$ is infinitely near. Since $\mathfrak{q}$ is a point of C, there exists a branch A of C at 0 passing through $\mathfrak{q}$, and $\mathfrak{q}_0, \dots, \mathfrak{q}_j$ are the first $j+1$ points of A infinitely near to 0, starting with 0.

We show that $L(E) \le M+1$ by distinguishing three cases.

- If A is smooth, then after any blow-up at a point of the strict transform of A, the intersection number of the strict transform of A with the reduced exceptional curve is 1. Thus any point of A infinitely near to 0 is contained in exactly one exceptional curve, all blow-ups in θ_E after the first are free and centered on the exceptional curve created last, the dual graph of θ_E is a chain, and $L(E) = 2$.
- If A is singular and $\mathfrak{q}$ is blown up in the minimal embedded resolution of A, let E_A be the last exceptional curve of the minimal embedded resolution of A. The strict transform of A on the minimal embedded resolution intersects the exceptional curve transversally at one point of E_A, so A is a curvetta of E_A at θ_{E_A}, and θ_{E_A} is θ_E followed by further blow-ups, so $L(E) \le L(E_A) = g(A)+1 \le M$ by (ii), (iii) with $\ell_{E_A} = 0$, and (iv).
- If A is singular and $\mathfrak{q}$ is not blown up in the minimal embedded resolution of A, then θ_E is the minimal embedded resolution of A followed by $\ell_E > 0$ blow-ups at the points of the strict transforms of A. As in the previous case, the strict transform of A on the minimal embedded resolution intersects the exceptional curve transversally at one point of the last exceptional curve, and after any further blow-up at the point of the strict transform of A, the strict transform of A intersects the exceptional curve transversally at one point of the exceptional curve created last. Thus A is a curvetta of E at θ_E. By (iii) and (iv), $L(E) = g(A)+2 \le M+1$.

In case (a), $L(\nu) = L(E) \le M+1$. Assume that we are in case (b). Order the blow-ups of π so that $\mathfrak{q}_0, \dots, \mathfrak{q}_j$ come first, let T_E be the surface obtained by blowing up $\mathfrak{q}_0, \dots, \mathfrak{q}_j$, and let $p_E \in E$ be the image of p. The induced morphism $T' \to T_E$ is the composition of the blow-ups of the remaining points of $\mathcal{K}$. If one of the remaining points of $\mathcal{K}$ were equal or infinitely near to p_E, then p would be contained in a curve $E_{\mathfrak{q}''}$ with $\mathfrak{q}''$ infinitely near to $\mathfrak{q}$, contrary to the choice of $\mathfrak{q}$. Thus $T' \to T_E$ is an isomorphism near p.

Hence θ_ν is θ_E followed by the blow-ups of $p_0, \dots, p_{\kappa-1}$ of **Step 2**, where we identify a neighborhood of p in T' with a neighborhood of p_E in T_E. By the last sentence of **Step 2**, the blow-up of p_i is satellite if both components of D_i through p_i are exceptional, and free otherwise.

- If $(u_2 = 0)$ is exceptional, then all components of D_i are exceptional for any i, all blow-ups of $p_0, \dots, p_{\kappa-1}$ are satellite, and $L(\nu) = L(E)$ by (ii).
- If $(u_2 = 0)$ is not exceptional, then $(u_1 = 0) = E$. If p_i is not contained in the strict transform of $(u_2 = 0)$, then neither is p_{i+1}, since p_{i+1} is an intersection point of E'_{i+1} and the strict transform of a component of D_i through p_i. Thus there exists $0 \le i_0 \le \kappa$ such that the blow-ups of $p_0, \dots, p_{i_0-1}$ are free and the blow-ups of $p_{i_0}, \dots, p_{\kappa-1}$ are satellite. By (ii), $L(\nu) \le L(E)+1$.

In all cases $L(\nu) \le M+2$. By (i), ν has a minimal generating sequence with at most $M+2 \le M+4$ elements, so (3) holds.

Step 4. In this step we embed X into an affine space in which F becomes a toric divisor, and prove (4).

Recall from **Step 1** that $\operatorname{ord}_F|_{\mathbb{C}(T)} = \varrho \operatorname{ord}_\nu$, $\operatorname{ord}_F x = k$, $\operatorname{ord}_F y = \varrho \operatorname{ord}_\nu g - k$, and that $I = I_G(F)$ fixes ord_ν and acts trivially on the residue field of ord_ν. The group I is cyclic and acts diagonally on T. By (3), ν has a minimal generating sequence with $m+1 \le M+4$ elements for some integer $m \ge 0$. Let $N := m+3 \le M+6$. By Proposition 18.7 applied to $H := I$, there exist I-semi-invariant polynomials $q_0, \dots, q_m \in \mathbb{C}[z,t]$, the closed embedding $\iota = (z, t, q_0, \dots, q_m) : T \to \mathbb{A}^N$, I-semi-invariant polynomials $f_1, \dots, f_{N-2}$ on $\mathbb{A}^N$, and $v \in \mathbb{Z}^N_{>0}$ with $\gcd(v_1, \dots, v_N) = 1$ and $v = (\operatorname{ord}_\nu z, \operatorname{ord}_\nu t, \operatorname{ord}_\nu q_0, \dots, \operatorname{ord}_\nu q_m)$, satisfying (1)–(3) of Proposition 18.7. Since $\operatorname{ord}_\nu q_i > 0$ and the center of ν is 0, we have $q_i(0) = 0$ for any i, so

$\iota(0) = 0$. By Proposition 18.7(2), there exist an I-semi-invariant polynomial $\tilde{g}$ on $\mathbb{A}^N$ with $\chi_{\tilde{g}} = \chi_g$, $\tilde{g} \circ \iota = g$, and $\mathrm{wt}_v(\tilde{g}) = \mathrm{ord}_\nu g$, and, for any j with $c_j > 0$, an I-semi-invariant polynomial $\tilde{h}_j$ on $\mathbb{A}^N$ with $\chi_{\tilde{h}_j} = \chi_{h_j}$, $\tilde{h}_j \circ \iota = h_j$, and $\mathrm{wt}_v(\tilde{h}_j) = \mathrm{ord}_\nu h_j$.

Let $n := N+2$, and let $x_1 := x$, $x_2 := y$, and $x_3, \dots, x_n$ be the coordinates of $\mathbb{A}^n = \mathbb{A}^2_{x,y} \times \mathbb{A}^N$, with $x_3, \dots, x_n$ the coordinates of $\mathbb{A}^N$. We regard $f_1, \dots, f_{N-2}$, $\tilde{g}$, and $\tilde{h}_j$ as polynomials on $\mathbb{A}^n$ not involving x, y. Let $f_{N-1} := xy + \tilde{g}$, $r := N-1$, $X' := (f_1 = \cdots = f_r = 0) \subset \mathbb{A}^n$, and $B' := \sum_{j:c_j>0} c_j \,\mathrm{div}(\tilde{h}_j|_{X'})$. The group I acts diagonally on $\mathbb{A}^n$ through χ_x, χ_y and the characters of $z, t, q_0, \dots, q_m$, and $f_1, \dots, f_r$, $\tilde{h}_j$ are I-semi-invariant. By Proposition 18.7(1), $f_1, \dots, f_{N-2}$ generate the ideal of $\mathbb{A}^2_{x,y} \times \iota(T)$ near 0, and f_{N-1} restricts to $xy + g$ on $\mathbb{A}^2_{x,y} \times \iota(T) \cong \mathbb{A}^4$. Thus the I-equivariant closed embedding $\mathrm{id}_{\mathbb{A}^2} \times \iota : \mathbb{A}^4 \to \mathbb{A}^n$ maps (X, B) isomorphically onto (X', B') near 0. The scheme X' may have other components away from 0, which does not matter since Set-up 18.4(1) concerns X' near 0 only. Let $\chi := \lambda$. Then n, r, $H := I$, $x_1, \dots, x_{n+1}$, X', $f_1, \dots, f_r$, the polynomials $\tilde{h}_j$ with $c_j > 0$, their coefficients c_j, B', and χ satisfy the conditions of Set-up 18.4, with $n \geq 5$, $2 \leq r < n$, $n+1 = N+3 \leq M+9$, $r = N-1 \leq M+5$, and at most s polynomials $\tilde{h}_j$ with coefficients $c_j \in \Gamma$.

We verify the hypotheses of Lemma 18.6 for F, regarded as a prime divisor over X' with center 0. We have $I_I(F) = I$, and $a(F, X', B')/|I| = a(F, X, B)/e = a = \mathrm{mld}((X' \ni 0, B')/I)$ by (2). The character of I on $\mathfrak{m}_F/\mathfrak{m}_F^2$ is $\lambda = \chi$. Let

$$\tilde{w} := (k,\ \varrho\,\mathrm{ord}_\nu g - k,\ \varrho v_1, \dots, \varrho v_N) \in \mathbb{Z}^n_{>0}.$$

Since $\gcd(v_1, \dots, v_N) = 1$, the greatest common divisor of $\varrho v_1, \dots, \varrho v_N$ is ϱ, and $\gcd(\varrho, k) = 1$, so $\gcd(\tilde{w}_1, \dots, \tilde{w}_n) = 1$. By the recalled values of $\mathrm{ord}_F x$, $\mathrm{ord}_F y$, and $\mathrm{ord}_F|_{\mathbb{C}(T)}$, and by Proposition 18.7(4), $\mathrm{ord}_F x_i = \tilde{w}_i$ for $1 \leq i \leq n$. For a polynomial h on $\mathbb{A}^N$, regarded on $\mathbb{A}^n$, we have $\mathrm{wt}_{\tilde{w}}(h) = \varrho\,\mathrm{wt}_v(h)$. Hence, for any j with $c_j > 0$,

$$\mathrm{ord}_F \tilde{h}_j = \mathrm{ord}_F h_j = \varrho\,\mathrm{ord}_\nu h_j = \mathrm{wt}_{\tilde{w}}(\tilde{h}_j),$$

$\mathrm{wt}_{\tilde{w}}(f_l) = \varrho\,\mathrm{wt}_v(f_l)$ for $l \leq N-2$, and, since the monomials of $\tilde{g}$ do not involve x, y,

$$\mathrm{wt}_{\tilde{w}}(f_{N-1}) = \min\{\tilde{w}_1 + \tilde{w}_2, \varrho\,\mathrm{wt}_v(\tilde{g})\} = \varrho\,\mathrm{ord}_\nu g.$$

By Proposition 18.7(3) and (18.18),

$$\begin{aligned}\sum_{i=1}^{n} \tilde{w}_i - \sum_{l=1}^{r} \mathrm{wt}_{\tilde{w}}(f_l) &= \varrho\,\mathrm{ord}_\nu g + \varrho\Big(\sum_{i=1}^{N} v_i - \sum_{l=1}^{N-2} \mathrm{wt}_v(f_l)\Big) - \varrho\,\mathrm{ord}_\nu g \\ &= \varrho a(\nu, T, 0) = a(F, X, 0) = a(F, X', 0).\end{aligned}$$

Thus Lemma 18.6 implies that $a = \mathrm{mld}((X' \ni 0, B')/I) = \psi_\chi(X', B')$. Thus (4) holds.

Step 5. In this step we prove (1), and show that it also holds when s varies.

Let $\Lambda := \Gamma\backslash\{0\}$, a DCC subset of $(0, +\infty)$, and let

$$n_0 := M+8, \qquad r_0 := M+5, \qquad s_0 := \max\{s, 1\}.$$

The numbers n_0, r_0, and s_0 depend only on Γ and s. By (4), any number $a = \mathrm{mld}((X \ni 0, B)/G) \in (0, 1)$ as in (1) is a finite number $\psi_\chi(X', B')$ for data as in Set-up 18.4 with $n \leq n_0$, $r \leq r_0$, at most s_0 polynomials $\tilde{h}_j$, and coefficients in Λ. By Corollary 18.5, the set of such numbers satisfies the ACC. Thus (1) holds.

Finally, we show that the set in (1) satisfies the ACC also when s varies. The terms with $h_j(0) \neq 0$ or $c_j = 0$ do not change B near 0. We show that fewer than $2/\epsilon$ other terms remain. Let E be the prime divisor over X with center 0 given by the weighted blow-up with weights $(1, \mathrm{mult}_0 g - 1, 1, 1)$ in x, y, z, t. Since the leading form of $xy + g$ is irreducible, $a(E, X, 0) = 1 + (\mathrm{mult}_0 g - 1) + 1 + 1 - \mathrm{mult}_0 g = 2$ (this is Lemma 18.2(1) for $\nu = E_0$ and $\varrho = k = 1$, with E_0 the exceptional curve of the blow-up of $0 \in T$ from **Step 1**). Moreover, $\mathrm{ord}_E h_j \geq \mathrm{mult}_0 h_j \geq 1$ whenever $h_j(0) = 0$, so the klt condition and Lemma 2.10(2) leave fewer

than $2/\epsilon$ of the h_j with $c_j > 0$ that vanish at 0. Hence the union over all s of the sets in (1) is a finite union of sets satisfying the ACC, and so it satisfies the ACC. □

18.5. **A fiber-linear boundary.** In this subsection, we prove Theorem 18.9, which extends Theorem 18.8 to boundaries with components $\operatorname{div}_X(x+\phi_i)$ of total coefficient at most one.

Theorem 18.9. *Fix non-negative integers s, r and a DCC set $\Gamma \subset [0,1]$. Let $\epsilon := \gamma(\Gamma)$ if $\Gamma\backslash\{0\} \neq \emptyset$ and $\epsilon := 1$ otherwise, and let $M := 2 + \lceil 4/\epsilon \rceil$. Let $X := (xy + g(z,t) = 0) \subset \mathbb{A}^4$ with $0 \neq g \in (z,t)^2$ be a normal klt threefold germ at 0, and G a finite diagonal abelian group acting faithfully on X and preserving the equation up to a character. Let $p_1, \dots, p_s$ be non-zero G-semi-invariant polynomials in z, t, let $\phi_1, \dots, \phi_r \in (z,t)$ be polynomials such that any non-zero ϕ_i has the character of x, let $b_j, c_i \in \Gamma$ with $\sum_i c_i \leq 1$, let*

$$B := \sum_j b_j \operatorname{div}_X(p_j) + \sum_i c_i \operatorname{div}_X(x+\phi_i)$$

with (X,B) klt, and let $a := \operatorname{mld}((X \ni 0, B)/G)$. Let $T := \mathbb{A}^2_{z,t}$ and $B_T := \sum_j b_j \operatorname{div}_T(p_j)$. For a prime divisor ν over T with center 0, let $J_\nu := I_G(\nu)$ for the action of G on T, which need not be faithful, and let α_ν be the character of J_ν on $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$. Let β be the character of x. Then:

(1) the set of such numbers $a \in (0,1)$ satisfies the ACC;

(2) for any prime divisor ν over T whose center contains 0,

$$a(\nu, T, B_T) - \sum_i c_i \min\{\operatorname{ord}_\nu g, \operatorname{ord}_\nu \phi_i\} > 0,$$

where $\operatorname{ord}_\nu 0 := +\infty$;

(3) the number a satisfies

$$a = \min \frac{\varrho a(\nu, T, B_T) - \sum_i c_i \min\{k, \varrho \operatorname{ord}_\nu \phi_i\}}{|\operatorname{Ker}(\beta^\varrho \alpha_\nu^{-k} : J_\nu \to \mathbb{C}^*)|},$$

where the minimum is attained and runs over the prime divisors ν over T with center 0 and the coprime positive integers ϱ, k with $k < \varrho \operatorname{ord}_\nu g$; the term of (ν, ϱ, k) equals $a(F, X, B)/|I_G(F)|$ for the prime divisor F over X given by the Gauss extension of $\varrho \operatorname{ord}_\nu$ to $\mathbb{C}(X) = \mathbb{C}(T)(x)$ with x-value k (Definition-Proposition 2.11), for which $\operatorname{ord}_F y = \varrho \operatorname{ord}_\nu g - k$ and $I_G(F) = \operatorname{Ker}(\beta^\varrho \alpha_\nu^{-k})$;

(4) if $a < 1$, then $a = \psi_\chi(X', B')$ for some data n, H, X', B', and χ as in Set-up 18.4 such that $n + 1 \leq M + 9$, X' is defined by at most $M + 5$ equations, and B' is defined by at most $s + r$ polynomials with coefficients in Γ.

Proof of Theorem 18.9. The proof is long, so we divide it into several steps. Let $\sigma := \sum_i c_i$, so $\sigma \leq 1$, and for $1 \leq i \leq r$ let $\mathfrak{a}_i := (g, \phi_i) \subset \mathbb{C}[z,t]$. For a prime divisor ν over T whose center contains 0, we have $\operatorname{mult}_\nu \mathfrak{a}_i = \min\{\operatorname{ord}_\nu g, \operatorname{ord}_\nu \phi_i\}$, and we let

$$b^*_\nu := a(\nu, T, B_T) - \sum_i c_i \operatorname{mult}_\nu \mathfrak{a}_i.$$

Step 1. In this step we prove (2), that is, $b^*_\nu > 0$ for any prime divisor ν over T whose center contains 0.

Let $h : T' \to T$ be a proper birational morphism from a smooth surface T' such that ν is a prime divisor E on T'. Since $x = -g/y$ on $X \cap (y \neq 0)$, the functions z, t, y define an isomorphism $X \cap (y \neq 0) \to T \times \mathbb{G}_m$. Let F_ν be the prime divisor over X given by the prime divisor $E \times \mathbb{G}_m$ of $T' \times \mathbb{G}_m$. The function y is a unit along $E \times \mathbb{G}_m$, and $y|_{E \times \mathbb{G}_m}$ is transcendental over the function field of E. Thus $\operatorname{ord}_{F_\nu}$ is the Gauss extension of ord_ν to $\mathbb{C}(X) = \mathbb{C}(T)(y)$ with y-value 0 (Definition-Proposition 2.11). Since $K_{T' \times \mathbb{G}_m}$ is the sum of the pullbacks of $K_{T \times \mathbb{G}_m}$ and $K_{T'/T}$, we have $a(F_\nu, X, 0) = a(\nu, T, 0)$. As $x + \phi_i = (\phi_i y - g)/y$ and $\operatorname{ord}_{F_\nu} y = 0$, Definition-Proposition 2.11 implies

$$\operatorname{ord}_{F_\nu}(x + \phi_i) = \min\{\operatorname{ord}_\nu \phi_i, \operatorname{ord}_\nu g\} = \operatorname{mult}_\nu \mathfrak{a}_i$$

for any i, also when $\phi_i = 0$. Moreover, $\operatorname{ord}_{F_\nu} p_j = \operatorname{ord}_\nu p_j$ for any j. Hence $a(F_\nu, X, B) = b^*_\nu$.

We show that $\operatorname{center}_X F_\nu$ contains 0. The center $\operatorname{center}_X F_\nu$ is the closure of the set V° of the points $(-g(z,t)/y, y, z, t)$ with $(z,t) \in h(E)$ and $y \neq 0$, and $0 \in h(E)$. If g vanishes on $h(E)$, then $(0, y, 0, 0) \in V^\circ$ for any $y \neq 0$, so $0 \in \operatorname{center}_X F_\nu$. Otherwise, $h(E)$ is a curve through 0, and $(y, y, z, t) \in V^\circ$ for any $(z,t) \in h(E)$ with $g(z,t) \neq 0$ and any y with $y^2 = -g(z,t)$. Letting (z,t) tend to 0 along $h(E)$, we get $0 \in \operatorname{center}_X F_\nu$. Since (X, B) is klt near 0, $b^*_\nu = a(F_\nu, X, B) > 0$. This proves (2).

Let E_0 be the exceptional divisor of the blow-up of T at 0, and let

$$\operatorname{mult}_0 \mathfrak{a}_i := \operatorname{mult}_{E_0} \mathfrak{a}_i = \min\{\operatorname{mult}_0 g, \operatorname{mult}_0 \phi_i\},$$

where $\operatorname{mult}_0 0 := +\infty$. Since $a(E_0, T, B_T) = 2 - \sum_j b_j \operatorname{mult}_0 p_j$ [KM98, Lemma 2.29], (2) for $\nu = E_0$ reads

$$\sum_j b_j \operatorname{mult}_0 p_j + \sum_i c_i \operatorname{mult}_0 \mathfrak{a}_i < 2. \tag{18.23}$$

Step 2. In this step we prove (3), and we find a prime divisor F over X computing a such that ord_F is a Gauss extension.

By Lemma 2.6(0)(2)(3), $(X, B)/G$ is a pair whose log discrepancies are the numbers $a(F, X, B)/|I_G(F)|$, where F runs over the prime divisors over X. Since (X, B) is klt near 0, $(X, B)/G$ is klt near the image of 0. Hence $a > 0$. Let $\bar{0} \in X/G$ be the image of 0. By [KM98, Theorem 0.2], there exists a projective birational morphism $f : W \to X/G$ from a smooth variety W such that $\mathfrak{m}_{\bar{0}}\mathcal{O}_W$ is invertible and $f^{-1}(\bar{0}) \cup \operatorname{Exc}(f) \cup \operatorname{Supp} f_*^{-1} B_{X/G}$ is an snc divisor. As $(X, B)/G$ is klt near $\bar{0}$, a is attained by a component of $f^{-1}(\bar{0})$ [Amb99, Section 1, the second paragraph after Definition 1.1]. As G fixes 0, the inverse image of $\bar{0}$ in X is $\{0\}$. Thus Lemma 2.6(2)(3) and the last formula of Lemma 2.6, at the point 0, imply

$$a = \min_F \frac{a(F, X, B)}{|I_G(F)|}, \tag{18.24}$$

where F runs over the prime divisors over X with $\operatorname{center}_X F = \{0\}$, and the minimum is attained.

For a prime divisor ν over T with center 0 and positive integers ϱ, k, let

$$\mathfrak{A}(\nu, \varrho, k) := \varrho a(\nu, T, B_T) - \sum_i c_i \min\{k, \varrho \operatorname{ord}_\nu \phi_i\}.$$

Assume that ϱ, k are coprime and $k < \varrho \operatorname{ord}_\nu g$. By Lemma 18.2(1), the Gauss extension of $\varrho \operatorname{ord}_\nu$ to $\mathbb{C}(X) = \mathbb{C}(T)(x)$ with x-value k is $\operatorname{ord}_{F_{\nu,\varrho,k}}$ for a prime divisor $F_{\nu,\varrho,k}$ over X with $\operatorname{center}_X F_{\nu,\varrho,k} = \{0\}$, and

$$\operatorname{ord}_{F_{\nu,\varrho,k}} y = \varrho \operatorname{ord}_\nu g - k, \qquad a(F_{\nu,\varrho,k}, X, 0) = \varrho a(\nu, T, 0), \qquad I_G(F_{\nu,\varrho,k}) = \operatorname{Ker}(\beta^\varrho \alpha_\nu^{-k}).$$

By Definition-Proposition 2.11, $\operatorname{ord}_{F_{\nu,\varrho,k}} p_j = \varrho \operatorname{ord}_\nu p_j$ and $\operatorname{ord}_{F_{\nu,\varrho,k}}(x + \phi_i) = \min\{k, \varrho \operatorname{ord}_\nu \phi_i\}$ for any i, j. Thus

$$\frac{a(F_{\nu,\varrho,k}, X, B)}{|I_G(F_{\nu,\varrho,k})|} = \frac{\mathfrak{A}(\nu, \varrho, k)}{|\operatorname{Ker}(\beta^\varrho \alpha_\nu^{-k})|}, \tag{18.25}$$

which is the term of (ν, ϱ, k) in (3), and the term of (ν, ϱ, k) is at least a by (18.24). It remains to show that some term equals a.

Let F be a prime divisor over X with $\operatorname{center}_X F = \{0\}$. By Lemma 18.2(2), $\operatorname{ord}_F|_{\mathbb{C}(T)} = \varrho \operatorname{ord}_\nu$ for a positive integer ϱ and a prime divisor ν over T with center 0, the number $k := \operatorname{ord}_F x$ satisfies $0 < k < \varrho \operatorname{ord}_\nu g$, and the integer $d_F = a(F, X, 0) - \varrho a(\nu, T, 0)$ of Lemma 18.2(2) is non-negative. For $1 \le i \le r$, since $\operatorname{ord}_F \phi_i = \varrho \operatorname{ord}_\nu \phi_i$, the number

$$\ell_i := \operatorname{ord}_F(x + \phi_i) - \min\{k, \varrho \operatorname{ord}_\nu \phi_i\}$$

is non-negative.

We show that $\ell_i \le d_F$. We may assume that $\ell_i > 0$. Then $k = \varrho \operatorname{ord}_\nu \phi_i$, and $w := x + \phi_i$ satisfies $\operatorname{ord}_F w = k + \ell_i > 0$. Let $Y := T \times \mathbb{A}^1$ with coordinates z, t, w. The functions z, t, w define a birational morphism $X \to Y$, as $\mathbb{C}(X) = \mathbb{C}(T)(x)$. Near 0, ω_X is generated by the Poincaré residue Ω of $dx \wedge dy \wedge dz \wedge dt/(xy + g)$, and $dw \wedge dz \wedge dt = dx \wedge dz \wedge dt = \pm x\Omega$ on X because $\phi_i \in \mathbb{C}[z, t]$. Thus

$$a(F, X, 0) = a(F, Y, 0) - \operatorname{ord}_F x = a(F, Y, 0) - k.$$

Let $h : T' \to T$ and E be as in **Step 1** for the present ν, let $Y' := T' \times \mathbb{A}^1$, and let η be the generic point of $V := \operatorname{center}_{Y'} F$. Since $\operatorname{ord}_F|_{\mathbb{C}(T)} = \varrho \operatorname{ord}_E$, V is contained in $E \times \mathbb{A}^1$ and dominates E. As $\operatorname{ord}_F w > 0$, $V \subset E \times \{0\}$, so $V = E \times \{0\}$. Let e be a local equation of E at the generic point of E. Then $\operatorname{ord}_F e = \varrho$, and e, w are regular parameters of $\mathcal{O}_{Y',\eta}$. Moreover, $K_{Y'}$ is the sum of the pullbacks of K_Y and $K_{T'/T}$, the coefficient of E in $K_{T'/T}$ is $a(\nu, T, 0) - 1$, and the other components of $K_{T'/T}$ do not contain the generic point of E. Thus Lemma 2.13, applied with $b_1 = b_2 = 0$ on a neighborhood of η on which $\operatorname{div}(e) + \operatorname{div}(w)$ is snc, implies

$$a(F, Y, 0) = a(F, Y', 0) + \varrho\big(a(\nu, T, 0) - 1\big) \ge \operatorname{ord}_F e + \operatorname{ord}_F w + \varrho\big(a(\nu, T, 0) - 1\big) = \varrho a(\nu, T, 0) + k + \ell_i.$$

Hence $a(F, X, 0) \ge \varrho a(\nu, T, 0) + \ell_i$, that is, $\ell_i \le d_F$.

Since $a(F, X, 0) = \varrho a(\nu, T, 0) + d_F$, $\operatorname{ord}_F p_j = \varrho \operatorname{ord}_\nu p_j$, and $\sum_i c_i \ell_i \le \sigma d_F$, we get

$$a(F, X, B) = \mathfrak{A}(\nu, \varrho, k) + d_F - \sum_i c_i \ell_i \ge \mathfrak{A}(\nu, \varrho, k) + (1 - \sigma) d_F \ge \mathfrak{A}(\nu, \varrho, k). \tag{18.26}$$

The hypothesis $\sigma \le 1$ is used only in the second inequality of (18.26). Since $k < \varrho \operatorname{ord}_\nu g$, we have $\min\{k, \varrho \operatorname{ord}_\nu \phi_i\} \le \varrho \operatorname{mult}_\nu \mathfrak{a}_i$ for any i, so **Step 1** implies

$$\mathfrak{A}(\nu, \varrho, k) \ge \varrho b^*_\nu > 0. \tag{18.27}$$

Let $I := I_G(F)$, $e_F := |I|$, and λ the character of I on $\mathfrak{m}_F/\mathfrak{m}_F^2$. By Lemma 2.6(1), I is cyclic and λ is faithful, so λ has order e_F. By Lemma 18.2(2), $I \subset J_\nu$. By Lemma 9.2(1), $\beta|_I = \lambda^k$. Let ϖ_0 be a uniformizer of ord_ν. The projections of ϖ_0 to the J_ν-isotypic components of $\mathbb{C}(T)$ have order at least 1 along ν and sum to ϖ_0, so one of them, say ϖ, satisfies $\operatorname{ord}_\nu \varpi = 1$. Then α_ν is the character of ϖ on J_ν, and Lemma 9.2(1) for the action of I on X implies $\alpha_\nu|_I = \lambda^{\operatorname{ord}_F \varpi} = \lambda^\varrho$.

Let $q := \gcd(\varrho, k)$, $\varrho_0 := \varrho/q$, $k_0 := k/q$, and $F_0 := F_{\nu,\varrho_0,k_0}$, which is defined as $k_0 < \varrho_0 \operatorname{ord}_\nu g$. Then $\mathfrak{A}(\nu, \varrho_0, k_0) = \mathfrak{A}(\nu, \varrho, k)/q$. On I, $\beta^{\varrho_0} \alpha_\nu^{-k_0} = \lambda^{k\varrho_0 - \varrho k_0} = 1$, so $I \subset I_G(F_0)$. Together with (18.25) and (18.26), we get

$$\frac{a(F_0, X, B)}{|I_G(F_0)|} \le \frac{\mathfrak{A}(\nu, \varrho, k)}{q e_F} \le \frac{a(F, X, B)}{q e_F}. \tag{18.28}$$

Now assume that F computes a, that is, $a(F, X, B) = e_F a$, which is possible by (18.24). By (18.24) and (18.28), $a \le a/q$. As $a > 0$, we get $q = 1$, and both inequalities in (18.28) are equalities. Thus $|I_G(F_0)| = e_F$, so $I_G(F_0) = I$, and $a(F_0, X, B) = a(F, X, B)$. In particular, the term of (ν, ϱ, k) in (3) equals a, which proves (3). Moreover, F_0 computes a, so possibly replacing F with F_0, we may assume that ord_F is the Gauss extension of $\varrho \operatorname{ord}_\nu$ with x-value k, where ϱ, k are coprime, that $I = \operatorname{Ker}(\beta^\varrho \alpha_\nu^{-k})$, and that

$$a = \frac{a(F, X, B)}{e_F} = \frac{\mathfrak{A}(\nu, \varrho, k)}{e_F}. \tag{18.29}$$

The relations $I \subset J_\nu$, $\beta|_I = \lambda^k$, and $\alpha_\nu|_I = \lambda^\varrho$ hold for this F, since they were proved for any prime divisor over X with center $\{0\}$.

Step 3. In this step we construct a G-equivariant sequence of blow-ups $\pi : S' \to T$ which principalizes the ideals $\mathfrak{a}_i$ with $c_i > 0$, and a reduced curve C on T with $\operatorname{mult}_0 C \le M$ such that the center of any blow-up in π is on the strict transform of C. Here $M = 2 + \lceil 4/\epsilon \rceil$ is the constant of Theorem 18.9, which differs from the constant M of Theorem 18.8.

Let D_T be the reduced curve $\operatorname{Supp} \operatorname{div}_T(zt \prod_{b_j > 0} p_j)$. Let $S_0 := T$, and let S_1 be the blow-up of T at 0. For $m \ge 1$, write $\mathfrak{a}_i \mathcal{O}_{S_m} = \mathcal{O}_{S_m}(-Z_{i,m}) \cdot \mathfrak{b}_{i,m}$ near the inverse image of 0, where

$Z_{i,m} \geq 0$ is a divisor and $\mathfrak{b}_{i,m}$ is an ideal with finite cosupport, and let D_m be the reduced divisor whose support is the union of the exceptional locus of $S_m \to T$, the strict transform of D_T, and the supports of $Z_{i,m}$ for $c_i > 0$. Let $\mathcal{B}_m$ be the set of closed points $P \in S_m$ over 0 such that $\mathfrak{b}_{i,m} \subset \mathfrak{m}_P$ for some i with $c_i > 0$, or D_m is not snc at P, where $\mathfrak{m}_P$ is the ideal of P. Then $\mathcal{B}_m$ is finite, and we let S_{m+1} be the blow-up of S_m at $\mathcal{B}_m$.

We show that $\mathcal{B}_m = \emptyset$ for some m. Let $\mathfrak{c} := \mathfrak{m}_0 \cdot \prod_{c_i>0} \mathfrak{a}_i \cdot \mathcal{O}_T(-D_T)$, where $\mathfrak{m}_0$ is the ideal of 0. By [KM98, Theorem 0.2], there exists a projective birational morphism $f : Y \to T$ from a smooth surface Y such that $\mathfrak{c}\mathcal{O}_Y = \mathcal{O}_Y(-D_Y)$ for a divisor D_Y and $\operatorname{Exc}(f) \cup \operatorname{Supp} D_Y$ is snc. Any $\mathfrak{a}_i\mathcal{O}_Y$ with $c_i > 0$ is invertible, since a factor of an invertible ideal of a local domain is invertible. Since $\mathfrak{m}_0$ is a factor of $\mathfrak{c}$, $\mathfrak{m}_0\mathcal{O}_Y$ is invertible, so f factors through S_1 [Har77, Proposition II.7.14].

Assume that f factors through S_m, and let $P \in \mathcal{B}_m$. If $Y \to S_m$ were an isomorphism over a neighborhood of P, then near P any $\mathfrak{a}_i\mathcal{O}_{S_m}$ with $c_i > 0$ would be invertible and $\operatorname{Supp} D_m$ would be contained in the image of $\operatorname{Exc}(f) \cup \operatorname{Supp} D_Y$, so $P \notin \mathcal{B}_m$, a contradiction. Since f is projective and $S_m \to T$ is separated, $Y \to S_m$ is a projective birational morphism of smooth surfaces [Har77, Exercises II.4.8(e) and II.4.9], and it is a composite of blow-ups of points. To see this, let $\bar{S}$ be the surface given by blowing up $\mathbb{P}^2 \supset T$ at 0, then at $\mathcal{B}_1$, ..., then at $\mathcal{B}_{m-1}$, which is a smooth projective surface containing S_m as an open subset. Fix a closed embedding $Y \subset \mathbb{P}^N \times S_m$ over S_m, and let $\bar{Y}$ be a resolution [KM98, Theorem 0.2] of the normalization of the closure of Y in $\mathbb{P}^N \times \bar{S}$ that is an isomorphism over the complement of the singular points of the normalization, all of which are outside Y. Then $\bar{Y} \to \bar{S}$ is a composite of blow-ups of points by [Har77, Corollary V.5.4], and so is the restriction $Y \to S_m$ of $\bar{Y} \to \bar{S}$ over S_m. The first blow-up in such a composite whose center maps to P is the blow-up at P. Thus $\mathfrak{m}_P\mathcal{O}_Y$ is invertible, and $Y \to S_m$ factors through the blow-up of S_m at P [Har77, Proposition II.7.14]. Hence f factors through S_{m+1}. By induction on m, if $\mathcal{B}_1, \dots, \mathcal{B}_{m-1} \neq \emptyset$, then f factors through S_m and $S_m \to T$ has at least m exceptional curves. As the number of exceptional curves of $S_m \to T$ is at most the number of exceptional curves of f, $\mathcal{B}_m = \emptyset$ for some m. Let m_* be the least such m, $S' := S_{m_*}$, and $\pi : S' \to T$ the composite morphism. Then $\mathfrak{a}_i\mathcal{O}_{S'} = \mathcal{O}_{S'}(-Z_{i,m_*})$ near $\pi^{-1}(0)$ for $c_i > 0$, and $D' := D_{m_*}$ is snc near $\pi^{-1}(0)$.

As g, ϕ_i, z, t, and p_j are G-semi-invariant, the ideals $\mathfrak{a}_i$ and the curve D_T are G-invariant. Thus, by induction on m, G acts on S_m, $\mathcal{B}_m$ is G-invariant, and $S_{m+1} \to S_m$ is G-equivariant. In particular, π is G-equivariant. Moreover, we have the following property $(*)$: if a closed point $P' \in S_m$ does not belong to $\mathcal{B}_m$, then $S_{m+1} \to S_m$ is an isomorphism near the inverse image of P', and since the conditions defining $\mathcal{B}_m$ and $\mathcal{B}_{m+1}$ depend only on a neighborhood, the inverse image does not belong to $\mathcal{B}_{m+1}$.

Let i be an index such that $c_i > 0$, and let E be an exceptional prime divisor of π with uniformizer ϖ_E of ord_E. Since $\operatorname{mult}_E \mathfrak{a}_i = \min\{\operatorname{ord}_E g, \operatorname{ord}_E \phi_i\}$, the residues of $g/\varpi_E^{\operatorname{mult}_E \mathfrak{a}_i}$ and $\phi_i/\varpi_E^{\operatorname{mult}_E \mathfrak{a}_i}$ in the residue field of ord_E are not both zero. Thus $\operatorname{ord}_E(\xi g + \xi'\phi_i) > \operatorname{mult}_E \mathfrak{a}_i$ for at most one point $(\xi : \xi') \in \mathbb{P}^1$. As π has finitely many exceptional prime divisors, there exist $\xi_{i,l}, \xi'_{i,l} \in \mathbb{C}$ for $l = 1, 2$ such that

$$g_{i,l} := \xi_{i,l} g + \xi'_{i,l}\phi_i \neq 0 \quad \text{and} \quad \operatorname{ord}_E g_{i,l} = \operatorname{mult}_E \mathfrak{a}_i \quad \text{for any exceptional prime divisor } E \text{ of } \pi \text{ (cf. [Cas00, Theo}$$

If $\phi_i = 0$ or ϕ_i is proportional to g, then $g_{i,1}$ and $g_{i,2}$ are non-zero multiples of g. Since the strict transform of E_0 on S' is an exceptional prime divisor of π, $\operatorname{mult}_0 g_{i,l} = \operatorname{mult}_0 \mathfrak{a}_i$.

Let C be the reduced curve

$$\operatorname{Supp} \operatorname{div}_T \Big(zt \prod_{b_j>0} p_j \prod_{c_i>0} g_{i,1} g_{i,2} \Big).$$

If $\Gamma \backslash \{0\} \neq \emptyset$, then the non-zero coefficients b_j, c_i are at least $\gamma(\Gamma) = \epsilon$, and $\epsilon > 0$ by Lemma 2.10(2). Otherwise all b_j, c_i are zero. Thus (18.23) implies

$$\sum_{b_j>0} \operatorname{mult}_0 p_j + \sum_{c_i>0} \operatorname{mult}_0 \mathfrak{a}_i \leq \frac{1}{\epsilon} \Big(\sum_j b_j \operatorname{mult}_0 p_j + \sum_i c_i \operatorname{mult}_0 \mathfrak{a}_i \Big) < \frac{2}{\epsilon},$$

and hence

$$\operatorname{mult}_0 C \le 2 + \sum_{b_j>0} \operatorname{mult}_0 p_j + 2 \sum_{c_i>0} \operatorname{mult}_0 \mathfrak{a}_i < 2 + \frac{4}{\epsilon}. \tag{18.30}$$

As $\operatorname{mult}_0 C$ is an integer, $\operatorname{mult}_0 C \le M$.

We show that the center of any blow-up in π is on the strict transform of C. The first center 0 belongs to C. Let $m \ge 1$, $P \in \mathcal{B}_m$, and $\pi_m : S_m \to T$ the composite morphism. Assume first that $\mathfrak{b}_{i,m} \subset \mathfrak{m}_P$ for some i with $c_i > 0$, and let e_P be a local equation of $Z_{i,m}$ at P. Then $\pi_m^* g_{i,1} = e_P \zeta_P$ for some $\zeta_P \in \mathfrak{b}_{i,m} \subset \mathfrak{m}_P$. For any exceptional curve E of π_m, the strict transform of E on S' is an exceptional prime divisor of π, so $\operatorname{ord}_E g_{i,1} = \operatorname{mult}_E \mathfrak{a}_i$, which is the coefficient of E in $Z_{i,m}$. Thus, near P, $\operatorname{div}(\zeta_P)$ is the strict transform of $\operatorname{div}(g_{i,1})$ minus the non-exceptional part of $Z_{i,m}$. As $\operatorname{div}(\zeta_P) \ge 0$ and $P \in \operatorname{Supp} \operatorname{div}(\zeta_P)$, the point P is on the strict transform of $\operatorname{div}(g_{i,1})$, hence on the strict transform of C. Assume next that $\mathfrak{b}_{i,m}\mathcal{O}_{S_m,P} = \mathcal{O}_{S_m,P}$ for any i with $c_i > 0$. Then D_m is not snc at P. The exceptional locus of π_m is an snc divisor, as blowing up a point of a smooth surface preserves the snc property of the exceptional locus. Thus a non-exceptional component of D_m passes through P. Such a component is the strict transform of a component of $D_T \subset C$, or a non-exceptional component of some $Z_{i,m}$, which is the strict transform of a common component of $\operatorname{div}(g)$ and $\operatorname{div}(\phi_i)$, hence of a component of $\operatorname{div}(g_{i,1}) \subset C$. In both cases P is on the strict transform of C.

Since any center of a blow-up in π is over 0, it is on the strict transform of an analytic branch of C at 0.

Step 4. From now on, we assume that $a < 1$, and we let F, ν, ϱ, k, I, e_F, and λ be as at the end of **Step 2**, so that (18.29) holds. In this step we show that ν has a minimal generating sequence [dFGPM23, Definition 2.9] with at most $M + 2$ elements.

Recall that $I \subset J_\nu$ (**Step 2**), so I stabilizes ν. Assume that ν is not a prime divisor on S', and let $p \in \pi^{-1}(0)$ be the center of ν on S'. As π is I-equivariant, I fixes p. Since D' is snc near p and $\pi^{-1}(0)$ is contained in the exceptional locus of π, at most two components of D' pass through p, and at least one of them is exceptional. Each of them is I-invariant: I maps any exceptional curve created by the blow-up of S_m at $\mathcal{B}_m$ to a curve created by the blow-up of S_m at $\mathcal{B}_m$, two distinct such curves are disjoint, and I does not map an exceptional curve to a non-exceptional one. Since I is abelian, projecting local equations at p of the components of D' through p and lifts of a basis of $\mathfrak{m}_p/\mathfrak{m}_p^2$ to I-isotypic components, we get I-semi-invariant regular parameters u_1, u_2 of $\mathcal{O}_{S',p}$ such that $(u_1 = 0)$ is an exceptional component of D' and any component of D' through p is $(u_1 = 0)$ or $(u_2 = 0)$ near p.

Let Θ be the $\mathbb{R}$-divisor on a neighborhood of $\pi^{-1}(0)$ in S' such that

$$K_{S'} + \Theta = \pi^*(K_T + B_T) + \sum_{c_i>0} c_i Z_{i,m_*}.$$

Since $\mathfrak{a}_i \mathcal{O}_{S'} = \mathcal{O}_{S'}(-Z_{i,m_*})$ near $\pi^{-1}(0)$ for $c_i > 0$, we have $a(\mu, S', \Theta) = b^*_\mu$ for any prime divisor μ over S' whose center on S' intersects $\pi^{-1}(0)$. Moreover, $\operatorname{Supp} \Theta \subset \operatorname{Supp} D'$ near $\pi^{-1}(0)$. For $l = 1, 2$, let $\theta_l := b^*_{\Sigma_l}$ if $(u_l = 0)$ is a component Σ_l of D' near p, and $\theta_l := 1$ otherwise. Then

$$\Theta = (1 - \theta_1) \operatorname{div}(u_1) + (1 - \theta_2) \operatorname{div}(u_2) \quad \text{near } p.$$

Any Σ_l is a prime divisor over T whose center contains 0, so $\theta_1, \theta_2 > 0$ by **Step 1**. Let

$$\tau := a(\nu, S', 0) - \operatorname{ord}_\nu u_1 - \operatorname{ord}_\nu u_2.$$

Let $U \subset S'$ be an open neighborhood of p on which u_1, u_2 are regular, $D_U := \operatorname{div}(u_1) + \operatorname{div}(u_2)$ is snc, and p is the only point of $(u_1 = 0) \cap (u_2 = 0)$. Then $\tau \ge 0$ by Lemma 2.13 with $b_1 = b_2 = 0$, and

$$b^*_\nu = a(\nu, S', \Theta) = \theta_1 \operatorname{ord}_\nu u_1 + \theta_2 \operatorname{ord}_\nu u_2 + \tau. \tag{18.31}$$

We show that e_F divides $\varrho\tau$. For a semi-invariant rational function or form h, let χ_h be the character of h. Near p, the exceptional locus of π is contained in $(u_1u_2=0)$, so

$$\pi^*(dz\wedge dt)=vu_1^{m_1}u_2^{m_2}du_1\wedge du_2$$

for non-negative integers m_1,m_2 and a unit v at p, and

$$a(\nu,T,0)=a(\nu,S',0)+m_1\operatorname{ord}_\nu u_1+m_2\operatorname{ord}_\nu u_2.$$

As a quotient of I-semi-invariant forms, v is I-semi-invariant, and since $v(p)\neq 0$ and I fixes p, $\chi_v|_I=1$. Thus, on I,

$$\chi_z\chi_t=\chi_{u_1}^{m_1+1}\chi_{u_2}^{m_2+1}.$$

By Lemma 9.2(1) for the action of I on X, for $l=1,2$,

$$\chi_{u_l}|_I=\lambda^{\operatorname{ord}_F u_l}=\lambda^{\varrho\operatorname{ord}_\nu u_l}.$$

By Lemma 9.2(2) for the action of I on $X\subset\mathbb{A}^4$ with eigenparameters x,y,z,t and $f=xy+g$, whose character is $\beta\chi_y$, and by Lemma 18.2(1), $(\chi_z\chi_t)|_I=\lambda^{a(F,X,0)}=\lambda^{\varrho a(\nu,T,0)}$. Hence

$$\lambda^{\varrho a(\nu,T,0)}=\lambda^{\varrho((m_1+1)\operatorname{ord}_\nu u_1+(m_2+1)\operatorname{ord}_\nu u_2)},$$

that is, $\lambda^{\varrho\tau}=1$. As λ has order e_F, e_F divides $\varrho\tau$.

By (18.29), (18.27), and (18.31),

$$1>a\geq\frac{\varrho b_\nu^*}{e_F}=\frac{\varrho\theta_1\operatorname{ord}_\nu u_1+\varrho\theta_2\operatorname{ord}_\nu u_2+\varrho\tau}{e_F}>\frac{\varrho\tau}{e_F},$$

because $\theta_l>0$ and $\operatorname{ord}_\nu u_l>0$ for $l=1,2$. Since $\varrho\tau/e_F$ is a non-negative integer, $\tau=0$.

Let $w_l:=\operatorname{ord}_\nu u_l$ for $l=1,2$. We show that ν is the monomial prime divisor in u_1,u_2 with weights (w_1,w_2). Let $r_{U,D_U}(\operatorname{ord}_\nu)$ be the image of ord_ν under the retraction of [JM12, §4.3] onto the set $\mathrm{QM}(U,D_U)$ of quasi-monomial valuations of the snc pair (U,D_U), that is, the monomial valuation in u_1,u_2 with weights (w_1,w_2). Writing A_U for the log discrepancy function of [JM12], [JM12, Proposition 5.1] implies $A_U(r_{U,D_U}(\operatorname{ord}_\nu))=w_1+w_2$. As $\tau=0$, $A_U(\operatorname{ord}_\nu)=a(\nu,S',0)=w_1+w_2$. By the equality case of [JM12, Corollary 5.4], $\operatorname{ord}_\nu\in\mathrm{QM}(U,D_U)$, so $\operatorname{ord}_\nu=r_{U,D_U}(\operatorname{ord}_\nu)$. Since $\operatorname{ord}_\nu(\mathbb{C}(T)^*)=\mathbb{Z}$ and the values of ord_ν on $\mathcal{O}_{S',p}\backslash\{0\}$ belong to $w_1\mathbb{Z}_{\geq 0}+w_2\mathbb{Z}_{\geq 0}$, $\gcd(w_1,w_2)=1$.

We describe the blow-ups which extract ν from S'. Let $p_0:=p$, $u_l^{(0)}:=u_l$, and $w_l^{(0)}:=w_l$. Assume that ν is the monomial prime divisor in regular parameters $u_1^{(h)},u_2^{(h)}$ at a point p_h with coprime weights $(w_1^{(h)},w_2^{(h)})$, and blow up p_h. If $w_1^{(h)}=w_2^{(h)}$, then $w_1^{(h)}=w_2^{(h)}=1$ and ν is the exceptional curve. If $w_1^{(h)}<w_2^{(h)}$, then the center p_{h+1} of ν is the origin of the chart with coordinates $u_1^{(h+1)}:=u_1^{(h)}$ and $u_2^{(h+1)}:=u_2^{(h)}/u_1^{(h)}$, in which ν is monomial with weights $(w_1^{(h)},w_2^{(h)}-w_1^{(h)})$ (by [JM12, Proposition 3.1(iii)] at p_h and p_{h+1}), $(u_1^{(h+1)}=0)$ is the new exceptional curve, and $(u_2^{(h+1)}=0)$ is the strict transform of $(u_2^{(h)}=0)$. If $w_1^{(h)}>w_2^{(h)}$, then p_{h+1} is the origin of the chart with coordinates $u_1^{(h+1)}:=u_1^{(h)}/u_2^{(h)}$ and $u_2^{(h+1)}:=u_2^{(h)}$, in which ν is monomial with weights $(w_1^{(h)}-w_2^{(h)},w_2^{(h)})$ (by [JM12, Proposition 3.1(iii)] at p_h and p_{h+1}), $(u_2^{(h+1)}=0)$ is the new exceptional curve, and $(u_1^{(h+1)}=0)$ is the strict transform of $(u_1^{(h)}=0)$. By induction on $w_1^{(h)}+w_2^{(h)}$, after finitely many blow-ups at $p_0,\dots,p_{h_\nu}$, ν appears as an exceptional curve. For any h, each of the curves $(u_1^{(h)}=0)$ and $(u_2^{(h)}=0)$ is the strict transform of $(u_1=0)$ or $(u_2=0)$, or an exceptional curve over p, and any exceptional curve over T through p_h is one of them. Thus, if $(u_2=0)$ is exceptional over T, then p_h is on two exceptional curves over T for any h. If $(u_2=0)$ is not exceptional over T, there exists $h_0\geq 0$ such that p_h is on exactly one exceptional curve over T for $h\leq h_0$, namely the curve created by the blow-up at p_{h-1}, or the curve $(u_1=0)$ if $h=0$, and p_h is on two exceptional curves over T for $h>h_0$.

We now count ends. For a prime divisor E' over T with center 0, the minimal resolution of $\mathrm{ord}_{E'}$ is the sequence $T = T_0 \leftarrow T_1 \leftarrow \cdots \leftarrow T_{n+1}$, where T_{h+1} is the blow-up of T_h at the center o_h of $\mathrm{ord}_{E'}$ on T_h, and E' is the exceptional curve of $T_{n+1} \to T_n$ [dFGPM23, Definition 2.8]. Let $\mathcal{G}_{E'}$ be the dual graph of the exceptional curves of $T_{n+1} \to T$, whose vertices are the exceptional curves of $T_{n+1} \to T$ and whose edges join intersecting curves. An end of a graph is a vertex of valence at most one. For $h \geq 1$, o_h is on one or two exceptional curves of $T_h \to T$, and we call o_h *free* in the first case and *satellite* in the second. If o_h is free, the dual graph of the exceptional curves of $T_{h+1} \to T$ is the dual graph for T_h together with a new vertex joined to the vertex of the exceptional curve through o_h. If o_h is satellite, it is obtained by subdividing the edge between both exceptional curves through o_h, as the exceptional curves of $T_h \to T$ are projective lines forming an snc divisor whose dual graph is a tree [dFGPM23, Section 2.1]. A new vertex joined to a vertex of valence one does not change the number of ends, and a new vertex joined to a vertex of valence zero or at least two adds one end. Thus the number of ends increases by zero or one at a free point and does not change at a satellite point.

Let E be an exceptional prime divisor of π. We show that $\mathcal{G}_E$ has at most $M+1$ ends. Let $P \in S_m$ be the center of the blow-up creating E, and A an analytic branch of C at 0 whose strict transform contains P (**Step 3**). For $m' \leq m$, let $P_{m'} \in S_{m'}$ be the image of P, which is on the strict transform of A. As $P_m \in \mathcal{B}_m$, $(*)$ in **Step 3** implies $P_{m'} \in \mathcal{B}_{m'}$ for any $m' \leq m$, where $\mathcal{B}_0 := \{0\}$. Let $Q_0 := 0$, and let Q_{h+1} be the point of the strict transform of A on the exceptional curve of the blow-up at Q_h, which is unique since A is a branch. By induction on m', $P_{m'} = Q_{m'}$. Hence the minimal resolution of ord_E is the sequence of blow-ups at $Q_0, \ldots, Q_m$. If A is smooth, then the strict transforms of A are smooth and transversal to the exceptional curve of the blow-up at Q_h, so Q_{h+1} is not on the strict transform of the exceptional curve of the blow-up at Q_{h-1}. Thus any Q_h with $h \geq 1$ is free and on the exceptional curve of the blow-up at Q_{h-1}, $\mathcal{G}_E$ is a chain, and $\mathcal{G}_E$ has at most 2 ends.

Assume that A is singular. Let $g(A)$ be the number of characteristic exponents of A, so that the semigroup of A has minimal generators $\bar{\beta}_0 < \cdots < \bar{\beta}_{g(A)}$ [dFGPM23, Notation 2.2], and let the minimal embedded resolution of A consist of the blow-ups at $Q_0, \ldots, Q_{n_A}$ (a point at which the total transform of A is not snc is on the strict transform of A, as the exceptional divisor is snc [dFGPM23, Section 2.1]). After the blow-up at Q_{n_A}, the strict transform of A is smooth and transversal to the exceptional curve E_A of the blow-up at Q_{n_A}, and does not intersect any other exceptional curve, since the total transform of A is snc. As A is singular, $n_A \geq 1$. The blow-ups at $Q_0, \ldots, Q_{n_A}$ form the minimal resolution of ord_{E_A} [dFGPM23, Definition 2.8], and A is a curvetta of E_A on it. Thus, by [dFGPM23, Proposition 2.14] applied to E_A and its curvetta A, a minimal generating sequence of ord_{E_A} has $g(A)+1$ elements, and by [dFGPM23, Definition 2.11 and Proposition 2.12], $\mathcal{G}_{E_A}$ has $g(A)+1$ ends. As in the smooth case, Q_h is free and on the exceptional curve of the blow-up at Q_{h-1} for $h > n_A$, so the exceptional curves of the blow-ups at $Q_{n_A+1}, \ldots, Q_m$ form a chain attached to one vertex, which adds at most one end. Since the number of ends does not decrease along the blow-ups at $Q_0, Q_1, \ldots$, $\mathcal{G}_E$ has at most $g(A)+1$ ends if $m \leq n_A$, and at most $g(A)+2$ ends if $m > n_A$.

By [dFGPM23, Notation 2.2 and Proposition 2.3(1)], $n_j \geq 2$ for $1 \leq j \leq g(A)$, where $e_0 = \bar{\beta}_0$, $e_j = \gcd(e_{j-1}, \bar{\beta}_j)$, and $n_j = e_{j-1}/e_j$, so $2^{g(A)} \leq n_1 \cdots n_{g(A)} = e_0/e_{g(A)} \leq \bar{\beta}_0$. As $\bar{\beta}_0$ is the least positive element of the semigroup of A, $\bar{\beta}_0 \leq (L \cdot A)_0 \leq (L \cdot C)_0 = \mathrm{mult}_0 C$ for a general line L through 0, where the equality follows from [Har77, Exercise I.5.4(b)] applied to each irreducible component of C. By (18.30), $2^{g(A)} \leq \bar{\beta}_0 \leq \mathrm{mult}_0 C \leq M$, so $g(A) \leq M-1$. In both cases $\mathcal{G}_E$ has at most $M+1$ ends.

If ν is a prime divisor on S', then $\mathcal{G}_\nu$ has at most $M+1$ ends. Otherwise, let p, u_1, u_2 be as defined before (18.31), and $p_0, \ldots, p_{h_\nu}$ as defined after (18.31). The centers of ν on $S_0, \ldots, S_{m_*}$ are the images of p. By $(*)$ in **Step 3**, the images of p blown up in π are the images in $S_0, \ldots, S_{m_p}$ for some m_p, and the blow-up at the image in S_{m_p} creates the newest exceptional curve E_p through p, which is $(u_1 = 0)$ or $(u_2 = 0)$. If $E_p = (u_2 = 0)$, then both $(u_1 = 0)$ and $(u_2 = 0)$

are exceptional, and we exchange u_1 and u_2. Thus we may assume that $E_p = (u_1 = 0)$. Any exceptional curve of π through p is created by the blow-up at an image of p. Thus the minimal resolution of ord_ν is the minimal resolution of ord_{E_p} followed by the blow-ups at $p_0, \dots, p_{h_\nu}$, and any exceptional curve over T through p_h is an exceptional curve of the minimal resolution of ord_ν. By the description of $p_0, \dots, p_{h_\nu}$ above, either all of $p_0, \dots, p_{h_\nu}$ are satellite, or $p_0, \dots, p_{h_0}$ are free with p_h on the exceptional curve of the blow-up at p_{h-1} for $1 \le h \le h_0$ and $p_{h_0+1}, \dots, p_{h_\nu}$ are satellite. In the second case the exceptional curves of the blow-ups at $p_0, \dots, p_{h_0}$ form a chain attached to E_p, which adds at most one end. As E_p is an exceptional prime divisor of π, $\mathcal{G}_\nu$ has at most $M+2$ ends.

By [dFGPM23, Definition 2.11 and Proposition 2.12], a minimal generating sequence of ord_ν has one element for each end of $\mathcal{G}_\nu$ if $\mathcal{G}_\nu$ has at least two vertices, and two elements if $\nu = E_0$ [dFGPM23, Example 2.13]. Hence ν has a minimal generating sequence with $m_\nu + 1$ elements for an integer m_ν with $0 \le m_\nu \le M+1$.

Step 5. In this step we prove (4). Recall that $a < 1$ and that F is the prime divisor over X of (18.29), given by the Gauss extension of $\varrho\,\mathrm{ord}_\nu$ with x-value k.

We first show that $\mathrm{mld}((X \ni 0, B)/I) = a$. For any prime divisor F' over X with $\mathrm{center}_X F' = \{0\}$, $I_I(F') = I \cap I_G(F')$, so (18.24) implies

$$\frac{a(F', X, B)}{|I_I(F')|} \ge \frac{a(F', X, B)}{|I_G(F')|} \ge a,$$

and $a(F, X, B)/|I_I(F)| = a$ by (18.29). As I acts faithfully on X and fixes 0, the last formula of Lemma 2.6, at the point 0, applied to I, implies $\mathrm{mld}((X \ni 0, B)/I) = a$.

The group I is cyclic, acts diagonally on T, stabilizes ν, and acts trivially on the residue field of ord_ν, since $I \subset J_\nu$. Let $N := m_\nu + 3 \le M + 4$, where m_ν is as in **Step 4**. By Proposition 18.7 with $H := I$, there exist I-semi-invariant polynomials $q_0, \dots, q_{m_\nu} \in \mathbb{C}[z, t]$, the closed embedding $\iota = (z, t, q_0, \dots, q_{m_\nu}) : T \to \mathbb{A}^N$, I-semi-invariant polynomials $f_1, \dots, f_{N-2}$ on $\mathbb{A}^N$, and $\omega \in \mathbb{Z}^N_{>0}$ with $\gcd(\omega_1, \dots, \omega_N) = 1$ satisfying Proposition 18.7(1)–(4) with ω in place of v. By Proposition 18.7(4), $\mathrm{ord}_\nu q_h > 0$, so $q_h(0) = 0$ for any h, and $\iota(0) = 0$. By Proposition 18.7(2), there exist I-semi-invariant polynomials $\tilde{g}$, $\tilde{p}_j$, and $\tilde{\phi}_i$ on $\mathbb{A}^N$ with the characters of g, p_j, and ϕ_i, such that $\tilde{g} \circ \iota = g$, $\tilde{p}_j \circ \iota = p_j$, $\tilde{\phi}_i \circ \iota = \phi_i$, $\mathrm{wt}_\omega(\tilde{g}) = \mathrm{ord}_\nu g$, $\mathrm{wt}_\omega(\tilde{p}_j) = \mathrm{ord}_\nu p_j$, and $\mathrm{wt}_\omega(\tilde{\phi}_i) = \mathrm{ord}_\nu \phi_i$, where $\tilde{\phi}_i := 0$ if $\phi_i = 0$.

Let $n := N + 2$, let $\mathbb{A}^n = \mathbb{A}^2 \times \mathbb{A}^N$ with coordinates x, y on $\mathbb{A}^2$ and the coordinates of $\mathbb{A}^N$, and let $f_{N-1} := xy + \tilde{g}$. Since G preserves $xy + g$ up to a character, the character of xy equals the character of g, so f_{N-1} is I-semi-invariant. As any non-zero ϕ_i has the character β of x, $x + \tilde{\phi}_i$ is I-semi-invariant. Let $X' \subset \mathbb{A}^n$ be the common zero scheme of $f_1, \dots, f_{N-1}$. Since $f_1, \dots, f_{N-2}$ generate the ideal of $\iota(T)$ near 0 and $\tilde{g} \circ \iota = g$, the morphism $\mathrm{id}_{\mathbb{A}^2} \times \iota$ restricts to an I-equivariant isomorphism from a neighborhood of 0 in X onto a neighborhood of 0 in X'. Thus, near 0, X' is a normal integral complete intersection of codimension $N-1$ with klt singularities, on which I acts faithfully. Let

$$B' := \sum_{b_j > 0} b_j \,\mathrm{div}(\tilde{p}_j|_{X'}) + \sum_{c_i > 0} c_i \,\mathrm{div}\left((x + \tilde{\phi}_i)|_{X'}\right),$$

which corresponds to B under $\mathrm{id}_{\mathbb{A}^2} \times \iota$. Then $n \ge 2$, $1 \le N - 1 < n$, $H := I$, X', the polynomials $f_1, \dots, f_{N-1}$, the polynomials $\tilde{p}_j$ with $b_j > 0$ and $x + \tilde{\phi}_i$ with $c_i > 0$, their coefficients, B', and $\chi := \lambda$ form data as in Set-up 18.4, where I acts on the coordinate x_{n+1} of Set-up 18.4 by λ.

We check the conditions of Lemma 18.6 for F, viewed as a prime divisor over X'. We have $I_I(F) = I$, and $a(F, X', B')/|I| = a = \mathrm{mld}((X' \ni 0, B')/I)$. Let

$$\tilde{w} := (k, \varrho\,\mathrm{ord}_\nu g - k, \varrho\omega_1, \dots, \varrho\omega_N) \in \mathbb{Z}^n_{>0}.$$

A common divisor of the entries of $\tilde{w}$ divides k and $\varrho \gcd(\omega_1, \dots, \omega_N) = \varrho$, so $\gcd(\tilde{w}_1, \dots, \tilde{w}_n) = 1$ as ϱ, k are coprime. By Lemma 18.2(1), $\mathrm{ord}_F x = k$ and $\mathrm{ord}_F y = \varrho\,\mathrm{ord}_\nu g - k$. By Proposition 18.7(4), the coordinates of $\mathbb{A}^N$ restrict to $z, t, q_0, \dots, q_{m_\nu}$, whose orders along F

are $\varrho\omega_1, \dots, \varrho\omega_N$. As $\tilde{p}_j$ and $\tilde{\phi}_i$ involve only the coordinates of $\mathbb{A}^N$, Definition-Proposition 2.11 implies

$$\operatorname{ord}_F \tilde{p}_j = \varrho \operatorname{ord}_\nu p_j = \operatorname{wt}_{\tilde{w}}(\tilde{p}_j), \qquad \operatorname{ord}_F(x + \tilde{\phi}_i) = \min\{k, \varrho \operatorname{ord}_\nu \phi_i\} = \operatorname{wt}_{\tilde{w}}(x + \tilde{\phi}_i).$$

Moreover, $\operatorname{wt}_{\tilde{w}}(f_l) = \varrho \operatorname{wt}_\omega(f_l)$ for $l \le N-2$, and $\operatorname{wt}_{\tilde{w}}(f_{N-1}) = \min\{\varrho \operatorname{ord}_\nu g, \varrho \operatorname{wt}_\omega(\tilde{g})\} = \varrho \operatorname{ord}_\nu g$. Thus Lemma 18.2(1) and Proposition 18.7(3) imply

$$a(F, X', 0) = a(F, X, 0) = \varrho a(\nu, T, 0) = \varrho\Big(\sum_{h=1}^{N} \omega_h - \sum_{l=1}^{N-2} \operatorname{wt}_\omega(f_l)\Big).$$

Since $\sum_{h=1}^{n} \tilde{w}_h = \varrho \operatorname{ord}_\nu g + \varrho \sum_{h=1}^{N} \omega_h$ and $\sum_{l=1}^{N-1} \operatorname{wt}_{\tilde{w}}(f_l) = \varrho \sum_{l=1}^{N-2} \operatorname{wt}_\omega(f_l) + \varrho \operatorname{ord}_\nu g$, we get

$$a(F, X', 0) = \sum_{h=1}^{n} \tilde{w}_h - \sum_{l=1}^{N-1} \operatorname{wt}_{\tilde{w}}(f_l).$$

Finally, λ is the character of I on $\mathfrak{m}_F/\mathfrak{m}_F^2$. By Lemma 18.6, $\operatorname{mld}((X' \ni 0, B')/I) = \psi_\lambda(X', B')$, so $a = \psi_\lambda(X', B')$. Here $n+1 = N+3 \le M+7 \le M+9$, X' is defined by $N-1 \le M+3 \le M+5$ equations, and B' is defined by at most $s+r$ polynomials with coefficients in Γ. This proves (4).

Step 6. In this step we prove (1). Let $\Lambda := \Gamma\backslash\{0\}$, which is a DCC subset of $(0,1]$. By (4), any number $a \in (0,1)$ as in (1) is $\psi_\chi(X', B')$ for data as in Set-up 18.4 with $n \le M+8$, at most $M+5$ equations, and at most $s+r$ boundary polynomials (the h_j of Set-up 18.4) with coefficients in Λ, and M depends only on Γ. By Corollary 18.5 with $n_0 := M+8$, $r_0 := M+5$, $s_0 := \max\{s+r, 1\}$, and Λ, the set of such numbers a satisfies the ACC. This proves (1). □

Remark 18.10. Fixing s and r in Theorem 18.9 is not a restriction for (1). The terms with $b_j = 0$, with $p_j(0) \ne 0$, or with $c_i = 0$ do not change B on a G-invariant neighborhood of 0, the hypotheses, or the number a. Each remaining p_j has $\operatorname{mult}_0 p_j \ge 1$, each remaining ϕ_i satisfies $\min\{\operatorname{mult}_0 g, \operatorname{mult}_0 \phi_i\} \ge 1$ as $g, \phi_i \in (z,t)$, where $\operatorname{mult}_0 0 := +\infty$, and the remaining coefficients are at least ϵ. By Theorem 18.9(2) for the exceptional divisor E_0 of the blow-up of T at 0 and $a(E_0, T, B_T) = 2 - \sum_j b_j \operatorname{mult}_0 p_j$ [KM98, Lemma 2.29],

$$\sum_j b_j \operatorname{mult}_0 p_j + \sum_i c_i \min\{\operatorname{mult}_0 g, \operatorname{mult}_0 \phi_i\} < 2,$$

so fewer than $2/\epsilon$ polynomials p_j and fewer than $2/\epsilon$ polynomials $x + \phi_i$ remain. Hence the union over all s, r of the sets in (1) is a finite union of sets satisfying the ACC, and so it satisfies the ACC by Lemma 2.10(4).

18.6. **Tail stability and products with a line.** In this subsection, we prove that the quotient mld of a smooth threefold germ does not change when the equations of the boundary are perturbed by multiples of a high power of a semi-invariant parameter of bounded character, and we compute the quotient mld of the product of a surface pair with a line (Proposition 18.11).

Proposition 18.11. *(1) Fix an integer $n_0 \ge 1$, real numbers $\lambda > 1$ and $\gamma > 0$, and an integer N with $N(\lambda-1)\gamma > 6n_0$. Let $(X \ni x)$ be a smooth threefold germ with a faithful action of a finite group G fixing x, and t a regular G-semi-invariant parameter at x whose character has order at most n_0, and let $T := (t = 0)$. For $1 \le j \le s$ let f_j, h_j be non-zero regular G-semi-invariant functions with equal characters, not divisible by t, with $h_j - f_j \in (t^N)$, and let $c_j \ge \gamma$. Let $B_0 := \sum_j c_j \operatorname{div}(f_j)$, $B_1 := \sum_j c_j \operatorname{div}(h_j)$, and $C := \sum_j c_j \operatorname{div}_T(f_j|_T)$, and assume that $(T, \lambda C)$ is lc near x. Then, for any subgroup $L \subset G$, the pairs (X, B_0) and (X, B_1) are klt near x and*

$$\operatorname{mld}((X \ni x, B_0)/L) = \operatorname{mld}((X \ni x, B_1)/L).$$

More precisely, any prime divisor E over X/L with center the image of x and with $a(E, X/L, B_{0,X/L}) \le 3$ or $a(E, X/L, B_{1,X/L}) \le 3$ satisfies $a(E, X/L, B_{0,X/L}) =$

$a(E, X/L, B_{1,X/L})$, *and* $\operatorname{ord}_F f_j = \operatorname{ord}_F h_j$ *for any prime divisor* F *over* X *such that* $\operatorname{ord}_F|_{\mathbb{C}(X/L)}$ *is a positive multiple of* ord_E *and any* j.

(2) *Let* $(U \ni u)$ *be a smooth surface germ with a faithful action of a finite group* G *fixing* u, *let* $X := U \times \mathbb{A}^1_t$, *where* G *acts on* t *by a character* χ *of order* q, *let* $f_1, \dots, f_s$ *be non-zero regular* G*-semi-invariant functions on* U *near* u *and* $c_1, \dots, c_s \geq 0$, *and let* $C_U := \sum_j c_j \operatorname{div}_U(f_j)$ *with* (U, C_U) *klt near* u, $B_0 := \operatorname{pr}_U^* C_U$, *and* $(U/G, C_{U/G}) := (U, C_U)/G$. *For a prime divisor* P *over* U/G *with center the image of* u, *a prime divisor* P' *over* U *such that* $\operatorname{ord}_{P'}|_{\mathbb{C}(U/G)}$ *is a positive multiple of* ord_P, *and the character* α *of* $I_G(P')$ *on* $\mathfrak{m}_{P'}/\mathfrak{m}_{P'}^2$, *let* $b \in \{0, \dots, |I_G(P')| - 1\}$ *with* $\chi|_{I_G(P')} = \alpha^b$, $\tau_P := b/|I_G(P')|$, d_P *the order of* $e^{2\pi\sqrt{-1}\tau_P}$, *and* $a_P := a(P, U/G, C_{U/G})$. *Then* τ_P *does not depend on the choice of* P', $d_P \leq q$, *and*

$$\operatorname{mld}((X \ni (u,0), B_0)/G) = \min_P \min_{1 \leq j \leq d_P} \left(j a_P + \langle j\tau_P \rangle_+\right),$$

where $\langle z \rangle_+ \in (0, 1]$ *is the representative of* z *modulo* $\mathbb{Z}$, *and the minimum is attained by the prime divisor* E *over* X/G *such that* $\operatorname{ord}_F|_{\mathbb{C}(X/G)}$ *is a positive multiple of* ord_E, *for some* P *and* P' *as in the definition of* τ_P, *coprime positive integers* p *and* l, *and the prime divisor* F *over* X *such that* ord_F *is the Gauss extension of* $p \operatorname{ord}_{P'}$ *to* $\mathbb{C}(U)(t)$ *with* t*-value* l *(Definition-Proposition 2.11). If* $L \subset \operatorname{Ker} \chi$ *is a subgroup, then*

$$\operatorname{mld}((X \ni (u,0), B_0)/L) = 1 + \operatorname{mld}((U \ni u, C_U)/L).$$

Proof. (1) All assertions are local near x, and G fixes x. Possibly replacing X with a G-invariant open neighborhood of x, we may assume that T is smooth, the functions t, f_j, and h_j are regular on X, and $(T, \lambda C)$ is lc. Since $N(\lambda - 1)\gamma > 6n_0 > 0$, we have $N \geq 1$, hence $h_j|_T = f_j|_T$ for any j.

Fix $i \in \{0, 1\}$. Since t divides neither f_j nor h_j, the divisor T is not a component of B_i. Since X and T are smooth, $(K_X + T + \lambda B_i)|_T = K_T + (\lambda B_i)|_T$ by adjunction, and

$$(\lambda B_i)|_T = \lambda \sum_j c_j \operatorname{div}_T(f_j|_T) = \lambda C.$$

By Theorem 2.23, $(X, T + \lambda B_i)$ is lc near T. Possibly replacing X with a smaller G-invariant open neighborhood of x, we may assume that $(X, T + \lambda B_i)$ is lc for $i = 0, 1$, hence $(X, \lambda B_0)$ and $(X, \lambda B_1)$ are lc. Let F be a prime divisor over X. Since X is smooth, $a(F, X, 0) > 0$, and since $(X, \lambda B_i)$ is lc, $a(F, X, 0) - \lambda \operatorname{mult}_F B_i \geq 0$. Since $c_{j'} \operatorname{div}(f_{j'}) \geq 0$ and $c_{j'} \operatorname{div}(h_{j'}) \geq 0$ for any j', we obtain, for any j and $i \in \{0, 1\}$,

$$a(F, X, 0) \geq \lambda c_j \operatorname{ord}_F f_j, \qquad a(F, X, 0) \geq \lambda c_j \operatorname{ord}_F h_j, \tag{18.32}$$

and

$$a(F, X, B_i) = a(F, X, 0) - \operatorname{mult}_F B_i \geq \left(1 - \frac{1}{\lambda}\right) a(F, X, 0) > 0. \tag{18.33}$$

Thus (X, B_0) and (X, B_1) are klt. For any subgroup $L \subset G$, the group L acts faithfully on X, and Lemma 2.6(0)(2)(3), applied to L, implies that $(X, B_0)/L$ and $(X, B_1)/L$ are klt.

Fix a subgroup $L \subset G$, and let $\pi_L : X \to X/L$ be the quotient morphism. Let E be a prime divisor over X/L with $\operatorname{center}_{X/L} E = \{\pi_L(x)\}$, and let F be a prime divisor over X such that $\operatorname{ord}_F|_{\mathbb{C}(X/L)}$ is a positive multiple of ord_E. Such F exists by Lemma 2.6(3). By Lemma 2.6(1)(2), $\operatorname{ord}_F|_{\mathbb{C}(X/L)} = |I_L(F)| \operatorname{ord}_E$ and $\pi_L(\operatorname{center}_X F) = \{\pi_L(x)\}$. Since $\pi_L^{-1}(\pi_L(x)) = \{x\}$, we have $\operatorname{center}_X F = \{x\}$, hence $\operatorname{ord}_F t \geq 1$. Let q_L be the order of the restriction to L of the character of t. Then $q_L \leq n_0$, and t^{q_L} is an L-invariant regular function on X, hence a regular function on X/L vanishing at $\pi_L(x)$. Thus $\operatorname{ord}_E(t^{q_L}) \geq 1$ and

$$q_L \operatorname{ord}_F t = \operatorname{ord}_F(t^{q_L}) = |I_L(F)| \operatorname{ord}_E(t^{q_L}) \geq |I_L(F)|. \tag{18.34}$$

Suppose that $\operatorname{ord}_F f_j \neq \operatorname{ord}_F h_j$ for some j. Since $\operatorname{ord}_F(h_j - f_j) \geq N \operatorname{ord}_F t$, the equality $\operatorname{ord}_F h_j = \operatorname{ord}_F f_j$ holds whenever $\operatorname{ord}_F f_j < N \operatorname{ord}_F t$ or $\operatorname{ord}_F h_j < N \operatorname{ord}_F t$. Thus $\operatorname{ord}_F f_j \geq$

$N \operatorname{ord}_F t$, and (18.32) and $c_j \geq \gamma$ imply $a(F, X, 0) \geq \lambda\gamma N \operatorname{ord}_F t$. By Lemma 2.6(2), (18.33), and (18.34), for $i = 0, 1$,

$$a(E, X/L, B_{i,X/L}) = \frac{a(F, X, B_i)}{|I_L(F)|} \geq \frac{(\lambda-1)a(F, X, 0)}{\lambda|I_L(F)|} \geq \frac{(\lambda-1)\gamma N \operatorname{ord}_F t}{|I_L(F)|} \geq \frac{(\lambda-1)\gamma N}{n_0} > 6.$$

Therefore, if $a(E, X/L, B_{0,X/L}) \leq 3$ or $a(E, X/L, B_{1,X/L}) \leq 3$, then $\operatorname{ord}_F f_j = \operatorname{ord}_F h_j$ for any j. In this case $\operatorname{mult}_F B_0 = \operatorname{mult}_F B_1$, so $a(F, X, B_0) = a(F, X, B_1)$, and Lemma 2.6(2) implies $a(E, X/L, B_{0,X/L}) = a(E, X/L, B_{1,X/L})$.

It remains to prove the equality of the minimal log discrepancies. Let F_0 be the exceptional divisor of the blow-up of X at x, and E_0 the prime divisor over X/L such that $\operatorname{ord}_{F_0}|_{\mathbb{C}(X/L)}$ is a positive multiple of ord_{E_0}. Then $a(F_0, X, 0) = 3$ [KM98, Lemma 2.29], $\operatorname{center}_{X/L} E_0 = \{\pi_L(x)\}$, and, since $B_i \geq 0$ and $|I_L(F_0)| \geq 1$, Lemma 2.6(2) implies

$$a(E_0, X/L, B_{i,X/L}) = \frac{a(F_0, X, B_i)}{|I_L(F_0)|} \leq a(F_0, X, 0) = 3 \quad \text{for } i = 0, 1.$$

Let $m_i := \operatorname{mld}((X \ni x, B_i)/L)$. Since $a(E_0, X/L, B_{0,X/L}) \leq 3$, m_0 is the infimum of $a(E, X/L, B_{0,X/L})$ over the prime divisors E over X/L with $\operatorname{center}_{X/L} E = \{\pi_L(x)\}$ and $a(E, X/L, B_{0,X/L}) \leq 3$. For any such E, we have shown that

$$a(E, X/L, B_{1,X/L}) = a(E, X/L, B_{0,X/L}).$$

Hence $m_1 \leq m_0$. Exchanging the roles of B_0 and B_1, we obtain $m_0 \leq m_1$, so $m_0 = m_1$.

(2) The proof is long, so we divide it into several steps. Let $\pi_U : U \to U/G$ be the quotient morphism. Since $\mathbb{C}(X) = \mathbb{C}(U)(t)$ and G acts on t by χ, we have $\sigma^* t = \chi(\sigma)t$ for any $\sigma \in G$. Since (U, C_U) is klt near u, Lemma 2.6(2) implies $a_P = a(P', U, C_U)/|I_G(P')| > 0$ for any P and P' as in the statement.

Step 1. In this step, for P and P' as in the statement, we show that b is well defined, that τ_P does not depend on P', and that d_P divides q, and we choose a semi-invariant uniformizer of $\operatorname{ord}_{P'}$.

Let $e_{P'} := |I_G(P')|$. By Lemma 2.6(1), $I_G(P')$ is cyclic and α is faithful. Let $\sigma_0 \in I_G(P')$ be the generator with $\alpha(\sigma_0) = \exp(2\pi\sqrt{-1}/e_{P'})$. A character ψ of $I_G(P')$ is determined by $\psi(\sigma_0)$, which is an $e_{P'}$-th root of unity. Hence there exists a unique $b \in \{0, \dots, e_{P'} - 1\}$ with $\chi|_{I_G(P')} = \alpha^b$, and

$$\exp(2\pi\sqrt{-1}\tau_P) = \alpha(\sigma_0)^b = \chi(\sigma_0).$$

Thus d_P is the order of $\chi(\sigma_0)$, which divides the order q of χ. In particular, $d_P \leq q$ and $d_P\tau_P \in \mathbb{Z}$.

Let P'' be another prime divisor over U such that $\operatorname{ord}_{P''}|_{\mathbb{C}(U/G)}$ is a positive multiple of ord_P. The extension $\mathbb{C}(U/G) \subset \mathbb{C}(U)$ is Galois with group G, and $\operatorname{ord}_{P'}$ and $\operatorname{ord}_{P''}$ extend positive multiples of ord_P. By [ZS60, Chapter VI, §7, Corollary 3 to Theorem 12], the valuation rings of $\operatorname{ord}_{P'}$ and $\operatorname{ord}_{P''}$ are conjugate under G, that is, there exists $\sigma \in G$ with $\sigma^* R'' = R'$, where R' and R'' are the valuation rings of $\operatorname{ord}_{P'}$ and $\operatorname{ord}_{P''}$. Since the value groups of $\operatorname{ord}_{P'}$ and $\operatorname{ord}_{P''}$ are both $\mathbb{Z}$, we obtain $\operatorname{ord}_{P''} = \operatorname{ord}_{P'} \circ \sigma^*$. Conjugation by σ maps $I_G(P')$ onto $I_G(P'')$ and α onto the character of $I_G(P'')$ on $\mathfrak{m}_{P''}/\mathfrak{m}_{P''}^2$. Since χ is constant on conjugacy classes, the integers $|I_G(P'')|$ and b defined by P'' are equal to $e_{P'}$ and b. Thus τ_P does not depend on P'.

Let ϖ_0 be a uniformizer of $\operatorname{ord}_{P'}$, and let

$$\varpi := \frac{1}{e_{P'}} \sum_{\sigma \in I_G(P')} \alpha(\sigma)^{-1} \sigma^* \varpi_0.$$

Since $\sigma^*\varpi_0 - \alpha(\sigma)\varpi_0 \in \mathfrak{m}_{P'}^2$ for any $\sigma \in I_G(P')$, we have $\varpi - \varpi_0 \in \mathfrak{m}_{P'}^2$. Thus ϖ is a uniformizer of $\operatorname{ord}_{P'}$ with $\sigma^*\varpi = \alpha(\sigma)\varpi$ for any $\sigma \in I_G(P')$.

Step 2. In this step, for a prime divisor F over X with center $(u, 0)$, we find the prime divisor P' over U underlying ord_F, and we bound $a(F, X, B_0)$ from below.

Let F be a prime divisor over X with $\operatorname{center}_X F = \{(u,0)\}$. Since $\mathcal{O}_{U,u} \subset \mathcal{O}_{X,(u,0)}$, the restriction $\operatorname{ord}_F|_{\mathbb{C}(U)}$ is non-negative on $\mathcal{O}_{U,u}$ and positive on $\mathfrak{m}_u$. In particular, $\operatorname{ord}_F|_{\mathbb{C}(U)}$ is a non-trivial valuation with center u on U. By Lemma 2.12, applied to $\mathbb{C}(U) \subset \mathbb{C}(X)$ and the point $u \in U$, there exist a positive integer ϱ and a prime divisor P' over U with center u such that

$$\operatorname{ord}_F|_{\mathbb{C}(U)} = \varrho \operatorname{ord}_{P'}. \tag{18.35}$$

Let P be the prime divisor over U/G such that $\operatorname{ord}_{P'}|_{\mathbb{C}(U/G)}$ is a positive multiple of ord_P. By Lemma 2.6(2), $\operatorname{center}_{U/G} P = \{\pi_U(u)\}$. Let $e_{P'} := |I_G(P')|$, and let α, b, and ϖ be as in **Step 1**.

Any $\sigma \in I_G(F)$ preserves ord_F, hence $\operatorname{ord}_{P'}$ by (18.35), and acts trivially on the residue field of $\operatorname{ord}_{P'}$, which is a subfield of the residue field of ord_F. Thus $I_G(F) \subset I_G(P')$. Let β be the character of $I_G(F)$ on $\mathfrak{m}_F/\mathfrak{m}_F^2$, which is faithful by Lemma 2.6(1). Since $I_G(F) \subset I_G(P')$, the functions ϖ and t are $I_G(F)$-semi-invariant with characters $\alpha|_{I_G(F)}$ and $\chi|_{I_G(F)}$, and $\operatorname{ord}_F \varpi = \varrho$ by (18.35). By Lemma 9.2(1), applied to the action of the cyclic group $I_G(F)$ on X, whose inertia group of F is $I_G(F)$, and to the functions ϖ and t, we obtain

$$\alpha|_{I_G(F)} = \beta^{\varrho} \quad \text{and} \quad \chi|_{I_G(F)} = \beta^{\operatorname{ord}_F t}. \tag{18.36}$$

Let $h : U' \to U$ be a proper birational morphism from a normal variety such that P' is a prime divisor on U', and let $C_{U'}$ be the $\mathbb{R}$-divisor with $K_{U'} + C_{U'} = h^*(K_U + C_U)$ and $h_* C_{U'} = C_U$. Let $X' := U' \times \mathbb{A}^1_t$ and $B' := \operatorname{pr}_{U'}^* C_{U'}$. Then $K_{X'} + B' = (h \times \mathrm{id}_{\mathbb{A}^1_t})^*(K_X + B_0)$, so $a(F, X, B_0) = a(F, X', B')$ [KM98, Lemma 2.30]. The center of F on X' is contained in $P' \times \{0\}$, because $\operatorname{ord}_F \varpi > 0$ and $\operatorname{ord}_F t > 0$, and it maps onto P', because the center of $\operatorname{ord}_{P'}$ on U' is P' by (18.35). Hence the center of F on X' is $P' \times \{0\}$. Let η be the generic point of $P' \times \{0\}$. Since U' is normal, it is smooth at the generic point of P', so X' is smooth at η, and ϖ, t are regular parameters of $\mathcal{O}_{X',\eta}$. Moreover $\operatorname{mult}_{P'} C_{U'} = 1 - a(P', U, C_U) = 1 - e_{P'} a_P$. Thus there exists an open neighborhood V' of η in X' such that V' is smooth, ϖ and t are regular on V', $\operatorname{div}(\varpi) + \operatorname{div}(t)$ is an snc divisor on V', and $B'|_{V'} = (1 - e_{P'} a_P) \operatorname{div}(\varpi)$. By Lemma 2.13,

$$a(F, X, B_0) = a(F, V', B'|_{V'}) \geq e_{P'} a_P \operatorname{ord}_F \varpi + \operatorname{ord}_F t = \varrho e_{P'} a_P + \operatorname{ord}_F t. \tag{18.37}$$

Step 3. In this step, we introduce a lattice minimum m_P for any P as in the statement and show that $a(E, X/G, B_{0,X/G}) \geq m_P$ for F as in **Step 2** and the prime divisor E over X/G such that $\operatorname{ord}_F|_{\mathbb{C}(X/G)}$ is a positive multiple of ord_E.

For P and P' as in the statement, with $e_{P'} := |I_G(P')|$ and b as in **Step 1**, let

$$\Lambda_P := \mathbb{Z}^2 + \frac{1}{e_{P'}}(1, b)\mathbb{Z} \subset \mathbb{R}^2, \qquad \phi_P(w) := e_{P'} a_P w_1 + w_2, \qquad m_P := \min_{1 \leq j \leq d_P} \left(j a_P + \langle j \tau_P \rangle_+ \right).$$

By **Step 1**, $e_{P'}$ and b, hence Λ_P, ϕ_P, and m_P, do not depend on P'. We show that

$$m_P = \min\{\phi_P(w) \mid w \in \Lambda_P \cap \mathbb{R}^2_{>0}\}. \tag{18.38}$$

Let $w \in \Lambda_P \cap \mathbb{R}^2_{>0}$, and write $w = \frac{k}{e_{P'}}(1, b) + (n_1, n_2)$ with $k, n_1, n_2 \in \mathbb{Z}$. Then $w_1 = \frac{j}{e_{P'}}$ with $j := k + n_1 e_{P'} \geq 1$, and $w_2 \equiv \frac{kb}{e_{P'}} \equiv \frac{jb}{e_{P'}} = j\tau_P$ modulo $\mathbb{Z}$, so $w_2 \geq \langle j\tau_P \rangle_+$. Hence $\phi_P(w) \geq j a_P + \langle j \tau_P \rangle_+$. Let $j' \in \{1, \ldots, d_P\}$ with $j' \equiv j$ modulo d_P. Since $d_P \tau_P \in \mathbb{Z}$ and $a_P > 0$, we have $\langle j\tau_P \rangle_+ = \langle j' \tau_P \rangle_+$ and $j a_P \geq j' a_P$. Thus $\phi_P(w) \geq m_P$. Conversely, for $1 \leq j \leq d_P$, the vector

$$w_j := \left(\frac{j}{e_{P'}}, \langle j\tau_P \rangle_+ \right) = \frac{j}{e_{P'}}(1, b) + \left(0, \langle j\tau_P \rangle_+ - j\tau_P\right)$$

belongs to $\Lambda_P \cap \mathbb{R}^2_{>0}$, and $\phi_P(w_j) = j a_P + \langle j \tau_P \rangle_+$. This proves (18.38).

Now let F, ϱ, P, P', $e_{P'}$, β be as in **Step 2**, and let E be the prime divisor over X/G such that $\operatorname{ord}_F|_{\mathbb{C}(X/G)}$ is a positive multiple of ord_E. Since $I_G(F) \subset I_G(P')$, $|I_G(F)|$ divides $e_{P'}$.

By (18.36) and $\chi|_{I_G(P')} = \alpha^b$, we have $\beta^{\mathrm{ord}_F t} = \beta^{\varrho b}$, and since β is faithful, $|I_G(F)|$ divides $\mathrm{ord}_F t - \varrho b$. Hence

$$\frac{1}{|I_G(F)|}(\varrho, \mathrm{ord}_F t) = \frac{\varrho e_{P'}}{|I_G(F)|} \cdot \frac{1}{e_{P'}}(1, b) + \Big(0, \frac{\mathrm{ord}_F t - \varrho b}{|I_G(F)|}\Big) \in \Lambda_P \cap \mathbb{R}^2_{>0}.$$

By Lemma 2.6(2), (18.37), and (18.38),

$$a(E, X/G, B_{0,X/G}) = \frac{a(F, X, B_0)}{|I_G(F)|} \geq \phi_P\Big(\frac{1}{|I_G(F)|}(\varrho, \mathrm{ord}_F t)\Big) \geq m_P. \tag{18.39}$$

Step 4. In this step, for any P and P' as in the statement, we construct a Gauss extension of a multiple of $\mathrm{ord}_{P'}$ which computes m_P on X/G.

Fix P and P', let $e_{P'} := |I_G(P')|$, and let α, b, and ϖ be as in **Step 1**. Since $\Lambda_P \subset \frac{1}{e_{P'}}\mathbb{Z}^2$ and both coefficients of ϕ_P are positive, the minimum in (18.38) is attained by some $w \in \Lambda_P \cap \mathbb{R}^2_{>0}$. Since $\phi_P(w) = m_P > 0$ and $\phi_P(\frac{1}{k}w) = \frac{1}{k}\phi_P(w)$, the vector w is primitive in Λ_P. Let e' be the least positive integer with $e'w \in \mathbb{Z}^2$, and write $w = \frac{1}{e'}(p, l)$ with positive integers p and l, so that $\gcd(p, l, e') = 1$.

We show that $\gcd(p, l) = 1$. Suppose that $\delta' := \gcd(p, l) > 1$. Then $\gcd(\delta', e') = 1$, so there exists an integer k with $k\delta' \equiv 1$ modulo e', and

$$\frac{1}{\delta'}w - kw = \frac{1 - k\delta'}{e'}\Big(\frac{p}{\delta'}, \frac{l}{\delta'}\Big) \in \mathbb{Z}^2.$$

Hence $\frac{1}{\delta'}w \in \Lambda_P$, which contradicts the primitivity of w.

We show that $e' = \gcd(e_{P'}, pb - l)$. Since $w \in \frac{1}{e_{P'}}\mathbb{Z}^2$, e' divides $e_{P'}$. Write $w = \frac{k}{e_{P'}}(1, b) + (n_1, n_2)$ with $k, n_1, n_2 \in \mathbb{Z}$. Then $bw_1 - w_2 = bn_1 - n_2 \in \mathbb{Z}$, that is, e' divides $pb - l$. Let $e'' := \gcd(e_{P'}, pb - l)$, so that e' divides e''. Then

$$\frac{1}{e''}(p, l) = \frac{pe_{P'}}{e''} \cdot \frac{1}{e_{P'}}(1, b) - \Big(0, \frac{pb - l}{e''}\Big) \in \Lambda_P \quad \text{and} \quad w = \frac{e''}{e'} \cdot \frac{1}{e''}(p, l),$$

so the primitivity of w implies $e'' = e'$.

Let F be the prime divisor over X such that ord_F is the Gauss extension of $p\,\mathrm{ord}_{P'}$ to $\mathbb{C}(U)(t)$ with t-value l. By Lemma 18.1, applied with U, C_U, P', p, and l in place of X, B, E, ϱ, and c, the prime divisor F exists, $\mathrm{center}_X F = \mathrm{center}_U P' \times \{0\} = \{(u, 0)\}$, and

$$a(F, X, B_0) = pa(P', U, C_U) + l = pe_{P'}a_P + l.$$

We show that $I_G(F)$ is the kernel of the character α^{pb-l} of $I_G(P')$. By Definition-Proposition 2.11, $\mathrm{ord}_F|_{\mathbb{C}(U)} = p\,\mathrm{ord}_{P'}$, so $I_G(F) \subset I_G(P')$ by **Step 2**. The function $t^p\varpi^{-l}$ satisfies $\mathrm{ord}_F(t^p\varpi^{-l}) = pl - pl = 0$, and for $\sigma \in I_G(P')$,

$$\sigma^*(t^p\varpi^{-l}) = \chi(\sigma)^p\alpha(\sigma)^{-l}t^p\varpi^{-l} = \alpha(\sigma)^{pb-l}t^p\varpi^{-l}.$$

If $\sigma \in I_G(F)$, then σ fixes the residue of $t^p\varpi^{-l}$, which is non-zero, hence $\alpha(\sigma)^{pb-l} = 1$. Conversely, let $\sigma \in I_G(P')$ with $\alpha(\sigma)^{pb-l} = 1$. Since σ preserves $\mathrm{ord}_{P'}$ and $\sigma^*t = \chi(\sigma)t$, the formula of Definition-Proposition 2.11 shows that σ preserves ord_F. For an integer m and integers k, i with $pk + li = m$, let $\epsilon_m := \alpha(\sigma)^k\chi(\sigma)^i = \alpha(\sigma)^{k+bi}$. Since $\gcd(p, l) = 1$, two such pairs (k, i) differ by $s(-l, p)$ for some $s \in \mathbb{Z}$, and $\alpha(\sigma)^{s(pb-l)} = 1$. Thus ϵ_m depends only on m. Let $g = \sum_i a_i t^i \in \mathbb{C}(U)[t]$ be non-zero with $\mathrm{ord}_F g = m$. For any i with $a_i \neq 0$ and $p\,\mathrm{ord}_{P'}(a_i) + li = m$, write $a_i = \varpi^{k_i}u_i$ with $\mathrm{ord}_{P'} u_i = 0$. Then

$$\sigma^*(a_it^i) = \epsilon_m\sigma^*(u_i)\varpi^{k_i}t^i,$$

and $\sigma^*u_i - u_i \in \mathfrak{m}_{P'}$ because $\sigma \in I_G(P')$, so $\mathrm{ord}_F(\sigma^*(a_it^i) - \epsilon_m a_it^i) > m$. For the other i with $a_i \neq 0$, both $\sigma^*(a_it^i)$ and a_it^i have ord_F greater than m. Hence $\mathrm{ord}_F(\sigma^*g - \epsilon_m g) > m$. Any

element of $\mathbb{C}(X)$ with ord_F equal to 0 is of the form g_1/g_2 with non-zero $g_1, g_2 \in \mathbb{C}(U)[t]$ and $\operatorname{ord}_F g_1 = \operatorname{ord}_F g_2 = m$ for some m, and

$$\sigma^*\Big(\frac{g_1}{g_2}\Big) - \frac{g_1}{g_2} = \frac{(\sigma^* g_1 - \epsilon_m g_1)g_2 - g_1(\sigma^* g_2 - \epsilon_m g_2)}{g_2 \sigma^* g_2} \in \mathfrak{m}_F.$$

Thus σ acts trivially on the residue field of ord_F, that is, $\sigma \in I_G(F)$.

Since $I_G(P')$ is cyclic of order $e_{P'}$ and α is faithful, the kernel of α^{pb-l} has order $\gcd(e_{P'}, pb-l) = e'$. Let E be the prime divisor over X/G such that $\operatorname{ord}_F|_{\mathbb{C}(X/G)}$ is a positive multiple of ord_E. By Lemma 2.6(2), $\operatorname{center}_{X/G} E$ is the image of $(u, 0)$ and

$$a(E, X/G, B_{0,X/G}) = \frac{pe_{P'}a_P + l}{e'} = \phi_P(w) = m_P. \tag{18.40}$$

Step 5. In this step, we conclude the proof.

Let $\pi : X \to X/G$ be the quotient morphism, and let E be a prime divisor over X/G with $\operatorname{center}_{X/G} E = \{\pi(u, 0)\}$. By Lemma 2.6(2)(3), there exists a prime divisor F over X such that $\operatorname{ord}_F|_{\mathbb{C}(X/G)}$ is a positive multiple of ord_E, and $\operatorname{center}_X F = \{(u, 0)\}$ because $\pi^{-1}(\pi(u, 0)) = \{(u, 0)\}$. By **Steps 2 and 3**, $a(E, X/G, B_{0,X/G}) \geq m_P$ for some P. By **Step 4**, for any P there exists such an E with $a(E, X/G, B_{0,X/G}) = m_P$. Hence

$$\operatorname{mld}((X \ni (u, 0), B_0)/G) = \inf_P m_P.$$

Since (U, C_U) is klt near u, for a log resolution $h : U' \to U$ of (U, C_U) near u and $C_{U'}$ with $K_{U'} + C_{U'} = h^*(K_U + C_U)$, the morphism $h \times \mathrm{id}_{\mathbb{A}^1_t}$ is a log resolution of (X, B_0) near $(u, 0)$ with

$$K_{U' \times \mathbb{A}^1_t} + \operatorname{pr}_{U'}^* C_{U'} = (h \times \mathrm{id}_{\mathbb{A}^1_t})^*(K_X + B_0).$$

Let F be a prime divisor over X whose center on X intersects a neighborhood of $(u, 0)$ over which $h \times \mathrm{id}_{\mathbb{A}^1_t}$ is a log resolution and (U, C_U) is klt. By [KM98, Lemma 2.30], $a(F, X, B_0) = a(F, U' \times \mathbb{A}^1_t, \operatorname{pr}_{U'}^* C_{U'})$. Let η be the generic point of the center of F on $U' \times \mathbb{A}^1_t$, and c its codimension. Since the support of $\operatorname{pr}_{U'}^* C_{U'}$ is snc near η, there exist regular parameters $x_1, \dots, x_c$ of $\mathcal{O}_{U' \times \mathbb{A}^1_t, \eta}$ such that the components of $\operatorname{pr}_{U'}^* C_{U'}$ through η are $\operatorname{div}(x_1), \dots, \operatorname{div}(x_k)$ for some $k \leq c$. Let b_i be the coefficient of $\operatorname{div}(x_i)$ in $\operatorname{pr}_{U'}^* C_{U'}$ for $i \leq k$, and $b_i := 0$ for $i > k$. Any coefficient of $\operatorname{pr}_{U'}^* C_{U'}$ is the coefficient $1 - a(P'', U, C_U) < 1$ of $C_{U'}$ along some prime divisor P'' on U', since (U, C_U) is klt. Thus $b_i < 1$ for any i. Let V be an open neighborhood of η in $U' \times \mathbb{A}^1_t$ such that V is smooth, every x_i is regular on V, $\sum_{i=1}^c \operatorname{div}(x_i)$ is an snc divisor on V, and $(\operatorname{pr}_{U'}^* C_{U'})|_V = \sum_{i=1}^c b_i \operatorname{div}(x_i)$. By Lemma 2.13,

$$a(F, X, B_0) = a\Big(F, V, \sum_{i=1}^c b_i \operatorname{div}(x_i)\Big) \geq \sum_{i=1}^c (1 - b_i) \operatorname{ord}_F x_i > 0.$$

Thus (X, B_0) is klt near $(u, 0)$. By Lemma 2.6(0)(2)(3), $(X, B_0)/G$ is klt near $\pi(u, 0)$, so $\operatorname{mld}((X \ni (u, 0), B_0)/G) = a(E_*, X/G, B_{0,X/G})$ for some prime divisor E_* over X/G with center $\pi(u, 0)$ [Amb99, Section 1, the second paragraph after Definition 1.1]. Let P_* be a prime divisor associated with E_* as in the previous paragraph. Then

$$\operatorname{mld}((X \ni (u, 0), B_0)/G) = a(E_*, X/G, B_{0,X/G}) \geq m_{P_*} \geq \inf_P m_P = \operatorname{mld}((X \ni (u, 0), B_0)/G).$$

Thus $\operatorname{mld}((X \ni (u, 0), B_0)/G) = m_{P_*} = \min_P m_P$, and by (18.40), the minimum is attained by the prime divisor E over X/G constructed in **Step 4** for P_* and some P', which is of the form in the statement. Together with **Step 1**, this proves the formula.

Finally, let $L \subset \operatorname{Ker} \chi$ be a subgroup. Then L acts faithfully on U, fixes u, and acts trivially on t. Applying the formula with L in place of G, the character of t has order 1, so $d_P = 1$ and $\tau_P = 0$ for any prime divisor P over U/L with center the image of u, and $\langle \tau_P \rangle_+ = 1$. Hence

$$\operatorname{mld}((X \ni (u, 0), B_0)/L) = \min_P(a_P + 1) = 1 + \operatorname{mld}((U \ni u, C_U)/L),$$

where now $a_P = a(P, U/L, C_{U/L})$ and $(U/L, C_{U/L}) := (U, C_U)/L$. □

The following example is in the setting of Theorem 18.16 and shows that, when χ is non-trivial, the minimum in Proposition 18.11(2) can be less than one plus the surface mld.

Example 18.12. For an integer $n \geq 2$, let $r := 2n$, $U := \mathbb{A}^2_{x,y}$, $X := U \times \mathbb{A}^1_t$, let $G := \mu_r$ act on X with weights $(1, -1)$ on U and weight n on t, and let $B := \frac{1}{4}\operatorname{div}(x)$. Then (X, B) is terminal, $X \to X/G$ is quasi-étale, $\operatorname{mld}((X \ni 0, B)/G) = \frac{5}{4} + \frac{1}{4r}$, and $\operatorname{mld}((U \ni 0, \frac{1}{4}\operatorname{div}_U(x))/G) = \frac{3}{4} + \frac{1}{4r}$.

Proof. Let $\zeta := e^{2\pi\sqrt{-1}/r}$. The element $\zeta^k \in G$ acts by $x \to \zeta^k x$, $y \to \zeta^{-k} y$, and $t \to \zeta^{kn} t$. If $\zeta^k \neq 1$, then neither of the eigenvalues ζ^k and ζ^{-k} on U is 1, so ζ^k fixes no prime divisor on X pointwise. Thus $I_G(D) = \{1\}$ for any prime divisor D on X, and $X \to X/G$ is quasi-étale by Lemma 2.6(1).

Let F be an exceptional prime divisor over X, η the generic point of $\operatorname{center}_X F$, and $c \geq 2$ the codimension of η. If $\eta \in (x = 0)$, we choose regular parameters $x, x_2, \dots, x_c$ of $\mathcal{O}_{X,\eta}$, and Lemma 2.13 implies $a(F, X, B) \geq \frac{3}{4}\operatorname{ord}_F x + \sum_{i \geq 2} \operatorname{ord}_F x_i \geq \frac{3}{4} + c - 1 > 1$. If $\eta \notin (x = 0)$, then $B = 0$ near η and Lemma 2.13 implies $a(F, X, B) \geq c > 1$. Thus (X, B) is terminal.

For the surface, by Lemma 3.4 with $d = 2$, $k = \mathbb{C}$, eigenparameters x, y, and $b = (\frac{1}{4}, 0)$, the number $\operatorname{mld}((U \ni 0, \frac{1}{4}\operatorname{div}_U(x))/G)$ is the minimum of $\frac{3}{4}w_1 + w_2$ over $w \in N_G \cap \mathbb{R}^2_{>0}$, where $N_G = \mathbb{Z}^2 + \frac{1}{r}(1, -1)\mathbb{Z}$. A vector of N_G is congruent to $\frac{k}{r}(1, -1)$ modulo $\mathbb{Z}^2$ for some $0 \leq k \leq r - 1$. For $k = 0$ the least value of $\frac{3}{4}w_1 + w_2$ on positive vectors is $\frac{7}{4}$, at $(1, 1)$, and for $1 \leq k \leq r - 1$ it is $\frac{3k}{4r} + 1 - \frac{k}{r} = 1 - \frac{k}{4r}$, at $(\frac{k}{r}, 1 - \frac{k}{r})$. Hence

$$\operatorname{mld}\Big(\Big(U \ni 0, \frac{1}{4}\operatorname{div}_U(x)\Big)/G\Big) = \min_{1 \leq k \leq r-1} \Big(1 - \frac{k}{4r}\Big) = \frac{3}{4} + \frac{1}{4r},$$

attained at $k = r - 1$, by the prime divisor P_0 over U/G such that $\operatorname{ord}_{P_0'}|_{\mathbb{C}(U/G)}$ is a positive multiple of ord_{P_0}, where P_0' is the monomial prime divisor in x, y with weights $(r - 1, 1)$.

For the threefold, let $C_U := \frac{1}{4}\operatorname{div}_U(x)$, which is klt, so that $B = \operatorname{pr}_U^* C_U$. By Proposition 18.11(2),

$$\operatorname{mld}((X \ni 0, B)/G) = \min_P \min_{1 \leq j \leq d_P} \big(j a_P + \langle j\tau_P \rangle_+\big),$$

and we call the inner minimum the *term* of P. The character χ of t is $\zeta^k \to \zeta^{kn}$, of order $q = 2$. Thus $d_P \leq 2$ and $\tau_P \in \{0, \frac{1}{2}\}$ for any P. If $\tau_P = 0$, then $d_P = 1$ and the term of P is $a_P + 1$. If $\tau_P = \frac{1}{2}$, then $d_P = 2$ and the term of P is $\min(a_P + \frac{1}{2}, 2a_P + 1) = a_P + \frac{1}{2}$. Since $a_P \geq \frac{3}{4} + \frac{1}{4r}$, any term is at least $\frac{5}{4} + \frac{1}{4r}$. For P_0 and P_0', the group G preserves $\operatorname{ord}_{P_0'}$. Let $z := x/y^{r-1}$. Then $\mathbb{C}(x, y) = \mathbb{C}(y)(z)$ and $x^i y^j = z^i y^{(r-1)i+j}$, so $\operatorname{ord}_{P_0'}$ and the Gauss extension of ord_y to $\mathbb{C}(y)(z)$ with z-value 0 (Definition-Proposition 2.11) both send $\sum_{i,j} a_{ij} x^i y^j \in \mathbb{C}[x, y]$ to $\min\{(r - 1)i + j \mid a_{ij} \neq 0\}$, hence are equal. By [Bou72, Chapter VI, §10, no. 1, Proposition 2], the residue field of $\operatorname{ord}_{P_0'}$ is $\mathbb{C}(\bar{z})$, where $\bar{z}$ is the residue of z. The function z is G-invariant because $\zeta^k \cdot \zeta^{k(r-1)} = 1$. Hence $I_G(P_0') = G$. Since $\operatorname{ord}_{P_0'} y = 1$, the function y is a uniformizer, so $\alpha(\zeta^k) = \zeta^{-k}$ and $\chi = \alpha^{-n} = \alpha^n$. Thus $b = n$, $\tau_{P_0} = \frac{n}{r} = \frac{1}{2}$, and the term of P_0 is $a_{P_0} + \frac{1}{2} = \frac{5}{4} + \frac{1}{4r}$. Therefore

$$\operatorname{mld}((X \ni 0, B)/G) = \frac{5}{4} + \frac{1}{4r} < \frac{7}{4} + \frac{1}{4r} = 1 + \operatorname{mld}\Big(\Big(U \ni 0, \frac{1}{4}\operatorname{div}_U(x)\Big)/G\Big).$$

Since $X \to X/G$ is quasi-étale, Lemma 2.6(4) implies $B_{X/G} = \frac{1}{4}\pi((x = 0))$, where $\pi : X \to X/G$ is the quotient morphism, so $\operatorname{mld}((X \ni 0, B)/G)$ is the minimal log discrepancy of the threefold pair $(X/G \ni \pi(0), B_{X/G})$ with the single coefficient $\frac{1}{4}$, although (X, B) is terminal. □

When G acts trivially on t, Proposition 18.11 and Theorem 2.18(1) imply the following.

Corollary 18.13. *Fix real numbers $\lambda > 1$ and $\gamma > 0$ and an integer N with $N(\lambda-1)\gamma > 6$. Let $(U \ni u)$ be a smooth surface germ with a faithful action of a finite group G fixing u, and let $X := U \times \mathbb{A}^1_t$ with trivial action on t. Let $f_1, \dots, f_s$ be non-zero regular G-semi-invariant functions near u, $c_j \geq \gamma$, and $C := \sum_j c_j \operatorname{div}_U(f_j)$ with $(U, \lambda C)$ lc near u. Let g_j be regular functions near $(u,0)$ such that $h_j := f_j + t^N g_j$ has the character of f_j, and let $B_0 := \operatorname{pr}_U^* C$ and $B := \sum_j c_j \operatorname{div}(h_j)$. Then, for any subgroup $L \subset G$,*

$$\operatorname{mld}((X \ni (u,0), B)/L) = \operatorname{mld}((X \ni (u,0), B_0)/L) = 1 + \operatorname{mld}((U \ni u, C)/L) > 1.$$

In particular, let $\Gamma \subset [0,1]$ be a DCC set with $\Gamma\backslash\{0\} \neq \emptyset$, and take $\gamma := \gamma(\Gamma)$. Then, for fixed λ and N, the set of the numbers $\operatorname{mld}((X \ni (u,0), B)/L)$, for all $(U \ni u)$, G, L, f_j, g_j, and $c_j \in \Gamma$ with $c_j \geq \gamma$ such that $(U, \lambda C)$ is lc near u and each h_j has the character of f_j, satisfies the ACC.

Proof. Let $T := U \times \{0\}$. The projection pr_U restricts to an isomorphism $T \to U$, and we write f_j also for $f_j \circ \operatorname{pr}_U$. The function t is a G-invariant regular parameter at $(u,0)$, so the character of t has order 1. Since $N(\lambda-1)\gamma > 6$, we have $N \geq 1$, hence $h_j|_T = f_j|_T$, and under the isomorphism $T \to U$ the function $f_j|_T$ corresponds to the non-zero function f_j. In particular, t divides neither f_j nor h_j. Thus f_j and h_j are non-zero regular G-semi-invariant functions near $(u,0)$ with equal characters, not divisible by t, and $h_j - f_j = t^N g_j \in (t^N)$. Moreover $\sum_j c_j \operatorname{div}(f_j \circ \operatorname{pr}_U) = B_0$, and $\sum_j c_j \operatorname{div}_T(f_j|_T)$ corresponds to C, so $(T, \lambda \sum_j c_j \operatorname{div}_T(f_j|_T))$ is lc near $(u,0)$. By Proposition 18.11(1) with $n_0 = 1$ and $B_1 = B$, for any subgroup $L \subset G$,

$$\operatorname{mld}((X \ni (u,0), B)/L) = \operatorname{mld}((X \ni (u,0), B_0)/L).$$

Since $(U, \lambda C)$ is lc near u, for any prime divisor P over a neighborhood of u, we have $a(P,U,C) = a(P,U,0) - \operatorname{mult}_P C \geq (1 - \frac{1}{\lambda}) a(P,U,0) > 0$, so (U,C) is klt near u. Since G acts trivially on t, any subgroup $L \subset G$ is contained in the kernel of the character of t, and Proposition 18.11(2) implies

$$\operatorname{mld}((X \ni (u,0), B_0)/L) = 1 + \operatorname{mld}((U \ni u, C)/L).$$

By Lemma 2.6(0)(2)(3), $(U,C)/L$ is klt near the image of u, so $\operatorname{mld}((U \ni u, C)/L)$ is the log discrepancy of some prime divisor [Amb99, Section 1, the second paragraph after Definition 1.1], and in particular $\operatorname{mld}((U \ni u, C)/L) > 0$. This proves the equalities and the inequality in the statement.

Now let $\Gamma \subset [0,1]$ be a DCC set with $\Gamma\backslash\{0\} \neq \emptyset$, so that $\gamma(\Gamma) > 0$ by Lemma 2.10(2), and assume that $c_j \in \Gamma$ for any j. Since $(U, \lambda C)$ is lc near u, so is (U,C). By Lemma 2.10(5), applied to U, u, L, $f_1, \dots, f_s$, and $c_1, \dots, c_s$, there exists an L-invariant affine open neighborhood U' of u such that $(U', C|_{U'})$ is lc and the coefficients of $(C|_{U'})_{U'/L}$ belong to the DCC set $\Psi(\Gamma_+ \cap [0,1])$, which is contained in $[0,1]$. Since $\operatorname{mld}((U \ni u, C)/L) = \operatorname{mld}((U' \ni u, C|_{U'})/L)$ and $(U', C|_{U'})/L$ is an lc surface pair, Theorem 2.18(1), applied to the DCC set $\Psi(\Gamma_+ \cap [0,1])$, implies that the numbers $\operatorname{mld}((U \ni u, C)/L)$ belong to a set satisfying the ACC. Hence the numbers $1 + \operatorname{mld}((U \ni u, C)/L)$ satisfy the ACC. □

18.7. **Products of plane pairs with a line.** In this subsection, we prove Theorem 18.16: for products of plane pairs with a line, the positive quotient minimal log discrepancies satisfy the ACC, for any finite diagonal abelian group and any character of the group on the line. Proposition 18.14 is not used in the proof of Theorem 18.16. It treats perturbations of the boundary by terms divisible by t^N when the group acts trivially on the line, where, in contrast with Corollary 18.13, the order N depends only on the coefficient set and (U,C) is only assumed to be klt.

Proposition 18.14. *Let $\Gamma \subset [0,1]$ be a DCC set with $\Gamma\backslash\{0\} \neq \emptyset$, $\gamma := \gamma(\Gamma)$, $\Sigma := \Gamma_+ \cap [0,1]$, $\Xi := \Psi(\Sigma)$, l the constant of Theorem 2.18(2) for Ξ, and $N > l/\gamma$ an integer. Let $U := \mathbb{A}^2_{x,y}$, $X := U \times \mathbb{A}^1_t$, and G a finite diagonal abelian group acting faithfully on U and trivially on t. Let $f_1, \dots, f_s$ be non-zero G-semi-invariant polynomials on U, $c_j \in \Gamma$ positive, $C := \sum_j c_j \operatorname{div}_U(f_j)$*

with (U, C) klt near 0, and g_j polynomials on X such that $h_j := f_j + t^N g_j$ has the character of f_j. Let $B_0 := \mathrm{pr}_U^* C$ and $B := \sum_j c_j \operatorname{div}_X(h_j)$. Then, for any subgroup $H \subset G$, the following hold.

(1) The pair (X, B) is klt near 0.

(2) We have

$$\operatorname{mld}((X \ni 0, B)/H) = \operatorname{mld}((X \ni 0, B_0)/H) = 1 + \operatorname{mld}((U \ni 0, C)/H) > 1,$$

and both minimal log discrepancies are attained by a prime divisor E over X/H with $\operatorname{ord}_E t = 1$ and $a(E, X/H, 0) \le l + 1$.

(3) $\operatorname{mld}((X \ni 0, B + (t = 0))/H) = \operatorname{mld}((U \ni 0, C)/H)$.

In particular, the set of such numbers $\operatorname{mld}((X \ni 0, B)/H)$ satisfies the ACC.

Proof. The proof is long, so we divide it into several steps. Fix a subgroup $H \subset G$. Let $\pi : U \to U/H$ be the quotient morphism, $(U/H, C_{U/H}) := (U, C)/H$, $\bar{0} := \pi(0)$, $\bar{x} \in X/H$ the image of $0 \in X$, and

$$q := \operatorname{mld}((U \ni 0, C)/H) = \operatorname{mld}(U/H \ni \bar{0}, C_{U/H}).$$

Since H acts trivially on t, we have $X/H = (U/H) \times \mathbb{A}^1_t$ and $\mathbb{C}(X/H) = \mathbb{C}(U/H)(t)$.

Step 1. In this step, we construct a prime divisor F over U with $a(F, U, C) = |I_H(F)|q$ and bound the orders $\operatorname{ord}_F f_j$.

By Lemma 2.10(5), there exists an H-invariant affine open neighborhood U° of 0 such that $(U^\circ, C|_{U^\circ})$ is lc and the coefficients of $(C|_{U^\circ})_{U^\circ/H}$ belong to $\Psi(\Sigma) = \Xi$. Since (U, C) is klt near 0, after shrinking U° to an H-invariant affine open neighborhood of 0, the pair $(U^\circ, C|_{U^\circ})$ is klt. By Lemma 2.6(0)(2)(3), the pair $(U^\circ/H, (C|_{U^\circ})_{U^\circ/H})$ is klt, and in particular $q > 0$, since the minimal log discrepancy of a klt pair at a closed point is attained [Amb99, Section 1, the second paragraph after Definition 1.1]. Since Ξ satisfies the DCC (Lemma 2.10(1)), Theorem 2.18(2) [HL23, Theorem 1.2] implies that there exists a prime divisor P over U/H with $\operatorname{center}_{U/H} P = \{\bar{0}\}$, $a(P, U/H, C_{U/H}) = q$, and $a(P, U/H, 0) \le l$. By Lemma 2.6(1)(3), there exists a prime divisor F over U with $\operatorname{ord}_F|_{\mathbb{C}(U/H)} = e \operatorname{ord}_P$, where $e := |I_H(F)|$, and by Lemma 2.6(2), $\operatorname{center}_U F = \{0\}$ and $a(F, U, C) = eq$.

Let $R_0 := \frac{1}{|H|}\pi_* R(\pi)$, so that $(U, 0)/H = (U/H, R_0)$. Since U/H is $\mathbb{Q}$-factorial [KM98, Lemma 5.16] and $R_0 \ge 0$, Lemma 2.6(2) and [KM98, Lemma 2.27] imply

$$a(F, U, 0) = e a(P, U/H, R_0) \le e a(P, U/H, 0) \le el.$$

Since $C \ge c_j \operatorname{div}_U(f_j)$ and $a(F, U, C) > 0$, we obtain

$$c_j \operatorname{ord}_F f_j \le \operatorname{mult}_F C = a(F, U, 0) - a(F, U, C) < el \quad \text{for any } j.$$

As $c_j \ge \gamma$ and $N > l/\gamma$,

$$\operatorname{ord}_F f_j < \frac{el}{\gamma} < eN \quad \text{for any } j. \tag{18.41}$$

Step 2. In this step, we show that any prime divisor F_1 over X with $\operatorname{center}_X F_1 = \{0\}$ satisfies

$$a(F_1, X, B + (t = 0)) \ge |I_H(F_1)|q. \tag{18.42}$$

The argument only uses $h_j|_{(t=0)} = f_j$, so (18.42) also holds with B_0 in place of B.

Let H act on $X \times \mathbb{A}^2_{w_1,w_2} = \mathbb{A}^5_{x,y,t,w_1,w_2}$ through the action on X, by the character of x on w_1, and by the character of y on w_2. We identify $U \times \mathbb{A}^2_{w_1,w_2}$ with the hypersurface $(t = 0)$ of $X \times \mathbb{A}^2_{w_1,w_2}$. Any $1 \ne \sigma \in H$ acts non-trivially on U, so σ multiplies both x and w_1, or both y and w_2, by a common root of unity $\zeta_\sigma \ne 1$. Thus the fixed locus of σ is contained in $(x = w_1 = 0)$ or in $(y = w_2 = 0)$, the fixed locus has codimension at least two in $X \times \mathbb{A}^2_{w_1,w_2}$, and the intersection of the fixed locus with $U \times \mathbb{A}^2_{w_1,w_2}$ has codimension at least two in $U \times \mathbb{A}^2_{w_1,w_2}$. Since the fixed locus of any $1 \ne \sigma \in H$ has codimension at least two in $X \times \mathbb{A}^2_{w_1,w_2}$, H contains no pseudoreflection of $X \times \mathbb{A}^2_{w_1,w_2}$. Since the quotient morphism of a smooth variety by a finite

group is étale exactly at the points with trivial stabilizer (the quotient morphism by a free action is étale, and if the quotient morphism is étale at a point, then it is étale along the orbit of the point, so the fiber through the point has $|H|$ points and the stabilizer is trivial), the image of the locus where $X \times \mathbb{A}^2_{w_1,w_2} \to (X \times \mathbb{A}^2_{w_1,w_2})/H$ is not étale intersects $(U \times \mathbb{A}^2_{w_1,w_2})/H$ in codimension at least two, and both quotient morphisms of $X \times \mathbb{A}^2_{w_1,w_2}$ and of $U \times \mathbb{A}^2_{w_1,w_2}$ by H are quasi-étale.

Let $r := |H|$ and let $\bar{0}'$ be the image of 0 in $(X \times \mathbb{A}^2_{w_1,w_2})/H$. The polynomials h_j^r and $(w_1w_2)^r$ are H-invariant, so

$$\mathfrak{a} := \prod_j (h_j^r)^{c_j/r} \cdot \big((w_1w_2)^r\big)^{1/r}$$

is an $\mathbb{R}$-ideal on $(X \times \mathbb{A}^2_{w_1,w_2})/H$, and the restriction $\mathfrak{a}|_{(U\times\mathbb{A}^2_{w_1,w_2})/H}$ to $(U \times \mathbb{A}^2_{w_1,w_2})/H$ is non-zero because $h_j|_{(t=0)} = f_j \neq 0$. Theorem 2.22, applied to the H-invariant equation t of $U \times \mathbb{A}^2_{w_1,w_2}$, implies

$$\operatorname{mld}\big((U \times \mathbb{A}^2_{w_1,w_2})/H \ni \bar{0}', \mathfrak{a}|_{(U\times\mathbb{A}^2_{w_1,w_2})/H}\big) = \operatorname{mld}\big((X \times \mathbb{A}^2_{w_1,w_2})/H \ni \bar{0}', (t^r)^{\frac{1}{r}}\mathfrak{a}\big).$$

Let

$$D := \operatorname{pr}^* B + (t = 0) + (w_1 = 0) + (w_2 = 0)$$

on $X \times \mathbb{A}^2_{w_1,w_2}$. Both quotient morphisms are quasi-étale, so their ramification divisors are zero. Thus Lemma 2.6(2), applied with the zero boundary and the $\mathbb{R}$-ideals $\mathfrak{a}|_{(U\times\mathbb{A}^2_{w_1,w_2})/H}$ and $(t^r)^{1/r}\mathfrak{a}$, and Lemma 2.6(3) imply that the left-hand side and the right-hand side are

$$\operatorname{mld}(((U \times \mathbb{A}^2_{w_1,w_2}) \ni 0, \operatorname{pr}^* C + (w_1 = 0) + (w_2 = 0))/H) \quad \text{and} \quad \operatorname{mld}(((X \times \mathbb{A}^2_{w_1,w_2}) \ni 0, D)/H),$$

respectively. Applying the last sentence of Lemma 8.1 twice, first to (U, C) with the character of x and then to $(U \times \mathbb{A}^1_{w_1}, \operatorname{pr}^* C + (w_1 = 0))$ with the character of y, the left-hand side is equal to q. Hence

$$\operatorname{mld}(((X \times \mathbb{A}^2_{w_1,w_2}) \ni 0, D)/H) = q. \tag{18.43}$$

Now let F_1 be a prime divisor over X with $\operatorname{center}_X F_1 = \{0\}$. By Lemma 8.1(2), applied twice, first to $(X, B + (t = 0))$ with the character of x and then to $(X \times \mathbb{A}^1_{w_1}, \operatorname{pr}^*(B + (t = 0)) + (w_1 = 0))$ with the character of y, there exists a prime divisor F_3 over $X \times \mathbb{A}^2_{w_1,w_2}$ with center 0 such that $I_H(F_3) = I_H(F_1)$ and

$$a(F_3, X \times \mathbb{A}^2_{w_1,w_2}, D) = a(F_1, X, B + (t = 0)).$$

Thus (18.43) and Lemma 2.6 imply (18.42).

Step 3. In this step, we show that $\operatorname{mld}((X \ni 0, B + (t = 0))/H) = q$, and that any prime divisor E_1 over X/H with $\operatorname{center}_{X/H} E_1 = \{\bar{x}\}$ satisfies

$$a(E_1, X/H, B_{X/H}) \geq q + \operatorname{ord}_{E_1} t \geq q + 1, \tag{18.44}$$

and that (18.44) also holds with B_0 in place of B.

By (18.42) and Lemma 2.6, we have

$$\operatorname{mld}((X \ni 0, B + (t = 0))/H) \geq q.$$

Conversely, let $\epsilon > 0$. By (18.43) and Lemma 2.6, there exists a prime divisor F_3 over $X \times \mathbb{A}^2_{w_1,w_2}$ with center 0 such that $a(F_3, X \times \mathbb{A}^2_{w_1,w_2}, D) < |I_H(F_3)|(q + \epsilon)$. By Lemma 8.1(1), applied twice, first to $(X \times \mathbb{A}^1_{w_1}, \operatorname{pr}^*(B + (t = 0)) + (w_1 = 0))$ with the character of y and then to $(X, B + (t = 0))$ with the character of x, there exist positive integers ϱ_1, ϱ_2 and a prime divisor F_1 over X with center 0 such that $I_H(F_3) \subset I_H(F_1)$ and

$$a(F_3, X \times \mathbb{A}^2_{w_1,w_2}, D) \geq \varrho_1\varrho_2 a(F_1, X, B + (t = 0)).$$

By (18.42), $a(F_1, X, B + (t = 0)) > 0$. Hence

$$\frac{a(F_1, X, B + (t = 0))}{|I_H(F_1)|} \leq \frac{\varrho_1\varrho_2 a(F_1, X, B + (t = 0))}{|I_H(F_3)|} \leq \frac{a(F_3, X \times \mathbb{A}^2_{w_1,w_2}, D)}{|I_H(F_3)|} < q + \epsilon.$$

Since $\epsilon > 0$ is arbitrary, $\mathrm{mld}((X \ni 0, B + (t = 0))/H) = q$.

Let E_1 be a prime divisor over X/H with $\mathrm{center}_{X/H} E_1 = \{\bar{x}\}$. By Lemma 2.6(1)(3), there exists a prime divisor F_1 over X with $\mathrm{ord}_{F_1}|_{\mathbb{C}(X/H)} = |I_H(F_1)| \mathrm{ord}_{E_1}$, and $\mathrm{center}_X F_1 = \{0\}$ by Lemma 2.6(2). Since $t \in \mathbb{C}(X/H)$ vanishes at $\bar{x}$, $\mathrm{ord}_{E_1} t$ is a positive integer and $\mathrm{ord}_{F_1} t = |I_H(F_1)| \mathrm{ord}_{E_1} t$. Thus Lemma 2.6(2) and (18.42) imply

$$a(E_1, X/H, B_{X/H}) = \frac{a(F_1, X, B + (t = 0)) + \mathrm{ord}_{F_1} t}{|I_H(F_1)|} \geq q + \mathrm{ord}_{E_1} t,$$

which is (18.44). Replacing B with B_0, Lemma 2.6(2) and (18.42) for B_0 (**Step 2**) imply (18.44) for B_0.

Step 4. In this step, we construct the prime divisor E and prove the equalities of minimal log discrepancies.

Let F and e be as in **Step 1**, and let λ be the character of $I_H(F)$ on $\mathfrak{m}_F/\mathfrak{m}_F^2$. By Lemma 2.6(1), λ is faithful of order e, so e is the unique integer $k \in \{1, \dots, e\}$ such that λ^k is trivial, that is, the t-value of Lemma 8.1(2) for the prime divisor F and the trivial character is e. Since H acts trivially on t, Lemma 8.1(2), applied to (U, C) with the trivial character, implies that the Gauss extension of ord_F to $\mathbb{C}(U)(t)$ with t-value e (Definition-Proposition 2.11) is $\mathrm{ord}_{F'}$ for a prime divisor F' over X with $\mathrm{center}_X F' = \{0\}$, $I_H(F') = I_H(F)$, and $a(F', X, B_0 + (t = 0)) = a(F, U, C) = eq$. Since $\mathrm{ord}_{F'} t = e$, we obtain $a(F', X, B_0) = e(q+1)$. Each g_j is regular at 0, so $\mathrm{ord}_{F'}(t^N g_j) \geq eN$, while $\mathrm{ord}_{F'} f_j = \mathrm{ord}_F f_j < eN$ by (18.41). Thus $\mathrm{ord}_{F'} h_j = \mathrm{ord}_{F'} f_j$ for any j, and

$$a(F', X, B) = a(F', X, B_0) = e(q + 1).$$

Let E be the prime divisor over X/H with $\mathrm{ord}_{F'}|_{\mathbb{C}(X/H)} = e \mathrm{ord}_E$. By Lemma 2.6(2), $a(E, X/H, B_{X/H}) = a(E, X/H, B_{0,X/H}) = q + 1$, where $(X/H, B_{0,X/H}) := (X, B_0)/H$. Together with (18.44) and Lemma 2.6, we obtain

$$\mathrm{mld}((X \ni 0, B)/H) = \mathrm{mld}((X \ni 0, B_0)/H) = 1 + q > 1,$$

and E attains both minimal log discrepancies. Moreover, $\mathrm{ord}_E t = \mathrm{ord}_{F'} t/e = 1$. For $a_i \in \mathbb{C}(U/H)$, we have

$$\mathrm{ord}_{F'}\Big(\sum_i a_i t^i\Big) = \min_i(\mathrm{ord}_F a_i + ie) = e \min_i(\mathrm{ord}_P a_i + i),$$

so ord_E is the Gauss extension of ord_P to $\mathbb{C}(U/H)(t)$ with t-value 1. Thus Lemma 18.1, applied to U/H with the zero boundary, P, and $\varrho = c = 1$, implies $a(E, X/H, 0) = a(P, U/H, 0) + 1 \leq l + 1$.

Step 5. In this step, we show that (X, B) is klt near 0 and prove the ACC.

On a log resolution of $(U, \mathrm{Supp}\, C)$, the coefficients of the crepant pullback of κC are affine functions of κ, and over a neighborhood of 0 they are less than one for $\kappa = 1$. Thus there exists $\kappa_0 > 1$ such that, over a neighborhood of 0, they are at most one for $\kappa = \kappa_0$, and $(U, \kappa_0 C)$ is lc near 0 by [KM98, Corollary 2.31(3) and Lemma 2.30]. Let $S := (t = 0) \subset X$. Since $h_j|_S = f_j \neq 0$, the $\mathbb{R}$-divisor $\kappa_0 B$ has no common component with S, and since X and S are smooth, the different of $(X, S + \kappa_0 B)$ on S is $\kappa_0 B|_S = \kappa_0 C$. Thus, after shrinking X around 0, $(X, S + \kappa_0 B)$ is lc near S by Theorem 2.23, and $(X, \kappa_0 B)$ is lc near 0. Hence any prime divisor F_2 over X whose center contains a point of a small neighborhood of 0 satisfies

$$a(F_2, X, B) = \Big(1 - \frac{1}{\kappa_0}\Big) a(F_2, X, 0) + \frac{1}{\kappa_0} a(F_2, X, \kappa_0 B) \geq \Big(1 - \frac{1}{\kappa_0}\Big) a(F_2, X, 0) > 0,$$

and (X, B) is klt near 0.

By **Step 1**, q is the minimal log discrepancy of the lc surface germ $(U^\circ/H \ni \bar{0}, (C|_{U^\circ})_{U^\circ/H})$, whose coefficients belong to the DCC set Ξ. Thus the numbers q satisfy the ACC by Theorem 2.18(1). Since $\mathrm{mld}((X \ni 0, B)/H) = 1 + q$ by **Step 4**, the set of such numbers $\mathrm{mld}((X \ni 0, B)/H)$ satisfies the ACC. □

The following lemma bounds the number of elements of a minimal generating sequence of a prime divisor over a plane whose log discrepancy is small compared with its inertia group. In the proofs of Theorems 18.16 and 18.17, it is applied to the restriction to the plane of the valuation of a prime divisor computing the minimal log discrepancy.

Lemma 18.15. *Let $\Lambda \subset [0,1]$ be a DCC set, let $\epsilon := \gamma(\Lambda)$ if $\Lambda\backslash\{0\} \neq \emptyset$ and $\epsilon := 1$ otherwise, and let $M := 2 + \lceil 2/\epsilon \rceil$. Let $T := \mathbb{A}^2_{z,t}$, let H be a finite cyclic group acting faithfully and diagonally on T, let $h_1, \dots, h_s$ be non-zero H-semi-invariant polynomials, let $c_1, \dots, c_s \in \Lambda$, and let $C := \sum_j c_j \operatorname{div}_T(h_j)$ with (T, C) klt near 0. Let ν be a prime divisor over T with center 0 such that $I_H(\nu) = H$ and $a(\nu, T, C) \leq 3|H|$. Then ν has a minimal generating sequence* [dFGPM23, Definition 2.9] *with at most $M + 12$ elements.*

Proof. The proof is long, so we divide it into several steps. Let $e := |H|$.

Step 1. In this step, we fix notation from [dFGPM23] and bound the multiplicity of the plane boundary.

Let S be a smooth surface, $o \in S$ a closed point, and E' a prime divisor over S with $\operatorname{center}_S E' = \{o\}$. Let $\psi_{E'}$ be the minimal resolution of $\operatorname{ord}_{E'}$ [dFGPM23, Definition 2.8], that is, the shortest sequence of blow-ups of closed points over o such that E' is a prime divisor on the last surface. The centers of the blow-ups of $\psi_{E'}$ are the centers of $\operatorname{ord}_{E'}$ on the intermediate surfaces. Let $G(\psi_{E'})$ be the dual tree of the exceptional prime divisors of $\psi_{E'}$, without vertices for strict transforms of curves on S, and let $g(E') + 1$ be the number of elements of a minimal generating sequence of $\operatorname{ord}_{E'}$ [dFGPM23, Definition 2.9].

For a finite tree G, let $\operatorname{end}(G)$ be the number of vertices of G of valence one. For a plane branch A on S through o, let $g(A) + 1$ be the number of minimal generators of the semigroup of A [dFGPM23, Notation 2.2]. We use the following facts.

(a) If $\psi_{E'}$ consists of at least two blow-ups, then $g(E') + 1 = \operatorname{end}(G(\psi_{E'}))$ [dFGPM23, Definition 2.11 and Proposition 2.12]. If $\psi_{E'}$ is one blow-up, then $g(E') + 1 = 2$ [dFGPM23, Example 2.13].

(b) For any curvetta A of E' at $\psi_{E'}$ [dFGPM23, Definition 2.5], $g(E') + 1 \leq g(A) + 2$ [dFGPM23, Proposition 2.14]. Moreover, for any $\theta \in \mathfrak{m}_o$ with $A \not\subset Z(\theta)$, we have

$$2^{g(A)} \leq n_1 \cdots n_{g(A)} \leq \bar{\beta}_0 \leq (Z(\theta) \cdot A)_o,$$

where $\bar{\beta}_0$ and the integers $n_j \geq 2$ are as in [dFGPM23, Notation 2.2, Proposition 2.3(1)], and $n_1 \cdots n_{g(A)}$ divides $\bar{\beta}_0$ by the definition of the n_j in [dFGPM23, Notation 2.2]; the last inequality holds because $(Z(\theta) \cdot A)_o$ is a positive element of the semigroup of A [dFGPM23, Definition 1.15] and $\bar{\beta}_0$ is its least positive element.

(c) For any $\theta \in \mathcal{O}_{S,o}\backslash\{0\}$ and any curvetta A of E' at $\psi_{E'}$, we have $\operatorname{ord}_{E'} \theta \leq (Z(\theta) \cdot A)_o$ [dFGPM23, Remark 2.6], with equality if the strict transforms of A and of the curve $Z(\theta) = (\theta = 0)$ on the last surface of $\psi_{E'}$ have no common point on E'.

For the equality, write $\psi^*_{E'} Z(\theta) = Z(\theta)' + \sum_i \operatorname{ord}_{E_i}(\theta) E_i$, where $Z(\theta)'$ is the strict transform and the E_i are the exceptional prime divisors of $\psi_{E'}$. The strict transform A' of A intersects $\psi^{-1}_{E'}(o)$ at one point of E', transversally and at no other E_i [dFGPM23, Definition 2.5], so the projection formula gives $(Z(\theta) \cdot A)_o = \operatorname{ord}_{E'} \theta + (Z(\theta)' \cdot A')$, and the last term vanishes if $Z(\theta)'$ and A' have no common point on E'; any common point maps to o, hence is contained in E'.

(d) If G_1 and G_2 are disjoint finite trees and G is the tree consisting of G_1, G_2, and one edge joining a vertex of G_1 to a vertex of G_2, then $\operatorname{end}(G)$ is at most $\operatorname{end}(G_1) + \operatorname{end}(G_2)$ plus the number of $i \in \{1, 2\}$ such that G_i has one vertex, because the added edge raises the valence of one vertex of each G_i by one, so a vertex of valence zero acquires valence one and a vertex of valence one loses it.

Deleting the h_j with $c_j = 0$ or $h_j(0) \neq 0$ does not change C near 0. Thus we may assume that $c_j > 0$ and $h_j(0) = 0$ for any j. Let E_0 be the exceptional divisor of the blow-up of $0 \in T$. Since

(T, C) is klt near 0, [KM98, Lemma 2.29] implies

$$a(E_0, T, C) = 2 - \sum_j c_j \operatorname{mult}_0 h_j > 0.$$

If $s \geq 1$, then $\Lambda \backslash \{0\} \neq \emptyset$ and $c_j \geq \epsilon$ for any j, so $\sum_j \operatorname{mult}_0 h_j < 2/\epsilon$. Let Z be the reduced curve $\operatorname{Supp} C \cup (z = 0) \cup (t = 0)$. Then Z is H-invariant and

$$\operatorname{mult}_0 Z \leq 2 + \sum_j \operatorname{mult}_0 h_j \leq 2 + \lceil 2/\epsilon \rceil = M,$$

which also holds if $s = 0$.

Step 2. In this step, we resolve Z and show that $\operatorname{end}(G(\psi_{E'})) \leq M + 1$ and $g(E') + 1 \leq M + 1$ for any exceptional prime divisor E' of the resolution.

Let $f_1 : T_1 \to T$ be the minimal embedded resolution of Z over a neighborhood of 0 on which Z is snc outside 0 if Z is not snc at 0, and the blow-up of 0 otherwise. By uniqueness, f_1 is H-equivariant. The exceptional curves of a sequence of point blow-ups of a smooth surface form an snc divisor, so any center of f_1 other than 0 is a point of the strict transform of Z.

Let E' be an f_1-exceptional prime divisor, and let $p_0 = 0, p_1, \dots, p_k$ be the centers of the blow-ups of $\psi_{E'}$, which are centers of f_1. If $k = 0$, then $g(E') + 1 = 2$ by (a), and $G(\psi_{E'})$ has no vertex of valence one. Assume that $k \geq 1$. Then p_k is a point of the strict transform of a branch A of Z at 0, so $\operatorname{mult}_0 A \leq \operatorname{mult}_0 Z \leq M$. Since p_i is the image of p_k, the strict transform of A passes through p_i for any i. Let $\phi : T' \to T$ be the composite of the blow-ups of $p_0, \dots, p_{k-1}$, so that $p_k \in T'$, let $E_1, \dots, E_k$ be the ϕ-exceptional prime divisors, and let $\tilde{A} \subset T'$ be the strict transform of A. Fix a curvetta $A_{E'}$ of E' at $\psi_{E'}$, and let L be the line defined by a general linear form ξ in z, t. Then L is tangent neither to A nor to $A_{E'}$, and the strict transform L' of L on T' intersects $\phi^{-1}(0)$ only at a general point of the strict transform of the first exceptional curve, so $p_k \notin L'$. Write $\phi^* L = L' + \sum_i \operatorname{ord}_{E_i}(\xi) E_i$. At most two E_i pass through p_k, and they are smooth and transverse at p_k. Thus

$$\operatorname{ord}_{E'} \xi = \operatorname{mult}_{p_k}(\phi^* L) = \sum_{i : p_k \in E_i} \operatorname{ord}_{E_i} \xi.$$

Since $(\tilde{A} \cdot L')_y \geq 0$ and $(\tilde{A} \cdot E_i)_y \geq 0$ for any $y \in \phi^{-1}(0)$ and any i, and $(\tilde{A} \cdot E_i)_{p_k} \geq 1$ if $p_k \in E_i$, the projection formula implies

$$\operatorname{mult}_0 A = (A \cdot L)_0 = \sum_{y \in \phi^{-1}(0)} (\tilde{A} \cdot \phi^* L)_y \geq \sum_{i : p_k \in E_i} \operatorname{ord}_{E_i}(\xi)(\tilde{A} \cdot E_i)_{p_k} \geq \operatorname{ord}_{E'} \xi.$$

On the other hand, the strict transform of L on the last surface of $\psi_{E'}$ does not intersect E', since $p_k \notin L'$. Thus (c) implies $\operatorname{ord}_{E'} \xi = (L \cdot A_{E'})_0$, so $(L \cdot A_{E'})_0 \leq M$, and (b) with $\theta = \xi$ gives $g(A_{E'}) \leq \log_2 M \leq M - 1$. Hence (a) and (b) imply

$$\operatorname{end}(G(\psi_{E'})) = g(E') + 1 \leq g(A_{E'}) + 2 \leq M + 1.$$

Step 3. In this step, assuming that the center of ν on T_1 is a closed point, we construct a monomial divisor E with $\operatorname{end}(G(\psi_E)) \leq M + 2$ and locate the center of ν over E.

If the center of ν on T_1 is the generic point of an f_1-exceptional prime divisor E', then $\nu = E'$ and $g(\nu) + 1 \leq M + 1$ by **Step 2**. Assume that the center of ν on T_1 is a closed point p. Since $I_H(\nu) = H$, the prime divisor ν is H-invariant, so p is fixed by H, and $p \in f_1^{-1}(0)$. The reduced total transform of Z on T_1 is snc, and one or two components of the reduced total transform contain p, at least one of them f_1-exceptional. Since H fixes p, it permutes the components of the reduced total transform of Z through p. If at most one of them is f_1-exceptional, then H fixes each of them, because H maps f_1-exceptional curves to f_1-exceptional curves. If two f_1-exceptional curves E' and E'' contain p, then, possibly swapping them, the blow-up creating E' is one of the blow-ups of $\psi_{E''}$, so $G(\psi_{E'})$ has fewer vertices than $G(\psi_{E''})$. Since H preserves the number of vertices, H fixes each of them. Let E^* be the f_1-exceptional curve containing p for

which $G(\psi_{E^*})$ has the most vertices. Since H is abelian and $|H|$ is invertible, any H-stable ideal of $\mathcal{O}_{T_1,p}$ is generated by semi-invariants, and $\mathfrak{m}_p/\mathfrak{m}_p^2$ is a direct sum of characters. Thus there exist H-semi-invariant regular parameters u_0, v_0 at p such that $E^* = (u_0 = 0)$ near p and the other component of the reduced total transform of Z through p, if it exists, is $(v_0 = 0)$ near p.

Let $m_1 := \operatorname{ord}_\nu u_0$, $m_2 := \operatorname{ord}_\nu v_0$, $d := \gcd(m_1, m_2)$, and let μ be the monomial valuation in u_0, v_0 with weights $(m_1/d, m_2/d)$ (Definition 2.16). Let $f_2' : T_2 \to T_1$ be obtained by repeatedly blowing up the center of μ until the center is a prime divisor E, and $f_2 := f_1 \circ f_2'$. Since u_0 and v_0 are semi-invariant, μ is H-invariant, the centers are fixed by H, and f_2 is H-equivariant. In the toric description with respect to u_0, v_0, the curves $(u_0 = 0)$ and $(v_0 = 0)$ correspond to the rays of $(1, 0)$ and $(0, 1)$, each blow-up inserts the sum of the primitive generators of the two-dimensional cone of the current fan containing (m_1, m_2) in the interior of the cone, and E corresponds to $(m_1, m_2)/d$. Thus each center of f_2' is the intersection point of two curves, and it is contained in a non-exceptional curve only if it is the intersection point of the last exceptional curve with the strict transform of $(v_0 = 0)$. Such a center occurs only for the first blow-ups of f_2', as long as the cone containing (m_1, m_2) has the ray of $(0, 1)$ as an edge.

No f_1-exceptional curve through p has more vertices than E^*, so no blow-up of f_1 after the creation of E^* has center the image of p. Thus ψ_E consists of ψ_{E^*} followed by the blow-ups of f_2', and $G(\psi_E)$ has at least two vertices. If p is contained in an f_1-exceptional curve $E'' \neq E^*$, then each center of f_2' is the intersection point of two exceptional curves, each blow-up subdivides an edge, and $\operatorname{end}(G(\psi_E)) = \operatorname{end}(G(\psi_{E^*})) \leq M + 1$ by **Step 2**. Otherwise, the blow-up of p attaches a vertex to E^*, which raises end by at most one if $G(\psi_{E^*})$ has at least two vertices. If $G(\psi_{E^*})$ is one vertex, then after the blow-up of p the tree is one edge, with two vertices of valence one. Each later blow-up either subdivides an edge or attaches a vertex to the last vertex, which then has valence one, and neither changes end. Thus, by **Step 2**,

$$\operatorname{end}(G(\psi_E)) \leq \max\{M + 2, 2\} = M + 2.$$

Since $\nu(u_0^i v_0^j) = im_1 + jm_2$ for any $i, j \geq 0$, the valuation ν is non-negative on the monomial coordinates of the affine toric open subset of T_2 over a neighborhood of p corresponding to a two-dimensional cone with edge the ray of (m_1, m_2), positive on the coordinate vanishing on E, and zero on the other coordinate. Hence the center of ν on T_2 is either the generic point of E or a closed point $p' \in E$ not contained in any other component of the reduced total transform of Z on T_2. In the first case, $\nu = E$ and $g(\nu) + 1 = \operatorname{end}(G(\psi_E)) \leq M + 2$ by (a). In the second case, p' is contained in the open torus orbit of E, and there exists an H-semi-invariant Laurent monomial φ in u_0, v_0 which restricts to an isomorphism from the open torus orbit of E onto $\mathbb{C}^*$.

Step 4. In this step, we assume that the center of ν on T_2 is a closed point p' and show that the order of ν along a transverse parameter at p' is $e\ell$ with $\ell \in \{1, 2\}$.

Since H fixes p' and $\varphi(p') \neq 0$, the function φ is H-invariant, so H acts trivially on E. Let $v := \varphi - \varphi(p')$, and let u be an H-semi-invariant local equation of E at p'. Then u, v are regular parameters at p' and v is H-invariant. If $\sigma \in H$ acts trivially on $\mathfrak{m}_{p'}/\mathfrak{m}_{p'}^2$, then $\sigma^* u = u$ and $\sigma^* v = v$, since u and v are H-semi-invariant regular parameters at p'. Hence σ acts trivially on $\widehat{\mathcal{O}}_{T_2,p'} = \mathbb{C}[[u, v]]$, hence on $\mathbb{C}(T_2) \subset \operatorname{Frac} \widehat{\mathcal{O}}_{T_2,p'}$, and $\sigma = 1$ because H acts faithfully on T. Since v is H-invariant, the character of u is faithful of order e. Thus $\widehat{\mathcal{O}}_{T_2,p'} = \mathbb{C}[[u, v]]$ with a generator of H acting by $u \to \zeta u$ and $v \to v$ for a primitive e-th root of unity ζ, the surface T_2/H is smooth at the image $\bar{p}'$ of p' with regular parameters $U_0 := u^e$ and $V_0 := v$, and near p' the quotient morphism $T_2 \to T_2/H$ is étale outside E.

Let C_2 be the $\mathbb{R}$-divisor with $K_{T_2} + C_2 = f_2^*(K_T + C)$. Since $\operatorname{Supp} C \subset Z$, near p' the only component of C_2 is E, with coefficient $1 - a(E, T, C)$, and $a(E, T, C) > 0$ since (T, C) is klt near 0. The group H acts trivially on E, so the ramification index of E over the image of E in T_2/H is e. Thus, by Lemma 2.6(4), near $\bar{p}'$ the boundary of $(T_2, C_2)/H = (T_2/H, C_{2,T_2/H})$ is $(1 - a(E, T, C)/e) \operatorname{div}(U_0)$. Let $\bar{\nu}$ be the prime divisor over T_2/H with $\operatorname{ord}_\nu |_{\mathbb{C}(T_2/H)} = e \operatorname{ord}_{\bar{\nu}}$ (Lemma 2.6(1), $I_H(\nu) = H$). By Lemma 2.6(2), [KM98, Lemma 2.30], Lemma 2.13, and the

assumption $a(\nu,T,C)\le 3e$,

$$3\ge\frac{a(\nu,T,C)}{e}=a(\bar\nu,T_2/H,C_{2,T_2/H})\ge\frac{a(E,T,C)}{e}\operatorname{ord}_{\bar\nu}U_0+\operatorname{ord}_{\bar\nu}V_0.$$

Since $\operatorname{ord}_{\bar\nu}U_0$ and $\operatorname{ord}_{\bar\nu}V_0$ are positive integers and $a(E,T,C)>0$, the integer $\ell:=\operatorname{ord}_{\bar\nu}V_0$ is 1 or 2, and $\operatorname{ord}_\nu v=e\ell$.

Let λ be the character of H on $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$, which is faithful by Lemma 2.6(1). Since $I_H(\nu)=H$, Lemma 9.2(1) implies that the character of any H-semi-invariant function f is $\lambda^{\operatorname{ord}_\nu f}$. Applied to u, this implies that $\lambda^{\operatorname{ord}_\nu u}$ is faithful, hence

$$\gcd(\operatorname{ord}_\nu u,e)=1,\qquad \operatorname{ord}_\nu v=e\ell,\qquad \ell\in\{1,2\}. \tag{18.45}$$

Step 5. In this step, we keep the assumption of **Step 4** and bound the minimal generating sequence of ν over p' by an H-invariant curvetta.

Let ψ' be the minimal resolution of ord_ν over $(T_2\ni p')$, let R be the last exceptional prime divisor of ψ', so that $\nu=R$, and let $g'+1$ be the number of elements of a minimal generating sequence of ord_ν as a valuation of $\mathcal{O}_{T_2,p'}$. The centers of ψ' are centers of ν, so they are fixed by H and ψ' is H-equivariant. Since H acts trivially on the residue field $\mathbb{C}(R)$ of ord_ν, it acts trivially on R. Let $p''\in R$ be a general point, not contained in any other exceptional curve of ψ' or in the strict transforms of $E=(u=0)$ and $(v=0)$. Let ϖ_R be an H-semi-invariant local equation of R at p'', whose character is λ. Let y_1 be a local function at p'' such that $y_1|_R$ is a local parameter of R at p'', and let y' be the projection of y_1 to the H-invariant functions. Since H acts trivially on R, $y'|_R=y_1|_R$, so ϖ_R,y' are regular parameters at p'' with y' invariant. The curve $(y'=0)$ is smooth, H-invariant, and transverse to R at p''. Thus the image A' of $(y'=0)$ in T_2 is a curvetta of R at ψ', the morphism $(y'=0)\to A'$ is the normalization, and (c) implies

$$(A'\cdot(u=0))_{p'}=\operatorname{ord}_\nu u,\qquad (A'\cdot(v=0))_{p'}=\operatorname{ord}_\nu v=e\ell.$$

Let $\tau:=\varpi_R|_{(y'=0)}$, a formal parameter of the normalization with $\sigma^*\tau=\lambda(\sigma)\tau$ for $\sigma\in H$, and let $k:=\operatorname{ord}_\nu u$. The function u has character λ^k, so only the powers τ^i with $i\equiv k \bmod e$ occur in $u|_{(y'=0)}\in\mathbb{C}[[\tau]]$, and $\operatorname{ord}_\tau u=k$. Hence $u|_{(y'=0)}=\tau^k w(\tau^e)$ for a power series w with $w(0)\ne0$. Similarly, v is invariant, so $v|_{(y'=0)}=\sum_{j\ge\ell}\beta_j\tau^{ej}$ with $\beta_\ell\ne0$. The power series $\tau w(\tau^e)^{1/k}$ is again a parameter with character λ, and τ^e is a power series in $(\tau w(\tau^e)^{1/k})^e$. Thus, possibly replacing τ with $\tau w(\tau^e)^{1/k}$, we may assume that

$$u=\tau^k,\qquad v=\sum_{j\ge\ell}\beta_j\tau^{ej},\qquad \beta_\ell\ne0$$

on $(y'=0)$. Let $d_0:=k$, and, as long as $d_{i-1}>1$, let $m_i':=\min\{ej\mid\beta_j\ne0,\ d_{i-1}\nmid ej\}$ and $d_i:=\gcd(d_{i-1},m_i')$. Since τ parametrizes the normalization, $\gcd(\{k\}\cup\{ej\mid\beta_j\ne0\})=1$, so the m_i' exist. Let g_u be the number of indices $i\ge1$ for which m_i' is defined above. This is the number of characteristic pairs of A' with respect to u in the sense of [Abh67], computed from the expansion of v as a power series in $u^{1/k}$. By [Abh67], for any regular parameter at p' not vanishing identically on A', transversal to A' or not, the number of characteristic pairs of A' with respect to it is at least $g(A')$. Thus $g(A')\le g_u$. By (18.45), $\gcd(k,e\ell)=\gcd(k,\ell)\le2$. If $k\nmid e\ell$, then $m_1'=e\ell$ and $d_1\le2$. If $k\mid e\ell$, then $k\mid\ell$ and $d_0=k\le2$. In both cases, the d_i are at most two from the index one on, and they decrease strictly, so $g_u\le2$. Hence $g(A')\le2$, and (b) implies $g'+1\le g(A')+2\le4$. By (a), if ψ' consists of at least two blow-ups, then $\operatorname{end}(G(\psi'))=g'+1\le4$.

Step 6. In this step, we glue the trees of **Steps 3** and 5 and conclude the proof.

We keep the assumption of **Step 4**. The point $p'\in E$ is mapped to the centers of ord_E. The blow-ups of f_1 that are not blow-ups of ψ_{E^*} have centers different from the images of p (after the creation of E^* by **Step 3**, and before it because the image of p there is the center of ord_{E^*}), so near p' the surface T_2 is isomorphic to the last surface of ψ_E, and ψ' is the minimal resolution of ord_ν over the last surface of ψ_E near p'. Hence ψ_ν consists of ψ_E followed by ψ'. The exceptional

curves of ψ_E other than E intersect E only at torus-invariant points, so p' is contained in no exceptional curve of ψ_E other than E. The strict transform of E together with the exceptional curves of ψ' forms an snc divisor, and the strict transform of E intersects the preimage of p' at one point. Thus the strict transform of E intersects exactly one exceptional curve of ψ', and $G(\psi_\nu)$ consists of $G(\psi_E)$, $G(\psi')$, and one edge joining a vertex of $G(\psi_E)$ to a vertex of $G(\psi')$. Since $G(\psi_E)$ has at least two vertices, (d) and **Steps 3** and 5 imply

$$\operatorname{end}(G(\psi_\nu)) \le \begin{cases} (M+2)+4 = M+6 & \text{if } \psi' \text{ consists of at least two blow-ups,} \\ (M+2)+0+1 = M+3 & \text{if } \psi' \text{ is one blow-up.} \end{cases}$$

Since ψ_ν consists of at least three blow-ups, (a) implies $g(\nu)+1 = \operatorname{end}(G(\psi_\nu)) \le M+6$.

By **Steps 3** and 6, in any case $g(\nu)+1 \le M+6$. Hence ν has a minimal generating sequence with at most $M+12$ elements. □

We now prove the main result of this subsection.

Theorem 18.16. *Let $\Gamma \subset [0,1]$ be a DCC set. Let $U := \mathbb{A}^2_{x,y}$, $X := U \times \mathbb{A}^1_t$, and G a finite diagonal abelian group acting faithfully on U and on t by a character. Let $f_1, \dots, f_s$ be non-zero G-semi-invariant polynomials on U, $c_j \in \Gamma$, and $C := \sum_j c_j \operatorname{div}_U(f_j)$ with (U, C) klt near 0, and let $B := \operatorname{pr}_U^* C$. Then the set of such positive numbers $\operatorname{mld}((X \ni 0, B)/G)$ satisfies the ACC. More precisely, any such number a is attained by the prime divisor E over X/G with $\operatorname{ord}_F|_{\mathbb{C}(X/G)} \in \mathbb{N}^+ \operatorname{ord}_E$ for a prime divisor F over X such that ord_F is the Gauss extension of $\varrho \operatorname{ord}_\nu$ to $\mathbb{C}(U)(t)$ with t-value k (Definition-Proposition 2.11), where ν is a prime divisor over U with center 0 and ϱ and k are coprime positive integers. Moreover, $\operatorname{mld}((X \ni 0, B)/I_G(F)) = a$, and $a = \psi_\chi(X', B')$ for some data n, H, X', B', and χ as in Set-up 18.4 with $H = I_G(F)$, where $X' \subset \mathbb{A}^n$ coincides near 0 with the image of an H-equivariant closed embedding of X, and n and the numbers of equations defining X' and of polynomials defining B' are bounded in terms of Γ.*

Proof. The proof is long, so we divide it into several steps. Deleting the f_j with $c_j = 0$ or $f_j(0) \ne 0$ does not change B near 0, hence changes no minimal log discrepancy at 0 or at the image of 0 in a quotient of X. Thus we may assume that $c_j > 0$ and $f_j(0) = 0$ for any j. Let $\epsilon := \gamma(\Gamma)$ if $\Gamma \setminus \{0\} \ne \emptyset$ and $\epsilon := 1$ otherwise, and let $M := 2 + \lceil 2/\epsilon \rceil$. If $s \ge 1$, then $c_j \ge \epsilon$ for any j, and the blow-up of $0 \in U$ and [KM98, Lemma 2.29] imply $\sum_j c_j \operatorname{mult}_0 f_j < 2$, so $s < 2/\epsilon$. In any case,

$$s \le \lfloor 2/\epsilon \rfloor. \tag{18.46}$$

Let $\phi : W \to U$ be a log resolution of (U, C) and $K_W + C_W = \phi^*(K_U + C)$. Since $\phi \times \mathrm{id}_{\mathbb{A}^1_t}$ is a log resolution of (X, B) with

$$K_{W \times \mathbb{A}^1_t} + \operatorname{pr}_W^* C_W = (\phi \times \mathrm{id}_{\mathbb{A}^1_t})^*(K_X + B),$$

the pair (X, B) is klt near 0 by [KM98, Corollary 2.31(3)].

Step 1. In this step, we fix a Gauss extension computing $a := \operatorname{mld}((X \ni 0, B)/G)$, replace G with an inertia group, and show $a(\nu, U, C) < 3|I|$ for the data ν and I below.

Fix such a positive number a, and let β be the character of G on t. By Proposition 18.11(2), applied to $(U \ni 0)$, G, the f_j, and the c_j, with the character χ there taken to be β, there exist a prime divisor ν over U with center 0, coprime positive integers ϱ and k, and the prime divisor F over X such that ord_F is the Gauss extension of $\varrho \operatorname{ord}_\nu$ to $\mathbb{C}(U)(t)$ with t-value k (Definition-Proposition 2.11), and the prime divisor over X/G under F attains a. Here ν, ϱ, and k are the P', p, and l of Proposition 18.11(2), and $\operatorname{center}_U \nu = \{0\}$ because 0 is the only point of U over the image of 0 in U/G. By Lemma 18.1, $\operatorname{center}_X F = \{0\}$ and $a(F, X, B) = \varrho a(\nu, U, C) + k$. Let $I := I_G(F)$, $e := |I|$, and λ the character of I on $\mathfrak{m}_F/\mathfrak{m}_F^2$. By Lemma 2.6(1)(2), I is cyclic, λ is faithful, and

$$a = \frac{\varrho a(\nu, U, C) + k}{e}. \tag{18.47}$$

Lemma 3.1, applied to the quotient morphism $X \to X/G$, the pair $(X/G, B_{X/G})$, the prime divisor F, the subgroup I, and the identity of X, implies that $\mathrm{mld}((X \ni 0, B)/I) = a$ and that the prime divisor over X/I under F attains $\mathrm{mld}((X \ni 0, B)/I)$. Any element of I preserves ord_F, hence preserves $\mathrm{ord}_\nu = \frac{1}{\varrho}\,\mathrm{ord}_F|_{\mathbb{C}(U)}$, and the residue field of ord_ν embeds into that of ord_F. Thus I acts trivially on the residue field of ord_ν, and $I_I(\nu) = I$.

Let E_0 be the exceptional divisor of the blow-up of $0 \in X$. By [KM98, Lemma 2.29], $a(E_0, X, B) = 3 - \sum_j c_j \,\mathrm{mult}_0 f_j \le 3$. The prime divisor E_0 is G-invariant, and by Lemma 2.6(2) the prime divisor $\bar{E}_0$ over X/G under E_0 satisfies $a(\bar{E}_0, X/G, B_{X/G}) = a(E_0, X, B)/|I_G(E_0)| \le 3$. Hence $a \le 3$. Since $\varrho \ge 1$ and $k \ge 1$, (18.47) implies

$$a(\nu, U, C) = \frac{ea - k}{\varrho} < 3e = 3|I|. \tag{18.48}$$

Step 2. In this step, we embed X equivariantly into an affine space and construct data as in Set-up 18.4.

By (18.48), Lemma 18.15, applied to $\Lambda = \Gamma$, $T = U$ with the coordinates (z, t) replaced by (x, y), the group I, the f_j, the c_j, and ν, implies that ν has a minimal generating sequence with $m+1 \le M+12$ elements. By Proposition 18.7, applied to $T = U$ with the coordinates (z, t) replaced by (x, y), the group I, ν, and m, there exist I-semi-invariant polynomials $q_0, \dots, q_m \in \mathbb{C}[x, y]$, the closed embedding $\iota = (x, y, q_0, \dots, q_m) : U \to \mathbb{A}^{m+3}$, I-semi-invariant polynomials $f'_1, \dots, f'_{m+1}$ on $\mathbb{A}^{m+3}$, and $v \in \mathbb{Z}^{m+3}_{>0}$ with $\gcd(v_1, \dots, v_{m+3}) = 1$ satisfying Proposition 18.7(1)–(4). By Proposition 18.7(4), $\mathrm{ord}_\nu q_i > 0$, so $q_i(0) = 0$ and $\iota(0) = 0$. By Proposition 18.7(2), for any j there exists an I-semi-invariant polynomial h'_j on $\mathbb{A}^{m+3}$ with the character of f_j such that $h'_j \circ \iota = f_j$ and $\mathrm{wt}_v(h'_j) = \mathrm{ord}_\nu f_j$.

Let $n := m + 4$ and $r := m + 1$, let $x_1, \dots, x_{m+3}$ be the coordinates of $\mathbb{A}^{m+3}$, and let $x_n := t$. Then $\iota \times \mathrm{id}_{\mathbb{A}^1_t} : X \to \mathbb{A}^n$ is an I-equivariant closed embedding, where I acts on $x_1, \dots, x_{m+3}$ by the characters of $x, y, q_0, \dots, q_m$ and on x_n by $\beta|_I$. We regard f'_l and h'_j as polynomials on $\mathbb{A}^n$ not involving x_n. Let $X' \subset \mathbb{A}^n$ be the common zero scheme of $f'_1, \dots, f'_r$, so that $X' = (\iota \times \mathrm{id}_{\mathbb{A}^1_t})(X)$ near 0 by Proposition 18.7(1), let $B' := \sum_j c_j \,\mathrm{div}(h'_j|_{X'})$, which corresponds to B under $\iota \times \mathrm{id}_{\mathbb{A}^1_t}$, and let $\chi := \lambda$. Then n, r, s, $H := I$, $x_1, \dots, x_{n+1}$, X', f'_l, h'_j, c_j, B', and χ are as in Set-up 18.4, because $n \ge 2$ and $1 \le r < n$, the f'_l vanish at 0 since they generate the ideal of $\iota(U)$ near 0 and $\iota(0) = 0$ (Proposition 18.7(1)), X' is smooth near 0 of codimension r, I acts faithfully on X' since I acts faithfully on U, $h'_j|_{X'} \ne 0$, $c_j > 0$, and (X', B') is klt near 0 since (X, B) is. By (18.46),

$$n \le M + 15, \qquad r \le M + 12, \qquad s \le \lfloor 2/\epsilon \rfloor. \tag{18.49}$$

Step 3. In this step, we check the assumptions of Lemma 18.6 for F and $\tilde{w} := (\varrho v, k) \in \mathbb{Z}^n_{>0}$, and conclude $a = \psi_\chi(X', B')$.

Through $\iota \times \mathrm{id}_{\mathbb{A}^1_t}$, we regard F as a prime divisor over X' with center 0. By **Step 1**, $I_H(F) = H$ and $a(F, X', B')/|H| = a(F, X, B)/e = a = \mathrm{mld}((X' \ni 0, B')/H)$, and $\chi = \lambda$ is the character of H on $\mathfrak{m}_F/\mathfrak{m}_F^2$. Since $\gcd(v_1, \dots, v_{m+3}) = 1$ and $\gcd(\varrho, k) = 1$, we have $\gcd(\tilde{w}_1, \dots, \tilde{w}_n) = 1$. For $i \le m+3$, the restriction of x_i to X' is one of $x, y, q_0, \dots, q_m \in \mathbb{C}(U)$, so Proposition 18.7(4) and $\mathrm{ord}_F|_{\mathbb{C}(U)} = \varrho\,\mathrm{ord}_\nu$ imply $\mathrm{ord}_F x_i = \varrho v_i = \tilde{w}_i$. Moreover, $\mathrm{ord}_F x_n = \mathrm{ord}_F t = k = \tilde{w}_n$. Since h'_j and f'_l do not involve x_n, we have $\mathrm{wt}_{\tilde{w}}(h'_j) = \varrho\,\mathrm{wt}_v(h'_j)$ and $\mathrm{wt}_{\tilde{w}}(f'_l) = \varrho\,\mathrm{wt}_v(f'_l)$. Thus

$$\mathrm{ord}_F h'_j = \mathrm{ord}_F f_j = \varrho\,\mathrm{ord}_\nu f_j = \varrho\,\mathrm{wt}_v(h'_j) = \mathrm{wt}_{\tilde{w}}(h'_j)$$

for any j. By Lemma 18.1 with the zero boundary and Proposition 18.7(3),

$$a(F, X', 0) = \varrho a(\nu, U, 0) + k = \varrho\Big(\sum_{i=1}^{m+3} v_i - \sum_{l=1}^{r} \mathrm{wt}_v(f'_l)\Big) + k = \sum_{i=1}^{n} \tilde{w}_i - \sum_{l=1}^{r} \mathrm{wt}_{\tilde{w}}(f'_l).$$

Hence Lemma 18.6 implies

$$a = \mathrm{mld}((X' \ni 0, B')/H) = \psi_\chi(X', B'),$$

and the infimum defining $\psi_\chi(X', B')$ is attained at $\frac{1}{|H|}(\tilde{w}, 1)$.

Step 4. In this step, we prove the ACC.

By **Steps 2** and 3, any positive number $\mathrm{mld}((X \ni 0, B)/G)$ as in the theorem is equal to $\psi_\chi(X', B')$ for data as in Set-up 18.4 satisfying (18.49), with coefficients c_j in $\Gamma\backslash\{0\}$, which is a DCC subset of $(0, +\infty)$. Thus Corollary 18.5, applied to $n_0 := M+15$, $r_0 := M+12$, $s_0 := \lfloor 2/\epsilon \rfloor$, and $\Gamma\backslash\{0\}$, implies that the set of such numbers satisfies the ACC. Together with **Steps 1–3**, this proves the theorem. □

18.8. **Binomial hypersurfaces.** In this subsection, we prove Theorem 18.17, the ACC for the positive minimal log discrepancies of finite quotients of the hypersurfaces $(x^q y^p = g(z,t)) \subset \mathbb{A}^4$ with boundary along x, y, and curves in the (z,t)-plane.

Theorem 18.17. *Fix an integer $P \geq 1$ and a DCC set $\Gamma \subset [0,1]$. Let $p, q \leq P$ be coprime positive integers, $X := (x^q y^p - g(z,t) = 0) \subset \mathbb{A}^4$ a normal klt threefold germ at 0, where g is a non-zero polynomial vanishing at 0, and G a finite diagonal abelian group preserving the equation up to a character and acting faithfully on X. Let $h_1, \dots, h_s$ be non-zero G-semi-invariant polynomials in z, t, $a_x, a_y, c_j \in \Gamma$, and $B := a_x \operatorname{div}_X(x) + a_y \operatorname{div}_X(y) + \sum_j c_j \operatorname{div}_X(h_j)$ with (X, B) klt near 0. Let $T := \mathbb{A}^2_{z,t}$, let m_x, m_y be integers with $qm_y - pm_x = 1$, and let $\xi := x^{m_x} y^{m_y}$, so that $\mathbb{C}(X) = \mathbb{C}(T)(\xi)$. Then the set of such positive numbers $\mathrm{mld}((X \ni 0, B)/G)$ satisfies the ACC. More precisely, any such number a is attained by the prime divisor over X/G under a prime divisor F over X for which there exist a prime divisor ν over T with center 0 and coprime integers $\varrho \geq 1$ and k such that $\mathrm{ord}_F(\sum_i u_i \xi^i) = \min_i(\varrho \,\mathrm{ord}_\nu(u_i) + ik)$ for any $u_i \in \mathbb{C}(T)$. Moreover, the following hold.*

(1) $\mathrm{mld}((X \ni 0, B)/I_G(F)) = a$.

(2) There exist data n, H, X', B', and χ as in Set-up 18.4 with $H = I_G(F)$ and $a = \psi_\chi(X', B')$, where $X' \subset \mathbb{A}^n$ coincides near 0 with the image of an H-equivariant closed embedding of X, the coefficients of B' belong to $\{a_x, a_y, c_1, \dots, c_s\}$, the infimum defining $\psi_\chi(X', B')$ is attained, and n and the numbers of equations defining X' and of polynomials defining B' are bounded in terms of P and Γ.

In particular, for fixed coprime $p, q \leq P$, the set of positive numbers $\mathrm{mld}((\mathbb{A}^3 \ni 0, B)/G)$ satisfies the ACC, where G is a finite diagonal abelian group acting faithfully on $\mathbb{A}^3_{x,y,t}$ and $B = \sum_j c_j \operatorname{div}(h_j)$ with $c_j \in \Gamma$, $(\mathbb{A}^3, B)$ klt near 0, and $h_j = x^{i_j} y^{i'_j} \varphi_j(x^q y^p, t)$ G-semi-invariant polynomials, where i_j, i'_j are non-negative integers and φ_j are polynomials in two variables.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we reduce to the case when $h_j(0) = 0$ and $c_j > 0$ for any j, we show that $a_x, a_y < 1$, and we construct a klt pair on T whose coefficients belong to a DCC set depending only on P and Γ.

Since G acts diagonally and preserves $x^q y^p - g$ up to a character, g is G-semi-invariant with the character of $x^q y^p$. If $h_j(0) \neq 0$, then $\operatorname{div}_X(h_j)$ is disjoint from the G-invariant open neighborhood $(h_j \neq 0)$ of 0. Thus removing the polynomials h_j with $c_j = 0$ or $h_j(0) \neq 0$ does not change B near 0, and we may assume that $h_j(0) = 0$ and $c_j > 0$ for any j.

The function x vanishes at 0, and x does not vanish identically on X near 0, since x does not divide $x^q y^p - g$. Thus there exists a prime divisor S_x on X through 0 with $\mathrm{mult}_{S_x} \operatorname{div}_X(x) \geq 1$. As $B - a_x \operatorname{div}_X(x) \geq 0$ and (X, B) is klt near 0, we have $a_x \leq \mathrm{mult}_{S_x} B < 1$. Likewise, $a_y < 1$. Let

$$c_{g,x} := 1 - \frac{1 - a_x}{q}, \qquad c_{g,y} := 1 - \frac{1 - a_y}{p}, \qquad c_g := \max\{c_{g,x}, c_{g,y}\},$$

and let $B_T := \sum_j c_j \operatorname{div}_T(h_j)$ and $C := B_T + c_g \operatorname{div}_T(g)$. We show that (T, C) is klt near 0.

Let $X_x := X \cap (x \neq 0)$ and $\varphi_x : X_x \to T \times \mathbb{G}_m$, $(x,y,z,t) \to (z,t,x)$. Since $X_x = \operatorname{Spec} \mathbb{C}[z,t,x,x^{-1}][y]/(y^p - x^{-q}g)$, the morphism φ_x is finite and surjective. By the adjunction formula, ω_{X_x} is generated by $dx \wedge dz \wedge dt/(px^qy^{p-1})$, and x is a unit on X_x. Hence $K_{X_x} = \varphi_x^* K_{T\times\mathbb{G}_m} + (p-1)\operatorname{div}_{X_x}(y)$. Since $\varphi_x^* \operatorname{div}(g) = \operatorname{div}_{X_x}(x^q y^p) = p \operatorname{div}_{X_x}(y)$ and $p(1-c_{g,y}) = 1 - a_y$, we obtain

$$K_{X_x} + B|_{X_x} = \varphi_x^*\big(K_{T\times\mathbb{G}_m} + \operatorname{pr}_T^*(B_T + c_{g,y}\operatorname{div}_T(g))\big).$$

As $g(0,0) = 0$, the curve $(y = z = t = 0)$ is contained in X. Since (X,B) is klt on a neighborhood of 0, there exists $x_0 \neq 0$ such that (X,B) is klt on a neighborhood U_0 of $o := (x_0,0,0,0)$. Since φ_x is finite and $\varphi_x^{-1}(\varphi_x(o)) = \{o\}$, there exists a neighborhood V_0 of $\varphi_x(o) = (0,0,x_0)$ with $\varphi_x^{-1}(V_0) \subset U_0$. By [KM98, Proposition 5.20], $(T \times \mathbb{G}_m, \operatorname{pr}_T^*(B_T + c_{g,y}\operatorname{div}_T(g)))$ is klt on V_0. If $\mu : W \to T$ is a log resolution of $(T, B_T + c_{g,y}\operatorname{div}_T(g))$, then $\mu \times \mathrm{id}_{\mathbb{G}_m}$ is a log resolution of $(T \times \mathbb{G}_m, \operatorname{pr}_T^*(B_T + c_{g,y}\operatorname{div}_T(g)))$ on which, for any prime divisor E on W, the prime divisor $E \times \mathbb{G}_m$ has log discrepancy $a(E, T, B_T + c_{g,y}\operatorname{div}_T(g))$. Thus, by [KM98, Corollaries 2.31(3) and 2.32], $(T, B_T + c_{g,y}\operatorname{div}_T(g))$ is klt near 0. Let $X_y := X \cap (y \neq 0)$ and $\varphi_y : X_y \to T \times \mathbb{G}_m$, $(x,y,z,t) \to (z,t,y)$. The formula $K_{X_y} = \varphi_y^* K_{T\times\mathbb{G}_m} + (q-1)\operatorname{div}_{X_y}(x)$, the equality $q(1-c_{g,x}) = 1 - a_x$, and the curve $(x = z = t = 0)$ imply, by the argument for φ_x with $(x,p,a_y,c_{g,y})$ and $(y,q,a_x,c_{g,x})$ interchanged, that $(T, B_T + c_{g,x}\operatorname{div}_T(g))$ is klt near 0. Since $c_g \in \{c_{g,x}, c_{g,y}\}$, the pair (T,C) is klt near 0.

Let

$$\Lambda := \Gamma \cup \Big\{1 - \frac{1-c}{n} \;\Big|\; c \in \Gamma,\ 1 \le n \le P\Big\}.$$

Then $\Lambda \subset \Psi(\Gamma)$, so Λ is a DCC subset of $[0,1]$ by Lemma 2.10(1). The coefficients $c_1, \dots, c_s, c_g$ of C belong to Λ, as $a_x, a_y \in \Gamma$ and $p, q \le P$. Let $\epsilon := \gamma(\Lambda)$ if $\Lambda \backslash \{0\} \neq \emptyset$ and $\epsilon := 1$ otherwise, and let $M := 2 + \lceil 2/\epsilon \rceil$. Let E_0 be the exceptional divisor of the blow-up of $0 \in T$. By [KM98, Lemma 2.29],

$$0 < a(E_0, T, C) = 2 - \sum_j c_j \operatorname{mult}_0 h_j - c_g \operatorname{mult}_0 g.$$

Since $c_j \ge \epsilon$ and $\operatorname{mult}_0 h_j \ge 1$ for any j, we have $s\epsilon < 2$, hence $s \le \lfloor 2/\epsilon \rfloor$.

Step 2. In this step, we bound the log discrepancy of a prime divisor F over X with center 0 from below in terms of a prime divisor over T, with equality for Gauss extensions.

Since $qm_y - pm_x = 1$ and $x^q y^p = g$ on X, we have $\xi^q = g^{m_x} y$ and $\xi^p = g^{m_y} x^{-1}$, hence

$$x = g^{m_y}\xi^{-p} \qquad \text{and} \qquad y = g^{-m_x}\xi^q.$$

Thus $\mathbb{C}(X) = \mathbb{C}(T)(\xi)$, and ξ is transcendental over $\mathbb{C}(T)$ as $\dim X = 3$. By the adjunction formula, ω_X is generated near 0 by $dx \wedge dz \wedge dt/(px^qy^{p-1})$. Since $dx/x = m_y dg/g - p d\xi/\xi$ and $dg \wedge dz \wedge dt = 0$,

$$\Omega := \frac{dx \wedge dz \wedge dt}{px^q y^{p-1}} = -\frac{xy}{g} \cdot \frac{d\xi}{\xi} \wedge dz \wedge dt. \tag{18.50}$$

Let F be a prime divisor over X with center 0. Since the center of F is 0 and $0 \in X$ maps to $0 \in T$, ord_F is non-negative on $\mathcal{O}_{T,0}$ and positive on the maximal ideal of $\mathcal{O}_{T,0}$. By Lemma 2.12, $\operatorname{ord}_F|_{\mathbb{C}(T)} = \varrho \operatorname{ord}_\nu$ for a positive integer ϱ and a prime divisor ν over T with center 0. Let $l_x := \operatorname{ord}_F x$, $l_y := \operatorname{ord}_F y$, and $k := \operatorname{ord}_F \xi$. Then

$$l_x = m_y \varrho \operatorname{ord}_\nu g - pk > 0, \qquad l_y = -m_x \varrho \operatorname{ord}_\nu g + qk > 0, \qquad q l_x + p l_y = \varrho \operatorname{ord}_\nu g. \tag{18.51}$$

We show that

$$a(F, X, 0) \ge \varrho\big(a(\nu, T, 0) - \operatorname{ord}_\nu g\big) + l_x + l_y, \tag{18.52}$$

with equality if ord_F is the Gauss extension of $\varrho \operatorname{ord}_\nu$ to $\mathbb{C}(T)(\xi)$ with ξ-value k (Definition-Proposition 2.11) and $\gcd(\varrho, k) = 1$ (here $\gcd(\varrho, 0) := \varrho$, so that $\gcd(\varrho, 0) = 1$ means $\varrho = 1$).

Let $Y_\nu \to T$ be a proper birational morphism from a smooth surface Y_ν such that ν is a prime divisor E_ν on Y_ν, let η_ν be the generic point of E_ν, let $\pi_\nu \in \mathcal{O}_{Y_\nu,\eta_\nu}$ be a local equation of E_ν, and let

ω_0 be a generator of ω_{Y_ν} near η_ν. By the definition of $a(\nu,T,0)$, we have $dz\wedge dt = u\pi_\nu^{a(\nu,T,0)-1}\omega_0$ for a unit $u\in\mathcal{O}_{Y_\nu,\eta_\nu}$. Let $\sigma := \xi$ if $k\geq 0$ and $\sigma := \xi^{-1}$ if $k<0$, so that $\operatorname{ord}_F\sigma = |k|$ and $d\xi/\xi = \pm d\sigma/\sigma$. The smooth variety $V := Y_\nu\times\mathbb{P}^1_\xi$ has function field $\mathbb{C}(X)$. Since the center of $\varrho\operatorname{ord}_\nu$ on Y_ν is η_ν and V is proper over Y_ν, F has a center θ on V over η_ν, and σ is regular at θ. Near θ, the form $\Theta := d\sigma\wedge\omega_0$ generates ω_V, and (18.50) implies

$$\Omega = \pm u\frac{xy}{g}\pi_\nu^{a(\nu,T,0)-1}\sigma^{-1}\Theta.$$

Since Ω generates ω_X near 0 and Θ generates ω_V near θ, the definition of log discrepancies implies

$$\begin{aligned} a(F,X,0) &= a(F,V,0) + \operatorname{ord}_F\frac{\Omega}{\Theta} \\ &= a(F,V,0) + l_x + l_y - \varrho\operatorname{ord}_\nu g + \varrho\big(a(\nu,T,0)-1\big) - |k|. \end{aligned} \tag{18.53}$$

As $V\to Y_\nu$ is smooth, π_ν is part of a regular system of parameters of $\mathcal{O}_{V,\theta}$. If $k\neq 0$, then θ is the generic point of $E_\nu\times(\sigma=0)$, and π_ν,σ form a regular system of parameters of $\mathcal{O}_{V,\theta}$. Thus there exists a regular system of parameters $x_1,\dots,x_c$ of $\mathcal{O}_{V,\theta}$ with $x_1=\pi_\nu$, and with $x_2=\sigma$ if $k\neq 0$. Lemma 2.13 implies

$$a(F,V,0)\geq\sum_{i=1}^{c}\operatorname{ord}_F x_i\geq\varrho+|k|.$$

Together with (18.53), we obtain (18.52).

Assume now that ord_F is the Gauss extension of $\varrho\operatorname{ord}_\nu$ with ξ-value k and $\gcd(\varrho,k)=1$. If $k=0$, then $\varrho=1$, ord_F is the valuation of the prime divisor $E_\nu\times\mathbb{P}^1$ of V, and $a(F,V,0)=1$. If $k\neq 0$, then the Gauss extension of $\varrho\operatorname{ord}_\nu$ to $\mathbb{C}(T)(\xi)$ with ξ-value k is the Gauss extension of $\varrho\operatorname{ord}_\nu$ to $\mathbb{C}(T)(\sigma)$ with σ-value $|k|$. For $k<0$, this follows from $\sum_i u_i\xi^i=\sum_i u_i\sigma^{-i}$ and $\min_i(\varrho\operatorname{ord}_\nu u_i+ik) = \min_i(\varrho\operatorname{ord}_\nu u_i - i|k|)$. We have $\operatorname{ord}_{E_\nu}=\operatorname{ord}_\nu$ on $\mathbb{C}(Y_\nu)=\mathbb{C}(T)$, and $a(E_\nu,Y_\nu,0)=1$ as E_ν is a prime divisor on the smooth surface Y_ν. Lemma 18.1 applied to Y_ν with the zero boundary, the prime divisor E_ν on Y_ν, the coprime positive integers $\varrho,|k|$, and the coordinate σ of $Y_\nu\times\mathbb{A}^1_\sigma$ implies $a(F,Y_\nu\times\mathbb{A}^1_\sigma,0)=\varrho+|k|$. As $Y_\nu\times\mathbb{A}^1_\sigma$ is an open subset of V containing θ, we obtain $a(F,V,0)=\varrho+|k|$. In both cases $a(F,V,0)=\varrho+|k|$, and (18.53) implies equality in (18.52).

For a prime divisor F over X with center 0 and the data ν,ϱ,l_x,l_y defined for F before (18.51), let

$$R(F) := \varrho\big(a(\nu,T,B_T)-\operatorname{ord}_\nu g\big) + (1-a_x)l_x + (1-a_y)l_y.$$

Since $\operatorname{mult}_F B = a_xl_x+a_yl_y+\varrho\sum_j c_j\operatorname{ord}_\nu h_j$ and $a(\nu,T,B_T)=a(\nu,T,0)-\sum_j c_j\operatorname{ord}_\nu h_j$, (18.52) implies

$$a(F,X,B)\geq R(F), \tag{18.54}$$

with equality if ord_F is the Gauss extension of $\varrho\operatorname{ord}_\nu$ with ξ-value k and $\gcd(\varrho,k)=1$. Since $ql_x+pl_y=\varrho\operatorname{ord}_\nu g$ by (18.51), $1-c_g=\min\{\frac{1-a_x}{q},\frac{1-a_y}{p}\}$, and (T,C) is klt near 0 (**Step 1**),

$$\begin{aligned} R(F) &= \varrho a(\nu,T,C) + \big(1-a_x-q(1-c_g)\big)l_x + \big(1-a_y-p(1-c_g)\big)l_y \\ &\geq \varrho a(\nu,T,C) > 0. \end{aligned} \tag{18.55}$$

Step 3. In this step, we show that $a := \operatorname{mld}((X\ni 0,B)/G)$ is computed by a Gauss extension F, that $a\leq 3$, and that $\operatorname{mld}((X\ni 0,B)/I_G(F)) = a$.

By Lemma 2.6(0)(2)(3), the pair $(X,B)/G$ is klt near the image of 0. Thus $a>0$, and a is attained by a prime divisor E over X/G with center the image of 0 [Amb99, Section 1, the second paragraph after Definition 1.1]. By Lemma 2.6(1)(2)(3), there exists a prime divisor F over X with $\operatorname{ord}_F|_{\mathbb{C}(X/G)} = |I_G(F)|\operatorname{ord}_E$ and $a(F,X,B)/|I_G(F)| = a$, and $\operatorname{center}_X F=\{0\}$ as G fixes 0.

Let F be any prime divisor over X with center 0, let $\nu, \varrho, k, l_x, l_y$ be as in **Step 2**, let $I := I_G(F)$, $e := |I|$, and let λ be the character of I on $\mathfrak{m}_F/\mathfrak{m}_F^2$, which is faithful by Lemma 2.6(1). Any $\tau \in I$ stabilizes ord_F, hence $\operatorname{ord}_F|_{\mathbb{C}(T)} = \varrho \operatorname{ord}_\nu$, hence ν. The residue field $\kappa(\nu)$ of ord_ν is a subfield of the residue field $\kappa(F)$ of ord_F, so I acts trivially on $\kappa(\nu)$, and I acts on $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$ by a character α. The function ξ is G-semi-invariant. Let χ_ξ be the character of ξ. For $\tau \in I$ and $f \in \mathbb{C}(X)^*$, the residue of $\tau^* f/f$ in $\kappa(F)$ is $\lambda(\tau)^{\operatorname{ord}_F f}$: write $f = \pi_F^{\operatorname{ord}_F f} w$ with π_F a uniformizer and w a unit of the valuation ring of ord_F, and use that τ acts trivially on $\kappa(F)$. Applying the last statement to π_ν and to ξ, we obtain

$$\alpha = \lambda^\varrho \qquad \text{and} \qquad \chi_\xi = \lambda^k \qquad \text{on } I. \tag{18.56}$$

Assume now that $a(F, X, B)/|I_G(F)| = a$. Let $d_0 := \gcd(\varrho, k)$, and let F_0 be the valuation given by the Gauss extension of $\frac{\varrho}{d_0} \operatorname{ord}_\nu$ with ξ-value $\frac{k}{d_0}$. By Definition-Proposition 2.11, the value group of ord_{F_0} is $\mathbb{Z}$. Let $\zeta := \xi^{\varrho/d_0} \pi_\nu^{-k/d_0}$. For $u_i \in \mathbb{C}(T)$, the powers of ξ in the terms $u_i \zeta^i = u_i \pi_\nu^{-ik/d_0} \xi^{i\varrho/d_0}$ are pairwise distinct, so the formula of the Gauss extension gives

$$\operatorname{ord}_{F_0}\Big(\sum_i u_i \zeta^i\Big) = \min_i \frac{\varrho}{d_0} \operatorname{ord}_\nu(u_i).$$

Thus $\operatorname{ord}_{F_0} \zeta = 0$, and $\operatorname{ord}_{F_0}(\sum_i u_i \zeta^i) = 0$ if $\min_i \operatorname{ord}_\nu(u_i) = 0$. Hence the residue $\bar{\zeta}$ of ζ is transcendental over $\kappa(\nu)$. By [Bou72, Chapter VI, §10, no. 1, Proposition 2], the restriction of ord_{F_0} to $\mathbb{C}(T)(\zeta)$ has value group $\frac{\varrho}{d_0}\mathbb{Z}$ and residue field $\kappa(\nu)(\bar{\zeta})$. As $\xi^{\varrho/d_0} = \zeta \pi_\nu^{k/d_0}$, we have $[\mathbb{C}(T)(\xi) : \mathbb{C}(T)(\zeta)] \le \frac{\varrho}{d_0}$. As the value group of ord_{F_0} is $\mathbb{Z}$, the ramification index of ord_{F_0} over its restriction to $\mathbb{C}(T)(\zeta)$ is at least $\frac{\varrho}{d_0}$. By [Bou72, Chapter VI, §8, no. 1, Lemma 2], the residue degree is 1, that is, the residue field of ord_{F_0} is $\kappa(\nu)(\bar{\zeta})$. By (18.51), $\operatorname{ord}_{F_0} x = l_x/d_0 > 0$ and $\operatorname{ord}_{F_0} y = l_y/d_0 > 0$, and $\operatorname{ord}_{F_0} z, \operatorname{ord}_{F_0} t > 0$. Thus F_0 is a prime divisor over X with center 0 by [KM98, Lemma 2.45]. Any $\tau \in I$ stabilizes ord_{F_0}, since τ stabilizes ν and $\tau^*\xi = \chi_\xi(\tau)\xi$. Moreover, τ acts trivially on $\kappa(\nu)$ and sends $\bar{\zeta}$ to $\chi_\xi(\tau)^{\varrho/d_0} \alpha(\tau)^{-k/d_0} \bar{\zeta} = \bar{\zeta}$ by (18.56). Thus $I \subset I_G(F_0)$. By **Step 2** applied to F_0, whose data are ν, $\frac{\varrho}{d_0}$, $\frac{l_x}{d_0}$, $\frac{l_y}{d_0}$, we have $a(F_0, X, B) = R(F_0) = R(F)/d_0$. By Lemma 2.6 and (18.54),

$$a \le \frac{a(F_0, X, B)}{|I_G(F_0)|} \le \frac{R(F)}{d_0 e} \le \frac{a(F, X, B)}{d_0 e} = \frac{a}{d_0}.$$

As $a > 0$, we have $d_0 = 1$ and $a(F_0, X, B)/|I_G(F_0)| = a$. Possibly replacing F with F_0, we may assume that ord_F is the Gauss extension of $\varrho \operatorname{ord}_\nu$ with ξ-value k and $\gcd(\varrho, k) = 1$, that is, $\operatorname{ord}_F(\sum_i u_i \xi^i) = \min_i(\varrho \operatorname{ord}_\nu(u_i) + ik)$ for any $u_i \in \mathbb{C}(T)$. From now on, F denotes F_0, and I, e, λ, and α denote the objects defined for F_0 in the paragraph containing (18.56), so that (18.56) holds. Moreover, E denotes the prime divisor over X/G under F (Definition 2.7), so that $\operatorname{ord}_F|_{\mathbb{C}(X/G)} = e \operatorname{ord}_E$, $\operatorname{center}_{X/G} E$ is the image of 0, and $a(E, X/G, B_{X/G}) = a(F, X, B)/e = a$ by Lemma 2.6(1)(2). By **Step 2**, $a(F, X, B) = R(F)$.

We show that $a \le 3$. For any prime divisor F' over X with center 0, Lemma 2.6(2) and $a(F', X, B) > 0$ imply $a \le a(F', X, B)/|I_G(F')| \le a(F', X, B)$. Thus $a \le \operatorname{mld}(X \ni 0, B)$. Since x, y, and h_j do not vanish identically on X, [EMY03, Theorem 0.1] applied to the normal prime divisor X of $\mathbb{A}^4$ and the closed subset $\{0\}$ implies

$$\operatorname{mld}(X \ni 0, B) = \operatorname{mld}\Big(\mathbb{A}^4 \ni 0, X + a_x \operatorname{div}(x) + a_y \operatorname{div}(y) + \sum_j c_j \operatorname{div}(h_j)\Big).$$

By [KM98, Lemma 2.29], the exceptional divisor of the blow-up of $0 \in \mathbb{A}^4$ has log discrepancy at most $4 - \operatorname{mult}_0(x^q y^p - g) \le 3$ with respect to the pair on the right-hand side. Hence $a \le 3$.

Finally, Lemma 3.1, applied to the quotient morphism $X \to X/G$, the pair $(X/G, B_{X/G})$, the image of 0, the prime divisor E over X/G with $\operatorname{ord}_F|_{\mathbb{C}(X/G)} = e \operatorname{ord}_E$, the prime divisor F, the subgroup I, and the identity of X, implies $\operatorname{mld}((X \ni 0, B)/I) = a$. By Lemma 2.6(1), I is cyclic, and I acts diagonally on $\mathbb{A}^4$.

Step 4. In this step, we show that ν has a minimal generating sequence [dFGPM23, Definition 2.9] with at most $M+12$ elements.

Let I_0 be the kernel of the action of I on T, $e_1 := |I_0|$, $\bar{I} := I/I_0$, and $e_0 := |\bar{I}|$. The cyclic group $\bar{I}$ acts faithfully and diagonally on T, stabilizes ν, and acts trivially on $\kappa(\nu)$ (**Step 3**), so $I_{\bar{I}}(\nu) = \bar{I}$. By Lemma 2.6(1) applied to $\bar{I}$ acting on T, the group $\bar{I}$ acts faithfully on $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2$. Hence I_0 is the kernel of α. As $\alpha = \lambda^\varrho$ by (18.56) and λ is a faithful character of the cyclic group I of order e, we obtain $e_1 = \gcd(e,\varrho) \le \varrho$. By **Step 3** and (18.55), $ea = a(F,X,B) = R(F) \ge \varrho a(\nu,T,C)$. Since $a \le 3$ (**Step 3**),

$$a(\nu,T,C) \le \frac{ea}{\varrho} = \frac{e_1}{\varrho} e_0 a \le e_0 a \le 3e_0. \tag{18.57}$$

The polynomials $h_1,\dots,h_s,g$ are I-semi-invariant, hence $\bar{I}$-semi-invariant, the coefficients $c_1,\dots,c_s,c_g$ of C belong to Λ, and (T,C) is klt near 0 (**Step 1**). By (18.57) and Lemma 18.15 applied with $\bar{I}$ in place of H, ν has a minimal generating sequence with $m+1 \le M+12$ elements, where $m \ge 0$.

Step 5. In this step, we embed X I-equivariantly into an affine space of dimension at most $M+16$, and we show that $a = \psi_\lambda(X',B')$ for the image X' and the boundary B' defined below.

The group I is cyclic and acts diagonally on T, ν is I-invariant with center 0, I acts trivially on $\kappa(\nu)$ (**Step 3**), and ν has a minimal generating sequence with $m+1$ elements (**Step 4**). Let $N := m+3 \le M+14$. By Proposition 18.7 applied with I in place of H, there exist I-semi-invariant polynomials $q_0,\dots,q_m \in \mathbb{C}[z,t]$, the closed embedding $\iota = (z,t,q_0,\dots,q_m) : T \to \mathbb{A}^N$, I-semi-invariant polynomials $f_1,\dots,f_{N-2}$ on $\mathbb{A}^N$, and $v \in \mathbb{Z}^N_{>0}$ with $\gcd(v_1,\dots,v_N) = 1$ satisfying Proposition 18.7(1)–(4). By Proposition 18.7(2), there exist I-semi-invariant polynomials $\tilde{g},\tilde{h}_1,\dots,\tilde{h}_s$ on $\mathbb{A}^N$ with the characters of $g,h_1,\dots,h_s$ such that $\tilde{g}\circ\iota = g$, $\tilde{h}_j\circ\iota = h_j$, $\mathrm{wt}_v(\tilde{g}) = \mathrm{ord}_\nu g$, and $\mathrm{wt}_v(\tilde{h}_j) = \mathrm{ord}_\nu h_j$. By Proposition 18.7(4), $v_i > 0$ is the ν-value of the i-th coordinate of ι. In particular, $\iota(0) = 0$.

Let $n := N+2$, and let $\mathbb{A}^n$ have the coordinates x, y, and the N coordinates of $\mathbb{A}^N$, on which I acts diagonally by the characters of x, y, z, t, $q_0,\dots,q_m$. Let $\Phi := (x,y,\iota(z,t)) : X \to \mathbb{A}^n$, let $f_0 := x^q y^p - \tilde{g}$, and let $X' \subset \mathbb{A}^n$ be the common zero scheme of $f_0,f_1,\dots,f_{N-2}$, where $f_1,\dots,f_{N-2}$ are regarded as polynomials on $\mathbb{A}^n$. The polynomial f_0 is I-semi-invariant, as $x^q y^p$ and $\tilde{g}$ have the character of g. By Proposition 18.7(1), near 0 the common zero scheme of $f_1,\dots,f_{N-2}$ is $\mathbb{A}^2_{x,y}\times\iota(T)$, and $f_0\circ(\mathrm{id}_{\mathbb{A}^2_{x,y}}\times\iota) = x^q y^p - g$. Thus Φ is an I-equivariant closed embedding, and $X' = \Phi(X)$ near 0. Let

$$B' := a_x \operatorname{div}(x|_{X'}) + a_y \operatorname{div}(y|_{X'}) + \sum_j c_j \operatorname{div}(\tilde{h}_j|_{X'}),$$

so that $B' = \Phi_* B$ near 0. Then n, $r := N-1$, s replaced by the number of positive coefficients among $a_x,a_y,c_1,\dots,c_s$, $H := I$, the coordinates of $\mathbb{A}^n$, X', the equations $f_0,\dots,f_{N-2}$, the polynomials among $x,y,\tilde{h}_1,\dots,\tilde{h}_s$ with positive coefficients, B', and $\chi := \lambda$ are data as in Set-up 18.4: the f_l vanish at 0, since $f_1,\dots,f_{N-2}$ generate the ideal of $\iota(T)$ near 0 (Proposition 18.7(1)) and $\iota(0) = 0$, and $f_0(0) = -\tilde{g}(\iota(0)) = -g(0) = 0$. Near 0, $X' \cong X$ is a normal integral complete intersection of codimension $N-1 < n$ with klt singularities on which I acts faithfully, and $(X',B') \cong (X,B)$ is klt near 0.

Let

$$\tilde{w} := (l_x, l_y, \varrho v_1, \dots, \varrho v_N) \in \mathbb{Z}^n_{>0}.$$

We verify the hypotheses of Lemma 18.6 for F, regarded as a prime divisor over X', and $\tilde{w}$. First, $I_I(F) = I$, $a(F,X',B')/|I| = a = \mathrm{mld}((X' \ni 0, B')/I)$ by **Step 3**, and λ is the character of I on $\mathfrak{m}_F/\mathfrak{m}_F^2$. Second, $\mathrm{ord}_F x = l_x$, $\mathrm{ord}_F y = l_y$, and the i-th coordinate of ι has ord_F-value ϱv_i, since $\mathrm{ord}_F|_{\mathbb{C}(T)} = \varrho\,\mathrm{ord}_\nu$. Third, $\gcd(\varrho v_1,\dots,\varrho v_N) = \varrho$, and by (18.51), $l_x \equiv -pk$ and $l_y \equiv qk$ modulo ϱ. As $\gcd(p,q) = 1$,

$$\gcd(\tilde{w}_1,\dots,\tilde{w}_n) = \gcd(l_x,l_y,\varrho) = \gcd(k,\varrho) = 1.$$

Fourth, $\operatorname{ord}_F x = \operatorname{wt}_{\tilde{w}}(x)$, $\operatorname{ord}_F y = \operatorname{wt}_{\tilde{w}}(y)$, and

$$\operatorname{ord}_F(\tilde{h}_j|_{X'}) = \varrho \operatorname{ord}_\nu h_j = \varrho \operatorname{wt}_v(\tilde{h}_j) = \operatorname{wt}_{\tilde{w}}(\tilde{h}_j).$$

Fifth, $\operatorname{wt}_{\tilde{w}}(f_l) = \varrho \operatorname{wt}_v(f_l)$ for $1 \le l \le N-2$. The monomial $x^q y^p$ has $\tilde{w}$-weight $ql_x + pl_y = \varrho \operatorname{ord}_\nu g$ by (18.51), and it does not occur in $\tilde{g}$, whose monomials have $\tilde{w}$-weight at least $\varrho \operatorname{wt}_v(\tilde{g}) = \varrho \operatorname{ord}_\nu g$, with equality for some monomial. Hence $\operatorname{wt}_{\tilde{w}}(f_0) = \varrho \operatorname{ord}_\nu g$. By Proposition 18.7(3) and the equality in (18.52),

$$\sum_{i=1}^{n} \tilde{w}_i - \sum_{l=0}^{N-2} \operatorname{wt}_{\tilde{w}}(f_l) = l_x + l_y + \varrho\Big(\sum_{i=1}^{N} v_i - \sum_{l=1}^{N-2} \operatorname{wt}_v(f_l)\Big) - \varrho \operatorname{ord}_\nu g = a(F, X, 0) = a(F, X', 0).$$

By Lemma 18.6,

$$a = \operatorname{mld}((X' \ni 0, B')/I) = \psi_\lambda(X', B'),$$

and the infimum defining $\psi_\lambda(X', B')$ is attained at $\frac{1}{e}(\tilde{w}, 1)$. The coefficients of B' belong to $\{a_x, a_y, c_1, \dots, c_s\}$.

Step 6. In this step, we prove the ACC.

By **Steps 4 and 5**, $n = N + 2 \le M + 16$, X' is defined by $N - 1 \le M + 13$ equations, and B' is defined by at most $s + 2 \le \lfloor 2/\epsilon \rfloor + 2$ polynomials (**Step 1**) with coefficients in $\Gamma\backslash\{0\}$. The numbers M and ϵ depend only on Λ, hence only on P and Γ. By Corollary 18.5 applied with $n_0 := M + 16$, $r_0 := M + 13$, $s_0 := \lfloor 2/\epsilon \rfloor + 2$, and the DCC subset $\Gamma\backslash\{0\}$ of $(0, +\infty)$ as the set of coefficients, the set of finite numbers $\psi_\chi(X', B')$ for data as in Set-up 18.4 with $n \le n_0$, $r \le r_0$, $s \le s_0$, and coefficients in $\Gamma\backslash\{0\}$ satisfies the ACC. By **Step 5**, any positive number $\operatorname{mld}((X \ni 0, B)/G)$ as in the theorem is such a number $\psi_\chi(X', B')$. Thus the set of such numbers satisfies the ACC.

Step 7. In this step, we prove the ACC for the numbers $\operatorname{mld}((\mathbb{A}^3 \ni 0, B)/G)$ in the theorem.

Let G, c_j, $h_j = x^{i_j} y^{i'_j} \varphi_j(x^q y^p, t)$, and B be as in the theorem, and let χ_x and χ_y be the characters of x and y. Let G act on $\mathbb{A}^4_{x,y,z,t}$ through the action on $\mathbb{A}^3_{x,y,t}$ and by the character $\chi_x^q \chi_y^p$ on z. Let $X_1 := (x^q y^p - z = 0) \subset \mathbb{A}^4$. Then $x^q y^p - z$ is G-semi-invariant, and the projection from X_1 to $\mathbb{A}^3_{x,y,t}$ is a G-equivariant isomorphism with inverse $(x, y, t) \to (x, y, x^q y^p, t)$. On X_1, $\operatorname{div}(h_j) = i_j \operatorname{div}(x) + i'_j \operatorname{div}(y) + \operatorname{div}(\varphi_j(z, t))$, so B corresponds to

$$a_x \operatorname{div}_{X_1}(x) + a_y \operatorname{div}_{X_1}(y) + \sum_j c_j \operatorname{div}_{X_1}(\varphi_j(z, t)), \qquad a_x := \sum_j c_j i_j, \quad a_y := \sum_j c_j i'_j.$$

Since h_j and $x^{i_j} y^{i'_j}$ are G-semi-invariant, so is $\varphi_j(x^q y^p, t)$. As G acts diagonally, any monomial $x^{qi} y^{pi} t^l$ of $\varphi_j(x^q y^p, t)$ is G-semi-invariant with the character of $\varphi_j(x^q y^p, t)$. The map $z^i t^l \to x^{qi} y^{pi} t^l$ is injective on monomials and preserves characters, so $\varphi_j(z, t)$ is G-semi-invariant. Since $(\mathbb{A}^3, B)$ is klt near 0 and $B - a_x \operatorname{div}(x) \ge 0$, we have $a_x < 1$. Likewise, $a_y < 1$. Thus $a_x, a_y \in \Gamma_+ \cap [0, 1]$, and $c_j \in \Gamma \subset \Gamma_+ \cap [0, 1]$. The set $\Gamma_+ \cap [0, 1]$ satisfies the DCC by Lemma 2.10(1). Applying the theorem with $g = z$ and with the DCC set $\Gamma_+ \cap [0, 1]$ in place of Γ, we obtain that the set of positive numbers $\operatorname{mld}((\mathbb{A}^3 \ni 0, B)/G) = \operatorname{mld}((X_1 \ni 0, B|_{X_1})/G)$ satisfies the ACC. □

Remark 18.18. For $p = q = 1$, Theorem 18.17 implies that the set of positive numbers $\operatorname{mld}((X \ni 0, B)/G)$ satisfies the ACC, where $X = (xy - g(z, t) = 0) \subset \mathbb{A}^4$ is a normal klt threefold germ at 0 for a non-zero polynomial g vanishing at 0, G is a finite diagonal abelian group preserving the equation up to a character and acting faithfully on X, $h_1, \dots, h_s$ are non-zero G-semi-invariant polynomials in z, t, and $B = a_x \operatorname{div}_X(x) + a_y \operatorname{div}_X(y) + \sum_j c_j \operatorname{div}_X(h_j)$ with $a_x, a_y, c_j \in \Gamma$ and (X, B) klt near 0. By contrast, Theorem 18.9(1) concerns only the numbers in $(0, 1)$, but the boundary there may contain the divisors $\operatorname{div}_X(x + \phi_i)$. Theorem 18.16 treats the products $U \times \mathbb{A}^1_t$.

19. Clean (-2) segments on surface quotients

In this section, we prove Theorem 19.2. The results of this section are not used in the proofs of the main theorems of this paper. Some of them are used in the sequels to this paper.

19.1. **Clean (-2) segments.** In this subsection, we introduce clean (-2) segments, state Theorem 19.2, and compare the log discrepancies of prime divisors whose centers on Y meet U_0.

Set-up 19.1. Let r, a, S, X, o, B, b_i, D_i, ρ, T, Y, f, B_Y, n_-, n_+, E_k, U_j, u, and v be as follows.

(1) r is a positive integer, a is an integer such that $\mu_r(1,a)$ acts on $\mathbb{A}^2_{x,y}$ without pseudoreflections, $S := \mathbb{A}^2_{x,y}/\mu_r(1,a)$, $X := S \times \mathbb{A}^1_z$, and $o \in X$ is the image of the origin of $\mathbb{A}^3_{x,y,z}$.
(2) $B = \sum_i b_i D_i$ with distinct prime divisors D_i and $0 < b_i < 1$, and (X, B) is klt near o.
(3) $\rho : T \to S$ is the minimal resolution, $Y := T \times \mathbb{A}^1_z$, $f : Y \to X$ is the induced morphism, and B_Y is the $\mathbb{R}$-divisor on Y with $K_Y + B_Y = f^*(K_X + B)$ and $f_*B_Y = B$ (X is $\mathbb{Q}$-factorial by [KM98, Proposition 5.15], so $f^*(K_X + B)$ is defined).
(4) $n_-, n_+ \geq 2$ are integers and E_k, $-n_- \leq k \leq n_+ + 1$, are consecutive exceptional curves of ρ such that, for any $-n_- + 1 \leq k \leq n_+$, $E_k^2 = -2$ and $(D_i^\eta \cdot E_k) = 0$ for any i, where D_i^η is the strict transform of D_i on the generic fiber of $Y \to \mathbb{A}^1_z$.
(5) For $-n_- \leq j \leq n_+$, $U_j \cong \mathbb{A}^3$ is the product with $\mathbb{A}^1_z$ of the affine toric open subset of T given by the cone spanned by the rays corresponding to E_j and E_{j+1}, with coordinates u, v, z, where u, v are toric coordinates such that $E_j \times \mathbb{A}^1_z = (u = 0)$ and $E_{j+1} \times \mathbb{A}^1_z = (v = 0)$ on U_j.

We call $E_{-n_-}, \dots, E_{n_++1}$ a *clean* (-2) *segment*.

The proof of the following theorem is given at the end of this section.

Theorem 19.2. *Let $\Gamma \subset [0,1]$ be a DCC set with $\Gamma\backslash\{0\} \neq \emptyset$, $\gamma := \gamma(\Gamma)$, $\delta := \delta(\Gamma)$, and $H_0 := 2 + \frac{1}{\delta}$. In Set-up 19.1, assume that $B \in \Gamma$, and let F be a prime divisor computing $c := \mathrm{mld}(X \ni o, B)$ with $0 < c < t \leq 1$ such that any other prime divisor over X with center o has log discrepancy at least t. If $\mathrm{center}_Y F \cap U_0 \neq \emptyset$, then $\min\{n_-, n_+\} < 1 + \frac{H_0}{\gamma}$.*

Remark 19.3. The gap hypothesis in Theorem 19.2 says that F is the only prime divisor over X with center o whose log discrepancy is less than t. The proof assumes $\min\{n_-, n_+\} \geq 1 + \frac{H_0}{\gamma}$ and derives a contradiction. The set $\mathrm{center}_Y F \cap U_0$ is $(u = z = 0)$, $(v = z = 0)$, a closed point with $z = 0$ at which exactly one of u, v vanishes, or the node $u = v = z = 0$. If $\mathrm{center}_Y F$ is $E_0 \times \{0\}$ or $E_1 \times \{0\}$, the isomorphisms $U_0 \cong U_j$ of the affine open subsets along the clean (-2) segment move F to prime divisors F_j over X with center o (Proposition 19.5), the inequalities $a(F_j, X, B) \geq t > c$ imply contact inequalities for the equations of the components of B along the segment, and $B \in \Gamma$ then bounds $\min\{n_-, n_+\}$ (Proposition 19.6). Closed points at which exactly one of u, v vanishes are excluded by Proposition 19.9. If the center is the node, the center of F on the product with $\mathbb{A}^1_z$ of the weighted blow-up of the origin of $\mathbb{A}^2_{u,v}$ with weights $(\mathrm{ord}_F u, \mathrm{ord}_F v)/\gcd(\mathrm{ord}_F u, \mathrm{ord}_F v)$ for (u, v) is a closed point of the exceptional divisor (Construction-Proposition 19.10 and Proposition 19.12), and after an étale cover near the point, Theorem 16.13(2) implies $a(F, X, B) \geq t + 1 > c$, a contradiction. The number t does not appear in the bound.

Construction-Proposition 19.4. Notation and conditions as in Set-up 19.1, and let $a_k := a(E_k \times \mathbb{A}^1_z, X, B)$ for $-n_- \leq k \leq n_+ + 1$. Then $a_k = a_0 + k(a_1 - a_0)$ and $0 < a_k \leq 1$ for any k, and $f^{-1}(o) \cap U_j = (z = 0) \cap (uv = 0)$ for any $-n_- \leq j \leq n_+$. Moreover, after shrinking X near o, for any i there exist a polynomial $P_i(w, z)$ with $P_i(0, z) \neq 0$ and, for any $-n_- \leq j \leq n_+$, a polynomial $Q_{i,j} \in (u^{n_- + j} v^{n_- + j + 1}, u^{n_+ - j + 1} v^{n_+ - j})$ such that the strict transform of D_i on U_j is

$\operatorname{div}(\Phi_{i,j})$, where $\Phi_{i,j} := P_i(uv, z) + Q_{i,j}$, and

$$B_Y|_{U_j} = (1 - a_j)(u = 0) + (1 - a_{j+1})(v = 0) + \sum_i b_i \operatorname{div}(\Phi_{i,j}).$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we describe X, Y, and U_j as toric varieties and construct, for any i and j, a polynomial in u, v, z whose divisor on U_j is the strict transform of D_i.

Let $G := \mu_r(1, a, 0)$, acting on $\mathbb{A}^3_{x,y,z}$, and let $\pi : \mathbb{A}^3 \to X$ be the quotient morphism. Since $\mu_r(1, a)$ acts on $\mathbb{A}^2_{x,y}$ without pseudoreflections, π is étale in codimension one. Thus $\pi^* D_i$ is a reduced G-invariant divisor on $\mathbb{A}^3$, and since $\mathbb{C}[x, y, z]$ is a unique factorization domain, $\pi^* D_i = \operatorname{div}(h_i)$ for a reduced polynomial h_i. For any $g \in G$, $\operatorname{div}(g^* h_i) = g^* \pi^* D_i = \pi^* D_i$, so $g^* h_i \in \mathbb{C}^* h_i$, that is, h_i is G-semi-invariant (Definition 2.14). For $\alpha \in \mathbb{Z}^3$ we write $x^\alpha := x^{\alpha_1} y^{\alpha_2} z^{\alpha_3}$, and we write

$$h_i = \sum_{\alpha \in \Lambda_i} c_{i,\alpha} x^\alpha,$$

where $\Lambda_i \subset \mathbb{Z}^3_{\geq 0}$ is finite and $c_{i,\alpha} \neq 0$ for any $\alpha \in \Lambda_i$.

Let $N_G \subset \mathbb{R}^3$ be the lattice generated by $\mathbb{Z}^3$ and $\frac{1}{r}(1, a, 0)$ (Definition 2.15), and let $M := \{m \in \mathbb{R}^3 \mid \langle m, w \rangle \in \mathbb{Z} \text{ for any } w \in N_G\} \subset \mathbb{Z}^3$. By [CLS11, Proposition 1.3.18], X is the affine toric variety of the cone $\mathbb{R}^3_{\geq 0}$ in N_G, and x^m, $m \in M$, are the characters of the torus of X. Since $\mu_r(1, a)$ contains no pseudoreflections, $e_x := (1, 0, 0)$, $e_y := (0, 1, 0)$, and $e_z := (0, 0, 1)$ are primitive in N_G. The group G acts on x^α through the character $\zeta^{\alpha_1 + a\alpha_2}$, $\zeta = e^{2\pi\sqrt{-1}/r}$, and h_i is G-semi-invariant, so

$$\alpha - \alpha' \in M \quad \text{for any } \alpha, \alpha' \in \Lambda_i. \tag{19.1}$$

By [Ful93, §2.6], there is a fan Σ subdividing $\mathbb{R}^2_{\geq 0}$ such that the toric morphism ρ_Σ from the toric surface of Σ to S is a resolution with the following properties: the primitive vectors of two consecutive rays form a basis of the lattice, the exceptional curves are smooth rational curves forming a chain in the order of the rays, and if $\mathbf{r}'$, $\mathbf{r}$, $\mathbf{r}''$ are the primitive vectors of three consecutive rays and E is the exceptional curve of $\mathbf{r}$, then $(E^2) \leq -2$ and $\mathbf{r}' + \mathbf{r}'' = -(E^2)\mathbf{r}$. Since the exceptional locus of ρ_Σ contains no (-1)-curves, ρ_Σ is the minimal resolution ρ, and T is the toric surface of Σ. For any k, let $\mathbf{r}_k \in N_G \cap (\mathbb{R}^2_{>0} \times \{0\})$ be the primitive vector of the ray corresponding to E_k. Then $\mathbf{r}_{k-1} + \mathbf{r}_{k+1} = 2\mathbf{r}_k$ whenever $E_k^2 = -2$. Thus Y is the toric variety whose maximal cones are spanned by e_z and two consecutive rays, $E_k \times \mathbb{A}^1_z$ is the toric divisor of $\mathbf{r}_k$, and

$$\mathbf{r}_k = \mathbf{r}_0 + k(\mathbf{r}_1 - \mathbf{r}_0) \quad \text{for } -n_- \leq k \leq n_+ + 1. \tag{19.2}$$

For $-n_- \leq j \leq n_+$, the vectors $\mathbf{r}_j$, $\mathbf{r}_{j+1}$, e_z form a basis of N_G. Thus the affine toric variety of the cone spanned by $\mathbf{r}_j$, $\mathbf{r}_{j+1}$, e_z is $\mathbb{A}^3$, whose coordinates are the characters of the dual basis [CLS11, Example 1.2.21], and by Set-up 19.1(5) the characters dual to $\mathbf{r}_j$, $\mathbf{r}_{j+1}$, e_z are u, v, z. Hence

$$x^m = u^{\langle m, \mathbf{r}_j \rangle} v^{\langle m, \mathbf{r}_{j+1} \rangle} z^{m_3} \quad \text{on } U_j \text{ for any } m \in M. \tag{19.3}$$

The set $f^{-1}(o)$ is the union of $E \times \{0\}$ over the exceptional curves E of ρ. By [CLS11, Theorem 3.2.6], the only torus-invariant curves of T meeting the affine toric open subset of T given by the cone spanned by $\mathbf{r}_j$ and $\mathbf{r}_{j+1}$ are E_j and E_{j+1}, which correspond to the rays spanned by $\mathbf{r}_j$ and $\mathbf{r}_{j+1}$, and $E_j \times \mathbb{A}^1_z = (u = 0)$ and $E_{j+1} \times \mathbb{A}^1_z = (v = 0)$ on U_j (Set-up 19.1(5)). Thus $f^{-1}(o) \cap U_j = (z = 0) \cap (uv = 0)$.

For $-n_- \leq k \leq n_+ + 1$ let

$$\mu_{i,k} := \operatorname{wt}_{\mathbf{r}_k}(h_i) = \min\{\langle \alpha, \mathbf{r}_k \rangle \mid \alpha \in \Lambda_i\}$$

(Definition 2.16). Since $\mathbf{r}_k \in \mathbb{R}^3_{\geq 0}$, $\mu_{i,k} \geq 0$. Fix i and $-n_- \leq j \leq n_+$, and pick $\alpha \in \Lambda_i$. By (19.1), $\langle \alpha, \mathbf{r}_j \rangle - \mu_{i,j}$ and $\langle \alpha, \mathbf{r}_{j+1} \rangle - \mu_{i,j+1}$ are integers. Since $\mathbf{r}_j, \mathbf{r}_{j+1}, e_z$ is a basis of N_G, there exists $m_0 \in M$ with

$$\langle m_0, \mathbf{r}_j \rangle = \langle \alpha, \mathbf{r}_j \rangle - \mu_{i,j}, \quad \langle m_0, \mathbf{r}_{j+1} \rangle = \langle \alpha, \mathbf{r}_{j+1} \rangle - \mu_{i,j+1}, \quad \langle m_0, e_z \rangle = \alpha_3.$$

Let $\eta_{i,j} := \alpha - m_0 \in \mathbb{Z}^3$. Then $\langle \eta_{i,j}, \mathbf{r}_j \rangle = \mu_{i,j}$, $\langle \eta_{i,j}, \mathbf{r}_{j+1} \rangle = \mu_{i,j+1}$, $\langle \eta_{i,j}, e_z \rangle = 0$, and $\alpha' - \eta_{i,j} \in M$ for any $\alpha' \in \Lambda_i$ by (19.1). By (19.3), the rational function $h_i / x^{\eta_{i,j}} \in \mathbb{C}(X)$ is, on U_j,

$$\Phi_{i,j} := \frac{h_i}{x^{\eta_{i,j}}} = \sum_{\alpha \in \Lambda_i} c_{i,\alpha} u^{\langle \alpha, \mathbf{r}_j \rangle - \mu_{i,j}} v^{\langle \alpha, \mathbf{r}_{j+1} \rangle - \mu_{i,j+1}} z^{\alpha_3}. \tag{19.4}$$

All exponents in (19.4) are non-negative integers, the least exponent of u and the least exponent of v are 0, and distinct α correspond to distinct monomials. Hence $\Phi_{i,j} \in \mathbb{C}[u, v, z]$ is divisible neither by u nor by v. In **Step 3** we show that $\Phi_{i,j} = P_i(uv, z) + Q_{i,j}$ for polynomials P_i and $Q_{i,j}$ as in the statement.

We show that $\operatorname{div}(\Phi_{i,j})$ is the strict transform of D_i on U_j. Since h_i is G-semi-invariant and G has order r, h_i^r is a regular function on X, and $\pi^* \operatorname{div}(h_i^r) = r \operatorname{div}(h_i) = \pi^*(rD_i)$. Since π^* is injective on divisors, $\operatorname{div}(h_i^r) = rD_i$. Moreover $r\eta_{i,j} \in r\mathbb{Z}^3 \subset M$, so by (19.3), $x^{r\eta_{i,j}} = u^{r\mu_{i,j}} v^{r\mu_{i,j+1}}$ on U_j. Thus, on U_j,

$$rf^*D_i = \operatorname{div}(f^* h_i^r) = r \operatorname{div}(\Phi_{i,j}) + r\mu_{i,j}(u = 0) + r\mu_{i,j+1}(v = 0).$$

The prime divisors on U_j that are exceptional over X are $(u = 0) = E_j \times \mathbb{A}^1_z$ and $(v = 0) = E_{j+1} \times \mathbb{A}^1_z$, and $\operatorname{div}(\Phi_{i,j})$ contains neither of them. Thus $\operatorname{div}(\Phi_{i,j})$ is the strict transform of D_i on U_j, and, taking $j = k$ or $j = k - 1$,

$$\operatorname{ord}_{E_k \times \mathbb{A}^1_z} f^* D_i = \mu_{i,k} \quad \text{for any } -n_- \leq k \leq n_+ + 1. \tag{19.5}$$

Step 2. In this step, we show that a_k and $\mu_{i,k}$ are affine functions of k, that $0 < a_k \leq 1$, and that $B_Y|_{U_j}$ is as in the statement.

Let ℓ be the linear function on $\mathbb{R}^3$ with $\ell(e_x) = \ell(e_y) = \ell(e_z) = 1$. Since e_x, e_y, e_z are the primitive generators of the rays of $\mathbb{R}^3_{\geq 0}$ in N_G, ℓ is the support function of K_X, and [CLS11, Lemma 11.4.10] implies $a(E_k \times \mathbb{A}^1_z, X, 0) = \ell(\mathbf{r}_k)$. By (19.5) and $K_Y + B_Y = f^*(K_X + B)$,

$$a_k = \ell(\mathbf{r}_k) - \sum_i b_i \mu_{i,k}. \tag{19.6}$$

Let Y_η be the generic fiber of $Y \to \mathbb{A}^1_z$, so that E_k, $-n_- \leq k \leq n_+ + 1$, are complete curves on Y_η. The restriction of $rf^*D_i = \operatorname{div}(f^* h_i^r)$ to Y_η is principal, so $(f^*D_i|_{Y_\eta} \cdot E_k) = 0$. For $-n_- + 1 \leq k \leq n_+$, the only exceptional curves of ρ intersecting E_k are E_{k-1} and E_{k+1}, and $(E_{k-1} \cdot E_k) = (E_{k+1} \cdot E_k) = 1$. By (19.5), $E_k^2 = -2$, and $(D_i^\eta \cdot E_k) = 0$,

$$0 = (f^*D_i|_{Y_\eta} \cdot E_k) = (D_i^\eta \cdot E_k) + \mu_{i,k-1} - 2\mu_{i,k} + \mu_{i,k+1} = \mu_{i,k-1} - 2\mu_{i,k} + \mu_{i,k+1}.$$

Thus

$$\mu_{i,k} = \mu_{i,0} + k(\mu_{i,1} - \mu_{i,0}) \quad \text{for any } -n_- \leq k \leq n_+ + 1. \tag{19.7}$$

By (19.2), $\ell(\mathbf{r}_k) = \ell(\mathbf{r}_0) + k(\ell(\mathbf{r}_1) - \ell(\mathbf{r}_0))$. Together with (19.6) and (19.7), $a_k = a_0 + k(a_1 - a_0)$.

Since $K_Y - f^*K_X$ is the pullback of $K_T - \rho^* K_S$ under the projection $Y \to T$, $\ell(\mathbf{r}_k) = a(E_k, S, 0)$. By [Ful93, §2.6], any exceptional curve E of ρ is a smooth rational curve with $(E^2) \leq -2$, so $(K_T \cdot E) = -2 - (E^2) \geq 0$ by adjunction. Thus K_T is ρ-nef, and [KM98, Corollary 4.3] implies $a(E, S, 0) \leq 1$ for any exceptional curve E of ρ. Since $b_i > 0$ and $\mu_{i,k} \geq 0$, (19.6) implies $a_k \leq \ell(\mathbf{r}_k) \leq 1$. The center of $E_k \times \mathbb{A}^1_z$ on X contains o and (X, B) is klt near o, so $a_k > 0$.

By (19.5), the coefficient of $E_k \times \mathbb{A}^1_z$ in B_Y is $1 - a_k$, and the other components of B_Y intersecting U_j are the strict transforms of the D_i with coefficients b_i. By **Step 1**,

$$B_Y|_{U_j} = (1 - a_j)(u = 0) + (1 - a_{j+1})(v = 0) + \sum_i b_i \operatorname{div}(\Phi_{i,j}).$$

Step 3. In this step, we construct P_i and $Q_{i,j}$.

Fix i. For $\alpha \in \Lambda_i$, let $A_\alpha := \langle \alpha, \mathbf{r}_0 \rangle - \mu_{i,0}$ and $\lambda_\alpha := \langle \alpha, \mathbf{r}_1 \rangle - \mu_{i,1} - A_\alpha$. They are integers and $A_\alpha \geq 0$. By (19.2) and (19.7), for any $-n_- \leq j \leq n_+$,

$$\langle \alpha, \mathbf{r}_j \rangle - \mu_{i,j} = A_\alpha + j\lambda_\alpha \quad \text{and} \quad \langle \alpha, \mathbf{r}_{j+1} \rangle - \mu_{i,j+1} = A_\alpha + (j+1)\lambda_\alpha,$$

so by (19.4),

$$\Phi_{i,j} = \sum_{\alpha \in \Lambda_i} c_{i,\alpha} u^{A_\alpha + j\lambda_\alpha} v^{A_\alpha + (j+1)\lambda_\alpha} z^{\alpha_3}.$$

Let

$$P_i(w,z) := \sum_{\alpha \in \Lambda_i,\ \lambda_\alpha = 0} c_{i,\alpha} w^{A_\alpha} z^{\alpha_3},$$

$$Q_{i,j} := \Phi_{i,j} - P_i(uv,z) = \sum_{\alpha \in \Lambda_i,\ \lambda_\alpha \neq 0} c_{i,\alpha} u^{A_\alpha + j\lambda_\alpha} v^{A_\alpha + (j+1)\lambda_\alpha} z^{\alpha_3}.$$

Let $\alpha \in \Lambda_i$ with $\lambda_\alpha \neq 0$. If $\lambda_\alpha > 0$, then the exponent of u in $\Phi_{i,-n_-}$ is non-negative, so $A_\alpha \geq n_- \lambda_\alpha$. Thus

$$A_\alpha + j\lambda_\alpha \geq (n_- + j)\lambda_\alpha \geq n_- + j \quad \text{and} \quad A_\alpha + (j+1)\lambda_\alpha \geq (n_- + j + 1)\lambda_\alpha \geq n_- + j + 1,$$

and the term of α in $Q_{i,j}$ is divisible by $u^{n_- + j} v^{n_- + j + 1}$. If $\lambda_\alpha < 0$, then the exponent of v in Φ_{i,n_+} is non-negative, so $A_\alpha \geq (n_+ + 1)|\lambda_\alpha|$. Thus

$$A_\alpha + j\lambda_\alpha \geq (n_+ - j + 1)|\lambda_\alpha| \geq n_+ - j + 1 \quad \text{and} \quad A_\alpha + (j+1)\lambda_\alpha \geq (n_+ - j)|\lambda_\alpha| \geq n_+ - j,$$

and the term of α in $Q_{i,j}$ is divisible by $u^{n_+ - j + 1} v^{n_+ - j}$. Therefore

$$Q_{i,j} \in (u^{n_- + j} v^{n_- + j + 1}, u^{n_+ - j + 1} v^{n_+ - j}) \quad \text{and} \quad \Phi_{i,j} = P_i(uv,z) + Q_{i,j}.$$

Since $n_-, n_+ \geq 2$, $Q_{i,0}$ is divisible by u. Since $\Phi_{i,0}$ is not divisible by u, $P_i(uv,z)$ is not divisible by u, that is, P_i contains a term with $A_\alpha = 0$. Hence $P_i(0,z) \neq 0$. Together with **Step 1** and **Step 2**, we obtain Construction-Proposition 19.4. □

Proposition 19.5. *Notation and conditions as in Set-up 19.1 and Construction-Proposition 19.4. Let F be a prime divisor computing $c := \mathrm{mld}(X \ni o, B)$ with $0 < c < t \leq 1$ such that any other prime divisor over X with center o has log discrepancy at least t, and assume that $\mathrm{center}_Y F \cap U_0 \neq \emptyset$. Let $p := \mathrm{ord}_F u$ and $q := \mathrm{ord}_F v$ for the coordinates u, v of U_0, $d := p + q$, $\sigma \in \{1, -1\}$ with $\sigma(a_1 - a_0) \leq 0$, $m_j := \min\{(n_- + j)p + (n_- + j + 1)q, (n_+ - j + 1)p + (n_+ - j)q\}$ for $-n_- \leq j \leq n_+$, and $m_* := \min\{m_0, m_\sigma\}$, and let F_j be the valuation of $\mathbb{C}(X)$ with $\mathrm{ord}_{F_j} \phi(u,v,z) = \mathrm{ord}_F \phi(u,v,z)$ for any $\phi \in \mathbb{C}(u,v,z)$, where u, v, z are the coordinates of U_j on the left and those of U_0 on the right. Then:*

(1) $d > 0$, $F_0 = F$, and $F_{-n_-}, \ldots, F_{n_+}$ are pairwise distinct prime divisors over X with center o;

(2) for any $j \neq 0$ with $-n_- \leq j \leq n_+$,

$$\sum_i b_i \big(\mathrm{ord}_F \Phi_{i,0} - \mathrm{ord}_{F_j} \Phi_{i,j} \big) \geq t - c - j(a_1 - a_0)d;$$

(3) there exists i_0 with $\mathrm{ord}_F \Phi_{i_0,0} \geq m_*$, and if $\gamma > 0$ is a real number such that $b_i \geq \gamma$ for any i, then $a(F, Y, 0) - c \geq \gamma m_* \geq \gamma(\min\{n_-, n_+\} - 1)d$.

Proof. (1) Since F computes $\mathrm{mld}(X \ni o, B)$, $\mathrm{center}_X F = o$, so $\mathrm{center}_Y F \subset f^{-1}(o)$. By Construction-Proposition 19.4, $f^{-1}(o) \cap U_0 = (z = 0) \cap (uv = 0)$. Since $\mathrm{center}_Y F \cap U_0$ is non-empty and contained in $(z = 0) \cap (uv = 0)$, the functions u, v, z are regular at the generic point of $\mathrm{center}_Y F$, $p, q \geq 0$, and $d = \mathrm{ord}_F(uv) > 0$.

Let $\psi_j : U_0 \to U_j$ be the isomorphism given by $(u,v,z) \to (u,v,z)$. Then $\mathrm{ord}_{F_j} = \mathrm{ord}_F \circ \psi_j^*$ on $\mathbb{C}(X) = \mathbb{C}(U_0) = \mathbb{C}(U_j)$. If F is a prime divisor on a higher model $g : W \to U_0$, then ord_{F_j} is the valuation of the prime divisor $F \subset W$, with W regarded over U_j through $\psi_j \circ g$. Thus F_j is a prime divisor over U_j, hence over X, and $\mathrm{center}_{U_j} F_j = \psi_j(\mathrm{center}_{U_0} F)$. By definition, $F_0 = F$.

Since ψ_j maps $(z=0)\cap(uv=0)$ of U_0 onto $(z=0)\cap(uv=0)$ of U_j, which is $f^{-1}(o)\cap U_j$ by Construction-Proposition 19.4, $\operatorname{center}_X F_j = o$.

Let ξ_j be the generic point of $\operatorname{center}_Y F_j$. Then $\xi_j \in U_j$. Since $\operatorname{ord}_{F_j} u = p$ and $\operatorname{ord}_{F_j} v = q$ for the coordinates u, v of U_j, $\xi_j \in (u=0)$ if and only if $p>0$, and $\xi_j \in (v=0)$ if and only if $q>0$. The prime divisors of Y exceptional over X that intersect U_j are $E_j \times \mathbb{A}^1_z$ and $E_{j+1}\times \mathbb{A}^1_z$. Thus the set I_j of k with $\xi_j \in E_k \times \mathbb{A}^1_z$ is $\{j\}$ if $q=0$, $\{j+1\}$ if $p=0$, and $\{j, j+1\}$ if $p,q>0$. Since $d>0$, $I_j \neq I_{j'}$ for $j\neq j'$. Thus the centers of F_j on Y are pairwise distinct, and so are the F_j.

(2) Fix $-n_- \le j \le n_+$. By [KM98, Lemma 2.30], $a(F_j, X, B) = a(F_j, Y, B_Y)$. Since ψ_j is an isomorphism, $a(F_j, U_j, 0) = a(F, U_0, 0) = a(F, Y, 0)$. By the definition of F_j, $\operatorname{ord}_{F_j}\Phi_{i,j} = \operatorname{ord}_F \Phi_{i,j}(u,v,z)$, where on the right $\Phi_{i,j}$ is evaluated at the coordinates of U_0. Recall from (1) that $\operatorname{ord}_{F_j} u = p$ and $\operatorname{ord}_{F_j} v = q$ for the coordinates u, v of U_j. By Construction-Proposition 19.4,

$$a(F_j, X, B) = a(F, Y, 0) - (1-a_j)p - (1-a_{j+1})q - \sum_i b_i \operatorname{ord}_{F_j}\Phi_{i,j}. \tag{19.8}$$

For $j=0$, (19.8) reads

$$c = a(F, Y, 0) - (1-a_0)p - (1-a_1)q - \sum_i b_i \operatorname{ord}_F \Phi_{i,0}. \tag{19.9}$$

Since $a_j - a_0 = a_{j+1} - a_1 = j(a_1 - a_0)$, subtracting (19.9) from (19.8) implies

$$a(F_j, X, B) - c = j(a_1-a_0)d + \sum_i b_i\big(\operatorname{ord}_F\Phi_{i,0} - \operatorname{ord}_{F_j}\Phi_{i,j}\big).$$

For $j\neq 0$, F_j is a prime divisor over X with center o different from F by (1), so $a(F_j, X, B) \ge t$. This implies (2).

(3) Since $n_-, n_+ \ge 2$, σ is an index with $-n_- \le \sigma \le n_+$. By (2) and $\sigma(a_1-a_0)\le 0$,

$$\sum_i b_i\big(\operatorname{ord}_F\Phi_{i,0} - \operatorname{ord}_{F_\sigma}\Phi_{i,\sigma}\big) \ge t - c - \sigma(a_1-a_0)d \ge t - c > 0.$$

Since $b_i > 0$ for any i, there exists i_0 with

$$\operatorname{ord}_F \Phi_{i_0,0} > \operatorname{ord}_{F_\sigma}\Phi_{i_0,\sigma}. \tag{19.10}$$

For any $-n_- \le j \le n_+$, write $Q_{i_0,j} = g_1 u^{n_- + j} v^{n_- + j + 1} + g_2 u^{n_+ - j + 1} v^{n_+ - j}$ with polynomials g_1, g_2. Since u, v, z are regular at the generic point of $\operatorname{center}_Y F$, $\operatorname{ord}_F g_1, \operatorname{ord}_F g_2 \ge 0$ when evaluated at the coordinates of U_0, so

$$\operatorname{ord}_{F_j} Q_{i_0,j} \ge m_j.$$

Suppose that $\operatorname{ord}_F \Phi_{i_0,0} < m_*$. Since $\operatorname{ord}_F Q_{i_0,0} \ge m_0 \ge m_* > \operatorname{ord}_F \Phi_{i_0,0}$ and $\Phi_{i_0,0} = P_{i_0}(uv,z) + Q_{i_0,0}$, we have $\operatorname{ord}_F P_{i_0}(uv,z) = \operatorname{ord}_F \Phi_{i_0,0} < m_*$. Since $\operatorname{ord}_{F_\sigma} Q_{i_0,\sigma} \ge m_\sigma \ge m_* > \operatorname{ord}_F P_{i_0}(uv,z)$ and $\operatorname{ord}_{F_\sigma} P_{i_0}(uv,z) = \operatorname{ord}_F P_{i_0}(uv,z)$, we have

$$\operatorname{ord}_{F_\sigma}\Phi_{i_0,\sigma} = \operatorname{ord}_F P_{i_0}(uv,z) = \operatorname{ord}_F \Phi_{i_0,0},$$

which contradicts (19.10). Thus $\operatorname{ord}_F \Phi_{i_0,0} \ge m_*$.

Assume that $b_i \ge \gamma$ for any i. Since $1-a_0, 1-a_1 \ge 0$ (Construction-Proposition 19.4), $p, q \ge 0$, and $\operatorname{ord}_F \Phi_{i,0} \ge 0$ for any i, (19.9) implies

$$a(F, Y, 0) - c = (1-a_0)p + (1-a_1)q + \sum_i b_i \operatorname{ord}_F \Phi_{i,0} \ge b_{i_0} \operatorname{ord}_F \Phi_{i_0,0} \ge \gamma m_*.$$

For $j \in \{0, \sigma\} \subset \{-1, 0, 1\}$, the four integers $n_- + j$, $n_- + j + 1$, $n_+ - j + 1$, and $n_+ - j$ are at least $\min\{n_-, n_+\} - 1$, so $m_j \ge (\min\{n_-, n_+\} - 1)d$. Hence $m_* \ge (\min\{n_-, n_+\} - 1)d$, and (3) holds. □

Proposition 19.6. *Let $\Gamma \subset [0,1]$ be a DCC set, and let $H_0 := 2 + \frac{1}{\delta(\Gamma)}$ if $\Gamma\backslash\{0\} \neq \emptyset$ and $H_0 := 3$ otherwise.*

(1) *Let $(V \ni p)$ be a smooth surface germ, D a non-zero reduced curve through p with local equation d, $C \geq 0$ an $\mathbb{R}$-divisor supported on D with coefficients in $[0,1]$, and $B \geq 0$ an $\mathbb{R}$-divisor with $B \in \Gamma$ and without common components with D, such that $(V, B+C)$ is lc near p. Then $\operatorname{mld}(V \ni p, B+C)$ is attained, and any prime divisor E computing $\operatorname{mld}(V \ni p, B+C)$ with $a(E,V,0)$ least among such prime divisors satisfies $a(E,V,0) \leq H_0 \operatorname{ord}_E d$.*

(2) *In Set-up 19.1, assume that $B \in \Gamma$, and let F be a prime divisor computing $c := \operatorname{mld}(X \ni o, B)$ with $0 < c < t \leq 1$ such that any other prime divisor over X with center o has log discrepancy at least t. If $\operatorname{center}_Y F = E_0 \times \{0\}$ or $\operatorname{center}_Y F = E_1 \times \{0\}$, then $\Gamma\backslash\{0\} \neq \emptyset$ and $\min\{n_-, n_+\} < 1 + \frac{H_0}{\gamma(\Gamma)}$.*

Proof. (1) Let $c := \operatorname{mld}(V \ni p, B+C)$. Let $g : W \to V$ be a log resolution of $(V, \operatorname{Supp}(B+C))$ such that $g^{-1}(p)$ is a divisor, and write $K_W + B_W = g^*(K_V + B + C)$. Since $(V, B+C)$ is lc near p, c is the minimum of $a(E', V, B+C)$ over the finitely many components E' of $g^{-1}(p)$ [Amb99, Section 1, the second paragraph after Definition 1.1]. Thus c is attained. Since $a(E', V, 0)$ is a positive integer for any prime divisor E' over V with center p, there exists a prime divisor computing c with $a(\cdot, V, 0)$ least among such prime divisors. Let E be any such prime divisor.

Write $B = \sum_l b_l \operatorname{div}(f_l)$ near p, where $b_l \in \Gamma\backslash\{0\}$ and the f_l are local equations of distinct prime divisors, and $d = \prod_j g_j$, where the g_j are local equations of the distinct components of D, and $C = \sum_j \beta_j \operatorname{div}(g_j)$ with $\beta_j \in [0,1]$. Let $\mathfrak{a}$ be the $\mathbb{R}$-ideal $\prod_l f_l^{b_l} \prod_{\beta_j > 0} g_j^{\beta_j}$. Then $(V, \mathfrak{a})$ is lc, $\operatorname{mld}(V \ni p, \mathfrak{a}) = c$, and E computes $\operatorname{mld}(V \ni p, \mathfrak{a})$. By [Kaw17, Theorem 1(i)], E is obtained by a weighted blow-up: there exist regular parameters x_1, x_2 of $\mathcal{O}_{V,p}$ and coprime positive integers $w_1 \leq w_2$ such that E is the monomial prime divisor in x_1, x_2 with weights (w_1, w_2) (Definition 2.16). In particular $\operatorname{ord}_E x_i = w_i$.

For coprime positive integers w_1', w_2', let $E(w_1', w_2')$ be the monomial prime divisor in x_1, x_2 with weights (w_1', w_2'). Then $E(w_1', w_2')$ is the exceptional divisor of the weighted blow-up in x_1, x_2 [Kaw17, p. 188], $\operatorname{center}_V E(w_1', w_2') = p$, and $a(E(w_1', w_2'), V, 0) = w_1' + w_2'$ by [CLS11, Lemma 11.4.10], since the morphism to $\mathbb{A}^2$ given by x_1, x_2 is étale near p. In particular $E = E(w_1, w_2)$ and $a(E, V, 0) = w_1 + w_2$. Since D is non-zero and any g_j vanishes at p,

$$\operatorname{ord}_E d = \sum_j \operatorname{ord}_E g_j \geq w_1 \geq 1.$$

Let $\delta_0 := \delta(\Gamma)$ if $\Gamma\backslash\{0\} \neq \emptyset$ and $\delta_0 := 1$ otherwise. Then $0 < \delta_0 \leq 1$ (Lemma 2.10(2)) and $H_0 = 2 + \frac{1}{\delta_0}$. Moreover, for non-negative integers n_l,

$$\sum_l b_l n_l > 1 \quad \text{implies} \quad \sum_l b_l n_l \geq 1 + \delta_0. \tag{19.11}$$

If $\Gamma\backslash\{0\} = \emptyset$, then $B = 0$, so $\sum_l b_l n_l = 0$ and (19.11) holds vacuously. Otherwise $\sum_l b_l n_l$ is a sum of elements of $\Gamma\backslash\{0\}$, and (19.11) follows from the definition of $\delta(\Gamma)$.

For any real number $s > 0$, let ν_s be the monomial valuation in x_1, x_2 with weights (w_1, s) (Definition 2.16), and let

$$\theta(s) := w_1 + s - \sum_l b_l \nu_s(f_l) - \sum_j \beta_j \nu_s(g_j).$$

Let $[s_1, s_2] \subset (0, +\infty)$ be a compact interval and h one of the f_l, g_j. Fix a monomial $x_1^{e_1} x_2^{e_2}$ occurring in the expansion of h in $\kappa(p)[[x_1, x_2]]$. A monomial $x_1^{e_1'} x_2^{e_2'}$ with $e_1' w_1 + e_2' s_1 > e_1 w_1 + e_2 s_2$ does not compute $\nu_s(h)$ for any $s \in [s_1, s_2]$, and there are finitely many other monomials. Thus, on $[s_1, s_2]$, $\nu_s(h)$ is the minimum of finitely many affine functions of s. Hence θ is continuous and piecewise linear with finitely many pieces on any compact subinterval of $(0, +\infty)$. For any rational number $s > 0$, let (w_1', w_2') be the primitive integral vector on the ray of (w_1, s), and let

$E^{(s)} := E(w_1', w_2')$. Then $\nu_s = \frac{w_1}{w_1'} \operatorname{ord}_{E^{(s)}}$, so

$$a(E^{(s)}, V, B+C) = w_1' + w_2' - \sum_l b_l \operatorname{ord}_{E^{(s)}} f_l - \sum_j \beta_j \operatorname{ord}_{E^{(s)}} g_j = \frac{w_1'}{w_1}\theta(s).$$

Since $(V, B+C)$ is lc near p, $\theta(s) \geq 0$ for any rational number $s > 0$, hence for any real number $s > 0$ by continuity. Since $\gcd(w_1, w_2) = 1$, we have $E^{(w_2)} = E$, and hence $\theta(w_2) = a(E, V, B+C) = c$.

Suppose that

$$w_2 > 1 + \max\Big\{ \operatorname{ord}_E d, \frac{w_1}{\delta_0} \Big\}. \tag{19.12}$$

Fix j and $s \in [w_2 - 1, w_2]$. Since $s > \operatorname{ord}_E d \geq \operatorname{ord}_E g_j = \nu_{w_2}(g_j)$, a monomial occurring in g_j and divisible by x_2 has ν_s-weight and ν_{w_2}-weight at least $s > \nu_{w_2}(g_j)$. Hence $\nu_{w_2}(g_j)$ is computed by the monomials x_1^e of g_j not divisible by x_2, whose ν_s-weight and ν_{w_2}-weight are both ew_1. Thus

$$\nu_s(g_j) = \operatorname{ord}_E g_j \quad \text{for any } s \in [w_2 - 1, w_2].$$

Let $I \subset (w_2 - 1, w_2)$ be an open interval on which θ and all $\nu_s(f_l)$ are linear, and for any l pick a monomial $x_1^{e_l} x_2^{n_l}$ of f_l computing $\nu_s(f_l)$ for any $s \in I$. Then, for $s \in I$,

$$\theta(s) = w_1\Big(1 - \sum_l b_l e_l\Big) - \sum_j \beta_j \operatorname{ord}_E g_j + s\Big(1 - \sum_l b_l n_l\Big).$$

The constant term is at most w_1. If $1 - \sum_l b_l n_l < 0$, then $1 - \sum_l b_l n_l \leq -\delta_0$ by (19.11), and (19.12) implies

$$\theta(s) \leq w_1 - \delta_0 s < w_1 - \delta_0(w_2 - 1) < 0 \quad \text{for } s \in I,$$

which is not possible. Thus the slopes of θ on $[w_2 - 1, w_2]$ are non-negative, and

$$0 \leq \theta(w_2 - 1) \leq \theta(w_2) = c.$$

By (19.12), $w_2 - 1 > \operatorname{ord}_E d \geq 1$ is a positive integer. Let $e := \gcd(w_1, w_2 - 1)$ and $E' := E(\frac{w_1}{e}, \frac{w_2 - 1}{e}) = E^{(w_2 - 1)}$. Then

$$a(E', V, B+C) = \frac{\theta(w_2 - 1)}{e} \leq c \quad \text{and} \quad a(E', V, 0) = \frac{w_1 + w_2 - 1}{e} < w_1 + w_2 = a(E, V, 0).$$

Since E' has center p, $a(E', V, B+C) \geq c$. Thus E' computes c and $a(E', V, 0) < a(E, V, 0)$, which contradicts the choice of E. Therefore (19.12) does not hold. Since $\delta_0 \leq 1$ and $w_1 \leq \operatorname{ord}_E d$,

$$w_2 \leq 1 + \max\Big\{ \operatorname{ord}_E d, \frac{w_1}{\delta_0} \Big\} \leq 1 + \frac{\operatorname{ord}_E d}{\delta_0},$$

and since $\operatorname{ord}_E d \geq 1$,

$$a(E, V, 0) = w_1 + w_2 \leq \operatorname{ord}_E d + 1 + \frac{\operatorname{ord}_E d}{\delta_0} \leq \Big(2 + \frac{1}{\delta_0}\Big) \operatorname{ord}_E d = H_0 \operatorname{ord}_E d.$$

(2) Assume that $\operatorname{center}_Y F = E_0 \times \{0\}$. The case $\operatorname{center}_Y F = E_1 \times \{0\}$ is proved by the argument below with u and v, E_0 and E_1, and a_0 and a_1 exchanged. We use the coordinates u, v, z of U_0 and the notation of Construction-Proposition 19.4. Since $E_0 \times \{0\} = (u = z = 0)$ on U_0, $\operatorname{center}_Y F \cap U_0 \neq \emptyset$, $\operatorname{ord}_F u > 0$, and $\operatorname{ord}_F v = 0$. By Proposition 19.5(3), $\operatorname{ord}_F \Phi_{i,0} \geq m_*$ for some i, so $B \neq 0$. Since $B \in \Gamma$, $\Gamma \backslash \{0\} \neq \emptyset$, $H_0 = 2 + \frac{1}{\delta(\Gamma)}$, and $b_i \geq \gamma(\Gamma)$ for any i. By Proposition 19.5(3) again,

$$a(F, Y, 0) - c \geq \gamma(\Gamma)(\min\{n_-, n_+\} - 1)(\operatorname{ord}_F u + \operatorname{ord}_F v). \tag{19.13}$$

Let $K := \mathbb{C}(v)$, $V := \operatorname{Spec} K[u, z]$, and $0 \in V$ the origin, and let

$$B_V := \sum_i b_i \operatorname{div}(\Phi_{i,0}) \quad \text{and} \quad C_V := (1 - a_0) \operatorname{div}(u)$$

on V. Then V is the generic fiber of the projection $U_0 \to \mathbb{A}^1_v$, $\mathbb{C}(Y) = K(u,z)$, and $B_V + C_V$ is the restriction of $B_Y|_{U_0}$ to V, because v is a unit on V. A valuation of $\mathbb{C}(Y)$ with center $(u = z = 0)$ on U_0 is trivial on K, because $(u = z = 0)$ dominates $\mathbb{A}^1_v$. Thus base change to the generic point of $\mathbb{A}^1_v$ is a bijection from the set of prime divisors over Y with center $E_0 \times \{0\}$ to the set of prime divisors over V with center 0, which preserves the valuations of $\mathbb{C}(Y)$. For a higher model $W \to U_0$, the restriction of K_{W/U_0} to $W_K := W \times_{\mathbb{A}^1_v} \operatorname{Spec} K$ is the relative canonical divisor of W_K over V. Thus

$$a(P, Y, B_Y) = a(P, V, B_V + C_V) \quad \text{and} \quad a(P, Y, 0) = a(P, V, 0)$$

for any prime divisor P over Y with center $E_0 \times \{0\}$. Since $E_0 \times \{0\} \subset f^{-1}(o)$, any such P has center o on X, and $a(P, X, B) = a(P, Y, B_Y)$ [KM98, Lemma 2.30]. Hence $\operatorname{mld}(V \ni 0, B_V + C_V) = c$, and F is the only prime divisor over V with center 0 computing c. Since (X, B) is klt near o, (Y, B_Y) is sub-klt near $f^{-1}(o)$, so $(V, B_V + C_V)$ is klt near 0. Moreover $B_V \geq 0$, $0 \leq 1 - a_0 < 1$ (Construction-Proposition 19.4), and u does not divide any $\Phi_{i,0}$. Each $\Phi_{i,0}$ is either a non-zero constant or, up to a constant, the irreducible equation of the strict transform of D_i on U_0. If $\Phi_{i,0}$ is not constant, then $\Phi_{i,0} \notin \mathbb{C}[v]$, because $\Phi_{i,0}(0, v, z) = P_i(0, z)$. By the Gauss lemma, the non-constant $\Phi_{i,0}$ are irreducible and pairwise non-associate in $K[u, z]$.

Let $\overline{K}$ be an algebraic closure of K, and let $V_{\overline{K}}$, $B_{\overline{K}}$, and $C_{\overline{K}}$ be the base changes of V, B_V, and C_V. Since $\overline{K}/K$ is separable and the non-constant $\Phi_{i,0}$ are irreducible and pairwise non-associate in $K[u, z]$, the coefficients of $B_{\overline{K}}$ belong to $\{b_i\}$, so $B_{\overline{K}} \in \Gamma$, and $B_{\overline{K}}$ has no common components with $(u = 0)$. Any prime divisor E' over $V_{\overline{K}}$ is defined over a finite extension K'/K. Since $V_{K'} \to V$ is finite étale, on any higher model carrying E' the canonical divisor pulls back to the canonical divisor and the ramification index along E' is 1. Thus the restriction of $\operatorname{ord}_{E'}$ to $K(u, z)$ is $\operatorname{ord}_{E''}$ for a prime divisor E'' over V with $a(E'', V, 0) = a(E', V_{\overline{K}}, 0)$, $a(E'', V, B_V + C_V) = a(E', V_{\overline{K}}, B_{\overline{K}} + C_{\overline{K}})$, and $\operatorname{ord}_{E''} u = \operatorname{ord}_{E'} u$, and E'' has center 0 if E' has center the origin of $V_{\overline{K}}$. Conversely, any component F' of the base change of F satisfies $a(F', V_{\overline{K}}, 0) = a(F, V, 0)$, $a(F', V_{\overline{K}}, B_{\overline{K}} + C_{\overline{K}}) = a(F, V, B_V + C_V)$, and $\operatorname{ord}_{F'} u = \operatorname{ord}_F u$. Hence $(V_{\overline{K}}, B_{\overline{K}} + C_{\overline{K}})$ is lc near the origin 0 of $V_{\overline{K}}$, $\operatorname{mld}(V_{\overline{K}} \ni 0, B_{\overline{K}} + C_{\overline{K}}) = c$, and the restriction of any prime divisor computing $\operatorname{mld}(V_{\overline{K}} \ni 0, B_{\overline{K}} + C_{\overline{K}})$ is a prime divisor over V computing $\operatorname{mld}(V \ni 0, B_V + C_V)$, which is F.

Since $\overline{K}$ is an algebraically closed field of characteristic zero with the cardinality of $\mathbb{C}$, there exists a field isomorphism $\overline{K} \cong \mathbb{C}$ (cf. [Lan02, Theorem VIII.1.1 and Corollary V.2.9]). Through an isomorphism $\overline{K} \cong \mathbb{C}$, (1) holds for the germ $(V_{\overline{K}} \ni 0)$, with $D = (u = 0)$, the local equation u, the $\mathbb{R}$-divisor $C_{\overline{K}}$ supported on D, and the $\mathbb{R}$-divisor $B_{\overline{K}} \in \Gamma$. Let E' be a prime divisor computing $\operatorname{mld}(V_{\overline{K}} \ni 0, B_{\overline{K}} + C_{\overline{K}})$ with $a(E', V_{\overline{K}}, 0)$ least among such prime divisors. By (1), $a(E', V_{\overline{K}}, 0) \leq H_0 \operatorname{ord}_{E'} u$. The restriction of E' is F, so

$$a(F, Y, 0) = a(F, V, 0) = a(E', V_{\overline{K}}, 0) \leq H_0 \operatorname{ord}_{E'} u = H_0 \operatorname{ord}_F u = H_0(\operatorname{ord}_F u + \operatorname{ord}_F v).$$

Together with (19.13) and $c > 0$,

$$\gamma(\Gamma)(\min\{n_-, n_+\} - 1)(\operatorname{ord}_F u + \operatorname{ord}_F v) < a(F, Y, 0) \leq H_0(\operatorname{ord}_F u + \operatorname{ord}_F v).$$

Since $\operatorname{ord}_F u + \operatorname{ord}_F v > 0$, $\min\{n_-, n_+\} < 1 + \frac{H_0}{\gamma(\Gamma)}$. □

19.2. **Perturbations and non-node points.** In this subsection, we show that the minimal log discrepancy of a smooth surface germ with a marked curve does not change when the boundary equations are perturbed by multiples of a high power of the equation of the marked curve (Lemma 19.8). Then we show that in Set-up 19.1 with n_- and n_+ large, any closed point of U_0 with $z = 0$ at which exactly one of u, v vanishes has minimal log discrepancy bigger than one with respect to (Y, B_Y) (Proposition 19.9).

Lemma 19.7. *Let V be a smooth surface, $p \in V$ a closed point, and Δ an $\mathbb{R}$-divisor on V. If (V, Δ) is not sub-lc near p, then $a(E, V, \Delta) < 0$ for some prime divisor E over V with $\operatorname{center}_V E = \{p\}$. In particular, if $\operatorname{mld}(V \ni p, \Delta) \geq 0$, then (V, Δ) is sub-lc near p.*

Proof. The singular points of $\operatorname{Supp}\Delta$ and the intersection points of distinct components of $\operatorname{Supp}\Delta$ form a finite set, and the statement is local around p. Thus, possibly shrinking V around p, we may assume that any component of $\operatorname{Supp}\Delta$ passes through p and that $\operatorname{Supp}\Delta$ is snc on $V\backslash\{p\}$. Let $g : V' \to V$ be a log resolution of (V,Δ) which is a composition of blow-ups of closed points over p, the first of which is the blow-up of p (such g exists, since after the blow-up of p the total transform of $\operatorname{Supp}\Delta$ is snc outside the exceptional curve, and [Har77, Theorem V.3.9], whose construction is local, resolves it by blow-ups of points over p). Then g is an isomorphism over $V\backslash\{p\}$ and $g^{-1}(p) = \operatorname{Exc}(g)$ is a divisor. Let Δ' be the $\mathbb{R}$-divisor on V' such that $K_{V'} + \Delta' = g^*(K_V + \Delta)$ and $g_*\Delta' = \Delta$.

If the coefficients of Δ' are at most 1, then (V',Δ') is sub-lc by [KM98, Corollary 2.31(3)], so (V,Δ) is sub-lc by [KM98, Lemma 2.30], contrary to our assumption. If the coefficient of an exceptional curve E of g in Δ' is bigger than 1, then $a(E,V,\Delta) < 0$ and $\operatorname{center}_V E = \{p\}$, and we are done. Thus we may assume that any exceptional curve of g has coefficient at most 1 in Δ', and that some component C of $\operatorname{Supp}\Delta$ has coefficient $c > 1$ in Δ.

Let C_0 be the strict transform of C on $V_0 := V'$. Since C passes through p, there exists a point $q_0 \in C_0 \cap g^{-1}(p)$. Since $\operatorname{Exc}(g) \cup \operatorname{Supp} g_*^{-1}\Delta$ is snc, q_0 belongs to exactly one exceptional curve F_0 of g, the curves C_0 and F_0 intersect transversally at q_0, and near q_0 we have

$$\Delta' = cC_0 + (1 - e_0)F_0, \quad e_0 := a(F_0, V, \Delta) \geq 0.$$

For $k \geq 1$, let $V_k \to V_{k-1}$ be the blow-up of q_{k-1} with exceptional curve F_k, let C_k be the strict transform of C_{k-1}, and let q_k be the intersection point of C_k and F_k. Since C_{k-1} is smooth near q_{k-1} and intersects F_{k-1} transversally at q_{k-1}, the curve C_k intersects F_k transversally at the single point q_k [Har77, Proposition V.3.6, Corollary V.3.7] (the computation is local), and the strict transform of F_{k-1} does not pass through q_k, since C_{k-1} and F_{k-1} have distinct tangent directions at q_{k-1}. Let $e_k := a(F_k, V, \Delta)$. By induction on k and [KM98, Lemma 2.30], the crepant pullback of $K_V + \Delta$ to V_{k-1} has boundary $cC_{k-1} + (1 - e_{k-1})F_{k-1}$ near q_{k-1}. Thus [KM98, Lemma 2.29] implies

$$e_k = 2 - c - (1 - e_{k-1}) = e_{k-1} - (c - 1),$$

so $e_k = e_0 - k(c-1)$. Hence $a(F_k, V, \Delta) < 0$ for any integer $k > \frac{e_0}{c-1}$, and $\operatorname{center}_V F_k = \{p\}$.

The last statement follows, because $\operatorname{mld}(V \ni p, \Delta) \geq 0$ means that $a(E,V,\Delta) \geq 0$ for any prime divisor E over V with $\operatorname{center}_V E = \{p\}$. □

Lemma 19.8. *Let $\Gamma \subset [0,1]$ be a DCC set with $\Gamma\backslash\{0\} \neq \emptyset$, $\gamma := \gamma(\Gamma)$, $\delta := \delta(\Gamma)$, and $H_M := 2 + \frac{1+M}{\delta}$ for real numbers $M \geq 0$. Let $(V \ni p)$ be a smooth surface germ, D a non-zero reduced curve through p with local equation d, C an $\mathbb{R}$-divisor supported on D with coefficients in $[-M, 1]$, $N > \frac{H_M + M}{\gamma}$ an integer, $b_i \in \Gamma$ positive, f_i, h_i non-zero regular functions with $h_i - f_i \in (d^N)$, $B_f := \sum_i b_i \operatorname{div}(f_i)$, and $B_h := \sum_i b_i \operatorname{div}(h_i)$. Then $(V, C + B_f)$ is sub-lc near p if and only if $(V, C + B_h)$ is sub-lc near p. In this case $\operatorname{mld}(V \ni p, C + B_f) = \operatorname{mld}(V \ni p, C + B_h)$, attained by a common prime divisor E with $a(E,V,0) \leq H_M \operatorname{ord}_E d$ and $\operatorname{ord}_E f_i = \operatorname{ord}_E h_i$ for any i.*

Proof. Since the statement is local around p, possibly shrinking V around p, we may assume that V is affine. The condition $h_i - f_i \in (d^N)$ is symmetric in f_i and h_i. Thus it suffices to show that if $(V, C + B_f)$ is sub-lc near p, then $(V, C + B_h)$ is sub-lc near p, $\operatorname{mld}(V \ni p, C + B_h)$ equals $\operatorname{mld}(V \ni p, C + B_f)$, and a prime divisor E as in the statement computes both minimal log discrepancies.

Assume that $(V, C + B_f)$ is sub-lc near p, and let $\mu := \operatorname{mld}(V \ni p, C + B_f) \geq 0$. Regard f_i and h_i as functions on $V \times \mathbb{A}^1_\tau$ which do not depend on τ. Then $h_i - f_i \in (d^N)\mathcal{O}_{V\times\mathbb{A}^1_\tau}$, $\gamma N > H_M + M$, and

$$\operatorname{pr}_V^* C + \sum_i b_i \operatorname{div}(h_i) = \operatorname{pr}_V^*(C + B_h).$$

By Theorem 16.13(2),

$$a\big(F, V \times \mathbb{A}^1_\tau, \mathrm{pr}_V^*(C + B_h)\big) \geq \mu + 1$$

for any prime divisor F over $V \times \mathbb{A}^1_\tau$ with center $(p, 0)$.

Let E' be a prime divisor over V with $\mathrm{center}_V E' = \{p\}$. By Lemma 8.1(2), applied to $(V \ni p, C + B_h)$ with the trivial group and the trivial character, for which the integer q of that lemma equals 1, the Gauss extension of $\mathrm{ord}_{E'}$ to $\mathbb{C}(V)(\tau)$ with τ-value 1 (Definition-Proposition 2.11) is ord_F for a prime divisor F over $V \times \mathbb{A}^1_\tau$ with center $(p, 0)$, and

$$a\big(F, V \times \mathbb{A}^1_\tau, \mathrm{pr}_V^*(C + B_h) + (\tau = 0)\big) = a(E', V, C + B_h).$$

Since $\mathrm{ord}_F \tau = 1$, we have

$$a(E', V, C + B_h) = a\big(F, V \times \mathbb{A}^1_\tau, \mathrm{pr}_V^*(C + B_h)\big) - 1 \geq \mu.$$

Since E' is arbitrary, $\mathrm{mld}(V \ni p, C+B_h) \geq \mu \geq 0$, and $(V, C+B_h)$ is sub-lc near p by Lemma 19.7.

Let E be the prime divisor over V of Theorem 16.13(1). Then $\mathrm{center}_V E = \{p\}$, $a(E, V, C + B_f) = \mu$, $a(E, V, 0) \leq H_M \mathrm{ord}_E d$, and $\mathrm{ord}_E f_i < N \mathrm{ord}_E d$ for any i. Since

$$\mathrm{ord}_E(h_i - f_i) \geq N \mathrm{ord}_E d > \mathrm{ord}_E f_i,$$

we have $\mathrm{ord}_E h_i = \mathrm{ord}_E f_i$ for any i. Hence

$$a(E, V, C + B_h) = a(E, V, C) - \sum_i b_i \mathrm{ord}_E h_i = a(E, V, C) - \sum_i b_i \mathrm{ord}_E f_i = \mu.$$

Thus $\mathrm{mld}(V \ni p, C + B_h) = \mu$, and E computes both minimal log discrepancies. □

Proposition 19.9. *Let $\Gamma \subset [0, 1]$ be a DCC set with $\Gamma \backslash \{0\} \neq \emptyset$, $\gamma := \gamma(\Gamma)$, $\delta := \delta(\Gamma)$, $H_0 := 2 + \frac{1}{\delta}$, and $N > \frac{H_0}{\gamma}$ an integer.*

(1) Let $(V \ni p)$ be a smooth surface germ, D a non-zero reduced curve through p with local equation d, $C \geq 0$ an $\mathbb{R}$-divisor supported on D, $b_i \in \Gamma$ positive, and f_i, h_i non-zero regular functions with $h_i - f_i \in (d^N)$. Let $B_f := \sum_i b_i \mathrm{div}(f_i)$ and $B_h := \sum_i b_i \mathrm{div}(h_i)$. Then $(V, C + B_f)$ is lc near p if and only if $(V, C + B_h)$ is lc near p, and in this case $\mathrm{mld}(V \ni p, C + B_f) = \mathrm{mld}(V \ni p, C + B_h)$, attained by a common prime divisor E with $a(E, V, 0) \leq H_0 \mathrm{ord}_E d$ and $\mathrm{ord}_E f_i = \mathrm{ord}_E h_i$ for any i.

(2) In Set-up 19.1, assume that $B \in \Gamma$ and $n_-, n_+ \geq N$. Then $\mathrm{mld}(Y \ni y, B_Y) > 1$ for any closed point $y \in U_0$ with $z = 0$ at which exactly one of u, v vanishes.

(3) In Set-up 19.1, assume that $B \in \Gamma$ and $\min\{n_-, n_+\} \geq 1 + \frac{H_0}{\gamma}$, and let F be a prime divisor computing $c := \mathrm{mld}(X \ni o, B)$ with $0 < c < t \leq 1$ such that any other prime divisor over X with center o has log discrepancy at least t. If $\mathrm{center}_Y F \cap U_0 \neq \emptyset$, then $\mathrm{center}_Y F$ is the closed point $u = v = z = 0$ of U_0.

Proof. (1) Since the statement is local around p, possibly shrinking V around p, we may assume that any component of D passes through p. If the coefficient of C along some component of D is bigger than 1, then neither $(V, C + B_f)$ nor $(V, C + B_h)$ is lc near p, because $B_f, B_h \geq 0$. Thus we may assume that the coefficients of C belong to $[0, 1]$. Since $C + B_f \geq 0$ and $C + B_h \geq 0$, the pair $(V, C + B_f)$ (resp. $(V, C + B_h)$) is lc near p if and only if it is sub-lc near p. Hence (1) follows from Lemma 19.8 with $M = 0$, because $H_M = H_0$ and the condition $N > \frac{H_M + M}{\gamma}$ becomes $N > \frac{H_0}{\gamma}$.

(2) Exchanging u with v, a_0 with a_1, and n_- with n_+ preserves the form of $B_Y|_{U_0}$ in Construction-Proposition 19.4. Thus, possibly exchanging them, we may assume that $y = (0, v_0, 0)$ for some $v_0 \neq 0$.

Since (X, B) is klt near o and $y \in f^{-1}(o)$, possibly shrinking X around o, we may assume that (X, B) is klt. Then $a(E, Y, B_Y) = a(E, X, B) > 0$ for any prime divisor E over Y which is exceptional over Y [KM98, Lemma 2.30]. Let $W := U_0 \cap (v \neq 0)$ and $w := uv$, so that $u = wv^{-1}$ on W. Here U_0 is an open subset of $\mathbb{A}^3_{u,v,z}$ containing $(z = 0) \cap (uv = 0)$, since X has been shrunk around o. Since (X, B) is klt, so is $(W, B_Y|_W)$, and the crepant pullback of $K_W + B_Y|_W$

to a log resolution of $(W, B_Y|_W)$ has all coefficients less than 1. Hence $\operatorname{discrep}(W, B_Y|_W) > -1$ by [KM98, Corollary 2.31(3)] and [KM98, Lemma 2.30], since the resolution has finitely many exceptional divisors. By Construction-Proposition 19.4,

$$B_Y|_W = (1-a_0)(w=0) + \sum_i b_i \operatorname{div}(\Phi_{i,0}), \quad \Phi_{i,0} = P_i(w,z) + Q_{i,0}.$$

Since

$$u^{n_-}v^{n_-+1} = w^{n_-}v, \qquad u^{n_++1}v^{n_+} = w^{n_++1}v^{-1}$$

and $n_-, n_+ \geq N$, there exist $G_i \in \mathbb{C}[w, v, v^{-1}, z]$ such that

$$\Phi_{i,0} = P_i(w,z) + w^N G_i(w,v,z).$$

For $v_1 \neq 0$, let $V_{v_1} := (v = v_1) \subset W$ and $y_{v_1} := (0, v_1, 0)$, which belongs to W since W contains $(z=0) \cap (u=0) \cap (v \neq 0)$. The functions w and z restrict to coordinates on V_{v_1}, and

$$B_Y|_{V_{v_1}} = (1-a_0)(w=0) + \sum_i b_i \operatorname{div}\big(P_i(w,z) + w^N G_i(w,v_1,z)\big),$$

where $P_i(w,z) + w^N G_i(w,v_1,z) \neq 0$ because $\big(P_i(w,z) + w^N G_i(w,v_1,z)\big)|_{(w=0)} = P_i(0,z) \neq 0$.

The linear system on W spanned by 1 and v is free, and a general member of the linear system is V_{v_1} for a general $v_1 \neq 0$. By [KM98, Lemma 5.17(1)], which holds for $\mathbb{R}$-divisors by Remark 2.17, there exists $v_1 \neq 0$ such that

$$\operatorname{discrep}(V_{v_1}, B_Y|_{V_{v_1}}) \geq \operatorname{discrep}(W, B_Y|_W) > -1.$$

Since discrepancies are log discrepancies minus one, $a(E, V_{v_1}, B_Y|_{V_{v_1}}) \geq 1 + \operatorname{discrep}(W, B_Y|_W) > 0$ for any prime divisor E which is exceptional over V_{v_1}. Fix such a v_1. In particular, $\operatorname{mld}(V_{v_1} \ni y_{v_1}, B_Y|_{V_{v_1}}) \geq 0$, so $(V_{v_1}, B_Y|_{V_{v_1}})$ is sub-lc near y_{v_1} by Lemma 19.7, hence lc near y_{v_1} as $B_Y|_{V_{v_1}} \geq 0$. Let

$$\Delta_0 := (1-a_0)(w=0) + \sum_i b_i \operatorname{div}(P_i(w,z))$$

on $\mathbb{A}^2_{w,z}$, and note that Δ_0 does not depend on v_1. By (1), applied to the germ $(V_{v_1} \ni y_{v_1})$ with coordinates w, z, the curve $(w=0)$ with local equation w, the $\mathbb{R}$-divisor $(1-a_0)(w=0) \geq 0$, and the functions $P_i(w,z) + w^N G_i(w,v_1,z)$ and $P_i(w,z)$, whose differences belong to (w^N), the pair $(\mathbb{A}^2_{w,z}, \Delta_0)$ is lc near 0, and

$$\mu := \operatorname{mld}(\mathbb{A}^2_{w,z} \ni 0, \Delta_0) = \operatorname{mld}(V_{v_1} \ni y_{v_1}, B_Y|_{V_{v_1}})$$

is attained by a prime divisor E over V_{v_1} with $\operatorname{center}_{V_{v_1}} E = \{y_{v_1}\}$. Since E is exceptional over V_{v_1}, we have $\mu = a(E, V_{v_1}, B_Y|_{V_{v_1}}) > 0$.

Let $g : W \to \mathbb{A}^2_{w,z} \times \mathbb{A}^1_\tau$ be the morphism $(u,v,z) \to (uv, z, v - v_0)$. Then g is an open immersion, since it is the restriction to W of the isomorphism from $(v \neq 0) \subset \mathbb{A}^3_{u,v,z}$ onto $\mathbb{A}^2_{w,z} \times (\mathbb{A}^1_\tau \backslash \{-v_0\})$ whose inverse is $(w,z,\tau) \to (w(\tau+v_0)^{-1}, \tau + v_0, z)$, and $g(y) = (0,0,0)$. Let

$$h_i := P_i(w,z) + w^N G_i(w, \tau + v_0, z),$$

and let $\operatorname{pr} : \mathbb{A}^2_{w,z} \times \mathbb{A}^1_\tau \to \mathbb{A}^2_{w,z}$ be the projection. Then $h_i \neq 0$, $h_i - P_i(w,z) \in (w^N)$, and

$$g_*(B_Y|_W) = \Big(\operatorname{pr}^*\big((1-a_0)(w=0)\big) + \sum_i b_i \operatorname{div}(h_i)\Big)\Big|_{g(W)}.$$

The hypotheses of Theorem 16.13 hold for the germ $(\mathbb{A}^2_{w,z} \ni 0)$, the curve $(w=0)$ with local equation w, the $\mathbb{R}$-divisor $(1-a_0)(w=0)$ with coefficient in $[0,1]$, $M = 0$, the functions $P_i(w,z)$, and the functions h_i: $b_i \in \Gamma$ and $b_i > 0$ since $B \in \Gamma$, $(\mathbb{A}^2_{w,z}, \Delta_0)$ is lc near 0 with $\operatorname{mld}(\mathbb{A}^2_{w,z} \ni 0, \Delta_0) = \mu \geq 0$, and $\gamma N > H_0$. Hence, by Theorem 16.13(2),

$$\operatorname{mld}(Y \ni y, B_Y) = \operatorname{mld}\Big(\mathbb{A}^2_{w,z} \times \mathbb{A}^1_\tau \ni (0,0,0),\ \operatorname{pr}^*\big((1-a_0)(w=0)\big) + \sum_i b_i \operatorname{div}(h_i)\Big) = \mu + 1 > 1.$$

(3) Let $N_0 := \lfloor \frac{H_0}{\gamma} \rfloor + 1$. Then N_0 is an integer with $N_0 > \frac{H_0}{\gamma}$, and

$$N_0 \le 1 + \frac{H_0}{\gamma} \le \min\{n_-, n_+\}.$$

Since F computes $\mathrm{mld}(X \ni o, B)$, $\mathrm{center}_X F = \{o\}$, so $\mathrm{center}_Y F \subset f^{-1}(o)$. By Construction-Proposition 19.4, $f^{-1}(o) \cap U_0 = (z = 0) \cap (uv = 0)$. Thus $\mathrm{center}_Y F \cap U_0$ is a non-empty irreducible closed subset of $(z = 0) \cap (uv = 0)$.

If $\mathrm{center}_Y F \cap U_0$ is a curve, then it is $(u = z = 0)$ or $(v = z = 0)$, so $\mathrm{center}_Y F = E_0 \times \{0\}$ or $\mathrm{center}_Y F = E_1 \times \{0\}$. In this case Proposition 19.6(2) implies $\min\{n_-, n_+\} < 1 + \frac{H_0}{\gamma}$, a contradiction. Thus $\mathrm{center}_Y F$ is a closed point $y \in U_0$ with $z = 0$ and $uv = 0$. By [KM98, Lemma 2.30],

$$\mathrm{mld}(Y \ni y, B_Y) \le a(F, Y, B_Y) = a(F, X, B) = c < 1.$$

Since $N_0 > \frac{H_0}{\gamma}$ and $n_-, n_+ \ge N_0$, (2) applied with N_0 in place of N implies that y is not a point at which exactly one of u, v vanishes. Hence $u = v = 0$ at y, that is, $\mathrm{center}_Y F$ is the closed point $u = v = z = 0$ of U_0. □

19.3. **Node centers and the depth bound.** In this subsection, we prove Theorem 19.2. By Propositions 19.6 and 19.9, it remains to treat a prime divisor whose center on U_0 is the point $u = v = z = 0$. We treat such a prime divisor in two steps: we pass first to the product with $\mathbb{A}^1_z$ of a weighted blow-up of the origin of $\mathbb{A}^2_{u,v}$, and then to an étale cover of an open subset of the product intersecting the exceptional divisor.

We first describe the crepant boundary near the exceptional divisor of such a blow-up.

Construction-Proposition 19.10. Notation and conditions as in Set-up 19.1 and Construction-Proposition 19.4. Let $N \le \min\{n_-, n_+\}$ be a positive integer, p, q coprime positive integers, a', b' integers with $pb' - qa' = 1$, $m := p + q$, $\ell := a' + b'$, and $a_{p,q} := pa_0 + qa_1$. Let $h : Y' \to U_0$ be the restriction over $U_0 \subset \mathbb{A}^3_{u,v,z}$ of the product with $\mathbb{A}^1_z$ of the weighted blow-up of $\mathbb{A}^2_{u,v}$ with weights (p, q) for (u, v), $E_{p,q}$ the exceptional divisor of h, and $Y'^{\circ} \subset Y'$ the intersection of Y' with the product with $\mathbb{A}^1_z$ of the affine toric open subset of the weighted blow-up given by the ray through (p, q), with coordinates s, ς, z such that ς is invertible on Y'°, $u = s^p \varsigma^{a'}$, and $v = s^q \varsigma^{b'}$. Let $B_{Y'}$ be the $\mathbb{R}$-divisor on Y' with $K_{Y'} + B_{Y'} = h^*(K_{U_0} + B_Y|_{U_0})$ and $h_* B_{Y'} = B_Y|_{U_0}$. Then $E_{p,q} \cap Y'^{\circ} = (s = 0)$, $a(E_{p,q}, X, B) = a_{p,q}$, $0 < a_{p,q} \le m$, and there exist regular functions R_i on Y'° such that

$$B_{Y'}|_{Y'^{\circ}} = (1 - a_{p,q})(s = 0) + \sum_i b_i \,\mathrm{div}\left(P_i(s^m \varsigma^{\ell}, z) + s^{Nm} R_i\right).$$

Proof. The affine toric open subset of the weighted blow-up given by the ray through (p, q) is $\mathrm{Spec}\, \mathbb{C}[u^{\alpha} v^{\beta} \mid p\alpha + q\beta \ge 0]$, and Y'° is its product with $\mathbb{A}^1_z$ intersected with Y'. Moreover, $\mathrm{ord}_{E_{p,q}}(u^{\alpha} v^{\beta}) = p\alpha + q\beta$ for any integers α, β (see [CLS11, Theorem 1.2.18 and Proposition 4.1.1]). Since $p \cdot b' + q \cdot (-a') = 1$, $p \cdot (-q) + q \cdot p = 0$, and the vectors $(b', -a')$ and $(-q, p)$ form a basis of $\mathbb{Z}^2$, the integer vectors (α, β) with $p\alpha + q\beta \ge 0$ are the vectors $i(b', -a') + j(-q, p)$ with $i \in \mathbb{Z}_{\ge 0}$ and $j \in \mathbb{Z}$, and then $p\alpha + q\beta = i$. Thus $s = u^{b'} v^{-a'}$, $\varsigma = u^{-q} v^p$, Y'° is an open subset of $\mathrm{Spec}\, \mathbb{C}[s, \varsigma, \varsigma^{-1}, z]$,

$$\mathrm{ord}_{E_{p,q}} s = 1, \qquad \mathrm{ord}_{E_{p,q}} \varsigma = 0,$$

and $E_{p,q} \cap Y'^{\circ} = (s = 0)$. Moreover, on Y'°,

$$uv = s^{p+q} \varsigma^{a'+b'} = s^m \varsigma^{\ell}.$$

Since $u = s^p \varsigma^{a'}$ and $v = s^q \varsigma^{b'}$, we have $\frac{du}{u} = p\frac{ds}{s} + a'\frac{d\varsigma}{\varsigma}$ and $\frac{dv}{v} = q\frac{ds}{s} + b'\frac{d\varsigma}{\varsigma}$, hence

$$\frac{du}{u} \wedge \frac{dv}{v} \wedge dz = (pb' - qa')\frac{ds}{s} \wedge \frac{d\varsigma}{\varsigma} \wedge dz = \frac{ds}{s} \wedge \frac{d\varsigma}{\varsigma} \wedge dz.$$

Since ς is invertible on Y'°, it follows that $h^*(K_{U_0}+(u=0)+(v=0))|_{Y'^{\circ}} = K_{Y'^{\circ}}+(s=0)$. On Y'° we have $h^*(u=0)=\operatorname{div}(s^p\varsigma^{a'})=p(s=0)$ and $h^*(v=0)=q(s=0)$. Hence

$$h^*\big(K_{U_0}+(1-a_0)(u=0)+(1-a_1)(v=0)\big)\big|_{Y'^{\circ}} = K_{Y'^{\circ}}+(1-a_{p,q})(s=0).$$

By Construction-Proposition 19.4,

$$B_Y|_{U_0} = (1-a_0)(u=0)+(1-a_1)(v=0)+\sum_i b_i \operatorname{div}(\Phi_{i,0})$$

with $\Phi_{i,0}=P_i(uv,z)+Q_{i,0}$. Since $N\le\min\{n_-,n_+\}$,

$$Q_{i,0}\in(u^{n_-}v^{n_-+1},u^{n_++1}v^{n_+})\subset\big((uv)^N\big),$$

so $Q_{i,0}=(uv)^NQ'_i$ for some polynomial $Q'_i\in\mathbb{C}[u,v,z]$. Let $R_i:=\varsigma^{N\ell}h^*Q'_i$, a regular function on Y'°. Then $h^*\Phi_{i,0}=P_i(s^m\varsigma^\ell,z)+s^{Nm}R_i$, and the formula for $B_{Y'}|_{Y'^{\circ}}$ follows from the last two formulas.

The restriction of $h^*\Phi_{i,0}$ to $(s=0)$ is $P_i(0,z)\ne0$, so $(s=0)$ is not a component of $\operatorname{div}(h^*\Phi_{i,0})$. Thus $\operatorname{mult}_{E_{p,q}}B_{Y'}=1-a_{p,q}$, that is, $a(E_{p,q},U_0,B_Y|_{U_0})=a_{p,q}$. Since $\operatorname{center}_Y E_{p,q}\subset U_0$, [KM98, Lemma 2.30], applied to the proper morphism f, implies $a(E_{p,q},X,B)=a(E_{p,q},Y,B_Y)=a(E_{p,q},U_0,B_Y|_{U_0})$. Finally, $0<a_0,a_1\le1$ by Construction-Proposition 19.4, hence $0<pa_0+qa_1\le p+q=m$. □

In the next lemma, we bound from below the log discrepancies of two surface pairs built from the polynomials P_i of Construction-Proposition 19.4. The coefficient $1-a_*$ of $(w=0)$ is between $1-a_0$ and $1-a_1$.

Lemma 19.11. *Let $\Gamma\subset[0,1]$ be a DCC set with $\Gamma\backslash\{0\}\ne\emptyset$, $\gamma:=\gamma(\Gamma)$, $\delta:=\delta(\Gamma)$, $H_0:=2+\frac{1}{\delta}$, and $N>\frac{H_0}{\gamma}$ an integer. In Set-up 19.1, assume that $B\in\Gamma$ and $n_-,n_+\ge N$, and let a_0, a_1, and P_i be as in Construction-Proposition 19.4. Let F be a prime divisor computing $c:=\operatorname{mld}(X\ni o,B)$ with $0<c<t\le1$ such that any other prime divisor over X with center o has log discrepancy at least t, and assume that $\operatorname{center}_{U_0}F$ is the point $u=v=z=0$. Let a_* be a real number with $\min\{a_0,a_1\}\le a_*\le\max\{a_0,a_1\}$, m a positive integer, and $\lambda\in\mathbb{C}^*$, and let Δ on $\mathbb{A}^2_{w,z}$ and $\Delta_{m,\lambda}$ on $\mathbb{A}^2_{s,z}$ be*

$$\Delta:=(1-a_*)(w=0)+\sum_i b_i\operatorname{div}(P_i(w,z)),$$

$$\Delta_{m,\lambda}:=(1-ma_*)(s=0)+\sum_i b_i\operatorname{div}(P_i(\lambda s^m,z)).$$

Then:

(1) $(\mathbb{A}^2_{w,z},\Delta)$ is lc near 0, and $a(E,\mathbb{A}^2_{w,z},\Delta)\ge t$ for any prime divisor E over $\mathbb{A}^2_{w,z}$ with center 0;

(2) $(\mathbb{A}^2_{s,z},\Delta_{m,\lambda})$ is sub-lc near 0, and $\operatorname{mld}(\mathbb{A}^2_{s,z}\ni0,\Delta_{m,\lambda})\ge t$.

Proof. (1) Let $\Delta_j:=(1-a_j)(w=0)+\sum_i b_i\operatorname{div}(P_i(w,z))$ for $j=0,1$. We first show that $(\mathbb{A}^2_{w,z},\Delta_0)$ is lc near 0 and $\operatorname{mld}(\mathbb{A}^2_{w,z}\ni0,\Delta_0)\ge t$.

Let η_0 be the generic point of $E_0\times\{0\}$. Since $\operatorname{center}_{U_0}F$ is the closed point $u=v=z=0$, $\operatorname{center}_Y F$ is the point $u=v=z=0$. Any prime divisor E over Y with $\operatorname{center}_Y E=E_0\times\{0\}$ satisfies $\operatorname{center}_X E=\{o\}$ and $E\ne F$, hence $a(E,Y,B_Y)=a(E,X,B)\ge t$. Thus $\operatorname{mld}(Y\ni\eta_0,B_Y)\ge t$.

The curve $E_0\times\{0\}$ intersects U_0 in the curve $(u=z=0)$, and the restriction of v to $(u=z=0)$ is not constant. Let Ω be an algebraic closure of $\mathbb{C}(v)$. By Lemma 3.6(2), applied with (X,B) and C there equal to $(U_0,B_Y|_{U_0})$ and $(u=z=0)$, and with the rational function of that lemma equal to v,

$$\operatorname{mld}\Big(\mathbb{A}^2_{u,z,\Omega}\ni0,(1-a_0)(u=0)+\sum_i b_i\operatorname{div}\big(P_i(uv,z)+Q_{i,0}(u,v,z)\big)\Big)=\operatorname{mld}(Y\ni\eta_0,B_Y)\ge t,$$

where v is regarded as an element of Ω^*. Let $w := uv$, a coordinate on $\mathbb{A}^2_{u,z,\Omega}$ in place of u. Since $N \le \min\{n_-, n_+\}$, we have $Q_{i,0} \in (u^{n_-}v^{n_-+1}, u^{n_++1}v^{n_+}) \subset ((uv)^N)$, so $Q_{i,0} = w^N G_i$ for some $G_i \in \Omega[w,z]$. Hence

$$\operatorname{mld}\Big(\mathbb{A}^2_{w,z,\Omega} \ni 0, (1-a_0)(w=0) + \sum_i b_i \operatorname{div}\big(P_i(w,z) + w^N G_i\big)\Big) \ge t.$$

Since Ω is an algebraically closed field of characteristic zero with the cardinality of $\mathbb{C}$, it is isomorphic to $\mathbb{C}$ as an abstract field by Steinitz's theorem (both fields are algebraic closures of purely transcendental extensions of $\mathbb{Q}$ whose transcendence bases have the cardinality of $\mathbb{C}$, see [Lan02, Theorem VIII.1.1 and Corollary V.2.9], and [Lan02, V, §2] for the cardinality of algebraic extensions), so Lemma 19.7 and Proposition 19.9(1) hold over Ω. By Lemma 19.7, the last pair is sub-lc near 0, and it is lc near 0 since $a_0 \le 1$ and $b_i > 0$. The functions $P_i(w,z)$ and $P_i(w,z) + w^N G_i$ are non-zero, because their restrictions to $(w=0)$ are $P_i(0,z) \ne 0$. By Proposition 19.9(1) over Ω, applied with $(V \ni p)$, D, d, C, f_i, and h_i there equal to $(\mathbb{A}^2_{w,z,\Omega} \ni 0)$, $(w=0)$, w, $(1-a_0)(w=0)$, $P_i(w,z)$, and $P_i(w,z) + w^N G_i$, the base change $(\mathbb{A}^2_{w,z,\Omega}, \Delta_{0,\Omega})$ of $(\mathbb{A}^2_{w,z}, \Delta_0)$ to Ω is lc near 0, and $\operatorname{mld}(\mathbb{A}^2_{w,z,\Omega} \ni 0, \Delta_{0,\Omega}) \ge t$. The pair $(\mathbb{A}^2_{w,z}, \Delta_0)$ is defined over $\mathbb{C}$. A log resolution of $(\mathbb{A}^2_{w,z}, \Delta_0)$ over $\mathbb{C}$ on which the maximal ideal of 0 becomes invertible [KM98, Theorem 0.2] remains such a log resolution after base change to Ω, and base change to Ω preserves the log discrepancies of the exceptional curves. Hence $(\mathbb{A}^2_{w,z}, \Delta_0)$ is lc near 0 if and only if $(\mathbb{A}^2_{w,z,\Omega}, \Delta_{0,\Omega})$ is lc near 0 (see [KM98, Corollaries 2.31(3) and 2.32] and Remark 2.17), and

$$\operatorname{mld}(\mathbb{A}^2_{w,z} \ni 0, \Delta_0) = \operatorname{mld}(\mathbb{A}^2_{w,z,\Omega} \ni 0, \Delta_{0,\Omega})$$

by Lemma 3.5, applied to the log resolution above and to its base change to Ω. Thus $(\mathbb{A}^2_{w,z}, \Delta_0)$ is lc near 0 and $\operatorname{mld}(\mathbb{A}^2_{w,z} \ni 0, \Delta_0) \ge t$.

The curve $E_1 \times \{0\}$ intersects U_0 in $(v = z = 0)$, and the inclusion $Q_{i,0} \in ((uv)^N)$ and the polynomials $P_i(uv,z)$ are symmetric in u and v. Thus, replacing $E_0 \times \{0\}$, η_0, $(u = z = 0)$, $\mathbb{A}^2_{u,z,\Omega}$, $(u=0)$, $\mathbb{C}(v)$, v, a_0, Δ_0, and $\Delta_{0,\Omega}$ with $E_1 \times \{0\}$, the generic point η_1 of $E_1 \times \{0\}$, $(v = z = 0)$, $\mathbb{A}^2_{v,z,\Omega}$, $(v=0)$, $\mathbb{C}(u)$, u, a_1, Δ_1, and $\Delta_{1,\Omega}$, the argument of the last three paragraphs shows that $(\mathbb{A}^2_{w,z}, \Delta_1)$ is lc near 0 and $\operatorname{mld}(\mathbb{A}^2_{w,z} \ni 0, \Delta_1) \ge t$.

Let E be a prime divisor over $\mathbb{A}^2_{w,z}$ with center 0. Then

$$a(E, \mathbb{A}^2_{w,z}, \Delta) = a(E, \mathbb{A}^2_{w,z}, 0) - (1-a_*)\operatorname{ord}_E w - \sum_i b_i \operatorname{ord}_E P_i(w,z)$$

is an affine function of a_*, and it is at least t for $a_* = a_0$ and for $a_* = a_1$. Hence $a(E, \mathbb{A}^2_{w,z}, \Delta) \ge t$ for any a_* between a_0 and a_1. By Lemma 19.7, $(\mathbb{A}^2_{w,z}, \Delta)$ is sub-lc near 0. Since $a_* \le \max\{a_0, a_1\} \le 1$, we have $\Delta \ge 0$, so $(\mathbb{A}^2_{w,z}, \Delta)$ is lc near 0.

(2) Let ζ be a primitive m-th root of unity, and let μ_m act on $\mathbb{A}^2_{s,z}$ by $(s,z) \to (\zeta s, z)$. The morphism $(s,z) \to (w,z) = (\lambda s^m, z)$ is the quotient morphism $\mathbb{A}^2_{s,z} \to \mathbb{A}^2_{w,z}$, which is étale over the complement of $(w=0)$ and has ramification index m along $(s=0)$. The pullback of $\operatorname{div}(P_i(w,z))$ is $\operatorname{div}(P_i(\lambda s^m, z))$, which does not contain $(s=0)$ since $P_i(0,z) \ne 0$. Since

$$1 - \frac{1-(1-ma_*)}{m} = 1 - a_*,$$

Lemma 2.6(4) implies that $(\mathbb{A}^2_{s,z}, \Delta_{m,\lambda})/\mu_m = (\mathbb{A}^2_{w,z}, \Delta)$. Let E' be a prime divisor over $\mathbb{A}^2_{s,z}$ with center 0, and E the prime divisor over $\mathbb{A}^2_{w,z}$ with $\operatorname{ord}_{E'}|_{\mathbb{C}(w,z)} = e \operatorname{ord}_E$ for a positive integer e. By Lemma 2.6(2), $\operatorname{center}_{\mathbb{A}^2_{w,z}} E = \{0\}$ and

$$a(E', \mathbb{A}^2_{s,z}, \Delta_{m,\lambda}) = |I_{\mu_m}(E')| \cdot a(E, \mathbb{A}^2_{w,z}, \Delta) \ge t$$

by (1). Thus $\operatorname{mld}(\mathbb{A}^2_{s,z} \ni 0, \Delta_{m,\lambda}) \ge t$, and $(\mathbb{A}^2_{s,z}, \Delta_{m,\lambda})$ is sub-lc near 0 by Lemma 19.7. □

The next proposition locates the center of F on the blow-up Y' of Construction-Proposition 19.10 with weights determined by $(\operatorname{ord}_F u, \operatorname{ord}_F v)$, and shows that $a_{p,q} > 1$.

Proposition 19.12. *Let $\Gamma \subset [0,1]$ be a DCC set with $\Gamma\backslash\{0\} \neq \emptyset$, $\gamma := \gamma(\Gamma)$, $\delta := \delta(\Gamma)$, $H_0 := 2 + \frac{1}{\delta}$, and $N > \frac{H_0}{\gamma}$ an integer. In Set-up 19.1, assume that $B \in \Gamma$ and $n_-, n_+ \geq N$, and let F be a prime divisor computing $c := \operatorname{mld}(X \ni o, B)$ with $0 < c < t \leq 1$ such that any other prime divisor over X with center o has log discrepancy at least t. Assume that $\operatorname{center}_{U_0} F$ is the point $u = v = z = 0$. Let $g := \gcd(\operatorname{ord}_F u, \operatorname{ord}_F v)$, $p := \operatorname{ord}_F u/g$, and $q := \operatorname{ord}_F v/g$, choose integers a', b' with $pb' - qa' = 1$, and let Y', Y'°, s, ς, and $a_{p,q}$ be as in Construction-Proposition 19.10 for N, p, q, a', and b'. Then $\operatorname{center}_{Y'} F$ is the closed point of Y'° with $s = z = 0$ and $\varsigma = \lambda$ for some $\lambda \in \mathbb{C}^*$, and $a_{p,q} > 1$.*

Proof. The proof is long, so we divide it into several steps. Let $m := p + q$, $\ell := a' + b'$, and R_i be as in Construction-Proposition 19.10, so that

$$B_{Y'}|_{Y'^{\circ}} = (1 - a_{p,q})(s = 0) + \sum_i b_i \operatorname{div}\big(P_i(s^m \varsigma^\ell, z) + s^{Nm} R_i\big). \tag{19.14}$$

Since $\operatorname{center}_{U_0} F$ is the point $u = v = z = 0$, F is a prime divisor over U_0 and over Y'. Since $K_{Y'} + B_{Y'}$ is the pullback of $K_X + B$, [KM98, Lemma 2.30], applied to h and to f, implies that $a(F, Y', B_{Y'}) = a(F, X, B) = c$. Let $a_* := \frac{a_{p,q}}{m} = \frac{pa_0 + qa_1}{p+q}$, so that $\min\{a_0, a_1\} \leq a_* \leq \max\{a_0, a_1\}$ and $1 - ma_* = 1 - a_{p,q}$. Let $M := \max\{a_{p,q} - 1, 0\}$. Let $H_M := 2 + \frac{1+M}{\delta}$, which is the number H_M of Lemma 19.8. Then $1 - a_{p,q} \in [-M, 1]$ and $1 + M = \max\{a_{p,q}, 1\} \leq m$ by Construction-Proposition 19.10. Since $m \geq 1$ and $N > \frac{H_0}{\gamma}$,

$$H_M + M = 1 + (1 + M)\Big(1 + \frac{1}{\delta}\Big) \leq 1 + m\Big(1 + \frac{1}{\delta}\Big) \leq m\Big(2 + \frac{1}{\delta}\Big) = mH_0 < \gamma N m. \tag{19.15}$$

Step 1. In this step we show that $\operatorname{center}_{Y'} F$ is either the closure of the curve $(s = z = 0)$ of Y'° or a closed point of Y'° with $s = z = 0$ and $\varsigma = \lambda$ for some $\lambda \in \mathbb{C}^*$.

Since $\operatorname{center}_{U_0} F$ is the point $u = v = z = 0$, we have $\operatorname{ord}_F z > 0$, $\operatorname{ord}_F u = gp$, and $\operatorname{ord}_F v = gq$. Since $s = u^{b'} v^{-a'}$ and $\varsigma = u^{-q} v^p$,

$$\operatorname{ord}_F s = g(pb' - qa') = g > 0, \qquad \operatorname{ord}_F \varsigma = g(-qp + pq) = 0.$$

Thus ord_F is non-negative on $\mathbb{C}[s, \varsigma, \varsigma^{-1}, z]$, so $\operatorname{center}_{Y'} F$ intersects Y'°, since the center of F on $\operatorname{Spec} \mathbb{C}[s, \varsigma, \varsigma^{-1}, z]$ maps to $\operatorname{center}_{U_0} F$ in U_0 and Y' is separated (cf. [Har77, Exercise II.4.5]). Since $\operatorname{ord}_F s > 0$ and $\operatorname{ord}_F z > 0$, $\operatorname{center}_{Y'} F \cap Y'^{\circ}$ is contained in $(s = z = 0) \cong \mathbb{A}^1_\varsigma \backslash \{0\}$. This proves the claim of **Step 1**.

Step 2. In this step we show that $\operatorname{center}_{Y'} F$ is not the closure of the curve $(s = z = 0)$.

Suppose that $\operatorname{center}_{Y'} F$ is the closure of $(s = z = 0)$, and let η' be the generic point of $(s = z = 0)$. Then

$$c = a(F, Y', B_{Y'}) \geq \operatorname{mld}(Y'^{\circ} \ni \eta', B_{Y'}|_{Y'^{\circ}}).$$

Let Ω be an algebraic closure of $\mathbb{C}(\varsigma)$. The generic fiber of ς on Y'° is an open subset of $\mathbb{A}^2_{s,z}$ over $\mathbb{C}(\varsigma)$ containing the origin. Regard ς as an element of Ω^*, and let

$$f_i := P_i(\varsigma^\ell s^m, z), \qquad h_i := P_i(\varsigma^\ell s^m, z) + s^{Nm} R_i(s, \varsigma, z),$$

elements of $\Omega[s, z]$, which are non-zero since their restrictions to $(s = 0)$ are $P_i(0, z) \neq 0$. By Lemma 3.6(2), applied with (X, B) and C there equal to $(Y'^{\circ}, B_{Y'}|_{Y'^{\circ}})$ and $(s = z = 0)$, and with the rational function of that lemma equal to ς, and by (19.14),

$$\operatorname{mld}(Y'^{\circ} \ni \eta', B_{Y'}|_{Y'^{\circ}}) = \operatorname{mld}\Big(\mathbb{A}^2_{s,z,\Omega} \ni 0, (1 - a_{p,q})(s = 0) + \sum_i b_i \operatorname{div}(h_i)\Big).$$

Since Ω is isomorphic to $\mathbb{C}$ as an abstract field by Steinitz's theorem [Lan02, Theorem VIII.1.1 and Corollary V.2.9], Lemmas 19.7 and 19.8 hold over Ω, and Lemma 2.6 holds over Ω by Remark 2.17. Let $\Delta := (1 - a_*)(w = 0) + \sum_i b_i \operatorname{div}(P_i(w, z))$. By Lemma 19.11(1), $(\mathbb{A}^2_{w,z}, \Delta)$ is

lc near 0 and $\mathrm{mld}(\mathbb{A}^2_{w,z} \ni 0, \Delta) \geq t$. The pair $(\mathbb{A}^2_{w,z}, \Delta)$ is defined over $\mathbb{C}$, and a log resolution of it over $\mathbb{C}$ on which the maximal ideal of 0 becomes invertible [KM98, Theorem 0.2] remains such a log resolution after base change to Ω, and base change to Ω preserves the log discrepancies of the exceptional curves. Thus the base change $(\mathbb{A}^2_{w,z,\Omega}, \Delta_\Omega)$ of $(\mathbb{A}^2_{w,z}, \Delta)$ is lc near 0 (see [KM98, Corollaries 2.31(3) and 2.32] and Remark 2.17), and $\mathrm{mld}(\mathbb{A}^2_{w,z,\Omega} \ni 0, \Delta_\Omega) = \mathrm{mld}(\mathbb{A}^2_{w,z} \ni 0, \Delta) \geq t$ by Lemma 3.5, applied to the log resolution above and to its base change to Ω.

Let ζ be a primitive m-th root of unity, and let μ_m act on $\mathbb{A}^2_{s,z,\Omega}$ by $(s,z) \to (\zeta s, z)$, with quotient morphism $(s,z) \to (w,z) = (\varsigma^\ell s^m, z)$. The quotient morphism has ramification index m along $(s=0)$ and is étale over the complement of $(w=0)$, and the pullback of $\mathrm{div}(P_i(w,z))$ is $\mathrm{div}(f_i)$, which does not contain $(s=0)$. Let $\Delta^f := (1-a_{p,q})(s=0) + \sum_i b_i \,\mathrm{div}(f_i)$. Since

$$1 - \frac{1-(1-a_{p,q})}{m} = 1 - a_*,$$

Lemma 2.6(4) implies that $(\mathbb{A}^2_{s,z,\Omega}, \Delta^f)/\mu_m = (\mathbb{A}^2_{w,z,\Omega}, \Delta_\Omega)$. By Lemma 2.6(2), any prime divisor E' over $\mathbb{A}^2_{s,z,\Omega}$ with center 0 satisfies

$$a(E', \mathbb{A}^2_{s,z,\Omega}, \Delta^f) = |I_{\mu_m}(E')| \cdot a(E, \mathbb{A}^2_{w,z,\Omega}, \Delta_\Omega) \geq t,$$

where E is the prime divisor over $\mathbb{A}^2_{w,z,\Omega}$ induced by E', with center 0. Thus

$$\mathrm{mld}(\mathbb{A}^2_{s,z,\Omega} \ni 0, \Delta^f) \geq t,$$

and $(\mathbb{A}^2_{s,z,\Omega}, \Delta^f)$ is sub-lc near 0 by Lemma 19.7.

Since $h_i - f_i \in (s^{Nm})$ and $\gamma Nm > H_M + M$ by (19.15), Lemma 19.8 over Ω, applied with $(V \ni p)$, D, d, C, M, N, f_i, and h_i there equal to $(\mathbb{A}^2_{s,z,\Omega} \ni 0)$, $(s=0)$, s, $(1-a_{p,q})(s=0)$, M, Nm, f_i, and h_i (recall that $1 - a_{p,q} \in [-M, 1]$), implies that

$$\mathrm{mld}\Big(\mathbb{A}^2_{s,z,\Omega} \ni 0, (1-a_{p,q})(s=0) + \sum_i b_i \,\mathrm{div}(h_i)\Big) = \mathrm{mld}(\mathbb{A}^2_{s,z,\Omega} \ni 0, \Delta^f) \geq t.$$

Hence $c \geq t$, a contradiction.

Step 3. In this step we show that $a_{p,q} > 1$.

By **Steps 1 and 2**, $\mathrm{center}_{Y'} F$ is the closed point y' of Y'° with $s = z = 0$ and $\varsigma = \lambda$ for some $\lambda \in \mathbb{C}^*$. Suppose that $a_{p,q} \leq 1$. Then $M = 0$, and $B_{Y'}|_{Y'^\circ} \geq 0$ by (19.14). Let $\Sigma_\lambda := (\varsigma = \lambda) \subset Y'^\circ$, a smooth surface with coordinates s, z. By (19.14),

$$B_{Y'}|_{\Sigma_\lambda} = (1-a_{p,q})(s=0) + \sum_i b_i \,\mathrm{div}(h_{i,\lambda}), \qquad h_{i,\lambda} := P_i(\lambda^\ell s^m, z) + s^{Nm} R_i(s, \lambda, z),$$

and $h_{i,\lambda} \neq 0$ since the restriction of $h_{i,\lambda}$ to $(s=0)$ is $P_i(0,z)$. By Lemma 19.11(2), applied with a_*, m, and the constant $\lambda^\ell \in \mathbb{C}^*$,

$$\Delta_{m,\lambda^\ell} = (1-a_{p,q})(s=0) + \sum_i b_i \,\mathrm{div}(P_i(\lambda^\ell s^m, z))$$

is sub-lc near 0 with $\mathrm{mld}(\mathbb{A}^2_{s,z} \ni 0, \Delta_{m,\lambda^\ell}) \geq t$. Since $h_{i,\lambda} - P_i(\lambda^\ell s^m, z) \in (s^{Nm})$ and $\gamma Nm > H_0$ by (19.15), Lemma 19.8, applied with $(V \ni p)$, D, d, C, M, N, f_i, and h_i there equal to $(\Sigma_\lambda \ni y')$, $(s=0)$, s, $(1-a_{p,q})(s=0)$, 0, Nm, $P_i(\lambda^\ell s^m, z)$, and $h_{i,\lambda}$, implies that

$$\mathrm{mld}(\Sigma_\lambda \ni y', B_{Y'}|_{\Sigma_\lambda}) = \mathrm{mld}(\mathbb{A}^2_{s,z} \ni 0, \Delta_{m,\lambda^\ell}) \geq t.$$

The variety Y'° is an open subset of $\mathbb{A}^3_{s,\varsigma,z}$, $B_{Y'}|_{Y'^\circ} \geq 0$ near y', and Σ_λ is not a component of $B_{Y'}$. Only a neighborhood of y' in $\mathbb{A}^3_{s,\varsigma,z}$ enters [NS26b, Theorem 7.1] below, since the minimal log discrepancy at y' only depends on such a neighborhood. Since $\varsigma(y') = \lambda \neq 0$, near y' we have $P_i(s^m \varsigma^\ell, z) + s^{Nm} R_i = \epsilon_i \phi_i$ for a unit ϵ_i and a polynomial $\phi_i \in \mathbb{C}[s, \varsigma, z]$. Thus, on an open neighborhood of y', $B_{Y'}|_{Y'^\circ}$ is the divisor of an $\mathbb{R}$-ideal on $\mathbb{A}^3_{s,\varsigma,z}$, with exponent $1 - a_{p,q} \geq 0$ on s and exponent b_i on ϕ_i, and the restriction of this $\mathbb{R}$-ideal to $\Sigma_\lambda = (\varsigma - \lambda = 0)$ is non-zero since

Σ_λ is not a component of $B_{Y'}$. By precise inversion of adjunction [NS26b, Theorem 7.1], applied with the trivial group, with Z there empty, c and d there equal to 1, and Y there equal to Σ_λ,

$$\mathrm{mld}(Y'^{\circ} \ni y', B_{Y'}|_{Y'^{\circ}} + \Sigma_\lambda) = \mathrm{mld}(\Sigma_\lambda \ni y', B_{Y'}|_{\Sigma_\lambda}) \geq t.$$

Any prime divisor J over Y'° with center y' satisfies $\mathrm{ord}_J(\varsigma - \lambda) \geq 1$, hence

$$a(J, Y', B_{Y'}) = a(J, Y'^{\circ}, B_{Y'}|_{Y'^{\circ}} + \Sigma_\lambda) + \mathrm{ord}_J(\varsigma - \lambda) \geq t + 1 > 1.$$

Taking $J = F$, we obtain $c \geq t + 1 > 1$, contrary to $c < t \leq 1$. Thus $a_{p,q} > 1$. □

Proof of Theorem 19.2. The proof is long, so we divide it into several steps. Suppose that $\min\{n_-, n_+\} \geq 1 + \frac{H_0}{\gamma}$, and let $n_0 := \lfloor \frac{H_0}{\gamma} \rfloor + 1$. Then $n_0 > \frac{H_0}{\gamma}$ and $n_0 \leq 1 + \frac{H_0}{\gamma} \leq \min\{n_-, n_+\}$.

Step 1. In this step we show that $\mathrm{center}_{U_0} F$ is the point $u = v = z = 0$.

By Proposition 19.9(3), whose hypotheses hold since $B \in \Gamma$, $\min\{n_-, n_+\} \geq 1 + \frac{H_0}{\gamma}$, and $\mathrm{center}_Y F \cap U_0 \neq \emptyset$, $\mathrm{center}_Y F$ is the closed point $u = v = z = 0$ of U_0.

Step 2. In this step we pass to an étale cover near the center of F on the blow-up Y' of Construction-Proposition 19.10.

Let $g := \gcd(\mathrm{ord}_F u, \mathrm{ord}_F v)$, $p := \mathrm{ord}_F u/g$, and $q := \mathrm{ord}_F v/g$, choose integers a', b' with $pb' - qa' = 1$, let $m := p + q$ and $\ell := a' + b'$, and let Y', Y'°, s, ς, $a_{p,q}$, $B_{Y'}$, and R_i be as in Construction-Proposition 19.10 for $N := n_0$, p, q, a', and b'. By Proposition 19.12, applied with $N := n_0$, $\mathrm{center}_{Y'} F$ is the closed point y' of Y'° with $s = z = 0$ and $\varsigma = \lambda$ for some $\lambda \in \mathbb{C}^*$, and $a_{p,q} > 1$. By Construction-Proposition 19.10,

$$B_{Y'}|_{Y'^{\circ}} = (1 - a_{p,q})(s = 0) + \sum_i b_i \,\mathrm{div}\left(P_i(s^m \varsigma^\ell, z) + s^{n_0 m} R_i\right), \qquad a_{p,q} \leq m.$$

Since $\mathrm{center}_Y F \subset U_0$, [KM98, Lemma 2.30], applied to f and to the proper morphism h, implies

$$a(F, X, B) = a(F, Y, B_Y) = a(F, U_0, B_Y|_{U_0}) = a(F, Y', B_{Y'}),$$

and $a(F, Y'^{\circ}, B_{Y'}|_{Y'^{\circ}}) = a(F, Y', B_{Y'})$ since $\mathrm{center}_{Y'} F = \{y'\} \subset Y'^{\circ}$. Thus

$$a(F, Y'^{\circ}, B_{Y'}|_{Y'^{\circ}}) = c.$$

Let π be the morphism

$$\pi : \mathbb{A}^1_s \times (\mathbb{A}^1_\xi \backslash \{0\}) \times \mathbb{A}^1_z \to \mathbb{A}^1_s \times (\mathbb{A}^1_\varsigma \backslash \{0\}) \times \mathbb{A}^1_z$$

given by $\varsigma = \xi^m$, and let $Y'' := \pi^{-1}(Y'^{\circ})$. Let ζ be a primitive m-th root of unity. Then $\pi|_{Y''} : Y'' \to Y'^{\circ}$ is the quotient morphism of the action of μ_m on Y'' by $(s, \xi, z) \to (s, \zeta\xi, z)$, which is free since $\xi \neq 0$ on Y''. In particular, π is finite and étale. Let $B_{Y''} := \pi^*(B_{Y'}|_{Y'^{\circ}})$. Then $K_{Y''} + B_{Y''} = \pi^*(K_{Y'^{\circ}} + B_{Y'}|_{Y'^{\circ}})$, and Lemma 2.6(4), with all ramification indices equal to 1, implies $(Y'', B_{Y''})/\mu_m = (Y'^{\circ}, B_{Y'}|_{Y'^{\circ}})$.

By Lemma 2.6(3), applied with X, G, and E there equal to Y'', μ_m, and F, there exist a prime divisor F' over Y'' and a positive integer e such that $\mathrm{ord}_{F'}|_{\mathbb{C}(Y'^{\circ})} = e\,\mathrm{ord}_F$. By Lemma 2.6(2), $\pi(\mathrm{center}_{Y''} F') = \mathrm{center}_{Y'^{\circ}} F = \{y'\}$. Since π is finite, $\mathrm{center}_{Y''} F'$ is a closed point y'' with $\pi(y'') = y'$. Any element of $I_{\mu_m}(F')$ maps F' to itself, hence fixes y''. Since μ_m acts freely on Y'', $I_{\mu_m}(F')$ is trivial. By Lemma 2.6(1)(2), $e = 1$ and

$$a(F', Y'', B_{Y''}) = a(F, Y'^{\circ}, B_{Y'}|_{Y'^{\circ}}) = c.$$

Let ξ_0 be the ξ-coordinate of y''. Then $\xi_0^m = \lambda$ and $y'' = (0, \xi_0, 0)$ in the coordinates (s, ξ, z).

Let $\tilde{s} := s\xi^\ell$. The morphism

$$Y'' \to \mathbb{A}^3_{\tilde{s}, z, \xi - \xi_0}, \qquad (s, \xi, z) \to (\tilde{s}, z, \xi - \xi_0) = (s\xi^\ell, z, \xi - \xi_0),$$

is an isomorphism onto an open subset of $\mathbb{A}^3_{\tilde{s}, z, \xi - \xi_0}$ that contains the origin and is contained in $\{\xi - \xi_0 \neq -\xi_0\}$, with inverse given by $\xi = (\xi - \xi_0) + \xi_0$ and $s = \tilde{s}\xi^{-\ell}$, and it maps y'' to the origin. On Y'',

$$s^m \xi^{m\ell} = \tilde{s}^m, \qquad (s = 0) = (\tilde{s} = 0).$$

Let $\tilde{R}_i := \xi^{-n_0 m\ell} R_i(s, \xi^m, z)$, a regular function on Y''. Then $s^{n_0 m} R_i(s, \xi^m, z) = \tilde{s}^{n_0 m}\tilde{R}_i$, hence

$$B_{Y''} = (1 - a_{p,q})(\tilde{s} = 0) + \sum_i b_i \operatorname{div}\big(P_i(\tilde{s}^m, z) + \tilde{s}^{n_0 m}\tilde{R}_i\big). \tag{19.16}$$

Step 3. In this step we bound $a(F', Y'', B_{Y''})$ from below and conclude.

Recall that y'' is the center of F' on Y'' and $a(F', Y'', B_{Y''}) = c$. Let $a_* := \frac{a_{p,q}}{m} = \frac{pa_0+qa_1}{p+q}$. Then $\min\{a_0, a_1\} \le a_* \le \max\{a_0, a_1\}$ and $1 - ma_* = 1 - a_{p,q}$. By Lemma 19.11(2), applied with a_*, m, the constant 1, and the coordinate $\tilde{s}$ in place of s, the pair

$$(\mathbb{A}^2_{\tilde{s},z}, \Delta_{m,1}), \qquad \Delta_{m,1} = (1 - a_{p,q})(\tilde{s} = 0) + \sum_i b_i \operatorname{div}(P_i(\tilde{s}^m, z)),$$

is sub-lc near 0 and $\operatorname{mld}(\mathbb{A}^2_{\tilde{s},z} \ni 0, \Delta_{m,1}) \ge t$. Let $M := a_{p,q} - 1 > 0$ and $H_M := 2 + \frac{1+M}{\delta}$, which is the number H_M of Theorem 16.13. Since $a_{p,q} \le m$ and $n_0 > \frac{H_0}{\gamma}$,

$$H_M + M = 1 + a_{p,q}\Big(1 + \frac{1}{\delta}\Big) \le 1 + m\Big(1 + \frac{1}{\delta}\Big) \le mH_0 < \gamma n_0 m.$$

The functions $P_i(\tilde{s}^m, z)$ and $P_i(\tilde{s}^m, z) + \tilde{s}^{n_0 m}\tilde{R}_i$ are non-zero, regular near the origin, and differ by an element of $(\tilde{s}^{n_0 m})$. By Theorem 16.13(2), applied with $(S \ni p)$, D, d, C, and M there equal to $(\mathbb{A}^2_{\tilde{s},z} \ni 0)$, $(\tilde{s} = 0)$, $\tilde{s}$, $(1 - a_{p,q})(\tilde{s} = 0)$, and $a_{p,q} - 1$, with the coordinate of the factor $\mathbb{A}^1$ of that theorem equal to $\xi - \xi_0$, with N, f_l, and h_l there equal to $n_0 m$, $P_i(\tilde{s}^m, z)$, and $P_i(\tilde{s}^m, z) + \tilde{s}^{n_0 m}\tilde{R}_i$, any prime divisor over $\mathbb{A}^3_{\tilde{s},z,\xi-\xi_0}$ with center the origin has log discrepancy at least

$$\operatorname{mld}(\mathbb{A}^2_{\tilde{s},z} \ni 0, \Delta_{m,1}) + 1 \ge t + 1$$

with respect to $(\mathbb{A}^3_{\tilde{s},z,\xi-\xi_0}, B_{\mathbb{A}^3})$, where, near the origin,

$$B_{\mathbb{A}^3} := (1 - a_{p,q})(\tilde{s} = 0) + \sum_i b_i \operatorname{div}\big(P_i(\tilde{s}^m, z) + \tilde{s}^{n_0 m}\tilde{R}_i\big).$$

By (19.16), the restriction of $(\mathbb{A}^3_{\tilde{s},z,\xi-\xi_0}, B_{\mathbb{A}^3})$ to the image of Y'' is $(Y'', B_{Y''})$. Thus

$$c = a(F', Y'', B_{Y''}) \ge t + 1 > 1 \ge t > c,$$

a contradiction. Hence $\min\{n_-, n_+\} < 1 + \frac{H_0}{\gamma}$. □

20. Proofs of the main theorems

In this section, we prove Theorems 1.2, 1.3, and 1.5 and Corollary 1.4. We first prove Proposition 20.1, which concerns $\mathbb{Q}$-factorial klt pairs.

Proposition 20.1. *Let $\Lambda \subset [0, 1]$ be a DCC set. Then $\operatorname{MLD}_{\mathrm{Qf}}(3, \Lambda) \cap [1, +\infty)$ satisfies the ACC.*

Proof. Suppose that the proposition does not hold. Then there exist $\mathbb{Q}$-factorial klt threefold pairs (X_i, B_i) and closed points $x_i \in X_i$, $i \ge 1$, such that $B_i \in \Lambda$ and $a_i := \operatorname{mld}(X_i \ni x_i, B_i) \ge 1$ is strictly increasing (Definition 2.3). By Lemma 6.5, applied with $\Gamma := \Lambda$, after shrinking X_i to an affine neighborhood of x_i and passing to a subsequence, the conditions of Set-up 6.4 hold for $\Gamma := \Lambda$, with $B_i = \sum_{j=1}^r b_{ij} D_{ij}$. Since $D_{i1}, \dots, D_{ir}$ are distinct prime divisors and $B_i \in \Lambda$, we have $b_{ij} \in \Lambda$. Moreover, $b_{ij} \ge \frac{b_j}{2} > 0$ for any i and j, where $b_j = \lim_i b_{ij} \in (0, 1]$ (Set-up 6.4(2)).

We apply Construction-Theorem 6.7 and use the notation there. Namely, $\pi_i : X'_i \to X_i$ is the finite quasi-étale abelian Galois morphism with group G_i and $\pi_i^{-1}(x_i) = \{x'_i\}$, q_{ij} is a regular G_i-semi-invariant equation of $\pi_i^* D_{ij}$, $I_i = I_{G_i}(F_i)$ is the inertia group of a prime divisor F_i over X'_i lying over the prime divisor E_i over X_i computing a_i (Set-up 6.4(4)), $g_i : Y_i \to X'_i$ is the I_i-equivariant projective birational morphism of Construction-Theorem 6.7(2), $p_{ij} = g_i^* q_{ij}$, $C_i = \sum_j b_{ij} \operatorname{div}(p_{ij})$, $y_i = \operatorname{center}_{Y_i} F_i$, $(Z_i, \Delta_i) = (Y_i, C_i)/I_i$, and z_i is the image of y_i. After

shrinking X_i near x_i and deleting finitely many terms, the following hold. The group $I_i \subset G_i$ is cyclic and fixes the closed point $y_i \in Y_i$, the pair (Y_i, C_i) is klt, Y_i is terminal, K_{Y_i} is Cartier, and

$$a_i = \mathrm{mld}(Z_i \ni z_i, \Delta_i) = \mathrm{mld}((Y_i \ni y_i, C_i)/I_i).$$

Since I_i acts faithfully on X_i' and g_i is I_i-equivariant and birational, I_i acts faithfully on Y_i. Since q_{ij} is a non-zero regular G_i-semi-invariant function, p_{ij} is a non-zero regular I_i-semi-invariant function on Y_i. Thus

$$C_i = \sum_{j=1}^{r} b_{ij} \operatorname{div}(p_{ij}), \quad \text{where } b_{ij} \in \Lambda,\ b_{ij} > 0, \tag{20.1}$$

and p_{ij} is a non-zero regular I_i-semi-invariant function on Y_i.

By Corollary 6.8(1), for any $i \gg 0$, either Y_i is smooth at y_i, or there exist an I_i-equivariant closed embedding of $(Y_i \ni y_i)$ into a smooth fourfold germ $(W_i \ni y_i)$ with a faithful action of I_i and a regular I_i-semi-invariant function f_i on W_i such that $Y_i = (f_i = 0)$ and $(Y_i \ni y_i)$ is an isolated cDV singularity. Passing to a subsequence, we may assume that either Y_i is smooth at y_i for any i, or Y_i is singular at y_i for any i.

Case 1. Y_i is smooth at y_i for any i. Then $(Y_i \ni y_i)$ is a smooth threefold germ. As shown above, the cyclic, hence abelian, group I_i acts faithfully on Y_i and fixes y_i, and C_i is given by (20.1). By Theorem 14.2(2), applied with $\Gamma := \Lambda$, the set $\{a_i \mid i \geq 1\} \subset [1, +\infty)$ satisfies the ACC. This contradicts the strict increase of a_i.

Case 2. Y_i is singular at y_i for any i. By Corollary 6.8(1), $(Y_i \ni y_i)$ is an isolated cDV singularity, and $Y_i = (f_i = 0)$ in the smooth fourfold germ W_i, where f_i is a regular I_i-semi-invariant function. Since K_{Y_i} is Cartier and Y_i is terminal, $(Y_i \ni y_i)$ is an isolated Gorenstein terminal cDV threefold singularity. Since Y_i is singular at y_i, we have $f_i \in \mathfrak{m}^2_{W_i, y_i}$: otherwise f_i is part of a regular system of parameters of $\mathcal{O}_{W_i, y_i}$, and $\mathcal{O}_{Y_i, y_i} = \mathcal{O}_{W_i, y_i}/(f_i)$ is a regular local ring. As shown above, the cyclic, hence abelian, group I_i acts faithfully on Y_i and fixes y_i, and the klt pair (Y_i, C_i) has C_i given by (20.1). By Proposition 15.1, applied to $(Y_i \ni y_i, C_i)$, the group I_i, the closed embedding $Y_i \subset W_i$, and the function f_i, the set $\{a_i \mid i \geq 1\} \subset [1, +\infty)$ satisfies the ACC. This contradicts the strict increase of a_i. □

Proof of Theorem 1.2. Let $\Lambda := (\Gamma \cap [0, 1)) \cup \{\frac{1}{2}\}$. Then Λ satisfies the DCC: a strictly decreasing sequence in Λ has at most one term equal to $\frac{1}{2}$, and the other terms form a strictly decreasing sequence in Γ.

Suppose that the theorem does not hold. Then there exists a strictly increasing sequence $a_i \in \mathrm{MLD}(3, \Gamma) \cap [1, +\infty)$. Since $a_i > 0$, by Lemma 6.1(2), we may write $a_i = b_i - d_i$ with $b_i \in \mathrm{MLD}_{\mathrm{Qf}}(3, \Lambda)$ and $d_i \in \{0, 1\}$. Passing to a subsequence, we may assume that $d_i = d$ for any i. Then

$$b_i = a_i + d \geq 1$$

is strictly increasing, so $\mathrm{MLD}_{\mathrm{Qf}}(3, \Lambda) \cap [1, +\infty)$ does not satisfy the ACC. This contradicts Proposition 20.1. □

Proof of Theorem 1.3. Since $\frac{1}{2} = 1 - \frac{1}{2} \in \Gamma_{\mathrm{std}}$, we have

$$\Lambda := (\Gamma_{\mathrm{std}} \cap [0, 1)) \cup \{\tfrac{1}{2}\} = \Gamma_{\mathrm{std}} \cap [0, 1).$$

The set Λ satisfies the DCC: a strictly decreasing sequence $1 - \frac{1}{m_k}$ in Λ would yield a strictly decreasing sequence of positive integers m_k. Let $S := \mathrm{MLD}_{\mathrm{Qf}}(3, \Lambda)$. We first show that S satisfies the ACC.

Let $b = \mathrm{mld}(X \ni x, B) \in S$, where (X, B) is a $\mathbb{Q}$-factorial klt threefold pair, $x \in X$ is a closed point, and $B \in \Lambda$. Since (X, B) is klt, $b > 0$. Since the elements of Λ are rational, B is a $\mathbb{Q}$-divisor. Since X is $\mathbb{Q}$-factorial, $K_X + B$ is $\mathbb{Q}$-Cartier.

Suppose that $b < 1$. By Construction-Theorem 5.3, applied to $(X \ni x, B)$ and the trivial group G, there exist a positive integer N and a finite morphism $p : X' \to X$, Galois with group μ_N, such that $p^{-1}(x)$ is a single point x', X' is klt, and $K_{X'}$ is Cartier. By Construction-Theorem 5.3(2), applied with the trivial group G, there exists a finite abelian group $\hat{G}$ of order $N|G| = N$ acting faithfully on X' and fixing x', with an exact sequence $1 \to \mu_N \to \hat{G} \to G \to 1$, such that $(X, B) = (X, B)/G = (X', 0)/\hat{G}$. Thus $\hat{G} = \mu_N$, $(X, B) = (X', 0)/\mu_N$, and

$$b = \operatorname{mld}((X' \ni x', 0)/\mu_N) \in (0, 1).$$

Since μ_N is a finite abelian group acting faithfully on X' and fixing x', Theorem 13.1 implies that the set of such b satisfies the ACC.

Suppose that $b \geq 1$. Since $\Lambda \subset [0, 1]$ satisfies the DCC, Proposition 20.1 implies that $S \cap [1, +\infty)$ satisfies the ACC.

By Lemma 2.10(4), $S = (S \cap (0, 1)) \cup (S \cap [1, +\infty))$ satisfies the ACC.

Now suppose that the theorem does not hold. Then there exists a strictly increasing sequence $a_i \in \operatorname{MLD}(3, \Gamma_{\mathrm{std}})$. Since a_1 is the minimal log discrepancy of an lc pair, $a_1 \geq 0$, and since a_i is strictly increasing, $a_i > 0$ for any $i \geq 2$. By Lemma 6.1(2), applied with $\Gamma := \Gamma_{\mathrm{std}}$, we may write $a_i = b_i - d_i$ for any $i \geq 2$, with $b_i \in S$ and $d_i \in \{0, 1\}$. Passing to a subsequence, we may assume that $d_i = d$ for any i. Then $b_i = a_i + d$ is a strictly increasing sequence in S, which contradicts the ACC for S. □

Proof of Corollary 1.4. Let X be a $\mathbb{Q}$-factorial klt threefold and $x \in X$ a closed point. Since X is $\mathbb{Q}$-factorial, K_X is $\mathbb{Q}$-Cartier, so $(X, 0)$ is an lc pair with $0 \in \Gamma_{\mathrm{std}}$ and $\operatorname{mld}(X \ni x) \in \operatorname{MLD}(3, \Gamma_{\mathrm{std}})$. Thus Corollary 1.4 follows from Theorem 1.3. □

Proof of Theorem 1.5. We first show that the last statement follows from the first one. Let $(X \ni x)$ and G be as in the theorem, and let $B \geq 0$ be a G-invariant $\mathbb{R}$-divisor with $B \in \Gamma$ such that (X, B) is klt near x. Since G fixes x, the set of components of B not passing through x is G-invariant. Possibly replacing X with the complement of the union of the components of B not passing through x, we may assume that any component of B passes through x. Since B is G-invariant, we may write $B = \sum_{k=1}^{l} c_k C_k$, where $c_k \in \Gamma$ and C_k is the reduced union of a G-orbit of components of B. Fix k. Since $\mathcal{O}_{X,x}$ is a regular local ring, it is a unique factorization domain by the theorem of Auslander and Buchsbaum, and any height-one prime ideal of $\mathcal{O}_{X,x}$ is principal [Mat89, Theorems 20.1 and 20.3]. Thus the ideal $J_k \subset \mathcal{O}_{X,x}$ of C_k is generated by one element f_k (in the UFD $\mathcal{O}_{X,x}$, J_k is generated by the product of generators of the height-one primes of the components of C_k), and $\dim J_k/\mathfrak{m}_{X,x}J_k = 1$. Since G preserves J_k and G is abelian, there exists a character χ_k of G such that $\sigma^* f_k - \chi_k(\sigma) f_k \in \mathfrak{m}_{X,x} J_k$ for any $\sigma \in G$. Let

$$h_k := \frac{1}{|G|} \sum_{\sigma \in G} \chi_k(\sigma)^{-1} \sigma^* f_k.$$

A direct calculation shows that $\tau^* h_k = \chi_k(\tau) h_k$ for any $\tau \in G$, since G is abelian. Moreover, $h_k - f_k \in \mathfrak{m}_{X,x} J_k = f_k \mathfrak{m}_{X,x}$, so h_k/f_k is a unit of $\mathcal{O}_{X,x}$ and $\operatorname{div}(h_k) = C_k$ near x. Hence $B = \sum_k c_k \operatorname{div}(h_k)$ near x, where $c_k \in \Gamma$ and h_k is a non-zero regular G-semi-invariant function near x. Thus the last statement follows from the first one.

Now we prove the first statement. Suppose that it does not hold. Then for any $i \geq 1$ there exist a smooth threefold germ $(X_i \ni x_i)$, a finite abelian group G_i acting faithfully on X_i and fixing x_i, and $B_i = \sum_j c_{ij} \operatorname{div}(h_{ij})$ with $c_{ij} \in \Gamma$ and non-zero regular G_i-semi-invariant functions h_{ij} near x_i, such that (X_i, B_i) is klt near x_i and $a_i := \operatorname{mld}((X_i \ni x_i, B_i)/G_i)$ is positive and strictly increasing. Removing the terms with $c_{ij} = 0$, we may assume that $c_{ij} > 0$ for any i and j.

Since $(X_i \ni x_i)$ is smooth, it is terminal and K_{X_i} is Cartier. By Lemma 5.4, there exist G_i-semi-invariant regular parameters x_{i1}, x_{i2}, x_{i3} of X_i at x_i. By Construction-Theorem 7.2, applied to the germ $(X_i \ni x_i)$ with $n = 3$ (so that the ambient germ is X_i itself), the group G_i, the parameters x_{i1}, x_{i2}, x_{i3}, the functions h_{ij} with coefficients c_{ij}, $r = 0$, and $l = 0$, there exist a

faithful diagonal action of G_i on $\mathbb{A}^3$ and non-zero G_i-semi-invariant polynomials $h_{ij,0}$ such that, for $B_{i,0} := \sum_j c_{ij} \operatorname{div}(h_{ij,0})$, the pair $(\mathbb{A}^3, B_{i,0})$ is klt near 0 and

$$\operatorname{mld}((\mathbb{A}^3 \ni 0, B_{i,0})/G_i) = \operatorname{mld}((X_i \ni x_i, B_i)/G_i) = a_i$$

(take $H' = G_i$ in Construction-Theorem 7.2(1)). Possibly replacing $(X_i \ni x_i, B_i)$ with $(\mathbb{A}^3 \ni 0, B_{i,0})$, we may assume that $X_i = \mathbb{A}^3$, $x_i = 0$, G_i is a finite diagonal group acting faithfully on $\mathbb{A}^3$, and the h_{ij} are non-zero G_i-semi-invariant polynomials.

By Theorem 16.1, after passing to a subsequence and permuting the coordinates, there exist coprime positive integers p, q and a positive integer d such that

$$[G_i : G_i \cap \mathbf{T}_{(p,-q,0)}] = d$$

for any i, where $\mathbf{T}_{(p,-q,0)} = \{(s^p, s^{-q}, 1) \mid s \in \mathbb{C}^*\}$ (Definition 2.15). By Theorem 14.3, applied with p, q, $e_0 := d$, and Γ, and with the coordinates of $\mathbb{A}^3$ in the order for which $[G_i : G_i \cap \mathbf{T}_{(p,-q,0)}] = d$, the set $\{a_i \mid i \geq 1\}$ satisfies the ACC. This contradicts the strict increase of a_i. □

References

[Abh56] S. S. Abhyankar, *On the valuations centered in a local domain*, Amer. J. Math. **78** (1956), 321–348.

[Abh67] S. S. Abhyankar, *Inversion and invariance of characteristic pairs*, Amer. J. Math. **89** (1967), no. 2, 363–372.

[Ale93] V. Alexeev, *Two two-dimensional terminations*, Duke Math. J. **69** (1993), no. 3, 527–545.

[AF93] P. Aluffi and C. Faber, *Linear orbits of smooth plane curves*, J. Algebraic Geom. **2** (1993), no. 1, 155–184.

[Amb99] F. Ambro, *On minimal log discrepancies*, Math. Res. Lett. **6** (1999), no. 5, 573–580.

[Amb03] F. Ambro, *Inversion of adjunction for non-degenerate hypersurfaces*, Manuscripta Math. **111** (2003), no. 1, 43–49.

[Amb06] F. Ambro, *The set of toric minimal log discrepancies*, Cent. Eur. J. Math. **4** (2006), no. 3, 358–370.

[AGV85] V. I. Arnold, S. M. Gusein-Zade, and A. N. Varchenko, *Singularities of differentiable maps. Vol. I. The classification of critical points, caustics and wave fronts*, Monographs in Mathematics **82**, Birkhäuser Boston, Boston, MA (1985).

[Art68] M. Artin, *On the solutions of analytic equations*, Invent. Math. **5** (1968), no. 4, 277–291.

[AM69] M. F. Atiyah and I. G. Macdonald, *Introduction to commutative algebra*, Addison-Wesley, Reading, MA (1969).

[Bar03] V. Baranovsky, *Orbifold cohomology as periodic cyclic homology*, Internat. J. Math. **14** (2003), no. 8, 791–812.

[BCHM10] C. Birkar, P. Cascini, C. D. Hacon, and J. McKernan, *Existence of minimal models for varieties of log general type*, J. Amer. Math. Soc. **23** (2010), no. 2, 405–468.

[Blu18] H. Blum, *Existence of valuations with smallest normalized volume*, Compos. Math. **154** (2018), no. 4, 820–849.

[Bou72] N. Bourbaki, *Commutative algebra*, Elements of Mathematics, Hermann, Paris; Addison-Wesley, Reading, MA (1972).

[Bou98] N. Bourbaki, *General topology. Chapters 5–10*, Elements of Mathematics, Springer-Verlag, Berlin (1998).

[BH98] W. Bruns and J. Herzog, *Cohen–Macaulay rings*, revised edition, Cambridge Studies in Advanced Mathematics **39**, Cambridge University Press, Cambridge (1998).

[BBI01] D. Burago, Yu. Burago, and S. Ivanov, *A course in metric geometry*, Graduate Studies in Mathematics **33**, American Mathematical Society, Providence, RI (2001).

[Cas00] E. Casas-Alvero, *Singularities of plane curves*, London Mathematical Society Lecture Note Series **276**, Cambridge University Press, Cambridge (2000).

[CL26] W. Chen and K.-H. S. Lin, *On the ACC for minimal log discrepancies for bounded generalized sub-pairs*, arXiv:2609.20515.

[CLO15] D. Cox, J. Little, and D. O'Shea, *Ideals, varieties, and algorithms*, fourth ed., Undergraduate Texts in Mathematics, Springer, Cham (2015).

[CLS11] D. A. Cox, J. B. Little, and H. K. Schenck, *Toric varieties*, Graduate Studies in Mathematics **124**, American Mathematical Society, Providence, RI (2011).

[dFGPM23] A. B. de Felipe, P. D. González Pérez, and H. Mourtada, *Resolving singularities of curves with one toric morphism*, Math. Ann. **387** (2023), no. 3-4, 1853–1902.

[dFEM10] T. de Fernex, L. Ein, and M. Mustaţă, *Shokurov's ACC conjecture for log canonical thresholds on smooth varieties*, Duke Math. J. **152** (2010), no. 1, 93–114.

[dFEM11] T. de Fernex, L. Ein, and M. Mustaţă, *Log canonical thresholds on varieties with bounded singularities*, Classification of algebraic varieties, EMS Ser. Congr. Rep., Eur. Math. Soc., Zürich, 2011, pp. 221–257.

[dFM09] T. de Fernex and M. Mustaţă, *Limits of log canonical thresholds*, Ann. Sci. Éc. Norm. Supér. (4) **42** (2009), no. 3, 491–515.

[EMY03] L. Ein, M. Mustaţă, and T. Yasuda, *Jet schemes, log discrepancies and inversion of adjunction*, Invent. Math. **153** (2003), no. 3, 519–535.

[Eis95] D. Eisenbud, *Commutative algebra with a view toward algebraic geometry*, Graduate Texts in Mathematics **150**, Springer-Verlag, New York (1995).

[Fuj11] O. Fujino, *Fundamental theorems for the log minimal model program*, Publ. Res. Inst. Math. Sci. **47** (2011), no. 3, 727–789.

[FST11] O. Fujino, K. Schwede, and S. Takagi, *Supplements to non-lc ideal sheaves*, Higher dimensional algebraic geometry, RIMS Kôkyûroku Bessatsu **B24**, Res. Inst. Math. Sci. (RIMS), Kyoto, 2011, pp. 1–46.

[Ful93] W. Fulton, *Introduction to toric varieties*, Annals of Mathematics Studies **131**, Princeton University Press, Princeton, NJ (1993).

[GLS07] G.-M. Greuel, C. Lossen, and E. Shustin, *Introduction to singularities and deformations*, Springer Monographs in Mathematics, Springer, Berlin (2007).

[GP19] G.-M. Greuel and T. H. Pham, *Finite determinacy of matrices and ideals*, J. Algebra **530** (2019), 195–214.

[Gro65] A. Grothendieck, *Éléments de géométrie algébrique. IV. Étude locale des schémas et des morphismes de schémas. II*, Inst. Hautes Études Sci. Publ. Math. **24** (1965), 5–231.

[Gro66] A. Grothendieck, *Éléments de géométrie algébrique. IV. Étude locale des schémas et des morphismes de schémas. III*, Inst. Hautes Études Sci. Publ. Math. **28** (1966), 5–255.

[Gro67] A. Grothendieck, *Éléments de géométrie algébrique. IV. Étude locale des schémas et des morphismes de schémas. IV*, Inst. Hautes Études Sci. Publ. Math. **32** (1967), 5–361.

[Gro71] A. Grothendieck, *Revêtements étales et groupe fondamental (SGA 1)*, Lecture Notes in Mathematics **224**, Springer-Verlag, Berlin–New York (1971).

[Han26] J. Han, *A sharp inequality between local volumes and minimal log discrepancies*, arXiv:2608.16726.

[HL25] J. Han and J. Liu, *On termination of flips and exceptionally noncanonical singularities*, Geom. Topol. **29** (2025), no. 1, 399–441.

[HLL25] J. Han, J. Liu, and Y. Luo, *ACC for minimal log discrepancies of terminal threefolds*, Adv. Math. **480** (2025), Paper No. 110457.

[HLQ24] J. Han, J. Liu, and L. Qi, *ACC for local volumes*, arXiv:2408.15090.

[HLS26] J. Han, J. Liu, and V. V. Shokurov, *ACC for minimal log discrepancies of exceptional singularities*, Peking Math. J. **9** (2026), no. 2, 225–257.

[HL23] J. Han and Y. Luo, *On boundedness of divisors computing minimal log discrepancies for surfaces*, J. Inst. Math. Jussieu **22** (2023), no. 6, 2907–2930.

[HL24] J. Han and Y. Luo, *A simple proof of ACC for minimal log discrepancies for surfaces*, Acta Math. Sin. (Engl. Ser.) **40** (2024), 425–434.

[Har77] R. Hartshorne, *Algebraic geometry*, Graduate Texts in Mathematics **52**, Springer-Verlag, New York–Heidelberg (1977).

[HR79] E. Hewitt and K. A. Ross, *Abstract harmonic analysis. Vol. I*, 2nd ed., Grundlehren der Mathematischen Wissenschaften **115**, Springer-Verlag, Berlin (1979).

[IP01] S. Ishii and Y. G. Prokhorov, *Hypersurface exceptional singularities*, Internat. J. Math. **12** (2001), no. 6, 661–687.

[Jac85] N. Jacobson, *Basic algebra. I*, 2nd ed., W. H. Freeman and Company, New York (1985).

[Jia21] C. Jiang, *A gap theorem for minimal log discrepancies of noncanonical singularities in dimension three*, J. Algebraic Geom. **30** (2021), no. 4, 759–800.

[JM12] M. Jonsson and M. Mustaţă, *Valuations and asymptotic invariants for sequences of ideals*, Ann. Inst. Fourier (Grenoble) **62** (2012), no. 6, 2145–2209.

[Kaw07] M. Kawakita, *Inversion of adjunction on log canonicity*, Invent. Math. **167** (2007), no. 1, 129–133.

[Kaw15a] M. Kawakita, *A connectedness theorem over the spectrum of a formal power series ring*, Internat. J. Math. **26** (2015), no. 11, 1550088, 27 pp.

[Kaw15b] M. Kawakita, *The index of a threefold canonical singularity*, Amer. J. Math. **137** (2015), no. 1, 271–280.

[Kaw17] M. Kawakita, *Divisors computing the minimal log discrepancy on a smooth surface*, Math. Proc. Cambridge Philos. Soc. **163** (2017), no. 1, 187–192.

[Kaw21] M. Kawakita, *On equivalent conjectures for minimal log discrepancies on smooth threefolds*, J. Algebraic Geom. **30** (2021), no. 1, 97–149.

[Kaw24] M. Kawakita, *Minimal log discrepancies on a fixed threefold*, arXiv:2412.03006.

[Kaw26] M. Kawakita, *Minimal log discrepancies on smooth threefolds*, J. Lond. Math. Soc. (2) **113** (2026), no. 3, Paper No. e70510.

[Kaw88] Y. Kawamata, *Crepant blowing-up of* 3*-dimensional canonical singularities and its application to degenerations of surfaces*, Ann. of Math. (2) **127** (1988), no. 1, 93–163.

[Kaw93] Y. Kawamata, *The minimal discrepancy of a* 3*-fold terminal singularity*, appendix to V. V. Shokurov, 3*-fold log flips*, Russian Acad. Sci. Izv. Math. **40** (1993), no. 1, 95–202.

[Kol05] J. Kollár, *Resolution of singularities – Seattle lecture*, arXiv:math/0508332.

[Kol13] J. Kollár, *Singularities of the minimal model program*, Cambridge Tracts in Mathematics **200**, Cambridge University Press, Cambridge (2013).

[KM98] J. Kollár and S. Mori, *Birational geometry of algebraic varieties*, Cambridge Tracts in Math. **134**, Cambridge Univ. Press, Cambridge (1998).

[Kol92] J. Kollár et al., *Flips and abundance for algebraic threefolds*, Astérisque **211** (1992).

[Lan02] S. Lang, *Algebra*, revised third edition, Graduate Texts in Mathematics **211**, Springer-Verlag, New York (2002).

[Laz04] R. Lazarsfeld, *Positivity in algebraic geometry. II. Positivity for vector bundles, and multiplier ideals*, Ergeb. Math. Grenzgeb. (3) **49**, Springer-Verlag, Berlin (2004).

[Liu18] J. Liu, *Toward the equivalence of the ACC for a-log canonical thresholds and the ACC for minimal log discrepancies*, arXiv:1809.04839.

[LL22] J. Liu and Y. Luo, *Second largest accumulation point of minimal log discrepancies of threefolds*, arXiv:2207.04610.

[LX21] J. Liu and L. Xiao, *An optimal gap of minimal log discrepancies of threefold non-canonical singularities*, J. Pure Appl. Algebra **225** (2021), no. 9, 106674, 23 pp.

[Mac01] D. Maclagan, *Antichains of monomial ideals are finite*, Proc. Amer. Math. Soc. **129** (2001), no. 6, 1609–1615.

[Mat89] H. Matsumura, *Commutative ring theory*, Cambridge Studies in Advanced Mathematics **8**, Cambridge University Press, Cambridge (1989).

[Mor23a] J. Moraga, *Minimal log discrepancies of regularity one*, Int. Math. Res. Not. IMRN (2023), 15976–16014.

[Mor23b] J. Moraga, *Small quotient minimal log discrepancies*, Michigan Math. J. **73** (2023), 593–619.

[Mum74] D. Mumford, *Abelian varieties*, 2nd ed., Tata Inst. Fund. Res. Stud. Math. **5**, Oxford University Press, London (1974).

[MN18] M. Mustaţă and Y. Nakamura, *A boundedness conjecture for minimal log discrepancies on a fixed germ*, in: *Local and global methods in algebraic geometry*, Contemp. Math. **712** (2018), 287–306.

[Nak16] Y. Nakamura, *On minimal log discrepancies on varieties with fixed Gorenstein index*, Michigan Math. J. **65** (2016), no. 1, 165–187.

[NS22a] Y. Nakamura and K. Shibata, *Inversion of adjunction for quotient singularities*, Algebr. Geom. **9** (2022), no. 2, 214–251.

[NS22b] Y. Nakamura and K. Shibata, *Shokurov's index conjecture for quotient singularities*, arXiv:2209.04845.

[NS24] Y. Nakamura and K. Shibata, *A counterexample to the PIA conjecture for minimal log discrepancies*, arXiv:2404.06164.

[NS25] Y. Nakamura and K. Shibata, *Inversion of adjunction for quotient singularities II: non-linear actions*, Algebr. Geom. **12** (2025), no. 4, 443–496.

[NS26a] Y. Nakamura and K. Shibata, *A non-klt counterexample to Shokurov's index conjecture*, arXiv:2608.28038.

[NS26b] Y. Nakamura and K. Shibata, *Inversion of adjunction for quotient singularities III: semi-invariant case*, J. Lond. Math. Soc. (2) **113** (2026), no. 5, Paper No. e70545.

[Ram30] F. P. Ramsey, *On a problem of formal logic*, Proc. London Math. Soc. (2) **30** (1930), no. 4, 264–286.

[Rei83] M. Reid, *Minimal models of canonical* 3*-folds*, Algebraic varieties and analytic varieties (Tokyo, 1981), Adv. Stud. Pure Math. **1**, North-Holland, Amsterdam, 1983, pp. 131–180.

[Rei87] M. Reid, *Young person's guide to canonical singularities*, Algebraic geometry, Bowdoin, 1985 (Brunswick, Maine, 1985), Proc. Sympos. Pure Math. **46**, Part 1, Amer. Math. Soc., Providence, RI, 1987, pp. 345–414.

[Ser77] J.-P. Serre, *Linear representations of finite groups*, Graduate Texts in Mathematics **42**, Springer-Verlag, New York–Heidelberg (1977).

[Ser79] J.-P. Serre, *Local Fields*, Graduate Texts in Mathematics **67**, Springer-Verlag, New York (1979).

[Shi21] K. Shibata, *Characterization of two-dimensional semi-log canonical hypersurfaces in arbitrary characteristic*, Eur. J. Math. **7** (2021), no. 3, 931–951.

[Sho88] V. V. Shokurov, *Problems about Fano varieties*, in: *Birational geometry of algebraic varieties: open problems*, The XXIIIrd International Symposium, Division of Mathematics, The Taniguchi Foundation, Katata (1988), pp. 30–32.

[Sho92] V. V. Shokurov, *Threefold log flips*, with an appendix in English by Y. Kawamata, Izv. Ross. Akad. Nauk Ser. Mat. **56** (1992), no. 1, 105–203.

[Sho94] V. V. Shokurov, *A.c.c. in codimension 2*, preprint (1994).

[Sho04] V. V. Shokurov, *Letters of a bi-rationalist. V. Minimal log discrepancies and termination of log flips*, Tr. Mat. Inst. Steklova **246** (2004), 328–351; translation in Proc. Steklov Inst. Math. **246** (2004), 315–336.

[Ste05] D. A. Stepanov, *Non-rational divisors over non-degenerate cDV-points*, Sb. Math. **196** (2005), no. 7, 1075–1088.

[XZ21] C. Xu and Z. Zhuang, *Uniqueness of the minimizer of the normalized volume function*, Camb. J. Math. **9** (2021), no. 1, 149–176.

[ZS60] O. Zariski and P. Samuel, *Commutative algebra. Vol. II*, The University Series in Higher Mathematics, D. Van Nostrand Co., Princeton, NJ (1960).

[Zhu24] Z. Zhuang, *On boundedness of singularities and minimal log discrepancies of Kollár components*, J. Algebraic Geom. **33** (2024), no. 3, 521–565.

School of Mathematical Sciences, Peking University, No. 5 Yiheyuan Road, Haidian District, Beijing 100871, China

Beijing International Center for Mathematical Research, Peking University, No. 5 Yiheyuan Road, Haidian District, Beijing 100871, China

Email address: liujihao@math.pku.edu.cn